Statistical Modelling

Class notes — Bachelor in Applied Mathematics, IE University

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Preface

These notes accompany the course **Statistical Modelling** in the Bachelor in Applied Mathematics at IE University. They follow the 32 sessions of the course, but they go further than the slides: every result used in class is derived here step by step, every method is run in R on real data, and every chapter ends with exercises whose solutions are explained in full.

How to use these notes

- **Before a session**, read the sections listed in the session guide (`bam-statmod-r-examples` repository) and skim the chapter introduction.
- **During a session**, the slides give the main line of argument. The notes fill in the steps the slides skip.
- **After a session**, work through the worked examples with R open, then attempt the exercises at the end of the chapter *before* reading the solutions.
- **Each block** ends with a problem set that combines theory, R and interpretation. Problem sets are distributed separately, with solutions released after the deadline.

What you need

Calculus, linear algebra and probability at the level of the first year of the degree, and basic R. Chapter 1 reviews the tools we use. Install the packages once:

```
install.packages(c("tidyverse", "MASS", "faraway", "car", "glmnet", "nlme",
                   "lme4", "mgcv", "gamm4", "gratia", "survival",
                   "pscl", "brglm2", "leaps", "lmerTest", "pbkrtest"))
```

Notation

Object	Notation
Sample	$y_1, \dots, y_n$, index i
Likelihood	$L(\theta)$, $\ell(\theta) = \log L(\theta)$, score $S(\theta) = \partial \ell / \partial \theta$
Information	observed $I(\theta) = -\partial^2 \ell / \partial \theta^2$; expected $\mathcal{I}(\theta) = \mathrm{E}[I(\theta)]$; $\mathcal{I}(\theta) = n\,\mathcal{I}_1(\theta)$ for iid data
Standard error	$\mathrm{se}(\hat\theta) = I(\hat\theta)^{-1/2}$
Linear model	$y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$, X is $n \times p$ including the intercept; $H = X(X^\top X)^{-1}X^\top$; $\hat\sigma^2 = \mathrm{RSS}/(n-p)$

Object	Notation
GLM	$f(y;\theta,\phi)=\exp\{[y\theta-b(\theta)]/a(\phi)+c(y,\phi)\}$, $\mu=b'(\theta)$, $V(\mu)=b''(\theta)$, $g(\mu)=\eta=X\beta$
Mixed model	$y=X\beta+Zb+\varepsilon$, $b\sim N(0,G)$, $V=ZGZ^\top+\sigma^2 I$; groups i, observations j; BLUP $\tilde{b}$
Smoothing	$\hat{f}=S_\lambda y$, edf $=\operatorname{tr}(S_\lambda)$, bandwidth h, knots κ_k, B-spline basis B, difference matrix D_d
Survival	T, C, δ_i; $S(t)$, $h(t)$, $H(t)$; Cox $h(t\mid x)=h_0(t)\exp(x^\top\beta)$

Appendix A collects the notation in more detail.

Acknowledgement

The examples use datasets distributed with R packages (`datasets`, `MASS`, `faraway`, `lme4`, `nlme`, `survival`); copyright in the data remains with their authors. Recommended textbooks: Pawitan (2001), Faraway (2014), Wood (2017), Eilers and Marx (2021) and Kutner et al. (2005).

Part I

Likelihood

Chapter 1

Mathematical and probabilistic toolkit

Every method in this course is built from a short list of mathematical tools. We use a probability distribution to describe how a response varies; we use expectations and covariances to describe how an estimator behaves; we use the multivariate normal distribution because it is the one distribution for which exact finite-sample theory is available; we use matrices and their calculus to write a model with p parameters as one equation; and we use Taylor expansions and numerical optimisation because almost every estimator we meet from Chapter 3 onwards is the solution of an optimisation problem that has no closed form.

This chapter collects these tools in one place, with proofs. It is a reference as much as a chapter: you are expected to have seen most of the statements before, but not necessarily to remember the derivations. Later chapters cite results of this chapter by number.

Learning objectives. After working through this chapter you should be able to:

1. state the mean and variance of the Bernoulli, binomial, Poisson, normal, exponential, gamma and negative binomial distributions, and derive them for the Poisson and gamma;
2. compute $\mathrm{E}(AY+c)$ and $\mathrm{Var}(AY+c) = A\Sigma A^\top$ for a random vector Y, and the expectation of a quadratic form;
3. define the multivariate normal distribution, and prove that linear maps, marginals and conditionals of a normal vector are normal;
4. prove that $X^\top X$ is invertible if and only if X has full column rank, and characterise positive definite matrices and projection matrices;
5. differentiate linear and quadratic forms with respect to a vector, and derive the normal equations of least squares;
6. write a second-order Taylor expansion, classify stationary points, apply the delta method, and implement Newton–Raphson.

1.1 A motivating data set: great discoveries per year

Throughout the chapter we return to `discoveries`, a built-in data set with the number of "great" scientific discoveries and inventions in each year from 1860 to 1959. It is a count series, and a count is not a normal variable: it is a non-negative integer, small and skewed. A first look already raises questions that the tools below will answer.

```
y <- as.numeric(discoveries)
tab <- table(factor(y, levels = 0:12))
dd <- data.frame(count = 0:12, observed = as.numeric(tab) / length(y),
                 poisson = dpois(0:12, mean(y)))
```

```
p1 <- ggplot(data.frame(year = 1860:1959, y), aes(year, y)) +
  geom_col(fill = ie_navy, width = 0.8) + labs(x = "Year", y = "Discoveries")
p2 <- ggplot(dd, aes(count)) + geom_col(aes(y = observed), fill = "#D2E6F4") +
  geom_line(aes(y = poisson), colour = ie_coral) + geom_point(aes(y =
  ↪  poisson), colour = ie_coral) +
  labs(x = "Count", y = "Relative frequency")
gridExtra::grid.arrange(p1, p2, ncol = 2)
```

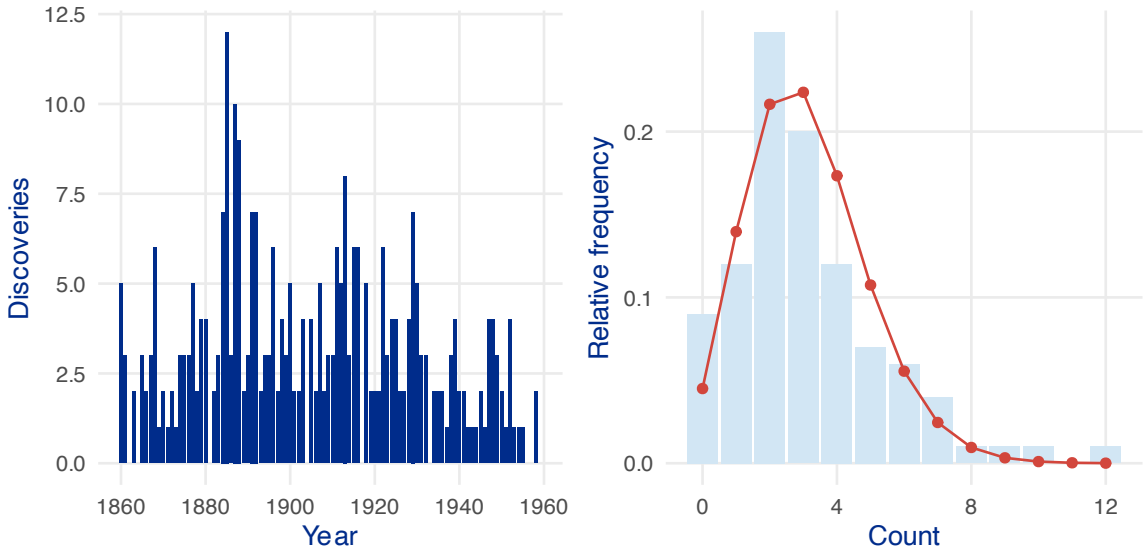


Figure 1.1: Number of great discoveries per year, 1860–1959 (left) and the empirical distribution of the counts with the Poisson probabilities that have the same mean (right).

```
c(n = length(y), mean = mean(y), variance = var(y), ratio = var(y) / mean(y))
```

```
#>         n      mean variance     ratio
#>   100.000     3.100    5.081     1.639
```

The mean is 3.1 but the variance is 5.08, about 1.64 times larger. For a Poisson distribution the mean and the variance are equal (Theorem 1.1). Is the discrepancy a fluke of a sample of 100 years, or evidence that the Poisson model is inadequate? To answer we need to know the distribution of the sample variance, which takes us through expectations of quadratic forms (Section 1.3) and, later, to the negative binomial distribution (Section 1.2.2) that allows the variance to exceed the mean.

1.2 Probability distributions

1.2.1 Notation and the distributions of the course

A random variable Y has a *distribution function* $F(y) = P(Y \leq y)$; it is discrete with probability mass function $f(y) = P(Y = y)$ or continuous with density $f(y) = F'(y)$. Its expectation and variance are

$$\mathrm{E}(Y) = \sum_y y f(y) \text{ or } \int y f(y)\, dy, \qquad \mathrm{Var}(Y) = \mathrm{E}\{(Y - \mathrm{E}Y)^2\} = \mathrm{E}(Y^2) - \{\mathrm{E}(Y)\}^2.$$

The distributions used in the course are in Table 1.1. The last column gives the **R parametrisation**, which is not always the one in textbooks: `dnorm` takes the standard deviation, while

`dexp` and `dgamma` take the *rate* (`dgamma` also accepts `scale = 1/rate`). Mixing up rate and scale is one of the most frequent sources of silent errors in R.

Table 1.1: Distributions used in the course, with R names. Rate β means density $\propto y^{\alpha-1}e^{-\beta y}$.

Distribution	Support	Mean	Variance	R
Bernoulli(π)	$\{0,1\}$	π	$\pi(1-\pi)$	‘dbinom(y,1,p)‘
Binomial(m,π)	$\{0,\dots,m\}$	$m\pi$	$m\pi(1-\pi)$	‘dbinom(y,m,p)‘
Poisson(λ)	$\{0,1,\dots\}$	λ	λ	‘dpois(y,lambda)‘
Neg. binomial(μ,k)	$\{0,1,\dots\}$	μ	$\mu+\mu^2/k$	‘dnbinom(y,mu=,size=k)‘
Normal(μ,σ^2)	$\mathbb{R}$	μ	σ^2	‘dnorm(y,mean,sd)‘
Exponential(λ)	$(0,\infty)$	$1/\lambda$	$1/\lambda^2$	‘dexp(y,rate)‘
Gamma(α,β)	$(0,\infty)$	α/β	α/β^2	‘dgamma(y,shape,rate)‘
Chi-squared(ν)	$(0,\infty)$	ν	2ν	‘dchisq(y,df)‘

The Poisson and the gamma deserve a proof, because their moments are obtained by two techniques that we reuse: *shifting the index of a sum* (the factorial moment) and *recognising a normalising constant* (the gamma integral).

Theorem 1.1 (Moments of the Poisson distribution)**.** *If $Y \sim$ Poisson(λ), i.e. $P(Y=y) = e^{-\lambda}\lambda^y/y!$ for $y = 0,1,2,\dots$, then $\mathrm{E}(Y) = \lambda$, $\mathrm{E}\{Y(Y-1)\} = \lambda^2$ and $\mathrm{Var}(Y) = \lambda$.*

Proof. The terms with $y = 0$ do not contribute to $\mathrm{E}(Y)$, and $y/y! = 1/(y-1)!$, so with the substitution $u = y-1$

$$\mathrm{E}(Y) = \sum_{y=1}^{\infty} y\,\frac{e^{-\lambda}\lambda^y}{y!} = \lambda\sum_{y=1}^{\infty}\frac{e^{-\lambda}\lambda^{y-1}}{(y-1)!} = \lambda\sum_{u=0}^{\infty}\frac{e^{-\lambda}\lambda^{u}}{u!} = \lambda,$$

because the last sum adds all the probabilities of a Poisson(λ) variable. For the factorial moment, $y(y-1)/y! = 1/(y-2)!$ for $y \geq 2$, so

$$\mathrm{E}\{Y(Y-1)\} = \lambda^2\sum_{y=2}^{\infty}\frac{e^{-\lambda}\lambda^{y-2}}{(y-2)!} = \lambda^2.$$

Then $\mathrm{E}(Y^2) = \mathrm{E}\{Y(Y-1)\} + \mathrm{E}(Y) = \lambda^2 + \lambda$ and $\mathrm{Var}(Y) = \lambda^2 + \lambda - \lambda^2 = \lambda$. □

Theorem 1.2 (Moments of the gamma distribution)**.** *Let $Y \sim$ Gamma(α,β) with density $f(y) = \beta^\alpha y^{\alpha-1}e^{-\beta y}/\Gamma(\alpha)$ for $y > 0$. Then for every $k > -\alpha$,*

$$\mathrm{E}(Y^k) = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)\,\beta^k},$$

so $\mathrm{E}(Y) = \alpha/\beta$, $\mathrm{E}(Y^2) = \alpha(\alpha+1)/\beta^2$ and $\mathrm{Var}(Y) = \alpha/\beta^2$.

Proof. Write

$$\mathrm{E}(Y^k) = \int_0^\infty y^k\,\frac{\beta^\alpha y^{\alpha-1}e^{-\beta y}}{\Gamma(\alpha)}dy = \frac{\beta^\alpha}{\Gamma(\alpha)}\int_0^\infty y^{\alpha+k-1}e^{-\beta y}dy.$$

The integrand is the kernel of a Gamma($\alpha+k,\beta$) density, whose integral is $\Gamma(\alpha+k)/\beta^{\alpha+k}$. Hence $\mathrm{E}(Y^k) = \beta^\alpha\Gamma(\alpha+k)/\{\Gamma(\alpha)\beta^{\alpha+k}\}$, which is the claim. Using $\Gamma(z+1) = z\Gamma(z)$ we get $\mathrm{E}(Y) = \alpha/\beta$ and $\mathrm{E}(Y^2) = \alpha(\alpha+1)/\beta^2$, so $\mathrm{Var}(Y) = \alpha(\alpha+1)/\beta^2 - \alpha^2/\beta^2 = \alpha/\beta^2$. □

Two special cases are used constantly: the exponential distribution is $\text{Gamma}(1, \lambda)$, and the chi-squared distribution with ν degrees of freedom is $\text{Gamma}(\nu/2, 1/2)$, so its mean is ν and its variance 2ν. Another useful property is the scaling rule: if $Y \sim \text{Gamma}(\alpha, \beta)$ and $c > 0$ then $cY \sim \text{Gamma}(\alpha, \beta/c)$. The *shape* α is unchanged and the coefficient of variation $\sqrt{\text{Var}(Y)}/\text{E}(Y) = 1/\sqrt{\alpha}$ depends on the shape only; this is why the gamma is the natural distribution for positive data whose standard deviation is proportional to the mean (Chapter 8).

1.2.2 Mixtures and the negative binomial

Two identities for conditioning are used throughout the course; the second is the key to variance decompositions in mixed models (Chapter 11).

Theorem 1.3 (Laws of total expectation and variance)**.** *For random variables Y and X with finite second moments,*

$$\text{E}(Y) = \text{E}\{\text{E}(Y \mid X)\}, \qquad \text{Var}(Y) = \text{E}\{\text{Var}(Y \mid X)\} + \text{Var}\{\text{E}(Y \mid X)\}.$$

Proof. The first identity is the tower property of conditional expectation. For the second, write $m(X) = \text{E}(Y \mid X)$ and expand $Y - \text{E}Y = (Y - m(X)) + (m(X) - \text{E}Y)$. Squaring and taking expectations, the cross term is $2\text{E}\{(Y - m(X))(m(X) - \text{E}Y)\} = 2\text{E}\big[(m(X) - \text{E}Y)\,\text{E}\{Y - m(X) \mid X\}\big] = 0$, because $\text{E}\{Y - m(X) \mid X\} = 0$. The first square has expectation $\text{E}\{\text{E}[(Y - m(X))^2 \mid X]\} = \text{E}\{\text{Var}(Y \mid X)\}$ and the second is $\text{Var}\{m(X)\}$. □

Example 1.1 (The negative binomial as a gamma mixture of Poissons)**.** Suppose that each year has its own rate Λ, drawn from a Gamma distribution with shape k and rate k/μ (so $\text{E}\Lambda = \mu$ and $\text{Var}\Lambda = \mu^2/k$), and that given Λ the count is $Y \mid \Lambda \sim \text{Poisson}(\Lambda)$. By Theorem 1.3,

$$\text{E}(Y) = \text{E}(\Lambda) = \mu, \qquad \text{Var}(Y) = \text{E}(\Lambda) + \text{Var}(\Lambda) = \mu + \frac{\mu^2}{k}.$$

The marginal distribution of Y is the **negative binomial** with mean μ and size k. The variance exceeds the mean by μ^2/k, so the data are *overdispersed* relative to the Poisson distribution, and the Poisson is recovered as $k \to \infty$. For `discoveries` the method of moments, which equates the sample mean and variance to μ and $\mu + \mu^2/k$, gives $\hat{k} = \bar{y}^2/(s^2 - \bar{y})$:

```
mu_hat <- mean(y); s2 <- var(y)
k_hat <- mu_hat^2 / (s2 - mu_hat)
c(mu = mu_hat, k = k_hat)
```

```
#>    mu     k
#> 3.100 4.852
```

```
# Expected relative frequencies under the two models
rbind(poisson = dpois(0:3, mu_hat), negbin = dnbinom(0:3, mu = mu_hat, size =
↪  k_hat),
      observed = as.numeric(table(factor(y, levels = 0:3))) / length(y))
```

```
#>             [,1]   [,2]   [,3]   [,4]
#> poisson  0.04505 0.1397 0.2165 0.2237
#> negbin   0.09099 0.1721 0.1963 0.1748
```

```
#> observed 0.09000 0.1200 0.2600 0.2000
```

The negative binomial reproduces the proportion of years with no discovery (0.09 observed, 0.091 fitted) far better than the Poisson (0.045). This informal comparison is made rigorous by likelihood methods in Chapters 3 and 10.

1.2.3 In R: d, p, q, r

Every distribution has four functions: `d` (density or mass), `p` (distribution function), `q` (quantile function, the inverse of `p`) and `r` (random generation).

```
dpois(3, lambda = 3.1)                          # P(Y = 3)
```

```
#> [1] 0.2237
```

```
ppois(3, lambda = 3.1)                          # P(Y <= 3)
```

```
#> [1] 0.6248
```

```
qpois(0.5, lambda = 3.1)                        # median
```

```
#> [1] 3
```

```
set.seed(1); rpois(5, 3.1)                      # five random counts
```

```
#> [1] 2 2 3 6 2
```

```
pnorm(1.96); qnorm(0.975)                       # normal: Phi and Phi^{-1}
```

```
#> [1] 0.975
```

```
#> [1] 1.96
```

```
pgamma(2, shape = 3, rate = 2)                  # rate parametrisation
```

```
#> [1] 0.7619
```

```
pgamma(2, shape = 3, scale = 1/2)               # same, with scale = 1/rate
```

```
#> [1] 0.7619
```

```
pexp(1, rate = 2) - pgamma(1, shape = 1, rate = 2)   # exponential is Gamma(1,
↪  rate): 0
```

```
#> [1] 0
```

```
integrate(function(t) t^2 * dgamma(t, 3, 2), 0, Inf)$value   # E(Y^2) =
↪  alpha(alpha+1)/beta^2 = 3
```

```
#> [1] 3
```

1.3 Expectations and variances of linear forms

1.3.1 Random vectors

Let $Y = (Y_1, \dots, Y_n)^\top$ be a random vector. Its **mean vector** is $\mu = \mathrm{E}(Y)$ with entries $\mu_i = \mathrm{E}(Y_i)$, and its **covariance matrix** is

$$\Sigma = \mathrm{Var}(Y) = \mathrm{E}\{(Y-\mu)(Y-\mu)^\top\}, \qquad \Sigma_{ij} = \mathrm{Cov}(Y_i, Y_j),\ \Sigma_{ii} = \mathrm{Var}(Y_i).$$

Σ is symmetric, and we will see in Proposition 1.1 that it is positive semidefinite. When the Y_i are uncorrelated with common variance σ^2, $\Sigma = \sigma^2 I_n$: this is the assumption on the errors of the linear model.

Theorem 1.4 (Moments of a linear transformation)**.** *Let Y be an n-vector with mean μ and covariance matrix Σ, let A be a fixed $k \times n$ matrix and c a fixed k-vector. Then*

$$\mathrm{E}(AY + c) = A\mu + c, \qquad \mathrm{Var}(AY + c) = A\Sigma A^\top.$$

More generally, for a fixed $l \times n$ matrix B, $\mathrm{Cov}(AY, BY) = A\Sigma B^\top$.

Proof. Expectation is linear, entry by entry: the ith entry of $AY + c$ is $\sum_j a_{ij} Y_j + c_i$ with expectation $\sum_j a_{ij}\mu_j + c_i$. For the covariance, the constant c cancels in $AY + c - \mathrm{E}(AY + c) = A(Y - \mu)$, so

$$\mathrm{Var}(AY + c) = \mathrm{E}\{A(Y-\mu)(Y-\mu)^\top A^\top\} = A\,\mathrm{E}\{(Y-\mu)(Y-\mu)^\top\}A^\top = A\Sigma A^\top,$$

since A is constant and can be taken out of the expectation. The same argument gives $\mathrm{Cov}(AY, BY) = \mathrm{E}\{A(Y-\mu)(Y-\mu)^\top B^\top\} = A\Sigma B^\top$. □

Setting $A = a^\top$ (a single row) gives the scalar formula that is used all the time:

$$\mathrm{Var}\Big(\sum_{i=1}^n a_i Y_i\Big) = a^\top \Sigma a = \sum_{i=1}^n a_i^2 \mathrm{Var}(Y_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(Y_i, Y_j). \tag{1.1}$$

Proposition 1.1 (Covariance matrices are positive semidefinite)**.** *For every $a \in \mathbb{R}^n$, $a^\top \Sigma a = \mathrm{Var}(a^\top Y) \geq 0$.*

Proof. Apply (1.1): $a^\top \Sigma a$ is the variance of the scalar $a^\top Y$, and variances are non-negative. □

Example 1.2 (The sample mean)**.** Let $Y_1, \dots, Y_n$ be uncorrelated with mean μ and variance σ^2. With $a = (1/n, \dots, 1/n)^\top$, $\mathrm{E}(\bar{Y}) = \mu$ and $\mathrm{Var}(\bar{Y}) = a^\top(\sigma^2 I)a = \sigma^2/n$. If instead neighbouring observations have correlation ρ (and all others none), formula (1.1) has $2(n-1)$ cross terms equal to $\rho\sigma^2/n^2$, and

$$\mathrm{Var}(\bar{Y}) = \frac{\sigma^2}{n}\Big\{1 + 2\rho\,\frac{n-1}{n}\Big\},$$

which is larger than σ^2/n when $\rho > 0$. Ignoring positive correlation therefore *understates* the uncertainty of a mean; this is why the independence assumption matters (Chapter 2).

1.3.2 A worked example with matrices

Let $Y = (Y_1, Y_2, Y_3)^\top$ have mean $\mu = (1, 2, 3)^\top$ and

$$\Sigma = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

Then $AY = (Y_1 - Y_2,\ Y_1 + Y_2 + Y_3)^\top$ has mean $A\mu = (-1, 6)^\top$ and covariance matrix $A\Sigma A^\top$. The dimension check is a good habit: A is 2×3, Σ is 3×3, so $A\Sigma A^\top$ is 2×2. Entry by entry, $\mathrm{Var}(Y_1 - Y_2) = 4 + 5 - 2 \cdot 2 = 5$, $\mathrm{Var}(Y_1 + Y_2 + Y_3) = 4 + 5 + 3 + 2(2 + 0 + 1) = 18$ and $\mathrm{Cov}(Y_1 - Y_2, Y_1 + Y_2 + Y_3) = (4 + 2 + 0) - (2 + 5 + 1) = -2$. We check this in R, first exactly and then by simulation from a normal distribution with these moments (Section 1.4).

```
mu <- c(1, 2, 3)
Sigma <- matrix(c(4, 2, 0,  2, 5, 1,  0, 1, 3), 3, byrow = TRUE)
A <- matrix(c(1, -1, 0,  1, 1, 1), 2, byrow = TRUE)
A %*% mu
```

```
#>      [,1]
#> [1,]   -1
#> [2,]    6
```

```
A %*% Sigma %*% t(A)
```

```
#>      [,1] [,2]
#> [1,]    5   -2
#> [2,]   -2   18
```

```
set.seed(2027)
Ysim <- MASS::mvrnorm(1e5, mu, Sigma)          # 100000 x 3 matrix, one draw
↪  per row
AY <- Ysim %*% t(A)                            # apply A to every draw
colMeans(AY); cov(AY)
```

```
#> [1] -0.9966  5.9920
```

```
#>         [,1]    [,2]
#> [1,]  5.020 -2.036
#> [2,] -2.036 18.090
```

The simulated mean and covariance agree with $A\mu$ and $A\Sigma A^\top$ to about two decimals. Common wrong answers are $A\Sigma$, $A^2\Sigma$ and $\Sigma A A^\top$; the dimension check rules them out.

1.3.3 Expectation of a quadratic form

Many estimators of variances are quadratic forms $Y^\top A Y$ with A symmetric.

Theorem 1.5 (Expectation of a quadratic form). *If* $\mathrm{E}(Y) = \mu$, $\mathrm{Var}(Y) = \Sigma$ *and* A *is a fixed symmetric* $n \times n$ *matrix, then*

$$\mathrm{E}(Y^\top A Y) = \mathrm{tr}(A\Sigma) + \mu^\top A \mu.$$

Proof. Because $Y^\top AY$ is a scalar it equals its trace, and $\operatorname{tr}(MN) = \operatorname{tr}(NM)$. Write $Y = \mu + (Y - \mu)$:

$$Y^\top AY = \mu^\top A\mu + 2\mu^\top A(Y-\mu) + (Y-\mu)^\top A(Y-\mu).$$

The middle term has expectation zero. For the last, $\mathrm{E}\{\operatorname{tr}((Y-\mu)^\top A(Y-\mu))\} = \mathrm{E}\{\operatorname{tr}(A(Y-\mu)(Y-\mu)^\top)\} = \operatorname{tr}(A\Sigma)$, since trace and expectation are both linear and commute. □

Example 1.3 (The sample variance is unbiased)**.** Let $Y_1, \dots, Y_n$ be uncorrelated with mean μ and variance σ^2, and write $\mathbf{1}$ for the vector of ones. The vector of deviations is $Y - \bar{Y}\mathbf{1} = CY$ with the *centring matrix* $C = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. This matrix is symmetric and idempotent ($C^2 = C$), so $\sum_i (Y_i - \bar{Y})^2 = Y^\top CY$. Since $C\mu\mathbf{1} = 0$, Theorem 1.5 gives $\mathrm{E}(Y^\top CY) = \sigma^2 \operatorname{tr}(C) + 0 = \sigma^2(n-1)$, because $\operatorname{tr}(C) = n - \frac{1}{n}\cdot n = n - 1$. Hence $s^2 = \sum(Y_i - \bar{Y})^2/(n-1)$ satisfies $\mathrm{E}(s^2) = \sigma^2$. The divisor $n-1$ is the trace of the matrix C: the *degrees of freedom*. In the linear model the same argument gives the divisor $n - p$ (Chapter 5).

1.4 The multivariate normal distribution

1.4.1 Definition

The multivariate normal distribution is the only one for which most finite-sample results of the linear model are exact. We define it through its moment generating function, which is valid even when Σ is singular.

Definition 1.1 (Multivariate normal)**.** A random vector $Y \in \mathbb{R}^n$ has the multivariate normal distribution $N_n(\mu, \Sigma)$, with $\mu \in \mathbb{R}^n$ and Σ symmetric positive semidefinite, if its moment generating function is

$$M_Y(t) = \mathrm{E}\, e^{t^\top Y} = \exp\left(t^\top \mu + \tfrac{1}{2} t^\top \Sigma t\right), \qquad t \in \mathbb{R}^n.$$

The distribution exists for every p.s.d. Σ: take a matrix L with $LL^\top = \Sigma$ (for instance the Cholesky factor or the symmetric square root), let $Z = (Z_1, \dots, Z_n)^\top$ with Z_i iid $N(0,1)$, and set $Y = \mu + LZ$. Then

$$\mathrm{E}e^{t^\top Y} = e^{t^\top \mu}\mathrm{E}e^{(L^\top t)^\top Z} = e^{t^\top\mu}\exp\left(\tfrac{1}{2}\|L^\top t\|^2\right) = \exp\left(t^\top\mu + \tfrac{1}{2}t^\top\Sigma t\right),$$

using the moment generating function $e^{s^2/2}$ of a standard normal variable for each coordinate. This is also how multivariate normal samples are generated in R. If Σ is positive definite, the density exists.

Theorem 1.6 (Density of the multivariate normal)**.** *If Σ is positive definite, $Y \sim N_n(\mu, \Sigma)$ has density*

$$f(y) = (2\pi)^{-n/2}\,|\Sigma|^{-1/2}\exp\Big\{-\tfrac{1}{2}(y-\mu)^\top\Sigma^{-1}(y-\mu)\Big\}.$$

Proof. Take L invertible with $LL^\top = \Sigma$ and $Y = \mu + LZ$, so $Z = L^{-1}(Y-\mu)$. The density of Z is $(2\pi)^{-n/2}\exp(-\frac{1}{2}z^\top z)$ and the Jacobian of the map $y \mapsto z$ is $|\det L^{-1}| = |\det L|^{-1} = |\Sigma|^{-1/2}$, because $|\Sigma| = |L|^2$. Finally $z^\top z = (y-\mu)^\top L^{-\top}L^{-1}(y-\mu) = (y-\mu)^\top\Sigma^{-1}(y-\mu)$. □

The proof also shows that $(Y-\mu)^\top\Sigma^{-1}(Y-\mu) = Z^\top Z = \sum Z_i^2 \sim \chi^2_n$. This is the multivariate generalisation of "the square of a standardised normal variable is χ^2_1" and is the origin of the Wald statistics of Chapter 4.

1.4.2 Properties

Theorem 1.7 (Linear maps, marginals, independence and conditioning)**.** *Let $Y \sim N_n(\mu, \Sigma)$.*

1. *For a fixed $k \times n$ matrix A and k-vector c, $AY + c \sim N_k(A\mu + c,\ A\Sigma A^\top)$.*
2. *Partition $Y = (Y_1^\top, Y_2^\top)^\top$, $\mu = (\mu_1^\top, \mu_2^\top)^\top$ and $\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$. Then $Y_1 \sim N(\mu_1, \Sigma_{11})$, and Y_1 and Y_2 are independent if and only if $\Sigma_{12} = 0$.*
3. *If Σ_{22} is positive definite, the conditional distribution of Y_1 given $Y_2 = y_2$ is normal with*

$$\mathrm{E}(Y_1 \mid Y_2 = y_2) = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \mu_2), \qquad \mathrm{Var}(Y_1 \mid Y_2 = y_2) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

Proof.

(1) For $t \in \mathbb{R}^k$, $\mathrm{E}e^{t^\top(AY+c)} = e^{t^\top c} M_Y(A^\top t) = \exp\{t^\top(A\mu + c) + \frac{1}{2} t^\top A\Sigma A^\top t\}$, the m.g.f. of $N_k(A\mu + c, A\Sigma A^\top)$; m.g.f.s determine the distribution.

(2) The marginal result is (1) with $A = (I \quad 0)$. If $\Sigma_{12} = 0$ the quadratic form splits, $t^\top \Sigma t = t_1^\top \Sigma_{11} t_1 + t_2^\top \Sigma_{22} t_2$, so $M_Y(t) = M_{Y_1}(t_1) M_{Y_2}(t_2)$ and the m.g.f. of the pair factorises, which is equivalent to independence. Independence always implies $\Sigma_{12} = 0$.

(3) Define $W = Y_1 - \Sigma_{12}\Sigma_{22}^{-1} Y_2$. By (1) (W, Y_2) is jointly normal, and $\mathrm{Cov}(W, Y_2) = \Sigma_{12} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22} = 0$, so by (2) W and Y_2 are independent. Then $Y_1 = W + \Sigma_{12}\Sigma_{22}^{-1} Y_2$, and given $Y_2 = y_2$ the second term is the constant $\Sigma_{12}\Sigma_{22}^{-1} y_2$ while W keeps its marginal distribution, which has mean $\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2$ and, by Theorem 1.4, variance $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

□

Warning. *Uncorrelated does not imply independent* unless the variables are **jointly** normal. If $Z \sim N(0,1)$, then Z and Z^2 are uncorrelated, because $\mathrm{Cov}(Z, Z^2) = \mathrm{E}Z^3 = 0$, yet Z^2 is a function of Z. Two normal marginals do not make a jointly normal vector either. A related consequence of Theorem 1.7: if Y_1, Y_2 are jointly normal, $Y_1 + Y_2$ and $Y_1 - Y_2$ are independent if and only if $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_2)$, since their covariance is $\mathrm{Var}(Y_1) - \mathrm{Var}(Y_2)$.

Example 1.4 (The bivariate normal and regression to the mean)**.** For (Y_1, Y_2) bivariate normal with means μ_1, μ_2, standard deviations σ_1, σ_2 and correlation ρ, Theorem 1.7(3) gives

$$\mathrm{E}(Y_1 \mid Y_2 = y_2) = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(y_2 - \mu_2), \qquad \mathrm{Var}(Y_1 \mid Y_2) = \sigma_1^2(1 - \rho^2).$$

The conditional mean is a *straight line* in y_2 with slope $\rho\sigma_1/\sigma_2$, which is the population version of the simple linear regression of Chapter 5. Because $|\rho| < 1$, if Y_2 is d standard deviations above its mean, the best prediction of Y_1 is only ρd standard deviations above its own: *regression to the mean*. The conditional variance does not depend on y_2 and is smaller than σ_1^2; knowing Y_2 explains the fraction ρ^2 of the variance of Y_1. We verify this on simulated data.

```
set.seed(2027)
rho <- 0.6; s1 <- 15; s2 <- 10; m1 <- 100; m2 <- 50
S <- matrix(c(s1^2, rho*s1*s2, rho*s1*s2, s2^2), 2)
xy <- as.data.frame(MASS::mvrnorm(2000, c(m1, m2), S)); names(xy) <- c("y1",
↪  "y2")
```

```
b <- rho * s1 / s2
ggplot(xy, aes(y2, y1)) + geom_point(alpha = 0.3, colour = ie_navy, size =
↪  0.8) +
  geom_abline(intercept = m1 - b * m2, slope = b, colour = ie_coral) +
  geom_abline(intercept = m1 - (s1/s2) * m2, slope = s1/s2, linetype = 2)
```

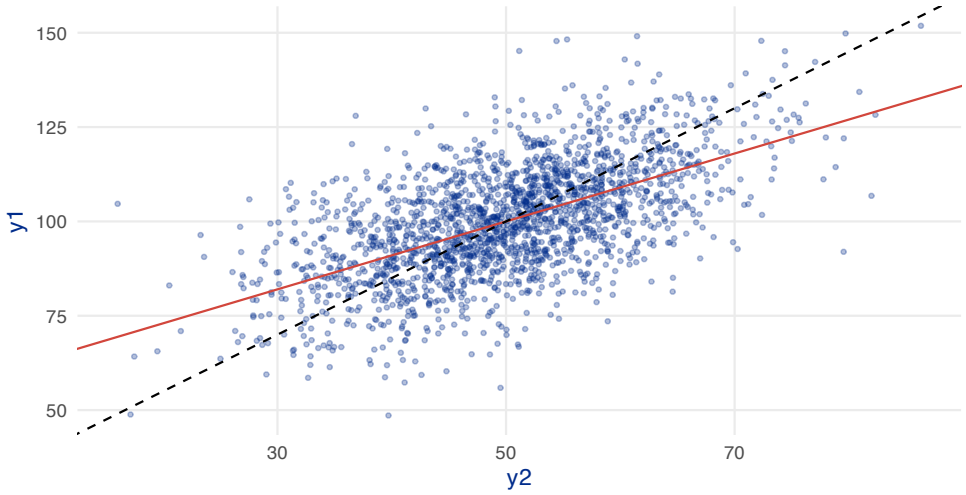


Figure 1.2: Simulated bivariate normal data with correlation 0.6. The solid line is the conditional mean of Y_1 given Y_2; the dashed line is the line with slope σ_1/σ_2.

```
# Condition on Y2 in a narrow window around 65
sub <- subset(xy, abs(y2 - 65) < 1)
c(n = nrow(sub), mean = mean(sub$y1), theory_mean = m1 + b * (65 - m2),
  sd = sd(sub$y1), theory_sd = s1 * sqrt(1 - rho^2))
```

```
#>           n        mean theory_mean          sd   theory_sd
#>       54.00      110.81      113.50       11.41       12.00
```

1.5 Linear algebra for models

A model with n observations and p parameters is written $y = X\beta + \varepsilon$, with X an $n \times p$ **design matrix that includes the intercept column**. Here p counts the columns of X, i.e. the parameters in β, not the number of predictors.

1.5.1 Rank and the invertibility of $X^\top X$

The *rank* of X is the maximum number of linearly independent columns. X has *full column rank* if $\operatorname{rank}(X) = p$, i.e. $Xv = 0$ implies $v = 0$.

Theorem 1.8 (Invertibility of $X^\top X$)**.** *For an $n \times p$ matrix X, $\operatorname{rank}(X^\top X) = \operatorname{rank}(X)$. In particular $X^\top X$ is invertible if and only if X has full column rank (which requires $n \geq p$).*

Proof. We show that $X^\top X$ and X have the same null space; the rank-nullity theorem then gives equal ranks since both have p columns. If $Xv = 0$ then $X^\top Xv = 0$. Conversely, if $X^\top Xv = 0$, then $0 = v^\top X^\top Xv = \|Xv\|^2$, so $Xv = 0$. □

The *dummy variable trap* is the most frequent way of losing full rank. With a factor of two levels and an intercept, the columns "intercept", "level 1" and "level 2" satisfy intercept = level 1 + level 2. In the `mtcars` data, with `am` the transmission indicator:

```
data(mtcars)
X_ok  <- model.matrix(~ wt + hp + am, data = mtcars)
mtcars$manual <- 1 - mtcars$am
X_bad <- cbind(X_ok, manual = mtcars$manual)
c(p_ok = ncol(X_ok), rank_ok = qr(X_ok)$rank, p_bad = ncol(X_bad), rank_bad =
↪  qr(X_bad)$rank)
```

```
#>     p_ok  rank_ok    p_bad rank_bad
#>        4        4        5        4
```

```
# the null vector v: intercept - am - manual = 0
range(X_bad %*% c(1, 0, 0, -1, -1))
```

```
#> [1] 0 0
```

```
coef(lm(mpg ~ wt + hp + am + manual, data = mtcars))   # R drops one column:
↪  NA
```

```
#> (Intercept)          wt          hp          am      manual
#>    34.00288    -2.87858    -0.03748     2.08371          NA
```

Adding any multiple of $v = (1, 0, 0, -1, -1)^\top$ to β leaves $X\beta$ unchanged, so the parameters are not identifiable; R reports `NA` for the aliased coefficient.

1.5.2 Quadratic forms and positive definiteness

Definition 1.2 (Definiteness)**.** A symmetric $p \times p$ matrix M is **positive semidefinite** (p.s.d.) if $v^\top M v \geq 0$ for all v, and **positive definite** (p.d.) if $v^\top M v > 0$ for all $v \neq 0$.

Theorem 1.9 (Spectral decomposition)**.** *A real symmetric M can be written $M = Q\Lambda Q^\top$ with Q orthogonal ($Q^\top Q = I$) whose columns $q_1, \dots, q_p$ are eigenvectors, and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_p)$ real. Moreover, M is p.s.d. (p.d.) if and only if all $\lambda_j \geq 0$ (> 0), and then $|M| = \prod_j \lambda_j$, $\operatorname{tr}(M) = \sum_j \lambda_j$.*

Proof. The existence of the decomposition is the spectral theorem of linear algebra, which we take as known. For the characterisation, $v^\top M v = \sum_j \lambda_j (q_j^\top v)^2$, because $v^\top Q \Lambda Q^\top v = \sum_j \lambda_j u_j^2$ with $u = Q^\top v$, and u ranges over all of $\mathbb{R}^p$ as v does. If all $\lambda_j > 0$, this is positive unless $u = 0$, i.e. $v = 0$. Conversely, taking $v = q_j$ gives $q_j^\top M q_j = \lambda_j$. The determinant and trace statements follow from $|Q\Lambda Q^\top| = |Q|^2 |\Lambda|$ and $\operatorname{tr}(Q\Lambda Q^\top) = \operatorname{tr}(\Lambda Q^\top Q)$. □

Two consequences are used repeatedly. First, $X^\top X$ is always p.s.d. ($v^\top X^\top X v = \|Xv\|^2 \geq 0$), and by Theorem 1.8 it is p.d. exactly when X has full column rank. Second, a p.d. matrix has a *Cholesky factor*: $M = LL^\top$ with L lower triangular, which is how multivariate normal variables are simulated and how linear systems with p.d. matrices are solved efficiently.

1.5.3 Projection matrices

Definition 1.3 (Projection matrix)**.** A matrix P is an (orthogonal) **projection matrix** if $P = P^\top$ and $P^2 = P$.

Theorem 1.10 (The hat matrix)**.** *Let X be $n \times p$ with full column rank and $H = X(X^\top X)^{-1}X^\top$. Then H and $I - H$ are projection matrices, $HX = X$, $\operatorname{rank}(H) = \operatorname{tr}(H) = p$ and $\operatorname{tr}(I - H) = n - p$. Every $y \in \mathbb{R}^n$ splits orthogonally as $y = Hy + (I - H)y$ with $Hy \in \operatorname{col}(X)$.*

Proof. $H^\top = X\{(X^\top X)^{-1}\}^\top X^\top = H$ as $X^\top X$ is symmetric, and $H^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = H$. Also $HX = X$ is immediate. For $I-H$: it is symmetric, and $(I-H)^2 = I-2H+H^2 = I-H$. The trace is $\operatorname{tr}(H) = \operatorname{tr}\{(X^\top X)^{-1}X^\top X\} = \operatorname{tr}(I_p) = p$, using $\operatorname{tr}(MN) = \operatorname{tr}(NM)$ and then $\operatorname{tr}(I - H) = n - p$. Finally $\langle Hy, (I - H)y\rangle = y^\top H(I - H)y = y^\top(H - H^2)y = 0$. □

Every eigenvalue λ of a projection matrix satisfies $\lambda^2 = \lambda$ (from $Pv = \lambda v$ and $P^2 = P$), so it is 0 or 1; the rank equals the trace, which counts the eigenvalues equal to one. In Chapter 5, $\hat{y} = Hy$ is the fitted vector, $e = (I - H)y$ the residual vector, and the identity $\operatorname{tr}(I - H) = n - p$ gives the degrees of freedom of the residual sum of squares, as in Example 1.3.

1.5.4 In R: matrices

```
X <- model.matrix(~ wt + hp, data = mtcars); ymp <- mtcars$mpg
XtX <- crossprod(X)                    # t(X) %*% X, faster and symmetric
Xty <- crossprod(X, ymp)                # t(X) %*% y
H <- X %*% solve(XtX) %*% t(X)         # hat matrix (inverting is fine for a
↪  demonstration)
c(trace = sum(diag(H)), idempotent = max(abs(H %*% H - H)))
```

```
#>      trace idempotent
#>  3.000e+00  1.721e-15
```

```
eigen(XtX)$values                      # all positive: X'X is positive
↪  definite
```

```
#> [1] 8.346e+05 3.884e+01 2.416e+00
```

```
chol(XtX)                              # upper triangular R with R'R = X'X
```

```
#>             (Intercept)     wt    hp
#> (Intercept)       5.657 18.200 829.8
#> wt                0.000  5.448 251.5
#> hp                0.000  0.000 287.2
```

```
qr(X)$rank                             # numerical rank
```

```
#> [1] 3
```

```
kappa(X, exact = TRUE)                 # condition number: large values signal
↪  near-collinearity
```

```
#> [1] 587.7
```

Remark. In practice, we never form $(X^\top X)^{-1}$ to solve a least-squares problem. `solve(XtX, Xty)` solves the linear system directly and `lm()` uses the QR decomposition of X, which avoids squaring the condition number. Inversion is acceptable for small demonstrations such as the hat matrix above.

1.6 Matrix calculus and least squares

1.6.1 Gradients and two rules

For a scalar function $f(\beta)$ of $\beta \in \mathbb{R}^p$, the **gradient** $\nabla f = \partial f/\partial \beta$ is the column vector with entries $\partial f/\partial \beta_j$, and the **Hessian** $\nabla^2 f$ is the symmetric matrix with entries $\partial^2 f/\partial\beta_j\partial\beta_k$.

Lemma 1.1 (Two rules)**.** *For a fixed vector a and a fixed symmetric matrix M,*

$$\frac{\partial}{\partial \beta}\, a^\top \beta = a, \qquad \frac{\partial}{\partial \beta}\, \beta^\top M \beta = 2M\beta, \qquad \frac{\partial^2}{\partial \beta\, \partial \beta^\top}\, \beta^\top M \beta = 2M.$$

Proof. For the first rule $a^\top\beta = \sum_j a_j\beta_j$ and $\partial/\partial\beta_k$ picks a_k. For the second, $\beta^\top M\beta = \sum_{j,l} M_{jl}\beta_j\beta_l$; differentiating with respect to β_k gives two contributions, from $j=k$ and from $l=k$:

$$\sum_{l} M_{kl}\beta_l + \sum_{j} M_{jk}\beta_j = 2\sum_{l} M_{kl}\beta_l = 2(M\beta)_k$$

by symmetry of M. Differentiating again with respect to β_l gives $2M_{kl}$. □

There is also a chain rule used in every generalised linear model: if $\eta = X\beta$ and f depends on β only through η, then $\partial f/\partial\beta = X^\top\, \partial f/\partial\eta$, because $\partial\eta_i/\partial\beta_j = x_{ij}$.

1.6.2 The normal equations

The least-squares estimator minimises the residual sum of squares

$$\mathrm{RSS}(\beta) = (y - X\beta)^\top(y - X\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta.$$

Theorem 1.11 (Least squares)**.** *Let X be $n \times p$. (1) Any minimiser $\hat\beta$ of $\mathrm{RSS}(\beta)$ satisfies the* ***normal equations***

$$X^\top X\hat\beta = X^\top y.$$

(2) If X has full column rank, the equations have the unique solution $\hat\beta = (X^\top X)^{-1}X^\top y$, which is the global minimiser, and for all β

$$\mathrm{RSS}(\beta) = \mathrm{RSS}(\hat\beta) + (\beta - \hat\beta)^\top X^\top X(\beta - \hat\beta).$$

Proof.

(1) By Lemma 1.1 the gradient is $\nabla\mathrm{RSS} = -2X^\top y + 2X^\top X\beta = -2X^\top(y - X\beta)$, and it vanishes at a minimum. (2) With full column rank, $X^\top X$ is invertible by Theorem 1.8 so the solution is unique. Let $\hat e = y - X\hat\beta$, so $X^\top\hat e = 0$ by the normal equations. For any β, $y - X\beta = \hat e + X(\hat\beta - \beta)$ and

$$\mathrm{RSS}(\beta) = \hat e^\top\hat e + 2(\hat\beta-\beta)^\top X^\top\hat e + (\hat\beta-\beta)^\top X^\top X(\hat\beta-\beta) = \mathrm{RSS}(\hat\beta) + (\beta-\hat\beta)^\top X^\top X(\beta-\hat\beta).$$

The last term is positive for $\beta \neq \hat\beta$ because $X^\top X$ is p.d., so $\hat\beta$ is the unique global minimiser. (The Hessian is $2X^\top X$, consistent with strict convexity.)

□

Warning. The *normal equations* are $X^\top X\hat\beta = X^\top y$; the expression $(X^\top X)^{-1}X^\top y$ is their *solution*. The word "normal" refers to orthogonality: the residuals are orthogonal (normal) to the columns of X, $X^\top \hat e = 0$. If X is rank deficient, the equations still have solutions, but infinitely many.

Example 1.5 (Fuel consumption of 32 cars)**.** We regress `mpg` on weight `wt` (1000 lb) and gross horsepower `hp` for the 32 cars in `mtcars`. Here $n = 32$ and, with the intercept column, $p = 3$. We solve the normal equations by hand in R and compare with `lm()`.

```
fit <- lm(mpg ~ wt + hp, data = mtcars)
beta_hat <- solve(XtX, Xty)                  # solves X'X b = X'y without
↪  inverting
cbind(by_hand = drop(beta_hat), lm = coef(fit))
```

```
#>              by_hand       lm
#> (Intercept) 37.22727 37.22727
#> wt          -3.87783 -3.87783
#> hp          -0.03177 -0.03177
```

```
e <- ymp - X %*% beta_hat
crossprod(X, e)                              # X'e = 0 (up to rounding):
↪  residuals orthogonal to columns
```

```
#>                    [,1]
#> (Intercept)  1.794e-13
#> wt          -7.452e-13
#> hp          -2.309e-12
```

```
rss <- sum(e^2); c(RSS = rss, sigma2_hat = rss / (nrow(X) - ncol(X)))
```

```
#>        RSS sigma2_hat
#>    195.048      6.726
```

Each extra 1000 lb is associated with a decrease of 3.88 miles per gallon at the same horsepower, and each extra horsepower with a decrease of 0.032. The estimate $\hat\sigma^2 = \mathrm{RSS}/(n-p) = \mathrm{RSS}/29$; the divisor is the residual degrees of freedom $n - p = \mathrm{tr}(I - H)$ of Theorem 1.10, not n.

Example 1.6 (Chain rule: gradient of a Poisson log-likelihood)**.** Let $y_i \sim \mathrm{Poisson}(\mu_i)$ independent with $\mu_i = e^{\eta_i}$ and $\eta = X\beta$. The log-likelihood is $\ell(\beta) = \sum_i (y_i\eta_i - e^{\eta_i}) + \text{const}$, so $\partial\ell/\partial\eta_i = y_i - \mu_i$. By the chain rule $\nabla\ell(\beta) = X^\top(y - \mu)$: the score vector has the same form as the normal equations, $X^\top(y - X\beta) = 0$, with the fitted mean μ in place of $X\beta$. This observation is developed in Chapter 9.

1.7 Taylor expansions

1.7.1 The expansion

Theorem 1.12 (Taylor's theorem)**.** *If $f : \mathbb{R} \to \mathbb{R}$ has $m+1$ continuous derivatives then, for x near a,*

$$f(x) = \sum_{j=0}^{m} \frac{f^{(j)}(a)}{j!}(x-a)^j + R_m, \qquad R_m = \frac{f^{(m+1)}(\xi)}{(m+1)!}(x-a)^{m+1}$$

for some ξ between a and x. For $f : \mathbb{R}^p \to \mathbb{R}$ twice continuously differentiable,

$$f(\theta) = f(a) + \nabla f(a)^\top(\theta - a) + \tfrac{1}{2}(\theta-a)^\top \nabla^2 f(a)(\theta - a) + o(\|\theta - a\|^2).$$

The remainder form (Lagrange) is a standard calculus result and is not re-derived here. The second-order expansion is the single most-used approximation of the course: *any smooth function looks like a parabola near its maximum.* The log-likelihood is a smooth function, which explains why the normal distribution appears in the large-sample theory of Chapter 4.

1.7.2 Stationary points

A point a is **stationary** if $\nabla f(a) = 0$. Applying the second-order expansion at a stationary point leaves $f(\theta) - f(a) \approx \frac{1}{2}(\theta-a)^\top \nabla^2 f(a)(\theta-a)$, hence the following criterion.

Theorem 1.13 (Second-order conditions)**.** *Let a be a stationary point of a twice continuously differentiable f. If $\nabla^2 f(a)$ is positive definite, a is a strict local minimum; if it is negative definite, a is a strict local maximum; if it has eigenvalues of both signs, a is a saddle point. If it is only semidefinite, the test is inconclusive.*

Proof. Let $\lambda_{\min} > 0$ be the smallest eigenvalue of $\nabla^2 f(a)$, positive definite. By Theorem 1.9, $(\theta-a)^\top \nabla^2 f(a)(\theta-a) \geq \lambda_{\min}\|\theta-a\|^2$, and the $o(\|\theta-a\|^2)$ term is smaller than $\frac{1}{4}\lambda_{\min}\|\theta-a\|^2$ in a small enough neighbourhood, so $f(\theta) - f(a) \geq \frac{1}{4}\lambda_{\min}\|\theta-a\|^2 > 0$. The negative definite case applies this to $-f$. For mixed signs, move along an eigenvector with positive eigenvalue (where f increases) and one with negative eigenvalue (where it decreases). □

Example 1.7 (A function of one variable)**.** Let $f(x) = x^2 e^{-x}$ for $x \in \mathbb{R}$. Then $f'(x) = (2x - x^2)e^{-x} = x(2-x)e^{-x}$ with stationary points $x = 0$ and $x = 2$, and $f''(x) = (2 - 4x + x^2)e^{-x}$. Since $f''(0) = 2 > 0$ the point 0 is a local minimum with $f(0) = 0$; since $f''(2) = -2e^{-2} < 0$ the point 2 is a local maximum with $f(2) = 4e^{-2} \approx 0.541$. The second-order Taylor expansion at the maximum is

$$f(x) \approx 4e^{-2} - e^{-2}(x-2)^2.$$

```
f <- function(x) x^2 * exp(-x)
taylor2 <- function(x) 4 * exp(-2) - exp(-2) * (x - 2)^2
xs <- seq(0, 5, length.out = 300)
ggplot(data.frame(xs), aes(xs)) + stat_function(fun = f, colour = ie_navy,
↪  linewidth = 1) +
  stat_function(fun = taylor2, colour = ie_coral, linetype = 2) +
  ↪  coord_cartesian(ylim = c(-0.2, 0.7)) +
  labs(x = "x", y = "f(x)")
```

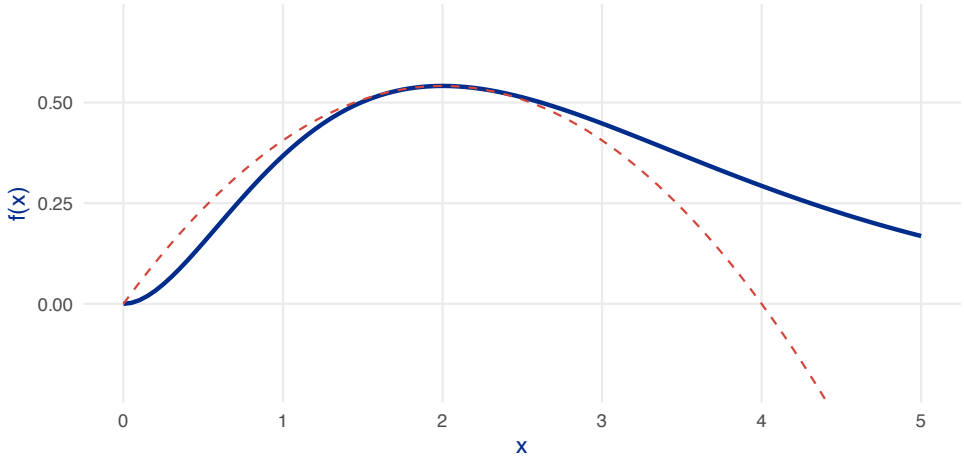


Figure 1.3: The function $x^2 e^{-x}$ (solid) and its second-order Taylor approximation at the maximum $x = 2$ (dashed).

```
c(max_value = f(2), four_exp_minus2 = 4 * exp(-2), f2_exact = f(2.3),
↪  f2_taylor = taylor2(2.3))
```

```
#>       max_value four_exp_minus2        f2_exact       f2_taylor
#>          0.5413          0.5413          0.5304          0.5292
```

Example 1.8 (A function of two variables)**.** Let $f(x, y) = x\, e^{-(x^2+y^2)}$. The gradient is $((1-2x^2)e^{-(x^2+y^2)},\ -2xy\, e^{-(x^2+y^2)})$. The second component is zero if $x = 0$ or $y = 0$; $x = 0$ makes the first component $e^{-y^2} \neq 0$, so $y = 0$ and $1 - 2x^2 = 0$: the stationary points are $(\pm 1/\sqrt{2}, 0)$. The Hessian entries are $f_{xx} = (4x^3 - 6x)e^{-(x^2+y^2)}$, $f_{xy} = -2y(1-2x^2)e^{-(x^2+y^2)}$ and $f_{yy} = (4xy^2 - 2x)e^{-(x^2+y^2)}$, so at $(1/\sqrt{2}, 0)$ the Hessian is diagonal with entries $-2\sqrt{2}e^{-1/2} < 0$ and $-\sqrt{2}e^{-1/2} < 0$: a local maximum with $f = e^{-1/2}/\sqrt{2} \approx 0.429$. By symmetry $f(-x, y) = -f(x, y)$, so $(-1/\sqrt{2}, 0)$ is a local minimum.

```
g <- function(p) p[1] * exp(-(p[1]^2 + p[2]^2))
opt <- optim(c(0.3, 0.3), g, method = "BFGS", control = list(fnscale = -1),
↪  hessian = TRUE)
rbind(numerical = opt$par, exact = c(1/sqrt(2), 0))
```

```
#>                [,1]      [,2]
#> numerical 0.7071 2.981e-06
#> exact     0.7071 0.000e+00
```

```
opt$hessian                                    # negative definite
```

```
#>              [,1]       [,2]
#> [1,] -1.716e+00 -1.943e-10
#> [2,] -1.943e-10 -8.578e-01
```

```
eigen(opt$hessian)$values
```

```
#> [1] -0.8578 -1.7156
```

1.7.3 The delta method

A first-order expansion tells us the variance of a *function* of an estimator.

Theorem 1.14 (Delta method)**.** *Let T have mean θ and a small variance τ^2, and let g be differentiable with $g'(\theta) \neq 0$. Then, to first order,*

$$\mathrm{E}\{g(T)\} \approx g(\theta), \qquad \mathrm{Var}\{g(T)\} \approx g'(\theta)^2 \tau^2.$$

If $\sqrt{n}(T_n - \theta) \to N(0, \tau^2)$ in distribution, then $\sqrt{n}\{g(T_n) - g(\theta)\} \to N(0, g'(\theta)^2\tau^2)$.

Proof. Expand $g(T) = g(\theta) + g'(\theta)(T - \theta) + R$. Taking expectation and variance of the linear part gives $\mathrm{E}g(T) \approx g(\theta)$ and $\mathrm{Var}g(T) \approx g'(\theta)^2 \mathrm{Var}(T)$. For the limit statement the remainder is of smaller order than $T_n - \theta = O_p(n^{-1/2})$ and the linear term $g'(\theta)\sqrt{n}(T_n - \theta)$ has the stated limit by Slutsky's theorem. □

For a Poisson sample of size n, $\bar{Y}$ has mean λ and variance λ/n. With $g(t) = \sqrt{t}$, $g'(\lambda) = 1/(2\sqrt{\lambda})$ and $\mathrm{Var}(\sqrt{\bar{Y}}) \approx \frac{1}{4} \cdot \frac{1}{n}$, which does not depend on λ: the square root *stabilises the variance*. With $g(t) = \log t$, $\mathrm{Var}(\log \bar{Y}) \approx 1/(n\lambda)$, so a standard error for $\log \hat{\lambda}$ is $1/\sqrt{n\hat{\lambda}}$ (a $1/\sqrt{\text{number of events}}$ rule that we use in survival analysis). We check the first claim by simulation.

```
set.seed(2027)
sim <- sapply(c(2, 5, 20, 80), function(lambda) {
  ybar <- replicate(5000, mean(rpois(30, lambda)))
  c(lambda = lambda, var_sqrt = var(sqrt(ybar)), delta = 1 / (4 * 30))
})
round(t(sim), 5)
```

```
#>      lambda var_sqrt   delta
#> [1,]      2  0.00834 0.00833
#> [2,]      5  0.00827 0.00833
#> [3,]     20  0.00808 0.00833
#> [4,]     80  0.00816 0.00833
```

All four simulated variances are within about 3% of $1/(4n) = 0.0083$ whatever λ is, as the variance-stabilising property predicts (the remaining differences are Monte Carlo error and higher-order terms, which matter more for small $n\lambda$).

1.8 Numerical optimisation

1.8.1 Newton–Raphson

Most estimators in this course (maximum likelihood, penalised likelihood) have no closed-form solution. To maximise a function f we look for a root of its gradient. Expanding the gradient at the current guess θ_t to first order, $\nabla f(\theta) \approx \nabla f(\theta_t) + \nabla^2 f(\theta_t)(\theta - \theta_t)$, and setting it equal to zero gives the **Newton–Raphson update**

$$\theta_{t+1} = \theta_t - \{\nabla^2 f(\theta_t)\}^{-1} \nabla f(\theta_t). \tag{1.2}$$

In one dimension, $x_{t+1} = x_t - f'(x_t)/f''(x_t)$. Equivalently, θ_{t+1} is the stationary point of the second-order Taylor approximation of f at θ_t: Newton's method fits a parabola and jumps to its vertex. Near a non-singular stationary point the convergence is *quadratic*: the number of correct digits roughly doubles with each step. For $f(\theta) = \ell(\theta)$, a log-likelihood, the Hessian is

minus the observed information, and (1.2) becomes $\theta_{t+1} = \theta_t + I(\theta_t)^{-1} S(\theta_t)$, studied in Chapter 3.

Warning. Newton's method finds a *stationary* point, not necessarily a maximum. Applied to $f(x) = x^2 e^{-x}$ from $x_0 = 3$, where $f'(3) = -3e^{-3}$ and $f''(3) = -e^{-3}$, the step is $3 - (-3)/(-1) = 0$: the method jumps to the *minimum*. A good starting value, step-halving and checking the sign of f'' are essential.

```
f1 <- function(x) (2 * x - x^2) * exp(-x)          # f'
f2 <- function(x) (2 - 4 * x + x^2) * exp(-x)      # f''
newton <- function(x0, iter = 6) {
  x <- numeric(iter + 1); x[1] <- x0
  for (t in seq_len(iter)) x[t + 1] <- x[t] - f1(x[t]) / f2(x[t])
  x
}
rbind(from_1.5 = newton(1.5), from_3 = newton(3))
```

```
#>          [,1]  [,2]  [,3] [,4] [,5] [,6] [,7]
#> from_1.5  1.5 1.929 1.998    2    2    2    2
#> from_3    3.0 0.000 0.000    0    0    0    0
```

Starting at 1.5 the iterates converge to the maximum at 2 within a few steps; from 3 the method lands exactly on the minimum at 0.

A statistical illustration. For the `discoveries` counts, the Poisson log-likelihood is $\ell(\lambda) = \sum y_i \log\lambda - n\lambda + \text{const}$ with $S(\lambda) = \sum y_i/\lambda - n$ and $I(\lambda) = \sum y_i/\lambda^2$. The maximiser is $\hat\lambda = \bar y$ and $\text{se}(\hat\lambda) = I(\hat\lambda)^{-1/2} = \sqrt{\bar y/n}$. Newton's method from $\lambda_0 = 1$ reaches it:

```
S <- function(l) sum(y) / l - length(y)
I <- function(l) sum(y) / l^2
l <- 1; path <- l
for (t in 1:8) { l <- l + S(l) / I(l); path <- c(path, l) }
round(path, 6)
```

```
#> [1] 1.000 1.677 2.447 2.963 3.094 3.100 3.100 3.100 3.100
```

```
c(lambda_hat = mean(y), se = sqrt(mean(y) / length(y)), se_from_I = 1 /
↪  sqrt(I(mean(y))))
```

```
#> lambda_hat         se  se_from_I
#>     3.1000     0.1761     0.1761
```

1.8.2 In R: optimisation

R has built-in tools for most needs:

- `optimize(f, interval, maximum = TRUE)` for a function of one variable;
- `uniroot(f, interval)` to find a root of a function;
- `optim(par, fn, method = "BFGS", hessian = TRUE)` for several variables; `optim` *minimises* by default, so use `control = list(fnscale = -1)` to maximise or pass the negative log-likelihood;
- `D()` and `deriv()` for symbolic derivatives, `integrate()` for one-dimensional integrals.

```
optimize(function(x) x^2 * exp(-x), c(0, 6), maximum = TRUE)          # 2 and
↪  4 exp(-2)
```

```
#> $maximum
#> [1] 2
#>
#> $objective
#> [1] 0.5413
```

```
uniroot(function(x) x^2 - 2, c(1, 2), tol = 1e-10)$root                #
↪  sqrt(2)
```

```
#> [1] 1.414
```

```
D(quote(x^2 * exp(-x)), "x")                                           #
↪  symbolic derivative
```

```
#> 2 * x * exp(-x) - x^2 * exp(-x)
```

```
# Rosenbrock's banana function: minimum at (1, 1)
rosen <- function(p) 100 * (p[2] - p[1]^2)^2 + (1 - p[1])^2
optim(c(-1.2, 1), rosen, method = "BFGS")[c("par", "value", "counts")]
```

```
#> $par
#> [1] 0.9998 0.9996
#>
#> $value
#> [1] 3.827e-08
#>
#> $counts
#> function gradient
#>      120       38
```

Remark. Optimise over an *unconstrained* parametrisation: for a variance use $\log \sigma$ rather than σ, for a probability use $\operatorname{logit} \pi$. Otherwise the optimiser may step outside the parameter space. The Hessian that `optim(..., hessian = TRUE)` returns is *numerical*; use it to check that the solution is a maximum (negative definite Hessian of the log-likelihood) and to obtain standard errors by inverting the information.

1.9 Common pitfalls

Warning. *Rate versus scale.* `dgamma(x, shape, rate)` and `dexp(x, rate)`; the mean of an exponential variable is 1/rate. `dnorm` takes the standard deviation, not the variance.

Warning. *Order of the matrices.* $\operatorname{Var}(AY) = A\Sigma A^\top$ and not $A\Sigma$, $A^2\Sigma$ or ΣA^2. Check dimensions: a $k \times n$ matrix A gives a $k \times k$ covariance.

Warning. *Uncorrelated versus independent.* They coincide only under joint normality.

Warning. *Normal equations versus their solution*, and *p versus the number of predictors*: p includes the intercept.

Warning. *Rank deficiency.* Redundant columns (a dummy for each level plus an intercept, a variable that is a sum of others) make $X^\top X$ singular. R reports `NA` rather than failing, so always inspect `coef()`.

Warning. *Stationary is not maximal.* Newton–Raphson, `uniroot` on a gradient, and the first-order conditions alone say nothing about a maximum: check the Hessian.

Warning. *Approximations have a range.* A Taylor expansion or the delta method is accurate near the expansion point; far from it (small counts, boundaries) check by simulation.

1.10 Chapter summary

- Poisson(λ): mean = variance = λ; Gamma(α, β) has mean α/β and variance α/β^2; a gamma mixture of Poissons is negative binomial with variance $\mu + \mu^2/k > \mu$ (overdispersion).
- $\mathrm{Var}(Y) = \mathrm{E}\{\mathrm{Var}(Y \mid X)\} + \mathrm{Var}\{\mathrm{E}(Y \mid X)\}$.
- $\mathrm{E}(AY + c) = A\mu + c$ and $\mathrm{Var}(AY + c) = A\Sigma A^\top$; Σ is p.s.d.; $\mathrm{E}(Y^\top AY) = \mathrm{tr}(A\Sigma) + \mu^\top A\mu$, which gives the divisor $n - 1$ of s^2.
- $N_n(\mu, \Sigma)$ is closed under linear maps, marginalisation and conditioning; jointly normal + uncorrelated $\Rightarrow$ independent; $(Y - \mu)^\top \Sigma^{-1}(Y - \mu) \sim \chi^2_n$.
- $X^\top X$ is invertible iff X has full column rank; p.s.d./p.d. are characterised by eigenvalues; $H = X(X^\top X)^{-1}X^\top$ is a projection with $\mathrm{tr}\, H = p$.
- $\partial(a^\top\beta)/\partial\beta = a$, $\partial(\beta^\top M\beta)/\partial\beta = 2M\beta$; minimising RSS gives the normal equations $X^\top X\beta = X^\top y$, with unique solution iff X has full column rank.
- Second-order Taylor expansion: $f(\theta) \approx f(a) + \nabla f(a)^\top(\theta - a) + \frac{1}{2}(\theta - a)^\top \nabla^2 f(a)(\theta - a)$; stationary points are classified by the Hessian; delta method $\mathrm{Var}\, g(T) \approx g'(\theta)^2 \mathrm{Var}(T)$.
- Newton–Raphson: $\theta_{t+1} = \theta_t - \{\nabla^2 f\}^{-1}\nabla f$, quadratic convergence but only to stationary points.

Next, Chapter 2 uses these tools to ask what a statistical model is, and Chapter 3 turns a model into a likelihood.

1.11 Exercises with explained solutions

Exercise 1.1. Let $Y_1, \ldots, Y_n$ be independent with $Y_i \sim \mathrm{Poisson}(\lambda_i)$.

(a) Show that the probability generating function of a Poisson(λ) variable is $\mathrm{E}(s^Y) = e^{\lambda(s-1)}$ and use it to prove that $\sum_i Y_i \sim \mathrm{Poisson}(\sum_i \lambda_i)$.

(b) Show that, given $\sum_i Y_i = m$, the vector $(Y_1, \ldots, Y_n)$ is multinomial with m trials and probabilities $\lambda_i / \sum_j \lambda_j$.

(c) Deduce that, if $Y_1, \ldots, Y_n$ are iid Poisson(λ), conditionally on $\sum Y_i = m$ each Y_i is Binomial($m, 1/n$).

Solution.

(a) $\mathrm{E}(s^Y) = \sum_y s^y e^{-\lambda}\lambda^y/y! = e^{-\lambda}\sum_y (\lambda s)^y/y! = e^{-\lambda}e^{\lambda s} = e^{\lambda(s-1)}$. For independent variables the p.g.f. of the sum is the product: $\prod_i e^{\lambda_i(s-1)} = e^{(\sum \lambda_i)(s-1)}$, the p.g.f. of Poisson($\sum_i \lambda_i$); p.g.f.s determine the distribution.

(b) Let $\Lambda = \sum_j \lambda_j$ and $\sum y_i = m$. Then

$$P(Y = y \mid \textstyle\sum Y_i = m) = \dfrac{\prod_i e^{-\lambda_i}\lambda_i^{y_i}/y_i!}{e^{-\Lambda}\Lambda^m/m!} = \dfrac{m!}{\prod_i y_i!}\prod_i \left(\dfrac{\lambda_i}{\Lambda}\right)^{y_i},$$

the multinomial mass function.

(c) With all $\lambda_i = \lambda$ the probabilities are $1/n$ and the marginal of a multinomial component is Binomial($m, 1/n$). This is why a Poisson sample, given its total, is a multinomial sample: a link between Poisson regression and multinomial models used in Chapter 10.

Exercise 1.2.
(a) Show that the m.g.f. of Gamma(α,β) is $(1-t/\beta)^{-\alpha}$ for $t<\beta$.
(b) Deduce that if $Y_1,\dots,Y_n$ are independent Gamma(α_i,β) then $\sum_i Y_i \sim$ Gamma$(\sum_i \alpha_i,\beta)$.
(c) Prove that a χ^2_1 variable (Z^2, $Z\sim N(0,1)$) is Gamma$(1/2,1/2)$, and conclude that $\chi^2_n=$ Gamma$(n/2,1/2)$, with mean n and variance $2n$.

Solution.

(a) $\mathrm{E}e^{tY}=\frac{\beta^\alpha}{\Gamma(\alpha)}\int_0^\infty y^{\alpha-1}e^{-(\beta-t)y}dy=\frac{\beta^\alpha}{(\beta-t)^\alpha}=(1-t/\beta)^{-\alpha}$ for $t<\beta$, recognising the Gamma$(\alpha,\beta-t)$ kernel as in the proof of Theorem 1.2.

(b) The m.g.f. of the sum is $\prod_i(1-t/\beta)^{-\alpha_i}=(1-t/\beta)^{-\sum\alpha_i}$, which is that of Gamma$(\sum\alpha_i,\beta)$.

(c) For $w>0$, $P(Z^2\le w)=P(-\sqrt{w}\le Z\le\sqrt{w})=2\Phi(\sqrt{w})-1$, and differentiating, $f(w)=2\phi(\sqrt{w})\frac{1}{2\sqrt{w}}=\frac{1}{\sqrt{2\pi}}w^{-1/2}e^{-w/2}$. This is $\beta^\alpha w^{\alpha-1}e^{-\beta w}/\Gamma(\alpha)$ with $\alpha=\beta=1/2$ because $\Gamma(1/2)=\sqrt{\pi}$ gives the constant $(1/2)^{1/2}/\sqrt{\pi}=1/\sqrt{2\pi}$. By (b) a sum of n independent χ^2_1 variables is Gamma$(n/2,1/2)$ with mean $\alpha/\beta=n$ and variance $\alpha/\beta^2=2n$.

Exercise 1.3. Let $Y\mid\Lambda\sim$ Poisson(Λ) with $\Lambda\sim$ Gamma$(k,k/\mu)$.
(a) Derive the marginal mass function $P(Y=y)$ and show it equals $\frac{\Gamma(y+k)}{\Gamma(k)\,y!}\big(\frac{k}{k+\mu}\big)^k\big(\frac{\mu}{k+\mu}\big)^y$.
(b) Verify $\mathrm{E}Y=\mu$ and $\mathrm{Var}Y=\mu+\mu^2/k$ with the laws of total expectation and variance, and compute the moment estimate of k for a sample with $\bar y=3.1$ and $s^2=5.08$.
(c) Using R, compare $P(Y=0)$ under the fitted Poisson and the fitted negative binomial for `discoveries` with the observed proportion of zeros.

Solution.

(a) $P(Y=y)=\int_0^\infty\frac{e^{-\lambda}\lambda^y}{y!}\cdot\frac{(k/\mu)^k\lambda^{k-1}e^{-k\lambda/\mu}}{\Gamma(k)}d\lambda=\frac{(k/\mu)^k}{y!\,\Gamma(k)}\int_0^\infty\lambda^{y+k-1}e^{-\lambda(1+k/\mu)}d\lambda$. The integral is $\Gamma(y+k)/(1+k/\mu)^{y+k}$, so

$$P(Y=y)=\frac{\Gamma(y+k)}{y!\,\Gamma(k)}\frac{(k/\mu)^k}{(1+k/\mu)^{y+k}}=\frac{\Gamma(y+k)}{y!\,\Gamma(k)}\Big(\frac{k}{k+\mu}\Big)^k\Big(\frac{\mu}{k+\mu}\Big)^y,$$

since $(k/\mu)/(1+k/\mu)=k/(k+\mu)$ and $1/(1+k/\mu)=\mu/(k+\mu)$.

(b) $\mathrm{E}Y=\mathrm{E}\{\mathrm{E}(Y\mid\Lambda)\}=\mathrm{E}\Lambda=\mu$, and $\mathrm{Var}Y=\mathrm{E}\Lambda+\mathrm{Var}\Lambda=\mu+\mu^2/k$ as in Example 1.1. From $s^2=\bar y+\bar y^2/k$, $\hat k=\bar y^2/(s^2-\bar y)=9.61/1.98=4.85$.

(c)

```
c(poisson = dpois(0, mean(y)), negbin = dnbinom(0, mu = mean(y), size =
↪  mean(y)^2 / (var(y) - mean(y))),
  observed = mean(y == 0))
```

```
#>  poisson   negbin observed
#>  0.04505  0.09099  0.09000
```

The observed proportion of zeros is 0.09; the Poisson model predicts 0.045 while the negative binomial gives 0.091, closer to the data.

Exercise 1.4. Let Y have mean $(2,0,1)^\top$ and covariance $\Sigma = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Define $U_1 = Y_1 + Y_2$, $U_2 = Y_1 - Y_2$, $U_3 = Y_2 + Y_3$.

(a) Write $U = AY$ and compute $\mathrm{E}U$ and $\mathrm{Var}U$ by hand.
(b) Are U_1 and U_2 uncorrelated? If Y is normal, are they independent?
(c) Verify with R.

Solution.

(a) $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, so $\mathrm{E}U = A\mu = (2,\,2,\,1)^\top$. The entries of $A\Sigma A^\top$ follow from (1.1): $\mathrm{Var}U_1 = 4+3+2(1) = 9$; $\mathrm{Var}U_2 = 4+3-2 = 5$; $\mathrm{Var}U_3 = 3+2+2(1) = 7$; $\mathrm{Cov}(U_1,U_2) = \mathrm{Var}Y_1 - \mathrm{Var}Y_2 = 1$; $\mathrm{Cov}(U_1,U_3) = \Sigma_{12}+\Sigma_{13}+\Sigma_{22}+\Sigma_{23} = 1+0+3+1 = 5$; $\mathrm{Cov}(U_2,U_3) = \Sigma_{12}+\Sigma_{13}-\Sigma_{22}-\Sigma_{23} = 1+0-3-1 = -3$. Therefore

$$\mathrm{Var}U = \begin{pmatrix} 9 & 1 & 5 \\ 1 & 5 & -3 \\ 5 & -3 & 7 \end{pmatrix}.$$

(b) $\mathrm{Cov}(U_1,U_2) = 1 \neq 0$, so they are correlated. For normal Y, correlated implies dependent. They would be independent iff $\mathrm{Var}Y_1 = \mathrm{Var}Y_2$.

(c)

```
Sg <- matrix(c(4,1,0, 1,3,1, 0,1,2), 3, byrow = TRUE)
Am <- rbind(c(1,1,0), c(1,-1,0), c(0,1,1))
Am %*% c(2, 0, 1); Am %*% Sg %*% t(Am)
```

```
#>      [,1]
#> [1,]    2
#> [2,]    2
#> [3,]    1
```

```
#>      [,1] [,2] [,3]
#> [1,]    9    1    5
#> [2,]    1    5   -3
#> [3,]    5   -3    7
```

Exercise 1.5.

(a) Let $Z \sim N(0,1)$. Show that Z and Z^2 are uncorrelated but not independent.
(b) Let Y_1, Y_2 be jointly normal with $\mathrm{Var}Y_1 = 4$, $\mathrm{Var}Y_2 = 3$. Show that $Y_1 + Y_2$ and $Y_1 - Y_2$ are dependent whatever $\mathrm{Cov}(Y_1,Y_2)$ is, and find a condition making them independent.
(c) Let $Z \sim N(0,1)$ and let S be ± 1 with probability $1/2$ each, independent of Z. Show that $SZ \sim N(0,1)$ but that (Z, SZ) is not bivariate normal.

Solution.

(a) $\mathrm{Cov}(Z,Z^2) = \mathrm{E}Z^3 - \mathrm{E}Z\,\mathrm{E}Z^2 = 0$ because odd moments of a symmetric distribution vanish. But $P(Z^2 \le 1,\ |Z| > 1) = 0 \neq P(Z^2 \le 1)P(|Z| > 1) > 0$.

(b) $\mathrm{Cov}(Y_1+Y_2, Y_1-Y_2) = \mathrm{Var}Y_1 - \mathrm{Var}Y_2 = 1$ regardless of $\mathrm{Cov}(Y_1, Y_2)$. The pair is jointly normal (a linear map of a normal vector, Theorem 1.7), so non-zero covariance means dependence. They are independent iff the two variances are equal.

(c) $P(SZ \le x) = \frac{1}{2}P(Z \le x) + \frac{1}{2}P(-Z \le x) = P(Z \le x)$ by symmetry, so $SZ \sim N(0,1)$. The two coordinates are uncorrelated ($\mathrm{E}(Z \cdot SZ) = \mathrm{E}S\,\mathrm{E}Z^2 = 0$) but $|SZ| = |Z|$, so they are dependent. If (Z, SZ) were bivariate normal, uncorrelated would imply independent, a contradiction. Normal marginals do not make a normal vector.

Exercise 1.6. Heights and weights of a population are modelled as bivariate normal: Y_1 (weight, kg) has mean 70 and sd 10; Y_2 (height, cm) has mean 170 and sd 8; the correlation is 0.6.

(a) Find the distribution of weight given height = 186 cm.
(b) Compute $P(Y_1 > 85 \mid Y_2 = 186)$ and compare with the unconditional $P(Y_1 > 85)$.
(c) Verify with a simulation from `MASS::mvrnorm` conditioning on $|Y_2 - 186| < 1$.

Solution.

(a) By Example 1.4 the conditional mean is $70 + 0.6 \cdot (10/8) \cdot (186 - 170) = 70 + 0.75 \cdot 16 = 82$ kg and the conditional variance is $10^2(1 - 0.36) = 64$ (sd 8 kg). So $Y_1 \mid Y_2 = 186 \sim N(82, 64)$. A height two standard deviations above the mean predicts a weight only $0.6 \times 2 = 1.2$ sd above the mean: regression to the mean.

(b) $P(Y_1 > 85 \mid Y_2 = 186) = 1 - \Phi\{(85 - 82)/8\} = 1 - \Phi(0.375) = 0.354$, against $P(Y_1 > 85) = 1 - \Phi(1.5) = 0.067$.

(c)

```
set.seed(2027)
Sg2 <- matrix(c(100, 0.6*80, 0.6*80, 64), 2)
d2 <- MASS::mvrnorm(4e5, c(70, 170), Sg2)
w <- d2[abs(d2[, 2] - 186) < 1, 1]
c(n = length(w), mean = mean(w), sd = sd(w), p85 = mean(w > 85),
  theory_p85 = 1 - pnorm(0.375), unconditional = 1 - pnorm(1.5))
```

```
#>             n          mean            sd           p85    theory_p85
#>     5.425e+03     8.206e+01     7.847e+00     3.495e-01     3.538e-01
#> unconditional
#>     6.681e-02
```

Exercise 1.7. Let $Y_1, \ldots, Y_n$ be iid $N(\mu, \sigma^2)$.

(a) Show that $\mathrm{Cov}(\bar{Y}, Y_i - \bar{Y}) = 0$ for each i, and deduce that $\bar{Y}$ and s^2 are independent.
(b) Using Theorem 1.5, compute $\mathrm{E}\{\sum_i (Y_i - \bar{Y})^2\}$ by a second method: expand the sum as $\sum Y_i^2 - n\bar{Y}^2$.
(c) For the `discoveries` data, test whether the variance exceeds the mean via the index of dispersion $D = (n-1)s^2/\bar{y}$, which is approximately χ^2_{n-1} under the Poisson model. Compute D and the p-value.

Solution.

(a) $\mathrm{Cov}(\bar{Y}, Y_i - \bar{Y}) = \mathrm{Cov}(\bar{Y}, Y_i) - \mathrm{Var}(\bar{Y}) = \sigma^2/n - \sigma^2/n = 0$. The vector $(\bar{Y}, Y_1 - \bar{Y}, \ldots, Y_n - \bar{Y})$ is a linear map of the normal vector Y, hence jointly normal, so $\bar{Y}$ is independent of every deviation, and s^2 is a function of the deviations.

(b) $\mathrm{E}\sum Y_i^2 = n(\sigma^2+\mu^2)$ and $\mathrm{E}(n\bar{Y}^2) = n(\sigma^2/n+\mu^2) = \sigma^2+n\mu^2$. The difference is $(n-1)\sigma^2$, as in Example 1.3.

(c)

```
D <- (length(y) - 1) * var(y) / mean(y)
c(D = D, df = length(y) - 1, p_value = pchisq(D, length(y) - 1, lower.tail =
↪  FALSE))
```

```
#>         D        df   p_value
#> 1.623e+02 9.900e+01 6.335e-05
```

\$D=,\$162.3 on 99 degrees of freedom has a p-value of about 6.3×10^{-5}: strong evidence of overdispersion relative to the Poisson model. (The χ^2 reference is approximate here; the exact test would use simulation.)

Exercise 1.8.
(a) Prove that $\operatorname{rank}(X^\top X) = \operatorname{rank}(X)$ by showing that X and $X^\top X$ have the same null space.
(b) A study has a factor with three levels A, B, C. Write the design matrix with an intercept and an indicator for each level. What is its rank? Which constraint does R impose by default?
(c) Using `mtcars`, build the design matrix with intercept and indicators for the three values of `cyl`, and show that `lm()` returns one `NA`.

Solution.

(a) See the proof of Theorem 1.8: $X^\top Xv = 0 \Rightarrow \|Xv\|^2 = 0 \Rightarrow Xv = 0$ and the converse is immediate; hence $\dim\ker(X^\top X) = \dim\ker X$ and rank-nullity gives equal ranks.

(b) With columns $(1, I_A, I_B, I_C)$ and $1 = I_A + I_B + I_C$ the four columns are linearly dependent, so $p = 4$ but the rank is 3 and $X^\top X$ is singular; β is not identifiable. By default R uses *treatment contrasts*: it drops the indicator of the first level, so that the intercept is the mean of level A and the other coefficients are differences from A.

(c)

```
Xc <- cbind(1, sapply(c(4, 6, 8), function(v) as.numeric(mtcars$cyl == v)))
c(p = ncol(Xc), rank = qr(Xc)$rank)
```

```
#>    p rank
#>    4    3
```

```
dd4 <- data.frame(mpg = mtcars$mpg, c4 = Xc[, 2], c6 = Xc[, 3], c8 = Xc[, 4])
coef(lm(mpg ~ c4 + c6 + c8, data = dd4))
```

```
#> (Intercept)          c4          c6          c8
#>      15.100      11.564       4.643          NA
```

The last coefficient is `NA`: it is aliased with the others.

Exercise 1.9. Let $n = 3$ and $x = (1, 2, 3)^\top$, and consider the simple linear regression design $X = (\mathbf{1}, x)$.
(a) Compute $X^\top X$, its inverse and the hat matrix H by hand.

(b) Check $\operatorname{tr}(H) = 2$ and $H^2 = H$.
(c) For $y = (1, 3, 2)^\top$ compute $\hat{y} = Hy$ and the residual $e = y - \hat{y}$; check that $X^\top e = 0$.

Solution.

(a) $X^\top X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}$ with determinant $42 - 36 = 6$, so $(X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}$. The diagonal element is $h_{ii} = \frac{1}{3} + \frac{(x_i - \bar{x})^2}{S_{xx}}$ with $S_{xx} = 2$, and in general $H_{ij} = \frac{1}{3} + \frac{(x_i - 2)(x_j - 2)}{2}$:

$$H = \begin{pmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & 5/6 \end{pmatrix}.$$

(b) $\operatorname{tr} H = 5/6 + 1/3 + 5/6 = 2 = p$. For idempotency, row 1 times column 1 of H^2 is $(25 + 4 + 1)/36 = 30/36 = 5/6 = H_{11}$, and similarly for the other entries.

(c) $\hat{y} = Hy = (5/6 + 1 - 2/6,\ 1/3 + 1 + 2/3,\ -1/6 + 1 + 5/3)^\top = (1.5,\ 2,\ 2.5)^\top$ and $e = (-0.5,\ 1,\ -0.5)^\top$. Then $\sum e_i = 0$ and $\sum x_i e_i = -0.5 + 2 - 1.5 = 0$, so $X^\top e = 0$. This is the line through the three points with slope $S_{xy}/S_{xx} = 1/2$.

```
Xh <- cbind(1, 1:3); Hh <- Xh %*% solve(crossprod(Xh)) %*% t(Xh)
Hh * 6; drop(Hh %*% c(1, 3, 2))
```

```
#>      [,1] [,2] [,3]
#> [1,]    5    2   -1
#> [2,]    2    2    2
#> [3,]   -1    2    5
```

```
#> [1] 1.5 2.0 2.5
```

Exercise 1.10. For a fixed $\lambda > 0$ consider the *penalised* residual sum of squares $Q_\lambda(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$.
(a) Derive the minimiser $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ and show that the matrix is invertible even if X is rank deficient.
(b) Show that the Hessian of Q_λ is positive definite, so the solution is the global minimum.
(c) With the rank-deficient design of the dummy trap in `mtcars` (`X_bad` above), compute $\hat{\beta}_\lambda$ for $\lambda = 10^{-4}$ and check that the fitted values are close to those of the unpenalised fit.

Solution.

(a) $Q_\lambda(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top (X^\top X + \lambda I)\beta$, so by Lemma 1.1 the gradient is $-2X^\top y + 2(X^\top X + \lambda I)\beta$, which vanishes at $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$. For $v \neq 0$, $v^\top (X^\top X + \lambda I) v = \|Xv\|^2 + \lambda\|v\|^2 \geq \lambda\|v\|^2 > 0$, so the matrix is p.d. and invertible for any X.

(b) The Hessian is $2(X^\top X + \lambda I)$, positive definite by the same calculation, so Q_λ is strictly convex and the stationary point is the unique global minimum. This is ridge regression, which resolves collinearity at the price of bias (Chapter 7).

(c)

```
yy <- mtcars$mpg
b_ridge <- solve(crossprod(X_bad) + 1e-4 * diag(ncol(X_bad)), crossprod(X_bad,
↪  yy))
```

```
fit_ok <- fitted(lm(mpg ~ wt + hp + am, data = mtcars))
c(max_abs_diff_fitted = max(abs(X_bad %*% b_ridge - fit_ok)))
```

```
#> max_abs_diff_fitted
#>           0.0008538
```

```
round(drop(b_ridge), 3)
```

```
#> (Intercept)          wt          hp          am      manual
#>      23.362      -2.878      -0.037      12.723      10.639
```

Among the infinitely many exact least-squares solutions, the penalty selects the one with the smallest norm, which spreads the effect of the redundant columns over the intercept, `am` and `manual`; the fitted values agree with those of the full-rank fit to within about 0.001 mpg.

Exercise 1.11. Let $f(x, y) = x^3 - 3xy + y^3$.

(a) Find all stationary points and classify them using the Hessian.

(b) Write the second-order Taylor expansion at the minimum.

(c) Confirm numerically with `optim` that the minimum is reached from a starting value near it.

Solution.

(a) $\nabla f = (3x^2 - 3y,\ 3y^2 - 3x)$. Setting both to zero, $y = x^2$ and $x = y^2 = x^4$, so $x(x^3 - 1) = 0$, i.e. $x = 0$ or $x = 1$, giving the points $(0, 0)$ and $(1, 1)$. The Hessian is $\begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$. At $(0, 0)$ it is $\begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$ with eigenvalues ± 3: a saddle point. At $(1, 1)$ it is $\begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$ with eigenvalues 3 and 9, positive definite: a local minimum, $f(1, 1) = -1$.

(b) With $u = x - 1$, $v = y - 1$: $f \approx -1 + \frac{1}{2}(6u^2 - 6uv + 6v^2) = -1 + 3u^2 - 3uv + 3v^2$. (The gradient term vanishes.)

(c)

```
fc <- function(p) p[1]^3 - 3 * p[1] * p[2] + p[2]^3
o <- optim(c(1.3, 0.7), fc, method = "BFGS", hessian = TRUE)
round(c(o$par, value = o$value), 4); round(eigen(o$hessian)$values, 3)
```

```
#>             value
#>     1     1    -1
```

```
#> [1] 9 3
```

Exercise 1.12.

(a) Apply Newton–Raphson to $g(x) = x^2 - 2$ with $x_0 = 1$ and show that $x_{t+1} = (x_t + 2/x_t)/2$. Compute three iterations by hand and the error after each.

(b) Show that if $e_t = x_t - \sqrt{2}$ then $e_{t+1} = e_t^2/(2x_t)$: convergence is quadratic.

(c) Apply Newton's method to maximise $f(x) = x^2 e^{-x}$ from $x_0 = 3$ and explain what happens.

Solution.

(a) With $g'(x) = 2x$, $x_{t+1} = x_t - (x_t^2 - 2)/(2x_t) = (x_t + 2/x_t)/2$. From $x_0 = 1$: $x_1 = 1.5$, $x_2 = (1.5 + 1.3333)/2 = 1.41667$, $x_3 = (1.41667 + 1.41176)/2 = 1.414216$. Errors: 0.0858, 0.00245, 2.1×10^{-6}.

(b) $x_{t+1} - \sqrt{2} = \frac{x_t^2 - 2\sqrt{2}x_t + 2}{2x_t} = \frac{(x_t - \sqrt{2})^2}{2x_t}$. Squaring the error and dividing by $2x_t \approx 2.83$ shows that the number of correct digits doubles; e.g. $0.00245^2/2.83 = 2.1 \times 10^{-6}$, matching the third error.

(c) The update is $x - f'(x)/f''(x)$. At $x = 3$: $f'(3) = (6 - 9)e^{-3} = -3e^{-3}$ and $f''(3) = (2 - 12 + 9)e^{-3} = -e^{-3}$, so $x_1 = 3 - (-3)/(-1) = 0$, the *minimum* of f. At $x = 0$, $f'(0) = 0$ and the iterations stop. Newton's method seeks a zero of f'; it does not know whether it is a maximum or a minimum. Starting at a point where $f'' < 0$ (here $f''(3) < 0$ means it is concave, as at a maximum) does not guarantee that the next iterate is better; one should use step-halving or check $f(x_{t+1}) > f(x_t)$.

```
x <- 1; for (t in 1:3) { x <- (x + 2 / x) / 2; cat(sprintf("x%d = %.7f  error
↪  = %.2e\n", t, x, x - sqrt(2))) }
```

```
#> x1 = 1.5000000  error = 8.58e-02
#> x2 = 1.4166667  error = 2.45e-03
#> x3 = 1.4142157  error = 2.12e-06
```

Exercise 1.13. Let $Y_1, \dots, Y_n$ be iid Poisson(λ) with $\bar{Y}$ their mean.
(a) Use the delta method to approximate $\mathrm{Var}(\sqrt{\bar{Y}})$ and $\mathrm{Var}(\log \bar{Y})$.
(b) Give an approximate 95% confidence interval for λ based on $\log \hat{\lambda} \pm 1.96\,\mathrm{se}(\log \hat{\lambda})$ for `discoveries`, and compare it with $\bar{y} \pm 1.96\sqrt{\bar{y}/n}$.
(c) Why is the log interval preferable for small counts?

Solution.

(a) $\mathrm{Var}(\bar{Y}) = \lambda/n$. For $g = \sqrt{\cdot}$, $g'(\lambda)^2 = 1/(4\lambda)$, so $\mathrm{Var}(\sqrt{\bar{Y}}) \approx 1/(4n)$. For $g = \log$, $g'(\lambda)^2 = 1/\lambda^2$ and $\mathrm{Var}(\log \bar{Y}) \approx 1/(n\lambda)$, with estimated standard error $1/\sqrt{n\bar{y}}$.

(b)

```
n <- length(y); ybar <- mean(y)
ci_wald <- ybar + c(-1, 1) * 1.96 * sqrt(ybar / n)
ci_log  <- exp(log(ybar) + c(-1, 1) * 1.96 / sqrt(n * ybar))
rbind(wald = ci_wald, log_scale = ci_log)
```

```
#>            [,1]  [,2]
#> wald      2.755 3.445
#> log_scale 2.773 3.465
```

The two intervals are close for $n = 100$ and $\bar{y} = 3.1$.

(c) The log-scale interval is $\bar{y} \cdot e^{\pm 1.96/\sqrt{n\bar{y}}}$, always positive, while the Wald interval can include negative values of λ when $n\bar{y}$ is small. The log scale is also where the sampling distribution of $\hat{\lambda}$ is closer to symmetric. Note that both intervals assume the Poisson variance; the discoveries data are overdispersed, so both are too narrow (Exercise 1.7).

1.12 Further reading

- Pawitan (2001), Chapter 1, for the role of probability models in inference and the likelihood

viewpoint.

- Faraway (2014), Chapter 1 and Section 2.1–2.4, for the same material in a regression setting with R.
- Kutner et al. (2005), Chapter 5, for matrix algebra applied to the linear model.

Chapter 2

Statistical models

A statistical model is a description of how data could have been generated, simple enough to be analysed and rich enough to be useful. Every later chapter of these notes defines a more sophisticated model, from the linear model to generalised additive mixed models, but they all share the same skeleton and the same limitations. This chapter builds that skeleton on the simplest possible example, a single time series, so that attention can stay on the ideas rather than on the algebra: what a model is, what its assumptions are, how to check them, how to *simulate* from it, how to separate the variability of the data from the uncertainty about parameters, how poor data quality undermines any model, and why a model that explains well may predict badly.

Learning objectives. After this chapter you should be able to:

1. formulate a statistical model as a systematic part plus a random part and list its assumptions;
2. derive the least-squares estimators of a constant mean and of a straight line, and their variances;
3. compare nested models and explain why the residual sum of squares cannot choose between them;
4. simulate data from a fitted model and use simulation to check it, using a line-up;
5. distinguish variability (σ) from uncertainty (se), and a confidence interval from a prediction interval;
6. analyse the effect of measurement error, missing data and selection on estimates;
7. contrast prediction and explanation, using a hold-out check;
8. organise an analysis so that anyone can reproduce it.

Chapter 1 supplies the probability and matrix tools used here; Chapter 3 makes the notion of a model operational through the likelihood.

2.1 A motivating data set: New Haven temperatures

The data set `nhtemp` contains the mean annual temperature in degrees Fahrenheit in New Haven, Connecticut, for the 60 years 1912–1971. It is a single series of $n = 60$ numbers. We will keep asking four questions of it:

1. What is a "typical" annual temperature, and how sure are we?
2. Is there a trend?
3. What will the temperature be in 1972?
4. How would a different 60-year record look, and does our model reproduce it?

```
data(nhtemp)
nh <- data.frame(year = as.numeric(time(nhtemp)), temp = as.numeric(nhtemp))
n <- nrow(nh)
ggplot(nh, aes(year, temp)) + geom_line(colour = ie_navy) + geom_point(colour
↪  = ie_navy, size = 1.2) +
  labs(x = "Year", y = "Mean annual temperature (deg F)")
```

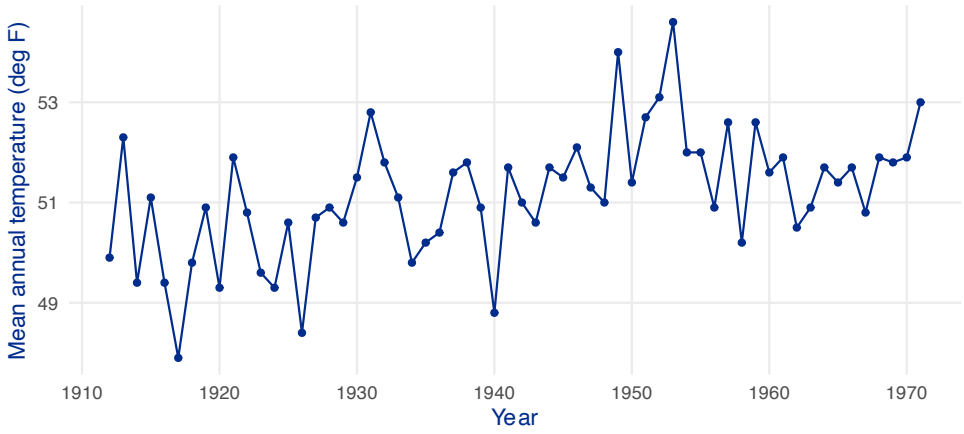


Figure 2.1: Mean annual temperature in New Haven, 1912–1971 (degrees Fahrenheit).

```
summary(nh$temp)
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>    47.9    50.6    51.2    51.2    51.9    54.6
```

Remark. We always read the data from the package, never type them by hand. A transcription error in a reproduced data set is not only a mistake; it is an invisible one. The data come from a single station, and changes in the instruments, siting or surroundings of a station can produce apparent trends; this *inductive* uncertainty is not reduced by collecting more years of the same record.

2.2 What is a statistical model?

2.2.1 Deterministic and stochastic descriptions

A *deterministic* model predicts the response exactly from the inputs, as in Newton's law $s = \tfrac{1}{2}gt^2$. Data on a real process never follow such a formula exactly: instruments have error, unmeasured factors intervene, and the units differ. A *statistical* model adds a probability distribution to describe what the formula does not.

Definition 2.1 (Statistical model)**.** A **statistical model** for data $y = (y_1, \dots, y_n)$ is a family of probability distributions $\{f(y;\theta) : \theta \in \Theta\}$ indexed by a parameter θ, such that the observed data are assumed to be a realisation from one member of the family. The most common form decomposes each observation into a **systematic part** and a **random part**,

$$y_i = f(x_i;\beta) + \varepsilon_i, \qquad \mathrm{E}(\varepsilon_i) = 0,\ \mathrm{Var}(\varepsilon_i) = \sigma^2, \tag{2.1}$$

where x_i are known explanatory variables ("covariates"), β are unknown parameters, and ε_i is a random error. If the distribution of ε_i is specified, e.g. $\varepsilon_i \sim N(0,\sigma^2)$ independently, the model is **parametric**.

Equation (2.1) is a statement about the *mean* of y_i: $\mathrm{E}(y_i) = f(x_i; \beta)$. Everything that is not captured by the mean is called noise, and the model's job is to describe the **signal** (the systematic part) while quantifying the **noise** (the random part). The two are separated by assumption, not by nature: what we call noise in one model (a trend) can be signal in a larger one.

The two models we will study for the New Haven series are:

- **Model 1 (constant mean):** $y_i = \mu + \varepsilon_i$.
- **Model 2 (linear trend):** $y_i = \beta_0 + \beta_1 t_i + \varepsilon_i$, with t_i the year.

with $\varepsilon_i \sim N(0, \sigma^2)$ independent in both. They are special cases of the linear model $y = X\beta + \varepsilon$, with $X = \mathbf{1}$ and $X = (\mathbf{1}, t)$, respectively, where the intercept column is included in X (notation of Appendix A).

2.2.2 All models are wrong

Box (1976) wrote that "all models are wrong, but some are useful". A model is an approximation designed for a purpose; it is not a claim that the world has this structure. The straight-line model is wrong for New Haven in the literal sense (temperature has no mechanism forcing a perfectly linear trend, and the errors are not exactly normal), but it may be useful for describing 1912–1971. The relevant question is not "is the model true?" but "is it adequate for this purpose?", and "useful for what?" has different answers for description, for prediction and for explanation (Section 2.8).

Two further distinctions will matter:

- A model can be **mechanistic** (derived from knowledge of the process, e.g. a differential equation) or **empirical** (chosen because it fits, e.g. a polynomial trend). The straight line is empirical.
- A model can be **parametric**, with a finite-dimensional θ, or **nonparametric/semi-parametric**, in which part of f is a smooth function estimated from the data (Chapters 14–16). The course moves along this ladder: constant, linear, generalised linear, mixed, additive.

2.2.3 The modelling cycle

Model building is a loop, not a line: *question* → *data* → *model* → *fit* → *check* → *revise*, finishing in a communicated conclusion with its uncertainty. The check step compares what the model predicts with what we see (Sections 2.4 and 2.5), and the loop closes when the model is adequate for the question.

2.3 Least squares for the two models

2.3.1 A constant mean

Given $y_1, \dots, y_n$, the least-squares criterion chooses μ to minimise $Q(\mu) = \sum_{i=1}^{n} (y_i - \mu)^2$.

Theorem 2.1 (Least squares for a constant mean). *$Q(\mu) = \sum_i (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$, so Q has the unique global minimum $\hat{\mu} = \bar{y}$. If $y_1, \dots, y_n$ are uncorrelated with mean μ and variance σ^2, then $\mathrm{E}(\bar{y}) = \mu$, $\mathrm{Var}(\bar{y}) = \sigma^2/n$, and $\hat{\sigma}^2 = s^2 = \sum_i (y_i - \bar{y})^2/(n-1)$ is unbiased for σ^2.*

Proof. Add and subtract $\bar{y}$: $\sum_i (y_i-\mu)^2 = \sum_i \{(y_i-\bar{y})+(\bar{y}-\mu)\}^2 = \sum_i (y_i-\bar{y})^2 + 2(\bar{y}-\mu)\sum_i (y_i-\bar{y}) + n(\bar{y}-\mu)^2$, and the cross term vanishes because $\sum_i (y_i-\bar{y}) = 0$. The first term does not depend on μ and the second is non-negative and zero only at $\mu = \bar{y}$. (Calculus gives the same answer: $Q'(\mu) = -2\sum(y_i-\mu) = 0$ and $Q'' = 2n > 0$.) The moments follow from Theorem 1.4 with $a = (1/n)\mathbf{1}$, and unbiasedness of s^2 from Example 1.3. □

The *standard error* of $\bar{y}$ is $\mathrm{se}(\bar{y}) = \sigma/\sqrt{n}$, estimated by $s/\sqrt{n}$. Notice where the independence assumption enters: $\mathrm{Var}(\bar{y}) = n^{-2}\{\sum_i \mathrm{Var}(y_i) + \sum_{i\neq j} \mathrm{Cov}(y_i, y_j)\}$, and the covariance terms are set to zero.

2.3.2 A straight line

For model 2, minimise $Q(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 t_i)^2$. Write $S_{tt} = \sum_i (t_i - \bar{t})^2$ and $S_{ty} = \sum_i (t_i - \bar{t})(y_i - \bar{y})$.

Theorem 2.2 (Least squares for a straight line)**.** *If $S_{tt} > 0$ (the t_i are not all equal), Q has the unique minimiser*

$$\hat{\beta}_1 = \frac{S_{ty}}{S_{tt}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{t}.$$

If $\mathrm{E}(\varepsilon_i) = 0$ and the errors are uncorrelated with variance σ^2, then both are unbiased, and

$$\mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{tt}}, \qquad \mathrm{Var}(\hat{\beta}_0) = \sigma^2\Big(\frac{1}{n} + \frac{\bar{t}^2}{S_{tt}}\Big), \qquad \mathrm{Cov}(\bar{y}, \hat{\beta}_1) = 0.$$

Proof. Setting the partial derivatives to zero gives the normal equations $\sum_i (y_i - \beta_0 - \beta_1 t_i) = 0$ and $\sum_i t_i (y_i - \beta_0 - \beta_1 t_i) = 0$. The first gives $\beta_0 = \bar{y} - \beta_1 \bar{t}$. Substituting into the second and using $\sum_i t_i (y_i - \bar{y}) = S_{ty}$ and $\sum_i t_i (t_i - \bar{t}) = S_{tt}$ gives $S_{ty} - \beta_1 S_{tt} = 0$. (The Hessian is $2X^\top X$, which is positive definite when $S_{tt} > 0$; compare Theorem 1.11.)

For the variance write $\hat{\beta}_1 = \sum_i c_i y_i$ with $c_i = (t_i - \bar{t})/S_{tt}$, using $\sum_i (t_i - \bar{t})\bar{y} = 0$. Then $\mathrm{E}\hat{\beta}_1 = \sum_i c_i(\beta_0 + \beta_1 t_i) = \beta_1 \sum_i c_i t_i = \beta_1$ because $\sum_i c_i = 0$ and $\sum_i c_i t_i = 1$, and $\mathrm{Var}(\hat{\beta}_1) = \sigma^2 \sum_i c_i^2 = \sigma^2/S_{tt}$. Next $\mathrm{Cov}(\bar{y}, \hat{\beta}_1) = \sigma^2 \sum_i \frac{1}{n} c_i = 0$. Finally $\mathrm{Var}(\hat{\beta}_0) = \mathrm{Var}(\bar{y}) + \bar{t}^2 \mathrm{Var}(\hat{\beta}_1)$, since the covariance vanishes, which is the stated formula; unbiasedness of $\hat{\beta}_0$ follows from $\mathrm{E}\bar{y} = \beta_0 + \beta_1 \bar{t}$. □

Centring the time variable ($t_i - \bar{t}$ or $t_i - 1912$) makes $\bar{t} = 0$ in the first case and makes the intercept interpretable in the second: it becomes the expected temperature in the reference year. The slope is unchanged. We now fit both models.

```
m1 <- lm(temp ~ 1, data = nh)
m2 <- lm(temp ~ year, data = nh)
ybar <- mean(nh$temp); s <- sd(nh$temp)
Stt <- sum((nh$year - mean(nh$year))^2); Sty <- sum((nh$year - mean(nh$year))
↪  * (nh$temp - ybar))
c(ybar = ybar, s = s, se_ybar = s / sqrt(n))
```

```
#>    ybar       s se_ybar
#> 51.1600  1.2656  0.1634
```

```
c(Stt = Stt, Sty = Sty, b1 = Sty / Stt, b0 = ybar - Sty / Stt * mean(nh$year))
```

```
#>         Stt         Sty          b1          b0
```

```
#>  1.800e+04  6.644e+02  3.692e-02 -2.052e+01
```

```
cbind(by_hand = c(ybar - Sty / Stt * mean(nh$year), Sty / Stt), lm = coef(m2))
```

```
#>                by_hand        lm
#> (Intercept) -20.52283 -20.52283
#> year          0.03692   0.03692
```

```
coef(lm(temp ~ I(year - 1912), data = nh))          # intercept = expected
↪  temperature in 1912
```

```
#>    (Intercept) I(year - 1912)
#>       50.07082        0.03692
```

```
10 * coef(m2)["year"]                                # degrees F per decade
```

```
#>   year
#> 0.3692
```

The sample mean is 51.16°F with standard error 0.163. The fitted slope is 0.0369°F per year, i.e. 0.37°F per decade, with standard error 0.0082 per year. Equivalently: the estimated slope is 4.5 standard errors from zero.

```
ggplot(nh, aes(year, temp)) + geom_point(colour = ie_navy, size = 1.3) +
  geom_hline(yintercept = ybar, linetype = 2, colour = ie_coral) +
  geom_line(aes(y = fitted(m2)), colour = ie_blue, linewidth = 1) +
  labs(x = "Year", y = "Temperature (deg F)")
```

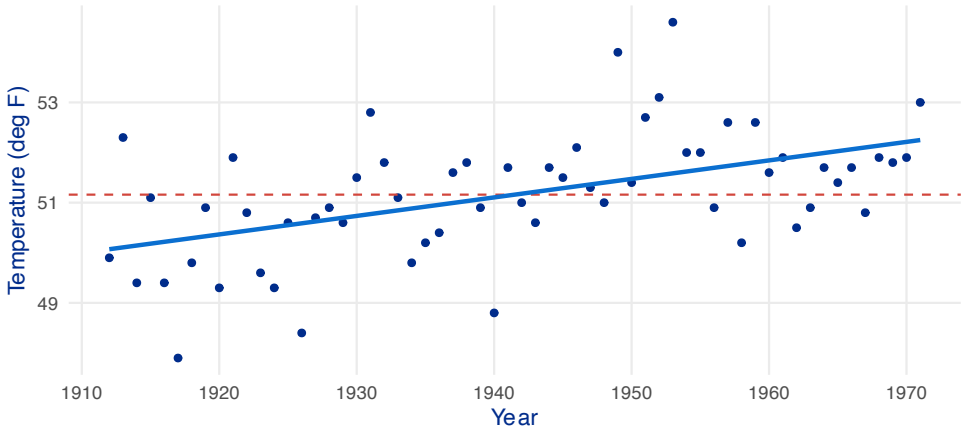


Figure 2.2: The two fitted models: constant mean (dashed) and linear trend (solid).

2.3.3 Comparing the fits

The residual sum of squares (RSS) of model k and the estimated error variance are

$$\mathrm{RSS}_k = \sum_i (y_i - \hat{y}_i)^2, \qquad \hat{\sigma}^2_k = \frac{\mathrm{RSS}_k}{n - p_k},$$

where p_k is the number of parameters in the mean ($p_1 = 1$, $p_2 = 2$).

```
rss1 <- sum(resid(m1)^2); rss2 <- sum(resid(m2)^2)
c(RSS1 = rss1, RSS2 = rss2, sigma1 = summary(m1)$sigma, sigma2 =
↪  summary(m2)$sigma, R2 = 1 - rss2 / rss1)
```

```
#>    RSS1    RSS2  sigma1  sigma2      R2
#> 94.5040 69.9734  1.2656  1.0984  0.2596
```

```
anova(m1, m2)
```

```
#> Analysis of Variance Table
#>
#> Model 1: temp ~ 1
#> Model 2: temp ~ year
#>   Res.Df  RSS Df Sum of Sq    F  Pr(>F)
#> 1     59 94.5
#> 2     58 70.0  1      24.5 20.3 3.2e-05 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Warning. *A smaller RSS does not mean a better model.* Model 1 is the special case $\beta_1 = 0$ of model 2, so $\mathrm{RSS}_2 \le \mathrm{RSS}_1$ always: minimising over a larger set cannot give a larger minimum. Adding any covariate, even a column of random numbers, lowers the RSS. The question is whether the reduction (94.5 to 70, i.e. a fraction $R^2 = 0.26$) is larger than what chance would produce; this is what the F test above addresses (Chapter 5), and what information criteria and hold-out checks address in a more general way (Section 2.8 and Chapter 7).

2.3.4 Signal and noise: how much does the model explain?

The decomposition of Theorem 2.1 has an analogue for the line. Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 t_i$ and $e_i = y_i - \hat{y}_i$.

Proposition 2.1 (Decomposition of the total sum of squares)**.** *For the least-squares line,*

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\textit{explained}} + \underbrace{\sum_i e_i^2}_{\text{RSS}}, \qquad R^2 = \frac{\textit{explained}}{\text{TSS}} = \hat{\beta}_1^2 \frac{S_{tt}}{\text{TSS}} = r_{ty}^2,$$

the squared sample correlation between t and y.

Proof. Write $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + e_i$ and square. The cross term is $2\sum_i (\hat{y}_i - \bar{y}) e_i = 2\hat{\beta}_1 \sum_i (t_i - \bar{t}) e_i = 0$, because $\hat{y}_i - \bar{y} = \hat{\beta}_1 (t_i - \bar{t})$ and the normal equations say that $\sum e_i = 0$ and $\sum t_i e_i = 0$. Also $\sum_i (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2 S_{tt} = S_{ty}^2 / S_{tt}$, so $R^2 = S_{ty}^2 / (S_{tt}\,\mathrm{TSS}) = r_{ty}^2$. □

For New Haven $R^2 = 0.26$: the trend explains about a quarter of the year-to-year variability, and the remaining three quarters is *noise* relative to this model. A low R^2 does not mean that the trend is unimportant or unreal; with $n = 60$ the slope is clearly different from zero. R^2 measures how much of the *scatter* is explained, not whether the trend *exists*, and a small slope with a tiny standard error and a large noise level can be both significant and low-R^2.

2.4 Assumptions and how to check them

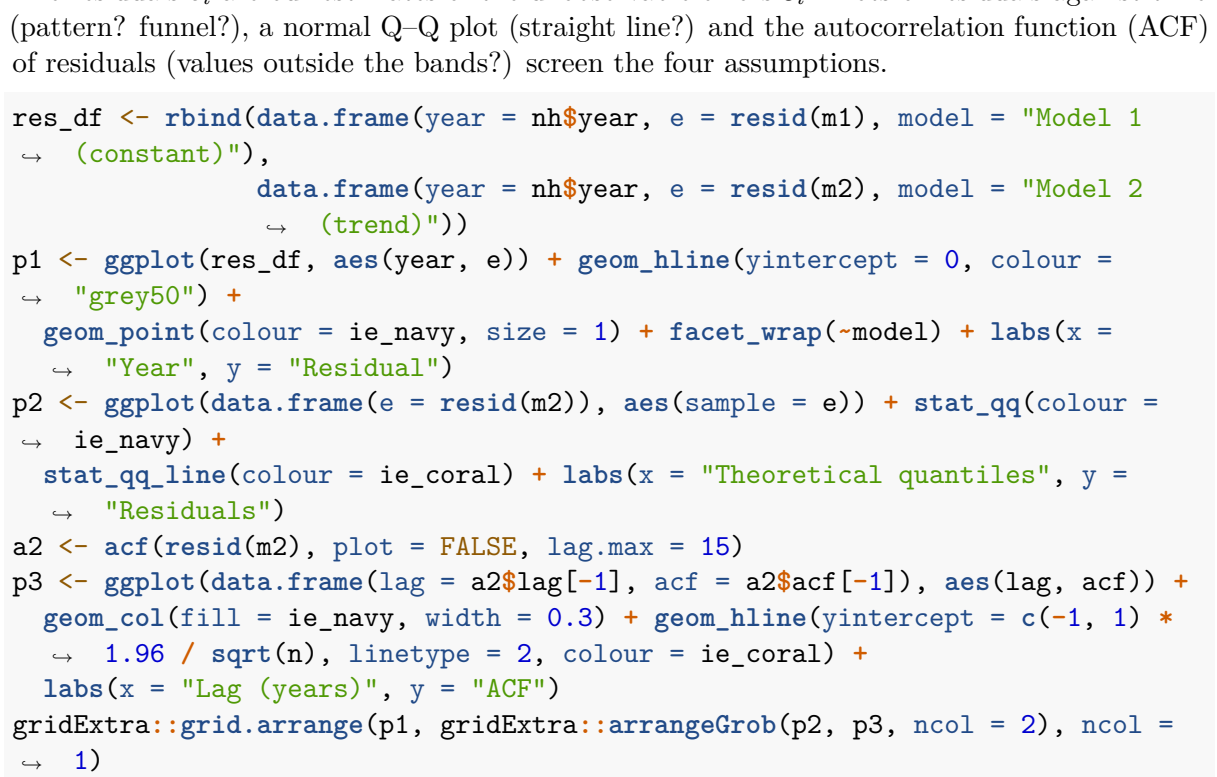
2.4.1 The assumptions

Model 2 assumes, in decreasing order of importance for most conclusions:

1. **Correct mean structure (linearity).** $\mathrm{E}(y_i) = \beta_0 + \beta_1 t_i$. If the true trend curves or has a jump, the slope is estimated *for the wrong model*, and everything else is affected.
2. **Independence.** The errors are uncorrelated (and for exact inference independent). Positive serial correlation makes standard errors too small (Example 1.2).
3. **Constant variance.** $\mathrm{Var}(\varepsilon_i) = \sigma^2$, the same for all i.
4. **Normality.** $\varepsilon_i \sim N(0, \sigma^2)$. This matters least for estimating the mean with moderate n (the central limit theorem protects us) but matters for prediction intervals.

The residuals e_i are our estimates of the unobservable errors ε_i. Plots of residuals against time (pattern? funnel?), a normal Q–Q plot (straight line?) and the autocorrelation function (ACF) of residuals (values outside the bands?) screen the four assumptions.

```
res_df <- rbind(data.frame(year = nh$year, e = resid(m1), model = "Model 1
↪  (constant)"),
                data.frame(year = nh$year, e = resid(m2), model = "Model 2
                ↪  (trend)"))
p1 <- ggplot(res_df, aes(year, e)) + geom_hline(yintercept = 0, colour =
↪  "grey50") +
  geom_point(colour = ie_navy, size = 1) + facet_wrap(~model) + labs(x =
  ↪  "Year", y = "Residual")
p2 <- ggplot(data.frame(e = resid(m2)), aes(sample = e)) + stat_qq(colour =
↪  ie_navy) +
  stat_qq_line(colour = ie_coral) + labs(x = "Theoretical quantiles", y =
  ↪  "Residuals")
a2 <- acf(resid(m2), plot = FALSE, lag.max = 15)
p3 <- ggplot(data.frame(lag = a2$lag[-1], acf = a2$acf[-1]), aes(lag, acf)) +
  geom_col(fill = ie_navy, width = 0.3) + geom_hline(yintercept = c(-1, 1) *
  ↪  1.96 / sqrt(n), linetype = 2, colour = ie_coral) +
  labs(x = "Lag (years)", y = "ACF")
gridExtra::grid.arrange(p1, gridExtra::arrangeGrob(p2, p3, ncol = 2), ncol =
↪  1)
```



```
c(lag1_m1 = acf(resid(m1), plot = FALSE)$acf[2], lag1_m2 = acf(resid(m2), plot
↪  = FALSE)$acf[2])
```

```
#> lag1_m1 lag1_m2
#>  0.3148  0.1070
```

```
shapiro.test(resid(m2))
```

```
#>
#>  Shapiro-Wilk normality test
#>
#> data:  resid(m2)
#> W = 0.98, p-value = 0.4
```

The residuals of model 1 show the trend (they are mostly negative early, positive late), and their lag-1 autocorrelation is 0.31. After fitting the trend the autocorrelation falls to 0.11, which is within the 95% bands $\pm 1.96/\sqrt{n} = \pm 0.25$.

Remark. The apparent dependence of model 1's residuals is mostly an *unmodelled trend*: a misspecified mean can masquerade as serial correlation. For this reason fix the systematic part before judging independence.

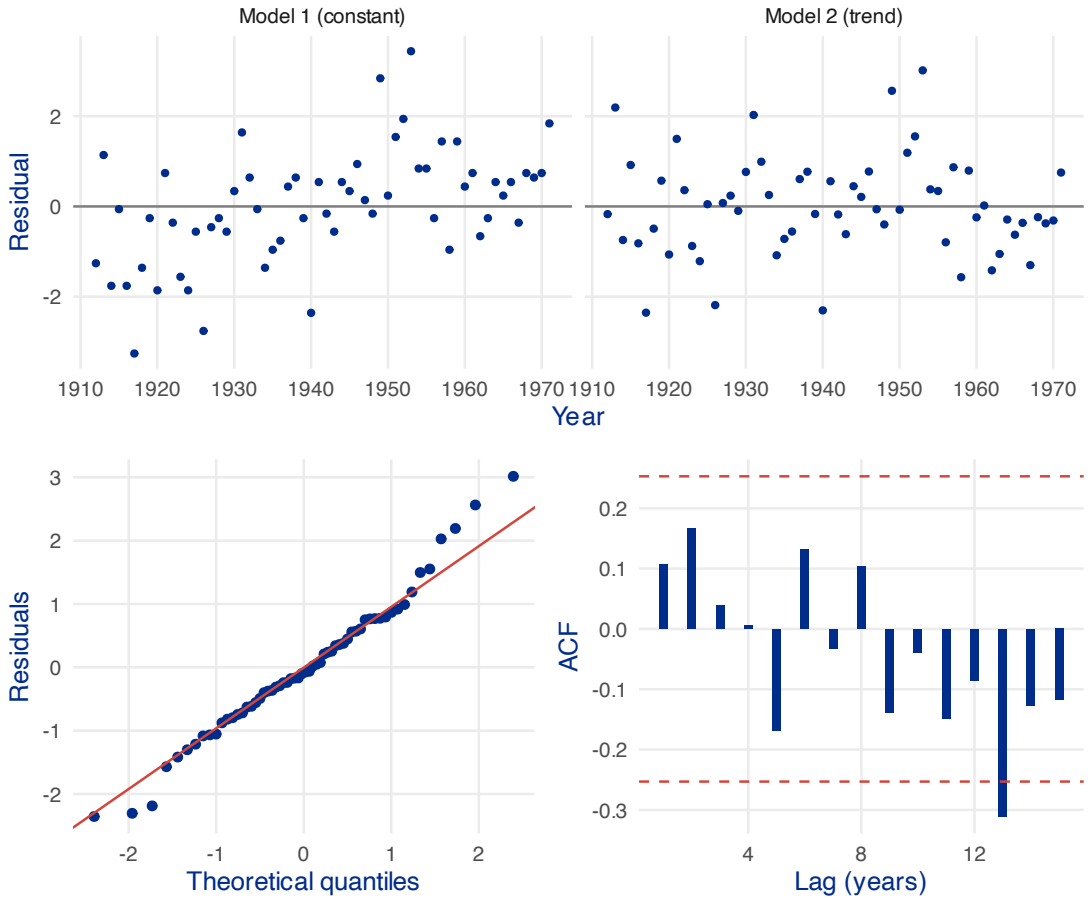


Figure 2.3: Residuals of the two models against year (top), normal Q–Q plot and ACF of the residuals of model 2 (bottom).

Warning. A Shapiro–Wilk p-value of 0.44 does not *prove* that the errors are normal; it says that 60 observations do not contradict normality. Conversely, with very large n the test rejects for trivial departures that do not matter. Use graphs and judgement.

2.4.2 Why independence matters for the standard error

Suppose in truth neighbouring years have correlation ρ. By Example 1.2, $\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}\{1+2\rho\frac{n-1}{n}\}$. With $n = 60$ and $\rho = 0.3$ the factor is $1 + 2 \cdot 0.3 \cdot 59/60 = 1.59$, so the true standard error of $\bar{y}$ is $\sqrt{1.59} = 1.26$ times the one computed under independence: 0.163 becomes 0.206. A confidence interval computed assuming independence is then about 20% too narrow: it covers less than 95% of the time.

2.5 Simulating from a model

A fitted parametric model is a *data-generating process*: given parameter values, we can draw new data sets from it. This simple fact has three uses: to **check** the model (does simulated data look like the real data?), to **study the sampling distribution** of an estimator when no formula is available, and to **validate** a method (does it recover known parameters?).

Definition 2.2 (Parametric simulation)**.** To simulate from the fitted model 2: compute the fitted values $\hat{y}_i$ and $\hat{\sigma}$, draw $y_i^* = \hat{y}_i + \hat{\sigma} z_i$ with $z_i \sim N(0,1)$ independently $(i = 1, \ldots, n)$, and repeat. In R: `fitted(m) + rnorm(n, 0, sigma(m))` or `simulate(m, nsim)`.

2.5.1 A line-up

In a *line-up* protocol the real data are hidden at a random position among simulated data sets, and a viewer is asked to pick out the real one. If the model reproduces the main features of the data, the real series is no more identifiable than the others; if the viewer picks the real one reliably, the model misses something.

```
set.seed(2027)
pos <- sample(1:9, 1)
sig2 <- summary(m2)$sigma
lu <- do.call(rbind, lapply(1:9, function(k) {
  data.frame(panel = k, year = nh$year,
             temp = if (k == pos) nh$temp else fitted(m2) + rnorm(n, 0, sig2))
}))
ggplot(lu, aes(year, temp)) + geom_line(colour = ie_navy, linewidth = 0.4) +
↪  facet_wrap(~panel, nrow = 3) +
  labs(x = NULL, y = NULL) + theme(axis.text = element_text(size = 6))
```

The real series is in panel 8. Most people cannot find it: model 2 reproduces the look of the data. Had we simulated from model 1 (constant mean), the real series, with its upward drift, would have stood out.

2.5.2 The sampling distribution of the slope

Suppose the truth were model 1, with no trend. How often would random noise produce a slope as large as the one we saw? We simulate series from fitted model 1, refit the slope, and compare. We also simulate from model 2 to see the variability of $\hat{\beta}_1$ when the trend is real.

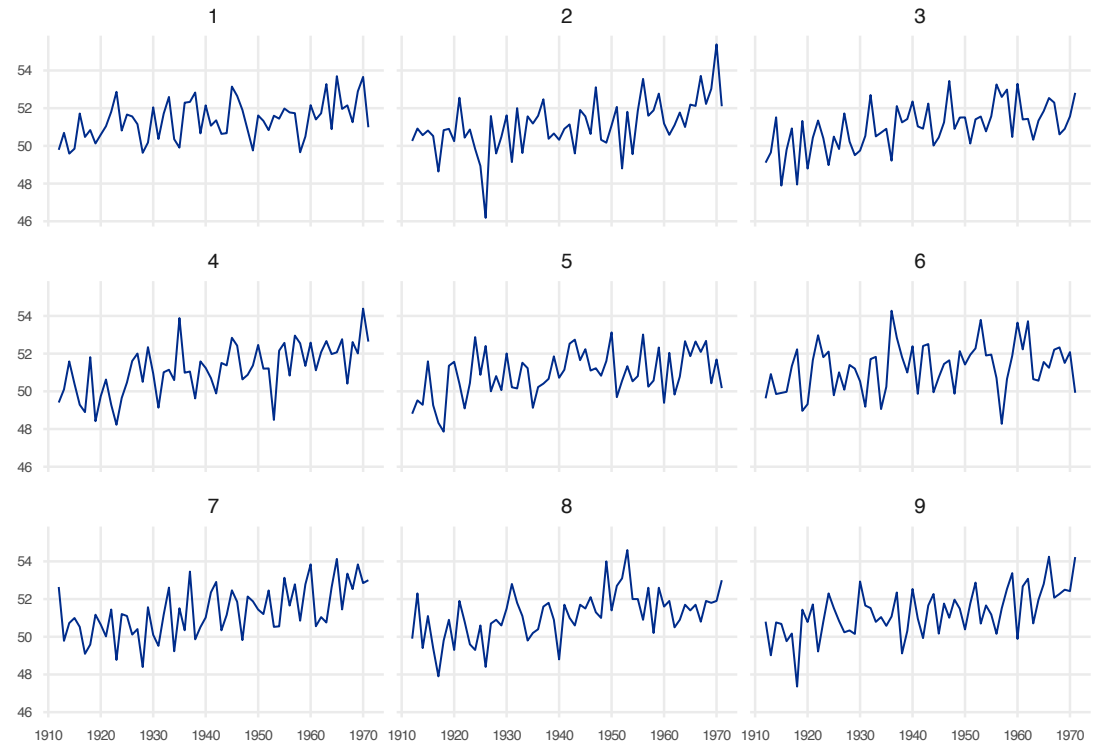


Figure 2.4: Line-up: one panel is the real New Haven series, the other eight are simulated from model 2. Which one is real? (Answer in the text.)

```
set.seed(2027)
B <- 1000
sig1 <- summary(m1)$sigma
slope1 <- replicate(B, coef(lm(rnorm(n, ybar, sig1) ~ nh$year))[2])
slope2 <- replicate(B, coef(lm(fitted(m2) + rnorm(n, 0, sig2) ~ nh$year))[2])
b1_obs <- unname(coef(m2)[2])
c(observed = b1_obs, max_under_m1 = max(slope1), share_m1_at_least_obs =
↪  mean(slope1 >= b1_obs))
```

```
#>              observed          max_under_m1 share_m1_at_least_obs
#>               0.03692               0.02758               0.00000
```

```
c(mean_m2 = mean(slope2), sd_m2 = sd(slope2), se_from_lm =
↪  summary(m2)$coefficients["year", "Std. Error"])
```

```
#>    mean_m2      sd_m2 se_from_lm
#>   0.036957   0.007971   0.008188
```

```
sl <- data.frame(slope = 10 * c(slope1, slope2), model = rep(c("Model 1: no
↪  trend", "Model 2: trend"), each = B))
ggplot(sl, aes(slope, fill = model)) + geom_histogram(bins = 40, alpha = 0.7,
↪  position = "identity") +
  geom_vline(xintercept = 10 * b1_obs, linewidth = 1) +
   ↪  scale_fill_manual(values = c(ie_coral, ie_navy)) +
  labs(x = "Slope (deg F per decade)", y = "Count", fill = NULL)
```

Under model 1 not one of the 1000 simulated slopes reaches the observed one: the maximum

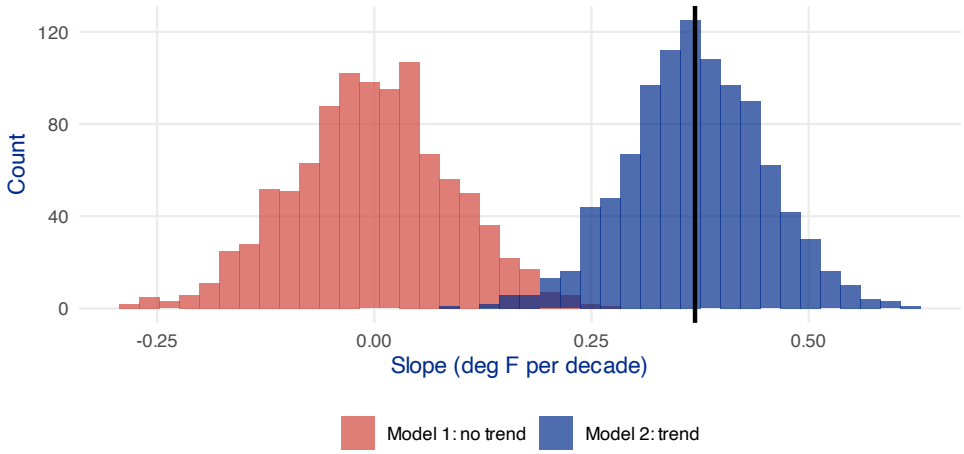


Figure 2.5: Distribution of the fitted slope (<U+00B0>F per decade) under model 1 (no trend) and model 2 (trend), 1000 simulated series each. The vertical line is the observed slope.

is 0.0276 against the observed 0.0369. This is a *Monte Carlo* p-value (strictly, "$< 1/1000$"), a simulation-based version of the t-test. Under model 2 the simulated slopes are centred at the true slope with a standard deviation (0.008) that matches the formula $\hat{\sigma}/\sqrt{S_{tt}} = 0.0082$ of Theorem 2.2; simulation reproduces the theory, and works when the theory is not available.

Remark. A Monte Carlo estimate of a probability q based on B draws has standard error $\sqrt{q(1-q)/B}$, so $B = 1000$ gives roughly ± 0.03 at $q = 0.5$ and much less near 0 or 1. Always report the seed: results from `set.seed(2027)` can be reproduced exactly.

2.6 Variability versus uncertainty

Two kinds of spread appear in every analysis, and confusing them is one of the most common errors.

- **Variability** is how much individual observations differ from the model's mean: it is measured by σ and does *not* shrink when we collect more data from the same process.
- **Uncertainty** is how much an *estimate* would differ from sample to sample: it is measured by the standard error, e.g. $\sigma/\sqrt{n}$ for the mean, and *does* shrink with n.

```
set.seed(2027)
ns <- c(10, 30, 100, 300, 1000)
vu <- t(sapply(ns, function(nn) {
  sims <- replicate(500, { z <- rnorm(nn, ybar, s); c(mean(z), sd(z)) })
  c(n = nn, variability = mean(sims[2, ]), uncertainty = sd(sims[1, ]), theory
   ↪  = s / sqrt(nn))
}))
round(vu, 3)
```

```
#>         n variability uncertainty theory
#> [1,]   10       1.236       0.399  0.400
#> [2,]   30       1.264       0.242  0.231
#> [3,]  100       1.265       0.125  0.127
#> [4,]  300       1.263       0.069  0.073
#> [5,] 1000       1.265       0.041  0.040
```

```
vl <- data.frame(n = rep(ns, 2), value = c(vu[, "variability"], vu[,
↪  "uncertainty"]),
                 what = rep(c("variability (SD of data)", "uncertainty (SE of
                  ↪  mean)"), each = length(ns)))
ggplot(vl, aes(n, value, colour = what)) + geom_line() + geom_point() +
↪  scale_x_log10() +
  scale_colour_manual(values = c(ie_coral, ie_navy)) + labs(x = "Sample size
   ↪  n", y = "deg F", colour = NULL)
```

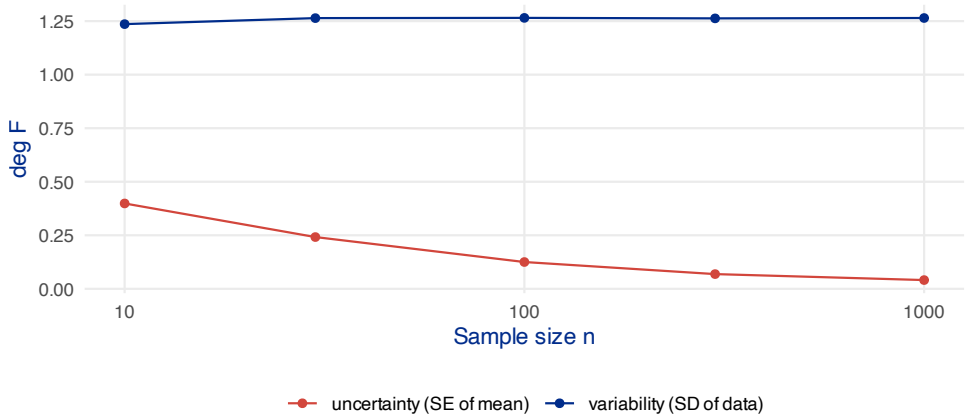


Figure 2.6: Variability (average sample standard deviation) stays near σ, while uncertainty (standard deviation of the sample mean) falls as $\sigma/\sqrt{n}$.

2.6.1 Confidence and prediction intervals

The distinction determines which interval we should report. A **confidence interval** for the mean μ expresses uncertainty about a parameter. A **prediction interval** for a *new observation* y_0 must in addition include the variability of that observation, which no amount of data removes.

Theorem 2.3 (Prediction error for the constant-mean model)**.** *Let y_0 be a new observation, independent of $y_1, \dots, y_n$, with the same mean and variance. Then* $\mathrm{E}(y_0 - \bar{y}) = 0$ *and*

$$\mathrm{Var}(y_0 - \bar{y}) = \sigma^2\Big(1 + \frac{1}{n}\Big).$$

Under normality, a $100(1-\alpha)\%$ ***confidence interval*** *for μ and* ***prediction interval*** *for y_0 are*

$$\bar{y} \pm t_{n-1,1-\alpha/2}\frac{s}{\sqrt{n}}, \qquad \bar{y} \pm t_{n-1,1-\alpha/2}\, s\sqrt{1 + \frac{1}{n}}.$$

Proof. y_0 is independent of $\bar{y}$, so the variances add: $\mathrm{Var}(y_0) + \mathrm{Var}(\bar{y}) = \sigma^2 + \sigma^2/n$. If the data are normal, $(y_0 - \bar{y})/\{s\sqrt{1 + 1/n}\}$ is a standard normal variable divided by an independent $\sqrt{\chi^2_{n-1}/(n-1)}$ (independence of $\bar{y}$ and s^2 is Exercise 1.7), hence t_{n-1}-distributed. The confidence interval follows in the same way from $(\bar{y} - \mu)/(s/\sqrt{n}) \sim t_{n-1}$. □

For model 2, the analogue at a covariate value t_0 is

$$\mathrm{Var}(\hat{y}_0) = \sigma^2\Big(\frac{1}{n} + \frac{(t_0 - \bar{t})^2}{S_{tt}}\Big), \qquad \mathrm{Var}(y_0 - \hat{y}_0) = \sigma^2\Big(1 + \frac{1}{n} + \frac{(t_0 - \bar{t})^2}{S_{tt}}\Big),$$

which follows from $\hat{y}_0 = \bar{y} + \hat{\beta}_1(t_0 - \bar{t})$ and Theorem 2.2 ($\bar{y}$ and $\hat{\beta}_1$ are uncorrelated). The width of both intervals increases with the distance of t_0 from the mean of the data, which is why extrapolation is risky.

```
tq <- qt(0.975, n - 1)
ci1 <- ybar + c(-1, 1) * tq * s / sqrt(n); pi1 <- ybar + c(-1, 1) * tq * s *
↪  sqrt(1 + 1 / n)
rbind(CI_mu = ci1, PI_y0 = pi1)
```

```
#>         [,1]  [,2]
#> CI_mu 50.83 51.49
#> PI_y0 48.61 53.71
```

```
new <- data.frame(year = 1972)
rbind(fit = predict(m2, new), predict(m2, new, interval = "confidence")[, -1],
↪  predict(m2, new, interval = "prediction")[, -1])
```

```
#>       lwr   upr
#> fit 52.29 52.29
#>     51.71 52.86
#>     50.01 54.56
```

The 95% confidence interval for the long-run mean temperature under model 1 is $(50.83, 51.49)$ whereas a 95% prediction interval for a single year is $(48.61, 53.71)$, nearly eight times wider. For 1972, model 2 predicts 52.29°F, with a confidence interval for the mean of $(51.71, 52.86)$ and a prediction interval for that year of $(50.01, 54.56)$.

Warning. Do not use the standard error of the mean for individual observations. Saying "95% of years lie in $\bar{y} \pm 2\,\mathrm{se}(\bar{y}) = 51.16 \pm 0.33$" is wrong: that interval is about the *mean*, and most individual years lie outside it. Equally, reporting the sample standard deviation s as the uncertainty of $\bar{y}$ overstates it by $\sqrt{n}$.

2.7 Data quality

A model, however carefully chosen, can only describe the data it is given. Three threats to quality are common: errors in measurement, missing values, and selection of the data.

2.7.1 Measurement error

Suppose the true model is $y = \beta_0 + \beta_1 x + \varepsilon$ but we observe $w = x + u$ in place of x, with $\mathrm{E}u = 0$, $\mathrm{Var}u = \sigma_u^2$, u independent of (x, ε), and $\mathrm{Var}x = \sigma_x^2$.

Proposition 2.2 (Attenuation). *The least-squares slope of y on w converges, as $n \to \infty$, to*

$$\beta_1 \frac{\sigma_x^2}{\sigma_x^2 + \sigma_u^2} = \lambda \beta_1, \qquad \lambda = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_u^2} \in (0, 1],$$

*the **reliability ratio**. If instead the* response *is measured with independent error, the slope is unbiased but the residual variance increases.*

Proof. The least-squares slope converges to $\mathrm{Cov}(y, w)/\mathrm{Var}(w)$. Now $\mathrm{Cov}(y, w) = \mathrm{Cov}(\beta_0 + \beta_1 x + \varepsilon,\ x + u) = \beta_1 \sigma_x^2$, since ε and u are independent of everything else, and $\mathrm{Var}(w) = \sigma_x^2 + \sigma_u^2$. For error v in the response, $y^{\mathrm{obs}} = y + v$ has $\mathrm{Cov}(y^{\mathrm{obs}}, x) = \beta_1 \sigma_x^2$ and $\mathrm{Var}(x) = \sigma_x^2$, so the slope is β_1; the residual variance becomes $\sigma^2 + \sigma_v^2$. □

Measurement error in a covariate therefore *biases the slope toward zero*, and more data do not repair it: the estimator is consistent for the wrong quantity. Error in the response only adds noise.

```
set.seed(2027)
N <- 1e5
x <- rnorm(N, 0, 1); yy <- 2 + 1 * x + rnorm(N, 0, 1); w <- x + rnorm(N, 0, 1)
↪  # sigma_x^2 = sigma_u^2 = 1
c(slope_true_x = unname(coef(lm(yy ~ x))[2]), slope_noisy_x =
↪  unname(coef(lm(yy ~ w))[2]), theory = 1 / (1 + 1))
```

```
#>  slope_true_x slope_noisy_x        theory
#>        1.0044        0.5034        0.5000
```

```
y_noisy <- yy + rnorm(N, 0, 1)
c(slope = unname(coef(lm(y_noisy ~ x))[2]), sigma_clean = summary(lm(yy ~
↪  x))$sigma, sigma_noisy = summary(lm(y_noisy ~ x))$sigma)
```

```
#>       slope sigma_clean sigma_noisy
#>      1.0023      0.9984      1.4145
```

2.7.2 Missing values

Let $R_i = 1$ if y_i is observed and $R_i = 0$ if it is missing. The way missingness arises determines whether ignoring it is harmful.

Definition 2.3 (Missing-data mechanisms)**.**

- **MCAR** (missing completely at random): $P(R_i = 0)$ depends on neither the observed nor the missing data. Discarding incomplete cases loses information but does not bias.
- **MAR** (missing at random): $P(R_i = 0)$ depends on *observed* data (e.g. a covariate) but, given those, not on the missing value. Complete-case analysis can be biased, but the bias can be corrected by modelling with the observed covariates.
- **MNAR** (missing not at random): $P(R_i = 0)$ depends on the missing value itself (e.g. high incomes not reported). No analysis of the observed data alone can remove the bias without extra assumptions.

The `airquality` data (daily air quality in New York, May–September 1973) have missing ozone values. We look at how missingness is related to what we observe.

```
data(airquality)
c(rows = nrow(airquality), complete = sum(complete.cases(airquality)))
```

```
#>     rows complete
#>      153      111
```

```
colSums(is.na(airquality))
```

```
#>   Ozone Solar.R    Wind    Temp   Month     Day
#>      37       7       0       0       0       0
```

```
miss <- with(airquality, tapply(is.na(Ozone), Month, sum)); days <-
↪  with(airquality, tapply(Ozone, Month, length))
rbind(missing = miss, days = days)
```

```
#>          5  6  7  8  9
#> missing  5 21  5  5  1
#> days    31 30 31 31 30
```

```
oz_month <- with(airquality, tapply(Ozone, Month, mean, na.rm = TRUE))
c(complete_case = mean(airquality$Ozone, na.rm = TRUE), month_weighted =
↪  sum(oz_month * days) / sum(days))
```

```
#>  complete_case month_weighted
#>          42.13          40.85
```

```
nobs(lm(Ozone ~ Solar.R + Wind + Temp, data = airquality))     # lm silently
↪  drops incomplete rows
```

```
#> [1] 111
```

Of the 153 days, 37 lack an ozone reading, and the missing values are *not* spread evenly across months: June has 21 of its 30 days missing, while the other months have between 1 and 5. The missingness depends on an observed variable (`Month`), so the data are not MCAR. The simple complete-case mean (42.1 ppb) differs from the mean reweighted so that each month counts in proportion to its length (40.9 ppb), because the months have different ozone levels. Under MAR one adjusts for the observed variable. `lm()` drops incomplete rows silently: the regression above uses 111 of 153 days.

Warning. Always check how many observations a fitted model used (`nobs()`), and ask *why* values are missing before choosing how to deal with them. Replacing missing values by the mean is not a solution: it understates the variance.

2.7.3 Selection

Selection occurs when the probability of inclusion in the sample depends on the outcome. Consider a thought experiment with the New Haven series: suppose that cold years (below 50°F) were never recorded, so that only years with $y \geq 50$ enter the data. This is MNAR by construction (not a claim about the real records).

```
sel <- subset(nh, temp >= 50)
c(n_all = nrow(nh), n_selected = nrow(sel))
```

```
#>      n_all n_selected
#>         60         49
```

```
c(mean_all = mean(nh$temp), mean_selected = mean(sel$temp))
```

```
#>      mean_all mean_selected
#>         51.16         51.59
```

```
c(slope_all = 10 * unname(coef(lm(temp ~ year, data = nh))[2]), slope_selected
↪  = 10 * unname(coef(lm(temp ~ year, data = sel))[2]))
```

```
#>      slope_all slope_selected
#>         0.3692         0.1409
```

The selected mean is biased upward, and the slope (per decade) drops from 0.37 to 0.14 because the cold early years have been removed. Selection on the *outcome* distorts both the mean and

the slope; selection on a *covariate* alone does not bias a correctly specified regression (Exercise 2.9).

2.8 Prediction and explanation

2.8.1 Two goals

A model can be built to **explain** (what is the effect of x on y? is $\beta_1 \neq 0$?) or to **predict** (what will y be for a new x?). Breiman (2001) contrasts a *data modelling culture*, which assumes a stochastic model and studies its parameters, with an *algorithmic modelling culture*, which judges models by their predictive accuracy; Shmueli (2010) argues that the two goals lead to different choices even within the same model class. In this course we mostly work in the first culture because of its interpretability, but we use predictive checks as a safeguard.

The two goals call for different criteria. For explanation we need the right *structure* and unbiased, interpretable coefficients, which requires that the model contains the relevant confounders: an association in observational data is not a causal effect. For prediction we need small *out-of-sample* error, to which an incorrect but simple model may contribute less error than a correct but complex one.

2.8.2 The anatomy of prediction error

Let $y_0 = \mu_0 + \varepsilon_0$ be a new observation at covariate value x_0 with true mean μ_0 and independent error of variance σ^2, and let $\hat{y}_0$ be a prediction computed from the data (so $\hat{y}_0$ is independent of ε_0).

Theorem 2.4 (Prediction error decomposition)**.**

$$\mathrm{E}(y_0 - \hat{y}_0)^2 = \underbrace{\sigma^2}_{\text{irreducible noise}} + \underbrace{\{\mathrm{E}(\hat{y}_0) - \mu_0\}^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}(\hat{y}_0)}_{\text{variance}}.$$

Proof. Write $y_0 - \hat{y}_0 = \varepsilon_0 + (\mu_0 - \hat{y}_0)$. The cross term $2\mathrm{E}\{\varepsilon_0(\mu_0 - \hat{y}_0)\} = 0$ because ε_0 is independent of $\hat{y}_0$ and has mean zero, so $\mathrm{E}(y_0 - \hat{y}_0)^2 = \sigma^2 + \mathrm{E}(\hat{y}_0 - \mu_0)^2$, and $\mathrm{E}(\hat{y}_0 - \mu_0)^2 = \mathrm{Var}(\hat{y}_0) + \{\mathrm{E}\hat{y}_0 - \mu_0\}^2$ by the usual bias–variance identity. □

The noise term cannot be reduced by any model. The other two terms trade off: a more flexible model reduces bias but increases variance. This is the principle behind regularisation, smoothing parameters and model selection in the later chapters. In the present setting, suppose the true model is the line $\mu_0 = \beta_0 + \beta_1 t_0$ and we predict with either the constant $\bar{y}$ (bias $\beta_1(t_0 - \bar{t})$, variance σ^2/n) or the fitted line (no bias, variance $\sigma^2\{1/n + (t_0 - \bar{t})^2/S_{tt}\}$). The constant has the smaller error exactly when $\beta_1^2(t_0 - \bar{t})^2 < \sigma^2(t_0 - \bar{t})^2/S_{tt}$, i.e. when $|\beta_1| < \sigma/\sqrt{S_{tt}} = \mathrm{se}(\hat{\beta}_1)$: the simpler model predicts better when the true slope is smaller than the standard error with which it can be estimated.

2.8.3 A hold-out check

To compare the predictive ability of the two models on data that was not used in fitting, we split the series: fit on 1912–1961 and predict 1962–1971.

```
train <- subset(nh, year <= 1961); test <- subset(nh, year > 1961)
h1 <- lm(temp ~ 1, data = train); h2 <- lm(temp ~ year, data = train)
rmse <- function(fit) sqrt(mean((test$temp - predict(fit, test))^2))
```

```
c(rmse_model1 = rmse(h1), rmse_model2 = rmse(h2))
```

```
#> rmse_model1 rmse_model2
#>      0.8287      1.1887
```

```
c(slope_train = unname(coef(h2)[2]), se = summary(h2)$coef[2, 2])
```

```
#> slope_train          se
#>     0.05081     0.01106
```

```
ggplot(nh, aes(year, temp)) + annotate("rect", xmin = 1961.5, xmax = 1971.5,
↪  ymin = -Inf, ymax = Inf, fill = "#D2E6F4") +
  geom_point(colour = ie_navy, size = 1.2) +
  geom_line(data = test, aes(y = predict(h1, test), colour = "Model 1")) +
  geom_line(data = test, aes(y = predict(h2, test), colour = "Model 2")) +
  geom_line(data = train, aes(y = fitted(h2), colour = "Model 2")) +
  ↪  geom_line(data = train, aes(y = fitted(h1), colour = "Model 1")) +
  scale_colour_manual(values = c(ie_coral, ie_blue)) + labs(x = "Year", y =
  ↪  "deg F", colour = NULL)
```

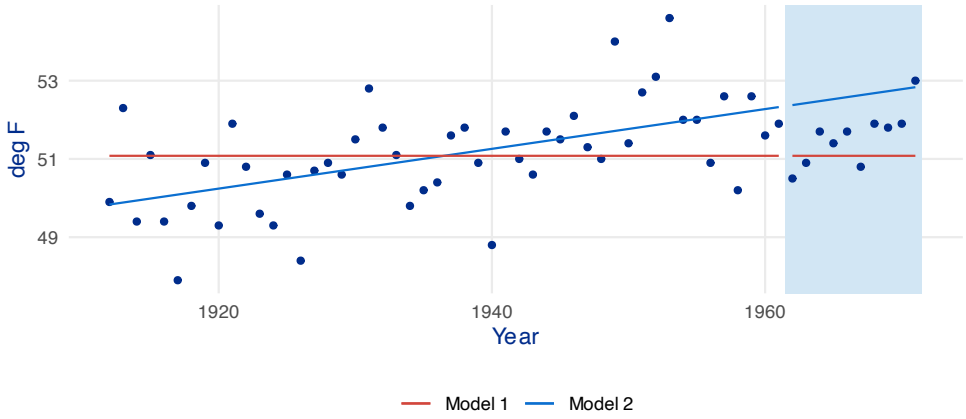


Figure 2.7: Hold-out check: models fitted to 1912–1961 predict 1962–1971 (shaded).

The result is a surprise: the constant model predicts the last ten years *better* (RMSE 0.83 against 1.19), although the trend is highly significant in the full series and in-sample the line always fits better. In these ten years the temperature happened to be below the extrapolated trend: with only 50 years of training data the estimated slope is 0.051 with a standard error of 0.011, and in the hold-out window the cost of extrapolating the line outweighs the cost of the bias of the constant. This is a single split of a single noisy series, so we should not conclude that the trend is spurious (the simulations of Section 2.5 show that it is not); we conclude that *in-sample fit and out-of-sample prediction answer different questions*, and that a model good for describing 1912–1971 is not necessarily good for extrapolating. Both models would be poor for the year 2050.

Remark. The honest way to assess prediction is to hold out data that played no role in model building (not in fitting, not in choosing between models), and to repeat the split in several ways (cross-validation, Chapter 7). For time series the split must respect the order in time.

2.9 A reproducible workflow

An analysis is *reproducible* if somebody else (or you, six months later) can rerun it and get the same numbers, tables and figures. This is not an extra: it is the only protection against silent errors, and it is a requirement of this course. Peng (2011) and Sandve et al. (2013) give the underlying reasons and rules. The minimal practice is:

1. **One project folder** with sub-folders `data/`, `R/` or `scripts/`, `output/`, and the report. Open it as an RStudio project so that all paths are *relative* to the project root; never use `setwd("C:/Users/...")`, which breaks on any other machine.
2. **Raw data are read-only.** Write cleaning steps as code, not as manual edits. Check the data on entry with `stopifnot(nrow(nh) == 60)` and `str()`; see Wickham (2014) for tidy data.
3. **Set a seed** before any random number generation: `set.seed(2027)`. The same seed gives the same numbers, and the generator is part of the record (`RNGkind()`).
4. **Literate programming:** write the report in R Markdown/Quarto so that numbers in the text come from code, as in these notes (`` `r round(mean(y), 2)` `` is replaced by the value).
5. **Record the software:** `sessionInfo()` at the end of the analysis, plus package versions.
6. **Version control** (git) and meaningful names, so that changes are tracked.

```
set.seed(2027); a <- rnorm(3)
set.seed(2027); b <- rnorm(3)
identical(a, b)                       # same seed, same numbers
```

```
#> [1] TRUE
```

```
identical(a, rnorm(3))                # no reset: different numbers
```

```
#> [1] FALSE
```

```
# A check that a stored copy of the data equals the package data
nh_csv <- tempfile(fileext = ".csv"); write.csv(nh, nh_csv, row.names = FALSE)
all.equal(read.csv(nh_csv)$temp, as.numeric(nhtemp))
```

```
#> [1] TRUE
```

```
R.version.string
```

```
#> [1] "R version 4.6.1 (2026-06-24)"
```

```
# A typical layout (relative paths from the project root):
# project/
#   data/nhtemp.csv         # raw data, never edited by hand
#   R/01-fit-models.R       # analysis scripts
#   report.Rmd              # report: reads data/, sources R/
#   output/                 # figures and tables written by code
nh <- read.csv("data/nhtemp.csv")      # relative path: works on any machine
stopifnot(nrow(nh) == 60, all(names(nh) == c("year", "temp")))
sessionInfo()                         # last line of every report
```

2.9.1 In R: the functions of this chapter

Task	Function
Fit constant mean / line	`lm(temp ~ 1)`, `lm(temp ~ year)`
Coefficients, standard errors, $\hat{\sigma}$	`coef()`, `summary()$coef`, `sigma()`
Fitted values and residuals	`fitted()`, `resid()`
Compare nested models	`anova(m1, m2)`, `logLik()`
Residual diagnostics	`qqnorm()`, `acf()`, `shapiro.test()`, `plot(m)`
Simulate from a fitted model	`simulate(m, nsim = 20)`, `rnorm()`, `replicate()`
Intervals	`confint()`, `predict(m, new, interval = "confidence" / "prediction")`
Missing values	`is.na()`, `complete.cases()`, `nobs()`, `na.action`

2.10 Common pitfalls

Warning. *Smaller RSS is not better.* A larger model always has the smaller RSS. Judge by the test, an information criterion or out-of-sample error.

Warning. *Mixing s and $s/\sqrt{n}$.* The sample standard deviation describes observations; the standard error describes the estimate.

Warning. *Confidence versus prediction.* A confidence interval for the mean is not a range for future observations.

Warning. *Checking the wrong thing first.* Misspecified means produce residual patterns that look like dependence or heteroscedasticity: fix the mean first.

Warning. *Absence of evidence.* A non-significant normality test does not show normality, and a non-significant slope does not show that there is no slope.

Warning. *More data does not fix bias.* Measurement error in covariates and selection on the outcome bias estimates for any sample size.

Warning. *Silent deletion.* `lm()` drops rows with missing values without a message; check `nobs()`.

Warning. *Extrapolation.* Prediction intervals widen away from the data and are valid only if the model holds there. A trend fitted to 1912–1971 says nothing reliable about 2050.

Warning. *Explaining is not predicting.* A good predictor need not have interpretable coefficients, and a significant coefficient in observational data is an association, not a causal effect.

2.11 Chapter summary

- A statistical model is a family of distributions; typically $y_i = f(x_i; \beta) + \varepsilon_i$ with a systematic part (signal) and a random part (noise). All models are approximations; the question is adequacy for a purpose.
- Least squares for the constant mean gives $\hat{\mu} = \bar{y}$ with $\mathrm{Var}(\bar{y}) = \sigma^2/n$ (requires uncorrelated errors); for the line $\hat{\beta}_1 = S_{ty}/S_{tt}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{t}$, $\mathrm{Var}(\hat{\beta}_1) = \sigma^2/S_{tt}$.
- A model containing another always has smaller RSS; $R^2 = r^2$ measures explained scatter, not existence of an effect.
- Assumptions in order of importance: mean structure, independence, constant variance, normality. Check with residual plots, Q–Q plots and ACF; fix the mean first.

- A fitted model can be simulated from: line-ups, Monte Carlo sampling distributions, p-values by simulation.
- Variability (σ) does not shrink with n; uncertainty ($\sigma/\sqrt{n}$) does. $\mathrm{Var}(y_0 - \bar{y}) = \sigma^2(1+1/n)$; prediction intervals are wider than confidence intervals.
- Data quality: covariate measurement error attenuates slopes by $\sigma_x^2/(\sigma_x^2+\sigma_u^2)$; missingness is MCAR, MAR or MNAR; selection on the outcome biases.
- Prediction error $= \sigma^2 + \text{bias}^2 + \text{variance}$; in-sample fit and out-of-sample prediction answer different questions; explanation and prediction have different goals.
- Reproducibility: project folder, relative paths, seed, literate report, `sessionInfo()`.

2.12 Exercises with explained solutions

Exercise 2.1. Let $y_1, \ldots, y_n$ be real numbers.

(a) Show that $Q(\mu) = \sum_i (y_i - \mu)^2$ is minimised at $\bar{y}$ by proving the decomposition $Q(\mu) = \sum_i (y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$.

(b) Show that $A(\mu) = \sum_i |y_i - \mu|$ is minimised at a sample median. (Hint: for μ not equal to any y_i, $A'(\mu) = \#\{y_i < \mu\} - \#\{y_i > \mu\}$.)

(c) For `nhtemp` suppose that the warmest year (the maximum) had been recorded as 10°F higher by mistake. Compute the change in the mean and in the median.

Solution.

(a) See the proof of Theorem 2.1.

(b) For μ not equal to any y_i, each term $|y_i - \mu|$ has derivative -1 if $y_i > \mu$ and $+1$ if $y_i < \mu$, so $A'(\mu) = \#\{y_i < \mu\} - \#\{y_i > \mu\}$. As μ increases from $-\infty$, A' starts at $-n$ and increases in steps of 2 at each data point, so A decreases until μ reaches the point where the number of observations below equals the number above, and increases afterwards. The minimum is at the median (for odd n the middle order statistic; for even n any point between the two middle ones, since A is flat there).

(c)

```
yy2 <- nh$temp; i <- which.max(yy2); yy3 <- yy2; yy3[i] <- yy3[i] + 10
c(mean_change = mean(yy3) - mean(yy2), median_change = median(yy3) -
↪  median(yy2))
```

```
#>   mean_change median_change
#>        0.1667        0.0000
```

The mean moves by $10/60 = 0.167$°F; the median does not move at all, because the changed value was already above the median. The mean is sensitive to outliers (its influence is unbounded), the median is robust, which is why the choice of loss function matters.

Exercise 2.2. Let $y_1, \ldots, y_n$ have $\mathrm{E}(y_i) = \mu$, $\mathrm{Var}(y_i) = \sigma^2$.

(a) Show that $\mathrm{E}(\bar{y}) = \mu$ whatever the dependence.

(b) If $\mathrm{Cov}(y_i, y_{i+1}) = \rho\sigma^2$ for $i = 1, \ldots, n-1$ and all other covariances are zero, show that $\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}\{1 + 2\rho\frac{n-1}{n}\}$.

(c) With $n = 60$, $s = 1.266$ and $\rho = 0.3$, compute the standard error with and without the dependence and the percentage by which the independent calculation understates it.

(d) A 95% interval $\bar{y} \pm 1.96\, s/\sqrt{n}$ is computed ignoring the dependence. What is its true coverage under the dependent model (use normality)?

Solution.

(a) $\mathrm{E}\bar{y} = \frac{1}{n}\sum \mathrm{E}y_i = \mu$ by linearity; no assumption on dependence is used.

(b) $\mathrm{Var}(\bar{y}) = n^{-2}\{\sum_i \mathrm{Var}(y_i) + 2\sum_{i<j}\mathrm{Cov}(y_i, y_j)\}$ and there are $n-1$ adjacent pairs, each with covariance $\rho\sigma^2$, so $\mathrm{Var}(\bar{y}) = n^{-2}\{n\sigma^2 + 2(n-1)\rho\sigma^2\} = \frac{\sigma^2}{n}\{1 + 2\rho\frac{n-1}{n}\}$.

(c) Independent: $1.266/\sqrt{60} = 0.1634$. With dependence the factor is $1+2(0.3)(59/60) = 1.59$ so the se is $0.1634\sqrt{1.59} = 0.2060$, i.e. 26% larger; the independent calculation understates it by $1 - 1/\sqrt{1.59} = 21\%$ (of the correct value).

(d) The interval half-width is 1.96×0.1634 while the true standard deviation of $\bar{y}$ is 0.2060, so the coverage is $P(|Z| < 1.96 \times 0.1634/0.2060) = P(|Z| < 1.555) = 0.880$. The nominal 95% interval covers only about 88% of the time.

```
se0 <- 1.266 / sqrt(60); f <- 1 + 2 * 0.3 * 59 / 60
c(se0 = se0, se_dep = se0 * sqrt(f), ratio = sqrt(f), coverage = 2 *
↪  pnorm(1.96 / sqrt(f)) - 1)
```

```
#>      se0   se_dep    ratio coverage
#>   0.1634   0.2061   1.2610   0.8799
```

Exercise 2.3. For the New Haven series under model 1: $n = 60$, $\bar{y} = 51.16$, $s = 1.266$, $t_{59,0.975} = 2.001$.

(a) Compute $\mathrm{se}(\bar{y})$ and a 95% confidence interval for μ.

(b) Compute a 95% prediction interval for a new year.

(c) How many years would be needed for $\mathrm{se}(\bar{y}) \le 0.05$? Could the prediction interval ever be narrower than ± 2°F? Explain with the words *variability* and *uncertainty*.

Solution.

(a) $\mathrm{se} = 1.266/\sqrt{60} = 0.1634$; the interval is $51.16 \pm 2.001 \times 0.1634 = 51.16 \pm 0.327 = (50.83, 51.49)$.

(b) By Theorem 2.3 the half-width is $2.001 \times 1.266\sqrt{1 + 1/60} = 2.001 \times 1.266 \times 1.0083 = 2.554$, giving $(48.61, 53.71)$.

(c) $\sigma/\sqrt{n} \le 0.05$ needs $n \ge (1.266/0.05)^2 = 641.1$, so $n = 642$ years (assuming the process is constant and independent). The prediction half-width is at least $1.96\sigma = 2.48$ even for $n = \infty$, since the factor $\sqrt{1 + 1/n} \to 1$; it can never be below ± 2. More data shrink the *uncertainty* about μ to zero, but the *variability* of an individual year around the mean remains.

```
c(se = 1.266 / sqrt(60), n_needed = (1.266 / 0.05)^2, limit_halfwidth =
↪  qnorm(0.975) * 1.266)
```

```
#>              se        n_needed limit_halfwidth
#>          0.1634        641.1024          2.4813
```

Exercise 2.4. Consider the model $y_i = \beta x_i + \varepsilon_i$ (a line through the origin), with uncorrelated errors of mean zero and variance σ^2, and fixed x_i not all zero.

(a) Derive the least-squares estimator.
(b) Show that it is unbiased with variance $\sigma^2 / \sum x_i^2$.
(c) Data: $x = (1, 2, 3, 4, 5)$, $y = (2.1, 3.9, 6.2, 7.8, 10.1)$. Compute $\hat\beta$, $\hat\sigma^2 = \mathrm{RSS}/(n-1)$ and $\mathrm{se}(\hat\beta)$.
(d) Name a situation in which forcing the line through the origin is a poor approximation.

Solution.

(a) $Q(\beta) = \sum (y_i - \beta x_i)^2$, $Q'(\beta) = -2\sum x_i(y_i - \beta x_i) = 0$ gives $\hat\beta = \sum x_i y_i / \sum x_i^2$; $Q'' = 2\sum x_i^2 > 0$.

(b) $\hat\beta = \sum c_i y_i$ with $c_i = x_i / \sum x_j^2$; $\mathrm{E}\hat\beta = \beta \sum c_i x_i = \beta$ and $\mathrm{Var}\hat\beta = \sigma^2 \sum c_i^2 = \sigma^2 / \sum x_j^2$.

(c) $\sum x_i y_i = 2.1 + 7.8 + 18.6 + 31.2 + 50.5 = 110.2$ and $\sum x_i^2 = 55$, so $\hat\beta = 2.0036$. Residuals $y_i - 2.0036 x_i$: 0.0964, -0.1073, 0.1891, -0.2145, 0.0818 with $\mathrm{RSS} = 0.0093 + 0.0115 + 0.0358 + 0.0460 + 0.0067 = 0.1093$. Dividing by $n - 1 = 4$ (one parameter): $\hat\sigma^2 = 0.0273$, $\mathrm{se}(\hat\beta) = \sqrt{0.0273/55} = 0.0223$.

```
xo <- 1:5; yo <- c(2.1, 3.9, 6.2, 7.8, 10.1)
fo <- lm(yo ~ 0 + xo); c(beta = coef(fo), RSS = sum(resid(fo)^2), sigma2 =
↪  sigma(fo)^2, se = summary(fo)$coef[1, 2])
```

```
#> beta.xo     RSS  sigma2      se
#> 2.00364 0.10927 0.02732 0.02229
```

(d) When the response has a non-zero baseline at $x = 0$, e.g. temperature versus altitude: the forced line distorts the slope. Note also that with no intercept the residuals no longer sum to zero and the usual R^2 is not comparable.

Exercise 2.5. For the straight-line model (Theorem 2.2):

(a) Show that $\mathrm{Var}(\hat\beta_0) = \sigma^2(1/n + \bar t^2 / S_{tt})$ and explain why centring t makes the estimators of the intercept and slope uncorrelated.
(b) For `nhtemp` compute S_{tt}, $\hat\sigma$, $\mathrm{se}(\hat\beta_1)$ and $\mathrm{se}(\hat\beta_0)$; compute them again with the year centred at 1912 and comment.
(c) Show $\mathrm{Var}(\hat y_0) = \sigma^2\{1/n + (t_0 - \bar t)^2 / S_{tt}\}$ and find the value of t_0 where the prediction is most precise.

Solution.

(a) $\hat\beta_0 = \bar y - \hat\beta_1 \bar t$ and $\mathrm{Cov}(\bar y, \hat\beta_1) = 0$ so $\mathrm{Var}(\hat\beta_0) = \sigma^2/n + \bar t^2 \sigma^2 / S_{tt}$. Also $\mathrm{Cov}(\hat\beta_0, \hat\beta_1) = -\bar t \mathrm{Var}(\hat\beta_1) = -\bar t \sigma^2 / S_{tt}$, which is zero exactly if $\bar t = 0$: with centred t the intercept is $\bar y$ and is uncorrelated with the slope.

(b)

```
Stt <- sum((nh$year - mean(nh$year))^2); sg <- sigma(m2)
c(Stt = Stt, sigma = sg, se_b1 = sg / sqrt(Stt), se_b0 = sg * sqrt(1 / n +
↪  mean(nh$year)^2 / Stt))
```

```
#>       Stt     sigma     se_b1     se_b0
#> 1.800e+04 1.098e+00 8.188e-03 1.590e+01
```

```
summary(m2)$coef[, 2]
```

```
#> (Intercept)        year
#>   15.897593    0.008188
```

```
m2c <- lm(temp ~ I(year - 1912), data = nh); summary(m2c)$coef[, 1:2]
```

```
#>                 Estimate Std. Error
#> (Intercept)    50.07082   0.280092
#> I(year - 1912)  0.03692   0.008188
```

$S_{tt} = 17995$ ($= n(n^2-1)/12$ for 60 consecutive years). The intercept of the uncentred model has a huge standard error (about 15) because it is an extrapolation to year 0; at the centred origin 1912 the standard error is only about 0.3. The slope and its standard error are identical.

(c) $\hat{y}_0 = \bar{y} + \hat{\beta}_1(t_0 - \bar{t})$ with $\bar{y}$ and $\hat{\beta}_1$ uncorrelated gives $\mathrm{Var}\hat{y}_0 = \sigma^2/n + (t_0-\bar{t})^2\sigma^2/S_{tt}$, minimised at $t_0 = \bar{t} = 1941.5$ where it equals σ^2/n. Precision falls quadratically away from the centre of the data.

Exercise 2.6.
(a) Prove that for the least-squares line $R^2 = r_{ty}^2$.
(b) For `nhtemp` check that $r_{ty}^2 = R^2$ and compute the F statistic $F = R^2(n-2)/(1-R^2)$ and its relation with t^2 for the slope.
(c) Explain, with a sentence, why a model with $R^2 = 0.26$ can nevertheless have a very significant slope.

Solution.

(a) See Proposition 2.1.

(b)

```
r <- cor(nh$year, nh$temp); R2 <- r^2
c(r = r, R2 = R2, R2_lm = summary(m2)$r.squared, F = R2 * (n - 2) / (1 - R2),
↪  t2 = summary(m2)$coef[2, 3]^2)
```

```
#>       r      R2   R2_lm       F      t2
#>  0.5095  0.2596  0.2596 20.3330 20.3330
```

$F = t^2$ for the slope in simple regression (one numerator degree of freedom).

(c) Significance depends on the size of the slope relative to its standard error, which decreases with n and with the spread of t; R^2 depends on how much of the total scatter the line removes. With much noise per observation but 60 observations the slope is well estimated though the scatter around the line stays large.

Exercise 2.7. The true model is $y = \beta_0 + \beta_1 x + \varepsilon$; we observe $w = x + u$ with u independent noise of variance σ_u^2.
(a) Derive the limit of the least-squares slope of y on w.
(b) With $\beta_1 = 2$, $\sigma_x^2 = 4$, $\sigma_u^2 = 1$, what slope do we expect with a huge sample? Verify by simulation.
(c) The error is now in the response, $y^{\mathrm{obs}} = y + v$. Is the slope biased? What happens to $\hat{\sigma}^2$ and to $\mathrm{se}(\hat{\beta}_1)$?

(d) A researcher says "the measurement error is random, so it averages out". Respond.

Solution.

(a) $\mathrm{Cov}(y, w) = \beta_1 \sigma_x^2$, $\mathrm{Var}(w) = \sigma_x^2 + \sigma_u^2$; the slope tends to $\beta_1 \sigma_x^2/(\sigma_x^2 + \sigma_u^2)$ (Proposition 2.2).

(b) $2 \times 4/(4+1) = 1.6$.

```
set.seed(2027); N <- 2e5
xs <- rnorm(N, 0, 2); ys <- 1 + 2 * xs + rnorm(N); ws <- xs + rnorm(N)
c(slope_on_w = unname(coef(lm(ys ~ ws))[2]), theory = 1.6, slope_on_x =
↪  unname(coef(lm(ys ~ xs))[2]))
```

```
#> slope_on_w     theory slope_on_x
#>      1.603      1.600      2.001
```

(c) No: $\mathrm{Cov}(y + v, x) = \beta_1 \sigma_x^2$ so the slope is unbiased. The residual variance increases to $\sigma^2 + \sigma_v^2$, so $\mathrm{se}(\hat{\beta}_1) = \sqrt{(\sigma^2 + \sigma_v^2)/S_{xx}}$ increases: less precision, no bias.

(d) The noise averages out in a *mean* of w but not in the *regression slope*: the variance of w contains σ_u^2 that is unrelated to y, and this dilutes the association. The bias does not shrink with n. A remedy needs information on the error (replicate measurements, a validation sample, instruments).

Exercise 2.8. Let $Y \sim N(0, 1)$ and suppose values above 1 are never observed.

(a) Show that $\mathrm{E}(Y \mid Y \leq 1) = -\phi(1)/\Phi(1)$ where ϕ, Φ are the standard normal density and distribution function. Compute it.

(b) Verify by simulation. What type of missingness is this?

(c) Now suppose that values are missing with probability 1/2 regardless of Y. What is the mean of the observed values, and what type of missingness is this?

Solution.

(a) $\mathrm{E}(Y \mid Y \leq 1) = \frac{1}{\Phi(1)} \int_{-\infty}^{1} y\phi(y)dy = \frac{1}{\Phi(1)}[-\phi(y)]_{-\infty}^{1} = -\phi(1)/\Phi(1)$ using $\phi'(y) = -y\phi(y)$. Numerically $-0.2420/0.8413 = -0.2876$.

(b)

```
set.seed(2027); z <- rnorm(1e6); obs <- z[z <= 1]
c(simulated = mean(obs), theory = -dnorm(1) / pnorm(1), share_observed =
↪  mean(z <= 1))
```

```
#>      simulated         theory share_observed
#>        -0.2876        -0.2876         0.8418
```

Missingness depends on the missing value itself: MNAR. The observed mean is biased downward (-0.29 instead of 0) whatever n is.

(c) If values are deleted with probability 1/2 independently of Y (MCAR), the observed values are a random half of the sample: the mean is still 0, with a larger standard error $(1/\sqrt{n/2})$.

Exercise 2.9. Let $y = 1 + 2x + \varepsilon$ with $x \sim N(0, 1)$, $\varepsilon \sim N(0, 1)$ independent.

(a) Simulate $n = 10^5$ observations and fit the slope in the whole sample, in the subsample

with $x > 0$ (selection on the covariate) and in the subsample with $y > 1$ (selection on the outcome).

(b) Explain the results. Why is selection on x harmless but selection on y not?

Solution.

(a)

```
set.seed(2027); N <- 1e5
xs <- rnorm(N); ys <- 1 + 2 * xs + rnorm(N)
c(all = unname(coef(lm(ys ~ xs))[2]), x_positive = unname(coef(lm(ys[xs > 0] ~
↪  xs[xs > 0]))[2]),
  y_above_1 = unname(coef(lm(ys[ys > 1] ~ xs[ys > 1]))[2]))
```

```
#>        all x_positive  y_above_1
#>      2.004      1.984      1.482
```

(b) Conditionally on x, the distribution of y is $N(1 + 2x, 1)$ whatever the subset of x values; restricting to $x > 0$ keeps the regression function unchanged (and reduces the spread of x, so the slope is less precise). Selecting on $y > 1$ removes observations according to the *error*: at a given x, only those with $\varepsilon > -2x$ remain, so the retained errors are no longer mean zero and depend on x, which flattens the slope (here to about 1.5).

Exercise 2.10.

(a) Prove Theorem 2.4.

(b) For prediction at t_0 when the truth is the line, show that the constant-mean predictor has smaller expected squared prediction error than the fitted line exactly when $|\beta_1| < \sigma/\sqrt{S_{tt}}$.

(c) For `nhtemp` compare $|\hat{\beta}_1|$ with $\hat{\sigma}/\sqrt{S_{tt}}$ and with the same quantities for the training period 1912–1961. What does the comparison say about the hold-out result of Section 2.8?

Solution.

(a) See the proof in the text.

(b) Constant: bias $= \beta_1(t_0 - \bar{t})$ (since $\mathrm{E}\bar{y} = \beta_0 + \beta_1\bar{t}$), variance σ^2/n. Line: bias 0, variance $\sigma^2\{1/n + (t_0 - \bar{t})^2/S_{tt}\}$. The constant is better iff $\beta_1^2(t_0 - \bar{t})^2 + \sigma^2/n < \sigma^2/n + \sigma^2(t_0 - \bar{t})^2/S_{tt}$, i.e. (for $t_0 \neq \bar{t}$) $\beta_1^2 < \sigma^2/S_{tt}$.

(c)

```
c(full_b1 = unname(coef(m2)[2]), full_ratio = unname(coef(m2)[2]) / (sigma(m2)
↪  / sqrt(Stt)))
```

```
#>    full_b1 full_ratio
#>    0.03692    4.50922
```

```
Stt_tr <- sum((train$year - mean(train$year))^2)
c(train_b1 = unname(coef(h2)[2]), train_ratio = unname(coef(h2)[2]) /
↪  (sigma(h2) / sqrt(Stt_tr)))
```

```
#>    train_b1 train_ratio
#>     0.05081     4.59450
```

The ratio $\hat{\beta}_1/\mathrm{se}(\hat{\beta}_1)$ is the t-statistic: about 4.5 on the full series and about 4.6 on the training period. The criterion of (b) concerns the *true* slope, which we do not know; the condition $|\beta_1| < \mathrm{se}$ is not met on either sample, so the line is expected to predict better on average. (The training slope, 0.051, is also steeper than the full-sample slope, 0.037, so the line extrapolated from 1912–61 overshoots.) The hold-out result goes the other way because the 1962–71 window is a single, short realisation of the noise (and there may be genuine departures from linearity): the hold-out check is itself uncertain, and decisions should rest on repeated validation rather than one split.

Exercise 2.11.
(a) Simulate $B = 2000$ series of length $n = 60$ from model 2 with the fitted parameters and compute the coverage of the 95% confidence interval for β_1.
(b) Repeat with errors that follow a first-order autoregression with coefficient 0.5 and the same marginal variance σ^2. What happens to the coverage? Why?

Solution. (a)–(b)

```
set.seed(2027)
tt <- nh$year; b1_true <- unname(coef(m2)[2]); mu_t <- fitted(m2); sg <-
↪  sigma(m2)
cover <- function(ar) mean(replicate(2000, {
  e <- if (ar == 0) rnorm(n, 0, sg) else as.numeric(arima.sim(list(ar = ar),
↪  n, sd = sg * sqrt(1 - ar^2)))
  f <- lm((mu_t + e) ~ tt); ci <- confint(f)[2, ]
  ci[1] <= b1_true && b1_true <= ci[2]
}))
c(iid = cover(0), ar1 = cover(0.5))
```

```
#>    iid    ar1
#> 0.9545 0.7370
```

With independent errors the coverage is close to the nominal 95% (up to Monte Carlo error ≈ 0.005). With positively autocorrelated errors it drops markedly (to about 74%): the residuals carry less independent information than $n = 60$ suggests, so the standard error computed under independence is too small and the interval too narrow. The remedy is a model for the dependence (Chapters 11 and 19 for the tools).

Exercise 2.12. A news article states: "Temperatures in New Haven rise by 0.037°F per year. Therefore in 2070 the mean annual temperature will be 55°F." (a) Identify three problems with the statement. (b) Compute what model 2 predicts for 2070 with a 95% prediction interval and comment. (c) Write a two-sentence correct statement for a non-statistician.

Solution.

(a) (i) It extrapolates 100 years beyond the data, assuming the straight line continues, but nothing in the model supports that. (ii) It gives a point prediction without uncertainty. (iii) It treats an association in one station's record as a law (and ignores the dependence and possible station changes); it also mixes the *mean trend* with the temperature of a single year.

(b)

```
predict(m2, data.frame(year = 2070), interval = "prediction")
```

```
#>    fit   lwr   upr
#> 1 55.9 52.85 58.96
```

```
predict(m2, data.frame(year = 2070), interval = "confidence")
```

```
#>    fit   lwr   upr
#> 1 55.9 53.78 58.03
```

The point prediction (about 55.9°F in our fit) comes with a 95% prediction interval of roughly ±3°F that only reflects sampling uncertainty *if the model were true*; the true uncertainty is larger because of model uncertainty.

(c) "Between 1912 and 1971, the average annual temperature in New Haven rose by about 0.37°F per decade (plausible range 0.2 to 0.5), although individual years varied by about 1.1°F around this trend. This describes the past record of one station; it does not tell us what will happen decades from now."

Exercise 2.13. The following script is sent to you by a colleague.

```
setwd("C:/Users/maria/Desktop/stats")
temp <- c(49.9, 52.3, 49.4, 51.1, 49.4)   # typed from a printed table
n <- 100
x <- rnorm(n)
fit <- lm(x ~ 1)
print(summary(fit))
```

(a) List five reproducibility problems. (b) Rewrite it.

Solution.

(a) (1) `setwd()` with an absolute path breaks on any other computer. (2) The data are typed by hand, with no source and no check (values of unknown origin; `nhtemp` has 60 values and loads with `data(nhtemp)`). (3) `temp` is never used: the analysis does not match the data (dead code). (4) `rnorm` without `set.seed()`: different numbers on every run. (5) No record of the software versions and no check of n; the number 100 is hard-coded rather than computed (`n <- nrow(...)`). Also: the output goes to the console only.

(b)

```
# Run from the project folder (RStudio project); data/ holds the raw data.
nh <- read.csv("data/nhtemp.csv")
stopifnot(nrow(nh) == 60, all(names(nh) == c("year", "temp")))
set.seed(2027)
fit <- lm(temp ~ year, data = nh)
summary(fit)
sessionInfo()
```

Relative path, data read from a file (or from `data(nhtemp)`), a check on the size, an explicit seed if randomness is used, an analysis of the actual data, and a record of the environment.

2.13 Further reading

- Box (1976), "Science and Statistics": the origin of "all models are wrong" and the iterative cycle of model building.
- Breiman (2001), "Statistical Modeling: The Two Cultures", and the discussion of Shmueli (2010), "To explain or to predict?".
- Pawitan (2001), Chapter 1, for a likelihood-oriented view of models, and Faraway (2014), Chapter 1, for the modelling process in regression.
- Peng (2011) and Sandve et al. (2013) for reproducible research.

Chapter 3

Likelihood and maximum likelihood

Almost every model in this book is fitted by maximum likelihood: the linear model (Chapter 5), generalised linear models (Chapters 8 and 9), mixed models (Chapters 11 to 13), smoothers and additive models (Chapters 14 to 19) and survival models (Chapter 20). This chapter builds the machinery once, in the simplest possible settings, so that later chapters can simply say "the MLE of β is approximately normal with covariance matrix $I(\hat{\beta})^{-1}$" and the reader knows exactly what that sentence means, where it comes from and when it fails.

The plan is the following. Section 3.1 introduces two data sets that we use throughout. Section 3.2 defines the likelihood and explains what it is *not*. Section 3.3 defines the maximum likelihood estimator and works out the standard examples. Section 3.4 introduces the score and the two information measures, and proves the identities that make them useful. Section 3.5 proves the main properties of the MLE: invariance, consistency, asymptotic normality and the Cramér–Rao bound. Section 3.6 extends the results to vector parameters, Section 3.8 shows what can go wrong when the likelihood has several maxima, and Section 3.9 collects the R functions we use. Chapter 4 continues with tests, intervals and numerical optimisation.

Notation follows Appendix A. For data $y_1, \ldots, y_n$ and a parameter θ, $L(\theta)$ is the likelihood, $\ell(\theta) = \log L(\theta)$ the log-likelihood, $S(\theta) = \partial \ell / \partial \theta$ the score, $I(\theta) = -\partial^2 \ell / \partial \theta^2$ the **observed** information of the whole sample, $\mathcal{I}(\theta) = \mathrm{E}[I(\theta)]$ the **expected** (Fisher) information, and $\mathcal{I}_1(\theta)$ the expected information of a single observation, so that $\mathcal{I}(\theta) = n\mathcal{I}_1(\theta)$ for iid data. The standard error of the MLE is $\mathrm{se}(\hat{\theta}) = I(\hat{\theta})^{-1/2}$. There is no extra factor σ: if σ is a parameter of the model, it is already inside I.

3.1 Two motivating data sets

Air-conditioning failures. The data set `aircondit` in package `boot` contains the times (in operating hours) between successive failures of the air-conditioning equipment in one aircraft (Pawitan, 2001). There are only twelve numbers.

```
data(aircondit, package = "boot")
y_ac <- aircondit$hours
n_ac <- length(y_ac)
y_ac
```

```
#>  [1]   3   5   7  18  43  85  91  98 100 130 230 487
```

```
c(n = n_ac, mean = mean(y_ac), median = median(y_ac), sd = sd(y_ac))
```

```
#>       n    mean median      sd
#>    12.0   108.1   88.0   136.2
```

The data are positive, strongly right-skewed (mean 108 hours, median 88, maximum 487), so a normal model is a poor description. A natural first model is the exponential distribution with mean μ. How should we estimate μ, how precisely, and is the exponential model itself adequate? These are the questions of this chapter and of Chapter 4.

Great discoveries. The built-in time series `discoveries` records the number of "great" scientific discoveries and inventions in each year from 1860 to 1959 ($n = 100$ counts). Counts of rare events suggest a Poisson model.

```
y_di <- as.numeric(discoveries)
n_di <- length(y_di)
obs <- table(factor(y_di, levels = 0:12)) / n_di
df <- data.frame(k = 0:12, obs = as.numeric(obs), fit = dpois(0:12,
↪  mean(y_di)))
ggplot(df, aes(k)) +
  geom_col(aes(y = obs), fill = ie_blue, alpha = .6) +
  geom_point(aes(y = fit), colour = ie_coral, size = 2) +
  geom_line(aes(y = fit), colour = ie_coral) +
  labs(x = "discoveries in a year", y = "relative frequency")
```

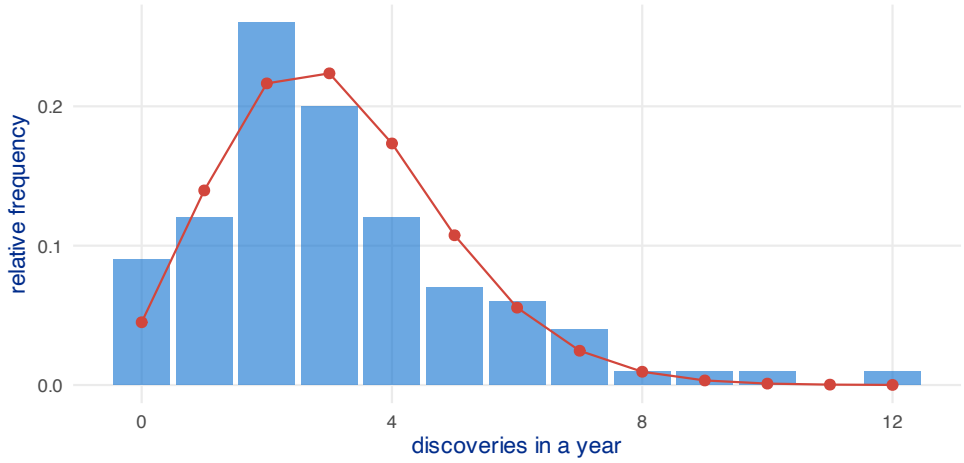


Figure 3.1: Great discoveries per year, 1860–1959, with the Poisson probabilities for the fitted rate $\hat{\lambda} = \bar{y} = 3.1$.

```
c(mean = mean(y_di), variance = var(y_di))
```

```
#>     mean variance
#>    3.100    5.081
```

The Poisson model describes the shape reasonably well, but the sample variance (5.08) is considerably larger than the mean (3.10), whereas a Poisson variable has equal mean and variance. We will see in Chapter 4 how to make this observation precise.

3.2 The likelihood function

Let $y = (y_1, \dots, y_n)$ be observed data, assumed to be a realisation of a random vector with joint density (or probability mass function) $f(y; \theta)$ depending on an unknown parameter $\theta \in \Theta \subseteq \mathbb{R}^p$.

Definition 3.1 (Likelihood and log-likelihood)**.** The **likelihood** is the function $L(\theta) = f(y;\theta)$ of θ, with the observed data y held fixed. Any function $c(y)\,L(\theta)$ with $c(y) > 0$ not depending on θ is equally a likelihood. The **log-likelihood** is $\ell(\theta) = \log L(\theta)$, defined up to an additive constant. If the observations are independent with densities $f_i(y_i;\theta)$,

$$L(\theta) = \prod_{i=1}^{n} f_i(y_i;\theta), \qquad \ell(\theta) = \sum_{i=1}^{n} \log f_i(y_i;\theta). \tag{3.1}$$

Two remarks are essential.

First, the likelihood is **not a probability density for** θ. Before the data are observed, $f(y;\theta)$ is a density in y and integrates to one over y. After the data are observed, we vary θ and the integral over θ has no reason to be one. For y successes in n Bernoulli trials, $L(\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$ and, by the Beta integral,

$$\int_0^1 L(\theta)\,d\theta = \binom{n}{y}\frac{y!\,(n-y)!}{(n+1)!} = \frac{1}{n+1}. \tag{3.2}$$

For $n = 10$ this is $1/11$, not 1. Statements such as "the probability that θ lies in $(0.6, 0.9)$ is the area under L" are meaningless in this framework; they require a prior distribution (Bayesian inference).

Second, only **ratios** $L(\theta_1)/L(\theta_2)$ are meaningful, since they do not depend on the arbitrary constant $c(y)$. They compare how well two parameter values explain the observed data. The **relative likelihood** $L(\theta)/L(\hat\theta) \in (0, 1]$ is the standard way to display this.

```
n <- 10; y <- 8
theta <- seq(0.01, 0.99, by = 0.005)
rel <- dbinom(y, n, theta) / dbinom(y, n, y / n)
ggplot(data.frame(theta, rel), aes(theta, rel)) + geom_line(colour = ie_navy,
↪  linewidth = 1) +
  geom_hline(yintercept = exp(-qchisq(0.95, 1) / 2), linetype = 3) +
  geom_vline(xintercept = y / n, linetype = 2) +
  labs(x = expression(theta), y = "relative likelihood")
```

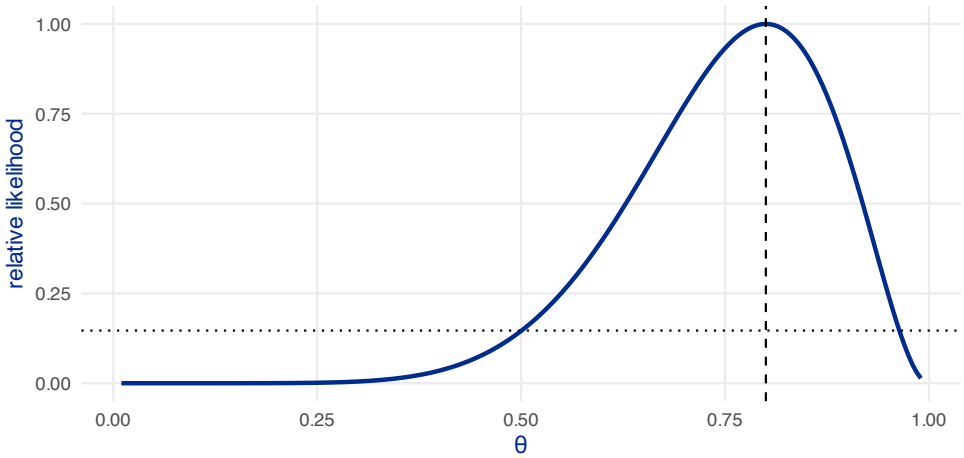


Figure 3.2: Relative likelihood for $y = 8$ successes in $n = 10$ trials. The dashed line is the 14.7% level $e^{-1.92}$ used in the next chapter to build a 95% likelihood interval.

```r
integrate(function(t) dbinom(y, n, t), 0, 1)$value   # 1/(n+1), not 1
```

```
#> [1] 0.09091
```

```r
dbinom(y, n, 0.5) / dbinom(y, n, 0.8)                # theta=0.5 vs theta=0.8
```

```
#> [1] 0.1455
```

The likelihood ratio is $L(0.8)/L(0.5) = 6.87$: the data are about four times better explained by $\theta = 0.8$ than by $\theta = 0.5$.

3.2.1 Reparametrisation

If $\psi = g(\theta)$ is a one-to-one reparametrisation, the likelihood of ψ is simply $L^*(\psi) = L(g^{-1}(\psi))$: we evaluate the same density at the same data point, only the label of the parameter changes. No Jacobian is involved, because the likelihood is not a density in θ. This is the origin of the invariance property (Theorem 3.2 below) and the reason likelihood-based intervals behave well under reparametrisation (Chapter 4).

3.3 The maximum likelihood estimator

Definition 3.2 (Maximum likelihood estimator)**.** A **maximum likelihood estimate** of θ is a value $\hat\theta = \hat\theta(y) \in \Theta$ with $L(\hat\theta) \geq L(\theta)$ for all $\theta \in \Theta$, that is $\hat\theta = \arg\max_{\theta\in\Theta} \ell(\theta)$. Seen as a function of the random data, $\hat\theta(Y)$ is the maximum likelihood **estimator** (MLE).

The MLE is "the parameter value under which the observed data are most probable". When ℓ is differentiable and the maximum lies in the interior of Θ, $\hat\theta$ solves the **likelihood equation** $S(\theta) = 0$, and a solution is a local maximum if $I(\hat\theta) = -\ell''(\hat\theta) > 0$. Three caveats: a solution of $S = 0$ may be a minimum or a saddle point; there may be several solutions (Section 3.8); and the maximum may be on the boundary of Θ, where $S = 0$ need not hold (Chapter 4).

3.3.1 Standard examples

Example 3.1 (Binomial)**.** Let $Y \sim \mathrm{Bin}(n, \theta)$ with y observed. Then $\ell(\theta) = y \log\theta + (n - y)\log(1-\theta)$ plus a constant, so

$$S(\theta) = \frac{y}{\theta} - \frac{n-y}{1-\theta}, \qquad I(\theta) = \frac{y}{\theta^2} + \frac{n-y}{(1-\theta)^2}.$$

Setting $S = 0$ gives $y(1-\theta) = (n-y)\theta$, i.e. $\hat\theta = y/n$. Since $I > 0$ for all $\theta \in (0,1)$ (when $0 < y < n$), ℓ is strictly concave and $\hat\theta$ is the unique maximum. The standard error is

$$\mathrm{se}(\hat\theta) = I(\hat\theta)^{-1/2} = \Big(\frac{n}{\hat\theta} + \frac{n}{1-\hat\theta}\Big)^{-1/2} = \sqrt{\frac{\hat\theta(1-\hat\theta)}{n}},$$

using $y = n\hat\theta$. For $y = 8, n = 10$: $\hat\theta = 0.8$ and $\mathrm{se} = \sqrt{0.8 \cdot 0.2/10} = 0.1265$.

Example 3.2 (Poisson)**.** Let $y_1, \ldots, y_n$ be iid Poisson(λ). Then $\ell(\lambda) = \sum_i y_i \log\lambda - n\lambda -$

$\sum_i \log y_i!$, so

$$S(\lambda) = \frac{\sum y_i}{\lambda} - n, \qquad I(\lambda) = \frac{\sum y_i}{\lambda^2}.$$

The unique root is $\hat\lambda = \bar y$, and $I(\hat\lambda) = n/\bar y > 0$, so $\mathrm{se}(\hat\lambda) = \sqrt{\bar y/n}$. For the discoveries data, $\hat\lambda = 3.1$ and se $= \sqrt{3.1/100} = 0.176$.

Example 3.3 (Exponential, with the air-conditioning data)**.** Let $y_1, \dots, y_n$ be iid exponential with mean μ, density $f(y;\mu) = \mu^{-1} e^{-y/\mu}$. Then

$$\ell(\mu) = -n\log\mu - \frac{\sum y_i}{\mu}, \quad S(\mu) = -\frac{n}{\mu} + \frac{\sum y_i}{\mu^2}, \quad I(\mu) = -\frac{n}{\mu^2} + \frac{2\sum y_i}{\mu^3}.$$

$S(\mu) = 0$ gives $\hat\mu = \bar y$, and $I(\hat\mu) = -n/\bar y^2 + 2n\bar y/\bar y^3 = n/\bar y^2 > 0$. Hence $\mathrm{se}(\hat\mu) = \bar y/\sqrt{n}$. By the invariance property below, the MLE of the failure rate $\lambda = 1/\mu$ is $\hat\lambda = 1/\bar y$.

```
mu_hat <- mean(y_ac)
se_mu  <- mu_hat / sqrt(n_ac)
loglik_exp <- function(mu) -n_ac * log(mu) - sum(y_ac) / mu
c(mu_hat = mu_hat, se = se_mu,
  numerical = optimize(loglik_exp, c(1, 1000), maximum = TRUE)$maximum)
```

```
#>    mu_hat        se numerical
#>     108.1      31.2     108.1
```

For the twelve failure times, $\hat\mu = 108.08$ hours with $\mathrm{se}(\hat\mu) = 31.2$: the mean time between failures is known only to within about a quarter of its value, which is what one expects from twelve skewed observations.

Example 3.4 (Normal with unknown mean and variance)**.** Let $y_1, \dots, y_n$ be iid $N(\mu, \sigma^2)$ with both parameters unknown; write $v = \sigma^2$. Then

$$\ell(\mu, v) = -\frac{n}{2}\log(2\pi v) - \frac{1}{2v}\sum_{i=1}^{n}(y_i - \mu)^2.$$

The two components of the score are

$$S_\mu = \frac{1}{v}\sum(y_i - \mu), \qquad S_v = -\frac{n}{2v} + \frac{1}{2v^2}\sum(y_i - \mu)^2.$$

$S_\mu = 0$ gives $\hat\mu = \bar y$. Substituting into $S_v = 0$ gives $\hat v = \frac{1}{n}\sum(y_i - \bar y)^2$. The MLE of the variance divides by n, not $n-1$. To verify that this is a maximum, note that for fixed v the log-likelihood is a downward parabola in μ with vertex $\bar y$, and the profile $\ell(\bar y, v) = -\frac{n}{2}\log(2\pi v) - \frac{n\hat v}{2v}$ has derivative $-\frac{n}{2v} + \frac{n\hat v}{2v^2}$, positive for $v < \hat v$ and negative for $v > \hat v$. The maximised log-likelihood is

$$\ell(\hat\mu, \hat v) = -\frac{n}{2}\log(2\pi\hat v) - \frac{n}{2}.$$

Its bias: since $\sum(y_i - \bar y)^2/\sigma^2 \sim \chi^2_{n-1}$, $\mathrm{E}(\hat v) = \frac{n-1}{n}\sigma^2$. The bias is $-\sigma^2/n$, of order $1/n$ (negligible next to the standard error, which is of order $1/\sqrt{n}$), but not zero: the MLE is not unbiased.

```
y_nh <- as.numeric(nhtemp); n_nh <- length(y_nh)   # New Haven mean annual
↪  temperature
mu_nh <- mean(y_nh); v_nh <- mean((y_nh - mu_nh)^2)
c(n = n_nh, mu_hat = mu_nh, v_hat = v_nh, s2 = var(y_nh))
```

```
#>      n mu_hat  v_hat     s2
#> 60.000 51.160  1.575  1.602
```

```
c(formula = -n_nh / 2 * log(2 * pi * v_nh) - n_nh / 2,
  dnorm_sum = sum(dnorm(y_nh, mu_nh, sqrt(v_nh), log = TRUE)))
```

```
#>   formula dnorm_sum
#>    -98.77    -98.77
```

For the $n = 60$ yearly mean temperatures at New Haven, $\hat{v} = 1.575$ against $s^2 = 1.602$; the ratio is $(n-1)/n = 0.983$, a small difference.

Example 3.5 (Gamma: a likelihood equation without closed-form solution)**.** Let $y_1, \dots, y_n$ be iid Gamma with shape α and rate β, $f(y) = \beta^\alpha y^{\alpha-1} e^{-\beta y}/\Gamma(\alpha)$. The log-likelihood is

$$\ell(\alpha, \beta) = n\alpha \log \beta - n \log \Gamma(\alpha) + (\alpha - 1) \sum \log y_i - \beta \sum y_i.$$

Let $\psi(\alpha) = \Gamma'(\alpha)/\Gamma(\alpha)$ denote the digamma function. Then $\partial \ell / \partial \beta = n\alpha/\beta - \sum y_i$ gives $\hat{\beta} = \hat{\alpha}/\bar{y}$ for any α. Substituting back and differentiating in α:

$$n\big[\log \alpha - \psi(\alpha) - \log \bar{y} + \overline{\log y}\big] = 0, \tag{3.3}$$

where $\overline{\log y} = n^{-1} \sum \log y_i$. This equation has no closed-form solution. It can be shown (Exercise 3.9) that the left side is strictly decreasing in α, so there is at most one root, but it must be found numerically (Chapter 4). Note that $\log \bar{y} - \overline{\log y} \geq 0$ by Jensen's inequality, with equality only if all y_i are equal.

For the air-conditioning data, $\log \bar{y} - \overline{\log y} = 0.8543$; the solution of (3.3) is $\hat{\alpha} = 0.706$, which we obtain in Chapter 4. A shape below one means a decreasing hazard: failures cluster early after a previous failure, a feature the exponential model ($\alpha = 1$) cannot capture.

3.4 Score and information

Definition 3.3 (Score and information)**.** For a scalar parameter, the **score** is $S(\theta) = \ell'(\theta)$. The **observed information** is $I(\theta) = -\ell''(\theta)$, a random quantity depending on the data and measuring the curvature of the log-likelihood at θ. The **expected (Fisher) information** is $\mathcal{I}(\theta) = \mathrm{E}_\theta[I(\theta)]$. For n iid observations, $\mathcal{I}(\theta) = n\,\mathcal{I}_1(\theta)$, where $\mathcal{I}_1$ is the expected information in one observation.

The observed information at the MLE, $I(\hat{\theta})$, is the curvature at the top of the log-likelihood: the sharper the peak, the more precisely the data pin down θ. This is why $\mathrm{se}(\hat{\theta}) = I(\hat{\theta})^{-1/2}$ is a measure of precision.

Regularity conditions (R). We say the model is regular if: (i) the support $\{y : f(y;\theta) > 0\}$ does not depend on θ; (ii) θ is an interior point of an open parameter space; (iii) f is three times continuously differentiable in θ; (iv) differentiation with respect to θ and integration over

y may be interchanged (it suffices that derivatives of f are dominated by an integrable function locally uniformly in θ); (v) $0 < \mathcal{I}_1(\theta) < \infty$. All the standard models in this book are regular; the uniform distribution on $(0, \theta)$ is the classical exception (Exercise 3.2).

Theorem 3.1 (Score identities)**.** *Under (R), for a single observation with* $S_1(\theta) = \partial \log f(Y;\theta)/\partial\theta$ *and true parameter* θ*,*

$$\mathrm{E}_\theta[S_1(\theta)] = 0, \qquad \mathrm{Var}_\theta[S_1(\theta)] = \mathrm{E}_\theta[S_1(\theta)^2] = \mathcal{I}_1(\theta) = -\mathrm{E}_\theta\Big[\frac{\partial^2 \log f(Y;\theta)}{\partial\theta^2}\Big]. \tag{3.4}$$

Consequently, for iid data the sample score $S(\theta) = \sum_i S_1^{(i)}(\theta)$ *satisfies* $\mathrm{E}[S(\theta)] = 0$ *and* $\mathrm{Var}[S(\theta)] = n\mathcal{I}_1(\theta) = \mathcal{I}(\theta)$*.*

Proof. Since $\int f(y;\theta)\,dy = 1$ for all θ, differentiating under the integral sign gives

$$0 = \int \frac{\partial f}{\partial\theta}\,dy = \int \frac{\partial \log f}{\partial\theta}\, f\,dy = \mathrm{E}[S_1(\theta)].$$

Differentiating $\int S_1(\theta) f\,dy = 0$ once more, and using the product rule,

$$0 = \int \frac{\partial^2 \log f}{\partial\theta^2}\, f\,dy + \int \frac{\partial \log f}{\partial\theta}\frac{\partial f}{\partial\theta}\,dy = -\mathcal{I}_1(\theta) + \int \Big(\frac{\partial \log f}{\partial\theta}\Big)^2 f\,dy,$$

where we used $\partial f/\partial\theta = f\,\partial \log f/\partial\theta$. Hence $\mathrm{E}[S_1^2] = \mathcal{I}_1$, and since $\mathrm{E}[S_1] = 0$, $\mathrm{Var}[S_1] = \mathcal{I}_1$. For n independent observations, expectations and variances of the sum add up. □

So the information has **two** interpretations that coincide: it is the expected curvature of the log-likelihood and the variance of the score. The second is what drives the asymptotic normality proof below: the score at the truth is a sum of iid mean-zero terms.

3.4.1 Observed versus expected information

The observed information $I(\theta)$ depends on the data; the expected information $\mathcal{I}(\theta) = \mathrm{E}[I(\theta)]$ averages over the data and is a function of θ alone. They are equal in expectation and, asymptotically, very close, but they differ for a given sample.

Example 3.6 (Binomial: observed and expected information differ)**.** For the binomial in Example 3.1, $\mathcal{I}(\theta) = \mathrm{E}[I(\theta)] = \dfrac{n\theta}{\theta^2} + \dfrac{n(1-\theta)}{(1-\theta)^2} = \dfrac{n}{\theta(1-\theta)}$ because $\mathrm{E}[Y] = n\theta$. For $y = 8$, $n = 10$:

$$I(0.8) = \frac{8}{0.64} + \frac{2}{0.04} = 12.5 + 50 = 62.5 = \mathcal{I}(0.8),$$
$$I(0.6) = \frac{8}{0.36} + \frac{2}{0.16} = 34.72, \qquad \mathcal{I}(0.6) = \frac{10}{0.24} = 41.67.$$

They coincide at $\hat\theta = y/n$ (because $y = n\hat\theta$ turns the observed information into the expected one), but differ elsewhere. They also coincide at $\theta = 1/2$, where $I(1/2) = 4y + 4(n-y) = 4n$ whatever y is.

```
obs_info <- function(t) y / t^2 + (n - y) / (1 - t)^2
exp_info <- function(t) n / (t * (1 - t))
tt <- seq(0.3, 0.95, by = 0.005)
dd <- data.frame(theta = rep(tt, 2), info = c(obs_info(tt), exp_info(tt)),
```

```
                type = rep(c("observed", "expected"), each = length(tt)))
ggplot(dd, aes(theta, info, linetype = type, colour = type)) +
↪  geom_line(linewidth = 1) +
  scale_colour_manual(values = c(ie_coral, ie_navy)) + scale_y_log10() +
  geom_vline(xintercept = 0.8, linetype = 3) +
  labs(x = expression(theta), y = "information (log scale)", colour = NULL,
  ↪  linetype = NULL)
```

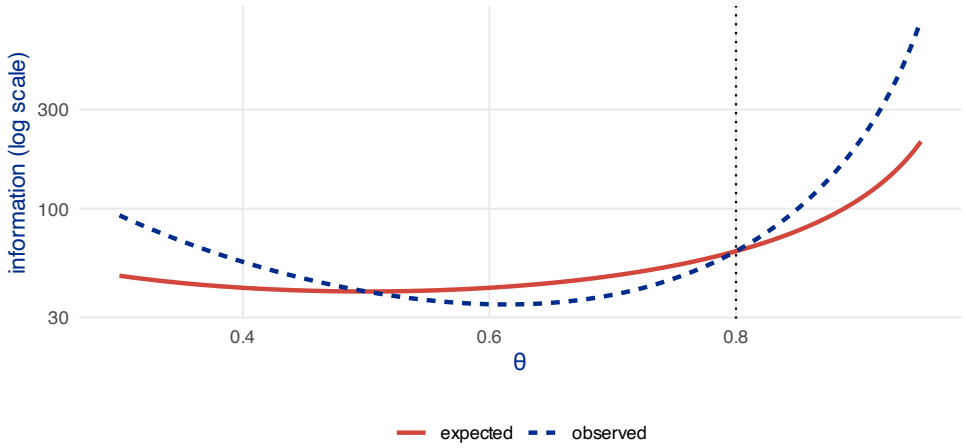


Figure 3.3: Observed information $I(\theta)$ (solid) and expected information $\mathcal{I}(\theta)$ (dashed) for $y = 8$ successes in $n = 10$ trials. They agree at $\hat\theta = 0.8$ and at $\theta = 1/2$.

```
c(obs = obs_info(0.6), exp = exp_info(0.6), obs_at_mle = obs_info(0.8), se = 1
↪  / sqrt(obs_info(0.8)))
```

```
#>       obs       exp obs_at_mle        se
#>   34.7222   41.6667    62.5000    0.1265
```

For exponential families in canonical form (Chapter 8) the observed and expected information coincide at *every* θ; the Poisson with the canonical parameter $\log\lambda$ is an example (Exercise 3.4). When they differ, there is no universal winner. Using $I(\hat\theta)$ is the convention in this course because it requires no integration and reflects the information actually present in the sample (Efron and Hinkley, 1978).

3.4.2 Reparametrisation of the information

Let $\psi = g(\theta)$ with g differentiable and monotone, and write $\theta = h(\psi)$. By the chain rule, the score transforms as $S_\psi(\psi) = S_\theta(h(\psi))\,h'(\psi)$. Differentiating again and evaluating at a point where $S_\theta = 0$ (for instance at the MLE),

$$I_\psi(\hat\psi) = I_\theta(\hat\theta)\,h'(\hat\psi)^2 = \frac{I_\theta(\hat\theta)}{g'(\hat\theta)^2}.$$

Hence $\mathrm{se}(\hat\psi) = |g'(\hat\theta)|\,\mathrm{se}(\hat\theta)$, the **delta method** formula. For example, if $\psi = \log\mu$ in the exponential model, $\mathrm{se}(\log\hat\mu) = \mathrm{se}(\hat\mu)/\hat\mu = 1/\sqrt{n}$, independent of the data. We will exploit this in Chapter 4.

3.5 Properties of the MLE

3.5.1 Invariance

Theorem 3.2 (Invariance of the MLE)**.** *If $\hat\theta$ is an MLE of θ and g is any function, then $g(\hat\theta)$ is an MLE of $\psi = g(\theta)$.*

Proof. For general g (not necessarily one-to-one) define the induced likelihood $L^*(\psi) = \sup\{L(\theta) : g(\theta) = \psi\}$. For every ψ in the range of g, $L^*(\psi) \le \sup_\theta L(\theta) = L(\hat\theta)$. On the other hand, $\hat\theta$ belongs to the set $\{\theta : g(\theta) = g(\hat\theta)\}$, so $L^*(g(\hat\theta)) \ge L(\hat\theta)$. Therefore $L^*(g(\hat\theta)) = \max_\psi L^*(\psi)$. When g is one-to-one, $L^*(\psi) = L(g^{-1}(\psi))$ and the claim is immediate. □

Invariance has an important practical consequence: the MLE of the rate $1/\mu$ is $1/\hat\mu$, the MLE of σ is $\sqrt{\hat\sigma^2}$, and so on. It also has a price: **unbiasedness is not invariant**. Since $1/x$ is convex, Jensen's inequality gives $\mathrm{E}(1/\bar y) > 1/\mathrm{E}(\bar y)$, so $1/\bar y$ is biased upwards for the rate λ (Exercise 3.6).

```
lam_hat <- 1 / mu_hat
num <- optimize(function(l) loglik_exp(1 / l), c(1e-4, 0.1), maximum = TRUE,
↪  tol = 1e-10)$maximum
c(rate_hat = lam_hat, numerical_in_rate = num)
```

```
#>          rate_hat numerical_in_rate
#>          0.009252          0.009252
```

```
# MLE of P(Y > 100) = exp(-100 / mu) by invariance
exp(-100 / mu_hat)
```

```
#> [1] 0.3964
```

3.5.2 Consistency

Definition 3.4 (Consistency)**.** An estimator $\hat\theta_n$ is **consistent** for θ_0 if $\hat\theta_n \to \theta_0$ in probability as $n \to \infty$.

Theorem 3.3 (Consistency of the MLE)**.** *Let $y_1, \dots, y_n$ be iid from $f(y;\theta_0)$, a regular model (R) which is **identifiable**: $f(\cdot;\theta) = f(\cdot;\theta_0)$ a.e. implies $\theta = \theta_0$. Then with probability tending to one the likelihood equation has a root $\hat\theta_n$ in any given neighbourhood of θ_0, hence there is a consistent sequence of roots. If, for each n, the likelihood equation has a unique root, this root is the MLE and the MLE is consistent.*

Proof. Fix $\delta > 0$ with $[\theta_0 - \delta, \theta_0 + \delta] \subset \Theta$ and consider, for $\theta \neq \theta_0$, the average log-likelihood ratio

$$\frac{1}{n}\big[\ell(\theta) - \ell(\theta_0)\big] = \frac{1}{n}\sum_{i=1}^{n} \log \frac{f(y_i;\theta)}{f(y_i;\theta_0)} \xrightarrow{\;p\;} \mathrm{E}_{\theta_0} \log \frac{f(Y;\theta)}{f(Y;\theta_0)} =: -K(\theta)$$

by the law of large numbers. The function $x \mapsto \log x$ is strictly concave, so Jensen's inequality gives

$$-K(\theta) < \log \mathrm{E}_{\theta_0} \frac{f(Y;\theta)}{f(Y;\theta_0)} = \log \int f(y;\theta)\, dy = \log 1 = 0,$$

where the inequality is strict because the ratio is not almost surely constant (identifiability). Thus $K(\theta) > 0$ is the Kullback–Leibler divergence of $f(\cdot;\theta)$ from $f(\cdot;\theta_0)$. Apply this at the two

points $\theta_0 \pm \delta$: with probability tending to one, $\ell(\theta_0 \pm \delta) < \ell(\theta_0)$. On that event the continuous function ℓ on $[\theta_0 - \delta, \theta_0 + \delta]$ attains its maximum at an interior point, where $S = 0$. So the likelihood equation has a root in $(\theta_0 - \delta, \theta_0 + \delta)$ with probability tending to one. Letting $\delta \downarrow 0$ along a sequence (and taking the root closest to θ_0) gives a consistent root. If the root is unique, it is the interior maximum found above, hence the MLE. □

The proof shows what consistency really rests on: the true parameter is the unique maximiser of the *expected* log-likelihood $\theta \mapsto \mathrm{E}_{\theta_0} \log f(Y;\theta)$ (equivalently, minimises the KL divergence), and the sample log-likelihood converges to it. Identifiability is indispensable: if two parameter values give the same distribution, no amount of data can distinguish them.

3.5.3 Asymptotic normality

Theorem 3.4 (Asymptotic normality of the MLE)**.** *Under the conditions of Theorem 3.3 and (R), if $\hat\theta_n$ is a consistent root of $S(\theta) = 0$, then*

$$\sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow{d} N\big(0, \mathcal{I}_1(\theta_0)^{-1}\big). \tag{3.5}$$

Equivalently, $\hat\theta_n \approx N\big(\theta_0, \mathcal{I}(\theta_0)^{-1}\big)$ with $\mathcal{I} = n\mathcal{I}_1$, and the same holds with $\mathcal{I}(\theta_0)$ replaced by $I(\hat\theta_n)$ or $\mathcal{I}(\hat\theta_n)$.

Proof. By the mean value theorem, there is a θ_n^* between $\hat\theta_n$ and θ_0 with

$$0 = S(\hat\theta_n) = S(\theta_0) + S'(\theta_n^*)(\hat\theta_n - \theta_0) = S(\theta_0) - I(\theta_n^*)(\hat\theta_n - \theta_0).$$

Solving, and dividing numerator and denominator by $\sqrt{n}$ and n respectively,

$$\sqrt{n}(\hat\theta_n - \theta_0) = \frac{n^{-1/2} S(\theta_0)}{n^{-1} I(\theta_n^*)}.$$

Numerator. $S(\theta_0) = \sum_i S_1^{(i)}(\theta_0)$ is a sum of iid variables with mean 0 and variance $\mathcal{I}_1(\theta_0)$ (Theorem 3.1), so by the central limit theorem $n^{-1/2} S(\theta_0) \xrightarrow{d} N(0, \mathcal{I}_1(\theta_0))$.

Denominator. By the law of large numbers, $n^{-1} I(\theta_0) \xrightarrow{p} \mathcal{I}_1(\theta_0)$. The third derivative is dominated, $|\partial^3 \log f / \partial\theta^3| \le M(y)$ with $\mathrm{E}M < \infty$ near θ_0, hence $|n^{-1} I(\theta_n^*) - n^{-1} I(\theta_0)| \le |\theta_n^* - \theta_0|\, n^{-1} \sum M(y_i) \xrightarrow{p} 0$, since $\theta_n^* \to \theta_0$ in probability (consistency) and $n^{-1}\sum M(y_i)$ is bounded in probability. So $n^{-1} I(\theta_n^*) \xrightarrow{p} \mathcal{I}_1(\theta_0) > 0$.

By Slutsky's lemma, the ratio converges in distribution to $N(0, \mathcal{I}_1)/\mathcal{I}_1 = N(0, \mathcal{I}_1^{-1})$. For the last statement, $n^{-1} I(\hat\theta_n) \to \mathcal{I}_1(\theta_0)$ in probability by the same argument, so the variance may be estimated by $I(\hat\theta_n)^{-1}$. □

The key message is: **the information is the reciprocal of the asymptotic variance**. In practice we use

$$\hat\theta \mathrel{\dot\sim} N\big(\theta,\, I(\hat\theta)^{-1}\big), \qquad \mathrm{se}(\hat\theta) = I(\hat\theta)^{-1/2}. \tag{3.6}$$

Where does n go? Formula (3.5) uses the per-observation information $\mathcal{I}_1$ together with the scaling $\sqrt{n}$. Formula (3.6) uses the total information $\mathcal{I} = n\mathcal{I}_1$ (or $I(\hat\theta)$) without any $\sqrt{n}$. Mixing the two (for instance, $\sqrt{n}(\hat\theta - \theta) \approx N(0, \mathcal{I}^{-1})$) is a frequent error.

```r
set.seed(2027)
lam <- 3.1; nn <- 25
est <- replicate(5000, mean(rpois(nn, lam)))
c(mean_of_mle = mean(est), sd_of_mle = sd(est), asymptotic_sd = sqrt(lam /
↪  nn))
```

```
#>   mean_of_mle     sd_of_mle asymptotic_sd
#>        3.0977        0.3531        0.3521
```

```r
ggplot(data.frame(est), aes(est)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = ie_blue,
  ↪  alpha = .6) +
  stat_function(fun = dnorm, args = list(mean = lam, sd = sqrt(lam / nn)),
  ↪  colour = ie_coral, linewidth = 1) +
  labs(x = expression(hat(lambda)), y = "density")
```

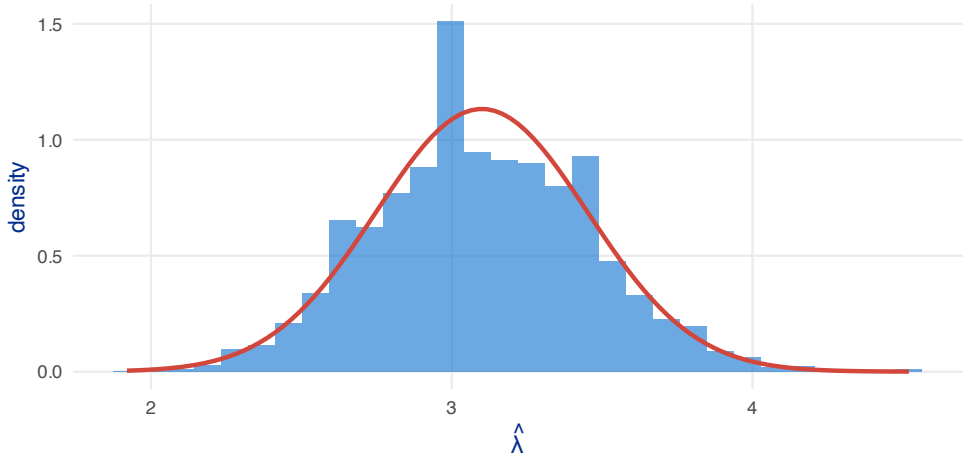


Figure 3.4: Sampling distribution of the Poisson MLE $\hat{\lambda} = \bar{y}$ for $\lambda = 3.1$ and $n = 25$ (histogram) with the asymptotic normal density $N(\lambda, \lambda/n)$ (curve).

3.5.4 The Cramér–Rao bound and efficiency

Theorem 3.5 (Cramér–Rao lower bound)**.** *Under (R), let $T = T(Y)$ be an unbiased estimator of $g(\theta)$ with finite variance, where $\mathrm{E}_\theta[T] = g(\theta)$ and Y has total expected information $\mathcal{I}(\theta)$. Then*

$$\mathrm{Var}_\theta(T) \;\geq\; \frac{g'(\theta)^2}{\mathcal{I}(\theta)}.$$

In particular, for unbiased estimators of θ, $\mathrm{Var}_\theta(T) \geq 1/\mathcal{I}(\theta)$.

Proof. Differentiate $g(\theta) = \int T(y) f(y;\theta)\, dy$ under the integral sign:

$$g'(\theta) = \int T(y) \frac{\partial \log f}{\partial \theta} f \, dy = \mathrm{E}[T\, S(\theta)] = \mathrm{Cov}(T, S(\theta)),$$

because $\mathrm{E}[S(\theta)] = 0$ (Theorem 3.1). By the Cauchy–Schwarz inequality, $\mathrm{Cov}(T,S)^2 \leq \mathrm{Var}(T)\mathrm{Var}(S) = \mathrm{Var}(T)\,\mathcal{I}(\theta)$. Rearranging gives the bound. □

An unbiased estimator attains the bound if and only if equality holds in Cauchy–Schwarz, i.e. $S(\theta) = a(\theta)\{T - g(\theta)\}$ almost surely. For the Poisson, $S(\lambda) = n(\bar{y} - \lambda)/\lambda$, so $\bar{y}$ attains the bound λ/n for every n. In general, the bound is not attained by any unbiased estimator, but the MLE attains it *asymptotically*: by Theorem 3.4 its asymptotic variance is exactly $1/\mathcal{I}(\theta_0)$. This is the sense in which the MLE is **asymptotically efficient**.

Warning. Efficiency is an asymptotic and regular-model statement. For finite n the MLE can be biased (the MLE of σ^2, of $1/\mu$), and unbiased estimators with smaller mean squared error often exist (a shrunken variance estimator has smaller MSE than s^2). Do not claim "no estimator has smaller variance than the MLE": the Cramér–Rao bound applies to *unbiased* estimators, and even then only to regular models.

3.6 Several parameters

Let $\theta = (\theta_1, \dots, \theta_p)^\top$. The **score** is the gradient $S(\theta) = \partial \ell / \partial \theta \in \mathbb{R}^p$, the **observed information** is the matrix $I(\theta) = -\partial^2 \ell / \partial\theta\, \partial\theta^\top$ with entries $I_{jk} = -\partial^2 \ell / \partial\theta_j \partial\theta_k$, and the expected information is $\mathcal{I}(\theta) = \mathrm{E}[I(\theta)]$. The identities of Theorem 3.1 hold in matrix form: $\mathrm{E}[S] = 0$ and $\mathrm{Var}[S] = \mathcal{I}(\theta)$, with identical proofs applied componentwise. The Cramér–Rao bound reads $\mathrm{Var}(T) \succeq \mathcal{I}(\theta)^{-1}$ for unbiased T of θ (Loewner order). The multi-parameter version of Theorem 3.4 is

$$\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N_p(0, \mathcal{I}_1(\theta_0)^{-1}), \qquad \text{in practice } \hat\theta \mathrel{\dot\sim} N_p(\theta, I(\hat\theta)^{-1}). \tag{3.7}$$

The proof is the same, using the multivariate Taylor expansion $0 = S(\theta_0) - I(\theta^*)(\hat\theta - \theta_0)$ and the multivariate CLT. The standard error of $\hat\theta_j$ is $\sqrt{[I(\hat\theta)^{-1}]_{jj}}$: the jth diagonal element of the *inverse* of the information matrix, which is generally not $1/I_{jj}$.

Example 3.7 (Information matrix of the normal)**.** Continuing Example 3.4, the second derivatives are

$$I_{\mu\mu} = \frac{n}{v}, \qquad I_{\mu v} = \frac{1}{v^2}\sum(y_i - \mu), \qquad I_{vv} = -\frac{n}{2v^2} + \frac{1}{v^3}\sum(y_i - \mu)^2.$$

At the MLE, $\sum(y_i - \hat\mu) = 0$ and $\sum(y_i - \hat\mu)^2 = n\hat v$, so

$$I(\hat\mu, \hat v) = \begin{pmatrix} n/\hat v & 0 \\ 0 & n/(2\hat v^2) \end{pmatrix}, \qquad \mathrm{se}(\hat\mu) = \sqrt{\hat v / n}, \quad \mathrm{se}(\hat v) = \hat v\sqrt{2/n}.$$

The expected information is $\mathcal{I} = \mathrm{diag}(n/v,\, n/(2v^2))$ and equals the observed information at the MLE. The off-diagonal element vanishes: $\hat\mu$ and $\hat v$ are asymptotically independent (in fact exactly independent). For the New Haven data, $\mathrm{se}(\hat\mu) = \sqrt{1.575/60} = 0.162$ and $\mathrm{se}(\hat v) = 1.575\sqrt{2/60} = 0.288$.

```
I_nh <- diag(c(n_nh / v_nh, n_nh / (2 * v_nh^2)))
sqrt(diag(solve(I_nh)))
```

```
#> [1] 0.1620 0.2876
```

Example 3.8 (Information matrix of the gamma)**.** For the gamma model in Example 3.5, $-\partial^2\ell/\partial\alpha^2 = n\psi'(\alpha)$ (with ψ' the trigamma function), $-\partial^2\ell/\partial\alpha\partial\beta = -n/\beta$ and $-\partial^2\ell/\partial\beta^2 = n\alpha/\beta^2$. None of these depends on the data, so observed and expected

information coincide:

$$\mathcal{I}(\alpha,\beta) = n\begin{pmatrix}\psi'(\alpha) & -1/\beta \\ -1/\beta & \alpha/\beta^2\end{pmatrix}, \qquad \det = \frac{n^2}{\beta^2}\{\alpha\psi'(\alpha) - 1\} > 0.$$

The determinant is positive because $\alpha\psi'(\alpha) > 1$ for all $\alpha > 0$. The off-diagonal term is non-zero: the estimates of shape and rate are correlated, and marginal standard errors must come from the *inverse* of the full matrix.

3.7 Likelihood with censored data

Many lifetimes are not fully observed: a study ends, a patient leaves, a device is still running. The lifetime is then **right-censored**: we only know that it exceeds the last observation time. Let T_i be the lifetime and C_i the censoring time, and observe

$$t_i = \min(T_i, C_i), \qquad \delta_i = \mathbf{1}\{T_i \le C_i\} \quad (\delta_i = 1 \text{ event},\ \delta_i = 0 \text{ censored}).$$

If C_i is independent of T_i and carries no information about θ (non-informative censoring), the likelihood contribution of unit i is the density $f(t_i;\theta)$ if $\delta_i = 1$ and the survival function $S(t_i;\theta) = P(T_i > t_i)$ if $\delta_i = 0$, so that

$$\ell(\theta) = \sum_{i=1}^{n}\{\delta_i \log f(t_i;\theta) + (1-\delta_i)\log S(t_i;\theta)\}. \tag{3.8}$$

Score, information and the Wald, score and likelihood-ratio machinery apply to (3.8) unchanged.

Example 3.9 (Exponential lifetimes with right censoring)**.** For $f(t) = \lambda e^{-\lambda t}$ and $S(t) = e^{-\lambda t}$, let $d = \sum_i \delta_i$ be the number of events. Then

$$\ell(\lambda) = d\log\lambda - \lambda\sum_{i=1}^{n} t_i, \qquad \frac{\partial\ell}{\partial\lambda} = \frac{d}{\lambda} - \sum_i t_i, \qquad \hat\lambda = \frac{d}{\sum_i t_i},$$

$$I(\lambda) = \frac{d}{\lambda^2} = \mathcal{I}(\lambda) \text{ (the observed information does not depend on the data given } d), \qquad \mathrm{se}(\hat\lambda) = \frac{\hat\lambda}{\sqrt{d}}.$$

The sum is over *all* n units (events and censored: total exposure time), the numerator counts *events only*, and the information is driven by d, not n. Without censoring $d = n$ and $\hat\lambda = 1/\bar t$.

Why the naive estimators fail. Suppose all units are followed for at most c (Type I censoring), true rate λ. Then $E[\min(T,c)] = (1-e^{-\lambda c})/\lambda$ and $P(\delta = 1) = 1 - e^{-\lambda c}$. Hence, as $n \to \infty$: (i) $1/\bar t \to \lambda/(1-e^{-\lambda c}) > \lambda$ (censored times are treated as failure times, so lifetimes look too short); (ii) $d/\sum_{\delta_i=1} t_i$ also overestimates λ, because it discards the exposure of the censored units, which were all long survivors; (iii) $d/n \to 1 - e^{-\lambda c}$ is a probability, not a rate, and depends on the follow-up length c. The MLE has ratio $E[d]/E[\sum t_i] = \lambda$ and is consistent.

Ovarian cancer data (`survival::ovarian`): 26 women, `futime` is follow-up in days and `fustat` the event indicator (12 deaths, 14 censored).

```
library(survival)
t_ov <- ovarian$futime;  delta_ov <- ovarian$fustat
d_ov <- sum(delta_ov);  T_ov <- sum(t_ov)
```

```
ll_cens <- function(l) d_ov * log(l) - l * T_ov
mle_ov <- d_ov / T_ov
c(d = d_ov, total_time = T_ov, mle = mle_ov, se = sqrt(d_ov) / T_ov,
  naive_1_over_mean = 1 / mean(t_ov))
```

```
#>                 d        total_time               mle                se
#>         1.200e+01         1.559e+04         7.698e-04         2.222e-04
#> naive_1_over_mean
#>         1.668e-03
```

```
# Wald 95% interval and LR test of H0: lambda = 1/1000 (mean survival 1000
↪  days)
mle_ov + c(-1, 1) * 1.96 * sqrt(d_ov) / T_ov
```

```
#> [1] 0.0003343 0.0012054
```

```
LR_ov <- 2 * (ll_cens(mle_ov) - ll_cens(1 / 1000));  c(LR = LR_ov, p =
↪  pchisq(LR_ov, 1, lower.tail = FALSE))
```

```
#>     LR      p
#> 0.8977 0.3434
```

```
# check against survreg (exponential model): same maximised log-likelihood
fit_ov <- survreg(Surv(futime, fustat) ~ 1, data = ovarian, dist =
↪  "exponential")
c(rate = unname(exp(-coef(fit_ov))), loglik = as.numeric(logLik(fit_ov)), ours
↪  = ll_cens(mle_ov))
```

```
#>       rate     loglik       ours
#>  7.698e-04 -9.803e+01 -9.803e+01
```

The MLE is $\hat\lambda = 12/15588 = 7.70 \times 10^{-4}$ per day (estimated median survival $\log 2/\hat\lambda \approx 900$ days; by invariance), with se $= 2.22 \times 10^{-4}$. The naive $1/\bar t = 1.67 \times 10^{-3}$ is more than twice as large. The test of $H_0 : \lambda = 1/1000$ gives LR $= 0.90$ ($p = 0.34$). Censored data return in Chapter 20, where the exponential assumption is relaxed (Kaplan–Meier, Cox).

3.8 A cautionary example: the Cauchy location

Theorems 3.3 and 3.4 say that a *consistent root* of the likelihood equation is asymptotically normal. They do not say that the likelihood equation has a single root. The Cauchy location model $f(y;\theta) = \{\pi[1+(y-\theta)^2]\}^{-1}$ is the standard counterexample. Its log-likelihood and score are

$$\ell(\theta) = -\sum_{i=1}^{n} \log\{1+(y_i-\theta)^2\} - n\log\pi, \qquad S(\theta) = \sum_{i=1}^{n} \frac{2(y_i-\theta)}{1+(y_i-\theta)^2}.$$

Take the five observations $y = (-2, -1, 0, 7, 8)$: three at the left, two far to the right.

```
yc <- c(-2, -1, 0, 7, 8)
llc <- function(t) -sapply(t, function(s) sum(log(1 + (yc - s)^2)))
scc <- function(t) sapply(t, function(s) sum(2 * (yc - s) / (1 + (yc - s)^2)))
tt <- seq(-5, 12, by = 0.02)
ggplot(data.frame(tt, ll = llc(tt)), aes(tt, ll)) + geom_line(colour =
↪  ie_navy, linewidth = 1) +
```

```
  geom_rug(data = data.frame(yc), aes(x = yc, y = NULL), sides = "b") +
  labs(x = expression(theta), y = "log-likelihood (up to a constant)")
```

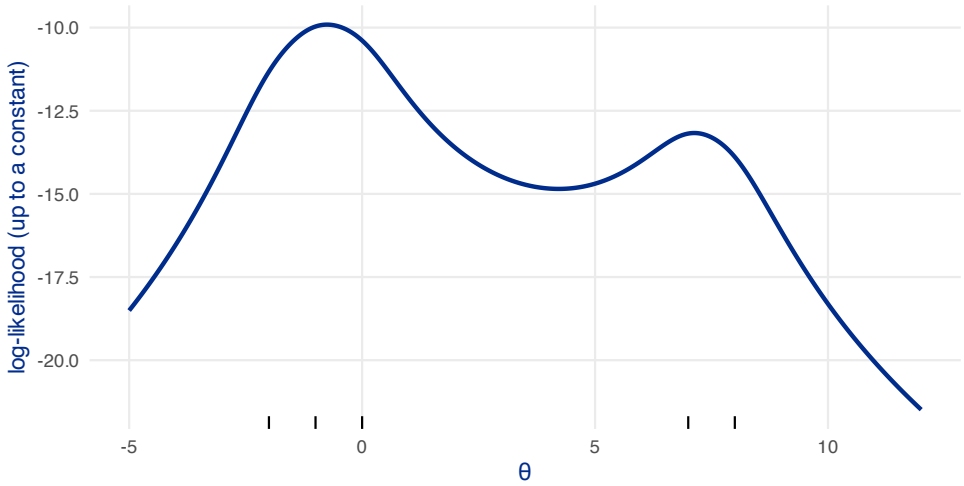


Figure 3.5: Cauchy location log-likelihood for $y = (-2, -1, 0, 7, 8)$ (ticks). There are two local maxima and one local minimum; only the left maximum is global.

```
# all roots of the score: sign changes on a fine grid
sg <- scc(tt); roots <- sapply(which(diff(sign(sg)) != 0), function(k)
↪  uniroot(scc, tt[k + 0:1])$root)
data.frame(root = roots, loglik = llc(roots), second_deriv = (llc(roots +
↪  1e-3) - 2 * llc(roots) + llc(roots - 1e-3)) / 1e-6)
```

```
#>      root  loglik second_deriv
#> 1 -0.7537  -9.911       -1.795
#> 2  4.2205 -14.850        0.499
#> 3  7.1304 -13.170       -1.970
```

The score has three roots; a negative second derivative of ℓ marks the two maxima, the positive one the minimum. A root-finder started at $\theta = 10$ converges to the local maximum at 7.13 with log-likelihood -13.17, whereas the MLE is -0.754 with -9.91. The expected information is constant, $\mathcal{I}_1(\theta) = 1/2$ for every θ (Exercise 3.10), so the asymptotic standard error of the MLE is $\sqrt{2/n}$, whatever θ is; but the *observed* information at a local maximum describes only the local curvature there, and says nothing about whether that maximum is the global one. Moral: a numerical optimiser can report convergence at a point that is not the MLE. Always plot the log-likelihood when you can, and use several starting values when you cannot (Chapter 4).

3.9 In R

The workflow for a one-parameter model, using the same functions as in Lab 01:

1. Write the **log-likelihood** as an R function of the parameter(s), using the vectorised `d*` density functions with `log = TRUE`, e.g. `sum(dpois(y, lambda, log = TRUE))`.
2. Plot it over a grid, to see its shape and the approximate location of the maximum.
3. Maximise it with `optimize()` (one parameter) or `optim()` (several), remembering that `optim()` **minimises** by default; return the negative log-likelihood or set `control = list(fnscale = -1)`.

4. Extract the curvature with `hessian = TRUE`; the Hessian of the *negative* log-likelihood at the optimum is the observed information $I(\hat\theta)$, so `sqrt(diag(solve(fit$hessian)))` are the standard errors.

```
loglik_pois <- function(lambda) sum(dpois(y_di, lambda, log = TRUE))
opt <- optimize(loglik_pois, c(0.1, 20), maximum = TRUE)
c(numerical = opt$maximum, closed_form = mean(y_di))
```

```
#>   numerical closed_form
#>         3.1         3.1
```

```
# Two parameters: normal, parametrised on the log-scale of the variance
nll <- function(p) -sum(dnorm(y_nh, p[1], exp(p[2] / 2), log = TRUE))   # p =
↪  (mu, log v)
fit <- optim(c(50, 0), nll, method = "BFGS", hessian = TRUE)
c(mu = fit$par[1], v = exp(fit$par[2]), loglik = -fit$value)
```

```
#>      mu       v  loglik
#>  51.160   1.575 -98.765
```

```
sqrt(diag(solve(fit$hessian)))     # se(mu_hat), se(log v_hat)
```

```
#> [1] 0.1620 0.1826
```

The second standard error is that of $\log\hat v$, not of $\hat v$: by the delta method $\mathrm{se}(\log\hat v) = \mathrm{se}(\hat v)/\hat v = \sqrt{2/n} = 0.183$, which is what R returns. Parametrisations on the whole real line (log for positive parameters, logit for probabilities) avoid constrained optimisation and give better-behaved likelihoods.

3.10 Common pitfalls

Warning. *The likelihood is not a probability distribution for θ.* Its integral over θ is generally not one, and it changes under reparametrisation in a way no density would (there is no Jacobian).

Warning. *Standard error formulas.* $\mathrm{se}(\hat\theta) = I(\hat\theta)^{-1/2}$. A formula such as $\sigma\, I(\hat\theta)^{-1/2}$ is wrong in this course; for the linear model, the familiar $\sigma\sqrt{(X^\top X)^{-1}_{jj}}$ is $I(\hat\beta)^{-1/2}$ with σ already inside the information.

Warning. *Total versus per-observation information.* With $\sqrt{n}$, use $\mathcal{I}_1$; with $\hat\theta$ itself, use the total $\mathcal{I} = n\mathcal{I}_1$.

Warning. *Observed versus expected.* The curve $n/\{\theta(1-\theta)\}$ is the expected information; it is not "the" observed information.

Warning. *`optim()` minimises.* Passing a log-likelihood without `fnscale = -1` finds the *minimum*. Its Hessian is the observed information only on the scale on which you optimised: if you optimised $\log\sigma$, you get $\mathrm{se}(\log\hat\sigma)$, not $\mathrm{se}(\hat\sigma)$.

Warning. *Convergence code 0 is not a proof of a global maximum.* Check the Hessian is positive definite (all eigenvalues of the Hessian of $-\ell$ are positive) and try different starts.

Remark. "MLEs are unbiased" is false (variance, rate). The correct claim is: in regular models the MLE is consistent, asymptotically normal and asymptotically efficient, with bias of order $1/n$. Do not say "the mean of $\hat\theta$ converges to θ": the MLE can have an infinite or undefined mean (for instance the logit of $\hat p$, which is infinite when $\hat p \in \{0, 1\}$).

3.11 Chapter summary

- The likelihood $L(\theta) = f(y;\theta)$ is a function of θ for fixed data; it is defined up to a constant factor, is not a density in θ, and only ratios matter. Under reparametrisation $L^*(\psi) = L(g^{-1}(\psi))$.
- $\hat\theta = \arg\max \ell(\theta)$; in regular problems it solves $S(\theta) = 0$ with $I(\hat\theta) > 0$. Closed forms: binomial y/n, Poisson $\bar y$, exponential mean $\bar y$, normal $(\bar y,\ \frac{1}{n}\sum(y_i - \bar y)^2)$; gamma needs numerical solution.
- Under regularity, $\mathrm{E}[S] = 0$ and $\mathrm{Var}[S] = \mathcal{I} = \mathrm{E}[I]$; $\mathcal{I} = n\mathcal{I}_1$ for iid data. Observed and expected information coincide at the MLE for the binomial but not in general.
- Invariance: $g(\hat\theta)$ is the MLE of $g(\theta)$. Consistency follows from identifiability and Jensen's inequality. Asymptotic normality: $\sqrt{n}(\hat\theta - \theta_0) \to N(0, \mathcal{I}_1^{-1})$, so $\hat\theta \dot\sim N(\theta, I(\hat\theta)^{-1})$ and $\mathrm{se}(\hat\theta) = I(\hat\theta)^{-1/2}$.
- Cramér–Rao: $\mathrm{Var}(T) \geq g'(\theta)^2/\mathcal{I}(\theta)$ for unbiased T; the MLE attains $1/\mathcal{I}$ asymptotically.
- With several parameters, standard errors come from the diagonal of $I(\hat\theta)^{-1}$.
- A likelihood may have several maxima (Cauchy); non-regular models (uniform on $(0,\theta)$) break the theory.

3.12 Exercises with explained solutions

Exercise 3.1. Let $y_1, \dots, y_n$ be iid Poisson(λ). (a) Derive S, I, $\mathcal{I}$ and the MLE. (b) Verify directly that $\mathrm{E}[S(\lambda)] = 0$ and $\mathrm{Var}[S(\lambda)] = \mathcal{I}(\lambda)$. (c) For the discoveries data ($n = 100$, $\sum y_i = 310$) compute $\mathrm{se}(\hat\lambda)$ and a 95% Wald interval $\hat\lambda \pm 1.96\,\mathrm{se}(\hat\lambda)$.

Solution.

(a) $\ell = \sum y_i \log\lambda - n\lambda + \text{const}$, so $S = \sum y_i/\lambda - n$, $I = \sum y_i/\lambda^2$, $\mathcal{I} = \mathrm{E}[\sum Y_i]/\lambda^2 = n/\lambda$. $S = 0$ gives $\hat\lambda = \bar y$.

(b) $S = (\sum Y_i - n\lambda)/\lambda$. Since $\sum Y_i \sim \text{Poisson}(n\lambda)$, $\mathrm{E}[S] = 0$ and $\mathrm{Var}[S] = n\lambda/\lambda^2 = n/\lambda = \mathcal{I}(\lambda)$.

(c) $\hat\lambda = 310/100 = 3.1$, $I(\hat\lambda) = 310/3.1^2 = 32.26$, $\mathrm{se} = 32.26^{-1/2} = 0.1761$. The interval is $3.1 \pm 1.96(0.1761) = (2.755,\ 3.445)$.

Exercise 3.2. Let $y_1, \dots, y_n$ be iid uniform on $(0,\theta)$. (a) Find the MLE. (b) Show that it is biased and compute its bias. (c) Explain which regularity condition fails and why the score identity $\mathrm{E}[S] = 0$ cannot be used.

Solution.

(a) $L(\theta) = \theta^{-n}\mathbf{1}\{\theta \geq y_{(n)}\}$, where $y_{(n)} = \max y_i$. It is zero for $\theta < y_{(n)}$ and decreasing for $\theta \geq y_{(n)}$, so $\hat\theta = y_{(n)}$. There is no root of $S = 0$: for $\theta > y_{(n)}$, $S(\theta) = -n/\theta < 0$.

(b) $P(y_{(n)} \leq t) = (t/\theta)^n$, density nt^{n-1}/θ^n, hence $\mathrm{E}[y_{(n)}] = \int_0^\theta nt^n\theta^{-n}dt = \frac{n}{n+1}\theta$. The bias is $-\theta/(n+1)$; $\frac{n+1}{n}y_{(n)}$ is unbiased.

(c) The support $(0,\theta)$ depends on θ, violating (i); differentiation under the integral sign picks up a boundary term, and $\mathrm{E}[S] = -n/\theta \neq 0$. Also the error $\theta - y_{(n)}$ is of order $1/n$ (not $1/\sqrt{n}$) and $n(\theta - \hat\theta)$ converges to an exponential law, not a normal one.

Exercise 3.3. For $y_1, \dots, y_n$ iid $N(\mu, \sigma^2)$ with μ **known**, find the MLE of σ^2, its expected value, and its observed and expected information. Compare $\mathrm{se}(\hat\sigma^2)$ with the case where μ is unknown.

Solution. With $v = \sigma^2$: $\ell = -\frac{n}{2}\log v - \frac{1}{2v}\sum(y_i-\mu)^2$, $S = -\frac{n}{2v} + \frac{1}{2v^2}\sum(y_i-\mu)^2$, giving $\hat v = \frac{1}{n}\sum(y_i-\mu)^2$. As $\mathrm{E}(y_i-\mu)^2 = v$, $\mathrm{E}\hat v = v$: unbiased. The observed information is $I(v) = -\frac{n}{2v^2} + \frac{1}{v^3}\sum(y_i-\mu)^2$; at $\hat v$ it equals $-\frac{n}{2\hat v^2} + \frac{n}{\hat v^2} = \frac{n}{2\hat v^2}$, and since $\mathrm{E}I = \frac{n}{2v^2}$ the expected information is $n/(2v^2)$. Hence $\mathrm{se}(\hat v) = \hat v\sqrt{2/n}$, identical to the case with unknown mean (the information matrix is diagonal). The difference is only the divisor of $\hat v$ (n versus the bias factor $(n-1)/n$).

Exercise 3.4. Let $y_1, \dots, y_n$ be iid Poisson(λ) and let $\eta = \log\lambda$. (a) Write $\ell(\eta)$, $S(\eta)$, $I(\eta)$. (b) Compute the observed and the expected information of η and compare them. (c) Verify $\mathrm{se}(\hat\eta) = \mathrm{se}(\hat\lambda)/\hat\lambda$ for $n = 100$, $\bar y = 3.1$.

Solution.

(a) Substituting $\lambda = e^\eta$, $\ell(\eta) = \eta\sum y_i - ne^\eta + \text{const}$, $S(\eta) = \sum y_i - ne^\eta$, $I(\eta) = ne^\eta$.

(b) $I(\eta) = ne^\eta$ does not depend on the data, so $\mathcal{I}(\eta) = \mathrm{E}[I(\eta)] = ne^\eta = I(\eta)$ for every η: observed and expected information coincide everywhere, a property of the canonical parametrisation of an exponential family (Chapter 8).

(c) $\hat\eta = \log\bar y$, $I(\hat\eta) = n\bar y = 310$, $\mathrm{se}(\hat\eta) = 310^{-1/2} = 0.0568$. And $\mathrm{se}(\hat\lambda)/\hat\lambda = 0.1761/3.1 = 0.0568$. They agree, as the delta method (or the chain rule for information) requires.

Exercise 3.5. Let $y_1, \dots, y_n$ be iid exponential with *mean* μ. (a) Compute the expected information $\mathcal{I}(\mu)$. (b) Show that $I(\hat\mu) = \mathcal{I}(\hat\mu)$ but $I(\mu) \neq \mathcal{I}(\mu)$ in general. (c) For the aircondit data compute $I(\mu)$ and $\mathcal{I}(\mu)$ at $\mu = 150$.

Solution.

(a) $I(\mu) = -n/\mu^2 + 2\sum y_i/\mu^3$. As $\mathrm{E}[\sum Y_i] = n\mu$, $\mathcal{I}(\mu) = -n/\mu^2 + 2n/\mu^2 = n/\mu^2$.

(b) $I(\hat\mu) = -n/\bar y^2 + 2n\bar y/\bar y^3 = n/\bar y^2 = \mathcal{I}(\hat\mu)$. For general μ, $I(\mu) = n/\mu^2\,(2\bar y/\mu - 1)$, which differs from n/μ^2 unless $\bar y = \mu$.

(c) With $n = 12$ and $\bar y = 108.08$: $\mathcal{I}(150) = 12/150^2 = 5.33\times10^{-4}$ and $I(150) = 5.33\times10^{-4}(2\cdot108.08/150 - 1) = 5.33\times10^{-4}\times0.4411 = 2.35\times10^{-4}$. The observed information is less than half the expected one at $\mu = 150$: the data, which suggest $\mu \approx 108$, carry less curvature at 150 than the model predicts under $\mu = 150$.

Exercise 3.6. Let $y_1, \dots, y_n$ be iid exponential with rate λ and $n \geq 2$. (a) Find the MLE of λ. (b) Show $\mathrm{E}[\hat\lambda] = \frac{n}{n-1}\lambda$ using the Gamma distribution of $\sum y_i$. (c) Construct an unbiased estimator and compare its variance with the Cramér–Rao bound. (d) Comment on invariance versus unbiasedness.

Solution.

(a) $\ell = n\log\lambda - \lambda\sum y_i$, $\hat\lambda = n/\sum y_i = 1/\bar y$.

(b) $T = \sum y_i \sim \mathrm{Gamma}(n, \lambda)$, so $\mathrm{E}[1/T] = \int_0^\infty t^{-1}\frac{\lambda^n t^{n-1}e^{-\lambda t}}{\Gamma(n)}dt = \frac{\lambda^n\Gamma(n-1)}{\Gamma(n)\lambda^{n-1}} = \frac{\lambda}{n-1}$. Thus $\mathrm{E}\hat\lambda = n\lambda/(n-1)$.

(c) $\tilde{\lambda} = (n-1)/T$ is unbiased. Its variance: $\mathrm{E}[T^{-2}] = \lambda^2/\{(n-1)(n-2)\}$ so $\mathrm{Var}(\tilde{\lambda}) = (n-1)^2\{\lambda^2/((n-1)(n-2)) - \lambda^2/(n-1)^2\} = \lambda^2/(n-2)$ for $n \geq 3$. The Cramér–Rao bound is $1/\mathcal{I} = \lambda^2/n$. The ratio is $n/(n-2) > 1$: the unbiased estimator does not attain the bound (none does), but approaches it as $n \to \infty$.

(d) $\bar{y}$ is unbiased for $\mu = 1/\lambda$ but $1/\bar{y}$ is biased for λ (Jensen): invariance holds for the MLE, unbiasedness does not transform.

Exercise 3.7. In the `aml` data (`survival::aml`), the maintained group has 11 patients with times $9, 13, 13^+, 18, 23, 28^+, 31, 34, 45, 48, 161^+$ weeks ($^+$ censored). Assume exponential survival. (a) Compute d, $\sum t_i$, $\hat{\lambda}$, $I(\hat{\lambda})$ and a 95% Wald interval for λ. (b) Test $H_0 : \lambda = 0.01$ with the likelihood-ratio statistic. (c) What would you get from $1/\bar{t}$, and why is it biased? (d) Show that if a new unit is censored at time c, its contribution to the score is $-c$, and explain why adding censored units can never increase $\hat{\lambda}$ above $d/\sum t_i$ computed without them.

Solution.

(a) $d = 7$, $\sum t_i = 423$, $\hat{\lambda} = 7/423 = 0.01655$, $I(\hat{\lambda}) = d/\hat{\lambda}^2 = 25\,561$ (equivalently $423^2/7$) and $\mathrm{se} = \sqrt{7}/423 = 0.00625$; Wald $0.01655 \pm 1.96(0.00625) = (0.0043,\ 0.0288)$.

(b) $\ell(\hat{\lambda}) = 7\log 0.01655 - 0.01655 \times 423 = -35.710$ and $\ell(0.01) = 7\log 0.01 - 0.01 \times 423 = -36.466$, hence LR $= 1.51$ on 1 df, $p = 0.22$: no evidence against $\lambda = 0.01$.

(c) $\bar{t} = 38.45$, so $1/\bar{t} = 0.0260$, about 57% too high: the 4 censored patients are treated as deaths at their last-seen times, although they survived at least that long.

(d) A censored unit adds $\log S(c) = -\lambda c$ to ℓ: the score changes by $-c < 0$ and d is unchanged, so $\hat{\lambda} = d/(\sum t_i + c)$ can only decrease: censored units add exposure but no events.

Exercise 3.8. Type I censoring at a fixed time c: $t_i = \min(T_i, c)$, $T_i \sim \mathrm{Exp}(\lambda)$. (a) Show $P(\delta_i = 1) = 1 - e^{-\lambda c}$ and $E[t_i] = (1 - e^{-\lambda c})/\lambda$. (b) Find the limits of $d/\sum t_i$ and of $1/\bar{t}$. (c) Evaluate them for $\lambda = 0.1$ and $c = 10$ and check by simulation.

Solution.

(a) $P(T \leq c) = 1 - e^{-\lambda c}$; $E[\min(T, c)] = \int_0^c P(T > u)du = \int_0^c e^{-\lambda u}du = (1 - e^{-\lambda c})/\lambda$.

(b) By the law of large numbers $d/n \to 1 - e^{-\lambda c}$ and $\bar{t} \to (1 - e^{-\lambda c})/\lambda$, so $d/\sum t_i \to \lambda$ (the MLE is consistent) while $1/\bar{t} \to \lambda/(1 - e^{-\lambda c}) > \lambda$.

(c) $1/\bar{t} \to 0.1/0.632 = 0.158$, $d/n \to 0.632$. In R: `x <- rexp(1e6, 0.1); y <- pmin(x, 10); d <- x <= 10; c(1/mean(y), sum(d)/sum(y), mean(d))` gives 0.158, 0.0999, 0.632.

Exercise 3.9. For the gamma model of Example 3.5: (a) show that the profile score $g(\alpha) = \log\alpha - \psi(\alpha) - \log\bar{y} + \overline{\log y}$ is strictly decreasing, hence at most one root; (b) compute by hand one Newton–Raphson step from $\alpha_0 = 1$ for the aircondit data, using $\psi(1) = -0.5772$, $\psi'(1) = 1.6449$ and the profile information $I(\alpha) = n(\psi'(\alpha) - 1/\alpha)$; check in R.

Solution.

(a) $g'(\alpha) = 1/\alpha - \psi'(\alpha)$. Using the series $\psi'(\alpha) = \sum_{k\geq 0}(\alpha + k)^{-2} > \int_0^\infty (\alpha + x)^{-2}dx = 1/\alpha$ (a decreasing sum exceeds the integral from 0), $g' < 0$.

(b) For the aircondit data, $\log \bar{y} = \log 108.083 = 4.6829$ and $\overline{\log y} = 3.8286$, so $-\log \bar{y} + \overline{\log y} = -0.8543$. At $\alpha_0 = 1$, $\log 1 - \psi(1) = 0.5772$, so $S(1) = 12(0.5772 - 0.8543) = -3.325$. The information is $I(1) = 12(1.6449 - 1) = 7.739$. Newton–Raphson: $\alpha_1 = \alpha_0 + S/I = 1 - 3.325/7.739 = 0.570$. Continuing: 0.678, 0.705, 0.7065, 0.7065 (quadratic convergence).

Exercise 3.10. Let y be one observation from the Cauchy location model with scale 1. (a) Show that $\mathcal{I}_1(\theta) = \mathrm{E}[S_1^2] = 1/2$. (b) What is the asymptotic standard error of the MLE based on $n = 25$ observations? (c) Why is the sample mean a poor estimator here?

Solution.

(a) With $z = y - \theta$, $S_1 = 2z/(1 + z^2)$, and z has density $\{\pi(1 + z^2)\}^{-1}$:

$$\mathcal{I}_1 = \int_{-\infty}^{\infty} \frac{4z^2}{(1+z^2)^2} \frac{1}{\pi(1+z^2)} dz = \frac{4}{\pi} \int \frac{z^2}{(1+z^2)^3} dz.$$

Substituting $z = \tan u$: $\int_{-\pi/2}^{\pi/2} \sin^2 u \cos^2 u \, du = \int \frac{1}{4} \sin^2(2u) du = \frac{\pi}{8}$. So $\mathcal{I}_1 = \frac{4}{\pi} \cdot \frac{\pi}{8} = \frac{1}{2}$.

(b) $\mathrm{se} \approx (n\mathcal{I}_1)^{-1/2} = \sqrt{2/n} = \sqrt{2/25} = 0.283$.

(c) The sample mean of n Cauchy variables is again standard Cauchy: it does not converge as n grows, so it is not even consistent. The MLE (essentially a robust estimator close to the median) has variance $\approx 2/n$.

Exercise 3.11. Using the aircondit data and the exponential model (mean μ): (a) give the MLE and the delta-method standard error of $p = P(Y > 100) = \exp(-100/\mu)$; (b) obtain a 95% interval for p by transforming a Wald interval for $\log \mu$.

Solution.

(a) By invariance $\hat{p} = \exp(-100/108.08) = 0.3965$. $dp/d\mu = p \cdot 100/\mu^2$, so With $\mathrm{se}(\hat{\mu}) = \hat{\mu}/\sqrt{n}$ the delta method gives $\mathrm{se}(\hat{p}) = \hat{p} \, \frac{100}{\hat{\mu}^2} \frac{\hat{\mu}}{\sqrt{n}} = \hat{p} \, \frac{100}{\hat{\mu}\sqrt{n}} = 0.3965 \times 0.9253 \times 0.2887 = 0.1059$.

(b) $\mathrm{se}(\log \hat{\mu}) = 1/\sqrt{12} = 0.2887$, so a 95% interval for $\log \mu$ is $\log 108.08 \pm 1.96(0.2887) = (4.117, 5.249)$, i.e. $\mu \in (61.4, 190.3)$. Since p is increasing in μ, the interval for p is $(\exp(-100/61.4), \exp(-100/190.3)) = (0.196, 0.591)$.

Exercise 3.12. (Neyman–Scott.) Observe pairs (y_{i1}, y_{i2}), $i = 1, \ldots, m$, with $y_{ij} \sim N(\mu_i, \sigma^2)$ independent, each pair having its own mean μ_i. (a) Find the MLE of σ^2. (b) Show that it converges to $\sigma^2/2$ as $m \to \infty$. (c) Which assumption of the consistency theorem fails?

Solution.

(a) $\ell = -m \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i,j} (y_{ij} - \mu_i)^2 + \text{const}$ (there are $2m$ observations, hence $-\frac{2m}{2} \log \sigma^2$). The MLE of μ_i is $\bar{y}_i = (y_{i1} + y_{i2})/2$, and then $\hat{\sigma}^2 = \frac{1}{2m} \sum_i \sum_j (y_{ij} - \bar{y}_i)^2 = \frac{1}{2m} \sum_i \frac{(y_{i1} - y_{i2})^2}{2}$.

(b) $(y_{i1} - y_{i2})^2/(2\sigma^2) \sim \chi^2_1$ has mean 1, so $\mathrm{E}(y_{i1} - y_{i2})^2 = 2\sigma^2$, and by the law of large numbers $\hat{\sigma}^2 \to \frac{1}{2m} \cdot m \cdot \sigma^2 = \sigma^2/2$. (The unbiased version divides by m instead of $2m$.)

(c) The number of parameters $(m + 1)$ grows with the sample size, so the fixed-dimensional parameter space and the iid framework behind consistency fail. The remedy is a likelihood that eliminates μ_i (the residual maximum likelihood of Chapter 12).

Exercise 3.13. In R, fit by maximum likelihood a Poisson and a negative-binomial model (`dnbinom(mu=, size=)`) to the `discoveries` data using `optim()` with the log-scale parametrisation. Report the estimates, their standard errors (through the Hessian), and the maximised log-likelihoods. Interpret the dispersion.

Solution.

```
nll_nb <- function(p) -sum(dnbinom(y_di, mu = exp(p[1]), size = exp(p[2]), log
↪  = TRUE))
k0 <- mean(y_di)^2 / (var(y_di) - mean(y_di))   # moment estimate of the size
↪  as starting value
fnb <- optim(c(log(mean(y_di)), log(k0)), nll_nb, method = "BFGS", hessian =
↪  TRUE)
est <- c(mu = exp(fnb$par[1]), size = exp(fnb$par[2]))
se_log <- sqrt(diag(solve(fnb$hessian)))
rbind(estimate = est, se_logscale = se_log)
```

```
#>                  mu   size
#> estimate    3.10000 5.4597
#> se_logscale 0.07112 0.4001
```

```
c(loglik_nb = -fnb$value, loglik_pois = loglik_pois(mean(y_di)))
```

```
#>   loglik_nb loglik_pois
#>      -210.8      -216.8
```

The mean is the same, $\hat{\mu} = 3.1$ (the Poisson MLE), but the size parameter $\hat{k} = 5.5$ is finite, meaning variance $\mu + \mu^2/k = 4.86$, close to the sample variance 5.08. The log-likelihood improves by about 6.1 units for one extra parameter, which is strong evidence of overdispersion (the formal test is in Chapter 4; the standard errors are on the log scale because we optimised on that scale).

Exercise 3.14. Critique each statement and correct it. (a) "The likelihood function integrates to one, so it is the posterior probability of the parameter." (b) "The standard error of $\hat{\theta}$ is $\sigma I(\hat{\theta})^{-1/2}$." (c) "$\sqrt{n}(\hat{\theta} - \theta) \approx N(0, \mathcal{I}(\theta)^{-1})$, where $\mathcal{I} = n\mathcal{I}_1$." (d) "The MLE is unbiased, as shown by the normal variance estimator."

Solution.

(a) False: $\int_0^1 L(\theta)d\theta = 1/(n+1)$ for the binomial; L is a function of the data not of θ in a probabilistic sense; a posterior needs a prior and normalisation.

(b) The correct standard error is $I(\hat{\theta})^{-1/2}$; if σ is a parameter it already enters I; multiplying by σ double counts it.

(c) With the $\sqrt{n}$ scaling the variance is $\mathcal{I}_1^{-1}$; $\mathcal{I}^{-1} = 1/(n\mathcal{I}_1)$ would be the variance of $\hat{\theta}$ itself, not of $\sqrt{n}(\hat{\theta} - \theta)$.

(d) The normal variance MLE $\hat{\sigma}^2$ has expectation $\frac{n-1}{n}\sigma^2$: biased. In regular models the bias of the MLE is of order $1/n$, so it is negligible relative to the standard error, but the mean of an MLE need not exist, so "asymptotically unbiased'' should not be claimed in general.

3.13 Further reading

The classic and most readable account of the likelihood as the basis of inference is Pawitan (2001), Chapters 2, 3, 8 and 9, which our treatment follows; Fisher (1922) is the original source of the concept and the information; Kalbfleisch and Prentice (2002) gives a rigorous treatment at a slightly higher level; and Efron and Hinkley (1978) discusses observed versus expected information.

Chapter 4

Likelihood-based inference and numerical optimisation

Chapter 3 showed how to *estimate* a parameter by maximum likelihood and how precise the estimate is, in the sense of an asymptotic standard error. This chapter turns the estimate into a decision or a statement of uncertainty. It answers three practical questions:

1. **Testing and intervals.** Is a particular value θ_0 compatible with the data? Which values are? Three classical answers, the Wald, score and likelihood-ratio procedures, are derived from the same asymptotic theory and compared (Sections 4.1 to 4.3).
2. **Nuisance parameters.** How do we make statements about one parameter when the model has others? The profile likelihood (Section 4.4).
3. **Computation.** Few likelihood equations can be solved by hand. We study Newton–Raphson, Fisher scoring and the quasi-Newton routines behind `optim()` (Section 4.5), and the problems that can arise: boundary estimates, several maxima, non-identifiable parameters (Section 4.6). An optional section presents the EM algorithm (Section 4.7).

We use the same data as in Chapter 3: the air-conditioning failure times (is the exponential model adequate, or is a gamma model needed?), the great discoveries (Poisson or overdispersed?), and, for the numerical part, the waiting times between eruptions of the Old Faithful geyser (`MASS::geyser`), which are visibly a mixture of two groups. Notation as in Appendix A: $I(\theta)$ is the observed and $\mathcal{I}(\theta)$ the expected information.

```
data(aircondit, package = "boot")
y_ac <- aircondit$hours; n_ac <- length(y_ac)
y_di <- as.numeric(discoveries); n_di <- length(y_di)
w_gy <- MASS::geyser$waiting
```

4.1 Wald, score and likelihood-ratio tests

Let θ be a scalar parameter with true value θ_0 and consider $H_0 : \theta = \theta_0$. The data are summarised by the log-likelihood ℓ, and three quantities measure how far θ_0 is from the data (Figure 4.1): the horizontal distance $\hat\theta - \theta_0$, the slope $S(\theta_0)$ of the log-likelihood at the null value, and the vertical drop $\ell(\hat\theta) - \ell(\theta_0)$.

Definition 4.1 (Wald, score and likelihood-ratio statistics)**.** For $H_0 : \theta = \theta_0$,

$$W = (\hat\theta - \theta_0)^2\, I(\hat\theta) \qquad \text{(Wald)}, \tag{4.1}$$

$$S_c = \frac{S(\theta_0)^2}{\mathcal{I}(\theta_0)} \qquad \text{(score, or Rao)}, \tag{4.2}$$

$$LR = 2\{\ell(\hat\theta) - \ell(\theta_0)\} \qquad \text{(likelihood ratio)}. \tag{4.3}$$

(In the score statistic the observed information $I(\theta_0)$ may replace $\mathcal{I}(\theta_0)$; the two versions are asymptotically equivalent. The square roots $z_W = (\hat\theta - \theta_0)\sqrt{I(\hat\theta)}$ and $z_S = S(\theta_0)/\sqrt{\mathcal{I}(\theta_0)}$ have a sign, and $r = \text{sign}(\hat\theta - \theta_0)\sqrt{LR}$ is the signed likelihood-ratio root; all can be used for one-sided tests.)

```
n <- 10; y <- 8; th0 <- 0.5; thh <- y / n
ll <- function(t) y * log(t) + (n - y) * log(1 - t)
sc <- function(t) y / t - (n - y) / (1 - t)
tt <- seq(0.25, 0.97, by = 0.005)
quad <- ll(thh) - 0.5 * n / (thh * (1 - thh)) * (tt - thh)^2
ggplot(data.frame(tt, l = ll(tt), q = quad, tg = ll(th0) + sc(th0) * (tt -
↪  th0)), aes(tt)) +
  geom_line(aes(y = l), colour = ie_navy, linewidth = 1) +
  geom_line(aes(y = q), colour = ie_green, linetype = 2) +
  geom_line(aes(y = tg), colour = ie_coral, linetype = 2) +
  geom_vline(xintercept = c(th0, thh), linetype = 3) +
  coord_cartesian(ylim = c(ll(thh) - 4, ll(thh) + 0.5)) +
  annotate("text", x = 0.9, y = ll(thh) - 0.6, label = "quadratic (Wald)",
  ↪  colour = ie_green, hjust = 0.2) +
  annotate("text", x = 0.3, y = ll(th0) + 1.9, label = "tangent at the null",
  ↪  colour = ie_coral, hjust = 0) +
  labs(x = expression(theta), y = "log-likelihood")
```

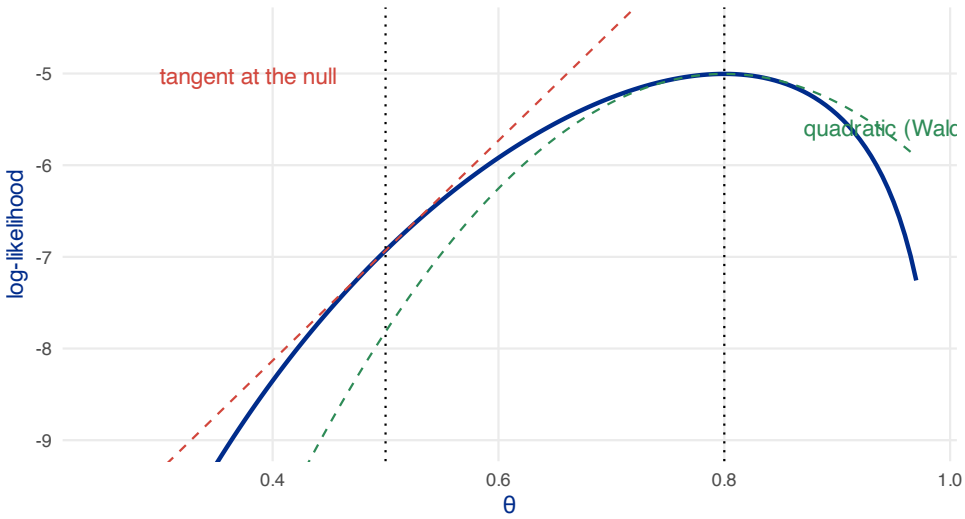


Figure 4.1: The three tests for $H_0 : \theta = 0.5$ with $y = 8$ successes in $n = 10$ trials. LR is twice the vertical gap; Wald is the squared horizontal gap scaled by the curvature at the MLE; the score test uses the slope at θ_0 (tangent line, dashed) scaled by the curvature at the null.

Theorem 4.1 (Asymptotic null distribution of the three statistics)**.** *In a regular model (conditions R of Chapter 3), under $H_0 : \theta = \theta_0$, each of W, S_c and LR converges in distribution to χ^2_1 as $n \to \infty$. Moreover $W - LR \to 0$ and $W - S_c \to 0$ in probability: the three are asymptotically equivalent.*

Proof. Wald. By the asymptotic normality of the MLE (Chapter 3), $\sqrt{\mathcal{I}(\theta_0)}(\hat\theta - \theta_0) \to N(0,1)$, and $I(\hat\theta)/\mathcal{I}(\theta_0) \to 1$ in probability. By Slutsky's lemma $z_W = (\hat\theta - \theta_0)\sqrt{I(\hat\theta)} \to N(0,1)$ and $W = z_W^2 \to \chi^2_1$.

Score. $S(\theta_0)$ is a sum of n iid terms with mean 0 and variance $\mathcal{I}_1(\theta_0)$, so by the central limit theorem $z_S = S(\theta_0)/\sqrt{n\mathcal{I}_1(\theta_0)} \to N(0,1)$ and $S_c = z_S^2 \to \chi^2_1$.

Likelihood ratio. Expand $\ell(\theta_0)$ around $\hat\theta$, where $S(\hat\theta) = 0$:

$$\ell(\theta_0) = \ell(\hat\theta) + S(\hat\theta)(\theta_0 - \hat\theta) - \tfrac{1}{2} I(\theta^*)(\theta_0 - \hat\theta)^2,$$

with θ^* between θ_0 and $\hat\theta$. Hence $LR = I(\theta^*)(\hat\theta - \theta_0)^2$. Since $\theta^* \to \theta_0$ and $I(\theta^*)/I(\hat\theta) \to 1$ in probability (the argument in the proof of asymptotic normality, Chapter 3), $LR = W\{1 + o_p(1)\}$, and $W = O_p(1)$, so $LR - W \to 0$.

Equivalence of S_c and W. From $0 = S(\hat\theta) = S(\theta_0) - I(\theta^{**})(\hat\theta - \theta_0)$ we get $S(\theta_0) = I(\theta^{**})(\hat\theta - \theta_0)$, so $S_c = I(\theta^{**})^2(\hat\theta - \theta_0)^2/\mathcal{I}(\theta_0) = W\{1 + o_p(1)\}$. □

So for large samples it does not matter which statistic we use; for small samples it does. The three are exactly equal when ℓ is exactly quadratic (a normal mean with known variance, Exercise 4.3), and differ in proportion to how non-quadratic ℓ is between θ_0 and $\hat\theta$. A level-α test rejects when the statistic exceeds $\chi^2_{1,1-\alpha}$ (3.84 for $\alpha = 0.05$).

Example 4.1 (A coin: 8 heads in 10 tosses)**.** Test $H_0 : \theta = 0.5$ with $y = 8$, $n = 10$. Recall $\hat\theta = 0.8$, $I(\hat\theta) = \mathcal{I}(\hat\theta) = n/\{\hat\theta(1-\hat\theta)\} = 62.5$, $S(\theta_0) = y/\theta_0 - (n-y)/(1-\theta_0) = 16 - 4 = 12$ and $\mathcal{I}(\theta_0) = 10/0.25 = 40$. Then

$$W = (0.3)^2 \cdot 62.5 = 5.625, \qquad S_c = \frac{12^2}{40} = 3.600, \qquad LR = 2\{8 \log \tfrac{0.8}{0.5} + 2 \log \tfrac{0.2}{0.5}\} = 3.855.$$

The p-values are 0.018, 0.058 and 0.050: the three tests disagree at the 5% level. This disagreement is informative: with $n = 10$, the log-likelihood is clearly not quadratic (Figure 4.1), and the Wald statistic, which uses the curvature at $\hat\theta = 0.8$ where the likelihood is steeper than at 0.5, exaggerates the evidence. The score statistic uses the curvature at the null and is conservative here. The LR statistic, which uses the whole curve, lies in between and is usually the most trustworthy.

```
W  <- (thh - th0)^2 * n / (thh * (1 - thh))
Sc <- sc(th0)^2 / (n / (th0 * (1 - th0)))
LR <- 2 * (ll(thh) - ll(th0))
stats <- c(Wald = W, Score = Sc, LR = LR)
round(rbind(statistic = stats, p_value = pchisq(stats, df = 1, lower.tail =
↪  FALSE)), 4)
```

```
#>             Wald  Score     LR
#> statistic 5.6250 3.6000 3.8549
#> p_value   0.0177 0.0578 0.0496
```

For comparison, the exact two-sided binomial test gives `binom.test(8, 10)$p.value` $= 0.1094$, and the score test is what `prop.test(8, 10, correct = FALSE)` computes (with p-value 0.0578).

Example 4.2 (Poisson rate: a hand calculation)**.** A Poisson sample has $n = 25$ and $\sum y_i = 50$, so $\hat\lambda = 2$. Test $H_0 : \lambda = 1.5$. We have $I(\hat\lambda) = \sum y_i/\hat\lambda^2 = 12.5$, $S(1.5) = 50/1.5 - 25 = 8.333$, $\mathcal{I}(1.5) = 25/1.5 = 16.667$. Therefore

$$W = (0.5)^2(12.5) = 3.125 \ (p = 0.077), \qquad S_c = \frac{8.333^2}{16.667} = 4.167 \ (p = 0.041),$$
$$LR = 2\{50 \log \tfrac{2}{1.5} - 25(2 - 1.5)\} = 3.768 \ (p = 0.052).$$

Again the three statistics bracket the decision: this is a borderline case, to be reported as such rather than dichotomised.

One-sided tests. For $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$ use z_W, z_S or r with the standard normal upper tail. For example, for the coin $z_S = 12/\sqrt{40} = 1.90$ and $p = 0.029$.

4.2 Confidence intervals by inverting the tests

A $100(1-\alpha)\%$ confidence interval is the set of θ_0 that the test does not reject. Inverting the three statistics gives three intervals.

Definition 4.2 (Wald, score and likelihood intervals)**.** Let $z = z_{1-\alpha/2}$ be the standard normal quantile and $c = \chi^2_{1,1-\alpha} = z^2$.

$$\text{Wald:} \quad \hat\theta \pm z\,\mathrm{se}(\hat\theta) = \hat\theta \pm z\, I(\hat\theta)^{-1/2}, \tag{4.4}$$
$$\text{score:} \quad \{\theta : S(\theta)^2/\mathcal{I}(\theta) \le c\}, \tag{4.5}$$
$$\text{likelihood (LR):} \quad \{\theta : 2[\ell(\hat\theta) - \ell(\theta)] \le c\} = \{\theta : \ell(\theta) \ge \ell(\hat\theta) - c/2\}. \tag{4.6}$$

The 95% LR interval is the set of parameter values whose relative likelihood exceeds $e^{-1.92} = 0.147$ (the dotted line in the relative-likelihood figure of Chapter 3); it needs only root finding (`uniroot()`), is **transformation-invariant**, and need not be symmetric around $\hat\theta$.

Theorem 4.2 (Invariance of likelihood intervals)**.** *If $\psi = g(\theta)$ is monotone and (θ_L, θ_U) is the LR interval for θ, then the LR interval for ψ is $(g(\theta_L), g(\theta_U))$ (for increasing g). The Wald interval does not have this property.*

Proof. The likelihood of ψ is $L^*(\psi) = L(g^{-1}(\psi))$ and $\hat\psi = g(\hat\theta)$ (invariance of the MLE, Chapter 3). Thus $2[\ell^*(\hat\psi) - \ell^*(\psi)] = 2[\ell(\hat\theta) - \ell(g^{-1}(\psi))] \le c$ if and only if $g^{-1}(\psi)$ is in the LR interval for θ. For the Wald interval, $\hat\psi \pm z\,|g'(\hat\theta)|\mathrm{se}(\hat\theta)$ is a symmetric interval around $g(\hat\theta)$, whereas $g(\hat\theta \pm z\,\mathrm{se}(\hat\theta))$ is generally not symmetric. The two coincide only for linear g. □

Example 4.3 (Three intervals for the binomial)**.** For $y = 8$, $n = 10$: the Wald interval is $0.8 \pm 1.96(0.1265) = (0.552, 1.048)$, which **exceeds one**. The score interval solves $(\hat\theta - \theta)^2 = z^2\theta(1-\theta)/n$, a quadratic equation in θ (Exercise 4.2); it is the **Wilson interval** $(0.490, 0.943)$. The LR interval is $(0.501, 0.964)$.

```
z <- qnorm(0.975); se <- sqrt(thh * (1 - thh) / n)
wald <- thh + c(-1, 1) * z * se
score_ci <- as.numeric(prop.test(y, n, correct = FALSE)$conf.int)   # Wilson =
↪  inverted score test
cut <- qchisq(0.95, df = 1) / 2                                     # 1.92
f <- function(t) ll(t) - ll(thh) + cut
lr_ci <- c(uniroot(f, c(1e-8, thh), tol = 1e-10)$root, uniroot(f, c(thh, 1 -
↪  1e-8), tol = 1e-10)$root)
round(rbind(Wald = wald, Score = score_ci, LR = lr_ci), 3)
```

```
#>        [,1]  [,2]
#> Wald  0.552 1.048
#> Score 0.490 0.943
#> LR    0.501 0.964
```

Example 4.4 (Mean time between failures: parametrisation matters for Wald, not for LR)**.** For the exponential mean μ with $\hat{\mu} = \bar{y} = 108.08$, $\mathrm{se}(\hat{\mu}) = \hat{\mu}/\sqrt{n} = 31.2$. The Wald interval is $108.1 \pm 1.96(31.2) = (46.9, 169.2)$. Since μ must be positive and the sampling distribution of $\hat{\mu}$ is right-skewed, it is better to work with $\log \mu$, whose standard error is $1/\sqrt{n} = 0.289$ whatever the data. Back-transforming, $\exp(\log \hat{\mu} \pm 1.96/\sqrt{n}) = (61.4, 190.3)$. The LR interval, which is automatically invariant, is $(64.4, 201.8)$. For the exponential there is an *exact* answer: $2\sum y_i/\mu \sim \chi^2_{2n}$ gives $(65.9, 209.2)$. The LR interval is the closest to exact and it correctly shifts to the right to reflect the skewness.

```
mu_hat <- mean(y_ac)
llmu <- function(mu) -n_ac * log(mu) - sum(y_ac) / mu
wald_mu    <- mu_hat + c(-1, 1) * z * mu_hat / sqrt(n_ac)
wald_logmu <- exp(log(mu_hat) + c(-1, 1) * z / sqrt(n_ac))
g <- function(mu) llmu(mu) - llmu(mu_hat) + cut
lr_mu <- c(uniroot(g, c(1, mu_hat), tol = 1e-10)$root, uniroot(g, c(mu_hat,
↪  5000), tol = 1e-10)$root)
exact <- 2 * sum(y_ac) / qchisq(c(0.975, 0.025), df = 2 * n_ac)
round(rbind(Wald_mu = wald_mu, Wald_log_mu = wald_logmu, LR = lr_mu, Exact =
↪  exact), 1)
```

```
#>             [,1]  [,2]
#> Wald_mu     46.9 169.2
#> Wald_log_mu 61.4 190.3
#> LR          64.4 201.8
#> Exact       65.9 209.2
```

```
# Invariance: the LR interval for the rate 1/mu is the reciprocal interval
rbind(rate_from_LR_mu = rev(1 / lr_mu))
```

```
#>                     [,1]    [,2]
#> rate_from_LR_mu 0.004954 0.01552
```

```
mm <- seq(40, 330, by = 1)
rl <- exp(sapply(mm, llmu) - llmu(mu_hat))
ggplot(data.frame(mm, rl), aes(mm, rl)) + geom_line(colour = ie_navy,
↪  linewidth = 1) +
```

```
  geom_hline(yintercept = exp(-cut), linetype = 3) +
  annotate("segment", x = lr_mu[1], xend = lr_mu[2], y = 0.04, yend = 0.04,
  ↪  colour = ie_coral, linewidth = 2) +
  annotate("segment", x = wald_mu[1], xend = wald_mu[2], y = 0.01, yend =
  ↪  0.01, colour = ie_green, linewidth = 2) +
  annotate("text", x = 300, y = 0.08, label = "LR", colour = ie_coral) +
  annotate("text", x = 300, y = 0.0, label = "Wald", colour = ie_green) +
  labs(x = expression(mu), y = "relative likelihood")
```

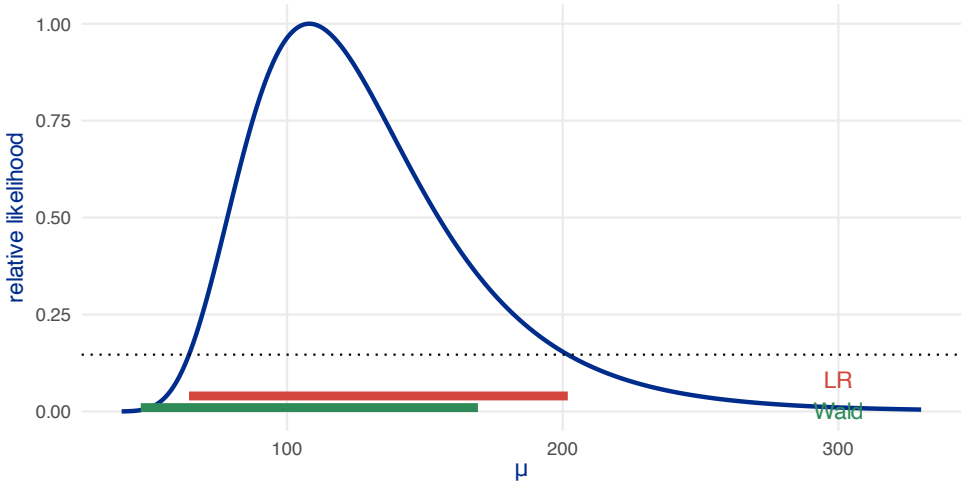


Figure 4.2: Relative likelihood of the exponential mean for the aircondit data with the LR cut-off (dotted) and the Wald (dashed) and LR intervals (bars). The likelihood is skewed to the right, the Wald interval is symmetric.

Warning. The Wald interval is the simplest, and it is the one printed by `summary()` and `confint.default()` for models in this book. It is only as good as the quadratic approximation of ℓ. For parameters that are bounded (probabilities, variances) or skewed (rates, odds ratios) compute it on a transformed scale (logit, log) and transform back, or use the LR interval (`confint()` for a `glm`, Chapter 9).

4.3 Several parameters and composite hypotheses

Let $\theta \in \mathbb{R}^p$ and consider testing $H_0 : \theta = \theta_0$ (a simple null). The statistics generalise to

$$W = (\hat\theta - \theta_0)^\top I(\hat\theta)(\hat\theta - \theta_0), \quad S_c = S(\theta_0)^\top \mathcal{I}(\theta_0)^{-1} S(\theta_0), \quad LR = 2\{\ell(\hat\theta) - \ell(\theta_0)\},$$

and each converges to χ^2_p under H_0. For the Wald statistic this follows from $\hat\theta \dot\sim N_p(\theta_0, I(\hat\theta)^{-1})$ and the fact that if $X \sim N_p(0, \Sigma)$ then $X^\top \Sigma^{-1} X \sim \chi^2_p$; for the score statistic from the multivariate CLT for $S(\theta_0)$; and for the LR statistic from the Taylor expansion $LR = (\hat\theta - \theta_0)^\top I(\theta^*)(\hat\theta - \theta_0)$ as in the scalar case.

More commonly, H_0 restricts only some parameters. Partition $\theta = (\psi, \lambda)$ with $\psi \in \mathbb{R}^q$ the parameters of interest and λ nuisance parameters, and let $H_0 : \psi = \psi_0$. Let $\hat\theta$ be the unrestricted MLE and $\hat{\hat\theta} = (\psi_0, \hat{\hat\lambda})$ the MLE under H_0.

Theorem 4.3 (Wilks' theorem)**.** *Under the regularity conditions and $H_0 : \psi = \psi_0$ true (with ψ_0 an interior point), $LR = 2\{\ell(\hat\theta) - \ell(\hat{\hat\theta})\} \xrightarrow{d} \chi^2_q$, where q is the number of restrictions,*

that is, the difference in the dimensions of the full and the null model (Wilks, 1938). The Wald version is $W = (\hat{\psi} - \psi_0)^\top \{[I(\hat{\theta})^{-1}]_{\psi\psi}\}^{-1} (\hat{\psi} - \psi_0) \to \chi^2_q$*, with* $[I(\hat{\theta})^{-1}]_{\psi\psi}$ *the block of the inverse information matrix belonging to* ψ *(not the inverse of the* $\psi\psi$ *block of* I*).*

Sketch of the proof. Expanding ℓ around the full MLE, as before, $2\{\ell(\hat{\theta}) - \ell(\theta_0)\} \approx (\hat{\theta} - \theta_0)^\top I\,(\hat{\theta} - \theta_0)$ where $\theta_0 = (\psi_0, \lambda_0)$ is the true value. The same expansion for the restricted problem, in which only λ varies, gives $2\{\ell(\hat{\hat{\theta}}) - \ell(\theta_0)\} \approx (\hat{\hat{\lambda}} - \lambda_0)^\top I_{\lambda\lambda} (\hat{\hat{\lambda}} - \lambda_0)$. Writing the first as the square of the norm $\|Z\|^2$ of a vector $Z = I^{1/2}(\hat{\theta} - \theta_0) \approx N_p(0, \mathbf{1}_p)$ and the second as the squared norm of the projection of Z onto the $(p-q)$-dimensional column space of $I^{1/2}$ corresponding to λ, the difference LR is the squared norm of the projection onto the orthogonal complement, which has dimension q, hence χ^2_q. □

Therefore, **degrees of freedom = number of independent restrictions**: this is the number you compare to a χ^2 table, for example 1 for $H_0 : \alpha = 1$ in the gamma model (exponential against gamma).

Example 4.5 (Exponential or gamma? Aircondit)**.** The gamma model with shape α and rate β contains the exponential at $\alpha = 1$. We fit both models, then test $H_0 : \alpha = 1$ with $q = 1$ restriction.

```
# Full gamma fit, parametrised on the log scale
nll_g <- function(p) -sum(dgamma(y_ac, shape = exp(p[1]), rate = exp(p[2]),
↪  log = TRUE))
fit_g <- optim(c(0, log(1 / mean(y_ac))), nll_g, method = "BFGS", hessian =
↪  TRUE)
c(alpha_hat = exp(fit_g$par[1]), beta_hat = exp(fit_g$par[2]), loglik =
↪  -fit_g$value, conv = fit_g$convergence)
```

```
#>  alpha_hat   beta_hat     loglik       conv
#>   0.706471   0.006536 -67.645425   0.000000
```

```
ll_exp   <- sum(dexp(y_ac, rate = 1 / mean(y_ac), log = TRUE))
LR_g <- 2 * (-fit_g$value - ll_exp)
c(ll_exp = ll_exp, LR = LR_g, p_value = pchisq(LR_g, df = 1, lower.tail =
↪  FALSE))
```

```
#>   ll_exp       LR  p_value
#> -68.1948   1.0988   0.2945
```

```
# Wald test of alpha = 1 via the delta method: se(alpha) = alpha * se(log
↪  alpha)
se_log <- sqrt(diag(solve(fit_g$hessian)))
a_hat <- exp(fit_g$par[1]); se_a <- a_hat * se_log[1]
z_w <- (a_hat - 1) / se_a
c(se_alpha = se_a, z = z_w, p_value = 2 * pnorm(-abs(z_w)))
```

```
#> se_alpha        z  p_value
#>   0.2460  -1.1934   0.2327
```

The maximum likelihood estimates are $\hat{\alpha} = 0.706$ and $\hat{\beta} = 0.0065$ (so $\hat{\alpha}/\hat{\beta} = \bar{y}$), and the log-likelihood increases from -68.19 to -67.65: $LR = 1.10$ on one degree of freedom, $p = 0.29$. The Wald test gives $z = -1.19$, $p = 0.23$. There is **no evidence against the exponential**

model, even though $\hat{\alpha}$ is below one. With twelve observations the data cannot tell $\alpha = 0.7$ from $\alpha = 1$; the likelihood simply is not informative enough. Absence of evidence is not evidence of absence: the sensible statement is that a decreasing hazard is plausible but not established.

Example 4.6 (Poisson against negative binomial: a boundary problem)**.** The negative binomial with mean μ and size k has variance $\mu + \mu^2/k$ and tends to the Poisson as $k \to \infty$. Testing Poisson against negative binomial is thus a test of $1/k = 0$ on the **boundary** of the parameter space $1/k \geq 0$. Wilks' theorem is for interior points; at a boundary the null distribution of LR is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (a point mass at zero with probability 1/2 plus a half-χ^2_1) (Self and Liang, 1987), so the correct p-value is *half* the usual χ^2_1 p-value.

```
ll_pois <- sum(dpois(y_di, mean(y_di), log = TRUE))
nll_nb  <- function(p) -sum(dnbinom(y_di, mu = exp(p[1]), size = exp(p[2]),
↪  log = TRUE))
k0 <- mean(y_di)^2 / (var(y_di) - mean(y_di))          # moment estimate of
↪  the size, used as starting value
fit_nb  <- optim(c(log(mean(y_di)), log(k0)), nll_nb, method = "BFGS", hessian
↪  = TRUE)
LR_nb   <- 2 * (-fit_nb$value - ll_pois)
c(mu_hat = exp(fit_nb$par[1]), k_hat = exp(fit_nb$par[2]), loglik_nb =
↪  -fit_nb$value,
  loglik_pois = ll_pois, LR = LR_nb)
```

```
#>      mu_hat       k_hat   loglik_nb loglik_pois          LR
#>        3.10        5.46     -210.79     -216.85       12.10
```

```
c(p_naive = pchisq(LR_nb, 1, lower.tail = FALSE), p_boundary = 0.5 *
↪  pchisq(LR_nb, 1, lower.tail = FALSE))
```

```
#>    p_naive p_boundary
#>  0.0005035  0.0002518
```

```
# the same model from the naive start log(k) = 0 (k = 1): BFGS runs off
↪  towards k = Inf, the Poisson limit
bad <- optim(c(log(mean(y_di)), 0), nll_nb, method = "BFGS")
c(k_hat = exp(bad$par[2]), loglik = -bad$value, convergence = bad$convergence)
```

```
#>       k_hat      loglik convergence
#>   3.241e+10  -2.168e+02   0.000e+00
```

For the discoveries data, $\hat{k} = 5.5$ and the log-likelihood increases by 6.1, so $LR = 12.1$, with $p < 0.001$ either way (2.5×10^{-4} with the boundary correction): there is clear overdispersion. Note the last lines of the code: started from $k = 1$ (a seemingly harmless choice), BFGS reports convergence at $\hat{k} \approx 3 \times 10^{10}$, i.e. the Poisson model, with an LR of zero. The likelihood in k is very flat for large k, so a bad first step can send the optimiser to the flat end; the starting value from the method of moments, $\hat{k}_0 = \bar{y}^2/(s^2 - \bar{y})$, avoids the problem. Remember, however, that these likelihood tests are only valid if the *rest* of the model is correct (independence across years, in particular).

Warning. Wilks' theorem fails when the null value is on the boundary of the parameter space (variance components equal to zero, Chapter 12; Poisson against negative binomial), when a parameter is not identified under H_0 (the number of components of a mixture, Section 4.7), and with very small samples. In such cases use a mixture-χ^2 reference or a parametric bootstrap.

4.4 Profile likelihood

When the model has parameters $\theta = (\psi, \lambda)$ and we care about ψ only, λ is a **nuisance parameter**. We cannot read $L(\psi, \lambda)$ as a function of ψ alone; we need to remove λ.

Definition 4.3 (Profile likelihood)**.** The **profile log-likelihood** of ψ is $\ell_p(\psi) = \max_\lambda \ell(\psi, \lambda) = \ell(\psi, \hat\lambda(\psi))$, where $\hat\lambda(\psi)$ is the MLE of λ for fixed ψ.

The profile likelihood is maximised at the overall MLE $\hat\psi$, with $\ell_p(\hat\psi) = \ell(\hat\theta)$, and $2\{\ell_p(\hat\psi) - \ell_p(\psi_0)\} = 2\{\ell(\hat\theta) - \ell(\psi_0, \hat{\hat\lambda})\}$ is exactly the LR statistic for $H_0 : \psi = \psi_0$. By Wilks' theorem, a $100(1-\alpha)\%$ profile-likelihood interval is $\{\psi : \ell_p(\psi) \geq \ell_p(\hat\psi) - \chi^2_{q,1-\alpha}/2\}$. Profile intervals inherit the invariance of Theorem 4.2 and are the intervals `confint()` computes for `glm` fits (via `MASS`) and for `stats4::mle` fits.

Example 4.7 (Profile likelihood for a normal mean)**.** For iid $N(\mu, \sigma^2)$, for fixed μ the MLE of σ^2 is $\hat\sigma^2(\mu) = n^{-1}\sum(y_i - \mu)^2$ and

$$\ell_p(\mu) = -\frac{n}{2}\log \hat\sigma^2(\mu) - \frac{n}{2} + \text{const.}$$

Since $\sum(y_i - \mu)^2 = \sum(y_i - \bar y)^2 + n(\bar y - \mu)^2 = n\hat\sigma^2\{1 + (\bar y - \mu)^2/\hat\sigma^2\}$ we get $\ell_p(\mu) = -\frac{n}{2}\log\{1 + (\bar y - \mu)^2/\hat\sigma^2\} + \text{const}$ and

$$LR(\mu_0) = n \log\left\{1 + \frac{t^2}{n-1}\right\}, \qquad t = \frac{\bar y - \mu_0}{s/\sqrt{n}},\ s^2 = \frac{n}{n-1}\hat\sigma^2.$$

The LR test is a monotone function of the t statistic, so it is the usual t-test; the χ^2_1 approximation to LR is the large-sample version of the t_{n-1} distribution.

```
y_nh <- as.numeric(nhtemp); n_nh <- length(y_nh)
lp <- function(m) -n_nh / 2 * log(mean((y_nh - m)^2))
m_hat <- mean(y_nh)
k <- function(m) lp(m) - lp(m_hat) + cut
prof_ci <- c(uniroot(k, c(m_hat - 3, m_hat), tol = 1e-10)$root, uniroot(k,
↪  c(m_hat, m_hat + 3), tol = 1e-10)$root)
s2_hat <- mean((y_nh - m_hat)^2)
round(rbind(profile_LR = prof_ci, Wald = m_hat + c(-1, 1) * z * sqrt(s2_hat /
↪  n_nh),
            t_interval = as.numeric(t.test(y_nh)$conf.int)), 3)
```

```
#>             [,1]  [,2]
#> profile_LR 50.84 51.48
#> Wald       50.84 51.48
#> t_interval 50.83 51.49
```

```
mm <- seq(50.6, 51.7, length.out = 200)
ggplot(data.frame(mm, p = sapply(mm, lp) - lp(m_hat)), aes(mm, p)) +
↪  geom_line(colour = ie_navy, linewidth = 1) +
  geom_hline(yintercept = -cut, linetype = 2) + geom_vline(xintercept =
  ↪  prof_ci, linetype = 3) +
  labs(x = expression(mu), y = "profile log-likelihood")
```

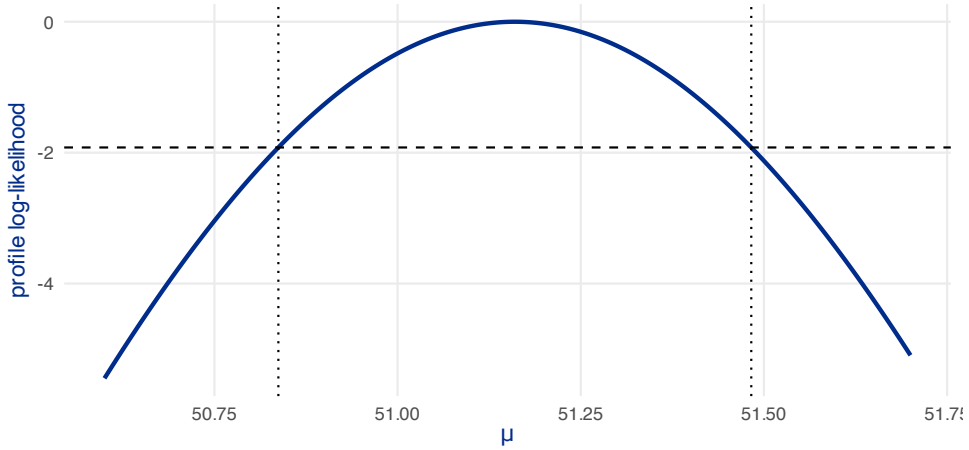


Figure 4.3: Profile log-likelihood of the mean annual temperature at New Haven (maximum scaled to zero), with the 95% cut-off.

The profile interval $(50.84, 51.48)$ is essentially the Wald interval and the exact t interval $(50.83, 51.49)$: with $n = 60$ the three agree to the second decimal. For small n the profile and Wald intervals, which rely on the χ^2_1 calibration, are slightly too short, and the t interval is the correct one (Example 4.7 shows that the LR statistic is a function of t).

4.4.1 Profile information

How does the curvature of the profile likelihood relate to the information matrix of the full model? The answer shows why the standard error of one parameter must be read from the *inverse* of the full information matrix.

Proposition 4.1 (Curvature of the profile log-likelihood)**.** *For $\theta = (\psi, \lambda)$ with ψ scalar, at the MLE*

$$-\ell_p''(\hat\psi) = I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi} = \frac{1}{[I(\hat\theta)^{-1}]_{\psi\psi}},$$

so the profile standard error $(-\ell_p''(\hat\psi))^{-1/2}$ equals $\mathrm{se}(\hat\psi)$ from the full model.

Proof. By definition $\partial \ell / \partial \lambda \,(\psi, \hat\lambda(\psi)) = 0$ for all ψ. Differentiating with respect to ψ gives $\ell_{\lambda\psi} + \ell_{\lambda\lambda}\hat\lambda'(\psi) = 0$, i.e. $\hat\lambda' = -\ell_{\lambda\lambda}^{-1}\ell_{\lambda\psi} = -I_{\lambda\lambda}^{-1} I_{\lambda\psi}$ (using $I = -\ell''$, signs cancel). Now $\ell_p'(\psi) = \ell_\psi(\psi, \hat\lambda(\psi))$ (the term involving $\hat\lambda'$ is multiplied by $\ell_\lambda = 0$) and

$$\ell_p''(\psi) = \ell_{\psi\psi} + \ell_{\psi\lambda}\hat\lambda' = -I_{\psi\psi} + I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi}.$$

The right-hand expression is minus the Schur complement of $I_{\lambda\lambda}$, and by the block-inverse formula the Schur complement equals $1/[I^{-1}]_{\psi\psi}$. □

For the gamma model with parameters (α, β) of Chapter 3 (information matrix of the gamma), the Schur complement is $n\psi'(\alpha) - (n/\beta)^2/(n\alpha/\beta^2) = n\{\psi'(\alpha) - 1/\alpha\}$: this is the quantity $I(\alpha)$ used in the Newton iteration below.

Example 4.8 (Profile interval for the gamma shape)**.** Maximising over β for fixed α gives $\hat\beta(\alpha) = \alpha/\bar y$, so $\ell_p(\alpha) = \sum \log f(y_i; \alpha, \alpha/\bar y)$.

```
lp_a <- function(a) sum(dgamma(y_ac, shape = a, rate = a / mean(y_ac), log =
↪  TRUE))
gg <- function(a) lp_a(a) - lp_a(a_hat) + cut
prof_a <- c(uniroot(gg, c(0.05, a_hat), tol = 1e-10)$root, uniroot(gg,
↪  c(a_hat, 10), tol = 1e-10)$root)
I_a <- n_ac * (trigamma(a_hat) - 1 / a_hat)          # profile information
wald_a <- exp(log(a_hat) + c(-1, 1) * z / (a_hat * sqrt(I_a)))   # Wald on
↪  log(alpha)
round(rbind(profile_LR = prof_a, Wald_log_scale = wald_a), 3)
```

```
#>                 [,1]  [,2]
#> profile_LR     0.334 1.317
#> Wald_log_scale 0.357 1.398
```

```
c(se_alpha_profile = 1 / sqrt(I_a), se_alpha_hessian = se_a)
```

```
#> se_alpha_profile se_alpha_hessian
#>            0.246            0.246
```

```
aa <- seq(0.15, 2.2, length.out = 200)
ggplot(data.frame(aa, p = sapply(aa, lp_a) - lp_a(a_hat)), aes(aa, p)) +
↪  geom_line(colour = ie_navy, linewidth = 1) +
  geom_hline(yintercept = -cut, linetype = 2) + geom_vline(xintercept =
  ↪  c(a_hat, 1), linetype = c(3, 1), colour = c("grey40", ie_coral)) +
  coord_cartesian(ylim = c(-4, 0.2)) + labs(x = expression(alpha), y =
  ↪  "profile log-likelihood")
```

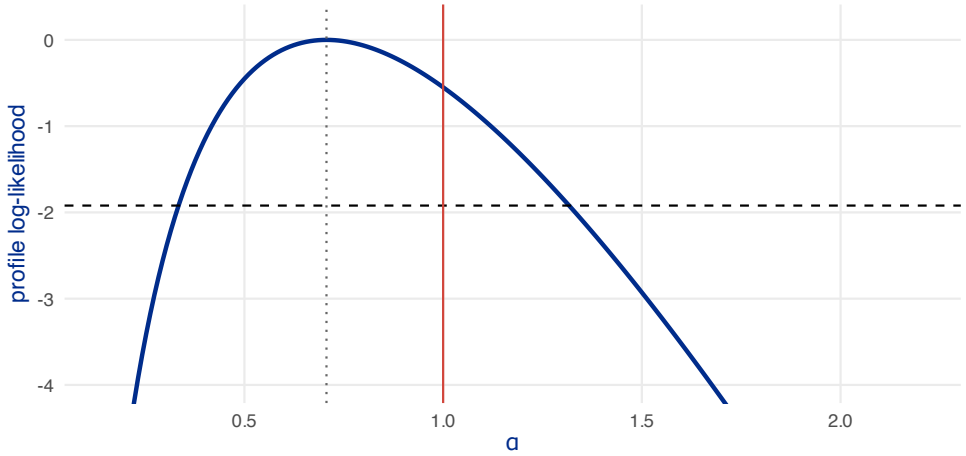


Figure 4.4: Profile relative log-likelihood of the gamma shape α for the aircondit data. The 95% interval includes $\alpha = 1$ (the exponential model).

The 95% profile interval for the shape is $(0.33, 1.32)$, strongly asymmetric around $\hat{\alpha} = 0.706$, whereas the Wald interval on the log scale is $(0.36, 1.40)$. Both contain one, in agreement with the LR test. The two ways of computing the standard error of $\hat{\alpha}$ (from the profile information and from the Hessian of the full two-parameter problem on the log scale, with the delta method) agree, as Proposition 4.1 predicts: 0.246.

4.5 Numerical maximisation

Only a few likelihoods (those of Chapter 3, except the gamma) have closed-form maximisers. In general we need an iterative algorithm that produces a sequence $\theta^{(0)}, \theta^{(1)}, \dots$ converging to $\hat\theta$.

4.5.1 Newton–Raphson

Suppose $\theta^{(k)}$ is the current guess. Replace S by its linear approximation at $\theta^{(k)}$, $S(\theta) \approx S(\theta^{(k)}) - I(\theta^{(k)})(\theta - \theta^{(k)})$, and solve for the zero:

$$\theta^{(k+1)} = \theta^{(k)} + \frac{S(\theta^{(k)})}{I(\theta^{(k)})}, \qquad \theta^{(k+1)} = \theta^{(k)} + I(\theta^{(k)})^{-1} S(\theta^{(k)}) \text{ (vector case).} \tag{4.7}$$

Equivalently, we maximise the quadratic Taylor approximation of ℓ at $\theta^{(k)}$. The update moves uphill (in the direction of the score) by an amount that is large when the curvature is small and vice versa.

Proposition 4.2 (Quadratic convergence)**.** *Let $\hat\theta$ be a root of S with $I(\hat\theta) > 0$ and ℓ three times continuously differentiable. Then there is a neighbourhood of $\hat\theta$ such that Newton–Raphson started in it converges, and $|\theta^{(k+1)} - \hat\theta| \le C\,|\theta^{(k)} - \hat\theta|^2$.*

Proof. Let $e_k = \theta^{(k)} - \hat\theta$. Taylor expansion of S around $\theta^{(k)}$, evaluated at $\hat\theta$: $0 = S(\hat\theta) = S(\theta^{(k)}) - I(\theta^{(k)})(\hat\theta - \theta^{(k)}) + \tfrac12 S''(\xi)(\hat\theta - \theta^{(k)})^2$ for some ξ between $\theta^{(k)}$ and $\hat\theta$, i.e. $S(\theta^{(k)})/I(\theta^{(k)}) = -e_k - \tfrac12\{S''(\xi)/I(\theta^{(k)})\}e_k^2$ (note $\hat\theta - \theta^{(k)} = -e_k$). Then $e_{k+1} = e_k + S(\theta^{(k)})/I(\theta^{(k)}) = -\tfrac12\{S''(\xi)/I(\theta^{(k)})\}e_k^2$. Near $\hat\theta$, $I \ge I(\hat\theta)/2 > 0$ and $|S''| = |\ell'''|$ is bounded, so $|e_{k+1}| \le C e_k^2$, with $C < \infty$. For $|e_0| < 1/(2C)$ the errors decrease, hence the convergence. □

Quadratic convergence means the number of correct digits roughly doubles at each step: once close, 5–6 iterations suffice. The proof also shows the limitation: the guarantee is **local**.

Example 4.9 (Newton–Raphson for the gamma shape)**.** For the aircondit data, the (profile) score and information of the shape are

$$S(\alpha) = n\{\log\alpha - \psi(\alpha) - \log\bar y + \overline{\log y}\}, \qquad I(\alpha) = n\{\psi'(\alpha) - 1/\alpha\}.$$

Start at $\alpha_0 = 1$ (the exponential model): $S(1) = -3.325$, $I(1) = 7.739$ and $\alpha_1 = 1 - 3.325/7.739 = 0.570$.

```
S_a <- function(a) n_ac * (log(a) - digamma(a) - log(mean(y_ac)) +
↪  mean(log(y_ac)))
I_a_fun <- function(a) n_ac * (trigamma(a) - 1 / a)
a <- 1; iter <- data.frame()
for (k in 0:5) {
  iter <- rbind(iter, data.frame(k = k, alpha = a, score = S_a(a), info =
↪  I_a_fun(a)))
  a <- a + S_a(a) / I_a_fun(a)
}
print(iter, digits = 6, row.names = FALSE)
```

```
#>  k    alpha        score     info
#>  0 1.000000 -3.32518e+00  7.73921
```

```
#>  1 0.570346  2.84662e+00 26.40735
#>  2 0.678142  4.90086e-01 18.07836
#>  3 0.705251  2.05655e-02 16.59211
#>  4 0.706491  3.94754e-05 16.52848
#>  5 0.706493  1.46022e-10 16.52835
```

The distance to the solution $\hat\alpha = 0.706493$ is $0.294,\ 0.136,\ 0.028,\ 1.2\times10^{-3},\ 1.9\times10^{-6},\ 10^{-11}$: after the first steps each error is roughly proportional to the square of the previous one ($0.028^2 = 7.8\times10^{-4}$, $0.00124^2 = 1.5\times10^{-6}$), as the proposition says.

Example 4.10 (Newton–Raphson can diverge)**.** For the exponential *rate* λ, $S = n/\lambda - \sum y_i$, $I = n/\lambda^2$, and so $\lambda^{(k+1)} = \lambda^{(k)}(2 - \bar y\lambda^{(k)})$. The error obeys $\lambda^{(k+1)} - \hat\lambda = -\bar y(\lambda^{(k)} - \hat\lambda)^2$ with $\hat\lambda = 1/\bar y$: convergence requires $|\bar y(\lambda^{(0)} - \hat\lambda)| < 1$, i.e. $0 < \lambda^{(0)} < 2/\bar y$. For the aircondit data, $2/\bar y = 0.0185$; starting at $\lambda^{(0)} = 0.02$ gives $\lambda^{(1)} = 0.02(2 - 2.16) < 0$, outside the parameter space, and the next iterations diverge to $-\infty$.

```
nr_rate <- function(l0, K = 6) { l <- numeric(K + 1); l[1] <- l0; for (k in
↪  1:K) l[k + 1] <- l[k] * (2 - mean(y_ac) * l[k]); l }
rbind(start_0.005 = nr_rate(0.005), start_0.02 = nr_rate(0.02)) |> signif(4)
```

```
#>              [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
#> start_0.005 0.005  0.007298  0.008839  0.009234  0.009252  0.009252
#> start_0.02  0.020 -0.003233 -0.007597 -0.021430 -0.092500 -1.110000
#>                   [,7]
#> start_0.005  9.252e-03
#> start_0.02  -1.353e+02
```

```
1 / mean(y_ac)
```

```
#> [1] 0.009252
```

The remedies are: a good starting value (for example, the method-of-moments estimate), step-halving (replace the update by $\theta^{(k)} + \delta\, S/I$ with $\delta \in (0,1]$ chosen so that ℓ increases), and a good parametrisation: in the log-scale $\eta = \log\lambda$, the problem is much better behaved (try it: Exercise 4.7).

4.5.2 Fisher scoring

Fisher scoring replaces the observed information in (4.7) by the expected information:

$$\theta^{(k+1)} = \theta^{(k)} + \mathcal{I}(\theta^{(k)})^{-1} S(\theta^{(k)}).$$

It is usually more stable, because $\mathcal{I}$ is positive definite by construction whereas $I(\theta)$ can be negative away from the maximum; it needs the expectation to be computable; and the convergence is linear rather than quadratic in general. When the model is an exponential family in canonical form, $I = \mathcal{I}$ and the two algorithms are identical; the IRLS algorithm of Chapter 9 is Fisher scoring.

Example 4.11 (Cauchy location: Newton–Raphson against Fisher scoring)**.** For the Cauchy location model with $y = (-2,-1,0,7,8)$, $I(\theta) = \sum \frac{2\{1-(y_i-\theta)^2\}}{\{1+(y_i-\theta)^2\}^2}$ can be negative, but $\mathcal{I}(\theta) = n/2$ is a positive constant, so Fisher scoring is $\theta^{(k+1)} = \theta^{(k)} + (2/n)S(\theta^{(k)})$.

```r
yc <- c(-2, -1, 0, 7, 8); nc <- length(yc)
Sc_ <- function(t) sum(2 * (yc - t) / (1 + (yc - t)^2))
Ic_ <- function(t) sum(2 * (1 - (yc - t)^2) / (1 + (yc - t)^2)^2)
run <- function(t0, info, K = 25) { t <- t0; for (k in 1:K) t <- t + Sc_(t) /
 ↪  info(t); t }
starts <- c(-3, 0, 3, 4, 6, 10)
data.frame(start = starts,
           newton   = sapply(starts, function(s) run(s, Ic_)),
           fisher   = sapply(starts, function(s) run(s, function(t) nc / 2)))
```

```
#>   start     newton  fisher
#> 1    -3 -1.722e+08 -0.7538
#> 2     0 -7.538e-01 -0.7538
#> 3     3  4.221e+00 -0.7538
#> 4     4  4.221e+00 -0.7537
#> 5     6  4.221e+00  7.1304
#> 6    10  1.663e+08  7.1304
```

Fisher scoring converges to a local maximum from every start, and which one depends on the start; Newton–Raphson may jump to a minimum or away, since I is negative near the middle of the data. Neither guarantees the global maximum (Section 4.6).

4.5.3 General-purpose optimisers: `optim()`

In practice we rarely code Newton–Raphson. `optim(par, fn, method = "BFGS", hessian = TRUE)` minimises `fn` with a **quasi-Newton** method (BFGS) which builds an approximation to the Hessian from successive gradients (so no second derivatives are required, and if no gradient is given, it is computed numerically). Alternatives: `method = "Nelder-Mead"` (the default, derivative-free, robust but slow and unreliable in more than a few dimensions), `"L-BFGS-B"` (box constraints), `"CG"`, and `nlm()`, `nlminb()`. `optimize()` and `uniroot()` handle one-dimensional problems.

Practical recipe:

1. Code the **negative** log-likelihood, with all parameters unconstrained: log for positive parameters (variances, rates, shapes), logit for probabilities, so that BFGS never leaves the parameter space.
2. Choose starting values from simple estimates (moment estimates, a fit of a simpler model).
3. Call `optim(..., hessian = TRUE)` and check `$convergence == 0`.
4. The Hessian of the negative log-likelihood at the optimum is the observed information *on the scale optimised*; the standard errors are `sqrt(diag(solve(hessian)))`. Transform with the delta method: $\mathrm{se}(\hat\alpha) = \hat\alpha\,\mathrm{se}(\log\hat\alpha)$.
5. Check that the Hessian has only positive eigenvalues (a true maximum) and re-run from several starting points.

```r
fit_mle <- function(nll, start, ...) {
  fit <- optim(start, nll, method = "BFGS", hessian = TRUE, ...)
  if (fit$convergence != 0) warning("optim() did not report convergence")
  ev <- eigen(fit$hessian, symmetric = TRUE, only.values = TRUE)$values
  if (min(ev) <= 0) warning("Hessian not positive definite: not a proper
   ↪  maximum")
  list(par = fit$par, se = sqrt(diag(solve(fit$hessian))), loglik =
   ↪  -fit$value, eigen = ev)
```

```
}
fit <- fit_mle(nll_g, c(0, log(1 / mean(y_ac))))
rbind(estimate_log = fit$par, se_log = fit$se)
```

```
#>                 [,1]    [,2]
#> estimate_log -0.3475 -5.0304
#> se_log        0.3482  0.4891
```

```
fit$eigen
```

```
#> [1] 22.030  3.175
```

The same model through `stats4::mle()`, which wraps `optim()` and provides `summary()`, `vcov()`, `logLik()`, and profile-likelihood intervals through `confint()`:

```
nll_exp_log <- function(logmu) -sum(dexp(y_ac, rate = exp(-logmu), log =
↪  TRUE))
m1 <- mle(nll_exp_log, start = list(logmu = 4), method = "BFGS")
coef(summary(m1))
```

```
#>       Estimate Std. Error
#> logmu    4.683     0.2887
```

```
exp(confint(m1))     # profile-likelihood interval for mu: it matches the LR
↪  interval above
```

```
#> Profiling...
```

```
#>  2.5 % 97.5 %
#>  64.45 201.85
```

The interval is $(64.4, 201.8)$, the LR interval computed manually in Example 4.4.

4.6 Practical problems

4.6.1 An estimate on the boundary

If the maximum of ℓ is on the boundary of Θ, the likelihood equation $S(\theta) = 0$ has no solution and the usual theory (asymptotic normality, Wald intervals) fails. A binomial sample with $y = 0$ successes in $n = 20$ trials has $\hat{\theta} = 0$; $S(0) = -n/(1-0) = -20 \neq 0$, and $\text{se} = \sqrt{\hat{\theta}(1-\hat{\theta})/n} = 0$, so the Wald interval is the useless $[0, 0]$. The LR interval, which does not use the curvature at $\hat{\theta}$, is still meaningful: $\ell(\theta) = 20\log(1-\theta)$ and $\ell(\hat{\theta}) = 0$, so the upper limit solves $20\log(1-\theta) = -1.92$, i.e. $\theta_U = 1 - e^{-1.92/20} = 0.0916$.

```
n0 <- 20
c(LR_upper = 1 - exp(-cut / n0),
  exact_Clopper_Pearson = binom.test(0, n0)$conf.int[2],
  Wilson_upper = prop.test(0, n0, correct = FALSE)$conf.int[2],
  rule_of_three = 3 / n0)
```

```
#>              LR_upper exact_Clopper_Pearson          Wilson_upper
#>               0.09157               0.16843               0.16113
#>         rule_of_three
```

```
#>                 0.15000
```

The upper limits 0.092 (LR), 0.161 (Wilson), 0.168 (exact) and 0.15 ("rule of three") differ substantially; for rare events, the exact or Wilson interval is preferred. The lesson is to **never report a zero standard error**; the second lesson is that, near a boundary, the choice of interval matters.

4.6.2 Several maxima

The Cauchy example of Chapter 3 is one. A second, more realistic case is the **two-component normal mixture**, which we meet on the geyser data: the waiting time between eruptions is bimodal, with short waits (about 54 min) and long waits (about 80 min).

```
ggplot(data.frame(w = w_gy), aes(w)) + geom_histogram(aes(y =
↪  after_stat(density)), bins = 30, fill = ie_blue, alpha = .6) +
  labs(x = "waiting time (min)", y = "density")
```

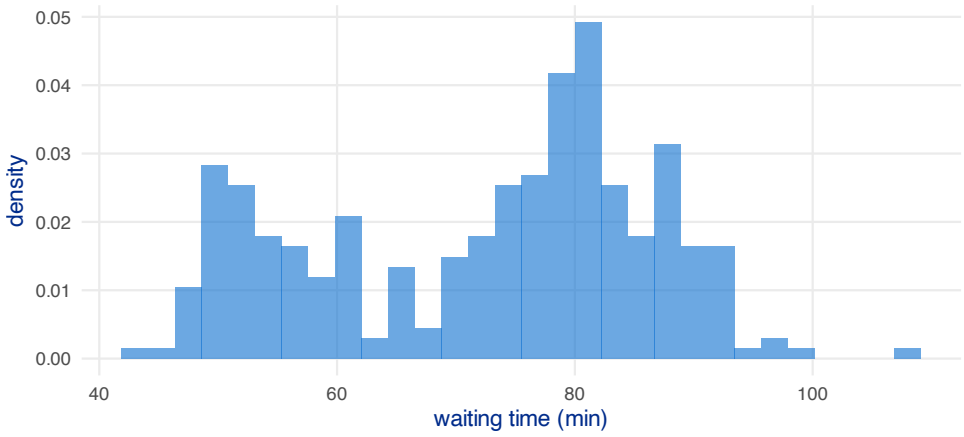


Figure 4.5: Waiting times between geyser eruptions (min): two groups are visible.

The density is $f(y) = p\,\phi(y;\mu_1,\sigma_1^2) + (1-p)\,\phi(y;\mu_2,\sigma_2^2)$ with five parameters $(p,\mu_1,\mu_2,\sigma_1,\sigma_2)$. We code the negative log-likelihood on unconstrained scales (logit for p, log for the standard deviations) and try two starting values.

```
nll_mix <- function(par, y) {
  p <- plogis(par[1])
  -sum(log(p * dnorm(y, par[2], exp(par[4])) + (1 - p) * dnorm(y, par[3],
  ↪  exp(par[5]))))
}
show_fit <- function(fit) round(c(p = plogis(fit$par[1]), mu1 = fit$par[2],
↪  mu2 = fit$par[3],
                                  sigma1 = exp(fit$par[4]), sigma2 =
                                  ↪  exp(fit$par[5]),
                                  loglik = -fit$value, conv =
                                  ↪  fit$convergence), 3)
good <- optim(c(0, 50, 80, log(5), log(5)), nll_mix, y = w_gy, method =
↪  "BFGS", hessian = TRUE)
sym  <- optim(c(0, mean(w_gy), mean(w_gy), log(sd(w_gy)), log(sd(w_gy))),
↪  nll_mix, y = w_gy, method = "BFGS", hessian = TRUE)
rbind(sensible_start = show_fit(good), symmetric_start = show_fit(sym))
```

```
#>                      p   mu1   mu2 sigma1 sigma2 loglik conv
#> sensible_start   0.308 54.20 80.36  4.952  7.507  -1158    0
#> symmetric_start 0.500 72.31 72.31 13.867 13.867  -1210    0
```

```
rbind(eigen_good = eigen(good$hessian)$values, eigen_sym =
↪  eigen(sym$hessian)$values)
```

```
#>              [,1]  [,2]    [,3]       [,4]     [,5]
#> eigen_good 314.6 126.7 54.4994  3.623e+00  1.67134
#> eigen_sym  299.0 150.9  0.7774 -9.704e-12 -0.08866
```

From a sensible start, BFGS finds the two-group solution ($\hat{p} = 0.308$, means 54.2 and 80.4, log-likelihood -1157.5). From a symmetric start, in which both components are identical, the gradient with respect to the *difference* of the two components is exactly zero, so BFGS never separates them and "converges" (code 0) to a single normal, with log-likelihood -1210.5. The Hessian exposes it: it has an eigenvalue equal to zero and another one negative, so this is a saddle point, not a maximum.

Warning. `convergence = 0` only says that the algorithm stopped because its tolerance criteria were met. **Always** inspect the Hessian eigenvalues, compare several starting values, and plot the fit.

Two further features of mixtures: the labels are arbitrary (swapping the components together with $p \mapsto 1 - p$ gives the same likelihood, so there are two equivalent maxima and one must **relabel** before reporting), and the likelihood is **unbounded**: centring one component exactly at a data point and letting $\sigma_1 \to 0$ makes $\ell \to \infty$, because that single term $p\,\phi(y_j; y_j, \sigma_1^2) = p/(\sigma_1\sqrt{2\pi}) \to \infty$ while the other terms stay bounded below by the second component. The sensible "MLE" is the best local maximum in the interior, away from these degenerate spikes.

```
spike <- function(s) -nll_mix(c(0, w_gy[1], 80, log(s), log(7.5)), w_gy)
sapply(c(1e-2, 1e-4, 1e-6, 1e-8), spike)
```

```
#> [1] -1666 -1607 -1547 -1487
```

Each time σ_1 is divided by 100 the log-likelihood increases by $\log 100 = 4.6$. (From any reasonable start BFGS does not find these spikes, which here only exceed the interior maximum when $\sigma_1 < 10^{-200}$; but a search that starts with a component centred on one observation and a tiny σ will.)

4.6.3 Non-identifiability and flat likelihoods

A parametrisation is **non-identifiable** if different parameter values give the same distribution. Take $y_i \sim N(a + b, 1)$: only the sum $a + b$ can be estimated. The log-likelihood is constant along the ridge $a + b = c$; the Hessian $I = \begin{pmatrix} n & n \\ n & n \end{pmatrix}$ has a zero eigenvalue and is not invertible; and an optimiser returns different answers from different starting values.

```
set.seed(2027)
yn <- rnorm(30, mean = 3, sd = 1)
nll_ab <- function(p) -sum(dnorm(yn, mean = p[1] + p[2], sd = 1, log = TRUE))
o1 <- optim(c(0, 0), nll_ab, method = "BFGS", hessian = TRUE)
o2 <- optim(c(5, -5), nll_ab, method = "BFGS", hessian = TRUE)
rbind(start_0_0 = c(o1$par, sum = sum(o1$par)), start_5_m5 = c(o2$par, sum =
↪  sum(o2$par)))
```

```
#>                             sum
#> start_0_0  1.534  1.534 3.068
#> start_5_m5 6.534 -3.466 3.068
```

```
eigen(o1$hessian)$values
```

```
#> [1] 60  0
```

Both runs end at different (a, b) with the same sum $a + b = \bar{y} = 3.068$. The signs of trouble: a Hessian that cannot be inverted (or has a very large condition number), huge standard errors, a profile likelihood that is flat. The remedy is to reparametrise (use $c = a + b$) or to add information (constraints, priors, more data). Near non-identifiability (for example, highly correlated predictors), the likelihood is not exactly flat but very elongated, and the standard errors are very large: this is *weak identifiability*.

Warning. Do not interpret individual standard errors in the presence of strongly correlated estimates; inspect the covariance matrix `solve(fit$hessian)` or plot the joint profile.

4.6.4 Other practical advice

- **Scale.** If parameters differ by orders of magnitude (for example $\hat{\beta} = 0.0065$ and $\hat{\alpha} = 0.7$), use `control = list(parscale = ...)` or reparametrise to comparable scales.
- **Analytical gradient.** If the likelihood is expensive, give `gr =`; this often makes BFGS faster and more reliable.
- **Step tolerance.** Report whether the log-likelihood has stabilised, not only the parameters: `control = list(reltol = 1e-12)` for a stricter criterion.
- **Wald fails far from normality.** If ℓ is visibly skewed (Figure 4.2), switch to profile LR intervals.

4.7 The EM algorithm (optional)

When the likelihood is awkward because of **missing or latent data**, the EM algorithm of Dempster et al. (1977) offers an alternative to direct optimisation. Suppose the complete data (y, u), with u unobserved, has a log-likelihood $\ell_c(\theta; y, u)$ that is easy to maximise, whereas the observed-data log-likelihood $\ell(\theta) = \log \int f(y, u; \theta)\, du$ is not. Starting from $\theta^{(k)}$:

- **E-step:** compute $Q(\theta \mid \theta^{(k)}) = \mathrm{E}[\ell_c(\theta; y, U) \mid y, \theta^{(k)}]$.
- **M-step:** $\theta^{(k+1)} = \arg\max_\theta Q(\theta \mid \theta^{(k)})$.

Theorem 4.4 (Ascent property of EM). *$\ell(\theta^{(k+1)}) \geq \ell(\theta^{(k)})$ with equality only if $\theta^{(k)}$ is a stationary point of the likelihood (under regularity).*

Proof. For any θ, $\log f(y; \theta) = \log f(y, u; \theta) - \log f(u \mid y; \theta)$ for every u. Taking the expectation with respect to $u \sim f(u \mid y; \theta^{(k)})$,

$$\ell(\theta) = Q(\theta \mid \theta^{(k)}) - H(\theta \mid \theta^{(k)}), \qquad H(\theta \mid \theta^{(k)}) = \mathrm{E}\big[\log f(U \mid y; \theta) \mid y, \theta^{(k)}\big].$$

By Jensen's inequality (or the non-negativity of the KL divergence), $H(\theta \mid \theta^{(k)}) \leq H(\theta^{(k)} \mid \theta^{(k)})$ for all θ. Hence

$$\ell(\theta^{(k+1)}) - \ell(\theta^{(k)}) = \underbrace{Q(\theta^{(k+1)} \mid \theta^{(k)}) - Q(\theta^{(k)} \mid \theta^{(k)})}_{\geq 0 \text{ (M-step)}} + \underbrace{H(\theta^{(k)} \mid \theta^{(k)}) - H(\theta^{(k+1)} \mid \theta^{(k)})}_{\geq 0} \geq 0. \quad \square$$

Two-component normal mixture. Let $u_i \in \{0,1\}$ indicate the component of y_i. The complete-data log-likelihood is linear in the u_i, so the E-step replaces them by the **responsibilities** $\gamma_i = P(u_i = 1 \mid y_i; \theta^{(k)}) = p\phi_1(y_i)/\{p\phi_1(y_i) + (1-p)\phi_2(y_i)\}$ and the M-step gives weighted versions of the usual MLEs:

$$\hat p = \frac{1}{n}\sum \gamma_i, \quad \hat\mu_1 = \frac{\sum \gamma_i y_i}{\sum \gamma_i}, \quad \hat\mu_2 = \frac{\sum (1-\gamma_i) y_i}{\sum (1-\gamma_i)}, \quad \hat\sigma_1^2 = \frac{\sum \gamma_i (y_i - \hat\mu_1)^2}{\sum \gamma_i},$$

and $\hat\sigma_2^2$ analogously with weights $1-\gamma_i$.

```
em_mix <- function(y, p, m, s, maxit = 500, tol = 1e-10) {
  ll <- numeric(0)
  for (k in 1:maxit) {
    d1 <- p * dnorm(y, m[1], s[1]); d2 <- (1 - p) * dnorm(y, m[2], s[2])
    ll[k] <- sum(log(d1 + d2))                          # observed-data
↪   log-likelihood BEFORE the update
    g <- d1 / (d1 + d2)                                 # E-step:
↪   responsibilities
    p <- mean(g)                                        # M-step
    m <- c(sum(g * y) / sum(g), sum((1 - g) * y) / sum(1 - g))
    s <- sqrt(c(sum(g * (y - m[1])^2) / sum(g), sum((1 - g) * (y - m[2])^2) /
↪   sum(1 - g)))
    if (k > 1 && abs(ll[k] - ll[k - 1]) < tol) break
  }
  list(p = p, mu = m, sigma = s, ll = ll)
}
fit_em <- em_mix(w_gy, 0.5, c(50, 80), c(5, 5))
c(p = fit_em$p, mu = fit_em$mu, sigma = fit_em$sigma, loglik = tail(fit_em$ll,
↪   1), iterations = length(fit_em$ll))
```

```
#>          p        mu1        mu2     sigma1     sigma2     loglik
#>     0.3076    54.2027    80.3603     4.9520     7.5076 -1157.5420
#> iterations
#>    42.0000
```

```
all(diff(fit_em$ll) >= -1e-9)                           # monotone increase
```

```
#> [1] TRUE
```

```
ggplot(data.frame(k = seq_along(fit_em$ll), ll = fit_em$ll), aes(k, ll)) +
↪   geom_line(colour = ie_navy) + geom_point(size = 1) +
  labs(x = "iteration", y = "log-likelihood")
```

EM and BFGS give the same estimates (within 10^{-3}) and the same log-likelihood, -1157.54. EM needs 42 iterations: each is cheap and guaranteed to increase ℓ, but convergence is only linear and slow when the components overlap. EM does not produce standard errors; compute the observed information at the solution with `optim(..., hessian = TRUE)` started from the EM estimate, or by numerical differentiation.

Warning. The log-likelihood used to monitor EM must be that of the *observed* data, $\sum_i \log\{p\phi_1 + (1-p)\phi_2\}$ computed with the parameters *before* the update, as in the code above. Computing it after normalising the responsibilities, or with the responsibilities as if they were data, gives meaningless values, stalls the stopping rule and hides bugs. Likewise, do not use the LR statistic with a χ^2_3 reference to decide between one and two components:

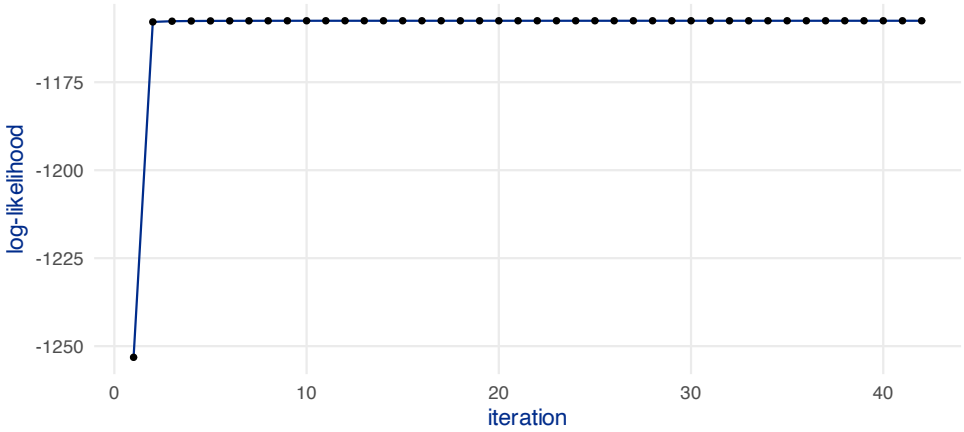


Figure 4.6: EM for the geyser mixture: observed-data log-likelihood at each iteration (increasing, as the theorem states).

the parameters (μ_2, σ_2) are not identified when $p = 1$, and $p = 1$ is on the boundary, so Wilks' theorem fails; use a parametric bootstrap or BIC (Exercise 4.12).

4.8 In R

Task	Function
Maximise in one dimension	`optimize(f, interval, maximum = TRUE)`
Maximise in several dimensions	`optim(par, nll, method = "BFGS", hessian = TRUE)` (minimises); `nlm()`, `nlminb()`
Standard errors	`sqrt(diag(solve(fit$hessian)))`, then the delta method
Solve $\ell(\theta) = \ell(\hat\theta) - 1.92$	`uniroot()` on each side of $\hat\theta$
Likelihood object with `summary`, `vcov`, `confint`	`stats4::mle()`
Binomial score (Wilson) interval / exact interval	`prop.test(x, n, correct = FALSE)` / `binom.test(x, n)`
LR test of nested models	`2 * (logLik(m1) - logLik(m0))` with `pchisq(., df, lower.tail = FALSE)`; `anova(m0, m1, test = "Chisq")` for `glm`
Profile intervals for GLMs	`confint(fit)` (profile likelihood via MASS) versus `confint.default(fit)` (Wald)

A generic helper for one-parameter LR intervals, used in the labs:

```
lr_interval <- function(loglik, mle, lower, upper, level = 0.95) {
  cut <- qchisq(level, df = 1) / 2
  f <- function(t) loglik(t) - loglik(mle) + cut
  c(uniroot(f, c(lower, mle), tol = 1e-10)$root, uniroot(f, c(mle, upper), tol
  ↪  = 1e-10)$root)
}
lr_interval(function(l) sum(dpois(y_di, l, log = TRUE)), mean(y_di), 1, 10)
↪  # Poisson, discoveries
```

```
#> [1] 2.768 3.458
```

4.9 Common pitfalls

Warning. *Wald, score and LR are asymptotically equivalent, not equal.* Report which you used. With small n, or when ℓ is asymmetric, prefer LR (or exact) methods.

Warning. *Degrees of freedom* of an LR test equal the number of restrictions (difference in the number of free parameters), not the number of parameters.

Warning. *Boundary nulls* (variances, mixing proportions, $k \to \infty$) need non-standard reference distributions.

Warning. *Wald p-values and intervals depend on the parametrisation*; LR ones do not.

Warning. *`optim()` minimises*; *convergence code 0 is not proof of a maximum*; its Hessian is on the optimised scale.

Warning. *A failure to reject is not a proof*: the exponential model is not "confirmed" by a p-value of 0.29 with twelve observations.

Remark. The coverage of a 95% interval is a property of the *procedure* over repeated samples, not a probability for the single interval computed.

4.10 Chapter summary

- $W = (\hat\theta - \theta_0)^2 I(\hat\theta)$, $S_c = S(\theta_0)^2/\mathcal{I}(\theta_0)$, $LR = 2\{\ell(\hat\theta) - \ell(\theta_0)\}$ are each asymptotically χ^2_1 under H_0, and asymptotically equivalent (Taylor expansion). In small samples they differ; LR is the most reliable.
- Intervals are obtained by inversion: Wald $\hat\theta \pm z\, I(\hat\theta)^{-1/2}$, Wilson/score, and the LR interval $\{\theta : \ell(\theta) \geq \ell(\hat\theta) - \chi^2_{1,1-\alpha}/2\}$. LR intervals are invariant under monotone reparametrisation, Wald intervals are not.
- Wilks: LR for q restrictions is asymptotically χ^2_q, provided the null is an interior point and parameters are identified.
- The profile likelihood $\ell_p(\psi) = \max_\lambda \ell(\psi, \lambda)$ removes nuisance parameters; its curvature gives the correct standard error $[I^{-1}]^{1/2}_{\psi\psi}$.
- Newton–Raphson $\theta \leftarrow \theta + I^{-1}S$ converges quadratically near $\hat\theta$ but may diverge; Fisher scoring uses $\mathcal{I}$ and is more stable; `optim(method = "BFGS", hessian = TRUE)` on an unconstrained parametrisation is the default tool.
- Check: Hessian positive definite, several starts, plot. Know the three typical problems: boundary, multiple maxima (mixtures, Cauchy), non-identifiability.
- EM increases the observed-data likelihood at each step (proof via Jensen), is slow but robust, and gives no standard errors.

4.11 Exercises with explained solutions

Exercise 4.1. A Poisson sample has $n = 25$ and $\sum y_i = 50$. Test $H_0 : \lambda = 1.5$ with the Wald, score and likelihood-ratio statistics. Give p-values and the three 95% intervals for λ (Wald, LR; the score interval by solving numerically).

Solution. The three statistics were computed in Example 4.2: $W = 3.125$ ($p = 0.077$), $S_c = 4.167$ ($p = 0.041$), $LR = 3.768$ ($p = 0.052$). For the intervals:

```
s <- 50; n <- 25; lam <- s / n
ll <- function(l) s * log(l) - n * l
wald <- lam + c(-1, 1) * qnorm(0.975) * sqrt(lam / n)
lr <- lr_interval(ll, lam, 0.5, 6)
sc <- uniroot(function(l) (s / l - n)^2 / (n / l) - qchisq(.95, 1), c(0.5,
↪  lam))$root
sc2 <- uniroot(function(l) (s / l - n)^2 / (n / l) - qchisq(.95, 1), c(lam,
↪  6))$root
round(rbind(Wald = wald, Score = c(sc, sc2), LR = lr), 3)
```

```
#>        [,1]  [,2]
#> Wald  1.446 2.554
#> Score 1.517 2.636
#> LR    1.496 2.607
```

The Wald interval is symmetric, $(1.446, 2.554)$; the LR and score intervals are shifted to the right, reflecting the skewness of the Poisson likelihood.

Exercise 4.2. Show that inverting the score test for the binomial gives the Wilson interval. Specifically, show that $\{\theta : S(\theta)^2/\mathcal{I}(\theta) \le z^2\}$ is the set of solutions of $(\hat\theta - \theta)^2 \le z^2\theta(1-\theta)/n$, and solve the quadratic equation.

Solution. $S(\theta) = y/\theta - (n-y)/(1-\theta) = \dfrac{y - n\theta}{\theta(1-\theta)}$ and $\mathcal{I}(\theta) = \dfrac{n}{\theta(1-\theta)}$, so $S^2/\mathcal{I} = \dfrac{(y-n\theta)^2}{n\theta(1-\theta)} = \dfrac{n(\hat\theta-\theta)^2}{\theta(1-\theta)}$. The condition $\le z^2$ is $(\hat\theta-\theta)^2 \le z^2\theta(1-\theta)/n$. Expand: $(1+z^2/n)\theta^2 - (2\hat\theta + z^2/n)\theta + \hat\theta^2 \le 0$. The roots are

$$\theta = \frac{\hat\theta + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat\theta(1-\hat\theta)}{n} + \frac{z^2}{4n^2}}}{1 + z^2/n},$$

the Wilson interval, centred at a weighted average of $\hat\theta$ and $1/2$. With $y = 8$, $n = 10$, $z = 1.96$: numerator centre $0.8 + 0.1921 = 0.9921$, divided by 1.3842 gives 0.7168; half-width $1.96\sqrt{0.016 + 0.0096}/1.3842 = 0.2266$, so $(0.490, 0.943)$. The lower end is far from the Wald lower limit and the upper end never exceeds one.

Exercise 4.3. Let $y_1, \dots, y_n$ be iid $N(\mu, \sigma^2)$ with σ^2 known. Show that W, S_c and LR for $H_0 : \mu = \mu_0$ are all equal to $n(\bar y - \mu_0)^2/\sigma^2$ and exactly χ^2_1 under H_0.

Solution. $\ell(\mu) = -\frac{1}{2\sigma^2}\sum(y_i - \mu)^2 + \text{const} = -\frac{n}{2\sigma^2}(\bar y - \mu)^2 + \text{const}'$, a quadratic. $S(\mu) = n(\bar y - \mu)/\sigma^2$, $I(\mu) = \mathcal{I}(\mu) = n/\sigma^2$, $\hat\mu = \bar y$. Then $W = (\bar y - \mu_0)^2 n/\sigma^2$; $S_c = \{n(\bar y - \mu_0)/\sigma^2\}^2\sigma^2/n = n(\bar y - \mu_0)^2/\sigma^2$; $LR = 2\{\ell(\bar y) - \ell(\mu_0)\} = n(\bar y - \mu_0)^2/\sigma^2$. Under H_0, $\sqrt{n}(\bar y - \mu_0)/\sigma \sim N(0,1)$ exactly, so the statistic is exactly χ^2_1. The Taylor expansion in the proof of Theorem 4.1 is exact because ℓ is quadratic with constant curvature.

Exercise 4.4. In a clinical study, 0 of 20 patients had an adverse event. (a) Why does the Wald interval fail? (b) Compute the 95% LR upper limit by hand and verify the numerical value in R. (c) Which interval would you report?

Solution.

(a) $\hat\theta = 0$, so $\text{se} = \sqrt{\hat\theta(1-\hat\theta)/n} = 0$ and the interval is $[0, 0]$, claiming certainty. The MLE is on the boundary, so $S(\hat\theta) \neq 0$, and the quadratic approximation, which relies on the interior stationary point, does not apply.

(b) $\ell(\theta) = 20\log(1-\theta)$, $\ell(\hat\theta) = 0$; the upper limit solves $20\log(1-\theta) = -1.921$ ($= -\chi^2_{1,0.95}/2$), so $\theta_U = 1 - e^{-0.0960} = 0.0916$. In R: `1 - exp(-qchisq(.95, 1) / 2 / 20)`, giving 0.0916 (see Section 4.6).

(c) The exact (Clopper–Pearson) upper limit is 0.168, Wilson 0.161. The LR upper limit 0.092 is much shorter than the exact one: the exact interval is conservative by construction, whereas the χ^2_1 calibration of the LR is only approximate for such small counts. In a regulatory setting the exact interval is the conservative standard; the rule of three ($3/n = 0.15$) is a good back-of-envelope approximation.

Exercise 4.5. For the aircondit data, derive the exact 95% confidence interval for the **failure rate** $\lambda = 1/\mu$ from the pivot $2\lambda\sum y_i \sim \chi^2_{2n}$. Compare with the Wald intervals on the λ and $\log\lambda$ scales and with the LR interval obtained by inverting the LR interval for μ.

Solution. $P(\chi^2_{24,0.025} \le 2\lambda\sum y_i \le \chi^2_{24,0.975}) = 0.95$ gives $\lambda \in (\chi^2_{24,0.025}, \chi^2_{24,0.975})/(2\sum y_i)$.

```
s <- sum(y_ac); lam <- 1 / mean(y_ac)
exact <- qchisq(c(0.025, 0.975), 2 * n_ac) / (2 * s)
wald_l <- lam + c(-1, 1) * z * lam / sqrt(n_ac)             # se(lambda_hat) =
↪  lambda_hat / sqrt(n)
wald_ll <- exp(log(lam) + c(-1, 1) * z / sqrt(n_ac))
lr_l <- rev(1 / lr_mu)                                       # invariance
signif(rbind(Exact = exact, Wald = wald_l, Wald_log = wald_ll, LR = lr_l), 3)
```

```
#>              [,1]   [,2]
#> Exact    0.00478 0.0152
#> Wald     0.00402 0.0145
#> Wald_log 0.00525 0.0163
#> LR       0.00495 0.0155
```

The exact interval is $(0.00478, 0.0152)$. The Wald interval on the λ scale, $(0.0040, 0.0145)$, is shifted to the left of it (both limits are too small), whereas the log-scale Wald interval $(0.00525, 0.0163)$ is shifted slightly to the right and the LR interval $(0.00495, 0.0155)$ is the closest. Because $\hat\lambda = 1/\bar y$ is skewed and biased upwards, a symmetric interval on the λ scale is inappropriate.

Exercise 4.6. For iid $N(\mu, \sigma^2)$ with μ unknown, (a) derive the profile log-likelihood of $v = \sigma^2$ and the LR statistic for $H_0 : v = v_0$. (b) Compute it for the New Haven temperatures with $v_0 = 1$ and compare with χ^2_1. (c) Find the 95% profile interval for v.

Solution.

(a) For any v the MLE of μ is $\bar y$, so $\ell_p(v) = -\frac{n}{2}\log v - \frac{n\hat v}{2v} + \text{const}$, with $\hat v = \frac{1}{n}\sum(y_i - \bar y)^2$. Then $\ell_p(\hat v) = -\frac{n}{2}\log\hat v - \frac{n}{2}$ and

$$LR = 2\{\ell_p(\hat v) - \ell_p(v_0)\} = n\Big\{\frac{\hat v}{v_0} - 1 - \log\frac{\hat v}{v_0}\Big\}.$$

This is zero at $\hat v = v_0$ and asymmetric in $\hat v/v_0$.

(b) With $n = 60$, $\hat{v} = 1.575$: $LR = 60\{1.575 - 1 - \log 1.575\} = 60\{0.575 - 0.4542\} = 7.25$, $p = 0.0071$: the variance is significantly larger than 1 (the data would be incompatible with $\sigma = 1$).

(c) Solve $LR(v) = 3.84$ numerically.

```
vh <- mean((y_nh - mean(y_nh))^2)
LRv <- function(v0) n_nh * (vh / v0 - 1 - log(vh / v0))
c(LR = LRv(1), p = pchisq(LRv(1), 1, lower.tail = FALSE))
```

```
#>       LR        p
#> 7.246144 0.007105
```

```
c(uniroot(function(v) LRv(v) - qchisq(.95, 1), c(0.5, vh))$root,
  uniroot(function(v) LRv(v) - qchisq(.95, 1), c(vh, 5))$root)
```

```
#> [1] 1.124 2.304
```

The interval is $(1.12, 2.30)$ for σ^2, asymmetric around $\hat{v} = 1.575$ (the right arm is longer); the Wald interval $\hat{v} \pm 1.96\hat{v}\sqrt{2/n} = (1.01, 2.14)$ is symmetric and shifted to the left.

Exercise 4.7. Let y_i be iid exponential with rate λ, $\bar{y} = 2$. (a) Show that the Newton–Raphson iteration is $\lambda^{(k+1)} = \lambda^{(k)}(2 - \bar{y}\lambda^{(k)})$ and that the error satisfies $\lambda^{(k+1)} - \hat{\lambda} = -\bar{y}(\lambda^{(k)} - \hat{\lambda})^2$. (b) Compute three iterations from $\lambda^{(0)} = 0.2$ and discuss what happens from $\lambda^{(0)} = 1.2$. (c) Write the iteration for the log-rate $\eta = \log\lambda$ and check numerically that it converges from $\lambda^{(0)} = 1.2$.

Solution.

(a) $S = n/\lambda - n\bar{y}$, $I = n/\lambda^2$, so $\lambda + S/I = \lambda + \lambda - \bar{y}\lambda^2 = \lambda(2 - \bar{y}\lambda)$. With $\hat{\lambda} = 1/\bar{y}$: $\lambda(2 - \bar{y}\lambda) - \hat{\lambda} = -\bar{y}(\lambda^2 - 2\lambda\hat{\lambda} + \hat{\lambda}^2) = -\bar{y}(\lambda - \hat{\lambda})^2$.

(b) $\hat{\lambda} = 0.5$. From 0.2: $0.2(2-0.4) = 0.32$; $0.32(2-0.64) = 0.4352$; $0.4352(2-0.8704) = 0.4916$: monotone convergence, errors $0.3, 0.18, 0.0648, 0.0084$ (each about $2\times$ the square of the previous). From 1.2: $\lambda^{(0)} > 2/\bar{y} = 1$, so $\lambda^{(1)} = 1.2(2 - 2.4) = -0.48 < 0$; then $-0.48(2 + 0.96) = -1.42$, and the sequence diverges to $-\infty$.

(c) For $\eta = \log\lambda$: $\ell = n\eta - ne^{\eta}\bar{y}$, $S = n(1 - \bar{y}e^{\eta})$, $I = n\bar{y}e^{\eta}$, so $\eta^{(k+1)} = \eta^{(k)} + (e^{-\eta^{(k)}}/\bar{y}) - 1$. The iteration is well defined for any real $\eta^{(0)}$ (no boundary), with $e^{-\eta}/\bar{y} = \hat{\lambda}/\lambda$. Numerically:

```
eta <- numeric(8); eta[1] <- log(1.2)
for (k in 1:7) eta[k + 1] <- eta[k] + exp(-eta[k]) / 2 - 1
rbind(lambda = exp(eta))
```

```
#>         [,1]   [,2]   [,3]   [,4] [,5] [,6] [,7] [,8]
#> lambda  1.2 0.6696 0.5198 0.5004  0.5  0.5  0.5  0.5
```

Starting from $\lambda^{(0)} = 1.2$ the log-scale iteration moves monotonically from 1.2 to 0.5, where the raw-scale iteration diverged. (It can overshoot from very small $\lambda^{(0)}$, because $e^{-\eta}$ grows fast, but the iterates stay positive.)

Exercise 4.8. For the Cauchy data $y = (-2, -1, 0, 7, 8)$ run `optim(start, nll, method = "BFGS")` from starts $-3, 0, 4, 6, 10$ and tabulate where it converges. (a) What does the

table tell you about the reliability of a single run? (b) Describe a strategy to find the MLE reliably.

Solution.

```
nll_c <- function(t) sum(log(1 + (yc - t)^2))
res <- t(sapply(c(-3, 0, 4, 6, 10), function(s) { o <- optim(s, nll_c, method
↪  = "BFGS"); c(start = s, theta_hat = o$par, loglik = -o$value, conv =
↪  o$convergence) }))
round(res, 3)
```

```
#>      start theta_hat  loglik conv
#> [1,]    -3    -0.754  -9.911    0
#> [2,]     0    -0.754  -9.911    0
#> [3,]     4    -0.754  -9.911    0
#> [4,]     6     7.130 -13.170    0
#> [5,]    10     7.130 -13.170    0
```

(a) All runs report convergence (code 0) but the result depends on the start: starts at -3, 0 and 4 reach the global maximum -0.754 (log-likelihood -9.91 up to the constant), starts at 6 and 10 reach the local one at 7.13 (-13.17). The start at 4 is next to the local minimum at 4.22, and the first step decides which maximum is reached. Convergence is a statement about the algorithm, not about the global optimum.

(b) Evaluate the log-likelihood on a grid and start the optimiser from the best grid point(s), or use many random/data-based starts (e.g. each observation, the median) and keep the best; verify with a plot. With n large the multimodality typically disappears, since the sample median is a good starting point.

Exercise 4.9. Using the aircondit data, test the exponential against the gamma model with the LR and Wald statistics. (a) Interpret. (b) Does the conclusion change if we use the profile LR interval for α? (c) Is the χ^2_1 reference valid here?

Solution.

(a) $LR = 1.10$ ($p = 0.29$) and $z_W = -1.19$ ($p = 0.23$); both fail to reject $\alpha = 1$ (Example 4.5). (b) The profile 95% interval $(0.33, 1.32)$ contains 1, in agreement with the LR test (inverting the LR test *is* the profile interval, so the two necessarily agree). (c) The null $\alpha = 1$ is an interior point of the parameter space $\alpha > 0$ and all parameters are identified under H_0, so Wilks' theorem applies, although $n = 12$ is a small sample and the χ^2 approximation is only approximate. A parametric bootstrap check is easy:

```
set.seed(2027)
boot_LR <- replicate(300, {
  yb <- rexp(n_ac, rate = 1 / mean(y_ac))
  nl <- function(p) -sum(dgamma(yb, shape = exp(p[1]), rate = exp(p[2]), log =
↪  TRUE))
  o <- optim(c(0, log(1 / mean(yb))), nl, method = "BFGS")
  2 * (-o$value - sum(dexp(yb, 1 / mean(yb), log = TRUE)))
})
c(boot_p_value = mean(boot_LR >= LR_g), chisq_p = pchisq(LR_g, 1, lower.tail =
↪  FALSE))
```

```
#> boot_p_value      chisq_p
#>       0.3433       0.2945
```

The bootstrap p-value is of the same order as the χ^2 one, supporting the conclusion.

Exercise 4.10. Let $y_i \sim N(a+b, 1)$, $i = 1, \ldots, n$. (a) Write the information matrix for (a, b) and show it is singular. (b) Which functions of (a, b) can be estimated? (c) How would the optimisation output reveal the problem?

Solution.

(a) $\ell = -\frac{1}{2}\sum(y_i - a - b)^2$, so $\partial\ell/\partial a = \partial\ell/\partial b = \sum(y_i - a - b)$ and $-\partial^2\ell/\partial a^2 = -\partial^2\ell/\partial a\partial b = -\partial^2\ell/\partial b^2 = n$ in absolute value: $I = \begin{pmatrix} n & n \\ n & n \end{pmatrix}$, with eigenvalues $2n$ and 0 and determinant 0. (b) Only the **estimable** functions, those that are functions of $a + b$ (for example $a + b$ itself: $\widehat{a+b} = \bar{y}$ with standard error $1/\sqrt{n}$); $a - b$ cannot be estimated. (c) The Hessian from `optim()` has a (numerically) zero eigenvalue, `solve(hessian)` fails or yields enormous values, and different starting values give different estimates with the same log-likelihood, as in the example of Section 4.6.

Exercise 4.11. For a two-component normal mixture, derive the E-step and the M-step for p and μ_1. Then show, using the structure of the proof of Theorem 4.4, why $\ell(\theta^{(k+1)}) \geq \ell(\theta^{(k)})$ even if the M-step only *increases* Q rather than maximising it (generalised EM).

Solution. Complete-data log-likelihood with indicators $u_i \in \{0, 1\}$ ($u_i = 1$ for component 1): $\ell_c = \sum_i \{u_i[\log p + \log \phi_1(y_i)] + (1 - u_i)[\log(1-p) + \log \phi_2(y_i)]\}$. E-step: $\mathrm{E}[u_i \mid y_i, \theta^{(k)}] = \gamma_i = \dfrac{p\phi_1(y_i)}{p\phi_1(y_i) + (1-p)\phi_2(y_i)}$ (Bayes' rule). So $Q = \sum_i \{\gamma_i[\log p + \log \phi_1(y_i)] + (1 - \gamma_i)[\log(1-p) + \log \phi_2(y_i)]\}$. M-step for p: $\partial Q/\partial p = \sum \gamma_i/p - \sum(1-\gamma_i)/(1-p) = 0$ gives $\hat{p} = \frac{1}{n}\sum \gamma_i$. For μ_1: only $\sum \gamma_i \log \phi_1(y_i; \mu_1, \sigma_1^2) = -\sum \gamma_i (y_i - \mu_1)^2/(2\sigma_1^2) + \ldots$ depends on it, which is a weighted least-squares problem with solution $\hat{\mu}_1 = \sum \gamma_i y_i / \sum \gamma_i$.

For the generalised EM, the proof only used $Q(\theta^{(k+1)} \mid \theta^{(k)}) \geq Q(\theta^{(k)} \mid \theta^{(k)})$ (any increase suffices) and $H(\theta^{(k+1)} \mid \theta^{(k)}) \leq H(\theta^{(k)} \mid \theta^{(k)})$ (which holds for every θ by Jensen). Adding, ℓ cannot decrease.

Exercise 4.12. (R analysis.) For the geyser waiting times: (a) fit the one-component normal and the two-component mixture and compare log-likelihoods, AIC and BIC; (b) explain why $LR \sim \chi^2_3$ is not valid; (c) approximate the null distribution of LR by a parametric bootstrap from the fitted single normal with $B = 100$ replications (use EM for speed, started from sensible values) and give a p-value; (d) report the fitted proportions and means in words.

Solution. (a),(d)

```
ll1 <- sum(dnorm(w_gy, mean(w_gy), sqrt(mean((w_gy - mean(w_gy))^2)), log =
↪  TRUE))
ll2 <- tail(fit_em$ll, 1)
n <- length(w_gy)
data.frame(model = c("1 normal", "2-component mixture"), npar = c(2, 5),
↪  loglik = c(ll1, ll2),
           AIC = -2 * c(ll1, ll2) + 2 * c(2, 5), BIC = -2 * c(ll1, ll2) +
           ↪  log(n) * c(2, 5))
```

```
#>                 model npar loglik  AIC  BIC
#> 1             1 normal    2  -1210 2425 2432
#> 2 2-component mixture    5  -1158 2325 2344
```

The mixture improves the log-likelihood by 53 units, and AIC and BIC both prefer it strongly (BIC difference about 88). About 31% of the waits belong to the short group with mean 54 min ($\sigma \approx 5$), and 69% to the long group with mean 80 min ($\sigma \approx 7.5$).

(b) The mixture with $p = 1$ (or with $\mu_1 = \mu_2, \sigma_1 = \sigma_2$) reduces to a single normal: under H_0 the parameters (μ_2, σ_2) are not identified (any value gives the same likelihood when $p = 1$), and $p = 1$ lies on the boundary. Both violate the regularity conditions of Wilks' theorem, so the reference is not χ^2_3 and its true distribution has heavier tails.

(c) Bootstrap:

```
set.seed(2027)
LR_obs <- 2 * (ll2 - ll1)
LR_boot <- replicate(100, {
  yb <- rnorm(n, mean(w_gy), sd(w_gy))
  l1 <- sum(dnorm(yb, mean(yb), sqrt(mean((yb - mean(yb))^2)), log = TRUE))
  f <- em_mix(yb, 0.5, mean(yb) + c(-1, 1) * sd(yb), rep(sd(yb) / 2, 2), maxit
↪  = 300, tol = 1e-6)
  2 * (tail(f$ll, 1) - l1)
})
c(LR_obs = LR_obs, boot_max = max(LR_boot), boot_95 = unname(quantile(LR_boot,
↪  0.95)), p_value = (1 + sum(LR_boot >= LR_obs)) / 101)
```

```
#>    LR_obs  boot_max   boot_95   p_value
#> 1.059e+02 9.656e+00 7.822e+00 9.901e-03
```

The observed $LR \approx 106$ is far above every bootstrap value (their 95th percentile is about 8 and their maximum about 10), so the bootstrap p-value is at the minimum possible with $B = 100$, $1/101 \approx 0.01$; more replications would only tighten it. The evidence for two groups is overwhelming.

4.12 Further reading

The tests and intervals follow Pawitan (2001) (Chapters 3, 5 and 9); the original contributions are Wald (1943), Rao (1948) and Wilks (1938). Non-standard null distributions at the boundary are treated by Self and Liang (1987). Numerical optimisation is covered in the first chapters of Venables and Ripley (2002) for R practitioners; the EM algorithm is due to Dempster et al. (1977). The profile likelihood is developed in Kalbfleisch and Prentice (2002); Efron and Hinkley (1978) discusses observed versus expected information, which matters for Wald-type standard errors.

Part II

Linear models

Chapter 5

The linear model in theory

Almost everything in this course is a variation on one idea: the mean of a response is a function of predictors, and the function has parameters that enter *linearly* (linear models and, after a link, generalized linear models), or whose shape is penalized (smoothing, GAMs), or that vary from group to group (mixed models). The linear model is the place where the whole theory can be written down in closed form. Least squares has an explicit solution, its sampling distribution is known exactly under normality, and the tests we use every day (t tests, F tests, confidence intervals) are exact, not asymptotic. In this chapter we derive all of it. The next chapter (Chapter 6) asks what to do when the assumptions fail, and Chapter 7 asks how to build a good model from many candidate predictors.

We learn the theory on one data set, `savings` from the `faraway` package (Faraway, 2014). It records, for $n = 50$ countries averaged over 1960–1970, the aggregate personal savings rate `sr` (percentage of disposable income) and four predictors: the percentage of the population under 15 (`pop15`) and over 75 (`pop75`), real per-capita disposable income (`dpi`) and its growth rate (`ddpi`). The life-cycle hypothesis of economics predicts that countries with many young or old (non-earning) people save less, and that faster growth raises savings.

```
data(savings, package = "faraway")
str(savings)
```

```
#> 'data.frame':    50 obs. of  5 variables:
#>  $ sr   : num  11.43 12.07 13.17 5.75 12.88 ...
#>  $ pop15: num  29.4 23.3 23.8 41.9 42.2 ...
#>  $ pop75: num  2.87 4.41 4.43 1.67 0.83 2.85 1.34 0.67 1.06 1.14 ...
#>  $ dpi  : num  2330 1508 2108 189 728 ...
#>  $ ddpi : num  2.87 3.93 3.82 0.22 4.56 2.43 2.67 6.51 3.08 2.8 ...
```

```
library(tidyr)
long <- pivot_longer(savings, c(pop15, pop75, dpi, ddpi), names_to =
↪  "predictor", values_to = "x")
ggplot(long, aes(x, sr)) +
  geom_point(colour = ie_navy, alpha = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, colour = ie_coral,
  ↪  linewidth = 0.7) +
  facet_wrap(~ predictor, scales = "free_x", nrow = 1) +
  labs(x = NULL, y = "Savings rate sr (%)")
```

Figure 5.1 raises the four questions that drive the chapter. By how much does `sr` change per

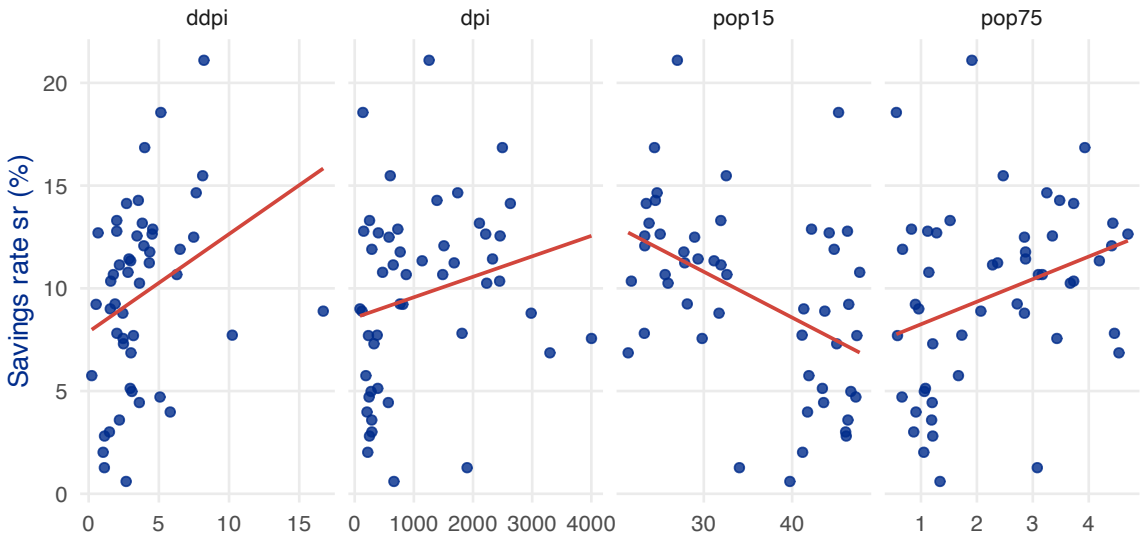


Figure 5.1: Savings rate against each predictor, 'savings' data, with the simple least-squares line in each panel.

unit of `pop15` *when the other variables are held fixed*, and how sure are we? Do `pop75` and `dpi` add anything once `pop15` and `ddpi` are in the model? Are a few countries driving the fit? What savings rate should we expect for a country not in the sample? One model answers all four.

5.1 The model and its assumptions

Throughout, $y = (y_1, \dots, y_n)^\top$ is the response vector, and X is the $n \times p$ **design matrix**. Its first column is the constant $\mathbf{1}$ (the intercept) and the others contain the predictors, so that p counts *all* columns of X, intercept included. For `savings`, $p = 5$.

Definition 5.1 (Linear model)**.** The linear model for y given X is

$$y = X\beta + \varepsilon, \qquad \text{i.e.} \quad y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i, \quad i = 1, \dots, n, \tag{5.1}$$

where $\beta \in \mathbb{R}^p$ is a vector of unknown constants and ε is a random error vector. The model is *linear in* β: the columns of X may be any fixed functions of the predictors (x, x^2, $\log x$, dummy variables, products).

We use four assumptions, introduced one at a time because each one is needed for a different purpose.

	Assumption	What it buys
A1	$\operatorname{rank}(X) = p \le n$	unique least-squares estimate $\hat\beta$
A2	$\mathrm{E}[\varepsilon] = 0$	$\hat\beta$ is unbiased: the mean is correctly specified
A3	$\operatorname{Var}(\varepsilon) = \sigma^2 I_n$	formula for $\operatorname{Var}(\hat\beta)$, Gauss–Markov theorem
A4	$\varepsilon \sim N(0, \sigma^2 I_n)$	exact t and F distributions; least squares = maximum likelihood

X is treated as fixed: all expectations and variances are *conditional on* X. Assumption A3 packs two statements, equal variances (homoscedasticity) and zero correlation between errors. Under A4, zero correlation implies independence. Least squares itself needs only A1, and the *properties* of least squares need A2–A4. With a large sample, A4 can be relaxed because of the central limit theorem, but A2 and A3 cannot: if the mean is wrong or the variance is not constant, the standard errors we compute are wrong even for large n.

```
fit <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
X <- model.matrix(fit)
y <- savings$sr
dim(X)
```

```
#> [1] 50  5
```

```
head(X, 3)
```

```
#>           (Intercept) pop15 pop75  dpi ddpi
#> Australia           1 29.35  2.87 2330 2.87
#> Austria             1 23.32  4.41 1508 3.93
#> Belgium             1 23.80  4.43 2108 3.82
```

5.2 Least squares

The idea of least squares is to choose the coefficient vector for which the fitted mean $X\beta$ is as close as possible to the data in Euclidean distance.

Definition 5.2 (Least-squares estimate)**.** The **residual sum of squares** is $S(\beta) = \|y - X\beta\|^2 = (y - X\beta)^\top (y - X\beta)$. The least-squares estimate is $\hat\beta = \arg\min_{\beta \in \mathbb{R}^p} S(\beta)$. Fitted values are $\hat y = X\hat\beta$, residuals are $e = y - \hat y$, and $\mathrm{RSS} = S(\hat\beta) = e^\top e$.

Theorem 5.1 (Normal equations)**.** *If* $\operatorname{rank}(X) = p$*, then* $S(\beta)$ *has a unique minimiser* $\hat\beta$*, which is the unique solution of the* ***normal equations***

$$X^\top X\,\hat\beta = X^\top y, \qquad \textit{that is,} \qquad \hat\beta = (X^\top X)^{-1} X^\top y. \tag{5.2}$$

Proof. Expand $S(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta$. Using the rules $\partial(\beta^\top a)/\partial\beta = a$ and $\partial(\beta^\top A\beta)/\partial\beta = 2A\beta$ for symmetric A, the gradient is $-2X^\top y + 2X^\top X\beta$. Setting it to zero gives the normal equations. Since $\operatorname{rank}(X) = p$, we have $v^\top X^\top X v = \|Xv\|^2 > 0$ for every $v \neq 0$, so $X^\top X$ is positive definite, hence invertible, and the solution is unique.

To see that this stationary point is the *global* minimum, take any β and write $y - X\beta = (y - X\hat\beta) + X(\hat\beta - \beta) = e + X(\hat\beta - \beta)$. The normal equations say $X^\top e = 0$, so the cross term vanishes and

$$S(\beta) = \|e\|^2 + \|X(\hat\beta - \beta)\|^2 = \mathrm{RSS} + (\beta - \hat\beta)^\top X^\top X (\beta - \hat\beta) \;\geq\; \mathrm{RSS}, \tag{5.3}$$

with equality if and only if $\beta = \hat\beta$ (positive definiteness again). □

Remark. The *normal equations* are the linear system $X^\top X\beta = X^\top y$. The *least-squares estimate* is its solution $(X^\top X)^{-1} X^\top y$. They are different objects, and a common slip is to call the formula for $\hat\beta$ "the normal equations". The name comes from the equivalent form

$$X^\top (y - X\hat\beta) = X^\top e = 0, \tag{5.4}$$

which says the residual vector is *normal* (orthogonal) to every column of X. Because $\mathbf{1}$ is a column of X when there is an intercept, $\sum_i e_i = 0$.

Equation (5.3) is worth remembering: the excess of $S(\beta)$ over its minimum is the *quadratic form* $(\beta - \hat\beta)^\top X^\top X (\beta - \hat\beta)$. Its level sets are ellipsoids centred at $\hat\beta$ with shape given by $X^\top X$. We use this again when we build confidence regions.

Example 5.1 (Simple linear regression)**.** For $p = 2$, $X = [\mathbf{1},\ x]$, so $X^\top X = \begin{pmatrix} n & n\bar x \\ n\bar x & \sum x_i^2 \end{pmatrix}$ and $X^\top y = \begin{pmatrix} n\bar y \\ \sum x_i y_i \end{pmatrix}$. The first normal equation is $n\hat\beta_0 + n\bar x \hat\beta_1 = n\bar y$, so $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$. Substituting in the second, $\hat\beta_1 (\sum x_i^2 - n\bar x^2) = \sum x_i y_i - n\bar x \bar y$, that is

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}}, \qquad S_{xx} = \sum (x_i - \bar x)^2, \quad S_{xy} = \sum (x_i - \bar x)(y_i - \bar y).$$

The fitted line passes through $(\bar x, \bar y)$. The determinant of $X^\top X$ is $n S_{xx}$, which vanishes when all x_i are equal: the predictor must vary.

In R: least squares by hand

```
XtX <- crossprod(X); Xty <- crossprod(X, y)
beta_hat <- solve(XtX, Xty)           # solves the normal equations
cbind(by_hand = drop(beta_hat), lm = coef(fit))
```

```
#>                by_hand         lm
#> (Intercept) 28.5660865 28.5660865
#> pop15       -0.4611931 -0.4611931
#> pop75       -1.6914977 -1.6914977
#> dpi         -0.0003369 -0.0003369
#> ddpi         0.4096949  0.4096949
```

```
e <- y - X %*% beta_hat
max(abs(crossprod(X, e)))             # X'e = 0 up to rounding
```

```
#> [1] 2.339e-10
```

```
sum(e)                                # residuals sum to 0 (intercept in the
↪  model)
```

```
#> [1] 2.665e-15
```

5.3 Geometry: projection and the hat matrix

The fitted vector is $\hat y = X\hat\beta = X(X^\top X)^{-1}X^\top y$. The matrix that maps y to $\hat y$ has a name.

Definition 5.3 (Hat matrix)**.** $H = X(X^\top X)^{-1}X^\top$ is the $n \times n$ **hat matrix**, so that $\hat y = Hy$ ("H puts the hat on y") and $e = (I - H)y$. Its diagonal entries h_{ii} are the **leverages**.

Proposition 5.1 (Properties of the hat matrix)**.** *Let* $\operatorname{rank}(X) = p$*. Then*

1. $H^\top = H$ *(symmetric) and* $H^2 = H$ *(idempotent);*
2. $HX = X$ *and* $(I - H)X = 0$*;*
3. $I - H$ *is also symmetric and idempotent, and* $H(I - H) = 0$*;*
4. $\operatorname{tr} H = p$ *and* $\operatorname{tr}(I - H) = n - p$*;*
5. *the eigenvalues of* H *are* 0 *or* 1*, with* p *eigenvalues equal to 1, so* $\operatorname{rank} H = p$*.*

Proof.

(1) $H^\top = X\big((X^\top X)^{-1}\big)^\top X^\top = X(X^\top X)^{-1}X^\top = H$, because the inverse of a symmetric matrix is symmetric. Also $H^2 = X(X^\top X)^{-1}\underbrace{X^\top X(X^\top X)^{-1}}_{I_p}X^\top = H$.

(2) $HX = X(X^\top X)^{-1}X^\top X = X$.

(3) $(I - H)^\top = I - H$, $(I - H)^2 = I - 2H + H^2 = I - H$, and $H(I - H) = H - H^2 = 0$.

(4) By the cyclic property of the trace, $\operatorname{tr} H = \operatorname{tr}\big((X^\top X)^{-1}X^\top X\big) = \operatorname{tr} I_p = p$, and $\operatorname{tr}(I-H) = n - p$.

(5) If $Hv = \lambda v$ with $v \neq 0$ then $\lambda v = Hv = H^2 v = \lambda^2 v$, so $\lambda \in \{0, 1\}$. H is symmetric, hence diagonalisable, and the sum of its eigenvalues is $\operatorname{tr} H = p$, so exactly p of them equal 1.

□

Property (2) says that H leaves every vector of the column space $\mathcal{C}(X) = \{Xb : b \in \mathbb{R}^p\}$ unchanged, and properties (1) and (3) say that H is the *orthogonal projection* onto $\mathcal{C}(X)$ and $I - H$ is the orthogonal projection onto its orthogonal complement $\mathcal{C}(X)^\perp$, which has dimension $n - p$.

Theorem 5.2 (Least squares as orthogonal projection)**.** *The fitted vector* $\hat{y} = Hy$ *is the point of* $\mathcal{C}(X)$ *closest to* y*; the residual* $e = (I - H)y$ *lies in* $\mathcal{C}(X)^\perp$*; and*

$$\|y\|^2 = \|\hat{y}\|^2 + \|e\|^2, \qquad \hat{y}^\top e = 0.$$

Proof. $\hat{y}^\top e = y^\top H(I - H)y = 0$ by property (3). For any $z = Xb \in \mathcal{C}(X)$, $\|y - z\|^2 = \|e\|^2 + \|\hat{y} - z\|^2 \geq \|e\|^2$ because $e \perp (\hat{y} - z) \in \mathcal{C}(X)$. This is equation (5.3) written in terms of fitted vectors. □

The picture (Figure 5.2) is the one to keep in mind. The data vector y lives in $\mathbb{R}^n$; the model says its mean lies in the p-dimensional subspace $\mathcal{C}(X)$; least squares drops a perpendicular. The projection is Hy, **not** $X^\top y$ (which is a p-vector of inner products). Two consequences deserve emphasis. First, $\hat{y}$ depends on X only through the subspace $\mathcal{C}(X)$: if $X^* = XA$ with A invertible (for instance, we centre a predictor or change its units), the column space is the same, so $\hat{y}$, e and RSS are unchanged, and only the coefficients change, $\hat{\beta}^* = A^{-1}\hat{\beta}$. Second, the residuals live in an $(n - p)$-dimensional space, which is where the "degrees of freedom" in $\hat{\sigma}^2 = \mathrm{RSS}/(n - p)$ come from.

Sums of squares and R^2

When the model has an intercept, $\mathbf{1} \in \mathcal{C}(X)$, hence $\mathbf{1}^\top e = 0$ and the fitted values have the same mean as the data, $\bar{\hat{y}} = \bar{y}$. Orthogonality of $\hat{y} - \bar{y}\mathbf{1} \in \mathcal{C}(X)$ and e gives the decomposition

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{TSS}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{ESS}} + \underbrace{\sum_i e_i^2}_{\text{RSS}}. \tag{5.5}$$

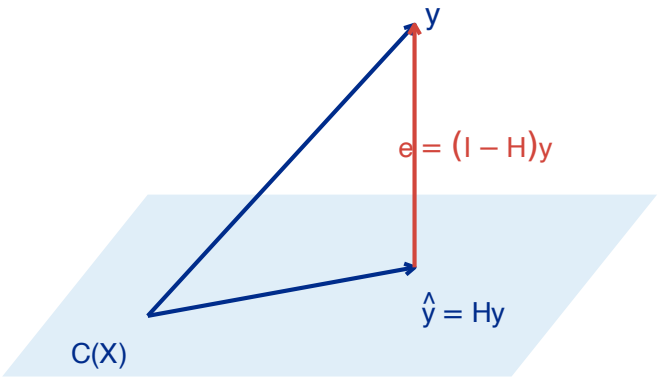


Figure 5.2: Least squares as an orthogonal projection of y onto the column space of X.

The **coefficient of determination** and its degrees-of-freedom-adjusted version are

$$R^2 = \frac{\mathrm{ESS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad R^2_{\mathrm{adj}} = 1 - \frac{\mathrm{RSS}/(n-p)}{\mathrm{TSS}/(n-1)}.$$

Since $\sum_i (\hat{y}_i - \bar{y})(y_i - \bar{y}) = \sum_i (\hat{y}_i - \bar{y})^2 = \mathrm{ESS}$ (the cross term with e vanishes), we get $\mathrm{Corr}(y, \hat{y})^2 = \mathrm{ESS}^2/(\mathrm{TSS} \cdot \mathrm{ESS}) = R^2$. R^2 never decreases when a column is added to X (a larger subspace is at least as close to y); R^2_{adj} can. Neither is a test, and a low R^2 can coexist with precisely estimated slopes if σ^2 is large.

5.4 Moments of the estimators and the Gauss–Markov theorem

Theorem 5.3 (Mean and variance of the least-squares estimator)**.** *Under A1–A3,* $\mathrm{E}[\hat{\beta}] = \beta$ *and* $\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$*. For any fixed* $c \in \mathbb{R}^p$*,* $\mathrm{Var}(c^\top \hat{\beta}) = \sigma^2 c^\top (X^\top X)^{-1} c$*.*

Proof. Write $\hat{\beta} = Ay$ with the fixed $p \times n$ matrix $A = (X^\top X)^{-1} X^\top$, so $AX = I_p$. By linearity of expectation and A2, $\mathrm{E}[\hat{\beta}] = A X\beta = \beta$. For the variance use $\mathrm{Var}(Ay) = A\mathrm{Var}(y)A^\top$ and $\mathrm{Var}(y) = \mathrm{Var}(\varepsilon) = \sigma^2 I$ (A3):

$$\mathrm{Var}(\hat{\beta}) = \sigma^2 AA^\top = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}.$$

The last statement follows from $\mathrm{Var}(c^\top \hat{\beta}) = c^\top \mathrm{Var}(\hat{\beta}) c$. □

Notice what this says: the precision of $\hat{\beta}_j$ depends on the design X only. Spreading the predictor values makes $X^\top X$ large and the variance small; strongly correlated predictors make the diagonal of $(X^\top X)^{-1}$ large (Chapter 6). The residuals and fitted values have simple moments too.

Proposition 5.2 (Moments of fitted values and residuals)**.** *Under A1–A3,* $\mathrm{E}[\hat{y}] = X\beta$*,* $\mathrm{Var}(\hat{y}) = \sigma^2 H$*,* $\mathrm{E}[e] = 0$*,* $\mathrm{Var}(e) = \sigma^2 (I - H)$ *and* $\mathrm{Cov}(\hat{y}, e) = 0$*. In particular* $\mathrm{Var}(e_i) = \sigma^2 (1 - h_{ii})$ *and the residuals are* correlated *(the off-diagonal entries of* $I - H$ *are* $-h_{ij}$*).*

Proof. $\hat{y} = Hy$, so $\mathrm{Var}(\hat{y}) = H(\sigma^2 I)H^\top = \sigma^2 H$. From $(I - H)X = 0$ we get $e = (I - H)(X\beta + \varepsilon) = (I - H)\varepsilon$, so $\mathrm{E}[e] = 0$ and $\mathrm{Var}(e) = \sigma^2 (I - H)(I - H)^\top = \sigma^2 (I - H)$. Finally $\mathrm{Cov}(\hat{y}, e) = H(\sigma^2 I)(I - H)^\top = \sigma^2 H(I - H) = 0$. □

The identity $e = (I - H)\varepsilon$ is the key to everything about $\hat{\sigma}^2$. It says that residuals are the part of the error that is *not* explained by the model: ε is projected onto $\mathcal{C}(X)^{\perp}$, and only $n - p$ dimensions of the n survive.

Lemma 5.1 (Expectation of a quadratic form)**.** *If* $\mathrm{E}[\varepsilon] = 0$ *and* $\mathrm{Var}(\varepsilon) = \sigma^2 I_n$*, then for a fixed* $n \times n$ *matrix* A*,* $\mathrm{E}[\varepsilon^\top A \varepsilon] = \sigma^2 \operatorname{tr}(A)$*.*

Proof. $\varepsilon^\top A \varepsilon$ is a scalar, so it equals its trace, and $\operatorname{tr}(\varepsilon^\top A \varepsilon) = \operatorname{tr}(A \varepsilon \varepsilon^\top)$ by the cyclic property. Taking expectations (trace and expectation commute, both are linear), $\mathrm{E}[\varepsilon^\top A \varepsilon] = \operatorname{tr}(A \mathrm{E}[\varepsilon\varepsilon^\top]) = \operatorname{tr}(A\, \sigma^2 I) = \sigma^2 \operatorname{tr} A$. □

Theorem 5.4 (Unbiased estimation of the error variance)**.** *Under A1–A3,* $\mathrm{E}[\mathrm{RSS}] = \sigma^2(n - p)$*, so* $\hat{\sigma}^2 = \mathrm{RSS}/(n - p)$ *is unbiased for* σ^2*.*

Proof. $\mathrm{RSS} = e^\top e = \varepsilon^\top (I-H)^\top (I-H)\varepsilon = \varepsilon^\top (I-H)\varepsilon$. By Lemma 5.1, $\mathrm{E}[\mathrm{RSS}] = \sigma^2 \operatorname{tr}(I-H) = \sigma^2(n - p)$. □

Normality was not used. We "lose" p degrees of freedom because the n residuals satisfy the p linear constraints $X^\top e = 0$. Dividing by n instead gives an estimator that is too small by the factor $(n - p)/n$: with $p = 5$ and $n = 50$ the relative bias is -10%.

The Gauss–Markov theorem

Why least squares? Among all reasonable ways of estimating β it has the smallest variance, provided we restrict attention to unbiased estimators that are linear in y.

Theorem 5.5 (Gauss–Markov)**.** *Under A1–A3, let* $\tilde{\beta} = Cy$ *be any linear estimator (*C *a fixed* $p \times n$ *matrix) that is unbiased for every* β*. Then* $\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta})$ *is positive semi-definite. Equivalently, for every* $c \in \mathbb{R}^p$*,* $\mathrm{Var}(c^\top \tilde{\beta}) \geq \mathrm{Var}(c^\top \hat{\beta})$*:* $\hat{\beta}$ *is the* ***best linear unbiased estimator*** *(BLUE).*

Proof. Unbiasedness for every β requires $\mathrm{E}[Cy] = CX\beta = \beta$ for all β, so $CX = I_p$. Write $C = A + D$ with $A = (X^\top X)^{-1} X^\top$. Then $AX = I_p$ forces $DX = 0$. The cross terms in $CC^\top$ vanish:

$$AD^\top = (X^\top X)^{-1} X^\top D^\top = (X^\top X)^{-1} (DX)^\top = 0,$$

and likewise $DA^\top = 0$. Hence

$$\mathrm{Var}(\tilde{\beta}) = \sigma^2 CC^\top = \sigma^2 AA^\top + \sigma^2 DD^\top = \mathrm{Var}(\hat{\beta}) + \sigma^2 DD^\top.$$

The matrix $DD^\top$ is positive semi-definite ($v^\top DD^\top v = \|D^\top v\|^2 \geq 0$), which is the claim. For a fixed c, $\mathrm{Var}(c^\top \tilde{\beta}) - \mathrm{Var}(c^\top \hat{\beta}) = \sigma^2 \|D^\top c\|^2 \geq 0$, with equality iff $D^\top c = 0$, i.e. iff $c^\top \tilde{\beta} = c^\top \hat{\beta}$. □

Remark. What the theorem does *not* say. (i) It is silent about *biased* estimators: ridge regression and the lasso (Chapter 7) are biased, and can have smaller mean squared error than $\hat{\beta}$. (ii) It is silent about non-linear estimators; under normality, however, $\hat{\beta}$ has the smallest variance among *all* unbiased estimators, linear or not (Section 5.7). (iii) It needs A3. If the variances differ or the errors are correlated, $\hat{\beta}$ is still unbiased but no longer best, and weighted or generalised least squares is better.

5.5 Distribution theory under normality

From now on we add A4, $\varepsilon \sim N(0, \sigma^2 I_n)$, so that $y \sim N_n(X\beta, \sigma^2 I_n)$.

Theorem 5.6 (Sampling distributions). *Under A1 and A4,*

1. $\hat{\beta} \sim N_p(\beta,\ \sigma^2 (X^\top X)^{-1})$;
2. $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p}$;
3. $\hat{\beta}$ *and* RSS *are independent.*

Proof.

(1) $\hat{\beta} = \beta + (X^\top X)^{-1} X^\top \varepsilon$ is a linear function of a Gaussian vector, hence Gaussian, with the mean and variance from Theorem 5.3.

(2) The matrix $I - H$ is symmetric and idempotent of rank $n - p$, so by the spectral theorem $I - H = QQ^\top$ for an $n \times (n-p)$ matrix Q with orthonormal columns, $Q^\top Q = I_{n-p}$ (its columns are the eigenvectors with eigenvalue 1). Then $\mathrm{RSS} = \varepsilon^\top (I-H)\varepsilon = \|Q^\top \varepsilon\|^2$. The vector $Q^\top \varepsilon$ is Gaussian with mean 0 and variance $\sigma^2 Q^\top Q = \sigma^2 I_{n-p}$, so its $n-p$ components are iid $N(0, \sigma^2)$, and the sum of their squares divided by σ^2 is χ^2_{n-p}.

(3) From $(I-H)X = 0$ we get $QQ^\top X = 0$; multiplying by $Q^\top$ and using $Q^\top Q = I$ gives $Q^\top X = 0$. The Gaussian vectors $\hat{\beta} - \beta = (X^\top X)^{-1} X^\top \varepsilon$ and $Q^\top \varepsilon$ are jointly Gaussian with covariance $\sigma^2 (X^\top X)^{-1} X^\top Q = 0$, hence independent. $\mathrm{RSS} = \|Q^\top \varepsilon\|^2$ is a function of $Q^\top \varepsilon$ alone.

□

The independence in part (3) is what makes the t statistic work: numerator and denominator come from orthogonal pieces of ε, the part in $\mathcal{C}(X)$ and the part in $\mathcal{C}(X)^\perp$. It is also special to the normal distribution (uncorrelated does not imply independent otherwise).

Corollary 5.1 (The t pivot). *Under A1 and A4, for any $c \in \mathbb{R}^p$,*

$$\frac{c^\top \hat{\beta} - c^\top \beta}{\hat{\sigma}\sqrt{c^\top (X^\top X)^{-1} c}} \sim t_{n-p}.$$

In particular, with $\mathrm{se}(\hat{\beta}_j) = \hat{\sigma}\sqrt{[(X^\top X)^{-1}]_{jj}}$, $(\hat{\beta}_j - \beta_j)/\mathrm{se}(\hat{\beta}_j) \sim t_{n-p}$.

Proof. The numerator divided by $\sigma\sqrt{c^\top (X^\top X)^{-1} c}$ is $Z \sim N(0,1)$ by Theorem 5.6(1). Write $\hat{\sigma}/\sigma = \sqrt{U/(n-p)}$ with $U = \mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p}$, independent of Z by (3). The ratio $Z/\sqrt{U/(n-p)}$ has the t_{n-p} distribution by definition. Note that the unknown σ cancels. □

5.6 Inference for the coefficients

Corollary 5.1 gives everything for a single linear combination:

- **Test** of $H_0 : \beta_j = 0$: $t_j = \hat{\beta}_j / \mathrm{se}(\hat{\beta}_j)$, with p-value $2P(T_{n-p} > |t_j|)$.
- **Confidence interval**: $\hat{\beta}_j \pm t_{n-p,\,1-\alpha/2}\,\mathrm{se}(\hat{\beta}_j)$ has coverage exactly $1-\alpha$ under A1–A4.
- **Interpretation**: each t test is *conditional on the other predictors being in the model*. It asks whether x_j adds anything given the rest, not whether x_j is related to y.

Example 5.2 (Savings: full model)**.** We fit the model of the introduction and rebuild every number of `summary()` from $\hat{\beta}$, $\hat{\sigma}^2$ and $(X^\top X)^{-1}$.

```
summary(fit)
```

```
#>
#> Call:
#> lm(formula = sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
#>
#> Residuals:
#>    Min     1Q Median     3Q    Max
#> -8.242 -2.686 -0.249  2.428  9.751
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept) 28.566087   7.354516    3.88  0.00033 ***
#> pop15       -0.461193   0.144642   -3.19  0.00260 **
#> pop75       -1.691498   1.083599   -1.56  0.12553
#> dpi         -0.000337   0.000931   -0.36  0.71917
#> ddpi         0.409695   0.196197    2.09  0.04247 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 3.8 on 45 degrees of freedom
#> Multiple R-squared:  0.338,  Adjusted R-squared:  0.28
#> F-statistic: 5.76 on 4 and 45 DF,  p-value: 0.00079
```

```
n <- nrow(X); p <- ncol(X)
RSS <- sum(resid(fit)^2)
sigma2_hat <- RSS / (n - p)
V <- sigma2_hat * solve(crossprod(X))          # estimated Var(beta_hat)
se <- sqrt(diag(V))
tval <- coef(fit) / se
pval <- 2 * pt(abs(tval), n - p, lower.tail = FALSE)
round(cbind(estimate = coef(fit), se = se, t = tval, p = pval), 4)
```

```
#>             estimate     se       t      p
#> (Intercept)  28.5661 7.3545  3.8842 0.0003
#> pop15        -0.4612 0.1446 -3.1885 0.0026
#> pop75        -1.6915 1.0836 -1.5610 0.1255
#> dpi          -0.0003 0.0009 -0.3618 0.7192
#> ddpi          0.4097 0.1962  2.0882 0.0425
```

```
c(RSS = RSS, sigma_hat = sqrt(sigma2_hat))
```

```
#>       RSS sigma_hat
#>   650.713     3.803
```

```
coef(fit)["pop15"] + c(-1, 1) * qt(0.975, n - p) * se["pop15"]     # CI by
↪  hand
```

```
#> [1] -0.7525 -0.1699
```

```
confint(fit)["pop15", ]
```

```
#>   2.5 %  97.5 %
#> -0.7525 -0.1699
```

For `pop15`, $\hat\beta = -0.461$, se $= 0.145$, $t = -3.19$ on 45 degrees of freedom ($p = 0.0026$) and the 95% CI is $(-0.753, -0.170)$. In words: countries whose population under 15 is one percentage point larger save between 0.17 and 0.75 points *less* of their disposable income, *on average and with `pop75`, `dpi`, `ddpi` held fixed.* The same table shows that `pop75` and `dpi` are not individually significant given the others; we return to this in Chapter 6, where we find that `pop15` and `pop75` are strongly correlated.

The F test for nested models

Often we want to test several coefficients at once, for instance that a *group* of predictors adds nothing. Let $M_0 \subset M_1$ be two linear models with design matrices X_0 ($n \times p_0$) and X_1 ($n \times p_1$), $p_1 > p_0$, such that $\mathcal{C}(X_0) \subset \mathcal{C}(X_1)$ (every column of X_0 is a linear combination of the columns of X_1; typically M_0 is M_1 with $q = p_1 - p_0$ coefficients set to zero). Write H_0, H_1 for the hat matrices and $\mathrm{RSS}_0 \geq \mathrm{RSS}_1$.

Theorem 5.7 (F test for nested models)**.** *Under A1 and A4, if M_0 is true,*

$$F = \frac{(\mathrm{RSS}_0 - \mathrm{RSS}_1)/(p_1 - p_0)}{\mathrm{RSS}_1/(n - p_1)} \;\sim\; F_{p_1 - p_0,\, n - p_1}.$$

Proof. Since $\mathcal{C}(X_0) \subset \mathcal{C}(X_1)$, we have $H_1 H_0 = H_0 = H_0 H_1$ (projecting onto the small space and then the big one is the same as projecting onto the small one). Hence $P = H_1 - H_0$ is symmetric and idempotent: $P^2 = H_1 - H_1H_0 - H_0H_1 + H_0 = H_1 - H_0 = P$, with $\operatorname{tr} P = p_1 - p_0 = q$, so $\operatorname{rank} P = q$. Moreover $P(I - H_1) = H_1 - H_1^2 - H_0 + H_0H_1 = 0$.

By the Pythagorean identity of Theorem 5.2 applied twice, $\mathrm{RSS}_0 - \mathrm{RSS}_1 = \|(I - H_0)y\|^2 - \|(I - H_1)y\|^2 = \|Py\|^2$. If M_0 holds, $y = X_0\beta_0 + \varepsilon$ and $PX_0 = H_1X_0 - H_0X_0 = X_0 - X_0 = 0$, so $Py = P\varepsilon$ and $\mathrm{RSS}_0 - \mathrm{RSS}_1 = \varepsilon^\top P \varepsilon$. As in the proof of Theorem 5.6, $\varepsilon^\top P\varepsilon/\sigma^2 \sim \chi^2_q$ (a projection of rank q applied to a spherical Gaussian vector). The vectors $P\varepsilon$ and $(I - H_1)\varepsilon$ are jointly Gaussian and uncorrelated because $P(I - H_1) = 0$, so they are independent, and $\mathrm{RSS}_1/\sigma^2 \sim \chi^2_{n-p_1}$. The ratio of two independent χ^2 variables divided by their degrees of freedom is $F_{q,n-p_1}$; σ^2 cancels. □

Special cases worth knowing: (i) when $q = 1$, $F = t^2$, the square of the t statistic for the single extra coefficient (Exercise 5.9); (ii) when M_0 is the intercept-only model ($p_0 = 1$, $\mathrm{RSS}_0 = \mathrm{TSS}$), F is the **overall F statistic** printed by `summary()`, and

$$F = \frac{(\mathrm{TSS} - \mathrm{RSS})/(p - 1)}{\mathrm{RSS}/(n - p)} = \frac{R^2/(p-1)}{(1 - R^2)/(n - p)}. \tag{5.6}$$

```
fit0 <- lm(sr ~ pop15 + ddpi, data = savings)      # M0: drops pop75 and dpi
anova(fit0, fit)
```

```
#> Analysis of Variance Table
#>
#> Model 1: sr ~ pop15 + ddpi
#> Model 2: sr ~ pop15 + pop75 + dpi + ddpi
```

```
#>   Res.Df RSS Df Sum of Sq    F Pr(>F)
#> 1     47 701
#> 2     45 651  2      49.8 1.72   0.19
```

```
RSS0 <- sum(resid(fit0)^2)
Fstat <- ((RSS0 - RSS) / 2) / (RSS / (n - p))
c(F = Fstat, p.value = pf(Fstat, 2, n - p, lower.tail = FALSE))
```

```
#>       F p.value
#>   1.723   0.190
```

$F = 1.72$ on $(2, 45)$ degrees of freedom, $p = 0.19$: no evidence that `pop75` and `dpi` *jointly* improve the fit once `pop15` and `ddpi` are in the model. Two separate t tests would not answer this question, because the two statistics are correlated and the individual p-values do not control the joint error rate.

5.7 Least squares and maximum likelihood

Under A4 the likelihood is available, and we can place least squares in the framework of Chapters 3 and 4.

Theorem 5.8 (Least squares is maximum likelihood)**.** *Under A1 and A4, the log-likelihood of* (β, σ^2) *is*

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta), \tag{5.7}$$

and its maximiser is $\hat\beta_{ML} = \hat\beta$ *and* $\hat\sigma^2_{ML} = \mathrm{RSS}/n$*, with maximised value*

$$\ell(\hat\beta, \hat\sigma^2_{ML}) = -\frac{n}{2}\Big\{\log\left(2\pi\,\mathrm{RSS}/n\right) + 1\Big\}.$$

Moreover $\mathrm{E}[\hat\sigma^2_{ML}] = \sigma^2(n-p)/n < \sigma^2$.

Proof. For any fixed σ^2, maximising (5.7) over β is minimising $S(\beta)$, so $\hat\beta_{\mathrm{ML}} = \hat\beta$ whatever σ^2 is. Substituting, the profile log-likelihood is $\ell_P(\sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \mathrm{RSS}/(2\sigma^2)$ with derivative $-n/(2\sigma^2) + \mathrm{RSS}/(2\sigma^4)$, which vanishes at $\sigma^2 = \mathrm{RSS}/n$; it is positive to the left and negative to the right, so this is the maximum. Plugging in gives the maximised value. The bias follows from Theorem 5.4. □

Two consequences. First, -2ℓ has the factor $1/\sigma^2$, not $1/(2\sigma^2)$: $-2\ell(\beta, \sigma^2) = n\log(2\pi\sigma^2) + \sigma^{-2}\|y - X\beta\|^2$. Second, **the likelihood estimator of σ^2 is biased** and the divisor $n - p$ replaces n to repair it. The same issue returns for variance components in mixed models, where it motivates REML (Chapter 12). Information theory completes the picture. The score for β is $S(\beta) = \sigma^{-2}X^\top(y - X\beta)$ and the information is $\mathcal{I}(\beta) = \sigma^{-2}X^\top X$ (it does not depend on β, so observed = expected), whose inverse is exactly $\mathrm{Var}(\hat\beta)$: the Cramér–Rao bound is attained, so under normality $\hat\beta$ is the minimum-variance unbiased estimator among *all* unbiased estimators, not only the linear ones. Note also that ℓ has $p + 1$ parameters, which is the number that enters AIC and BIC.

Theorem 5.9 (Likelihood ratio and F)**.** *For nested linear models as above, the likelihood-ratio statistic is*

$$2\{\ell_1 - \ell_0\} = n\log\frac{\mathrm{RSS}_0}{\mathrm{RSS}_1} = n\log\left(1 + \frac{q}{n - p_1}F\right),$$

an increasing function of F. Thus the LR test and the F test are the same test, but the F test has an exact null distribution while the LR test relies on the asymptotic χ^2_q result.

Proof. By Theorem 5.8, $2\ell_k = -n\log(2\pi \mathrm{RSS}_k/n) - n$ for $k = 0, 1$; subtract. Next, $F = \frac{(\mathrm{RSS}_0 - \mathrm{RSS}_1)(n-p_1)}{q\,\mathrm{RSS}_1}$, so $\mathrm{RSS}_0/\mathrm{RSS}_1 = 1 + qF/(n-p_1)$. □

```
sigma2_ml <- RSS / n
c(by_hand = -n / 2 * (log(2 * pi * sigma2_ml) + 1), logLik =
↪  as.numeric(logLik(fit)))
```

```
#> by_hand  logLik
#>  -135.1  -135.1
```

```
c(LR = n * log(RSS0 / RSS), from_F = n * log(1 + 2 / (n - p) * Fstat),
  chisq_p = pchisq(n * log(RSS0 / RSS), 2, lower.tail = FALSE))
```

```
#>      LR  from_F chisq_p
#>   3.690   3.690   0.158
```

The LR p-value (0.16) is slightly smaller than the exact F p-value (0.19), the usual direction: the asymptotic χ^2 approximation is liberal in small samples.

5.8 Computation

The formula $\hat\beta = (X^\top X)^{-1}X^\top y$ is for thinking, not for computing. Forming $X^\top X$ costs $O(np^2)$ operations and solving the $p \times p$ system $O(p^3)$; for $n \gg p$ the cost is dominated by $O(np^2)$. (Nothing in least squares costs $O(n^3)$: we never invert an $n \times n$ matrix, and H is never formed in practice.) Numerically, the condition number of $X^\top X$ is the *square* of that of X, so solving the normal equations can lose twice as many digits as necessary. `lm()` therefore uses the **QR decomposition** $X = QR$ with $Q^\top Q = I_p$ and R upper triangular. The normal equations become $R^\top R\beta = R^\top Q^\top y$, i.e. $R\beta = Q^\top y$, solved by back-substitution. Moreover $(X^\top X)^{-1} = R^{-1}R^{-\top}$ and $h_{ii} = \|Q_{i\cdot}\|^2$.

```
qrX <- qr(X)
beta_qr <- backsolve(qr.R(qrX), crossprod(qr.Q(qrX), y))
cbind(qr = drop(beta_qr), normal_eq = drop(beta_hat), lm = coef(fit))
```

```
#>                     qr  normal_eq         lm
#> (Intercept) 28.5660865 28.5660865 28.5660865
#> pop15       -0.4611931 -0.4611931 -0.4611931
#> pop75       -1.6914977 -1.6914977 -1.6914977
#> dpi         -0.0003369 -0.0003369 -0.0003369
#> ddpi         0.4096949  0.4096949  0.4096949
```

```
H <- X %*% solve(crossprod(X), t(X))
c(symmetric = max(abs(H - t(H))), idempotent = max(abs(H %*% H - H)), trace =
↪  sum(diag(H)))
```

```
#>  symmetric idempotent      trace
#>  4.025e-16  2.331e-15  5.000e+00
```

```
c(kappa_X = kappa(X, exact = TRUE), kappa_XtX = kappa(crossprod(X), exact =
↪  TRUE))
```

```
#>   kappa_X kappa_XtX
#>     20375 415155098
```

(The condition number of `savings` is large because `dpi` is of the order of 10^3 while the other columns are of order 10. Standardising columns reduces it without changing the fit, a use of invariance to $X \mapsto XA$.)

Rank deficiency. If A1 fails (a column is a linear combination of the others, for instance a full set of dummy variables plus an intercept) then $X^\top X$ is singular, the normal equations have infinitely many solutions, and *no* algorithm can return a unique $\hat\beta$. The fitted vector $\hat y = Hy$ is still unique, because $\mathcal{C}(X)$ does not depend on how we parametrise it, but the individual coefficients are not identifiable. `lm()` detects this and reports `NA` for the aliased coefficients.

```
# share of the population aged 15-75 is the complement of the other two
↪  shares
sav <- transform(savings, pop_mid = 100 - pop15 - pop75)
fit_def <- lm(sr ~ pop15 + pop75 + pop_mid, data = sav)
coef(fit_def)   # pop_mid is aliased: pop15 + pop75 + pop_mid = 100 *
↪  (intercept column)
```

```
#> (Intercept)       pop15       pop75     pop_mid
#>     30.6277     -0.4708     -1.9341          NA
```

5.9 A simulation check

Theorems 5.4 and 5.6 are statements about repeated sampling. We can check them by simulating data from a known model that shares the design of `savings`.

```
set.seed(505)
beta_true <- coef(fit); sigma_true <- 4; B <- 4000
sim <- replicate(B, {
  ysim <- drop(X %*% beta_true) + rnorm(n, sd = sigma_true)
  f <- lm(ysim ~ X - 1)
  ci <- confint(f)["Xpop15", ]
  c(rss = sum(resid(f)^2), b = coef(f)["Xpop15"],
    cover = unname(ci[1] < beta_true["pop15"] & beta_true["pop15"] < ci[2]))
})
c(mean_RSS = mean(sim["rss", ]), theory = sigma_true^2 * (n - p),
  mean_RSS_over_n = mean(sim["rss", ]) / n, mean_RSS_over_np = mean(sim["rss",
  ↪  ]) / (n - p),
  coverage = mean(sim["cover", ]))
```

```
#>         mean_RSS           theory  mean_RSS_over_n mean_RSS_over_np
#>         719.4854         720.0000          14.3897          15.9886
#>         coverage
#>           0.9483
```

```
c(cor_beta_RSS = cor(sim["b.Xpop15", ], sim["rss", ]),
  var_beta = var(sim["b.Xpop15", ]), theory = sigma_true^2 *
  ↪  solve(crossprod(X))["pop15", "pop15"])
```

```
#> cor_beta_RSS     var_beta       theory
#>    -0.006018     0.023325     0.023149
```

The average RSS matches $\sigma^2(n-p) = 720$, so RSS$/(n-p)$ is unbiased and RSS$/n$ underestimates by 10%; the 95% interval covers the truth in about 95% of data sets; $\hat{\beta}_j$ and RSS are uncorrelated (as they should, being independent); and the empirical variance of $\hat{\beta}_{\texttt{pop15}}$ matches $\sigma^2[(X^\top X)^{-1}]_{jj}$.

5.10 In R: the functions used in class

Task	Code
Fit	`fit <- lm(y ~ x1 + x2, data = d)`
Design matrix, response	`model.matrix(fit)`, `model.response(model.frame(fit))`
Coefficients, SEs, t, p	`summary(fit)$coefficients`, `coef(fit)`, `vcov(fit)`
Confidence intervals	`confint(fit, level = 0.95)`
$\hat{\sigma}$, df, RSS	`sigma(fit)`, `df.residual(fit)`, `deviance(fit)`
Nested F test	`anova(fit0, fit1)`
Hat values, residuals	`hatvalues(fit)`, `resid(fit)`
Log-likelihood, AIC, BIC	`logLik(fit)`, `AIC(fit)`, `BIC(fit)`
QR	`qr(X)`, `qr.R()`, `qr.Q()`, `backsolve()`

5.11 Common pitfalls

Warning. Calling $(X^\top X)^{-1}X^\top y$ "the normal equations" (it is their solution), or saying that $X^\top y$ is the projection of y (the projection is Hy).

Warning. Dividing RSS by n and calling it unbiased, or forgetting that p counts the intercept: the residual degrees of freedom of `savings` are $50 - 5 = 45$, not $50 - 4$.

Warning. Reading a non-significant t test as "x_j has no effect". The test is conditional on the other predictors, and a large p-value is absence of evidence, not evidence of absence. Look at the confidence interval.

Warning. Running several t tests when the question is about a group of coefficients. Use the nested F test.

Remark. Exact t and F results need A4. With moderate or large n and no heavy-tailed errors they are good approximations; they do *not* repair a wrong mean (A2) or non-constant variance (A3). Chapter 6 is about checking these.

5.12 Chapter summary

- The linear model $y = X\beta + \varepsilon$ with X of full column rank p (intercept included). Least squares minimises $\|y - X\beta\|^2$; the normal equations $X^\top X\hat{\beta} = X^\top y$ have the unique solution $\hat{\beta} = (X^\top X)^{-1}X^\top y$.
- $\hat{y} = Hy$ is the orthogonal projection of y onto $\mathcal{C}(X)$. H is symmetric, idempotent, $\operatorname{tr} H = p$; $e = (I - H)y \perp \mathcal{C}(X)$; TSS $=$ ESS $+$ RSS when there is an intercept.

- $\mathrm{E}[\hat\beta] = \beta$, $\mathrm{Var}(\hat\beta) = \sigma^2(X^\top X)^{-1}$, $\mathrm{Var}(e) = \sigma^2(I-H)$, $\mathrm{E}[\mathrm{RSS}] = \sigma^2(n-p)$ (A1–A3). Gauss–Markov: $\hat\beta$ is BLUE.
- Under normality: $\hat\beta \sim N_p(\beta, \sigma^2(X^\top X)^{-1})$, $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p}$, independent. Hence the t pivot with $n-p$ degrees of freedom, exact confidence intervals, and the nested-model F test.
- $\hat\beta_{\mathrm{LS}} = \hat\beta_{\mathrm{ML}}$; $\hat\sigma^2_{\mathrm{ML}} = \mathrm{RSS}/n$ is biased; the LR statistic is a monotone function of F.
- Compute with QR, not by inverting $X^\top X$; rank deficiency makes $\hat\beta$ non-unique but not $\hat y$.

5.13 Exercises

Exercise 5.1. Simple regression. For $X = [\mathbf{1}, x]$ with $S_{xx} > 0$, show that (a) $\mathrm{Var}(\hat\beta_1) = \sigma^2/S_{xx}$, (b) $\mathrm{Var}(\hat\beta_0) = \sigma^2(1/n + \bar x^2/S_{xx})$, (c) $\mathrm{Cov}(\hat\beta_0, \hat\beta_1) = -\sigma^2\bar x/S_{xx}$. When is $\hat\beta_0$ uncorrelated with $\hat\beta_1$?

Solution. $\det(X^\top X) = n\sum x_i^2 - n^2\bar x^2 = nS_{xx}$, using $\sum x_i^2 = S_{xx} + n\bar x^2$. Hence

$$(X^\top X)^{-1} = \frac{1}{nS_{xx}}\begin{pmatrix} \sum x_i^2 & -n\bar x \\ -n\bar x & n \end{pmatrix}.$$

Multiply by σ^2: (a) $\mathrm{Var}(\hat\beta_1) = \sigma^2 n/(nS_{xx}) = \sigma^2/S_{xx}$; (b) $\mathrm{Var}(\hat\beta_0) = \sigma^2\sum x_i^2/(nS_{xx}) = \sigma^2(S_{xx}+n\bar x^2)/(nS_{xx}) = \sigma^2(1/n + \bar x^2/S_{xx})$; (c) $\mathrm{Cov}(\hat\beta_0,\hat\beta_1) = -\sigma^2 n\bar x/(nS_{xx}) = -\sigma^2\bar x/S_{xx}$. The covariance is zero iff $\bar x = 0$: if the predictor is centred, the intercept and slope estimators are uncorrelated (and under normality independent). This is one reason for centring.

Exercise 5.2. Regression through the origin. Consider $y_i = \beta x_i + \varepsilon_i$ (no intercept) with A1–A3. Derive $\hat\beta$, $\mathrm{Var}(\hat\beta)$ and $\hat\sigma^2$. Do the residuals sum to zero? Is $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$ still guaranteed to lie in $[0,1]$?

Solution. Here $X = x$ is $n \times 1$ and $p = 1$, so $\hat\beta = x^\top y/x^\top x = \sum x_i y_i/\sum x_i^2$, $\mathrm{Var}(\hat\beta) = \sigma^2/\sum x_i^2$ and $\hat\sigma^2 = \mathrm{RSS}/(n-1)$. The normal equation is $x^\top e = 0$, but $\mathbf{1} \notin \mathcal{C}(x)$ in general, so $\sum e_i \neq 0$. The decomposition $\mathrm{TSS} = \mathrm{ESS} + \mathrm{RSS}$ relied on $\mathbf{1} \in \mathcal{C}(X)$, so it fails: RSS can exceed $\mathrm{TSS} = \sum(y_i - \bar y)^2$ (the model βx_i may fit worse than the constant $\bar y$) and $1 - \mathrm{RSS}/\mathrm{TSS}$ can be negative. R's `summary()` therefore uses the *uncentred* $R^2 = 1 - \mathrm{RSS}/\sum y_i^2$ for models without intercept, which is not comparable with the usual R^2.

Exercise 5.3. Fitted values. Under A1–A3 show that $\mathrm{Var}(\hat y_i) = \sigma^2 h_{ii}$ and $\mathrm{Cov}(\hat y, e) = 0$. Using $\sum_i h_{ii} = p$, find $\sum_i \mathrm{Var}(\hat y_i)$ and interpret the result for the average variance of the fitted values.

Solution. $\hat y = Hy$ gives $\mathrm{Var}(\hat y) = H\sigma^2 I H^\top = \sigma^2 H$, so $\mathrm{Var}(\hat y_i) = \sigma^2 h_{ii}$. $\mathrm{Cov}(\hat y, e) = H\sigma^2 I(I-H)^\top = \sigma^2 H(I-H) = 0$. Summing, $\sum_i \mathrm{Var}(\hat y_i) = \sigma^2 \operatorname{tr} H = p\sigma^2$, so the average variance of the fitted values is $p\sigma^2/n$: it grows linearly in the number of parameters. This is the variance part of the bias–variance trade-off that motivates AIC and C_p in Chapter 7.

Exercise 5.4. BLUE of a mean. Let $y_i = \mu + \varepsilon_i$ $(i = 1, \dots, n)$ with A2–A3. (a) Show that $\hat\mu = \bar y$ is the least-squares estimator and find its variance. (b) Let $\tilde\mu = \sum a_i y_i$ with $\sum a_i = 1$. Show it is unbiased and that $\mathrm{Var}(\tilde\mu) \geq \mathrm{Var}(\bar y)$, with equality iff $a_i = 1/n$. (c) Compare numerically $\bar y$ with $\tilde\mu = 0.5y_1 + 0.5y_n$ for $n = 10$.

Solution.

(a) $X = \mathbf{1}$, $X^\top X = n$, $\hat\mu = \mathbf{1}^\top y/n = \bar y$, $\mathrm{Var}(\bar y) = \sigma^2/n$. (b) $\mathrm{E}[\tilde\mu] = \mu\sum a_i = \mu$. $\mathrm{Var}(\tilde\mu) = \sigma^2\sum a_i^2$. Write $a_i = 1/n + d_i$ with $\sum d_i = 0$; then $\sum a_i^2 = 1/n + 2\sum d_i/n + \sum d_i^2 = 1/n + \sum d_i^2 \geq 1/n$, with equality iff all $d_i = 0$. This is the Gauss–Markov theorem for $p = 1$. (c) $\mathrm{Var}(\tilde\mu) = \sigma^2(0.25 + 0.25) = 0.5\sigma^2$ against $\mathrm{Var}(\bar y) = 0.1\sigma^2$: five times larger.

Exercise 5.5. Invariance. Let $X^* = XA$ with A a $p \times p$ invertible matrix. Show that $H^* = H$, $\hat\beta^* = A^{-1}\hat\beta$ and $\mathrm{RSS}^* = \mathrm{RSS}$. Apply this to simple regression with the centred predictor $x_i^c = x_i - \bar x$: write A and find $\hat\beta_0^c$, $\hat\beta_1^c$ and their standard errors.

Solution. $H^* = XA(A^\top X^\top XA)^{-1}A^\top X^\top = XAA^{-1}(X^\top X)^{-1}A^{-\top}A^\top X^\top = H$. Then $\hat\beta^* = (A^\top X^\top XA)^{-1}A^\top X^\top y = A^{-1}(X^\top X)^{-1}X^\top y = A^{-1}\hat\beta$, and $\hat y$, e, RSS depend on X only through H. For centring, $X^* = [\mathbf{1}, x - \bar x\mathbf{1}] = X\begin{pmatrix}1 & -\bar x\\ 0 & 1\end{pmatrix} = XA$, so $\hat\beta^* = A^{-1}\hat\beta = \begin{pmatrix}1 & \bar x\\ 0 & 1\end{pmatrix}\begin{pmatrix}\hat\beta_0\\ \hat\beta_1\end{pmatrix} = \begin{pmatrix}\hat\beta_0 + \hat\beta_1\bar x\\ \hat\beta_1\end{pmatrix} = \begin{pmatrix}\bar y\\ \hat\beta_1\end{pmatrix}$. The slope and its standard error ($\hat\sigma/\sqrt{S_{xx}}$) are unchanged; the new intercept is $\bar y$, the fitted mean at $\bar x$, with se $= \hat\sigma/\sqrt{n}$, which is *smaller* than the old intercept's standard error and is also meaningful.

Exercise 5.6. By hand. Five observations: $x = (1, 2, 3, 4, 5)$, $y = (2.1, 3.9, 6.2, 7.8, 10.1)$. (a) Compute $\hat\beta$, RSS, $\hat\sigma^2$, se$(\hat\beta_1)$. (b) Test $H_0 : \beta_1 = 0$ and give a 95% CI for β_1 ($t_{3,0.975} = 3.182$). (c) Compute R^2 and the overall F and check $F = t^2$. (d) Verify with `lm()`.

Solution.

(a) $\bar x = 3$, $\bar y = 6.02$, $S_{xx} = 10$, $S_{xy} = \sum(x_i - 3)(y_i - 6.02) = (-2)(-3.92) + (-1)(-2.12) + 0 + (1)(1.78) + (2)(4.08) = 7.84 + 2.12 + 1.78 + 8.16 = 19.9$. So $\hat\beta_1 = 1.99$, $\hat\beta_0 = 6.02 - 1.99 \times 3 = 0.05$. Fitted values $2.04, 4.03, 6.02, 8.01, 10.00$, residuals $0.06, -0.13, 0.18, -0.21, 0.10$ (sum 0), RSS $= 0.0036 + 0.0169 + 0.0324 + 0.0441 + 0.01 = 0.107$, $\hat\sigma^2 = 0.107/3 = 0.03567$, $\hat\sigma = 0.1889$, se$(\hat\beta_1) = 0.1889/\sqrt{10} = 0.0597$.
(b) $t = 1.99/0.0597 = 33.3$ on 3 df, $p \approx 6 \times 10^{-5}$. CI: $1.99 \pm 3.182 \times 0.0597 = (1.80, 2.18)$.
(c) TSS $= \sum(y_i - 6.02)^2 = 15.3664 + 4.4944 + 0.0324 + 3.1684 + 16.6464 = 39.708$, so $R^2 = 1 - 0.107/39.708 = 0.9973$; $F = (39.708 - 0.107)/(0.107/3) = 1110$, and $t^2 = 33.3^2 = 1110$.
(d) See the output below.

```
x5 <- 1:5; y5 <- c(2.1, 3.9, 6.2, 7.8, 10.1)
f5 <- lm(y5 ~ x5); summary(f5)$coefficients; c(RSS = deviance(f5), R2 =
↪  summary(f5)$r.squared)
```

```
#>             Estimate Std. Error t value  Pr(>|t|)
#> (Intercept)     0.05    0.19807  0.2524 8.170e-01
#> x5              1.99    0.05972 33.3213 5.942e-05
```

```
#>    RSS     R2
#> 0.1070 0.9973
```

```
summary(f5)$fstatistic
```

```
#> value numdf dendf
#>  1110     1     3
```

Exercise 5.7. Which divisor? Under A1–A4, $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p}$. For the estimators $\hat\sigma^2_k = \mathrm{RSS}/k$ compute the bias, variance and mean squared error as functions of k and find the k that minimises the MSE. Compare it with $k = n-p$ and $k = n$.

Solution. With $m = n-p$, $\mathrm{E}[\mathrm{RSS}] = m\sigma^2$ and $\mathrm{Var}(\mathrm{RSS}) = 2m\sigma^4$ (variance of a χ^2_m is $2m$). Then $\mathrm{E}[\hat\sigma^2_k] = m\sigma^2/k$, bias $= (m-k)\sigma^2/k$, $\mathrm{Var} = 2m\sigma^4/k^2$, so

$$\mathrm{MSE}(k) = \sigma^4\,\frac{(m-k)^2 + 2m}{k^2}.$$

Differentiating in k and setting to zero: $\frac{d}{dk}\left[(m-k)^2k^{-2} + 2mk^{-2}\right] = -2(m-k)k^{-2} - 2(m-k)^2k^{-3} - 4mk^{-3} = 0$, which simplifies to $-2(m-k)k - 2(m-k)^2 - 4m = 0$, i.e. $-2(m-k)(k+m-k) - 4m = -2m(m-k) - 4m = 0$, so $m-k = -2$ and $k = m+2 = n-p+2$. At $k = m$ (unbiased) MSE $= 2\sigma^4/m$; at $k = m+2$, MSE $= 2\sigma^4/(m+2)$, smaller. For $n = 50$, $p = 5$: unbiased $0.0444\sigma^4$, minimum $0.0425\sigma^4$, and $k = n$: $\sigma^4(25+90)/2500 = 0.046\sigma^4$. Unbiasedness is not the same as small error, but the differences are minor here.

Exercise 5.8. LR versus F. (a) Show that the LR statistic for nested linear models equals $n\log\{1 + qF/(n-p_1)\}$. (b) For `savings`, test $H_0 : \beta_{\texttt{pop75}} = \beta_{\texttt{dpi}} = 0$ with the F test, the LR test (chi-square reference) and compare the p-values. (c) Explain which one you would report with $n = 50$ and why they agree as $n \to \infty$.

Solution.

(a) Shown in Theorem 5.9. (b) The output below gives $F = 1.72$ ($p = 0.19$) and LR $= 3.69$ ($p = 0.16$). (c) Report the F test: it is exact under A1–A4. As $n \to \infty$ with q fixed, $qF \to \chi^2_q$ in distribution (the denominator converges to σ^2) and $n\log(1+qF/(n-p_1)) \approx qF$, so the two statistics have the same limit.

```
m0 <- lm(sr ~ pop15 + ddpi, data = savings); m1 <- lm(sr ~ pop15 + pop75 + dpi
↪  + ddpi, data = savings)
a <- anova(m0, m1); LR <- 2 * (as.numeric(logLik(m1)) -
↪  as.numeric(logLik(m0)))
c(F = a$F[2], p_F = a$`Pr(>F)`[2], LR = LR, p_LR = pchisq(LR, 2, lower.tail =
↪  FALSE))
```

```
#>     F   p_F    LR  p_LR
#> 1.723 0.190 3.690 0.158
```

Exercise 5.9. $F = t^2$. Let M_1 have design $X_1 = [X_0,\, x]$ with one extra column x and let $z = (I - H_0)x \neq 0$. (a) Show $H_1 - H_0 = zz^\top/\|z\|^2$. (b) Show that the coefficient of x in M_1 is $\hat\beta_x = z^\top y/\|z\|^2$, with $\mathrm{Var}(\hat\beta_x) = \sigma^2/\|z\|^2$. (c) Conclude that the F statistic for $H_0 : \beta_x = 0$ equals the square of the t statistic.

Solution.

(a) $z \perp \mathcal{C}(X_0)$ and $\mathcal{C}(X_1) = \mathcal{C}(X_0) \oplus \mathrm{span}(z)$ (because $x = H_0x + z$ with $H_0x \in \mathcal{C}(X_0)$). Projection onto an orthogonal direct sum is the sum of projections, so $H_1 = H_0 + zz^\top/\|z\|^2$.

(b) $\hat y = H_1y = H_0y + \frac{z^\top y}{\|z\|^2}z = H_0y - \frac{z^\top y}{\|z\|^2}H_0x + \frac{z^\top y}{\|z\|^2}x$, using $z = x - H_0x$. The first two terms are in $\mathcal{C}(X_0)$ and the last is a multiple of x; by uniqueness of coordinates (A1) the coefficient of x is $z^\top y/\|z\|^2$. Since z is fixed, $\mathrm{Var}(\hat\beta_x) = \sigma^2 z^\top z/\|z\|^4 = \sigma^2/\|z\|^2$, hence $[(X_1^\top X_1)^{-1}]_{xx} = 1/\|z\|^2$.

(c) $\mathrm{RSS}_0 - \mathrm{RSS}_1 = \|(H_1 - H_0)y\|^2 = (z^\top y)^2/\|z\|^2 = \hat\beta_x^2\|z\|^2$, so $F = \hat\beta_x^2\|z\|^2/\hat\sigma^2 = \hat\beta_x^2/(\hat\sigma^2[(X_1^\top X_1)^{-1}]_{xx}) = t^2$.

Exercise 5.10. R analysis: tree volume. The data `trees` (package `datasets`) contain the girth (diameter, inches), height (feet) and volume (cubic feet) of 31 black cherry trees. (a) Fit `Volume ~ Girth + Height`, compute $\hat\beta$ from the normal equations and check against `lm()`. (b) Test $H_0 : \beta_{\texttt{Height}} = 0$ and give a 95% CI for the Girth coefficient; interpret both. (c) Compare with the model `log(Volume) ~ log(Girth) + log(Height)`. Which is more plausible from geometry (volume of a cone or cylinder), and what does the estimated exponent of Girth suggest?

Solution.

```
data(trees)
Xt <- cbind(1, trees$Girth, trees$Height); yt <- trees$Volume
b_hand <- solve(crossprod(Xt), crossprod(Xt, yt))
ft <- lm(Volume ~ Girth + Height, data = trees)
cbind(hand = drop(b_hand), lm = coef(ft))
```

```
#>                 hand       lm
#> (Intercept) -57.9877 -57.9877
#> Girth         4.7082   4.7082
#> Height        0.3393   0.3393
```

```
summary(ft)$coefficients; confint(ft)["Girth", ]
```

```
#>             Estimate Std. Error t value  Pr(>|t|)
#> (Intercept) -57.9877     8.6382  -6.713 2.750e-07
#> Girth         4.7082     0.2643  17.816 8.223e-17
#> Height        0.3393     0.1302   2.607 1.449e-02
```

```
#>  2.5 % 97.5 %
#>  4.167  5.249
```

```
flog <- lm(log(Volume) ~ log(Girth) + log(Height), data = trees)
summary(flog)$coefficients; confint(flog)
```

```
#>             Estimate Std. Error t value  Pr(>|t|)
#> (Intercept)   -6.632    0.79979  -8.292 5.057e-09
#> log(Girth)     1.983    0.07501  26.432 2.423e-21
#> log(Height)    1.117    0.20444   5.464 7.805e-06
```

```
#>               2.5 % 97.5 %
#> (Intercept) -8.2699 -4.993
#> log(Girth)   1.8290  2.136
#> log(Height)  0.6984  1.536
```

```
c(R2_linear = summary(ft)$r.squared, R2_log = summary(flog)$r.squared)
```

```
#> R2_linear    R2_log
#>    0.9480    0.9777
```

(a) The normal-equation solution and `lm()` agree (output above). (b) t for `Height` is 2.61 on 28 df, $p = 0.014$: given girth, taller trees have more volume. The Girth coefficient is 4.71

with 95% CI $(4.17, 5.25)$ cubic feet per extra inch of girth, trees of the same height. (c) A cone or cylinder has volume proportional to girth$^2\times$height; so on the log scale we expect slopes near 2 for log(Girth) and 1 for log(Height). The fit gives 1.98 (95% CI 1.83–2.14) and 1.12 (CI 0.70–1.54): both intervals contain the geometric values. The log model also removes the curvature that the linear model leaves, and has the higher R^2 (0.978 vs 0.948). Here the transformation is suggested by the substantive model, not by data-fishing.

Exercise 5.11. Simulation. Generate $x_i \sim U(0, 10)$, $n = 20$, and $y_i = 1+0.5x_i+\varepsilon_i$. Over 5000 replications estimate (a) $\mathrm{E}[\hat{\sigma}^2]$ and $\mathrm{E}[\mathrm{RSS}/n]$ and (b) the coverage of the nominal 95% t-interval for β_1, in two settings: normal errors $\varepsilon_i \sim N(0, 2^2)$, and skewed errors $\varepsilon_i = \sqrt{2}\,(Z_i^2 - 1)$ with $Z_i \sim N(0, 1)$ (mean 0, variance 4). Interpret.

Solution.

```
set.seed(11)
one <- function(skew) {
  x <- runif(20, 0, 10)
  eps <- if (skew) sqrt(2) * (rnorm(20)^2 - 1) else rnorm(20, sd = 2)
  f <- lm(I(1 + 0.5 * x + eps) ~ x)
  ci <- confint(f)["x", ]
  c(s2 = sigma(f)^2, s2ml = deviance(f) / 20, cover = ci[1] < 0.5 & 0.5 <
  ↪  ci[2])
}
rbind(normal = rowMeans(replicate(5000, one(FALSE))), skewed =
↪  rowMeans(replicate(5000, one(TRUE))))
```

```
#>           s2  s2ml cover.2.5 %
#> normal 3.995 3.596      0.9502
#> skewed 3.935 3.541      0.9552
```

(a) $\mathrm{E}[\hat{\sigma}^2] \approx 4$ and $\mathrm{E}[\mathrm{RSS}/n] \approx 4 \times 18/20 = 3.6$ in both settings: unbiasedness of $\hat{\sigma}^2$ needs only A2–A3. (b) Coverage with normal errors is 0.95. With skewed errors it is still close to 0.95 (about 0.955 here): the slope estimator is a linear combination of the errors and the CLT helps even at $n = 20$, so the t-interval is robust to non-normality of the errors when the design is not extreme. It would not be robust to heavy tails with a few high-leverage points, or to non-constant variance.

Exercise 5.12. Rank deficiency. Let x be a factor with three levels A, B, C and four observations per level. Take $X = [\mathbf{1}, d_A, d_B, d_C]$ with the three indicator columns. (a) Show that $X^\top X$ is singular. (b) Show that $\hat{y}$ is nevertheless unique and equal to the group means. (c) Give two different solutions of the normal equations. (d) Show what `lm()` returns.

Solution.

```
d <- data.frame(g = factor(rep(c("A", "B", "C"), each = 4)), y = c(4, 5, 6, 5,
↪  7, 8, 9, 8, 3, 2, 4, 3))
Xd <- cbind(1, model.matrix(~ g - 1, d)); qr(Xd)$rank
```

```
#> [1] 3
```

```
coef(lm(y ~ g, d)); coef(lm(y ~ 1 + g - 1 + I(1 * (g == "A")) + I(1 * (g ==
↪  "B")) + I(1 * (g == "C")), d))
```

```
#> (Intercept)          gB          gC
#>           5           3          -2
```

```
#>                  gA                  gB                  gC I(1 * (g == "A"))
#>                   5                   8                   3                  NA
#> I(1 * (g == "B")) I(1 * (g == "C"))
#>                  NA                  NA
```

```
tapply(d$y, d$g, mean)
```

```
#> A B C
#> 5 8 3
```

(a) $d_A + d_B + d_C = \mathbf{1}$, so the columns are linearly dependent, $\operatorname{rank} X = 3 < 4$, and $X^\top X$ is singular (it has a zero eigenvalue with eigenvector $(1, -1, -1, -1)$). (b) $\mathcal{C}(X)$ is the space of vectors constant within groups, whatever the parametrisation; the projection of y onto it replaces each observation by its group mean. (c) With the group means $(5, 8, 3)$: $(\beta_0, \beta_A, \beta_B, \beta_C) = (0, 5, 8, 3)$ and also $(1, 4, 7, 2)$; adding any constant c to β_0 and subtracting it from the three group coefficients gives another solution. (d) `lm(y ~ g)` uses *treatment coding*, dropping the first dummy: intercept 5 (the mean of A), $gB = 3$, $gC = -2$. With all four columns given, `lm()` sets the last aliased coefficient to `NA`, as in the output above. Estimable functions such as the group means, or differences of group coefficients, are the same for every solution.

5.14 Further reading

The standard reference at the level of this course is Faraway (2014) (Chapters 2–4), which pairs the theory with R; Kutner et al. (2005) (Chapters 5–6) gives the matrix treatment with many worked numerical examples. The geometric view of least squares as projection is developed at greater length in Faraway (2014) (Chapter 2) and, for the Gauss–Markov theorem under non-spherical errors, in Kutner et al. (2005). For the likelihood link see Pawitan (2001). The ASA statement on p-values (Wasserstein and Lazar, 2016) is worth reading before using any t test as a decision rule.

Chapter 6

Diagnostics and prediction

The theory of Chapter 5 holds *if* the model is right: the mean is linear in the columns of X (A2), the errors have constant variance and are uncorrelated (A3), and, for exact small-sample inference, they are normal (A4). Real data never satisfy these exactly, and the question is whether they are close enough for the conclusions to be trusted. **Diagnostics** answer that question. They examine the residuals, which are our only window onto the unobservable errors; they ask which observations have unusual predictor values (leverage), which are poorly fitted (outliers), and which change the answer when removed (influence); and they check whether the predictors are so entangled that individual coefficients cannot be separated (collinearity). The chapter closes with the final use of a fitted model, **prediction**, and shows why a confidence interval for a mean response and a prediction interval for a new observation are different objects.

Throughout we keep the notation $H = X(X^\top X)^{-1}X^\top$, h_{ii} for its diagonal, $e = (I - H)y$ for the residuals, p for the number of columns of X and $\hat\sigma^2 = \mathrm{RSS}/(n-p)$. We reuse the `savings` data of Chapter 5 and add `gala` (species on the Galápagos islands), `LakeHuron` and `teengamb` where they teach something `savings` cannot.

```
fit <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
n <- nrow(savings); p <- length(coef(fit))
c(n = n, p = p, sigma_hat = sigma(fit))
```

```
#>         n         p sigma_hat
#>    50.000     5.000     3.803
```

6.1 Residuals

Recall from Chapter 5 that under A1–A3 the residual vector is $e = (I - H)\varepsilon$ with $\mathrm{Var}(e) = \sigma^2(I - H)$. Two facts follow. The residuals do *not* have constant variance even when the errors do, $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$; and they are *correlated*, $\mathrm{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$. For large n and moderate leverages the h_{ij} are small and the residuals behave almost like the errors, which is why the plots below work. But in small samples, or when some h_{ii} are close to 1, the raw residuals can mislead, so we rescale them.

Definition 6.1 (Standardized and studentized residuals)**.** The **(internally) standardized residual** is

$$r_i = \frac{e_i}{\hat\sigma\sqrt{1 - h_{ii}}}.$$

The **(externally) studentized residual** replaces $\hat\sigma$ by $\hat\sigma_{(i)}$, the estimate of σ from the

model fitted *without* observation i:

$$t_i = \frac{e_i}{\hat{\sigma}_{(i)}\sqrt{1 - h_{ii}}}.$$

In R, `rstandard()` and `rstudent()`.

Both have variance close to 1 when the model holds. The reason for $\hat{\sigma}_{(i)}$ is that a gross outlier inflates $\hat{\sigma}$ and hides itself; t_i is not contaminated by case i and, as we show in Section 6.3, has an exact t_{n-p-1} distribution.

The four residual plots

Each plot targets one assumption.

1. **Residuals against fitted values** $\hat{y}_i$ (or r_i against $\hat{y}_i$). Under A2–A3 there should be no pattern: a curve signals a wrong mean (A2), a funnel signals non-constant variance (A3). We plot against $\hat{y}$ and not against y because $\mathrm{Cov}(\hat{y}, e) = 0$ whereas e and y are necessarily correlated (the residual is the part of y not in $\hat{y}$), which would create a spurious slope.
2. **Normal QQ plot** of the r_i: the points should follow the line (A4). S-shapes mean skewness, a curve at both ends heavy or light tails.
3. **Scale–location**: $\sqrt{|r_i|}$ against $\hat{y}_i$; a trend indicates changing variance, and the absolute value makes the trend easier to see than a funnel.
4. **Residuals against each predictor and against time or order**: curvature points to a missing nonlinear term, runs of the same sign point to correlated errors (A3).

```
d <- data.frame(fitted = fitted(fit), res = resid(fit), r = rstandard(fit),
↪  country = rownames(savings))
p1 <- ggplot(d, aes(fitted, res)) + geom_hline(yintercept = 0, linetype = 2,
↪  colour = "grey50") +
  geom_point(colour = ie_navy) + geom_smooth(method = "loess", formula = y ~
  ↪  x, se = FALSE, colour = ie_coral, linewidth = 0.7) +
  labs(x = "Fitted values", y = "Residuals", title = "Residuals vs fitted")
p2 <- ggplot(d, aes(sample = r)) + stat_qq_line(colour = ie_coral) +
↪  stat_qq(colour = ie_navy) +
  labs(x = "N(0,1) quantiles", y = "Standardized residuals", title = "Normal
  ↪  QQ plot")
p3 <- ggplot(d, aes(fitted, sqrt(abs(r)))) + geom_point(colour = ie_navy) +
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, colour =
  ↪  ie_coral, linewidth = 0.7) +
  labs(x = "Fitted values", y = expression(sqrt(abs(r[i]))), title =
  ↪  "Scale-location")
gridExtra::grid.arrange(p1, p2, p3, nrow = 1)
```

For `savings` (Figure 6.1) there is no clear curvature or funnel, and the QQ plot is close to the line: no strong evidence against A2–A4. With $n = 50$, smoothers wiggle even for perfect data; a diagnostic plot *suggests* but never proves. We now turn to what the plots cannot easily show: unusual X values and influential observations.

6.2 Leverage

The fitted value is a weighted sum of all responses, $\hat{y}_i = \sum_j h_{ij} y_j$. The weight h_{ii} that observation i gives to *itself* measures how much the fit is pulled towards y_i.

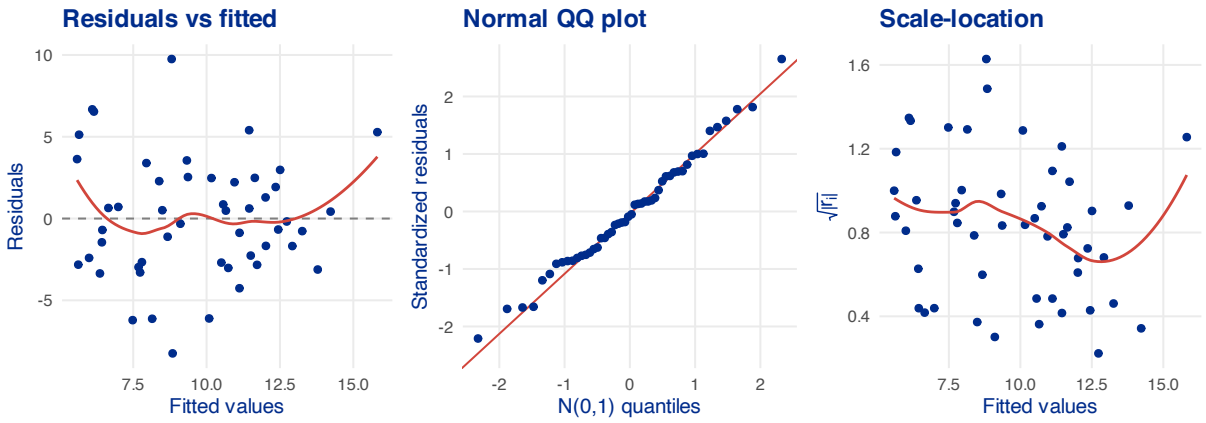


Figure 6.1: Diagnostic plots for the savings model: residuals against fitted values, normal QQ plot of the standardized residuals, and scale–location plot.

Proposition 6.1 (Properties of leverage)**.** *Let* $\operatorname{rank}(X) = p$*. Then*

1. $\partial \hat{y}_i / \partial y_i = h_{ii}$ *and* $0 \le h_{ii} \le 1$*;*
2. $\sum_{i=1}^{n} h_{ii} = p$*, so the average leverage is* p/n*;*
3. *if the model has an intercept,* $h_{ii} \ge 1/n$*; more precisely, writing* $X = [\mathbf{1}, X_1]$ *and* X_c *for* X_1 *with centred columns,* $h_{ii} = \dfrac{1}{n} + (x_i - \bar{x})^\top (X_c^\top X_c)^{-1} (x_i - \bar{x})$*, where* x_i *is here the* i*-th row of the predictors without the intercept;*
4. $\operatorname{Var}(e_i) = \sigma^2 (1 - h_{ii})$ *and* $h_{ii} = 1$ *implies* $e_i = 0$*.*

Proof.

(1) $\hat{y} = Hy$ is linear in y, so $\partial \hat{y}_i / \partial y_i = h_{ii}$. By symmetry and idempotence, $h_{ii} = (H^2)_{ii} = \sum_j h_{ij} h_{ji} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \ne i} h_{ij}^2 \ge h_{ii}^2$, so $h_{ii}(1 - h_{ii}) \ge 0$ and $0 \le h_{ii} \le 1$.
(2) $\sum_i h_{ii} = \operatorname{tr} H = p$ (Chapter 5).
(3) The vector $\mathbf{1}$ is orthogonal to the columns of X_c, and $\mathcal{C}(X) = \operatorname{span}(\mathbf{1}) \oplus \mathcal{C}(X_c)$ because $X_c = X_1 - \mathbf{1}\bar{x}^\top$. The projection on an orthogonal direct sum is the sum of the projections, so $H = \frac{1}{n}\mathbf{1}\mathbf{1}^\top + X_c (X_c^\top X_c)^{-1} X_c^\top$, whose (i,i) entry is the formula. The second term is a non-negative quadratic form, so $h_{ii} \ge 1/n$.
(4) $\operatorname{Var}(e_i)$ was derived in Chapter 5. If $h_{ii} = 1$ then $\operatorname{Var}(e_i) = 0$ and $e_i = 0$: the fit passes through y_i whatever its value.

□

For simple regression (3) gives the familiar $h_{ii} = 1/n + (x_i - \bar{x})^2 / S_{xx}$. Leverage is a *Mahalanobis-type distance of the row* x_i *from the centre of the predictors*; it depends only on X, not on y. The average is p/n, and a common rule flags $h_{ii} > 2p/n$ (some authors use $3p/n$). A high-leverage point is not necessarily bad, but it has the power to determine the fit, and since $\operatorname{Var}(e_i) = \sigma^2(1 - h_{ii})$ is small, its raw residual looks small even if the point is unusual: the fit is pulled through it.

```
h <- hatvalues(fit)
c(sum_h = sum(h), average = mean(h), cutoff_2p_over_n = 2 * p / n)
```

```
#>            sum_h          average cutoff_2p_over_n
#>              5.0              0.1              0.2
```

```
round(sort(h, decreasing = TRUE)[1:5], 3)
```

```
#>          Libya  United States          Japan        Ireland
#>          0.531          0.334          0.223          0.212
#> South Rhodesia
#>          0.161
```

```
savings["Libya", ]
```

```
#>         sr pop15 pop75   dpi  ddpi
#> Libya 8.89 43.69  2.07 123.6 16.71
```

Libya has $h = 0.531$, more than five times the average: its income growth, `ddpi`$= 16.7\%$, is far above all other countries (the median is 3%).

6.3 Outliers and the studentized residual

An **outlier** is an observation whose response is unusual *given its predictors*: it has a large residual. To judge "large" we need the distribution of a standardized residual, and to get it exactly we use a device that will also give us the influence measures: a model with one extra parameter for the suspect case.

Let $d_i \in \mathbb{R}^n$ be the i-th unit vector and consider the **mean-shift model**

$$y = X\beta + d_i\gamma + \varepsilon, \tag{6.1}$$

in which observation i has its own mean: γ is the shift of case i relative to the model. If $\gamma = 0$, case i is not an outlier.

Lemma 6.1 (Leave-one-out identities)**.** *Assume A1 and $h_{ii} < 1$. Fit* (M) *the model $y = X\beta + \varepsilon$ and* (M_i) *the mean-shift model* (6.1)*. Then, in* (M_i)*,*

1. $\hat\gamma = e_i/(1-h_{ii}) = y_i - x_i^\top \hat\beta_{(i)}$ *and* $\mathrm{Var}(\hat\gamma) = \sigma^2/(1-h_{ii})$*;*
2. $\hat\beta_{(i)} = \hat\beta - (X^\top X)^{-1}x_i\, e_i/(1-h_{ii})$*, where $\hat\beta_{(i)}$ is the least-squares estimate computed* without *observation i;*
3. $\mathrm{RSS}_{(i)} = \mathrm{RSS} - e_i^2/(1-h_{ii})$ *is the residual sum of squares of the fit without observation i, and $\hat\sigma^2_{(i)} = \mathrm{RSS}_{(i)}/(n-p-1)$.*

Proof. Let $z = (I-H)d_i$, the part of d_i orthogonal to $\mathcal{C}(X)$. Then $\|z\|^2 = d_i^\top(I-H)d_i = 1 - h_{ii} > 0$ and $z^\top y = d_i^\top(I-H)y = e_i$. The column space of the extended design is $\mathcal{C}(X) \oplus \mathrm{span}(z)$ (since $d_i = Hd_i + z$), so the extended hat matrix is $H + zz^\top/\|z\|^2$ and

$$\hat y = Hy + \frac{z^\top y}{\|z\|^2} z = Hy + \frac{e_i}{1-h_{ii}}(d_i - Hd_i).$$

Reading off the coefficient of d_i gives $\hat\gamma = e_i/(1-h_{ii})$, with variance $\sigma^2\|z\|^{-2} = \sigma^2/(1-h_{ii})$ because $\hat\gamma = z^\top y/\|z\|^2$ and z is fixed. The part in $\mathcal{C}(X)$ is $X\hat\beta_{\mathrm{aug}} = Hy - \hat\gamma Hd_i$; since $Hd_i = X(X^\top X)^{-1}x_i$ we get $\hat\beta_{\mathrm{aug}} = \hat\beta - \hat\gamma(X^\top X)^{-1}x_i$. The residual sum of squares drops by $(z^\top y)^2/\|z\|^2 = e_i^2/(1-h_{ii})$ (Pythagoras).

It remains to see that the extended fit is the leave-one-out fit. In the extended model the i-th observation is fitted exactly whatever $\hat\beta$ is (the free γ absorbs its residual), so it contributes zero to the residual sum of squares, and $\hat\beta_{\mathrm{aug}}$ minimises the sum of squares over the other $n-1$

observations: it is $\hat\beta_{(i)}$, and $\hat\gamma = y_i - x_i^\top \hat\beta_{(i)}$. Its residual sum of squares is $\mathrm{RSS}_{(i)}$ and has $n-1-p$ degrees of freedom. □

Part 1 is the **leave-one-out prediction error**: $y_i - x_i^\top \hat\beta_{(i)} = e_i/(1-h_{ii})$. The prediction error for case i when the model is fitted *without* it is obtained from the ordinary residual without refitting n times. We use it for cross-validation in Chapter 7.

Theorem 6.1 (Distribution of the studentized residual)**.** *Under A1–A4, if observation i follows the model (no outlier), $t_i \sim t_{n-p-1}$. Also*

$$t_i = r_i \sqrt{\frac{n-p-1}{n-p-r_i^2}}. \tag{6.2}$$

Proof. t_i is the t statistic for $H_0 : \gamma = 0$ in the mean-shift model, which has $p+1$ columns and $n-p-1$ residual degrees of freedom: by the t pivot of Chapter 5, $\hat\gamma / \{\hat\sigma_{(i)}/\sqrt{1-h_{ii}}\} \sim t_{n-p-1}$ under $\gamma = 0$, and by Lemma 6.1 this equals $\dfrac{e_i/(1-h_{ii})}{\hat\sigma_{(i)}/\sqrt{1-h_{ii}}} = \dfrac{e_i}{\hat\sigma_{(i)}\sqrt{1-h_{ii}}} = t_i$. For (6.2), $\mathrm{RSS}_{(i)} = \mathrm{RSS} - e_i^2/(1-h_{ii}) = \hat\sigma^2\{(n-p) - r_i^2\}$, so $\hat\sigma^2_{(i)} = \hat\sigma^2(n-p-r_i^2)/(n-p-1)$; substitute in t_i. □

Equation (6.2) shows that t_i is an increasing function of $|r_i|$, so the two flag the same cases but t_i has a known exact distribution.

Testing for an outlier. If we look at the *largest* $|t_i|$ among n cases, comparing it with the t_{n-p-1} quantile of an individual test is wrong: in 50 cases, a $|t_i|$ above 2 happens by chance in about 2 or 3 of them. The Bonferroni correction compares $\max_i |t_i|$ with $t_{n-p-1,\,1-\alpha/(2n)}$, which controls the probability of *any* false outlier flag at level α. `car::outlierTest()` reports the corrected p-value.

```
ti <- rstudent(fit)
cutoff <- qt(1 - 0.05 / (2 * n), df = n - p - 1)
c(max_abs_t = max(abs(ti)), bonferroni_cutoff = cutoff)
```

```
#>         max_abs_t bonferroni_cutoff
#>             2.854             3.526
```

```
round(sort(abs(ti), decreasing = TRUE)[1:3], 3)
```

```
#>      Zambia       Chile Philippines
#>       2.854       2.313       1.864
```

```
outlierTest(fit)
```

```
#> No Studentized residuals with Bonferroni p < 0.05
#> Largest |rstudent|:
#>         rstudent unadjusted p-value Bonferroni p
#> Zambia    2.854           0.006567       0.3283
```

```
# the identities of Lemma and Theorem, checked directly
i <- which(rownames(savings) == "Zambia")
ri <- rstandard(fit)[i]; hi <- h[i]; ei <- resid(fit)[i]
loo <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings[-i, ])
c(t_i = ti[i], from_r = ri * sqrt((n - p - 1) / (n - p - ri^2)),
```

```
  loo_prediction_error = savings$sr[i] - predict(loo, savings[i, ]),
  ↪  e_over_1mh = ei / (1 - hi))
```

```
#>                   t_i.Zambia                 from_r.Zambia
#>                        2.854                         2.854
#> loo_prediction_error.Zambia           e_over_1mh.Zambia
#>                       10.421                        10.421
```

The largest $|t_i|$ is Zambia's 2.85, below the Bonferroni cut-off 3.53: with 50 countries a value of 2.85 is not unusual, and there is no outlier at the 5% level.

6.4 Influence

An observation is **influential** if removing it changes the fit appreciably. A point can be an outlier but not influential (a large residual at the centre of the data), high-leverage but not influential (it lies on the line the others define), or both. Lemma 6.1 gives the exact effect of deleting case i without refitting:

$$\hat\beta - \hat\beta_{(i)} = \frac{(X^\top X)^{-1} x_i\, e_i}{1 - h_{ii}}, \qquad \hat y_i - x_i^\top \hat\beta_{(i)} = \frac{h_{ii}\, e_i}{1 - h_{ii}}. \tag{6.3}$$

Both are products of a residual term and a leverage term. The first is the `dfbeta` vector (divided by the standard error: `dfbetas`) and the second is the *change in the fitted value*, `dffits` when scaled. For a summary of the effect on all coefficients at once we use Cook's distance.

Definition 6.2 (Cook's distance)**.** Cook (1977) measured the influence of case i by how far the estimate moves, in the metric of the confidence ellipsoid of $\hat\beta$:

$$D_i = \frac{(\hat\beta_{(i)} - \hat\beta)^\top X^\top X (\hat\beta_{(i)} - \hat\beta)}{p\,\hat\sigma^2} = \frac{\|\hat y_{(i)} - \hat y\|^2}{p\,\hat\sigma^2},$$

where $\hat y_{(i)} = X\hat\beta_{(i)}$ are the fitted values for *all* cases when case i is left out of the fit. The two forms agree because $\|Xv\|^2 = v^\top X^\top X v$.

Theorem 6.2 (Cook's distance from the leverage and the residual)**.**

$$D_i = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}.$$

Proof. By (6.3), $(\hat\beta_{(i)} - \hat\beta)^\top X^\top X (\hat\beta_{(i)} - \hat\beta) = \dfrac{e_i^2}{(1-h_{ii})^2} x_i^\top (X^\top X)^{-1} X^\top X (X^\top X)^{-1} x_i = \dfrac{e_i^2\, h_{ii}}{(1-h_{ii})^2}$, because $x_i^\top (X^\top X)^{-1} x_i = h_{ii}$. Divide by $p\hat\sigma^2$ and use $r_i^2 = e_i^2/\{\hat\sigma^2(1-h_{ii})\}$: $D_i = \dfrac{e_i^2}{\hat\sigma^2(1-h_{ii})} \cdot \dfrac{h_{ii}}{p(1-h_{ii})} = \dfrac{r_i^2}{p}\dfrac{h_{ii}}{1-h_{ii}}$. □

So influence = outlyingness × leverage. Because D_i measures the distance in the metric of the $(1-\alpha)$ confidence ellipsoid $\{\beta : (\beta-\hat\beta)^\top X^\top X(\beta-\hat\beta) \le p\hat\sigma^2 F_{p,n-p,1-\alpha}\}$, a natural scale is the quantiles of $F_{p,n-p}$: D_i at the median of $F_{p,n-p}$ (about 0.9 for our data) means that deleting case i moves $\hat\beta$ to the edge of a 50% confidence region. In practice, $D_i > 4/n$ is used as a

screening level and D_i near or above 0.5–1 as a serious warning. **None of these thresholds is a test**; look at what actually changes.

```
dd <- data.frame(h = hatvalues(fit), t = rstudent(fit), D =
↪  cooks.distance(fit), country = rownames(savings))
ggplot(dd, aes(h, t)) +
  geom_vline(xintercept = 2 * p / n, linetype = 2, colour = "grey50") +
  geom_hline(yintercept = c(-1, 1) * cutoff, linetype = 3, colour = ie_coral)
  ↪  +
  geom_point(aes(size = D), colour = ie_navy, alpha = 0.7) +
  geom_text_repel(data = subset(dd, h > 2 * p / n | D > 4 / n | abs(t) > 2.5),
  ↪  aes(label = country),
                  colour = ie_coral, size = 3.2, seed = 1) +
  scale_size_area(max_size = 8, name = "Cook's D") +
  labs(x = "Leverage", y = "Studentized residual")
```

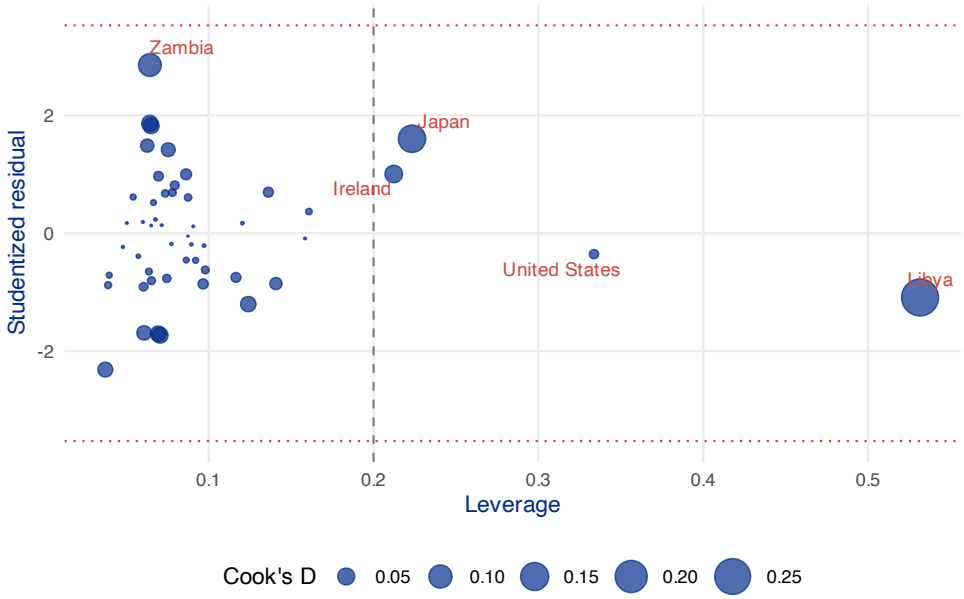


Figure 6.2: Leverage against studentized residual for the savings model; point area is proportional to Cook's distance. Vertical line: $2p/n$; horizontal lines: Bonferroni cut-off.

```
round(sort(cooks.distance(fit), decreasing = TRUE)[1:4], 3)
```

```
#>   Libya   Japan  Zambia Ireland
#>   0.268   0.143   0.097   0.054
```

```
qf(0.5, p, n - p)
```

```
#> [1] 0.8835
```

Libya has the largest Cook's distance (0.268) because of its leverage, although its studentized residual is unremarkable (-1.09). To see what it changes, we refit without it.

```
fit_noL <- update(fit, subset = rownames(savings) != "Libya")
tab <- cbind(all = coef(fit), without_Libya = coef(fit_noL),
             diff_in_se = (coef(fit_noL) - coef(fit)) /
              ↪  summary(fit)$coefficients[, "Std. Error"])
```

```
round(tab, 4)
```

```
#>                 all without_Libya diff_in_se
#> (Intercept) 28.5661       24.5240    -0.5496
#> pop15       -0.4612       -0.3914     0.4822
#> pop75       -1.6915       -1.2809     0.3790
#> dpi         -0.0003       -0.0003     0.0193
#> ddpi         0.4097        0.6103     1.0224
```

The `ddpi` slope moves from 0.41 to 0.61, about one standard error; the signs and the story are unchanged. The right response is to *report both fits* and understand why the case is different (here: income growth driven by oil), not to delete the case silently, and not to delete it merely because a threshold was crossed. When a point is both influential and doubtful, robust regression (`MASS::rlm`) is an alternative that downweights it automatically.

A more extreme example is the `gala` data: the 30 Galápagos islands with the number of plant species as response and geographic predictors (Johnson and Raven, 1973).

```
data(gala, package = "faraway")
g <- lm(Species ~ Area + Elevation + Nearest + Scruz + Adjacent, data = gala)
round(sort(cooks.distance(g), decreasing = TRUE)[1:3], 2)
```

```
#>    Isabela  SantaCruz Fernandina
#>      68.08       0.39       0.27
```

```
round(sort(hatvalues(g), decreasing = TRUE)[1:3], 2)
```

```
#>    Isabela Fernandina     Darwin
#>       0.97       0.95       0.47
```

Isabela, by far the largest island, has $h = 0.97$ and Cook's distance 68: the whole fit is determined by it. In such a case the problem is usually not the observation but the scale: `Area` ranges over four orders of magnitude, and a transformation (Section 6.5.1) is the right remedy.

6.5 Checking the assumptions in more detail

Non-linearity

If the mean is not linear in x_j, the plot of e against $\hat{y}$ or against x_j shows curvature. The **partial residual plot** (component-plus-residual plot) sharpens the picture for one predictor: plot $e_i + \hat{\beta}_j x_{ij}$ against x_{ij}. If the relationship is linear the points scatter around a line of slope $\hat{\beta}_j$; curvature shows the shape of the missing nonlinearity. In R: `car::crPlots(fit)`. The **added-variable plot** plots the residuals of y on all predictors but x_j against the residuals of x_j on the same predictors; its slope is exactly $\hat{\beta}_j$ (Chapter 7). A simulated example shows the logic.

```
set.seed(606)
x <- runif(80, 0, 10); y <- 2 + 0.4 * x + 0.25 * (x - 5)^2 + rnorm(80, sd = 1)
m_lin <- lm(y ~ x); m_quad <- lm(y ~ x + I(x^2))
q1 <- ggplot(data.frame(x, y), aes(x, y)) + geom_point(colour = ie_navy) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, colour = ie_coral)
q2 <- ggplot(data.frame(f = fitted(m_lin), r = resid(m_lin)), aes(f, r)) +
↪  geom_point(colour = ie_navy) +
```

```
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, colour =
  ↪  ie_coral) + labs(x = "Fitted", y = "Residual")
gridExtra::grid.arrange(q1, q2, nrow = 1)
```

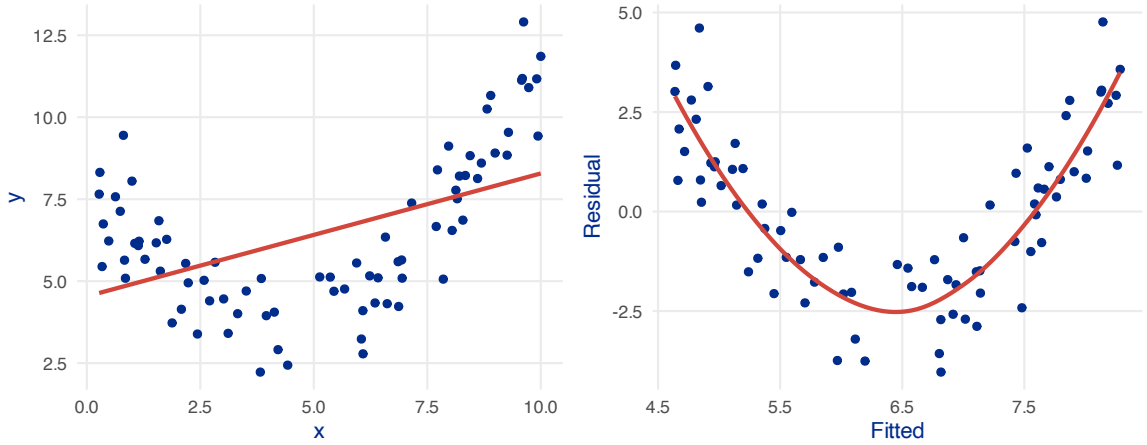


Figure 6.3: Fitting a straight line to a quadratic relationship. Left: data and fit. Right: residuals against fitted values show the curvature; the quadratic term removes it.

```
anova(m_lin, m_quad)
```

```
#> Analysis of Variance Table
#>
#> Model 1: y ~ x
#> Model 2: y ~ x + I(x^2)
#>   Res.Df RSS Df Sum of Sq   F Pr(>F)
#> 1     78 343
#> 2     77  83  1       260 241 <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The nested-model F test of Chapter 5 for the added quadratic term is decisive. The pattern of the residuals tells us *that* the mean is misspecified, and the shape tells us *how*; the test is only a confirmation. Remedies are polynomial or spline terms (Chapters 14–16) or a transformation.

Non-constant variance

A funnel in the residual plot or a trend in the scale–location plot means $\mathrm{Var}(\varepsilon_i)$ depends on the mean or on a predictor. Least squares is still unbiased, but the standard errors are wrong and it is no longer best (Gauss–Markov needs A3). A score test (`car::ncvTest()`, a version of the Breusch–Pagan test) regresses the squared standardized residuals on the fitted values and tests whether they are related. Typical remedies are a variance-stabilising transformation of y (Section 6.5.1) or **weighted least squares** (Exercise 6.7): if $\mathrm{Var}(\varepsilon_i) = \sigma^2 / w_i$ with known weights, minimise $\sum_i w_i (y_i - x_i^\top \beta)^2$, which gives $\hat{\beta}_W = (X^\top W X)^{-1} X^\top W y$ with $W = \mathrm{diag}(w_i)$.

Non-normality

The QQ plot of r_i examines A4. Normality matters little for the *coefficients* in moderate samples: $\hat{\beta}$ is a linear combination of the errors, so the central limit theorem makes the t and F procedures approximately valid when n is large and no observation dominates. It matters a

great deal for **prediction intervals**, which rest on the distribution of a single new error (Section 6.7), and heavy tails create outliers and influential points. Formal tests such as Shapiro–Wilk (`shapiro.test(rstandard(fit))`) have high power for large n against trivial departures and low power for small n; use the plot first.

Correlated errors

When observations are ordered in time or space, the errors are often positively correlated, which makes standard errors too small (the effective sample size is smaller than n). Plot residuals against the order of observation and examine their autocorrelation. The water level of Lake Huron (`LakeHuron`, 1875–1972, in feet above a reference level) regressed on year is a standard example.

```
lake <- data.frame(level = as.numeric(LakeHuron), year =
↪  as.numeric(time(LakeHuron)))
lh <- lm(level ~ year, data = lake)
op <- par(mfrow = c(1, 2), mar = c(4, 4, 2, 1)); plot(lake$year, resid(lh),
↪  type = "b", pch = 16, cex = 0.6,
  col = ie_navy, xlab = "Year", ylab = "Residual", main = "Residuals in time
  ↪  order"); abline(h = 0, lty = 2)
acf(resid(lh), main = "Autocorrelation of residuals"); par(op)
```

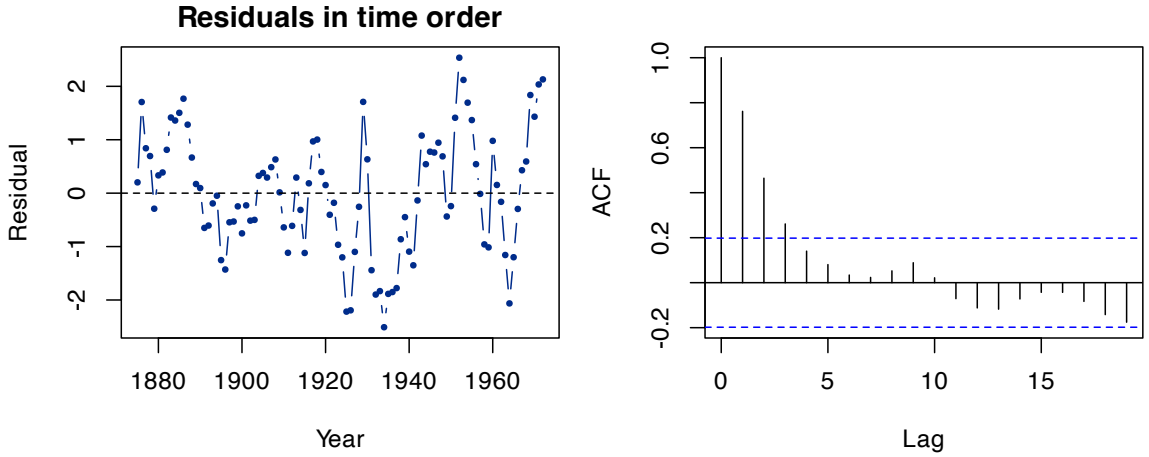


Figure 6.4: Residuals of the regression of Lake Huron level on year, and their autocorrelation function.

```
c(lag1_autocorrelation = acf(resid(lh), plot = FALSE)$acf[2])
```

```
#> lag1_autocorrelation
#>               0.7616
```

```
durbinWatsonTest(lh)
```

```
#>  lag Autocorrelation D-W Statistic p-value
#>    1          0.7616        0.4395       0
#>  Alternative hypothesis: rho != 0
```

The residuals come in long runs, the lag-1 autocorrelation is 0.76 and the Durbin–Watson statistic (0.44, where 2 means no autocorrelation) is far from 2. The model-based standard error of the slope is therefore too small. The remedy is a model for the correlation (generalised

least squares, `nlme::gls`, or the mixed models of Chapter 11).

6.5.1 Transformations

A transformation of the response can fix several problems at once: non-constant variance, skewness and some nonlinearity. For positive y, the **Box–Cox family** $y^{(\lambda)} = (y^\lambda - 1)/\lambda$ ($\log y$ for $\lambda = 0$) is indexed by λ, and λ can be chosen by maximising the profile log-likelihood of the linear model for $y^{(\lambda)}$ (`MASS::boxcox`). On the `gala` data:

```
bc <- MASS::boxcox(g, plotit = FALSE)
lambda_hat <- bc$x[which.max(bc$y)]
gs <- lm(sqrt(Species) ~ Area + Elevation + Nearest + Scruz + Adjacent, data =
↪  gala)
b1 <- ggplot(data.frame(f = fitted(g), r = resid(g)), aes(f, r)) +
↪  geom_point(colour = ie_navy) +
  geom_hline(yintercept = 0, linetype = 2) + labs(x = "Fitted", y =
  ↪  "Residual", title = "Species")
b2 <- ggplot(data.frame(f = fitted(gs), r = resid(gs)), aes(f, r)) +
↪  geom_point(colour = ie_navy) +
  geom_hline(yintercept = 0, linetype = 2) + labs(x = "Fitted", y =
  ↪  "Residual", title = "sqrt(Species)")
gridExtra::grid.arrange(b1, b2, nrow = 1)
```

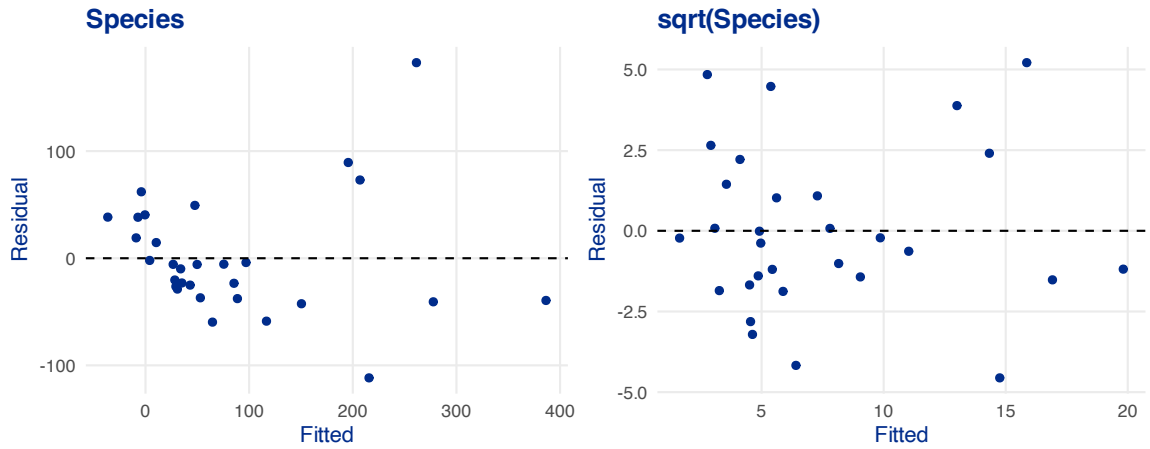


Figure 6.5: Gal<U+00E1>pagos species counts: residuals against fitted values before and after a square-root transformation of the response.

```
c(lambda_hat = lambda_hat)
```

```
#> lambda_hat
#>        0.3
```

```
rbind(original = unlist(ncvTest(g)[c("ChiSquare", "p")]), sqrt =
↪  unlist(ncvTest(gs)[c("ChiSquare", "p")]))
```

```
#>          ChiSquare         p
#> original   15.0718 0.0001035
#> sqrt        0.9122 0.3395390
```

The profile likelihood peaks at $\hat\lambda = 0.3$, close to $\frac{1}{2}$; the square root is a natural choice for counts (the variance of a Poisson count is proportional to its mean, and the delta method shows

that $\sqrt{y}$ has approximately constant variance). The funnel disappears and the score test goes from $p = 0.0001$ to $p = 0.34$. The price is interpretation: coefficients now describe $\mathrm{E}[\sqrt{y}]$, and back-transformed predictions are not means of y. For counts, the Poisson and negative-binomial GLMs of Chapters 8–10 model the mean and the variance directly and are preferable.

Log transformation and percentage changes. If $\log y = \beta_0 + \beta_1 x + \varepsilon$ then a unit increase in x multiplies the *median* response by e^{β_1}, a $100(e^{\beta_1} - 1)\%$ change, approximately $100\beta_1\%$ for small β_1. If both y and x are logged, β_1 is an elasticity: a 1% increase in x goes with a $\beta_1\%$ change in y.

6.6 Multicollinearity

When two predictors are strongly correlated, the data can tell that "they together matter" but not which one: the coefficients are poorly determined, the t tests are individually non-significant even though F is significant, and signs can be counter-intuitive. In the extreme case where a column of X is an exact linear combination of the others, A1 fails (Chapter 5). Here we quantify the milder, near-collinear case.

Theorem 6.3 (Variance inflation)**.** *Let x_j be a column of X (not the intercept),* $\operatorname{rank}(X) = p$*, $s_j^2 = \frac{1}{n-1}\sum_i (x_{ij} - \bar{x}_j)^2$ and R_j^2 the R^2 of the regression of x_j on all the other columns of X (including the intercept). Then*

$$\operatorname{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(n-1)s_j^2} \cdot \mathrm{VIF}_j, \qquad \mathrm{VIF}_j = \frac{1}{1 - R_j^2}. \tag{6.4}$$

Proof. Let H_{-j} be the hat matrix of the design without column j and $z_j = (I - H_{-j})x_j$, the residual of regressing x_j on the other columns, so $\|z_j\|^2 = (1 - R_j^2)\sum_i (x_{ij} - \bar{x}_j)^2$ (the residual sum of squares of that regression, because it contains an intercept). As in the proof of Lemma 6.1 (with x_j in place of d_i) the column space of X is $\mathcal{C}(X_{-j}) \oplus \operatorname{span}(z_j)$, and the coefficient of x_j in the full model is $\hat{\beta}_j = z_j^\top y / \|z_j\|^2$. Hence $\operatorname{Var}(\hat{\beta}_j) = \sigma^2/\|z_j\|^2 = \sigma^2/\{(1 - R_j^2)(n-1)s_j^2\}$. □

$\mathrm{VIF}_j = 1$ if x_j is orthogonal (after centring) to the other columns, and $\mathrm{VIF}_j = 10$ means a standard error $\sqrt{10} \approx 3.2$ times larger than with orthogonal columns. Common warnings: $\mathrm{VIF} > 5$ (moderate) or > 10 (severe), a rule of thumb and not a test. Four points deserve emphasis.

- Collinearity does **not bias** $\hat{\beta}$ and does not violate any assumption. It inflates variances.
- It does not harm *prediction inside the cloud of observed data*, since $\hat{y}$ is well determined even when individual coefficients are not. It does harm extrapolation along directions where predictors are no longer correlated in the same way.
- The remedy depends on the goal: for interpretation, report F tests for groups, combine or drop redundant variables, or collect data that break the correlation; for prediction, regularise (Chapter 7).
- Polynomial and interaction terms create collinearity by construction (x and x^2 are nearly collinear for positive x); centring x first removes most of it without changing the fit.

For a factor with K levels (several columns), the VIF of a single dummy is not meaningful; Fox and Monette (1992) define the **generalised VIF**, and `car::vif()` reports $\mathrm{GVIF}^{1/(2\,\mathrm{df})}$, comparable to a $\sqrt{\mathrm{VIF}}$. Eigenvalue-based diagnostics (condition indices of the column-scaled X; Belsley et al. (1980)) complement the VIFs by identifying *which* sets of columns are involved.

```
round(cor(savings[, c("pop15", "pop75", "dpi", "ddpi")]), 2)
```

```
#>       pop15 pop75   dpi  ddpi
#> pop15  1.00 -0.91 -0.76 -0.05
#> pop75 -0.91  1.00  0.79  0.03
#> dpi   -0.76  0.79  1.00 -0.13
#> ddpi  -0.05  0.03 -0.13  1.00
```

```
v <- vif(fit); round(v, 2)
```

```
#> pop15 pop75   dpi  ddpi
#>  5.94  6.63  2.88  1.07
```

```
# VIF by hand for pop75: regress it on the other predictors
R2_75 <- summary(lm(pop75 ~ pop15 + dpi + ddpi, data = savings))$r.squared
c(R2_75 = R2_75, VIF_75 = 1 / (1 - R2_75))
```

```
#>  R2_75 VIF_75
#> 0.8492 6.6291
```

The shares of the population under 15 and over 75 have correlation -0.91, and VIF ≈ 6 for both. Their standard errors are about $\sqrt{6} \approx 2.5$ times what they would be with orthogonal columns. This is why `pop75` is not significant given the others in Chapter 5, although it is marginally, and why the *joint* F test is more informative. The confidence ellipse for $(\beta_{\texttt{pop15}}, \beta_{\texttt{pop75}})$ shows the correlated uncertainty:

```
car::confidenceEllipse(fit, which.coef = c("pop15", "pop75"), levels = 0.95,
↪  col = ie_navy,
                       xlab = "coefficient of pop15", ylab = "coefficient of
                       ↪  pop75", fill = TRUE, fill.alpha = 0.1)
abline(v = confint(fit)["pop15", ], h = confint(fit)["pop75", ], lty = 2, col
↪  = "grey40")
points(0, 0, pch = 3, col = ie_coral, lwd = 2)
```

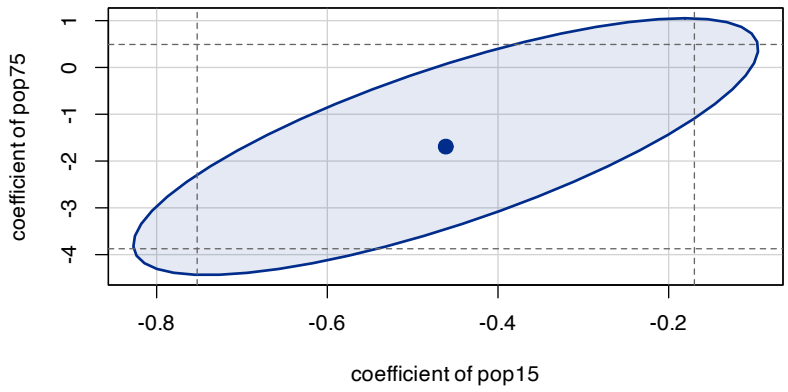


Figure 6.6: Joint 95% confidence ellipse for the coefficients of pop15 and pop75, with the marginal 95% intervals as dashed lines.

The ellipse is tilted: the data determine well a certain *combination* of the two coefficients, while each one separately is uncertain. The cross at the origin ($H_0 : \beta_{15} = \beta_{75} = 0$) lies well outside the ellipse, consistent with the joint significance of the age shares even though one of the marginal intervals contains 0.

A simulation confirms the variance formula: two standard-normal predictors with correlation ρ, $n = 50$, $y = 1 + x_1 + x_2 + \varepsilon$ with $\varepsilon \sim N(0, 1)$, and the standard deviation of $\hat\beta_1$ over 3000 data sets.

```
set.seed(1)
sd_beta1 <- sapply(c(0, 0.9, 0.99), function(rho) {
  sd(replicate(3000, {
    x1 <- rnorm(50); x2 <- rho * x1 + sqrt(1 - rho^2) * rnorm(50)
    coef(lm(1 + x1 + x2 + rnorm(50) ~ x1 + x2))["x1"]
  }))
})
rbind(rho = c(0, 0.9, 0.99), sd = sd_beta1, ratio_to_rho0 = sd_beta1 /
↪  sd_beta1[1],
      theory = 1 / sqrt(1 - c(0, 0.9, 0.99)^2))
```

```
#>                  [,1]   [,2]  [,3]
#> rho            0.0000 0.9000 0.990
#> sd             0.1479 0.3416 1.055
#> ratio_to_rho0 1.0000 2.3098 7.133
#> theory         1.0000 2.2942 7.089
```

The ratios agree with $1/\sqrt{1-\rho^2}$, the square root of the VIF.

6.7 Prediction

We have fitted a model; now we use it at a new covariate vector $x_0 \in \mathbb{R}^p$ (with a leading 1 for the intercept). Two different questions arise.

1. **Estimating the mean response** $\mu_0 = x_0^\top \beta$ at x_0: what is the average savings rate of countries with these characteristics?
2. **Predicting a new observation** $y_0 = x_0^\top \beta + \varepsilon_0$: what will the savings rate of *one* new country be? Here $\varepsilon_0 \sim N(0, \sigma^2)$ is independent of the data (the new unit is not in the sample).

The point estimate is the same, $\hat y_0 = x_0^\top \hat\beta$, but the uncertainty differs because predicting an individual also has to account for its own error ε_0.

Theorem 6.4 (Confidence and prediction intervals)**.** *Under A1–A4, let* $q_0 = x_0^\top (X^\top X)^{-1} x_0$. *Then*

$$\operatorname{Var}(\hat y_0) = \sigma^2 q_0, \qquad \operatorname{Var}(y_0 - \hat y_0) = \sigma^2 (1 + q_0),$$

and the following intervals have exact coverage $1 - \alpha$*:*

$$\text{CI for } \mathrm{E}[y_0]: \quad \hat y_0 \pm t_{n-p,\,1-\alpha/2}\, \hat\sigma \sqrt{q_0}, \tag{6.5}$$

$$\text{PI for } y_0: \quad \hat y_0 \pm t_{n-p,\,1-\alpha/2}\, \hat\sigma \sqrt{1 + q_0}. \tag{6.6}$$

Proof. $\operatorname{Var}(\hat y_0) = x_0^\top \operatorname{Var}(\hat\beta) x_0 = \sigma^2 q_0$ by the variance formula of Chapter 5. Since $y_0 = x_0^\top \beta + \varepsilon_0$ with ε_0 independent of $\hat\beta$ (a function of the observed y only), $\operatorname{Var}(y_0 - \hat y_0) = \operatorname{Var}(\varepsilon_0) + \operatorname{Var}(x_0^\top \hat\beta) = \sigma^2(1 + q_0)$, and $\mathrm{E}[y_0 - \hat y_0] = 0$. Under A4, $(y_0 - \hat y_0)/\{\sigma\sqrt{1+q_0}\} \sim N(0,1)$. The estimate $\hat\sigma^2$ is independent of $\hat\beta$ (Chapter 5) and of ε_0, and $(n-p)\hat\sigma^2/\sigma^2 \sim \chi^2_{n-p}$, so the ratio with $\hat\sigma$ in place of σ is t_{n-p}: this gives (6.6). The argument for (6.5) is the same without ε_0, using the t pivot of Chapter 5 with $c = x_0$. □

Note that q_0 is the leverage that the new point *would* have if it were in the sample: $q_0 = x_0^\top (X^\top X)^{-1} x_0$ is h_{ii} with x_i replaced by x_0. Prediction is therefore most precise near the centre of the data, and for simple regression $q_0 = 1/n + (x_0 - \bar{x})^2 / S_{xx}$: the interval widens quadratically away from $\bar{x}$ (Figure 6.7). Extrapolating is risky twice over: the interval is wide, and the *linear form of the model* is untested there.

Compare the two formulas as $n \to \infty$: $q_0 \to 0$, the CI shrinks to a point (we learn the mean exactly), but the PI converges to $\hat{y}_0 \pm z_{1-\alpha/2}\sigma$: even with infinite data, the next unit has its own error. The share of the prediction variance due to estimating β is $q_0/(1+q_0)$, usually small.

```
x0 <- data.frame(pop15 = 35, pop75 = 2, dpi = 1000, ddpi = 3)
rbind(CI = predict(fit, x0, interval = "confidence"), PI = predict(fit, x0,
↪  interval = "prediction"))
```

```
#>     fit   lwr   upr
#> 1 9.934 8.646 11.22
#> 1 9.934 2.167 17.70
```

```
X <- model.matrix(fit); x0v <- c(1, 35, 2, 1000, 3)
q0 <- drop(t(x0v) %*% solve(crossprod(X)) %*% x0v)
tq <- qt(0.975, n - p)
c(q0 = q0, CI_half = tq * sigma(fit) * sqrt(q0), PI_half = tq * sigma(fit) *
↪  sqrt(1 + q0))
```

```
#>      q0 CI_half PI_half
#> 0.02825 1.28733 7.76640
```

For a country with these characteristics the estimated mean savings rate is 9.93%, with 95% CI (8.65, 11.22): the mean is pinned down to about ± 1.3 points. A *single* such country has a savings rate in (2.17, 17.70) with 95% probability, a width six times larger: $\hat{\sigma} = 3.8$ dominates.

```
f1 <- lm(sr ~ pop15, data = savings)
grid <- data.frame(pop15 = seq(20, 50, length.out = 100))
ci <- predict(f1, grid, interval = "confidence"); pi <- predict(f1, grid,
↪  interval = "prediction")
b <- cbind(grid, fit = ci[, 1], cl = ci[, 2], cu = ci[, 3], pl = pi[, 2], pu =
↪  pi[, 3])
ggplot(b, aes(pop15)) +
  geom_ribbon(aes(ymin = pl, ymax = pu, fill = "95% prediction interval"),
  ↪  alpha = 0.35) +
  geom_ribbon(aes(ymin = cl, ymax = cu, fill = "95% CI for the mean"), alpha =
  ↪  0.6) +
  geom_line(aes(y = fit), colour = ie_navy) + geom_point(data = savings,
  ↪  aes(pop15, sr), colour = ie_navy, size = 1.5) +
  scale_fill_manual(values = c("95% CI for the mean" = ie_blue, "95%
  ↪  prediction interval" = "#D2E6F4"), name = NULL) +
  labs(x = "pop15 (% population under 15)", y = "Savings rate sr (%)")
```

The prediction interval relies on A3 and A4 more than anything else. The CI for the mean can be justified asymptotically even for non-normal errors, but the PI uses the distribution of one error. If the variance grows with the mean or the errors are skewed, the interval has the wrong shape. The `teengamb` data (47 British teenagers; response: yearly gambling expenditure in pounds) show this: the prediction interval for a boy with typical covariates has a negative

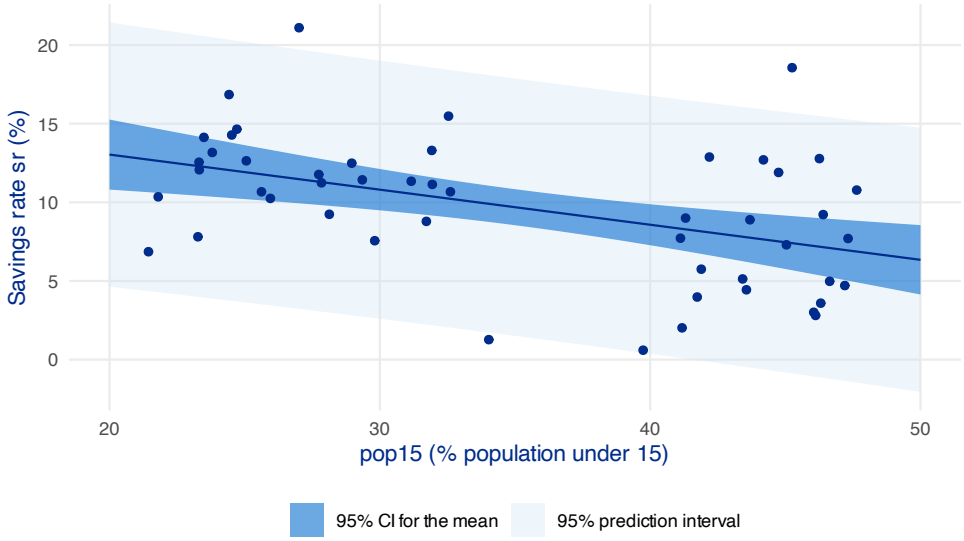


Figure 6.7: Simple regression of savings rate on pop15: 95% confidence band for the mean and 95% prediction band for a new observation.

lower limit, which is impossible. A square-root transformation (Section 6.5.1) repairs it, after back-transformation of the end-points.

```
data(teengamb, package = "faraway")
tg <- lm(gamble ~ sex + status + income + verbal, data = teengamb)
ts <- lm(sqrt(gamble) ~ sex + status + income + verbal, data = teengamb)
x0 <- data.frame(sex = 0, status = 40, income = 5, verbal = 6)
pr_raw <- predict(tg, x0, interval = "prediction")
pr_sqrt <- predict(ts, x0, interval = "prediction")
rbind(raw_scale = pr_raw, sqrt_model_back_transformed = pmax(pr_sqrt, 0)^2)
```

```
#>     fit    lwr   upr
#> 1 31.70 -15.26 78.65
#> 1 18.55   0.00 74.31
```

```
c(ncv_p_raw = ncvTest(tg)$p, ncv_p_sqrt = ncvTest(ts)$p)
```

```
#>  ncv_p_raw ncv_p_sqrt
#>  8.285e-07  2.290e-02
```

The back-transformed interval (0, 74.3) respects the constraint and is asymmetric around the point prediction, as the data suggest; the score test shows that the variance is much more stable on the square-root scale (the p-value moves from 10^{-6} to 0.02). Intervals for the *mean* on the back-transformed scale describe the *median* of y, not its mean.

Out-of-sample prediction error. The residual sum of squares measures fit to the data used to fit it and is optimistic. Lemma 6.1 gives an honest alternative at no extra cost: the **leave-one-out** (LOO) prediction error for case i is $e_i/(1-h_{ii})$, and

$$\text{PRESS} = \sum_{i=1}^{n} \left(\frac{e_i}{1-h_{ii}}\right)^2, \qquad \text{LOOCV-MSE} = \text{PRESS}/n.$$

```
loo_mse <- mean((resid(fit) / (1 - hatvalues(fit)))^2)
c(RSS_over_n = mean(resid(fit)^2), sigma2_hat = sigma(fit)^2, LOOCV_MSE =
↪  loo_mse)
```

```
#> RSS_over_n sigma2_hat  LOOCV_MSE
#>      13.01      14.46      15.98
```

The apparent error RSS/n = 13.0 understates the leave-one-out error (16.0), as expected; we use PRESS in Chapter 7 to compare models.

6.8 In R: the functions used in class

Task	Code
Standardized, studentized residuals	`rstandard(fit)`, `rstudent(fit)`
Leverage, Cook's distance	`hatvalues(fit)`, `cooks.distance(fit)`
All influence measures	`influence.measures(fit)`, `dfbetas(fit)`, `dffits(fit)`
Four default plots	`par(mfrow = c(2, 2)); plot(fit)`
Outlier test (Bonferroni)	`car::outlierTest(fit)`
Partial residual / added-variable plots	`car::crPlots(fit)`, `car::avPlots(fit)`
Non-constant variance test	`car::ncvTest(fit)`
Autocorrelation	`acf(resid(fit))`, `car::durbinWatsonTest(fit)`
Box–Cox	`MASS::boxcox(fit)`
Variance inflation factors	`car::vif(fit)`
Confidence / prediction intervals	`predict(fit, newdata, interval = "confidence" or "prediction")`

6.9 Common pitfalls

Warning. Flagging every $|t_i| > 2$ as an outlier. With n cases, about $0.05n$ will exceed 2 by chance. Use the Bonferroni cut-off or look for gross departures.

Warning. Treating $h_{ii} > 2p/n$ or $D_i > 4/n$ as proof that a point is wrong, and deleting it. These are screening rules. Investigate the case, report the results with and without it, and consider robust methods.

Warning. "Removing outliers until none remain": iterated deletion changes the sample and the target, and inflates apparent precision.

Warning. Reporting a confidence interval for the mean when the question is about a new unit. Check which of $\mathrm{E}[y_0]$ and y_0 is the object of interest.

Remark. Collinearity is not a violation of the assumptions: the estimates are unbiased, the intervals are valid, but wide. Dropping a variable to "fix" a large VIF introduces omitted-variable bias if it is a confounder (Chapter 7).

Remark. Diagnostics are exploratory; do not choose a model by passing a battery of tests on the same data and then report the usual p-values as if the model had been fixed in advance.

6.10 Chapter summary

- $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$: residuals are not errors. Use standardized r_i and studentized t_i residuals; $t_i \sim t_{n-p-1}$ under the model (exact, via the mean-shift model), and $t_i = r_i\sqrt{(n-p-1)/(n-p-r_i^2)}$.
- Leverage $h_{ii} \in [1/n, 1]$, $\sum_i h_{ii} = p$; it depends only on X; flag $h_{ii} > 2p/n$.
- Deleting case i changes $\hat{\beta}$ by $(X^\top X)^{-1}x_i e_i/(1 - h_{ii})$; Cook's distance is $D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}$: outlyingness times leverage. The leave-one-out error is $e_i/(1 - h_{ii})$.
- Plots check A2 (curvature), A3 (funnel, scale–location, residuals in time order) and A4 (QQ). Transformations (Box–Cox) and weighted least squares address non-constant variance.
- $\mathrm{Var}(\hat{\beta}_j) = \sigma^2 \mathrm{VIF}_j/\{(n-1)s_j^2\}$; collinearity inflates variance, does not bias, and does not hurt prediction inside the data cloud.
- CI for the mean: $\hat{y}_0 \pm t\,\hat{\sigma}\sqrt{q_0}$; PI for a new observation: $\hat{y}_0 \pm t\,\hat{\sigma}\sqrt{1+q_0}$, with $q_0 = x_0^\top(X^\top X)^{-1}x_0$. The PI never shrinks to zero width and is sensitive to A3–A4.

6.11 Exercises

Exercise 6.1. Diagnostics by hand. Four observations: $x = (1, 2, 3, 4)$, $y = (2, 3, 5, 6)$. The fitted line is $\hat{y} = 0.5 + 1.4x$ with RSS $= 0.2$. (a) Compute the leverages h_{ii} and verify $\sum h_{ii} = 2$. (b) Compute the standardized residuals r_i, the studentized residuals t_i and Cook's distances D_i. (c) Which observation is most influential, and why?

Solution.

(a) $h_{ii} = 1/4 + (x_i - 2.5)^2/5$, so $(0.7, 0.3, 0.3, 0.7)$, which add up to $2 = p$.

(b) Residuals $e = (0.1, -0.3, 0.3, -0.1)$ and $\hat{\sigma}^2 = 0.2/2 = 0.1$, $\hat{\sigma} = 0.3162$. $r_i = e_i/(\hat{\sigma}\sqrt{1-h_{ii}})$: $r_1 = 0.1/(0.3162 \times 0.5477) = 0.577$, $r_2 = -0.3/(0.3162 \times 0.8367) = -1.134$, $r_3 = 1.134$, $r_4 = -0.577$. With $n - p = 2$: $t_i = r_i\sqrt{1/(2 - r_i^2)}$: $t_1 = 0.577 \times 0.7746 = 0.447$, $t_2 = -1.134 \times 1.183 = -1.342$, $t_3 = 1.342$, $t_4 = -0.447$. $D_i = \frac{r_i^2}{2}\frac{h_{ii}}{1-h_{ii}}$: $D_1 = \frac{0.3333}{2} \cdot \frac{0.7}{0.3} = 0.389$, $D_2 = \frac{1.2857}{2} \cdot \frac{0.3}{0.7} = 0.276$, and by symmetry $D_3 = 0.276$, $D_4 = 0.389$.

(c) The end points ($i = 1, 4$), which have the highest leverage, are the most influential even though their residuals are the smallest. This is the leverage effect: the line is pulled towards them. Check with `lm()`, `hatvalues()`, `cooks.distance()`.

```
f4 <- lm(y ~ x, data.frame(x = 1:4, y = c(2, 3, 5, 6)))
round(cbind(h = hatvalues(f4), r = rstandard(f4), t = rstudent(f4), D =
↪  cooks.distance(f4)), 3)
```

```
#>     h      r      t     D
#> 1 0.7  0.577  0.447 0.389
#> 2 0.3 -1.134 -1.342 0.276
#> 3 0.3  1.134  1.342 0.276
#> 4 0.7 -0.577 -0.447 0.389
```

Exercise 6.2. Leverage in simple regression. (a) Show $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$ directly from $H = X(X^\top X)^{-1}X^\top$ with $X = [\mathbf{1}, x]$. (b) Show that $\sum_i h_{ii} = 2$. (c) For $n = 10$ with nine points at $x = 0, 1, \ldots, 8$ and one at $x = 30$, compute the leverage of the

extreme point. What does it imply about the fit and about its residual variance?

Solution.

(a) With $(X^\top X)^{-1} = \frac{1}{nS_{xx}}\begin{pmatrix} \sum x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix}$ (Exercise 1 of Chapter 5) and row $(1, x_i)$: $h_{ii} = \frac{\sum x_k^2 - 2n\bar{x}x_i + nx_i^2}{nS_{xx}} = \frac{S_{xx} + n\bar{x}^2 - 2n\bar{x}x_i + nx_i^2}{nS_{xx}} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

(b) $\sum_i h_{ii} = 1 + \sum_i (x_i - \bar{x})^2 / S_{xx} = 1 + 1 = 2$.

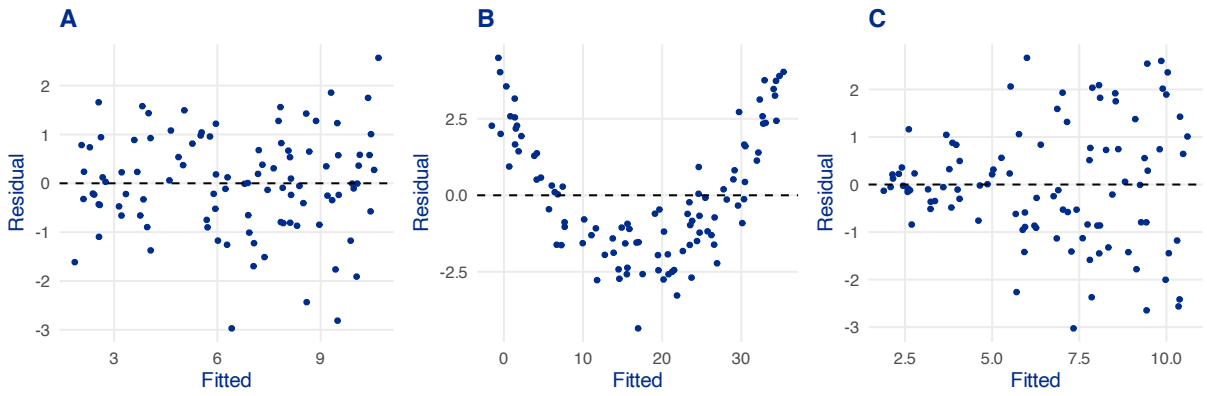
(c) $\bar{x} = (36 + 30)/10 = 6.6$; $S_{xx} = \sum_{k=0}^{8}(k - 6.6)^2 + (30 - 6.6)^2 = \{60 + 9(6.6 - 4)^2\} + 547.56 = 60 + 60.84 + 547.56 = 668.4$ (using $\sum_{k=0}^{8}(k-4)^2 = 60$). So $h = 0.1 + 547.56/668.4 = 0.919$. This single point has 92% of the "weight" on its own fitted value: the line passes almost exactly through it, its residual variance is $\sigma^2(1 - 0.919) = 0.081\sigma^2$, and the slope is determined essentially by the line from the extreme point to the other nine. The high leverage is not harmful if the point follows the pattern of the rest, but it is where any lack of linearity does most damage.

Exercise 6.3. Leave-one-out identities. Prove that $\hat{\sigma}^2_{(i)} = \hat{\sigma}^2 \dfrac{n - p - r_i^2}{n - p - 1}$ and that $D_i = \dfrac{\|\hat{y} - \hat{y}_{(i)}\|^2}{p\hat{\sigma}^2}$ equals $\dfrac{r_i^2}{p}\dfrac{h_{ii}}{1 - h_{ii}}$ by computing $\hat{y} - \hat{y}_{(i)}$ explicitly.

Solution. From Lemma 6.1, $\mathrm{RSS}_{(i)} = \mathrm{RSS} - e_i^2/(1-h_{ii}) = (n-p)\hat{\sigma}^2 - \hat{\sigma}^2 r_i^2$, because $e_i^2/(1-h_{ii}) = \hat{\sigma}^2 r_i^2$. Dividing by $n - p - 1$ gives the first claim. For the second, $\hat{y} - \hat{y}_{(i)} = X(\hat{\beta} - \hat{\beta}_{(i)}) = X(X^\top X)^{-1} x_i \, e_i/(1 - h_{ii}) = H d_i \, e_i/(1 - h_{ii})$, since $X(X^\top X)^{-1}x_i = X(X^\top X)^{-1}X^\top d_i = H d_i$. Then $\|H d_i\|^2 = d_i^\top H^\top H d_i = d_i^\top H d_i = h_{ii}$, so $\|\hat{y} - \hat{y}_{(i)}\|^2 = h_{ii} e_i^2/(1 - h_{ii})^2$. Divide by $p\hat{\sigma}^2$ and use $e_i^2 = \hat{\sigma}^2 r_i^2 (1 - h_{ii})$.

Exercise 6.4. Reading residual plots. Simulate three data sets with $n = 100$: (A) $y = 1 + x + \varepsilon$ with $\varepsilon \sim N(0, 1)$; (B) $y = 1 + x + 0.3x^2 + \varepsilon$; (C) $y = 1 + x + \varepsilon$ with $\varepsilon \sim N(0, (0.2x)^2)$, $x \sim U(1, 10)$. Fit straight lines, plot residuals against fitted values for each, and name the problem and a remedy in B and C.

```
set.seed(63)
x <- runif(100, 1, 10)
dA <- data.frame(x, y = 1 + x + rnorm(100))
dB <- data.frame(x, y = 1 + x + 0.3 * x^2 + rnorm(100))
dC <- data.frame(x, y = 1 + x + rnorm(100, sd = 0.2 * x))
plots <- lapply(list(A = dA, B = dB, C = dC), function(d) {
  f <- lm(y ~ x, d)
  ggplot(data.frame(f = fitted(f), r = resid(f)), aes(f, r)) +
  ↪  geom_point(colour = ie_navy, size = 1) +
    geom_hline(yintercept = 0, linetype = 2) + labs(x = "Fitted", y =
    ↪  "Residual")
})
gridExtra::grid.arrange(plots$A + ggtitle("A"), plots$B + ggtitle("B"),
↪  plots$C + ggtitle("C"), nrow = 1)
```



Solution.

(A) No pattern: the straight-line model is adequate. (B) A U-shaped pattern: the mean is quadratic, not linear (A2 fails); add x^2 (or a spline); the F test for the quadratic term confirms. (C) A funnel: the spread of the residuals grows with the fitted value (A3 fails). Remedies: model the variance (weights $w_i \propto 1/x_i^2$, Exercise 6.7), or transform; since the standard deviation is proportional to x and hence to the mean, $\log y$ is the variance-stabilising choice when $y > 0$. Least squares remains unbiased in (C) but the usual standard errors and intervals are not valid.

Exercise 6.5. Two correlated predictors. Let $X = [\mathbf{1}, x_1, x_2]$ and let r be the sample correlation between x_1 and x_2. (a) Show that $\mathrm{VIF}_1 = \mathrm{VIF}_2 = 1/(1-r^2)$. (b) Compute the VIF and the inflation of the standard error for $r = 0.9$ and $r = 0.99$. (c) If $\mathrm{se}(\hat{\beta}_1) = 0.2$ when $r = 0$, what is it when $r = 0.99$?

Solution.

(a) With a single other predictor, the regression of x_1 on $(1, x_2)$ has $R^2 = r^2$, so $\mathrm{VIF}_1 = 1/(1-r^2)$; the same holds for x_2. (b) $r = 0.9$: VIF $= 1/0.19 = 5.26$ and se inflation $\sqrt{5.26} = 2.29$; $r = 0.99$: $1/0.0199 = 50.3$, inflation 7.09. (c) $0.2 \times 7.09 = 1.42$. A coefficient that is detected at $t = 5$ with orthogonal predictors would have $t \approx 0.7$.

Exercise 6.6. Intervals by hand. A simple linear regression with $n = 27$ has $\bar{x} = 10$, $S_{xx} = 200$, $\hat{\sigma} = 3$, and fitted line $\hat{y} = 5 + 2x$. (a) Compute q_0 at $x_0 = 10$ and at $x_0 = 20$. (b) Give 95% CI and PI at both points ($t_{25,0.975} = 2.060$). (c) At which x_0 is the ratio PI width/CI width larger, and why?

Solution.

(a) $q_0 = 1/27 + (x_0 - 10)^2/200$: at $x_0 = 10$, $q_0 = 0.0370$; at $x_0 = 20$, $q_0 = 0.0370 + 0.5 = 0.5370$.
(b) At $x_0 = 10$: $\hat{y}_0 = 25$; CI half-width $2.060 \times 3 \times \sqrt{0.0370} = 1.189$, so $(23.81, 26.19)$; PI half-width $2.060 \times 3\sqrt{1.0370} = 6.293$, so $(18.71, 31.29)$. At $x_0 = 20$: $\hat{y}_0 = 45$; CI half-width $2.060 \times 3\sqrt{0.5370} = 4.529$, so $(40.47, 49.53)$; PI half-width $2.060 \times 3\sqrt{1.5370} = 7.662$, so $(37.34, 52.66)$.
(c) Ratio $\sqrt{(1+q_0)/q_0}$: 5.29 at $x_0 = 10$ and 1.69 at $x_0 = 20$. Near the centre the CI is narrow and the PI is dominated by σ; far from the centre, the uncertainty in the line dominates and the two intervals become comparable.

Exercise 6.7. Weighted least squares. Suppose $y = X\beta + \varepsilon$ with $\mathrm{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2 W^{-1}$, $W = \mathrm{diag}(w_1, \ldots, w_n)$ known, $w_i > 0$. (a) Show that applying ordinary least

squares to $W^{1/2}y$ on $W^{1/2}X$ gives $\hat\beta_W = (X^\top WX)^{-1}X^\top Wy$, unbiased with variance $\sigma^2(X^\top WX)^{-1}$. (b) Why is it BLUE? (c) Each y_i is the average of m_i independent observations with variance σ^2 taken at the same x_i. What are the weights?

Solution.

(a) Let $y^* = W^{1/2}y$, $X^* = W^{1/2}X$, $\varepsilon^* = W^{1/2}\varepsilon$. Then $y^* = X^*\beta + \varepsilon^*$, $\mathrm{E}[\varepsilon^*] = 0$ and $\mathrm{Var}(\varepsilon^*) = W^{1/2}\sigma^2 W^{-1}W^{1/2} = \sigma^2 I$. OLS on the starred model gives $\hat\beta = (X^{*\top}X^*)^{-1}X^{*\top}y^* = (X^\top WX)^{-1}X^\top Wy$, with variance $\sigma^2(X^{*\top}X^*)^{-1} = \sigma^2(X^\top WX)^{-1}$; it is unbiased because $\mathrm{E}[y^*] = X^*\beta$. (b) The starred model satisfies A1–A3, so Gauss–Markov applies to it: among estimators linear in y^* (equivalently in y, since $W^{1/2}$ is invertible) $\hat\beta_W$ is best. OLS on the original data is still unbiased but has larger variance. (c) $\mathrm{Var}(y_i) = \sigma^2/m_i$, so $w_i = m_i$. The weighted fit gives the groups with more observations more influence.

Exercise 6.8. R analysis: stack loss. The `stackloss` data (package `datasets`; $n = 21$ days of operation of a plant oxidising ammonia to nitric acid) record `stack.loss` (percentage of ammonia lost) with `Air.Flow`, `Water.Temp` and `Acid.Conc.`. Fit the linear model with all three predictors and (a) report coefficients, $\hat\sigma$, R^2; (b) compute leverages, studentized residuals and Cook's distances: is there an outlier at the 5% level? is any point influential? (c) compute VIFs; (d) refit without the most influential case and describe the change; (e) give a 95% CI and PI for `Air.Flow`=58, `Water.Temp`=20, `Acid.Conc.`=85 and explain the difference.

```
sl <- lm(stack.loss ~ ., data = stackloss)
round(summary(sl)$coefficients, 4); c(sigma = sigma(sl), R2 =
↪  summary(sl)$r.squared)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept) -39.9197    11.8960 -3.3557   0.0038
#> Air.Flow      0.7156     0.1349  5.3066   0.0001
#> Water.Temp    1.2953     0.3680  3.5196   0.0026
#> Acid.Conc.   -0.1521     0.1563 -0.9733   0.3440

#>  sigma     R2
#> 3.2434 0.9136
```

```
nn <- nrow(stackloss); pp <- length(coef(sl))
infl <- data.frame(h = hatvalues(sl), t = rstudent(sl), D =
↪  cooks.distance(sl))
round(infl[order(-infl$D)[1:4], ], 3); c(two_p_over_n = 2 * pp / nn,
↪  bonferroni = qt(1 - 0.05 / (2 * nn), nn - pp - 1))
```

```
#>        h      t     D
#> 21 0.285 -3.330 0.692
#> 1  0.302  1.209 0.154
#> 4  0.129  2.052 0.131
#> 3  0.175  1.618 0.126

#> two_p_over_n   bonferroni
#>        0.381        3.604
```

```
round(vif(sl), 2)
```

```
#>   Air.Flow Water.Temp Acid.Conc.
#>       2.91       2.57       1.33
```

```
sl2 <- update(sl, subset = -21); round(cbind(all = coef(sl), without_21 =
↪  coef(sl2)), 3)
```

```
#>                 all without_21
#> (Intercept) -39.920    -43.704
#> Air.Flow      0.716      0.889
#> Water.Temp    1.295      0.817
#> Acid.Conc.   -0.152     -0.107
```

```
new <- data.frame(Air.Flow = 58, Water.Temp = 20, Acid.Conc. = 85)
rbind(CI = predict(sl, new, interval = "confidence"), PI = predict(sl, new,
↪  interval = "prediction"))
```

```
#>     fit    lwr   upr
#> 1 14.56 12.967 16.16
#> 1 14.56  7.536 21.59
```

Solution.

(a) The model explains $R^2 = 0.91$ of the variability, $\hat{\sigma} = 3.24$ on 17 df; `Air.Flow` (0.72) and `Water.Temp` (1.30) are significant, `Acid.Conc.` (-0.15) is not. (b) The largest studentized residual is day 21 with $t = -3.33$, below the Bonferroni cut-off 3.60: no outlier at the 5% level (unadjusted $p = 0.004$ would mislead; the adjusted one is 0.089). Day 21 also has the largest Cook's distance, 0.69 (the median of $F_{4,17}$ is 0.87, so it is notable but not extreme), while its leverage 0.285 is below $2p/n = 0.38$; the highest leverage is day 17 (0.41) with a small residual. (c) All VIFs are below 3: no collinearity problem. (d) Without day 21 the `Air.Flow` coefficient rises from 0.72 to 0.89 ($+1.3$ standard errors) and that of `Water.Temp` falls from 1.30 to 0.82 (-1.3 standard errors); $\hat{\sigma}$ drops from 3.24 to 2.57 and R^2 rises from 0.91 to 0.95. Keep the case in the main analysis and report the sensitivity. (e) The mean stack loss at those conditions is estimated as 14.6 with 95% CI $(13.0, 16.2)$, the PI for one day is $(7.5, 21.6)$: the CI describes the average over many days of operation at those settings, the PI a single future day, which also carries $\sigma = 3.2$.

Exercise 6.9. R analysis: transformation. For `faraway::gala`, (a) fit `Species ~ Area + Elevation + Nearest + Scruz + Adjacent` and show the residual-versus-fitted plot; (b) choose a power transformation with `MASS::boxcox`; (c) refit with the square-root response and with the log response, and compare the diagnostic plots and the score test for non-constant variance; (d) interpret the coefficient of `Elevation` in the log model as a percentage change.

```
gl <- lm(log(Species) ~ Area + Elevation + Nearest + Scruz + Adjacent, data =
↪  gala)
round(summary(gl)$coefficients, 4)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   2.3074     0.3306  6.9788   0.0000
#> Area         -0.0007     0.0004 -1.8997   0.0696
```

```
#> Elevation     0.0045     0.0009  4.8602   0.0001
#> Nearest       0.0166     0.0182  0.9104   0.3716
#> Scruz        -0.0039     0.0037 -1.0481   0.3050
#> Adjacent     -0.0008     0.0003 -2.6187   0.0151
```

```
c(ncv_p_raw = ncvTest(g)$p, ncv_p_sqrt = ncvTest(gs)$p, ncv_p_log =
↪  ncvTest(gl)$p)
```

```
#>  ncv_p_raw ncv_p_sqrt  ncv_p_log
#>  0.0001035  0.3395390  0.0262703
```

```
c(pct_per_100m = 100 * (exp(100 * coef(gl)["Elevation"]) - 1))
```

```
#> pct_per_100m.Elevation
#>                  56.86
```

Solution.

(a) The raw fit shows a funnel (score test $p = 0.0001$). (b) The Box–Cox profile has its maximum at $\hat{\lambda} \approx 0.3$, i.e. between the log ($\lambda = 0$) and the square root ($\lambda = 0.5$). (c) The square-root model removes the non-constant variance ($p = 0.34$). The log model still leaves some ($p = 0.026$); both transformations make the residuals compatible with normality (Shapiro–Wilk $p = 0.60$ and 0.62). The square root is the better choice for the variance. (d) In the log model $\hat{\beta}_{\texttt{Elevation}} = 0.0045$ per metre, so each extra 100 m of elevation multiplies the median number of species by $e^{0.45} = 1.57$: about 57% more species, holding the other geographic variables fixed. (A Poisson or negative-binomial regression, Chapters 8 and 10, would be the more natural model for counts.)

Exercise 6.10. Critique. An analyst reports: "I fitted a regression of house prices on 12 predictors. $R^2 = 0.93$. I removed the 8 observations with Cook's distance above $4/n$ and refitted; no observation is now above the cut-off, and all coefficients are significant. The VIF of two predictors is 14 so I dropped the one with the larger VIF, which was the main variable of interest (square metres). The 95% confidence interval for the mean price of a new flat is (£281k, £294k), so we can tell clients that its price will be in this range." Identify at least four problems.

Solution.

(1) *Iterated deletion.* $4/n$ is a screening level, not a test, and after removing points, the Cook's distances are recomputed on a different sample and new points appear; stopping when "no observation is above the cut-off" is circular. Deleting 8 points without investigation changes the population being described and makes the significance levels meaningless. Report the fit with all the data, investigate the flagged cases, and show the sensitivity. (2) *Dropping a variable because of its VIF.* Collinearity inflates variance but does not bias; dropping the variable of interest answers a different question, and if the other predictor is a proxy for it, the remaining coefficient absorbs its effect (omitted-variable bias). Instead report the joint test, use a combined variable, or regularise. (3) *Confusing the CI for the mean with a prediction interval.* The interval is for the *average* price of such flats; one flat's price has the much wider PI, which also depends on A3–A4 (prices are right-skewed with variance increasing with the mean; model log price). (4) $R^2 = 0.93$ *and significant coefficients* after selecting the data say nothing about whether the assumptions hold: no residual plots, no check for non-constant variance or nonlinearity, no out-of-sample assessment. (5) After data-driven deletion, the reported p-values and intervals ignore the

selection.

Exercise 6.11. PRESS. (a) Show that the leave-one-out prediction error is $e_i/(1-h_{ii})$. (b) For `savings`, compute PRESS for the full model and for `sr ~ pop15 + ddpi` and decide which predicts better in the leave-one-out sense. (c) Verify the formula by brute-force deletion for three countries.

```
loo <- function(f) mean((resid(f) / (1 - hatvalues(f)))^2)
m_small <- lm(sr ~ pop15 + ddpi, data = savings)
c(full = loo(fit), small = loo(m_small), RSS_n_full = mean(resid(fit)^2),
↪  RSS_n_small = mean(resid(m_small)^2))
```

```
#>        full       small  RSS_n_full RSS_n_small
#>       15.98       16.68       13.01       14.01
```

```
sapply(c("Libya", "Japan", "Zambia"), function(cn) {
  i <- which(rownames(savings) == cn)
  brute <- savings$sr[i] - predict(update(fit, subset = -i), savings[i, ])
  c(brute = unname(brute), formula = unname(resid(fit)[i] / (1 -
  ↪  hatvalues(fit)[i])))
})
```

```
#>          Libya Japan Zambia
#> brute   -6.039   6.8  10.42
#> formula -6.039   6.8  10.42
```

Solution.

(a) Lemma 6.1(1): the prediction error $y_i - x_i^\top \hat\beta_{(i)}$ equals $\hat\gamma = e_i/(1-h_{ii})$. (b) The full model has the lower apparent error (RSS$/n = 13.0$ against 14.0; adding columns cannot increase RSS) and also the lower leave-one-out error (16.0 against 16.7). The gain from `pop75` and `dpi` is small once the optimism is removed (the apparent gain is 7%, the honest gain 4%) and the nested F test ($p = 0.19$) does not find it significant: the three criteria say "little difference". (c) The brute-force and formula columns agree to rounding.

6.12 Further reading

Faraway (2014) (Chapters 6–7) covers residuals, leverage, influence, transformations and collinearity with the same R functions. Belsley et al. (1980) is the classic treatise on influence and collinearity diagnostics, and Cook (1977) introduces the distance that bears his name. The mean-shift derivation of the studentized residual follows Kutner et al. (2005) (Chapter 10). For the generalised VIF see Fox and Monette (1992); for transformations, Box (1976) gives the philosophy ("all models are wrong, but some are useful") and Faraway Chapter 7 the Box–Cox method.

Chapter 7

Linear models in practice and model selection

Chapters 5 and 6 gave us the estimator, its properties and the tools to check the assumptions. Real work starts with questions that the theory leaves open. *What does a coefficient mean* when many predictors are in the model, and why can it change sign when we add another variable? How do we put a *categorical* predictor, such as the number of cylinders or a treatment group, into X, and what happens when the effect of one predictor *depends on another* (an interaction)? Among many candidate predictors, *which ones should be in the model*, and does the answer depend on whether we want to explain or to predict? And what can we do when there are so many correlated predictors that least squares becomes unstable (ridge, lasso, elastic net)?

These are questions about **specification**, and the central message of the chapter is that the answer depends on the purpose. When the goal is *explanation* (estimating and interpreting a particular coefficient), the model should be fixed in advance from subject knowledge, adjusting for confounders; selecting variables from the data then invalidates the usual inference. When the goal is *prediction*, the model can be chosen from the data, but must be judged by error on data that played no part in the choice (Shmueli, 2010; Breiman, 2001). Most of the chapter develops tools for one or the other, with the algebra needed to understand why they work.

We use three data sets. `savings` (Chapter 5) illustrates confounding; `mtcars` (32 cars, `datasets` package) and `PlantGrowth` illustrate factors and interactions; and `prostate` (Stamey et al., 1989) is the main example for selection and shrinkage: 97 men about to undergo radical prostatectomy, with the response `lpsa` (log of the prostate-specific antigen) and eight clinical predictors: log cancer volume (`lcavol`), log prostate weight (`lweight`), `age`, log benign prostatic hyperplasia (`lbph`), seminal vesicle invasion (`svi`, 0/1), log capsular penetration (`lcp`), Gleason score (`gleason`) and percentage of Gleason scores 4 or 5 (`pgg45`).

7.1 Interpreting the coefficients of a multiple regression

In the model $\mathrm{E}[y \mid x] = \beta_0 + \beta_1 x_1 + \cdots + \beta_{p-1} x_{p-1}$, the coefficient $\beta_j = \partial \mathrm{E}[y \mid x] / \partial x_j$ is the difference in the *mean* response per unit of x_j **with the other predictors held fixed**. "Held fixed" is a statement about the model, not about the world: in observational data we usually cannot move one variable and leave the others alone, and β_j depends on *which other predictors are in the model*. The next theorem gives the exact meaning of "adjusting for the others".

Theorem 7.1 (Frisch–Waugh–Lovell)**.** *Partition $X = [X_1, X_2]$ with X of full column rank, let $H_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$, $M_1 = I - H_1$, and let $\hat\beta = (\hat\beta_1^\top, \hat\beta_2^\top)^\top$ be the corresponding blocks of the least-squares estimate. Then*

$$\hat\beta_2 = (X_2^\top M_1 X_2)^{-1} X_2^\top M_1 y, \qquad \operatorname{Var}(\hat\beta_2) = \sigma^2 (X_2^\top M_1 X_2)^{-1},$$

i.e. $\hat\beta_2$ is the least-squares coefficient of the regression of $M_1 y$ (the residuals of y on X_1) on $M_1 X_2$ (the residuals of X_2 on X_1). The residuals of that regression are the residuals e of the full fit.

Proof. $M_1 X_2$ is orthogonal to $\mathcal{C}(X_1)$ and $\mathcal{C}(X) = \mathcal{C}(X_1) \oplus \mathcal{C}(M_1 X_2)$, because $X_2 = H_1 X_2 + M_1 X_2$ with $H_1 X_2 \in \mathcal{C}(X_1)$. Projection onto an orthogonal direct sum is the sum of the projections, so

$$\hat y = H_1 y + M_1 X_2\, b, \qquad b = (X_2^\top M_1 X_2)^{-1} X_2^\top M_1 y,$$

where we used $M_1^\top M_1 = M_1$ to write the projection onto $\mathcal{C}(M_1 X_2)$. Now $M_1 X_2 b = X_2 b - H_1 X_2 b$ and $H_1 X_2 b \in \mathcal{C}(X_1)$, so $\hat y = X_2 b + (\text{something in } \mathcal{C}(X_1))$. Because the coordinates of $\hat y$ in the columns of X are unique, the coefficient of X_2 is b: $\hat\beta_2 = b$. The variance follows from $\hat\beta_2 = Ay$ with $A = (X_2^\top M_1 X_2)^{-1} X_2^\top M_1$ and $AA^\top = (X_2^\top M_1 X_2)^{-1}$. Finally $e = y - \hat y = M_1 y - M_1 X_2 b$. □

The theorem has three practical consequences. (i) A multiple-regression coefficient is a *simple-regression slope between residuals*: the part of y, and the part of x_j, that are not explained by the other predictors. This is the added-variable plot. (ii) The standard error of $\hat\beta_j$ is large if $M_1 x_j$ is small, i.e. if x_j is nearly a linear combination of the other columns (collinearity, Chapter 6). (iii) Coefficients depend on the *set* of other predictors in a precise way, studied next.

```
fit_full <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data = savings)
ry <- resid(lm(sr ~ pop75 + dpi + ddpi, data = savings))     # y adjusted for
↪  the others
rx <- resid(lm(pop15 ~ pop75 + dpi + ddpi, data = savings))  # pop15 adjusted
↪  for the others
c(multiple_regression = coef(fit_full)[["pop15"]], residual_regression =
↪  coef(lm(ry ~ rx))[["rx"]])
```

```
#> multiple_regression residual_regression
#>             -0.4612             -0.4612
```

Units and scale. Rescaling $x_j \mapsto a x_j$ rescales $\beta_j \mapsto \beta_j / a$ and its standard error in the same way, so t, p-values and fitted values do not change. "Standardised coefficients" (those from the regression of y/s_y on x_j/s_{x_j}) express the effect in standard deviations and put predictors on a comparable footing, but comparing their magnitudes as "importance" is risky when predictors are correlated. Centring a predictor changes the intercept (it becomes the mean response at the average value of the predictor) and nothing else, unless interactions or polynomial terms are present (Section 7.4). When y or x is logged, the interpretation is in percentages (Chapter 6).

Language. "Countries with a population under 15 one point higher and the same `pop75`, `dpi` and `ddpi` have on average 0.46 points lower savings rates" is correct. "Raising `pop15` by one point *lowers* a country's savings rate by 0.46 points" is a causal statement that the data cannot support.

Table 7.1: The coefficient of pop75 with and without pop15 in the model.

	Estimate	Std. Error	t value	Pr(>|t|)
sr ~ pop75	1.099	0.475	2.312	0.025
sr ~ pop75 + pop15	-1.934	1.041	-1.858	0.069

7.2 Confounding and omitted-variable bias

Table 7.1 shows a classical puzzle in `savings`. The coefficient of `pop75`, the share of the population over 75, is *positive* and significant when it is the only predictor, and *negative* when `pop15` (the share under 15) is included.

```
short <- lm(sr ~ pop75, data = savings)
long  <- lm(sr ~ pop75 + pop15, data = savings)
knitr::kable(round(rbind(`sr ~ pop75` = summary(short)$coefficients["pop75",
↪  ],
                         `sr ~ pop75 + pop15` =
                         ↪  summary(long)$coefficients["pop75", ]), 3),
             caption = "The coefficient of pop75 with and without pop15 in the
             ↪  model.")
```

The two shares are strongly negatively correlated (-0.91): countries with many old people have few children. The marginal slope of `pop75` mixes its own effect with the effect of `pop15`. The following theorem quantifies this.

Theorem 7.2 (Omitted-variable bias)**.** *Suppose the true model is* $y = X_1\beta_1 + X_2\beta_2 + \varepsilon$ *with* $\mathrm{E}[\varepsilon] = 0$*, but we fit only* X_1*, obtaining* $\tilde\beta_1 = (X_1^\top X_1)^{-1}X_1^\top y$*. Then*

$$\mathrm{E}[\tilde\beta_1] = \beta_1 + \Delta\,\beta_2, \qquad \Delta = (X_1^\top X_1)^{-1}X_1^\top X_2.$$

Δ *is the matrix of coefficients of the regressions of the* omitted *columns on the* included *ones. The bias* $\Delta\beta_2$ *vanishes if and only if* $\beta_2 = 0$ *(the omitted variables are irrelevant) or* $\Delta\beta_2 = 0$ *(in particular if* X_2 *is orthogonal to* X_1*). Moreover, in the sample, if* $\hat\beta_1, \hat\beta_2$ *are the estimates of the* long *regression and* $\hat\Delta$ *is the matrix of coefficients of the regression of* X_2 *on* X_1*, then exactly*

$$\tilde\beta_1 = \hat\beta_1 + \hat\Delta\,\hat\beta_2.$$

Proof. $\mathrm{E}[\tilde\beta_1] = (X_1^\top X_1)^{-1}X_1^\top \mathrm{E}[y] = (X_1^\top X_1)^{-1}X_1^\top(X_1\beta_1 + X_2\beta_2) = \beta_1 + \Delta\beta_2$. For the sample identity, substitute $y = X_1\hat\beta_1 + X_2\hat\beta_2 + e$ into $\tilde\beta_1 = (X_1^\top X_1)^{-1}X_1^\top y$ and use $X_1^\top e = 0$: $\tilde\beta_1 = \hat\beta_1 + (X_1^\top X_1)^{-1}X_1^\top X_2\hat\beta_2 = \hat\beta_1 + \hat\Delta\hat\beta_2$. □

A **confounder** is a variable related both to the response and to the predictor of interest. Leaving it out biases the coefficient of interest, with a sign given by the product of the two associations. In `savings`: $\hat\beta_{\texttt{pop15}} = -0.471$ (more children go with lower savings) and $\hat\delta = -6.44$ (more over-75s go with fewer children), so the bias is $(-0.471)(-6.44) = +3.03$, which overturns the true -1.93 into $+1.10$:

```
aux <- lm(pop15 ~ pop75, data = savings)
c(marginal = coef(short)[["pop75"]],
  adjusted_plus_bias = coef(long)[["pop75"]] + coef(long)[["pop15"]] *
  ↪  coef(aux)[["pop75"]],
  adjusted = coef(long)[["pop75"]], delta = coef(aux)[["pop75"]], beta15 =
  ↪  coef(long)[["pop15"]])
```

```
#>           marginal adjusted_plus_bias           adjusted
#>             1.0987             1.0987            -1.9341
#>              delta              beta15
#>            -6.4412            -0.4708
```

A simulation shows the bias in repeated sampling. We generate x_2 correlated with x_1 and $y = 1 + x_1 + 2x_2 + \varepsilon$, and fit with and without x_2:

```
set.seed(707)
sim <- replicate(2000, {
  x1 <- rnorm(100); x2 <- 0.6 * x1 + 0.8 * rnorm(100)     # Delta = 0.6
  y <- 1 + x1 + 2 * x2 + rnorm(100)
  c(short = coef(lm(y ~ x1))[["x1"]], long = coef(lm(y ~ x1 + x2))[["x1"]])
})
rbind(mean = rowMeans(sim), sd = apply(sim, 1, sd))       # theory: 1 + 0.6 * 2
↪  = 2.2 vs 1
```

```
#>       short   long
#> mean 2.1970 1.0020
#> sd   0.1925 0.1261
```

The short regression has mean $2.2 = 1 + 0.6 \times 2$, the long one 1. Note also that the standard deviation is *larger* for the short regression (0.19 against 0.13): the effect of the omitted variable is absorbed into the error term and inflates the residual variance.

Remark (the opposite error). Including a variable that is *irrelevant* ($\beta_2 = 0$) does not bias $\hat{\beta}_1$, but it inflates its variance when x_2 is correlated with x_1 (Chapter 6). Omission biases; over-inclusion costs variance. Both errors are about *which variables to adjust for*, and that is not decided by the data alone: one should adjust for confounders (common causes of the predictor of interest and the response) but *not* for consequences of the predictor of interest (mediators) or common effects of the predictor and the response (colliders), which would induce bias. The regression coefficient is a **conditional association**; interpreting it causally requires a causal model, not a better fit.

7.3 Categorical predictors

A categorical predictor (a **factor**) with K levels enters the design matrix through $K-1$ indicator (dummy) columns; the omitted level is the **baseline** (reference). With R's default *treatment coding*, for a factor `group` with levels A, B, C and baseline A:

level	intercept	$\mathbb{1}\{B\}$	$\mathbb{1}\{C\}$
A	1	0	0
B	1	1	0
C	1	0	1

Why $K-1$ and not K? If we included all K indicators and an intercept, the indicators would add up to the intercept column, $\operatorname{rank}(X) < p$ and the normal equations would have infinitely many solutions (the "dummy variable trap", Exercise 5 of Chapter 5).

Example 7.1 (One-way layout: PlantGrowth)**.** The data `PlantGrowth` contain the dried weight of 30 plants, 10 in each of three conditions: a control (`ctrl`) and two treatments (`trt1`, `trt2`). The model $y_{ij} = \mu_j + \varepsilon_{ij}$ is a linear model with a factor.

```
pg_fit <- lm(weight ~ group, data = PlantGrowth)
round(summary(pg_fit)$coefficients, 3)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)    5.032      0.197  25.527    0.000
#> grouptrt1     -0.371      0.279  -1.331    0.194
#> grouptrt2      0.494      0.279   1.772    0.088
```

```
tapply(PlantGrowth$weight, PlantGrowth$group, mean)
```

```
#>  ctrl  trt1  trt2
#> 5.032 4.661 5.526
```

```
anova(pg_fit)
```

```
#> Analysis of Variance Table
#>
#> Response: weight
#>           Df Sum Sq Mean Sq F value Pr(>F)
#> group      2   3.77   1.883    4.85  0.016 *
#> Residuals 27  10.49   0.389
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The intercept is the mean of the baseline (`ctrl`, 5.032); `grouptrt1` is the mean difference between `trt1` and `ctrl` (-0.371) and `grouptrt2` the difference between `trt2` and `ctrl` ($+0.494$). The individual t tests ($p = 0.19$, $p = 0.09$) compare each treatment with the control. The question "does the condition matter at all?" involves *both* dummies at once and is answered by the F test of the nested comparison with the intercept-only model: $F = 4.85$ on $(2, 27)$ degrees of freedom, $p = 0.016$. This is the one-way analysis of variance.

Proposition 7.1 (The one-way model fits group means)**.** *In the model with a factor only, the column space of X is the space of vectors that are constant within each group, so $\hat{y}_i$ is the mean of the group of observation i, and $\hat{\beta}$ is determined by those means: with treatment coding, $\hat{\beta}_0 = \bar{y}_{baseline}$ and $\hat{\beta}_k = \bar{y}_k - \bar{y}_{baseline}$.*

Proof. Every column of X is constant within groups and the K group indicators span exactly that space (dimension K, equal to p in the full-rank coding), so $\mathcal{C}(X)$ is the space of group-constant vectors. The projection of y onto it is the vector of group means: this vector is in the space and the residual $y_i - \bar{y}_{g(i)}$ sums to zero within each group, hence is orthogonal to every group indicator. Solving $\mu_{\text{baseline}} = \hat{\beta}_0$ and $\mu_k = \hat{\beta}_0 + \hat{\beta}_k$ for the fitted group means gives the stated formulas. □

Coding and contrasts

The coding is a *choice of parametrisation*: the model space $\mathcal{C}(X)$ does not depend on it. If two full-rank designs X and X^* have the same column space, then $X^* = XA$ for an invertible matrix A, and by the invariance exercise of Chapter 5 the fitted values, residuals, RSS, the

overall F test and the F test of the whole factor are identical; only the meaning of the individual coefficients changes. The main codings:

- `contr.treatment` (default): each level against the baseline. Change the baseline with `relevel(f, ref = "B")`.
- `contr.sum`: the coefficients are deviations of the level means from the *unweighted* average of the level means, and sum to zero; the last level's coefficient is minus the sum of the others. The intercept is the mean of the group means.
- `contr.poly`: orthogonal polynomial contrasts (linear, quadratic, ...) for *ordered* factors.
- Custom contrasts, e.g. "average of treatments minus control", $\frac{1}{2}(\mu_{trt1} + \mu_{trt2}) - \mu_{ctrl}$.

```
pg2 <- PlantGrowth; contrasts(pg2$group) <- contr.sum(3)
round(summary(lm(weight ~ group, data = pg2))$coefficients, 3)
↪  # sum coding
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)    5.073      0.114  44.573    0.000
#> group1        -0.041      0.161  -0.255    0.801
#> group2        -0.412      0.161  -2.560    0.016
```

```
pg3 <- transform(PlantGrowth, group = relevel(group, ref = "trt1"))
round(coef(lm(weight ~ group, data = pg3)), 3)
↪  # baseline trt1
```

```
#> (Intercept)   groupctrl   grouptrt2
#>       4.661       0.371       0.865
```

```
c(deviance_treatment = deviance(pg_fit), deviance_sum = deviance(lm(weight ~
↪  group, data = pg2)))
```

```
#> deviance_treatment       deviance_sum
#>              10.49              10.49
```

With sum coding, the intercept (5.073) is the average of the three group means and `group1` (-0.041) is the control's deviation from it. The same fitted values and RSS arise from every coding.

Remark. Always ask of a factor coefficient: *difference from what?* A t test on one dummy compares that level with the baseline and changes when the baseline changes; test a factor as a whole with the nested F test. When many pairwise comparisons between levels are done, the family-wise error rate must be controlled (for instance Tukey's honest significant differences, `TukeyHSD`): for `PlantGrowth` only `trt2` against `trt1` is significantly different (adjusted $p = 0.012$).

A factor can be combined with numeric predictors. In `mpg ~ wt + cyl`, treating the number of cylinders as a factor gives *parallel* lines: one slope for weight and three intercepts.

```
mt <- mtcars; mt$cyl <- factor(mt$cyl)
fit_cyl <- lm(mpg ~ wt + cyl, data = mt)
round(summary(fit_cyl)$coefficients, 3)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   33.991      1.888  18.006    0.000
#> wt            -3.206      0.754  -4.252    0.000
#> cyl6          -4.256      1.386  -3.070    0.005
```

```
#> cyl8          -6.071      1.652  -3.674    0.001
```

```
anova(lm(mpg ~ wt, data = mt), fit_cyl)                    # 2-df test for
↪  the factor
```

```
#> Analysis of Variance Table
#>
#> Model 1: mpg ~ wt
#> Model 2: mpg ~ wt + cyl
#>   Res.Df RSS Df Sum of Sq    F Pr(>F)
#> 1     30 278
#> 2     28 183  2      95.3 7.29 0.0028 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

At the same weight, six-cylinder cars average 4.26 mpg less than four-cylinder cars and eight-cylinder cars 6.07 mpg less; the joint F test gives $F = 7.29$ on $(2, 28)$ df, $p = 0.003$.

7.4 Interactions

So far each predictor has a *parallel* effect: the coefficient of x_1 does not depend on x_2. An **interaction** allows it to. For a continuous predictor x and a binary factor g (coded 0/1):

$$\mathrm{E}[y] = \beta_0 + \beta_1 x + \gamma g + \delta\, x g. \tag{7.1}$$

For $g = 0$ the line is $\beta_0 + \beta_1 x$; for $g = 1$ it is $(\beta_0 + \gamma) + (\beta_1 + \delta)x$. Thus δ is the *difference in slopes*, γ the difference in intercepts (the gap at $x = 0$), and $\gamma + \delta x$ is the gap at any x. The main effect γ is therefore **not** "the effect of g" when $\delta \neq 0$; it is the effect of g at $x = 0$, often an extrapolation. The lines cross at $x^* = -\gamma/\delta$.

```
mt$am <- factor(mt$am, labels = c("automatic", "manual"))
fit_am <- lm(mpg ~ wt * am, data = mt)
round(summary(fit_am)$coefficients, 3)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   31.416      3.020  10.402    0.000
#> wt            -3.786      0.786  -4.819    0.000
#> ammanual      14.878      4.264   3.489    0.002
#> wt:ammanual   -5.298      1.445  -3.667    0.001
```

```
b <- coef(fit_am)
c(slope_auto = b[["wt"]], slope_manual = b[["wt"]] + b[["wt:ammanual"]],
  crossing_weight = -b[["ammanual"]] / b[["wt:ammanual"]],
  gap_at_2 = b[["ammanual"]] + b[["wt:ammanual"]] * 2, gap_at_3 =
  ↪  b[["ammanual"]] + b[["wt:ammanual"]] * 3,
  gap_at_4 = b[["ammanual"]] + b[["wt:ammanual"]] * 4)
```

```
#>      slope_auto    slope_manual crossing_weight         gap_at_2
#>          -3.786          -9.084           2.808           4.282
#>        gap_at_3        gap_at_4
#>          -1.017          -6.315
```

```
grid <- expand.grid(wt = seq(1.5, 5.5, length.out = 60), am = levels(mt$am))
grid$mpg <- predict(fit_am, grid)
ggplot(mt, aes(wt, mpg, colour = am)) + geom_point() + geom_line(data = grid)
↪  +
  scale_colour_manual(values = c(automatic = ie_navy, manual = ie_coral), name
  ↪  = NULL) +
  labs(x = "Weight (1000 lb)", y = "Miles per gallon")
```

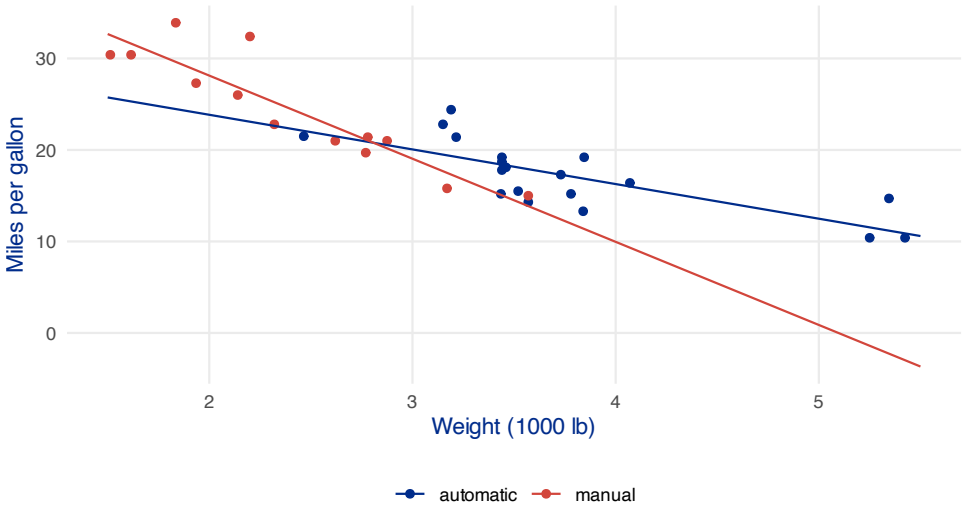


Figure 7.1: Fuel consumption against weight for automatic and manual cars, with the fitted lines of the interaction model. The lines cross at about 2.8 (2800 lb).

```
range(mtcars$wt[mtcars$am == 0]); range(mtcars$wt[mtcars$am == 1])
```

```
#> [1] 2.465 5.424
```

```
#> [1] 1.513 3.570
```

The fitted slope is -3.79 for automatics and -9.08 for manuals; the interaction is significant ($t = -3.67$, $p = 0.001$; as a nested test $F = 13.45 = t^2$). Reading the table correctly matters: the coefficient `ammanual` $= 14.88$ does *not* say that manual cars get 15 more mpg. The gap at weight w is $14.88 - 5.30w$: $+4.3$ mpg at 2000 lb, -1.0 at 3000 lb, -6.3 at 4000 lb. The lines cross at 2.81 (2810 lb). Manuals have higher fitted mpg only for lighter cars, and the observed weight ranges overlap only between 2.47 and 3.57 (the manuals are light: 1.5–3.6; the automatics heavy: 2.5–5.4), so "same weight" comparisons rest on few cars. Centring `wt` at 3 changes the main effect of `am` into the gap at 3000 lb (-1.02) and leaves the fitted lines unchanged.

For two continuous predictors, $\mathrm{E}[y] = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ gives $\partial \mathrm{E}[y]/\partial x_1 = \beta_1 + \beta_3 x_2$: the slope of x_1 is linear in x_2 and vice versa.

```
fit_wh <- lm(mpg ~ wt * hp, data = mtcars)
round(summary(fit_wh)$coefficients, 4)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  49.8084     3.6052  13.816    0e+00
#> wt           -8.2166     1.2697  -6.471    0e+00
#> hp           -0.1201     0.0247  -4.863    0e+00
#> wt:hp         0.0278     0.0074   3.753    8e-04
```

```
b <- coef(fit_wh)
round(setNames(b[["wt"]] + b[["wt:hp"]] * c(100, 150, 250), paste0("hp=",
↪  c(100, 150, 250))), 2)
```

```
#> hp=100 hp=150 hp=250
#>  -5.43  -4.04  -1.25
```

The weight slope is -5.43 mpg per 1000 lb at 100 hp, -4.04 at 150 hp and -1.25 at 250 hp: since $\hat{\beta}_3 > 0$, horsepower *weakens* the weight penalty. The "main effect" $\hat{\beta}_1 = -8.22$ is the slope at $\texttt{hp} = 0$, which does not exist. Report slopes at meaningful values of the other variable.

The hierarchical principle

Definition 7.1 (Hierarchical principle)**.** If a model contains a term of higher order (x^2, x_1x_2) it should also contain the lower-order terms (x; x_1 and x_2).

The reason is *invariance to the arbitrary origin of the predictors*, which has a clean mathematical form.

Proposition 7.2 (Shift invariance)**.** *For* $c \in \mathbb{R}$, $\operatorname{span}\{1, x, \ldots, x^d\} = \operatorname{span}\{1, x-c, \ldots, (x-c)^d\}$, *and* $\operatorname{span}\{1, x_1, x_2, x_1x_2\} = \operatorname{span}\{1, x_1 - c_1, x_2 - c_2, (x_1 - c_1)(x_2 - c_2)\}$. *In contrast, for generic data and* $c \neq 0$, $\operatorname{span}\{1, x^2\} \neq \operatorname{span}\{1, (x-c)^2\}$ *and* $\operatorname{span}\{1, x_1, x_1x_2\} \neq \operatorname{span}\{1, x_1 - c, (x_1 - c)x_2\}$ *(the interaction without the main effect of* x_2*).*

Proof. By the binomial theorem $(x-c)^k = \sum_{j=0}^{k} \binom{k}{j}(-c)^{k-j}x^j$ lies in $\operatorname{span}\{1, \ldots, x^d\}$ for $k \leq d$; exchanging the roles of x and $x - c$ (take $-c$) gives the reverse inclusion. For the interaction, $(x_1 - c_1)(x_2 - c_2) = x_1x_2 - c_2x_1 - c_1x_2 + c_1c_2$ is in the span of $\{1, x_1, x_2, x_1x_2\}$ and conversely. For the failures: $(x-c)^2 = x^2 - 2cx + c^2$ contains the term $-2cx$, and the function x is not a combination of 1 and x^2; likewise $(x_1 - c)x_2 = x_1x_2 - cx_2$ contains x_2, which is not a combination of $1, x_1, x_1x_2$ for generic data. A different span means different fitted values and a different RSS. □

When the spans are equal, the fitted values, RSS and tests are the same for any c; when they are not, they change with the units (degrees Celsius or Fahrenheit) or the centring, an arbitrary choice. The table (computed below) shows it for `mtcars`.

```
rss <- function(f, d = mtcars) round(deviance(lm(f, d)), 1)
data.frame(
  model = c("I(hp^2)", "hp + I(hp^2)", "wt + wt:hp (no hp)", "wt * hp"),
  RSS = c(rss(mpg ~ I(hp^2)), rss(mpg ~ hp + I(hp^2)), rss(mpg ~ wt + wt:hp),
  ↪  rss(mpg ~ wt * hp)),
  RSS_shifted = c(rss(mpg ~ I((hp - 150)^2)), rss(mpg ~ I(hp - 150) + I((hp -
  ↪  150)^2)),
                  rss(mpg ~ I(wt - 3) + I(wt - 3):hp), rss(mpg ~ I(wt - 3) *
                  ↪  I(hp - 150)))
)
```

```
#>                 model   RSS RSS_shifted
#> 1            I(hp^2) 628.7      1125.1
#> 2       hp + I(hp^2) 274.6       274.6
#> 3 wt + wt:hp (no hp) 239.3       236.9
```

```
#> 4            wt * hp 129.8       129.8
```

Non-hierarchical models change with the origin (628.7 against 1125.1 for the quadratic without the linear term); hierarchical ones do not. The exception is when $x = 0$ has a physical meaning and theory says the lower-order term is exactly zero (a dose with no effect at dose zero); this should be justified explicitly.

7.5 Comparing nested models

For $M_0 \subset M_1$ the F statistic of Chapter 5 is the tool; in R it is `anova(m0, m1)`. Two related functions answer *different* questions and are easily confused.

- `anova(m)` on one model gives **sequential** (Type I) sums of squares: each term is tested after the terms *above* it, so the order matters when predictors are correlated.
- `drop1(m, test = "F")` and `car::Anova(m)` give **marginal** (Type II) tests: each term is tested after all the others, respecting hierarchy (the main effect of `wt` is not dropped while `wt:am` is in the model).

```
m <- lm(mpg ~ wt + hp + am, data = mt)
anova(m)                                    # sequential
```

```
#> Analysis of Variance Table
#>
#> Response: mpg
#>           Df Sum Sq Mean Sq F value  Pr(>F)
#> wt         1    848     848  131.66 4.3e-12 ***
#> hp         1     83      83   12.93  0.0012 **
#> am         1     15      15    2.29  0.1413
#> Residuals 28    180       6
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(lm(mpg ~ hp + wt + am, data = mt))    # different order, different rows
↪  (first two)
```

```
#> Analysis of Variance Table
#>
#> Response: mpg
#>           Df Sum Sq Mean Sq F value  Pr(>F)
#> hp         1    678     678  105.35 5.4e-11 ***
#> wt         1    253     253   39.23 9.0e-07 ***
#> am         1     15      15    2.29    0.14
#> Residuals 28    180       6
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(m)                                    # each term after the others (Type
↪  II)
```

```
#> Anova Table (Type II tests)
#>
#> Response: mpg
#>           Sum Sq Df F value  Pr(>F)
```

```
#> wt          65.1  1   10.12 0.00357 **
#> hp          98.0  1   15.22 0.00055 ***
#> am          14.8  1    2.29 0.14127
#> Residuals  180.3 28
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F tests only compare *nested* models. For non-nested candidates (different predictors) we need information criteria or out-of-sample error.

7.6 Model selection

Suppose now that many candidate predictors are available and the aim is a good model. Three ideas organise the rest of the chapter: (i) fit to the data used for fitting *always improves* when columns are added, so we need a criterion that penalises complexity; (ii) there are two broad ways to do this, estimating the prediction error of each candidate or maximising a penalised likelihood; (iii) searching over many candidates and then using the selected model for inference is dangerous.

The bias–variance trade-off and Mallows' C_p

Let $\mu = \mathrm{E}[y]$ be the true mean (which need not lie in any model's column space) and consider a candidate submodel with design X_p of p columns and hat matrix H_p. The fitted vector is $\hat{y}_p = H_p y$. We measure its quality by the **total squared error** $\mathrm{E}\|\hat{y}_p - \mu\|^2$, the expected squared distance to the truth at the n design points.

Theorem 7.3 (Bias and variance of a submodel; Mallows' C_p)**.** *Let $y = \mu + \varepsilon$ with $\mathrm{E}[\varepsilon] = 0$, $\mathrm{Var}(\varepsilon) = \sigma^2 I$ and write $b_p^2 = \|(I - H_p)\mu\|^2$ for the squared bias of the submodel. Then*

$$\mathrm{E}\|\hat{y}_p - \mu\|^2 = b_p^2 + p\sigma^2, \qquad \mathrm{E}[\mathrm{RSS}_p] = b_p^2 + (n-p)\sigma^2.$$

Consequently $\mathrm{E}\|\hat{y}_p - \mu\|^2 = \mathrm{E}[\mathrm{RSS}_p] - (n-2p)\sigma^2$ and, replacing σ^2 by the estimate $\hat{\sigma}^2_{full}$ from the model with all candidate predictors,

$$C_p = \frac{\mathrm{RSS}_p}{\hat{\sigma}^2_{full}} - n + 2p$$

estimates $\mathrm{E}\|\hat{y}_p - \mu\|^2/\sigma^2$. If the submodel is unbiased ($b_p = 0$), then $\mathrm{E}[C_p] \approx p$.

Proof. $\hat{y}_p - \mu = (H_p\mu - \mu) + H_p\varepsilon = -(I - H_p)\mu + H_p\varepsilon$. The two terms are orthogonal in the sense that their inner product is $-\mu^\top (I - H_p) H_p \varepsilon = 0$ (as $(I - H_p)H_p = 0$), so $\|\hat{y}_p - \mu\|^2 = b_p^2 + \varepsilon^\top H_p \varepsilon$ and the expectation of the second term is $\sigma^2 \operatorname{tr} H_p = p\sigma^2$ by the quadratic-form lemma of Chapter 5. Next, $e_p = (I - H_p)y = (I - H_p)\mu + (I - H_p)\varepsilon$ and $\mathrm{RSS}_p = b_p^2 + 2\mu^\top (I - H_p)\varepsilon + \varepsilon^\top (I - H_p)\varepsilon$; the middle term has mean zero and the last has mean $\sigma^2(n - p)$. Subtracting the two expectations gives $(n - 2p)\sigma^2$. Dividing by σ^2 and substituting $\hat{\sigma}^2_{\mathrm{full}}$ (unbiased for σ^2 if the full model is unbiased) gives C_p. For $b_p = 0$, $\mathrm{E}[\mathrm{RSS}_p] \approx \sigma^2(n - p)$ makes $C_p \approx (n - p) - n + 2p = p$. □

The theorem is the bias–variance decomposition in matrix form: a bigger model has smaller bias b_p^2 but larger variance $p\sigma^2$. C_p is computed with p counting the intercept, and subsets with C_p close to p (or below) are the good ones; a C_p well above p signals bias. A C_p *below* p is not evidence of overfitting; it is sampling variation around p.

Information criteria: AIC and BIC

The log-likelihood $\ell(\hat\theta)$ of a model always improves with more parameters, so we subtract a penalty:

$$\mathrm{AIC} = -2\ell(\hat\theta) + 2k, \qquad \mathrm{BIC} = -2\ell(\hat\theta) + k\log n, \tag{7.2}$$

where k is the number of estimated parameters. For a Gaussian linear model with p columns, $k = p + 1$ (σ^2 counts) and by the maximum-likelihood result of Chapter 5 $-2\ell(\hat\theta) = n\log(2\pi) + n\log(\mathrm{RSS}/n) + n$, so

$$\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2p + \text{constant}, \qquad \mathrm{BIC} = n\log(\mathrm{RSS}/n) + p\log n + \text{constant},$$

where the constant depends on n only. *Smaller is better*, only differences between models fitted to the **same response and the same rows** matter, and differences below about 2 are not informative.

The two criteria answer different questions.

- **AIC** estimates, up to a constant, the expected Kullback–Leibler divergence between the fitted model and the truth: the *predictive* quality of the model (Akaike, 1974). The penalty $2k$ is the asymptotic *optimism* of the maximised log-likelihood. In the linear model with known σ^2 this is exact: for an independent copy y^* of the data, $\mathrm{E}\|y^* - \hat y_p\|^2 - \mathrm{E}\|y - \hat y_p\|^2 = \{n\sigma^2 + b_p^2 + p\sigma^2\} - \{b_p^2 + (n-p)\sigma^2\} = 2p\sigma^2$ by Theorem 7.3. The in-sample error is optimistic by $2p\sigma^2$, and "$\mathrm{RSS}/\sigma^2 + 2p$" is therefore an unbiased estimate of the out-of-sample error, which is C_p again. AIC is asymptotically efficient for prediction but does not consistently select the true model: it keeps weak terms. For small n use the correction $\mathrm{AICc} = \mathrm{AIC} + 2k(k+1)/(n-k-1)$ (Burnham and Anderson, 2004).
- **BIC** (Schwarz, 1978) approximates $-2\log$ of the marginal likelihood of the model (a Laplace approximation), so it is a Bayesian model-comparison criterion, and is *consistent*: if the true model is among the candidates, it is selected with probability tending to one. Its penalty $\log n$ exceeds 2 as soon as $n \geq 8$, so BIC prefers smaller models than AIC; for $n = 97$ the penalty per parameter is 4.57.

In R, `AIC(fit)` and `BIC(fit)` include the full constants, while `extractAIC(fit)` (used by `step`) drops the terms that do not depend on the model: $n\log(\mathrm{RSS}/n) + 2p$. They differ by $n\log(2\pi) + n + 2$ and give the same ranking.

```
full <- lm(lpsa ~ ., data = prostate)
n <- nrow(prostate); p_full <- length(coef(full))
RSS <- deviance(full)
c(AIC = AIC(full), AIC_by_hand = n * log(2 * pi) + n * log(RSS / n) + n + 2 *
↪  (p_full + 1),
  extractAIC = extractAIC(full)[2], difference = AIC(full) -
  ↪  extractAIC(full)[2],
  const = n * log(2 * pi) + n + 2)
```

```
#>         AIC AIC_by_hand  extractAIC  difference       const
#>      218.95      218.95      -58.32      277.27      277.27
```

Cross-validation

The most direct way to estimate predictive performance is to *hold out* data. In K-fold cross-validation the sample is split into K folds; each fold is predicted by the model fitted to the others, and the squared errors are averaged. For least squares and $K = n$ (leave-one-out) the

result is available in closed form: the prediction error for case i is $e_i/(1-h_{ii})$, and

$$\text{PRESS} = \sum_i \left(\frac{e_i}{1-h_{ii}}\right)^2,$$

as derived in Chapter 6. For any *selection* procedure, honest assessment requires that **the selection itself is repeated inside each fold**; otherwise the held-out data have influenced the choice.

Search strategies and the caveats of stepwise selection

With q candidate predictors there are 2^q subsets. *Best-subset* search (`leaps::regsubsets`) is feasible for q up to about 30; *stepwise* procedures add (forward) or remove (backward) one variable at a time by the criterion, and `step()` does this by AIC (penalty `k = 2`) or BIC (`k = log(n)`).

```
m_aic <- step(full, trace = 0)
m_bic <- step(full, trace = 0, k = log(n))
sel <- function(m) paste(setdiff(names(coef(m)), "(Intercept)"), collapse = "
↪  + ")
data.frame(model = c("full", "step AIC", "step BIC"),
           predictors = c(sel(full), sel(m_aic), sel(m_bic)),
           k = c(10, 7, 5),
           AIC = round(c(AIC(full), AIC(m_aic), AIC(m_bic)), 2),
           BIC = round(c(BIC(full), BIC(m_aic), BIC(m_bic)), 2))
```

```
#>      model                                                        predictors
#> 1     full lcavol + lweight + age + lbph + svi + lcp + gleason + pgg45
#> 2 step AIC                          lcavol + lweight + age + lbph + svi
#> 3 step BIC                                    lcavol + lweight + svi
#>    k   AIC   BIC
#> 1 10 218.9 244.7
#> 2  7 215.9 233.9
#> 3  5 216.6 229.5
```

```
library(leaps)
rs <- summary(regsubsets(lpsa ~ ., data = prostate, nvmax = 8))
data.frame(p = 2:9, RSS = round(rs$rss, 2), Cp = round(rs$cp, 2), adjR2 =
↪  round(rs$adjr2, 3), BIC = round(rs$bic, 1))
```

```
#>   p   RSS    Cp adjR2   BIC
#> 1 2 58.91 24.39 0.535 -66.1
#> 2 3 52.97 14.54 0.577 -71.8
#> 3 4 47.78  6.22 0.614 -77.2
#> 4 5 46.48  5.63 0.621 -75.3
#> 5 6 45.53  5.72 0.625 -72.8
#> 6 7 44.87  6.40 0.626 -69.6
#> 7 8 44.20  7.08 0.627 -66.5
#> 8 9 44.16  9.00 0.623 -62.0
```

`regsubsets` reports for each model size the best subset: its C_p is smallest (5.6) at $p = 5$ and close to p for $p = 5$–7, and BIC is smallest at $p = 4$ (three predictors: `lcavol`, `lweight`, `svi`, the same model chosen by `step` with BIC). AIC chooses a larger model with five predictors, as expected.

Warning (stepwise selection). (i) *Greedy*: one variable at a time; it can miss the best subset and forward and backward searches may disagree. (ii) *Unstable*: small changes in the data change the selected set. (iii) **Post-selection inference is invalid**: the t statistics, p-values and confidence intervals printed for the selected model ignore the fact that the model was chosen by looking at the same data; p-values are too small and intervals too narrow (Berk et al., 2013).

The simulation below makes the point (iii) concrete. The response is pure noise, unrelated to 20 noise predictors, $n = 100$; we select by AIC and count how many of the selected predictors appear "significant" at 5% in the final model.

```
set.seed(2027)
res <- replicate(100, {
  d <- as.data.frame(matrix(rnorm(100 * 21), 100)); names(d)[1] <- "y"
  m <- step(lm(y ~ ., data = d), trace = 0)
  k <- length(coef(m)) - 1
  sig <- if (k > 0) sum(summary(m)$coefficients[-1, 4] < 0.05) else 0
  c(kept = k, significant = sig)
})
c(mean_kept = mean(res["kept", ]), mean_significant = mean(res["significant",
↪  ]),
  share_significant = sum(res["significant", ]) / sum(res["kept", ]))
```

```
#>         mean_kept  mean_significant share_significant
#>            3.9300            1.4800            0.3766
```

On average about four of the twenty noise predictors are kept, and more than a third of the kept predictors look significant at the 5% level, against the nominal 5%. The remedies are: *decide the model in advance* (explanation); *split the sample* (select on one half, estimate and test on the other); or judge the selected procedure by *out-of-sample error* (prediction).

7.7 Shrinkage: ridge, lasso and elastic net

Gauss–Markov says that $\hat{\beta}$ has the smallest variance among *unbiased* linear estimators. A biased estimator may have a smaller *mean squared error*, $\mathrm{MSE} = \mathrm{bias}^2 + \mathrm{variance}$, and the gain is largest when $X^\top X$ is nearly singular (collinear predictors) or when p is large relative to n. Shrinkage methods deliberately bias the coefficients towards zero by penalising their size.

Ridge regression

Definition 7.2 (Ridge regression)**.** For $\lambda \geq 0$ the ridge estimator minimises $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$ over β (with the predictors centred and standardised and the intercept left unpenalised, so that we can take y centred and drop the intercept). Setting the gradient to zero gives

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y.$$

For $\lambda > 0$ the matrix $X^\top X + \lambda I$ is positive definite, hence invertible *even when* $X^\top X$ *is singular* ($p > n$, exact collinearity), so ridge is defined where least squares is not. The singular value decomposition $X = UDV^\top$ with singular values $d_1 \geq \cdots \geq d_p > 0$ makes the effect transparent: $\hat{\beta}_\lambda = V\,\mathrm{diag}\big(\frac{d_j}{d_j^2+\lambda}\big)U^\top y$, while least squares is $V\mathrm{diag}(1/d_j)U^\top y$. In the coordinates $\alpha = V^\top\beta$, each component is shrunk by the factor $d_j^2/(d_j^2 + \lambda) \in (0, 1)$, most strongly for the directions with small singular value, which are precisely the poorly determined ones.

Theorem 7.4 (Ridge can beat least squares)**.** *For the model $y = X\beta + \varepsilon$ with A1–A3, $\hat\beta_\lambda$ has*

$$\text{bias}^2 = \lambda^2 \sum_{j=1}^{p} \frac{\alpha_j^2}{(d_j^2+\lambda)^2}, \qquad \textit{total variance} = \sigma^2 \sum_{j=1}^{p} \frac{d_j^2}{(d_j^2+\lambda)^2},$$

and there exists $\lambda > 0$ such that $\mathrm{E}\|\hat\beta_\lambda - \beta\|^2 < \mathrm{E}\|\hat\beta - \beta\|^2 = \sigma^2 \sum_j d_j^{-2}$.

Proof. In the V-coordinates $\hat\alpha_{\lambda,j} = \frac{d_j}{d_j^2+\lambda} u_j^\top y$ with $u_j^\top y \sim (d_j\alpha_j, \sigma^2)$ (the u_j are orthonormal and $u_j^\top X\beta = d_j\alpha_j$). Hence $\mathrm{E}[\hat\alpha_{\lambda,j}] = \frac{d_j^2}{d_j^2+\lambda}\alpha_j$, with bias $-\frac{\lambda}{d_j^2+\lambda}\alpha_j$ and variance $\sigma^2 \frac{d_j^2}{(d_j^2+\lambda)^2}$. Orthogonality of V preserves the squared norm of the error, giving the two formulas. At $\lambda = 0$ the total variance is $\sigma^2 \sum_j d_j^{-2}$ and the bias is zero. Differentiate in λ: $\frac{d}{d\lambda}\frac{d_j^2}{(d_j^2+\lambda)^2} = -\frac{2d_j^2}{(d_j^2+\lambda)^3}$, which equals $-2d_j^{-4}$ at $\lambda = 0$, while the squared bias is of order λ^2 and has derivative 0 at $\lambda = 0$. So the total MSE has derivative $-2\sigma^2 \sum_j d_j^{-4} < 0$ at $\lambda = 0$ and decreases for small $\lambda > 0$. □

The shrinkage is controlled by one number, the **effective degrees of freedom** $\mathrm{edf}(\lambda) = \mathrm{tr}\{X(X^\top X + \lambda I)^{-1} X^\top\} = \sum_j d_j^2/(d_j^2+\lambda)$, which equals p at $\lambda = 0$ and decreases to 0. The same quantity reappears for smoothers (Chapter 14). The theorem does not say how to find the λ that helps; this is done by cross-validation.

The lasso and elastic net

The **lasso** (Tibshirani, 1996) replaces the squared penalty by $\lambda\|\beta\|_1 = \lambda \sum_j |\beta_j|$. This penalty has a corner at zero, so the solution typically has some coefficients *exactly* zero: the lasso estimates and selects variables simultaneously. There is no closed form in general, but in the special case of orthonormal predictors, it is explicit.

Theorem 7.5 (Lasso with orthonormal predictors: soft thresholding)**.** *Suppose $X^\top X/n = I_p$ (centred, standardised and orthogonal columns). The minimiser of $\frac{1}{2n}\|y - X\beta\|^2 + \lambda\|\beta\|_1$ is*

$$\hat\beta_j^{\,lasso} = S_\lambda(z_j) = \mathrm{sign}(z_j)\,(|z_j| - \lambda)_+, \qquad z_j = \frac{x_j^\top y}{n} = \hat\beta_j^{\,LS}.$$

Proof. Expand $\frac{1}{2n}\|y - X\beta\|^2 = \frac{1}{2n}\|y\|^2 - \beta^\top \frac{X^\top y}{n} + \frac{1}{2}\beta^\top \frac{X^\top X}{n}\beta = \text{const} - \sum_j z_j\beta_j + \frac{1}{2}\sum_j \beta_j^2$. The objective separates over j: minimise $g(b) = \frac{1}{2}(b - z_j)^2 + \lambda|b|$ (up to a constant). For $b > 0$, $g'(b) = b - z_j + \lambda = 0$ gives $b = z_j - \lambda$, valid if $z_j > \lambda$. For $b < 0$, $g'(b) = b - z_j - \lambda = 0$ gives $b = z_j + \lambda$, valid if $z_j < -\lambda$. If $|z_j| \le \lambda$, then $g(b) \ge g(0)$ for all b (the right derivative at 0, $-z_j + \lambda$, is ≥ 0 and the left derivative $-z_j - \lambda$ is ≤ 0), so the minimum is at 0. □

So the lasso *shrinks every least-squares coefficient by λ and sets to zero those smaller than λ in absolute value*, whereas ridge would rescale them, $\hat\beta_j^{\text{ridge}} = \hat\beta_j^{\text{LS}}/(1+\lambda)$ in this orthonormal case (Exercise 7.9). The **elastic net** (Zou and Hastie, 2005) mixes the two penalties; it retains the sparsity of the lasso but spreads weight over groups of correlated predictors instead of picking one arbitrarily.

The glmnet objective

For `family = "gaussian"`, `glmnet` (Friedman et al., 2010) minimises over (β_0, β)

$$\frac{1}{2n}\sum_{i=1}^{n}\left(y_i - \beta_0 - x_i^\top \beta\right)^2 + \lambda\Big[\frac{1-\alpha}{2}\|\beta\|_2^2 + \alpha\|\beta\|_1\Big]. \tag{7.3}$$

Here $\alpha = 1$ is the lasso, $\alpha = 0$ ridge and $0 < \alpha < 1$ the elastic net; $\lambda \geq 0$ is the overall strength; the intercept is not penalised; and the loss is $\mathrm{RSS}/(2n)$, so λ is on a per-observation scale. By default predictors are standardised internally (`standardize = TRUE`) and the coefficients are reported on the original scale.

Standardisation matters because the penalty is not scale invariant: if a predictor is measured in different units, its coefficient and hence its penalty change. Standardising puts all predictors on the same footing (for dummy variables, think about their scale too). In a train/test analysis, the standardisation (means, standard deviations) must be computed from the *training* data only; passing only the training matrix to `glmnet` does this automatically.

Choosing λ. `cv.glmnet` performs K-fold cross-validation along a grid of λ values and reports `lambda.min` (smallest cross-validated error) and `lambda.1se` (the largest λ whose error is within one standard error of the minimum: a sparser model that is statistically indistinguishable). The folds are random, so **fix them with `foldid`**: this makes the results reproducible and, crucially, lets different values of α be compared on the *same* folds.

```
x <- as.matrix(prostate[, names(prostate) != "lpsa"]); y <- prostate$lpsa
set.seed(606)
foldid <- sample(rep(1:10, length.out = nrow(x)))
cv_lasso <- cv.glmnet(x, y, alpha = 1, foldid = foldid)
cv_ridge <- cv.glmnet(x, y, alpha = 0, foldid = foldid)
cv_enet  <- cv.glmnet(x, y, alpha = 0.5, foldid = foldid)
rbind(lasso = c(lambda.min = cv_lasso$lambda.min, lambda.1se =
↪  cv_lasso$lambda.1se),
      ridge = c(cv_ridge$lambda.min, cv_ridge$lambda.1se),
      enet  = c(cv_enet$lambda.min, cv_enet$lambda.1se))
```

```
#>       lambda.min lambda.1se
#> lasso    0.07508     0.2293
#> ridge    0.13430     1.1412
#> enet     0.12467     0.3807
```

```
round(cbind(OLS = coef(full), ridge = coef(cv_ridge, s = "lambda.1se")[, 1],
            lasso = coef(cv_lasso, s = "lambda.1se")[, 1], enet =
            ↪  coef(cv_enet, s = "lambda.1se")[, 1]), 3)
```

```
#>                OLS  ridge lasso  enet
#> (Intercept)  0.669  0.406 1.354 1.165
#> lcavol       0.587  0.250 0.456 0.395
#> lweight      0.454  0.290 0.121 0.187
#> age         -0.020 -0.001 0.000 0.000
#> lbph         0.107  0.050 0.000 0.000
#> svi          0.766  0.432 0.307 0.407
#> lcp         -0.105  0.079 0.000 0.000
#> gleason      0.045  0.086 0.000 0.000
#> pgg45        0.005  0.003 0.000 0.000
```

At `lambda.1se` the lasso keeps three predictors (`lcavol`, `lweight`, `svi`), the same as BIC; the elastic net keeps the same three, with coefficients between the lasso's and the ridge's, and ridge keeps all eight but shrinks them. Figure 7.2 shows the coefficient path and the cross-validation curve.

```
op <- par(mfrow = c(1, 2), mar = c(4.2, 4.2, 2, 1))
plot(cv_lasso$glmnet.fit, xvar = "lambda", label = TRUE)
plot(cv_lasso); par(op)
```

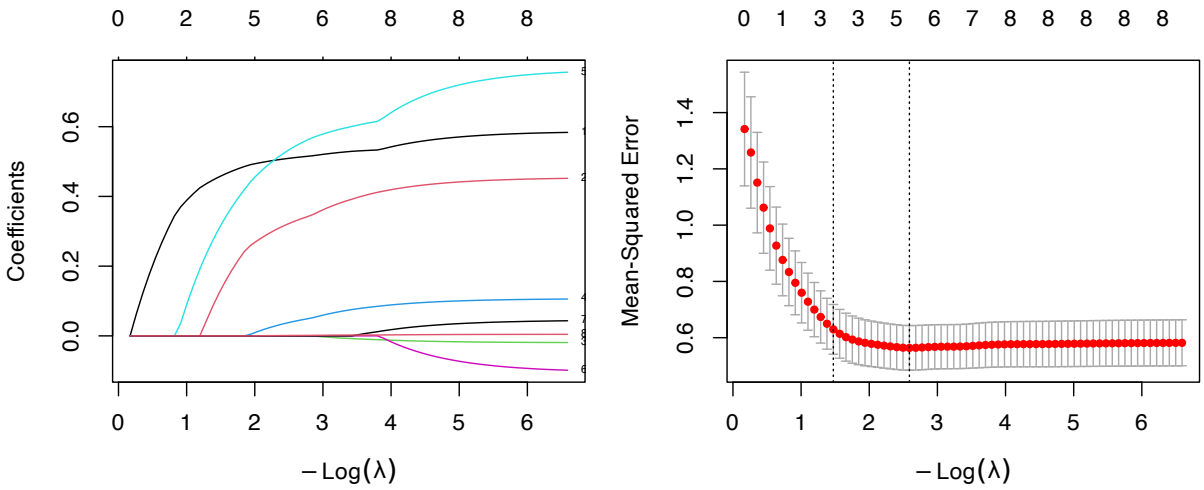


Figure 7.2: Lasso on the prostate data. Left: coefficient paths as lambda decreases (predictors enter one by one; lcavol first). Right: 10-fold cross-validation curve with lambda.min and lambda.1se.

How good is the prediction? One split is not enough

To judge the methods honestly we hold out data and use them *once*. A single split of 97 men is very noisy; repeating the split shows how noisy. In each of 50 random splits (67 training, 30 test) we select by `step` (AIC and BIC) and by the lasso with cross-validated λ, all using only the training rows, and compute the test error.

```
set.seed(1)
res <- t(sapply(1:50, function(r) {
  id <- sample(nrow(prostate), 30); tr <- prostate[-id, ]; te <- prostate[id,
↪  ]
  xtr <- as.matrix(tr[, names(tr) != "lpsa"]); xte <- as.matrix(te[, names(te)
↪  != "lpsa"])
  ft <- lm(lpsa ~ ., data = tr)
  fa <- step(ft, trace = 0); fb <- step(ft, trace = 0, k = log(nrow(tr)))
  cvr <- cv.glmnet(xtr, tr$lpsa, alpha = 1, foldid = sample(rep(1:10,
↪  length.out = nrow(tr))))
  mse <- function(pred) mean((te$lpsa - pred)^2)
  c(OLS = mse(predict(ft, te)), step_AIC = mse(predict(fa, te)), step_BIC =
  ↪  mse(predict(fb, te)),
    lasso_min = mse(predict(cvr, xte, s = "lambda.min")), lasso_1se =
    ↪  mse(predict(cvr, xte, s = "lambda.1se")))
}))
rbind(mean = colMeans(res), sd = apply(res, 2, sd))
```

```
#>          OLS step_AIC step_BIC lasso_min lasso_1se
#> mean 0.55878  0.57466   0.5822    0.5524    0.6145
#> sd   0.09886  0.09496   0.1002    0.1036    0.1527
```

The mean test errors differ by a few hundredths (0.55 to 0.61) while the split-to-split standard deviation is 0.10–0.15: with $n = 97$ and eight predictors, the methods are indistinguishable in prediction, and a single test set cannot rank them. Shrinkage and selection pay off when the number of predictors is large relative to n or the predictors are strongly collinear. In the exercises, `seatpos` has eight predictors for 38 cars, two of which are almost identical.

7.8 Explanation versus prediction

	Explanation	Prediction
Goal	Estimate and interpret specific β_j	Small error for new y
Model choice	Fixed in advance from theory; include confounders	Data-driven; anything that predicts
Bias	Must be small for the target β_j	May trade some bias for less variance
Inference	Confidence intervals and tests are central	Validated by held-out error
Tools	F tests, diagnostics, contextual knowledge	AIC, CV, ridge, lasso

Shmueli (2010) and Breiman (2001) argue that the two goals lead to different models, and that the habit of selecting by predictive criteria and then *interpreting* the selected coefficients mixes them in a way that serves neither. Decide which problem you have *before* selecting. In an explanatory study, the order of work is: (1) state the question and the variables that must be adjusted for; (2) fix the model; (3) check the assumptions (Chapter 6); (4) report estimates with intervals and the sensitivity to the specification. In a predictive study: (1) define the loss; (2) split or cross-validate; (3) tune everything (variables, λ, transformations) *inside* the training data; (4) evaluate once on held-out data; (5) report the expected error and its uncertainty.

7.9 In R: the functions used in class

Task	Code
Factors, baseline, contrasts	`factor()`, `relevel(f, ref =)`, `contrasts(f) <- contr.sum(K)`, `lm(..., contrasts = list(f = "contr.sum"))`
Interactions	`y ~ x * g` (main effects and interaction), `y ~ x:g` (interaction only)
Nested test, sequential vs marginal	`anova(m0, m1)`, `anova(m)`, `drop1(m, test = "F")`, `car::Anova(m)`
Pairwise comparisons	`TukeyHSD(aov(y ~ g))`
Criteria	`AIC(m)`, `BIC(m)`, `extractAIC(m)`
Stepwise	`step(full, trace = 0)` (AIC), `step(full, trace = 0, k = log(n))` (BIC)

Task	Code
Best subsets	`leaps::regsubsets(y ~ ., data, nvmax = q)`
Ridge, lasso, elastic net	`glmnet(x, y, alpha = 1)`, `cv.glmnet(x, y, alpha = 1, foldid = foldid)`, `coef(cv, s = "lambda.1se")`

7.10 Common pitfalls

Warning. Reading the main effect in a model with an interaction as "the" effect. It is the effect where the other variable is 0, which may not exist. Compute slopes or gaps at meaningful values, and say where lines cross.

Warning. Testing a factor with separate t tests, or reading a dummy coefficient without naming the baseline.

Warning. Dropping x but keeping x^2 or x_1x_2. The model then depends on the units and the origin of x.

Warning. Interpreting an adjusted coefficient causally, or adjusting for a consequence of the predictor of interest.

Warning. Comparing AIC or BIC across models fitted to different responses (y against $\log y$) or different rows (missing values). They are only comparable for the same data.

Warning. Reporting the p-values and confidence intervals of the model chosen by `step` as if it had been pre-specified.

Warning. Standardising the data, or choosing λ, using the test set; or not fixing `foldid` so that the chosen λ changes from run to run.

7.11 Chapter summary

- $\hat{\beta}_j$ is the slope of the residuals of y on the residuals of x_j after adjusting for the other predictors (Frisch–Waugh–Lovell). It depends on which other predictors are in the model.
- Omitting a relevant variable X_2 biases $\hat{\beta}_1$ by $\Delta\beta_2$, and in the sample $\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\Delta}\hat{\beta}_2$ exactly: this explains sign reversals. Over-inclusion costs variance, not bias.
- A factor with K levels uses $K - 1$ dummies. The column space, fitted values and the F test for the factor do not depend on the coding; the meaning of each coefficient does. Test factors with the nested F test.
- With an interaction the effect of one variable depends on the other; main effects are effects at the reference value. Keep lower-order terms (hierarchical principle): the span of $\{1, x, x^2\}$ is invariant to shifts of x, that of $\{1, x^2\}$ is not.
- Selection criteria: $C_p = \mathrm{RSS}_p/\hat{\sigma}^2_{\text{full}} - n + 2p$, AIC $= -2\ell + 2k$ (prediction), BIC $= -2\ell + k \log n$ (consistent). Stepwise selection is greedy, unstable and invalidates the usual inference.
- Ridge: $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ (can beat OLS in MSE); lasso: soft thresholding, exact zeros; elastic net: both. glmnet minimises $\mathrm{RSS}/(2n) + \lambda[\frac{1-\alpha}{2}\|\beta\|_2^2 + \alpha\|\beta\|_1]$; choose λ by cross-validation with fixed folds; standardise on training data.
- Explanation: fix the model from theory, report intervals. Prediction: choose by held-out error, tune inside the training data.

7.12 Exercises

Exercise 7.1. Frisch–Waugh–Lovell. (a) For $X = [\mathbf{1}, x_1, x_2]$, show that $\hat\beta_1 = \dfrac{\sum_i \tilde x_{i1} y_i}{\sum_i \tilde x_{i1}^2}$ where $\tilde x_1$ are the residuals of x_1 regressed on $(1, x_2)$. (b) Check numerically with `mtcars` for the coefficient of `wt` in `mpg ~ wt + hp`. (c) Deduce that $\mathrm{Var}(\hat\beta_1) = \sigma^2 / \sum_i \tilde x_{i1}^2$ and explain what happens when x_1 and x_2 are almost collinear.

Solution.

(a) By Theorem 7.1 with $X_1 = [\mathbf{1}, x_2]$ and $X_2 = x_1$: $\hat\beta_1 = (x_1^\top M_1 x_1)^{-1} x_1^\top M_1 y = \tilde x_1^\top y / \tilde x_1^\top \tilde x_1$, because M_1 is symmetric idempotent and $\tilde x_1 = M_1 x_1$ (so $x_1^\top M_1 y = \tilde x_1^\top y$ and $x_1^\top M_1 x_1 = \tilde x_1^\top \tilde x_1$). (b) See the code below: both give -3.878. (c) $\mathrm{Var}(\hat\beta_1) = \sigma^2 \tilde x_1^\top \tilde x_1 / (\tilde x_1^\top \tilde x_1)^2 = \sigma^2 / \sum \tilde x_{i1}^2$. If x_1 is almost a linear combination of $(1, x_2)$ its residual $\tilde x_1$ is tiny, so the variance is huge: this is collinearity, and $\sum \tilde x_{i1}^2 = (1 - R_1^2) \sum (x_{i1} - \bar x_1)^2$ links it to the VIF.

```
rx <- resid(lm(wt ~ hp, data = mtcars))
c(fwl = sum(rx * mtcars$mpg) / sum(rx^2), lm = coef(lm(mpg ~ wt + hp, data =
↪  mtcars))[["wt"]])
```

```
#>    fwl     lm
#> -3.878 -3.878
```

Exercise 7.2. Omitted-variable bias. In `mtcars`, the coefficient of `wt` is -5.344 in `mpg ~ wt` and -3.878 in `mpg ~ wt + hp`; the coefficient of `hp` in the second model is -0.0318. (a) Use Theorem 7.2 to compute the slope of the regression of `hp` on `wt` and check it. (b) Explain the sign of the bias in words. (c) Under which conditions would the two `wt` coefficients be equal?

Solution.

(a) $\tilde\beta_{\texttt{wt}} = \hat\beta_{\texttt{wt}} + \hat\delta \hat\beta_{\texttt{hp}}$, so $\hat\delta = (-5.344 + 3.878)/(-0.03177) = 46.16$ horsepower per 1000 lb (the code confirms). (b) The bias is $\hat\delta \hat\beta_{\texttt{hp}} = (46.16)(-0.0318) = -1.47 < 0$: heavier cars have more powerful engines ($\hat\delta > 0$) and more horsepower goes with lower mpg ($\hat\beta_{\texttt{hp}} < 0$), so the marginal weight slope mixes the weight effect with the engine effect and is more negative. (c) If $\hat\beta_{\texttt{hp}} = 0$ (horsepower adds nothing given weight), or $\hat\delta = 0$ (weight and horsepower uncorrelated in the sample).

```
m1 <- lm(mpg ~ wt, mtcars); m2 <- lm(mpg ~ wt + hp, mtcars); a <- lm(hp ~ wt,
↪  mtcars)
c(delta = coef(a)[["wt"]], marginal = coef(m1)[["wt"]], reconstructed =
↪  coef(m2)[["wt"]] + coef(m2)[["hp"]] * coef(a)[["wt"]])
```

```
#>         delta      marginal reconstructed
#>        46.160        -5.344        -5.344
```

Exercise 7.3. Coding of a factor. For `PlantGrowth` (three groups of 10; group means 5.032, 4.661, 5.526), write the design matrix for treatment coding and for sum coding. (a) Express the coefficients in each coding in terms of the group means. (b) Show that $\hat y$ is identical. (c) Compute the contrast "average of the two treatments minus the control" and

its standard error given $\hat{\sigma} = 0.623$ on 27 df.

Solution.

(a) Treatment coding (baseline ctrl): $\hat{\beta}_0 = 5.032$, $\hat{\beta}_{trt1} = -0.371$, $\hat{\beta}_{trt2} = +0.494$. Sum coding (columns $(1,0),(0,1),(-1,-1)$ for ctrl, trt1, trt2): $\hat{\beta}_0 = \frac{1}{3}(5.032 + 4.661 + 5.526) = 5.073$, $\hat{\beta}_1 = 5.032 - 5.073 = -0.041$, $\hat{\beta}_2 = 4.661 - 5.073 = -0.412$, and the third group's deviation is $-(\hat{\beta}_1 + \hat{\beta}_2) = +0.453$. (b) Both designs span the group-constant vectors, so $\hat{y}_i = \bar{y}_{g(i)}$ in either (Proposition 7.1). (c) $\hat{L} = \frac{1}{2}(4.661 + 5.526) - 5.032 = 0.0615$; with $n_j = 10$, $\mathrm{Var}(\hat{L}) = \sigma^2(\frac{1}{4}\cdot\frac{1}{10} + \frac{1}{4}\cdot\frac{1}{10} + \frac{1}{10}) = \sigma^2 \cdot 0.15$, so $\mathrm{se} = 0.623\sqrt{0.15} = 0.241$ and $t = 0.0615/0.241 = 0.25$ on 27 df: no evidence that the average treatment differs from the control.

Exercise 7.4. Reading an interaction. In `lm(mpg ~ wt * am)` on `mtcars` the estimates are: intercept 31.42, `wt` −3.79, `ammanual` 14.88, `wt:ammanual` −5.30. (a) Write the fitted line for each transmission. (b) Where do the lines cross? (c) Compute the fitted gap manual − automatic at 2000, 3000 and 4000 lb. (d) Refit with `wt` centred at 3 and state the new `ammanual` coefficient. (e) Write one correct sentence that a journalist could print.

Solution.

(a) Automatic: $31.42 - 3.79w$; manual: $(31.42 + 14.88) + (-3.79 - 5.30)w = 46.30 - 9.09w$. (b) $w^* = 14.88/5.30 = 2.81$ (2810 lb). (c) Gap $14.88 - 5.30w$: $+4.28$, -1.02, -6.32 mpg. (d) The gap at $w = 3$: -1.02; the slopes and fitted lines are unchanged. (e) "Light manual cars (under about 2800 lb) are more fuel-efficient than automatics of the same weight, but for heavier cars the advantage disappears and reverses; few heavy manuals were observed."

Exercise 7.5. Hierarchy. Take the data $x = (0,1,2,3)$, $y = (1.3, 0, 1.1, 4.2)$. (a) Fit by least squares $y \approx a + bx^2$ and compute RSS. (b) Fit $y \approx a + b(x-1)^2$ and compute RSS; are they equal? (c) Fit $y \approx a + bx + cx^2$ and its shifted version and compute both RSS. Explain.

```
d <- data.frame(x = 0:3, y = c(1.3, 0, 1.1, 4.2))
c(RSS_x2 = deviance(lm(y ~ I(x^2), d)), RSS_shift = deviance(lm(y ~ I((x -
↪  1)^2), d)),
  RSS_full = deviance(lm(y ~ x + I(x^2), d)), RSS_full_shift = deviance(lm(y ~
  ↪  I(x - 1) + I((x - 1)^2), d)))
```

```
#>         RSS_x2      RSS_shift       RSS_full RSS_full_shift
#>          2.205          0.040          0.008          0.008
```

Solution. (a)–(c) See the output: the two non-hierarchical fits have very different RSS (2.20 for $a + bx^2$ and 0.04 for $a + b(x-1)^2$: the data are nearly $(x-1)^2$, so the shift matters), while the full quadratic has the same RSS (0.008) in both parametrisations. Reason: $\mathrm{span}\{1, x^2\} \neq \mathrm{span}\{1, (x-1)^2\}$ (Proposition 7.2), but $\mathrm{span}\{1, x, x^2\} = \mathrm{span}\{1, x-1, (x-1)^2\}$.

Exercise 7.6. Nested F test by hand. In a regression with $n = 60$, the model with predictors x_1, x_2 has RSS = 480; adding x_3, x_4, x_5 gives RSS = 420. (a) Test at 5% whether the three added predictors jointly improve the fit ($F_{3,54,0.95} = 2.78$). (b) The t statistics of the three new coefficients are 1.1, 1.3 and 0.4 (all with $|t| < 2$). Is this consistent with your answer? (c) Compute the change in AIC (use $\mathrm{AIC} = n\log(\mathrm{RSS}/n) + 2p$ and count

the intercept).

Solution.

(a) $p_0 = 3$, $p_1 = 6$: $F = \frac{(480-420)/3}{420/54} = \frac{20}{7.78} = 2.57 < 2.78$; do not reject ($p \approx 0.063$). (b) Yes: none of the three is individually significant, and jointly the evidence is borderline; the joint test is the right one for a group, as the t statistics are correlated. (c) $\Delta\text{AIC} = 60\log(420/480) + 2 \times 3 = -8.01 + 6 = -2.01$: AIC prefers the larger model (by 2.0), while the F test at 5% does not reject the smaller: the two criteria have different thresholds. BIC would add $3\log 60 = 12.3$ and prefer the smaller model clearly.

Exercise 7.7. Mallows' C_p. (a) Prove $\mathrm{E}[\mathrm{RSS}_p] = b_p^2 + (n-p)\sigma^2$ where $b_p^2 = \|(I - H_p)\mu\|^2$. (b) For the prostate data, the full model has $\hat\sigma^2_{\text{full}} = 0.5019$ ($n = 97$), and the best 3-predictor subset has RSS = 47.785 ($p = 4$). Compute C_p and decide whether this subset is biased.

Solution.

(a) Proof of Theorem 7.3. (b) $C_p = 47.785/0.5019 - 97 + 8 = 95.20 - 97 + 8 = 6.2 > p = 4$: the value is about 2 above p, which suggests some bias (the dropped predictors carry a little signal), though differences of this size from p are within sampling noise; the best subset of size 5 ($p = 5$) has $C_p = 5.6$, the minimum. Compare with the BIC choice in the text.

Exercise 7.8. AIC, BIC and AICc. Three models fitted to the same $n = 40$ observations have $(p, \mathrm{RSS}) = (3, 52.0), (5, 45.0), (9, 41.5)$. (a) Compute AIC and BIC (use $n\log(\mathrm{RSS}/n) +$ penalty with $k = p$). (b) Which does each criterion choose? (c) Compute AICc for the third model (use $k = p + 1$ in the correction) and comment.

```
n <- 40; p <- c(3, 5, 9); rss <- c(52, 45, 41.5)
aic <- n * log(rss / n) + 2 * p; bic <- n * log(rss / n) + p * log(n)
aicc <- aic + 2 * (p + 1) * (p + 2) / (n - p - 2)
round(rbind(AIC = aic, BIC = bic, AICc_with_k = aicc), 2)
```

```
#>              [,1]  [,2]  [,3]
#> AIC         16.49 14.71 19.47
#> BIC         21.56 23.16 34.67
#> AICc_with_k 17.64 17.26 27.06
```

Solution.

(a) The output above gives AIC = 16.49, 14.71, 19.47 and BIC = 21.56, 23.16, 34.67. (b) AIC chooses the second model (5 columns), BIC the first (3 columns): the BIC penalty $\log 40 = 3.69$ per parameter is much heavier than 2, and the 2 extra parameters of the second model reduce $n\log(\mathrm{RSS}/n)$ by only 5.8. (c) With $k = p + 1 = 10$ the correction for the third model is $2 \cdot 10 \cdot 11/29 = 7.6$, so AICc = 27.06, far above the others; the AICc values are 17.64, 17.26, 27.06, and AICc still prefers the second model, narrowly. With small n and many parameters, AIC can favour overfitted models and AICc is preferable.

Exercise 7.9. Ridge in the orthonormal case. Suppose $X^\top X = nI_p$. (a) Show $\hat\beta_\lambda = \hat\beta/(1 + \lambda/n)$ for the ridge criterion $\|y - X\beta\|^2 + \lambda\|\beta\|^2$. (b) Compute the bias,

variance and MSE of $\hat{\beta}_{\lambda,j}$ as functions of β_j, and find the λ minimising the MSE if β_j and σ^2 were known. (c) Is that λ usable in practice?

Solution.

(a) $(X^\top X+\lambda I)^{-1}X^\top y = (n+\lambda)^{-1}X^\top y = \frac{n}{n+\lambda}\hat{\beta}$ since $\hat{\beta} = X^\top y/n$. Let $c = n/(n+\lambda) \in (0,1)$. (b) $\hat{\beta}_{\lambda,j} = c\hat{\beta}_j$, with $\mathrm{E} = c\beta_j$, bias $-(1-c)\beta_j$ and variance $c^2\sigma^2/n$ (since $\mathrm{Var}(\hat{\beta}_j) = \sigma^2/n$). $\mathrm{MSE}(c) = (1-c)^2\beta_j^2 + c^2\sigma^2/n$; setting the derivative to zero, $-2(1-c)\beta_j^2 + 2c\sigma^2/n = 0$ gives $c^* = \beta_j^2/(\beta_j^2 + \sigma^2/n)$, i.e. $\lambda_j^* = \sigma^2/\beta_j^2$. With this choice $\mathrm{MSE} = c^*\sigma^2/n < \sigma^2/n$: shrinkage strictly improves on least squares for every β_j. (c) No: λ^* depends on the unknown β_j and σ^2; it is estimated by cross-validation, and a single λ is used for all coordinates. The formula shows that the larger the signal-to-noise ratio $n\beta_j^2/\sigma^2$, the smaller the shrinkage.

Exercise 7.10. Soft thresholding. (a) For orthonormal predictors, the least-squares estimates are $(2.5, -1.2, 0.4, -0.1)$ and $\lambda = 0.5$. Compute the lasso estimates and the ridge estimates for the same λ (ridge criterion as in glmnet with $\alpha = 0$, i.e. $\hat{\beta}/(1+\lambda)$). (b) Which coefficients are exactly zero? (c) For which λ are all lasso coefficients zero?

Solution.

(a) Lasso: $\mathrm{sign}(z)(|z|-0.5)_+ = (2.0, -0.7, 0, 0)$. Ridge: $z/(1+0.5) = (1.667, -0.8, 0.267, -0.067)$. (b) Lasso: the third and fourth. Ridge: none. (c) $\lambda \geq \max_j |z_j| = 2.5$. Above this, the lasso path starts at the zero model; `glmnet` computes $\lambda_{\max} = \max_j |x_j^\top y|/n$ (with standardised x) as the start of its grid.

Exercise 7.11. R analysis: collinearity and shrinkage. The data `seatpos` (package `faraway`) give the driver seat position `hipcenter` for $n = 38$ drivers with eight body measurements: `Age`, `Weight`, `HtShoes` (height with shoes), `Ht` (height), `Seated`, `Arm`, `Thigh`, `Leg`. (a) Fit the full model. Report R^2, the overall F test and the individual t tests; what looks wrong? (b) Compute VIFs and the correlation of `Ht` and `HtShoes`. (c) Compare 10-fold cross-validated MSE (same folds) of OLS with the full model, OLS with `Age + Weight + Ht`, ridge and lasso. (d) Recommend a model for prediction and one for interpreting the effect of `Ht`.

```
data(seatpos, package = "faraway")
sf <- lm(hipcenter ~ ., data = seatpos)
c(R2 = summary(sf)$r.squared, F = summary(sf)$fstatistic[[1]],
  p_F = pf(summary(sf)$fstatistic[[1]], 8, 29, lower.tail = FALSE))
```

```
#>        R2         F       p_F
#> 6.866e-01 7.940e+00 1.306e-05
```

```
round(summary(sf)$coefficients, 3)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  436.432    166.572   2.620    0.014
#> Age            0.776      0.570   1.360    0.184
#> Weight         0.026      0.331   0.080    0.937
#> HtShoes       -2.692      9.753  -0.276    0.784
#> Ht             0.601     10.130   0.059    0.953
```

```
#> Seated          0.534      3.762   0.142    0.888
#> Arm            -1.328      3.900  -0.341    0.736
#> Thigh          -1.143      2.660  -0.430    0.671
#> Leg            -6.439      4.714  -1.366    0.182
```

```
round(vif(sf), 1); cor(seatpos$Ht, seatpos$HtShoes)
```

```
#>     Age  Weight HtShoes      Ht  Seated     Arm   Thigh     Leg
#>     2.0     3.6   307.4   333.1     9.0     4.5     2.8     6.7
```

```
#> [1] 0.9981
```

```
xs <- as.matrix(seatpos[, names(seatpos) != "hipcenter"]); ys <-
↪  seatpos$hipcenter
set.seed(7); fid <- sample(rep(1:10, length.out = nrow(xs)))
cv_err <- function(form) mean(sapply(1:10, function(k) {
  f <- lm(form, seatpos[fid != k, ]); mean((ys[fid == k] - predict(f,
↪  seatpos[fid == k, ]))^2) }))
cr <- cv.glmnet(xs, ys, alpha = 0, foldid = fid); cl <- cv.glmnet(xs, ys,
↪  alpha = 1, foldid = fid)
c(OLS_full = cv_err(hipcenter ~ .), OLS_small = cv_err(hipcenter ~ Age +
↪  Weight + Ht),
  ridge = min(cr$cvm), lasso_min = min(cl$cvm))
```

```
#>  OLS_full OLS_small     ridge lasso_min
#>      2179      1563      1438      1634
```

```
round(cbind(ridge = coef(cr, s = "lambda.min")[, 1], lasso = coef(cl, s =
↪  "lambda.1se")[, 1]), 3)
```

```
#>                ridge   lasso
#> (Intercept) 400.177 213.098
#> Age           0.474   0.000
#> Weight       -0.106   0.000
#> HtShoes      -0.738   0.000
#> Ht           -0.737  -1.604
#> Seated       -1.204   0.000
#> Arm          -1.357   0.000
#> Thigh        -1.305   0.000
#> Leg          -3.113  -2.943
```

Solution.

(a) $R^2 = 0.687$ and the overall $F = 7.94$ ($p = 1.3 \times 10^{-5}$) show that the predictors jointly explain much of `hipcenter`, but *no* individual t test is significant (smallest $p = 0.18$ for `Leg`) and several coefficients have implausible signs and huge standard errors: the classic signature of collinearity. (b) The VIFs of `HtShoes` and `Ht` are about 307 and 333 (correlation 0.998); `Seated` (9), `Leg` (6.7) are also high. The two heights measure the same thing, so the data cannot separate their effects, though together they are well determined. (c) 10-fold CV MSE: OLS full 2179, OLS with `Age + Weight + Ht` 1563, ridge 1438, lasso 1634: removing redundant predictors or shrinking them reduces the prediction error by 25–35%, and ridge (which keeps all the correlated measurements and shares the effect among them) is best. (d) For prediction: ridge (or the 3-predictor OLS), chosen by the CV error. For interpreting the effect of height: height is the one measurement of interest;

fix `Ht` (not `HtShoes`) with `Age` and `Weight` as covariates *before* looking at the response. In that model the coefficient of `Ht` is -4.2 (se 1.0): about 4 mm lower hip centre per extra cm of height, and it is estimated precisely once the duplicate measurement is dropped. The regularised fits shrink the two heights towards each other and share the effect.

Exercise 7.12. Critique. A paper reports the following for a prostate-cancer data set: "We selected the best subset of predictors by stepwise AIC from 8 candidates. In the selected model (`lcavol`, `lweight`, `age`, `lbph`, `svi`), the coefficient of `lbph` is significant ($p = 0.03$) and its 95% CI excludes zero, so benign prostatic hyperplasia *causes* higher PSA, and reducing it would lower PSA. The cross-validated R^2 of the final model was 0.62, computed by 10-fold CV of the final model." Identify at least four problems.

Solution.

(1) *Post-selection inference.* The p-value and CI of `lbph` are computed in a model chosen by looking at the same data; they ignore the selection and are too optimistic (Section 7.6). Either pre-specify the model or select and test on separate data. (2) *Causal language.* This is observational data; a conditional association (given the other four predictors) does not show that reducing `lbph` would lower PSA, and the other three variables, or others not measured, may confound. (3) *Cross-validation after selection.* If the five predictors were chosen on all the data, then cross-validating only the final model leaks information: the held-out folds influenced the selection. The selection (step) must be inside each fold. (4) *AIC for explanation.* AIC targets prediction; for estimating the effect of one variable one should adjust for the confounders chosen by subject knowledge, and AIC may drop or keep a variable for reasons unrelated to confounding. (5) *Reliance on a p-value near 0.05* in a data set of 97 cases with eight correlated predictors; report the estimate and interval and the sensitivity to the model specification.

7.13 Further reading

Faraway (2014) Chapters 5 (problems with the predictors), 10 (model selection), 11 (shrinkage) and 14 (categorical predictors) cover the same ground in R. The treatment of AIC as an estimate of Kullback–Leibler information is in Akaike (1974) and Burnham and Anderson (2004); BIC in Schwarz (1978); C_p in Mallows (1973). Shrinkage is developed in Hastie et al. (2009) (Chapter 3) and the original papers Hoerl and Kennard (1970) (ridge), Tibshirani (1996) (lasso) and Zou and Hastie (2005) (elastic net); Friedman et al. (2010) describes the algorithm in `glmnet`. Berk et al. (2013) discusses valid inference after selection, and Gelman and Hill (2006) is a good source on interpretation of regression coefficients, interactions and centring.

Part III

Generalized linear models

Chapter 8

Exponential families and generalized linear models

The linear model of Chapters 5–7 assumes a response that can take any real value, whose mean is a linear function of the covariates and whose variance does not depend on the mean. Many responses violate all three assumptions: a binary outcome (a baby is born with low weight or not), a count (the number of damage incidents of a ship), a positive and skewed measurement (the volume of a tree). The **generalized linear model** (GLM) of Nelder and Wedderburn (1972) keeps what is useful in the linear model (a linear predictor, maximum likelihood, a unified theory of estimation and testing) and changes two ingredients: the distribution of the response, which may be any member of the *exponential family*, and the relation between the mean and the linear predictor, which goes through a *link function*.

This chapter builds the theory. We define the exponential family, derive its moments from the cumulant function $b(\theta)$, define the GLM and the canonical link, discuss the choice of the link, and work three examples (binomial, Poisson, gamma). Chapter 9 shows how GLMs are fitted and tested, and Chapter 10 applies them to binary and count data. The standard references are McCullagh and Nelder (1989), Wood (2017) (Chapter 3) and Pawitan (2001) (Chapters 4 and 6); Faraway (2016) is a practical companion.

8.1 Why the linear model is not enough

Two data sets from the `MASS` package run through the three GLM chapters. `birthwt` has 189 births at a hospital in Springfield (Massachusetts) in 1986; the response is `low`, 1 if the birth weight is below 2.5 kg. `ships` records damage incidents for cargo ships, in cells defined by ship type, year of construction and period of operation, together with the aggregate months of service (the *exposure*).

```
bw <- with(birthwt, data.frame(low = low, age = age, lwt = lwt,
  race = factor(race, labels = c("white", "black", "other")),
  smoke = smoke, ptd = as.integer(ptl > 0), ht = ht, ui = ui))
table(bw$low)
```

```
#>
#>   0   1
#> 130  59
```

```
sh <- subset(ships, service > 0)          # 6 of the 40 cells have no service
sh$year <- factor(sh$year); sh$period <- factor(sh$period)
```

```
c(cells = nrow(sh), incidents = sum(sh$incidents), months = sum(sh$service))
```

```
#>     cells incidents    months
#>        34       356    163574
```

Suppose we ignore the nature of `low` and fit a linear model, the *linear probability model*:

```
lpm <- lm(low ~ age + lwt + race + smoke + ptd + ht + ui, data = bw)
logit_fit <- glm(low ~ age + lwt + race + smoke + ptd + ht + ui, family =
↪  binomial, data = bw)
c(negative = sum(fitted(lpm) < 0), min = min(fitted(lpm)), max =
↪  max(fitted(lpm)))
```

```
#> negative      min      max
#>   8.0000  -0.1396   0.8898
```

```
range(fitted(logit_fit))
```

```
#> [1] 0.02334 0.87997
```

```
d <- rbind(data.frame(model = "Linear model", fitted = fitted(lpm)),
           data.frame(model = "Logistic GLM", fitted = fitted(logit_fit)))
ggplot(d, aes(fitted)) +
  geom_histogram(binwidth = 0.05, boundary = 0, fill = ie_navy, colour =
  ↪  "white") +
  geom_vline(xintercept = c(0, 1), colour = ie_coral, linetype = 2) +
  facet_wrap(~model, ncol = 1) + labs(x = "fitted value of P(low = 1)", y =
  ↪  "births")
```

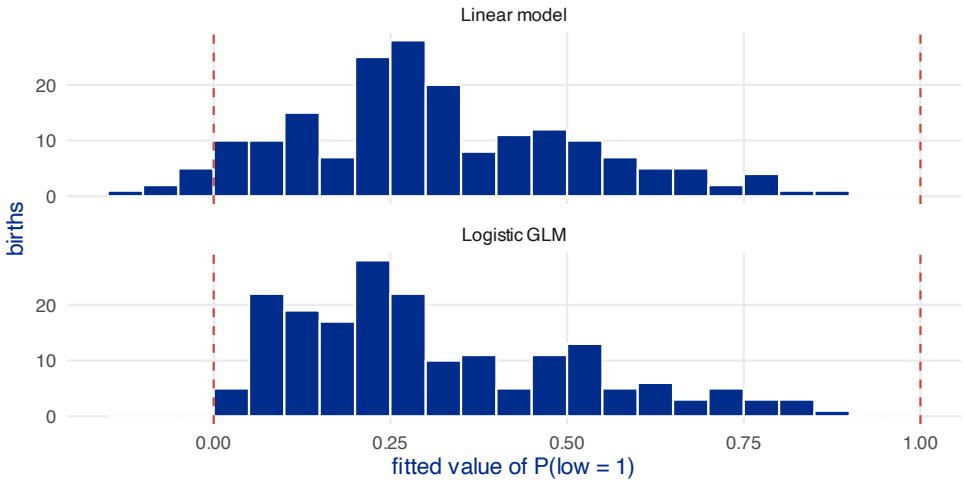


Figure 8.1: Fitted values of the linear probability model and of the logistic GLM for the birth-weight data. The linear model produces negative 'probabilities'.

Three things go wrong with $y_i = \boldsymbol{x}_i^\top \beta + \varepsilon_i$, $\varepsilon_i \sim N(0, \sigma^2)$, for such data.

1. **Range.** $\boldsymbol{x}_i^\top \beta$ can be any real number, but the mean $\mu_i = \mathrm{E}[Y_i]$ of a binary response lies in $(0, 1)$ and the mean of a count lies in $(0, \infty)$.
2. **Variance.** For $Y_i \in \{0, 1\}$ with $P(Y_i = 1) = \mu_i$ we have $\mathrm{Var}(Y_i) = \mu_i(1 - \mu_i)$, which changes with μ_i. For counts the variance typically grows with the mean. The constant variance σ^2 is wrong by construction.

3. **Distribution.** A 0/1 or integer-valued response is not normal, however large n is; exact t and F tests are lost.

Transforming the response does not repair this: $\log(y)$ does not exist when $y = 0$, and $\log(y+1)$ changes the target, because in general $\mathrm{E}[\log(Y+1)] \neq \log(\mathrm{E}[Y]+1)$. A GLM instead transforms the **mean**, $g(\mu) = \boldsymbol{x}^\top \beta$, and leaves y on its own scale, with a distribution suited to it.

8.2 The exponential family

Definition 8.1 (Exponential dispersion family)**.** A random variable Y belongs to the **exponential (dispersion) family** if its density (or probability mass function) can be written

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}, \tag{8.1}$$

for functions a, b, c. The parameter θ is the **canonical** (natural) parameter, $\phi > 0$ is the **dispersion** parameter, $b(\theta)$ is the **cumulant function**, and $c(y,\phi)$ does not involve θ. Usually $a(\phi) = \phi/w$ with a known prior weight w (for instance $w = m$ for a mean of m observations).

Everything about the dependence of the distribution on its mean is in b. The function $c(y,\phi)$ normalises the density, but it plays no role in estimating θ (for known ϕ): this is what makes the likelihood of a GLM tractable. We assume that the support of Y does not depend on θ, that b is twice differentiable, and that the set of θ for which (8.1) integrates to 1 is an open interval.

Example 8.1 (Poisson, Bernoulli, binomial, normal, gamma)**.** *Poisson with mean μ.* $f(y) = e^{-\mu}\mu^y/y! = \exp\{y\log\mu - \mu - \log y!\}$. Hence $\theta = \log\mu$, $b(\theta) = e^\theta$, $a(\phi) = 1$, $c(y) = -\log y!$.
Bernoulli with $P(Y=1) = \mu$. $f(y) = \mu^y(1-\mu)^{1-y} = \exp\{y\log\frac{\mu}{1-\mu} + \log(1-\mu)\}$. Take $\theta = \log\frac{\mu}{1-\mu} = \operatorname{logit}\mu$; then $\mu = e^\theta/(1+e^\theta)$ and $\log(1-\mu) = -\log(1+e^\theta)$. So $b(\theta) = \log(1+e^\theta)$, $a(\phi) = 1$, $c = 0$.
Binomial proportion. If $Z \sim \mathrm{Bin}(m,\mu)$ and $Y = Z/m$, then $P(Y=y) = \binom{m}{my}\exp\{my\,\theta - m\log(1+e^\theta)\} = \exp\left\{\frac{y\theta - \log(1+e^\theta)}{1/m} + \log\binom{m}{my}\right\}$. The same θ and b as the Bernoulli, with $a(\phi) = 1/m$: the prior weight is m.
Normal $N(\mu,\sigma^2)$. $f(y) = \exp\{(y\mu - \mu^2/2)/\sigma^2 - y^2/(2\sigma^2) - \frac{1}{2}\log(2\pi\sigma^2)\}$, so $\theta = \mu$, $b(\theta) = \theta^2/2$, $a(\phi) = \phi = \sigma^2$.
Gamma with mean μ and shape ν. The density $\frac{1}{\Gamma(\nu)}\left(\frac{\nu}{\mu}\right)^\nu y^{\nu-1}e^{-\nu y/\mu}$ equals

$$\exp\left\{\frac{y(-1/\mu) - \log\mu}{1/\nu} + \nu\log\nu + (\nu-1)\log y - \log\Gamma(\nu)\right\}.$$

So $\theta = -1/\mu < 0$, $b(\theta) = -\log(-\theta)$ (indeed $-\log(-\theta) = \log\mu$), $a(\phi) = \phi = 1/\nu$.

The Bernoulli, binomial and Poisson distributions have no free dispersion ($a(\phi) = 1$ for the counts): their variance is fully determined by the mean. The normal and the gamma have a free ϕ. Chapter 10 shows what to do when the data are more variable than the Poisson or binomial allows.

8.3 Moments from the cumulant function

The mean and variance of any member of the family come from derivatives of b. We give two proofs; the first uses the score and is the one used again in Chapter 9, the second uses the moment generating function and gives all cumulants.

Lemma 8.1 (Score identities). *Let $f(y;\theta)$ be a density whose support does not depend on θ, differentiable in θ, and such that differentiation under the integral sign is allowed. With $\ell(\theta) = \log f(Y;\theta)$ and $S(\theta) = \partial \ell / \partial \theta$,*

$$\mathrm{E}[S(\theta)] = 0, \qquad \mathrm{E}[S(\theta)^2] = \mathrm{E}\left[-\frac{\partial^2 \ell}{\partial \theta^2}\right] \quad \text{so that} \quad \mathrm{Var}(S) = \mathcal{I}_1(\theta).$$

Proof. Differentiate $\int f(y;\theta)\,dy = 1$. Since $\partial f/\partial\theta = S\,f$,

$$0 = \int \frac{\partial f}{\partial \theta} dy = \int S\, f\, dy = \mathrm{E}[S].$$

Differentiate once more, using the product rule on $S\,f$:

$$0 = \int \frac{\partial}{\partial \theta}(S\,f)\,dy = \int \left(\frac{\partial S}{\partial \theta} + S^2\right) f\, dy = \mathrm{E}\left[\frac{\partial^2 \ell}{\partial \theta^2}\right] + \mathrm{E}[S^2].$$

□

Theorem 8.1 (Mean and variance in the exponential family). *If Y has density (8.1), then*

$$\mathrm{E}[Y] = \mu = b'(\theta), \qquad \mathrm{Var}(Y) = b''(\theta)\, a(\phi). \tag{8.2}$$

Proof. For one observation, $\ell(\theta) = \{y\theta - b(\theta)\}/a(\phi) + c(y,\phi)$, so

$$S(\theta) = \frac{y - b'(\theta)}{a(\phi)}, \qquad \frac{\partial^2 \ell}{\partial \theta^2} = -\frac{b''(\theta)}{a(\phi)}.$$

By Lemma 8.1, $\mathrm{E}[S] = 0$ gives $\mathrm{E}[Y] = b'(\theta)$. The second identity gives $\mathrm{Var}(Y)/a(\phi)^2 = \mathrm{E}[S^2] = b''(\theta)/a(\phi)$, that is, $\mathrm{Var}(Y) = b''(\theta)\,a(\phi)$ (note that $\mathrm{E}[S^2] = \mathrm{Var}(S)$ because $\mathrm{E}[S] = 0$, and $\mathrm{Var}(S) = \mathrm{Var}(Y)/a(\phi)^2$ because $b'(\theta)$ is a constant). □

Theorem 8.2 (Cumulant generating function). *For t such that $\theta + t\,a(\phi)$ lies in the parameter space, the moment generating function of Y is*

$$M_Y(t) = \mathrm{E}[e^{tY}] = \exp\left\{\frac{b(\theta + t\,a(\phi)) - b(\theta)}{a(\phi)}\right\},$$

so the rth cumulant of Y is $\kappa_r = a(\phi)^{r-1}\, b^{(r)}(\theta)$.

Proof. Write $a = a(\phi)$. Then

$$\begin{aligned} M_Y(t) &= \int e^{ty} \exp\left\{\frac{y\theta - b(\theta)}{a} + c(y,\phi)\right\} dy \\ &= e^{\{b(\theta+ta) - b(\theta)\}/a} \int \exp\left\{\frac{y(\theta+ta) - b(\theta+ta)}{a} + c(y,\phi)\right\} dy. \end{aligned}$$

The last integral is 1, because the integrand is the density (8.1) with θ replaced by $\theta + ta$. The cumulant generating function is $K(t) = \log M_Y(t) = \{b(\theta + ta) - b(\theta)\}/a$, and $\kappa_r = K^{(r)}(0) = a^r b^{(r)}(\theta)/a$. □

The moment formulae give the *variance function*. Because $\mathrm{Var}(Y) = a(\phi)b''(\theta) \geq 0$ (and > 0 unless Y is degenerate), b' is strictly increasing, so $\theta = (b')^{-1}(\mu)$ is a one-to-one function of the mean, and

$$\mathrm{Var}(Y) = a(\phi)\, V(\mu), \qquad V(\mu) = b''((b')^{-1}(\mu)). \tag{8.3}$$

The variance is a known function of the mean, times a dispersion factor. This single relation is what a GLM adds to a linear model, and it is all that quasi-likelihood (Chapter 9) retains.

Table 8.1: Members of the exponential family used in this course. The inverse Gaussian is Exercise 8.2.

Family	θ	$b(\theta)$	$a(\phi)$	$V(\mu)$	canonical link $g(\mu)$
Normal	μ	$\theta^2/2$	σ^2	1	identity, μ
Bernoulli / binomial Z/m	$\mathrm{logit}\,\mu$	$\log(1+e^\theta)$	$1/m$	$\mu(1-\mu)$	logit, $\log\frac{\mu}{1-\mu}$
Poisson	$\log\mu$	e^θ	1	μ	log
Gamma (shape ν)	$-1/\mu$	$-\log(-\theta)$	$1/\nu$	μ^2	inverse, $-1/\mu$
Inverse Gaussian	$-1/(2\mu^2)$	$-\sqrt{-2\theta}$	$1/\lambda$	μ^3	$-1/(2\mu^2)$

Example 8.2 (Moments of the four families)**.** Check Theorem 8.1 on the families of Example 8.1.
Poisson: $b'(\theta) = e^\theta = \mu$ and $b''(\theta) = e^\theta = \mu$, so $\mathrm{Var}(Y) = \mu$.
Bernoulli: $b'(\theta) = e^\theta/(1+e^\theta) = \mu$ and $b''(\theta) = e^\theta/(1+e^\theta)^2 = \mu(1-\mu)$. For the binomial proportion, $\mathrm{Var}(Y) = \mu(1-\mu)/m$, so $\mathrm{Var}(Z) = m^2\mathrm{Var}(Y) = m\mu(1-\mu)$.
Normal: $b'(\theta) = \theta = \mu$, $b''(\theta) = 1$, $\mathrm{Var}(Y) = \sigma^2$.
Gamma: $b'(\theta) = -1/\theta = \mu$ (the sign matters: $b'(\theta) = 1/\theta$ would be negative) and $b''(\theta) = 1/\theta^2 = \mu^2$, so $\mathrm{Var}(Y) = \mu^2/\nu$. The coefficient of variation $\sqrt{\mathrm{Var}(Y)}/\mu = 1/\sqrt{\nu}$ is constant: the gamma family is the natural one for positive responses whose standard deviation is proportional to the mean.

We can check such formulae numerically with finite differences of b.

```
d1 <- function(b, th, h = 1e-4) (b(th + h) - b(th - h)) / (2 * h)
d2 <- function(b, th, h = 1e-4) (b(th + h) - 2 * b(th) + b(th - h)) / h^2
c(mean = d1(exp, log(3)), var = d2(exp, log(3)))                    # Poisson,
↪  mu = 3
```

```
#> mean  var
#>    3    3
```

```
b_gamma <- function(th) -log(-th)                                    # gamma,
↪  mu = 2, nu = 4
c(mean = d1(b_gamma, -1/2), var = d2(b_gamma, -1/2) * (1/4))        # a(phi) =
↪  1/nu
```

```
#> mean  var
#>    2    1
```

```
set.seed(1); x <- rgamma(1e5, shape = 4, rate = 2)              # mean 2,
↪  variance 1
c(mean = mean(x), var = var(x))
```

```
#>   mean    var
#> 1.9999 0.9978
```

The third cumulant gives the skewness, $\kappa_3/\kappa_2^{3/2}$. For the Poisson, $b'''(\theta) = e^\theta = \mu$, so the skewness is $\mu/\mu^{3/2} = 1/\sqrt{\mu}$, which shrinks as the mean grows: the reason why a Poisson with large mean looks normal (Exercise 8.3).

8.4 Generalized linear models

Definition 8.2 (Generalized linear model)**.** A GLM for independent responses $Y_1, \dots, Y_n$ and a covariate matrix X ($n \times p$, rows $\boldsymbol{x}_i^\top$, including the intercept column) has three components.

1. **Random component.** The Y_i are independent, each with density (8.1) of the same family, with canonical parameter θ_i (hence mean $\mu_i = b'(\theta_i)$) and with $a(\phi)$ replaced by ϕ/w_i for known prior weights w_i.
2. **Systematic component.** The **linear predictor** $\eta_i = \boldsymbol{x}_i^\top \beta$, that is, $\boldsymbol{\eta} = X\beta$.
3. **Link.** A smooth strictly monotone function g with $g(\mu_i) = \eta_i$, that is, $\mu_i = g^{-1}(\boldsymbol{x}_i^\top \beta)$.

The covariates enter only through η_i. The distribution enters only through the variance function $V(\mu)$ and the dispersion $a(\phi)$. The *linear model* is the special case "normal family, identity link": $\mu_i = \boldsymbol{x}_i^\top \beta$, $\mathrm{Var}(Y_i) = \sigma^2$.

Remark. What keeps μ_i in its range for every β and every $\boldsymbol{x}$ is the **inverse link** g^{-1}: the logistic function $g^{-1}(\eta) = e^\eta/(1+e^\eta)$ maps $\mathbb{R}$ into $(0,1)$, and $g^{-1}(\eta) = e^\eta$ maps $\mathbb{R}$ into $(0,\infty)$. The *link* g is the function that takes you the other way.

8.4.1 Link functions

The link is a modelling choice. The common ones are:

Table 8.2: Link functions.

Link	$\eta = g(\mu)$	$\mu = g^{-1}(\eta)$	range of μ	typical use
identity	μ	η	$\mathbb{R}$	normal; counts far from 0
log	$\log \mu$	e^η	$(0,\infty)$	counts, positive responses
logit	$\log \frac{\mu}{1-\mu}$	$\frac{e^\eta}{1+e^\eta}$	$(0,1)$	binary, proportions
probit	$\Phi^{-1}(\mu)$	$\Phi(\eta)$	$(0,1)$	binary (latent normal)
complementary log-log	$\log\{-\log(1-\mu)\}$	$1-\exp(-e^\eta)$	$(0,1)$	binary from counts

Link	$\eta = g(\mu)$	$\mu = g^{-1}(\eta)$	range of μ	typical use
inverse	$1/\mu$	$1/\eta$	needs $\eta > 0$	gamma (canonical up to sign)

For binary data the logit, probit and complementary log-log links have a common origin. Suppose $Y = 1$ exactly when a latent variable $Z = \eta + \varepsilon$ is positive, with ε independent of the covariates and with a symmetric distribution function F. Then $P(Y = 1) = P(\varepsilon > -\eta) = F(\eta)$ and $g = F^{-1}$. A standard logistic ε gives the logit link, a standard normal ε the probit link. The complementary log-log link arises from counts: if $N \sim \text{Poisson}(\lambda)$ with $\log \lambda = \eta$ and we only observe $Y = \mathbf{1}\{N > 0\}$, then $\mu = P(N > 0) = 1 - e^{-\lambda}$, so $\log\{-\log(1 - \mu)\} = \log \lambda = \eta$. It is asymmetric: μ approaches 1 faster than it leaves 0.

```
eta <- seq(-4, 4, length.out = 300)
d <- data.frame(eta = rep(eta, 3), mu = c(plogis(eta), pnorm(eta), 1 -
↪  exp(-exp(eta))),
                link = rep(c("logit", "probit", "cloglog"), each = 300))
ggplot(d, aes(eta, mu, colour = link)) + geom_line(linewidth = 1) +
  scale_colour_manual(values = ie_pal[1:3]) + labs(x = expression(eta), y =
  ↪  expression(mu), colour = NULL)
```

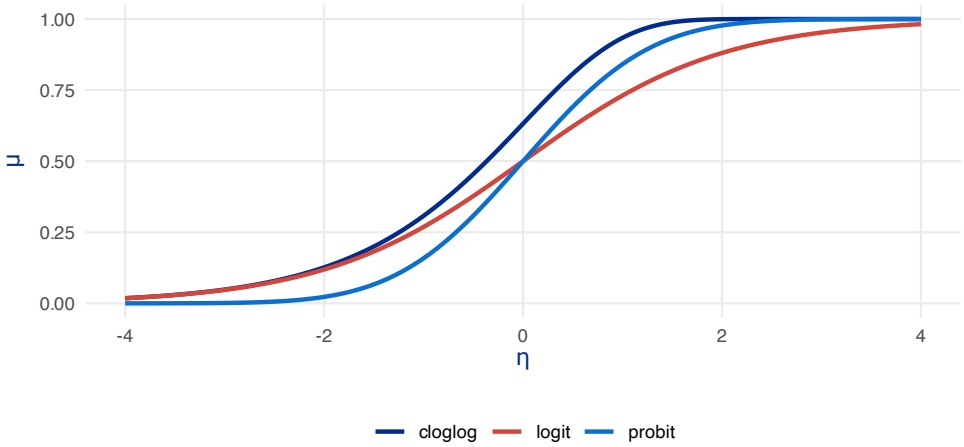


Figure 8.2: Three inverse links for binary data. Logit and probit are symmetric about $\mu = 1/2$; the complementary log-log is not.

8.4.2 The canonical link

Definition 8.3 (Canonical link)**.** The **canonical link** is $g = (b')^{-1}$, the function that maps the mean to the canonical parameter, $g(\mu) = \theta$. With it, $\theta_i = \eta_i = \boldsymbol{x}_i^\top \beta$.

The canonical links of Table 8.1 are the identity (normal), logit (binomial), log (Poisson) and $-1/\mu$ (gamma; R uses $1/\mu$ and absorbs the sign in β). For the gamma the canonical inverse link is not guaranteed to give a positive mean: $\mu = 1/\eta$ requires $\eta > 0$ for every observation, so in practice the gamma is mostly used with the log link.

Theorem 8.3 (Likelihood with a canonical link)**.** *With the canonical link and* $a(\phi) = \phi/w_i$

(prior weights w_i), the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \frac{w_i\{y_i\, \boldsymbol{x}_i^\top \beta - b(\boldsymbol{x}_i^\top \beta)\}}{\phi} + \sum_i c(y_i, \phi),$$

which is strictly concave in β if X has full column rank. Its score and Hessian are

$$S(\beta) = \frac{1}{\phi} X^\top W_0 (\boldsymbol{y} - \boldsymbol{\mu}), \qquad \frac{\partial^2 \ell}{\partial \beta\, \partial \beta^\top} = -\frac{1}{\phi} X^\top W_0 \operatorname{diag}\{b''(\theta_i)\} X, \tag{8.4}$$

where $W_0 = \operatorname{diag}\{w_i\}$. In particular $X^\top W_0 \boldsymbol{y}$ is a sufficient statistic for β, and the likelihood equations are $X^\top W_0 \boldsymbol{y} = X^\top W_0 \hat{\boldsymbol{\mu}}$. If the model has an intercept and $w_i \equiv 1$, the fitted totals equal the observed totals, $\sum_i \hat{\mu}_i = \sum_i y_i$.

Proof. Substitute $\theta_i = \boldsymbol{x}_i^\top \beta$ in (8.1). The terms involving β are $\beta^\top X^\top W_0 \boldsymbol{y} - \sum_i w_i b(\boldsymbol{x}_i^\top \beta)$, so $X^\top W_0 \boldsymbol{y}$ is sufficient (factorization theorem). Differentiating, $\partial \ell / \partial \beta = \phi^{-1} \sum_i w_i \{y_i - b'(\boldsymbol{x}_i^\top \beta)\} \boldsymbol{x}_i = \phi^{-1} X^\top W_0 (\boldsymbol{y} - \boldsymbol{\mu})$ since $b'(\theta_i) = \mu_i$. A second derivative gives $-\phi^{-1} \sum_i w_i b''(\theta_i) \boldsymbol{x}_i \boldsymbol{x}_i^\top$. Because $b'' = V > 0$, this matrix is negative definite when X has full column rank, hence ℓ is strictly concave. The constant column of X gives $\sum_i w_i (y_i - \hat{\mu}_i) = 0$. □

Two consequences matter in practice. First, the Hessian does not involve $\boldsymbol{y}$: observed and expected information coincide, and Newton–Raphson and Fisher scoring are the same algorithm (Chapter 9). Second, the likelihood has a unique maximum if it is attained at a finite $\hat{\beta}$; for binary data it may fail to be (Chapter 10). A simple example of the likelihood equations is a Poisson model with an intercept and one binary covariate (Exercise 8.5): the fitted means are the two group averages.

8.5 The mean–variance relationship

A linear model has $\operatorname{Var}(Y_i) = \sigma^2$ whatever μ_i is. In a GLM the variance is $a(\phi) V(\mu_i)$, so observations with larger means are noisier, and the likelihood equations weight them less (Chapter 9). Figure 8.3 shows the three standard cases by simulation.

```
set.seed(2027)
mus <- c(1, 2, 4, 6, 8, 10)
sim_var <- function(fam, mu) var(switch(fam,
  Normal = rnorm(4000, mu, sqrt(2)), Poisson = rpois(4000, mu),
  Gamma = rgamma(4000, shape = 2, rate = 2 / mu)))
pts <- expand.grid(family = c("Normal", "Poisson", "Gamma"), mu = mus,
 ↪  stringsAsFactors = FALSE)
pts$var <- mapply(sim_var, pts$family, pts$mu)
mm <- seq(0.5, 10.5, length.out = 100)
lines_df <- rbind(data.frame(family = "Normal", mu = mm, var = 2),
                  data.frame(family = "Poisson", mu = mm, var = mm),
                  data.frame(family = "Gamma", mu = mm, var = mm^2 / 2))
ggplot(pts, aes(mu, var, colour = family)) + geom_line(data = lines_df,
 ↪  linewidth = 1) + geom_point(size = 2.5) +
  scale_colour_manual(values = ie_pal[c(3, 1, 2)]) + labs(x = expression(mu),
   ↪  y = "Var(Y)", colour = NULL)
```

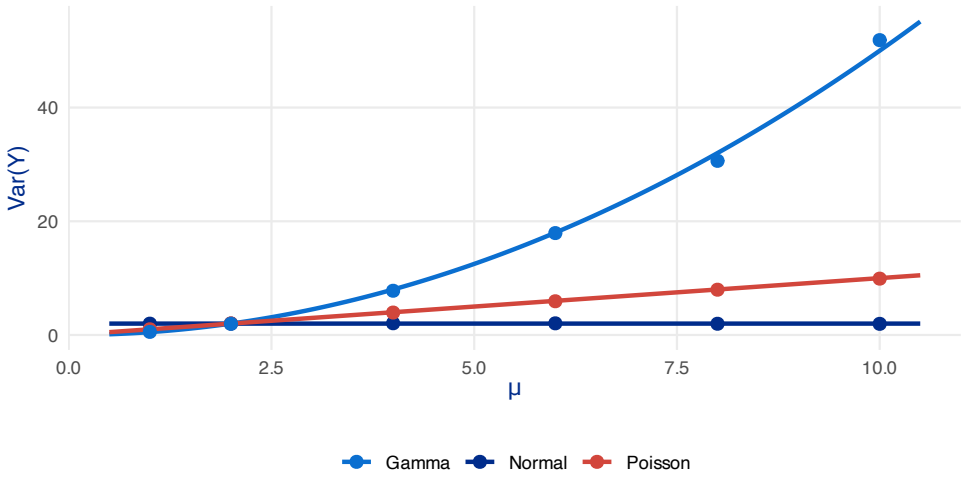


Figure 8.3: Variance against mean for simulated normal ($\phi = 2$), Poisson and gamma ($\nu = 2$, so $\phi = 1/2$) samples of size 4000. The lines are $a(\phi)V(\mu)$.

When a response has a variance that grows with the mean, a GLM models this directly. The alternative, a linear model for a transformed response, changes the quantity being modelled. The next example shows the difference.

Example 8.3 (Transforming y is not the same as transforming the mean)**.** Let Y be lognormal, $\log Y \sim N(m, s^2)$. Then $\mathrm{E}[\log Y] = m$ but $\log \mathrm{E}[Y] = m + s^2/2$. A linear model of $\log y$ on x estimates the first; a gamma or Poisson GLM with log link estimates the second, a quantity on the original scale. For counts the situation is worse, since $y = 0$ must be shifted before taking logs. We simulate 200 data sets of $n = 200$ Poisson counts with $\mathrm{E}[Y] = \exp(1 + 0.5x)$ and compare the slope of $\log(y + 1)$ on x with the Poisson GLM slope.

```
set.seed(2027)
res <- replicate(200, { x <- runif(200); y <- rpois(200, exp(1 + 0.5 * x))
  c(`lm(log(y+1))` = unname(coef(lm(log(y + 1) ~ x))[2]),
    `Poisson glm` = unname(coef(glm(y ~ x, family = poisson))[2])) })
round(rbind(mean = rowMeans(res), sd = apply(res, 1, sd), truth = c(0.5,
↪  0.5)), 3)
```

```
#>       lm(log(y+1)) Poisson glm
#> mean         0.403       0.491
#> sd           0.116       0.135
#> truth        0.500       0.500
```

The Poisson GLM is centred on the true slope 0.5 of $\log \mathrm{E}[Y]$. The model for $\log(y + 1)$ is centred on a different number (about 0.4 here), because it targets $\mathrm{E}[\log(Y + 1)]$, which depends on the whole distribution of Y and not only on its mean (Exercise 8.12).

8.6 Three worked examples

8.6.1 Binomial: O-ring damage

The `faraway::orings` data record, for 23 shuttle launches before the Challenger accident, the temperature on the launch pad and the number of the 6 primary O-rings that showed damage.

The model is $Z_i \sim \text{Bin}(6, \mu_i)$ with $\text{logit}\,\mu_i = \beta_0 + \beta_1 \text{temp}_i$.

```
o <- glm(cbind(damage, 6 - damage) ~ temp, family = binomial, data = orings)
round(summary(o)$coefficients, 4)
```

```
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept)  11.6630     3.2963   3.538    4e-04
#> temp         -0.2162     0.0532  -4.066    0e+00
```

The random component is binomial (proportion $Y_i = Z_i/6$ with weight $m_i = 6$), the linear predictor is $\beta_0 + \beta_1 \text{temp}$, and the link is the canonical logit. Each additional degree Fahrenheit multiplies the odds of damage to a given O-ring by $e^{\hat\beta_1} = 0.806$; ten degrees multiply it by 0.115. At the launch temperature of the accident, 31°F, the model gives

```
pr <- predict(o, data.frame(temp = 31), type = "link", se.fit = TRUE)
c(prob = plogis(pr$fit), lower = plogis(pr$fit - 1.96 * pr$se.fit), upper =
↪  plogis(pr$fit + 1.96 * pr$se.fit))
```

```
#>  prob.1 lower.1 upper.1
#>  0.9930  0.8445  0.9997
```

```
range(orings$temp)
```

```
#> [1] 53 81
```

The point estimate is a damage probability of 0.993 per O-ring, with a 95% interval from 0.84 to 0.9997 (the interval is computed on the log-odds scale and then mapped through g^{-1}, so it stays in $(0, 1)$). More importantly, 31°F is far outside the observed temperatures (53 to 81°F): this is an extrapolation, and the model has no way of knowing whether the logistic relation continues to hold there.

8.6.2 Poisson: ship incidents

For `ships`, the number of incidents Y_i in a cell with t_i months of service is Poisson with mean $\mu_i = t_i \lambda_i$; the rate is modelled with the log link, $\log \lambda_i = \boldsymbol{x}_i^\top \beta$, so that $\log \mu_i = \log t_i + \boldsymbol{x}_i^\top \beta$. The term $\log t_i$ is an **offset**: a covariate whose coefficient is fixed at 1 (Chapter 10).

```
fs <- glm(incidents ~ type + year + period + offset(log(service)), family =
↪  poisson, data = sh)
round(coef(summary(fs)), 4)
```

```
#>             Estimate Std. Error  z value Pr(>|z|)
#> (Intercept)  -6.4059     0.2174 -29.4600   0.0000
#> typeB        -0.5433     0.1776  -3.0595   0.0022
#> typeC        -0.6874     0.3290  -2.0891   0.0367
#> typeD        -0.0760     0.2906  -0.2614   0.7938
#> typeE         0.3256     0.2359   1.3803   0.1675
#> year65        0.6971     0.1496   4.6587   0.0000
#> year70        0.8184     0.1698   4.8207   0.0000
#> year75        0.4534     0.2332   1.9446   0.0518
#> period75      0.3845     0.1183   3.2507   0.0012
```

Here the random component is Poisson, the linear predictor includes ship type (4 dummy variables), construction year (3) and period of operation (1), and the link is the canonical log.

The coefficients are log rate ratios: `period75` has $e^{\hat\beta} = 1.469$, so the incident rate in 1975–79 is about 47% higher than in 1960–74, for ships of the same type and year of construction.

8.6.3 Gamma: tree volume

The `trees` data have the girth, height and volume of 31 black cherry trees. The volume is positive, and trees of larger size vary more in absolute terms, which suggests a gamma model with the log link, $\log \mu_i = \beta_0 + \beta_1 \log(\text{Girth}_i) + \beta_2 \log(\text{Height}_i)$, i.e. $\mu_i = e^{\beta_0}\,\text{Girth}_i^{\beta_1}\,\text{Height}_i^{\beta_2}$. A tree is roughly a cylinder or a cone, with volume proportional to $\text{Girth}^2 \times \text{Height}$, so we expect $\beta_1 \approx 2$ and $\beta_2 \approx 1$.

```
fg <- glm(Volume ~ log(Girth) + log(Height), family = Gamma(link = "log"),
↪  data = trees)
fl <- lm(log(Volume) ~ log(Girth) + log(Height), data = trees)
round(rbind(`gamma, log link` = coef(fg), `lm on log(y)` = coef(fl)), 3)
```

```
#>                 (Intercept) log(Girth) log(Height)
#> gamma, log link      -6.691      1.980       1.133
#> lm on log(y)         -6.632      1.983       1.117
```

```
c(phi_hat = summary(fg)$dispersion, nu_hat = 1 / summary(fg)$dispersion)
```

```
#>   phi_hat    nu_hat
#> 6.427e-03 1.556e+02
```

```
round(confint(fg), 3)
```

```
#>              2.5 % 97.5 %
#> (Intercept) -8.236 -5.139
#> log(Girth)   1.836  2.125
#> log(Height)  0.736  1.528
```

The two fits are close but not identical, because they model different things: the gamma GLM models $\log \mathrm{E}[Y]$ with $\mathrm{Var}(Y) = \phi\mu^2$, the linear model models $\mathrm{E}[\log Y]$ with constant variance on the log scale. The estimated dispersion $\hat\phi = 0.0064$ means a coefficient of variation of about $\sqrt{\hat\phi} = 0.08$, i.e. 8% around the mean at any size. The 95% profile intervals contain $\beta_1 = 2$ and $\beta_2 = 1$, consistent with a volume proportional to $\text{Girth}^2 \times \text{Height}$.

8.7 In R

`glm()` takes a model formula, a `family` and the data. A `family` object bundles the link and the variance function.

```
fam <- poisson()
c(fam$family, fam$link)
```

```
#> [1] "poisson" "log"
```

```
fam$linkfun(3); fam$linkinv(log(3)); fam$mu.eta(log(3)); fam$variance(3)
```

```
#> [1] 1.099
```

```
#> [1] 3
```

```
#> [1] 3
```

```
#> [1] 3
```

```
binomial()$linkfun(0.3); Gamma()$link; binomial(link = "probit")$link
```

```
#> [1] -0.8473
```

```
#> [1] "inverse"
```

```
#> [1] "probit"
```

`linkfun` is g, `linkinv` is g^{-1}, `mu.eta` is $d\mu/d\eta = 1/g'(\mu)$ and `variance` is $V(\mu)$. The families used in this course are `binomial`, `poisson`, `Gamma`, `gaussian`, `inverse.gaussian` and the "quasi" versions `quasibinomial` and `quasipoisson`. A binomial response is given either as 0/1 (or a factor), or as `cbind(successes, failures)`, or as a proportion with the argument `weights = m`. An offset enters as `offset(log(t))` in the formula or as the argument `offset =`.

```
glm(low ~ smoke + age, family = binomial, data = bw)                    # 0/1
↪  response
glm(cbind(damage, 6 - damage) ~ temp, family = binomial, data = orings)  #
↪  counts out of 6
glm(damage / 6 ~ temp, family = binomial, weights = rep(6, 23), data = orings)
↪  # same fit
glm(incidents ~ type + offset(log(service)), family = poisson, data = sh)
glm(Volume ~ log(Girth), family = Gamma(link = "log"), data = trees)
```

8.8 Common pitfalls

Warning. Do not say that "the logit link keeps the probability in $(0, 1)$". It is the *inverse* link that maps the real line into the valid range of the mean.

Warning. For the gamma, $b'(\theta) = -1/\theta$, not $1/\theta$. Many formulas in the literature absorb the sign by using $\theta = 1/\mu$ with $b(\theta) = \log\theta$ and a negative a; the family is the same, but be consistent.

Remark. $\mathrm{Var}(Y) = a(\phi)V(\mu)$, not $V(\mu)$: forgetting $a(\phi)$ gives the wrong variance for the binomial proportion ($a = 1/m$), the normal and the gamma.

Remark. A GLM is *not* a linear model for a transformed response (Example 8.3), and e^{β_j} is not, in general, a ratio of means: for the log link it is a ratio of means (a rate ratio); for the logit link it is a ratio of *odds*.

8.9 Chapter summary

- The exponential family $f(y;\theta,\phi) = \exp\{[y\theta - b(\theta)]/a(\phi) + c(y,\phi)\}$ contains the normal, Bernoulli/binomial, Poisson, gamma and inverse Gaussian distributions.
- From the score identities, $\mathrm{E}[Y] = b'(\theta)$ and $\mathrm{Var}(Y) = a(\phi)b''(\theta) = a(\phi)V(\mu)$; the cumulant generating function is $\{b(\theta + ta) - b(\theta)\}/a$ and $\kappa_r = a^{r-1}b^{(r)}(\theta)$.
- A GLM has a random component (exponential family), a linear predictor $\eta = X\beta$ and a link $g(\mu) = \eta$. The inverse link guarantees a valid mean.
- The canonical link $g = (b')^{-1}$ makes $\theta = \eta$, gives a concave log-likelihood with score $X^\top(\boldsymbol{y} - \boldsymbol{\mu})/\phi$, sufficient statistic $X^\top\boldsymbol{y}$, and $\sum \hat{\mu}_i = \sum y_i$ with an intercept.

- Logit, probit and complementary log-log links for binary data come from latent-variable or counting arguments; the log link for counts and positive data gives multiplicative effects.
- The variance depends on the mean through $V(\mu)$: 1 (normal), $\mu(1-\mu)$ (binomial), μ (Poisson), μ^2 (gamma), μ^3 (inverse Gaussian).

8.10 Exercises

Exercise 8.1. Let Y be exponential with rate λ, $f(y) = \lambda e^{-\lambda y}$, $y > 0$. (a) Write f in the form (8.1) and give θ, b, $a(\phi)$. (b) Use b to find $\mathrm{E}[Y]$ and $\mathrm{Var}(Y)$. (c) What is the canonical link for a regression model with exponential responses?

Solution.

(a) $f(y) = \exp\{-\lambda y + \log\lambda\}$. Take $\theta = -\lambda < 0$, $b(\theta) = -\log(-\theta)$ (so that $-b(\theta) = \log\lambda$), $a(\phi) = 1$, $c = 0$. This is the gamma family with shape $\nu = 1$.

(b) $b'(\theta) = -1/\theta = 1/\lambda = \mathrm{E}[Y]$ and $b''(\theta) = 1/\theta^2 = 1/\lambda^2 = \mathrm{Var}(Y)$, as expected for an exponential distribution.

(c) $\theta = -1/\mu$, so the canonical link is $g(\mu) = -1/\mu$ (or $1/\mu$ in R). It requires $\eta = \boldsymbol{x}^\top\beta < 0$ for every observation, which is restrictive. The log link is used in practice.

Exercise 8.2. The inverse Gaussian density with mean μ and shape λ is $f(y) = \sqrt{\lambda/(2\pi y^3)}\exp\{-\lambda(y-\mu)^2/(2\mu^2 y)\}$, $y > 0$. Write it in the form (8.1) and find θ, $b(\theta)$, $a(\phi)$, $V(\mu)$ and the canonical link.

Solution. Expand the exponent: $-\lambda(y-\mu)^2/(2\mu^2 y) = \lambda\{-y/(2\mu^2) + 1/\mu\} - \lambda/(2y)$. Hence

$$f(y) = \exp\left\{\frac{y\cdot(-\frac{1}{2\mu^2}) - (-\frac{1}{\mu})}{1/\lambda} - \frac{\lambda}{2y} + \tfrac{1}{2}\log\frac{\lambda}{2\pi y^3}\right\}.$$

So $\theta = -1/(2\mu^2) < 0$, $a(\phi) = \phi = 1/\lambda$, and, since $1/\mu = \sqrt{-2\theta}$, $b(\theta) = -\sqrt{-2\theta}$, $c(y,\phi) = -\lambda/(2y) + \frac{1}{2}\log\{\lambda/(2\pi y^3)\}$. Then $b'(\theta) = (-2\theta)^{-1/2} = \mu$ and $b''(\theta) = (-2\theta)^{-3/2} = \mu^3$. So $V(\mu) = \mu^3$, $\mathrm{Var}(Y) = \mu^3/\lambda$, and the canonical link is $g(\mu) = \theta = -1/(2\mu^2)$.

Exercise 8.3. Use Theorem 8.2 to (a) find the skewness $\kappa_3/\kappa_2^{3/2}$ of the Poisson distribution with mean μ and of the gamma distribution with mean μ and shape ν; (b) show that the standardised Poisson variable $(Y-\mu)/\sqrt{\mu}$ has cumulants $\kappa_r\mu^{-r/2}$ that tend to those of a standard normal as $\mu\to\infty$ for $r \geq 3$.

Solution.

(a) Poisson: $a = 1$, $b'''(\theta) = e^\theta = \mu$, so $\kappa_3 = \mu = \kappa_2$ and the skewness is $\mu/\mu^{3/2} = 1/\sqrt{\mu}$. Gamma: $b'(\theta) = -1/\theta$, $b'' = 1/\theta^2$, $b'''(\theta) = -2/\theta^3 = 2\mu^3$ (since $\theta = -1/\mu$). With $a = 1/\nu$, $\kappa_3 = a^2 b''' = 2\mu^3/\nu^2$ and $\kappa_2 = \mu^2/\nu$, so the skewness is $\{2\mu^3/\nu^2\}/\{\mu^3/\nu^{3/2}\} = 2/\sqrt{\nu}$.

(b) For the Poisson all cumulants equal μ. Standardising divides the rth cumulant by $\mu^{r/2}$, giving $\mu^{1-r/2}\to 0$ for $r \geq 3$, while $\kappa_1 = 0$ and $\kappa_2 = 1$ after standardisation. A normal distribution has all cumulants of order three and higher equal to 0.

Exercise 8.4. Let Y have the negative binomial distribution $P(Y=y) = \frac{\Gamma(y+k)}{y!\,\Gamma(k)}\left(\frac{k}{\mu+k}\right)^k\left(\frac{\mu}{\mu+k}\right)^y$, $y=0,1,\ldots$, with $k>0$ known. Show that it is in the exponential family with $a(\phi)=1$, and find $b(\theta)$, the mean and the variance function.

Solution. Taking logs, $\log P = y\log\frac{\mu}{\mu+k} + k\log\frac{k}{\mu+k} + \log\frac{\Gamma(y+k)}{y!\Gamma(k)}$. Put $\theta = \log\frac{\mu}{\mu+k} < 0$, so $e^\theta = \frac{\mu}{\mu+k}$ and $1-e^\theta = \frac{k}{\mu+k}$. Then $k\log\frac{k}{\mu+k} = k\log(1-e^\theta)$, so $b(\theta) = -k\log(1-e^\theta)$, $a=1$, $c(y)=\log\frac{\Gamma(y+k)}{y!\Gamma(k)}$. Differentiating, $b'(\theta) = \frac{ke^\theta}{1-e^\theta} = k\frac{\mu/(\mu+k)}{k/(\mu+k)} = \mu$ and $b''(\theta) = \frac{ke^\theta}{(1-e^\theta)^2} = k\frac{\mu}{\mu+k}\frac{(\mu+k)^2}{k^2} = \frac{\mu(\mu+k)}{k} = \mu + \frac{\mu^2}{k}$. So $V(\mu) = \mu + \mu^2/k$, which exceeds the Poisson variance μ. For *unknown* k the family is not an exponential family, but one estimates k separately (Chapter 10).

Exercise 8.5. A Poisson GLM with log link has an intercept and one binary covariate $x_i \in \{0,1\}$, with n_0 observations at $x=0$ and n_1 at $x=1$. Use the likelihood equations of Theorem 8.3 to show that $\hat\mu_i$ equals the sample mean of y in the group of observation i, and deduce $\hat\beta_1$ in closed form. Check it with $y=(2,3,6,7)$ for $x=0$ and $y=(10,12,15)$ for $x=1$.

Solution. The columns of X are $\mathbf{1}$ and x. The likelihood equations $X^\top(\boldsymbol{y}-\hat{\boldsymbol{\mu}}) = 0$ are $\sum_i(y_i-\hat\mu_i)=0$ and $\sum_{i:x_i=1}(y_i-\hat\mu_i)=0$. Subtracting, $\sum_{i:x_i=0}(y_i-\hat\mu_i)=0$ also. Within a group, $\hat\mu_i = \exp(\hat\beta_0+\hat\beta_1 x_i)$ is constant, so $\hat\mu = \bar y_{(0)}$ in group 0 and $\hat\mu = \bar y_{(1)}$ in group 1. Therefore $\hat\beta_0 = \log\bar y_{(0)}$ and $\hat\beta_1 = \log(\bar y_{(1)}/\bar y_{(0)})$. For the data, $\bar y_{(0)} = 18/4 = 4.5$ and $\bar y_{(1)} = 37/3 = 12.33$, so $\hat\beta_0 = \log 4.5 = 1.504$ and $\hat\beta_1 = \log(12.33/4.5) = 1.008$.

```
y <- c(2, 3, 6, 7, 10, 12, 15); x <- c(0, 0, 0, 0, 1, 1, 1)
coef(glm(y ~ x, family = poisson))
```

```
#> (Intercept)           x
#>       1.504       1.008
```

```
c(log(mean(y[x == 0])), log(mean(y[x == 1]) / mean(y[x == 0])))
```

```
#> [1] 1.504 1.008
```

Exercise 8.6. For each response give the random component, a link and, where needed, an offset or weights; justify in one sentence. (a) The number of customer complaints per month for 30 call centres of different sizes (number of calls received is known). (b) Whether a patient is readmitted within 30 days. (c) The number of seedlings out of 40 planted per plot that survive, as a function of shade. (d) The time in minutes to complete a task, where the standard deviation is about 40% of the mean. (e) The number of goals in a football match.

Solution.

(a) Poisson with log link and offset log(calls): a count with exposure, so the model is for the complaint *rate* per call. Check overdispersion. (b) Bernoulli (binomial with $m=1$), logit link; probit or complementary log-log are alternatives. (c) Binomial with $m=40$, `cbind(survived, 40 - survived)`, logit link; check overdispersion between plots. (d) Gamma with log link: a positive response with constant coefficient of variation $0.4 = 1/\sqrt{\nu}$ means $\nu \approx 6$; the log link keeps $\mu>0$. (e) Poisson with log link (the number of goals has no upper bound and no exposure; matches played enter as an offset if matches differ in length).

Exercise 8.7. A field is divided into plots; the number N_i of weeds in plot i of area A_i is Poisson with mean $A_i\lambda_i$, $\log\lambda_i = \boldsymbol{x}_i^\top\beta$. We only record whether a plot contains any weed, $Y_i = \mathbf{1}\{N_i > 0\}$. (a) Derive the link for $\mu_i = P(Y_i = 1)$ and show that the area appears as an offset. (b) How do you interpret e^{β_j}? (c) Is the link canonical for the Bernoulli family?

Solution.

(a) $\mu_i = 1 - P(N_i = 0) = 1 - \exp(-A_i\lambda_i)$, so $\log\{-\log(1-\mu_i)\} = \log A_i + \boldsymbol{x}_i^\top\beta$: the complementary log-log link with offset $\log A_i$. (b) e^{β_j} is the ratio of the *weed densities* λ (rate ratio of the underlying Poisson process) per unit increase in x_j; it is not an odds ratio. (c) No: the canonical link of the Bernoulli is the logit. For small μ, $-\log(1-\mu) \approx \mu$ and $\operatorname{logit}\mu \approx \log\mu$, so the two links and coefficients are close.

Exercise 8.8. Suppose $V(\mu) = \mu^p$ for some $p \neq 1, 2$. (a) Show that the canonical parameter is $\theta = \mu^{1-p}/(1-p)$ and find $b(\theta)$. (b) Verify that $p = 0, 1, 2, 3$ give the normal, Poisson, gamma and inverse Gaussian families (for $p = 1, 2$ use the limits).

Solution.

(a) Since $\mu = b'(\theta)$, $d\mu/d\theta = b''(\theta) = V(\mu) = \mu^p$, hence $d\theta/d\mu = \mu^{-p}$ and $\theta = \mu^{1-p}/(1-p)$ (choosing the constant 0). Solving, $\mu = \{(1-p)\theta\}^{1/(1-p)}$. Then $b'(\theta) = \mu = \{(1-p)\theta\}^{1/(1-p)}$ and $b(\theta) = \frac{1}{2-p}\{(1-p)\theta\}^{(2-p)/(1-p)} = \frac{\mu^{2-p}}{2-p}$.

(b) $p = 0$: $\theta = \mu$, $b = \theta^2/2$, normal. $p = 3$: $\theta = \mu^{-2}/(-2) = -1/(2\mu^2)$ and $b = \mu^{-1}/(-1) = -1/\mu$, the inverse Gaussian of Exercise 8.2. In the limits, $\mu^{1-p}/(1-p) \to \log\mu$ as $p \to 1$ (Poisson) and $\to -1/\mu$ when $p = 2$ (gamma). The members with $1 < p < 2$ (compound Poisson–gamma, Tweedie) have a point mass at 0 and are used for insurance claims.

Exercise 8.9. A logistic model for default of a loan gives $\hat\beta_0 = -3.2$, $\hat\beta_{\text{income}} = -0.015$ (income in thousand euros) and $\hat\beta_{\text{debt}} = 0.04$ (debt-to-income ratio in per cent). (a) Compute the probability of default for income 40 and debt ratio 30. (b) The odds ratio for an extra 10 thousand euros of income, and for a 5-point rise of the debt ratio. (c) Use the "divide by 4" rule to bound the change in probability for a one-point rise of the debt ratio. (d) Compute the actual change in probability at the profile of (a), and compare.

Solution.

(a) $\hat\eta = -3.2 - 0.015(40) + 0.04(30) = -2.6$, $\hat\mu = e^{-2.6}/(1+e^{-2.6}) = 0.069$. (b) $e^{-0.15} = 0.861$ for income (odds multiplied by 0.86, a 14% fall) and $e^{0.20} = 1.221$ for the debt ratio (22% higher odds). (c) $\partial\mu/\partial x = \mu(1-\mu)\beta \le \beta/4$, so at most $0.04/4 = 0.010$ per point, i.e. one percentage point. (d) At $\mu = 0.069$, $\mu(1-\mu)\beta = 0.069 \times 0.931 \times 0.04 = 0.0026$ per point: much smaller than the bound, because the probability is far from $1/2$. (Exact change from 30 to 31: $\text{plogis}(-2.56) - \text{plogis}(-2.6) = 0.0718 - 0.0691 = 0.0027$.)

Exercise 8.10. Using `trees`, fit the gamma log-link model of Section 8.6.3 and (a) test $H_0 : \beta_1 = 2$ with a Wald statistic; (b) compare the fitted volume for a tree with girth 15 and height 75 under the gamma GLM and under `lm(log(Volume) ~ ...)` (back-transformed); (c) explain in two sentences why the two answers differ.

Solution.

```
fg <- glm(Volume ~ log(Girth) + log(Height), family = Gamma(link = "log"),
↪  data = trees)
fl <- lm(log(Volume) ~ log(Girth) + log(Height), data = trees)
est <- coef(summary(fg))["log(Girth)", ]
c(z = (est[1] - 2) / est[2])
```

```
#> z.Estimate
#>    -0.2651
```

```
nd <- data.frame(Girth = 15, Height = 75)
c(gamma = predict(fg, nd, type = "response"), lm_backtransformed =
↪  exp(predict(fl, nd)),
  lm_smearing = exp(predict(fl, nd)) * mean(exp(residuals(fl))))
```

```
#>              gamma.1 lm_backtransformed.1        lm_smearing.1
#>                35.27                35.19                35.29
```

(a) $z = (\hat\beta_1 - 2)/\text{se} = -0.27$ (reference: t with 28 df, as in the summary, or $N(0,1)$): no evidence against $\beta_1 = 2$. (b) The predictions are close: gamma 35.3, back-transformed linear model 35.2 (35.3 with Duan's smearing correction). (c) Back-transforming $\mathrm{E}[\log Y]$ estimates the *median* of a lognormal response, not its mean, which is larger by the factor $\exp(s^2/2)$ (about 0.3% here, since $\hat\sigma^2 \approx 0.0066$); the gamma GLM models the mean directly. With such a small variance on the log scale the difference is tiny.

Exercise 8.11. With the O-ring model of Section 8.6.1: (a) Estimate the temperature at which the damage probability of a single O-ring is 1/2 (the "LD50") and give a delta-method standard error. (b) Estimate the probability that *none* of the 6 O-rings is damaged at 70°F, assuming independence. (c) State two reasons why the extrapolation to 31°F is not reliable.

Solution.

```
b <- coef(o); V <- vcov(o)
ld50 <- -b[1] / b[2]
g <- c(-1 / b[2], b[1] / b[2]^2)                    # gradient of -b0/b1
c(ld50 = ld50, se = sqrt(drop(t(g) %*% V %*% g)))
```

```
#> ld50.(Intercept)               se
#>           53.937            2.516
```

```
(1 - plogis(b[1] + b[2] * 70))^6
```

```
#> (Intercept)
#>      0.8326
```

(a) $\hat\beta_0 + \hat\beta_1 T = 0$ gives $T = -\hat\beta_0/\hat\beta_1 = 53.9$°F, with delta-method standard error, from the gradient $(-1/\beta_1,\ \beta_0/\beta_1^2)$, of 2.5°F. (b) $(1-\hat\mu)^6 = 0.83$ at 70°F. (c) 31°F is 22°F below the lowest observed temperature, and the logistic shape of the dose–response curve is an assumption that the data cannot check outside their range; moreover the O-rings of one launch are not independent (shared conditions), which the binomial model ignores and which would make the true uncertainty larger.

Exercise 8.12. *Critique.* A student writes: "To model the number of claims I took $\log(y+1)$ and fitted a linear model. The coefficient 0.5 of `x` means that the mean number of claims increases by 50% per unit of `x`." Use simulations like the one in Example 8.3, with the intercept 1 and with the intercept 4 (large means), to explain what is wrong and how to fix it.

Solution.

```
sim <- function(b0, reps = 200) {
  set.seed(2027)
  res <- replicate(reps, { x <- runif(200); y <- rpois(200, exp(b0 + 0.5 * x))
    c(unname(coef(lm(log(y + 1) ~ x))[2]), unname(coef(glm(y ~ x, family =
    ↪  poisson))[2])) })
  setNames(rowMeans(res), c("lm(log(y+1))", "Poisson glm"))
}
round(rbind(`intercept 1` = sim(1), `intercept 4` = sim(4)), 3)
```

```
#>             lm(log(y+1)) Poisson glm
#> intercept 1        0.403       0.491
#> intercept 4        0.492       0.495
```

The slope of $\log(y+1)$ on x is a slope for $\mathrm{E}[\log(Y+1)]$, not for $\log \mathrm{E}[Y]$. With small means (intercept 1) it is too small, because $\log(y+1)$ compresses the small counts; with large means (intercept 4) $\log(y+1) \approx \log y$ and the two approaches agree. The bias therefore depends on the level of the counts, which makes the coefficients hard to interpret. Moreover, even for the Poisson GLM a coefficient of 0.5 does not mean a 50% increase: for the log link $e^{0.5} = 1.65$, a 65% increase of the *mean* count per unit of `x`, and the percentage change is $100(e^{\hat\beta} - 1)$. The fix is `glm(y ~ x, family = poisson)` (or quasi-Poisson or negative binomial if the counts are overdispersed).

8.11 Further reading

Nelder and Wedderburn (1972) introduced GLMs; McCullagh and Nelder (1989) is the classic monograph (Chapters 1–2 for the exponential family and the canonical link). Wood (2017), Section 3.1, covers the same material concisely with R, and Pawitan (2001), Chapter 4, derives the exponential family from the likelihood point of view. For the choice of link functions and diagnostics, Faraway (2016) and Hosmer et al. (2013) are readable accounts.

Chapter 9

Fitting and inference for GLMs

Chapter 8 defined the GLM. In this chapter we show how it is fitted and how to draw conclusions from it. The coefficients $\hat{\beta}$ are maximum likelihood estimates, but unlike the linear model they have no closed form: they are obtained by an iterative algorithm, *iteratively reweighted least squares* (IRLS), which turns out to be Fisher scoring and, for canonical links, Newton–Raphson. Once $\hat{\beta}$ is available, the likelihood theory of Chapter 4 provides standard errors, Wald tests and confidence intervals, and likelihood-ratio tests based on the *deviance*. We also study the estimation of the dispersion, several kinds of residuals (including randomized quantile residuals, which work for discrete data), and model comparison by AIC. The references are McCullagh and Nelder (1989) (Chapters 2–4), Wood (2017) (Section 3.1) and Dunn and Smyth (1996) for residuals.

9.1 A motivating example: what does `glm()` do?

We use the ships data of Chapter 8: Y_i incidents in t_i months of service for the cell i defined by ship type, year of construction and period of operation. A Poisson model with offset $\log t_i$ is

```
sh <- subset(ships, service > 0)
sh$year <- factor(sh$year); sh$period <- factor(sh$period)
fs <- glm(incidents ~ type + year + period + offset(log(service)), family =
↪  poisson, data = sh)
summary(fs)
```

```
#>
#> Call:
#> glm(formula = incidents ~ type + year + period + offset(log(service)),
#>     family = poisson, data = sh)
#>
#> Coefficients:
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept)   -6.406      0.217  -29.46  < 2e-16 ***
#> typeB         -0.543      0.178   -3.06   0.0022 **
#> typeC         -0.687      0.329   -2.09   0.0367 *
#> typeD         -0.076      0.291   -0.26   0.7938
#> typeE          0.326      0.236    1.38   0.1675
#> year65         0.697      0.150    4.66  3.2e-06 ***
#> year70         0.818      0.170    4.82  1.4e-06 ***
#> year75         0.453      0.233    1.94   0.0518 .
```

```
#> period75       0.385      0.118    3.25   0.0012 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> (Dispersion parameter for poisson family taken to be 1)
#>
#>     Null deviance: 146.328  on 33  degrees of freedom
#> Residual deviance:  38.695  on 25  degrees of freedom
#> AIC: 154.6
#>
#> Number of Fisher Scoring iterations: 5
```

The printed output contains the coefficients with their standard errors and Wald z statistics, the *null* and *residual deviance*, the AIC, and the number of "Fisher scoring iterations". This chapter explains each of them. We will also see that the residual deviance 38.7 on 25 degrees of freedom is larger than we would expect from a Poisson model with $\phi = 1$, a first sign of overdispersion, taken up in Chapter 10.

9.2 The likelihood equations

Let $Y_1, \dots, Y_n$ be independent with densities of the form (8.1), $a_i(\phi) = \phi / m_i$ for known prior weights m_i (for binomial proportions, m_i is the number of trials; usually $m_i = 1$). Write $\mu_i = b'(\theta_i)$, $g(\mu_i) = \eta_i = \boldsymbol{x}_i^\top \beta$ and $V(\mu_i) = b''(\theta_i)$. The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \ell_i, \qquad \ell_i = \frac{m_i \{ y_i \theta_i - b(\theta_i) \}}{\phi} + c(y_i, \phi). \tag{9.1}$$

The parameter β enters through θ_i, which depends on β via μ_i and η_i.

Theorem 9.1 (Score and information of a GLM)**.** *Let $G = \operatorname{diag}\{g'(\mu_i)\}$ and $W = \operatorname{diag}\{w_i\}$ with*

$$w_i = \frac{m_i}{V(\mu_i)\, g'(\mu_i)^2}. \tag{9.2}$$

Then the score and the expected (Fisher) information of β are

$$S(\beta) = \frac{1}{\phi} X^\top W G (\boldsymbol{y} - \boldsymbol{\mu}), \qquad \mathcal{I}(\beta) = \frac{1}{\phi} X^\top W X. \tag{9.3}$$

The likelihood equations $S(\hat\beta) = 0$ can be written $\sum_i \frac{m_i (y_i - \hat\mu_i)\, x_{ij}}{V(\hat\mu_i)\, g'(\hat\mu_i)} = 0$, $j = 1, \dots, p$.

Proof. By the chain rule,

$$\frac{\partial \ell_i}{\partial \beta_j} = \frac{\partial \ell_i}{\partial \theta_i} \frac{d\theta_i}{d\mu_i} \frac{d\mu_i}{d\eta_i} \frac{\partial \eta_i}{\partial \beta_j}.$$

Now $\partial \ell_i / \partial \theta_i = m_i \{ y_i - b'(\theta_i) \} / \phi = m_i (y_i - \mu_i)/\phi$; since $d\mu_i / d\theta_i = b''(\theta_i) = V(\mu_i)$, $d\theta_i / d\mu_i = 1 / V(\mu_i)$; $d\mu_i / d\eta_i = 1 / g'(\mu_i)$; and $\partial \eta_i / \partial \beta_j = x_{ij}$. Hence

$$\frac{\partial \ell}{\partial \beta_j} = \frac{1}{\phi} \sum_i \frac{m_i (y_i - \mu_i)\, x_{ij}}{V(\mu_i)\, g'(\mu_i)} = \frac{1}{\phi} \sum_i w_i \, g'(\mu_i) (y_i - \mu_i)\, x_{ij},$$

because $w_i g'(\mu_i) = m_i / \{ V(\mu_i) g'(\mu_i) \}$. This is the jth component of $\phi^{-1} X^\top W G (\boldsymbol{y} - \boldsymbol{\mu})$. For the information, $\mathcal{I}(\beta) = \operatorname{Cov}(S)$ (Lemma 8.1 applied to the whole sample). With $\operatorname{Var}(\boldsymbol{y}) =$

$\phi\,\mathrm{diag}\{V(\mu_i)/m_i\}$ we get $G\mathrm{Var}(\boldsymbol{y})G = \phi\,\mathrm{diag}\{g'(\mu_i)^2 V(\mu_i)/m_i\} = \phi W^{-1}$. Therefore

$$\mathrm{Cov}(S) = \frac{1}{\phi^2} X^\top W G\,\mathrm{Var}(\boldsymbol{y})\,GWX = \frac{1}{\phi^2} X^\top W (\phi W^{-1}) W X = \frac{1}{\phi} X^\top W X.$$

□

Three observations follow at once.

1. ϕ **cancels from the likelihood equations**: $\hat\beta$ does not depend on the dispersion, whether it is known or not. Only the standard errors do.
2. Only the **mean and the variance function** appear in the equations, not the full distribution. This is the basis of quasi-likelihood (Section 9.6).
3. The weight w_i is large for observations whose mean is determined precisely (V small relative to the link slope) and shrinks for noisy ones. For the canonical link, $g'(\mu) = 1/V(\mu)$ and $w_i = m_i V(\mu_i)$; for the Poisson with the log link $w_i = \mu_i$, for the logistic $w_i = m_i\mu_i(1-\mu_i)$.

If the link is canonical, $WG = \mathrm{diag}\{m_i\}$ and the equations reduce to Theorem 8.3: $X^\top M(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) = 0$. For a general link the equations remain nonlinear in β and must be solved iteratively.

9.2.1 Observed information

The **observed** information is $I(\beta) = -\partial^2\ell/\partial\beta\partial\beta^\top$. For a non-canonical link it differs from the expected information.

Proposition 9.1 (Observed information)**.** *For any link,*

$$I(\beta) = \frac{1}{\phi}\left[X^\top W X + \sum_{i=1}^{n} m_i\,(y_i - \mu_i)\,\frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{V(\mu_i)^2\,g'(\mu_i)^3}\,\boldsymbol{x}_i\boldsymbol{x}_i^\top\right]. \tag{9.4}$$

The second term vanishes when $g'(\mu) = 1/V(\mu)$, that is, for the canonical link.

Proof. Write the ith score term as $\phi^{-1} m_i u_i \boldsymbol{x}_i$ with $u_i = (y_i - \mu_i)/\{V(\mu_i)g'(\mu_i)\}$. Its derivative with respect to $\beta^\top$ is $\phi^{-1} m_i\,(du_i/d\eta_i)\,\boldsymbol{x}_i\boldsymbol{x}_i^\top$, and by the chain rule $du_i/d\eta_i = (du_i/d\mu_i)/g'(\mu_i)$, where

$$\frac{du_i}{d\mu_i} = -\frac{1}{Vg'} - (y_i - \mu_i)\frac{V'g' + Vg''}{(Vg')^2}.$$

Hence $-\,du_i/d\eta_i = \dfrac{1}{Vg'^2} + (y_i - \mu_i)\dfrac{V'g' + Vg''}{V^2 g'^3}$, which gives (9.4). For $g' = 1/V$ we have $g'' = -V'/V^2$ and $V'g' + Vg'' = V'/V - V'/V = 0$. □

Because $\mathrm{E}[y_i - \mu_i] = 0$, the expectation of the second term is zero, so $\mathrm{E}[I(\beta)] = \mathcal{I}(\beta)$, as it should. A numerical check for the probit model of the birth-weight data (non-canonical link) compares (9.4), computed at $\hat\beta$, with a numerical Hessian of the log-likelihood.

```
bw <- with(birthwt, data.frame(low = low, age = age, lwt = lwt,
  race = factor(race, labels = c("white", "black", "other")),
  smoke = smoke, ptd = as.integer(ptl > 0), ht = ht, ui = ui))
form <- low ~ age + lwt + race + smoke + ptd + ht + ui
fpr <- glm(form, family = binomial(link = "probit"), data = bw)
X <- model.matrix(form, bw); y <- bw$low; mu <- fitted(fpr); eta <- drop(X %*%
↪  coef(fpr))
loglik <- function(b) sum(dbinom(y, 1, pnorm(drop(X %*% b)), log = TRUE))
```

```r
H <- -optimHess(coef(fpr), loglik)                          # numerical observed
↪  information
g1 <- 1 / dnorm(eta); g2 <- eta / dnorm(eta)^2              # g'(mu), g''(mu) for
↪  g = qnorm
V <- mu * (1 - mu); V1 <- 1 - 2 * mu; w <- 1 / (V * g1^2)
I_exp <- crossprod(X, w * X)
I_obs <- crossprod(X, (w + (y - mu) * (V1 * g1 + V * g2) / (V^2 * g1^3)) * X)
rel <- function(A, B) max(abs(A - B)) / max(abs(B))
c(formula_vs_numerical = rel(I_obs, H), expected_vs_numerical = rel(I_exp, H))
```

```
#>  formula_vs_numerical expected_vs_numerical
#>             6.352e-05             1.682e-03
```

The formula reproduces the numerical Hessian to the accuracy of the finite differences, while the expected information differs from the observed one at the percent level: the two give slightly different standard errors for a non-canonical link.

9.3 Iteratively reweighted least squares

Newton–Raphson solves $S(\beta)=0$ by the iteration $\beta^{(t+1)}=\beta^{(t)}+I(\beta^{(t)})^{-1}S(\beta^{(t)})$. **Fisher scoring** replaces the observed information by its expectation:

$$\beta^{(t+1)}=\beta^{(t)}+\mathcal{I}(\beta^{(t)})^{-1}S(\beta^{(t)}). \tag{9.5}$$

Fisher scoring is often preferred because $\mathcal{I}=\phi^{-1}X^\top WX$ is always positive semidefinite (so the step is an ascent direction), and it does not involve $\boldsymbol{y}$ in the matrix.

Theorem 9.2 (Fisher scoring is iterated weighted least squares)**.** *Let $\boldsymbol{z}$ be the **working response**, with components $z_i=\eta_i+(y_i-\mu_i)\,g'(\mu_i)$, computed at $\beta^{(t)}$ (so W, $\boldsymbol{\eta}$, $\boldsymbol{\mu}$ and G are all at $\beta^{(t)}$). Then*

$$\beta^{(t+1)}=(X^\top WX)^{-1}X^\top W\boldsymbol{z}.$$

That is, $\beta^{(t+1)}$ is the weighted least-squares estimate from regressing $\boldsymbol{z}$ on X with weights w_i.

Proof. Substitute (9.3) in (9.5); the ϕ cancels:

$$\begin{aligned}\beta^{(t+1)}&=\beta^{(t)}+(X^\top WX)^{-1}X^\top WG(\boldsymbol{y}-\boldsymbol{\mu})\\&=(X^\top WX)^{-1}X^\top W\{X\beta^{(t)}+G(\boldsymbol{y}-\boldsymbol{\mu})\}=(X^\top WX)^{-1}X^\top W\boldsymbol{z},\end{aligned}$$

since $X\beta^{(t)}=\boldsymbol{\eta}$ and $\boldsymbol{\eta}+G(\boldsymbol{y}-\boldsymbol{\mu})$ has components $\eta_i+g'(\mu_i)(y_i-\mu_i)=z_i$. □

The working response has a nice interpretation: a first-order Taylor expansion of $g(y_i)$ around μ_i gives $g(y_i)\approx g(\mu_i)+(y_i-\mu_i)g'(\mu_i)=z_i$. The working response is a *linearized* version of the transformed data $g(y_i)$, and the weight w_i is the inverse of its approximate variance, $\operatorname{Var}(z_i)\approx g'(\mu_i)^2\operatorname{Var}(y_i)=\phi/w_i$. Each step is therefore exactly a weighted linear regression, which explains the name: the weights are recomputed at every iteration.

Definition 9.1 (IRLS algorithm)**.**

1. Choose starting values $\hat\mu_i$ in the range of the mean (for instance $(y_i+\bar y)/2$, which avoids $\log 0$ and $\operatorname{logit}\{0,1\}$) and set $\hat\eta_i=g(\hat\mu_i)$.

2. Compute $z_i = \hat\eta_i - o_i + (y_i - \hat\mu_i) g'(\hat\mu_i)$ and $w_i = m_i / \{V(\hat\mu_i) g'(\hat\mu_i)^2\}$, where o_i is a known offset.
3. Solve the weighted least-squares problem $\hat\beta = (X^\top W X)^{-1} X^\top W \boldsymbol{z}$.
4. Update $\hat\eta = X\hat\beta + \boldsymbol{o}$, $\hat\mu = g^{-1}(\hat\eta)$ and the deviance D (Section 9.5).
5. Stop if $|D^{\text{new}} - D^{\text{old}}| / (|D^{\text{new}}| + 0.1) < \varepsilon$ (R: `epsilon = 1e-8`); otherwise return to step 2.

Two details are easy to get wrong in an implementation: the coefficient vector returned must be the one of the **last** update (the one that matches the final $\hat\mu$), and the convergence flag must record whether the stopping rule was actually met. At convergence the weights are evaluated at $\hat\beta$, which gives $\widehat{\operatorname{Var}}(\hat\beta) = \hat\phi\,(X^\top \hat W X)^{-1}$.

9.3.1 Newton–Raphson versus Fisher scoring

Theorem 9.3 (Fisher scoring equals Newton–Raphson for canonical links)**.** *If g is the canonical link, then $I(\beta) = \mathcal{I}(\beta)$ for all β, and the two iterations coincide. For any other link they have the same fixed point $\hat\beta$ but different paths.*

Proof. For the canonical link, the second term of (9.4) is 0 (Proposition 9.1), so $I = \mathcal{I}$. Both algorithms stop where $S(\beta) = 0$ because the update is $\beta + \mathcal{I}^{-1} S(\beta)$ in both cases, which equals β iff $S(\beta) = 0$. □

Newton–Raphson converges quadratically near $\hat\beta$ (the number of correct digits roughly doubles); Fisher scoring with a non-canonical link converges linearly, with a rate that depends on how far I is from $\mathcal{I}$. The following implementation follows Definition 9.1; it uses the pieces stored in every R `family` object.

```
irls <- function(X, y, family = binomial(), offset = rep(0, length(y)),
↪  weights = rep(1, length(y)),
                tol = 1e-10, maxit = 50) {
 mu <- (y + mean(y)) / 2;  eta <- family$linkfun(mu)       # starting values
↪  inside the range
 dev <- sum(family$dev.resids(y, mu, weights));  trace <- numeric(0);
↪  converged <- FALSE
 for (iter in 1:maxit) {
   d    <- family$mu.eta(eta)                              # d mu / d eta = 1
↪  / g'(mu)
   z    <- eta - offset + (y - mu) / d                     # working
↪  response
   w    <- weights * d^2 / family$variance(mu)             # working weights
   beta <- solve(crossprod(X, w * X), crossprod(X, w * z))
   eta  <- drop(X %*% beta) + offset                       # update with the
↪  NEW beta
   mu   <- family$linkinv(eta)
   dev_new <- sum(family$dev.resids(y, mu, weights));  trace <- c(trace,
↪  dev_new)
   if (abs(dev_new - dev) / (abs(dev_new) + 0.1) < tol) { converged <- TRUE;
    ↪  dev <- dev_new; break }
   dev <- dev_new
 }
```

```
  w <- weights * family$mu.eta(eta)^2 / family$variance(mu)  # weights at the
↪  final beta
  list(coefficients = drop(beta), deviance = dev, iter = iter, converged =
↪  converged,
       trace = trace, cov_unscaled = solve(crossprod(X, w * X)))
}
my  <- irls(X, y, binomial())
fit <- glm(form, family = binomial, data = bw, control = glm.control(epsilon =
↪  1e-10))
c(iterations = my$iter, converged = my$converged, max_abs_diff =
↪  max(abs(my$coefficients - coef(fit))))
```

```
#>   iterations    converged max_abs_diff
#>    5.000e+00    1.000e+00    5.096e-14
```

```
round(my$trace, 4)
```

```
#> [1] 198.0 196.8 196.8 196.8 196.8
```

```
max(abs(sqrt(diag(my$cov_unscaled)) - sqrt(diag(vcov(fit)))))
```

```
#> [1] 1.941e-09
```

The hand-made IRLS reproduces the coefficients of `glm()` to about 10^{-13} and the standard errors to about 10^{-9} (the tolerance of `glm()` itself). The deviance (printed after each update) drops from 198.0 to 196.8337 in three steps and then does not change.

For the ships model, with an offset, the same function works:

```
Xs <- model.matrix(~ type + year + period, data = sh)
my_s <- irls(Xs, sh$incidents, poisson(), offset = log(sh$service))
c(iterations = my_s$iter, max_abs_diff = max(abs(my_s$coefficients -
↪  coef(fs))))
```

```
#>   iterations max_abs_diff
#>    6.000e+00    2.904e-10
```

Figure 9.1 shows the distance to the MLE at each iteration for the logistic (canonical, Newton–Raphson) and the probit (non-canonical, Fisher scoring) models.

```
path <- function(link, maxit = 40, show = 9) {
  fam <- binomial(link = link); mu <- (y + mean(y)) / 2; eta <-
↪  fam$linkfun(mu)
  B <- matrix(NA, maxit, ncol(X))
  for (t in 1:maxit) {
    d <- fam$mu.eta(eta); z <- eta + (y - mu) / d; w <- d^2 / fam$variance(mu)
    B[t, ] <- solve(crossprod(X, w * X), crossprod(X, w * z)); eta <- drop(X
↪  %*% B[t, ]); mu <- fam$linkinv(eta)
  }
  data.frame(link = link, iter = 1:show, err = pmax(apply(B[1:show, ], 1,
↪  function(b) max(abs(b - B[maxit, ]))), 1e-16))
}
ggplot(rbind(path("logit"), path("probit")), aes(iter, err, colour = link)) +
  geom_line(linewidth = 1) + geom_point(size = 2.2) + scale_y_log10() +
```

```
scale_colour_manual(values = ie_pal[1:2]) + labs(x = "IRLS iteration", y =
↪  "max |beta(t) - beta-hat|", colour = NULL)
```

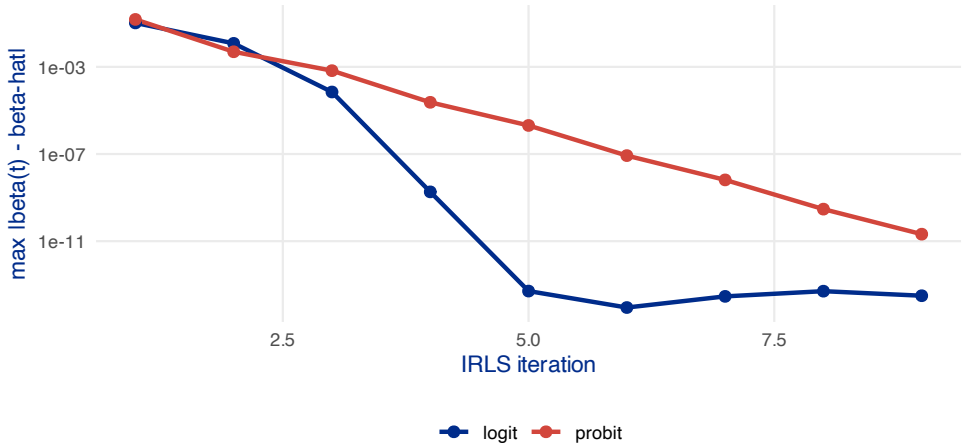


Figure 9.1: Distance of the IRLS iterate from the MLE on the birth-weight data: quadratic convergence for the canonical logit link, linear for the probit link.

The logistic model reaches machine precision after five steps; the probit model gains about 1.5 orders of magnitude per step. Both stop at the same $\hat\beta$.

Remark. **Practical issues.** (i) *Starting values* must lie in the range of μ; R uses family-specific values such as $(y_i + 0.5)/2$ for Bernoulli data and $y_i + 0.1$ for Poisson data. (ii) *Offsets* enter the working response as $z_i = \eta_i - o_i + \dots$ and are added back when updating η. (iii) *Prior weights* multiply w_i. (iv) A step may overshoot and increase the deviance; R then halves the step. (v) For binary data the likelihood may have no maximum (*separation*, Chapter 10): the iterations then diverge, and `glm()` warns that "fitted probabilities numerically 0 or 1 occurred".

9.4 Inference for the coefficients

Under the regularity conditions of Chapter 4, the MLE $\hat\beta$ is consistent and asymptotically normal, with covariance equal to the inverse information:

$$\hat\beta \;\dot\sim\; N\!\left(\beta,\; \mathcal{I}(\beta)^{-1}\right) = N\!\left(\beta,\; \phi\,(X^\top W X)^{-1}\right). \tag{9.6}$$

This is a large-sample statement: the "sample size" that matters is the amount of information per parameter, which for grouped binomial data means that both m_i and n may need to be large. For a normal model with identity link it is exact. In practice W is evaluated at $\hat\beta$ and, if ϕ is unknown, replaced by an estimate $\hat\phi$ (Section 9.6).

Wald test and interval. For a single coefficient, $\mathrm{se}(\hat\beta_j) = \sqrt{\hat\phi\,[(X^\top \hat W X)^{-1}]_{jj}}$ and

$$z = \frac{\hat\beta_j}{\mathrm{se}(\hat\beta_j)} \;\dot\sim\; N(0,1) \quad \text{under } H_0 : \beta_j = 0, \qquad \hat\beta_j \pm z_{1-\alpha/2}\,\mathrm{se}(\hat\beta_j).$$

When ϕ is estimated, `glm()` reports a t statistic with $n - p$ degrees of freedom instead (for the gamma, quasi-Poisson, quasi-binomial and normal families). A Wald interval for β_j transforms to an interval for e^{β_j} by transforming the endpoints.

Likelihood-ratio test and profile interval. For nested models $M_0 \subset M_1$ with $p_0 < p_1$ parameters, $2\{\ell(\hat\beta_1) - \ell(\hat\beta_0)\} \;\dot\sim\; \chi^2_{p_1 - p_0}$ under M_0 (Wilks). The *profile likelihood* interval collects

the β_j for which $2\{\ell(\hat\beta) - \ell_p(\beta_j)\} \le \chi^2_{1;1-\alpha}$, where $\ell_p(\beta_j)$ is the likelihood maximized over the other parameters. It does not rely on a quadratic log-likelihood and is the interval that `confint()` computes for a `glm`. **Score test.** The score statistic $S(\tilde\beta)^\top \mathcal{I}(\tilde\beta)^{-1} S(\tilde\beta)$, evaluated at the estimate $\tilde\beta$ under M_0, needs only the fit of the smaller model; `anova(fit, test = "Rao")` computes it.

Example 9.1 (Wald and profile intervals on birth weight)**.** Is the effect of smoking on low birth weight significant, and how different are the two kinds of interval?

```
fit <- glm(form, family = binomial, data = bw)
est <- coef(summary(fit))
wald <- exp(cbind(OR = est[, 1], lower = est[, 1] - 1.96 * est[, 2], upper =
↪  est[, 1] + 1.96 * est[, 2]))
prof <- exp(confint(fit))
round(cbind(wald, prof_lower = prof[, 1], prof_upper = prof[, 2])[c("smoke",
↪  "ht", "ptd"), ], 3)
```

```
#>          OR lower  upper prof_lower prof_upper
#> smoke 2.331 1.048  5.187      1.058      5.293
#> ht    6.288 1.585 24.954      1.640     27.258
#> ptd   3.393 1.369  8.409      1.382      8.602
```

```
est["smoke", ]
```

```
#>   Estimate Std. Error    z value   Pr(>|z|)
#>    0.84640    0.40806    2.07419    0.03806
```

For smoking, $z = 2.074$, $p = 0.038$. The Wald interval for the odds ratio is $(1.05, 5.19)$ and the profile interval is $(1.06, 5.29)$: similar. For hypertension, with only 12 hypertensive mothers, the log-likelihood in β_{ht} is visibly skewed and the profile interval reaches farther to the right. When the two intervals disagree, prefer the profile one.

9.5 The deviance

The **saturated model** has one parameter per observation, $\tilde\mu_i = y_i$ (the best possible fit, a perfect one); its log-likelihood ℓ_{sat} is the maximum possible for the family. The deviance measures how far the fitted model is from it.

Definition 9.2 (Deviance)**.** Let $\hat\theta_i$ and $\tilde\theta_i$ be the canonical parameters of the fitted and the saturated model. The **scaled deviance** and the **deviance** are

$$D^* = 2\{\ell_{\text{sat}} - \ell(\hat\beta)\} = \frac{2}{\phi}\sum_{i=1}^{n} m_i\{y_i(\tilde\theta_i - \hat\theta_i) - b(\tilde\theta_i) + b(\hat\theta_i)\}, \qquad D = \phi\, D^* = \sum_{i=1}^{n} d_i,$$

with $d_i \ge 0$ the unit deviances. The **null deviance** is the deviance of the model with only an intercept, the **residual deviance** that of the fitted model.

The deviance generalizes the residual sum of squares. The unit deviances of the main families are (with $0 \log 0 = 0$):

Table 9.1: Unit deviances.

Family	unit deviance d_i
Normal	$(y_i - \hat\mu_i)^2$
Poisson	$2\{y_i \log(y_i/\hat\mu_i) - (y_i - \hat\mu_i)\}$
Binomial (y_i of m_i)	$2\{y_i \log \frac{y_i}{m_i\hat\mu_i} + (m_i - y_i)\log\frac{m_i - y_i}{m_i(1-\hat\mu_i)}\}$
Gamma	$2\{-\log(y_i/\hat\mu_i) + (y_i - \hat\mu_i)/\hat\mu_i\}$

Proof. Poisson. Here $\theta = \log\mu$, $b(\theta) = e^\theta$, $a = 1$, so $d = 2\{y(\log y - \log\hat\mu) - y + \hat\mu\} = 2\{y\log(y/\hat\mu) - (y - \hat\mu)\}$. *Normal.* $\theta = \mu$, $b = \theta^2/2$: $d = 2\{y(y - \hat\mu) - y^2/2 + \hat\mu^2/2\} = (y - \hat\mu)^2$. *Gamma.* $\theta = -1/\mu$, $b = -\log(-\theta) = \log\mu$: $d = 2\{y(-1/y + 1/\hat\mu) - \log y + \log\hat\mu\} = 2\{-1 + y/\hat\mu - \log(y/\hat\mu)\}$. The binomial case is the Poisson computation applied to successes and failures. □

Three points to remember about the deviance.

1. **Deviance differences are likelihood-ratio statistics.** For nested models, the saturated term cancels,
$$2\{\ell(\hat\beta_1) - \ell(\hat\beta_0)\} = D_0^* - D_1^* \mathrel{\dot\sim} \chi^2_{p_1 - p_0} \quad \text{under } M_0. \tag{9.7}$$
If ϕ is known (Poisson, binomial) use $D_0 - D_1$ directly. If ϕ is estimated, use the F statistic $F = \{(D_0 - D_1)/(p_1 - p_0)\}/\hat\phi \mathrel{\dot\sim} F_{p_1 - p_0,\, n - p_1}$.
2. **The residual deviance is not generally χ^2_{n-p}.** The saturated model has n parameters, which grow with the sample, so Wilks's theorem does not apply to D^* itself; the approximation $D^* \approx \chi^2_{n-p}$ holds only if each observation carries a lot of information (Poisson means that are not small, binomial m_i large). With small counts $\mathrm{E}[D] < n - p$. For *binary* data it is useless, as the next proposition shows.
3. **The null and residual deviances give a pseudo-R^2,** McFadden's $1 - D/D_{\text{null}}$, which is the proportional reduction of the deviance relative to the intercept-only model. It is not a proportion of variance explained.

Proposition 9.2 (The deviance of a binary model carries no information about fit)**.** *For Bernoulli data with the logit link and an intercept in the model, $\ell_{sat} = 0$ and*

$$D = -2\sum_{i=1}^{n}\{\hat\mu_i \operatorname{logit}\hat\mu_i + \log(1 - \hat\mu_i)\},$$

which is a function of the fitted probabilities only.

Proof. With $y_i \in \{0, 1\}$ the saturated model has $\tilde\mu_i = y_i$ and $\ell_{\text{sat}} = \sum\{y_i \log y_i + (1 - y_i)\log(1 - y_i)\} = 0$. Thus $D = -2\ell(\hat\beta) = -2\sum\{y_i\hat\eta_i + \log(1 - \hat\mu_i)\}$. The likelihood equations $X^\top(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) = 0$ give $\boldsymbol{y}^\top X\hat\beta = \hat{\boldsymbol{\mu}}^\top X\hat\beta$, that is, $\sum_i y_i\hat\eta_i = \sum_i \hat\mu_i\hat\eta_i$, and $\hat\eta_i = \operatorname{logit}\hat\mu_i$. □

Two data sets with the same fitted probabilities have the same deviance however badly the model fits: there is no reference distribution to compare it with.

Example 9.2 (A likelihood-ratio test for race)**.** In the birth-weight model, `race` has three levels and so contributes two parameters. Test H_0: no race effect.

```
fit0 <- update(fit, . ~ . - race)
anova(fit0, fit, test = "Chisq")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: low ~ age + lwt + smoke + ptd + ht + ui
#> Model 2: low ~ age + lwt + race + smoke + ptd + ht + ui
#>   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
#> 1       182        203
#> 2       180        197  2     6.41    0.041 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
LR <- deviance(fit0) - deviance(fit); df <- df.residual(fit0) -
↪  df.residual(fit)
c(LR = LR, df = df, p = pchisq(LR, df, lower.tail = FALSE), p_by_hand =
↪  exp(-LR / 2))
```

```
#>        LR        df         p p_by_hand
#>   6.40659   2.00000   0.04063   0.04063
```

```
c(AIC_without = AIC(fit0), AIC_with = AIC(fit), McFadden = 1 - deviance(fit) /
↪  fit$null.deviance)
```

```
#> AIC_without    AIC_with    McFadden
#>    217.2403    214.8337      0.1612
```

The statistic is $D_0 - D_1 = 6.41$ on 2 df, $p = 0.041$; for 2 degrees of freedom $P(\chi^2_2 > x) = e^{-x/2}$ gives the same number by hand. Race is a *block* of two dummies: test it jointly rather than by two separate Wald z values (which give $p = 0.023$ and 0.073). `drop1(fit, test = "Chisq")` produces such a test for every term at once.

For the ships model, the deviance table is sequential:

```
anova(fs, test = "Chisq")
```

```
#> Analysis of Deviance Table
#>
#> Model: poisson, link: log
#>
#> Response: incidents
#>
#> Terms added sequentially (first to last)
#>
#>
#>        Df Deviance Resid. Df Resid. Dev Pr(>Chi)
#> NULL                      33      146.3
#> type    4     55.4        29       90.9  2.6e-11 ***
#> year    3     41.5        26       49.4  5.0e-09 ***
#> period  1     10.7        25       38.7   0.0011 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Each line adds one term to the previous ones. For a Poisson model with known $\phi = 1$ the χ^2 test is appropriate; the next section shows why it may be too optimistic for these data.

9.6 Estimating the dispersion

For families with unknown ϕ (gamma, normal), or to check that $\phi = 1$ for the Poisson and binomial, we need an estimate $\hat\phi$. Since $\mathrm{E}[(Y_i - \mu_i)^2 / V(\mu_i)] = \phi / m_i$, a natural estimator is the **Pearson estimator**

$$X^2 = \sum_{i=1}^{n} \frac{m_i (y_i - \hat\mu_i)^2}{V(\hat\mu_i)}, \qquad \hat\phi = \frac{X^2}{n - p}. \tag{9.8}$$

The division by $n - p$ corrects for estimating β: $\mathrm{E}[X^2] \approx \phi(n - p)$ (Section 9.7: the Pearson residuals have variance close to $\phi(1 - h_{ii})$ and $\sum_i h_{ii} = p$). An alternative is the **deviance estimator** $D/(n - p)$, which is biased for small counts and should not be confused with the Pearson one: R's `summary()` of `quasipoisson`, `quasibinomial` and `Gamma` fits uses (9.8).

Proposition 9.3 (Quasi-likelihood: same estimates, rescaled standard errors)**.** *Suppose only* $\mathrm{E}[Y_i] = \mu_i$ *and* $\mathrm{Var}(Y_i) = \phi V(\mu_i)/m_i$ *are assumed, with* ϕ *unknown and not necessarily 1. The estimating equations* $U(\beta) = \sum_i \frac{m_i (y_i - \mu_i) \boldsymbol{x}_i}{V(\mu_i) g'(\mu_i)} = 0$ *have the same solution* $\hat\beta$ *as the likelihood equations, and* $\hat\beta$ *is asymptotically normal with covariance* $\phi (X^\top W X)^{-1}$.

Since ϕ does not enter the equations, the quasi-likelihood estimate $\hat\beta$ is identical to the maximum likelihood estimate under the corresponding family, but the standard errors are multiplied by $\sqrt{\hat\phi}$ and tests become t or F tests. The "quasi" families do not specify a full distribution, hence there is no likelihood and no AIC (Wedderburn, 1974). For the ships model:

```
X2 <- sum(residuals(fs, type = "pearson")^2)
c(X2 = X2, D = deviance(fs), df = df.residual(fs), phi_pearson = X2 /
↪  df.residual(fs),
  phi_deviance = deviance(fs) / df.residual(fs))
```

```
#>           X2            D           df  phi_pearson phi_deviance
#>       42.275       38.695       25.000        1.691        1.548
```

```
fq <- update(fs, family = quasipoisson)
round(cbind(est = coef(fs), se_poisson = coef(summary(fs))[, 2], se_quasi =
↪  coef(summary(fq))[, 2],
            ratio = coef(summary(fq))[, 2] / coef(summary(fs))[, 2]), 3)[1:4,
            ↪  ]
```

```
#>                est se_poisson se_quasi ratio
#> (Intercept) -6.406      0.217    0.283   1.3
#> typeB       -0.543      0.178    0.231   1.3
#> typeC       -0.687      0.329    0.428   1.3
#> typeD       -0.076      0.291    0.378   1.3
```

```
anova(update(fq, . ~ . - type), fq, test = "F")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: incidents ~ year + period + offset(log(service))
#> Model 2: incidents ~ type + year + period + offset(log(service))
#>   Resid. Df Resid. Dev Df Deviance   F Pr(>F)
#> 1        29       62.4
#> 2        25       38.7  4     23.7 3.5  0.021 *
```

```
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Pearson estimate $\hat\phi = 1.69$ exceeds 1, so the quasi-Poisson standard errors are $\sqrt{\hat\phi} = 1.3$ times the Poisson ones, and the test for `type` has $F = 3.5$, $p = 0.021$, to be compared with $p < 10^{-4}$ from the Poisson likelihood-ratio test.

9.7 Residuals

A residual should be a transformation of $(y_i, \hat\mu_i)$ that behaves like a standard normal variable if the model is correct, so that plots against fitted values and covariates reveal what the model misses. For non-normal data several definitions are in use.

Pearson residuals $r_i^P = (y_i - \hat\mu_i)\sqrt{m_i / V(\hat\mu_i)}$ standardize by the fitted standard deviation; $\sum_i (r_i^P)^2 = X^2$. **Deviance residuals** $r_i^D = \operatorname{sign}(y_i - \hat\mu_i)\sqrt{d_i}$ split the deviance, $\sum_i (r_i^D)^2 = D$; they are closer to normal for skewed families such as the Poisson. **Working residuals** are $(y_i - \hat\mu_i) g'(\hat\mu_i)$, the residuals of the final IRLS regression. *Standardized* versions divide by $\sqrt{\hat\phi(1 - h_{ii})}$ where h_{ii} are the diagonal elements of the hat matrix of the final weighted least-squares fit,

$$H = W^{1/2} X (X^\top W X)^{-1} X^\top W^{1/2}, \qquad \operatorname{tr} H = p.$$

The leverage h_{ii} measures the influence of y_i on its own fitted value, and **Cook's distance** is $D_i = (r_i^P)^2 h_{ii} / \{\hat\phi\, p\, (1 - h_{ii})^2\}$.

For discrete responses these residuals are themselves discrete: for a binary response y_i they take only two values for each $\hat\mu_i$, and the plot against $\hat\mu_i$ shows two curves, even if the model is correct (left panel of Figure 9.2). The **randomized quantile residual** of Dunn and Smyth (1996) repairs this.

Definition 9.3 (Randomized quantile residual)**.** Let $F_i(\cdot) = F(\cdot; \hat\mu_i, \hat\phi)$ be the fitted distribution function of Y_i. If Y_i is continuous, $r_i^Q = \Phi^{-1}\{F_i(y_i)\}$. If Y_i is discrete, draw u_i uniformly on $(F_i(y_i^-), F_i(y_i))$, where $F_i(y_i^-) = P_i(Y < y_i)$ (for counts, $F_i(y_i - 1)$), and set $r_i^Q = \Phi^{-1}(u_i)$.

Theorem 9.4 (Distribution of the randomized quantile residual)**.** *If the fitted distribution is the true distribution of Y_i (with known parameters), then $u_i \sim U(0,1)$ and $r_i^Q \sim N(0,1)$ exactly, for continuous and discrete responses.*

Proof. Continuous case: this is the probability integral transform, $P\{F(Y) \le t\} = t$. Discrete case with support $\{0, 1, 2, \dots\}$: let $F_{-1} = 0$ and $p_y = P(Y = y) = F(y) - F(y-1)$. Conditional on $Y = y$, u is uniform on $(F(y-1), F(y))$. For $t \in (0,1)$,

$$\begin{aligned} P(u \le t) &= \sum_y p_y\, P(u \le t \mid Y = y) = \sum_y p_y \min\left\{1, \max\left(0, \frac{t - F(y-1)}{p_y}\right)\right\} \\ &= \sum_y \min\{p_y,\ \max(0,\ t - F(y-1))\}. \end{aligned}$$

The intervals $(F(y-1), F(y)]$ tile $(0,1]$, so t falls in exactly one interval, say with index y^*: all $y < y^*$ contribute p_y (their sum is $F(y^* - 1)$), index y^* contributes $t - F(y^* - 1)$, and larger y contribute 0. The total is t. □

The residual is random, so fix the seed (or look at several draws). The function below works for any family with a distribution function in R (`ppois`, `pbinom`, `pnbinom`); it uses the upper tail when the lower one is numerically 1.

```
rqr <- function(y, F, ...) {                    # F: fitted CDF, its parameters
↪  go in "..."
  lo <- F(y - 1, ...); hi <- F(y, ...); u <- runif(length(y))
  up_lo <- F(y, ..., lower.tail = FALSE); up_hi <- F(y - 1, ..., lower.tail =
↪  FALSE)
  ifelse(lo < 0.5, qnorm(lo + u * (hi - lo)), qnorm(up_lo + u * (up_hi -
   ↪  up_lo), lower.tail = FALSE))
}
set.seed(2027)
r_ships <- rqr(sh$incidents, ppois, lambda = fitted(fs))
r_bin   <- rqr(bw$low, pbinom, size = 1, prob = fitted(fit))
c(shapiro_ships = shapiro.test(r_ships)$p.value, shapiro_binary =
↪  shapiro.test(r_bin)$p.value)
```

```
#>  shapiro_ships shapiro_binary
#>         0.3780         0.0813
```

```
op <- par(mfrow = c(1, 2), mar = c(4, 4, 1, 1))
plot(fitted(fit), residuals(fit, "pearson"), col = ifelse(bw$low == 1,
↪  ie_coral, ie_navy), pch = 16,
     xlab = "fitted probability", ylab = "Pearson residual")
legend("topright", c("y = 1", "y = 0"), col = c(ie_coral, ie_navy), pch = 16,
↪  bty = "n")
qqnorm(r_bin, main = "", pch = 16, col = ie_navy, xlab = "N(0,1) quantile",
↪  ylab = "randomized quantile residual"); qqline(r_bin, col = ie_coral)
```

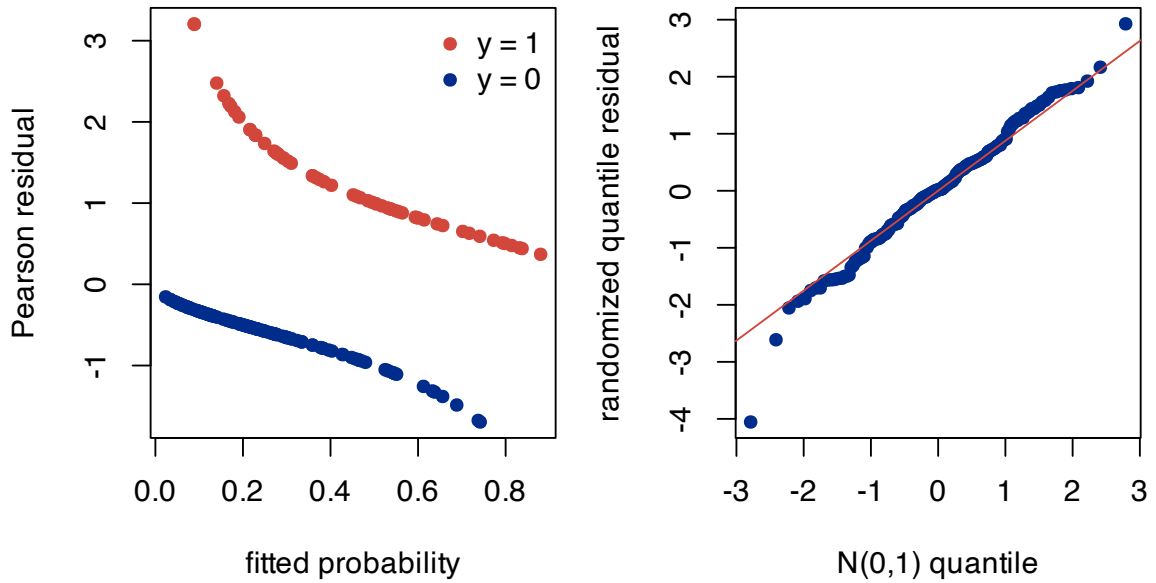


Figure 9.2: Residuals of the birth-weight logistic model. Left: Pearson residuals form two curves (upper curve: $y = 1$, lower: $y = 0$). Right: randomized quantile residuals follow the normal reference line.

```
par(op)
```

For the ships model (counts, a few of them small) we compare the three residuals and look at influence.

```
rs <- data.frame(y = sh$incidents, mu = fitted(fs), pearson = residuals(fs,
↪  "pearson"),
                 deviance = residuals(fs, "deviance"), rstd = rstandard(fs,
                 ↪  type = "pearson"),
                 h = hatvalues(fs), cook = cooks.distance(fs), rq = r_ships)
round(head(rs[order(-abs(rs$pearson)), ], 4), 3)
```

```
#>     y    mu pearson deviance  rstd     h  cook    rq
#> 21  6 1.474   3.727    2.791 4.018 0.140 0.291 2.952
#> 36  7 2.948   2.360    2.001 2.536 0.134 0.111 2.231
#> 30 11 6.158   1.951    1.755 2.558 0.418 0.522 1.677
#> 35  7 3.624   1.773    1.570 1.940 0.164 0.082 1.498
```

```
c(sum_pearson2 = sum(rs$pearson^2), sum_deviance2 = sum(rs$deviance^2),
↪  trace_H = sum(rs$h))
```

```
#>  sum_pearson2 sum_deviance2       trace_H
#>         42.28         38.70          9.00
```

The largest Pearson residual belongs to a cell with 6 incidents where the model expects 1.5. The identities $\sum (r^P)^2 = X^2$, $\sum (r^D)^2 = D$ and $\operatorname{tr} H = p = 9$ hold exactly.

9.8 Model comparison: AIC

Nested models are compared by likelihood-ratio (χ^2 or F) tests; models that are not nested, or many models at once, by the **Akaike information criterion** (Akaike, 1974)

$$\mathrm{AIC} = -2\ell(\hat\beta) + 2p,$$

where p counts all estimated parameters (including ϕ for the gamma and normal families, which R does). AIC estimates, up to a constant, the expected Kullback–Leibler discrepancy between the fitted model and the truth; the penalty $2p$ corrects the optimism of the maximized likelihood. Only differences between models fitted to the *same data* with the same response are meaningful; a difference below about 2 means the models are indistinguishable. AIC needs a likelihood: it does not exist for quasi families. For binomial data in grouped form, AIC $\neq D + 2p$, because D subtracts the saturated log-likelihood, which is a constant that cancels in comparisons.

```
fit_probit <- update(fit, family = binomial(link = "probit"))
fit_cll <- update(fit, family = binomial(link = "cloglog"))
sapply(list(logit = fit, probit = fit_probit, cloglog = fit_cll), AIC)
```

```
#>   logit  probit cloglog
#>   214.8   214.5   214.6
```

The three links for the binary response differ by less than 0.4 in AIC: with 189 observations, the data cannot tell them apart. We choose the logit link for its odds-ratio interpretation.

9.9 In R

Task	Code
Fit	`fit <- glm(y ~ x, family = poisson, data = d)`
Coefficients, standard errors, deviances	`summary(fit)`, `coef(summary(fit))`
Wald and profile intervals	`confint.default(fit)`, `confint(fit)`
Covariance matrix	`vcov(fit)` $(= \hat{\phi}(X^\top \hat{W} X)^{-1})$
LR test of nested models	`anova(fit0, fit1, test = "Chisq")` (or `"F"` if ϕ is estimated)
One test per term	`drop1(fit, test = "Chisq")`, `anova(fit, test = "Chisq")` (sequential). With `test = "F"`, `anova(fit0, fit1)` estimates ϕ by the Pearson statistic of the larger model, `drop1()` by $D/(n-p)$
Residuals	`residuals(fit, type = "pearson" / "deviance" / "working" / "response")`, `rstandard(fit)`
Leverage, Cook	`hatvalues(fit)`, `cooks.distance(fit)`, `influence.measures(fit)`
Fitted values	`fitted(fit)` $(\hat{\mu})$, `predict(fit, nd, type = "link" / "response", se.fit = TRUE)`
Dispersion	`sum(residuals(fit, "pearson")^2) / df.residual(fit)`, `summary(fit)$dispersion`
Criteria	`AIC(fit)`, `BIC(fit)`, `logLik(fit)`
Convergence	`glm.control(epsilon = 1e-8, maxit = 25)`, `fit$converged`, `fit$iter`

9.10 Common pitfalls

Warning. IRLS is always Fisher scoring. It equals Newton–Raphson only for the canonical link. The "number of Fisher scoring iterations" in `summary()` counts IRLS steps.

Warning. The residual deviance of a binary model is not a goodness-of-fit statistic (Proposition 9.2); a "deviance close to the degrees of freedom" says nothing about the fit.

Warning. Estimate ϕ by the Pearson statistic $X^2/(n-p)$, which is what the quasi families do; $D/(n-p)$ is a different, biased estimator for small counts.

Remark. McFadden's pseudo-R^2 is not "the proportion of variance explained". Test a factor with several levels by one likelihood-ratio test, not by its separate Wald statistics. Wald intervals can be poor when the log-likelihood is far from quadratic (small counts, near separation); use profile intervals.

Remark. Standard errors and tests assume that ϕ and the variance function are right. A model with overdispersion has too small standard errors, even though $\hat{\beta}$ is still consistent when the mean is right (Chapter 10).

9.11 Chapter summary

- Score $S(\beta) = \phi^{-1} X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$ and expected information $\mathcal{I}(\beta) = \phi^{-1} X^\top W X$ with $w_i = m_i / \{V(\mu_i) g'(\mu_i)^2\}$; ϕ cancels from the likelihood equations, which depend on the family only through μ_i and $V(\mu_i)$.
- IRLS is Fisher scoring: $\beta^{(t+1)} = (X^\top W X)^{-1} X^\top W \boldsymbol{z}$, $z_i = \eta_i + (y_i - \mu_i) g'(\mu_i)$. It equals Newton–Raphson iff the link is canonical, where observed and expected information coincide.
- $\hat\beta \dot\sim N(\beta, \phi (X^\top W X)^{-1})$: Wald tests and intervals, LR tests, profile intervals, score tests.
- Deviance $D = 2\phi\{\ell_{\text{sat}} - \ell(\hat\beta)\} = \sum d_i$; deviance differences are LR statistics, χ^2 if ϕ is known, F if estimated. The residual deviance is not χ^2_{n-p} in general and is uninformative for binary data.
- Dispersion: $\hat\phi = X^2/(n-p)$ (Pearson); quasi-likelihood keeps $\hat\beta$ and multiplies standard errors by $\sqrt{\hat\phi}$.
- Residuals: Pearson, deviance, working, standardized; randomized quantile residuals are exactly $N(0,1)$ under the model, also for discrete data.
- AIC $= -2\ell + 2p$ compares models fitted to the same data.

9.12 Exercises

Exercise 9.1. A Poisson model uses the square-root link, $g(\mu) = \sqrt{\mu}$, so $\eta_i = \sqrt{\mu_i} = \boldsymbol{x}_i^\top \beta$. (a) Give $g'(\mu)$, the IRLS weight w_i and the working response z_i. (b) Is the link canonical? What does this imply for Newton–Raphson versus Fisher scoring? (c) Compute w_i and z_i at $\mu_i = 4$, $y_i = 6$.

Solution.

(a) $g'(\mu) = 1/(2\sqrt{\mu})$, $V(\mu) = \mu$, so $w_i = 1/\{\mu_i \cdot \frac{1}{4\mu_i}\} = 4$: constant weights, and $z_i = \eta_i + (y_i - \mu_i)/(2\sqrt{\mu_i})$. (b) The canonical link of the Poisson is the log, so the square-root link is not canonical; IRLS is Fisher scoring and is not Newton–Raphson: the observed information has the extra term of Proposition 9.1, with $V'g' + Vg'' = \frac{1}{2\sqrt{\mu}} - \frac{\mu}{4\mu^{3/2}} = \frac{1}{4\sqrt{\mu}} \neq 0$. (c) $w = 4$; $\eta = \sqrt{4} = 2$, $z = 2 + (6-4)/(2 \cdot 2) = 2.5$.

Exercise 9.2. Gamma regression with the log link: $V(\mu) = \mu^2$, $g(\mu) = \log \mu$. (a) Show that the IRLS weights are $w_i \equiv 1$ (take $m_i = 1$) and give z_i. (b) Use Proposition 9.1 to show that the observed information is $\phi^{-1} \sum_i (y_i/\mu_i)\, \boldsymbol{x}_i \boldsymbol{x}_i^\top$ while the expected information is $\phi^{-1} X^\top X$. (c) What does this tell you about Newton–Raphson and Fisher scoring?

Solution.

(a) $g' = 1/\mu$ so $w = 1/(\mu^2 \cdot \mu^{-2}) = 1$ and $z_i = \eta_i + (y_i - \mu_i)/\mu_i = \eta_i + y_i/\mu_i - 1$. (b) With $V = \mu^2$, $V' = 2\mu$, $g' = 1/\mu$, $g'' = -1/\mu^2$: $V'g' + Vg'' = 2 - 1 = 1$ and $V^2 g'^3 = \mu^4 \mu^{-3} = \mu$. So the weight in (9.4) is $1 + (y_i - \mu_i)/\mu_i = y_i/\mu_i$, and the observed information is $\phi^{-1} \sum (y_i/\mu_i) \boldsymbol{x}_i \boldsymbol{x}_i^\top$. Since $\mathrm{E}[y_i/\mu_i] = 1$, the expected information is $\phi^{-1} X^\top X$. (c) They differ by the random weights y_i/μ_i: Newton–Raphson and Fisher scoring are different algorithms for the gamma log-link model, although both converge to the same $\hat\beta$. Fisher scoring here is an *ordinary* least-squares fit of $\boldsymbol{z}$ on X at each step.

Exercise 9.3. Six observations: $x = (0, 1, 2, 3, 4, 5)$, $y = (0, 0, 1, 0, 1, 1)$. Fit a logistic regression logit $\mu_i = \beta_0 + \beta_1 x_i$. (a) Starting from $\beta^{(0)} = (0, 0)^\top$, perform one IRLS step by hand. (b) Verify with R and run the algorithm to convergence. (c) Why does a finite MLE exist here?

Solution.

(a) At $\beta = 0$: $\hat\mu_i = 1/2$, $w_i = \mu(1-\mu) = 1/4$, $z_i = 0 + (y_i - \tfrac12)/\tfrac14 = 4y_i - 2 \in \{-2, -2, 2, -2, 2, 2\}$. The weighted LS normal equations with $W = \tfrac14 I$ are $X^\top X\beta = X^\top \boldsymbol{z}$. Now $X^\top X = \begin{pmatrix} 6 & 15 \\ 15 & 55 \end{pmatrix}$, $X^\top \boldsymbol{z} = (0,\ \sum x_i z_i)^\top$ with $\sum x_i z_i = 0 - 2 + 4 - 6 + 8 + 10 = 14$ and $\sum z_i = 0$. The determinant is $330 - 225 = 105$, so $\beta^{(1)} = \frac{1}{105}\begin{pmatrix} 55 & -15 \\ -15 & 6 \end{pmatrix}\begin{pmatrix} 0 \\ 14 \end{pmatrix} = \frac{1}{105}(-210,\ 84)^\top = (-2,\ 0.8)^\top$. (b) See the code (the function `irls()` of the chapter uses the starting values $(y_i + \bar y)/2$, so its first step differs from the step from $\beta = 0$ computed above; the converged values agree). (c) The 0/1 labels overlap in x (the observation at $x = 2$ is a 1 below a 0 at $x = 3$), so no value of β_1 separates them perfectly, and the log-likelihood has a finite maximizer.

```
xx <- 0:5; yy <- c(0, 0, 1, 0, 1, 1); Xh <- cbind(1, xx)
zz <- 4 * yy - 2                                          # working
↪  response at beta = 0, w = 1/4
solve(crossprod(Xh, 0.25 * Xh), crossprod(Xh, 0.25 * zz))        # one step
↪  from beta = 0
```

```
#>    [,1]
#>    -2.0
#> xx  0.8
```

```
full <- irls(Xh, yy, binomial()); rbind(irls = full$coefficients, glm =
↪  coef(glm(yy ~ xx, binomial)))
```

```
#>                xx
#> irls -3.035 1.214
#> glm  -3.035 1.214
```

```
full$iter
```

```
#> [1] 5
```

Exercise 9.4. Derive the unit deviance (a) of the binomial family for y_i successes out of m_i trials; (b) of the gamma family; (c) show that for the normal family with identity link, D equals the residual sum of squares and that the F statistic reduces to that of the linear model.

Solution.

(a) Per observation the log-likelihood in terms of μ is $y\log\mu + (m-y)\log(1-\mu)$+const; the saturated value has $\tilde\mu = y/m$. So $d = 2\{y\log\frac{y/m}{\hat\mu} + (m-y)\log\frac{1-y/m}{1-\hat\mu}\}$, which equals the formula of Table 9.1. (b) With $\theta = -1/\mu$, $b(\theta) = \log\mu$ and $a = 1/\nu$, the log-likelihood of one observation is $\nu\{-y/\mu - \log\mu\} + c(y)$. At the saturated value $\mu = y$ it is $\nu\{-1 - \log y\} + c(y)$. Hence $D^* = 2\nu\{y/\hat\mu - 1 - \log(y/\hat\mu)\}$ and the unit

deviance is $d = 2\{y/\hat\mu - 1 - \log(y/\hat\mu)\} \geq 0$ (since $\log x \leq x - 1$), as in Table 9.1. (c) With $d = (y-\hat\mu)^2$, $D = \sum(y_i - \hat\mu_i)^2 = \text{RSS}$, $a(\phi) = \sigma^2$, $\hat\phi = \text{RSS}/(n-p)$, so $F = \{(\text{RSS}_0 - \text{RSS}_1)/(p_1 - p_0)\}/\hat\sigma^2$, the usual F statistic, exactly F-distributed under normality.

Exercise 9.5.
(a) For the intercept-only logistic model with y successes in n Bernoulli trials ($0 < y < n$), find $\hat\mu$ and the null deviance as a function of y and n. (b) Evaluate for the birth-weight data ($y = 59$, $n = 189$) and compare with the value printed by `glm()` (234.67). (c) Compute McFadden's R^2 for the full model (residual deviance 196.83).

Solution.

(a) $\hat\mu = y/n$ (MLE of a proportion), and $D_{\text{null}} = -2\ell(\hat\mu) = -2\{y\log(y/n) + (n-y)\log(1-y/n)\}$ because $\ell_{\text{sat}} = 0$. (b) $D_{\text{null}} = -2\{59\log(59/189) + 130\log(130/189)\}$; in R:

```
yy <- 59; n <- 189
c(by_hand = -2 * (yy * log(yy / n) + (n - yy) * log(1 - yy / n)), glm =
↪  fit$null.deviance)
```

```
#> by_hand     glm
#>   234.7   234.7
```

```
1 - deviance(fit) / fit$null.deviance
```

```
#> [1] 0.1612
```

(c) $R^2_{\text{McF}} = 1 - 196.83/234.67 = 0.161$: the covariates reduce the deviance by 16% relative to the intercept-only model. This is a proportional reduction in deviance, not "16% of the variance".

Exercise 9.6. For the ships data the sequential residual deviances are: intercept and offset only, 146.33 on 33 df; + type, 90.89 on 29; + year, 49.36 on 26; + period, 38.70 on 25. (a) Test each added term with a χ^2 test. (b) The Pearson statistic of the full model is 42.28: estimate ϕ and redo the test for `type` as an F test. (c) Do the conclusions change? Which do you report?

Solution.

(a) type: $146.33 - 90.89 = 55.44$ on 4 df, $p = 2.6 \times 10^{-11}$; year: $90.89 - 49.36 = 41.53$ on 3 df, $p = 5 \times 10^{-9}$; period: $49.36 - 38.70 = 10.66$ on 1 df, $p = 0.0011$. (b) $\hat\phi = 42.28/25 = 1.69$. The *sequential* test of `type` (entered first) is $F = (55.44/4)/1.69 = 8.2$ on $(4, 25)$ df, $p = 0.0002$; the test of `type` *after* year and period, which compares the models with and without `type` in the presence of the other two terms, is $F = (23.67/4)/1.69 = 3.50$, $p = 0.021$ (computed in the chapter). (c) All terms remain significant at 5%, but the p-values become much larger than the Poisson ones (for `type` given the others, $p = 9 \times 10^{-5}$ with $\phi = 1$ against 0.021 with $\hat\phi = 1.69$): the evidence is moderate rather than overwhelming. We report the quasi-Poisson (or negative binomial) result, because $\hat\phi = 1.69$ shows that the assumption $\phi = 1$ is not supported. Sequential tests depend on the order of the terms; the tests of a term given all the others (marginal) do not.

```
c(F_seq_type = (55.44 / 4) / (42.28 / 25), p = pf((55.44 / 4) / (42.28 / 25),
↪  4, 25, lower.tail = FALSE))
```

```
#> F_seq_type          p
#>  8.1953642  0.0002291
```

```
c(F_marginal_type = (23.67 / 4) / (42.28 / 25), p = pf((23.67 / 4) / (42.28 /
↪  25), 4, 25, lower.tail = FALSE))
```

```
#> F_marginal_type               p
#>         3.49899         0.02117
```

Exercise 9.7. A Poisson regression with $n = 40$ observations and $p = 4$ has residual deviance $D = 52.0$ and Pearson statistic $X^2 = 63.0$. (a) Estimate ϕ by the two methods. (b) A coefficient has Poisson standard error 0.10 and estimate 0.30; compute the quasi-Poisson standard error, t statistic and approximate two-sided p-value. (c) Explain why $\hat{\beta}$ is unchanged.

Solution.

(a) $\hat{\phi}_P = 63/36 = 1.75$ and $\hat{\phi}_D = 52/36 = 1.44$. (b) The quasi-Poisson standard error is $0.10\sqrt{1.75} = 0.1323$; $t = 0.30/0.1323 = 2.27$, which with 36 df gives $p = 0.029$ (the Poisson $z = 3.0$ gives $p = 0.003$). (c) The likelihood (quasi-score) equations of Theorem 9.1 do not involve ϕ.

Exercise 9.8. In a Poisson model, an observation has $y = 0$ and $\hat{\mu} = 1.3$, and another has $y = 6$ and $\hat{\mu} = 1.5$. (a) Compute the Pearson and deviance residuals of both. (b) Give the interval in which the randomized quantile residual of each lies, using $P(Y \le 5) = 0.9955$ and $P(Y \le 6) = 0.9991$ for the second ($\hat{\mu} = 1.5$), and $P(Y = 0) = e^{-1.3}$ for the first.

Solution.

```
f <- function(y, mu) c(pearson = (y - mu) / sqrt(mu),
  deviance = sign(y - mu) * sqrt(2 * ifelse(y == 0, mu, y * log(y / mu) - (y -
  ↪  mu))))
round(rbind(`y=0, mu=1.3` = f(0, 1.3), `y=6, mu=1.5` = f(6, 1.5)), 3)
```

```
#>              pearson deviance
#> y=0, mu=1.3  -1.140   -1.612
#> y=6, mu=1.5   3.674    2.763
```

```
c(qnorm(exp(-1.3)), qnorm(ppois(5, 1.5)), qnorm(ppois(6, 1.5)))
```

```
#> [1] -0.6052  2.6154  3.1130
```

(a) For $y = 0$: $r^P = -1.3/\sqrt{1.3} = -1.14$ and $r^D = -\sqrt{2\mu} = -1.61$. For $y = 6$: $r^P = 4.5/\sqrt{1.5} = 3.67$, $r^D = \sqrt{2(6\log 4 - 4.5)} = 2.76$. (b) For $y = 0$ the randomized quantile residual lies in $(-\infty,\ \Phi^{-1}(e^{-1.3})) = (-\infty, -0.61)$. For $y = 6$ it lies in $(\Phi^{-1}(0.9955), \Phi^{-1}(0.9991)) = (2.62,\ 3.11)$: far in the upper tail, so this observation is unusual under the fitted model.

Exercise 9.9. Prove directly that the randomized quantile residual of a Bernoulli(μ) variable is exactly $N(0, 1)$.

Solution. $F(0) = 1-\mu$, $F(1) = 1$, $F(-1) = 0$. If $y = 0$, $u \sim U(0, 1-\mu)$; if $y = 1$, $u \sim U(1-\mu, 1)$. $P(y = 0) = 1 - \mu$. The mixture density of u is $(1 - \mu) \cdot \frac{1}{1-\mu} = 1$ on $(0, 1 - \mu)$ and $\mu \cdot \frac{1}{\mu} = 1$ on

$(1-\mu, 1)$: uniform on $(0,1)$. Hence $\Phi^{-1}(u) \sim N(0,1)$. The Pearson residual, in contrast, takes only two values.

Exercise 9.10.
(R) Use `irls()` of this chapter to fit the gamma log-link model `Volume ~ log(Girth) + log(Height)` for the `trees` data (family `Gamma(link = "log")`). (a) Check agreement with `glm()`. (b) Compute the Pearson dispersion estimate from the IRLS output and compare with `summary()`. (c) Compute the scaled standard errors $\sqrt{\hat\phi\,\mathrm{diag}(X^\top \hat W X)^{-1}}$ and compare with `summary()`.

Solution.

```
Xt <- model.matrix(~ log(Girth) + log(Height), trees)
mg <- irls(Xt, trees$Volume, Gamma(link = "log"))
gg <- glm(Volume ~ log(Girth) + log(Height), family = Gamma(link = "log"),
↪  data = trees)
max(abs(mg$coefficients - coef(gg)))
```

```
#> [1] 1.557e-06
```

```
mu_g <- exp(drop(Xt %*% mg$coefficients))
phi <- sum((trees$Volume - mu_g)^2 / mu_g^2) / (nrow(trees) - 3)
c(phi_irls = phi, phi_summary = summary(gg)$dispersion)
```

```
#>    phi_irls phi_summary
#>    0.006427    0.006427
```

```
cbind(se_irls = sqrt(phi * diag(mg$cov_unscaled)), se_summary =
↪  coef(summary(gg))[, 2])
```

```
#>             se_irls se_summary
#> (Intercept) 0.78784    0.78784
#> log(Girth)  0.07389    0.07389
#> log(Height) 0.20138    0.20138
```

(a) The agreement is to about 10^{-6}, the convergence tolerance of the deviance criterion. (b) With $V(\mu)=\mu^2$ the Pearson statistic is $\sum (y_i-\hat\mu_i)^2/\hat\mu_i^2$, divided by $n-p=28$. (c) Since $W=I$ for this model (Exercise 9.2), $\widehat{\mathrm{Var}}(\hat\beta)=\hat\phi(X^\top X)^{-1}$, exactly what `summary()` reports.

Exercise 9.11.
(R) For `orings` (binomial, `cbind(damage, 6 - damage) ~ temp`) compare the Wald interval and the profile-likelihood interval for the slope, and for the odds ratio of a 10°F rise. Which would you report, and why do they differ?

Solution.

```
o <- glm(cbind(damage, 6 - damage) ~ temp, family = binomial, data = orings)
b <- coef(summary(o))["temp", ]
wald <- b[1] + c(-1, 1) * 1.96 * b[2]
prof <- confint(o)["temp", ]
round(rbind(wald = wald, profile = prof, `OR per 10F, Wald` = exp(10 * wald),
↪  `OR per 10F, profile` = exp(10 * prof)), 3)
```

```
#>                        2.5 % 97.5 %
#> wald                 -0.320 -0.112
#> profile              -0.333 -0.120
#> OR per 10F, Wald      0.041  0.326
#> OR per 10F, profile   0.036  0.301
```

The profile interval is slightly asymmetric and shifted to the left (more negative values): with only 23 launches and a few damaged O-rings, the log-likelihood is not quadratic. The difference is small here but it is in the direction in which the Wald interval is too optimistic about a weaker effect. Report the profile interval (computed by `confint()`).

Exercise 9.12. *Critique.* A colleague fits a logistic model with 8 covariates on 500 patients. The output reads "Residual deviance: 480.2 on 491 degrees of freedom" and she writes: "The residual deviance is close to its degrees of freedom, so the model fits well." Another adds: "the model with the lower AIC is the true model." Comment on both statements.

Solution. For binary data the residual deviance does not follow a χ^2_{n-p} distribution (Proposition 9.2: it is a function of the fitted probabilities only), so comparing it with the degrees of freedom says nothing about fit. A calibration check (binning the fitted probabilities and comparing with observed frequencies), randomized quantile residuals for groups, or the Hosmer–Lemeshow test is needed. (The deviance can still be used to compare nested models.) The second statement is also wrong: AIC estimates predictive discrepancy and selects the *best approximating* model among the candidates; the true model need not be among them, and differences below 2 units are not distinguishable. AIC also cannot compare models for different responses or on different data.

9.13 Further reading

McCullagh and Nelder (1989) (Chapters 2–4) derive IRLS, the deviance and quasi-likelihood. Wood (2017), Section 3.1, gives a compact treatment closely aligned with this chapter. The residuals are from Dunn and Smyth (1996); Faraway (2016) gives further diagnostics. For the choice of ε-criteria and the geometry of the likelihood, Pawitan (2001) (Chapters 2 and 9) is helpful, and Efron and Hinkley (1978) compare observed and expected information.

Chapter 10

Binary and count data in practice

Chapters 8 and 9 gave the theory of GLMs. This chapter applies it to the two most frequent kinds of responses: **binary** outcomes (logistic regression) and **counts** (Poisson regression). For each we show how to interpret the coefficients on the scale of interest (odds ratios, rate ratios), how to check the model, and what goes wrong in practice: *separation* in logistic regression, the choice of a classification threshold *without leakage*, and *overdispersion* and *excess zeros* in count data, with their remedies (quasi-Poisson, negative binomial, zero-inflated and hurdle models). The references are Hosmer et al. (2013) for logistic regression, Cameron and Trivedi (1990) and Lambert (1992) for overdispersion and zero inflation, and Fawcett (2006) for ROC analysis. The packages `pscl` (zero-inflated and hurdle models) and `brglm2` (Firth's penalized likelihood) are used in a few places; install them once with `install.packages(c("pscl", "brglm2"))`.

10.1 Logistic regression

10.1.1 The model and its interpretation

For a binary response $Y_i \in \{0, 1\}$ with $P(Y_i = 1) = \mu_i$, the **logistic regression** model is the GLM with Bernoulli family and its canonical logit link,

$$\operatorname{logit} \mu_i = \log \frac{\mu_i}{1-\mu_i} = \boldsymbol{x}_i^\top \beta \quad \Longleftrightarrow \quad \mu_i = \frac{\exp(\boldsymbol{x}_i^\top \beta)}{1+\exp(\boldsymbol{x}_i^\top \beta)}. \tag{10.1}$$

Responses given as successes out of m_i trials (grouped data) are modelled in the same way with `cbind(successes, failures)`.

Proposition 10.1 (The coefficient is a log odds ratio)**.** *Let $\boldsymbol{e}_j$ be the jth unit vector. Whatever the covariate vector $\boldsymbol{x}$,*

$$\frac{odds(\boldsymbol{x}+\boldsymbol{e}_j)}{odds(\boldsymbol{x})} = e^{\beta_j}, \qquad where\ odds(\boldsymbol{x}) = \frac{\mu(\boldsymbol{x})}{1-\mu(\boldsymbol{x})} = e^{\boldsymbol{x}^\top \beta}.$$

Proof. $\operatorname{odds}(\boldsymbol{x}) = \mu/(1-\mu) = \exp(\boldsymbol{x}^\top\beta)$ by (10.1), so the ratio is $\exp\{(\boldsymbol{x}+\boldsymbol{e}_j)^\top\beta - \boldsymbol{x}^\top\beta\} = e^{\beta_j}$. □

A unit increase of x_j multiplies the odds of $Y = 1$ by e^{β_j}, a constant *multiplicative* effect on the odds. For a binary covariate e^{β_j} compares the two groups. Two warnings: the odds ratio is not a ratio of probabilities (the *risk ratio* μ_1/μ_0 is close to the odds ratio only when both

probabilities are small), and the effect on the **probability** scale is not constant:

$$\frac{\partial \mu}{\partial x_j} = \mu(1-\mu)\,\beta_j, \qquad \left|\frac{\partial \mu}{\partial x_j}\right| \le \frac{|\beta_j|}{4},$$

with the maximum at $\mu = 1/2$ (the "divide by 4" rule gives an upper bound for the change in probability per unit). Near $\mu = 0$ or 1 the same β_j moves the probability very little. Report odds ratios and a few predicted probabilities for typical profiles.

Example 10.1 (Risk factors for low birth weight)**.** We model `low` (birth weight below 2.5 kg) in the 189 births of `MASS::birthwt` by the mother's age and weight at the last menstrual period (`lwt`, pounds), race, smoking, a history of premature labour (`ptd`), hypertension (`ht`) and uterine irritability (`ui`). The variable `bwt` (the weight itself) is not used as a predictor: `low` is defined from it.

```
bw <- with(birthwt, data.frame(low = low, age = age, lwt = lwt,
  race = factor(race, labels = c("white", "black", "other")),
  smoke = smoke, ptd = as.integer(ptl > 0), ht = ht, ui = ui))
fit <- glm(low ~ age + lwt + race + smoke + ptd + ht + ui, family = binomial,
↪  data = bw)
round(coef(summary(fit)), 4)
```

```
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept)   0.6369     1.2303  0.5177   0.6047
#> age          -0.0377     0.0378 -0.9984   0.3181
#> lwt          -0.0149     0.0070 -2.1179   0.0342
#> raceblack     1.2127     0.5325  2.2775   0.0228
#> raceother     0.8041     0.4484  1.7932   0.0729
#> smoke         0.8464     0.4081  2.0742   0.0381
#> ptd           1.2218     0.4630  2.6387   0.0083
#> ht            1.8387     0.7032  2.6146   0.0089
#> ui            0.7111     0.4631  1.5355   0.1247
```

```
round(exp(cbind(OR = coef(fit), confint(fit))), 3)
```

```
#>                OR 2.5 % 97.5 %
#> (Intercept) 1.891 0.175 22.277
#> age         0.963 0.892  1.036
#> lwt         0.985 0.971  0.998
#> raceblack   3.363 1.186  9.734
#> raceother   2.235 0.938  5.501
#> smoke       2.331 1.058  5.293
#> ptd         3.393 1.382  8.602
#> ht          6.288 1.640 27.258
#> ui          2.036 0.812  5.058
```

The coefficients are log odds ratios. Adjusted for the other covariates, the odds of a low-weight baby are about 2.3 times higher for smoking mothers (95% profile interval 1.06 to 5.29), and ten more pounds of weight multiply the odds by $e^{10\hat\beta_{lwt}} = 0.86$. Since the data are observational, these are associations, not causal effects. The profile interval is not symmetric on the odds-ratio scale, and for hypertension (12 mothers only) it is very wide.

Two mothers. Take a white mother, 25 years old, 120 lb, with no premature labour, hypertension or irritability, who does not smoke versus one who does:

```
nd <- data.frame(age = 25, lwt = 120, race = "white", smoke = 0:1, ptd = 0, ht
↪  = 0, ui = 0)
eta <- predict(fit, nd); p <- plogis(eta)
rbind(eta = eta, prob = p, odds = p / (1 - p))
```

```
#>            1       2
#> eta  -2.0961 -1.2497
#> prob  0.1095  0.2228
#> odds  0.1229  0.2866
```

```
c(odds_ratio = unname((p[2] / (1 - p[2])) / (p[1] / (1 - p[1]))), risk_ratio =
↪  unname(p[2] / p[1]),
  exp_beta_smoke = unname(exp(coef(fit)["smoke"])))
```

```
#>     odds_ratio     risk_ratio exp_beta_smoke
#>          2.331          2.035          2.331
```

The probabilities are 0.109 and 0.223. The ratio of odds is exactly $e^{\hat\beta_{\text{smoke}}} = 2.33$, while the ratio of probabilities is 2.03: the odds ratio overstates the risk ratio here. At this profile the change in probability per extra pound of weight is $\hat\beta_{lwt}\,\hat\mu(1-\hat\mu) = -0.0015$ for the non-smoker, well within the bound $|\hat\beta_{lwt}|/4 = 0.0037$.

```
grid <- expand.grid(age = 25, lwt = 80:250, race = "white", smoke = 0:1, ptd =
↪  0, ht = 0:1, ui = 0)
pr <- predict(fit, grid, se.fit = TRUE)
grid$p <- plogis(pr$fit); grid$lo <- plogis(pr$fit - 1.96 * pr$se.fit);
↪  grid$hi <- plogis(pr$fit + 1.96 * pr$se.fit)
grid$smoke <- factor(grid$smoke, labels = c("non-smoker", "smoker"))
grid$ht <- factor(grid$ht, labels = c("no hypertension", "hypertension"))
ggplot(grid, aes(lwt, p, colour = smoke, fill = smoke)) + geom_ribbon(aes(ymin
↪  = lo, ymax = hi), alpha = 0.15, colour = NA) +
  geom_line(linewidth = 1) + facet_wrap(~ht) + scale_colour_manual(values =
  ↪  ie_pal[1:2]) + scale_fill_manual(values = ie_pal[1:2]) +
  labs(x = "mother's weight (lb)", y = "P(low birth weight)", colour = NULL,
  ↪  fill = NULL)
```

10.1.2 Latent variables, other links and retrospective sampling

The logit link is the canonical one, but the probit and complementary log-log links of Chapter 8 are also used. If $Y = 1$ exactly when $\eta + \varepsilon > 0$ with ε standard logistic (variance $\pi^2/3$) the model is logistic; with $\varepsilon \sim N(0,1)$ it is probit. Matching the two inverse links by their slope at $\eta = 0$ (density of $N(0,1)$ at 0: 0.399; density of the logistic at 0: 1/4) suggests $\hat\beta_{\text{logit}} \approx 1.6\,\hat\beta_{\text{probit}}$; matching the standard deviations gives $\pi/\sqrt{3} = 1.81$. In practice the ratio lies between 1.6 and 1.8, and the fitted probabilities hardly differ:

```
fit_probit <- update(fit, family = binomial(link = "probit"))
round(coef(fit) / coef(fit_probit), 2)
```

```
#> (Intercept)         age         lwt   raceblack   raceother       smoke
#>        1.72        1.57        1.73        1.71        1.69        1.65
```

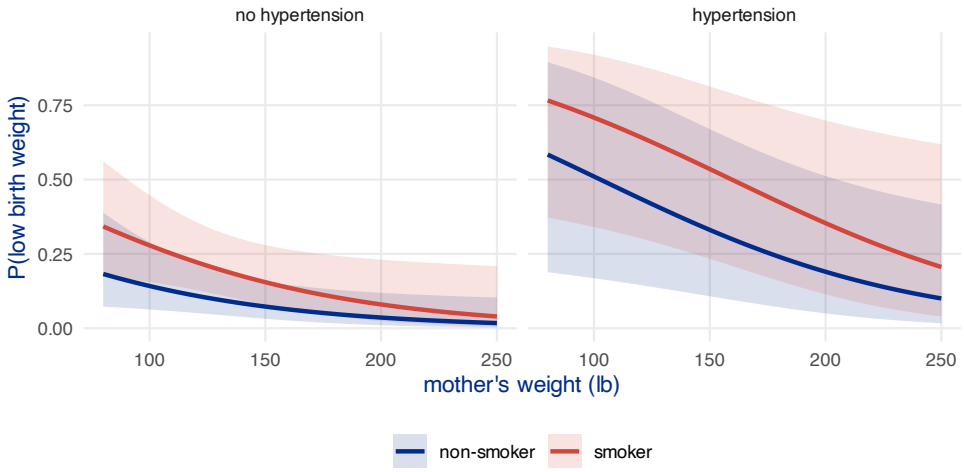


Figure 10.1: Predicted probability of low birth weight against the mother's weight, with 95% Wald bands computed on the log-odds scale and mapped through the inverse link (so they stay inside $(0, 1)$).

```
#>         ptd          ht          ui
#>        1.64        1.67        1.61
```

```
c(max_prob_diff = max(abs(fitted(fit) - fitted(fit_probit))), AIC_logit =
↪  AIC(fit), AIC_probit = AIC(fit_probit))
```

```
#> max_prob_diff     AIC_logit    AIC_probit
#>       0.02041     214.83370     214.45204
```

With 189 observations the data cannot distinguish the two; we prefer the logit link because e^{β} is an odds ratio (probit coefficients are not log odds ratios).

A second reason for the popularity of the logit link is **retrospective (case–control) sampling**. Suppose cases ($Y = 1$) and controls ($Y = 0$) are sampled with probabilities π_1 and π_0 from a population in which the model (10.1) holds, and let $S = 1$ denote "selected".

Proposition 10.2 (Case–control sampling shifts only the intercept)**.** *In the selected sample,* $\operatorname{logit} P(Y = 1 \mid \boldsymbol{x}, S = 1) = \beta_0 + \log(\pi_1/\pi_0) + \boldsymbol{x}_{-0}^{\top}\beta_{-0}$*, where* β_0 *is the intercept and the slopes* β_{-0} *are unchanged.*

Proof. By Bayes' theorem, $\dfrac{P(Y=1 \mid \boldsymbol{x}, S=1)}{P(Y=0 \mid \boldsymbol{x}, S=1)} = \dfrac{P(S=1 \mid Y=1)\,P(Y=1 \mid \boldsymbol{x})}{P(S=1 \mid Y=0)\,P(Y=0 \mid \boldsymbol{x})} = \dfrac{\pi_1}{\pi_0}\, e^{\boldsymbol{x}^{\top}\beta}$, because selection depends only on Y. □

Hence the odds ratios from a case–control study are estimated consistently by ordinary logistic regression, but the fitted *probabilities* are not those of the population. This is also why predicted probabilities must be re-calibrated when a classifier is trained on a re-balanced sample (Section 10.3).

10.1.3 Checking a logistic model

The residual deviance of a binary model is not a goodness-of-fit statistic (Proposition 9.2), and Pearson residuals lie on two curves (Figure 9.2). Useful checks are: randomized quantile residuals; **calibration**, comparing the fitted probabilities with the observed frequencies in

groups; and likelihood-ratio tests against richer models (interactions, nonlinear terms). The **Hosmer–Lemeshow** statistic sorts the observations by $\hat{\mu}_i$, forms G groups (often 10) and compares the observed and expected numbers of events, $\sum_g (O_g - n_g \bar{\mu}_g)^2 / \{n_g \bar{\mu}_g (1 - \bar{\mu}_g)\}$, which is referred approximately to χ^2_{G-2}. Its power is low for moderate n and it depends on the grouping, so a calibration table is as informative.

```
p <- fitted(fit); g <- cut(p, quantile(p, 0:5 / 5), include.lowest = TRUE)
cal <- data.frame(n = as.vector(table(g)), mean_pred = tapply(p, g, mean),
↪  observed = tapply(bw$low, g, mean))
round(cal, 3)
```

```
#>                  n mean_pred observed
#> [0.0233,0.123] 38      0.080    0.053
#> (0.123,0.225]  38      0.180    0.211
#> (0.225,0.301]  37      0.262    0.297
#> (0.301,0.504]  38      0.396    0.368
#> (0.504,0.88]   38      0.641    0.632
```

```
g10 <- cut(p, quantile(p, 0:10 / 10), include.lowest = TRUE)
O <- tapply(bw$low, g10, sum); n <- tapply(bw$low, g10, length); pb <-
↪  tapply(p, g10, mean)
HL <- sum((O - n * pb)^2 / (n * pb * (1 - pb)))
c(HL = HL, p = pchisq(HL, 8, lower.tail = FALSE))
```

```
#>     HL      p
#> 4.6762 0.7916
```

The predicted and observed proportions agree within each quintile and the Hosmer–Lemeshow statistic is 4.68 ($p = 0.79$): no evidence of miscalibration, although with 189 births the check has little power.

10.1.4 Grouped binary data

When covariates are discrete, observations with the same covariate pattern can be pooled into counts out of m_i. The model is the same, the information is the same, but the deviance becomes meaningful when the m_i are large (each cell then carries a lot of information). The `faraway::babyfood` data record, by sex and type of feeding, the number of infants who developed a respiratory disease in their first year of life.

```
bf <- glm(cbind(disease, nondisease) ~ sex + food, family = binomial, data =
↪  babyfood)
round(cbind(OR = exp(coef(bf)), exp(confint(bf))), 3)
```

```
#>                OR 2.5 % 97.5 %
#> (Intercept) 0.199 0.159  0.247
#> sexGirl     0.732 0.554  0.963
#> foodBreast  0.512 0.378  0.690
#> foodSuppl   0.842 0.556  1.246
```

```
c(deviance = deviance(bf), df = df.residual(bf), p = pchisq(deviance(bf),
↪  df.residual(bf), lower.tail = FALSE))
```

```
#> deviance       df        p
#>   0.7219   2.0000   0.6970
```

Breast-fed infants have 49% lower odds of disease than bottle-fed ones (same sex), and the residual deviance 0.72 on 2 df shows no lack of fit: here the cells have between about 130 and 500 infants, so the χ^2 approximation is reasonable.

10.2 Separation

The log-likelihood of a logistic regression is concave but need not have a maximizer. If a combination of covariates predicts the response perfectly, the likelihood increases without bound.

Definition 10.1 (Separation)**.** The data exhibit **complete separation** if there is a vector $\boldsymbol{b}$ with $\boldsymbol{x}_i^\top \boldsymbol{b} > 0$ whenever $y_i = 1$ and $\boldsymbol{x}_i^\top \boldsymbol{b} < 0$ whenever $y_i = 0$. They exhibit **quasi-complete separation** if the inequalities hold weakly (with ≥ 0 and ≤ 0) with equality for some observations, but not for all.

Theorem 10.1 (No maximum under complete separation)**.** *Under complete separation the log-likelihood $\ell(\beta)$ of the logistic model has no maximizer: $\ell(t\boldsymbol{b}) \uparrow 0$ as $t \to \infty$ while $\ell(\beta) < 0$ for every finite β.*

Proof. For every observation $\ell_i(\beta) = y_i\eta_i - \log(1 + e^{\eta_i}) = -\log(1 + e^{-s_i\eta_i})$ with $s_i = 2y_i - 1 \in \{\pm 1\}$, which is strictly negative and equals $-\log(1 + e^{-s_i \boldsymbol{x}_i^\top \beta})$. At $\beta = t\boldsymbol{b}$, $s_i \boldsymbol{x}_i^\top \boldsymbol{b} > 0$ for all i, so every term tends to 0 as $t \to \infty$, hence $\ell(t\boldsymbol{b}) \to 0$. Since each term is strictly negative for finite β, the supremum 0 is not attained. □

In R the symptoms are the warning "fitted probabilities numerically 0 or 1 occurred", coefficients of absurd size (here 47) with astronomical standard errors (so Wald $z \approx 0$), and a residual deviance near 0. The IRLS iterations stop at the iteration limit rather than converging.

```
x <- 1:6; y <- c(0, 0, 0, 1, 1, 1)                                  # complete
↪  separation at x = 3.5
fsep <- suppressWarnings(glm(y ~ x, family = binomial))
c(coef(fsep), se = coef(summary(fsep))[2, 2], deviance = deviance(fsep),
↪  iterations = fsep$iter)
```

```
#> (Intercept)           x          se    deviance  iterations
#>  -1.653e+02   4.723e+01   1.153e+05   2.215e-10   2.500e+01
```

```
x2 <- c(1, 2, 3, 3, 4, 5, 6); y2 <- c(0, 0, 0, 1, 1, 1, 1)          #
↪  quasi-complete: both outcomes at x = 3
round(coef(summary(suppressWarnings(glm(y2 ~ x2, family = binomial)))), 4)
```

```
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept)   -58.78      23139 -0.0025    0.998
#> x2             19.59       7713  0.0025    0.998
```

The likelihood-ratio test and the profile interval remain informative (the likelihood is flat in one direction), but the Wald statistics are useless. Remedies: report the separation and say that the effect is "at least large" (a one-sided statement); drop or merge the offending covariate if it is a coding artefact; or use **Firth's penalized likelihood**, which maximizes $\ell(\beta) + \frac{1}{2}\log\det \mathcal{I}(\beta)$ (a Jeffreys-prior penalty). It always gives finite estimates and removes the first-order bias of the MLE:

```
fb <- glm(y ~ x, family = binomial, method = brglm2::brglmFit)
round(coef(summary(fb)), 3)
```

```
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept)   -3.951      3.187  -1.240    0.215
#> x              1.129      0.855   1.321    0.187
```

Firth estimates (package `brglm2`) are finite, with usable standard errors; the price is a mild shrinkage towards 0.

10.3 Classification: thresholds and decisions

A fitted logistic model gives probabilities $\hat{\mu}(\boldsymbol{x})$. A *decision* ("treat", "approve the loan", "flag") requires a threshold: predict $\hat{y} = 1$ if $\hat{\mu}(\boldsymbol{x}) \geq c$. We use the `MASS::Pima.tr` (200 women, for training) and `Pima.te` (332 women, held out) data: the response `type` indicates diabetes, with `npreg`, `glu`, `bmi`, `ped` (pedigree function) and `age` as covariates.

```
fp <- glm(type ~ npreg + glu + bmi + ped + age, family = binomial, data =
↪  Pima.tr)
round(coef(summary(fp)), 4)
```

```
#>             Estimate Std. Error z value Pr(>|z|)
#> (Intercept)  -9.9381     1.5416  -6.447   0.0000
#> npreg         0.1031     0.0645   1.599   0.1099
#> glu           0.0318     0.0067   4.771   0.0000
#> bmi           0.0797     0.0326   2.440   0.0147
#> ped           1.8114     0.6610   2.740   0.0061
#> age           0.0393     0.0210   1.874   0.0610
```

```
p_tr <- fitted(fp); y_tr <- Pima.tr$type == "Yes"
p_te <- predict(fp, Pima.te, type = "response"); y_te <- Pima.te$type == "Yes"
```

10.3.1 Confusion matrix, sensitivity, specificity, ROC

For a threshold c, the **confusion matrix** counts true positives (TP), false positives (FP), false negatives (FN) and true negatives (TN). **Sensitivity** (true-positive rate) is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, the share of true cases caught; **specificity** is $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$, the share of non-cases correctly cleared (so 1−specificity is the false-positive rate; high specificity means *few false positives*). Raising c increases specificity and lowers sensitivity. The **ROC curve** plots sensitivity against 1−specificity as c varies, and the **AUC** summarizes it.

Theorem 10.2 (AUC as a probability)**.** *Let $\hat{\mu}^+$ and $\hat{\mu}^-$ be the fitted scores of a case and a control, drawn independently at random from the cases and controls. Then* $\mathrm{AUC} = P(\hat{\mu}^+ > \hat{\mu}^-) + \frac{1}{2}P(\hat{\mu}^+ = \hat{\mu}^-)$.

Proof. Let F^+, F^- be the distribution functions of the scores in the two groups. At threshold c, sensitivity is $1 - F^+(c^-)$ and the false-positive rate is $1 - F^-(c^-)$. The area under the curve of sensitivity against false-positive rate is $\int s_+(c)\, d\{-s_-(c)\}$ with $s_\pm(c) = P(\hat{\mu}^\pm \geq c)$, which is $P(\hat{\mu}^+ \geq \hat{\mu}^-)$ for independent scores in the continuous case (condition on $\hat{\mu}^- = c$: the probability that the case exceeds it is $s_+(c)$, and $-ds_-(c)$ is the density of $\hat{\mu}^-$). With ties the empirical curve joins the points linearly, which counts a tie as $\frac{1}{2}$. □

The AUC thus equals the Mann–Whitney statistic: the proportion of (case, control) pairs that the model ranks correctly. It does not depend on the threshold nor on the calibration of the probabilities, only on the ranking.

```
auc <- function(p, y) (sum(rank(p)[y]) - sum(y) * (sum(y) + 1) / 2) / (sum(y)
↪  * sum(!y))
c(train = auc(p_tr, y_tr), test = auc(p_te, y_te),
  pairs = mean(outer(p_te[y_te], p_te[!y_te], ">")))
```

```
#>  train   test  pairs
#> 0.8509 0.8652 0.8652
```

```
rates <- function(p, y, cut) { yhat <- p >= cut; c(sens = mean(yhat[y]), spec
↪  = mean(!yhat[!y])) }
conf <- function(p, y, cut) { yhat <- p >= cut; c(TP = sum(yhat & y), FP =
↪  sum(yhat & !y), FN = sum(!yhat & y), TN = sum(!yhat & !y)) }
rbind(`c = 0.50` = conf(p_te, y_te, 0.5), `c = 0.27` = conf(p_te, y_te, 0.27))
```

```
#>          TP FP FN  TN
#> c = 0.50 67 24 42 199
#> c = 0.27 91 60 18 163
```

10.3.2 The cost-optimal threshold

Theorem 10.3 (Cost-optimal threshold)**.** *Let $p = P(Y = 1 \mid \boldsymbol{x})$ be the true conditional probability, and let a false positive cost C_{FP}, a false negative C_{FN}, and correct decisions cost 0. The decision that minimizes the expected cost is $\hat{y} = 1$ if and only if*

$$p > c^* = \frac{C_{FP}}{C_{FP} + C_{FN}}.$$

Proof. Predicting 1 costs C_{FP} when $Y = 0$, with probability $1 - p$: expected cost $(1 - p)C_{FP}$. Predicting 0 costs C_{FN} when $Y = 1$: expected cost $p\,C_{FN}$. Predict 1 iff $(1 - p)C_{FP} < p\,C_{FN}$, i.e. $p > C_{FP}/(C_{FP} + C_{FN})$. □

With equal costs $c^* = 1/2$; if a missed case is three times as costly as a false alarm, $c^* = 1/4$. Two points matter. (i) c^* does **not** depend on the prevalence: the prior probability of $Y = 1$ is already inside p. (ii) It requires **calibrated** probabilities. If the model was trained on a re-balanced sample (case–control sampling, oversampling the minority class), $\hat{\mu}$ is not calibrated: by Proposition 10.2 the odds must be corrected by the factor (π_0/π_1) before comparing with c^* (Exercise 10.5). A calibration table on held-out data is the check.

```
b <- cut(p_te, c(0, 0.1, 0.25, 0.5, 0.75, 1))
round(data.frame(n = as.vector(table(b)), mean_pred = tapply(p_te, b, mean),
↪  observed = tapply(y_te, b, mean)), 3)
```

```
#>             n mean_pred observed
#> (0,0.1]    86     0.053    0.012
#> (0.1,0.25] 90     0.163    0.167
#> (0.25,0.5] 65     0.373    0.400
#> (0.5,0.75] 44     0.631    0.659
#> (0.75,1]   47     0.867    0.809
```

```
cost <- function(p, y, cut, cfp = 1, cfn = 3) { yh <- p >= cut; cfp * sum(yh &
↪  !y) + cfn * sum(!yh & y) }
sapply(c(`c = 0.50` = 0.5, `c = 0.27` = 0.27, `c* = 0.25` = 0.25), function(k)
↪  cost(p_te, y_te, k))
```

```
#>  c = 0.50  c = 0.27 c* = 0.25
#>       150       114       111
```

The probabilities are well calibrated except in the lowest bin (mean prediction 0.053 against an observed frequency of 0.012). With $C_{FN} = 3C_{FP}$, the threshold $c^* = 0.25$ gives a test cost of 111, against 150 for the conventional $c = 0.5$.

10.3.3 Choosing a threshold without leakage

Often the costs are not known, and the threshold is chosen to optimize a criterion, for instance **Youden's index** $J(c) = \text{sensitivity}(c) + \text{specificity}(c) - 1$. The cut-off is a tuning parameter and must be chosen *without using the test labels*. If the test set is used both to choose c and to report sensitivity and specificity, the reported figures are the maximum of many noisy estimates: they are optimistically biased. This is **leakage**. The honest procedures are: choose c on the training data; or, better, by K-fold cross-validation on the training data; or from costs; then evaluate once on the test set.

```
cuts <- seq(0.05, 0.95, by = 0.01)
J <- function(p, y) sapply(cuts, function(k) sum(rates(p, y, k)) - 1)
c_train <- cuts[which.max(J(p_tr, y_tr))]                          # honest:
↪  training data only
c_test  <- cuts[which.max(J(p_te, y_te))]                          # LEAKAGE:
↪  uses the test labels
c(c_train = c_train, c_test = c_test)
```

```
#> c_train  c_test
#>    0.27    0.22
```

```
rbind(`chosen on training` = rates(p_te, y_te, c_train), `chosen on test
↪  (leaky)` = rates(p_te, y_te, c_test))
```

```
#>                          sens   spec
#> chosen on training     0.8349 0.7309
#> chosen on test (leaky) 0.8899 0.6816
```

Choosing on the training data gives cut-off 0.27, with test sensitivity 0.835 and specificity 0.731; choosing on the test data gives 0.22 and higher sensitivity but lower specificity. The second pair of numbers is *not* an estimate of how the classifier would perform on new women: the test labels have been used twice. A single split cannot show how large the optimism is, so we repeat the exercise on 200 random splits of the whole Pima sample (532 women), recording the value of Youden's J on the test part when the cut-off is chosen honestly (on the training part) and with leakage (on the test part).

```
al <- rbind(Pima.tr, Pima.te)
set.seed(2027)
res <- t(replicate(200, {
  idx <- sample(nrow(al), 200); tr <- al[idx, ]; te <- al[-idx, ]
  f <- glm(type ~ npreg + glu + bmi + ped + age, family = binomial, data = tr)
```

```
  ptr <- fitted(f); pte <- predict(f, te, type = "response"); ytr <- tr$type
↪   == "Yes"; yte <- te$type == "Yes"
  c(honest = J(pte, yte)[which.max(J(ptr, ytr))],
    leaked = max(J(pte, yte)))
}))
round(rbind(mean = colMeans(res), sd = apply(res, 2, sd)), 3)
```

```
#>      honest leaked
#> mean  0.508  0.550
#> sd    0.040  0.031
```

```
mean(res[, "leaked"] - res[, "honest"])
```

```
#> [1] 0.04285
```

On average, the leaky procedure reports a test Youden index 0.043 higher than the honest one, for the *same* classifier; the honest value 0.508 is the one that estimates performance on new data. K-fold cross-validation on the training data makes the choice less variable than a single training split:

```
set.seed(2027); K <- 5; fold <- sample(rep(1:K, length.out = nrow(Pima.tr)))
pcv <- numeric(nrow(Pima.tr))
for (k in 1:K) {
  f <- glm(type ~ npreg + glu + bmi + ped + age, family = binomial, data =
↪   Pima.tr[fold != k, ])
  pcv[fold == k] <- predict(f, Pima.tr[fold == k, ], type = "response")
}
c_cv <- cuts[which.max(J(pcv, y_tr))]
c(c_cv = c_cv); rates(p_te, y_te, c_cv)
```

```
#> c_cv
#> 0.22
```

```
#>   sens   spec
#> 0.8899 0.6816
```

The cross-validated cut-off is 0.22. (It happens to coincide with the value that the test labels would have chosen; what matters is the *procedure*: the test set was not consulted.)

10.4 Poisson regression

For a count response, the Poisson family with the canonical log link, $\log \mu_i = \boldsymbol{x}_i^\top \beta$, gives **multiplicative effects** on the mean: e^{β_j} is the ratio of expected counts for a unit increase in x_j. Counts are often accumulated over different exposures (months of service, person-years, number of policies, area). Let Y_i be the number of events over an exposure t_i with event *rate* λ_i per unit exposure.

Proposition 10.3 (Poisson regression for rates and the offset)**.** *If* $Y_i \sim \text{Poisson}(\mu_i)$ *with* $\mu_i = t_i \lambda_i$ *and* $\log \lambda_i = \boldsymbol{x}_i^\top \beta$*, then*

$$\log \mu_i = \log t_i + \boldsymbol{x}_i^\top \beta.$$

The term $\log t_i$ *is an* ***offset****: a covariate whose coefficient is fixed at 1. The rate ratio for a*

unit increase in x_j *is* $\lambda(\boldsymbol{x}+\boldsymbol{e}_j)/\lambda(\boldsymbol{x}) = e^{\beta_j}$.

The offset says that the expected count is proportional to exposure: twice the months of service, twice the expected incidents. It must be $\log t_i$, not t_i ($\mu_i \propto e^{\gamma t_i}$ would be an exponential, not a proportional, relation). Cells with $t_i = 0$ have $\mu_i = 0$ and carry no information: they must be removed. In R, `offset(log(t))` in the formula, or the argument `offset =`.

Example 10.2 (Rates of damage incidents for cargo ships)**.** We model the incident rate of `MASS::ships` by ship type (A–E), year of construction and period of operation. The 6 cells with no service months are dropped; year and period are *codes* and must be converted to factors.

```
sh <- subset(ships, service > 0)
sh$year <- factor(sh$year); sh$period <- factor(sh$period)
fs <- glm(incidents ~ type + year + period + offset(log(service)), family =
↪  poisson, data = sh)
round(exp(cbind(RR = coef(fs), confint(fs))), 3)
```

```
#>                RR 2.5 % 97.5 %
#> (Intercept) 0.002 0.001  0.003
#> typeB       0.581 0.414  0.832
#> typeC       0.503 0.252  0.928
#> typeD       0.927 0.511  1.608
#> typeE       1.385 0.866  2.193
#> year65      2.008 1.503  2.705
#> year70      2.267 1.628  3.170
#> year75      1.574 0.988  2.469
#> period75    1.469 1.166  1.854
```

```
c(deviance = deviance(fs), df = df.residual(fs), Pearson = sum(residuals(fs,
↪  "pearson")^2))
```

```
#> deviance       df  Pearson
#>    38.70    25.00    42.28
```

```
ci <- confint(fs); rr <- data.frame(term = names(coef(fs)), est =
↪  exp(coef(fs)), lo = exp(ci[, 1]), hi = exp(ci[, 2]))[-1, ]
rr$term <- factor(rr$term, levels = rev(rr$term))
ggplot(rr, aes(est, term)) + geom_vline(xintercept = 1, colour = "grey60",
↪  linetype = 2) +
  geom_errorbar(aes(xmin = lo, xmax = hi), width = 0.2, colour = ie_navy,
  ↪  orientation = "y") +
  geom_point(colour = ie_coral, size = 2.5) + scale_x_log10() + labs(x = "rate
  ↪  ratio", y = NULL)
```

Operating in 1975–79 goes with an incident rate 1.47 times that of 1960–74 (95% interval 1.17 to 1.85), for ships of the same type and year of construction; type B has a rate 0.58 times that of type A. For a type B ship built 1965–69 operating in 1975–79 with 1000 months of service, the model predicts

```
nd <- data.frame(type = "B", year = "65", period = "75", service = c(1000,
↪  2000))
```

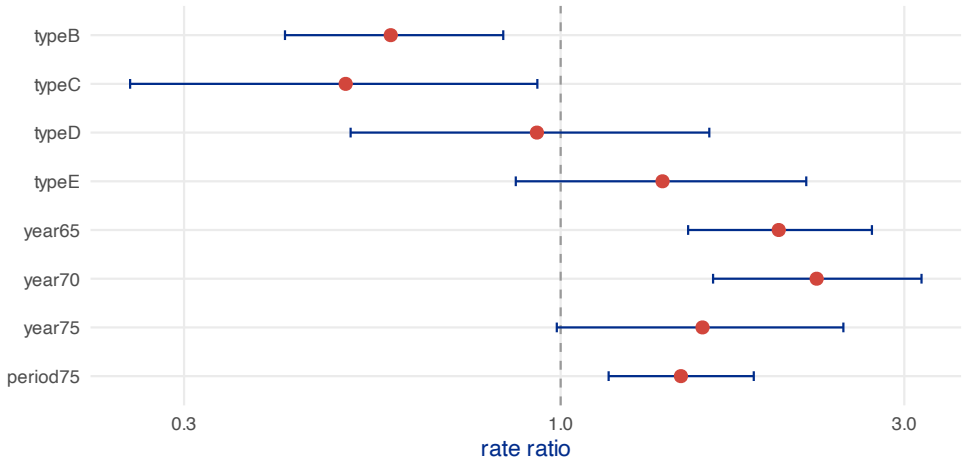


Figure 10.2: Rate ratios (log scale) with 95% profile intervals for the ships data. Baseline: type A, built 1960–64, operating 1960–74.

```
predict(fs, nd, type = "response")
```

```
#>     1     2
#> 2.830 5.659
```

incidents; doubling the exposure doubles the expected count. We can *test* the offset assumption by letting the coefficient of $\log t_i$ be free:

```
fs_free <- glm(incidents ~ type + year + period + log(service), family =
↪  poisson, data = sh)
round(coef(summary(fs_free))["log(service)", ], 3)
```

```
#>   Estimate Std. Error    z value   Pr(>|z|)
#>      0.903      0.102      8.867      0.000
```

The estimated exponent $\hat{\gamma} = 0.9$ (standard error 0.1) is consistent with 1: the offset is reasonable. The residual deviance of 38.7 on 25 df and the Pearson statistic 42.3 suggest, however, more variation than the Poisson model with $\phi = 1$ allows.

Example 10.3 (Claims per policy)**.** `MASS::Insurance` has, for each combination of district, car group and age group of the policyholder, the number of policyholders (`Holders`) and of claims. The model for the claim rate per holder is Poisson with offset log(Holders).

```
ins <- Insurance; ins$Group <- factor(ins$Group, ordered = FALSE); ins$Age <-
↪  factor(ins$Age, ordered = FALSE)
fi <- glm(Claims ~ District + Group + Age + offset(log(Holders)), family =
↪  poisson, data = ins)
round(exp(cbind(RR = coef(fi), confint(fi))), 3)[c("Age>35", "Group>2l",
↪  "District4"), ]
```

```
#>                RR 2.5 % 97.5 %
#> Age>35      0.585 0.511  0.672
#> Group>2l    1.757 1.523  2.022
#> District4 1.264 1.118  1.424
```

```
c(phi_hat = sum(residuals(fi, "pearson")^2) / df.residual(fi), deviance =
↪  deviance(fi), df = df.residual(fi))
```

```
#>  phi_hat deviance       df
#>   0.9005  51.4200  54.0000
```

Drivers older than 35 have a claim rate 0.58 times that of drivers under 25 in the same district and car group. Here the Pearson dispersion 0.9 is below 1: there is *no* overdispersion, and the Poisson model is adequate. The deviance 51.4 on 54 df is also unremarkable (large counts per cell, so the χ^2 reference is usable).

10.5 Overdispersion

The Poisson (and the binomial) family fixes $\phi = 1$: $\mathrm{Var}(Y) = \mu$. Real counts are often more variable.

Definition 10.2 (Overdispersion)**.** The response is **overdispersed** relative to the family if $\mathrm{Var}(Y_i) > \phi V(\mu_i)$ with $\phi = 1$ assumed, that is, if the true dispersion exceeds the assumed one.

Common causes are unobserved heterogeneity (omitted covariates make the rate vary between units with the same $\boldsymbol{x}$), clustering and positive correlation (events come in clumps), a misspecified mean or link, outliers, and excess zeros. The *consequence* is that, if the mean is correctly specified, $\hat{\beta}$ is still consistent (the equations of Theorem 9.1 are unbiased), but the standard errors computed from $(X^\top \hat{W} X)^{-1}$ are too small, and p-values and intervals are too optimistic.

10.5.1 Detecting overdispersion

1. **The Pearson dispersion** $\hat{\phi} = X^2/(n-p)$. A value clearly above 1 is a warning (above about 1.5 with moderate n); it is only reliable when the counts are not tiny.
2. **The Cameron–Trivedi test** (Cameron and Trivedi, 1990) against the alternative $\mathrm{Var}(Y) = \mu + \alpha\mu^2$, $\alpha > 0$. Under this alternative $\mathrm{E}[(Y-\mu)^2 - Y] = \mathrm{Var}(Y) - \mu = \alpha\mu^2$ so $\mathrm{E}\{[(Y-\mu)^2 - Y]/\mu\} = \alpha\mu$. One regresses $z_i = \{(y_i - \hat{\mu}_i)^2 - y_i\}/\hat{\mu}_i$ on $\hat{\mu}_i$ **without intercept**, and tests $\alpha = 0$ with a one-sided t-test. For the alternative $\mathrm{Var}(Y) = (1+\alpha)\mu$ (the quasi-Poisson shape) one regresses z_i on a constant.
3. **Residuals and mean–variance plots.** Randomized quantile residuals far outside ± 3 and an S-shaped QQ plot indicate overdispersion. Grouping observations with similar covariates and plotting the sample variance against the sample mean shows which variance function fits.

Example 10.4 (Absence from school in Walgett)**.** `MASS::quine` records the days absent from school for 146 children in Walgett (New South Wales), with ethnic background (`Eth`), sex, age group and learner status. The mean is 16.5 days and the variance 264.2: the variance is about 16 times the mean.

```
fp <- glm(Days ~ Eth + Sex + Age + Lrn, family = poisson, data = quine)
phi_hat <- sum(residuals(fp, "pearson")^2) / df.residual(fp)
mu <- fitted(fp); yq <- quine$Days
aux <- lm(((yq - mu)^2 - yq) / mu ~ 0 + mu)
c(phi_pearson = phi_hat, phi_deviance = deviance(fp) / df.residual(fp))
```

```
#>  phi_pearson phi_deviance
#>        13.17        12.21
```

```
ct <- coef(summary(aux)); c(alpha = ct[1, 1], t = ct[1, 3], p_one_sided =
↪  pt(ct[1, 3], df.residual(aux), lower.tail = FALSE))
```

```
#>       alpha           t p_one_sided
#>   6.362e-01   5.327e+00   1.865e-07
```

The Pearson dispersion is 13.2, and the Cameron–Trivedi test gives $\hat{\alpha} = 0.64$, $t = 5.3$, a one-sided p-value of 2×10^{-7}, i.e. $\mathrm{Var}(Y) \approx \mu + 0.64\mu^2$. The Poisson model badly underestimates the variation.

10.5.2 Remedy 1: quasi-Poisson

Quasi-Poisson specifies only $\mathrm{E}[Y] = \mu$ and $\mathrm{Var}(Y) = \phi\mu$. By Proposition 9.3, $\hat{\beta}$ equals the Poisson MLE, the standard errors are multiplied by $\sqrt{\hat{\phi}}$, with $\hat{\phi} = X^2/(n-p)$, and tests are F tests. There is no likelihood, hence no AIC. The variance is a *constant multiple* of the mean.

10.5.3 Remedy 2: the negative binomial

A second remedy is a full parametric model with a variance that grows faster than the mean. Suppose that the Poisson rate itself varies between units: $Y \mid \lambda \sim \mathrm{Poisson}(\lambda)$ and λ is gamma distributed with mean μ and shape k (so $\mathrm{Var}(\lambda) = \mu^2/k$).

Theorem 10.4 (Gamma–Poisson mixture)**.** *The marginal distribution of Y is the negative binomial*

$$P(Y = y) = \frac{\Gamma(y+k)}{y!\,\Gamma(k)} \left(\frac{k}{\mu+k}\right)^k \left(\frac{\mu}{\mu+k}\right)^y, \qquad y = 0, 1, \ldots$$

with $\mathrm{E}[Y] = \mu$ and $\mathrm{Var}(Y) = \mu + \mu^2/k$.

Proof. With the gamma density $f(\lambda) = \frac{(k/\mu)^k}{\Gamma(k)} \lambda^{k-1} e^{-k\lambda/\mu}$,

$$P(Y = y) = \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!} f(\lambda)\, d\lambda = \frac{(k/\mu)^k}{y!\,\Gamma(k)} \int_0^\infty \lambda^{y+k-1} e^{-\lambda(1+k/\mu)} d\lambda = \frac{(k/\mu)^k\,\Gamma(y+k)}{y!\,\Gamma(k)\,(1+k/\mu)^{y+k}}.$$

Simplifying $(k/\mu)^k (\mu/(\mu+k))^{y+k} = k^k \mu^y/(\mu+k)^{y+k}$ gives the stated mass function. By the laws of total expectation and variance, $\mathrm{E}[Y] = \mathrm{E}[\lambda] = \mu$ and $\mathrm{Var}(Y) = \mathrm{E}[\lambda] + \mathrm{Var}(\lambda) = \mu + \mu^2/k$. □

For known k the negative binomial is an exponential family with $V(\mu) = \mu + \mu^2/k$ (Exercise 8.4), so it is a GLM; k is estimated by maximum likelihood, alternating IRLS for β and a one-dimensional optimization for k (this is what `MASS::glm.nb()` does). The variance exceeds the Poisson one by μ^2/k; $k \to \infty$ recovers the Poisson. **Notation:** `glm.nb` calls k "`theta`"; in this course θ is the canonical parameter, so we write k. k is not "the dispersion": *small* k means *more* overdispersion. The test $\alpha = 1/k = 0$ of Cameron and Trivedi is the test of the Poisson against the negative binomial.

The two remedies differ in the assumed **variance function**: quasi-Poisson has $\mathrm{Var}(Y) = \phi\mu$ (linear in the mean), the negative binomial has $\mu + \mu^2/k$ (quadratic). Plotting the variance against the mean for groups of similar covariates shows which is better.

```
fq  <- update(fp, family = quasipoisson)
fnb <- glm.nb(Days ~ Eth + Sex + Age + Lrn, data = quine)
c(phi = summary(fq)$dispersion, k = fnb$theta, se_k = fnb$SE.theta)
```

```
#>    phi      k   se_k
#> 13.167  1.275  0.161
```

```
tab <- cbind(est_poisson = coef(fp), se_poisson = coef(summary(fp))[, 2],
  se_quasi = coef(summary(fq))[, 2],
             est_nb = coef(fnb), se_nb = coef(summary(fnb))[, 2])
round(tab, 3)
```

```
#>             est_poisson se_poisson se_quasi est_nb se_nb
#> (Intercept)       2.715      0.065    0.235  2.895 0.228
#> EthN             -0.534      0.042    0.152 -0.569 0.153
#> SexM              0.162      0.043    0.154  0.082 0.160
#> AgeF1            -0.334      0.070    0.254 -0.448 0.240
#> AgeF2             0.258      0.062    0.226  0.088 0.236
#> AgeF3             0.428      0.068    0.246  0.357 0.248
#> LrnSL             0.349      0.052    0.189  0.292 0.186
```

```
c(AIC_poisson = AIC(fp), AIC_nb = AIC(fnb))
```

```
#> AIC_poisson      AIC_nb
#>        2299        1109
```

The estimates of the Poisson and quasi-Poisson fits are identical, and the standard errors of the latter are $\sqrt{13.17} = 3.63$ times larger. The negative binomial estimates differ slightly (it weights observations differently), with standard errors close to the quasi-Poisson ones. The AIC drops from 2299 to 1109. Under the Poisson model every term is highly significant; under either remedy, `Sex` and learner status are no longer significant at the 5% level. F tests for quasi-Poisson and likelihood-ratio tests for the NB confirm this:

```
anova(update(fq, . ~ . - Lrn), fq, test = "F")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: Days ~ Eth + Sex + Age
#> Model 2: Days ~ Eth + Sex + Age + Lrn
#>   Resid. Df Resid. Dev Df Deviance    F Pr(>F)
#> 1       140       1743
#> 2       139       1697  1     45.8 3.48  0.064 .
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(update(fnb, . ~ . - Lrn), fnb)
```

```
#> Likelihood ratio tests of Negative Binomial Models
#>
#> Response: Days
#>                   Model theta Resid. df    2 x log-lik.   Test    df
#> 1       Eth + Sex + Age 1.251       140           -1096
#> 2 Eth + Sex + Age + Lrn 1.275       139           -1093 1 vs 2     1
```

```
#>   LR stat. Pr(Chi)
#> 1
#> 2    2.502  0.1137
```

```
cells <- aggregate(Days ~ Eth + Sex + Age + Lrn, data = quine, FUN =
↪  function(d) c(m = mean(d), v = var(d), n = length(d)))
cells <- data.frame(cells[, 1:4], cells$Days); cells <- subset(cells, n >= 3)
mm <- exp(seq(log(min(cells$m) * 0.9), log(max(cells$m) * 1.1), length.out =
↪  100))
curves <- rbind(data.frame(model = "Poisson", m = mm, v = mm),
                data.frame(model = "quasi-Poisson", m = mm, v =
                ↪  summary(fq)$dispersion * mm),
                data.frame(model = "negative binomial", m = mm, v = mm + mm^2
                ↪  / fnb$theta))
ggplot(cells, aes(m, v)) + geom_line(data = curves, aes(colour = model),
↪  linewidth = 1) + geom_point(size = 2) +
  scale_colour_manual(values = ie_pal[c(3, 2, 1)]) + scale_x_log10() +
  ↪  scale_y_log10() +
  labs(x = "cell mean", y = "cell variance", colour = NULL)
```

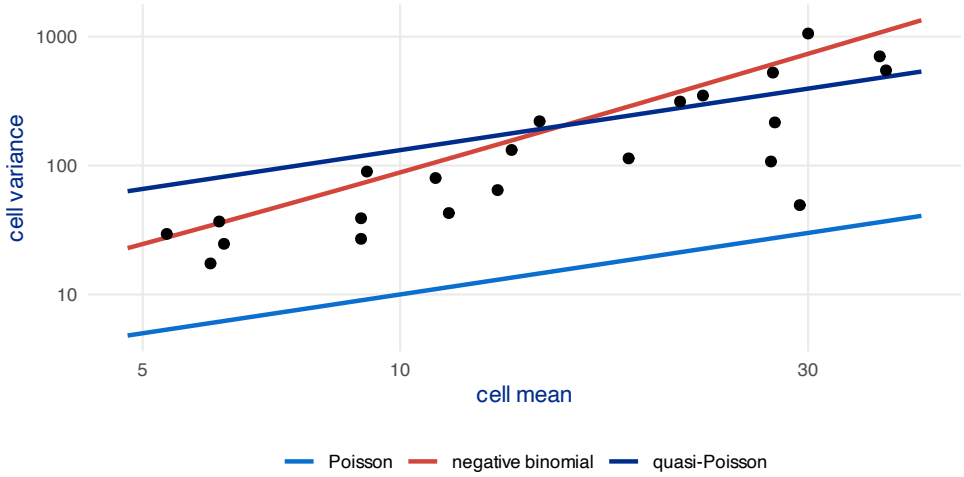


Figure 10.3: Sample variance against sample mean for covariate cells of the quine data (cells with at least 3 children). The quasi-Poisson line $\hat{\phi}\mu$ and the negative binomial curve $\mu + \mu^2/\hat{k}$.

The cell variances grow faster than linearly in the mean, and the negative binomial curve tracks them better than the straight line $\hat{\phi}\mu$. Randomized quantile residuals (Chapter 9) confirm the improvement:

```
rqr <- function(y, F, ...) {
  lo <- F(y - 1, ...); hi <- F(y, ...); u <- runif(length(y))
  up_lo <- F(y, ..., lower.tail = FALSE); up_hi <- F(y - 1, ..., lower.tail =
↪  FALSE)
  ifelse(lo < 0.5, qnorm(lo + u * (hi - lo)), qnorm(up_lo + u * (up_hi -
  ↪  up_lo), lower.tail = FALSE))
}
set.seed(2027)
r_p <- rqr(quine$Days, ppois, lambda = fitted(fp))
r_n <- rqr(quine$Days, pnbinom, size = fnb$theta, mu = fitted(fnb))
d <- rbind(data.frame(model = "Poisson", lm = log(fitted(fp)), r = r_p),
↪  data.frame(model = "negative binomial", lm = log(fitted(fnb)), r = r_n))
```

```
ggplot(d, aes(lm, r)) + geom_hline(yintercept = c(-3, 3), linetype = 2, colour
↪  = "grey60") + geom_hline(yintercept = 0, colour = "grey60") +
  geom_point(colour = ie_navy, alpha = 0.7) + facet_wrap(~model) + labs(x =
  ↪  expression(log(hat(mu))), y = "randomized quantile residual")
```

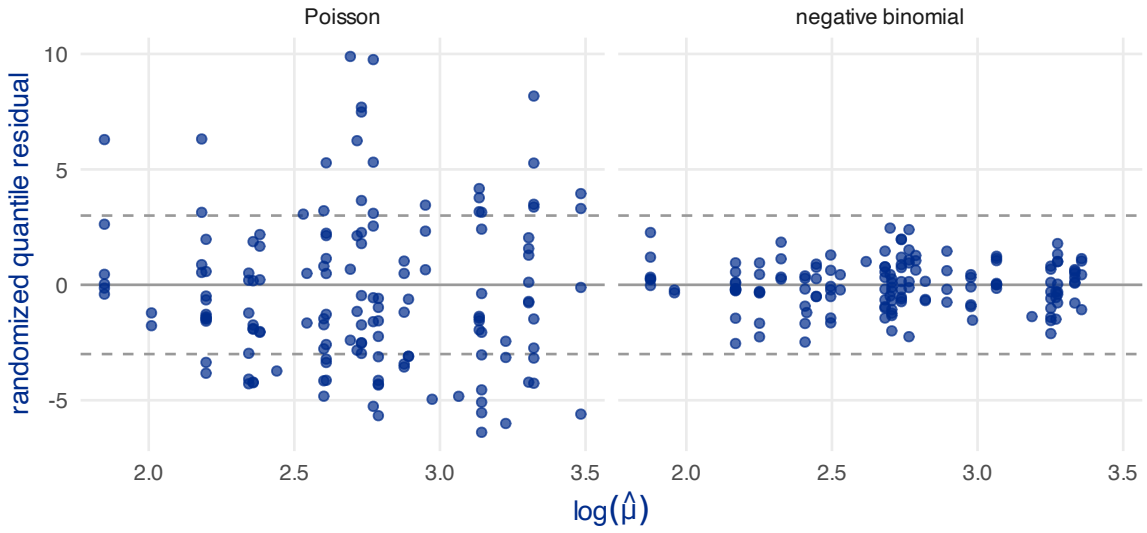


Figure 10.4: Randomized quantile residuals against $\log \hat{\mu}$ for the Poisson (left) and negative binomial (right) fits to the quine data.

```
round(rbind(Poisson = range(r_p), NB = range(r_n)), 2)
```

```
#>          [,1] [,2]
#> Poisson -6.39 9.89
#> NB      -2.54 2.46
```

The Poisson residuals range far beyond ± 3 (from -6 to about 10); the negative binomial residuals stay close to $N(0, 1)$.

Remark. **Remark.** Sandwich (robust) standard errors, `sandwich::vcovHC()`, are a third remedy: they keep the Poisson $\hat{\beta}$ and correct the variance under any variance function, as long as the mean is correct. Quasi-Poisson and negative binomial standard errors, which assume a particular variance function, are more efficient when that assumption holds.

10.6 Excess zeros: zero-inflated and hurdle models

Count data often contain many zeros. Before reaching for a special model, remember that **overdispersion alone produces extra zeros**: a negative binomial has more zeros than a Poisson with the same mean. The proper check is the *expected number of zeros under the fitted model*, $\sum_i P(Y_i = 0 \mid \hat{\mu}_i)$, which takes the covariates into account (and is not `dpois(0, mean(y))`).

```
c(observed = sum(quine$Days == 0), Poisson = sum(dpois(0, fitted(fp))),
  NB = sum(dnbinom(0, size = fnb$theta, mu = fitted(fnb))))
```

```
#> observed  Poisson       NB
#>  9.00000  0.01411  6.21356
```

For `quine`, 9 zeros are observed, the Poisson model expects 0.01 and the negative binomial 6.2: most of the "excess" was variance, not a separate zero process. If zeros are still in excess after fixing the variance, two model classes are used.

Definition 10.3 (Zero-inflated Poisson (ZIP))**.** With probability π the observation is a *structural zero*; otherwise it is Poisson(μ):

$$P(Y=0) = \pi + (1-\pi)e^{-\mu}, \qquad P(Y=y) = (1-\pi)\frac{e^{-\mu}\mu^y}{y!}, \quad y \geq 1.$$

Typically $\operatorname{logit}\pi_i = \boldsymbol{z}_i^\top\gamma$ and $\log\mu_i = \boldsymbol{x}_i^\top\beta$. Replacing the Poisson by a negative binomial gives the ZINB.

Proposition 10.4 (Moments of the ZIP)**.** $\mathrm{E}[Y] = (1-\pi)\mu$ *and* $\mathrm{Var}(Y) = (1-\pi)\mu(1+\pi\mu)$, *so* $\mathrm{Var}(Y)/\mathrm{E}[Y] = 1+\pi\mu > 1$ *whenever* $\pi > 0$.

Proof. $\mathrm{E}[Y] = (1-\pi)\mu$. $\mathrm{E}[Y^2] = (1-\pi)(\mu+\mu^2)$. Therefore $\mathrm{Var}(Y) = (1-\pi)(\mu+\mu^2) - (1-\pi)^2\mu^2 = (1-\pi)\mu\{1+\mu-(1-\pi)\mu\} = (1-\pi)\mu(1+\pi\mu)$. □

A ZIP is itself overdispersed, so zero inflation and overdispersion are hard to distinguish by looking at the variance only. The **hurdle model** treats *all* zeros as coming from a binary model, and the positive counts from a zero-truncated count distribution:

$$P(Y=0) = \pi_0, \qquad P(Y=y) = (1-\pi_0)\frac{f(y;\mu)}{1-f(0;\mu)}, \quad y \geq 1.$$

Its likelihood factorizes into a binary part (zero versus positive) and a truncated-count part, which can be fitted separately; ZIP, in contrast, mixes zeros from both processes. The ZIP likelihood is

$$\ell(\pi,\mu) = \sum_{i:y_i=0}\log\{\pi+(1-\pi)e^{-\mu}\} + \sum_{i:y_i>0}\{\log(1-\pi) - \mu + y_i\log\mu - \log y_i!\}.$$

We fit the intercept-only ZIP by hand for the number of visits to a doctor in the last two weeks (`faraway::dvisits`, 5190 Australians) and compare with `pscl::zeroinfl`.

```
y <- dvisits$doctorco
nll <- function(par) { pi <- plogis(par[1]); mu <- exp(par[2])
  -sum(ifelse(y == 0, log(pi + (1 - pi) * exp(-mu)), log(1 - pi) + dpois(y,
  ↪  mu, log = TRUE))) }
o <- optim(c(0, 0), nll)
z <- zeroinfl(y ~ 1 | 1, dist = "poisson")
rbind(by_hand = c(pi = plogis(o$par[1]), mu = exp(o$par[2]), loglik =
↪  -o$value),
      pscl = c(pi = plogis(coef(z)[2]), mu = exp(coef(z)[1]), loglik =
      ↪  as.numeric(logLik(z))))
```

```
#>             pi     mu loglik
#> by_hand 0.6503 0.8626  -3714
#> pscl    0.6504 0.8631  -3714
```

```
c(mean = mean(y), variance = var(y), zeros = mean(y == 0), poisson_zeros =
↪  dpois(0, mean(y)))
```

```
#>          mean      variance         zeros poisson_zeros
#>        0.3017        0.6370        0.7979        0.7395
```

The two fits agree. The data have mean 0.30 and variance 0.64; the share of zeros (80%) exceeds that of a Poisson with the same mean (74%).

Example 10.5 (Doctor visits: which model for the zeros?)**.** We add covariates (sex, age, income, private insurance `levyplus`, `freepoor`, illness count, days of reduced activity, general health score) and compare Poisson, negative binomial, ZIP, ZINB and hurdle models.

```
fm <- doctorco ~ sex + age + income + levyplus + freepoor + illness + actdays
↪  + hscore
m1 <- glm(fm, family = poisson, data = dvisits)
m2 <- glm.nb(fm, data = dvisits)
m3 <- zeroinfl(fm, data = dvisits)                          # same covariates in
↪  both parts
m4 <- zeroinfl(fm, data = dvisits, dist = "negbin")
m5 <- hurdle(fm, data = dvisits)
m6 <- hurdle(fm, data = dvisits, dist = "negbin")
ez <- c(Poisson = sum(dpois(0, fitted(m1))), NB = sum(dnbinom(0, size =
↪  m2$theta, mu = fitted(m2))),
        ZIP = sum(predict(m3, type = "prob")[, 1]), ZINB = sum(predict(m4,
        ↪  type = "prob")[, 1]),
        Hurdle = sum(predict(m5, type = "prob")[, 1]), HurdleNB =
        ↪  sum(predict(m6, type = "prob")[, 1]))
round(data.frame(df = sapply(list(m1, m2, m3, m4, m5, m6), function(m)
↪  attr(logLik(m), "df")),
                 AIC = sapply(list(m1, m2, m3, m4, m5, m6), AIC),
                 ↪  expected_zeros = ez,
                 row.names = names(ez)), 1)
```

```
#>          df   AIC expected_zeros
#> Poisson   9 6735          4013
#> NB       10 6423          4172
#> ZIP      18 6421          4112
#> ZINB     19 6279          4181
#> Hurdle   18 6492          4141
#> HurdleNB 19 6351          4141
```

```
sum(dvisits$doctorco == 0)
```

```
#> [1] 4141
```

The observed number of zeros is 4141. The Poisson model expects only 4013 zeros; the negative binomial 4172, the ZIP 4112 and the ZINB 4181, while the hurdle models reproduce the number of zeros exactly (the zero part is a logistic regression with an intercept, whose score equation forces the fitted number of zeros to equal the observed one). In terms of AIC the Poisson model is clearly worst (6735); the negative binomial (6423) and the ZIP (6421) are almost tied, and the best fit is the ZINB (6279), although with 9 more parameters than the NB. This illustrates the identifiability problem: the data can be described as an overdispersed count (NB) *or* as a mixture with structural zeros (ZIP); choose between them on substantive grounds (is there a

subpopulation that can never visit a doctor?) rather than by a significance test.

Testing. The Poisson model is **nested** in the ZIP ($\pi = 0$, a value on the boundary of the parameter space). The Vuong test is for *non-nested* models, so it is not appropriate here. Use AIC, the expected-zeros check, and, if a test is wanted, the likelihood-ratio statistic referred to a mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ for one boundary parameter (the usual χ^2 is conservative). Vuong's test is for comparing ZIP with the hurdle model or with a non-nested alternative.

10.7 In R

Task	Code
Logistic regression	`glm(y ~ x, family = binomial)`; grouped: `cbind(succ, fail) ~ x`
Odds ratios with profile CIs	`exp(cbind(OR = coef(fit), confint(fit)))`
Predicted probability with CI	`predict(fit, nd, type = "link", se.fit = TRUE)`, then `plogis()`
Separation / Firth	`glm(..., method = brglm2::brglmFit)`
ROC and AUC	`pROC::roc(y, p)`, or the rank formula of this chapter
Poisson with exposure	`glm(y ~ x + offset(log(t)), family = poisson)`
Overdispersion	`sum(residuals(fit, "pearson")^2) / df.residual(fit)`; regress $((y-\hat\mu)^2 - y)/\hat\mu$ on $\hat\mu$
Quasi-Poisson, NB	`family = quasipoisson`; `MASS::glm.nb(y ~ x)` (reports k as `theta`)
Zero inflation, hurdle	`pscl::zeroinfl(y ~ x | z, dist = "poisson" / "negbin")`, `pscl::hurdle(...)`
Expected zeros	`sum(dpois(0, fitted(fit)))`; `sum(predict(m, type = "prob")[, 1])` for `pscl` models

10.8 Common pitfalls

Warning. Do not choose a classification threshold (or a model) by looking at the test set. Choose on training data (or by cross-validation) or from costs, and evaluate once.

Warning. e^{β_j} is an *odds* ratio in logistic regression and a *rate* ratio in Poisson regression with offset; it is not a ratio of probabilities. "Smokers are 2.3 times as likely to have a low-weight baby" misreports an odds ratio.

Warning. A Poisson model with exposure needs `offset(log(t))`, not `t`, and cells with $t = 0$ must be removed. `year` and `period` coded as numbers (60, 65, 70, 75) must be turned into factors.

Remark. Specificity controls *false positives*. A cost-optimal threshold $c^* = C_{FP}/(C_{FP} + C_{FN})$ needs calibrated probabilities and does not depend on prevalence. The residual deviance of a binary model is not a goodness-of-fit test.

Remark. Fix overdispersion before adding zero inflation. The expected number of zeros is $\sum_i P(Y_i = 0 \mid \hat{\mu}_i)$, not `dpois(0, mean(y))`. `glm.nb` reports the size k as `theta`; small k means more overdispersion. When `glm.nb` warns that the iteration limit was reached and $\hat{k}$ is huge, the negative binomial has collapsed to the Poisson: the data show no extra variance of the μ^2 type.

10.9 Chapter summary

- Logistic regression: $\operatorname{logit} \mu = \boldsymbol{x}^\top \beta$; e^{β_j} is an odds ratio, constant over the covariate space; the effect on the probability is at most $|\beta_j|/4$. Case–control sampling changes only the intercept.
- The binary deviance is not a goodness-of-fit statistic; check calibration. Complete separation makes the MLE infinite: use LR tests, Firth, or reformulate.
- AUC $= P(\text{case ranks above control})$. The cost-optimal threshold is $c^* = C_{FP}/(C_{FP} + C_{FN})$ for calibrated probabilities. Choose thresholds without the test labels; leakage biases the reported performance upwards.
- Poisson regression: e^{β_j} is a rate ratio; exposure enters as $\log t_i$ with coefficient 1 (an offset); drop zero-exposure cells.
- Overdispersion leaves $\hat{\beta}$ consistent but standard errors too small. Detect with $\hat{\phi} = X^2/(n-p)$, the Cameron–Trivedi regression and residuals. Remedies: quasi-Poisson ($\text{Var} = \phi\mu$), negative binomial ($\text{Var} = \mu + \mu^2/k$, from the gamma–Poisson mixture), robust standard errors.
- Zero-inflated and hurdle models describe excess zeros; first check expected zeros under the fitted model and fix overdispersion. Poisson is nested in ZIP (boundary): no Vuong test.

10.10 Exercises

Exercise 10.1. In `birthwt`, the cross-classification of smoking and low birth weight is: smokers 30 low / 44 not low; non-smokers 29 low / 86 not low. (a) Compute the odds ratio, its logarithm, the standard error $\sqrt{1/a + 1/b + 1/c + 1/d}$ and a 95% Wald interval for the odds ratio. (b) Show that the logistic regression `low ~ smoke` reproduces these numbers exactly. (c) Compute the risk ratio and explain why it differs from the odds ratio.

Solution.

(a) $\widehat{\text{OR}} = \frac{30 \times 86}{44 \times 29} = \frac{2580}{1276} = 2.022$, $\log \widehat{\text{OR}} = 0.704$, $\text{se} = \sqrt{\frac{1}{30} + \frac{1}{44} + \frac{1}{29} + \frac{1}{86}} = 0.320$; the interval is $\exp(0.704 \pm 1.96(0.320)) = (1.08,\ 3.78)$. (b) The model with one binary covariate is saturated in the two groups, so $\hat{\beta}_0 = \operatorname{logit}(29/115)$ and $\hat{\beta}_1$ is the difference of the two sample logits, the log odds ratio; the information matrix gives the same standard error.

```
tb <- matrix(c(30, 44, 29, 86), 2, byrow = TRUE)
f1 <- glm(low ~ smoke, family = binomial, data = birthwt)
c(by_hand = log((30 * 86) / (44 * 29)), glm = unname(coef(f1)[2]))
```

```
#> by_hand     glm
#>  0.7041  0.7041
```

```
c(se_hand = sqrt(sum(1 / tb)), se_glm = coef(summary(f1))[2, 2])
```

```
#> se_hand  se_glm
```

```
#>  0.3196  0.3196
```

```
c(risk_ratio = (30 / 74) / (29 / 115))
```

```
#> risk_ratio
#>      1.608
```

(c) The risk ratio is $(30/74)/(29/115) = 1.61$, smaller than the odds ratio 2.02, because the outcome is not rare: the proportions 0.41 and 0.25 are far from 0, and odds $= p/(1-p)$ stretches large probabilities. They agree only when both probabilities are small.

Exercise 10.2.
(R) With the `babyfood` data (Section 10.1.4): (a) compute by hand, from the counts, the odds of respiratory disease for bottle-fed boys and for breast-fed boys, and their ratio; (b) compare with the fitted odds ratio of the model with `sex + food`; (c) test whether `food` is needed with a likelihood-ratio test; (d) is there evidence of an interaction between sex and food?

Solution.

```
(77 / 381) / (47 / 447)                          # bottle vs breast, boys:
↪  crude OR
```

```
#> [1] 1.922
```

```
exp(-coef(bf)["foodBreast"])                     # model: bottle vs breast,
↪  same sex
```

```
#> foodBreast
#>      1.953
```

```
anova(update(bf, . ~ . - food), bf, test = "Chisq")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: cbind(disease, nondisease) ~ sex
#> Model 2: cbind(disease, nondisease) ~ sex + food
#>   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
#> 1         4      20.90
#> 2         2       0.72  2     20.2  4.2e-05 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
bf2 <- glm(cbind(disease, nondisease) ~ sex * food, family = binomial, data =
↪  babyfood)
anova(bf, bf2, test = "Chisq")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: cbind(disease, nondisease) ~ sex + food
#> Model 2: cbind(disease, nondisease) ~ sex * food
#>   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
#> 1         2      0.722
#> 2         0      0.000  2    0.722      0.7
```

(a) Odds: bottle boys $77/381 = 0.202$, breast boys $47/447 = 0.105$; ratio 1.92. (b) The model assumes the same odds ratio for boys and girls and adjusts for them: $e^{0.669} = 1.95$, close to the crude 1.92 for boys. (c) Dropping `food` (2 df) increases the deviance by 20.2 ($p = 4 \times 10^{-5}$): strong evidence of a feeding effect. (d) The interaction (2 df) has deviance 0.72 and $p = 0.70$: no evidence that the feeding effect differs between boys and girls.

Exercise 10.3. Let $Y = \mathbf{1}\{\eta + \varepsilon > 0\}$ where ε has distribution function F symmetric about 0. (a) Show $P(Y = 1) = F(\eta)$. (b) For $F = \Phi$, the standard normal, and $F = \Lambda$, the standard logistic, show that $F'(0) = 0.399$ and 0.25, and deduce the approximate ratio of the logit to the probit coefficients. (c) If the latent error is $N(0, \sigma^2)$ with σ unknown, why can β and σ not be identified separately?

Solution.

(a) $P(Y = 1) = P(\varepsilon > -\eta) = 1 - F(-\eta) = F(\eta)$ by symmetry. (b) $\Phi'(0) = 1/\sqrt{2\pi} = 0.399$; $\Lambda(x) = 1/(1 + e^{-x})$ has $\Lambda'(0) = 1/4$. Near $\eta = 0$ the two models have the same slope when $\beta_{\text{logit}} \cdot 0.25 = \beta_{\text{probit}} \cdot 0.399$, i.e. $\beta_{\text{logit}}/\beta_{\text{probit}} \approx 1.6$ (matching standard deviations, $\pi/\sqrt{3} = 1.81$, gives 1.8). (c) $P(Y = 1) = P(\varepsilon/\sigma > -\eta/\sigma) = \Phi(\eta/\sigma)$: only β/σ appears, so the scale is fixed by convention ($\sigma = 1$ for probit, $\pi/\sqrt{3}$ for logit). This is why logistic coefficients cannot be compared across models with different covariates in the same way as linear-model coefficients.

Exercise 10.4.

(a) Show that for the data $x = (1, 2, 3, 4, 5, 6)$, $y = (0, 0, 0, 1, 1, 1)$ the logistic log-likelihood along the line $\beta = t(-3.5, 1)^\top$ equals $-\sum_i \log(1 + e^{-t|x_i - 3.5|})$ and increases to 0. (b) Give the quasi-complete separation condition for a 2×2 table with a zero cell, and say what happens to the odds ratio estimate. (c) Why is the likelihood-ratio test still usable?

Solution.

(a) $\eta_i = t(x_i - 3.5)$, and $s_i \eta_i = t|x_i - 3.5|$ for every i because $y_i = 1$ iff $x_i > 3.5$. By Theorem 10.1, $\ell(t\boldsymbol{b}) = -\sum \log(1 + e^{-t|x_i - 3.5|})$, an increasing function of t with limit 0.

```
ell <- function(t) -sum(log(1 + exp(-t * abs(1:6 - 3.5))))
sapply(c(0, 1, 2, 5, 10), ell)
```

```
#> [1] -4.15888 -1.50876 -0.73713 -0.15889 -0.01343
```

(b) A 2×2 table with one zero cell (say $a = 0$) has $\widehat{\text{OR}} = 0$ (or ∞), so $\hat\beta = \mp\infty$: quasi-complete separation. (c) The likelihood is flat in one direction but its *profile* is informative: $2\{\ell(\hat\beta_{\text{full}}) - \ell(\hat\beta_{\text{null}})\}$ is finite and valid to test whether the effect is zero, whereas the Wald standard error (computed at an infinite $\hat\beta$) is meaningless.

Exercise 10.5. A classifier for a disease with population prevalence 5% was trained on a sample in which all cases and an equal number of controls were included (so cases are over-represented by a factor 19: $\pi_1/\pi_0 = 19$). (a) For a patient the model gives probability 0.6. What is the population probability? (b) With $C_{FN} = 4C_{FP}$, should this patient be flagged? (c) What if we had (wrongly) compared the 0.6 with c^*?

Solution.

(a) By Proposition 10.2, the odds in the sample are the population odds multiplied by

$\pi_1/\pi_0 = 19$, so the population odds are $(0.6/0.4)/19 = 1.5/19 = 0.0789$ and $p = 0.0789/1.0789 = 0.073$. (b) $c^* = 1/(1+4) = 0.2$. The corrected probability $0.073 < 0.2$, so we do *not* flag. (c) $0.6 > 0.2$ would flag: the re-balanced probability is not calibrated for the population, and the decision would be wrong (more false positives than the costs justify).

Exercise 10.6. Six patients have scores $0.9, 0.8, 0.6, 0.5, 0.3, 0.2$ with true labels $1, 0, 1, 1, 0, 0$ (1 = case). (a) Count the concordant (case, control) pairs and compute the AUC. (b) Find the threshold that maximizes Youden's index and the corresponding sensitivity and specificity. (c) Verify with `auc()` of the chapter.

Solution.

(a) Cases have scores $0.9, 0.6, 0.5$; controls $0.8, 0.3, 0.2$. Of the 9 pairs, the case exceeds the control in: for case 0.9: 3 of 3; for 0.6: vs 0.8 no, 0.3 yes, 0.2 yes: 2; for 0.5: vs 0.8 no, 0.3 yes, 0.2 yes: 2. Total 7 of 9, AUC $= 7/9 = 0.778$. (b) Thresholds between the scores: $c = 0.55$ gives cases 0.9, 0.6 caught (sens 2/3), controls: 0.8 flagged, 0.3, 0.2 not (spec 2/3): $J = 1/3$. $c = 0.4$: cases 0.9, 0.6, 0.5 (sens 1), controls 0.8 flagged only (spec 2/3): $J = 2/3$. $c = 0.85$: sens 1/3, spec 1, $J = 1/3$. The maximum $J = 2/3$ at $c \in (0.3, 0.5]$ (e.g. 0.4): sens 1, spec 2/3.

```
sc <- c(0.9, 0.8, 0.6, 0.5, 0.3, 0.2); lab <- c(1, 0, 1, 1, 0, 0) == 1
auc(sc, lab)
```

```
#> [1] 0.7778
```

```
sapply(c(0.55, 0.4, 0.85), function(k) sum(rates(sc, lab, k)) - 1)
```

```
#> [1] 0.3333 0.6667 0.3333
```

Exercise 10.7. Generalize Theorem 10.3 to a full cost matrix: costs C_{TP}, C_{FP}, C_{FN}, C_{TN} for the four outcomes (with $C_{FP} > C_{TN}$, $C_{FN} > C_{TP}$). Derive the cost-optimal threshold, and compute it for a screening program in which a true positive costs 20 (treatment), a false positive 30 (needless follow-up), a false negative 200 (late treatment) and a true negative 0.

Solution. Predict 1: expected cost $pC_{TP} + (1-p)C_{FP}$. Predict 0: $pC_{FN} + (1-p)C_{TN}$. Predict 1 iff $pC_{TP} + (1-p)C_{FP} < pC_{FN} + (1-p)C_{TN}$, i.e. $(1-p)(C_{FP} - C_{TN}) < p(C_{FN} - C_{TP})$, i.e. $p > c^* = \dfrac{C_{FP} - C_{TN}}{(C_{FP} - C_{TN}) + (C_{FN} - C_{TP})}$. With the given costs: $c^* = \dfrac{30 - 0}{30 + (200 - 20)} = \dfrac{30}{210} = 0.143$. Treating patients with a probability above 14% is optimal for this program; note that the cost of a true positive enters through the *difference* $C_{FN} - C_{TP}$ (the regret of missing a case).

Exercise 10.8. Two groups of workers are followed: group A has 10 events in 500 person-years, group B has 25 events in 1000 person-years. (a) Write the Poisson model with offset, and find $\hat\beta$ for the rate ratio B vs A in closed form. (b) Give a 95% Wald interval for the rate ratio (the standard error of $\hat\beta$ is $\sqrt{1/10 + 1/25}$). (c) Verify with `glm`.

Solution.

(a) $\log \mu_i = \log t_i + \beta_0 + \beta_1 \mathbf{1}\{B\}$. By Exercise 8.5 (with the offset the likelihood equations

give $\hat\mu_A = y_A$, $\hat\mu_B = y_B$), $e^{\hat\beta_0} = 10/500 = 0.02$ and $e^{\hat\beta_0+\hat\beta_1} = 25/1000 = 0.025$, so $\widehat{\text{RR}} = 0.025/0.02 = 1.25$, $\hat\beta_1 = 0.223$. (b) se $= \sqrt{0.1 + 0.04} = 0.374$; the interval is $\exp(0.223 \pm 1.96 \times 0.374) = (0.60,\ 2.60)$: no evidence that the rates differ. (c)

```
d <- data.frame(events = c(10, 25), py = c(500, 1000), grp = c("A", "B"))
f <- glm(events ~ grp + offset(log(py)), family = poisson, data = d)
c(exp(coef(f)), se = coef(summary(f))[2, 2])
```

```
#> (Intercept)        grpB          se
#>      0.0200      1.2500      0.3742
```

```
exp(confint.default(f)[2, ])
```

```
#>  2.5 % 97.5 %
#> 0.6004 2.6026
```

Exercise 10.9. Let Y have the negative binomial distribution with mean μ and variance $\mu + \alpha\mu^2$ ($\alpha = 1/k$). (a) Show that $\mathrm{E}[(Y-\mu)^2 - Y] = \alpha\mu^2$. (b) A Poisson model gives, for four observations, $y = (0, 3, 1, 8)$ and fitted means $\hat\mu = (1, 2, 2, 5)$. Compute the values $z_i = \{(y_i - \hat\mu_i)^2 - y_i\}/\hat\mu_i$ and the least-squares estimate of α in the regression of z_i on $\hat\mu_i$ without intercept. (c) Compute the Pearson estimate of ϕ (with $p = 1$) for the same data.

Solution.

(a) $\mathrm{E}[(Y-\mu)^2] = \mathrm{Var}(Y) = \mu + \alpha\mu^2$ and $\mathrm{E}[Y] = \mu$, so $\mathrm{E}[(Y-\mu)^2 - Y] = \alpha\mu^2$. (b) $z_1 = \{(0-1)^2 - 0\}/1 = 1$; $z_2 = \{(3-2)^2 - 3\}/2 = -1$; $z_3 = \{(1-2)^2 - 1\}/2 = 0$; $z_4 = \{(8-5)^2 - 8\}/5 = 1/5 = 0.2$. The no-intercept least-squares slope is $\hat\alpha = \sum \hat\mu_i z_i / \sum \hat\mu_i^2 = (1-2+0+1)/(1+4+4+25) = 0/34 = 0$. (c) $X^2 = \sum (y_i - \hat\mu_i)^2/\hat\mu_i = 1+0.5+0.5+1.8 = 3.8$, $\hat\phi = 3.8/(4-1) = 1.27$. With so few observations neither statistic is conclusive.

```
yy <- c(0, 3, 1, 8); m <- c(1, 2, 2, 5); z <- ((yy - m)^2 - yy) / m
c(z, alpha = sum(m * z) / sum(m^2), phi = sum((yy - m)^2 / m) / 3)
```

```
#>                              alpha    phi
#>  1.000 -1.000  0.000  0.200  0.000  1.267
```

Exercise 10.10.

(R) For the `Insurance` model of Example 10.3: (a) test the three factors with likelihood-ratio tests (`drop1`); (b) find the expected number of claims for 1000 holders in district 1, car group `<1l`, age `<25`; (c) estimate the exponent when `log(Holders)` is a free covariate and decide whether the offset is acceptable; (d) the deviance is 51.4 on 54 df and $\hat\phi = 0.90$: what does this say, and which model would you report?

Solution.

```
drop1(fi, test = "Chisq")
```

```
#> Single term deletions
#>
#> Model:
#> Claims ~ District + Group + Age + offset(log(Holders))
#>          Df Deviance AIC  LRT Pr(>Chi)
#> <none>         51.4 389
```

```
#> District  3     65.3 397 13.9   0.0031 **
#> Group     3    140.1 471 88.7   <2e-16 ***
#> Age       3    136.3 468 84.9   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
predict(fi, data.frame(District = "1", Group = "<1l", Age = "<25", Holders =
↪  1000), type = "response")
```

```
#>     1
#> 161.7
```

```
fi_free <- glm(Claims ~ District + Group + Age + log(Holders), family =
↪  poisson, data = ins)   # no offset
b <- coef(summary(fi_free))["log(Holders)", ]
c(exponent = unname(b[1]), se = unname(b[2]), z_vs_1 = unname((b[1] - 1) /
↪  b[2]))
```

```
#> exponent       se   z_vs_1
#>   1.2017   0.1441   1.3993
```

(a) All three factors are significant: district ($\chi^2_3 = 13.9$, $p = 0.003$), car group ($\chi^2_3 = 88.7$) and age ($\chi^2_3 = 84.9$), the last two with $p < 10^{-15}$. (b) About 162 claims (a rate of 0.162 per holder). (c) With `log(Holders)` as a free covariate its coefficient is 1.2 with standard error 0.14; the test of proportionality is $H_0 : \gamma = 1$, $z = (1.2 - 1)/0.14 = 1.4$, so there is no evidence against the offset. (d) $\hat{\phi} = 0.90 < 1$ and a deviance of 51.4 on 54 df: no sign of overdispersion, so we report the Poisson model with offset (quasi-Poisson would shrink the standard errors by $\sqrt{0.9}$, which is not justified: when $\hat{\phi} < 1$ one keeps the Poisson).

Exercise 10.11.

(R) With `dvisits`: (a) compare the expected number of zeros under the Poisson and the negative binomial models of Example 10.5 with the observed 4141; (b) compute the Pearson $\hat{\phi}$ of the Poisson model; (c) fit the negative binomial with only `illness + actdays + hscore` and compare AIC; (d) a colleague says "the ZIP is better than the Poisson, so there is zero inflation". Comment.

Solution.

```
c(zeros_obs = sum(dvisits$doctorco == 0), pois = sum(dpois(0, fitted(m1))), nb
↪  = sum(dnbinom(0, size = m2$theta, mu = fitted(m2))))
```

```
#> zeros_obs      pois        nb
#>      4141      4013      4172
```

```
c(phi_pearson = sum(residuals(m1, "pearson")^2) / df.residual(m1))
```

```
#> phi_pearson
#>       1.325
```

```
m2s <- glm.nb(doctorco ~ illness + actdays + hscore, data = dvisits)
c(AIC_full = AIC(m2), AIC_small = AIC(m2s))
```

```
#>  AIC_full AIC_small
#>      6423      6468
```

(a) The Poisson model expects about 4013 zeros (fewer than the 4141 observed) and the negative binomial 4172 (slightly more). (b) $\hat{\phi} = 1.32$: mild overdispersion. (c) The smaller negative binomial model has AIC 6468 against 6423 for the full one: the dropped covariates (sex, age, income, insurance) are jointly informative, and the larger model is preferred. (d) Both ZIP and NB improve on the Poisson (AIC 6421 and 6423 against 6735), but the ZIP is itself an *overdispersed* model: its structural zeros inflate the variance. An improvement over the Poisson therefore cannot prove that the zeros are structural. The fair comparison is ZIP against NB, which are almost tied, and the choice between them rests on subject-matter reasoning and on the expected-zeros check.

Exercise 10.12. *Critique.* A report on emergency-room visits states: "The data contain 62% zeros, so a standard Poisson model is invalid. We therefore fitted a ZIP model and it has a lower AIC than the Poisson (by 300 units), which proves that the data are zero-inflated. Also, the Vuong test rejected the Poisson in favour of the ZIP ($p < 0.001$)." List the errors and say what you would do.

Solution.

(1) A large share of zeros is not by itself a problem for a Poisson model: with a mean of 0.5 the Poisson itself gives 61% zeros. What matters is the number of zeros *expected under the fitted model* given the covariates. (2) A lower AIC for ZIP shows only that the ZIP describes the data better; overdispersion alone (without a structural-zero subpopulation) produces the same improvement, so it does not prove zero inflation. Compare with the negative binomial and check the expected zeros. (3) The Vuong test is for non-nested models; Poisson is nested in ZIP ($\pi = 0$, on the boundary), so a boundary likelihood-ratio test (or AIC) is the appropriate tool. (4) Practical steps: fit Poisson and NB, check $\hat{\phi}$, compute expected zeros under each, fit ZIP/ZINB or a hurdle model if zeros remain in excess, compare by AIC and by substantive plausibility, and report which mechanism (structural zeros or heterogeneity) is supported by the context.

10.11 Further reading

Hosmer et al. (2013) is the standard text on logistic regression (model building, goodness of fit, case–control studies). Fawcett (2006) explains ROC analysis and the pitfalls of thresholds. For counts, Cameron and Trivedi (1990) develop the tests of overdispersion, Lambert (1992) introduced zero-inflated Poisson regression and Mullahy (1986) the hurdle model; Venables and Ripley (2002) discusses `glm.nb` in R. Wood (2017), Section 3.1, and Faraway (2016) treat binary and count regression with R; Pawitan (2001), Chapter 6, gives the likelihood perspective.

Part IV

Mixed models

Chapter 11

Linear mixed models

The linear model of Chapters 5–7 assumes that the n observations are independent given the covariates. Many data sets violate this assumption by design: a patient is measured on several days, pupils are taught in classes, batches of a product are sampled several times. Observations that share a *group* share unmeasured influences, they are more alike than observations of different groups, and treating them as independent produces standard errors that are wrong, sometimes by a large factor.

The **linear mixed model** (LMM) repairs this by adding *random effects* to the linear predictor: group-specific deviations that are modelled as random draws from a distribution. The model keeps the linear structure of Chapter 5 but its covariance matrix is no longer $\sigma^2 I$; it is a structured matrix V with a handful of parameters. This chapter introduces the model and its implications for the *distribution* of the data; Chapter 12 studies estimation, prediction and inference, and Chapter 13 extends the idea to non-Gaussian responses.

The plan is the following. Section 11.1 presents the running example and shows why grouping matters. Section 11.2 contrasts complete, no and partial pooling. Section 11.3 asks when a group effect should be fixed and when random, and proves a decomposition that makes the assumptions of each choice transparent. Section 11.4 studies the random-intercept model in detail: marginal moments, the intraclass correlation, compound symmetry and the variance of a group mean. Section 11.5 adds random slopes. Section 11.6 writes the general model in matrix form, including crossed and nested effects, and Section 11.7 compares the mixed model with direct models for the covariance matrix. Section 11.8 collects the R code.

Notation follows Appendix A: groups $i = 1, \dots, m$, observations $j = 1, \dots, n_i$, $n = \sum_i n_i$; $y = X\beta + Zb + \varepsilon$, $b \sim N(0, G)$, $\varepsilon \sim N(0, \sigma^2 I)$, $V = ZGZ^\top + \sigma^2 I$. For a random intercept the variance of b_{0i} is σ_b^2 and the intraclass correlation is $\mathrm{ICC} = \sigma_b^2/(\sigma_b^2 + \sigma^2)$. Fixed effects are **estimated**; random effects are **predicted**.

11.1 Grouped data: the sleep-deprivation study

The data set `sleepstudy` in package `lme4` comes from a study of sleep restriction (Belenky et al., 2003). Eighteen long-distance truck drivers were restricted to three hours in bed per night; their mean reaction time (ms) on a vigilance task was recorded on each of ten days (`Days` $= 0, \dots, 9$). So there are $m = 18$ groups (subjects), $n_i = 10$ observations each, and $n = 180$ rows.

```
data(sleepstudy, package = "lme4")
ggplot(sleepstudy, aes(Days, Reaction, group = Subject)) +
```

```
  geom_line(colour = ie_navy, alpha = .45) + geom_point(colour = ie_navy, size
  ↪  = .8, alpha = .6) +
  geom_smooth(aes(group = 1), method = "lm", formula = y ~ x, se = FALSE,
  ↪  colour = ie_coral, linewidth = 1.2) +
  scale_x_continuous(breaks = 0:9) + labs(x = "days of sleep restriction", y =
  ↪  "reaction time (ms)")
```

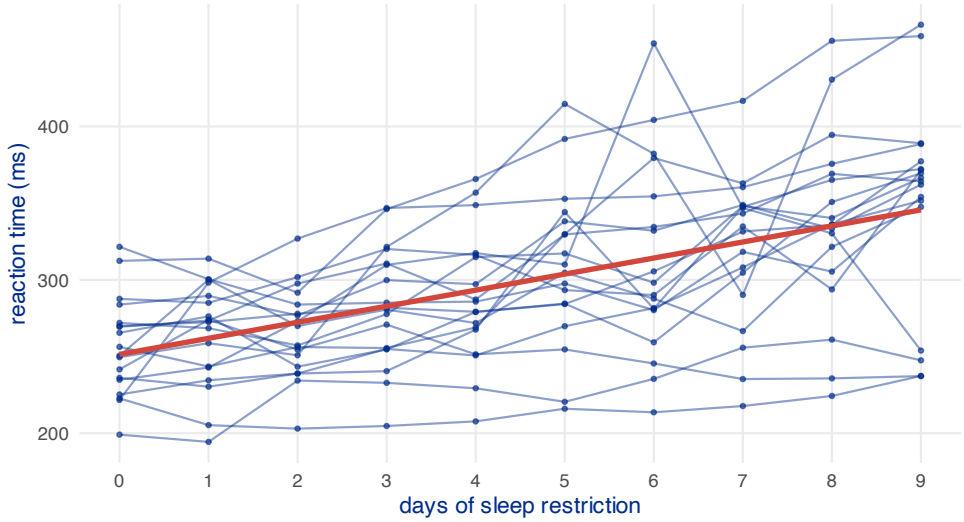


Figure 11.1: Reaction time against days of sleep restriction. Each thin line is one subject; the thick coral line is a single least-squares line fitted to all 180 rows.

Two questions are of interest. (i) By how much does reaction time increase, *on average*, per day of restriction? (ii) How much do subjects *differ*, both in their baseline level and in their rate of deterioration? The second question concerns a **variance**, not a mean, and the ordinary linear model has no way to ask it.

Fit the pooled line and look at its residuals. If the errors of the linear model were independent, residuals of two different days would be uncorrelated *even for the same subject*. Arrange the residuals in a 10×18 matrix (days by subjects) and average the correlations between days:

```
fit_cp <- lm(Reaction ~ Days, data = sleepstudy)
r  <- matrix(resid(fit_cp), nrow = 10)      # column = subject, row = day
cr <- cor(t(r))                             # 10 x 10 day-by-day correlations
↪  over the 18 subjects
mean(cr[upper.tri(cr)])
```

```
#> [1] 0.6135
```

```
summary(fit_cp)$coefficients
```

```
#>             Estimate Std. Error t value  Pr(>|t|)
#> (Intercept)   251.41      6.610  38.033 2.157e-87
#> Days           10.47      1.238   8.454 9.894e-15
```

The average correlation is about 0.61, far from zero: a subject that is slow on Day 3 tends to be slow on Day 7. The 180 rows are *not* 180 independent pieces of information, and the standard error of the pooled slope (1.238) is not to be trusted. We will see in Section 11.8 that the correct value is about 25% larger.

11.2 Complete, no and partial pooling

There are three ways to fit a line to grouped data.

- **Complete pooling.** Ignore the groups and fit one line, $y_{ij} = \beta_0 + \beta_1 x_{ij} + \varepsilon_{ij}$. It uses all data for the two parameters but assumes that every subject has the same intercept and slope.
- **No pooling.** Fit a separate line per group, $y_{ij} = \alpha_{0i} + \alpha_{1i} x_{ij} + \varepsilon_{ij}$: $2m = 36$ parameters. Each subject is learned only from its own ten points; subjects with noisy data get noisy lines, and nothing is learned about the *population* of subjects.
- **Partial pooling.** Treat the subject lines as draws from a population of lines,

$$y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i}) x_{ij} + \varepsilon_{ij}, \qquad \begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N(0, G), \quad \varepsilon_{ij} \sim N(0, \sigma^2). \tag{11.1}$$

  The population line (β_0, β_1) and the spread G are estimated from all subjects; each subject's line is a compromise between its own data and the population line.

```
fit_np <- lme4::lmList(Reaction ~ Days | Subject, data = sleepstudy)       #
↪  no pooling
fit_pp <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)      #
↪  partial pooling
sub <- c("330", "335", "370")
tab <- cbind(coef(fit_np)[sub, ], coef(fit_pp)$Subject[sub, ])
colnames(tab) <- c("int (no pool)", "slope (no pool)", "int (partial)", "slope
↪  (partial)")
round(tab, 2)
```

```
#>     int (no pool) slope (no pool) int (partial) slope (partial)
#> 330         289.7            3.01         275.1            5.65
#> 335         263.0           -2.88         251.1           -0.28
#> 370         210.4           18.06         225.8           15.29
```

```
rbind(complete = coef(fit_cp), population = fixef(fit_pp))
```

```
#>            (Intercept)  Days
#> complete         251.4 10.47
#> population       251.4 10.47
```

Subject 335 has a *negative* slope on its own (−2.88 ms/day), which is implausible for a sleep-deprived driver; the mixed model pulls it to −0.28. Subject 370 has a very steep line (18.1) that is pulled down to 15.3. In the design of `sleepstudy` the estimated population line equals the complete-pooling line because every subject was measured on the same ten days; with unbalanced data they differ.

```
subs <- c("308", "309", "330", "335", "350", "370")
ln <- rbind(
  data.frame(Subject = rownames(coef(fit_np)), a = coef(fit_np)[, 1], b =
  ↪  coef(fit_np)[, 2], fit = "no pooling"),
  data.frame(Subject = rownames(coef(fit_pp)$Subject), a =
  ↪  coef(fit_pp)$Subject[, 1],
             b = coef(fit_pp)$Subject[, 2], fit = "partial pooling"),
  data.frame(Subject = levels(sleepstudy$Subject), a = coef(fit_cp)[[1]], b =
  ↪  coef(fit_cp)[[2]], fit = "complete pooling"))
```

```
ln <- ln[ln$Subject %in% subs, ]
ggplot(sleepstudy[sleepstudy$Subject %in% subs, ], aes(Days, Reaction)) +
  geom_point(colour = "grey30", size = 1) +
  geom_abline(data = ln, aes(intercept = a, slope = b, colour = fit, linetype
  ↪  = fit), linewidth = .8) +
  scale_colour_manual(values = c("complete pooling" = "grey45", "no pooling" =
  ↪  ie_coral, "partial pooling" = ie_navy), name = NULL) +
  scale_linetype_manual(values = c("complete pooling" = "dotted", "no pooling"
  ↪  = "dashed", "partial pooling" = "solid"), name = NULL) +
  facet_wrap(~ Subject, nrow = 2) + labs(x = "days", y = "reaction time (ms)")
```

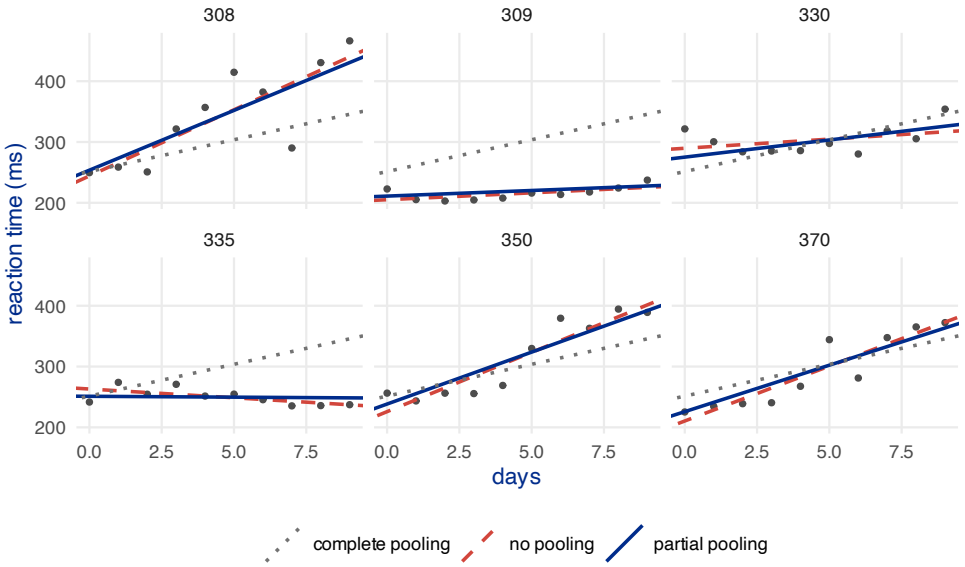


Figure 11.2: Six subjects: complete pooling (dotted grey), no pooling (dashed coral) and partial pooling (solid navy).

How strongly a subject is pulled towards the population line, and why, is the subject of Chapter 12 (shrinkage, Section 12.4). Here we only note the principle: **partial pooling borrows strength across groups** and the amount borrowed depends on how much information the group has and how different groups really are.

11.3 Fixed or random group effects?

Consider the simplest grouped model, with a single covariate. If the group effects are treated as **fixed**,

$$y_{ij} = \alpha_i + \beta x_{ij} + \varepsilon_{ij}, \qquad \alpha_1, \ldots, \alpha_m \text{ unknown constants}, \tag{11.2}$$

we estimate m separate levels α_i and β. If they are treated as **random**,

$$y_{ij} = \beta_0 + \beta x_{ij} + b_i + \varepsilon_{ij}, \qquad b_i \sim N(0, \sigma_b^2), \quad b_i \perp \varepsilon_{ij}, \tag{11.3}$$

the m levels cost a single parameter σ_b^2.

Definition 11.1 (Fixed and random effects)**.** An effect is **fixed** if its levels are unknown constants, estimated separately; inference is about *these* levels. An effect is **random** if its levels are regarded as a sample from a distribution (here $N(0, \sigma_b^2)$); inference is about the *distribution* of effects (the variance) and about predictions of individual levels.

Table 11.1: Guide to fixed versus random group effects.

Random	Fixed
levels are a sample from a larger population (subjects, herds, schools)	levels are the whole population of interest (the 3 fertilisers tested)
we want conclusions beyond the observed levels	we care about these specific levels
many levels (say 5 or more)	very few levels
group effects plausibly unrelated to the covariates	group effects may be related to the covariates (confounding)

The practical guidance is summarised in Table 11.1. It is a guide, not a rule: the decisive criterion is the **assumption** behind the choice, which the next result makes precise.

11.3.1 The within and between decomposition

Suppose all groups have the same size $n_i = n$. Write $\bar{x}_i, \bar{y}_i$ for group means and $\bar{x}, \bar{y}$ for the grand means, and define the within-group and between-group sums of squares and products

$$S_W^{xx} = \sum_{i,j} (x_{ij} - \bar{x}_i)^2, \qquad S_W^{xy} = \sum_{i,j} (x_{ij} - \bar{x}_i)(y_{ij} - \bar{y}_i),$$
$$S_B^{xx} = n \sum_{i} (\bar{x}_i - \bar{x})^2, \qquad S_B^{xy} = n \sum_{i} (\bar{x}_i - \bar{x})(\bar{y}_i - \bar{y}).$$

The **within estimator** $\hat{\beta}_W = S_W^{xy}/S_W^{xx}$ is the least-squares estimate of β in the fixed-effects model (11.2) (it is obtained by subtracting group means, which removes the α_i). The **between estimator** $\hat{\beta}_B = S_B^{xy}/S_B^{xx}$ is the slope of the regression of $\bar{y}_i$ on $\bar{x}_i$.

Theorem 11.1 (Random-effects estimator as a weighted average)**.** *In the balanced random-intercept model (11.3), let $\theta = \sigma^2/(\sigma^2 + n\sigma_b^2) \in (0, 1]$. The generalised least-squares estimator of β (Chapter 12) is*

$$\hat{\beta}_{GLS} = \frac{S_W^{xy} + \theta\, S_B^{xy}}{S_W^{xx} + \theta\, S_B^{xx}} = \frac{S_W^{xx}\,\hat{\beta}_W + \theta\, S_B^{xx}\,\hat{\beta}_B}{S_W^{xx} + \theta\, S_B^{xx}}. \tag{11.4}$$

Proof. Let $P = \mathrm{blockdiag}(P_1, \dots, P_m)$ with $P_i = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, a projection that replaces each observation by its group mean. The covariance matrix of group i is $V_i = \sigma_b^2 \mathbf{1}\mathbf{1}^\top + \sigma^2 I = \sigma^2(I - P_i) + (\sigma^2 + n\sigma_b^2)P_i$, because $\mathbf{1}\mathbf{1}^\top = nP_i$. Since $I - P_i$ and P_i are orthogonal projections with $(I - P_i)P_i = 0$,

$$V^{-1} = \sigma^{-2}\big[(I - P) + \theta P\big].$$

Because $P\mathbf{1} = \mathbf{1}$, $V^{-1}\mathbf{1} = (\sigma^2 + n\sigma_b^2)^{-1}\mathbf{1}$. Replace the covariate by the grand-mean-centred $\tilde{x} = x - \bar{x}\mathbf{1}$; the columns $(\mathbf{1}, \tilde{x})$ span the same space as $(\mathbf{1}, x)$ and $\mathbf{1}^\top V^{-1}\tilde{x} \propto \mathbf{1}^\top \tilde{x} = 0$. So the intercept and the slope decouple in the GLS normal equations, and the GLS slope is $\tilde{x}^\top V^{-1} y / \tilde{x}^\top V^{-1}\tilde{x}$. Now

$$\tilde{x}^\top V^{-1} y = \sigma^{-2}\big[\tilde{x}^\top (I - P) y + \theta\, \tilde{x}^\top P y\big] = \sigma^{-2}\big[S_W^{xy} + \theta S_B^{xy}\big],$$

since $\tilde{x}^\top (I - P) y = \sum_{i,j}(x_{ij} - \bar{x}_i)(y_{ij} - \bar{y}_i)$ and $\tilde{x}^\top P y = n \sum_i (\bar{x}_i - \bar{x})\bar{y}_i = S_B^{xy}$. The same computation with y replaced by x gives the denominator, and the factors σ^{-2} cancel. □

Three consequences follow at once.

1. If $\sigma_b^2 \to \infty$ then $\theta \to 0$ and $\hat{\beta}_{GLS} \to \hat{\beta}_W$: with an unrestricted group effect only within-group variation identifies β (the **fixed-effects** estimator).
2. If $\sigma_b^2 = 0$ then $\theta = 1$ and $\hat{\beta}_{GLS}$ is ordinary least squares on all data (**complete pooling**).
3. In between, the random-effects estimator uses *both* kinds of variation, so it is more efficient than the within estimator. But the between estimator is only valid if b_i is independent of $\bar{x}_i$: if the group effect is correlated with the group-mean covariate, $\hat{\beta}_B$ estimates $\beta + \mathrm{Cov}(b_i, \bar{x}_i)/\mathrm{Var}(\bar{x}_i)$ and contaminates $\hat{\beta}_{GLS}$.

This is the exact meaning of the slogan **fixed effects *allow* the group effect to be correlated with the covariates; random effects *assume* that it is not**. Fixed effects make *no* assumption about that correlation. A simulation makes the point. We generate 60 groups of 5 observations with true $\beta = 1$; the group effect b_i contains the unobserved component u_i that also drives x_{ij}, so b_i and $\bar{x}_i$ are correlated.

```
set.seed(2027)
m <- 60; n <- 5; id <- rep(1:m, each = n)
one_sim <- function() {
  u <- rnorm(m); x <- u[id] + rnorm(m * n)       # covariate partly driven by
↪  u
  b <- u + rnorm(m, 0, .5)                       # group effect correlated
↪  with the covariate
  y <- 1 * x + b[id] + rnorm(m * n)              # true slope beta = 1
  d <- data.frame(y, x, id = factor(id)); d$xbar <- ave(d$x, d$id)
  c(pooled = coef(lm(y ~ x, d))[[2]],
    random = fixef(lmer(y ~ x + (1 | id), d))[[2]],
    fixed  = coef(lm(y ~ x + id, d))[[2]],
    random_plus_group_mean = fixef(lmer(y ~ x + xbar + (1 | id), d))[[2]])
}
sims <- replicate(200, one_sim())
round(rbind(mean = rowMeans(sims), sd = apply(sims, 1, sd)), 3)
```

```
#>      pooled random fixed random_plus_group_mean
#> mean  1.484  1.179 1.003                 1.003
#> sd    0.071  0.065 0.059                 0.059
```

The fixed-effects estimator is unbiased (mean 1.003); the random-effects estimator is biased upwards (1.179) and so is the pooled one (1.484). A common remedy, due to Mundlak, keeps the random effect but adds the group mean $\bar{x}_i$ as an extra *fixed* covariate: the coefficient of x_{ij} then reproduces the within estimator (last row; identical to the fixed-effects row in this balanced design) while still allowing group-level covariates and a variance component.

Warning. The number of levels is not the only criterion. "200 shops, so random" is wrong if shops in rich areas were more likely to receive the campaign whose effect we study: the shop effect is correlated with the covariate and a random shop effect biases the campaign effect. Conversely, a covariate that is constant within groups (sex of a patient, size of a class) cannot be estimated if the groups carry fixed effects, because it is a linear combination of the group dummies.

11.4 The random-intercept model

The simplest mixed model has one random effect per group, a shift of the whole regression line.

Definition 11.2 (Random-intercept model)**.** For groups $i = 1, \dots, m$ and observations $j = 1, \dots, n_i$,

$$y_{ij} = x_{ij}^\top \beta + b_i + \varepsilon_{ij}, \qquad b_i \overset{iid}{\sim} N(0, \sigma_b^2), \quad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2), \quad b_i \perp \varepsilon_{ij}. \tag{11.5}$$

The parameters are the fixed effects β and the **variance components** σ_b^2 (between groups) and σ^2 (within groups). The number of parameters does not depend on m.

Theorem 11.2 (Moments of the random-intercept model)**.** *Under* (11.5),

$$\mathrm{E}[y_{ij}] = x_{ij}^\top \beta, \qquad \mathrm{Var}(y_{ij}) = \sigma_b^2 + \sigma^2, \tag{11.6}$$

$$\mathrm{Cov}(y_{ij}, y_{ik}) = \sigma_b^2 \ (j \neq k), \qquad \mathrm{Cov}(y_{ij}, y_{i'k}) = 0 \ (i \neq i'). \tag{11.7}$$

Hence two observations of the same group have the ***intraclass correlation***

$$\mathrm{ICC} = \mathrm{Corr}(y_{ij}, y_{ik}) = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2} \in [0, 1), \tag{11.8}$$

the share of the total variance that lies between groups, and the covariance matrix of group i is the ***compound-symmetry*** *matrix $V_i = \sigma_b^2 \mathbf{1}\mathbf{1}^\top + \sigma^2 I_{n_i}$.*

Proof. $y_{ij} - x_{ij}^\top \beta = b_i + \varepsilon_{ij}$ has mean zero because both terms do. By independence of b_i and ε_{ij}, $\mathrm{Var}(y_{ij}) = \mathrm{Var}(b_i) + \mathrm{Var}(\varepsilon_{ij})$. For $j \neq k$, bilinearity of the covariance gives $\mathrm{Cov}(b_i + \varepsilon_{ij}, b_i + \varepsilon_{ik}) = \mathrm{Var}(b_i) + \mathrm{Cov}(b_i, \varepsilon_{ik}) + \mathrm{Cov}(\varepsilon_{ij}, b_i) + \mathrm{Cov}(\varepsilon_{ij}, \varepsilon_{ik}) = \sigma_b^2$. Observations of different groups involve independent $b_i, b_{i'}$ and independent errors, so their covariance is zero. The correlation is the ratio of σ_b^2 to $\sigma_b^2 + \sigma^2$. In matrix form V_i has $\sigma_b^2 + \sigma^2$ on the diagonal and σ_b^2 elsewhere, i.e. $\sigma_b^2 \mathbf{1}\mathbf{1}^\top + \sigma^2 I$, and $\mathrm{Var}(y) = \mathrm{blockdiag}(V_1, \dots, V_m)$. □

Compound symmetry says that all pairs of observations within a group are equally correlated, however far apart in time or space; it also says that the correlation is **non-negative**.

Warning. The random-intercept model cannot produce a negative within-group correlation. A compound-symmetric matrix $\sigma_t^2\{(1-\rho)I + \rho \mathbf{1}\mathbf{1}^\top\}$ is positive definite for $-1/(n_i - 1) < \rho < 1$, but the mixed model restricts $\rho = \sigma_b^2/(\sigma_b^2 + \sigma^2) \geq 0$ (Exercise 11.6).

11.4.1 Averages and the design effect

Averaging removes the within-group noise but not the group effect. For a group with n_i observations and a common mean μ (no covariates),

$$\mathrm{Var}(\bar{y}_{i\cdot}) = \sigma_b^2 + \frac{\sigma^2}{n_i}. \tag{11.9}$$

For a balanced design with m groups of size n the grand mean has $\mathrm{Var}(\bar{y}) = (\sigma_b^2 + \sigma^2/n)/m$, whereas the naive formula for mn independent observations gives $(\sigma_b^2 + \sigma^2)/(mn)$. Their ratio is the **design effect**

$$\mathrm{deff} = \frac{\sigma_b^2 + \sigma^2/n}{(\sigma_b^2 + \sigma^2)/n} = 1 + (n-1)\,\mathrm{ICC}, \tag{11.10}$$

and mn/deff is the *effective sample size*. Large groups with a strong ICC carry much less information about group-level quantities than their size suggests: more observations per group cannot remove the between-group variation; only more groups can.

11.4.2 Example: variance components of a one-way layout

Example 11.1 (Yield of a dyestuff intermediate)**.** The data set `Dyestuff` in `lme4` records the yield (g) of a dyestuff from $m = 6$ batches of raw material, with $n = 5$ samples per batch. This is the classical balanced one-way random-effects layout. Let MSB and MSW be the between- and within-batch mean squares of the analysis of variance. Then $\mathrm{E}[\mathrm{MSW}] = \sigma^2$ and $\mathrm{E}[\mathrm{MSB}] = \sigma^2 + n\sigma_b^2$, which suggests the **ANOVA estimators** $\hat\sigma^2 = \mathrm{MSW}$ and $\hat\sigma_b^2 = (\mathrm{MSB} - \mathrm{MSW})/n$. (Proof of the second expectation: Exercise 11.2.)

```
data(Dyestuff, package = "lme4")
tab_aov <- summary(aov(Yield ~ Batch, Dyestuff))[[1]]
tab_aov
```

```
#>             Df Sum Sq Mean Sq F value Pr(>F)
#> Batch        5  56357   11271     4.6 0.0044 **
#> Residuals   24  58830    2451
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
MSB <- tab_aov[1, "Mean Sq"]; MSW <- tab_aov[2, "Mean Sq"]; n_s <- 5
c(sigma2 = MSW, sigma_b2 = (MSB - MSW) / n_s)
```

```
#>   sigma2 sigma_b2
#>     2451     1764
```

```
fit_dy <- lmer(Yield ~ 1 + (1 | Batch), data = Dyestuff)
VarCorr(fit_dy)
```

```
#>  Groups   Name        Std.Dev.
#>  Batch    (Intercept) 42.0
#>  Residual             49.5
```

```
vc_dy <- as.data.frame(VarCorr(fit_dy))$vcov
c(ICC = vc_dy[1] / sum(vc_dy), deff = 1 + (n_s - 1) * vc_dy[1] / sum(vc_dy))
```

```
#>    ICC   deff
#> 0.4185 2.6739
```

```
ggplot(Dyestuff, aes(Batch, Yield)) + geom_point(colour = ie_navy, size = 2,
↪  alpha = .8) +
  stat_summary(fun = mean, geom = "point", colour = ie_coral, shape = 18, size
  ↪  = 4) +
  geom_hline(yintercept = mean(Dyestuff$Yield), linetype = 2, colour =
  ↪  "grey45")
```

The mixed-model estimates (REML, the default) coincide with the ANOVA estimators: $\hat\sigma_b^2 = 1764.1$, $\hat\sigma^2 = 2451.2$. The intraclass correlation is 0.418: 42% of the variation of the yield lies *between* batches, and two samples from the same batch have correlation 0.42. The design effect is 2.67, so the 30 samples are worth about 11 independent ones for estimating the mean yield. The standard error of the mean, 19.38 g, is exactly $\sqrt{(\hat\sigma_b^2 + \hat\sigma^2/5)/6}$ (Section 12.2).

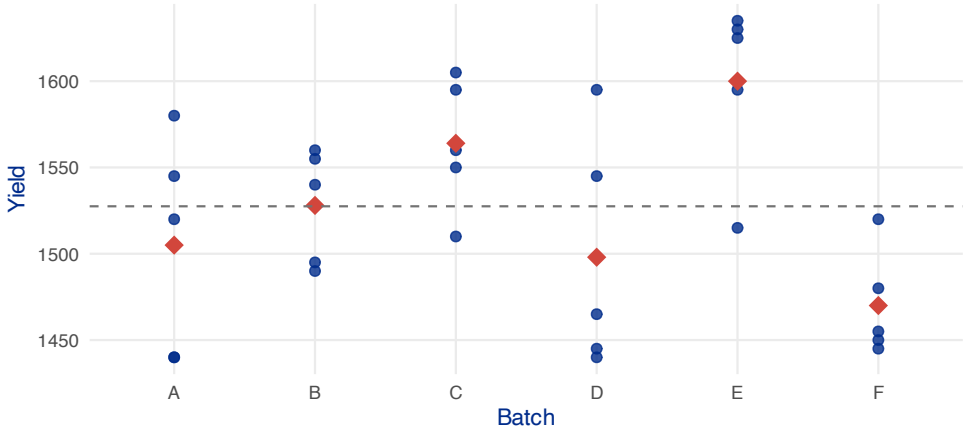


Figure 11.3: Dyestuff: yield of 6 batches, 5 samples each; coral diamonds are batch means.

11.4.3 Example: sleepstudy with a random intercept

```
m_ri <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)
summary(m_ri)$coefficients
```

```
#>             Estimate Std. Error t value
#> (Intercept)   251.41     9.7467   25.79
#> Days           10.47     0.8042   13.02
```

```
VarCorr(m_ri)
```

```
#>  Groups   Name        Std.Dev.
#>  Subject  (Intercept) 37.1
#>  Residual             31.0
```

```
vc_ri <- as.data.frame(VarCorr(m_ri))$vcov
c(ICC = vc_ri[1] / sum(vc_ri), deff = 1 + 9 * vc_ri[1] / sum(vc_ri))
```

```
#>    ICC   deff
#> 0.5893 6.3038
```

Subjects' baseline levels vary with standard deviation $\hat{\sigma}_b = 37.1$ ms around the population line, and the day-to-day scatter around a subject's own line has standard deviation $\hat{\sigma} = 31$ ms. The ICC is 0.589: 59% of the variance (given `Days`) is between subjects. The design effect for an overall mean is $1 + 9(0.589) = 6.3$, so the 180 rows are worth only about 29 independent observations for estimating the average level.

11.5 Random slopes

If subjects differ in the *rate* of change as well as in the level, we let the slope vary too.

Definition 11.3 (Random intercept and random slope)**.**

$$y_{ij} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})x_{ij} + \varepsilon_{ij}, \qquad \begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \overset{iid}{\sim} N(\mathbf{0}, G), \quad G = \begin{pmatrix} \sigma_{b0}^2 & \sigma_{b01} \\ \sigma_{b01} & \sigma_{b1}^2 \end{pmatrix}, \tag{11.11}$$

with $\sigma_{b01} = \rho\,\sigma_{b0}\sigma_{b1}$, independent of $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$. There are four variance parameters: σ^2_{b0}, σ^2_{b1}, ρ and σ^2.

Theorem 11.3 (Moments of the random-slope model)**.** *Under* (11.11),

$$\mathrm{Var}(y_{ij}) = \sigma^2_{b0} + 2x_{ij}\sigma_{b01} + x^2_{ij}\sigma^2_{b1} + \sigma^2, \tag{11.12}$$

$$\mathrm{Cov}(y_{ij}, y_{ik}) = \sigma^2_{b0} + (x_{ij} + x_{ik})\sigma_{b01} + x_{ij}x_{ik}\sigma^2_{b1} \qquad (j \neq k). \tag{11.13}$$

Proof. $y_{ij} - \mathrm{E}[y_{ij}] = b_{0i} + x_{ij}b_{1i} + \varepsilon_{ij}$. Expand $\mathrm{Var}(b_{0i} + x_{ij}b_{1i} + \varepsilon_{ij}) = \mathrm{Var}(b_{0i}) + x^2_{ij}\mathrm{Var}(b_{1i}) + 2x_{ij}\mathrm{Cov}(b_{0i}, b_{1i}) + \sigma^2$ and $\mathrm{Cov}(b_{0i} + x_{ij}b_{1i} + \varepsilon_{ij},\ b_{0i} + x_{ik}b_{1i} + \varepsilon_{ik})$ term by term; the errors contribute nothing for $j \neq k$. □

The variance is a **quadratic function of** x: with random slopes the group lines fan out (or in), the variance is not constant, and the correlation of two observations of the same group depends on where they are. In matrix form, $V_i = Z_i G Z_i^\top + \sigma^2 I$ with $Z_i = [\mathbf{1},\ x_i]$, which is not compound symmetric.

```
m_ris <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)
print(VarCorr(m_ris), comp = c("Variance", "Std.Dev."))
```

```
#>  Groups   Name        Variance Std.Dev. Corr
#>  Subject  (Intercept) 612.1    24.74
#>           Days         35.1     5.92    0.07
#>  Residual             654.9    25.59
```

```
fixef(m_ris)
```

```
#> (Intercept)        Days
#>      251.41       10.47
```

```
G <- unclass(VarCorr(m_ris)$Subject)[1:2, 1:2]
s2 <- sigma(m_ris)^2
Zi <- cbind(1, 0:9)
Vi <- Zi %*% G %*% t(Zi) + s2 * diag(10)          # V_i = Z_i G Z_i' + sigma^2
↪  I
round(diag(Vi))                                     # the variance grows with
↪  Days
```

```
#>  [1] 1267 1321 1446 1640 1905 2240 2645 3120 3665 4281
```

```
round(cov2cor(Vi)[c(1, 5, 10), c(1, 5, 10)], 2)     # correlation of Days 0, 4,
↪  9
```

```
#>      [,1] [,2] [,3]
#> [1,] 1.00 0.42  0.3
#> [2,] 0.42 1.00  0.7
#> [3,] 0.30 0.70  1.0
```

```
emp <- aggregate(Reaction ~ Days, sleepstudy, sd)
imp <- data.frame(Days = 0:9, ri = sqrt(sum(vc_ri)), rs = sqrt(diag(Vi)))
ggplot(imp, aes(Days)) +
```

```
  geom_line(aes(y = ri, colour = "random intercept"), linewidth = 1) +
  geom_line(aes(y = rs, colour = "intercept + slope"), linewidth = 1) +
  geom_point(data = emp, aes(y = Reaction)) +
  scale_colour_manual(values = c("random intercept" = ie_coral, "intercept +
  ↪  slope" = ie_navy), name = NULL) +
  scale_x_continuous(breaks = 0:9) + labs(y = "SD of reaction time (ms)")
```

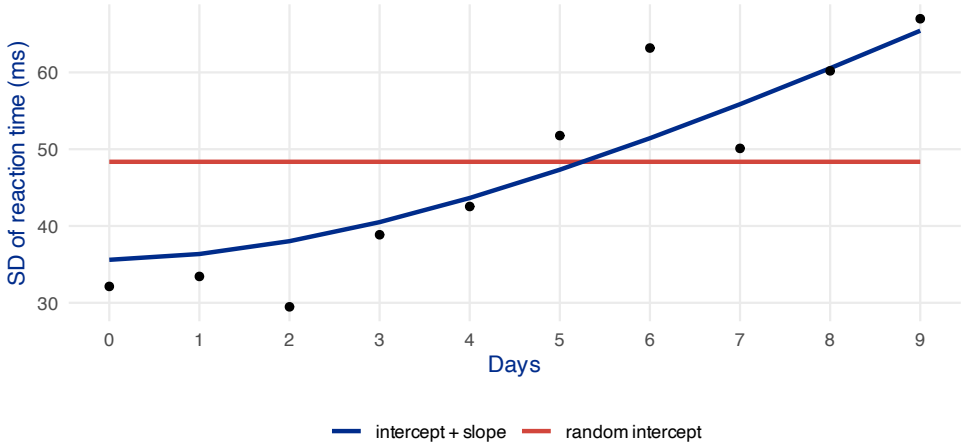


Figure 11.4: Observed standard deviation of reaction time by day (points) against the standard deviation implied by the random-intercept model (coral, constant) and by the random-slope model (navy).

The numbers are read on the data scale. The average subject has a reaction time of 251.4 ms at Day 0 and deteriorates by 10.47 ms per day. The standard deviation of the subject-specific slopes is $\hat{\sigma}_{b1} = 5.92$ ms/day, so about 95% of subjects have a slope in $10.47 \pm 1.96 \times 5.92 = [-1.1,\ 22.1]$ ms/day: some subjects hardly deteriorate, others deteriorate twice as fast as the average. The Day-0 levels vary with $\hat{\sigma}_{b0} = 24.7$ ms, and the correlation $\hat{\rho} = 0.07$ says that baseline and rate are almost unrelated. The residual standard deviation falls from 31 to 25.6 ms because the random slopes explain systematic subject-specific trends that the random-intercept model leaves in the error.

11.5.1 The origin of x matters for ρ and σ^2_{b0}

Proposition 11.1 (Change of origin). *Replace x by $x^* = x - c$. The model (11.11) is unchanged in distribution, with random effects $b^*_i = Ab_i$, $A = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$, so that $G^* = AGA^\top$:*

$$\sigma^{2*}_{b0} = \sigma^2_{b0} + 2c\,\sigma_{b01} + c^2\sigma^2_{b1}, \qquad \sigma^*_{b01} = \sigma_{b01} + c\,\sigma^2_{b1}, \qquad \sigma^{2*}_{b1} = \sigma^2_{b1}.$$

The fixed intercept becomes $\beta_0 + c\beta_1$. In particular $\sigma^*_{b01} = 0$ for $c = -\sigma_{b01}/\sigma^2_{b1}$, the value of x at which $\mathrm{Var}(y \mid x)$ is smallest.

Proof. $\beta_0 + b_{0i} + (\beta_1 + b_{1i})x = \beta_0 + c\beta_1 + (b_{0i} + c\,b_{1i}) + (\beta_1 + b_{1i})(x - c)$, so $b^*_{0i} = b_{0i} + cb_{1i}$ and $b^*_{1i} = b_{1i}$, i.e. $b^* = Ab$. Then $\mathrm{Var}(b^*) = AGA^\top$, which gives the displayed formulas. The variance σ^{2*}_{b0} is the quadratic of Theorem 11.3 evaluated at $x = c$; its derivative in c is $2(\sigma_{b01} + c\sigma^2_{b1}) = 2\sigma^*_{b01}$, which vanishes at the minimum. □

```
m_c <- lmer(Reaction ~ I(Days - 4.5) + (I(Days - 4.5) | Subject), data =
↪  sleepstudy)
print(VarCorr(m_c), comp = c("Variance", "Std.Dev."))
```

```
#>  Groups   Name          Variance Std.Dev. Corr
#>  Subject  (Intercept)   1408.7   37.53
#>           I(Days - 4.5)   35.1    5.92    0.75
#>  Residual                654.9   25.59
```

```
c(logLik_original = as.numeric(logLik(m_ris)), logLik_centred =
↪  as.numeric(logLik(m_c)))
```

```
#> logLik_original  logLik_centred
#>          -871.8          -871.8
```

```
-G[1, 2] / G[2, 2]    # centre that makes intercept and slope uncorrelated
```

```
#> [1] -0.2739
```

After centring at 4.5 days the correlation jumps from 0.07 to 0.75 and the intercept standard deviation from 24.7 to 37.5, yet the likelihood is identical. A reported correlation of random intercept and slope is therefore meaningful only together with the location of the origin of x; centring at a sensible value (here the middle of the study) makes σ_{b0} interpretable as the between-subject SD at that value of x.

11.6 The general linear mixed model

Definition 11.4 (Linear mixed model)**.** Let $y \in \mathbb{R}^n$ be the response vector. The linear mixed model is

$$y = X\beta + Zb + \varepsilon, \qquad b \sim N(0, G), \quad \varepsilon \sim N(0, \sigma^2 I_n), \quad b \perp \varepsilon, \tag{11.14}$$

where X is the $n \times p$ fixed-effects design matrix (including the intercept column), $\beta \in \mathbb{R}^p$, Z is the $n \times q$ random-effects design matrix, $b \in \mathbb{R}^q$ stacks the random effects of all groups, and $G = G(\theta)$ is a $q \times q$ covariance matrix that depends on a few parameters θ.

When groups are independent, $Z = \text{blockdiag}(Z_1, \dots, Z_m)$ and $G = \text{blockdiag}(G_0, \dots, G_0)$ with the same G_0 (the 2×2 matrix of the random-slope model) in each block. For `sleepstudy` with `(Days | Subject)`: $n = 180$, $p = 2$, $q = 2m = 36$ and $Z_i = [\mathbf{1}_{10},\, (0, \dots, 9)^\top]$.

```
X <- getME(m_ris, "X"); Z <- getME(m_ris, "Z")
c(n = nrow(X), p = ncol(X), q = ncol(Z))
```

```
#>   n   p   q
#> 180   2  36
```

```
as.matrix(Z[1:3, 1:4])          # first three rows: columns = (int_1,
↪  slope_1, int_2, slope_2)
```

```
#>   308 308 309 309
#> 1   1   0   0   0
#> 2   1   1   0   0
```

```
#> 3    1    2    0    0
```

Theorem 11.4 (Conditional, marginal and joint distributions)**.** *Under* (11.14),

$$y \mid b \sim N(X\beta + Zb,\ \sigma^2 I), \qquad y \sim N(X\beta,\ V), \quad V = ZGZ^\top + \sigma^2 I, \tag{11.15}$$

and $\mathrm{Cov}(b, y) = GZ^\top$*, so that* (b, y) *is jointly Gaussian with mean* $(0, X\beta)$ *and covariance* $\begin{pmatrix} G & GZ^\top \\ ZG & V \end{pmatrix}$.

Proof. Given b, the term $X\beta + Zb$ is fixed and y differs from it by $\varepsilon \sim N(0, \sigma^2 I)$, which gives the conditional distribution. The pair (b, ε) is jointly Gaussian (independent Gaussian vectors) and $y = X\beta + (Zb + \varepsilon)$ is an affine function of it, hence Gaussian. Then $\mathrm{E}[y] = X\beta + Z\mathrm{E}[b] + \mathrm{E}[\varepsilon] = X\beta$, and since $\mathrm{Cov}(b, \varepsilon) = 0$, $\mathrm{Var}(y) = Z\mathrm{Var}(b)Z^\top + \mathrm{Var}(\varepsilon) = ZGZ^\top + \sigma^2 I$, $\mathrm{Cov}(b, y) = \mathrm{Cov}(b, Zb + \varepsilon) = GZ^\top$. □

The random effects enter the **marginal** model only through V. An LMM is therefore a linear model $y \sim N(X\beta, V)$ whose covariance matrix is not $\sigma^2 I$ but a structured matrix with few parameters: this is the viewpoint of Chapter 12, where β is estimated by generalised least squares. We can verify Theorem 11.4 numerically: build V from the fitted G, σ^2 and Z and compare the Gaussian log-likelihood of the data with `logLik()` of an ML fit.

```
m_ml <- lmer(Reaction ~ Days + (Days | Subject), sleepstudy, REML = FALSE)
Zm <- as.matrix(getME(m_ml, "Z")); Xm <- getME(m_ml, "X"); ym <- getME(m_ml,
↪  "y")
Gm <- kronecker(diag(18), unclass(VarCorr(m_ml)$Subject)[1:2, 1:2])   # b is
↪  ordered (int_1, slope_1, int_2, ...)
Vm <- Zm %*% Gm %*% t(Zm) + sigma(m_ml)^2 * diag(180)
res <- ym - Xm %*% fixef(m_ml)
ll_by_hand <- -0.5 * (180 * log(2 * pi) + determinant(Vm)$modulus + t(res) %*%
↪  solve(Vm, res))
print(c(by_hand = ll_by_hand[1], lme4 = as.numeric(logLik(m_ml))), digits =
↪  10)
```

```
#>      by_hand         lme4
#> -875.9696722 -875.9696722
```

11.6.1 Crossed and nested random effects

A model may have several random effects, each with its own grouping factor and variance. The matrix $Z = [Z_1, Z_2, \dots]$ has one block of columns per factor, and $G = \mathrm{blockdiag}(\sigma_1^2 I, \sigma_2^2 I, \dots)$. Two grouping factors are **crossed** if every level of one can occur with every level of the other (each sample of penicillin is tested on each plate), and **nested** if each level of one occurs in a single level of the other (casks are sampled within batches).

```
data(Penicillin, package = "lme4")      # diameter of inhibition zone; plate x
↪  sample (crossed)
VarCorr(lmer(diameter ~ 1 + (1 | plate) + (1 | sample), Penicillin))
```

```
#>  Groups   Name        Std.Dev.
#>  plate    (Intercept) 0.847
#>  sample   (Intercept) 1.932
```

```
#>  Residual             0.550
```

```
data(Pastes, package = "lme4")          # strength of paste; cask nested in
↪  batch
VarCorr(lmer(strength ~ 1 + (1 | batch/cask), Pastes))
```

```
#>  Groups     Name        Std.Dev.
#>  cask:batch (Intercept) 2.904
#>  batch      (Intercept) 1.287
#>  Residual               0.823
```

In `Penicillin` the variance between samples (SD 1.93) dominates the variance between plates and the residual: the plate and sample effects add independent components of variance to every observation. In `Pastes`, `(1 | batch/cask)` is shorthand for `(1 | batch) + (1 | batch:cask)`: the response has three variance components, batch, cask within batch, and residual, and the ICC of two measurements of the same cask is $(\sigma^2_{\text{batch}} + \sigma^2_{\text{cask}})/(\sigma^2_{\text{batch}} + \sigma^2_{\text{cask}} + \sigma^2)$, while two casks of the same batch correlate only through σ^2_{batch}.

11.6.2 Number of parameters and identifiability

The matrix G_0 of a model with r random effects per group has $r(r+1)/2$ free parameters, so the variance part has $r(r+1)/2+1$ parameters in all. These parameters must be *identifiable*: a random intercept with one observation per group has $y_i = \mu + b_i + \varepsilon_i$ and only the sum $\sigma_b^2 + \sigma^2$ can be estimated. Likewise a random slope needs at least two (and preferably many) distinct values of x within groups. When the data are too sparse for the model, the optimiser returns a boundary solution (a variance of zero or a correlation of ± 1) and `lme4` prints the message *boundary (singular) fit*: simplify the random part.

11.7 Mixed models and direct covariance models

Because the random effects enter the marginal model only through V, one may skip them and model the covariance matrix of each group directly. The package `nlme` does this with `gls()`: correlation structures such as `corCompSymm` (equal correlation), `corAR1` (autoregressive) and variance functions such as `varIdent`. For the random-intercept model the two descriptions are the same:

```
g_cs  <- gls(Reaction ~ Days, sleepstudy, correlation = corCompSymm(form = ~ 1
↪  | Subject))
g_ar1 <- gls(Reaction ~ Days, sleepstudy, correlation = corAR1(form = ~ Days |
↪  Subject))
rbind(gls_compound_symmetry = c(coef(g_cs), logLik =
↪  as.numeric(logLik(g_cs))),
      lmer_random_intercept = c(fixef(m_ri), logLik =
      ↪  as.numeric(logLik(m_ri))))
```

```
#>                       (Intercept)  Days logLik
#> gls_compound_symmetry       251.4 10.47 -893.2
#> lmer_random_intercept       251.4 10.47 -893.2
```

```
round(c(rho_cs = coef(g_cs$modelStruct$corStruct, unconstrained = FALSE),
        phi_ar1 = coef(g_ar1$modelStruct$corStruct, unconstrained = FALSE)),
        ↪  3)
```

Table 11.2: Random-effects formula syntax in lme4.

lmer formula	Meaning
`y ~ x + (1 | g)`	random intercept
`y ~ x + (x | g)`	correlated random intercept and slope
`y ~ x + (x || g)`	uncorrelated intercept and slope ($\rho = 0$)
`y ~ x + (1 | g1) + (1 | g2)`	crossed random effects
`y ~ x + (1 | g1/g2)`	nested: g2 within g1, i.e. (1 | g1) + (1 | g1:g2)

```
#>  rho_cs.Rho phi_ar1.Phi
#>       0.589       0.800
```

```
c(gls_cs = AIC(g_cs), gls_ar1 = AIC(g_ar1), lmer_ri = AIC(m_ri), lmer_rs =
↪  AIC(m_ris))
```

```
#>  gls_cs gls_ar1 lmer_ri lmer_rs
#>    1794    1747    1794    1756
```

The compound-symmetry `gls()` fit and the random-intercept `lmer()` fit have the same coefficients and log-likelihood, and the estimated correlation 0.589 is the ICC. The comparison of the four REML-based AICs (same fixed effects, so they are comparable) is instructive: for these data an AR(1) correlation with constant variance (AIC 1747.2) beats the random-intercept model (1794.5) and even the random-slope model (1755.6). The random-intercept model is poor because correlation decays with the distance between days; the AR(1) model captures that but not the fan-out of the variance. The two descriptions answer different questions: the direct model describes the covariance and is enough for estimating the average slope, while the mixed model describes the *population of subject-specific lines* and allows us to **predict** an individual subject's trajectory (Chapter 12). Direct covariance models can also be combined with random effects (`lme(..., correlation = corAR1())`).

11.8 In R

The two main functions are `lme4::lmer()` and `nlme::lme()` (the function for *linear* mixed models in `nlme`; `nlme::nlme()` fits non-linear ones).

```
m_lme <- lme(Reaction ~ Days, random = ~ Days | Subject, data = sleepstudy)
↪  # same model in nlme
rbind(lmer = fixef(m_ris), lme = fixef(m_lme))
```

```
#>      (Intercept)  Days
#> lmer       251.4 10.47
#> lme        251.4 10.47
```

```
VarCorr(m_lme)
```

```
#> Subject = pdLogChol(Days)
#>             Variance StdDev Corr
#> (Intercept) 612.08   24.740 (Intr)
#> Days         35.07    5.922 0.066
#> Residual    654.94   25.592
```

```
head(ranef(m_ris)$Subject, 3)            # predicted random effects b~
↪  (BLUPs)
```

```
#>     (Intercept)   Days
#> 308       2.259  9.199
#> 309     -40.399 -8.620
#> 310     -38.960 -5.449
```

```
head(coef(m_ris)$Subject, 3)             # beta + b~: subject-specific lines
```

```
#>     (Intercept)   Days
#> 308       253.7 19.666
#> 309       211.0  1.848
#> 310       212.4  5.018
```

```
predict(m_ris, newdata = data.frame(Days = 5, Subject = "308"))          #
↪  subject-specific
```

```
#>   1
#> 352
```

```
predict(m_ris, newdata = data.frame(Days = 5, Subject = "new"),
↪  allow.new.levels = TRUE)  # population
```

```
#>     1
#> 303.7
```

Useful extractors: `fixef()` ($\hat{\beta}$), `ranef()` (predicted $\tilde{b}$), `coef()` ($\hat{\beta} + \tilde{b}_i$), `VarCorr()` (variance components), `sigma()` ($\hat{\sigma}$), `getME(fit, "X")`, `getME(fit, "Z")`, `fitted()`, `resid()`, `logLik()`, `predict(..., re.form = NA)` for population-level predictions. Grouping variables must be **factors**: a numeric `Subject` would be read as a covariate.

11.9 Common pitfalls

Warning. Fixed effects do not *assume* correlation between group effects and covariates; they *allow* it. Random effects assume independence from the covariates (Theorem 11.1).

Warning. Random effects b_i are not parameters to be estimated; they are random variables and are *predicted*. Writing "ML estimates the random effects" is a category error (Chapter 12).

Warning. Do not treat a factor with two or three levels (sex, treatment arm, three hospitals) as random: there is no basis to estimate a variance. Conversely, do not model school as fixed with 200 levels if you want to generalise to other schools.

Warning. ρ and σ^2_{b0} depend on where $x = 0$ is. Interpret them at a meaningful origin (Proposition 11.1).

Warning. Reporting `lm()` standard errors for clustered data. They treat correlated rows as independent; the error can go in either direction (smaller for covariates that vary within groups when a random intercept absorbs between-group noise, larger for group-level covariates).

11.10 Chapter summary

- Grouped data violate independence; observations in the same group are correlated through shared unmeasured effects. **Partial pooling** is a compromise between one line for all and one line per group.
- **Fixed** group effects estimate each level and allow any relation with the covariates; **random** effects model the levels as draws from $N(0, \sigma_b^2)$, cost few parameters, allow group-level covariates, and assume independence from the covariates. In balanced data the random-effects slope is a weighted average of the within and between estimators (Theorem 11.1).
- Random intercept: $\mathrm{Var}(y_{ij}) = \sigma_b^2 + \sigma^2$, $\mathrm{Cov}(y_{ij}, y_{ik}) = \sigma_b^2$, $\mathrm{ICC} = \sigma_b^2/(\sigma_b^2 + \sigma^2)$, compound symmetry; $\mathrm{Var}(\bar{y}_{i\cdot}) = \sigma_b^2 + \sigma^2/n_i$; design effect $1 + (n-1)\mathrm{ICC}$.
- Random slope: $\mathrm{Var}(y_{ij}) = \sigma_{b0}^2 + 2x\sigma_{b01} + x^2\sigma_{b1}^2 + \sigma^2$, a quadratic in x; ρ depends on the origin of x.
- General LMM: $y = X\beta + Zb + \varepsilon$; $y \mid b \sim N(X\beta + Zb, \sigma^2 I)$ and $y \sim N(X\beta, ZGZ^\top + \sigma^2 I)$, $\mathrm{Cov}(b, y) = GZ^\top$.
- Crossed, nested and several random effects add blocks to Z and G; too rich a random part gives singular fits.
- The marginal model is a covariance model; `gls()` with compound symmetry reproduces the random-intercept model; mixed models additionally predict subject-specific trajectories.

11.11 Exercises with explained solutions

Exercise 11.1. In the random-intercept model without covariates, $y_{ij} = \mu + b_i + \varepsilon_{ij}$, with $\sigma_b^2 = 9$, $\sigma^2 = 16$ and $m = 10$ groups of size $n = 4$. (a) Compute the ICC. (b) Compute $\mathrm{Var}(\bar{y}_{i\cdot})$ and $\mathrm{Var}(\bar{y})$ for the grand mean, and compare with the value obtained if the 40 observations were independent with variance 25. (c) Compute the design effect and the effective sample size. (d) How many groups m would be needed with $n = 4$ to reach the precision of 40 independent observations?

Solution.

(a) $\mathrm{ICC} = 9/25 = 0.36$. (b) $\mathrm{Var}(\bar{y}_{i\cdot}) = \sigma_b^2 + \sigma^2/n = 9 + 4 = 13$. Since the group means are independent with the same variance, $\mathrm{Var}(\bar{y}) = 13/10 = 1.3$. For 40 independent observations: $25/40 = 0.625$. (c) $\mathrm{deff} = 1.3/0.625 = 2.08 = 1 + (4-1)(0.36)$; the effective sample size is $40/2.08 = 19.2$. (d) We need $13/m' \le 0.625$, i.e. $m' \ge 13/0.625 = 20.8$, so $m = 21$ groups of four, **84 observations** instead of 40, because the between-group variance of 9 is unaffected by the group size. With very large n, $\mathrm{Var}(\bar{y}) \to \sigma_b^2/m = 0.9$ for $m = 10$: it cannot go below 0.9 however many observations per group we take.

Exercise 11.2. In the balanced one-way random-effects model $y_{ij} = \mu + b_i + \varepsilon_{ij}$ ($i = 1, \ldots, m$, $j = 1, \ldots, n$) define $\mathrm{MSW} = \sum_{i,j}(y_{ij} - \bar{y}_{i\cdot})^2/\{m(n-1)\}$ and $\mathrm{MSB} = n\sum_i(\bar{y}_{i\cdot} - \bar{y})^2/(m-1)$. Prove that $\mathrm{E[MSW]} = \sigma^2$ and $\mathrm{E[MSB]} = \sigma^2 + n\sigma_b^2$. Compute the ANOVA estimate of σ_b^2 for `Dyestuff` (MSB = 11271.4, MSW = 2451.25, $n = 5$).

Solution. Within group i, $y_{ij} - \bar{y}_{i\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}$ because b_i cancels, so $\sum_j(y_{ij} - \bar{y}_{i\cdot})^2 = \sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2$ has expectation $(n-1)\sigma^2$ (the usual result for the sample variance of n iid $N(0, \sigma^2)$ variables). Summing over m groups and dividing by $m(n-1)$ gives $\mathrm{E[MSW]} = \sigma^2$. For the group means, $\bar{y}_{i\cdot} = \mu + b_i + \bar{\varepsilon}_{i\cdot}$ are iid with variance $\sigma_b^2 + \sigma^2/n$. The sample variance $\sum_i(\bar{y}_{i\cdot} - \bar{y})^2/(m-1)$ of m iid variables is unbiased for that variance, so $\mathrm{E[MSB]} = n(\sigma_b^2 + \sigma^2/n) = \sigma^2 + n\sigma_b^2$. Hence

$\mathrm{E}[(\mathrm{MSB}-\mathrm{MSW})/n]=\sigma_b^2$. For `Dyestuff`: $(11271.4-2451.25)/5=1764.0$, which agrees with the `lmer` REML estimate; the estimate is negative whenever MSB $<$ MSW, a problem that REML handles by truncating at 0.

Exercise 11.3. Two classes: class 1 has 3 pupils, class 2 has 2. The model is $y_{ij}=\beta_0+b_i+\varepsilon_{ij}$ with $\sigma_b^2=9$, $\sigma^2=7$. (a) Write Z and G and compute $V=ZGZ^\top+\sigma^2 I$. (b) Give the correlation matrix of the observations. (c) The covariance of the class means $\bar y_{1\cdot}$ and $\bar y_{2\cdot}$ and $\mathrm{Var}(\bar y_{1\cdot}-\bar y_{2\cdot})$. (d) What is $\mathrm{Var}(y_{11}-y_{12})$ and why does it not depend on σ_b^2?

Solution.

(a) $Z=\begin{pmatrix}1&0\\1&0\\1&0\\0&1\\0&1\end{pmatrix}$, $G=9I_2$, $ZGZ^\top=9\,\mathrm{blockdiag}(J_3,J_2)$ with $J_k=\mathbf{1}_k\mathbf{1}_k^\top$. So $V=\mathrm{blockdiag}(V_1,V_2)$ with $V_1=\begin{pmatrix}16&9&9\\9&16&9\\9&9&16\end{pmatrix}$, $V_2=\begin{pmatrix}16&9\\9&16\end{pmatrix}$. (b) ICC $=9/16=0.5625$ within classes, 0 between classes. (c) Class means of different classes are independent, covariance 0; $\mathrm{Var}(\bar y_{1\cdot})=9+7/3=11.33$, $\mathrm{Var}(\bar y_{2\cdot})=9+7/2=12.5$, so $\mathrm{Var}(\bar y_{1\cdot}-\bar y_{2\cdot})=23.83$. (d) $\mathrm{Var}(y_{11}-y_{12})=16+16-2(9)=14=2\sigma^2$: the shared b_1 cancels in the difference. Within-group contrasts are free of the group effect, which is why they identify σ^2 and (Chapter 12) why REML works with contrasts.

Exercise 11.4. For `sleepstudy`, $\hat\sigma_{b0}^2=612.1$, $\hat\sigma_{b1}^2=35.07$, $\hat\sigma_{b01}=9.60$. (a) Use Proposition 11.1 to compute $\hat\sigma_{b0}$ and $\hat\rho$ after centring `Days` at $c=4.5$. (b) At which value of Days would intercept and slope be uncorrelated, and what is the minimum of $\mathrm{Var}(y\mid\mathrm{Days})-\sigma^2$? (c) A colleague writes "the negative correlation between baseline and rate shows that slow starters deteriorate more slowly" after fitting with `Days` centred at 9. Evaluate.

Solution.

(a) With $c=4.5$: $\sigma_{b0}^{2*}=612.1+2(4.5)(9.60)+4.5^2(35.07)=612.1+86.4+710.2=1408.7$, SD $=37.53$; $\sigma_{b01}^{*}=9.60+4.5(35.07)=167.4$; $\rho^*=167.4/(37.53\times5.922)=0.753$. This agrees with the refit in the text (37.53, 0.753). (b) $c=-9.60/35.07=-0.274$ days: outside the observed range, so the variance increases over the whole study. The minimum of $\sigma_{b0}^2(x)$ is $612.1-9.60^2/35.07=609.5$. (c) At the origin $c=9$ the "intercept" is the level on the *last* day. The correlation between level at Day 9 and slope is $\rho^*=(9.60+9\times35.07)/\{\sqrt{612.1+2(9)(9.60)+81(35.07)}\times5.922\}=325.2/(60.2\times5.922)=0.91$, strongly positive; in any case it is not about "starting". The correlation depends on the origin and says nothing about a causal relation unless the origin is the starting condition of interest.

Exercise 11.5. Show that in the fixed-effects model (11.2) with $n_i=n$ the least-squares estimator of β is the within estimator $\hat\beta_W=S_W^{xy}/S_W^{xx}$. Then use Theorem 11.1 to show that $\mathrm{Var}(\hat\beta_{GLS})\le\mathrm{Var}(\hat\beta_W)$ in the random-effects model, and say why the inequality is not a reason to choose random effects when b_i is correlated with $\bar x_i$.

Solution. For fixed β, the least-squares problem in α_i gives $\hat\alpha_i=\bar y_{i\cdot}-\beta\bar x_{i\cdot}$. Substituting, the residual sum of squares is $\sum_{i,j}\{(y_{ij}-\bar y_{i\cdot})-\beta(x_{ij}-\bar x_{i\cdot})\}^2$, minimised at $\hat\beta_W=S_W^{xy}/S_W^{xx}$. In the

random-effects model $\mathrm{Var}(S_W^{xy}/S_W^{xx}) = \sigma^2/S_W^{xx}$ (the within variation is free of b_i). By the proof of Theorem 11.1 the GLS estimator has $\mathrm{Var}(\hat\beta_{GLS}) = \{\tilde x^\top V^{-1}\tilde x\}^{-1} = \sigma^2/(S_W^{xx} + \theta S_B^{xx}) \le \sigma^2/S_W^{xx}$, with equality iff $\theta S_B^{xx} = 0$. So the random-effects estimator is more efficient. But efficiency is only worth having if the estimator is unbiased: if $\mathrm{Cov}(b_i, \bar x_i) \neq 0$, the between part is biased (simulation in the text), and a biased but precise estimator is a bad bargain; the Mundlak device restores unbiasedness.

Exercise 11.6. Let $V_i = \sigma_t^2\{(1-\rho)I_n + \rho \mathbf{1}\mathbf{1}^\top\}$. (a) Find the eigenvalues of V_i and prove that V_i is positive definite iff $-1/(n-1) < \rho < 1$. (b) Show that the random-intercept model gives $\rho \ge 0$. (c) Give a situation in which a negative ρ is natural (for example, a competition between n seedlings that share a pot).

Solution.

(a) $\mathbf{1}$ is an eigenvector with eigenvalue $\sigma_t^2\{(1-\rho) + n\rho\} = \sigma_t^2\{1 + (n-1)\rho\}$; any vector orthogonal to $\mathbf{1}$ has eigenvalue $\sigma_t^2(1-\rho)$, with multiplicity $n-1$. Positive definiteness requires both to be positive: $\rho < 1$ and $\rho > -1/(n-1)$. (b) $\rho = \sigma_b^2/(\sigma_b^2 + \sigma^2) \ge 0$ because variances are non-negative. (c) Competition for a fixed resource: when one seedling grows more, the others grow less, so within-pot correlations are negative. A random pot effect cannot represent this; use a marginal covariance model (`gls` with `corCompSymm`, which allows $\rho < 0$).

Exercise 11.7. (R analysis.) In a cluster-randomised trial 20 clusters of 10 subjects are randomised to treatment or control (10 clusters each). Simulate $y_{ij} = b_i + \varepsilon_{ij}$ with *no* treatment effect, $\sigma_b^2 = 0.3$, $\sigma^2 = 0.7$ (ICC $= 0.3$). Compare, over 500 simulated trials, the rejection rate at the nominal 5% level of the t test of treatment from `lm(y ~ trt)` and of the z test from `lmer(y ~ trt + (1 | cluster))`. Explain the result with the design effect.

Solution.

```
set.seed(1)
one_trial <- function() {
  m <- 20; n <- 10; g <- rep(1:m, each = n)
  trt <- rep(0:1, each = m / 2)[g]
  y <- rnorm(m, 0, sqrt(.3))[g] + rnorm(m * n, 0, sqrt(.7))
  d <- data.frame(y, trt, g = factor(g))
  c(lm   = summary(lm(y ~ trt, d))$coefficients[2, 4] < .05,
    lmer = abs(summary(lmer(y ~ trt + (1 | g), d))$coefficients[2, 3]) > 1.96)
}
rej <- rowMeans(replicate(500, one_trial()))
rej
```

```
#>    lm  lmer
#> 0.266 0.056
```

The ordinary linear model rejects the (true) null hypothesis in 27% of the trials instead of 5%, because it treats 200 correlated observations as independent while the treatment varies only *between* clusters: the design effect is $1 + 9(0.3) = 3.7$, so the true variance of the treatment contrast is 3.7 times the naive one (standard error inflated by $\sqrt{3.7} = 1.9$). The mixed model's rejection rate is close to the nominal 5% (the z test is slightly liberal with only 20 clusters; t approximations are discussed in Chapter 12).

Exercise 11.8. (R analysis.) The data `nlme::ergoStool` give the effort needed to rise from a stool of four types (`Type`) for each of nine subjects (`Subject`). (a) Which factor should be fixed and which random? (b) Fit the model, report the variance components and the ICC, and interpret the coefficients. (c) What is the estimated difference between types T2 and T1 and its standard error? Why is the standard error much smaller than it would be for a design in which each subject tried only one stool type?

Solution.

(a) `Type`: four specific stools of interest, so **fixed**. `Subject`: nine people sampled from a population, each measured under all types, so **random**. (b)

```
es <- as.data.frame(nlme::ergoStool); es$Subject <-
↪  factor(as.character(es$Subject))
m_es <- lmer(effort ~ Type + (1 | Subject), es)
summary(m_es)$coefficients
```

```
#>             Estimate Std. Error t value
#> (Intercept)   8.5556     0.5760  14.853
#> TypeT2        3.8889     0.5187   7.498
#> TypeT3        2.2222     0.5187   4.284
#> TypeT4        0.6667     0.5187   1.285
```

```
vc_es <- as.data.frame(VarCorr(m_es))$vcov
c(sigma_b2 = vc_es[1], sigma2 = vc_es[2], ICC = vc_es[1] / sum(vc_es))
```

```
#> sigma_b2   sigma2      ICC
#>   1.7755   1.2106   0.5946
```

The reference stool T1 needs an average effort of 8.56 on the rating scale; stool T2 needs 3.89 more, T3 2.22 more and T4 0.67 more (not clearly different from T1). Subjects differ in their general level of effort with SD $\hat{\sigma}_b = 1.33$, a bit more than the residual SD 1.1; ICC 0.59. (c) The difference T2 − T1 is 3.89 with standard error 0.519. Within a subject the difference between two stools is $(\mu_2 + \varepsilon_{i2}) - (\mu_1 + \varepsilon_{i1})$: the subject effect b_i cancels, so the variance of the within-subject difference is $2\sigma^2/m = 2(1.211)/9 = 0.269$, i.e. se $= 0.519$. If each subject tried only one stool the subject effect would stay in the contrast and the variance would be $2(\sigma_b^2 + \sigma^2)/m_{\text{per type}}$, much larger even with the same total of observations. The randomised complete block design uses each subject as its own control.

Exercise 11.9. (R analysis and interpretation.) The data `nlme::Oxboys` give the height (cm) of 26 boys measured on nine occasions; `age` is centred and scaled (years, roughly −1 to 1). Fit `height ~ age + (age | Subject)`. (a) Report and interpret the fixed effects and the four variance parameters. (b) Which fraction of the total variance at `age = 0` is between boys? (c) Give a 95% range for the growth rate of boys. (d) Is the correlation estimate 0.64 interpretable? What would change if `age` were centred at its minimum?

Solution.

```
ox <- as.data.frame(nlme::Oxboys); ox$Subject <-
↪  factor(as.character(ox$Subject))
m_ox <- lmer(height ~ age + (age | Subject), ox)
print(VarCorr(m_ox), comp = c("Variance", "Std.Dev."))
```

```
#>  Groups   Name        Variance Std.Dev. Corr
#>  Subject  (Intercept) 65.304   8.08
#>           age          2.825   1.68     0.64
#>  Residual              0.435   0.66
```

```
fixef(m_ox)
```

```
#> (Intercept)         age
#>     149.372       6.525
```

```
Gx <- unclass(VarCorr(m_ox)$Subject)[1:2, 1:2]; s2x <- sigma(m_ox)^2
Gx[1, 1] / (Gx[1, 1] + s2x)
```

```
#> [1] 0.9934
```

```
fixef(m_ox)[2] + c(-1.96, 1.96) * sqrt(Gx[2, 2])
```

```
#> [1] 3.231 9.820
```

(a) The average boy is 149.4 cm tall at the age origin and grows 6.53 cm per unit of age. Heights at the origin vary between boys with SD $\hat{\sigma}_{b0} = 8.1$ cm, growth rates with SD $\hat{\sigma}_{b1} = 1.68$ cm/unit, and a boy's measurements scatter around his own line with SD $\hat{\sigma} = 0.66$ cm (small: growth is nearly linear in this range). (b) At the origin, $\sigma_{b0}^2/(\sigma_{b0}^2 + \sigma^2) = 0.993$: almost all variance is between boys. (c) $6.53 \pm 1.96 \times 1.68 = [3.2,\ 9.8]$. (d) $\hat{\rho} = 0.64$ means that, at the origin, taller boys tend to grow faster. It refers to the origin `age = 0` (about the middle of the study); centring at the minimum age ($c = -1$ on this scale) would turn it into the correlation between *initial* height and growth, $\rho^* = (\sigma_{b01} - \sigma_{b1}^2)/(\sigma_{b0}^* \sigma_{b1})$, which is different: here $\rho^* \approx 0.49$ (refit with `I(age + 1)`).

Exercise 11.10. (Critique.) A student writes: "We have data on 3 hospitals. To account for the clustering of patients within hospitals, we fitted a random-intercept model. The estimated hospital variance was $\hat{\sigma}_b^2 = 0$, so there is no hospital effect and we pooled the data. Random effects were used because fixed effects assume that hospital is correlated with treatment." Identify the errors and give a better analysis plan.

Solution.

(1) With $m = 3$ hospitals the variance σ_b^2 is estimated from three numbers: the estimate is extremely imprecise, and a boundary estimate $\hat{\sigma}_b^2 = 0$ does **not** mean "no hospital effect", only that the between-hospital variation is not larger than expected from the within-hospital noise with so few hospitals. (2) Pooling after a non-significant variance test inflates type I error; the design is clustered and the analysis should respect it regardless of the estimate. (3) The reason given for random effects is backwards: fixed effects *allow* hospital to be correlated with treatment; random effects *assume* it is not. (4) Plan: treat hospital as **fixed** (three levels, interest in these hospitals, few levels), or random with a profile-likelihood or Bayesian interval for σ_b and a warning that three levels are too few for a stable variance; if treatment is randomised within hospital the hospital effect is balanced against treatment, and fixed effects are the safe choice; report an interval for the treatment effect that does not depend on $\hat{\sigma}_b^2 = 0$.

Exercise 11.11. (R analysis.) The data `lme4::Pastes` give the strength of a chemical paste: 10 batches, 3 casks sampled from each batch, and two measurements per cask (60

observations). Fit `strength ~ 1 + (1 | batch/cask)`. (a) Report the three variance components and the percentage of the total variance due to each. (b) What is the correlation between two measurements of the same cask? Between two casks of the same batch? Between two batches? (c) A new study can afford 60 measurements. Which of "more batches" or "more casks within batch" reduces the variance of the overall mean more? Use $\mathrm{Var}(\bar{y}) = \sigma^2_{\text{batch}}/m_b + \sigma^2_{\text{cask}}/(m_b m_c) + \sigma^2/(m_b m_c n)$.

Solution.

```
data(Pastes, package = "lme4")
m_pa <- lmer(strength ~ 1 + (1 | batch/cask), Pastes)
vcp <- as.data.frame(VarCorr(m_pa)); vcp[, c("grp", "vcov")]
```

```
#>          grp  vcov
#> 1 cask:batch 8.434
#> 2      batch 1.657
#> 3   Residual 0.678
```

```
v <- setNames(vcp$vcov, vcp$grp); round(100 * v / sum(v), 1)
```

```
#> cask:batch      batch   Residual
#>       78.3       15.4        6.3
```

(a) The variance components are cask-within-batch $\hat{\sigma}^2_{\text{cask}} = 8.43$ (78%), batch $\hat{\sigma}^2_{\text{batch}} = 1.66$ (15%) and residual 0.68 (6%). (b) Same cask: $(\sigma^2_{\text{batch}} + \sigma^2_{\text{cask}})/\text{total} = 0.94$; same batch, different casks: $\sigma^2_{\text{batch}}/\text{total} = 0.15$; different batches: 0. (c) With 60 measurements and $n = 2$ measurements per cask, the options are, e.g., $(m_b, m_c) = (10, 3)$ versus $(5, 6)$: the variance of the mean is $v_b/m_b + v_c/(m_b m_c) + v_e/(m_b m_c n)$:

```
vm <- function(mb, mc, n = 2) v["batch"] / mb + v["cask:batch"] / (mb * mc) +
↪  v["Residual"] / (mb * mc * n)
c(`10 batches x 3 casks` = unname(vm(10, 3)), `5 batches x 6 casks` =
↪  unname(vm(5, 6)))
```

```
#> 10 batches x 3 casks  5 batches x 6 casks
#>               0.4582               0.6239
```

More *batches* is better. Both designs have the same number of casks (30) and measurements (60), so the cask and residual terms are the same, but the batch term $\sigma^2_{\text{batch}}/m_b$ is halved when m_b doubles. A variance component of a given level is reduced only by sampling more units *of that level*.

Exercise 11.12. Consider the random-intercept model with one observation per group ($n_i = 1$ for all i). (a) Show that $\mathrm{Var}(y_i) = \sigma_b^2 + \sigma^2$ and that the likelihood depends on (σ_b^2, σ^2) only through their sum. (b) What happens in `lmer`? (c) What if half of the groups have two observations?

Solution.

(a) The y_i are independent $N(x_i^\top \beta, \sigma_b^2 + \sigma^2)$: the likelihood is a function of $\sigma_b^2 + \sigma^2$, so the two components are not separately identifiable (infinitely many pairs with the same sum give the same likelihood). (b) `lmer` stops with an error: *number of levels of each grouping factor must be < number of observations*; if it did not, the profiled likelihood would be

flat along $\sigma_b^2 + \sigma^2 = \text{const}$. (c) Groups with two observations contribute the within-group difference $y_{i1} - y_{i2} \sim N(0, 2\sigma^2)$, which identifies σ^2 alone; then σ_b^2 is identified from the totals. Identifiability requires *some* groups with more than one observation.

11.12 Further reading

Laird and Ware (1982) introduced the LMM for longitudinal data; Pinheiro and Bates (2000) is the reference for `nlme`; Bates et al. (2015) describes `lme4`. A readable applied account is Gelman and Hill (2006); Wood (2017, Ch. 2) and Pawitan (2001, Ch. 17) give the likelihood point of view followed here. Robinson (1991) discusses random effects and BLUPs.

Chapter 12

Estimation, prediction and inference for linear mixed models

Chapter 11 defined the linear mixed model $y = X\beta + Zb + \varepsilon$ and showed that its *marginal* distribution is $y \sim N(X\beta, V)$ with $V = ZGZ^\top + \sigma^2 I$. This chapter answers the questions that follow. How are the fixed effects β and the variance parameters ϑ estimated, and why are there *two* likelihood criteria, ML and REML, that give different variance estimates? How should the random effects b themselves be **predicted**, and why do the predictions shrink towards zero? How do we test hypotheses about fixed effects and about variance components, and why does the usual χ^2 result fail for the latter? And how do we check the assumptions of the model?

The plan is the following. Section 12.2 writes the likelihood, derives the generalised least-squares estimator of β and the profile likelihood of ϑ. Section 12.3 explains the bias of ML, defines REML through error contrasts, *derives the REML criterion* by two routes and shows that they agree, and works out the balanced one-way case in closed form. Section 12.4 derives the best linear unbiased predictor in three ways (conditional expectation, minimum mean squared error, Henderson's mixed-model equations) and studies shrinkage. Section 12.5 treats inference for fixed effects, Section 12.6 tests and intervals for variance components, including the boundary problem, Section 12.7 model comparison, and Section 12.8 diagnostics. Section 12.9 collects the R code.

Throughout, n is the number of observations, $p = \operatorname{rank}(X)$ the number of fixed effects, ϑ the vector of variance parameters, $V = V(\vartheta)$, and $r(\vartheta) = y - X\hat\beta(\vartheta)$. (We write ϑ for the vector of variance parameters to keep θ for the generic or canonical parameter of Appendix A; `lme4` calls this vector θ.)

12.1 Two running examples

We use the two data sets of Chapter 11: `sleepstudy` (18 subjects, 10 days each; random intercept and slope) and the small balanced layout `Dyestuff` (6 batches, 5 samples each), where everything can be computed by hand.

```
m_ri_reml <- lmer(Reaction ~ Days + (1 | Subject), sleepstudy)
m_ri_ml   <- lmer(Reaction ~ Days + (1 | Subject), sleepstudy, REML = FALSE)
m_dy_reml <- lmer(Yield ~ 1 + (1 | Batch), Dyestuff)
m_dy_ml   <- lmer(Yield ~ 1 + (1 | Batch), Dyestuff, REML = FALSE)
rbind(sleep_REML = as.data.frame(VarCorr(m_ri_reml))$vcov, sleep_ML =
↪  as.data.frame(VarCorr(m_ri_ml))$vcov,
      dye_REML = as.data.frame(VarCorr(m_dy_reml))$vcov,   dye_ML =
      ↪  as.data.frame(VarCorr(m_dy_ml))$vcov) |>
```

```
  `colnames<-`(c("sigma_b^2", "sigma^2")) |> round(1)
```

```
#>             sigma_b^2 sigma^2
#> sleep_REML       1378   960.5
#> sleep_ML         1297   954.5
#> dye_REML         1764  2451.2
#> dye_ML           1388  2451.2
```

ML and REML give *different* variance components: for `Dyestuff` the batch variance is 1764 under REML and 1388 under ML. Which is right, and why? That is the main question of Section 12.3.

12.2 The marginal likelihood and generalised least squares

Because b is not observed, the likelihood is built from the marginal distribution $y \sim N(X\beta, V(\vartheta))$:

$$\ell(\beta, \vartheta) = -\tfrac{1}{2}\Big[n\log 2\pi + \log|V| + (y - X\beta)^\top V^{-1}(y - X\beta)\Big]. \tag{12.1}$$

The random effects have been integrated out, so the parameters are β (p of them) and ϑ (a few), **not** the q random effects. We maximise in two steps.

Theorem 12.1 (Generalised least squares)**.** *For fixed ϑ, $\ell(\beta, \vartheta)$ is maximised over β by*

$$\hat\beta(\vartheta) = (X^\top V^{-1} X)^{-1} X^\top V^{-1} y, \tag{12.2}$$

which also minimises $(y-X\beta)^\top V^{-1}(y-X\beta)$. It is unbiased, $\mathrm{E}[\hat\beta(\vartheta)] = \beta$, with $\mathrm{Var}\{\hat\beta(\vartheta)\} = (X^\top V^{-1}X)^{-1}$, and it is the best linear unbiased estimator of β (Gauss–Markov for the covariance V). For $V = \sigma^2 I$ it is the ordinary least-squares estimator.

Proof. V does not depend on β, so $\partial\ell/\partial\beta = X^\top V^{-1}(y - X\beta)$; setting it to zero gives (12.2), and the Hessian $-X^\top V^{-1}X$ is negative definite, so this is the maximum. Writing $\hat\beta = Ly$ with $L = (X^\top V^{-1}X)^{-1}X^\top V^{-1}$ we get $\mathrm{E}[\hat\beta] = LX\beta = \beta$ and $\mathrm{Var}(\hat\beta) = LVL^\top = (X^\top V^{-1}X)^{-1}X^\top V^{-1}VV^{-1}X(X^\top V^{-1}X)^{-1} = (X^\top V^{-1}X)^{-1}$. For optimality, let $\tilde\beta = Ly + Dy$ be another linear unbiased estimator; unbiasedness requires $DX = 0$. Then $\mathrm{Var}(\tilde\beta) = LVL^\top + DVD^\top + LVD^\top + DVL^\top$ and $LVD^\top = (X^\top V^{-1}X)^{-1}X^\top D^\top = (X^\top V^{-1}X)^{-1}(DX)^\top = 0$, so $\mathrm{Var}(\tilde\beta) = \mathrm{Var}(\hat\beta) + DVD^\top \succeq \mathrm{Var}(\hat\beta)$. Finally $V = \sigma^2 I$ gives $(X^\top X)^{-1}X^\top y$. □

Equivalently, with a square root $V^{-1/2}$, $V^{-1/2}y = V^{-1/2}X\beta + e$ with $\mathrm{Var}(e) = I$: GLS is ordinary least squares on **whitened** data. In the random-intercept case it weights groups by their precision. For the mean-only model $y_{ij} = \mu + b_i + \varepsilon_{ij}$ the GLS estimator is a weighted average of group means,

$$\hat\mu = \frac{\sum_i w_i \bar y_{i\cdot}}{\sum_i w_i}, \qquad w_i = \frac{1}{\sigma_b^2 + \sigma^2/n_i}, \qquad \mathrm{Var}(\hat\mu) = \Big(\sum_i w_i\Big)^{-1}$$

(Exercise 12.1). With equal group sizes all weights are equal, $\hat\mu = \bar y$ and $\mathrm{Var}(\hat\mu) = (\sigma_b^2 + \sigma^2/n)/m$.

Substituting $\hat\beta(\vartheta)$ into (12.1) gives the **profile log-likelihood** of the variance parameters,

$$\ell_P(\vartheta) = \ell(\hat\beta(\vartheta), \vartheta) = -\tfrac{1}{2}\Big[n\log 2\pi + \log|V(\vartheta)| + r(\vartheta)^\top V(\vartheta)^{-1} r(\vartheta)\Big]. \tag{12.3}$$

The **ML estimator** maximises ℓ_P over ϑ (variances ≥ 0, G positive semi-definite) and then sets $\hat\beta_{ML} = \hat\beta(\hat\vartheta_{ML})$. Everything is explicit, so we can reproduce `lmer` by hand for the random-intercept model of `sleepstudy`: build $V = \sigma_b^2 ZZ^\top + \sigma^2 I$ and maximise (12.3).

```
y <- sleepstudy$Reaction; X <- model.matrix(~ Days, sleepstudy)
Z <- model.matrix(~ 0 + Subject, sleepstudy); n <- nrow(X); p <- ncol(X)
V_of  <- function(sb2, s2) sb2 * tcrossprod(Z) + s2 * diag(n)
gls   <- function(V) solve(t(X) %*% solve(V) %*% X, t(X) %*% solve(V) %*% y)
lP <- function(par) {                                  # profile
↪  log-likelihood, par = log-variances
  V <- V_of(exp(par[1]), exp(par[2])); r <- y - X %*% gls(V)
  -0.5 * (n * log(2 * pi) + determinant(V)$modulus + t(r) %*% solve(V, r))[1]
}
opt <- optim(c(log(1000), log(1000)), function(p) -lP(p), control =
↪  list(reltol = 1e-12))
rbind(by_hand = exp(opt$par), lme4 = as.data.frame(VarCorr(m_ri_ml))$vcov)
```

```
#>          [,1]  [,2]
#> by_hand 1297 954.5
#> lme4    1297 954.5
```

```
Vh <- V_of(exp(opt$par[1]), exp(opt$par[2]))
cbind(by_hand = drop(gls(Vh)), lme4 = fixef(m_ri_ml))
```

```
#>             by_hand   lme4
#> (Intercept)  251.41 251.41
#> Days          10.47  10.47
```

```
cbind(by_hand = sqrt(diag(solve(t(X) %*% solve(Vh) %*% X))), lme4 =
↪  sqrt(diag(vcov(m_ri_ml))))
```

```
#>             by_hand   lme4
#> (Intercept)  9.5062 9.5062
#> Days         0.8017 0.8017
```

The hand-made ML fit reproduces `lme4` (variance components, $\hat\beta$ and standard errors $\sqrt{[(X^\top \hat V^{-1} X)^{-1}]_{kk}}$). Two remarks. First, ML **estimates** β and ϑ; it does not estimate b, which is *predicted* afterwards (Section 12.4). Second, `lme4` does not optimise over σ_b^2, σ^2 directly: it writes $G = \sigma^2 \Lambda_\vartheta \Lambda_\vartheta^\top$, profiles out β and σ^2 analytically, and optimises a function of the entries of Λ_ϑ only, using sparse Cholesky factorisations so that V is never formed; the maximum is the same.

12.3 Why ML underestimates variances, and REML

In the linear model $y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$, the ML estimator of the variance is RSS/n, but $\mathrm{E}[\mathrm{RSS}] = \sigma^2(n-p)$ (Chapter 5; $\mathrm{E}[\varepsilon^\top (I-H)\varepsilon] = \sigma^2 \mathrm{tr}(I-H)$). So ML is biased down by the factor $(n-p)/n$: p degrees of freedom were used to estimate $\hat\beta$, and the residuals live in a space of dimension $n-p$. ML treats $\hat\beta$ as if it were the true β. The same happens in the mixed model: ML variance components are too small, the more so when p is large compared with the number of independent units.

The cure is to estimate ϑ from the part of the data that carries *no information about* β.

Definition 12.1 (Error contrasts and the REML likelihood)**.** Let A be an $n \times (n-p)$ matrix of full column rank with $A^\top X = 0$. The vector $w = A^\top y$ is an **error contrast**: $w = A^\top(Zb+\varepsilon) \sim N(0, A^\top V A)$ does not depend on β. The **REML** (restricted, or residual, maximum likelihood) criterion is the log-likelihood of w,

$$\ell_C(\vartheta) = -\tfrac{1}{2}\Big[(n-p)\log 2\pi + \log|A^\top V A| + y^\top A(A^\top V A)^{-1}A^\top y\Big],$$

and the REML estimator $\hat\vartheta_{REML}$ maximises it. Then $\hat\beta = \hat\beta(\hat\vartheta_{REML})$ by GLS.

This is well defined only if the answer does not depend on the arbitrary choice of A: the next two results show that different A change ℓ_C by a constant, and give a formula that does not involve A at all.

Lemma 12.1 (Two matrix identities)**.** *Let A be $n \times (n-p)$ with $A^\top X = 0$, $A^\top A = I_{n-p}$ (an orthonormal basis of the orthogonal complement of the column space of X), V positive definite, and*

$$P = V^{-1} - V^{-1}X(X^\top V^{-1}X)^{-1}X^\top V^{-1}.$$

Then (i) $A(A^\top V A)^{-1}A^\top = P$ and (ii) $|A^\top V A| = |V|\,|X^\top V^{-1}X|/|X^\top X|$.

Proof.

(i) The $n \times n$ matrix $C = [X,\ VA]$ is invertible: if $Xc_1 + VAc_2 = 0$, multiplying by $A^\top$ gives $A^\top V A c_2 = 0$, so $c_2 = 0$ because $A^\top V A$ is positive definite, and then $c_1 = 0$ because X has full column rank. Both P and $A(A^\top V A)^{-1}A^\top$ satisfy the same equations on C: $PX = V^{-1}X - V^{-1}X(X^\top V^{-1}X)^{-1}X^\top V^{-1}X = 0$ and $A(A^\top V A)^{-1}A^\top X = 0$; $PVA = A - V^{-1}X(X^\top V^{-1}X)^{-1}X^\top A = A$ and $A(A^\top V A)^{-1}A^\top V A = A$. Two matrices that agree on an invertible C are equal.

(ii) Let $S = (X^\top X)^{-1/2}$ and $M = [A,\ XS]$, an orthogonal $n \times n$ matrix ($A^\top XS = 0$, $S^\top X^\top X S = I$). Then $|M^\top V M| = |V|$ and $M^\top V^{-1} M = (M^\top V M)^{-1}$. For a partitioned matrix, the determinant is $|M^\top V M| = |A^\top V A| \cdot |\Sigma|$ where Σ is the Schur complement of the block $A^\top V A$, and Σ^{-1} is the lower-right block of the inverse $M^\top V^{-1} M$, namely $SX^\top V^{-1}XS$. Hence $|V| = |A^\top V A|/|SX^\top V^{-1}XS|$, i.e. $|A^\top V A| = |V|\,|S|^2|X^\top V^{-1}X| = |V||X^\top V^{-1}X|/|X^\top X|$.

□

Theorem 12.2 (The REML criterion)**.** *For any A as in the Lemma, the log-likelihood of the error contrasts is*

$$\ell_C(\vartheta) = \ell_R(\vartheta) + \tfrac{1}{2}\log|X^\top X|, \qquad \ell_R(\vartheta) = \ell_P(\vartheta) - \tfrac{1}{2}\log|X^\top V^{-1}X| + \tfrac{p}{2}\log 2\pi, \quad (12.4)$$

that is,

$$\ell_R(\vartheta) = -\tfrac{1}{2}\Big[(n-p)\log 2\pi + \log|V| + \log|X^\top V^{-1}X| + r(\vartheta)^\top V^{-1} r(\vartheta)\Big]. \quad (12.5)$$

Moreover, ℓ_R is the logarithm of the likelihood integrated over β with a flat weight: $\int_{\mathbb{R}^p} L(\beta,\vartheta)\,d\beta = \exp\{\ell_R(\vartheta)\}$. The constant $\frac{1}{2}\log|X^\top X|$ does not depend on ϑ, so ℓ_C and ℓ_R have the same maximiser.

Proof. Route 1: contrasts. With A orthonormal, ℓ_C has the determinant and the quadratic form of the Lemma: $y^\top A(A^\top V A)^{-1}A^\top y = y^\top P y = r^\top V^{-1} r$, because $Py = V^{-1}\{y - X\hat\beta(\vartheta)\} = V^{-1}r$

by the definition of the GLS estimator, and $y^\top P y = (y)^\top V^{-1} r = r^\top V^{-1} r$ since $X^\top V^{-1} r = 0$. Substituting (ii),

$$\ell_C = -\tfrac{1}{2}\left[(n-p)\log 2\pi + \log|V| + \log|X^\top V^{-1} X| - \log|X^\top X| + r^\top V^{-1} r\right] = \ell_R + \tfrac{1}{2}\log|X^\top X|.$$

For a different full-rank $A_2 = AB$ (B invertible), ℓ_C changes by the ϑ-free constant $-\log|B|$ (the Jacobian of $w \mapsto B^\top w$), so REML does not depend on the choice of A. *Route 2: integration.* Complete the square around the GLS estimate $\hat\beta = \hat\beta(\vartheta)$:

$$(y - X\beta)^\top V^{-1}(y - X\beta) = r^\top V^{-1} r + (\beta - \hat\beta)^\top X^\top V^{-1} X (\beta - \hat\beta),$$

because the cross term is $2(\hat\beta - \beta)^\top X^\top V^{-1} r$ and $X^\top V^{-1} r = 0$ (normal equations). Therefore

$$\int_{\mathbb{R}^p} e^{\ell(\beta,\vartheta)} d\beta = e^{\ell_P(\vartheta)} \int_{\mathbb{R}^p} \exp\{-\tfrac{1}{2}(\beta - \hat\beta)^\top X^\top V^{-1} X (\beta - \hat\beta)\} d\beta = e^{\ell_P(\vartheta)} (2\pi)^{p/2} |X^\top V^{-1} X|^{-1/2},$$

the integral of an unnormalised $N(\hat\beta, (X^\top V^{-1} X)^{-1})$ density. Taking logarithms gives (12.4); inserting ℓ_P from (12.3) gives (12.5). □

Four observations help to read the criterion.

- REML **subtracts** $\frac{1}{2}\log|X^\top V^{-1} X|$ from the profile log-likelihood. The matrix $X^\top V^{-1} X$ is the information about β; it *decreases* when variances grow, so the extra term *increases* with the variances and pushes the REML estimate of ϑ up compared with ML. (A common error is to write $+\frac{1}{2}\log|X^\top V^{-1} X|$.)
- The constant is $(n-p)\log 2\pi$, not $n \log 2\pi$: the contrast has $n - p$ components. The constant does not change $\hat\vartheta$ but matters when comparing with software output; `lme4` reports $-2\ell_R$ as `REMLcrit`.
- **Check,** $V = \sigma^2 I$**.** Then $|V| = \sigma^{2n}$, $|X^\top V^{-1} X| = |X^\top X|\sigma^{-2p}$ and $r = y - X\hat\beta$ is the OLS residual with $r^\top r = \mathrm{RSS}$, so $\ell_R = -\frac{1}{2}[(n-p)\log\sigma^2 + \mathrm{RSS}/\sigma^2] + \text{const}$, whence $\hat\sigma^2_{REML} = \mathrm{RSS}/(n-p)$, the unbiased estimator.
- The REML likelihood depends on X through the contrasts: **REML likelihoods of models with different fixed effects are likelihoods of different data** and cannot be compared (Section 12.7).

We verify (12.5) on `sleepstudy`: the hand-made $-2\ell_R$ equals `lme4`'s `REMLcrit`.

```
lR <- function(sb2, s2) {
  V <- V_of(sb2, s2); Vi <- solve(V); XtViX <- t(X) %*% Vi %*% X
  r <- y - X %*% solve(XtViX, t(X) %*% Vi %*% y)
  -0.5 * ((n - p) * log(2 * pi) + determinant(V)$modulus +
   ↪  determinant(XtViX)$modulus + t(r) %*% Vi %*% r)[1]
}
vc <- as.data.frame(VarCorr(m_ri_reml))$vcov
c(by_hand = -2 * lR(vc[1], vc[2]), REMLcrit = REMLcrit(m_ri_reml))
```

```
#>  by_hand REMLcrit
#>     1786     1786
```

```
opt_r <- optim(c(log(1000), log(1000)), function(p) -lR(exp(p[1]), exp(p[2])),
↪  control = list(reltol = 1e-12))
rbind(REML_by_hand = exp(opt_r$par), REML_lme4 = vc)
```

```
#>                [,1]  [,2]
#> REML_by_hand 1378 960.5
#> REML_lme4    1378 960.5
```

12.3.1 The balanced one-way layout in closed form

Proposition 12.1 (REML, ML and ANOVA in the balanced one-way model)**.** *In* $y_{ij} = \mu + b_i + \varepsilon_{ij}$ *with* m *groups of size* n*, let* $\mathrm{SSW} = \sum_{i,j}(y_{ij} - \bar{y}_{i\cdot})^2$, $\mathrm{SSB} = n\sum_i(\bar{y}_{i\cdot} - \bar{y})^2$, $\mathrm{MSW} = \mathrm{SSW}/\{m(n-1)\}$, $\mathrm{MSB} = \mathrm{SSB}/(m-1)$*, and* $\lambda = \sigma^2 + n\sigma_b^2$*. If* $\mathrm{MSB} \geq \mathrm{MSW}$ *then*

$$\hat{\sigma}^2_{REML} = \mathrm{MSW}, \qquad \hat{\sigma}^2_{b,REML} = \frac{\mathrm{MSB} - \mathrm{MSW}}{n},$$
$$\hat{\sigma}^2_{ML} = \mathrm{MSW}, \qquad \hat{\sigma}^2_{b,ML} = \frac{\frac{m-1}{m}\mathrm{MSB} - \mathrm{MSW}}{n}.$$

(If the expression is negative the estimate is 0*.) REML coincides with the unbiased ANOVA estimator; ML shrinks* MSB *by the factor* $(m-1)/m$*.*

Proof. The n observations of a group are transformed by an orthogonal matrix whose first row is $\mathbf{1}^\top/\sqrt{n}$: the first coordinate is $\sqrt{n}\,\bar{y}_{i\cdot} \sim N(\sqrt{n}\mu, \lambda)$ and the other $n-1$ coordinates are the within-group contrasts, iid $N(0,\sigma^2)$, independent of the first. Collect the $m(n-1)$ within contrasts (sum of squares SSW) and the m group-mean coordinates. The log-likelihood is

$$-\tfrac{1}{2}\Big[m(n-1)\log\sigma^2 + \frac{\mathrm{SSW}}{\sigma^2} + m\log\lambda + \frac{\mathrm{SSB} + mn(\bar{y}-\mu)^2}{\lambda}\Big] + \text{const.}$$

ML: $\hat{\mu} = \bar{y}$ kills the last term; maximising over (σ^2, λ) separately gives $\hat{\sigma}^2 = \mathrm{SSW}/\{m(n-1)\} = \mathrm{MSW}$ and $\hat{\lambda} = \mathrm{SSB}/m = \frac{m-1}{m}\mathrm{MSB}$ (provided $\hat{\lambda} \geq \hat{\sigma}^2$, the constraint $\lambda \geq \sigma^2$). *REML*: the contrasts remove μ, i.e. the coordinate $\sqrt{mn}\,\bar{y}$; the remaining $m-1$ group-mean coordinates have sum of squares SSB, and the restricted log-likelihood is $-\frac{1}{2}[m(n-1)\log\sigma^2 + \mathrm{SSW}/\sigma^2 + (m-1)\log\lambda + \mathrm{SSB}/\lambda]$, maximised at $\hat{\lambda} = \mathrm{SSB}/(m-1) = \mathrm{MSB}$. Then $\hat{\sigma}^2_b = (\hat{\lambda} - \hat{\sigma}^2)/n$. If $\hat{\lambda} < \hat{\sigma}^2$, the constrained maximum is on the boundary $\lambda = \sigma^2$ (i.e. $\sigma^2_b = 0$). □

For `Dyestuff` ($m = 6$, $n = 5$): $\mathrm{MSB} = 11271.4$, $\mathrm{MSW} = 2451.25$, so REML gives $\hat{\sigma}^2_b = (11271.4 - 2451.25)/5 = 1764.0$ and ML gives $(\frac{5}{6}\,11271.4 - 2451.25)/5 = 1388.3$, a 21% reduction. Both agree with `lmer`:

```
aov_t <- summary(aov(Yield ~ Batch, Dyestuff))[[1]]
MSB <- aov_t[1, "Mean Sq"]; MSW <- aov_t[2, "Mean Sq"]; m <- 6; nn <- 5
rbind(REML_formula = c(MSB - MSW, MSW * nn) / c(nn, nn), ML_formula = c((m -
↪  1) / m * MSB - MSW, MSW * nn) / c(nn, nn),
      REML_lme4 = as.data.frame(VarCorr(m_dy_reml))$vcov, ML_lme4 =
       ↪  as.data.frame(VarCorr(m_dy_ml))$vcov)
```

```
#>              [,1] [,2]
#> REML_formula 1764 2451
#> ML_formula   1388 2451
#> REML_lme4    1764 2451
#> ML_lme4      1388 2451
```

12.3.2 Less biased, not unbiased

Warning. REML is *less biased* than ML, not unbiased. It reproduces the unbiased ANOVA estimators in balanced designs when they are positive; in general (unbalanced data, truncation at zero, standard deviations instead of variances) REML estimates are still biased, only less. The difference matters when p is not small compared with the number of groups.

```
set.seed(2027)
g <- factor(rep(1:6, each = 4)); xx <- rep(1:4, 6)
est <- t(replicate(500, {
  yy <- 1 + 0.5 * xx + rnorm(6)[g] + rnorm(24); d <- data.frame(yy, xx, g)
  c(ML   = as.data.frame(VarCorr(suppressMessages(lmer(yy ~ xx + (1 | g), d,
  ↪  REML = FALSE))))$vcov[1],
    REML = as.data.frame(VarCorr(suppressMessages(lmer(yy ~ xx + (1 | g),
    ↪  d))))$vcov[1])
}))
round(colMeans(est), 3)
```

```
#>    ML  REML
#> 0.820 1.016
```

```
d_est <- data.frame(v = c(est), method = rep(colnames(est), each = nrow(est)))
ggplot(d_est, aes(v, fill = method)) + geom_histogram(bins = 30, alpha = .6,
↪  position = "identity") +
  geom_vline(xintercept = 1, linetype = 2) + scale_fill_manual(values = c(ML =
  ↪  ie_coral, REML = ie_navy)) +
  labs(x = expression(hat(sigma)[b]^2), y = "count", fill = NULL)
```

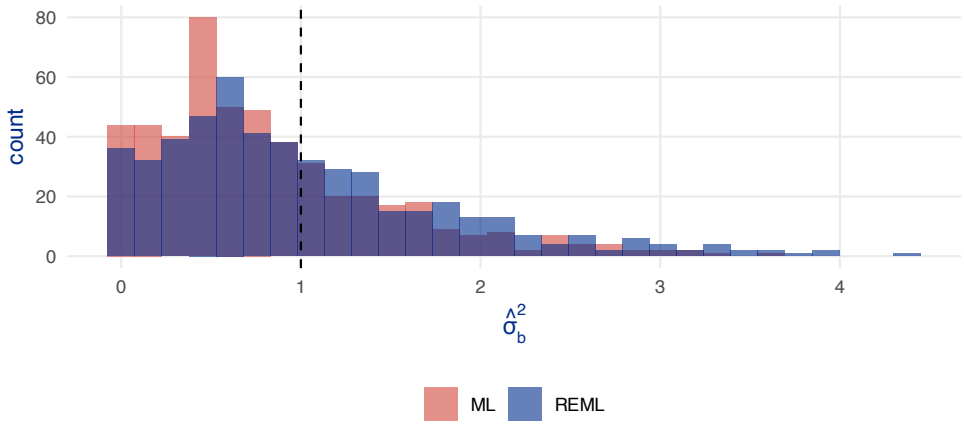


Figure 12.1: Sampling distributions of the estimated between-group variance (true value 1) from 500 simulated data sets with 6 groups of 4 observations. ML underestimates; REML is close to unbiased (slightly above 1 because negative solutions are truncated at 0).

12.4 Prediction of random effects: BLUPs and shrinkage

The random effects b are not parameters but random variables, so we do not *estimate* them; we **predict** them. Given the data, our best knowledge about b is its conditional distribution.

Theorem 12.3 (Conditional distribution and the BLUP)**.** *Under the LMM, $b \mid y \sim N(GZ^\top V^{-1}(y - X\beta),\ G - GZ^\top V^{-1}ZG)$. The **best linear unbiased predictor** (BLUP) of b is*

$$\tilde{b} = GZ^\top V^{-1}(y - X\hat{\beta}), \qquad \hat{\beta} = (X^\top V^{-1}X)^{-1}X^\top V^{-1}y, \tag{12.6}$$

the conditional mean with β replaced by its GLS estimate. It minimises the mean squared error $\mathrm{E}\|Ly - b\|^2$ among all linear predictors that are unbiased, $\mathrm{E}[Ly - b] = 0$ for every β (these are the L with $LX = 0$). If ϑ is replaced by $\hat{\vartheta}$ one speaks of the empirical BLUP; this

is what `ranef()` *returns.*

Proof. Conditional distribution. By Theorem 11.4, (b, y) is jointly Gaussian with $\mathrm{Cov}(b, y) = GZ^\top$. For a Gaussian pair, $\mathrm{E}[b \mid y] = \mathrm{E}[b] + \mathrm{Cov}(b, y)V^{-1}(y - \mathrm{E}[y]) = GZ^\top V^{-1}(y - X\beta)$ and $\mathrm{Var}(b \mid y) = \mathrm{Var}(b) - \mathrm{Cov}(b, y)V^{-1}\mathrm{Cov}(y, b) = G - GZ^\top V^{-1}ZG$. *Optimality.* Write $P = V^{-1} - V^{-1}X(X^\top V^{-1}X)^{-1}X^\top V^{-1}$, so $\tilde{b} = GZ^\top Py$ (indeed $Py = V^{-1}(y - X\hat{\beta})$) and $L_0 = GZ^\top P$ satisfies $L_0X = 0$ because $PX = 0$. For any L with $LX = 0$, put $D = L - L_0$ (so $DX = 0$). Since $\mathrm{Var}(y) = V$ and $\mathrm{Cov}(b, y) = GZ^\top$,

$$\mathrm{E}\|Ly - b\|^2 = \mathrm{tr}\{LVL^\top - LZG - GZ^\top L^\top + G\} = \mathrm{E}\|L_0y - b\|^2 + \mathrm{tr}(DVD^\top) + 2\,\mathrm{tr}\{D(VL_0^\top - ZG)\}.$$

Now $VL_0^\top - ZG = VPZG - ZG = -X(X^\top V^{-1}X)^{-1}X^\top V^{-1}ZG$, because $VP = I - X(X^\top V^{-1}X)^{-1}X^\top V^{-1}$. The cross term contains $DX = 0$ and vanishes, hence $\mathrm{E}\|Ly - b\|^2 = \mathrm{E}\|L_0y - b\|^2 + \mathrm{tr}(DVD^\top) \geq \mathrm{E}\|L_0y - b\|^2$. □

Note that $\tilde{b}$ is not the ML estimate of anything: it is a conditional mean, available in closed form. The **subject-specific** line of subject i in `sleepstudy` is $\hat{\beta} + \tilde{b}_i$ (`coef()` in `lme4`).

12.4.1 Henderson's mixed-model equations

Henderson (Henderson, 1975) obtained $\hat{\beta}$ and $\tilde{b}$ simultaneously, without inverting V.

Theorem 12.4 (Mixed-model equations)**.** *The pair $(\hat{\beta}, \tilde{b})$ minimises the penalised least-squares criterion*

$$\frac{1}{\sigma^2}\|y - X\beta - Zb\|^2 + b^\top G^{-1}b, \tag{12.7}$$

(the negative twice the joint log-density of (y, b), up to constants), i.e. it solves

$$\begin{pmatrix} X^\top X & X^\top Z \\ Z^\top X & Z^\top Z + \sigma^2 G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \tilde{b} \end{pmatrix} = \begin{pmatrix} X^\top y \\ Z^\top y \end{pmatrix}. \tag{12.8}$$

Its solution is the GLS estimator and the BLUP of Theorem 12.3.

Proof. Setting the derivatives of (12.7) with respect to β and b to zero gives the two block rows $X^\top(y - X\beta - Zb) = 0$ and $Z^\top(y - X\beta - Zb) = \sigma^2 G^{-1}b$, which is (12.8). To identify the solution, note first the push-through identity

$$(Z^\top Z + \sigma^2 G^{-1})\,GZ^\top = Z^\top ZGZ^\top + \sigma^2 Z^\top = Z^\top V \implies GZ^\top V^{-1} = (Z^\top Z + \sigma^2 G^{-1})^{-1}Z^\top.$$

The second block row gives $b = (Z^\top Z + \sigma^2 G^{-1})^{-1}Z^\top(y - X\beta) = GZ^\top V^{-1}(y - X\beta)$. Insert this into the first row: $X^\top\{I - ZGZ^\top V^{-1}\}(y - X\beta) = 0$, and $I - ZGZ^\top V^{-1} = (V - ZGZ^\top)V^{-1} = \sigma^2 V^{-1}$, so $X^\top V^{-1}(y - X\beta) = 0$: the GLS equation. Hence $\beta = \hat{\beta}$ and $b = \tilde{b}$. □

The criterion (12.7) is a **ridge-type** penalised least-squares problem: the random effects are shrunk by the penalty $b^\top G^{-1}b$, with penalty weight $\sigma^2 G^{-1}$ that is *estimated from the data*. This is the link to smoothing: a penalised spline is a mixed model (Chapters 15 and 19).

12.4.2 Shrinkage in the random-intercept model

For a random intercept, $Z_i = \mathbf{1}$ and $V_i = \sigma^2 I + \sigma_b^2 \mathbf{1}\mathbf{1}^\top$. The Sherman–Morrison formula gives $V_i^{-1} = \sigma^{-2}(I - \frac{\sigma_b^2}{\sigma^2 + n_i\sigma_b^2}\mathbf{1}\mathbf{1}^\top)$. With $r_i = y_i - X_i\hat\beta$, $\mathbf{1}^\top r_i = n_i \bar r_i$ and $\mathbf{1}^\top V_i^{-1} r_i = \sigma^{-2} n_i \bar r_i (1 - \frac{n_i\sigma_b^2}{\sigma^2 + n_i\sigma_b^2}) = \frac{n_i \bar r_i}{\sigma^2 + n_i\sigma_b^2}$. So

$$\tilde b_i = \sigma_b^2 \mathbf{1}^\top V_i^{-1} r_i = k_i\, \bar r_i, \qquad k_i = \frac{n_i\sigma_b^2}{n_i\sigma_b^2 + \sigma^2} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2/n_i} \in (0,1), \qquad \bar r_i = \bar y_{i\cdot} - \bar x_{i\cdot}^\top \hat\beta, \tag{12.9}$$

and the conditional variance is $\mathrm{Var}(b_i \mid y) = \sigma_b^2 - \sigma_b^4 \mathbf{1}^\top V_i^{-1}\mathbf{1} = \sigma_b^2(1 - k_i) = k_i\sigma^2/n_i$ (use $\mathbf{1}^\top V_i^{-1}\mathbf{1} = n_i/(\sigma^2 + n_i\sigma_b^2)$). Interpretation: $\bar r_i = b_i + \bar\varepsilon_i$ is the group's raw deviation, which has variance $\sigma_b^2 + \sigma^2/n_i$ of which σ_b^2 is "signal"; k_i is the **reliability** of the group mean. The raw deviation is multiplied by $k_i < 1$ (**shrinkage towards zero**): the less data a group has, or the noisier they are, the more its deviation is attributed to noise. $k_i \to 1$ if $n_i \to \infty$ or $\sigma_b^2 \gg \sigma^2$, and $k_i \to 0$ if $\sigma_b^2 \ll \sigma^2/n_i$.

```
vc_d <- as.data.frame(VarCorr(m_dy_reml))$vcov
k_dye <- 5 * vc_d[1] / (5 * vc_d[1] + vc_d[2])
raw_dye <- tapply(Dyestuff$Yield, Dyestuff$Batch, mean) - fixef(m_dy_reml)
round(rbind(raw = raw_dye, k_times_raw = k_dye * raw_dye, blup =
↪  ranef(m_dy_reml)$Batch[, 1]), 2)
```

```
#>                  A    B     C      D     E     F
#> raw          -22.50 0.50 36.50 -29.50 72.50 -57.5
#> k_times_raw -17.61 0.39 28.56 -23.08 56.73 -45.0
#> blup         -17.61 0.39 28.56 -23.08 56.73 -45.0
```

```
k_dye
```

```
#> [1] 0.7825
```

For `Dyestuff` the factor is $k = 5(1764)/(5(1764) + 2451) = 0.783$, the same for all batches (balanced data): batch E, with raw deviation 72.5 g, is predicted to deviate by 56.7 g. In unbalanced data the shrinkage differs by group:

```
sl_unb <- subset(sleepstudy, !(Subject %in% c("308", "309", "310")) | Days <=
↪  1)
m_unb <- lmer(Reaction ~ Days + (1 | Subject), sl_unb)
vcu <- as.data.frame(VarCorr(m_unb))$vcov
ni <- as.vector(table(sl_unb$Subject))
raw_u <- tapply(sl_unb$Reaction - model.matrix(~ Days, sl_unb) %*%
↪  fixef(m_unb), sl_unb$Subject, mean)
du <- data.frame(raw = as.numeric(raw_u), blup = ranef(m_unb)$Subject[, 1], n
↪  = factor(ni))
ku <- unique(round(ni * vcu[1] / (ni * vcu[1] + vcu[2]), 3)); ku
```

```
#> [1] 0.648 0.902
```

```
ggplot(du, aes(raw, blup, colour = n)) + geom_abline(slope = 1, intercept = 0,
↪  colour = "grey60") +
  geom_abline(slope = ku[1], intercept = 0, linetype = 2) + geom_abline(slope
  ↪  = ku[2], intercept = 0, linetype = 2) +
  geom_point(size = 2.3) + scale_colour_manual(values = c(ie_coral, ie_navy),
  ↪  name = "observations per subject") +
```

```
  labs(x = "raw deviation of the subject mean (ms)", y = "BLUP (ms)")
```

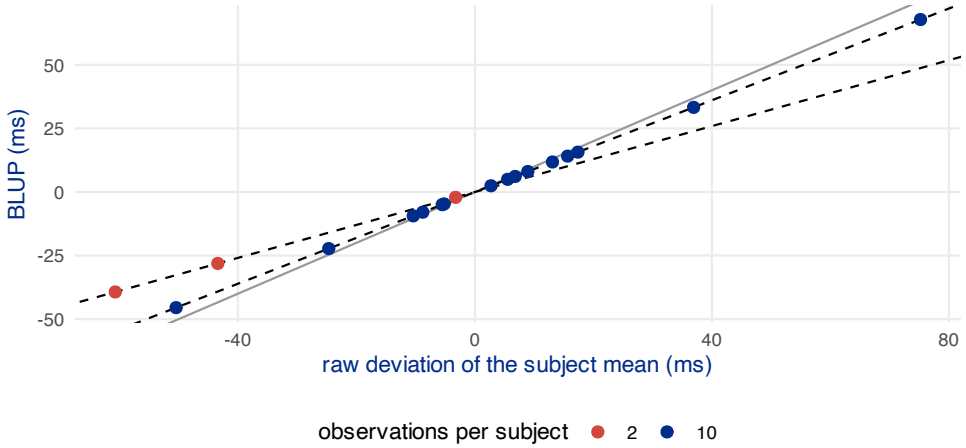


Figure 12.2: Unbalanced version of sleepstudy: subjects 308–310 keep only Days 0–1 (n = 2); the others keep all ten days. The BLUP equals k_i times the raw deviation, with $k_i = 0.76$ for $n_i = 2$ and 0.94 for $n_i = 10$ (dashed lines).

12.4.3 Shrinkage with random slopes

For a general group, $\tilde{b}_i = GZ_i^\top V_i^{-1} r_i = (Z_i^\top Z_i/\sigma^2 + G^{-1})^{-1} Z_i^\top r_i/\sigma^2$ by the push-through identity applied to one group. Let $\hat{b}_i^{OLS} = (Z_i^\top Z_i)^{-1} Z_i^\top r_i$ be the least-squares fit of group i alone to the deviations from the population line. Then

$$\tilde{b}_i = K_i \, \hat{b}_i^{OLS}, \qquad K_i = \left(Z_i^\top Z_i + \sigma^2 G^{-1}\right)^{-1} Z_i^\top Z_i, \tag{12.10}$$

a *matrix* shrinkage factor (for $Z_i = \mathbf{1}$ it reduces to k_i in (12.9)). The BLUP is the no-pooling estimate pulled towards zero, more along directions in which the group has little information ($Z_i^\top Z_i$ small) or in which the population variation G is small. We check it for subject 335 of `sleepstudy`:

```
m_pp <- lmer(Reaction ~ Days + (Days | Subject), sleepstudy)
G <- unclass(VarCorr(m_pp)$Subject)[1:2, 1:2]; s2 <- sigma(m_pp)^2
d335 <- subset(sleepstudy, Subject == "335"); Zi <- cbind(1, d335$Days)
b_ols <- solve(crossprod(Zi), crossprod(Zi, d335$Reaction - Zi %*%
↪  fixef(m_pp)))
K <- solve(crossprod(Zi) + s2 * solve(G), crossprod(Zi))
cbind(K_times_ols = drop(K %*% b_ols), ranef =
↪  unlist(ranef(m_pp)$Subject["335", ]))
```

```
#>             K_times_ols    ranef
#> (Intercept)     -0.3337  -0.3337
#> Days           -10.7522 -10.7522
```

```
round(K, 3)
```

```
#>              [,1]  [,2]
#> (Intercept) 0.751 0.680
#> Days        0.038 0.838
```

As a consequence the predictions have **less spread** than the population distribution: $\mathrm{Var}(\tilde{b}) = GZ^\top V^{-1} ZG \preceq G$. In `sleepstudy` the standard deviation of the predicted slope deviations is

5.46 ms/day, that of the no-pooling slopes 6.56, while the estimate of σ_{b1} is 5.92. Predicted random effects must therefore not be treated as a sample from $N(0, G)$: they are shrunk and correlated.

12.4.4 Uncertainty and prediction for new data

The conditional variance quantifies the uncertainty of the prediction: $\mathrm{Var}(b_i \mid y) = (Z_i^\top Z_i/\sigma^2 + G^{-1})^{-1}$ (take V_i^{-1} to the push-through form), available from `ranef(fit, condVar = TRUE)`. This accounts for the variability of b_i given y but not for the estimation of β and ϑ. To predict a *new observation* $y_{\text{new}} = x_0^\top \beta + z_0^\top b_{\text{new}} + \varepsilon$ from a **new subject** ($b_{\text{new}} \sim N(0, G)$ independent of the data), the point prediction is $x_0^\top \hat\beta$ and the variance

$$\mathrm{Var}(y_{\text{new}} - x_0^\top \hat\beta) = x_0^\top (X^\top V^{-1} X)^{-1} x_0 + z_0^\top G z_0 + \sigma^2. \tag{12.11}$$

For an **existing subject** one uses $x_0^\top \hat\beta + z_0^\top \tilde b_i$ and the conditional variance of b_i instead of G in the middle term (plus a covariance term between $\hat\beta$ and $\tilde b_i$ that we neglect).

12.5 Inference for the fixed effects

By Theorem 12.1, $\mathrm{Var}(\hat\beta) = (X^\top V^{-1} X)^{-1}$, estimated by plugging in $\hat\vartheta$; the Wald statistic is $t_k = \hat\beta_k / \mathrm{se}(\hat\beta_k)$. Two caveats:

- The standard error treats $\hat\vartheta$ as known, so it is slightly optimistic, notably with few groups. A Wald (z) interval $\hat\beta_k \pm 1.96\,\mathrm{se}$ is then **anti-conservative** (too short, actual coverage below 95%): with m groups the t interval with the appropriate degrees of freedom (exactly $m-1$ for the mean in a balanced one-way layout) is wider. For the six `Dyestuff` batches, $1527.5 \pm 1.96(19.4) = [1489.5, 1565.5]$ but $1527.5 \pm 2.571(19.4) = [1477.7, 1577.3]$.
- Even with the true V, $\hat\beta_k/\mathrm{se}$ is exactly normal; with estimated V its distribution is *not* exactly Student t with a known number of degrees of freedom (except in balanced layouts). This is why `lme4` prints t values but no p-values.

Options for tests and intervals are: (i) the normal approximation (many groups); (ii) degrees-of-freedom approximations, Satterthwaite or Kenward–Roger (Kenward and Roger, 1997) (packages `lmerTest`, `pbkrtest`); (iii) the **likelihood-ratio test with ML fits**, $T = 2\{\ell_{ML}(\text{full}) - \ell_{ML}(\text{null})\} \dot\sim \chi^2_d$; (iv) profile-likelihood intervals; (v) the **parametric bootstrap**: simulate from the fitted null model and refit.

```
cf <- summary(m_pp)$coefficients; cf
```

```
#>             Estimate Std. Error t value
#> (Intercept)   251.41      6.825  36.838
#> Days           10.47      1.546   6.771
```

```
c(t = cf["Days", "t value"], p_normal = 2 * pnorm(-abs(cf["Days", "t
↪  value"])))
```

```
#>         t  p_normal
#> 6.771e+00 1.275e-11
```

```
m1_ml <- update(m_pp, REML = FALSE); m0_ml <- update(m1_ml, . ~ . - Days)
anova(m0_ml, m1_ml)                          # LRT for Days, ML fits
```

```
#> Data: sleepstudy
```

```
#> Models:
#> m0_ml: Reaction ~ (Days | Subject)
#> m1_ml: Reaction ~ Days + (Days | Subject)
#>       npar  AIC  BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
#> m0_ml    5 1785 1801   -888      1775
#> m1_ml    6 1764 1783   -876      1752  23.5  1    1.2e-06 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
2 * (logLik(m_pp) - logLik(update(m_pp, . ~ . - Days)))   # NOT a valid LRT:
↪  two REML fits with different X
```

```
#> 'log Lik.' 26.22 (df=6)
```

```
confint(update(m_pp), parm = "Days", method = "profile")     # on a copy:
↪  profiling changes the internal state of a merMod fit
```

```
#>      2.5 % 97.5 %
#> Days 7.359  13.58
```

```
fit_t <- lmerTest::lmer(Reaction ~ Days + (Days | Subject), sleepstudy)
summary(fit_t)$coefficients                                  # Satterthwaite
↪  degrees of freedom
```

```
#>             Estimate Std. Error df t value  Pr(>|t|)
#> (Intercept)   251.41      6.825 17  36.838 1.172e-17
#> Days           10.47      1.546 17   6.771 3.264e-06
```

```
pbkrtest::KRmodcomp(m_pp, update(m_pp, . ~ . - Days))$stats    # Kenward-Roger
↪  F test
```

```
#> $ndf
#> [1] 1
#>
#> $ddf
#> [1] 17
#>
#> $Fstat
#> [1] 45.85
#>
#> $p.value
#> [1] 3.264e-06
#>
#> $F.scaling
#> [1] 1
#>
#> $FstatU
#> [1] 45.85
#>
#> $p.valueU
#> [1] 3.264e-06
#>
#> $aux
```

```
#>        A1        A2        V0        V1        V2       rho F.scaling
#>    0.1176    0.1176    0.9412    0.8824    0.7647    1.2308    1.0000
```

With 18 subjects, $t = 6.77$ and the Satterthwaite degrees of freedom are about 17, so the normal and the t approximations give essentially the same conclusion (the t_{17} interval for the slope is [7.2, 13.7] against [7.4, 13.5] for z): reaction time increases with sleep restriction (LRT $T = 23.5$ on 1 df; the Wald $t^2 = 45.8$ is a different approximation, and here the two differ noticeably: the null model of the LRT keeps the random slope, whose variance inflates to absorb part of the average slope). The 95% profile interval for the slope is [7.36, 13.58] ms/day, wider than the naive $10.47 \pm 1.96(1.55) = [7.4, 13.5]$ only slightly. Comparing two REML fits with different fixed effects (second line) gives 26.2, which is not a likelihood-ratio statistic, since the two REML likelihoods are for different error contrasts.

12.6 Inference for variance components

Wald intervals for variance components are poor: the sampling distribution of $\hat{\sigma}_b^2$ is skewed, bounded at 0, and the standard error of a variance estimate relies on a large number of groups. Likelihood-based intervals are better.

```
confint(update(m_dy_reml), method = "profile")      # Dyestuff: only 6
↪  batches
```

```
#>                2.5 %  97.5 %
#> .sig01         12.20   84.06
#> .sigma         38.23   67.66
#> (Intercept) 1486.45 1568.55
```

```
confint(update(m_pp), method = "profile")           # sleepstudy: sig01 (sd
↪  intercept), sig02 (corr), sig03 (sd slope)
```

```
#>                 2.5 %  97.5 %
#> .sig01        14.3814  37.717
#> .sig02         3.8012   8.753
#> .sig03        -0.4815   0.685
#> .sigma        22.8983  28.858
#> (Intercept) 237.6807 265.130
#> Days          7.3587  13.576
```

For `Dyestuff` the 95% interval for σ_b is about [12, 84] g: six batches carry little information on the batch standard deviation. The interval for the correlation ρ in `sleepstudy`, $[-0.48, 0.68]$, shows that nothing can be said about ρ.

12.6.1 The boundary problem

Test $H_0 : \sigma_b^2 = 0$ against $H_1 : \sigma_b^2 > 0$ in a random-intercept model. The null value lies on the **boundary** of the parameter space $[0, \infty)$, so the usual regularity conditions of Chapter 4 fail and the LRT is *not* χ^2_1.

Theorem 12.5 (Likelihood-ratio test on the boundary)**.** *Let $T = 2\{\ell(\hat{\sigma}_b^2, \hat{\gamma}) - \ell(0, \hat{\gamma}_0)\}$ be the LRT statistic for $H_0 : \sigma_b^2 = 0$ in a regular model with $\sigma_b^2 \geq 0$ and a nuisance parameter*

γ. *Under* H_0, *as the number of groups grows,*

$$T \xrightarrow{d} \left(\max\{Z,0\}\right)^2, \quad Z \sim N(0,1), \qquad \textit{i.e.} \qquad T \;\dot{\sim}\; \tfrac{1}{2}\chi^2_0 + \tfrac{1}{2}\chi^2_1,$$

so $P(T > t) = \frac{1}{2}P(\chi^2_1 > t)$ *for* $t > 0$ *and the 5% critical value is* $\chi^2_{1,0.90} = 2.71$, *not* 3.84.

Proof. (Sketch under the usual regularity.) Orthogonalise the nuisance parameter so that γ and σ_b^2 are asymptotically independent and consider the profile log-likelihood $\ell_P(s)$ of $s = \sigma_b^2$. At $s = 0$ let S be its score and $I > 0$ its information, so $S/\sqrt{I} \xrightarrow{d} Z \sim N(0,1)$. By the quadratic approximation, $\ell_P(s) - \ell_P(0) \approx Ss - \frac{1}{2}Is^2$ for small s. Maximising over $s \geq 0$: if $S > 0$ the maximum is at $\hat{s} = S/I$ and $T = S^2/I = Z^2$; if $S \leq 0$ the maximum over $[0, \infty)$ is at the boundary $\hat{s} = 0$ and $T = 0$. Hence $T \to \{\max(Z,0)\}^2$, which equals 0 with probability $\frac{1}{2}$ and equals $Z^2 \sim \chi^2_1$ conditionally on $Z > 0$: the mixture. □

The mixture is a statement about *one* variance component. When a correlated random slope is added to a random intercept, two parameters (σ^2_{b1} and ρ, or equivalently the extra row and column of G) are tested and the reference distribution is $\frac{1}{2}\chi^2_1 + \frac{1}{2}\chi^2_2$ (Self and Liang, 1987; Stram and Lee, 1994); in general, testing q against $q+1$ correlated random effects gives $\frac{1}{2}\chi^2_q + \frac{1}{2}\chi^2_{q+1}$. The naive χ^2 with the full degrees of freedom gives p-values that are too large (conservative).

```
m_lm <- lm(Reaction ~ Days, sleepstudy)
T_int <- as.numeric(2 * (logLik(m_ri_ml) - logLik(m_lm)))                    #
↪  random intercept vs none (ML)
c(T = T_int, p_naive = pchisq(T_int, 1, lower.tail = FALSE), p_mixture = .5 *
↪  pchisq(T_int, 1, lower.tail = FALSE))
```

```
#>         T   p_naive p_mixture
#> 1.062e+02 6.617e-25 3.309e-25
```

```
a <- anova(m_ri_reml, m_pp, refit = FALSE)                                   #
↪  random slope + corr vs intercept only
a
```

```
#> Data: sleepstudy
#> Models:
#> m_ri_reml: Reaction ~ Days + (1 | Subject)
#> m_pp: Reaction ~ Days + (Days | Subject)
#>           npar  AIC  BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
#> m_ri_reml    4 1794 1807   -893      1786
#> m_pp         6 1756 1775   -872      1744  42.8  2      5e-10 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
T_sl <- a$Chisq[2]
c(p_naive = pchisq(T_sl, 2, lower.tail = FALSE), p_mixture = .5 * pchisq(T_sl,
↪  1, lower.tail = FALSE) + .5 * pchisq(T_sl, 2, lower.tail = FALSE))
```

```
#>   p_naive p_mixture
#> 4.990e-10 2.793e-10
```

Here both tests reject overwhelmingly (the random intercept: $T = 106$; the random slope and correlation: $T = 42.8$), so the correction does not change the conclusion; it matters when T is

moderate (Exercise 12.8). The REML fits in `anova(..., refit = FALSE)` may be compared because the two models have the same fixed effects, hence the same error contrasts. A hypothesis such as $\rho = 0$ *alone* (`(Days || Subject)` versus `(Days | Subject)`) is an interior test ($\rho = 0$ lies inside $(-1, 1)$) and the ordinary χ^2_1 applies.

The asymptotic mixture can be checked by simulation: generate data under H_0 from a fitted model, refit with and without the random intercept, and look at the distribution of T (a *parametric bootstrap*). With 30 groups of 5 observations:

```
set.seed(2027)
g30 <- factor(rep(1:30, each = 5)); x30 <- rep(1:5, 30)
T_sim <- replicate(500, {
  ys <- 2 + .3 * x30 + rnorm(150); d <- data.frame(ys, x30, g30)
  f1 <- suppressMessages(lmer(ys ~ x30 + (1 | g30), d, REML = FALSE))
  max(0, as.numeric(2 * (logLik(f1) - logLik(lm(ys ~ x30, d)))))
})
c(share_zero = mean(T_sim < 1e-6), q95 = unname(quantile(T_sim, .95)),
↪  q95_mixture = qchisq(.90, 1), q95_naive = qchisq(.95, 1))
```

```
#>  share_zero         q95 q95_mixture   q95_naive
#>       0.582       1.789       2.706       3.841
```

```
ggplot(data.frame(T = T_sim), aes(T)) + geom_histogram(bins = 40, fill =
↪  ie_blue, alpha = .7) +
  geom_vline(xintercept = c(qchisq(.9, 1), qchisq(.95, 1)), linetype = 2) +
  ↪  labs(x = "LRT statistic", y = "count")
```

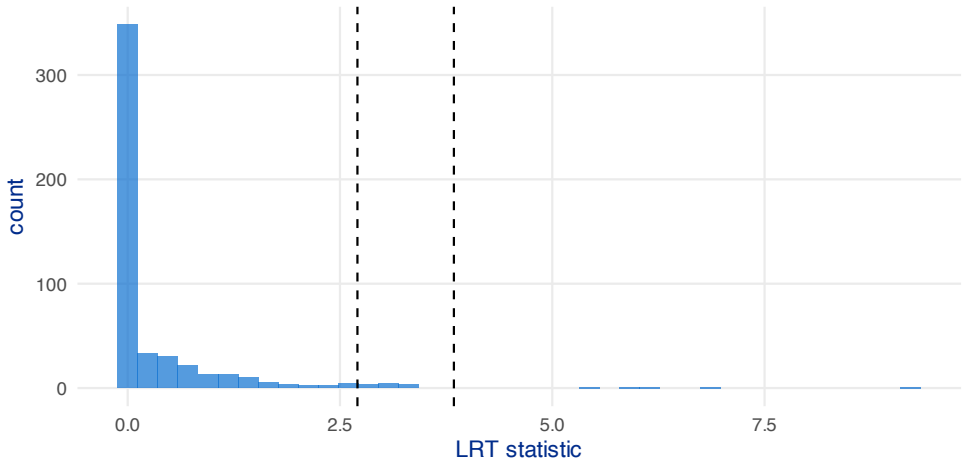


Figure 12.3: Parametric bootstrap distribution of the LRT statistic for $\sigma^2_b = 0$ (30 groups of 5 observations, 500 replicates, data generated under the null). Dashed: 5% critical values 2.71 (mixture) and 3.84 (naive chi-square).

A majority of the simulated statistics are exactly zero (58%, against 50% asymptotically) and the simulated 95% quantile (1.79) lies *below* both asymptotic values 2.71 and 3.84. The asymptotic mixture is thus itself conservative in samples of this size (the loss of degrees of freedom from estimating β and the discreteness at zero both matter); p-values obtained from it are on the safe side, and the exact alternative is the parametric bootstrap, as in Exercise 12.9.

Table 12.1: Choosing between ML and REML.

Task	Use
Report variance components	REML (default in lmer and lme)
Compare models with different fixed effects (LRT, AIC, BIC)	ML (REML = FALSE); anova() refits with ML automatically
Compare models with the same fixed effects, different random effects	REML or ML, the same for both models

12.7 Model comparison

Which likelihood to use depends on what is compared.

The reason for the second row is the last observation after Theorem 12.2: the REML likelihood is the likelihood of $A^\top y$ with $A^\top X = 0$, and when X changes so does A.

AIC and BIC. AIC $= -2\ell(\hat\beta, \hat\vartheta) + 2k$ with k the number of fixed effects plus variance parameters. AIC in this form is the *marginal* AIC, appropriate when the target is a new subject; it is not exactly justified when parameters lie on the boundary, so differences smaller than about 2 should be treated as ties. For the three random-effects structures of `sleepstudy`:

```
m_unc <- lmer(Reaction ~ Days + (Days || Subject), sleepstudy)       # rho =
↪  0
data.frame(model = c("(1 | Subject)", "(Days || Subject), rho = 0", "(Days |
↪  Subject)"),
           npar = c(4, 5, 6), AIC = AIC(m_ri_reml, m_unc, m_pp)$AIC, BIC =
           ↪  BIC(m_ri_reml, m_unc, m_pp)$BIC) |>
  transform(AIC = round(AIC, 1), BIC = round(BIC, 1))
```

```
#>                       model npar  AIC  BIC
#> 1               (1 | Subject)    4 1794 1807
#> 2 (Days || Subject), rho = 0    5 1754 1770
#> 3           (Days | Subject)    6 1756 1775
```

```
anova(m_unc, m_pp, refit = FALSE)
```

```
#> Data: sleepstudy
#> Models:
#> m_unc: Reaction ~ Days + ((1 | Subject) + (0 + Days | Subject))
#> m_pp: Reaction ~ Days + (Days | Subject)
#>       npar  AIC  BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
#> m_unc    5 1754 1770   -872      1744
#> m_pp     6 1756 1775   -872      1744  0.04  1       0.84
```

The random slope is essential (AIC drops by 41), while the correlation adds nothing: the likelihood-ratio statistic of $\rho = 0$ is 0.04 ($p = 0.84$) and the uncorrelated model has the smaller AIC. A practical strategy is to start from the *maximal* random-effects structure that the design supports, simplify only when the fit is singular or the optimiser does not converge, and keep a random slope for any covariate that varies within groups and whose effect is of interest (ignoring a real random slope makes the standard error of the fixed slope too small).

12.8 Diagnostics

The LMM has assumptions at two levels: $\varepsilon \sim N(0, \sigma^2 I)$ within groups and $b \sim N(0, G)$ between groups, plus the linearity of the mean. Correspondingly there are several residuals.

- **Marginal residuals** $y - X\hat\beta$ have covariance $\approx V$: they are correlated within groups and are used for the mean structure only.
- **Conditional residuals** $\hat e = y - X\hat\beta - Z\tilde b$ estimate ε (`resid()` in `lme4`, level 1).
- **Predicted random effects** $\tilde b$ (`ranef()`, level 2).

Proposition 12.2 (Conditional residuals)**.** *$\hat e = \sigma^2 P y$ with P as in Lemma 12.1, and $\mathrm{Var}(\hat e) = \sigma^4 P$ (with ϑ known). In particular the conditional residuals are **not** homoscedastic, $\mathrm{Var}(\hat e_j) = \sigma^4 P_{jj} \le \sigma^2$, and are correlated; the* studentised *residuals $\hat e_j / (\sigma^2 \sqrt{P_{jj}})$ have unit variance.*

Proof. $Z\tilde b = ZGZ^\top V^{-1}(y - X\hat\beta)$, so $\hat e = \{I - ZGZ^\top V^{-1}\}(y - X\hat\beta) = \sigma^2 V^{-1}(y - X\hat\beta) = \sigma^2 P y$, using $I - ZGZ^\top V^{-1} = \sigma^2 V^{-1}$. Also $PVP = P$ (because $X^\top P = 0$ and $PV = I - V^{-1}X(X^\top V^{-1}X)^{-1}X^\top$), so $\mathrm{Var}(\sigma^2 P y) = \sigma^4 PVP = \sigma^4 P$. Finally $\sigma^2 P = V^{-1}\sigma^2 - \cdots \preceq \sigma^2 I$ since $P \preceq V^{-1} \preceq \sigma^{-2} I$. □

The raw conditional residuals scaled by $\hat\sigma$ (`resid(fit, scaled = TRUE)`) are the standard diagnostics; the studentised version corrects for the heteroscedasticity above. The predicted random effects are *shrunk*, so their spread underestimates G and their QQ plot is only a rough check, especially with few groups.

```
res <- resid(m_pp); fit <- fitted(m_pp); re <- ranef(m_pp)$Subject
op <- par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
plot(fit, res, xlab = "fitted", ylab = "conditional residual", main =
↪  "Residuals vs fitted"); abline(h = 0, lty = 2)
qqnorm(res, main = "Residuals"); qqline(res)
qqnorm(re[, 1], main = "BLUPs: intercepts"); qqline(re[, 1])
qqnorm(re[, 2], main = "BLUPs: slopes"); qqline(re[, 2])
```

```
par(op)
head(sleepstudy[order(-abs(resid(m_pp, scaled = TRUE))), ], 3)
```

```
#>     Reaction Days Subject
#> 57    454.2    6     332
#> 8     290.1    7     308
#> 60    253.9    9     332
```

The residual QQ plot shows a heavy upper tail caused by a few observations (the largest, 454 ms for subject 332 on Day 6, has a standardised residual of 5.2); the BLUPs look roughly normal, as far as 18 points can tell. We verify Proposition 12.2:

```
Gm <- kronecker(diag(18), G); Zm <- as.matrix(getME(m_pp, "Z")); Xm <-
↪  getME(m_pp, "X")
Vm <- Zm %*% Gm %*% t(Zm) + s2 * diag(180); Vi <- solve(Vm)
Pm <- Vi - Vi %*% Xm %*% solve(t(Xm) %*% Vi %*% Xm, t(Xm) %*% Vi)
c(max_abs_diff = max(abs(resid(m_pp) - s2 * drop(Pm %*%
↪  sleepstudy$Reaction))))
```

```
#> max_abs_diff
```

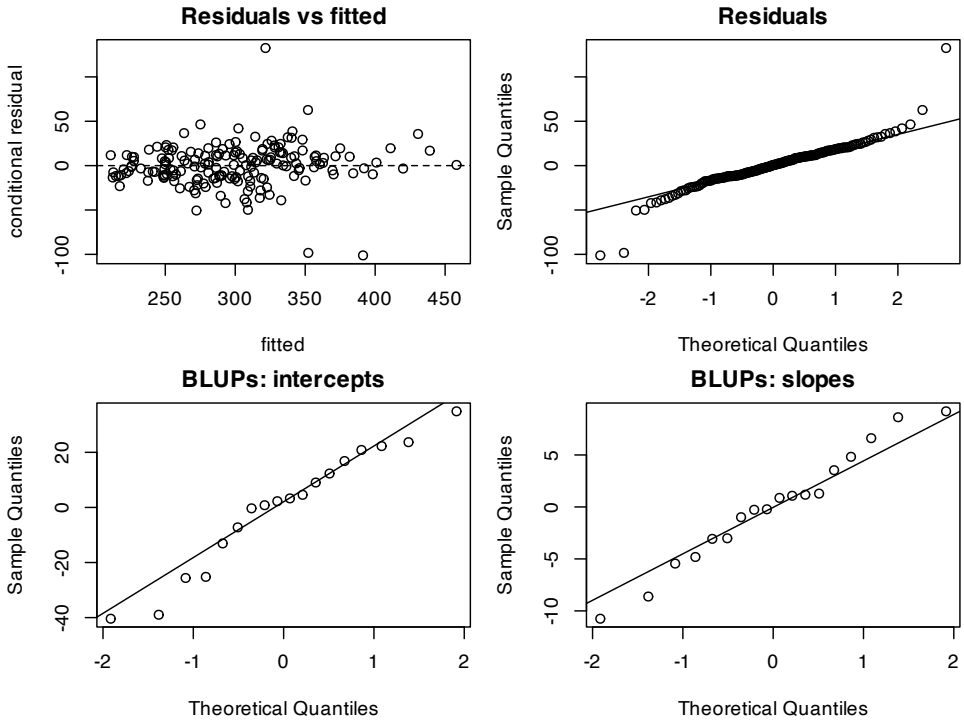


Figure 12.4: Diagnostics for the random-slope model of sleepstudy. Top: conditional residuals against fitted values and their normal QQ plot; bottom: normal QQ plots of the predicted random intercepts and slopes.

```
#>    1.071e-12
```

```
summary(sqrt(s2^2 * diag(Pm)) / sqrt(s2))      # sd(e_j)/sigma: below 1
```

```
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>   0.841   0.884   0.925   0.915   0.948   0.953
```

The ratio $\sqrt{\mathrm{Var}(\hat{e}_j)}/\sigma$ ranges between 0.84 and 0.95 in this data set: the conditional residuals are slightly, but not dramatically, heteroscedastic because every subject has ten observations and the random effects absorb a part of the variation.

12.9 In R

```
fit_reml <- lmer(y ~ x + (x | g), data = d)              # REML (default)
fit_ml   <- update(fit_reml, REML = FALSE)               # ML, for LRT / AIC
↪  with different fixed effects
VarCorr(fit_reml); fixef(fit_reml); vcov(fit_reml)       # variance
↪  components, beta-hat, Var(beta-hat)
ranef(fit_reml, condVar = TRUE)                          # BLUPs with
↪  conditional variances
anova(fit_small, fit_big)                                # LRT; refits with
↪  ML automatically
anova(fit_a, fit_b, refit = FALSE)                       # keep REML: same
↪  fixed effects
confint(fit_reml, method = "profile")                    # profile intervals
↪  (also "boot")
```

```
bootMer(fit_reml, FUN = fixef, nsim = 500)                  # parametric
↪  bootstrap
resid(fit_reml, scaled = TRUE); plot(fit_reml)              # level-1 residuals
lme(y ~ x, random = ~ x | g, data = d, method = "REML")   # same model in nlme;
↪  intervals(fit)
```

`nlme::lme()` reports degrees-of-freedom-based p-values using a "within/between" rule and has `intervals()`, `anova()`, `plot()` and `qqnorm(fit, ~ ranef(.))` for diagnostics. `lme4::lmer()` is faster for large and crossed designs; `nlme` supports correlation and variance functions (`corAR1`, `varIdent`) for the within-group errors.

12.10 Common pitfalls

Warning. Do not compare models with different fixed effects using REML likelihoods or REML-based AIC.

Warning. "REML is unbiased" is false: it is less biased than ML and unbiased only in balanced ANOVA cases.

Warning. Do not test a variance component with χ^2_1 (p-values too large) and do not conclude that $\sigma_b^2 = 0$ means "no group effect" when there are few groups.

Warning. Do not call b "estimated" or treat BLUPs as a random sample from $N(0, G)$; they are conditional means, shrunk towards zero.

Warning. Writing the REML criterion with $+\frac{1}{2}\log|X^\top V^{-1}X|$ or $n\log 2\pi$ instead of $(n-p)\log 2\pi$ is an error; so is reading `logLik()` of `nAGQ = 1` and `nAGQ > 1` GLMM fits as comparable (Chapter 13).

12.11 Chapter summary

- The marginal likelihood is $y \sim N(X\beta, V(\vartheta))$. For fixed ϑ, $\hat\beta = (X^\top V^{-1}X)^{-1}X^\top V^{-1}y$ (GLS, BLUE) with variance $(X^\top V^{-1}X)^{-1}$; ϑ is estimated from the profile likelihood.
- ML underestimates variances ($\mathrm{E}[\mathrm{RSS}] = \sigma^2(n-p)$). REML maximises the likelihood of error contrasts: $\ell_R = -\frac{1}{2}[(n-p)\log 2\pi + \log|V| + \log|X^\top V^{-1}X| + r^\top V^{-1} r]$, equal to the log of the β-integrated likelihood; it gives $\mathrm{RSS}/(n-p)$ in the linear model and the ANOVA estimators in balanced one-way layouts. It is less biased, not unbiased.
- Use REML to report variance components and to compare random-effects structures with the same fixed effects; use ML to compare different fixed effects.
- Random effects are predicted: $\tilde b = GZ^\top V^{-1}(y - X\hat\beta) = \mathrm{E}[b \mid y]$, the BLUP; Henderson's equations give $(\hat\beta, \tilde b)$ as a ridge-type penalised least-squares solution. In a random-intercept model $\tilde b_i = k_i \bar r_i$ with $k_i = n_i\sigma_b^2/(n_i\sigma_b^2 + \sigma^2)$: shrinkage towards zero, stronger for small or noisy groups.
- Fixed effects: Wald/t, LRT (ML), profile and bootstrap intervals, Satterthwaite or Kenward–Roger. Variance components: profile intervals; the LRT for $\sigma_b^2 = 0$ has the reference $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ (critical value 2.71); adding a correlated random slope gives $\frac{1}{2}\chi^2_1 + \frac{1}{2}\chi^2_2$.
- Check both levels: conditional residuals (within) and predicted random effects (between); BLUP plots are only a rough check.

12.12 Exercises with explained solutions

Exercise 12.1. Consider $y_{ij} = \mu + b_i + \varepsilon_{ij}$ with m groups of unequal sizes n_i. (a) Derive the GLS estimator of μ and its variance. (b) Compute the GLS and OLS estimates and their variances for two groups with $n_1 = 1$, $n_2 = 4$, group means $\bar{y}_1 = 10$, $\bar{y}_2 = 12$, $\sigma_b^2 = 2$, $\sigma^2 = 1$. (c) Which is more efficient and why?

Solution.

(a) The group i vector has covariance $V_i = \sigma^2 I + \sigma_b^2 \mathbf{1}\mathbf{1}^\top$, with $\mathbf{1}^\top V_i^{-1}\mathbf{1} = n_i/(\sigma^2 + n_i\sigma_b^2) = 1/(\sigma_b^2 + \sigma^2/n_i) = w_i$ and $\mathbf{1}^\top V_i^{-1} y_i = n_i\bar{y}_{i\cdot}/(\sigma^2 + n_i\sigma_b^2) = w_i\bar{y}_{i\cdot}$ (Sherman–Morrison). With $X = \mathbf{1}$, $\hat{\mu} = (\sum_i w_i)^{-1}\sum_i w_i\bar{y}_{i\cdot}$ and $\mathrm{Var}(\hat{\mu}) = (\sum w_i)^{-1}$. (b) $w_1 = 1/(2+1) = 0.333$, $w_2 = 1/(2+0.25) = 0.444$; $\hat{\mu}_{GLS} = (0.333\cdot 10 + 0.444\cdot 12)/0.778 = 11.14$, Var $= 1/0.778 = 1.286$. OLS is the mean of all five observations, $\hat{\mu}_{OLS} = (10 + 4\cdot 12)/5 = 11.6$ with Var $= \sum_i n_i^2(\sigma_b^2 + \sigma^2/n_i)/25 = \{1(3) + 16(2.25)\}/25 = 1.56$. (c) GLS is more efficient ($1.286 < 1.56$): the large group is worth *less* than n_2 times a single observation, because its b_2 is a single draw; OLS overweights it (weight 0.8 versus 0.571 for GLS).

Exercise 12.2. Show that the REML estimator of σ^2 in the linear model $y = X\beta + \varepsilon$ is $\mathrm{RSS}/(n-p)$, directly from the error contrasts of Definition 12.1.

Solution. Take A with orthonormal columns spanning $\{v : X^\top v = 0\}$, so $AA^\top = I - H$. Then $w = A^\top y = A^\top\varepsilon \sim N(0, \sigma^2 A^\top A) = N(0, \sigma^2 I_{n-p})$. The log-likelihood of w is $-\frac{n-p}{2}\log\sigma^2 - \frac{w^\top w}{2\sigma^2} + c$ and $w^\top w = y^\top AA^\top y = y^\top(I-H)y = \mathrm{RSS}$. Setting the derivative in σ^2 to zero gives $\hat{\sigma}^2 = \mathrm{RSS}/(n-p)$, which is unbiased because $\mathrm{E}[\mathrm{RSS}] = \sigma^2(n-p)$.

Exercise 12.3. A balanced one-way design has $m = 8$ groups of $n = 4$, MSB $= 60$, MSW $= 12$. (a) Give the REML and ML estimates of σ^2, σ_b^2 and of the ICC. (b) What do the formulas give if MSB $= 10$, MSW $= 12$? (c) Give $\hat{k}$ and the BLUP of a group with mean 9 above the grand mean (REML).

Solution.

(a) REML: $\hat{\sigma}^2 = 12$, $\hat{\sigma}_b^2 = (60-12)/4 = 12$, ICC $= 0.5$. ML: $\hat{\sigma}^2 = 12$, $\hat{\sigma}_b^2 = (\frac{7}{8}\cdot 60 - 12)/4 = (52.5 - 12)/4 = 10.125$, ICC $= 0.458$. (b) REML: $(10-12)/4 < 0$, so $\hat{\sigma}_b^2 = 0$ (boundary); the ANOVA estimator would be negative, which has no meaning as a variance. ML: $(\frac{7}{8}\cdot 10 - 12)/4 < 0$, also 0. Then $\hat{\sigma}^2$ should be recomputed as the pooled variance of all observations, $(\mathrm{SSW} + \mathrm{SSB})/(mn-1)$ for REML (with μ removed), since the model with $\sigma_b^2 = 0$ is the ordinary mean model. (c) $k = 4(12)/(4(12) + 12) = 0.8$, BLUP $= 0.8 \times 9 = 7.2$.

Exercise 12.4. (R.) Check Lemma 12.1 numerically: generate a random positive definite V ($n = 8$) and an 8×3 matrix X, build an orthonormal A with $A^\top X = 0$ (e.g. from the QR decomposition of X) and verify (i) $A(A^\top VA)^{-1}A^\top = P$ and (ii) $|A^\top VA| = |V||X^\top V^{-1}X|/|X^\top X|$. Then show that for another choice $A_2 = AB$, the criterion ℓ_C changes by a constant that does not depend on V.

Solution.

```
set.seed(11); n8 <- 8
W <- matrix(rnorm(n8^2), n8); V8 <- crossprod(W) + diag(n8)        # random
↪  positive definite V
```

```
X8 <- cbind(1, matrix(rnorm(n8 * 2), n8))
A8 <- qr.Q(qr(X8), complete = TRUE)[, -(1:3)]                    #
↪  orthonormal basis of the complement of col(X)
max(abs(t(A8) %*% X8))                                            # A'X = 0
```

```
#> [1] 1.232e-16
```

```
Vi8 <- solve(V8); P8 <- Vi8 - Vi8 %*% X8 %*% solve(t(X8) %*% Vi8 %*% X8, t(X8)
↪  %*% Vi8)
max(abs(A8 %*% solve(t(A8) %*% V8 %*% A8) %*% t(A8) - P8))          # identity
↪  (i)
```

```
#> [1] 5.551e-17
```

```
c(det_left = determinant(t(A8) %*% V8 %*% A8)$modulus,
  det_right = determinant(V8)$modulus + determinant(t(X8) %*% Vi8 %*%
  ↪  X8)$modulus - determinant(crossprod(X8))$modulus)  # (ii) on log scale
```

```
#>  det_left det_right
#>     10.21     10.21
```

Both identities hold to rounding error. For $A_2 = AB$ (with B $(n-p)\times(n-p)$ invertible), $A_2^\top V A_2 = B^\top (A^\top V A) B$ and $A_2(A_2^\top V A_2)^{-1} A_2^\top = ABB^{-1}(A^\top V A)^{-1} B^{-\top} B^\top A^\top = A(A^\top V A)^{-1} A^\top$, so the quadratic form is unchanged while $\log|A_2^\top V A_2| = \log|A^\top V A| + 2\log|B|$: the criterion shifts by the constant $-\log|B|$, free of V and hence of ϑ.

Exercise 12.5. In a random-intercept model, $\hat\sigma_b^2 = 36$ and $\hat\sigma^2 = 100$. Three groups all have raw deviation $\bar r_i = +12$, with $n_i = 1, 4, 25$. (a) Compute k_i and the three BLUPs. (b) Compute the conditional variance of b_i for each. (c) Explain why a group with $n_i = 1$ and $\bar r_i = -30$ gets a BLUP of only -7.9.

Solution.

(a) $k_1 = 36/136 = 0.265$, $k_4 = 144/244 = 0.590$, $k_{25} = 900/1000 = 0.9$; BLUPs 3.18, 7.08, 10.8. (b) $\mathrm{Var}(b_i \mid y) = k_i\sigma^2/n_i = 0.265\cdot 100/1 = 26.5$, $0.590\cdot 100/4 = 14.8$, $0.9\cdot 100/25 = 3.6$ (compared with the prior variance 36). The more data, the more certain. (c) $0.265\times(-30) = -7.9$: a single observation of a group is a noisy measurement, since $\mathrm{Var}(\bar r_1) = \sigma_b^2 + \sigma^2 = 136$ of which only 36 is genuine group variation; the extreme value is mostly attributed to noise (regression to the mean).

Exercise 12.6. Prove the push-through identity $GZ^\top(ZGZ^\top + \sigma^2 I)^{-1} = (Z^\top Z + \sigma^2 G^{-1})^{-1} Z^\top$ and use it to show that, for a single group with design matrix Z_i, the BLUP is $\tilde b_i = K_i \hat b_i^{OLS}$ with $K_i = (Z_i^\top Z_i + \sigma^2 G^{-1})^{-1} Z_i^\top Z_i$, and that $K_i \to I$ when $Z_i^\top Z_i \to \infty$.

Solution. $(Z^\top Z + \sigma^2 G^{-1}) GZ^\top = Z^\top ZGZ^\top + \sigma^2 Z^\top = Z^\top(ZGZ^\top + \sigma^2 I)$. Multiplying on the left by $(Z^\top Z + \sigma^2 G^{-1})^{-1}$ and on the right by $(ZGZ^\top + \sigma^2 I)^{-1}$ gives the identity. For one group, $\tilde b_i = GZ_i^\top V_i^{-1} r_i = (Z_i^\top Z_i + \sigma^2 G^{-1})^{-1} Z_i^\top r_i = (Z_i^\top Z_i + \sigma^2 G^{-1})^{-1}(Z_i^\top Z_i)(Z_i^\top Z_i)^{-1} Z_i^\top r_i = K_i \hat b_i^{OLS}$. If the information $Z_i^\top Z_i$ grows, the term $\sigma^2 G^{-1}$ becomes negligible and $K_i = (I + \sigma^2 G^{-1}(Z_i^\top Z_i)^{-1})^{-1} \to I$: with lots of data the BLUP equals the group's own least-squares fit.

Exercise 12.7. (R.) In `nlme::ergoStool`, test the effect of the stool `Type` (3 df) in `effort ~ Type + (1 | Subject)`: (a) by an LRT with the correct likelihood; (b) by the (wrong) LRT from REML fits; (c) compare with the F test of the classical randomised-block ANOVA, `aov(effort ~ Type + Error(Subject))`. Interpret.

Solution.

```
es <- as.data.frame(nlme::ergoStool); es$Subject <-
↪  factor(as.character(es$Subject))
f1 <- lmer(effort ~ Type + (1 | Subject), es); f0 <- lmer(effort ~ 1 + (1 |
↪  Subject), es)
a_ml <- anova(f0, f1)                                          # refits with ML
T_ml <- a_ml$Chisq[2]
T_reml <- 2 * (logLik(f1) - logLik(f0))                         # wrong: REML
↪  with different X
c(T_ML = T_ml, p_ML = a_ml$`Pr(>Chisq)`[2], T_REML_wrong = as.numeric(T_reml))
```

```
#>          T_ML          p_ML T_REML_wrong
#>    3.601e+01    7.468e-08    3.671e+01
```

```
aov_es <- summary(aov(effort ~ Type + Error(Subject), es)); aov_es
```

```
#>
#> Error: Subject
#>           Df Sum Sq Mean Sq F value Pr(>F)
#> Residuals  8   66.5    8.31
#>
#> Error: Within
#>           Df Sum Sq Mean Sq F value  Pr(>F)
#> Type       3   81.2   27.06    22.4 3.9e-07 ***
#> Residuals 24   29.1    1.21
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
F_es <- aov_es[[2]][[1]]['Type', 'F value']
```

(a) The ML LRT is $T = 36$ on 3 df (highly significant). (b) The REML-based "LRT" gives a different number (36.7) that is not valid. (c) The classical $F = 22.4$ on (3, 24) df, $p < 0.001$, is *exact* in this balanced design; the LRT agrees on the conclusion but is only asymptotic (it is liberal with 9 subjects), which is why F or Kenward–Roger tests are preferred for fixed effects in small balanced designs.

Exercise 12.8.
(a) The LRT statistic for adding a random intercept to a linear model is $T = 3.2$. Give the naive and boundary-corrected p-values and say whether they agree at the 5% level. (b) The same $T = 3.2$ is obtained when adding a correlated random slope to a random intercept. Give the p-value. (c) For `nlme::Oxboys`, compare `height ~ age + (1 | Subject)` with `height ~ age + (age | Subject)` and report the correct p-value.

Solution.

(a) Naive: $P(\chi^2_1 > 3.2) = 0.074$; corrected: $0.5 \times 0.074 = 0.037$. At 5% the naive test does not reject and the corrected one does. (b) $\frac{1}{2}\chi^2_1 + \frac{1}{2}\chi^2_2$: $\frac{1}{2}(0.074) + \frac{1}{2}(0.202) = 0.138$ (note $P(\chi^2_2 > 3.2) = e^{-1.6} = 0.202$). (c)

```
ox <- as.data.frame(nlme::Oxboys); ox$Subject <-
↪  factor(as.character(ox$Subject))
a_ox <- anova(lmer(height ~ age + (1 | Subject), ox), lmer(height ~ age + (age
↪  | Subject), ox), refit = FALSE)
T_ox <- a_ox$Chisq[2]
c(T = T_ox, p_naive = pchisq(T_ox, 2, lower.tail = FALSE),
  p_mixture = .5 * pchisq(T_ox, 1, lower.tail = FALSE) + .5 * pchisq(T_ox, 2,
  ↪  lower.tail = FALSE))
```

```
#>         T   p_naive p_mixture
#> 2.159e+02 1.287e-47 6.782e-48
```

The random slope is clearly needed for the heights of boys: $T = 216$ and the correction changes nothing practically; boys differ in both their level and their rate of growth.

Exercise 12.9. (R.) Using 400 parametric-bootstrap replicates for `Dyestuff` (6 batches), approximate the p-value of the LRT for $H_0 : \sigma_b^2 = 0$ and compare with the naive χ^2_1 and the mixture. Why can the mixture be inaccurate here?

Solution.

```
set.seed(5)
lm0 <- lm(Yield ~ 1, Dyestuff)
T_obs <- as.numeric(2 * (logLik(m_dy_ml) - logLik(lm0)))
B <- 600
T_b <- replicate(B, {
  d <- Dyestuff; d$Yield <- simulate(lm0)[[1]]
  max(0, as.numeric(2 * (logLik(suppressMessages(lmer(Yield ~ 1 + (1 | Batch),
  ↪  d, REML = FALSE))) - logLik(lm(Yield ~ 1, d)))))
})
c(T_obs = T_obs, p_boot = (1 + sum(T_b >= T_obs)) / (B + 1), p_mixture = .5 *
↪  pchisq(T_obs, 1, lower.tail = FALSE),
  p_naive = pchisq(T_obs, 1, lower.tail = FALSE), P_T_equals_0 = mean(T_b <
  ↪  1e-6))
```

```
#>       T_obs       p_boot    p_mixture      p_naive P_T_equals_0
#>    5.402826    0.003328     0.010052    0.020104    0.660000
```

The bootstrap p-value (0.0033) is smaller than the mixture value 0.010 and far below the naive 0.020. With only six batches the asymptotics are rough: the simulated probability of $T = 0$ is 0.66, well above $\frac{1}{2}$, so the mixture is conservative. The exact F test of the one-way ANOVA ($F = 4.60$, $p = 0.0044$) is the best tool in this balanced design, and the bootstrap value agrees with it much better than the asymptotic one.

Exercise 12.10. (R and derivation.) For the random-intercept-and-slope model of `sleepstudy`, compute the prediction and 95% prediction interval, from formula (12.11), for the reaction time of a *new* subject on Day 10, and compare it with the interval for the *average* reaction time at Day 10 (confidence interval). Why are the two so different?

Solution.

```
x0 <- c(1, 10); fe <- fixef(m_pp); Vb <- as.matrix(vcov(m_pp))
G <- unclass(VarCorr(m_pp)$Subject)[1:2, 1:2]; s2 <- sigma(m_pp)^2
mu0 <- sum(x0 * fe)
v_mean <- drop(t(x0) %*% Vb %*% x0)                    # uncertainty of the
↪  population mean
v_new  <- v_mean + drop(t(x0) %*% G %*% x0) + s2       # formula (c12-pred), z0
↪  = x0
rbind(mean_CI = mu0 + c(-1.96, 1.96) * sqrt(v_mean), new_subject_PI = mu0 +
↪  c(-1.96, 1.96) * sqrt(v_new))
```

```
#>                 [,1]  [,2]
#> mean_CI        324.7 387.5
#> new_subject_PI 214.4 497.7
```

```
c(sd_mean = sqrt(v_mean), sd_new = sqrt(v_new))
```

```
#> sd_mean  sd_new
#>   16.02   72.27
```

Both are centred at 356.1 ms. The confidence interval for the *population mean* only reflects the uncertainty of $\hat{\beta}$ (SD 16 ms). A new subject's reaction time adds the between-subject variation at Day 10, $z_0^\top G z_0 = 612.1 + 2(10)(9.60) + 100(35.07) = 4311$ (SD 65.7 ms) and the day-to-day noise $\sigma^2 = 655$, so the prediction interval is about 142 ms either side: individuals are much less predictable than the average. With 18 subjects the z quantile is nearly adequate ($t_{17} = 2.11$ would widen both intervals by 8%); with few groups, replace 1.96 by a t quantile with Satterthwaite or Kenward–Roger df.

Exercise 12.11. (Critique.) Evaluate these statements from a report. (a) "To decide whether to include the treatment effect we compared the REML log-likelihoods of the models with and without it: $2\Delta\ell = 9.1$, $p = 0.003$." (b) "A likelihood-ratio test of $H_0 : \sigma_b^2 = 0$ gave $p = 0.07$ from a χ^2_1 distribution, so the random effect is not significant and was dropped." (c) "The QQ plot of the BLUPs is straight, so the random effects are normally distributed." (d) "REML gives unbiased variance estimates."

Solution.

(a) Invalid: REML likelihoods of models with different fixed effects are for different error contrasts. Refit with ML (or use `anova()`, which does it automatically) and use χ^2_1. (b) The null is on the boundary: the correct reference is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ and the correct p-value is 0.035, which *is* significant at 5%. More generally, a random effect that follows from the design (clustering) should not be dropped on the basis of a significance test: it protects the standard errors of the fixed effects. (c) BLUPs are shrunk and are a conditional mean of the random effects, not a sample from $N(0, G)$; with few groups a straight line is weak evidence, and with groups of unequal size the BLUPs have unequal variances; the check is rough. (d) False: REML is less biased than ML and unbiased only in balanced ANOVA cases (and only before truncation at zero).

Exercise 12.12. (Communicate.) Write about 150 words for a non-statistician explaining why the predicted batch effects in the `Dyestuff` study are *smaller* than the observed batch deviations, and why this is desirable.

Solution. A model answer. “We sampled only five specimens from each of six batches. A batch whose five specimens happened to be high may really be a high-yield batch, but part of the difference is chance in the five measurements. The model weighs how variable batches are (SD about 42 g) against how variable specimens are within a batch (SD about 50 g, so the batch mean of five has an SD of about 22 g) and concludes that about 78% of a batch’s observed deviation reflects the batch and 22% reflects chance. Batch E averaged 72.5 g above the overall mean, so we predict that the batch itself is about 57 g above. Predictions that stay a little closer to the overall average are, on average, closer to the truth than the raw batch means: a batch with an extreme average is likely to be less extreme if sampled again (regression to the mean). The method also protects us from over-interpreting small differences between batches.”

12.13 Further reading

The foundations are Patterson and Thompson (1971) (REML), Harville (1977) (a review of ML and REML for variance components), Henderson (1975) (mixed-model equations) and Robinson (1991) (BLUP). Self and Liang (1987) and Stram and Lee (1994) give the boundary results; Kenward and Roger (1997) the small-sample F test; Pinheiro and Bates (2000) and Bates et al. (2015) the computational methods of `nlme` and `lme4`. Wood (2017, Ch. 2) and Pawitan (2001, Ch. 17) cover the likelihood treatment.

Chapter 13

Generalized linear mixed models

Chapters 11 and 12 dealt with Gaussian responses. Many grouped data are not Gaussian: counts of new infections in herds, presence or absence of bacteria in children examined repeatedly, numbers of parasites on chicks of the same brood. The generalized linear models of Chapters 8 to 10 handle such responses but assume independent observations. The **generalized linear mixed model** (GLMM) puts random effects in the *linear predictor* of a GLM. It keeps the exponential-family response and the link function, and it models the correlation within groups with Gaussian random effects.

The price is computational and conceptual. Computationally, the likelihood is an integral over the random effects that has no closed form (except for the Gaussian case), so we need approximations: the Laplace approximation, adaptive Gauss–Hermite quadrature, penalised quasi-likelihood. Conceptually, the coefficients of a GLMM are **conditional** on the random effects, and they differ from the **marginal** (population-averaged) effects that a GLM without random effects estimates.

The plan is the following. Section 13.1 introduces the data. Section 13.2 defines the model, and Section 13.3 derives its marginal moments and shows that random effects create overdispersion and within-group correlation. Section 13.4 writes the integrated likelihood. Section 13.5 derives the Laplace approximation, Section 13.6 adaptive Gauss–Hermite quadrature and Section 13.7 penalised quasi-likelihood. Section 13.8 fits and compares the three approaches for the data of Section 13.1 and shows the inference tools. Section 13.9 studies conditional versus marginal effects and proves the attenuation result. Section 13.10 treats overdispersed counts by observation-level random effects, and Section 13.11 diagnostics and practical issues.

13.1 Binomial data in groups: contagious bovine pleuropneumonia

The data set `cbpp` in `lme4` comes from a study of contagious bovine pleuropneumonia in zebu cattle in 15 commercial herds in Ethiopia (Lesnoff et al., 2004). Each herd was visited in up to four periods; for herd i and period j the data give the number of new serological cases `incidence` out of the `size` animals at risk. There are 56 herd-periods and 842 animal-periods.

```
cb <- cbpp; cb$prop <- cb$incidence / cb$size; cb$period_n <-
↪  as.integer(cb$period)
pool <- aggregate(cbind(incidence, size) ~ period, cbpp, sum); pool$prop <-
↪  pool$incidence / pool$size; pool$period_n <- 1:4
ggplot(cb, aes(period_n, prop, group = herd)) + geom_line(colour = ie_navy,
↪  alpha = .4) +
```

```
  geom_point(aes(size = size), colour = ie_navy, alpha = .6) +
  geom_line(data = pool, aes(group = 1), colour = ie_coral, linewidth = 1.3) +
  scale_size_area(max_size = 4, name = "herd size") + labs(x = "period", y =
  ↪  "proportion of new cases")
```

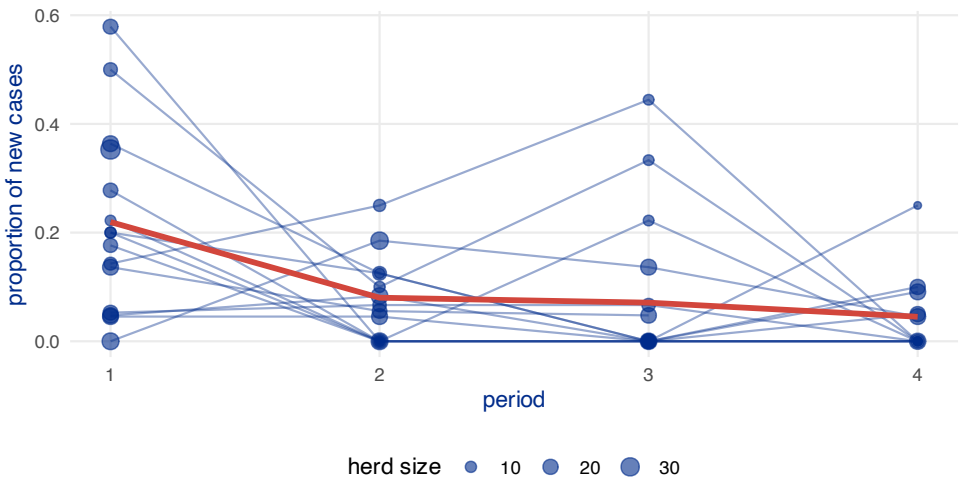


Figure 13.1: Proportion of new cases in each herd (thin lines) and pooled proportion per period (coral).

Two questions: how does the risk of a new case change over the periods, and how different are herds in their baseline risk? A binomial GLM with `period` as covariate treats the 56 counts as independent binomials. Its Pearson statistic divided by the residual degrees of freedom should be close to 1 if the model is adequate.

```
fit_glm <- glm(cbind(incidence, size - incidence) ~ period, family = binomial,
↪  data = cbpp)
sum(residuals(fit_glm, type = "pearson")^2) / df.residual(fit_glm)
```

```
#> [1] 2.19
```

The value 2.19 signals **overdispersion**: the binomial variance $s\mu(1-\mu)$ is too small. A plausible cause is that herds differ in their baseline risk (management, contact patterns), which a random herd effect can represent. Quasi-binomial inflation of the standard errors would not model this heterogeneity and would not allow us to estimate how much herds differ.

13.2 The generalized linear mixed model

Definition 13.1 (Generalized linear mixed model)**.** Let $b \sim N(0, G)$ be the vector of random effects. Conditionally on b, the responses y_{ij} are independent with a distribution from the exponential family of Chapter 8 with mean $\mu_{ij} = \mathrm{E}[y_{ij} \mid b]$ and variance $\mathrm{Var}(y_{ij} \mid b) = \phi V(\mu_{ij})/\text{weight}$, and

$$g(\mu_{ij}) = \eta_{ij} = x_{ij}^{\top}\beta + z_{ij}^{\top}b_i, \qquad \text{i.e.} \qquad \mathrm{E}[y \mid b] = g^{-1}(X\beta + Zb). \tag{13.1}$$

The parameters are β, ϑ (the parameters of G) and, if applicable, ϕ; the random effects are predicted.

Two special cases are used in this chapter.

- **Binomial, logit link.** For herd i and period j: $y_{ij} \mid b_i \sim \text{Bin}(s_{ij}, p_{ij})$ with $\text{logit}(p_{ij}) = \beta_0 + \beta_{\text{period}(j)} + b_i$, $b_i \sim N(0, \sigma_b^2)$.
- **Poisson, log link.** $y_{ij} \mid b_i \sim \text{Poisson}(\mu_{ij})$, $\log \mu_{ij} = x_{ij}^\top \beta + b_i$. This is a statement about the **conditional** mean $\text{E}[y_{ij} \mid b_i]$, not about $\text{E}[y_{ij}]$: Writing $\text{E}[y] = \exp(X\beta + Zb)$ is a common error (Section 13.3).

The LMM is the special case of the identity link and Gaussian errors. As before β and the variance components are *estimated* (by approximate maximum likelihood) and the random effects are *predicted* (by the conditional modes $\hat{b}_i$, returned by `ranef()`).

A latent-variable view of the logistic random intercept. Let $y_{ij}^* = \eta_{ij} + b_i + e_{ij}$ with e_{ij} standard logistic (variance $\pi^2/3$) and $y_{ij} = \mathbb{1}\{y_{ij}^* > 0\}$. Since the logistic distribution is symmetric, $P(y_{ij} = 1 \mid b_i) = \text{expit}(\eta_{ij} + b_i)$: this is the logistic GLMM. The latent variable has a random-intercept LMM structure with variance components σ_b^2 and $\pi^2/3$, so the **latent-scale intraclass correlation** is

$$\text{ICC}_{\text{latent}} = \frac{\sigma_b^2}{\sigma_b^2 + \pi^2/3}. \tag{13.2}$$

(for a probit link replace $\pi^2/3$ by 1). For `cbpp`, $\hat{\sigma}_b^2 = 0.412$ gives $0.412/(0.412 + 3.29) = 0.11$: on the latent scale about 11% of the variation of the risk is between herds.

13.3 Marginal moments: overdispersion and correlation

What does a random effect do to the *marginal* distribution of the data? Since g^{-1} is non-linear, the answer is not $g^{-1}(X\beta)$.

Theorem 13.1 (Moments of the Poisson random-intercept model)**.** *Let* $y_{ij} \mid b_i \sim \text{Poisson}(e^{\eta_{ij} + b_i})$, $b_i \sim N(0, \sigma_b^2)$, *and* $m_{ij} = \text{E}[y_{ij}]$. *Then*

$$m_{ij} = e^{\eta_{ij} + \sigma_b^2/2}, \tag{13.3}$$

$$\text{Var}(y_{ij}) = m_{ij} + m_{ij}^2 \, (e^{\sigma_b^2} - 1) > m_{ij}, \tag{13.4}$$

$$\text{Cov}(y_{ij}, y_{ik}) = m_{ij} m_{ik} \, (e^{\sigma_b^2} - 1) > 0 \quad (j \neq k). \tag{13.5}$$

Proof. By the moment generating function of the normal distribution, $\text{E}[e^{k b_i}] = e^{k^2 \sigma_b^2/2}$. Hence $m_{ij} = \text{E}\{\text{E}[y_{ij} \mid b_i]\} = e^{\eta_{ij}} \text{E}[e^{b_i}] = e^{\eta_{ij} + \sigma_b^2/2}$. By the law of total variance, $\text{Var}(y_{ij}) = \text{E}[\text{Var}(y_{ij} \mid b_i)] + \text{Var}(\text{E}[y_{ij} \mid b_i]) = \text{E}[e^{\eta_{ij} + b_i}] + e^{2\eta_{ij}} \text{Var}(e^{b_i})$, with $\text{Var}(e^{b_i}) = e^{2\sigma_b^2} - e^{\sigma_b^2} = e^{\sigma_b^2}(e^{\sigma_b^2} - 1)$, so the second term is $e^{2\eta_{ij} + \sigma_b^2}(e^{\sigma_b^2} - 1) = m_{ij}^2 (e^{\sigma_b^2} - 1)$. Given b_i the responses y_{ij}, y_{ik} are independent, so $\text{Cov}(y_{ij}, y_{ik}) = \text{Cov}(\mu_{ij}, \mu_{ik}) = e^{\eta_{ij} + \eta_{ik}} \text{Var}(e^{b_i}) = m_{ij} m_{ik} (e^{\sigma_b^2} - 1)$. □

Three consequences. (i) The marginal mean is larger than the conditional mean of the typical group ($b_i = 0$) by the factor $e^{\sigma_b^2/2}$; for the log link, only the *intercept* changes, and rate ratios are the same on both scales. (ii) The random effect creates **overdispersion**: $\text{Var}/\text{E} = 1 + m(e^{\sigma_b^2} - 1) > 1$ and growing with the mean (quadratic mean–variance relation, like the negative binomial). (iii) It creates **positive correlation** within groups, $\text{Corr}(y_{ij}, y_{ik}) = \dfrac{m_{ij} m_{ik} (e^{\sigma_b^2} - 1)}{\sqrt{\{m_{ij} + m_{ij}^2 (e^{\sigma_b^2} - 1)\}\{m_{ik} + m_{ik}^2 (e^{\sigma_b^2} - 1)\}}}$, which tends to 1 for large means. A simulation confirms the formulas for $e^\eta = 4$ and $\sigma_b^2 = 0.5$:

```
set.seed(1); B <- 1e6; sb2 <- .5
bb <- rnorm(B, 0, sqrt(sb2)); mu <- 4 * exp(bb); y1 <- rpois(B, mu); y2 <-
↪  rpois(B, mu)
```

```
m_th <- 4 * exp(sb2 / 2)
rbind(theory = c(mean = m_th, var = m_th + m_th^2 * (exp(sb2) - 1), cov =
↪  m_th^2 * (exp(sb2) - 1)),
      simulation = c(mean(y1), var(y1), cov(y1, y2)))
```

```
#>             mean   var   cov
#> theory     5.136 22.25 17.11
#> simulation 5.135 22.23 17.09
```

The mean is 5.14 instead of 4, the variance 22.2 instead of 5.1 (a variance-to-mean ratio of 4.3), and two counts of the same group have correlation 0.77.

Theorem 13.2 (Variance of the binomial random-intercept model)**.** *Let* $y \mid b \sim \mathrm{Bin}(s, p(b))$ *and* $\bar{p} = \mathrm{E}[p(b)]$*. Then* $\mathrm{E}[y] = s\bar{p}$ *and*

$$\mathrm{Var}(y) = s\bar{p}(1-\bar{p}) + s(s-1)\mathrm{Var}\{p(b)\} \;\geq\; s\bar{p}(1-\bar{p}).$$

Proof. $\mathrm{Var}(y) = \mathrm{E}[sp(1-p)] + \mathrm{Var}(sp) = s\bar{p} - s\mathrm{E}[p^2] + s^2\mathrm{Var}(p) = s\bar{p} - s\{\mathrm{Var}(p) + \bar{p}^2\} + s^2\mathrm{Var}(p) = s\bar{p}(1-\bar{p}) + s(s-1)\mathrm{Var}(p)$. □

The extra-binomial variation is proportional to $s(s-1)$: it vanishes for binary data ($s = 1$), where overdispersion is not identifiable, and grows quickly with the group size. In every case, **random effects in a GLM are a model for overdispersion and for within-group correlation at the same time.**

13.4 The integrated likelihood

Groups are independent and, given b_i, the y_{ij} are independent, so the likelihood of (β, ϑ) is the product over groups of integrals,

$$L(\beta, \vartheta) = \prod_{i=1}^{m} \int_{\mathbb{R}^q} \Big[\prod_{j=1}^{n_i} f(y_{ij} \mid b_i; \beta) \Big] \phi(b_i; 0, G_\vartheta)\, db_i, \tag{13.6}$$

where $\phi(\cdot; 0, G)$ is the $N(0, G)$ density. In the LMM, f is Gaussian and the integrand is a Gaussian density in (y, b); the integral is the closed-form $N(X\beta, V)$ density of Chapter 12. For binomial or Poisson f combined with a normal density there is generally **no closed form**. The integral has dimension q (the number of random effects per group) and must be evaluated for each of the m groups at every step of the numerical optimiser.

There are exceptions that use conjugate (non-normal) random effects. They are instructive because they show where the closed form comes from.

Example 13.1 (Poisson–gamma: the negative binomial)**.** Let $y \mid u \sim \mathrm{Poisson}(\mu u)$ with $u \sim \mathrm{Gamma}(\text{shape } k, \text{rate } k)$ (so $\mathrm{E}[u] = 1$, $\mathrm{Var}(u) = 1/k$). Then

$$\begin{aligned} P(y) &= \int_0^\infty \frac{e^{-\mu u}(\mu u)^y}{y!} \frac{k^k u^{k-1} e^{-ku}}{\Gamma(k)}\, du = \frac{\mu^y k^k}{y!\,\Gamma(k)} \int_0^\infty u^{y+k-1} e^{-(\mu+k)u}\, du \\ &= \frac{\Gamma(y+k)}{y!\,\Gamma(k)} \Big(\frac{k}{k+\mu}\Big)^k \Big(\frac{\mu}{k+\mu}\Big)^y, \end{aligned}$$

the negative binomial distribution with mean μ and variance $\mu + \mu^2/k$. Compare with the Poisson–lognormal of Theorem 13.1: the two coincide up to the second moment when

$1/k = e^{\sigma_b^2} - 1$. The gamma random effect is analytically convenient; the Gaussian random effect generalises to correlated effects, several grouping factors and random slopes, which is why GLMMs use it.

13.5 The Laplace approximation

The integrals in (13.6) have the form $\int e^{h(b)}db$ with h smooth and sharply peaked. The idea of the Laplace approximation is to replace h by its second-order Taylor expansion around the maximum.

Theorem 13.3 (Laplace approximation)**.** *Let $h : \mathbb{R}^q \to \mathbb{R}$ have a unique interior maximum at $\hat{b}$ with $h'(\hat{b}) = 0$ and $H = -h''(\hat{b})$ positive definite. Then*

$$\int_{\mathbb{R}^q} e^{h(b)}\,db \;\approx\; e^{h(\hat{b})}\,(2\pi)^{q/2}\,|H|^{-1/2}. \tag{13.7}$$

The approximation is exact if h is a quadratic function, and its relative error is $O(1/N)$ when h is of the form $h = N\bar{h}$ with N large (for GLMMs, N is the number of observations in a group).

Proof. (Derivation of the leading term.) Taylor-expand around $\hat{b}$: $h(b) \approx h(\hat{b}) + \tfrac{1}{2}(b-\hat{b})^\top h''(\hat{b})(b-\hat{b}) = h(\hat{b}) - \tfrac{1}{2}(b-\hat{b})^\top H (b-\hat{b})$, since $h'(\hat{b}) = 0$. Then $\int e^{h(b)}db \approx e^{h(\hat{b})}\int \exp\{-\tfrac{1}{2}(b-\hat{b})^\top H(b-\hat{b})\}db = e^{h(\hat{b})}(2\pi)^{q/2}|H|^{-1/2}$: the integral of an unnormalised $N(\hat{b}, H^{-1})$ density. If h is quadratic, the Taylor expansion is exact. The error term follows from expanding to higher order: the cubic term is odd and integrates to zero, and the next terms are of relative order $1/N$ (Tierney and Kadane, 1986). □

For a GLMM, group i contributes $\int e^{h_i(b)}db$ with

$$h_i(b) = \sum_{j=1}^{n_i} \log f(y_{ij} \mid b; \beta) - \tfrac{1}{2} b^\top G^{-1} b - \tfrac{1}{2}\log|2\pi G|, \tag{13.8}$$

and $\hat{b}_i$ is the **conditional mode** of b_i given the data of group i. For a canonical link, the Hessian is $-h_i''(\hat{b}_i) = Z_i^\top W_i Z_i + G^{-1}$ with $W_i = \operatorname{diag}\{V(\mu_{ij})/a_{ij}\}$ evaluated at $\hat{b}_i$ (Exercise 13.6): $W = \operatorname{diag}(\mu_{ij})$ for Poisson, $\operatorname{diag}(s_{ij}p_{ij}(1-p_{ij}))$ for binomial. Using $|G|\,|Z_i^\top W_i Z_i + G^{-1}| = |I + G Z_i^\top W_i Z_i|$, the Laplace log-likelihood is

$$\ell_{LA}(\beta, \vartheta) = \sum_{i=1}^{m}\Big[\sum_{j} \log f(y_{ij} \mid \hat{b}_i; \beta) - \tfrac{1}{2}\hat{b}_i^\top G^{-1}\hat{b}_i - \tfrac{1}{2}\log|I + G Z_i^\top W_i Z_i|\Big], \tag{13.9}$$

which is what `glmer` maximises with `nAGQ = 1` (its default), with $\hat{b}_i$ found by an inner iteration. In an LMM, $W = \sigma^{-2}I$, h_i is exactly quadratic, and (13.9) coincides with the exact log-likelihood of Chapter 12.

Example 13.2 (Laplace for one Poisson count, by hand)**.** A single count $y = 3$ with $\log\mu = b$, $b \sim N(0,1)$ has likelihood $L = \int \text{Pois}(3; e^b)\phi(b)\,db$. Here $h(b) = 3b - e^b - \log 6 - \tfrac{1}{2}b^2 - \tfrac{1}{2}\log 2\pi$, $h'(b) = 3 - e^b - b$ and $-h''(b) = e^b + 1$. The mode solves $3 = e^{\hat{b}} + \hat{b}$, which Newton's method finds in three steps from 0.5: $\hat{b} = 0.7921$. The curvature is $H = e^{0.7921} + 1 = 3.208$ and $h(\hat{b}) = -2.856$, so $L \approx e^{-2.856}\sqrt{2\pi/3.208} = 0.0575 \times 1.3995 =$

0.0805.

```
y0 <- 3
f0 <- function(b) dpois(y0, exp(b)) * dnorm(b)
exact <- integrate(f0, -Inf, Inf)$value
bhat <- uniroot(function(b) y0 - exp(b) - b, c(0, 3), tol = 1e-12)$root
H <- exp(bhat) + 1
lap <- f0(bhat) * sqrt(2 * pi / H)
c(mode = bhat, curvature = H, Laplace = lap, exact = exact, relative_error =
↪  lap / exact - 1)
```

```
#>           mode      curvature        Laplace          exact
#>       0.792060       3.207940       0.080459       0.080739
#> relative_error
#>      -0.003473
```

```
bg <- seq(-2.5, 3.5, length.out = 300)
ggplot(data.frame(b = rep(bg, 2), f = c(f0(bg), f0(bhat) * exp(-.5 * H * (bg -
↪  bhat)^2)),
                  what = rep(c("exact integrand", "Gaussian (Laplace)"), each
                  ↪  = length(bg))),
       aes(b, f, colour = what, linetype = what)) + geom_line(linewidth = 1.1)
       ↪  +
  scale_colour_manual(values = c(ie_coral, ie_navy), name = NULL) +
  ↪  scale_linetype_manual(values = c("dashed", "solid"), name = NULL) +
  labs(y = "integrand")
```

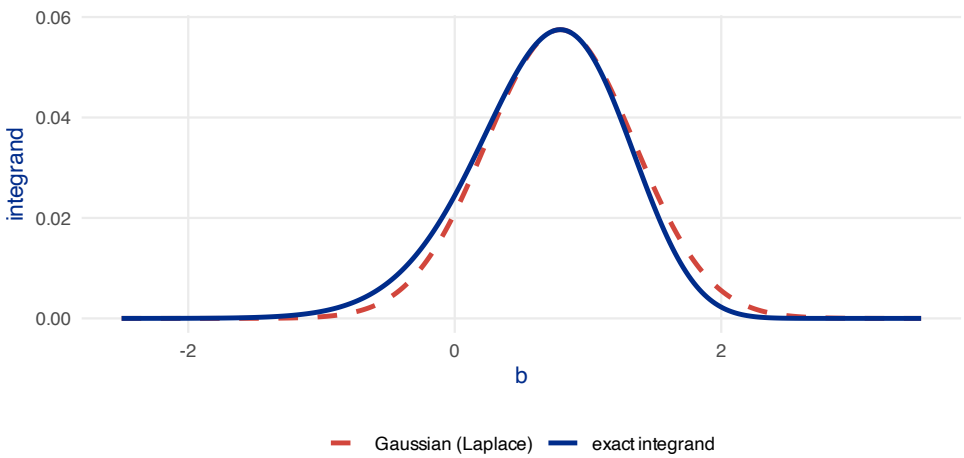


Figure 13.2: The integrand $f(y = 3 \mid b)\phi(b)$ (coral, dashed) and its Gaussian (Laplace) approximation (navy).

The Laplace value 0.08046 is within 0.35% of the exact 0.08074. The approximation is good when the integrand is close to Gaussian: counts that are not too small and several observations per group; it is worst for binary data with few observations per group.

The same computation for a real herd: the contribution of herd 1 of `cbpp` (four periods, 40 animals) at the fitted parameters.

```
fit_lap <- glmer(cbind(incidence, size - incidence) ~ period + (1 | herd),
↪  family = binomial, data = cbpp)
```

```
beta <- fixef(fit_lap); sb <- attr(VarCorr(fit_lap)$herd, "stddev")
h1 <- subset(cbpp, herd == "1"); eta1 <- drop(model.matrix(~ period,
↪  cbpp)[cbpp$herd == "1", ] %*% beta)
logh <- function(b) sapply(b, function(bb) sum(dbinom(h1$incidence, h1$size,
↪  plogis(eta1 + bb), log = TRUE)) + dnorm(bb, 0, sb, log = TRUE))
exact1 <- integrate(function(b) exp(logh(b)), -Inf, Inf)$value
mode1 <- optimize(logh, c(-4, 4), maximum = TRUE)$maximum
p1 <- plogis(eta1 + mode1); Hh <- sum(h1$size * p1 * (1 - p1)) + 1 / sb^2
↪  # Z'WZ + G^{-1}
lap1 <- unname(exp(logh(mode1)) * sqrt(2 * pi / Hh))
c(mode = mode1, exact = exact1, Laplace = lap1, ratio = lap1 / exact1)
```

```
#>      mode     exact   Laplace     ratio
#> 5.896e-01 6.132e-05 6.110e-05 9.963e-01
```

The conditional mode of herd 1 is $\hat{b}_1 = 0.590$ (the value of `ranef(fit_lap)`), and the Laplace approximation is accurate to 0.4%.

13.6 Adaptive Gauss–Hermite quadrature

A deterministic alternative to Laplace is numerical quadrature. The **Gauss–Hermite** rule with K nodes approximates integrals against the weight e^{-x^2}:

$$\int_{-\infty}^{\infty} e^{-x^2} u(x)\,dx \approx \sum_{k=1}^{K} w_k u(x_k), \tag{13.10}$$

where the nodes x_k are the roots of the Hermite polynomial H_K and the weights are positive; the rule is exact for all polynomials u of degree at most $2K-1$. (Nodes and weights are the eigenvalues and first-component weights of the Jacobi matrix; see the code below.) Applied naively to $\int g(b)\phi(b;0,\sigma_b^2)db$ with nodes centred at 0, the rule works for smooth g, but when the integrand $f(y \mid b)\phi(b)$ is concentrated far from 0 (large counts, large groups) the nodes miss it. The remedy is to *centre and scale the nodes at the integrand*, using the Laplace quantities.

Theorem 13.4 (Adaptive Gauss–Hermite quadrature)**.** *For a scalar random effect with h, $\hat{b}$ and $H = -h''(\hat{b})$ as in Theorem 13.3,*

$$\int e^{h(b)} db \approx \sqrt{\tfrac{2}{H}} \sum_{k=1}^{K} w_k\, e^{x_k^2}\, e^{h\left(\hat{b}+\sqrt{2/H}\,x_k\right)}. \tag{13.11}$$

With $K=1$ ($x_1=0$, $w_1=\sqrt{\pi}$) this is exactly the Laplace approximation (13.7); as $K\to\infty$ it converges to the integral.

Proof. Substitute $b = \hat{b} + \sqrt{2/H}\,x$, $db = \sqrt{2/H}\,dx$, and write $e^{h(b)} = e^{-x^2}\cdot e^{x^2 + h(\hat{b}+\sqrt{2/H}x)}$. Apply (13.10) to $u(x) = e^{x^2+h(\hat{b}+\sqrt{2/H}x)}$. For $K=1$, $x_1=0$ and $w_1=\sqrt{\pi}$ (the integral of e^{-x^2}), so the right-hand side is $\sqrt{2/H}\sqrt{\pi}e^{h(\hat{b})} = e^{h(\hat{b})}(2\pi/H)^{1/2}$. □

The substitution centres the nodes at the mode and scales them by the curvature: near the mode, $h(\hat{b}+\sqrt{2/H}x) \approx h(\hat{b}) - x^2$ and u is almost constant, which is why very few nodes suffice. The cost grows like K^q for q random effects per group; `lme4` provides it only for a *single scalar*

random effect per group (`nAGQ` > 1). We compare the plain and the adaptive rule on two integrals of the same type as above, a small count $y = 3$ and a large count $y = 30$:

```
gh_rule <- function(K) {                                  # Golub-Welsch:
↪  nodes and weights for exp(-x^2)
  J <- matrix(0, K, K); if (K > 1) J[cbind(1:(K - 1), 2:K)] <- J[cbind(2:K,
↪  1:(K - 1))] <- sqrt((1:(K - 1)) / 2)
  e <- eigen(J, symmetric = TRUE); list(x = e$values, w = sqrt(pi) *
↪  e$vectors[1, ]^2)
}
quad <- function(f, bhat, H, K, adaptive = TRUE) {
  r <- gh_rule(K)
  if (adaptive) sqrt(2 / H) * sum(r$w * exp(r$x^2) * f(bhat + sqrt(2 / H) *
   ↪  r$x))
  else          sum(r$w * exp(r$x^2) * f(r$x))                # nodes centred
   ↪  at 0, scale 1
}
tab <- do.call(rbind, lapply(c(3, 30), function(yy) {
  f <- function(b) dpois(yy, exp(b)) * dnorm(b)
  bh <- uniroot(function(b) yy - exp(b) - b, c(0, 6), tol = 1e-12)$root; Hh <-
↪  exp(bh) + 1
  Ks <- c(1, 3, 5, 9, 15)
  data.frame(y = yy, K = Ks, adaptive = sapply(Ks, function(K) quad(f, bh, Hh,
   ↪  K)),
             plain = sapply(Ks, function(K) quad(f, bh, Hh, K, adaptive =
              ↪  FALSE)),
             exact = integrate(f, -Inf, Inf, rel.tol = 1e-12)$value)
}))
print(tab, digits = 4)
```

```
#>     y  K  adaptive     plain     exact
#> 1   3  1 8.046e-02 4.336e-02 8.074e-02
#> 2   3  3 7.984e-02 8.421e-02 8.074e-02
#> 3   3  5 8.070e-02 8.122e-02 8.074e-02
#> 4   3  9 8.074e-02 8.075e-02 8.074e-02
#> 5   3 15 8.074e-02 8.074e-02 8.074e-02
#> 6  30  1 5.151e-05 9.807e-34 5.164e-05
#> 7  30  3 5.150e-05 2.834e-19 5.164e-05
#> 8  30  5 5.164e-05 2.568e-11 5.164e-05
#> 9  30  9 5.164e-05 1.005e-04 5.164e-05
#> 10 30 15 5.164e-05 2.813e-05 5.164e-05
```

For $y = 3$ both rules converge quickly. For $y = 30$ the integrand is concentrated around $b \approx 3.3$ with a standard deviation of only 0.19; the plain rule with $K = 15$ nodes is still wrong by a factor of two, while the adaptive rule with a *single* node (Laplace) is within 0.3% and with five nodes within 0.005%. This is why `glmer(..., nAGQ = k)` uses the adaptive version.

13.7 Penalised quasi-likelihood

A cheaper approach, which works for any G and many random effects, is **penalised quasi-likelihood** (Breslow and Clayton, 1993). Maximise the *joint* log-density of (y, b) over (β, b)

treating the random effects as parameters penalised by their prior:

$$\sum_{ij} \log f(y_{ij} \mid \beta, b_i) - \tfrac{1}{2} b^\top G^{-1} b. \tag{13.12}$$

The Fisher-scoring (IRLS) step for this criterion is derived exactly as in Chapter 9, with a ridge-type penalty on b. Define the **working response** $z = \hat\eta + (y - \hat\mu)\, g'(\hat\mu)$ and the **working weights** $W = \operatorname{diag}\{1/(a(\phi) V(\hat\mu) g'(\hat\mu)^2)\}$ at the current iterate. The score of (13.12) is $(X^\top W(z - X\beta - Zb),\ Z^\top W(z - X\beta - Zb) - G^{-1} b)$ and the update solves

$$\begin{pmatrix} X^\top W X & X^\top W Z \\ Z^\top W X & Z^\top W Z + G^{-1} \end{pmatrix} \begin{pmatrix} \beta \\ b \end{pmatrix} = \begin{pmatrix} X^\top W z \\ Z^\top W z \end{pmatrix}, \tag{13.13}$$

which are **Henderson's mixed-model equations** (Chapter 12) for the working LMM $z = X\beta + Zb + \varepsilon$, $\varepsilon \sim N(0, W^{-1})$. PQL therefore iterates: compute z and W, fit this LMM (estimating ϑ by ML or REML), update $\hat\eta$, repeat. It is fast and flexible (`MASS::glmmPQL`, and the engine of `mgcv::gamm`, Chapter 19), but it is not a likelihood method: it ignores the dependence of W on b that the Laplace formula (13.9) keeps in the determinant. It is biased towards zero for binary data and small clusters (variance components underestimated), and it provides no true likelihood, hence no LRT or AIC.

13.8 Fitting and inference in R

```
summary(fit_lap)$coefficients
```

```
#>             Estimate Std. Error z value  Pr(>|z|)
#> (Intercept)  -1.3983     0.2312  -6.048 1.468e-09
#> period2      -0.9919     0.3032  -3.272 1.068e-03
#> period3      -1.1282     0.3228  -3.495 4.745e-04
#> period4      -1.5797     0.4220  -3.743 1.818e-04
```

```
VarCorr(fit_lap)
```

```
#>  Groups Name        Std.Dev.
#>  herd   (Intercept) 0.642
```

```
fit_agq <- update(fit_lap, nAGQ = 25)
pql <- MASS::glmmPQL(cbind(incidence, size - incidence) ~ period, random = ~ 1
↪  | herd, family = binomial, data = cbpp, verbose = FALSE)
sd_of <- function(f) if (inherits(f, "lme")) as.numeric(VarCorr(f)[1, 2]) else
↪  attr(VarCorr(f)$herd, "stddev")
round(rbind(`Laplace (nAGQ = 1)` = c(fixef(fit_lap), sd = sd_of(fit_lap)),
            `AGQ (nAGQ = 25)`    = c(fixef(fit_agq), sd = sd_of(fit_agq)),
            PQL                  = c(fixef(pql), sd = sd_of(pql))), 3)
```

```
#>                     (Intercept) period2 period3 period4 sd.(Intercept)
#> Laplace (nAGQ = 1)      -1.398  -0.992  -1.128  -1.580          0.642
#> AGQ (nAGQ = 25)         -1.399  -0.991  -1.128  -1.579          0.648
#> PQL                     -1.327  -1.016  -1.150  -1.605          0.556
```

Laplace and adaptive quadrature agree closely (herd-periods contain between 2 and 34 animals), while PQL gives a smaller variance (0.556 against 0.648). Interpretation, on the log-odds scale: the baseline risks of herds vary with standard deviation $\hat\sigma_b = 0.64$; a herd one standard deviation

above average has odds $e^{0.64} = 1.9$ times those of an average herd in the same period. **For a given herd**, the odds of a new case in period 2 are $e^{-0.992} = 0.37$ times those in period 1, and in period 4 they are $e^{-1.580} = 0.21$ times. The latent intraclass correlation is 0.11.

Tests and intervals. `summary()` gives Wald z tests. Because the fit is (approximate) maximum likelihood, likelihood-ratio tests of fixed effects are valid (there is no REML for GLMMs in `glmer`); the variance component is tested on the boundary:

```
fit0 <- update(fit_lap, . ~ . - period)
anova(fit0, fit_lap)                                    # LRT for period, 3
↪  df
```

```
#> Data: cbpp
#> Models:
#> fit0: cbind(incidence, size - incidence) ~ (1 | herd)
#> fit_lap: cbind(incidence, size - incidence) ~ period + (1 | herd)
#>         npar AIC BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
#> fit0       2 214 218   -105       210
#> fit_lap    5 194 204    -92       184  25.6  3    1.2e-05 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
fit_nore <- glm(cbind(incidence, size - incidence) ~ period, family =
↪  binomial, data = cbpp)
T_h <- as.numeric(2 * (logLik(fit_lap) - logLik(fit_nore)))
c(T = T_h, p_naive = pchisq(T_h, 1, lower.tail = FALSE), p_mixture = .5 *
↪  pchisq(T_h, 1, lower.tail = FALSE))
```

```
#>         T   p_naive p_mixture
#> 1.401e+01 1.823e-04 9.115e-05
```

```
confint(update(fit_lap), method = "profile", parm = "beta_")        # profile
↪  intervals; done on a copy because profiling changes the internal state of
↪  a merMod object
```

```
#>                2.5 %  97.5 %
#> (Intercept) -1.901 -0.9478
#> period2     -1.617 -0.4077
#> period3     -1.801 -0.5115
#> period4     -2.501 -0.8007
```

Period has a clear effect ($T = 25.6$ on 3 df) and the herd variance is clearly positive ($T = 14.0$, boundary-corrected $p = 9 \times 10^{-5}$). Compare log-likelihoods only between fits with the same `nAGQ`: `lme4` evaluates `logLik()` on different scales for `nAGQ = 1` and `nAGQ > 1` (here the two differ by about 42 although the estimates agree), and never mix PQL with likelihood-based criteria. The **conditional modes** of the herd effects are

```
re <- ranef(fit_lap, condVar = TRUE)$herd
round(head(cbind(mode = re[, 1], se = sqrt(attr(ranef(fit_lap, condVar =
↪  TRUE)$herd, "postVar")[1, 1, ])), 5), 3)
```

```
#>        mode    se
#> [1,]  0.590 0.348
#> [2,] -0.299 0.365
```

```
#> [3,]  0.406 0.297
#> [4,]  0.039 0.416
#> [5,] -0.190 0.350
```

13.9 Conditional versus marginal effects

The coefficients of a GLMM answer a **subject-specific** question: *for a given herd*, how do the odds change from period 1 to period 2? A GLM without random effects, or a marginal model fitted by generalised estimating equations (Zeger et al., 1988), answers a **population-averaged** question: *across the population of herds*, how does the proportion of new cases change? The two are the same only for the identity link.

Definition 13.2 (Conditional and marginal effects)**.** The **conditional** effects β_C are the coefficients of the GLMM (13.1). The **marginal mean** is

$$\mu_M(x) = \mathrm{E}[y \mid x] = \int g^{-1}(x^\top \beta_C + z^\top b)\, \phi(b; 0, G)\, db, \tag{13.14}$$

and the **marginal effect** of a covariate describes how $\mu_M(x)$, or $g(\mu_M(x))$, changes with it.

Theorem 13.5 (Marginal effects: log, probit and logit links)**.** *Consider a random intercept $b \sim N(0, \sigma_b^2)$ and linear predictor $\eta = x^\top \beta_C$. (a)* Log link. *$\mu_M(x) = e^{\eta + \sigma_b^2/2}$: the marginal and conditional coefficients coincide, except for the intercept, which shifts by $\sigma_b^2/2$. (b)* Probit link, *$g^{-1} = \Phi$. $\mu_M(x) = \Phi(\eta/\sqrt{1+\sigma_b^2})$: the marginal model is again probit with coefficients $\beta_M = \beta_C/\sqrt{1+\sigma_b^2}$,* ***attenuated****. (c)* Logit link. *Approximately $\mu_M(x) \approx \mathrm{expit}(\eta/\sqrt{1 + c^2\sigma_b^2})$ with $c^2 = (16\sqrt{3}/(15\pi))^2 = 0.346$, so that*

$$\beta_M \approx \frac{\beta_C}{\sqrt{1 + 0.346\,\sigma_b^2}}. \tag{13.15}$$

Proof.

(a) $\mathrm{E}[e^{\eta+b}] = e^{\eta} e^{\sigma_b^2/2}$. (b) Let $U \sim N(0,1)$ be independent of b. Then $\Phi(\eta + b) = P(U \le \eta + b \mid b)$, so $\mu_M = P(U - b \le \eta)$, and $U - b \sim N(0, 1 + \sigma_b^2)$ gives $\mu_M = \Phi(\eta/\sqrt{1+\sigma_b^2})$. (c) The logistic function is well approximated by a scaled probit, $\mathrm{expit}(\eta) \approx \Phi(c\eta)$ with $c = 16\sqrt{3}/(15\pi) \approx 0.588$ (a classical approximation; the maximum absolute difference between the two functions is below 0.01). By (b) with $c\eta$ and $c\,b \sim N(0, c^2\sigma_b^2)$, $\mathrm{E}_b[\Phi(c(\eta + b))] = \Phi(c\eta/\sqrt{1 + c^2\sigma_b^2})$; apply the approximation backwards, $\Phi(c\eta') \approx \mathrm{expit}(\eta')$ with $\eta' = \eta/\sqrt{1 + c^2\sigma_b^2}$.

□

So averaging a logistic curve over cluster-specific intercepts gives a **flatter** curve: the typical cluster has slope β_C but the population average has the smaller slope β_M (the more variable the clusters, the flatter). With $\sigma_b = 2$ the factor is $1/\sqrt{1 + 0.346 \cdot 4} = 0.65$; with the $\hat\sigma_b = 0.64$ of `cbpp` it is 0.94. A simulation confirms it: 2000 clusters of 10 observations, $\beta_0 = -1$, $\beta_C = 1$, $\sigma_b = 2$.

```
set.seed(2027)
m <- 2000; n_i <- 10; id <- rep(1:m, each = n_i); x <- rnorm(m * n_i); b <-
↪  rnorm(m, 0, 2)[id]
d <- data.frame(y = rbinom(m * n_i, 1, plogis(-1 + 1 * x + b)), x = x, id =
↪  factor(id))
fit_cond <- glmer(y ~ x + (1 | id), family = binomial, data = d)
fit_pop  <- glm(y ~ x, family = binomial, data = d)
round(rbind(true_conditional = c(-1, 1), glmer_conditional = fixef(fit_cond),
↪  glm_marginal = coef(fit_pop),
            approximation = c(-1, 1) / sqrt(1 + 0.346 * 4)), 3)
```

```
#>                    (Intercept)     x
#> true_conditional        -1.000 1.000
#> glmer_conditional       -1.002 1.017
#> glm_marginal            -0.615 0.622
#> approximation           -0.648 0.648
```

```
xs <- seq(-3, 3, by = .05)
p_marg <- sapply(xs, function(x0) integrate(function(bb) plogis(-1 + x0 + bb)
↪  * dnorm(bb, 0, 2), -Inf, Inf)$value)
coef(lm(qlogis(p_marg) ~ xs))                  # the exact marginal curve is
↪  close to a logistic one
```

```
#> (Intercept)          xs
#>     -0.6382      0.6329
```

```
xs <- seq(-4, 4, by = .05); bq <- qnorm(c(.1, .25, .75, .9), 0, 2)
cond <- expand.grid(x = xs, b = bq); cond$p <- plogis(-1 + cond$x + cond$b)
marg <- data.frame(x = xs, p = sapply(xs, function(x0) integrate(function(bb)
↪  plogis(-1 + x0 + bb) * dnorm(bb, 0, 2), -Inf, Inf)$value))
ggplot() + geom_line(data = cond, aes(x, p, group = b), colour = "grey60") +
  geom_line(data = data.frame(x = xs, p = plogis(-1 + xs)), aes(x, p, colour =
  ↪  "typical cluster"), linewidth = 1.2) +
  geom_line(data = marg, aes(x, p, colour = "population average"), linewidth =
  ↪  1.2) +
  geom_line(data = data.frame(x = xs, p = plogis((-1 + xs) / sqrt(1 + .346 *
  ↪  4))), aes(x, p, colour = "approximation"), linetype = 2, linewidth = 1)
  ↪  +
  scale_colour_manual(values = c("typical cluster" = ie_green, "population
  ↪  average" = ie_coral, "approximation" = ie_navy), name = NULL) +
  labs(y = "P(y = 1 | x)")
```

For `cbpp`, the approximate marginal coefficient of period 2 is $-0.992/\sqrt{1+0.346(0.412)} = -0.928$, an odds ratio of 0.40 instead of 0.37. The typical herd ($b = 0$) has a *lower* risk than the population average, because expit is convex at low probabilities (Jensen's inequality):

```
eta <- fixef(fit_lap)[1] + c(0, fixef(fit_lap)[-1])
rbind(typical_herd = plogis(eta),
      population_average = sapply(eta, function(e) integrate(function(b)
      ↪  plogis(e + b) * dnorm(b, 0, sb), -Inf, Inf)$value),
      observed = tapply(cbpp$incidence, cbpp$period, sum) / tapply(cbpp$size,
      ↪  cbpp$period, sum))
```

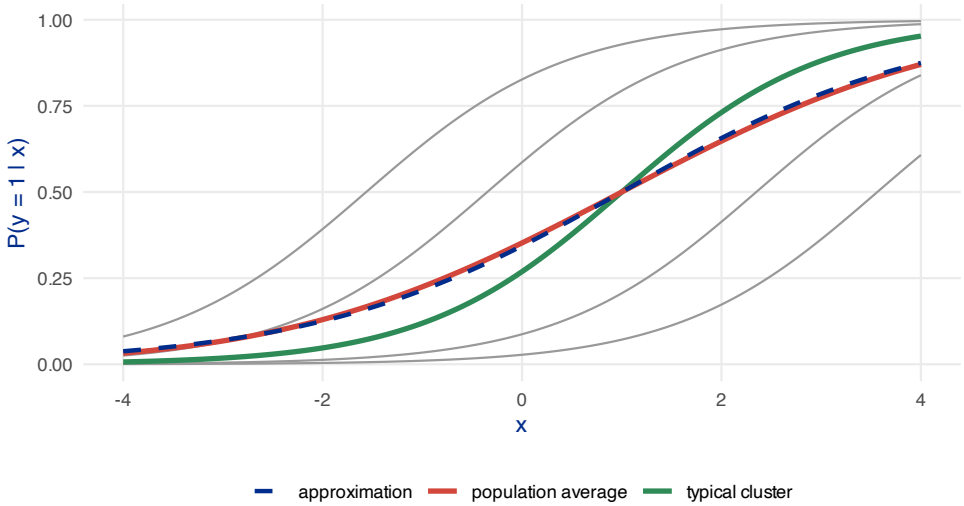


Figure 13.3: Logistic random intercept with $\beta_0 = -1$, $\beta_C = 1$ and $\sigma_b = 2$. Grey: clusters at the 10, 25, 75 and 90 percent quantiles of b; green: typical cluster (b = 0); coral: population average (exact integral); navy dashed: approximation with slope 0.65.

```
#>                          period2 period3 period4
#> typical_herd          0.1981 0.08392 0.07402 0.04843
#> population_average    0.2161 0.09709 0.08616 0.05733
#> observed              0.2194 0.08019 0.07107 0.04516
```

In period 1 the typical herd has a probability of new cases of 0.198, the population average is 0.216, and the pooled observed proportion 0.219. **Report the effect that answers the question.** A veterinarian advising one farmer needs the conditional odds ratio; a ministry planning vaccine stocks needs the population-average one (from the GLMM by integration, or from a marginal model).

Warning. `glm()` and `glmer()` do not estimate the same quantity for logistic models. The data `MASS::bacteria` record the presence of *H. influenzae* in 220 tests of 50 children with otitis media (`trt`: placebo, drug, drug+ with compliance encouragement; `week`):

```
data(bacteria, package = "MASS")
fit_b <- glmer(y ~ trt + I(week > 2) + (1 | ID), family = binomial, data =
↪  bacteria)
sb_b <- attr(VarCorr(fit_b)$ID, "stddev")
cbind(GLMM_conditional = fixef(fit_b), approx_marginal = fixef(fit_b) / sqrt(1
↪  + .346 * sb_b^2),
      glm_marginal = coef(glm(y ~ trt + I(week > 2), family = binomial, data =
      ↪  bacteria)))
```

```
#>                 GLMM_conditional approx_marginal glm_marginal
#> (Intercept)               3.5479          2.8646       2.8332
#> trtdrug                  -1.3667         -1.1034      -1.1187
#> trtdrug+                 -0.7826         -0.6319      -0.6372
#> I(week > 2)TRUE          -1.5985         -1.2906      -1.2949
```

For the drug contrast, $\hat{\beta}_C$ = -1.367 from the GLMM, −1.119 from the plain `glm`, and the approximation (13.15) gives $-1.367/\sqrt{1 + 0.346(1.543)}$ = -1.103. The `glm` is not wrong as an

estimate of the population-averaged effect, but its standard errors ignore the clustering.

13.10 Overdispersed counts and observation-level random effects

Counts often show variation beyond what the Poisson allows, even after accounting for known groupings. A convenient device is an **observation-level random effect** (OLRE): one additional random intercept *per observation*, $e_k \sim N(0, \sigma_e^2)$, so that $\log \mathrm{E}[y_k \mid b] = x_k^\top \beta + b_{\mathrm{group}(k)} + e_k$. By Theorem 13.1 the conditional-on-groups variance is $m_k + m_k^2(e^{\sigma_e^2} - 1)$: a Poisson–lognormal model, with the quadratic mean–variance relation of the negative binomial (Harrison, 2014). An OLRE is possible only because the response is not Gaussian: in an LMM it would be confounded with the residual ε. It is not identifiable for binary data.

The data set `grouseticks` (Elston et al., 2001) records the number of ticks on 403 red grouse chicks (`TICKS`) in 118 broods at 63 locations, with altitude (`HEIGHT`) and year as covariates.

```
gt <- grouseticks; gt$HEIGHT_s <- as.numeric(scale(gt$HEIGHT))          #
↪  standardised altitude
c(mean = mean(gt$TICKS), variance = var(gt$TICKS))
```

```
#>     mean variance
#>     6.37   172.66
```

```
pdisp <- function(fit) sum(residuals(fit, type = "pearson")^2) /
↪  df.residual(fit)
fit_g0 <- glm(TICKS ~ YEAR + HEIGHT_s, family = poisson, data = gt)
fit_g1 <- glmer(TICKS ~ YEAR + HEIGHT_s + (1 | LOCATION) + (1 | BROOD), family
↪  = poisson, data = gt)
fit_g2 <- update(fit_g1, . ~ . + (1 | INDEX))                                #
↪  INDEX = row number: the OLRE
tab_g <- rbind(`Poisson GLM` = c(AIC(fit_g0), pdisp(fit_g0),
↪  coef(fit_g0)["HEIGHT_s"], sqrt(vcov(fit_g0)["HEIGHT_s", "HEIGHT_s"])),
               `+ brood, location` = c(AIC(fit_g1), pdisp(fit_g1),
               ↪  fixef(fit_g1)["HEIGHT_s"], sqrt(vcov(fit_g1)["HEIGHT_s",
               ↪  "HEIGHT_s"])),
               `+ OLRE` = c(AIC(fit_g2), pdisp(fit_g2),
               ↪  fixef(fit_g2)["HEIGHT_s"], sqrt(vcov(fit_g2)["HEIGHT_s",
               ↪  "HEIGHT_s"])))
colnames(tab_g) <- c("AIC", "Pearson/df", "beta_height", "se")
round(tab_g, 3)
```

```
#>                     AIC Pearson/df beta_height    se
#> Poisson GLM        4383     12.484      -0.771 0.026
#> + brood, location  1988      1.741      -0.847 0.125
#> + OLRE             1795      0.412      -0.854 0.124
```

```
VarCorr(fit_g2)
```

```
#>  Groups   Name        Std.Dev.
#>  INDEX    (Intercept) 0.542
#>  BROOD    (Intercept) 0.750
#>  LOCATION (Intercept) 0.529
```

The variance (173) is 27 times the mean: massive overdispersion. The Poisson GLM has Pearson

dispersion 12.5, and its standard error for altitude (0.026) is about five times too small. Adding brood and location random effects brings the dispersion to 1.74 and moves the estimate from -0.771 to -0.847, with a standard error of 0.125; the OLRE lowers AIC by a further 193 units. Variation is mostly between broods ($\hat{\sigma} = 0.75$) and locations (0.53); the OLRE absorbs the rest (0.54). The **rate ratio** for one standard deviation (36 m) higher altitude is $e^{-0.854} = 0.43$, for chicks with the same random effects. Do not read the Pearson dispersion of the model *with* an OLRE (0.41) as underdispersion: the Pearson residuals are conditional on the fitted chick effects.

13.11 Diagnostics and practical issues

- **Overdispersion.** Compute the Pearson dispersion of the model *without* an OLRE; values well above 1 call for an OLRE, a negative binomial family or a richer random part. For `cbpp` it is 1.24 for the GLMM, against 2.19 for the GLM.
- **Random effects.** QQ plot of the conditional modes $\hat{b}_i$ (`ranef()`); they are shrunk (Chapter 12), so with few groups the plot is only a rough check.
- **Residuals.** Pearson residuals against the fitted values are hard to read for binary or small counts. **Randomised quantile residuals** are approximately $N(0,1)$ if the model is right: with F_{ij} the conditional distribution function at the fitted mean, $u_{ij} = F_{ij}(y_{ij}-1) + U\{F_{ij}(y_{ij}) - F_{ij}(y_{ij}-1)\}$, $U \sim U(0,1)$, and $r_{ij} = \Phi^{-1}(u_{ij})$.
- **Approximation.** Refit with `nAGQ` > 1 (scalar random effect); if estimates move, Laplace was not accurate enough, notably for binary data with small clusters.
- **Convergence.** Warnings often disappear after centring or scaling covariates (`scale()`), changing the optimiser (`glmerControl(optimizer = "bobyqa")`), or `allFit()`; a *singular fit* means a variance estimated at 0.

```
set.seed(3)
p_hat <- fitted(fit_lap)                                    # conditional
↪  probabilities
Fhi <- pbinom(cbpp$incidence, cbpp$size, p_hat); Flo <- pbinom(cbpp$incidence
↪  - 1, cbpp$size, p_hat)
u <- Flo + runif(nrow(cbpp)) * (Fhi - Flo); r_q <- qnorm(u)
ggplot(data.frame(r_q), aes(sample = r_q)) + stat_qq(colour = ie_navy) +
↪  stat_qq_line(colour = ie_coral) +
  labs(x = "theoretical quantiles", y = "randomised quantile residuals")
```

```
c(mean = mean(r_q), sd = sd(r_q))
```

```
#>     mean       sd
#> -0.08577  1.14581
```

Few clusters, small clusters, many random effects. The variance of a random effect needs enough groups; with fewer than about 5 levels fixed effects are safer. For binary responses with 2 or 3 observations per group, the Laplace approximation (and PQL even more) underestimates the variance component (Exercise 13.10). *Separation* (a covariate perfectly predicting the response) produces huge coefficients exactly as in logistic regression.

13.12 In R

```
fit  <- glmer(cbind(cases, size - cases) ~ x + (1 | g), family = binomial,
↪  data = d)   # Laplace (nAGQ = 1)
```

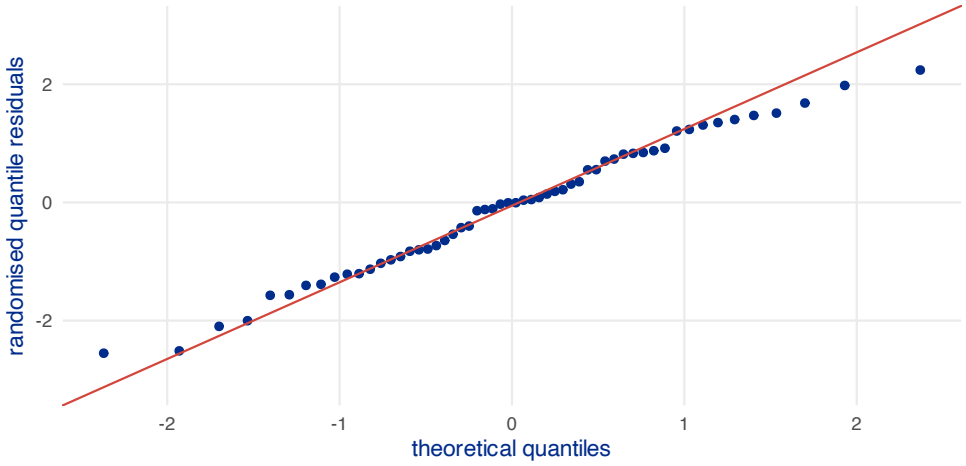


Figure 13.4: Randomised quantile residuals of the cbpp GLMM (conditional on the fitted herd effects).

```
fit25 <- update(fit, nAGQ = 25)                          # adaptive
↪  Gauss-Hermite, scalar random effect only
fitP <- MASS::glmmPQL(y ~ x, random = ~ 1 | g, family = poisson, data = d)
↪  # PQL
summary(fit); fixef(fit); VarCorr(fit); ranef(fit, condVar = TRUE)
↪  # estimates, predictions
anova(fit_small, fit_big)                                # LRT with the same
↪  nAGQ; boundary for variances
confint(fit, method = "profile")                         # also "Wald", "boot"
predict(fit, newdata = nd, re.form = NA, type = "response")    # typical group
↪  (b = 0), NOT the population average
simulate(fit, nsim = 1000)                               # parametric bootstrap,
↪  randomised quantile residuals
glmer(y ~ x + (1 | g) + (1 | obs), family = poisson, data = d)   # OLRE: obs =
↪  1:nrow(d)
glmer.nb(y ~ x + (1 | g), data = d)                      # negative binomial
↪  GLMM
```

13.13 Common pitfalls

Warning. The GLMM for counts is $\mathrm{E}[y \mid b] = \exp(X\beta + Zb)$; the marginal mean is $\exp(X\beta + \sigma_b^2/2)$ for a random intercept.

Warning. A GLMM odds ratio is **not** a population-averaged effect. Do not expect `glmer` and `glm` (or GEE) to give the same logistic slopes.

Warning. Do not compare `logLik()` across fits with different `nAGQ`, and never use PQL for likelihood-ratio tests or AIC.

Warning. Test a variance component with the 50:50 mixture, not with χ^2_1, and do not call `predict(..., re.form = NA)` a "population average": it is the typical group ($b = 0$).

Warning. Do not read the Pearson dispersion of a model with an observation-level random effect, and do not add an OLRE to binary data.

13.14 Chapter summary

- A GLMM is a GLM with Gaussian random effects in the linear predictor: $\mathrm{E}[y \mid b] = g^{-1}(X\beta + Zb)$, $b \sim N(0, G)$. Random effects create overdispersion and within-group correlation: Poisson: $\mathrm{E}[y] = e^{\eta + \sigma_b^2/2}$, $\mathrm{Var}(y) = m + m^2(e^{\sigma_b^2} - 1)$; binomial: extra variance $s(s-1)\mathrm{Var}\{p(b)\}$.
- The likelihood is $\prod_i \int [\prod_j f(y_{ij} \mid b_i)] \phi(b_i; 0, G)\, db_i$, with no closed form except in the Gaussian (and conjugate) case.
- **Laplace**: $\int e^h \approx e^{h(\hat b)} (2\pi)^{q/2} |H|^{-1/2}$, exact for quadratic h; for GLMMs $\ell_{LA} = \sum_i [\sum_j \log f(y_{ij} \mid \hat b_i) - \tfrac{1}{2}\hat b_i^\top G^{-1} \hat b_i - \tfrac{1}{2} \log |I + G Z_i^\top W_i Z_i|]$. **Adaptive Gauss–Hermite** centres and scales the nodes at $\hat b$; one node is Laplace. **PQL** iterates Henderson's equations on a working response; fast, biased for binary data, no likelihood.
- Conditional $\neq$ marginal for non-identity links. Logit: $\beta_M \approx \beta_C / \sqrt{1 + 0.346 \sigma_b^2}$ (attenuation); probit: exact with 1 instead of 0.346; log: only the intercept shifts by $\sigma_b^2/2$.
- An observation-level random effect models overdispersion of counts (Poisson–lognormal).
- Tests: Wald, LRT with ML (same `nAGQ`), boundary mixture for variances. Check overdispersion, random-effect plots, randomised quantile residuals, and the sensitivity to `nAGQ`.

13.15 Exercises with explained solutions

Exercise 13.1. In a Poisson random-intercept model, $\mathrm{E}[y \mid b] = 6e^b$ with $b \sim N(0, 1)$. (a) Compute $\mathrm{E}[y]$, $\mathrm{Var}(y)$ and the variance-to-mean ratio. (b) Two observations of the same group have the same η. Compute their covariance and correlation. (c) If x changes the linear predictor by 0.3, what is the rate ratio conditionally and marginally?

Solution.

(a) $\mathrm{E}[y] = 6e^{1/2} = 9.89$; $\mathrm{Var}(y) = 9.89 + 9.89^2(e-1) = 9.89 + 168.1 = 178.0$; ratio 18.0. (b) $\mathrm{Cov} = 9.89^2(e-1) = 168.1$; $\mathrm{Corr} = 168.1/178.0 = 0.944$. (c) $e^{0.3} = 1.35$ on both scales, since $\mu_M(x) = e^{\eta + \sigma_b^2/2}$ and the factor $e^{1/2}$ cancels in the ratio. Verify numerically:

```
m <- 6 * exp(.5); c(mean = m, var = m + m^2 * (exp(1) - 1), cov = m^2 *
↪  (exp(1) - 1), cor = m^2 * (exp(1) - 1) / (m + m^2 * (exp(1) - 1)))
```

```
#>     mean      var      cov      cor
#>   9.8923 178.0402 168.1479   0.9444
```

Exercise 13.2. Prove that if $y \mid u \sim \mathrm{Poisson}(\mu u)$ and $u \sim \mathrm{Gamma}(k, k)$ then y is negative binomial with $\mathrm{E}[y] = \mu$ and $\mathrm{Var}(y) = \mu + \mu^2/k$ (a) by the law of total variance and (b) from the pmf in Example 13.1. (c) Which k matches the variance of the lognormal random effect $\sigma_b^2 = 0.5$?

Solution.

(a) $\mathrm{E}[y] = \mathrm{E}[\mu u] = \mu$. $\mathrm{Var}(y) = \mathrm{E}[\mu u] + \mathrm{Var}(\mu u) = \mu + \mu^2/k$. (b) From the pmf $\Gamma(y + k)/(y!\Gamma(k))\, p^k (1-p)^y$ with $p = k/(k+\mu)$, the negative binomial has mean $k(1-p)/p = k \cdot \{\mu/(k+\mu)\}/\{k/(k+\mu)\} = \mu$ and variance $k(1-p)/p^2 = \mu(k+\mu)/k = \mu + \mu^2/k$. (c) Matching $\mu^2/k = \mu^2(e^{\sigma_b^2} - 1)$ requires $1/k = e^{0.5} - 1 = 0.649$, i.e. $k = 1.54$ (the lognormal variance is for the marginal mean $m = \mu e^{\sigma_b^2/2}$, so the matching is of the dispersion parameters, not the means).

Exercise 13.3. Apply the Laplace approximation to $n! = \int_0^\infty x^n e^{-x} dx$ and derive Stirling's formula. Compute the relative error for $n = 5$ and $n = 10$.

Solution. Write $x^n e^{-x} = e^{h(x)}$ with $h(x) = n \log x - x$. Then $h'(x) = n/x - 1 = 0$ gives $\hat{x} = n$, $h(\hat{x}) = n \log n - n$ and $-h''(\hat{x}) = n/\hat{x}^2 = 1/n$. The Laplace approximation (13.7) with $q = 1$ is $e^{n \log n - n}\sqrt{2\pi n} = \sqrt{2\pi n}\,(n/e)^n$, Stirling's formula.

```
st <- function(n) sqrt(2 * pi * n) * (n / exp(1))^n
rbind(n = c(5, 10), Stirling = st(c(5, 10)), factorial = factorial(c(5, 10)),
↪  rel_error = st(c(5, 10)) / factorial(c(5, 10)) - 1)
```

```
#>                [,1]       [,2]
#> n           5.00000  1.000e+01
#> Stirling  118.01917  3.599e+06
#> factorial 120.00000  3.629e+06
#> rel_error  -0.01651 -8.296e-03
```

The relative error is -1.7% for $n = 5$ and -0.8% for $n = 10$, approximately $-1/(12n)$: the error is $O(1/n)$, as claimed in Theorem 13.3.

Exercise 13.4. (By hand and R.) A single count $y = 1$ with $\log \mu = b$, $b \sim N(0, \tau^2)$, $\tau^2 = 0.5$. (a) Write $h(b)$ and $h'(b)$. (b) Find the mode and compute the Laplace approximation of L. (c) Compute the exact value with `integrate()` and the adaptive quadrature approximations with $K = 1, 3, 5$.

Solution.

(a) $h(b) = b - e^b - \log 1! - b^2/(2\tau^2) - \frac{1}{2}\log(2\pi\tau^2)$, $h'(b) = 1 - e^b - b/\tau^2$, $-h''(b) = e^b + 1/\tau^2$. (b) $h'(0) = 1 - e^0 - 0 = 0$, so the mode is $\hat{b} = 0$ exactly (at $y = 1$ the Poisson likelihood peaks at $\mu = 1$, i.e. $b = 0$, which is also the mode of the prior). $H = e^0 + 1/0.5 = 3$ and $h(0) = 0 - 1 - 0 - 0 - \frac{1}{2}\log(\pi) = -1.5724$. $L \approx e^{-1.5724}\sqrt{2\pi/3} = 0.2076 \times 1.4472 = 0.3004$. (c)

```
tau2 <- .5; fh <- function(b) dpois(1, exp(b)) * dnorm(b, 0, sqrt(tau2))
ex <- integrate(fh, -Inf, Inf, rel.tol = 1e-12)$value
bh <- 0; Hh <- exp(bh) + 1 / tau2
sapply(c(exact = 0, K1 = 1, K3 = 3, K5 = 5), function(K) if (K == 0) ex else
↪  quad(fh, bh, Hh, K))
```

```
#>  exact     K1     K3     K5
#> 0.2986 0.3004 0.2976 0.2986
```

The Laplace value ($K = 1$) is 0.3004 against the exact 0.2986; $K = 3$ and $K = 5$ are essentially exact. (The functions `quad` and `gh_rule` are defined in the text.)

Exercise 13.5. Show that adaptive Gauss–Hermite quadrature with $K = 1$ is the Laplace approximation, and that with $K = 1$ it is *exact* when h is a quadratic function (as in the LMM).

Solution. For $K = 1$ the Hermite polynomial $H_1(x) = 2x$ has the single root $x_1 = 0$ and the weight must integrate 1 exactly against e^{-x^2}, so $w_1 = \int e^{-x^2} dx = \sqrt{\pi}$. Then (13.11) gives $\sqrt{2/H}\sqrt{\pi}e^0 e^{h(\hat{b})} = e^{h(\hat{b})}\sqrt{2\pi/H}$, the Laplace formula. If $h(b) = h(\hat{b}) - \frac{1}{2}H(b - \hat{b})^2$ exactly, then

$u(x) = e^{x^2 + h(\hat b) - x^2} = e^{h(\hat b)}$ is constant, a polynomial of degree $0 \le 2K-1 = 1$, so the $K=1$ rule is exact. In the LMM, $\log f(y \mid b)$ is quadratic in b and so is $\log \phi(b)$.

Exercise 13.6. Show that for a canonical link and group design Z_i, $-h_i''(b) = Z_i^\top W_i Z_i + G^{-1}$ with $W_i = \operatorname{diag}\{b''(\theta_{ij})/a_{ij}\}$ in the notation $f(y;\theta,\phi) = \exp\{[y\theta - b(\theta)]/a(\phi) + c(y,\phi)\}$. Give W_i for the Poisson and the binomial.

Solution. With a canonical link, $\theta_{ij} = \eta_{ij} = x_{ij}^\top \beta + z_{ij}^\top b$, so $\log f(y_{ij} \mid b) = [y_{ij}\eta_{ij} - b(\eta_{ij})]/a_{ij} + c$. Its gradient in b is $z_{ij}\{y_{ij} - b'(\eta_{ij})\}/a_{ij} = z_{ij}(y_{ij} - \mu_{ij})/a_{ij}$ and its Hessian is $-z_{ij}z_{ij}^\top b''(\eta_{ij})/a_{ij}$, since $\partial\eta/\partial b = z$. Summing over j gives $-Z_i^\top W_i Z_i$ with $W_i = \operatorname{diag}\{b''(\eta_{ij})/a_{ij}\}$, and the prior term $-\tfrac12 b^\top G^{-1} b$ contributes $-G^{-1}$. For the Poisson, $b(\theta) = e^\theta$ and $b'' = \mu$, $a = 1$: $W = \operatorname{diag}(\mu_{ij})$. For the binomial with s_{ij} trials, modelling the *count*, $W = \operatorname{diag}\{s_{ij} p_{ij}(1-p_{ij})\}$. For non-canonical links the Hessian has an extra term; the usual practice is to use the expected version $Z^\top W Z$ with $W = \{a\, V(\mu) g'(\mu)^2\}^{-1}$ (the IRLS weight).

Exercise 13.7.
(a) Derive the latent-scale ICC (13.2) for the logistic random intercept. (b) Compute it for `cbpp` ($\hat\sigma_b^2 = 0.412$) and for the `MASS::bacteria` model `y ~ trt + I(week > 2) + (1 | ID)` ($\hat\sigma_b^2 = 1.543$). (c) Why is this ICC not the correlation between two binary responses of the same group?

Solution.

(a) In $y_{ij}^* = \eta_{ij} + b_i + e_{ij}$ the common term is b_i with variance σ_b^2, and $\operatorname{Var}(e_{ij}) = \pi^2/3$ for a standard logistic variable, so $\operatorname{Corr}(y_{ij}^*, y_{ik}^*) = \sigma_b^2/(\sigma_b^2 + \pi^2/3)$. (b) $0.412/(0.412 + 3.290) = 0.111$; $1.543/(1.543 + 3.290) = 0.319$. (c) The correlation of the observed 0/1 responses depends on the covariates and on the marginal probabilities (it is smaller than the latent correlation: e.g. for $\sigma_b = 2$ and $p \approx 0.2$, the binary correlation is much smaller), because dichotomising a latent variable loses information. The latent ICC is a scale-free measure of clustering comparable across models.

Exercise 13.8. (R.) For $\sigma_b = 0.5, 1, 2, 3$ compute, by numerical integration, the marginal curve $\mu_M(x) = \int \operatorname{expit}(x+b)\phi(b;0,\sigma_b^2)db$, fit a logistic regression slope to it over $x \in [-3,3]$ (e.g. with `glm(p ~ x, family = quasibinomial)`) and compare with the approximation $1/\sqrt{1 + 0.346\sigma_b^2}$ and with the probit factor $1/\sqrt{1+\sigma_b^2}$. Comment.

Solution.

```
sig <- c(.5, 1, 2, 3); xs <- seq(-3, 3, by = .05)
slope <- sapply(sig, function(s) {
  p <- sapply(xs, function(x0) integrate(function(b) plogis(x0 + b) * dnorm(b,
↪  0, s), -Inf, Inf)$value)
  coef(glm(p ~ xs, family = quasibinomial))[[2]]
})
round(cbind(sigma_b = sig, best_logistic_slope = slope, approx = 1 / sqrt(1 +
↪  .346 * sig^2), naive_probit = 1 / sqrt(1 + sig^2)), 3)
```

```
#>      sigma_b best_logistic_slope approx naive_probit
#> [1,]     0.5                0.955  0.959        0.894
#> [2,]     1.0                0.848  0.862        0.707
#> [3,]     2.0                0.622  0.648        0.447
```

```
#> [4,]     3.0                0.468  0.493        0.316
```

The approximation $1/\sqrt{1+0.346\sigma_b^2}$ is close for moderate σ_b (differences of 0.004 at $\sigma_b = 0.5$ and 0.03 at $\sigma_b = 2$) and slightly too large: the true attenuation is a bit stronger, because the exact marginal curve is not exactly logistic. The probit factor $1/\sqrt{1+\sigma_b^2}$, applied to a logistic model, over-attenuates badly (0.45 against 0.62 at $\sigma_b = 2$): the constant $c^2 = 0.346$ matters. The attenuation is negligible for $\sigma_b = 0.5$ (4%) and large for $\sigma_b = 3$ (more than 50%).

Exercise 13.9. (R analysis.) The data `MASS::epil` record the number of seizures `y` in four two-week periods for 59 epileptic patients randomised to placebo or progabide (`trt`), with baseline seizures `base` and `age`. Centre `log(base/4)` and `log(age)`, fit `y ~ lbase_c * trt + lage_c + V4 + (1 | subject)`, then add an observation-level random effect. (a) Report the Pearson dispersion of the Poisson GLM. (b) Compare the three models by AIC and the OLRE by LRT (boundary!). (c) Interpret the treatment effect with the interaction. (d) Compare the conditional and marginal interpretation of the rate ratio.

Solution.

```
data(epil, package = "MASS")
epil$lbase_c <- log(epil$base / 4) - mean(log(epil$base / 4)); epil$lage_c <-
↪  log(epil$age) - mean(log(epil$age))
epil$trt <- relevel(factor(epil$trt), "placebo"); epil$obs <- 1:nrow(epil)
ep0 <- glm(y ~ lbase_c * trt + lage_c + V4, family = poisson, data = epil)
ep1 <- glmer(y ~ lbase_c * trt + lage_c + V4 + (1 | subject), family =
↪  poisson, data = epil)
ep2 <- update(ep1, . ~ . + (1 | obs))
c(pearson_glm = sum(residuals(ep0, "pearson")^2) / ep0$df.residual)
```

```
#> pearson_glm
#>       4.412
```

```
AIC(ep0, ep1, ep2)
```

```
#>     df  AIC
#> ep0  6 1647
#> ep1  7 1345
#> ep2  8 1266
```

```
T_ol <- as.numeric(2 * (logLik(ep2) - logLik(ep1))); c(T = T_ol, p_mixture =
↪  .5 * pchisq(T_ol, 1, lower.tail = FALSE))
```

```
#>         T p_mixture
#> 8.099e+01 1.135e-19
```

```
round(rbind(glm = coef(ep0), subject = fixef(ep1), subject_plus_OLRE =
↪  fixef(ep2))[, c("trtprogabide", "lbase_c", "lbase_c:trtprogabide")], 3)
```

```
#>                   trtprogabide lbase_c lbase_c:trtprogabide
#> glm                     -0.347   0.950                0.562
#> subject                 -0.335   0.884                0.338
#> subject_plus_OLRE       -0.331   0.880                0.349
```

```
round(rbind(glm = sqrt(diag(vcov(ep0))), subject = sqrt(diag(vcov(ep1))), olre
↪  = sqrt(diag(vcov(ep2))))[, "trtprogabide"], 3)
```

```
#>     glm subject    olre
#>   0.061   0.147   0.145
```

```
VarCorr(ep2)
```

```
#>  Groups  Name        Std.Dev.
#>  obs     (Intercept) 0.357
#>  subject (Intercept) 0.459
```

(a) The Pearson dispersion of the Poisson GLM is 4.4: strong overdispersion, and the GLM standard error for the treatment effect (0.061) is far too small. (b) The patient random effect lowers AIC from 1647 to 1345 (the repeated measures on the same patient are highly correlated), and the OLRE to 1266; the LRT of the OLRE has $T = 81$ with boundary p-value $< 10^{-15}$. (c) With the covariates centred, the `trtprogabide` coefficient is the log rate ratio of progabide versus placebo for a patient with *average baseline seizures* and age: -0.33 (rate ratio 0.72), with standard error 0.15: a reduction by about 28% with a Wald p-value around 0.02; the interaction `lbase_c:trtprogabide` is positive (the benefit shrinks for patients with high baseline counts) but not significant. (d) With the log link, the conditional and the marginal rate ratios are the same: only the intercept shifts by $\sigma^2/2$ (total variance of patient and observation effects), which is another reason why GLMMs with log link are easy to interpret.

Exercise 13.10. (R simulation.) Generate 150 clusters of 3 binary observations with $\text{logit}\, P(y_{ij} = 1) = -1 + x_{ij} + b_i$, $b_i \sim N(0, 2^2)$. Over 40 replicates compare the estimates of the slope and of σ_b obtained by Laplace (`nAGQ = 1`), adaptive quadrature (`nAGQ = 15`) and PQL (`glmmPQL`).

Solution.

```
set.seed(2)
sdl <- function(f) if (inherits(f, "lme")) as.numeric(VarCorr(f)[1, 2]) else
↪  attr(VarCorr(f)$id, "stddev")[[1]]
res <- replicate(40, {
  m <- 150; n <- 3; id <- rep(1:m, each = n); x <- rnorm(m * n); b <- rnorm(m,
↪  0, 2)[id]
  d <- data.frame(y = rbinom(m * n, 1, plogis(-1 + x + b)), x = x, id =
↪  factor(id))
  f1 <- glmer(y ~ x + (1 | id), family = binomial, data = d); f2 <- update(f1,
↪  nAGQ = 15)
  f3 <- MASS::glmmPQL(y ~ x, ~ 1 | id, binomial, d, verbose = FALSE)
  c(slope_Laplace = fixef(f1)[[2]], slope_AGQ = fixef(f2)[[2]], slope_PQL =
   ↪  fixef(f3)[[2]],
    sd_Laplace = sdl(f1), sd_AGQ = sdl(f2), sd_PQL = sdl(f3))
})
round(rbind(mean = rowMeans(res), sd = apply(res, 1, sd)), 3)
```

```
#>      slope_Laplace slope_AGQ slope_PQL sd_Laplace sd_AGQ sd_PQL
#> mean         0.956     0.987     0.917      1.830  2.038  1.874
#> sd           0.181     0.189     0.200      0.304  0.334  0.289
```

The true values are 1 for the slope and 2 for σ_b. For binary data with three observations per cluster, the Laplace approximation and PQL **underestimate** the cluster standard deviation (about 1.8–1.9 on average), and the slope is attenuated too, while adaptive quadrature is closest to the truth for σ_b (it is slightly high on average with such small samples). The difference is a systematic bias, comparable to the sampling standard deviation.

Exercise 13.11. (R analysis.) For `grouseticks`, compare three ways to handle the overdispersion in `TICKS ~ YEAR + HEIGHT_s`: (i) `MASS::glm.nb` (no random effects), (ii) a Poisson GLMM with `(1 | BROOD)` only, (iii) the OLRE model of the text. Compare AIC and the standard error of the altitude effect.

Solution.

```
nb <- MASS::glm.nb(TICKS ~ YEAR + HEIGHT_s, data = gt)
g_b <- glmer(TICKS ~ YEAR + HEIGHT_s + (1 | BROOD), family = poisson, data =
↪  gt)
round(rbind(`NB GLM` = c(AIC(nb), coef(nb)["HEIGHT_s"],
↪  sqrt(vcov(nb)["HEIGHT_s", "HEIGHT_s"])),
            `Poisson + brood` = c(AIC(g_b), fixef(g_b)["HEIGHT_s"],
            ↪  sqrt(vcov(g_b)["HEIGHT_s", "HEIGHT_s"])),
            `Poisson + brood + location + OLRE` = c(AIC(fit_g2),
            ↪  fixef(fit_g2)["HEIGHT_s"], sqrt(vcov(fit_g2)["HEIGHT_s",
            ↪  "HEIGHT_s"]))), 3)
```

```
#>                                        HEIGHT_s
#> NB GLM                           1942   -0.896 0.071
#> Poisson + brood                  1988   -0.858 0.108
#> Poisson + brood + location + OLRE 1795   -0.854 0.124
```

The negative binomial GLM removes most of the overdispersion (AIC 1942 against 4383 for the Poisson GLM) but it cannot represent that chicks of the same brood are correlated: its standard error for altitude is the smallest of the three, and too small. The models with random effects have standard errors about 1.5 to 1.75 times as large; the best AIC (1795) belongs to the model with all three levels of variation. Overdispersion and correlation are different problems with a common cure when one needs both.

Exercise 13.12. (Critique.) Evaluate: (a) "The odds ratio for treatment from the logistic mixed model is 2.4, so in the whole population treatment multiplies the odds of the outcome by 2.4." (b) "The Pearson dispersion of the model with an observation-level random effect is 0.41, so the data are underdispersed." (c) "We chose the model with `nAGQ = 25` because its log-likelihood (-50.0) is higher than that of the Laplace fit (-92.0)." (d) "`predict(fit, re.form = NA, type = 'response')` gives the average probability in the population."

Solution.

(a) A GLMM odds ratio is conditional on the random effect: it applies to a given cluster (for subjects with the same random effect), not to the population as a whole. The marginal odds ratio is closer to 1 (attenuation). (b) Pearson residuals in the OLRE model are conditional on the estimated observation-level effects, which absorb the extra-Poisson variation by construction; the number is not interpretable as under-dispersion. (c) The two log-likelihoods are on different scales (`lme4` evaluates `logLik()` differently for `nAGQ = 1` and `nAGQ > 1`); compare estimates, not log-likelihoods across `nAGQ`. (d) `re.form`

`= NA` sets the random effects to 0, so it returns the probability for the typical cluster, $\text{expit}(x^\top \hat{\beta})$, which is below the population average at low probabilities (0.198 against 0.216 in `cbpp`, period 1). The population average requires integrating over the random-effect distribution.

Exercise 13.13. (Communicate.) Write about 150 words for a health agency that must plan vaccine stocks for a region and a veterinarian who advises single farmers, explaining why the same `cbpp` analysis gives two different "effects of period 2 versus period 1".

Solution. A model answer. "Herds differ in how exposed their animals are. Our model describes each herd by a baseline risk and a common change over time. For a given herd, the odds of a new case in the second period are about 37% of the odds in the first period (an odds ratio of 0.37). This is the number a veterinarian advising a single farmer should use. It is not the change in the *region-wide* proportion of sick animals. Because herds have very different baselines, averaging over herds mixes herds with low and high risks, and the region-wide change is a bit less steep: an odds ratio of about 0.40. The agency planning vaccine stocks needs the second number and the average proportion (21.6% in the first period, against 19.8% in a typical herd). Both numbers come from the same model, and neither is wrong: they answer different questions."

13.16 Further reading

Breslow and Clayton (1993) introduced PQL; Tierney and Kadane (1986) is the source of the accurate Laplace approximations; Zeger, Liang and Albert (1988) contrast mixed and marginal (GEE) models; Bolker et al. (2009) is a practical guide to GLMMs; Harrison (2014) discusses observation-level random effects; Elston et al. (2001) introduced the tick data. Wood (2017, Ch. 3.4), Pinheiro and Bates (2000) and Bates et al. (2015) describe the methods and software.

Part V

Smoothing

Chapter 14

Polynomial regression and scatterplot smoothing

Every model so far described the effect of a covariate through a handful of parameters: a slope, a log-odds ratio, a random-effect variance. That is the right thing to do when theory or experience tells us the *shape* of the relationship. When it does not, we would like the data to draw the curve. This chapter is about estimating a regression function f in

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(\varepsilon_i) = 0, \quad \mathrm{Var}(\varepsilon_i) = \sigma^2, \quad i = 1, \dots, n,$$

when we only know that f is *smooth*. We first push the linear model as far as it goes (polynomial regression) and see exactly where and why it fails. Then we study *local* methods: kernel smoothers, local polynomial regression and LOESS. All of them turn out to be **linear smoothers**, $\hat{\boldsymbol{f}} = S\boldsymbol{y}$, so one framework gives us effective degrees of freedom, a bias–variance decomposition and a fast leave-one-out cross-validation formula for choosing the amount of smoothing (Hastie and Tibshirani, 1990; Buja et al., 1989; Wand and Jones, 1995; Eilers and Marx, 2021). Chapter 15 continues with splines, which combine global and local ideas and lead to generalized additive models (Chapter 16).

```
library(MASS)
data(mcycle)
x <- mcycle$times
y <- mcycle$accel
n <- length(y)
```

14.1 A motivating example: the motorcycle data

The data frame `mcycle` (package `MASS`) records the head acceleration (`accel`, in units of g) of a crash-test dummy at `times` milliseconds after a simulated motorcycle impact (Silverman, 1985). There are $n = 133$ observations at only 94 distinct times, so some times are tied. The data are sorted by time.

```
ggplot(mcycle, aes(times, accel)) +
  geom_point(colour = ie_navy, size = 1.8) +
  labs(x = "Time after impact (ms)", y = "Head acceleration (g)")
```

No physical formula for f is available, and no simple parametric family describes the picture: a flat stretch, a sharp dip, an overshoot and a decay. A straight line explains only $R^2 = 0.088$ of the variation. Two further features will matter later: the noise is far from constant (it is

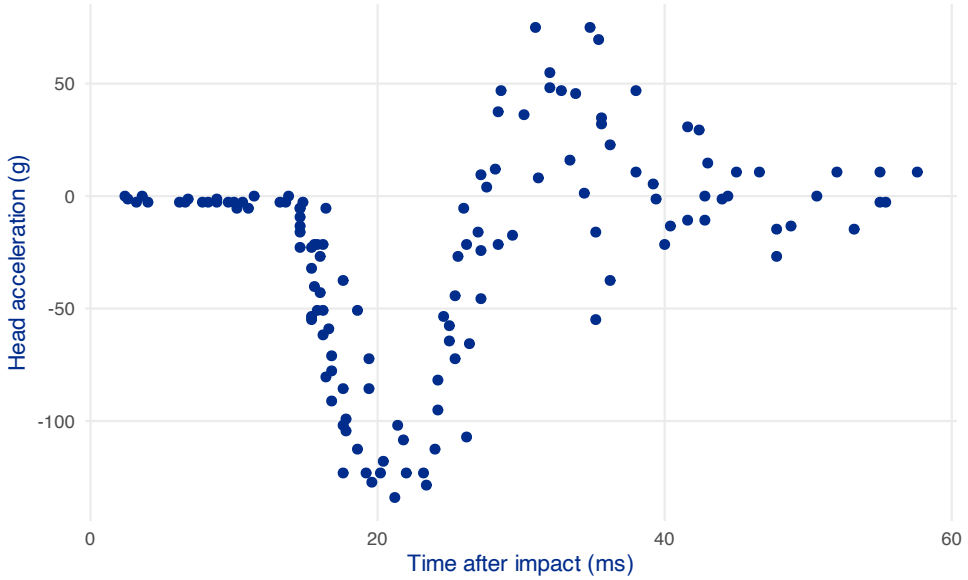


Figure 14.1: The motorcycle data. The head is almost still for 14 ms, then is thrown backwards, rebounds, and settles. The scatter around the trend is small at the beginning and large in the middle.

largest around 20–35 ms), and the design is uneven (many observations between 10 and 30 ms, few after 45 ms). We use `mcycle` throughout this chapter and the next, and a second data set (`faraway::exa`, a known function observed with noise) whenever we want to compare an estimate with the truth.

There are two strategies. A **global** method uses one flexible family of functions for the whole range of x and fits it by least squares (polynomials, Section 14.2). A **local** method estimates $f(x)$ from the observations near x (Sections 14.3–14.6). We shall see that the second strategy is much better behaved, and that both are special cases of a linear smoother (Section 14.7).

14.2 Polynomial regression

14.2.1 The model

A polynomial of degree q is nonlinear in x but linear in the parameters,

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_q x_i^q + \varepsilon_i, \qquad X = \begin{bmatrix} 1 & x_1 & \cdots & x_1^q \\ \vdots & & & \vdots \\ 1 & x_n & \cdots & x_n^q \end{bmatrix},$$

so everything in Chapters 5–7 applies: $\hat\beta = (X^\top X)^{-1} X^\top y$, fitted values $\hat y = Hy$ with $H = X(X^\top X)^{-1}X^\top$, and $\operatorname{tr}(H) = q + 1$ provided X has full column rank, which requires at least $q+1$ distinct values of x. The degree q is a tuning constant: the larger it is, the more flexible the fit.

```
degrees <- c(1, 3, 5, 8, 12, 20)
fits <- lapply(degrees, function(q) lm(accel ~ poly(times, q), data = mcycle))
tab <- data.frame(degree = degrees,
                  RSS = sapply(fits, function(f) round(sum(resid(f)^2))),
                  R2 = sapply(fits, function(f) round(summary(f)$r.squared,
                   ↪  3)),
```

```
                  AIC = sapply(fits, function(f) round(AIC(f), 1)))
tab
```

```
#>   degree    RSS    R2  AIC
#> 1      1 281144 0.088 1402
#> 2      3 206424 0.330 1365
#> 3      5 145962 0.526 1322
#> 4      8  88740 0.712 1262
#> 5     12  61693 0.800 1222
#> 6     20  60270 0.804 1235
```

The residual sum of squares can only decrease with q, and AIC is smallest near degree 12. Figure 14.2 shows why one should not stop there.

```
grid <- data.frame(times = seq(min(x), max(x), length.out = 300))
pf <- do.call(rbind, lapply(c(3, 8, 12, 20), function(q) {
  m <- lm(accel ~ poly(times, q), data = mcycle)
  data.frame(times = grid$times, fit = predict(m, grid), degree =
  ↪  paste("degree", q))
}))
pf$degree <- factor(pf$degree, levels = paste("degree", c(3, 8, 12, 20)))
ggplot(pf, aes(times, fit)) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey60", size = 1) +
  geom_line(colour = ie_coral, linewidth = 0.9) + facet_wrap(~degree, nrow =
  ↪  1) +
  coord_cartesian(ylim = c(-150, 100)) + labs(x = "Time (ms)", y =
  ↪  "Acceleration (g)")
```

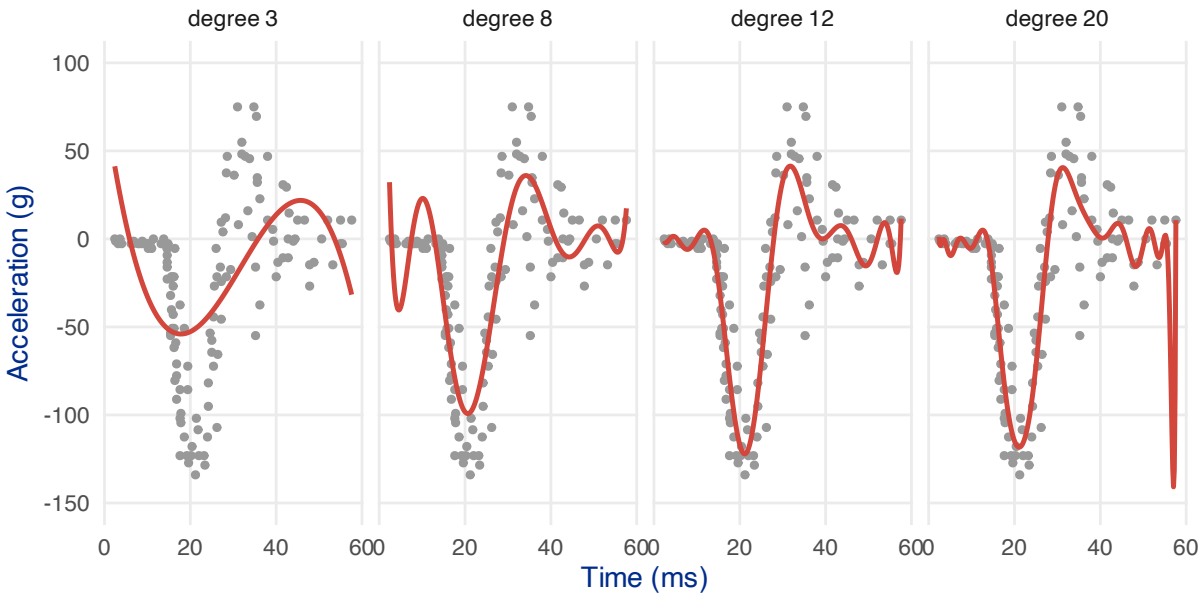


Figure 14.2: Polynomial fits of degree 3, 8, 12 and 20 to the motorcycle data. Higher degrees follow the dip but oscillate at the ends, and degree 20 is already visibly unstable at the last observations.

14.2.2 Why polynomials fail at the boundaries

Two different problems hide behind the wiggles of Figure 14.2. It is important to separate them, because only one of them can be cured by a better parametrisation.

A statistical problem: leverage. The variance of the fitted value is $\mathrm{Var}(\hat{y}_i) = \sigma^2 h_{ii}$ with $h_{ii} = H_{ii}$ the leverage, and $\sum_i h_{ii} = q + 1$. For a high-degree polynomial the leverages pile up at both ends of the x-range.

```
lev <- do.call(rbind, lapply(c(5, 12), function(q) {
  m <- lm(accel ~ poly(times, q), data = mcycle)
  data.frame(times = x, h = hatvalues(m), degree = paste("degree", q))
}))
ggplot(lev, aes(times, h, colour = degree)) + geom_point(size = 1.4) +
  scale_colour_manual(values = c(ie_navy, ie_coral), name = NULL) +
  labs(x = "Time (ms)", y = expression("leverage " * h[ii]))
```

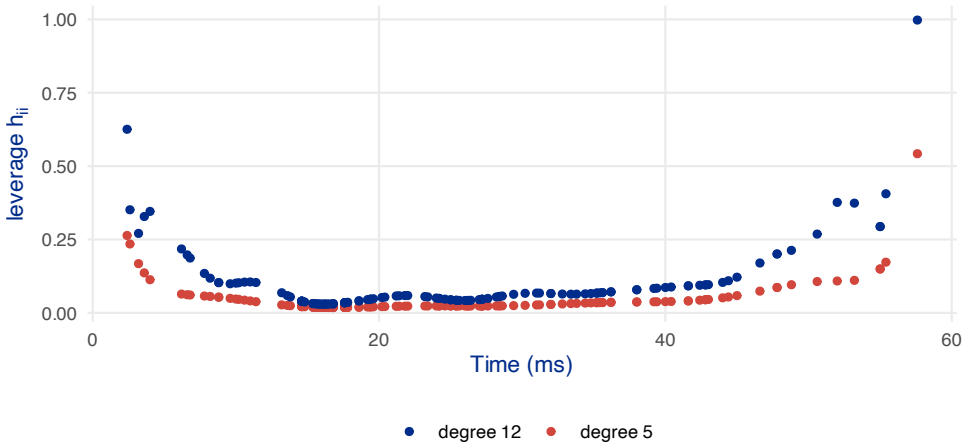


Figure 14.3: Leverage h_{ii} of polynomial fits of degree 5 and 12 on the motorcycle data. The variance of the fitted value at the last observation is almost the full error variance.

```
m12 <- lm(accel ~ poly(times, 12), data = mcycle)
c(last = unname(hatvalues(m12)[n]), median = median(hatvalues(m12)), average =
↪  13 / n)
```

```
#>    last  median average
#> 0.99770 0.05825 0.09774
```

For degree 12 the last observation has leverage 0.998: the fitted value at 57.6 ms is essentially the observation itself, with variance close to σ^2, whereas the typical leverage is only 0.058. The reason is structural. A polynomial of degree q is a *global* object: changing one response near the edge changes the whole curve, and the highest powers x^q are dominated by the extreme values of x. Buying flexibility around the dip at 20 ms is paid for with variance at 50 ms. No reparametrisation changes this, because it does not change the column space of X and hence not H.

Extrapolation. Outside the data the leading term $\hat{\beta}_q x^q$ takes over.

```
new <- data.frame(times = c(57.6, 60, 65))
sapply(c(3, 8, 12), function(q) round(predict(lm(accel ~ poly(times, q), data
↪  = mcycle), new), 1))
```

```
#>      [,1]   [,2]    [,3]
#> 1  -31.7   17.4    11.6
#> 2  -58.8  198.9   999.2
#> 3 -138.1 2483.1 36711.0
```

At 65 ms, only 7.4 ms after the last observation, the degree-12 fit predicts an acceleration of tens of thousands of g while the data never leave the interval $[-134, 75]$. A polynomial must never be used for extrapolation.

A numerical problem: conditioning. The columns $x, x^2, \dots, x^q$ are almost collinear on $[2.4, 57.6]$.

```
signif(sapply(c(3, 5, 8, 10, 12), function(q) kappa(outer(x, 0:q, `^`), exact
↪  = TRUE)), 3)
```

```
#> [1] 3.08e+05 1.58e+09 8.83e+14 2.08e+18 9.18e+22
```

```
sum(is.na(coef(lm(accel ~ poly(times, 15, raw = TRUE), data = mcycle))))
```

```
#> [1] 2
```

The condition number $\kappa(X) = \sigma_{\max}(X)/\sigma_{\min}(X)$ of the raw design grows by about four orders of magnitude for every two extra degrees, and with $q = 15$ the function `lm()` silently declares two coefficients non-estimable (`NA`) because $X^\top X$ is numerically singular. The normal equations make things worse: $\kappa(X^\top X) = \kappa(X)^2$, so a relative error of size ϵ in the data can be amplified by up to $\kappa(X)^2$ in $\hat\beta$. The cure for this second problem is a better basis.

14.2.3 Orthogonal polynomials

Let $X = [\mathbf{1}, x, \dots, x^q]$ have full column rank. Gram–Schmidt (equivalently, a QR decomposition) applied to the columns of X, with the usual inner product on $\mathbb{R}^n$ evaluated *at the data*, yields

$$X = \tilde Q R, \qquad \tilde Q = [\tilde q_0, \tilde q_1, \dots, \tilde q_q], \quad \tilde Q^\top \tilde Q = I_{q+1}, \quad R \text{ upper triangular.}$$

The first column is $\tilde q_0 = \mathbf{1}/\sqrt{n}$ and $\tilde q_k$ is a polynomial of degree k in x evaluated at the data, orthogonal to all polynomials of lower degree. The R function `poly(x, q)` returns $Q = [\tilde q_1, \dots, \tilde q_q]$ (up to sign); because the intercept is not included, each column also has mean zero.

Theorem 14.1 (Orthogonal polynomials)**.** *Let $X = \tilde Q R$ as in (14.2.3) and $\hat\gamma = \tilde Q^\top y$. Then*

(a) $\tilde Q R \hat\beta = \tilde Q \hat\gamma$ with $\hat\beta = R^{-1}\hat\gamma$: the fitted values of the raw and the orthogonal fits are identical, and the raw coefficients are an exact *linear function of the orthogonal ones;*

(b) $\hat\gamma \sim (\gamma, \sigma^2 I_{q+1})$: the orthogonal coefficients are uncorrelated, with equal variance, and adding a degree does not change the earlier ones;

(c) $\kappa(\tilde Q) = 1$ while $\kappa(X) = \kappa(R)$.

Proof.

(a) $\hat\beta = (X^\top X)^{-1} X^\top y = (R^\top \tilde Q^\top \tilde Q R)^{-1} R^\top \tilde Q^\top y = R^{-1} R^{-\top} R^\top \tilde Q^\top y = R^{-1}\hat\gamma$. Hence $X\hat\beta = \tilde Q R R^{-1}\hat\gamma = \tilde Q \hat\gamma$. (b) $\operatorname{Var}(\hat\gamma) = \tilde Q^\top (\sigma^2 I) \tilde Q = \sigma^2 I_{q+1}$. Since $\hat\gamma_k = \tilde q_k^\top y$ does not involve q, it does not change when higher columns are added. (c) The singular values of a matrix with orthonormal columns are all 1; and $X = \tilde Q R$ has the same singular values as R.

□

Part (b) explains why the sequential t-tests of `summary(lm(y ~ poly(x, q)))` are independent of the order in which degrees are added, and part (c) why `poly()` fixes the numerical problem. Two warnings. First, the columns of `poly()` are *uncorrelated* (orthogonal with mean zero), not *independent*: $\tilde{q}_2$ is a deterministic function of $\tilde{q}_1$. Second, orthogonal polynomials do **not** cure the leverage problem: $H = \tilde{Q}\tilde{Q}^\top$ is the same matrix as before.

Example 14.1 (From orthogonal to raw coefficients, exactly)**.** The identity $\hat{\beta} = R^{-1}\hat{\gamma}$ means that the raw cubic coefficients are recovered from the orthogonal fit without refitting.

```
Q <- poly(x, 3)
Xraw <- cbind(1, x, x^2, x^3)
A <- solve(crossprod(Xraw), crossprod(Xraw, cbind(1, Q)))   # [1, Q] = Xraw
↪  %*% A
fit_orth <- lm(y ~ poly(x, 3))
fit_raw  <- lm(y ~ poly(x, 3, raw = TRUE))
cbind(converted = drop(A %*% coef(fit_orth)), raw_fit = coef(fit_raw))
```

```
#>    converted    raw_fit
#>    78.582514  78.582514
#> x -17.113715 -17.113715
#>     0.667766   0.667766
#>    -0.007012  -0.007012
```

```
c(max_error_A = max(abs(Xraw %*% A - cbind(1, Q))),
  max_diff_fitted = max(abs(fitted(fit_orth) - fitted(fit_raw))))
```

```
#>     max_error_A max_diff_fitted
#>       7.006e-14       8.740e-13
```

The two columns agree to all printed digits. Here A is the upper triangular matrix with $[\mathbf{1}, Q] = X_{\text{raw}}A$ (a different normalisation of the same QR idea, since `poly()` does not scale the constant column), and the equality $\hat{\beta}_{\text{raw}} = A\hat{\gamma}$ holds because X_{raw} has full column rank, so the coefficient vector of a given fitted vector is unique. There is no need to "approximate" the raw coefficients by refitting on a grid.

14.2.4 Choosing the degree

With q as tuning constant, AIC and cross-validation are available. AIC for the Gaussian linear model is $n\log(\text{RSS}/n) + 2(q+2)$ up to constants, which `AIC()` reports. Cross-validation needs care because `mcycle` is sorted by time: contiguous folds would force each test fold to be an extrapolation. We shuffle.

```
set.seed(2027)
folds <- sample(rep(1:5, length.out = n))
cv_degree <- function(q) {
  sse <- 0
  for (k in 1:5) {
    tr <- folds != k
    m <- lm(accel ~ poly(times, q), data = mcycle[tr, ])
    sse <- sse + sum((mcycle$accel[!tr] - predict(m, mcycle[!tr, ]))^2)
  }
  sse / n
```

```
}
cv <- sapply(1:15, cv_degree)
round(cv, 0)
```

```
#>  [1] 2188 2065 1654 1683 1258 1139 1258  790 1310 1696 2200 1404 2009
#> [14] 8522 5113
```

```
which.min(cv)
```

```
#> [1] 8
```

The curve is jagged (8 is the minimiser for this particular split and would change with the seed), and even its minimum is not good: five-fold CV error 790 is far above the value of about 560 that the smoothers below reach. The conclusion is that a global polynomial is simply the wrong tool for this curve. The remedy is to let each point influence only its neighbourhood.

14.3 Local averaging

To estimate $f(x)$ we average the responses of the observations whose x_j is close to x.

Definition 14.1 (Local averaging estimator)**.** A **local averaging** estimator of f at x is

$$\hat{f}(x) = \sum_{j=1}^{n} w_j(x)\, y_j, \qquad \sum_{j=1}^{n} w_j(x) = 1,$$

where the weights $w_j(x)$ depend on x, on the design points $x_1, \dots, x_n$ and on a tuning constant, but **not** on y.

The classical choices differ only in the weights:

- **Bin smoother.** Cut the x-axis into intervals and take the mean of the y_j in the interval containing x. The fit is a step function.
- **Running mean / k nearest neighbours (kNN).** $w_j(x) = 1/k$ for the k points with x_j closest to x. The fit is a step function that jumps whenever the neighbourhood changes.
- **Kernel smoother.** $w_j(x) \propto K_h(x - x_j)$: weights that decrease smoothly with the distance $|x - x_j|$ (Section 14.4).

```
nw_S <- function(x0, x, h) {                          # Nadaraya-Watson weights,
↪  one row per x0
  K <- dnorm(outer(x0, x, "-") / h)
  K / rowSums(K)
}
i0 <- which.min(abs(x - 20))
w_bin <- as.numeric(cut(x, seq(0, 60, 5)) == cut(x[i0], seq(0, 60, 5))); w_bin
↪  <- w_bin / sum(w_bin)
w_knn <- numeric(n); w_knn[order(abs(x - x[i0]))[1:15]] <- 1 / 15
w_nw <- nw_S(x[i0], x, 2)[1, ]
d <- abs(x - x[i0]); Delta <- sort(d)[ceiling(0.3 * n)]
w_tri <- ifelse(d < Delta, (1 - (d / Delta)^3)^3, 0); w_tri <- w_tri /
↪  sum(w_tri)
wd <- data.frame(times = rep(x, 4), w = c(w_bin, w_knn, w_nw, w_tri),
                 scheme = factor(rep(c("bins (5 ms)", "kNN (k = 15)",
                 ↪  "Gaussian (h = 2)", "tricube (span 0.3)"),
```

```
                                        each = n), levels = c("bins (5 ms)", "kNN
                                        ↪  (k = 15)", "Gaussian (h = 2)",
                                        ↪  "tricube (span 0.3)")))
ggplot(wd, aes(times, w)) + geom_segment(aes(xend = times, yend = 0), colour =
↪  ie_navy) +
  geom_vline(xintercept = x[i0], colour = ie_coral, linetype = 2) +
  ↪  facet_wrap(~scheme, nrow = 1) +
  coord_cartesian(xlim = c(5, 35)) + labs(x = "Time (ms)", y = "weight")
```

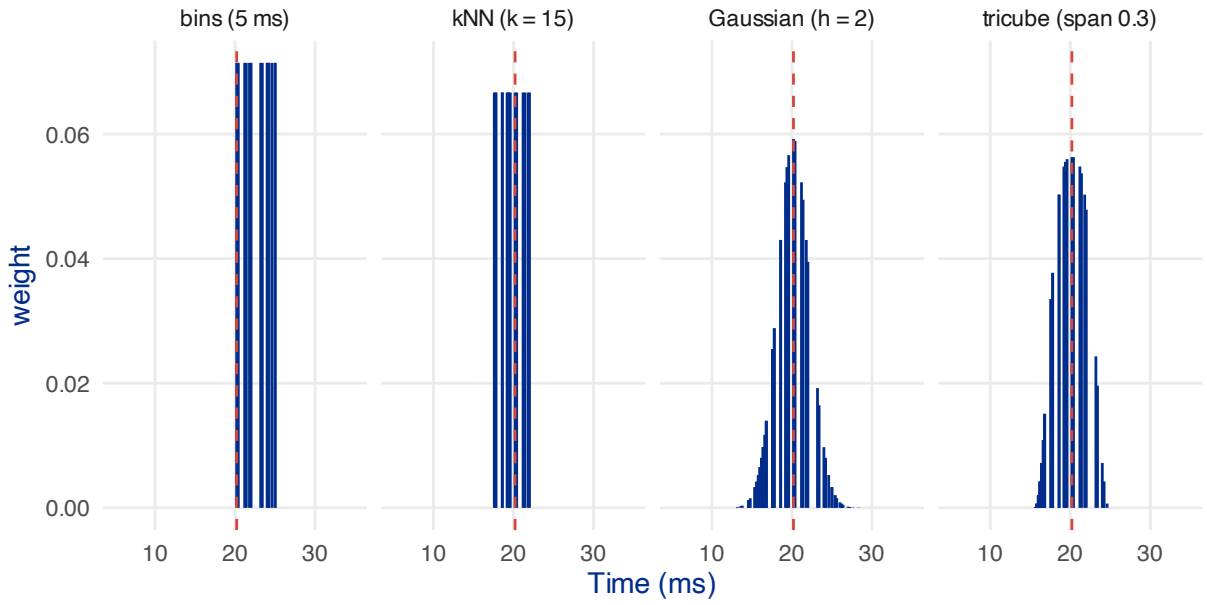


Figure 14.4: The weights of four local averaging schemes at the target point $x = 20$ ms (dashed line) on the motorcycle data. Bins and kNN give equal weights and jump; kernel weights fade smoothly.

Every local averaging estimator is *linear* in y: stacking $\hat{f}(x_i)$ for $i = 1, \dots, n$ gives $\hat{\boldsymbol{f}} = S\boldsymbol{y}$ with $S_{ij} = w_j(x_i)$. We return to this structure in Section 14.7. For the k-nearest-neighbour mean, row i of S has k entries equal to $1/k$ and one of them is the diagonal (a point is its own nearest neighbour, at distance 0), so $S_{ii} = 1/k$ and $\operatorname{tr}(S) = n/k$ (assuming no ties in the distances).

14.4 Kernel regression

14.4.1 The Nadaraya–Watson estimator

Definition 14.2 (Kernel and bandwidth)**.** A **kernel** is a symmetric density $K \geq 0$, $\int K(u)\,du = 1$, $K(u) = K(-u)$. For a **bandwidth** $h > 0$ write $K_h(u) = K(u/h)/h$. Its second moment and roughness are $\mu_2(K) = \int u^2 K(u)\,du$ and $R(K) = \int K(u)^2\,du$.

```
kernels <- list(
  Gaussian     = function(u) dnorm(u),
  Epanechnikov = function(u) 0.75 * (1 - u^2) * (abs(u) < 1),
  biweight     = function(u) 15 / 16 * (1 - u^2)^2 * (abs(u) < 1),
  tricube      = function(u) 70 / 81 * (1 - abs(u)^3)^3 * (abs(u) < 1),
  uniform      = function(u) 0.5 * (abs(u) < 1))
```

```
kmom <- t(sapply(kernels, function(K) c(
  integral = integrate(K, -Inf, Inf)$value,
  mu2 = integrate(function(u) u^2 * K(u), -Inf, Inf)$value,
  R = integrate(function(u) K(u)^2, -Inf, Inf)$value)))
round(kmom, 4)
```

```
#>              integral    mu2      R
#> Gaussian            1 1.0000 0.2821
#> Epanechnikov        1 0.2000 0.6000
#> biweight            1 0.1429 0.7143
#> tricube             1 0.1440 0.7085
#> uniform             1 0.3333 0.5000
```

Definition 14.3 (Nadaraya–Watson estimator)**.** The **Nadaraya–Watson** (NW) kernel regression estimator (Nadaraya, 1964; Watson, 1964) is

$$\hat{f}_h(x) = \frac{\sum_{j=1}^{n} K_h(x - x_j)\, y_j}{\sum_{j=1}^{n} K_h(x - x_j)}. \tag{14.1}$$

The sum runs over the *data points* j, and x is the target at which we estimate: the weights are $w_j(x) = K_h(x - x_j) / \sum_l K_h(x - x_l)$, and they are non-negative and sum to one. Writing y_i inside the sum over j is a common slip.

Proposition 14.1 (NW is a local constant fit)**.** *$\hat{f}_h(x)$ is the solution of* $\min_a \sum_{j=1}^{n} K_h(x - x_j)(y_j - a)^2$.

Proof. Differentiate with respect to a: $-2\sum_j K_h(x - x_j)(y_j - a) = 0$. Solving for a gives (14.1). The second derivative $2\sum_j K_h(x - x_j)$ is positive, so this is the minimum. □

The bandwidth controls everything. As $h \to 0$ the weight concentrates on the observations at distance zero from x: at a design point x_i, $\hat{f}_h(x_i) \to$ the mean of the responses observed at x_i (the fit interpolates, with ties averaged). As $h \to \infty$ with a Gaussian kernel, all weights $K(\,(x - x_j)/h\,) \to K(0)$ become equal and $\hat{f}_h(x) \to \bar{y}$.

```
gridx <- seq(min(x), max(x), length.out = 300)
hs3 <- c(0.5, 2, 6)
kf <- do.call(rbind, lapply(hs3, function(h)
  data.frame(times = gridx, fit = drop(nw_S(gridx, x, h) %*% y), h = paste("h
  ↪  =", h))))
ggplot(kf, aes(times, fit, colour = h)) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey65", size = 1,
  ↪  inherit.aes = FALSE) +
  geom_line(linewidth = 0.9) + scale_colour_manual(values = c(ie_coral,
  ↪  ie_navy, ie_green), name = NULL) +
  labs(x = "Time (ms)", y = "Acceleration (g)")
```

A quick check against base R: `ksmooth(x, y, kernel = "normal", bandwidth = b)` scales its kernel so that the quartiles lie at $\pm 0.25\, b$, which for a Gaussian means a standard deviation $h = 0.25\, b / z_{0.75} \approx 0.3706\, b$. The two implementations agree once this is taken into account.

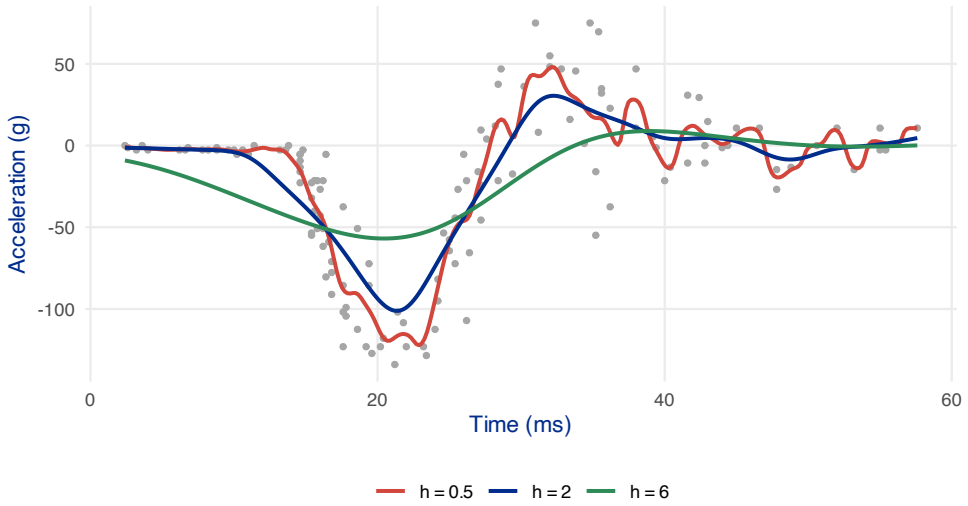


Figure 14.5: Nadaraya–Watson fits with a Gaussian kernel for $h = 0.5, 2, 6$ ms. A small h follows the noise, a large h flattens the dip.

```
h <- 2
b <- h / (0.25 / qnorm(0.75))                    # bandwidth argument that
↪  gives sd h
ks <- ksmooth(x, y, kernel = "normal", bandwidth = b, x.points = x)
c(b = b, max_abs_diff = max(abs(ks$y - drop(nw_S(x, x, h) %*% y))))
```

```
#>            b max_abs_diff
#>     5.395918     0.006136
```

(The small remaining difference, of order 10^{-2} in a response of scale 100, is due to `ksmooth` truncating the Gaussian kernel at ± 4 standard-deviations-equivalent.)

14.4.2 Bias and variance of the NW estimator

How good is $\hat{f}_h(x)$? We analyse its bias and variance at an interior point x, treating the design as random: $x_1, \dots, x_n$ iid with density p, and then, as usual in this literature, replace sums by integrals. This gives the leading terms and shows which features of f and p matter.

Theorem 14.2 (Bias and variance of Nadaraya–Watson)**.** *Let K be a kernel with finite $\mu_2(K)$, f twice and p once continuously differentiable near an interior point x with $p(x) > 0$, and let $h \to 0$ with $nh \to \infty$. Then, conditionally on the design,*

$$\mathrm{E}\{\hat{f}_h(x)\} - f(x) = h^2 \mu_2(K) \left\{ \tfrac{1}{2} f''(x) + \frac{f'(x) p'(x)}{p(x)} \right\} + o(h^2), \tag{14.2}$$

$$\mathrm{Var}\{\hat{f}_h(x)\} = \frac{\sigma^2 R(K)}{nh\, p(x)} \{1 + o(1)\}. \tag{14.3}$$

Proof. Averages of kernel weights. For a smooth function g, with $v = (x - u)/h$,

$$\frac{1}{n} \sum_{j=1}^{n} g(x_j) K_h(x - x_j) \;\approx\; \int g(u) p(u) K_h(x - u)\, du = \int g(x - hv)\, p(x - hv) K(v)\, dv.$$

Taylor-expanding $(gp)(x - hv) = (gp)(x) - hv(gp)'(x) + \tfrac{1}{2} h^2 v^2 (gp)''(x) + o(h^2)$ and using

$\int vK(v)\,dv = 0$ (symmetry), $\int v^2 K = \mu_2$,

$$\frac{1}{n}\sum_j g(x_j)K_h(x - x_j) = (gp)(x) + \tfrac{1}{2}h^2\mu_2\,(gp)''(x) + o(h^2). \tag{14.4}$$

Bias. Because $\hat f_h(x)$ is a weighted mean, $\hat f_h(x) - f(x) = \sum_j K_h(x-x_j)(y_j - f(x))/\sum_j K_h(x-x_j)$. Its conditional expectation uses $\mathrm{E}(y_j \mid x_j) = f(x_j)$. Apply (14.4) with $g = 1$ to the denominator: $p + \frac{1}{2}h^2\mu_2 p''$. With $g = f - f(x)$, which vanishes at x, and using $(gp)'' = g''p + 2g'p' + gp''$ with $g(x) = 0$, $g' = f'$, $g'' = f''$, the numerator is

$$\tfrac{1}{2}h^2\mu_2\{f''(x)p(x) + 2f'(x)p'(x)\} + o(h^2).$$

Dividing by $p(x)\{1 + O(h^2)\}$ gives (14.2).

Variance. Conditionally on the design the y_j are uncorrelated with variance σ^2 and $\hat f_h(x) = \sum_j w_j y_j$, so $\mathrm{Var}(\hat f_h(x) \mid x_1, \dots, x_n) = \sigma^2 \sum_j w_j^2 = \sigma^2 \sum_j K_h(x - x_j)^2 / \{\sum_j K_h(x - x_j)\}^2$. Since $K_h^2(u) = h^{-1}\{K^2\}_h(u)$ and $\int K^2 = R(K)$, (14.4) with $g = 1$ and the kernel K^2 gives $\sum_j K_h(x-x_j)^2 \approx n\,p(x)R(K)/h$, and $\sum_j K_h(x-x_j) \approx n\,p(x)$. Hence $\mathrm{Var} \approx \sigma^2\, np\, R/h \,/\, (np)^2 = \sigma^2 R(K)/\{nh\,p(x)\}$. □

Three lessons can be read off.

1. **The bandwidth trades bias for variance.** The squared bias grows like h^4 and the variance falls like $1/(nh)$. The first moves the estimate away from f systematically; the second is the noise.
2. **The design enters the bias.** The term $f'(x)p'(x)/p(x)$ says that where the design density is changing, the local average is pulled towards the side with more observations, *even if f is a straight line.* Local linear regression removes it (Section 14.5).
3. **The rate is slower than n^{-1}.** Minimise the asymptotic mean squared error $\mathrm{AMSE}(h) = B^2h^4 + V/(nh)$ with $B = \mu_2(K)\{\frac{1}{2}f'' + f'p'/p\}$ and $V = \sigma^2 R(K)/p$. Setting the derivative to zero, $4B^2h^3 = V/(nh^2)$, so

$$h_{\mathrm{opt}} = \left(\frac{V}{4B^2\,n}\right)^{1/5} \propto n^{-1/5}, \qquad \mathrm{AMSE}(h_{\mathrm{opt}}) = 5B^2h_{\mathrm{opt}}^4 \propto n^{-4/5}. \tag{14.5}$$

   At the optimum the variance term $V/(nh)$ equals $4B^2h^4$, i.e. four times the squared bias. The parametric rate n^{-1} is the price of not knowing the shape of f. The formula for h_{opt} is not a practical recipe, because it contains the unknown f'', f' and σ^2: Section 14.8 uses cross-validation instead.

Boundary effects. The theorem is for an interior point. At a boundary a of the support only one side of the kernel carries data, and the same calculation with integrals over $v \geq 0$ shows $\mathrm{E}\hat f_h(a) - f(a) \approx 2h\,f'(a)\int_0^\infty vK(v)\,dv$ for a symmetric kernel and locally constant p: the bias is of order h, not h^2.

```
ll_S <- function(x0, x, h) {                          # local linear smoother
↪  matrix (Section 14.5)
  vapply(seq_along(x0), function(i) {
    d <- x - x0[i]; w <- dnorm(d / h)
    s0 <- sum(w); s1 <- sum(w * d); s2 <- sum(w * d^2)
    w * (s2 - d * s1) / (s0 * s2 - s1^2)
  }, numeric(length(x)))|> t()
}
```

```
set.seed(2027)
xs <- seq(0, 1, length.out = 100); gx <- c(0, 0.05, 0.1, 0.2, 0.5)
SN <- nw_S(gx, xs, 0.1); SL <- ll_S(gx, xs, 0.1)
est <- replicate(300, { ys <- 1 + 3 * xs + rnorm(100, sd = 0.3); c(SN %*% ys,
↪  SL %*% ys) })
bias <- data.frame(x = rep(gx, 2), method = rep(c("NW", "local linear"), each
↪  = length(gx)),
                   bias = rowMeans(est) - rep(1 + 3 * gx, 2))
ggplot(bias, aes(x, bias, colour = method)) + geom_hline(yintercept = 0,
↪  linetype = 2) +
  geom_line() + geom_point() + scale_colour_manual(values = c(ie_coral,
  ↪  ie_navy), name = NULL) +
  labs(x = "x (the boundary is at 0)", y = "average estimate minus truth")
```

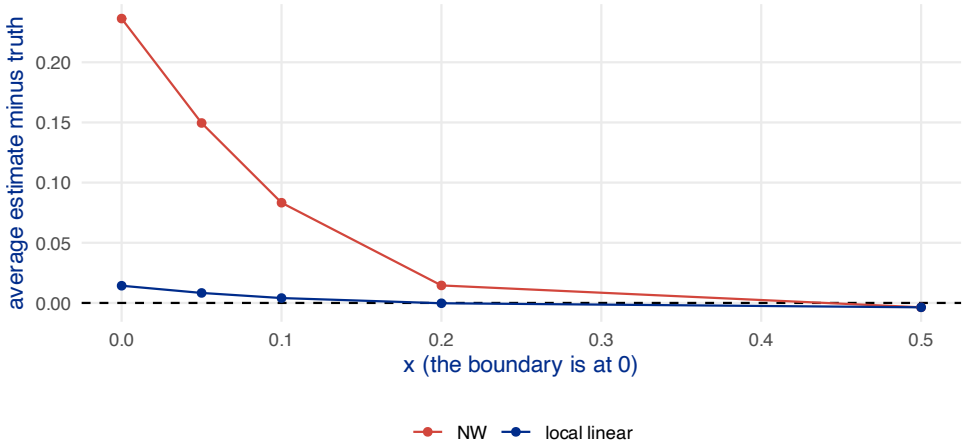


Figure 14.6: Nadaraya–Watson and local linear estimates at $x = 0$ for data from the exact line $f(x) = 1 + 3x$, averaged over 300 simulated data sets ($n = 100$, $\sigma = 0.3$, $h = 0.1$). NW is biased at the boundary, local linear is not.

The simulated NW bias at $x = 0$ is 0.236, in agreement with the formula $2hf'(0)\int_0^\infty vK = 2(0.1)(3)/\sqrt{2\pi} = 0.239$ for the Gaussian kernel (the exact value is smaller because the data stop at $x = 1$ and the design has finite spacing).

14.5 Local polynomial regression

14.5.1 Local linear regression

The NW estimator fits a *constant* locally. Fitting a *line* locally removes the boundary and design biases.

Definition 14.4 (Local linear regression)**.** At each target x, fit a straight line in $(x_j - x)$ by weighted least squares,

$$(\hat a, \hat b) = \arg\min_{a,b} \sum_{j=1}^{n} K_h(x_j - x)\{y_j - a - b(x_j - x)\}^2, \qquad \hat f_h(x) = \hat a.$$

Centring at x makes the intercept the fitted value at x (and $\hat b$ an estimate of $f'(x)$). With

$d_j = x_j - x$, $w_j = K_h(d_j)$ and $s_r = \sum_j w_j d_j^r$ the normal equations are

$$\begin{bmatrix} s_0 & s_1 \\ s_1 & s_2 \end{bmatrix} \begin{bmatrix} \hat a \\ \hat b \end{bmatrix} = \begin{bmatrix} \sum_j w_j y_j \\ \sum_j w_j d_j y_j \end{bmatrix}.$$

Proposition 14.2 (Local linear weights)**.** $\hat f_h(x) = \sum_j l_j(x) y_j$ *with*

$$l_j(x) = \frac{w_j\,(s_2 - d_j s_1)}{s_0 s_2 - s_1^2}, \qquad \sum_j l_j(x) = 1, \qquad \sum_j l_j(x)\,(x_j - x) = 0.$$

Proof. Invert the 2×2 matrix: $\hat a = \{s_2 \sum_j w_j y_j - s_1 \sum_j w_j d_j y_j\}/(s_0 s_2 - s_1^2)$, which is the stated form. Then $\sum_j w_j (s_2 - d_j s_1) = s_0 s_2 - s_1^2$ and $\sum_j w_j d_j (s_2 - d_j s_1) = s_1 s_2 - s_2 s_1 = 0$. The denominator is positive by the Cauchy–Schwarz inequality as soon as two design points have positive weight and are distinct. □

The two identities say that the weights reproduce constants *and* straight lines: if $y_j = \alpha + \beta x_j = (\alpha + \beta x) + \beta d_j$, then $\sum_j l_j y_j = \alpha + \beta x$ exactly, whatever the design and wherever x is. That is why the boundary and design biases vanish.

Theorem 14.3 (Bias and variance of local linear regression)**.** *Under the conditions of Theorem 14.2, for an interior point,*

$$\mathrm{E}\{\hat f_h(x)\} - f(x) = \tfrac12 h^2 \mu_2(K) f''(x) + o(h^2), \qquad \mathrm{Var}\{\hat f_h(x)\} = \frac{\sigma^2 R(K)}{nh\,p(x)}\{1 + o(1)\}.$$

The bias has the same order h^2 at the boundary*, with a constant that depends on the kernel.*

Proof. Conditionally on the design the bias is $\sum_j l_j \{f(x_j) - f(x)\}$. Expand $f(x_j) - f(x) = f'(x) d_j + \tfrac12 f''(x) d_j^2 + o(d_j^2)$. The first term vanishes by Proposition 14.2, so

$$\text{bias} \approx \tfrac12 f''(x) \sum_j l_j d_j^2 = \tfrac12 f''(x)\, \frac{s_2^2 - s_1 s_3}{s_0 s_2 - s_1^2}.$$

By (14.4), $s_r \approx nh^r p(x) \mu_r$ to leading order, with $\mu_0 = 1$, $\mu_1 = 0$; the terms involving s_1 and s_3 are of smaller order ($s_1 = O(nh^2)$, $s_3 = O(nh^4)$ versus $s_0 s_2 \asymp n^2 h^2$), so $\sum_j l_j d_j^2 \approx s_2/s_0 \approx h^2 \mu_2$. For the variance, $l_j = K_h(d_j)\{1 + O(h)\}/\{np(x)\}$, so $\sum_j l_j^2$ has the same leading term as for NW. □

Compared with NW, local linear regression has the same variance and a bias that is free of $f'p'/p$ and of boundary effects. It is a free improvement, at the cost of a 2×2 solve per target point.

14.5.2 Local polynomials of higher degree

Replace the line by a polynomial of degree r in $(x_j - x)$: $\hat f_h(x) = e_1^\top (X_x^\top W_x X_x)^{-1} X_x^\top W_x y$ with $X_x = [\,(x_j - x)^k\,]_{j,k=0,\dots,r}$ and $W_x = \mathrm{diag}\{K_h(x_j - x)\}$. This reproduces polynomials of degree $\le r$, and $k!\,\hat b_k$ estimates $f^{(k)}(x)$. Degrees $r = 0$ (NW), 1 and 2 are those used in practice; odd degrees are preferred because the extra bias term of an even degree depends on the design density (Fan, 1992; Wand and Jones, 1995).

```
cmp <- data.frame(times = rep(gridx, 2),
                  fit = c(drop(nw_S(gridx, x, 3) %*% y), drop(ll_S(gridx, x,
                  ↪  3) %*% y)),
                  method = rep(c("Nadaraya-Watson", "local linear"), each =
                  ↪  length(gridx)))
ggplot(cmp, aes(times, fit, colour = method)) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey65", size = 1,
  ↪  inherit.aes = FALSE) +
  geom_line(linewidth = 0.9) + scale_colour_manual(values = c(ie_navy,
  ↪  ie_coral), name = NULL) +
  labs(x = "Time (ms)", y = "Acceleration (g)")
```

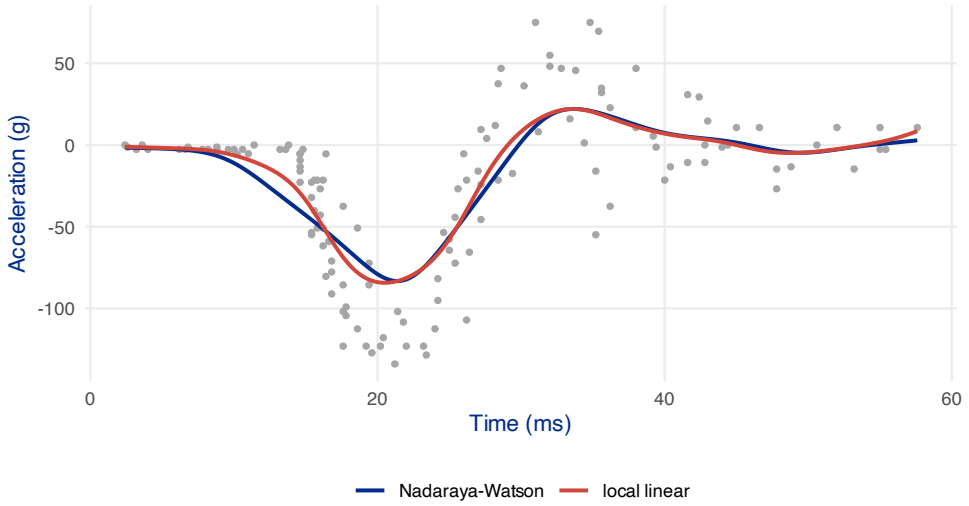


Figure 14.7: Nadaraya–Watson and local linear fits with the same Gaussian kernel and $h = 3$ ms. The local linear fit follows the steep descent and the bottom of the dip more closely.

14.6 LOESS

LOESS (locally estimated scatterplot smoothing) (Cleveland, 1979; Cleveland and Devlin, 1988) is local polynomial regression with a *nearest-neighbour bandwidth*.

Definition 14.5 (LOESS)**.** Fix a **span** $\alpha \in (0, 1]$ and a degree $r \in \{1, 2\}$. For each target x:

1. Let $k = \lceil \alpha n \rceil$ and $\Delta(x)$ the distance from x to its k-th nearest design point.
2. Give each x_j the tricube weight $w_j = (1 - |x_j - x|^3 / \Delta(x)^3)_+^3$.
3. Fit a polynomial of degree r in $(x_j - x)$ by weighted least squares and set $\hat{f}(x)$ equal to its value at x.

The bandwidth is $h(x) = \Delta(x)$: wide where the data are sparse, narrow where they are dense, so every local fit uses about αn points. The estimator is still of the form $\sum_j l_j(x) y_j$, so everything in Section 14.7 applies. In R, `loess(y ~ x, span = 0.75, degree = 2)` (the defaults) fits it; `lowess(x, y, f = 2/3)` is the older *robust* version that repeats the fit with weights that downweight points with large residuals (the function `lowess.smooth()` does not exist; `loess.smooth()` and `scatter.smooth()` evaluate a LOESS curve for plotting).

```
sp <- c(0.1, 0.3, 0.75)
lf <- do.call(rbind, lapply(sp, function(s) {
  m <- loess(accel ~ times, mcycle, span = s, degree = 2)
  data.frame(times = gridx, fit = predict(m, data.frame(times = gridx)),
             span = sprintf("span = %.2f (tr S = %.1f)", s, m$trace.hat))
}))
ggplot(lf, aes(times, fit, colour = span)) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey65", size = 1,
  ↪  inherit.aes = FALSE) +
  geom_line(linewidth = 0.9) + scale_colour_manual(values = c(ie_coral,
  ↪  ie_navy, ie_green), name = NULL) +
  coord_cartesian(ylim = c(-150, 100)) + labs(x = "Time (ms)", y =
  ↪  "Acceleration (g)")
```

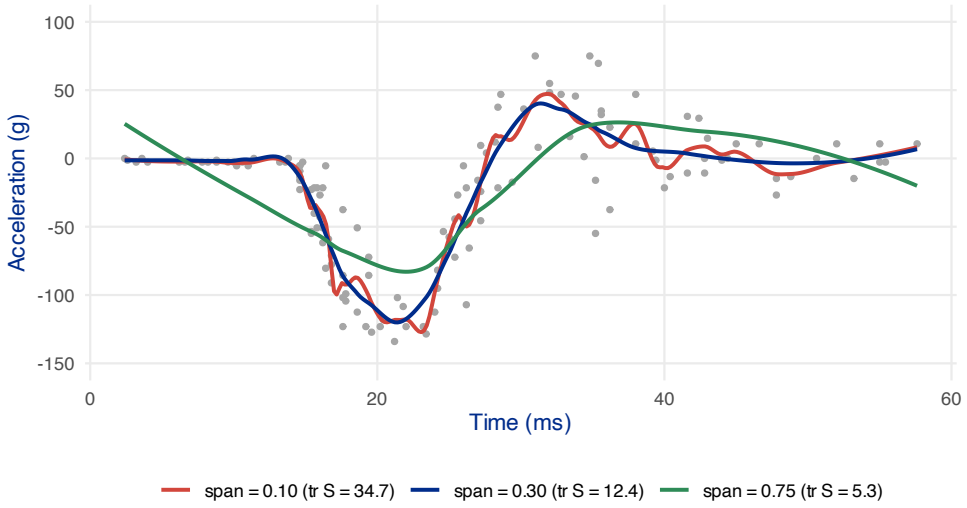


Figure 14.8: LOESS (local quadratic) with spans 0.1, 0.3 and 0.75. The default span misses the dip; span 0.1 is noisy.

```
lo <- loess(accel ~ times, data = mcycle, span = 0.3, degree = 2)
c(trace_S = lo$trace.hat, enp = lo$enp, RSS = sum(resid(lo)^2))
```

```
#>  trace_S      enp      RSS
#>    12.44    11.25 60673.20
```

The object returned by `loess()` contains two different numbers: `trace.hat` is $\operatorname{tr}(S)$, the effective degrees of freedom of the smoother in the sense of Section 14.7, while `enp` ("equivalent number of parameters") approximates $\operatorname{tr}(S^\top S)$ and is smaller. Use `trace.hat`. Note also that `predict()` for a `loess` fit returns `NA` outside the range of the training data unless `control = loess.control(surface = "direct")` is used.

14.7 Linear smoothers and effective degrees of freedom

Every estimator in this chapter, from the polynomial to LOESS, has the form $\hat{f}(x) = \sum_j w_j(x) y_j$ with weights that do not depend on y.

Definition 14.6 (Linear smoother)**.** A **linear smoother** is an estimator of the vector $\boldsymbol{f} = (f(x_1), \dots, f(x_n))^\top$ of the form

$$\hat{\boldsymbol{f}} = S\,\boldsymbol{y},$$

where the $n \times n$ **smoother matrix** S depends on $x_1, \dots, x_n$ and on the tuning constant (degree, bandwidth, span, …) but not on $\boldsymbol{y}$. The **effective degrees of freedom** are $\mathrm{edf} = \mathrm{tr}(S) = \sum_i S_{ii}$.

Table 14.1: The smoother matrix of the methods of this chapter.

Method	Row i of S
polynomial of degree q	row i of $H = X(X^\top X)^{-1}X^\top$
k nearest neighbours	$1/k$ on the k nearest neighbours of x_i
Nadaraya–Watson	$S_{ij} = K_h(x_i - x_j)/\sum_l K_h(x_i - x_l)$
local linear / LOESS	$l_j(x_i)$ of Proposition 14.2 (LOESS: with tricube weights and nearest-neighbour bandwidth)

In all cases the rows sum to one when the method reproduces constants, i.e. $S\mathbf{1} = \mathbf{1}$. For least squares H is symmetric and idempotent, a projection; a kernel S is neither symmetric nor idempotent: it *shrinks* instead of projecting.

14.7.1 Why trace?

For least squares $\mathrm{tr}(H) = \mathrm{tr}\{(X^\top X)^{-1}X^\top X\} = \mathrm{tr}(I_p) = p$ is the number of parameters. The trace generalises this in a way that is justified by a second argument.

Proposition 14.3 (edf as covariance)**.** *For $\boldsymbol{y}$ with $\mathrm{Var}(\boldsymbol{y}) = \sigma^2 I$ and any linear smoother,*

$$\frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{Cov}(\hat{f}_i, y_i) = \mathrm{tr}(S).$$

Proof. $\mathrm{Cov}(S\boldsymbol{y}, \boldsymbol{y}) = S\mathrm{Var}(\boldsymbol{y}) = \sigma^2 S$; take the trace. □

So $\mathrm{tr}(S)$ measures how strongly the fit follows its own data, in units of σ^2: this definition makes sense for any smoother, not only for parametric ones. It also puts bandwidths, spans, k and degrees on one scale. For the Gaussian NW smoother $h \to \infty$ gives $S \to \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ and edf $\to 1$, whereas $h \to 0$ gives $S \to I$ and edf $\to n$ (for distinct x_i). Other definitions exist ($\mathrm{tr}(S^\top S)$, $\mathrm{tr}(2S - SS^\top)$); they coincide for projections and differ slightly otherwise. We use $\mathrm{tr}(S)$.

```
edf_nw <- sapply(c(0.25, 0.5, 1, 2, 4, 8, 16, 100), function(h)
↪  sum(diag(nw_S(x, x, h))))
round(edf_nw, 2)
```

```
#> [1] 59.76 38.77 21.58 11.28  5.87  2.97  1.64  1.02
```

14.7.2 A five-point example by hand

Example 14.2 (Smoother matrix, edf and leave-one-out residuals)**.** Take $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 3, 7, 9)$ and the triangular kernel $K(u) = (1 - |u|)_+$ with $h = 2$, so the weight is 1 at distance 0, $\frac{1}{2}$ at distance 1 and 0 at distance 2. Row 1 has weights $(1, \frac{1}{2}, 0, 0, 0)$ normalised by 1.5, row 2 has $(\frac{1}{2}, 1, \frac{1}{2}, 0, 0)$ normalised by 2, and so on.

```
xt <- 1:5; yt <- c(2, 4, 3, 7, 9)
Kt <- pmax(1 - abs(outer(xt, xt, "-")) / 2, 0)
St <- Kt / rowSums(Kt)
fractions <- MASS::fractions
fractions(St)
```

```
#>      [,1] [,2] [,3] [,4] [,5]
#> [1,] 2/3  1/3    0    0    0
#> [2,] 1/4  1/2  1/4    0    0
#> [3,]   0  1/4  1/2  1/4    0
#> [4,]   0    0  1/4  1/2  1/4
#> [5,]   0    0    0  1/3  2/3
```

```
ft <- drop(St %*% yt); rt <- yt - ft
rbind(fitted = ft, residual = rt, loo = rt / (1 - diag(St)))
```

```
#>              [,1] [,2]  [,3] [,4]   [,5]
#> fitted    2.6667 3.25  4.25  6.5 8.3333
#> residual -0.6667 0.75 -1.25  0.5 0.6667
#> loo      -2.0000 1.50 -2.50  1.0 2.0000
```

```
c(edf = sum(diag(St)), RSS = sum(rt^2),
  LOOCV = mean((rt / (1 - diag(St)))^2), GCV = 5 * sum(rt^2) / (5 -
  ↪  sum(diag(St)))^2)
```

```
#>   edf   RSS LOOCV   GCV
#> 2.833 3.264 3.500 3.476
```

The effective degrees of freedom are $\operatorname{tr}(S) = \frac{2}{3} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{3} = 2.833$. For $i = 1$, deleting y_1 leaves only the point x_2 with positive weight at x_1, so $\hat{f}^{(-1)}(1) = y_2 = 4$ and the leave-one-out residual is $2 - 4 = -2$, equal to $(y_1 - \hat{f}_1)/(1 - S_{11}) = (-0.667)/(1/3)$. That is the shortcut of Section 14.8.

14.7.3 Bias and variance of a linear smoother

Theorem 14.4 (Mean squared error of a linear smoother)**.** *Let* $\boldsymbol{y} = \boldsymbol{f} + \boldsymbol{\varepsilon}$ *with* $\mathrm{E}\boldsymbol{\varepsilon} = \boldsymbol{0}$, $\mathrm{Var}\boldsymbol{\varepsilon} = \sigma^2 I$ *and* $\hat{\boldsymbol{f}} = S\boldsymbol{y}$*. Then*

(a) $\mathrm{E}\hat{\boldsymbol{f}} = S\boldsymbol{f}$ *and* $\mathrm{Var}\hat{\boldsymbol{f}} = \sigma^2 SS^\top$*;*

(b) the average mean squared error is

$$\frac{1}{n}\mathrm{E}\|\hat{\boldsymbol{f}} - \boldsymbol{f}\|^2 = \underbrace{\frac{1}{n}\|(S - I)\boldsymbol{f}\|^2}_{\text{squared bias}} + \underbrace{\frac{\sigma^2}{n}\operatorname{tr}(SS^\top)}_{\text{variance}};$$

(c) $\mathrm{E}\,\mathrm{RSS} = \|(I - S)\boldsymbol{f}\|^2 + \sigma^2\{n - 2\operatorname{tr}(S) + \operatorname{tr}(SS^\top)\}$*, with* $\mathrm{RSS} = \|\boldsymbol{y} - S\boldsymbol{y}\|^2$*;*

(d) the prediction risk for new responses $\boldsymbol{y}^* = \boldsymbol{f} + \boldsymbol{\varepsilon}^*$ independent of $\boldsymbol{y}$ satisfies

$$\frac{1}{n}\mathrm{E}\|\boldsymbol{y}^* - \hat{\boldsymbol{f}}\|^2 = \mathrm{E}\Big(\frac{\mathrm{RSS}}{n}\Big) + \frac{2\sigma^2}{n}\operatorname{tr}(S).$$

Proof.

(a) is linearity. (b) Write $\hat{\boldsymbol{f}} - \boldsymbol{f} = (S - I)\boldsymbol{f} + S\boldsymbol{\varepsilon}$, a fixed vector plus a zero-mean vector, so $\mathrm{E}\|\cdot\|^2 = \|(S-I)\boldsymbol{f}\|^2 + \mathrm{E}\|S\boldsymbol{\varepsilon}\|^2$ and $\mathrm{E}\|S\boldsymbol{\varepsilon}\|^2 = \sigma^2 \operatorname{tr}(S^\top S) = \sigma^2 \operatorname{tr}(SS^\top)$. (c) $\boldsymbol{y} - S\boldsymbol{y} = (I-S)\boldsymbol{f} + (I-S)\boldsymbol{\varepsilon}$, and $\mathrm{E}\|(I-S)\boldsymbol{\varepsilon}\|^2 = \sigma^2 \operatorname{tr}\{(I-S)^\top(I-S)\} = \sigma^2\{n - 2\operatorname{tr} S + \operatorname{tr}(S^\top S)\}$. (d) The new errors are independent of $\hat{\boldsymbol{f}}$, so $\frac{1}{n}\mathrm{E}\|\boldsymbol{y}^* - \hat{\boldsymbol{f}}\|^2 = \sigma^2 + \frac{1}{n}\mathrm{E}\|\hat{\boldsymbol{f}} - \boldsymbol{f}\|^2 = \sigma^2 + \frac{1}{n}\|(S-I)\boldsymbol{f}\|^2 + \frac{\sigma^2}{n}\operatorname{tr}(SS^\top)$. Subtracting $\frac{1}{n}\mathrm{E}\,\mathrm{RSS}$ from (c) leaves $\sigma^2\{1 - (n - 2\operatorname{tr} S + \operatorname{tr} SS^\top)/n + \operatorname{tr}(SS^\top)/n\} = 2\sigma^2 \operatorname{tr}(S)/n$.

□

Part (d) is Mallows' C_p idea for smoothers: the training error underestimates the prediction error by $2\sigma^2 \operatorname{edf}/n$. For least squares it reproduces the familiar $C_p = \mathrm{RSS}/n + 2\sigma^2 p/n$. Part (c) shows that $\hat{\sigma}^2 = \mathrm{RSS}/(n - \operatorname{edf})$ is a natural estimator of σ^2: if the bias is small and $\operatorname{tr}(SS^\top) \approx \operatorname{tr}(S)$, then $\mathrm{E}\,\mathrm{RSS} \approx \sigma^2(n - \operatorname{tr} S)$, exactly so for a projection.

Part (b) is exact; no asymptotics are needed once $\boldsymbol{f}$ and σ are fixed. We use it to compute the bias–variance trade-off for NW on `mcycle`, taking a heavily smoothed fit as the "true" $\boldsymbol{f}$ (a device used only to draw the picture).

```
f_true <- drop(ll_S(x, x, 2.5) %*% y)
hg <- seq(0.5, 5, by = 0.05); sigma <- 23
bv <- t(sapply(hg, function(h) {
  S <- nw_S(x, x, h)
  c(bias2 = mean((S %*% f_true - f_true)^2), var = sigma^2 *
   ↪  mean(rowSums(S^2)))
}))
mse <- rowSums(bv)
bvl <- data.frame(h = rep(hg, 3), value = c(bv, mse),
                  part = rep(c("squared bias", "variance", "MSE"), each =
                   ↪  length(hg)))
ggplot(bvl, aes(h, value, colour = part)) + geom_line(linewidth = 0.9) +
  geom_vline(xintercept = hg[which.min(mse)], linetype = 3) +
  scale_colour_manual(values = c(ie_coral, ie_navy, ie_blue), name = NULL) +
  labs(x = "bandwidth h (ms)", y = "average over design points")
```

```
hg[which.min(mse)]
```

```
#> [1] 1.2
```

Here $\operatorname{diag}(SS^\top)$ is computed as the row sums of S^2 (elementwise), which avoids forming the product.

14.8 Choosing the bandwidth

The tuning constant is chosen to minimise an estimate of the prediction error. The residual sum of squares is useless for this (it always falls as h falls); we need to leave something out.

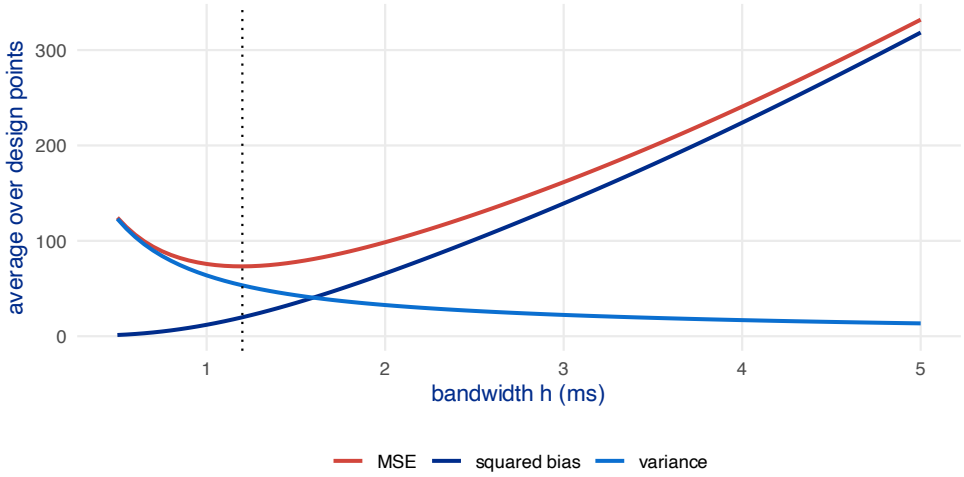


Figure 14.9: Exact squared bias, variance and mean squared error of the Nadaraya–Watson smoother as functions of h, for a known test function (a smooth fit to the motorcycle data) and $\sigma = 23$. The minimum of the MSE is near $h = 1.2$ ms.

14.8.1 Leave-one-out cross-validation without refitting

Definition 14.7 (Leave-one-out cross-validation)**.**

$$\mathrm{LOOCV}(h) = \frac{1}{n}\sum_{i=1}^{n}\{y_i - \hat{f}^{(-i)}(x_i)\}^2,$$

where $\hat{f}^{(-i)}$ is the estimate computed *without* observation i.

Computed literally, this needs n fits. For a large class of smoothers, one fit is enough.

Theorem 14.5 (Leave-one-out shortcut)**.** *Suppose the smoother has the* leave-one-out property*: if the i-th response is replaced by the leave-one-out prediction $\hat{f}^{(-i)}(x_i)$, the fit at x_i does not change. Then*

$$y_i - \hat{f}^{(-i)}(x_i) = \frac{y_i - \hat{f}_i}{1 - S_{ii}}. \tag{14.6}$$

The property holds for least squares, for Nadaraya–Watson and any local polynomial with a fixed *bandwidth, and for penalized least squares with fixed penalty (Chapter 15).*

Proof. Write $\hat{f}_i = \sum_j S_{ij} y_j$, and let $\hat{f}_i^{(-i)}$ stand for $\hat{f}^{(-i)}(x_i)$. By the leave-one-out property, replacing y_i by $\hat{f}_i^{(-i)}$ leaves the fit at x_i equal to $\hat{f}_i^{(-i)}$: $\hat{f}_i^{(-i)} = \sum_{j \neq i} S_{ij} y_j + S_{ii}\hat{f}_i^{(-i)}$. Replacing $\sum_{j \neq i} S_{ij} y_j = \hat{f}_i - S_{ii} y_i$ gives $\hat{f}_i^{(-i)}(1 - S_{ii}) = \hat{f}_i - S_{ii} y_i$. Hence

$$y_i - \hat{f}_i^{(-i)} = \frac{y_i(1 - S_{ii}) - \hat{f}_i + S_{ii} y_i}{1 - S_{ii}} = \frac{y_i - \hat{f}_i}{1 - S_{ii}}.$$

The property. Each of these estimators minimises a criterion $\sum_j w_j\{y_j - m_j(\theta)\}^2 + \mathrm{pen}(\theta)$ over a fixed set of fits, with weights $w_j \geq 0$ that do not depend on y. The leave-one-out fit minimises the same criterion without the i-th term. If we put $y_i = \hat{f}_i^{(-i)}$ in that term, it is zero at the leave-one-out minimiser, so adding it does not change the minimiser, and the fit at x_i is again $\hat{f}_i^{(-i)}$. □

For Nadaraya–Watson there is also a direct argument (this is the derivation used in class): write $K_{ij} = K_h(x_i - x_j)$, so $S_{ij} = K_{ij} / \sum_l K_{il}$. Dropping $j = i$ from both sums,

$$\hat{f}^{(-i)}(x_i) = \frac{\sum_{j \neq i} K_{ij} y_j}{\sum_{l \neq i} K_{il}} = \frac{\hat{f}_i - S_{ii} y_i}{1 - S_{ii}},$$

dividing numerator and denominator by $\sum_l K_{il}$. For kNN and LOESS with a nearest-neighbour bandwidth the leave-one-out property fails, because deleting x_i changes the neighbourhood itself, and (14.6) is only an approximation.

The shortcut is easily checked.

```
h <- 2
S <- nw_S(x, x, h)
r <- y - drop(S %*% y)
shortcut <- mean((r / (1 - diag(S)))^2)
brute <- mean(sapply(1:n, function(i) {
  w <- dnorm((x[-i] - x[i]) / h)
  (y[i] - sum(w * y[-i]) / sum(w))^2
}))
c(shortcut = shortcut, brute_force = brute)
```

```
#>    shortcut brute_force
#>       689.7       689.7
```

14.8.2 Generalized cross-validation

Replacing every S_{ii} in (14.6) by their average $\operatorname{tr}(S)/n$ gives

$$\mathrm{GCV}(h) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{f}_i}{1 - \operatorname{tr}(S)/n} \right)^2 = \frac{n \, \mathrm{RSS}(h)}{\{n - \mathrm{edf}(h)\}^2}, \tag{14.7}$$

introduced by Craven and Wahba (1978). It needs only the trace and not the diagonal of S, and is invariant to orthogonal transformations of the data; for smoothers whose diagonal is nearly constant it is almost identical to LOOCV. It is the same correction as $\mathrm{RSS}/(n-p)$ in the linear model, with edf in place of p. A slip to avoid: GCV is not "RSS divided by $n - \mathrm{edf}$"; the denominator is squared and the numerator carries a factor n.

```
crit <- function(S) {
  r <- y - drop(S %*% y); edf <- sum(diag(S))
  c(edf = edf, LOOCV = mean((r / (1 - diag(S)))^2), GCV = n * sum(r^2) / (n -
  ↪  edf)^2)
}
hs <- seq(0.5, 5, by = 0.01)
tab_nw <- t(sapply(hs, function(h) crit(nw_S(x, x, h))))
tab_ll <- t(sapply(hs, function(h) crit(ll_S(x, x, h))))
best <- function(tab, col) c(h = hs[which.min(tab[, col])],
↪  tab[which.min(tab[, col]), ])
opt <- rbind(NW_LOOCV = best(tab_nw, "LOOCV"), NW_GCV = best(tab_nw, "GCV"),
             LL_LOOCV = best(tab_ll, "LOOCV"), LL_GCV = best(tab_ll, "GCV"))
cvd <- data.frame(h = rep(hs, 4),
                  value = c(tab_nw[, "LOOCV"], tab_nw[, "GCV"], tab_ll[,
                  ↪  "LOOCV"], tab_ll[, "GCV"]),
```

```
                  method = rep(c("NW", "local linear"), each = 2 *
                   ↪  length(hs)),
                  criterion = rep(rep(c("LOOCV", "GCV"), each = length(hs)),
                   ↪  2))
ggplot(cvd, aes(h, value, colour = method, linetype = criterion)) +
 ↪  geom_line(linewidth = 0.9) +
  geom_point(data = data.frame(h = opt[, "h"], value = opt[, c("LOOCV")] *
   ↪  c(1, NA, 1, NA) + opt[, "GCV"] * c(NA, 1, NA, 1),
                               method = rep(c("NW", "local linear"), each =
                                ↪  2),
                               criterion = rep(c("LOOCV", "GCV"), 2)), size =
                                ↪  2.5) +
  scale_colour_manual(values = c(ie_coral, ie_navy), name = NULL) +
   ↪  coord_cartesian(ylim = c(540, 1000)) +
  labs(x = "bandwidth h (ms)", y = "criterion")
```

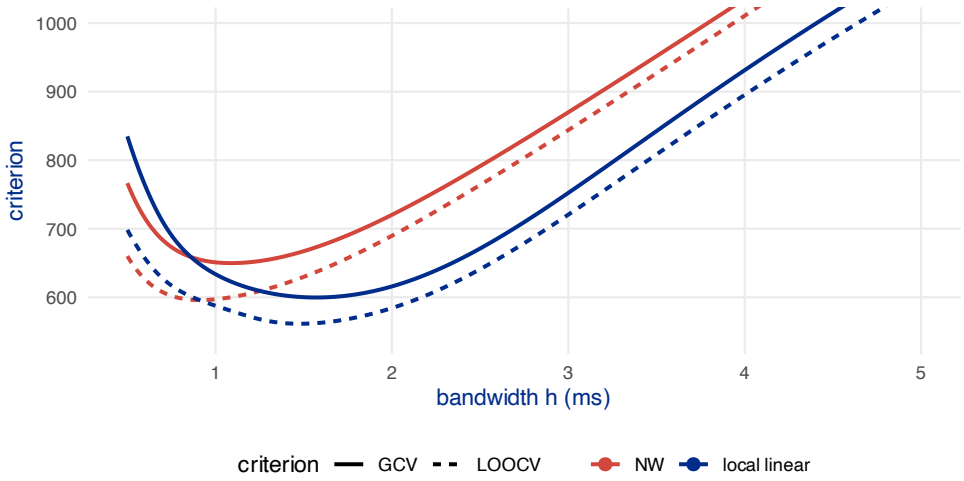


Figure 14.10: LOOCV and GCV against the bandwidth for local linear and Nadaraya–Watson smoothers on the motorcycle data. The dots mark the minima.

```
round(opt, 2)
```

```
#>              h   edf LOOCV   GCV
#> NW_LOOCV 0.91 23.50 595.9 654.9
#> NW_GCV   1.09 19.94 600.1 649.8
#> LL_LOOCV 1.48 16.57 561.3 600.4
#> LL_GCV   1.57 15.70 562.2 599.7
```

Local linear regression reaches a lower minimum (LOOCV 561.3 at $h = 1.48$, edf 16.6) than NW (LOOCV 595.9 at $h = 0.91$, edf 23.5), and with a larger bandwidth; GCV selects a very similar h.

Pitfalls of cross-validation. (i) With *tied* design points, leaving out one observation leaves its twin at the same x in the training set, which makes LOOCV optimistic and biased towards small h in `mcycle`; leaving out all observations at an x ("grouped" CV) mimics prediction at a new time. (ii) With data sorted by x and unshuffled folds, each test fold is an extrapolation (Exercise 14.10). (iii) The criterion is itself a random variable: its minimiser varies from sample to sample and can be flat; report a range or use repeated CV. (iv) The shortcut and GCV

assume constant variance; with the strongly heteroscedastic noise of `mcycle` the noisy middle section dominates the criterion.

14.9 Comparing smoothers

A bandwidth of 2 for a kernel and a span of 0.3 for LOESS do not mean the same, and RSS at different complexity is not comparable. A fair comparison matches the effective degrees of freedom.

```
target <- 12
h_nw <- uniroot(function(h) sum(diag(nw_S(x, x, h))) - target, c(0.5, 5))$root
h_ll <- uniroot(function(h) sum(diag(ll_S(x, x, h))) - target, c(0.5, 5))$root
span <- uniroot(function(s) loess(accel ~ times, mcycle, span = s)$trace.hat -
↪  target, c(0.2, 0.5))$root
lo12 <- loess(accel ~ times, mcycle, span = span)
ss12 <- smooth.spline(x, y, df = target)
f4 <- cbind(NW = drop(nw_S(x, x, h_nw) %*% y), LL = drop(ll_S(x, x, h_ll) %*%
↪  y),
            LOESS = fitted(lo12), spline = predict(ss12, x)$y)
rss4 <- colSums((y - f4)^2)
data.frame(tuning = round(c(h_nw, h_ll, span, NA), 3), RSS = round(rss4),
           GCV = round(n * rss4 / (n - target)^2, 1), min_fit =
           ↪  round(apply(f4, 2, min), 1))
```

```
#>         tuning   RSS   GCV min_fit
#> NW       1.873 77560 704.6  -103.6
#> LL       2.118 68881 625.7   -99.3
#> LOESS    0.301 60839 552.7  -115.5
#> spline      NA 62254 565.5  -114.9
```

```
gd <- data.frame(times = rep(gridx, 4),
                 fit = c(drop(nw_S(gridx, x, h_nw) %*% y), drop(ll_S(gridx, x,
                 ↪  h_ll) %*% y),
                         predict(lo12, data.frame(times = gridx)),
                         ↪  predict(ss12, gridx)$y),
                 method = rep(c("NW", "local linear", "LOESS (quadratic)",
                 ↪  "smoothing spline"), each = length(gridx)))
ggplot(gd, aes(times, fit, colour = method)) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey65", size = 1,
  ↪  inherit.aes = FALSE) +
  geom_line(linewidth = 0.8) + scale_colour_manual(values = ie_pal[1:4], name
  ↪  = NULL) +
  labs(x = "Time (ms)", y = "Acceleration (g)")
```

At equal complexity the NW fit leaves the most structure in the residuals; local linear is still too shallow at the dip, which is where the bias $\frac{1}{2}h^2\mu_2 f''$ is largest (f'' is large and negative there); local quadratic LOESS and the smoothing spline (Chapter 15) are practically the same curve. The amount of smoothing matters more than the method.

Confidence bands. The fit $\pm 2\,\mathrm{sd}(\text{residuals})$ is *not* a confidence band for f: the residual standard deviation describes the spread of a *new* observation around the curve. By Theorem 14.4 the variance of $\hat f_i$ is $\sigma^2 (SS^\top)_{ii}$, so a pointwise band is $\hat f_i \pm 2\hat\sigma\sqrt{(SS^\top)_{ii}}$ with $\hat\sigma^2 = \mathrm{RSS}/(n - \mathrm{edf})$.

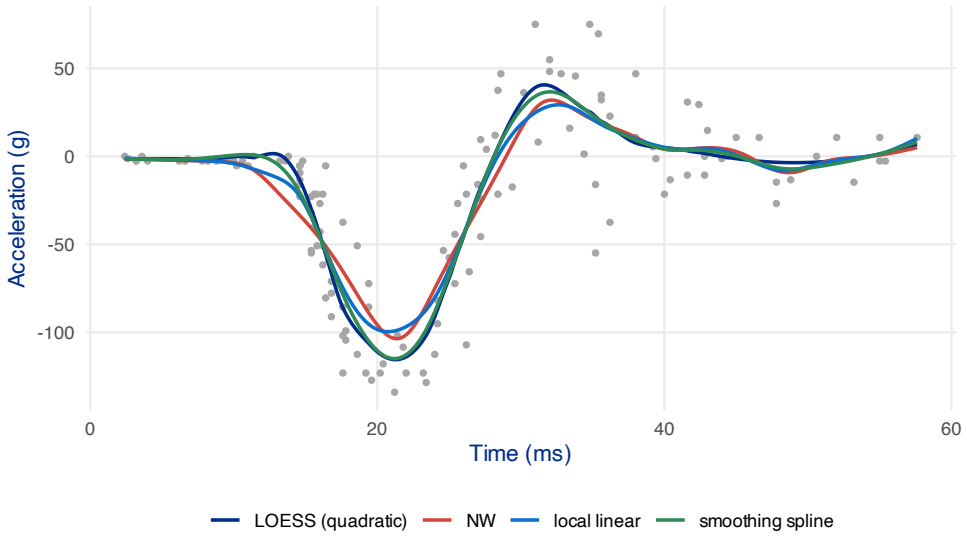


Figure 14.11: Four smoothers with the same effective degrees of freedom, $\mathrm{tr}(S) = 12$, on the motorcycle data. The NW and local linear fits are shallower at the bottom of the dip.

It covers $\mathrm{E}\hat{f}_i = (Sf)_i$, not $f(x_i)$, because it ignores the bias, and it assumes constant σ^2, which is violated here.

```
Sl <- ll_S(x, x, h_ll); fl <- drop(Sl %*% y)
sig_hat <- sqrt(sum((y - fl)^2) / (n - sum(diag(Sl))))
se_f <- sig_hat * sqrt(rowSums(Sl^2))
bd <- data.frame(times = x, f = fl, se = se_f, sdr = sd(y - fl))
ggplot(bd, aes(times)) +
  geom_ribbon(aes(ymin = f - 2 * sdr, ymax = f + 2 * sdr), fill = ie_coral,
  ↪  alpha = 0.18) +
  geom_ribbon(aes(ymin = f - 2 * se, ymax = f + 2 * se), fill = ie_navy, alpha
  ↪  = 0.35) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey55", size = 1,
  ↪  inherit.aes = FALSE) +
  geom_line(aes(y = f), colour = ie_navy, linewidth = 0.9) + labs(x = "Time
  ↪  (ms)", y = "Acceleration (g)")
```

```
c(sigma_hat = sig_hat, sd_resid = sd(y - fl), se_min = min(se_f), se_max =
↪  max(se_f))
```

```
#> sigma_hat  sd_resid    se_min    se_max
#>    23.859    22.838     3.778    21.762
```

14.10 In R: the functions used in this chapter

Task	Function
Orthogonal / raw polynomial	`poly(x, q)`, `poly(x, q, raw = TRUE)` in `lm()`
Kernel smoother	`ksmooth(x, y, kernel = "normal", bandwidth = b, x.points = )` ($h = 0.3706\,b$ for the normal kernel); own code via the matrix S

Task	Function
Local polynomial regression	`loess(y ~ x, span = , degree = )`; `predict(fit, newdata, se = TRUE)`
Robust local linear	`lowess(x, y, f = )` (no `predict`)
Smoother matrix / edf	own `S`; `fit$trace.hat` for `loess`
Smoothing spline	`smooth.spline(x, y, df = )` (Chapter 15)

`KernSmooth::locpoly()` and `KernSmooth::dpill()` provide fast local polynomials and a plug-in bandwidth in the spirit of (14.5); `ggplot2::geom_smooth(method = "loess")` draws a LOESS curve with its pointwise bands. The student-lab file for this material is Lab 05.

14.11 Pitfalls

Warning. *Using a high-degree polynomial and trusting it near or beyond the ends.* Leverage approaches 1 at the boundaries and the fit diverges outside the data (Section 14.2).

Warning. *Calling `poly()` columns independent.* They are orthogonal with mean zero, hence uncorrelated; they are deterministic functions of each other.

Warning. *Writing the kernel estimator with y_i inside the sum over j.* The target is x; the sum runs over the data (x_j, y_j).

Warning. *Comparing a bandwidth with a span.* Compare effective degrees of freedom, and compare smoothers by GCV or cross-validation, not by RSS.

Warning. *Reading `ksmooth(bandwidth = b)` as the kernel standard deviation.* It is $0.3706\,b$; and *using `enp` instead of `trace.hat` for* $\mathrm{tr}(S)$ in `loess`.

Warning. *Cross-validating sorted data with unshuffled folds*, and *drawing* $\hat f \pm 2\,\mathrm{sd}(\textit{resid})$ *as a confidence band.*

14.12 Summary

- Nonparametric regression estimates f in $y = f(x) + \varepsilon$ without assuming its form. Polynomial regression is the linear model with columns $1, x, \dots, x^q$; it has edf $= q + 1$ but is a *global* fit: leverage piles up at the ends, extrapolation explodes, and raw powers are ill-conditioned. Orthogonal polynomials (`poly()`) fix conditioning ($\hat\beta = R^{-1}\hat\gamma$ exactly) but not the leverage.
- A local averaging estimator is $\hat f(x) = \sum_j w_j(x) y_j$. The Nadaraya–Watson estimator is the weighted mean with kernel weights $K_h(x - x_j)$, the solution of a local constant weighted least squares problem.
- NW has bias $h^2\mu_2\{\frac{1}{2}f'' + f'p'/p\}$ and variance $\sigma^2 R(K)/(nh\,p)$; the optimal bandwidth is $\propto n^{-1/5}$ and the MSE decreases like $n^{-4/5}$. At boundaries the NW bias is of order h.
- Local linear regression reproduces straight lines, which removes the design term and the boundary bias: bias $\frac{1}{2}h^2\mu_2 f''$, same variance. LOESS is local polynomial regression with a nearest-neighbour bandwidth (span) and tricube weights.
- Every smoother is linear, $\hat{\boldsymbol f} = S\boldsymbol y$, with edf $= \mathrm{tr}(S) = \sigma^{-2}\sum \mathrm{Cov}(\hat f_i, y_i)$; $\mathrm{E}\hat{\boldsymbol f} = S\boldsymbol f$, $\mathrm{Var}\hat{\boldsymbol f} = \sigma^2 SS^\top$, and $\mathrm{MSE} = n^{-1}\|(S - I)\boldsymbol f\|^2 + n^{-1}\sigma^2\,\mathrm{tr}(SS^\top)$.
- Leave-one-out residuals are $(y_i - \hat f_i)/(1 - S_{ii})$ (exact under the leave-one-out property); $\mathrm{GCV} = n\,\mathrm{RSS}/(n - \mathrm{edf})^2$. Choose h by LOOCV/GCV, never by RSS, and compare smoothers at equal edf.

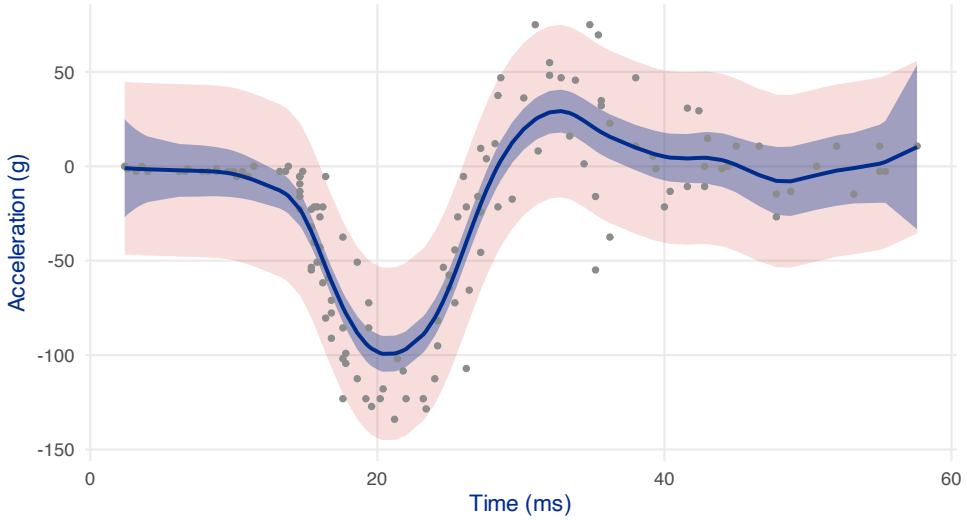


Figure 14.12: Local linear fit with edf 12. Light band: fit ± 2 sd(residuals). Dark band: pointwise ± 2 standard errors of the fitted curve. The dark band is far narrower, and it is the right object for the uncertainty of $\hat{f}$ (though it ignores the bias and assumes constant variance).

- Pointwise standard errors are $\hat{\sigma}\sqrt{(SS^\top)_{ii}}$ with $\hat{\sigma}^2 = \mathrm{RSS}/(n - \mathrm{edf})$; $\hat{f} \pm 2\,\mathrm{sd(resid)}$ is not a confidence band.

14.13 Exercises

Exercise 14.1. Orthogonal polynomials by hand. Let $x = (-2, -1, 0, 1, 2)$ and $y = (1, 3, 2, 5, 4)$. (a) Verify that $\tilde{q}_0 = \mathbf{1}/\sqrt{5}$, $\tilde{q}_1 = x/\sqrt{10}$ and $\tilde{q}_2 = (x^2 - 2)/\sqrt{14}$ are orthonormal. (b) Compute $\hat{\gamma}_k = \tilde{q}_k^\top y$ and the quadratic fit. (c) Convert to raw coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. (d) If a cubic term is added, which of the quantities in (b) change?

Solution.

(a) $\sum x = 0$, $\sum x^2 = 10$, $\sum (x^2 - 2) = 10 - 10 = 0$, $\sum x(x^2 - 2) = \sum x^3 - 2\sum x = 0$ (odd powers cancel), $\sum (x^2 - 2)^2 = (2, -1, -2, -1, 2) \cdot (2, -1, -2, -1, 2) = 4 + 1 + 4 + 1 + 4 = 14$. So the three vectors are orthogonal with squared norms 5, 10, 14, and dividing by the norms gives an orthonormal set.

(b) $\hat{\gamma}_0 = \sum y/\sqrt{5} = 15/\sqrt{5}$, i.e. the fit contains $\bar{y} = 3$. $\hat{\gamma}_1 = \sum xy/\sqrt{10} = (-2 - 3 + 0 + 5 + 8)/\sqrt{10} = 8/\sqrt{10}$ and $\hat{\gamma}_2 = \sum (x^2 - 2)y/\sqrt{14} = (2 - 3 - 4 - 5 + 8)/\sqrt{14} = -2/\sqrt{14}$. Then

$$\hat{y} = 3 + \frac{8}{10}x - \frac{2}{14}(x^2 - 2) = 3 + 0.8x - \tfrac{1}{7}(x^2 - 2).$$

(c) Expand: $\hat{y} = (3 + \frac{2}{7}) + 0.8x - \frac{1}{7}x^2$, so $(\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2) = (3.286,\ 0.8,\ -0.143)$. This is an exact linear transformation of $\hat{\gamma}$ (Theorem 14.1).

(d) Nothing in (b) changes: $\hat{\gamma}_0, \hat{\gamma}_1, \hat{\gamma}_2$ are the same because $\hat{\gamma}_k = \tilde{q}_k^\top y$ does not depend on the other columns; one extra coefficient $\hat{\gamma}_3$ appears. The *raw* coefficients all change, since x^3 enters the expansion of $\tilde{q}_3$. Check in R:

```
xe <- c(-2, -1, 0, 1, 2); ye <- c(1, 3, 2, 5, 4)
rbind(orth_q2 = coef(lm(ye ~ poly(xe, 2))), orth_q3 = coef(lm(ye ~ poly(xe,
↪  3)))[1:3])
```

```
#>          (Intercept) poly(xe, 2)1 poly(xe, 2)2
#> orth_q2           3         2.53      -0.5345
#> orth_q3           3         2.53      -0.5345
```

```
rbind(raw_q2 = coef(lm(ye ~ poly(xe, 2, raw = TRUE))), raw_q3 = coef(lm(ye ~
↪  poly(xe, 3, raw = TRUE)))[1:3])
```

```
#>        (Intercept) poly(xe, 2, raw = TRUE)1 poly(xe, 2, raw = TRUE)2
#> raw_q2       3.286                    0.800                   -0.1429
#> raw_q3       3.286                    1.083                   -0.1429
```

(The `poly()` coefficients are $\hat\gamma_1, \hat\gamma_2$ for the *unit-norm* columns: $8/\sqrt{10} = 2.530$, $-2/\sqrt{14} = -0.535$, identical in both fits, while the raw ones change.)

Exercise 14.2. Limits of Nadaraya–Watson. Let K be Gaussian. (a) Show that $\hat f_h(x) \to \bar y$ as $h \to \infty$ and hence $S \to \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, edf $\to 1$. (b) For distinct x_i, show that $\hat f_h(x_i) \to y_i$ as $h \to 0$ and edf $\to n$. (c) Compute the edf of the 5-point triangular-kernel smoother of Example 14.2 if the bandwidth is increased to $h = 10$.

Solution.

(a) $K_h(x - x_j) = \frac{1}{h}\phi\{(x - x_j)/h\}$. For fixed x and x_j, $\phi\{(x - x_j)/h\} \to \phi(0)$ as $h \to \infty$, so all weights tend to the same value, and after normalisation $w_j \to 1/n$. Thus $\hat f_h(x) \to \sum_j y_j/n = \bar y$ and every row of S tends to $(1/n, \dots, 1/n)$, whose trace is $n \cdot \frac{1}{n} = 1$.

(b) At $x = x_i$ the weights are proportional to $\exp\{-(x_i - x_j)^2/(2h^2)\}$. Divide numerator and denominator by $\exp(0) = 1$ (the weight of $j = i$): for $j \neq i$, $\exp\{-(x_i - x_j)^2/2h^2\} \to 0$ as $h \to 0$, so $w_i \to 1$ and $w_j \to 0$. Hence $S \to I$, $\operatorname{tr}(S) \to n$. (With tied x's the limit averages the responses at the same x.)

(c) With $h = 10$ the triangular kernel $(1 - |u|/10)_+$ gives positive weights $0.9, 0.8, 0.7, 0.6$ at distances 1, 2, 3, 4, and 1 at distance 0:

```
Kt10 <- pmax(1 - abs(outer(xt, xt, "-")) / 10, 0)
S10 <- Kt10 / rowSums(Kt10)
c(edf = sum(diag(S10)), edf_h2 = sum(diag(St)))
```

```
#>    edf edf_h2
#>  1.192  2.833
```

Solution. (Continued.) The edf is 1.192, close to the lower limit 1 (compare 2.833 for $h = 2$), as part (a) predicts: five points all within the kernel's reach are nearly averaged together.

Exercise 14.3. Boundary and design bias by hand. Data $x = (1, 2, 3, 4)$, $y = (2, 3, 5, 4)$, triangular kernel $K(u) = (1 - |u|)_+$, $h = 2$. (a) Compute the NW estimate at $x_0 = 1$ and the local linear estimate at $x_0 = 1$. (b) Compute both at $x_0 = 2.5$. (c) Explain the difference between the two cases.

Solution.

(a) At $x_0 = 1$ the weights are $1, \frac{1}{2}, 0, 0$ for $x = 1, 2, 3, 4$. NW: $(1 \cdot 2 + \frac{1}{2} \cdot 3)/1.5 = 3.5/1.5 = 2.333$. Local linear: only two points have positive weight, so the weighted line passes through $(1, 2)$ and $(2, 3)$ and takes the value 2 at $x_0 = 1$.

(b) At $x_0 = 2.5$ the distances are $1.5, 0.5, 0.5, 1.5$ and the weights $0.25, 0.75, 0.75, 0.25$ (sum 2). NW: $(0.25 \cdot 2 + 0.75 \cdot 3 + 0.75 \cdot 5 + 0.25 \cdot 4)/2 = 7.5/2 = 3.75$. Local linear: with $d_j = x_j - 2.5 = (-1.5, -0.5, 0.5, 1.5)$ we get $s_0 = 2$, $s_1 = \sum w_j d_j = 0$ (symmetric design and weights), $s_2 = 0.25(2.25)\cdot 2 + 0.75(0.25)\cdot 2 = 1.5$, so $\hat a = (s_2 \sum w_j y_j - s_1(\cdot))/(s_0 s_2 - s_1^2) = \sum w_j y_j / s_0 = 3.75$, the same as NW (the slope is $\hat b = \sum w_j d_j y_j / s_2 = 1.0$ but does not affect $\hat a$).

(c) At $x_0 = 2.5$ the neighbours are placed symmetrically around the target, so the local mean already sits at the right place: $s_1 = 0$ and the line adds nothing. At the boundary all neighbours lie on one side, $s_1 \neq 0$, and the NW estimate is biased by the slope of f (here it is pulled up to 2.333 when the data rise); the local line accounts for the slope. This is the content of the term $f'p'/p$ and of the $O(h)$ boundary bias.

```
#>              NW   LL
#> x0 = 1    2.333 2.00
#> x0 = 2.5  3.750 3.75
```

Exercise 14.4. The optimal bandwidth. Let p be uniform on $[0, 1]$, $f(x) = x^2$, $\sigma = 0.5$, Gaussian kernel, and consider the interior point $x = 0.5$. (a) Write down B and V of (14.5). (b) Compute h_{opt} and the optimal AMSE for $n = 100$ and $n = 1600$. (c) By what factor do h_{opt} and the AMSE change when n is multiplied by 16?

Solution.

(a) $p = 1$, $p' = 0$, $f'' = 2$, so $B = \mu_2 \cdot \frac{1}{2} \cdot 2 = 1$ (Gaussian: $\mu_2 = 1$) and $V = \sigma^2 R(K)/p = 0.25/(2\sqrt{\pi}) = 0.0705$.

(b) $h_{\text{opt}} = (V/(4B^2 n))^{1/5}$ and $\text{AMSE} = 5B^2 h_{\text{opt}}^4$:

```
B <- 1; V <- 0.25 / (2 * sqrt(pi))
res <- sapply(c(100, 1600), function(n) { h <- (V / (4 * B^2 * n))^(1 / 5);
↪  c(n = n, h_opt = h, AMSE = 5 * B^2 * h^4) })
round(t(res), 4)
```

```
#>         n  h_opt  AMSE
#> [1,]  100 0.1775 5e-03
#> [2,] 1600 0.1020 5e-04
```

Solution. (Continued.) $h_{\text{opt}} = 0.178$ for $n = 100$ and 0.102 for $n = 1600$; the AMSE is 0.005 and 5.4×10^{-4}. (c) Since $h_{\text{opt}} \propto n^{-1/5}$, it is multiplied by $16^{-1/5} = 0.574$; the AMSE $\propto n^{-4/5}$ is multiplied by $16^{-4/5} = 0.109$. A parametric estimator would gain a factor $1/16$ in variance; the nonparametric one gains only a factor $0.109 \approx 1/9.2$, the price of not knowing the shape of f.

Exercise 14.5. Mallows' C_p for smoothers. (a) Using Theorem 14.4 show that $\text{RSS}/n + 2\sigma^2 \operatorname{tr}(S)/n$ is an unbiased estimator of the prediction risk $\sigma^2 + \frac{1}{n}\mathrm{E}\|\hat{\boldsymbol{f}} - \boldsymbol{f}\|^2$. (b) Check that for $\hat{\boldsymbol{y}} = H\boldsymbol{y}$ it gives the usual C_p. (c) On `mcycle`, take $\hat\sigma^2$ from the degree-12 polynomial fit and compute C_p for polynomials of degree $1, \dots, 15$. Which degree is selected?

Solution.

(a) By Theorem 14.4(d), the risk equals $\mathrm{E}(\text{RSS}/n) + 2\sigma^2 \operatorname{tr}(S)/n$, so $\text{RSS}/n + 2\sigma^2 \operatorname{tr}(S)/n$ is unbiased. In practice σ^2 is replaced by an estimate from a low-bias fit. (b) $\operatorname{tr}(H) = p$, giving $C_p = \text{RSS}/n + 2\sigma^2 p/n$, the usual form (Mallows' C_p multiplied by σ^2/n). (c):

```
s2 <- summary(lm(accel ~ poly(times, 12), data = mcycle))$sigma^2
cp <- sapply(1:15, function(q) sum(resid(lm(accel ~ poly(times, q), data =
↪  mcycle))^2) / n + 2 * s2 * (q + 1) / n)
round(cp, 1)
```

```
#>  [1] 2129.3 2007.6 1583.0 1590.4 1143.8 1098.6  929.0  736.8  736.9
#> [10]  594.0  597.3  564.4  570.5  577.9  584.7
```

```
which.min(cp)
```

```
#> [1] 12
```

Solution. (Continued.) C_p is minimised at degree 12, with $\hat{\sigma} = 22.7$ taken from degree 12 (so degree 12 is favoured by construction when the criterion is flat). The curve is very flat between degrees 10 and 15, as for AIC: the criterion estimates *in-sample prediction error at the observed* x_i, which polynomials achieve well even though they fail at the edges and outside the data.

Exercise 14.6. Leave-one-out in the linear model. (a) Prove from Theorem 14.5 that for least squares $y_i - \hat{y}_{(i)} = e_i/(1 - h_{ii})$ with e_i the residual. (b) On `mcycle` with a polynomial of degree 5, compute LOOCV by the formula and by refitting n times.

Solution.

(a) Least squares minimises $\sum_j \{y_j - x_j^\top \beta\}^2$ with a fixed set of fits, so the leave-one-out property holds (the fit with y_i replaced by the leave-one-out prediction has zero residual at i, so it minimises the same criterion as the leave-one-out fit), and $S = H$ gives (14.6) with $S_{ii} = h_{ii}$.

(b)

```
m5 <- lm(accel ~ poly(times, 5), data = mcycle)
loo_formula <- mean((resid(m5) / (1 - hatvalues(m5)))^2)
loo_brute <- mean(sapply(1:n, function(i) {
  mi <- lm(accel ~ poly(times, 5), data = mcycle[-i, ])
  (y[i] - predict(mi, mcycle[i, ]))^2
}))
c(formula = loo_formula, brute_force = loo_brute)
```

```
#>     formula brute_force
#>        1245        1245
```

Solution. (Continued.) The two numbers agree: the shortcut replaces 133 refits by one. (Note that `poly()` is re-evaluated on the reduced data in each refit, which does not matter because the column space is the same.)

Exercise 14.7. kNN is not exactly leave-one-out. For $x = 1, \dots, 6$, $y = (2, 4, 3, 7, 9, 8)$ and the $k = 3$ nearest-neighbour mean (the point itself counts as one of the three): (a) write S and show $\mathrm{tr}(S) = n/k$ in general; (b) compute the shortcut residuals $(y_i - \hat{f}_i)/(1 - S_{ii})$; (c) compute the true leave-one-out residuals, where $\hat{f}^{(-i)}(x_i)$ is the mean of the 3 points nearest to x_i *among the other five*; (d) explain the discrepancy.

Solution.

(a) Row i has k entries equal to $1/k$ and the point is its own nearest neighbour, so $S_{ii} = 1/k$ and $\operatorname{tr}(S) = n/k = 2$ here. (b)–(c):

```
xk <- 1:6; yk <- c(2, 4, 3, 7, 9, 8); k <- 3
Sk <- t(sapply(xk, function(i) { w <- numeric(6); w[order(abs(xk - i),
↪  xk)[1:k]] <- 1 / k; w }))
fk <- drop(Sk %*% yk)
short <- (yk - fk) / (1 - diag(Sk))
loo <- sapply(1:6, function(i) { o <- setdiff(1:6, i); nb <- o[order(abs(xk[o]
↪  - xk[i]), xk[o])[1:k]]; yk[i] - mean(yk[nb]) })
rbind(shortcut = short, true_loo = loo)
```

```
#>              [,1] [,2]   [,3]  [,4] [,5]  [,6]
#> shortcut -1.500  1.5 -2.500 1.000  1.5 0.000
#> true_loo -2.667  0.0 -1.333 1.667  3.0 1.667
```

Solution. (Continued.) The shortcut residuals are -1.5, 1.5, -2.5, 1, 1.5, 0 while the true leave-one-out residuals are -2.67, 0, -1.33, 1.67, 3, 1.67: they differ for every i. Reason: the shortcut sets the weight of x_i to zero and renormalises the remaining weights, i.e. it predicts x_i by the mean of its *two* other neighbours (weights $\frac{1}{2}, \frac{1}{2}$). The true leave-one-out fit keeps $k = 3$ neighbours, so a *new* point (the next nearest) enters the neighbourhood. The deleted-point fit is therefore not obtained from row i of S, i.e. the leave-one-out property of Theorem 14.5 fails: the weights depend on which points are present. Under a fixed bandwidth (kernel, local polynomial) deleting a point only removes its weight and renormalises, so the shortcut is exact.

Exercise 14.8. Comparing smoothers against the truth. The data frame `faraway::exa` has $n = 256$ observations of $y = m(x) + \varepsilon$ and also contains the true mean `m`. (a) Choose h for NW and local linear by LOOCV and the span of LOESS (degree 2) by GCV, using trace.hat. (b) For each tuned smoother compute the mean squared error against the truth, $n^{-1}\sum\{\hat{f}(x_i) - m_i\}^2$, overall and separately for the 10% of points at each boundary and the interior. (c) Comment on where the methods differ.

Solution. (a)–(b):

```
data(exa, package = "faraway")
xe <- exa$x; ye <- exa$y; me <- exa$m; ne <- length(ye)
critE <- function(S) { r <- ye - drop(S %*% ye); edf <- sum(diag(S))
  c(edf = edf, LOOCV = mean((r / (1 - diag(S)))^2), GCV = ne * sum(r^2) / (ne
  ↪  - edf)^2) }
hE <- seq(0.01, 0.08, by = 0.002)
cN <- t(sapply(hE, function(h) critE(nw_S(xe, xe, h))))
cL <- t(sapply(hE, function(h) critE(ll_S(xe, xe, h))))
h_n <- hE[which.min(cN[, "LOOCV"])]; h_l <- hE[which.min(cL[, "LOOCV"])]
spE <- seq(0.05, 0.3, by = 0.01)
cLo <- sapply(spE, function(s) { m <- loess(ye ~ xe, span = s, degree = 2); ne
↪  * sum(resid(m)^2) / (ne - m$trace.hat)^2 })
s_lo <- spE[which.min(cLo)]
fE <- cbind(NW = drop(nw_S(xe, xe, h_n) %*% ye), LL = drop(ll_S(xe, xe, h_l)
↪  %*% ye),
            LOESS = fitted(loess(ye ~ xe, span = s_lo, degree = 2)))
reg <- ifelse(xe < 0.1 | xe > 0.9, "boundary", "interior")
mseE <- rbind(overall = colMeans((fE - me)^2),
```

```
              boundary = colMeans((fE[reg == "boundary", ] - me[reg ==
              ↪  "boundary"])^2),
              interior = colMeans((fE[reg == "interior", ] - me[reg ==
              ↪  "interior"])^2))
list(tuning = c(h_NW = h_n, h_LL = h_l, span = s_lo), mse = round(mseE * 1000,
↪  2))
```

```
#> $tuning
#>  h_NW  h_LL  span
#> 0.024 0.024 0.210
#>
#> $mse
#>              NW   LL LOESS
#> overall    8.30 8.44  8.70
#> boundary 10.08 9.46  8.81
#> interior  7.79 8.15  8.68
```

Solution. (Continued.) The table reports 1000× MSE. The selected bandwidths are $h = 0.024$ (NW), $h = 0.024$ (local linear), span 0.21 (LOESS). All three methods have an overall error of order 8.5×10^{-3}, small compared with $\sigma^2 = 0.098$. The differences are in the boundary rows: NW has the largest boundary error (bias of order h), the two local polynomials are better there; in the interior the three are alike. With a function that is flat near the edges the differences are modest here; they would be larger for a function with a steep slope at an end (Figure 14.6).

Exercise 14.9. Critique: a confidence band for the motorcycle curve. A report shows the local linear fit with $h = 2.12$ (edf 12) to `mcycle` and writes: "The grey band, the fit ±2 residual standard deviations, is a 95% confidence band for the true acceleration curve. Since 95% of the observations fall within it, the model is validated." (a) Identify two errors. (b) Compute the correct pointwise standard error band and state precisely what it covers. (c) Rewrite the sentence.

Solution.

(a) First, sd(residuals) estimates the scatter of a *single new observation* around the curve, not the uncertainty of the curve: a band of width ±2 sd is a rough *prediction* band, and it would not shrink as n grows, whereas the uncertainty of $\hat{f}(x)$ does. Second, the fraction of observations inside the band is a property of the residuals; it says nothing about whether $\hat{f}$ is close to f (any smoother, however biased, has about 95% of the residuals within 2 sd), so it does not validate the model. A third problem is that the noise is far from constant, so a constant-width band is wrong at both ends.

(b) The variance of the fitted value is $\sigma^2 (SS^\top)_{ii}$ (Theorem 14.4), estimated with $\hat{\sigma}^2 = \mathrm{RSS}/(n - \mathrm{edf})$:

```
Sl <- ll_S(x, x, h_ll); fl <- drop(Sl %*% y)
sig_hat <- sqrt(sum((y - fl)^2) / (n - sum(diag(Sl))))
se_f <- sig_hat * sqrt(rowSums(Sl^2))
c(inside_2sd = mean(abs(y - fl) < 2 * sd(y - fl)), sigma_hat = sig_hat,
  se_min = min(se_f), se_max = max(se_f), halfwidth_naive = 2 * sd(y - fl))
```

```
#>      inside_2sd       sigma_hat          se_min          se_max
#>          0.9398         23.8593         3.7777        21.7621
```

```
#> halfwidth_naive
#>         45.6756
```

Solution. (Continued.) The pointwise standard errors range from 3.8 to 21.8 g, so $\hat{f} \pm 2\widehat{\mathrm{se}}$ is between 8 and 44 g wide on each side, against 46 g for the naive band. It covers the *expected value of the smoother*, $(Sf)_i$, which equals $f(x_i)$ only if the bias is negligible (not true at the sharp dip, where the NW and local linear fits are too shallow), and it assumes constant variance. A proper treatment would use weights or a variance model (Chapter 16) and a bias correction or undersmoothing.

(c) "The curve is the local linear fit with 12 effective degrees of freedom. The shaded band shows ±2 pointwise standard errors of the fitted curve, computed under constant error variance; it ignores smoothing bias and is too narrow where the noise is largest. The scatter of single observations around the curve is much larger (about plus or minus 46 g)."

Exercise 14.10. LOESS span by cross-validation: shuffled against sorted folds. (a) Choose the span of a quadratic LOESS on `mcycle` by 10-fold CV with randomly assigned folds, for 10 different seeds. (b) Repeat with folds of consecutive observations (`rep(1:10, each = 14)[1:n]`). (c) Interpret.

Solution. (a)–(b): in R, `loess.control(surface = "direct")` is needed so that `predict()` can extrapolate (otherwise it returns `NA` outside the training range).

```
cv_loess <- function(span, folds) {
  sse <- 0
  for (k in unique(folds)) {
    te <- folds == k
    m <- loess(accel ~ times, data = mcycle[!te, ], span = span, degree = 2,
               control = loess.control(surface = "direct"))
    sse <- sse + sum((mcycle$accel[te] - predict(m, mcycle[te, ]))^2)
  }
  sse / n
}
spans <- seq(0.15, 0.6, by = 0.05)
best_shuf <- sapply(1:10, function(s) { set.seed(s); f <- sample(rep(1:10,
↪  length.out = n))
  spans[which.min(sapply(spans, cv_loess, folds = f))] })
fsort <- rep(1:10, each = 14)[1:n]
cv_sorted <- sapply(spans, cv_loess, folds = fsort)
set.seed(1); f1 <- sample(rep(1:10, length.out = n)); cv_shuf1 <-
↪  sapply(spans, cv_loess, folds = f1)
list(shuffled_best_spans = best_shuf, shuffled_min_seed1 = min(cv_shuf1),
     sorted_best_span = spans[which.min(cv_sorted)], sorted_min =
     ↪  min(cv_sorted))
```

```
#> $shuffled_best_spans
#>  [1] 0.35 0.35 0.35 0.35 0.35 0.40 0.35 0.35 0.35 0.30
#>
#> $shuffled_min_seed1
#> [1] 525.6
#>
#> $sorted_best_span
```

```
#> [1] 0.5
#>
#> $sorted_min
#> [1] 1207
```

Solution. (Continued.) With shuffled folds the chosen span varies between 0.3 and 0.4 across seeds, centred around 0.35: the CV criterion is flat and noisy, so report a range or average repeated CV. With sorted folds the criterion is minimised at a *larger* span (0.5) and its value (1207) is more than twice the shuffled one (526 for seed 1): every test fold is a contiguous block of times, so predicting it is an extrapolation from one side. The criterion then measures extrapolation, not interpolation, and rewards smoother fits. Folds must be randomised when data are sorted (and grouped by tied x if the aim is prediction at new times).

Exercise 14.11. Communicate. Write about 150 words for a non-statistician explaining what the bandwidth of a smoother does and how it should be chosen, using the motorcycle example.

Solution. A model answer: "A smoother draws a curve through noisy data by averaging nearby points. The *bandwidth* decides what 'nearby' means. With a very small bandwidth the curve follows every wiggle, including random noise: it fits the points we have very well but would predict new points badly. With a very large bandwidth it averages almost everything and flattens real features: in the crash-test data, a wide window erases the sharp dip in head acceleration at about 20 ms. We therefore choose the bandwidth to predict *new* data best. Cross-validation does this by repeatedly hiding one observation, predicting it from the others, and picking the bandwidth with the smallest prediction errors. Fits should be compared at the same 'effective number of parameters' (here about 12), not by how closely they hug the points, because a more flexible curve always hugs the points more closely." (About 150 words; it names the trade-off, the cure and the trap of comparing training error.)

Exercise 14.12. edf as an in-sample covariance. (a) Show that $\sum_i \mathrm{Cov}(\hat{f}_i, y_i) = \sigma^2 \operatorname{tr}(S)$. (b) For polynomial regression of degree q interpret the result. (c) For the $k = 10$ nearest-neighbour mean on `mcycle` give the edf, and compare it with the edf of NW with $h = 2$ (computed above). Which is the more flexible smoother?

Solution.

(a) $\mathrm{Cov}(\hat{f}_i, y_i) = \mathrm{Cov}(\sum_j S_{ij} y_j, y_i) = S_{ii}\sigma^2$ because the y_j are uncorrelated; summing over i gives $\sigma^2 \operatorname{tr}(S)$ (Proposition 14.3). (b) With $S = H$, $\operatorname{tr}(H) = q + 1$: each parameter makes the fit follow the data by exactly σ^2 in covariance; edf counts how many such "units of self-following" the smoother uses. (c) For kNN, edf $= n/k = 133/10 = 13.3$; for NW with $h = 2$ it is 11.3. The two are comparable (kNN with $k = 10$ is slightly *more* flexible, with 13.3 against 11.3 effective parameters), so a fair comparison of their RSS values needs both on the same scale, which is exactly what the edf provides.

14.14 Further reading

The motorcycle example and the smoothing-spline approach are in Silverman (1985); kernel smoothing theory in Wand and Jones (1995) and the local polynomial literature in Fan (1992) and Cleveland and Devlin (1988); LOESS in Cleveland (1979); linear smoothers and their degrees of freedom in Buja et al. (1989) and Hastie and Tibshirani (1990) (Chapter 3); cross-validation for smoothers in Craven and Wahba (1978). A modern, practical treatment with R code is

Eilers and Marx (2021) (Chapter 1), and a likelihood-based view in Pawitan (2001) (Chapter 18). Faraway's *Linear Models with R* (Faraway, 2014) has a short introduction to smoothing in the context of regression.

Chapter 15

Splines and penalized splines

Chapter 14 ended with a dilemma. A global polynomial has all the algebra of the linear model but cannot adapt locally; local smoothers adapt but are not parametric models and need one fit per target point. **Splines** sit in between: piecewise polynomials, joined smoothly at points called knots, form a linear space of functions with a basis, so that estimating f is again a linear-model problem, $y = B\beta + \varepsilon$, while the pieces give local flexibility (de Boor, 1978; Green and Silverman, 1994). The remaining difficulty is the choice of knots. This chapter shows how to avoid it by using many knots and a *penalty* on roughness, which leads to smoothing splines (Reinsch, 1967; Craven and Wahba, 1978) and P-splines (Eilers and Marx, 1996, 2021), and ends by showing that a penalized spline is a mixed model, the link to Chapters 11 and 19. The same machinery, one penalized spline per covariate, is the engine of generalized additive models (Chapters 16 and 17).

```
library(MASS)
library(splines)
library(mgcv)
data(mcycle)
x <- mcycle$times
y <- mcycle$accel
n <- length(y)
```

15.1 Regression splines

Definition 15.1 (Spline)**.** Let $\kappa_1 < \kappa_2 < \cdots < \kappa_K$ be **knots**. A **spline of degree** q with these knots is a function that is a polynomial of degree at most q on each of the intervals $(-\infty, \kappa_1], [\kappa_1, \kappa_2], \ldots, [\kappa_K, \infty)$ and has $q-1$ continuous derivatives on $\mathbb{R}$. The set of such functions is a vector space $\mathcal{S}_q(\kappa)$. A **cubic spline** has $q = 3$.

Cubic splines are the standard choice: the jump in the third derivative at a knot is invisible to the eye.

15.1.1 The truncated power basis

Write $u_+ = \max(u, 0)$. The function $(x-\kappa)_+^q$ is zero to the left of κ and a polynomial of degree q to the right, and it has $q-1$ continuous derivatives at κ (the q-th derivative jumps from 0 to $q!$).

Theorem 15.1 (Truncated power basis and dimension)**.** *The functions* $1, x, \dots, x^q, (x-\kappa_1)_+^q, \dots, (x-\kappa_K)_+^q$ *form a basis of* $\mathcal{S}_q(\kappa)$*, which therefore has dimension* $q+1+K$*. Every spline can be written*

$$f(x) = \sum_{k=0}^{q} \beta_k x^k + \sum_{k=1}^{K} \gamma_k (x-\kappa_k)_+^q, \tag{15.1}$$

and $f^{(q)}$ *jumps by* $q!\,\gamma_k$ *at* κ_k*.*

Proof. Each function belongs to $\mathcal{S}_q(\kappa)$. Conversely let $f \in \mathcal{S}_q(\kappa)$ and let p_0 be the polynomial equal to f on $(-\infty, \kappa_1]$. On $[\kappa_1, \kappa_2]$, $f = p_1$ is another polynomial of degree $\le q$; the difference $p_1 - p_0$ has $q-1$ continuous derivatives vanishing at κ_1, i.e. $(p_1-p_0)^{(m)}(\kappa_1) = 0$ for $m = 0, \dots, q-1$. A polynomial of degree $\le q$ with these properties is $\gamma_1(x-\kappa_1)^q$ (Taylor expansion at κ_1). Hence $f = p_0 + \gamma_1(x-\kappa_1)_+^q$ up to κ_2; repeating at each knot gives (15.1). The representation is unique (a spline with all β_k and γ_k zero is the zero function, and $q+1+K$ functions that represent every element uniquely are a basis). Differentiating q times, $f^{(q)} = q!\beta_q + q!\sum_k \gamma_k \mathbf{1}\{x > \kappa_k\}$. □

Another way to count: $K+1$ polynomial pieces with $q+1$ coefficients each, minus q continuity constraints at each of K knots, gives $(K+1)(q+1) - Kq = q+1+K$.

A **regression spline** fits (15.1) by least squares: it is a linear model with $q+1+K$ columns, so $\hat\beta = (X^\top X)^{-1} X^\top y$, $S = H$ and $\text{edf} = q+1+K$. In R the basis is rarely built by hand: `splines::bs(x, df)` generates cubic B-splines (Section 15.2) and `splines::ns(x, df)` natural cubic splines (Section 15.3); by default knots are placed at quantiles of x.

```
gridx <- seq(min(x), max(x), length.out = 300)
rs <- do.call(rbind, lapply(c(3, 8, 20), function(K) {
  m <- lm(accel ~ bs(times, df = K + 3), data = mcycle)
  data.frame(times = gridx, fit = predict(m, data.frame(times = gridx)), K =
  ↪  paste(K, "knots"))
}))
rs$K <- factor(rs$K, levels = paste(c(3, 8, 20), "knots"))
ggplot(rs, aes(times, fit)) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey60", size = 1) +
  geom_line(colour = ie_coral, linewidth = 0.9) + facet_wrap(~K, nrow = 1) +
  coord_cartesian(ylim = c(-150, 100)) + labs(x = "Time (ms)", y =
  ↪  "Acceleration (g)")
```

The number of knots is a tuning constant, chosen like any model size by AIC, BIC or (shuffled) cross-validation:

```
dfs <- c(5, 8, 10, 12, 15)
aic <- sapply(dfs, function(df) c(
  bs = AIC(lm(accel ~ bs(times, df = df), data = mcycle)),
  ns = AIC(lm(accel ~ ns(times, df = df), data = mcycle))))
colnames(aic) <- paste0("df=", dfs)
round(aic, 1)
```

```
#>    df=5 df=8 df=10 df=12 df=15
#> bs 1312 1222  1220  1220  1226
#> ns 1254 1216  1218  1220  1224
```

AIC is flat between 8 and 12 degrees of freedom: several different models fit equally well. This

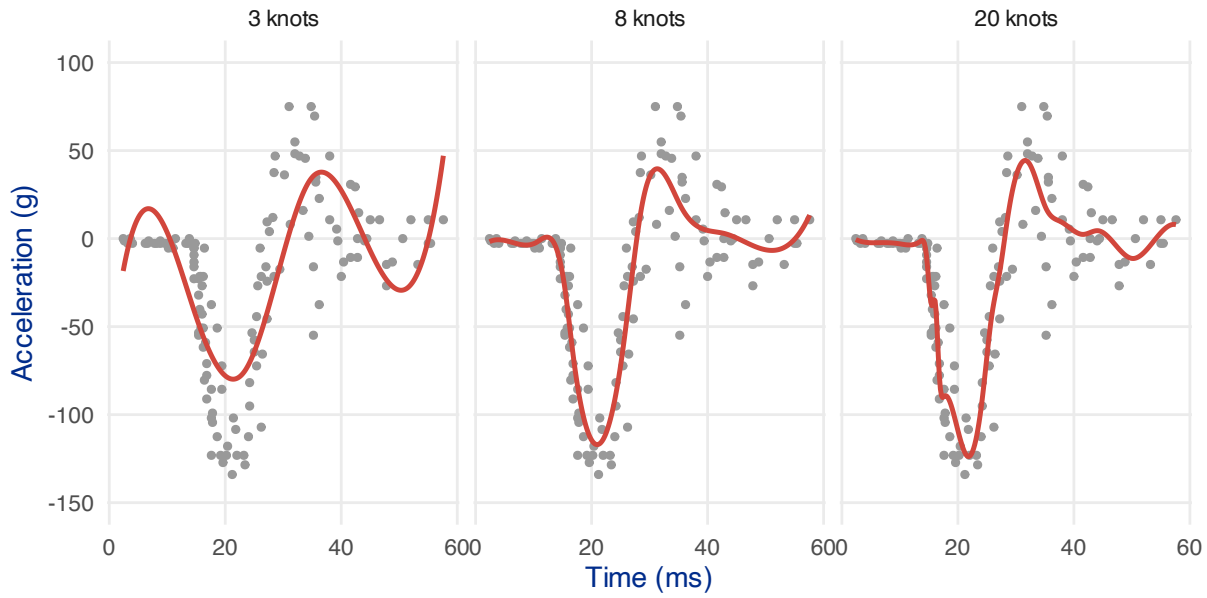


Figure 15.1: Cubic regression splines on the motorcycle data with 3, 8 and 20 interior knots (‘bs(times, df = K + 3)‘). Three knots miss the rebound; 20 knots chase the noise.

is unsatisfactory for three reasons. (i) The fit depends on *where* the knots are, not only on how many there are. (ii) Models with different knot sets are not nested, so the likelihood-ratio machinery does not apply. (iii) It is a discrete search, hard to automate when there are several covariates (Chapter 16). Penalization (Section 15.3) removes the problem. First, a better basis.

15.1.2 Why change the basis?

Truncated powers are not local (each stays non-zero to the right of its knot) and the design matrix is badly conditioned. On `mcycle` with eight interior knots:

```
tpower <- function(x, knots, q) outer(x, knots, function(u, k) pmax(u - k,
↪  0)^q)
knots8 <- quantile(x, probs = (1:8) / 9)
X_tp <- cbind(outer(x, 0:3, `^`), tpower(x, knots8, 3))
X_bs <- cbind(1, bs(x, knots = knots8, degree = 3))
c(dim = dim(X_tp), kappa_TP = kappa(X_tp, exact = TRUE), kappa_B = kappa(X_bs,
↪  exact = TRUE),
  max_diff_fitted = max(abs(fitted(lm(y ~ X_tp - 1)) - fitted(lm(y ~ X_bs -
  ↪  1)))))
```

```
#>            dim1            dim2        kappa_TP         kappa_B
#>       1.330e+02       1.200e+01       2.265e+06       3.072e+01
#> max_diff_fitted
#>       5.628e-11
```

The two designs span the same space (identical fitted values to 10^{-11}) but the condition numbers differ by a factor of 10^5.

15.2 B-splines

B-splines are a basis of the *same* space $\mathcal{S}_q(\kappa)$ made of non-negative bumps with small support (de Boor, 1978).

Definition 15.2 (B-splines)**.** Let $t_1 < t_2 < \cdots < t_N$ be a sequence of knots (the interior knots plus $q+1$ extra knots at each end). The B-splines are defined recursively. For degree 0,

$$B_{j,0}(x) = \mathbf{1}\{t_j \le x < t_{j+1}\},$$

and for $q \ge 1$ (the **Cox–de Boor recursion**)

$$B_{j,q}(x) = \frac{x - t_j}{t_{j+q} - t_j} B_{j,q-1}(x) + \frac{t_{j+q+1} - x}{t_{j+q+1} - t_{j+1}} B_{j+1,q-1}(x). \tag{15.2}$$

For equally spaced knots, $t_j = j\delta$, the recursion becomes $B_{j,q}(x) = \{(x - t_j)B_{j,q-1}(x) + (t_{j+q+1} - x)B_{j+1,q-1}(x)\}/(q\delta)$ and all B-splines are shifted copies of a single bump.

Theorem 15.2 (Properties of B-splines)**.** *For degree q and strictly increasing knots:*

(a) Local support and positivity: *$B_{j,q}(x) > 0$ for $t_j < x < t_{j+q+1}$ and $B_{j,q}(x) = 0$ outside $[t_j, t_{j+q+1})$; hence $B_{j,q} \ge 0$ and at most $q+1$ B-splines are non-zero at any x.*

(b) Partition of unity: *$\sum_j B_{j,q}(x) = 1$ for $t_{q+1} \le x < t_{N-q}$.*

(c) Smoothness and span: *each $B_{j,q}$ is a spline of degree q with $q-1$ continuous derivatives at the (simple) knots; the B-splines span $\mathcal{S}_q(\kappa)$ only on the interior of the knot range, $t_{q+1} \le x < t_{N-q}$, where they sum to one: this is the condition for the "same space as the truncated power basis", and it is met by extending the knot sequence with $q+1$ boundary knots at each end. On that interval every polynomial of degree $\le q$ is a combination of the B-splines (polynomial reproduction, Marsden's identity).*

Proof.

(a) Induction on q. For $q = 0$ it is the definition. For $q \ge 1$ the first term of (15.2) lives on $[t_j, t_{j+q})$ and the second on $[t_{j+1}, t_{j+q+1})$, and their coefficients $(x - t_j)/(t_{j+q} - t_j)$ and $(t_{j+q+1} - x)/(t_{j+q+1} - t_{j+1})$ are non-negative on those supports and positive inside; the union is $[t_j, t_{j+q+1})$.

(b) Shift the index in the second sum of $\sum_j B_{j,q}$:

$$\sum_j B_{j,q}(x) = \sum_j \left\{ \frac{x - t_j}{t_{j+q} - t_j} + \frac{t_{j+q} - x}{t_{j+q} - t_j} \right\} B_{j,q-1}(x) = \sum_j B_{j,q-1}(x),$$

because the two fractions add to one. Repeating q times reduces the sum to $\sum_j B_{j,0} = 1$ on the interval where every function contributing is present.

(c) Piecewise degree q follows from the recursion (each term multiplies a linear function by a polynomial of degree $q-1$); the continuity of $q-1$ derivatives, and that $N - q - 1$ B-splines are linearly independent and so span a space of the right dimension, is the theorem of Curry and Schoenberg (see de Boor (1978), Chapters IX–X).

□

The consequence of (a) is that $B^\top B$ is banded: row i of the design matrix $B = [B_j(x_i)]$ has at most $q+1$ non-zero entries, so the normal equations are cheap to solve and well conditioned.

Example 15.1 (Cubic B-splines at $x = 1.5$ by hand)**.** Take $t_j = j$. Only $B_{1,0}$ is non-zero at $x = 1.5$. Degree 1: $B_{0,1} = \frac{2-1.5}{1} \cdot 1 = \frac{1}{2}$ and $B_{1,1} = \frac{1.5-1}{1} \cdot 1 = \frac{1}{2}$. Degree 2: $B_{-1,2} = \frac{(2-1.5)\cdot\frac{1}{2}}{2} = \frac{1}{8}$, $B_{0,2} = \frac{1.5\cdot\frac{1}{2}+(3-1.5)\cdot\frac{1}{2}}{2} = \frac{3}{4}$, $B_{1,2} = \frac{1}{8}$. Degree 3: $B_{-2,3} = \frac{0.5\cdot\frac{1}{8}}{3} = \frac{1}{48}$, $B_{-1,3} = \frac{2.5\cdot\frac{1}{8}+1.5\cdot\frac{3}{4}}{3} = \frac{23}{48}$, and by symmetry $B_{0,3} = \frac{23}{48}$, $B_{1,3} = \frac{1}{48}$. They sum to one at every degree.

```
splineDesign(-3:8, 1.5, ord = 4)[, 2:5]
```

```
#> [1] 0.02083 0.47917 0.47917 0.02083
```

```
c(1, 23, 23, 1) / 48
```

```
#> [1] 0.02083 0.47917 0.47917 0.02083
```

A convenient family for smoothing is the one of Eilers and Marx: equally spaced knots on a domain slightly *wider* than the data, with `nseg` segments and degree `bdeg`, giving `nseg + bdeg` B-splines.

```
bbase <- function(x, xl = min(x) - 0.01, xr = max(x) + 0.01, nseg = 20, bdeg =
↪  3) {
  dx <- (xr - xl) / nseg
  knots <- seq(xl - bdeg * dx, xr + bdeg * dx, by = dx)        # bdeg extra
↪  segments at each end
  B <- splineDesign(knots, x, ord = bdeg + 1, outer.ok = TRUE)
  attr(B, "knots") <- knots
  B
}
B <- bbase(x)
c(dim(B), range(rowSums(B)), max(rowSums(B > 0)))
```

```
#> [1] 133  23   1   1   4
```

The matrix has 133 rows and $23 = \texttt{nseg} + \texttt{bdeg}$ columns, rows sum to one (partition of unity) and at most four entries per row are non-zero. (A frequent slip is `xl <- min(x) + 0.01`, which leaves the smallest observation outside the domain.)

```
Bg <- bbase(gridx)
Dm <- diff(diag(ncol(B)), differences = 2)
th <- solve(crossprod(B) + 0.64 * crossprod(Dm), crossprod(B, y))
bl <- data.frame(times = rep(gridx, ncol(B)), j = rep(1:ncol(B), each =
↪  length(gridx)),
                 val = as.vector(Bg), wval = as.vector(sweep(Bg, 2, th, "*")))
ggplot(bl, aes(times, wval, group = j)) + geom_line(colour = ie_blue,
↪  linewidth = 0.4) +
  geom_line(data = data.frame(times = gridx, fit = drop(Bg %*% th)),
  ↪  aes(times, fit, group = NULL),
            colour = ie_coral, linewidth = 1) +
  geom_point(data = mcycle, aes(times, accel, group = NULL), colour =
  ↪  "grey60", size = 0.9) +
  labs(x = "Time (ms)", y = expression(beta[j] * B[j](x)))
```

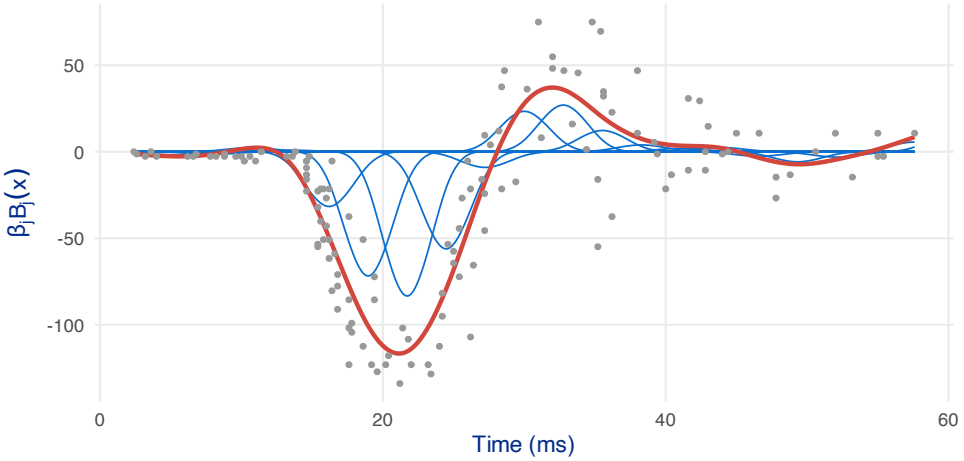


Figure 15.2: The 23 cubic B-splines of the basis used below, and the fit $B\hat{\beta}$ they produce when each is multiplied by its estimated coefficient (P-spline fit of Section 15.4).

Derivatives of B-spline curves. For equally spaced knots with spacing δ and $f = \sum_j \beta_j B_{j,q}$,

$$f'(x) = \frac{1}{\delta} \sum_j (\beta_j - \beta_{j-1})\, B_{j,q-1}(x), \tag{15.3}$$

a classical result (de Boor, 1978); applying it d times, $f^{(d)} = \delta^{-d} \sum_j (\Delta^d \beta)_j B_{j,q-d}$ where Δ is the difference operator. *The roughness of the curve is controlled by differences of its coefficients.* This is the key to P-splines (Section 15.4); a numerical check follows (the first derivative of f, computed directly, and from the coefficient differences):

```
tk <- attr(B, "knots"); dl <- diff(tk)[1]
thv <- drop(th)
f1_direct <- drop(splineDesign(tk, gridx, ord = 4, derivs = 1, outer.ok =
↪  TRUE) %*% thv)
f1_diff <- drop(splineDesign(tk[-c(1, length(tk))], gridx, ord = 3, outer.ok =
↪  TRUE) %*% (diff(thv) / dl))
c(dim_Bq1 = ncol(splineDesign(tk[-c(1, length(tk))], gridx, ord = 3, outer.ok
↪  = TRUE)),
  max_abs_diff = max(abs(f1_direct - f1_diff)))
```

```
#>      dim_Bq1 max_abs_diff
#>    2.200e+01    1.066e-14
```

15.3 Natural cubic splines and smoothing splines

Definition 15.3 (Natural cubic spline)**.** A cubic spline with knots $\kappa_1 < \cdots < \kappa_K$ is **natural** if it is linear on $(-\infty, \kappa_1]$ and on $[\kappa_K, \infty)$, equivalently $f'' = f''' = 0$ outside the knot range.

A cubic spline with K knots has $K + 4$ parameters (Theorem 15.1); linearity on each outer piece removes the coefficients of x^2 and x^3 there, i.e. two constraints at each end, so the natural splines form a space of dimension $K + 4 - 4 = K$. In R, `ns(x, df)` returns `df` basis functions (without intercept) with `df - 1` interior knots plus two boundary knots at the extremes of the data. Natural splines have less variance near and beyond the ends than ordinary cubic splines, where the extra x^2, x^3 terms are free:

```
gx2 <- seq(0, 75, length.out = 300)
cmp <- do.call(rbind, lapply(c("bs", "ns"), function(b) {
  m <- if (b == "bs") lm(accel ~ bs(times, df = 8), mcycle) else lm(accel ~
↪  ns(times, df = 8), mcycle)
  data.frame(times = gx2, fit = suppressWarnings(predict(m, data.frame(times =
  ↪  gx2))), basis = b)
}))
ggplot(cmp, aes(times, fit, colour = basis)) + geom_vline(xintercept =
↪  range(x), linetype = 3) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey65", size = 1,
  ↪  inherit.aes = FALSE) +
  geom_line(linewidth = 0.9) + scale_colour_manual(values = c(ie_coral,
  ↪  ie_navy), name = NULL) +
  coord_cartesian(ylim = c(-200, 100)) + labs(x = "Time (ms)", y =
  ↪  "Acceleration (g)")
```

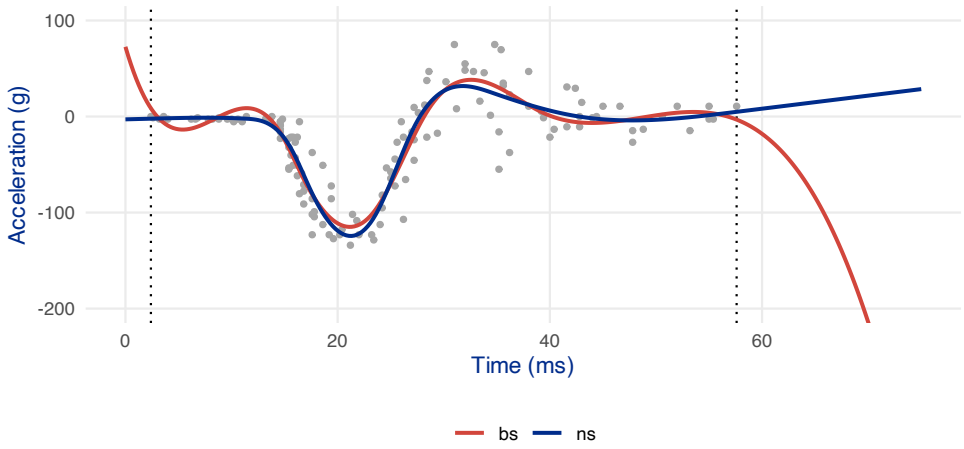


Figure 15.3: Cubic (‘bs‘) and natural cubic (‘ns‘) regression splines with 8 degrees of freedom, extended beyond the data. Beyond the last observation the cubic piece dives, while the natural spline continues as a straight line.

15.3.1 Smoothing splines

Instead of choosing knots, we may search over *all* twice-differentiable functions and penalise roughness. The roughness of f is measured by its total squared curvature $\int f''(u)^2 du$, which is zero exactly for straight lines.

Definition 15.4 (Smoothing spline)**.** For $\lambda \geq 0$, the **smoothing spline** is the minimiser over twice-differentiable f of

$$Q_\lambda(f) = \sum_{i=1}^{n} \{y_i - f(x_i)\}^2 + \lambda \int f''(u)^2 \, du. \tag{15.4}$$

The first term rewards fidelity, the second penalises curvature; λ is the exchange rate. As $\lambda \to 0$ the minimiser interpolates the data; as $\lambda \to \infty$ it tends to the least-squares straight line, the only functions with no penalty. Remarkably, the infinite-dimensional problem has a finite-dimensional solution.

Theorem 15.3 (Reinsch (1967)). *Let $x_1 < \cdots < x_m$ be the distinct values of the design, $m \geq 2$, and $\lambda > 0$. The minimiser of (15.4) is unique and is a natural cubic spline with knots at $x_1, \ldots, x_m$.*

Proof. Step 1: the natural interpolant has the least curvature. Let g be any twice-differentiable function and let s be the natural cubic spline with knots $x_1, \ldots, x_m$ that interpolates g there, $s(x_i) = g(x_i)$ (it exists and is unique: m interpolation conditions for a space of dimension m). Put $h = g - s$, so $h(x_i) = 0$ for all i. Since $s'' = 0$ outside $[x_1, x_m]$, integrate by parts on $[x_1, x_m]$:

$$\int_{x_1}^{x_m} s''h''\,du = \left[s''h'\right]_{x_1}^{x_m} - \int_{x_1}^{x_m} s'''h'\,du.$$

The boundary term vanishes because $s''(x_1) = s''(x_m) = 0$ (natural). On each (x_i, x_{i+1}) the third derivative of s is a constant c_i, so $\int_{x_1}^{x_m} s'''h' = \sum_{i=1}^{m-1} c_i\{h(x_{i+1}) - h(x_i)\} = 0$. Hence $\int s''h'' = 0$ and

$$\int_{x_1}^{x_m} g''^2 = \int_{x_1}^{x_m} (s'' + h'')^2 = \int_{x_1}^{x_m} s''^2 + \int_{x_1}^{x_m} h''^2 \;\geq\; \int_{x_1}^{x_m} s''^2,$$

with equality only if $h'' \equiv 0$ on $[x_1, x_m]$, i.e. h is linear there; a linear function with $m \geq 2$ zeros is $h \equiv 0$, so $g = s$ on $[x_1, x_m]$.

Step 2: conclusion. $\int g''^2 \geq \int_{x_1}^{x_m} g''^2 \geq \int_{x_1}^{x_m} s''^2 = \int s''^2$, because $s'' = 0$ outside the data range. The residual sum of squares of g and s is the same, since they agree at the design points. So $Q_\lambda(s) \leq Q_\lambda(g)$, with equality only if $g = s$ (for $\lambda > 0$). Thus the minimiser, which exists in this class, is a natural cubic spline with the stated knots. □

The proof also explains why the fit is *linear outside the data range*: any curvature there costs penalty and buys nothing.

Corollary 15.1 (The solution is a penalized regression). *Let $N_1, \ldots, N_m$ be a basis of the natural cubic splines with knots $x_1, \ldots, x_m$, $N = [N_j(x_i)]$ the $n \times m$ design matrix and $\Omega_{jk} = \int N_j'' N_k''$. Then $\int f''^2 = \beta^\top \Omega \beta$ for $f = \sum_j \beta_j N_j$, and the smoothing spline is $\hat{f} = N\hat{\beta}$ with*

$$\hat{\beta} = (N^\top N + \lambda\Omega)^{-1} N^\top y, \qquad S_\lambda = N(N^\top N + \lambda\Omega)^{-1} N^\top.$$

This is a *penalized least squares* problem of the form studied below: only the penalty matrix Ω is different. For distinct x_i ($m = n$, N invertible) $S_\lambda = (I + \lambda K)^{-1}$ with $K = N^{-\top}\Omega N^{-1}$ (Reinsch, 1967). The matrix K is symmetric and positive semi-definite with a two-dimensional null space (constants and straight lines, which have zero curvature), so S_λ has eigenvalues $1, 1$ and $1/(1 + \lambda d_k) \in (0, 1)$ for the positive eigenvalues d_k of K: the fit leaves linear functions untouched and shrinks the other components of y, high-frequency ones more. Consequently $\text{edf} = \text{tr}(S_\lambda) = 2 + \sum_k 1/(1 + \lambda d_k)$ decreases from n (at $\lambda = 0$) to 2 (at $\lambda = \infty$).

```
ss <- smooth.spline(x, y)                              # lambda chosen by GCV
ss_cv <- suppressWarnings(smooth.spline(x, y, cv = TRUE))
rbind(GCV = c(df = ss$df, lambda = ss$lambda), LOOCV = c(df = ss_cv$df, lambda
↪  = ss_cv$lambda))
```

```
#>           df    lambda
#> GCV   12.21 1.108e-04
#> LOOCV 12.74 9.147e-05
```

```
c(sum_leverage = sum(ss$lev), df = ss$df)
```

```
#> sum_leverage           df
#>        12.21        12.21
```

`smooth.spline()` chooses λ by GCV by default and reports $\mathrm{tr}(S_\lambda)$ as `df`, equal to the sum of the leverages `lev`. Its `lambda` refers to x rescaled to $[0,1]$ and a standardised penalty, so it is not comparable to the λ of other software. One can also fix the complexity (`df = 12`) instead of letting GCV choose.

15.4 P-splines

Smoothing splines have as many coefficients as distinct x values and need integrals of products of basis functions. Eilers and Marx (Eilers and Marx, 1996, 2021) kept the idea and simplified the algebra: use a *rich* B-spline basis with equally spaced knots, deliberately too many, and penalise the *differences of neighbouring coefficients*.

15.4.1 The estimator

Definition 15.5 (Difference matrix)**.** Let $\Delta\beta_j = \beta_j - \beta_{j-1}$ and $\Delta^d = \Delta(\Delta^{d-1})$. For c coefficients the $(c-d)\times c$ matrix D_d with $D_d\beta = \Delta^d\beta$ is the **difference matrix of order** d.

In R, `D <- diff(diag(c), differences = d)`. For $c = 5$,

$$D_1 = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & -2 & 1 \end{bmatrix}, \quad D_2^\top D_2 = \begin{bmatrix} 1 & -2 & 1 & 0 & 0 \\ -2 & 5 & -4 & 1 & 0 \\ 1 & -4 & 6 & -4 & 1 \\ 0 & 1 & -4 & 5 & -2 \\ 0 & 0 & 1 & -2 & 1 \end{bmatrix}.$$

```
crossprod(diff(diag(5), differences = 2))
```

```
#>      [,1] [,2] [,3] [,4] [,5]
#> [1,]    1   -2    1    0    0
#> [2,]   -2    5   -4    1    0
#> [3,]    1   -4    6   -4    1
#> [4,]    0    1   -4    5   -2
#> [5,]    0    0    1   -2    1
```

The penalty $\|D_d\beta\|^2$ has a direct meaning. For $d = 1$ it is $\sum_j(\beta_j - \beta_{j-1})^2$, the squared differences in *level* between neighbouring coefficients (not slopes); a large penalty forces all β_j equal and the fit becomes a constant. For $d = 2$ it is $\sum_j(\beta_j - 2\beta_{j-1} + \beta_{j-2})^2$, the squared changes of those differences; a large penalty forces β_j to be linear in j and the fit becomes a straight line. By (15.3), $\|D_d\beta\|^2$ is a discretised version of $\int\{f^{(d)}\}^2$, which links P-splines to the smoothing spline penalty with $d = 2$.

Theorem 15.4 (The P-spline estimator)**.** *Let B be the $n\times c$ B-spline matrix with full column rank and $\lambda > 0$. The criterion*

$$Q(\beta) = \|y - B\beta\|^2 + \lambda\|D_d\beta\|^2 \tag{15.5}$$

has the unique minimiser

$$\hat{\beta} = (B^\top B + \lambda D_d^\top D_d)^{-1} B^\top y, \qquad S_\lambda = B(B^\top B + \lambda D_d^\top D_d)^{-1} B^\top, \tag{15.6}$$

and $\mathrm{edf}(\lambda) = \mathrm{tr}(S_\lambda) = \mathrm{tr}\{(B^\top B + \lambda D_d^\top D_d)^{-1} B^\top B\}$. *The matrix* S_λ *is symmetric and not idempotent.*

Proof. $\partial Q/\partial\beta = -2B^\top(y - B\beta) + 2\lambda D_d^\top D_d\beta = 0$ gives $(B^\top B + \lambda D_d^\top D_d)\beta = B^\top y$. The Hessian $2(B^\top B + \lambda D_d^\top D_d)$ is positive definite because $\beta^\top(B^\top B + \lambda D_d^\top D_d)\beta = \|B\beta\|^2 + \lambda\|D_d\beta\|^2 > 0$ for $\beta \neq 0$ (B has full column rank), so Q is strictly convex with a unique minimiser. The trace follows from $\mathrm{tr}(B\,M\,B^\top) = \mathrm{tr}(M\,B^\top B)$. S_λ is symmetric by construction; $S_\lambda^2 = S_\lambda$ would force $S_\lambda B = B$ for B of full rank, i.e. $\lambda D_d^\top D_d = 0$. □

Remark. The inverse is essential: writing $S_\lambda = B(B^\top B + \lambda D_d^\top D_d)B^\top$ is a mistake that recurs in lecture notes. Also the penalty multiplies λ only once.

```
psfit <- function(B, y, lambda, d = 2) {
  D <- diff(diag(ncol(B)), differences = d)
  A <- crossprod(B) + lambda * crossprod(D)          # B'B + lambda D'D
  theta <- solve(A, crossprod(B, y))
  f <- drop(B %*% theta); n <- length(y)
  edf <- sum(diag(solve(A, crossprod(B)))); rss <- sum((y - f)^2)
  list(theta = drop(theta), f = f, edf = edf, rss = rss,
       gcv = n * rss / (n - edf)^2, aic = n * log(rss / n) + 2 * edf, sigma =
       ↪  sqrt(rss / (n - edf)))
}
sapply(c(1e-8, 0.01, 1, 100, 1e4, 1e10), function(l) psfit(B, y, l)$edf)
```

```
#> [1] 23.000 19.656 10.519  4.428  2.169  2.000
```

15.4.2 Limits and what the penalty leaves alone

Theorem 15.5 (Limits of the P-spline)**.** *(a) As* $\lambda \to 0$, $S_\lambda \to B(B^\top B)^{-1}B^\top$ *and* $\mathrm{edf} \to c$. *(b) As* $\lambda \to \infty$ *the penalty forces* $D_d\beta = 0$*:* β *is a polynomial of degree* $< d$ *in the index* j. *The fit tends to the least-squares fit within the null space* $\{B\beta : D_d\beta = 0\}$*, which has dimension* d*, and* $\mathrm{edf} \to d$.

Proof.

(a) is continuity. (b) $Q(\beta) \le Q(\beta_0)$ at the least-squares solution β_0 in the null space of D_d (where the penalty is zero) bounds $\lambda\|D_d\hat{\beta}_\lambda\|^2 \le Q(\beta_0)$, so $D_d\hat{\beta}_\lambda \to 0$ and every limit point is in the null space, where Q reduces to the residual sum of squares. The null space of D_d consists of the sequences with $\Delta^d\beta = 0$, polynomials in j of degree $\le d-1$: dimension d. The limit of S_λ is the projection onto $\{B\beta : D_d\beta = 0\}$, of trace d.

□

So the *null space* of the penalty is what the penalty does not touch. Choosing d chooses what "infinitely smooth" means.

Theorem 15.6 (Polynomial reproduction and conservation of moments)**.** *Assume (i) the knots are equally spaced and (ii) $q \geq d-1$ (degree of B-splines). Then:*

(a) if the data lie exactly on a polynomial of degree $\leq d-1$, $y_i = p(x_i)$, the P-spline fit reproduces it for ***every*** *λ;*

(b) for $k = 0, 1, \ldots, d-1$, $\sum_i x_i^k \hat{f}_i = \sum_i x_i^k y_i$ for every λ.

Proof. By (ii) a polynomial p of degree $\leq d-1 \leq q$ is a spline in the span of the B-splines, $p(x) = \sum_j \beta_j^* B_j(x)$ on the domain; by (i) the coefficients β_j^* are themselves a polynomial of the same degree in j (the B-splines are translates of one another), so $D_d\beta^* = 0$. (a) At β^* the criterion is $Q = 0$, its smallest possible value, and the minimiser is unique. (b) The normal equations read $B^\top(y - B\hat{\beta}) = \lambda D_d^\top D_d \hat{\beta}$. The vector $(x_i^k)_i$ equals Bu with $D_d u = 0$ (by the argument for $p(x) = x^k$), so

$$\sum_i x_i^k (y_i - \hat{f}_i) = u^\top B^\top (y - B\hat{\beta}) = \lambda\, (D_d u)^\top D_d \hat{\beta} = 0.$$

□

Both conditions matter (more precisely, the data degree must not exceed $\min(q, d-1)$). With $d = 1$ a straight line is *not* reproduced (it has degree $1 > d - 1 = 0$); with $d = 2$ it is. And moments of order $\geq d$ are not conserved.

```
ylin <- 3 + 2 * x                                    # exactly linear data
errs <- sapply(1:3, function(d) max(abs(psfit(B, ylin, 1000, d = d)$f -
↪  ylin)))
names(errs) <- paste0("d=", 1:3); signif(errs, 3)
```

```
#>      d=1      d=2      d=3
#> 5.67e+01 9.90e-12 1.04e-10
```

```
f2 <- psfit(B, y, 0.64, d = 2)$f
rbind(data = c(sum(y), sum(x * y), sum(x^2 * y)), fit = c(sum(f2), sum(x *
↪  f2), sum(x^2 * f2)))
```

```
#>       [,1]   [,2]     [,3]
#> data -3398 -60720 -1023707
#> fit  -3398 -60720 -1023809
```

For data exactly on $y = 3 + 2x$ the error is 10^{-11} for $d = 2, 3$ but of order 57 for $d = 1$. For the `mcycle` fit with $d = 2$, $\sum \hat{f}_i = \sum y_i$ and $\sum x_i \hat{f}_i = \sum x_i y_i$ hold exactly while $\sum x_i^2 \hat{f}_i \neq \sum x_i^2 y_i$.

15.4.3 The effective dimension

In terms of the generalised eigenvalues of the penalty, write $\mathrm{edf}(\lambda) = \sum_{k=1}^{c} 1/(1 + \lambda\gamma_k)$ where $\gamma_k \geq 0$ are the eigenvalues of $(B^\top B)^{-1/2} D_d^\top D_d (B^\top B)^{-1/2}$ (d of them are zero). Then the edf is strictly decreasing in λ, from c to d, and the smoother shrinks the k-th component of the data by the factor $1/(1 + \lambda\gamma_k)$: components in the null space ($\gamma_k = 0$) are not shrunk at all, wiggly ones are shrunk the most. This is the same structure as for the smoothing spline.

```
ll <- seq(-6, 8, by = 0.25)
ggplot(data.frame(ll, edf = sapply(ll, function(l) psfit(B, y, 10^l)$edf)),
↪  aes(ll, edf)) +
  geom_line(colour = ie_navy, linewidth = 1) + geom_hline(yintercept = c(2,
  ↪  23), linetype = 3) +
```

```
  labs(x = expression(log[10](lambda)), y = "edf")
```

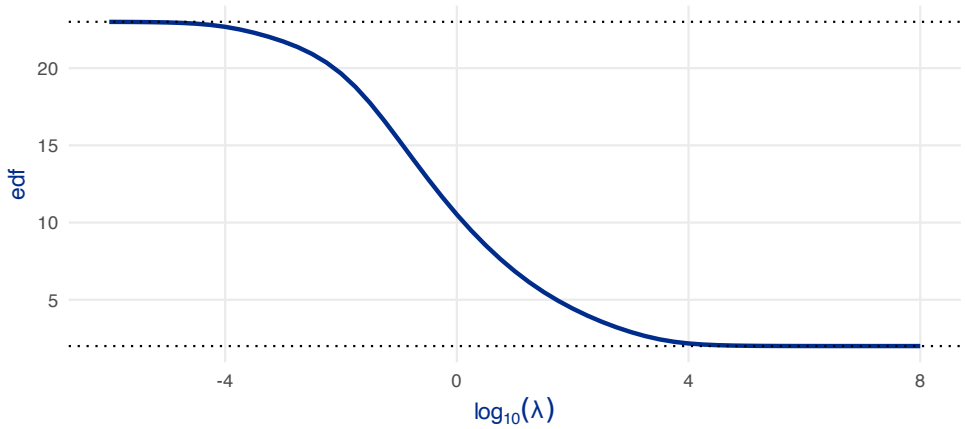


Figure 15.4: Effective degrees of freedom of the P-spline on the motorcycle data as a function of $\log_{10}\lambda$ ($d = 2$, 23 B-splines). The edf falls from 23 to the null-space dimension 2.

15.4.4 Choosing λ

The P-spline is a linear smoother, so Chapter 14 applies: leave-one-out residuals are $(y_i - \hat{f}_i)/(1 - S_{ii})$ (penalized least squares satisfies the leave-one-out property of Theorem 14.5), and

$$\mathrm{GCV}(\lambda) = \frac{n\,\mathrm{RSS}(\lambda)}{\{n - \mathrm{edf}(\lambda)\}^2}, \qquad \mathrm{AIC}(\lambda) = n\log\frac{\mathrm{RSS}(\lambda)}{n} + 2\,\mathrm{edf}(\lambda), \qquad \hat{\sigma}^2 = \frac{\mathrm{RSS}}{n - \mathrm{edf}}. \tag{15.7}$$

AIC here is the Gaussian log-likelihood with σ^2 profiled out and the number of parameters replaced by the edf. We search over $\log_{10}\lambda$.

```
loglam <- seq(-3, 6, by = 0.05)
crit <- t(sapply(loglam, function(l) { f <- psfit(B, y, 10^l); c(GCV = f$gcv,
↪  AIC = f$aic, edf = f$edf) }))
opt <- optimize(function(l) psfit(B, y, 10^l)$gcv, c(-3, 6))
fit <- psfit(B, y, 10^opt$minimum)
c(log10_lambda = opt$minimum, lambda = 10^opt$minimum, edf = fit$edf, GCV =
↪  fit$gcv, sigma_hat = fit$sigma,
  lambda_AIC_grid = 10^loglam[which.min(crit[, "AIC"])])
```

```
#>    log10_lambda          lambda             edf             GCV
#>         -0.1925          0.6419         11.3766        562.9759
#>       sigma_hat lambda_AIC_grid
#>         22.6896          0.6310
```

```
ggplot(data.frame(loglam, GCV = crit[, "GCV"]), aes(loglam, GCV)) +
↪  geom_line(colour = ie_navy, linewidth = 1) +
  geom_vline(xintercept = opt$minimum, linetype = 2) + coord_cartesian(ylim =
  ↪  c(550, 1500)) +
  labs(x = expression(log[10](lambda)), y = "GCV")
```

GCV selects $\hat{\lambda} = 0.64$ with edf $= 11.38$ and $\hat{\sigma} = 22.7$ g, and AIC the same value to the resolution of the grid. A check against `mgcv`, which implements the same estimator with the same B-spline basis (`bs = "ps"`: B-splines, penalty of order `m[2]`):

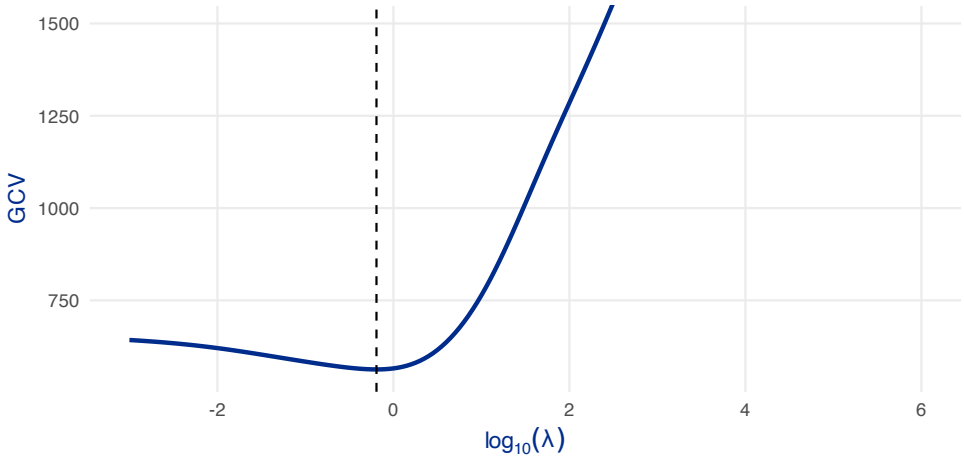


Figure 15.5: GCV and AIC (rescaled to the GCV range) against $\log_{10}\lambda$ for the P-spline on the motorcycle data. Both are minimised near $\log_{10}\lambda = -0.2$.

```
g <- gam(accel ~ s(times, bs = "ps", k = ncol(B), m = c(2, 2)), data = mcycle,
         knots = list(times = attr(B, "knots")), method = "GCV.Cp")
c(edf_mgcv = sum(g$edf), edf_hand = fit$edf, max_abs_diff_fit =
↪  max(abs(fitted(g) - fit$f)))
```

```
#>          edf_mgcv          edf_hand max_abs_diff_fit
#>         1.138e+01         1.138e+01         2.142e-04
```

```
g10 <- gam(accel ~ s(times, bs = "ps"), data = mcycle, method = "GCV.Cp")   #
↪  default k = 10
c(edf_default_k = sum(g10$edf), edf_k23 = sum(g$edf))
```

```
#> edf_default_k       edf_k23
#>         9.699        11.377
```

The edf agree to 10^{-4} and the fitted curves to 10^{-4} g (the remaining difference is the optimiser). mgcv rescales the penalty, so its smoothing parameter `sp` is *not* our λ: compare fits and edf, not `sp`. Note also that the default `k = 10` basis dimension caps the edf at 9.7: here the *basis*, not the data, limits the complexity (Exercise 15.13).

Knots barely matter once there are enough. Because the penalty controls smoothness, the number of segments is only required to be large enough to represent the curve. With 10, 20 and 40 segments, GCV selects different λ but nearly identical edf and fits (Exercise 15.9).

15.4.5 Pointwise uncertainty

With $A = B^\top B + \lambda D_d^\top D_d$, $\mathrm{Var}(\hat\beta) = \sigma^2 A^{-1} B^\top B A^{-1}$ in the frequentist sense, and $\mathrm{Var}(\hat{\boldsymbol{f}}) = \sigma^2 S_\lambda S_\lambda^\top$ (Theorem 14.4); in the Bayesian interpretation of Section 15.5, which `mgcv` uses for its bands, the posterior covariance is $\sigma^2 A^{-1}$, so $\mathrm{Var}(\boldsymbol{f} \mid y) = \sigma^2 B A^{-1} B^\top = \sigma^2 S_\lambda$. The Bayesian version accounts for the bias of the penalised estimate on average, and is the one `predict(gam, se.fit = TRUE)` returns. The frequentist band assumes the smoothing bias is negligible; both assume constant variance, which `mcycle` violates.

```
A <- crossprod(B) + 10^opt$minimum * crossprod(diff(diag(ncol(B)), differences
↪  = 2))
S_l <- B %*% solve(A, t(B))
```

```
se_bayes <- fit$sigma * sqrt(diag(S_l)); se_freq <- fit$sigma *
↪  sqrt(rowSums(S_l^2))
rbind(bayes = range(se_bayes), frequentist = range(se_freq))
```

```
#>              [,1]  [,2]
#> bayes       4.098 18.04
#> frequentist 3.838 16.26
```

```
bd <- data.frame(times = x, f = fit$f, lo = fit$f - 2 * se_bayes, hi = fit$f +
↪  2 * se_bayes)
ggplot(bd, aes(times)) + geom_ribbon(aes(ymin = lo, ymax = hi), fill =
↪  ie_navy, alpha = 0.25) +
  geom_point(data = mcycle, aes(times, accel), colour = "grey60", size = 1) +
  geom_line(aes(y = f), colour = ie_navy, linewidth = 1) + labs(x = "Time
  ↪  (ms)", y = "Acceleration (g)")
```

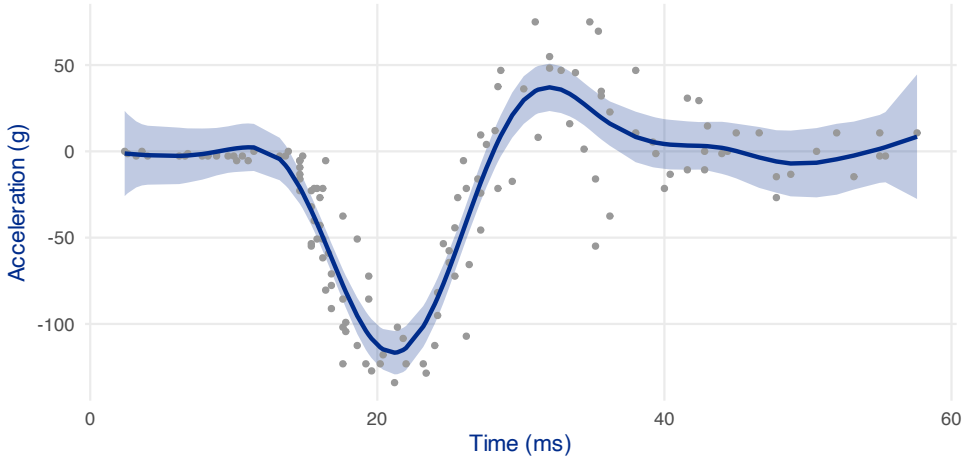


Figure 15.6: P-spline fit with pointwise 95% bands (Bayesian standard errors, $\sigma^2 S_\lambda$). The band is constant-variance-based: too wide at the start, too narrow at the dip.

15.5 Smoothing as regularization, and mixed models

15.5.1 Penalized least squares

Smoothing spline, P-spline, ridge regression (Chapter 7) and penalized truncated power splines all have the form

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \beta^\top P \beta = (X^\top X + \lambda P)^{-1} X^\top y, \qquad S_\lambda = X(X^\top X + \lambda P)^{-1} X^\top, \tag{15.8}$$

with a positive semi-definite penalty matrix P:

Table 15.1: Penalized least squares in this chapter and Chapter 7.

Method	X	P	shrinks towards
ridge regression	X	I	$\beta = 0$
P-spline	B	$D_d^\top D_d$	polynomial of degree $d-1$

Method	X	P	shrinks towards
smoothing spline	N	Ω	straight line
penalized truncated powers	$[1, x, \ldots, x^q, (x-\kappa_k)_+^q]$	$\mathrm{diag}(0, \ldots, 0, 1, \ldots, 1)$	polynomial of degree q

The **null space** of P is left unpenalised and is what survives as $\lambda \to \infty$; the penalty introduces bias on purpose to reduce variance, and λ sets the exchange rate. For truncated powers it is essential to penalise *only* the knot coefficients γ_k: penalising the polynomial coefficients as well would shrink the fit towards zero (not towards a polynomial) and make it depend on the origin and scale of x.

```
q <- 3; K <- 30
knots30 <- quantile(unique(x), probs = (1:K) / (K + 1))
xs <- x / max(x); ks <- knots30 / max(x)                    # rescale for
↪  conditioning
Xt <- cbind(outer(xs, 0:q, `^`), tpower(xs, ks, q)); Pt <- diag(c(rep(0, q +
↪  1), rep(1, K)))
tp_fit <- function(lambda) {
  A <- crossprod(Xt) + lambda * Pt; f <- drop(Xt %*% solve(A, crossprod(Xt,
↪  y)))
  edf <- sum(diag(solve(A, crossprod(Xt)))); c(edf = edf, GCV = n * sum((y -
↪  f)^2) / (n - edf)^2)
}
cr <- t(sapply(10^seq(-10, 0, by = 0.05), tp_fit))
c(cr[which.min(cr[, "GCV"]), ], edf_huge_lambda = unname(tp_fit(1e8)["edf"]))
```

```
#>             edf             GCV edf_huge_lambda
#>           11.04          560.98            4.00
```

With 30 cubic truncated-power knots at quantiles, GCV selects an edf of about 11 and a GCV value close to the P-spline's (563): the penalty matters, not the particular basis.

15.5.2 Penalized splines are mixed models

A penalty $\beta^\top P \beta$ with λ chosen from the data looks like a Bayesian prior with a variance to be estimated. It is, exactly, a mixed model (Chapter 11): the unpenalised null space becomes the fixed effects and the penalised part the random effects (Robinson, 1991; Verbyla et al., 1999; Ruppert et al., 2003; Wand, 2003).

Theorem 15.7 (Mixed-model representation of a P-spline)**.** *Let G be a $c \times d$ matrix whose columns span the null space of D_d ($D_d G = 0$) and put*

$$X_f = BG, \qquad Z = BD_d^\top (D_d D_d^\top)^{-1}.$$

Then $\beta = G\alpha + D_d^\top (D_d D_d^\top)^{-1} b$ with $b = D_d\beta$, $B\beta = X_f\alpha + Zb$ and $\|D_d\beta\|^2 = \|b\|^2$. The P-spline fit $B\hat\beta$ equals the fit of the linear mixed model

$$y = X_f\alpha + Zb + \varepsilon, \qquad b \sim N(0, \sigma_b^2 I), \quad \varepsilon \sim N(0, \sigma^2 I), \quad \lambda = \frac{\sigma^2}{\sigma_b^2}, \tag{15.9}$$

with α estimated (BLUE) and b predicted (BLUP).

Proof. The $c \times c$ matrix $[G,\ D_d^\top (D_d D_d^\top)^{-1}]$ is invertible: its second block has full column rank $c-d$ and its columns lie in the row space of D_d, which is orthogonal to the null space spanned by G. Hence every β has a unique decomposition $\beta = G\alpha + D_d^\top (D_d D_d^\top)^{-1} b$, and $D_d\beta = D_d G\alpha + D_d D_d^\top (D_d D_d^\top)^{-1} b = b$. Substituting, $\|y - B\beta\|^2 + \lambda \|D_d\beta\|^2 = \|y - X_f\alpha - Zb\|^2 + \lambda\|b\|^2$. For the mixed model (15.9) the BLUE/BLUP $(\hat\alpha, \tilde b)$ are the solution of Henderson's equations, which are the normal equations of $\|y - X_f\alpha - Zb\|^2/\sigma^2 + \|b\|^2/\sigma_b^2$ (Chapter 12). Multiplying by σ^2 gives the criterion with $\lambda = \sigma^2/\sigma_b^2$. □

Three consequences. (i) The smoothing parameter is a ratio of variance components: choosing λ means estimating σ_b^2 and σ^2, which REML does (Chapter 12); this is the "REML" option of `mgcv`. (ii) The Bayesian covariance $\sigma^2 S_\lambda$ above is the covariance of $\tilde b - b$ in the mixed model. (iii) Smooths can be combined with random effects and correlated errors in one model (`gamm()`, Chapter 19). A numerical illustration fits the representation with `nlme::lme()` (one dummy group, `pdIdent` for the independent random effects) and compares with the P-spline and with `mgcv`.

```
library(nlme)
cc <- ncol(B); D2 <- diff(diag(cc), differences = 2)
G <- cbind(1, 1:cc)                                          # null space of
↪  D2: constants and linear
Xf <- B %*% G; Zr <- B %*% t(D2) %*% solve(D2 %*% t(D2))
dd <- data.frame(y = y, g = factor(1))
m_lme <- lme(y ~ Xf - 1, random = list(g = pdIdent(~ Zr - 1)), data = dd,
↪  method = "REML")
s2 <- m_lme$sigma^2; sb2 <- as.numeric(VarCorr(m_lme)[2, 1]); lam_reml <- s2 /
↪  sb2
g_reml <- gam(accel ~ s(times, bs = "ps", k = cc, m = c(2, 2)), data = mcycle,
              knots = list(times = attr(B, "knots")), method = "REML")
c(sigma2 = s2, sigma_b2 = sb2, lambda_REML = lam_reml, edf_P_spline = psfit(B,
↪  y, lam_reml)$edf,
  edf_mgcv_REML = sum(g_reml$edf), max_diff_fit = max(abs(fitted(m_lme) -
   ↪  psfit(B, y, lam_reml)$f)))
```

```
#>        sigma2      sigma_b2   lambda_REML  edf_P_spline edf_mgcv_REML
#>     5.127e+02     1.302e+03     3.939e-01     1.237e+01     1.237e+01
#>  max_diff_fit
#>     2.957e-05
```

The REML estimate $\hat\lambda = \hat\sigma^2/\hat\sigma_b^2 = 0.394$ gives exactly the same curve as the P-spline with that λ (3×10^{-5} apart), and the edf agree with `mgcv`'s REML fit, 12.37. REML here is slightly *less* smooth than GCV (edf = 11.38): they are different estimators of the same quantity and will not coincide.

15.6 In R: the functions used in this chapter

Task	Function
B-spline / natural spline basis	`splines::bs(x, df, degree = 3)`, `splines::ns(x, df)`, in `lm()`
Evaluate B-splines and derivatives	`splines::splineDesign(knots, x, ord = q + 1, derivs = )`

Task	Function
Smoothing spline	`smooth.spline(x, y, df =, lambda =, cv = )`; `$lev`, `$df`, `predict()`
Difference matrix	`diff(diag(c), differences = d)`
P-spline by hand	`solve(crossprod(B) + lambda * crossprod(D), crossprod(B, y))`
P-spline in `mgcv`	`gam(y ~ s(x, bs = "ps", k = c, m = c(q - 1, d)), method = "GCV.Cp" or "REML")`
Mixed-model fit	`nlme::lme()` with `pdIdent`, or `mgcv::gamm()` (Chapter 19)

In `mgcv`'s `m = c(m1, m2)`, `m1 = 2` means cubic B-splines (the *order* minus one) and `m2` the penalty order. The package `JOPS` contains the original code of Eilers and Marx.

15.7 Pitfalls

Warning. *Stating polynomial reproduction without its conditions.* It requires equally spaced knots, $q \geq d-1$ and data of degree $\leq d-1$.

Warning. *Saying $d = 1$ penalises slopes.* It penalises differences between neighbouring *coefficients* (levels); the limit as $\lambda \to \infty$ is a constant.

Warning. *Penalising the polynomial part of a truncated power spline*, or counting the penalty twice (writing $\lambda \int (\lambda f'')^2$).

Warning. *GCV is* $n\,\mathrm{RSS}/(n-\mathrm{edf})^2$, not a sum of residuals divided by $n-\mathrm{edf}$.

Warning. *Comparing `sp` of `mgcv` with a hand-coded* λ, or reading the default `k = 10` as a finding about the data.

Warning. *Simulation-code slips*: `rnorm(n, 0.10)` is mean 0.1 and standard deviation 1; `xl <- min(x) + 0.01` leaves the smallest observation outside the basis domain.

15.8 Summary

- Splines of degree q with K knots form a space of dimension $q+1+K$ with truncated power basis (15.1); regression splines fit them by least squares, but the choice of knots is awkward. B-splines are a local, non-negative, well-conditioned basis of the same space on the interior of the knot range, with extended boundary knots (Cox–de Boor recursion, partition of unity, at most $q+1$ non-zero per x); coefficient differences control derivatives (15.3).
- Natural cubic splines are linear beyond the boundary knots (dimension K). The minimiser of $\sum\{y_i - f(x_i)\}^2 + \lambda \int f''^2$ is a natural cubic spline with knots at the distinct x_i (Reinsch), proved by integration by parts; $S_\lambda = (I + \lambda K)^{-1}$ has two unit eigenvalues (straight lines are not penalised).
- P-splines: rich equally-spaced B-spline basis B plus $\lambda \|D_d \beta\|^2$; $\hat\beta = (B^\top B + \lambda D_d^\top D_d)^{-1} B^\top y$, $S_\lambda = B(\cdots)^{-1} B^\top$, $\mathrm{edf} = \mathrm{tr}\, S_\lambda$ falling from c to d as λ grows; for equally spaced knots and $q \geq d-1$ polynomials of degree $< d$ are reproduced and their moments conserved.
- λ is chosen by GCV/AIC/LOOCV (exact shortcut) or REML; $\hat\sigma^2 = \mathrm{RSS}/(n-\mathrm{edf})$; the number of knots hardly matters if large enough.

- Penalized splines are penalised least squares; the null space of the penalty is the unpenalised part. They are mixed models with $\lambda = \sigma^2/\sigma_b^2$, which links smoothing to Chapters 11 and 19.

15.9 Exercises

Exercise 15.1. A cubic spline in truncated power form. Let $f(x) = 1 + 2x - x^2 + \frac{1}{2}x^3 - \frac{3}{2}(x-1)_+^3$. (a) Show that f, f' and f'' are continuous at the knot $\kappa = 1$ and compute the jump of f'''. (b) Evaluate $f(0), f(1), f(2)$. (c) What is the dimension of the space of cubic splines with knots at 1 and 3? Is f natural?

Solution.

(a) For $x < 1$, f is the cubic $p(x) = 1 + 2x - x^2 + \frac{1}{2}x^3$. For $x > 1$, $f = p(x) - \frac{3}{2}(x-1)^3$. The difference $\frac{3}{2}(x-1)^3$ and its first two derivatives $\frac{9}{2}(x-1)^2$, $9(x-1)$ vanish at $x = 1$, so f, f', f'' are continuous there (with $f'(1) = p'(1) = 2 - 2 + \frac{3}{2} = 1.5$ and $f''(1) = p''(1) = -2 + 3 = 1$). The third derivative is $p''' = 3$ for $x < 1$ and $3 - 9 = -6$ for $x > 1$: it jumps by $-9 = 3!\,\gamma$ with $\gamma = -\frac{3}{2}$ (Theorem 15.1).
(b) $f(0) = 1$; $f(1) = 1 + 2 - 1 + \frac{1}{2} = 2.5$; $f(2) = 1 + 4 - 4 + 4 - \frac{3}{2} = 3.5$.
(c) $q + 1 + K = 4 + 2 = 6$. The function f is not natural: on $(-\infty, 1]$ it is the cubic p with non-zero x^2 and x^3 terms. (A natural spline needs $f'' = f''' = 0$ left of the first knot.)

Exercise 15.2. B-splines and derivatives by hand. With integer knots, the four cubic B-splines that are non-zero on $[3, 4)$ take, for $u = x - 3 \in [0, 1)$, the values $\frac{(1-u)^3}{6}$, $\frac{3u^3-6u^2+4}{6}$, $\frac{-3u^3+3u^2+3u+1}{6}$, $\frac{u^3}{6}$ and the three quadratic ones $\frac{(1-u)^2}{2}$, $\frac{-2u^2+2u+1}{2}$, $\frac{u^2}{2}$. (a) Evaluate the four cubic B-splines at $x = 3.7$ and check that they sum to one. (b) For coefficients $\beta = (2, 5, 1, 4)$ compute $f(3.7)$. (c) Compute $f'(3.7)$ using (15.3) and confirm with R.

Solution.

(a) With $u = 0.7$: $0.3^3/6 = 0.0045$; $(3(0.343) - 6(0.49) + 4)/6 = 0.3482$; $(-1.029 + 1.47 + 2.1 + 1)/6 = 0.5902$; $0.343/6 = 0.0572$. They sum to 1.0001, i.e. 1 up to rounding.
(b) $f(3.7) = 2(0.0045) + 5(0.3482) + 1(0.5902) + 4(0.0572) = 2.569$.
(c) By (15.3) with $\delta = 1$, $f' = \sum_j (\beta_j - \beta_{j-1}) B_{j,2}$: the differences are $(3, -4, 3)$ and the quadratic B-splines at $u = 0.7$ are $(0.045, 0.71, 0.245)$, so $f'(3.7) = 3(0.045) - 4(0.71) + 3(0.245) = 0.135 - 2.84 + 0.735 = -1.97$.

```
tt <- 0:7; th4 <- c(2, 5, 1, 4)
c(f = sum(splineDesign(tt, 3.7, ord = 4) * th4), fprime = sum(splineDesign(tt,
↪  3.7, ord = 4, derivs = 1) * th4))
```

```
#>       f fprime
#>   2.569 -1.970
```

Solution. (Continued.) R returns $f(3.7) = 2.569$ and $f'(3.7) = -1.97$, in agreement with the hand calculation.

Exercise 15.3. Penalising the first derivative. Consider $\sum_i \{y_i - f(x_i)\}^2 + \lambda \int f'(u)^2\,du$ over differentiable f. Adapt the proof of Theorem 15.3 to show that the minimiser is a continuous *piecewise linear* function with knots at the distinct x_i (a linear spline, not a

cubic one), constant outside $[x_1, x_m]$.

Solution. Let s be the piecewise linear interpolant of g at $x_1, \dots, x_m$, constant outside $[x_1, x_m]$ and $h = g - s$. Then $h(x_i) = 0$ and

$$\int_{x_1}^{x_m} s'h' = \sum_{i=1}^{m-1} c_i\{h(x_{i+1}) - h(x_i)\} = 0,$$

because $s' = c_i$ is constant on (x_i, x_{i+1}): no integration by parts is needed. Therefore $\int g'^2 = \int s'^2 + \int h'^2 \geq \int s'^2$ on $[x_1, x_m]$; outside, $s' = 0$ so s costs nothing. The residual sum of squares is the same, so $Q(s) \leq Q(g)$, with equality only if $h' \equiv 0$, i.e. $h \equiv 0$ on $[x_1, x_m]$. The general principle: a penalty $\int f^{(k)2}$ leads to splines of degree $2k-1$ that are *natural* ($k = 2$: cubic).

Exercise 15.4. A P-spline by hand. Take $B = I_3$, $d = 2$, $y = (3, 0, 3)^\top$ and $\lambda = 1$. (a) Write $A = I + D_2^\top D_2$ and solve for $\hat\beta$, given $A^{-1} = \frac{1}{7}\begin{bmatrix} 6 & 2 & -1 \\ 2 & 3 & 2 \\ -1 & 2 & 6 \end{bmatrix}$. (b) Compute the edf. (c) Verify the conservation of moments of order 0 and 1 and explain why moments of order 2 are not conserved. (d) What are the limits of $\hat\beta$ and edf as $\lambda \to \infty$?

Solution.

(a) $D_2 = (1, -2, 1)$, $D_2^\top D_2 = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$, so $A = \begin{bmatrix} 2 & -2 & 1 \\ -2 & 5 & -2 \\ 1 & -2 & 2 \end{bmatrix}$ and

$$\hat\beta = A^{-1}y = \tfrac{1}{7}(18 - 3,\ 6 + 6,\ -3 + 18) = \tfrac{1}{7}(15, 12, 15) = (2.143,\ 1.714,\ 2.143).$$

The curve is pulled from the data $(3, 0, 3)$ towards a straight line.

(b) edf $= \operatorname{tr}(A^{-1}) = \frac{6+3+6}{7} = \frac{15}{7} = 2.143$.

(c) $\sum \hat\beta_j = 6 = \sum y_j$ and $\sum j\hat\beta_j = (15 + 24 + 45)/7 = 12 = \sum j y_j = 3 + 0 + 9$. Both conditions follow from Theorem 15.6(b) with $d = 2$ ($k = 0, 1$); $\sum j^2\hat\beta_j = (15 + 48 + 135)/7 = 28.29 \neq \sum j^2 y_j = 30$, since $k = 2 \geq d$.

(d) As $\lambda \to \infty$, $\hat\beta$ tends to the least-squares line through $(1, 3), (2, 0), (3, 3)$, which is the constant $\bar y = 2$ (slope 0), and edf $\to d = 2$.

Exercise 15.5. The edf is decreasing in λ. Let $M = B^\top B$ be positive definite and $P = D_d^\top D_d$. (a) Show $\text{edf}(\lambda) = \sum_{k=1}^{c}(1 + \lambda\gamma_k)^{-1}$ where $\gamma_k \geq 0$ are the eigenvalues of $M^{-1/2}PM^{-1/2}$. (b) Deduce that edf is strictly decreasing from c to the number of zero γ_k ($= d$). (c) Compute the γ_k for the `mcycle` basis and check the formula at $\lambda = 0.64$.

Solution.

(a) $(M + \lambda P)^{-1}M$ is similar to $M^{1/2}\{(M + \lambda P)^{-1}M\}M^{-1/2} = M^{1/2}(M + \lambda P)^{-1}M^{1/2} = (I + \lambda M^{-1/2}PM^{-1/2})^{-1}$, whose eigenvalues are $(1 + \lambda\gamma_k)^{-1}$; similar matrices have the same trace. (b) $\frac{d}{d\lambda}(1 + \lambda\gamma_k)^{-1} = -\gamma_k(1 + \lambda\gamma_k)^{-2} \leq 0$, strictly negative for $\gamma_k > 0$; the limits are c ($\lambda = 0$) and $\#\{k : \gamma_k = 0\}$, the nullity of P, which is d. (c):

```
M <- crossprod(B); P <- crossprod(diff(diag(cc), differences = 2))
eM <- eigen(M, symmetric = TRUE); Mih <- eM$vectors %*% diag(1 /
↪  sqrt(eM$values)) %*% t(eM$vectors)
gam_k <- eigen(Mih %*% P %*% Mih, symmetric = TRUE, only.values = TRUE)$values
```

```
c(smallest = sort(gam_k)[1:3], edf_formula = sum(1 / (1 + 0.64 * gam_k)),
↪  edf_psfit = psfit(B, y, 0.64)$edf)
```

```
#>   smallest1   smallest2   smallest3 edf_formula   edf_psfit
#>  -1.544e-14   2.677e-14   6.510e-04   1.138e+01   1.138e+01
```

Solution. (Continued.) Two eigenvalues are zero to rounding error (the null space of D_2), the next is 6.5×10^{-4}, and the formula gives the same edf as `psfit()`.

Exercise 15.6. The conditions of Theorem 15.6. Data are exactly quadratic, $y = 1 + 2x + 3x^2$ on 100 equally spaced points in $[0, 1]$. Fit P-splines with $\lambda = 10^3$, B-spline degree $q \in \{1, 2, 3\}$ and penalty order $d \in \{1, 2, 3, 4\}$ and report the maximum absolute error. Explain the pattern with the two conditions.

```
xx <- seq(0, 1, length.out = 100); yq <- 1 + 2 * xx + 3 * xx^2
err <- sapply(1:3, function(bd) sapply(1:4, function(d)
  max(abs(psfit(bbase(xx, bdeg = bd), yq, 1e3, d)$f - yq))))
dimnames(err) <- list(paste0("d=", 1:4), paste0("q=", 1:3)); signif(err, 2)
```

```
#>        q=1     q=2     q=3
#> d=1 2.6000 2.6e+00 2.6e+00
#> d=2 0.2200 2.2e-01 2.2e-01
#> d=3 0.0012 2.4e-12 1.1e-12
#> d=4 0.0013 5.9e-12 9.2e-12
```

Solution. The error is of order 10^{-12} in exactly two situations: when the data degree (2) is at most $d - 1$ (so $d \geq 3$) *and* the B-splines can represent a quadratic ($q \geq 2$). For $q = 1$ and $d = 3$ the error is 0.0012: linear B-splines cannot represent a quadratic, although the penalty would not object (condition (ii), $q \geq d - 1 = 2$, fails). For $d = 1, 2$ the data have degree $2 > d - 1$ and the fit is pulled towards a constant ($d = 1$, error 2.6) or a line ($d = 2$, error 0.22), whatever the B-spline degree. (For $d = 4$, $q = 2$ the error is again 10^{-12}: condition (ii) is sufficient to reproduce *every* polynomial of degree $\leq d - 1$, but a quadratic only needs $\min(q, d - 1) \geq 2$.) Equal spacing (condition (i)) holds by construction; with unequally spaced knots even $d = 2$, $q = 3$ would not reproduce a line exactly.

Exercise 15.7. Conservation of moments. For the `mcycle` P-spline with $\lambda = 0.64$ compute $\sum_i x_i^k (y_i - \hat{f}_i) / \sum_i |x_i^k y_i|$ for $k = 0, \ldots, 3$ and $d = 1, 2, 3, 4$. Which moments are conserved and why does the result match Theorem 15.6(b)?

```
mom <- t(sapply(1:4, function(d) { f <- psfit(B, y, 0.64, d)$f
  sapply(0:3, function(k) sum(x^k * (y - f)) / sum(abs(x^k * y))) }))
dimnames(mom) <- list(paste0("d=", 1:4), paste0("k=", 0:3)); signif(mom, 2)
```

```
#>          k=0      k=1      k=2      k=3
#> d=1 -1.2e-16  1.3e-04  1.4e-03  3.0e-03
#> d=2 -1.2e-16 -1.3e-16  3.2e-05  1.4e-04
#> d=3 -1.4e-16 -1.3e-16 -7.9e-17 -4.5e-06
#> d=4 -2.8e-16 -2.8e-16 -2.1e-16 -1.2e-16
```

Solution. The entries with $k < d$ are zero to rounding error (10^{-16}); those with $k \geq d$ are not (of order 10^{-3} to 10^{-6} relative to $\sum |x_i^k y_i|$; for $d = 4$ there are none, as $k \leq 3$): moments up to order $d - 1$ are conserved for every λ, as the theorem states, because $(x_i^k)_i$ lies in the null space

of the penalty for $k \leq d-1$. Practical meaning: with $d = 2$ the P-spline preserves the mean and the linear trend of the data exactly, whatever the smoothness (a property shared with the smoothing spline, whose null space is the straight lines).

Exercise 15.8. Cross-validation for the P-spline. On `mcycle` ($d = 2$, 23 B-splines) choose $\log_{10}\lambda$ on a grid in $[-3, 3]$ by (i) GCV, (ii) exact LOOCV via $(y_i - \hat{f}_i)/(1 - S_{ii})$, (iii) grouped leave-one-out CV in which *all observations with the same time* are left out and then predicted. Compare and interpret.

```
Pm <- crossprod(diff(diag(cc), differences = 2))
ll <- seq(-3, 3, by = 0.1)
crit_ps <- t(sapply(10^ll, function(l) {
  A <- crossprod(B) + l * Pm; S <- B %*% solve(A, t(B)); r <- y - drop(S %*%
↪  y)
  c(GCV = n * sum(r^2) / (n - sum(diag(S)))^2, LOOCV = mean((r / (1 -
  ↪  diag(S)))^2))
}))
grp <- split(seq_len(n), x)
cross <- lapply(grp, function(i) list(i = i, BtB = crossprod(B[-i, ]), Bty =
↪  crossprod(B[-i, ], y[-i])))
gcv_grp <- sapply(10^ll, function(l) {
  sum(sapply(cross, function(g) { th <- solve(g$BtB + l * Pm, g$Bty);
  ↪  sum((y[g$i] - B[g$i, , drop = FALSE] %*% th)^2) })) / n })
c(GCV = ll[which.min(crit_ps[, "GCV"])], LOOCV = ll[which.min(crit_ps[,
↪  "LOOCV"])], grouped = ll[which.min(gcv_grp)])
```

```
#>     GCV   LOOCV grouped
#>    -0.2    -0.3    -0.3
```

```
c(GCV = min(crit_ps[, "GCV"]), LOOCV = min(crit_ps[, "LOOCV"]), grouped =
↪  min(gcv_grp))
```

```
#>     GCV   LOOCV grouped
#>   563.0   540.7   549.2
```

Solution. GCV and exact LOOCV choose almost the same $\log_{10}\lambda$ (-0.2 and -0.3): the shortcut is exact and GCV replaces S_{ii} by its average. The grouped criterion also selects -0.3, but its minimum value (549.2) is larger than the ordinary LOOCV one (540.7). The reason is that the 133 observations sit at 94 distinct times: with ordinary CV a tied twin of the left-out point stays in the training set and helps to predict it, so LOOCV is slightly optimistic; leaving all replicates out mimics prediction at a *new* time. Here the effect on the selected λ is nil and on the error estimate about 1.6 percent, because the P-spline with 23 coefficients has little freedom to exploit the twins; for a more flexible smoother (small h or span) the optimism is larger.

Exercise 15.9. Do knots matter? Fit P-splines ($d = 2$, cubic) with 10, 20 and 40 segments, choosing λ by GCV in each case. Report $\hat{\lambda}$, edf and GCV, and the maximum difference between each fitted curve and the 20-segment fit.

```
res <- lapply(c(10, 20, 40), function(ns) {
  Bn <- bbase(x, nseg = ns); o <- optimize(function(l) psfit(Bn, y, 10^l)$gcv,
↪  c(-4, 8)); f <- psfit(Bn, y, 10^o$minimum)
  list(row = c(nseg = ns, columns = ncol(Bn), log10_lambda = o$minimum, edf =
  ↪  f$edf, GCV = f$gcv), f = f$f) })
```

```
tab9 <- do.call(rbind, lapply(res, `[[`, "row")); f20 <- res[[2]]$f
cbind(round(tab9, 2), max_diff_vs_20 = round(sapply(res, function(r)
↪  max(abs(r$f - f20))), 2))
```

```
#>      nseg columns log10_lambda   edf   GCV max_diff_vs_20
#> [1,]   10      13        -1.61 10.45 553.5           5.38
#> [2,]   20      23        -0.19 11.38 563.0           0.00
#> [3,]   40      43         0.81 11.93 564.2           1.09
```

Solution. The selected $\log_{10}\lambda$ changes a lot (-1.6, -0.2, 0.8: a basis with more columns needs a larger penalty for the same smoothness), but the edf moves only from 10.5 to 11.9 and the GCV from 553.5 to 564.2. The curves with 20 and 40 segments differ by at most 1.1 g, those with 10 and 20 segments by 5.4 g (the response has a standard deviation of about 48 g): with 10 segments the basis is a little coarse at the sharp dip. The lesson: use enough segments (20–40 here) and let the penalty do the work; the smoothing parameter is meaningful only relative to a given basis.

Exercise 15.10. Smoothing spline against P-spline. Fit `smooth.spline(x, y, df = 12)` and the P-spline with edf exactly 12 (find λ by `uniroot`). Compare the two fits (RSS, maximum difference) and explain why they are so close.

```
lam12 <- 10^uniroot(function(l) psfit(B, y, 10^l)$edf - 12, c(-3, 4))$root
ps12 <- psfit(B, y, lam12); ss12 <- smooth.spline(x, y, df = 12)
fit_ss <- predict(ss12, x)$y
c(lambda_PS = lam12, RSS_PS = ps12$rss, RSS_SS = sum((y - fit_ss)^2), max_diff
↪  = max(abs(fit_ss - ps12$f)),
  sd_y = sd(y))
```

```
#> lambda_PS    RSS_PS    RSS_SS  max_diff      sd_y
#> 4.714e-01 6.208e+04 6.225e+04 3.040e+00 4.832e+01
```

Solution. The two fits have almost the same RSS and differ by at most 3 g (the response has a standard deviation of 48 g). The smoothing spline penalises exactly $\int f''^2$ with knots at all distinct x_i; the P-spline penalises $\|D_2\beta\|^2$, a discretisation of the same integral by (15.3), on 23 B-splines. At the same effective df both are shrinking the same high-frequency components of the data by nearly the same factors $1/(1+\lambda\gamma_k)$; the differences are at the boundaries and in the dip, where the choice of basis matters a little.

Exercise 15.11. The mixed-model form. (a) For $d=1$, find G and Z in Theorem 15.7 and write the fixed effect. (b) Verify numerically, for $d=2$ and $\lambda=0.64$, that solving Henderson's equations

$$\begin{bmatrix} X_f^\top X_f & X_f^\top Z \\ Z^\top X_f & Z^\top Z + \lambda I \end{bmatrix} \begin{bmatrix} \hat\alpha \\ \tilde b \end{bmatrix} = \begin{bmatrix} X_f^\top y \\ Z^\top y \end{bmatrix}$$

reproduces the P-spline fit. (c) A REML fit gives $\hat\sigma^2 = 513$ and $\hat\sigma_b^2 = 1302$. What λ does this imply and which fit does it correspond to?

Solution.

(a) The null space of D_1 is spanned by $G = \mathbf{1}_c$ (constants), so $X_f = B\mathbf{1}_c = \mathbf{1}_n$ by the partition of unity: a single fixed intercept, and $Z = BD_1^\top(D_1D_1^\top)^{-1}$ carries the random effects

$b = D_1\beta$, the differences of the coefficients. (b):

```
D2 <- diff(diag(cc), differences = 2); G2 <- cbind(1, 1:cc)
Xf <- B %*% G2; Zr <- B %*% t(D2) %*% solve(D2 %*% t(D2)); lam <- 0.64
H <- rbind(cbind(crossprod(Xf), crossprod(Xf, Zr)), cbind(crossprod(Zr, Xf),
↪  crossprod(Zr) + lam * diag(cc - 2)))
sol <- solve(H, c(crossprod(Xf, y), crossprod(Zr, y)))
f_mm <- drop(Xf %*% sol[1:2] + Zr %*% sol[-(1:2)])
max(abs(f_mm - psfit(B, y, lam)$f))
```

```
#> [1] 4.221e-12
```

Solution. (Continued.) The difference is at rounding level (10^{-12}): the two computations solve the same normal equations in different coordinates (Theorem 15.7). (c) $\lambda = \hat{\sigma}^2/\hat{\sigma}_b^2 = 513/1302 = 0.394$, the REML smoothing parameter of the chapter, with edf 12.37: a slightly more flexible curve than the GCV choice ($\lambda = 0.64$, edf 11.38).

Exercise 15.12. Critique and communicate. A student writes: "The P-spline for the motorcycle data has 23 coefficients and edf = 11.4. Since 23 parameters for 133 points is too many, the model overfits; and an edf of 11.4 means that the curve is a polynomial of degree 10. The standard error band shows the uncertainty of the acceleration of a new crash dummy." (a) Identify the three errors. (b) Rewrite the paragraph in about 120 words for a non-statistician.

Solution.

(a) (i) The number of coefficients is not the complexity: the penalty shrinks the 23 coefficients to an effective 11.4; a P-spline with 23 coefficients and $\lambda = 0$ would overfit, with $\lambda = 0.64$ it does not. (ii) The edf is a *measure of flexibility* (the trace of the smoother matrix), not the degree of a polynomial; the fit is a piecewise cubic with 20 segments. (iii) A pointwise standard error band for $\hat{f}$ describes the uncertainty of the *estimated mean curve*; the spread of a new observation is much larger because it adds the error variance σ^2 (here $\hat{\sigma} = 22.7$ g) to the variance of $\hat{f}$.

(b) "We described head acceleration after impact by a flexible curve built from 23 building blocks. We did not use all their freedom: a penalty on roughness shrinks them so that the curve has about the flexibility of 11 free parameters, chosen by cross-validation so as to predict new data well. The curve is not a polynomial. The shaded band shows how uncertain our *average* curve is at each time. A single new test would scatter much more widely around that curve, by about plus or minus 45 g. The band assumes the noise is equal throughout, which is not the case here: the noise is largest around the impact."

Exercise 15.13. The basis dimension. Fit `gam(accel ~ s(times, bs = "ps", k = k), method = "GCV.Cp")` for $k = 10, 23, 40$ and compare edf. Use `mgcv::k.check()` for the $k = 10$ fit and say what it indicates.

```
ed <- sapply(c(10, 23, 40), function(k) sum(gam(accel ~ s(times, bs = "ps", k
↪  = k), data = mcycle, method = "GCV.Cp")$edf))
names(ed) <- paste0("k=", c(10, 23, 40)); round(ed, 2)
```

```
#>  k=10  k=23  k=40
#>  9.70 11.37 11.90
```

```
round(k.check(gam(accel ~ s(times, bs = "ps", k = 10), data = mcycle, method =
↪  "GCV.Cp")), 3)
```

```
#>          k'   edf k-index p-value
#> s(times)  9 8.699   0.835   0.013
```

Solution. With $k = 10$ the total edf is 9.7; subtracting the intercept, the smooth has edf 8.7 out of a maximum $k' = 9$ (the basis dimension minus the identifiability constraint): it hugs the ceiling, which means the basis, not the data, limits the complexity. With $k = 23$ and $k = 40$ the total edf is 11.37 and 11.9: stable at about 11.4–11.9, so the data need roughly 11 effective parameters for the smooth. `k.check()` reports $k' = 9$, edf 8.70, a k-index of 0.84 (below 1) and a small p-value (0.013): there is remaining structure in the residuals that a larger basis could capture. The rule: choose k clearly larger than the edf you expect and verify that the edf does not approach k'.

15.10 Further reading

The classical sources are de Boor (1978) for B-splines and Reinsch (1967) and Green and Silverman (1994) for smoothing splines. The P-spline idea is introduced in Eilers and Marx (1996) and developed, with code and many applications, in Eilers and Marx (2021) (Chapters 2–3); the mixed-model view is in Ruppert et al. (2003), Wand (2003), Currie and Durban (2002) and Verbyla et al. (1999), and its role in GAMs in Wood (2017) (Sections 4.2–4.3 and 5.1–5.3 on bases and penalties, Chapter 6 on smoothing-parameter estimation). Generalised cross-validation is due to Craven and Wahba (1978), and Silverman (1985) relates smoothing splines to kernel smoothing. The motorcycle data are discussed in Silverman (1985); Hastie et al. (2009) (Chapter 5) gives a statistical-learning perspective.

Part VI

Additive models

Chapter 16

Generalized additive models

Linear models and GLMs describe the effect of a covariate by one number, a slope. Chapters 14 and 15 taught us how to estimate a *curve* $f(x)$ without choosing its shape in advance. A **generalized additive model** (GAM) puts the two ideas together: each covariate enters the linear predictor through its own smooth function, the response keeps the exponential-family distribution and link of a GLM, and the amount of smoothness of every curve is estimated from the data (Hastie and Tibshirani, 1986, 1990; Wood, 2017).

This chapter introduces the model, shows how it can be fitted one smooth at a time (backfitting) and why that algorithm converges, explains in which sense a GAM *is* a GLM, and then uses the R package `mgcv` on real data. The theory behind `mgcv` (penalties, constraints, smoothing-parameter selection and inference) is the subject of Chapter 17.

```
library(mgcv)
library(MASS)
data(airquality)
aq <- na.omit(airquality[, c("Ozone", "Solar.R", "Wind", "Temp")])
```

16.1 A motivating example: ozone in New York

The data frame `airquality` (package `datasets`) holds daily air-quality measurements in New York from May to September 1973: mean ozone concentration between 13:00 and 15:00 (`Ozone`, parts per billion), solar radiation (`Solar.R`, Langleys), average wind speed (`Wind`, mph) and the daily maximum temperature (`Temp`, degrees Fahrenheit). The data frame has 153 rows. Ozone is missing on 37 days and `Solar.R` on 7; the complete cases for the model below are $n = 111$ days.

```
long <- rbind(data.frame(var = "Solar.R (Langleys)", x = aq$Solar.R, y =
↪  aq$Ozone),
              data.frame(var = "Wind (mph)", x = aq$Wind, y = aq$Ozone),
              data.frame(var = "Temp (F)", x = aq$Temp, y = aq$Ozone))
ggplot(long, aes(x, y)) + geom_point(colour = ie_navy, alpha = .6) +
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, colour =
  ↪  ie_coral) +
  facet_wrap(~var, scales = "free_x") + labs(x = NULL, y = "Ozone (ppb)")
```

A first attempt is the linear model $\text{Ozone}_i = \beta_0 + \beta_1 \text{Solar.R}_i + \beta_2 \text{Wind}_i + \beta_3 \text{Temp}_i + \varepsilon_i$.

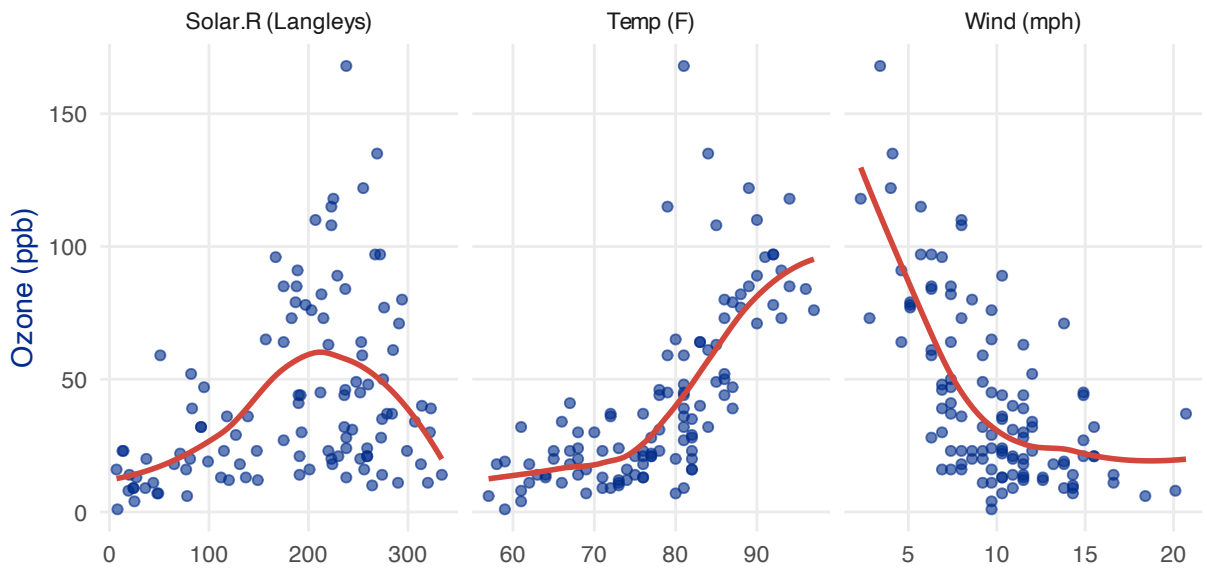


Figure 16.1: Ozone against each covariate in the 111 complete days, with a loess curve. All three relationships bend, and the scatter around the curve grows with the level of ozone.

```
lm_raw <- lm(Ozone ~ Solar.R + Wind + Temp, data = aq)
round(coef(summary(lm_raw)), 4)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept) -64.3421    23.0547  -2.791   0.0062
#> Solar.R       0.0598     0.0232   2.580   0.0112
#> Wind         -3.3336     0.6544  -5.094   0.0000
#> Temp          1.6521     0.2535   6.516   0.0000
```

```
c(R2 = summary(lm_raw)$r.squared, sigma = summary(lm_raw)$sigma)
```

```
#>      R2   sigma
#>  0.6059 21.1808
```

The coefficients are easy to read: one more degree Fahrenheit adds about 1.65 ppb of ozone, one more mph of wind removes about 3.33 ppb, and $R^2 = 0.61$. But the residuals tell a different story.

```
ggplot(data.frame(fit = fitted(lm_raw), res = resid(lm_raw)), aes(fit, res)) +
  geom_hline(yintercept = 0, linetype = 2) + geom_point(colour = ie_navy,
  ↪  alpha = .7) +
  geom_smooth(method = "loess", formula = y ~ x, se = FALSE, colour =
  ↪  ie_coral) +
  labs(x = "Fitted ozone (ppb)", y = "Residual (ppb)")
```

Two assumptions fail at once. The U-shape says that the *mean* is not linear in the covariates, the funnel that the *variance* increases with the mean. A log transformation of the response takes care of the funnel, a topic we know from Chapter 6. Does it also take care of the curvature?

```
lm_log <- lm(log(Ozone) ~ Solar.R + Wind + Temp, data = aq)
rl <- rbind(data.frame(var = "Solar.R", x = aq$Solar.R, r = resid(lm_log)),
            data.frame(var = "Wind", x = aq$Wind, r = resid(lm_log)),
            data.frame(var = "Temp", x = aq$Temp, r = resid(lm_log)))
```

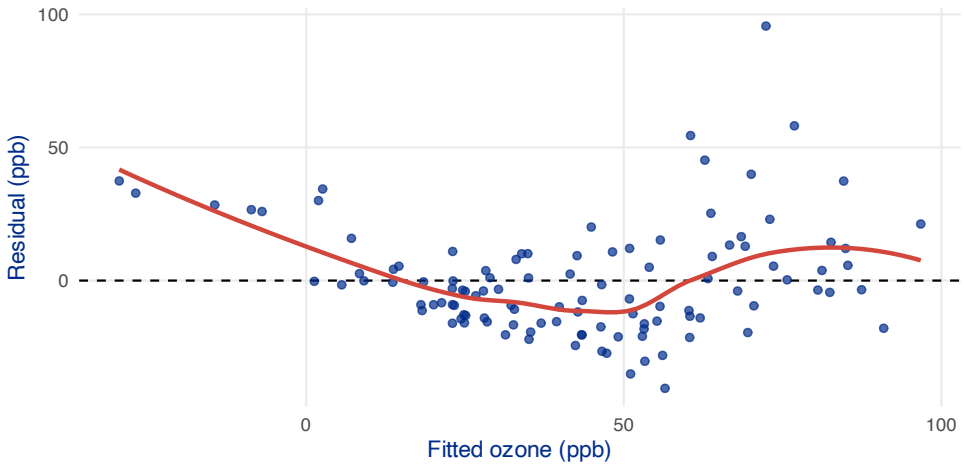


Figure 16.2: Residuals of the linear model for Ozone. The U-shape shows that the mean is not linear in the covariates; the funnel shows that the variance grows with the mean.

```
ggplot(rl, aes(x, r)) + geom_hline(yintercept = 0, linetype = 2) +
  geom_point(colour = ie_navy, alpha = .6) +
  geom_smooth(method = "loess", formula = y ~ x, colour = ie_coral, fill =
  ↪  "grey80") +
  facet_wrap(~var, scales = "free_x") + labs(x = NULL, y = "Residual, log
  ↪  scale")
```

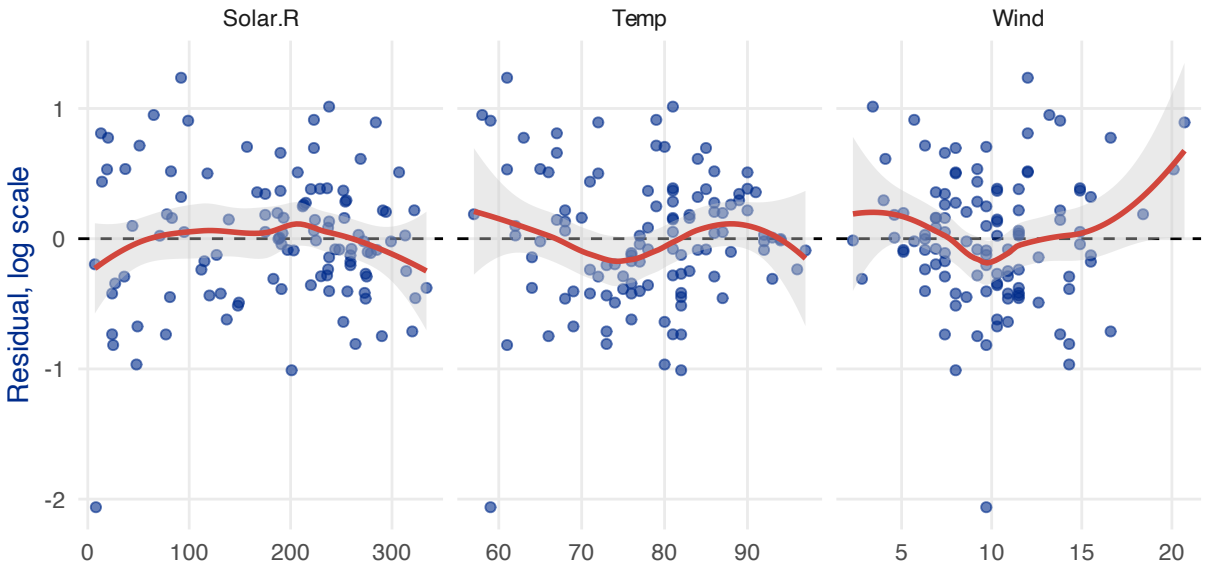


Figure 16.3: Residuals of the linear model for log(Ozone) against each covariate. The funnel is gone, but there is a hump in Solar.R and a bend in Wind: straight lines are too rigid.

It does not. We want a model that

1. keeps one effect per covariate, to be read and plotted one at a time (*additivity*);
2. lets every effect be a smooth function learned from the data;
3. lets the data decide how wiggly each function is;
4. keeps the likelihood machinery of GLMs: families, links, deviance, AIC.

That is the generalized additive model.

16.2 The additive model

Definition 16.1 (Additive model)**.** Let y_i, $i = 1, \dots, n$, be a response and $x_{i1}, \dots, x_{ip}$ the values of p covariates. The **additive model** is

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2), \tag{16.1}$$

where β_0 is an intercept and each f_j is an unknown **smooth function** of one covariate.

The linear model is the special case $f_j(x) = \beta_j x$. The word *additive* says that the effects of the covariates add on the scale of the linear predictor: there are no interactions between covariates unless we put them in on purpose (tensor-product smooths, Chapter 18). The function f_j is a **partial effect**: it shows how the mean changes with x_j when the other covariates are held fixed. This is why additive models remain interpretable even though the f_j are not parametric: every f_j can be drawn on its own.

16.2.1 Identifiability

The decomposition in (16.1) is not unique.

Proposition 16.1 (Non-identifiability of the additive predictor)**.** *For any constants* $c_1, \dots, c_p$,

$$\beta_0 + \sum_{j=1}^{p} f_j(x_j) = \Big(\beta_0 + \sum_{j=1}^{p} c_j\Big) + \sum_{j=1}^{p} \{f_j(x_j) - c_j\},$$

so $(\beta_0, f_1, \dots, f_p)$ *and* $(\beta_0 + \sum_j c_j,\ f_1 - c_1, \dots, f_p - c_p)$ *define the same model. The model is identifiable once we impose*

$$\sum_{i=1}^{n} f_j(x_{ij}) = 0, \qquad j = 1, \dots, p. \tag{16.2}$$

Proof. The identity is obvious. Conversely, suppose two parameter vectors satisfying (16.2) give the same predictor values at the observed points. Their difference is $\delta_0 + \sum_j g_j(x_{ij}) = 0$ for all i, with $\sum_i g_j(x_{ij}) = 0$. Summing over i gives $n\delta_0 = 0$, so $\delta_0 = 0$. If the covariates vary independently enough that $\sum_j g_j(x_{ij}) = 0$ for all i forces each $g_j \equiv 0$ (no exact functional relationship among the covariates, see "concurvity" in Chapter 18), the two parametrisations are identical. □

With (16.2) the intercept has a clear meaning: in the Gaussian additive model $\hat\beta_0 = \bar y$ (Theorem 16.1 below). The constraint is the *sample* version of $\mathrm{E}\{f_j(X_j)\} = 0$. Software imposes it by reparametrising the basis (Section 17.2). One consequence we shall use repeatedly: a centred partial effect is **not** a prediction. To predict you must add $\hat\beta_0$ and the other terms.

16.2.2 Smooth terms as basis expansions

Each smooth is written as a linear combination of K_j known basis functions, as in Chapter 15:

$$f_j(x) = \sum_{k=1}^{K_j} \beta_{jk}\, b_{jk}(x) \quad \Longrightarrow \quad \big(f_j(x_{1j}), \dots, f_j(x_{nj})\big)^\top = B_j \boldsymbol{\beta}_j,$$

where B_j is the $n \times K_j$ matrix with entries $b_{jk}(x_{ij})$ (B-splines, thin-plate splines, …). The additive model is now a *linear* model in the stacked coefficient vector, $\boldsymbol{\eta} = \beta_0 \mathbf{1} + B_1\boldsymbol{\beta}_1 + \cdots + B_p\boldsymbol{\beta}_p$,

but with so many coefficients that we must regularise. Each smooth gets a roughness penalty $\boldsymbol{\beta}_j^\top S_j \boldsymbol{\beta}_j$ measuring its wiggliness (for example $\int f_j''(x)^2 dx$, or $\|D_2 \boldsymbol{\beta}_j\|^2$ for P-splines) and a smoothing parameter $\lambda_j \geq 0$. We estimate the coefficients by **penalized least squares**,

$$Q(\beta_0, \boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_p) = \Big\| \boldsymbol{y} - \beta_0 \mathbf{1} - \sum_{j=1}^{p} B_j \boldsymbol{\beta}_j \Big\|^2 + \sum_{j=1}^{p} \lambda_j \, \boldsymbol{\beta}_j^\top S_j \boldsymbol{\beta}_j. \tag{16.3}$$

Remark. Count λ_j *once*: it multiplies the quadratic form $\boldsymbol{\beta}_j^\top S_j \boldsymbol{\beta}_j$ and nothing else. Writing $\lambda \int (\lambda f'')^2$, or a penalty $\lambda_j J(f_j)$ in which J already contains λ_j, is a common mistake in lecture notes. As $\lambda_j \to \infty$ the function f_j is forced into the null space of the penalty, a straight line for a second-order penalty; at $\lambda_j = 0$ it is an unpenalized regression spline. How the λ_j are chosen is the topic of Section 17.5.

16.3 Backfitting

Before software such as `mgcv`, additive models were fitted by an iterative method that only needs a smoother for *one* covariate at a time. It remains instructive: it shows what the equations of an additive model mean, it explains why strongly related covariates are a problem, and it is still used for very large problems and for non-spline smoothers.

16.3.1 Derivation

Write $\boldsymbol{f}_k = B_k \boldsymbol{\beta}_k \in \mathbb{R}^n$ for the vector of fitted values of the k-th smooth.

Theorem 16.1 (Normal equations of the penalized additive model)**.** *Let $\hat\beta_0, \hat{\boldsymbol{\beta}}_1, \ldots, \hat{\boldsymbol{\beta}}_p$ minimise Q in (16.3) and suppose the columns of every B_j sum to zero, $\mathbf{1}^\top B_j = \mathbf{0}^\top$ (this is the centring constraint (16.2)). Then*

(a) $\hat\beta_0 = \bar y$;

(b) for each j, with the partial residual $\boldsymbol{r}_j = \boldsymbol{y} - \hat\beta_0 \mathbf{1} - \sum_{k \neq j} \hat{\boldsymbol{f}}_k$,

$$\hat{\boldsymbol{\beta}}_j = (B_j^\top B_j + \lambda_j S_j)^{-1} B_j^\top \boldsymbol{r}_j, \qquad \hat{\boldsymbol{f}}_j = A_j \boldsymbol{r}_j, \qquad A_j = B_j (B_j^\top B_j + \lambda_j S_j)^{-1} B_j^\top.$$

Proof.

(a) $\partial Q / \partial \beta_0 = -2 \mathbf{1}^\top (\boldsymbol{y} - \beta_0 \mathbf{1} - \sum_k \boldsymbol{f}_k) = 0$ gives $n\beta_0 = \sum_i y_i - \sum_k \mathbf{1}^\top \boldsymbol{f}_k$. Since $\mathbf{1}^\top \boldsymbol{f}_k = \mathbf{1}^\top B_k \boldsymbol{\beta}_k = 0$, we get $\hat\beta_0 = \bar y$.

(b) Differentiating with respect to $\boldsymbol{\beta}_j$, the penalties of the other smooths do not matter:

$$\frac{\partial Q}{\partial \boldsymbol{\beta}_j} = -2 B_j^\top (\boldsymbol{r}_j - B_j \boldsymbol{\beta}_j) + 2 \lambda_j S_j \boldsymbol{\beta}_j = \mathbf{0} \iff (B_j^\top B_j + \lambda_j S_j) \boldsymbol{\beta}_j = B_j^\top \boldsymbol{r}_j.$$

The matrix on the left is positive definite as long as no nonzero vector lies in the null spaces of both B_j and S_j (true for the bases used here); multiplying the solution by B_j gives $\hat{\boldsymbol{f}}_j = A_j \boldsymbol{r}_j$.

□

The matrix A_j is the **smoother matrix** of Chapters 14 and 15: a linear smoother applied to the partial residuals, and its trace $\operatorname{tr}(A_j)$ is the effective degrees of freedom of the j-th smooth when it is fitted alone. Collecting the p equations,

$$\hat{\boldsymbol{f}}_j = A_j \Big(\boldsymbol{y} - \hat\beta_0 \mathbf{1} - \sum_{k \neq j} \hat{\boldsymbol{f}}_k \Big), \qquad j = 1, \ldots, p. \tag{16.4}$$

These are the *normal equations* of the additive model. The statement of the theorem is a statement about equations, not about an algorithm; each $\hat{\boldsymbol{f}}_j$ depends on all the others, so we solve them iteratively. Replace each unknown on the right-hand side by its current value, update one $\hat{\boldsymbol{f}}_j$ at a time and always use the newest values (the Gauss–Seidel method). Because a constant could drift between the intercept and the $\boldsymbol{f}_j$ whenever a smoother can reproduce constants ($A_j\mathbf{1} = \mathbf{1}$), we **centre** each update.

Backfitting algorithm (Gaussian additive model, fixed smoothers $A_1, \dots, A_p$).

1. Set $\hat{\beta}_0 = \bar{y}$ and $\hat{\boldsymbol{f}}_j = \mathbf{0}$, $j = 1, \dots, p$.
2. Repeat until $\max_j \|\hat{\boldsymbol{f}}_j - \hat{\boldsymbol{f}}_j^{\text{old}}\|$ is negligible: for $j = 1, \dots, p$
   a. compute the partial residual $\boldsymbol{r}_j = \boldsymbol{y} - \hat{\beta}_0\mathbf{1} - \sum_{k \neq j} \hat{\boldsymbol{f}}_k$ ($\hat{\beta}_0$ included);
   b. smooth: $\hat{\boldsymbol{f}}_j \leftarrow A_j \boldsymbol{r}_j$;
   c. centre: $\hat{\boldsymbol{f}}_j \leftarrow \hat{\boldsymbol{f}}_j - \bar{f}_j\mathbf{1}$ with $\bar{f}_j = n^{-1}\sum_i \hat{f}_j(x_{ij})$.

Any linear smoother can play the role of A_j: loess, a smoothing spline, a kernel smoother. The same loop can be used for GLMs: wrap it in the IRLS loop of Chapter 9 and backfit the *working response* z_i with the *working weights* w_i (this is called **local scoring**, Hastie and Tibshirani, 1990). `mgcv` does not use backfitting but solves the whole penalized problem at once (Chapter 17).

16.3.2 Backfitting in R

We backfit log(Ozone) on the three covariates with smoothing splines with 3 degrees of freedom each, as a stand-alone loop.

```
y  <- log(aq$Ozone); xs <- aq[, c("Solar.R", "Wind", "Temp")]
n  <- length(y); beta0 <- mean(y)
f  <- matrix(0, n, 3, dimnames = list(NULL, names(xs)))
for (it in 1:100) {
  f_old <- f
  for (j in 1:3) {
    r  <- y - beta0 - rowSums(f[, -j, drop = FALSE])      # partial residual
    fj <- predict(smooth.spline(xs[[j]], r, df = 3), xs[[j]])$y
    f[, j] <- fj - mean(fj)                               # centre
  }
  if (max(abs(f - f_old)) < 1e-8) break
}
c(sweeps = it, beta0 = beta0)
```

```
#> sweeps  beta0
#> 15.000  3.416
```

The loop stops after 15 sweeps. Note that $\hat{\beta}_0 = \bar{y} = 3.4159$ appears in every partial residual, and that each component has mean zero by construction. A frequent error is to leave β_0 out of the partial residuals: the smooths then absorb the constant, and the loop does not solve (16.4).

To see that backfitting solves exactly the same problem as `gam()`, take the smoother matrices A_j from an `mgcv` fit (same basis, same λ_j) and run the loop with them.

```
g  <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp), data = aq, method =
↪  "REML")
X  <- model.matrix(g)
A  <- lapply(1:3, function(j) {
```

```
  sm <- g$smooth[[j]]; ii <- sm$first.para:sm$last.para; Xj <- X[, ii]
  Xj %*% solve(crossprod(Xj) + g$sp[j] * sm$S[[1]], t(Xj))     # A_j = B_j
↪   (B_j'B_j + lambda_j S_j)^{-1} B_j'
})
f2 <- matrix(0, n, 3); tm <- predict(g, type = "terms")
err <- numeric(25)
for (it in 1:25) {
  for (j in 1:3) f2[, j] <- A[[j]] %*% (y - beta0 - rowSums(f2[, -j, drop =
↪   FALSE]))
  err[it] <- max(abs(f2 - tm))
}
signif(err[c(1, 5, 10, 15, 20, 25)], 3)
```

```
#> [1] 5.60e-01 1.07e-02 8.02e-06 1.63e-08 2.64e-11 1.61e-14
```

The maximal distance to the `gam()` fit falls geometrically, by a factor of about 0.28 per sweep, and reaches 10^{-14} after 25 sweeps. (The columns of `X` are already centred by `mgcv`, which is why no explicit centring step is needed here.) The two approaches differ only in how they solve the same equations.

```
d <- do.call(rbind, lapply(1:3, function(j) rbind(
  data.frame(var = names(xs)[j], x = xs[[j]], fit = tm[, j], how = "gam,
↪   REML"),
  data.frame(var = names(xs)[j], x = xs[[j]], fit = f[, j], how =
↪   "backfitting, 3 df"))))
d <- d[order(d$how, d$var, d$x), ]
ggplot(d, aes(x, fit, colour = how, linetype = how)) + geom_line(linewidth =
↪   1) +
  facet_wrap(~var, scales = "free_x") + scale_colour_manual(values =
↪   c(ie_navy, ie_coral)) +
  labs(x = NULL, y = "centred effect", colour = NULL, linetype = NULL)
```

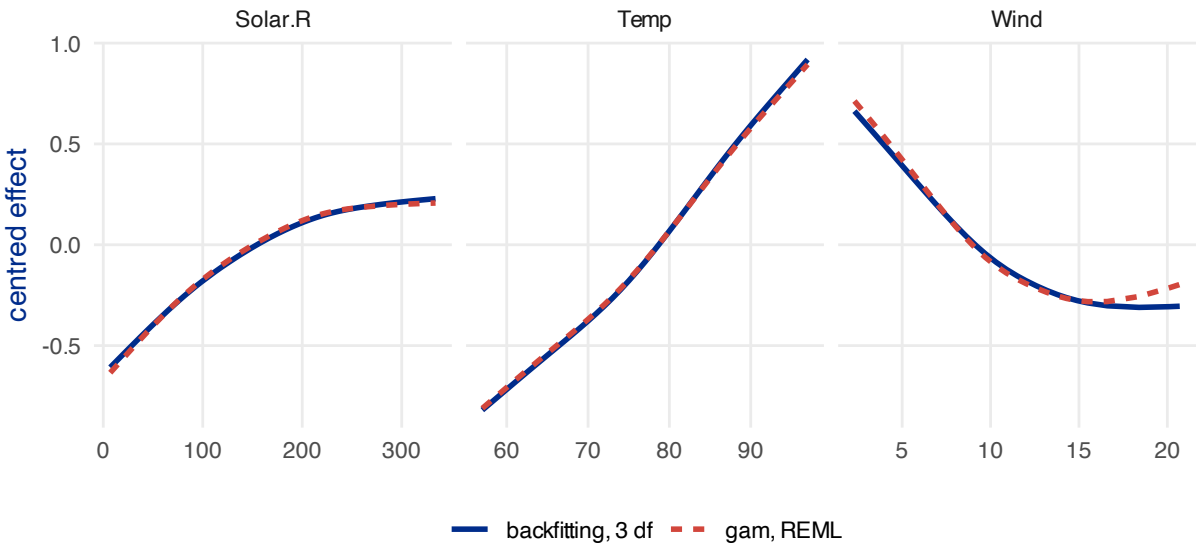


Figure 16.4: Centred effects on the log scale: gam() with REML smoothing (solid) and the stand-alone backfitting loop with 3-df smoothing splines (dashed). The shapes agree; the small differences come from different amounts of smoothing.

16.3.3 Does backfitting converge?

Backfitting is block-wise coordinate descent on the convex function Q, so we expect it to converge. Here is a complete argument, including what controls the speed.

Write all unknown coefficients as one vector $\boldsymbol{\gamma} = (\beta_0, \boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_p^\top)^\top$, $X = [\mathbf{1}, B_1, \dots, B_p]$ and $S_\lambda = \operatorname{blockdiag}(0, \lambda_1 S_1, \dots, \lambda_p S_p)$. Then $Q(\boldsymbol{\gamma}) = \|\boldsymbol{y} - X\boldsymbol{\gamma}\|^2 + \boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}$ and its gradient vanishes at the unique solution of $H\boldsymbol{\gamma} = X^\top \boldsymbol{y}$, $H = X^\top X + S_\lambda$. Partition H into blocks H_{jk} conforming to the $p+1$ coefficient blocks.

Theorem 16.2 (Convergence of backfitting)**.** *Suppose $H = X^\top X + S_\lambda$ is positive definite. Then, from any starting value, a sweep of exact block updates (one block at a time, the newest values of the others) produces a sequence $\boldsymbol{\gamma}^{(t)}$ such that*

(a) $Q(\boldsymbol{\gamma}^{(t)})$ is non-increasing in every block update;

*(b) $\boldsymbol{\gamma}^{(t)} \to \hat{\boldsymbol{\gamma}}$, the unique minimiser, **geometrically**: there is $c < 1$ with $Q(\boldsymbol{\gamma}^{(t+1)}) - Q(\hat{\boldsymbol{\gamma}}) \le c\,\{Q(\boldsymbol{\gamma}^{(t)}) - Q(\hat{\boldsymbol{\gamma}})\}$.*

Proof. Let $\boldsymbol{e} = \boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}}$. Because Q is quadratic with Hessian $2H$ and zero gradient at $\hat{\boldsymbol{\gamma}}$, $Q(\boldsymbol{\gamma}) - Q(\hat{\boldsymbol{\gamma}}) = \boldsymbol{e}^\top H \boldsymbol{e} =: E(\boldsymbol{e})$.

(a) Updating block j means minimising over $\boldsymbol{e}_j$ with the other blocks fixed: $\boldsymbol{e}_j^{\text{new}} = -H_{jj}^{-1} \sum_{k \neq j} H_{jk} \boldsymbol{e}_k$. Completing the square in $\boldsymbol{e}_j$,

$$E(\boldsymbol{e}) = E(\boldsymbol{e}^{\text{new}}) + (\boldsymbol{e}_j - \boldsymbol{e}_j^{\text{new}})^\top H_{jj} (\boldsymbol{e}_j - \boldsymbol{e}_j^{\text{new}}),$$

so E decreases by the non-negative quantity $\delta_j = (\boldsymbol{e}_j - \boldsymbol{e}_j^{\text{new}})^\top H_{jj} (\boldsymbol{e}_j - \boldsymbol{e}_j^{\text{new}})$ (H_{jj} is positive definite as a diagonal block of H).

(b) A full sweep is a *linear* map $\boldsymbol{e} \mapsto T\boldsymbol{e}$, and by (a) $E(T\boldsymbol{e}) \le E(\boldsymbol{e})$. Equality holds only if all block decrements δ_j vanish, i.e. if every block update leaves $\boldsymbol{e}$ unchanged: $H_{jj}\boldsymbol{e}_j + \sum_{k \neq j} H_{jk}\boldsymbol{e}_k = \mathbf{0}$ for all j, i.e. $H\boldsymbol{e} = \mathbf{0}$, hence $\boldsymbol{e} = \mathbf{0}$. So $E(T\boldsymbol{e}) < E(\boldsymbol{e})$ for every $\boldsymbol{e} \neq \mathbf{0}$. The continuous function $\boldsymbol{e} \mapsto E(T\boldsymbol{e})/E(\boldsymbol{e})$ is homogeneous of degree zero and so attains its maximum c on the compact set $\{E(\boldsymbol{e}) = 1\}$; $c < 1$ by what we just showed, and $E(T^t \boldsymbol{e}) \le c^t E(\boldsymbol{e})$.

□

The theorem says *that* backfitting converges. The constant c says *how fast*, and it is governed by how much the blocks overlap. The simplest case can be computed exactly.

Proposition 16.2 (Rate of backfitting for two projection smoothers)**.** *Let $p = 2$ and let the smoothers be the least-squares projections on the centred covariates, $A_j = \boldsymbol{x}_j (\boldsymbol{x}_j^\top \boldsymbol{x}_j)^{-1} \boldsymbol{x}_j^\top$, $\boldsymbol{x}_j^\top \mathbf{1} = 0$. Write $r = \operatorname{cor}(\boldsymbol{x}_1, \boldsymbol{x}_2)$. Then the error of the coefficient of the* first *smooth shrinks by the factor r^2 in every sweep, and backfitting needs about $\log(\varepsilon)/\log(r^2)$ sweeps to reduce the error by a factor ε.*

Proof. Write $\boldsymbol{f}_1 = a\boldsymbol{x}_1$, $\boldsymbol{f}_2 = b\boldsymbol{x}_2$. The updates are $a \leftarrow (\boldsymbol{x}_1^\top \boldsymbol{y} - b\,\boldsymbol{x}_1^\top \boldsymbol{x}_2)/\boldsymbol{x}_1^\top \boldsymbol{x}_1$ and $b \leftarrow (\boldsymbol{x}_2^\top \boldsymbol{y} - a\,\boldsymbol{x}_1^\top \boldsymbol{x}_2)/\boldsymbol{x}_2^\top \boldsymbol{x}_2$ (the intercept does not interfere because $\boldsymbol{x}_j^\top \mathbf{1} = 0$). The fixed point (a^*, b^*) solves the least-squares normal equations. Let e_a, e_b be the errors. Subtracting the fixed-point equations gives $e_a^{(t+1)} = -c_{12} e_b^{(t)}$ and $e_b^{(t+1)} = -c_{21} e_a^{(t+1)}$ with $c_{12} = \boldsymbol{x}_1^\top \boldsymbol{x}_2 / \boldsymbol{x}_1^\top \boldsymbol{x}_1$, $c_{21} = \boldsymbol{x}_1^\top \boldsymbol{x}_2 / \boldsymbol{x}_2^\top \boldsymbol{x}_2$. Hence $e_a^{(t+2)} = c_{12} c_{21}\, e_a^{(t+1)}$ and $c_{12} c_{21} = (\boldsymbol{x}_1^\top \boldsymbol{x}_2)^2 / (\boldsymbol{x}_1^\top \boldsymbol{x}_1\, \boldsymbol{x}_2^\top \boldsymbol{x}_2) = r^2$. □

For general smoothers A_1, A_2 the factor r^2 is replaced by the largest squared canonical correlation between the two smoothing spaces, which is the *concurvity* of the two terms (Buja et al., 1989). With $r = 0.5$ it takes $\lceil 6/(-2\log_{10} 0.5)\rceil = 10$ sweeps to gain six digits; with $r = 0.95$ it takes 135.

Example 16.1 (Backfitting slows down when covariates are related)**.** We simulate $y = \sin(2\pi x_1) + x_2 + \varepsilon$, $n = 200$, and make x_2 more and more related to x_1: $x_2 = r\,x_1 + \sqrt{1-r^2}\,u$ with $x_1, u \sim U(0,1)$. Both smooths use fixed, small penalties. We count the sweeps needed to reduce the maximal change of the $\hat{\boldsymbol{f}}_j$ below 10^{-8}.

```
set.seed(16)
n_s <- 200
sweeps <- function(r) {
  x1 <- runif(n_s); x2 <- r * x1 + sqrt(1 - r^2) * runif(n_s)
  yy <- sin(2 * pi * x1) + x2 + rnorm(n_s, sd = .3)
  gg <- gam(yy ~ s(x1) + s(x2), sp = c(.01, .01)); Xg <- model.matrix(gg)
  Ag <- lapply(1:2, function(j) {
    sm <- gg$smooth[[j]]; ii <- sm$first.para:sm$last.para; Xj <- Xg[, ii]
    Xj %*% solve(crossprod(Xj) + gg$full.sp[j] * sm$S[[1]], t(Xj))
  })
  ff <- matrix(0, n_s, 2)
  for (it in 1:5000) {
    old <- ff
    for (j in 1:2) ff[, j] <- Ag[[j]] %*% (yy - mean(yy) - rowSums(ff[, -j,
    ↪  drop = FALSE]))
    if (max(abs(ff - old)) < 1e-8) break
  }
  c(cor = cor(x1, x2), sweeps = it,
    concurvity = concurvity(gg, full = FALSE)$worst[2, 3])
}
res <- t(sapply(c(0, .5, .9, .99), sweeps))
data.frame(r = c(0, .5, .9, .99), round(res, 3))
```

```
#>      r   cor sweeps concurvity
#> 1 0.00 0.084      8      0.147
#> 2 0.50 0.499     20      0.408
#> 3 0.90 0.904     97      0.852
#> 4 0.99 0.988    706      0.983
```

The number of sweeps explodes as the correlation approaches one: the two smooths can trade off one another almost without changing the fit, and block coordinate descent crawls along that nearly flat valley of Q. The column `concurvity` is the `mgcv` measure of how well one smooth can be approximated by the other (Chapter 18); it rises together with the iteration counts.

16.3.4 Limits of backfitting

- The smoothing parameters (or degrees of freedom) must be fixed in advance; choosing them inside the loop is awkward and was done by ad hoc rules.
- It is slow under concurvity (Example 16.1).
- There is no direct covariance matrix for $\hat{\boldsymbol{\gamma}}$, so no easy standard errors or confidence bands.
- The modern alternative writes all smooths as *one* penalized regression, $\boldsymbol{\eta} = X\boldsymbol{\gamma}$ with penalty $\sum_j \lambda_j \boldsymbol{\gamma}^\top S_j \boldsymbol{\gamma}$, fits it jointly, and chooses the λ_j by REML or GCV. This is what

`mgcv` does.

16.4 GAMs as GLMs with smooth terms

The step from the Gaussian additive model to a GAM is exactly the step from the linear model to a GLM (Chapter 8).

Definition 16.2 (Generalized additive model)**.** A **generalized additive model** has a response y_i with an exponential-family distribution with mean $\mu_i = \mathrm{E}(y_i)$ and variance $\phi\, V(\mu_i)/\omega_i$ (Chapter 2), and

$$g(\mu_i) = \eta_i = \boldsymbol{x}_i^{*\top}\boldsymbol{\theta} + \sum_{j=1}^{p} f_j(x_{ij}), \tag{16.5}$$

where g is a link function, $\boldsymbol{x}_i^{*\top}\boldsymbol{\theta}$ collects the ordinary **parametric terms** (intercept, factors, linear effects) and each f_j is a smooth function with the identifiability constraint (16.2). Some smooths may be functions of several covariates (tensor products, Chapter 18), or of the form $f(x)$ with a random-effect penalty (Chapter 19).

Everything that was said about GLMs carries over: the random component, the link, the variance function, the deviance. What changes is the linear predictor and, with it, the estimation method and the meaning of "degrees of freedom".

	GLM (Chapters 8, 9)	GAM (this chapter)
Random part	$y_i \sim \mathrm{EF}(\mu_i, \phi)$	same
Link	$g(\mu_i) = \eta_i$	same
Predictor	$\eta_i = \boldsymbol{x}_i^{\top}\boldsymbol{\beta}$	$\eta_i = \boldsymbol{x}_i^{*\top}\boldsymbol{\theta} + \sum_j f_j(x_{ij})$
Estimation	maximum likelihood by IRLS	penalized maximum likelihood by PIRLS
Complexity	number of parameters p	$\sum_j \mathrm{edf}_j$ + parametric
Comparison	deviance, LR test, AIC	approximate tests, AIC

Given the basis expansions, a GAM is a GLM with a large model matrix $X = [X^*, B_1, \ldots, B_p]$ and a quadratic penalty. The **penalized log-likelihood** is

$$\ell_p(\boldsymbol{\gamma}) = \ell(\boldsymbol{\gamma}) - \tfrac{1}{2}\sum_{j=1}^{p} \lambda_j\, \boldsymbol{\gamma}^{\top} S_j \boldsymbol{\gamma}, \tag{16.6}$$

where S_j is padded with zeros to the full length of $\boldsymbol{\gamma}$ (equivalently, the penalty is $\lambda_j \boldsymbol{\beta}_j^{\top} S_j \boldsymbol{\beta}_j$ for the sub-vector $\boldsymbol{\beta}_j$ of smooth j) and ℓ is the GLM log-likelihood. **Notation change:** from here on $S_\lambda = \sum_j \lambda_j S_j$ is the *total penalty*; in Chapters 14 and 15 the symbol S_λ denoted the smoother matrix of a single smooth, a role now played by the influence matrix A (Appendix A). For a fixed $\boldsymbol{\lambda}$ it is maximised by an IRLS iteration in which every weighted least-squares step is replaced by a *penalized* weighted least-squares step (**PIRLS**):

$$\boldsymbol{\gamma}^{\text{new}} = (X^{\top} W X + S_\lambda)^{-1} X^{\top} W \boldsymbol{z},$$

with W and $\boldsymbol{z}$ the IRLS weights and working response of Chapter 9. We derive this in Section 17.3. The smoothing parameters $\boldsymbol{\lambda}$ are chosen in an outer loop by GCV or REML (Section 17.5).

16.4.1 Advantages and costs

Advantages. Shapes are learned instead of guessed (no hunt for the right x^2, $\log x$ or cut-point); the amount of smoothness is chosen from the data; the model stays additive and can be plotted term by term; all GLM families and links are available; a linear effect is recovered when it is the truth (edf ≈ 1). *Costs.* There are more choices (basis, dimension, penalty, selection criterion); inference is approximate and conditional on the estimated smoothing parameters; there is no single slope to report, effects are curves; extrapolation beyond the data is unreliable; correlated covariates make the individual f_j unstable; computation is heavier.

16.5 First analyses with `mgcv`

16.5.1 A Gaussian GAM for log-ozone

The function `gam()` has the same interface as `glm()`. A term `s(x)` declares a penalized smooth. The default is a thin-plate regression spline (Wood, 2003) with basis dimension $k = 10$, and `method = "REML"` selects the smoothing parameters by restricted maximum likelihood.

```
m_log <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp), data = aq, method =
↪  "REML")
summary(m_log)
```

```
#>
#> Family: gaussian
#> Link function: identity
#>
#> Formula:
#> log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp)
#>
#> Parametric coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   3.4159     0.0459    74.4   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Approximate significance of smooth terms:
#>             edf Ref.df     F p-value
#> s(Solar.R) 2.16   2.72  9.68 3.4e-05 ***
#> s(Wind)    2.46   3.12  6.47 0.00042 ***
#> s(Temp)    1.95   2.45 21.11 < 2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> R-sq.(adj) =  0.688   Deviance explained = 70.7%
#> -REML = 86.024  Scale est. = 0.23375   n = 111
```

Warning. The default of `gam()` is `method = "GCV.Cp"`, not REML. In this course we always write `method = "REML"`; Chapter 17 explains why.

16.5.2 Reading `summary()`

The output has three parts.

Parametric coefficients. A table exactly as in `lm()`. The intercept is 3.416 and equals $\overline{\log y} = 3.416$ because every smooth is centred and the model is Gaussian with the identity link (Theorem 16.1). It is the mean *log*-ozone, not the mean ozone.

Approximate significance of smooth terms. One row per smooth.

- `edf` is the **effective degrees of freedom** of the smooth. A value of 1 means a straight line (the penalty has pushed the smooth into its null space), about 2 to 3 a gentle curve, and a value close to $k-1$ means that the basis may be too small (Section 17.8). It does *not* mean "no effect": an edf of 1 can come with a very small p-value.
- `Ref.df`, `F` (or `Chi.sq` for families with known scale) and `p-value` belong to an approximate test of $H_0 : f_j \equiv 0$. The p-value treats the estimated smoothing parameters as known and is therefore somewhat too small; do not read values near 0.05 too literally (Section 17.7).

Model summaries. `R-sq.(adj)` is the adjusted R^2 on the response scale; `Deviance explained` is $1 - D/D_0$, which equals R^2 for a Gaussian model; `-REML` is the value of the criterion at the optimum; `Scale est.` is $\hat{\phi}$, here $\hat{\sigma}^2 = 0.234$ on the log scale.

16.5.3 Plotting the smooths

```
par(mfrow = c(1, 3), mar = c(4, 4, 1, 1))
plot(m_log, shade = TRUE, residuals = TRUE, pch = 16, cex = .5, seWithMean =
↪  TRUE,
     shade.col = "grey85")
```

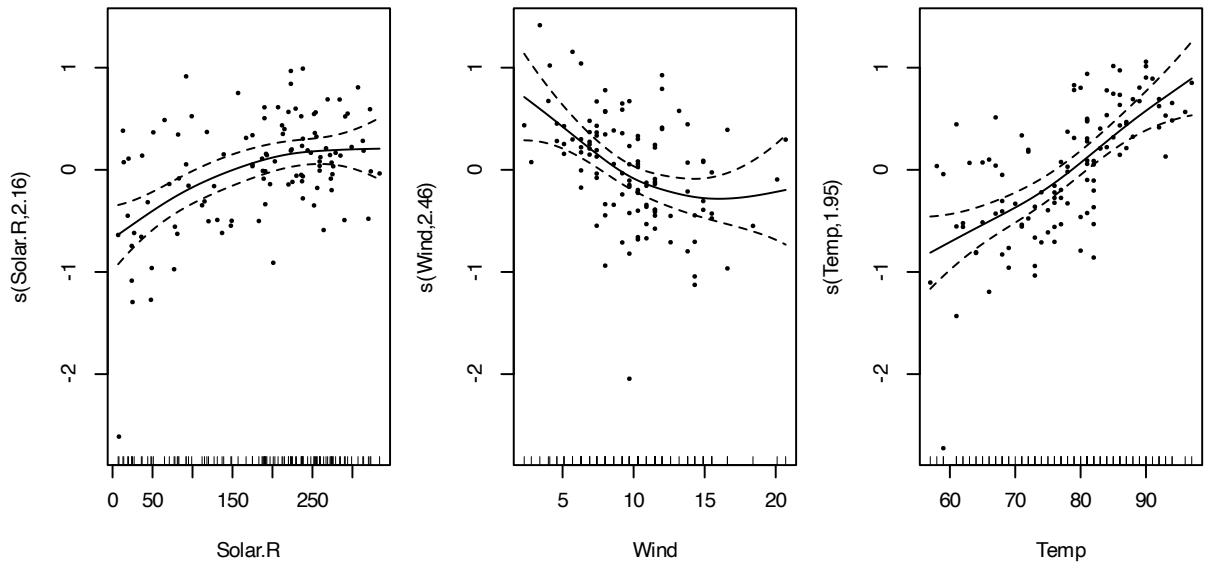


Figure 16.5: Partial effects of the Gaussian GAM for log(Ozone). Each curve is centred; the shaded band is the estimate plus and minus two standard errors, the points are partial residuals.

The points are **partial residuals**, $\hat{f}_j(x_{ij}) + \hat{\varepsilon}_i$: the data with the other terms subtracted. They show how well the curve describes what is left of the data. The band is a Bayesian credible interval (Section 17.6); `seWithMean = TRUE` adds the uncertainty of the intercept. The package `gratia` offers `gratia::draw(m_log)` for ggplot-based versions of the same plots.

What do the plots say? For `Temp` the effect is close to a straight line on the log scale (edf 1.95) and increases steeply above about 75 degrees. For `Wind`, log-ozone falls quickly up to about 12 mph and then flattens; the band widens where wind speeds are sparse. For `Solar.R` the effect

rises and levels off above about 250 Langleys. The vertical scale is log(Ozone): a change of $0.69 = \log 2$ doubles the median ozone. A band that contains a horizontal line everywhere is weak evidence of an effect.

16.5.4 Predictions and contrasts

Predictions use `predict()`, and the *linear-predictor matrix* (`type = "lpmatrix"`) gives access to any linear combination of the coefficients, hence to contrasts and their standard errors. We hold `Solar.R` and `Wind` at their medians and compare 70 and 90 degrees.

```
nd <- data.frame(Solar.R = median(aq$Solar.R), Wind = median(aq$Wind), Temp =
↪  c(70, 90))
pr <- predict(m_log, nd, se.fit = TRUE)
cbind(nd, eta = pr$fit, se = pr$se.fit, ppb = exp(pr$fit))
```

```
#>   Solar.R Wind Temp   eta     se   ppb
#> 1     207  9.7   70 3.117 0.1021 22.58
#> 2     207  9.7   90 4.065 0.1131 58.24
```

```
Xp <- predict(m_log, nd, type = "lpmatrix")
d  <- Xp[2, ] - Xp[1, ]
est <- sum(d * coef(m_log)); se <- sqrt(drop(t(d) %*% vcov(m_log) %*% d))
c(difference = est, se = se, ratio = exp(est),
  lower = exp(est - 1.96 * se), upper = exp(est + 1.96 * se))
```

```
#> difference         se      ratio      lower      upper
#>     0.9476     0.1384     2.5795     1.9667     3.3832
```

Going from 70 to 90 degrees multiplies the *median* ozone by about 2.58 (95% interval 1.97 to 3.38). Two points deserve emphasis. First, $e^{\hat{\eta}}$ estimates the median of ozone if the errors on the log scale are symmetric, not the mean. Second, the linear model gives the ratio $e^{20 \times \hat{\beta}_3} = 2.67$ for *any* 20-degree step, whereas the GAM's ratio depends on where the step starts: a curve has no single slope.

16.5.5 Gamma and binomial responses

Gamma. Instead of transforming the response, model the mean of ozone directly with a gamma distribution and a log link: $\log \mathrm{E}(y_i) = \beta_0 + f_1 + f_2 + f_3$, $\mathrm{Var}(y_i) = \phi \mu_i^2$. The constant coefficient of variation builds the funnel into the model, and e^{f_j} multiplies the *mean*.

```
m_gam <- gam(Ozone ~ s(Solar.R) + s(Wind) + s(Temp), family = Gamma(link =
↪  "log"),
             data = aq, method = "REML")
round(rbind(edf = summary(m_gam)$edf, `edf log-response model` =
↪  summary(m_log)$edf[1:3]), 2)
```

```
#>                          [,1] [,2] [,3]
#> edf                      2.04 2.51 3.29
#> edf log-response model 2.16 2.46 1.95
```

```
round(m_gam$scale, 3)
```

```
#> [1] 0.197
```

AIC is a function of the likelihood of the *response*. The Gaussian model for $\log y$ and the Gamma model for y model different random variables, so their AICs cannot be compared directly. If $U = \log Y$ has density f_U, then Y has density $f_Y(y) = f_U(\log y)/y$, so the log-likelihoods differ by $\sum_i \log y_i$ and, with the same number of parameters,

$$\mathrm{AIC}_Y = \mathrm{AIC}_{\log Y} + 2\sum_{i=1}^{n} \log y_i .$$

```
m_id <- gam(Ozone ~ s(Solar.R) + s(Wind) + s(Temp), data = aq, method =
↪  "REML")
c(gaussian_identity   = AIC(m_id),
  gaussian_on_log     = AIC(m_log) + 2 * sum(log(aq$Ozone)),
  gamma_log_link      = AIC(m_gam))
```

```
#> gaussian_identity   gaussian_on_log    gamma_log_link
#>             967.8             924.7             918.1
```

On the original scale the Gamma model with a log link is the best of the three. Comparing `AIC(m_log)` $= 166.4$ with `AIC(m_gam)` $= 918.1$ directly would be meaningless.

Binomial. The data frame `Pima.tr` and `Pima.te` (package `MASS`) describe 532 Pima Indian women; `type` records diabetes. We model the log-odds with smooth effects of plasma glucose, body mass index, age and the diabetes pedigree function.

```
pima <- rbind(Pima.tr, Pima.te)
m_glm <- glm(type ~ glu + bmi + age + ped, family = binomial, data = pima)
m_bin <- gam(type ~ s(glu) + s(bmi) + s(age) + s(ped), family = binomial,
             data = pima, method = "REML")
round(summary(m_bin)$s.table, 4)
```

```
#>          edf Ref.df Chi.sq p-value
#> s(glu) 1.000  1.000  66.25  0.0000
#> s(bmi) 3.208  4.076  20.25  0.0005
#> s(age) 3.579  4.486  36.47  0.0000
#> s(ped) 1.893  2.397  13.31  0.0022
```

```
AIC(m_glm, m_bin)
```

```
#>          df   AIC
#> m_glm  5.00 485.4
#> m_bin 12.96 467.2
```

For binomial and Poisson responses the scale parameter is known ($\phi = 1$) and the test statistic is χ^2. The effect of glucose has edf $= 1$: the data choose a straight line on the logit scale. The effect of age is clearly non-linear.

```
par(mfrow = c(1, 2), mar = c(4, 4, 1, 1))
plot(m_bin, select = 1, shade = TRUE, ylab = "effect on logit", shade.col =
↪  "grey85")
plot(m_bin, select = 3, shade = TRUE, ylab = "effect on logit", shade.col =
↪  "grey85")
```

A linear logistic model cannot represent the rise and fall in the age effect; the GAM improves the AIC from 485.4 to 467.2.

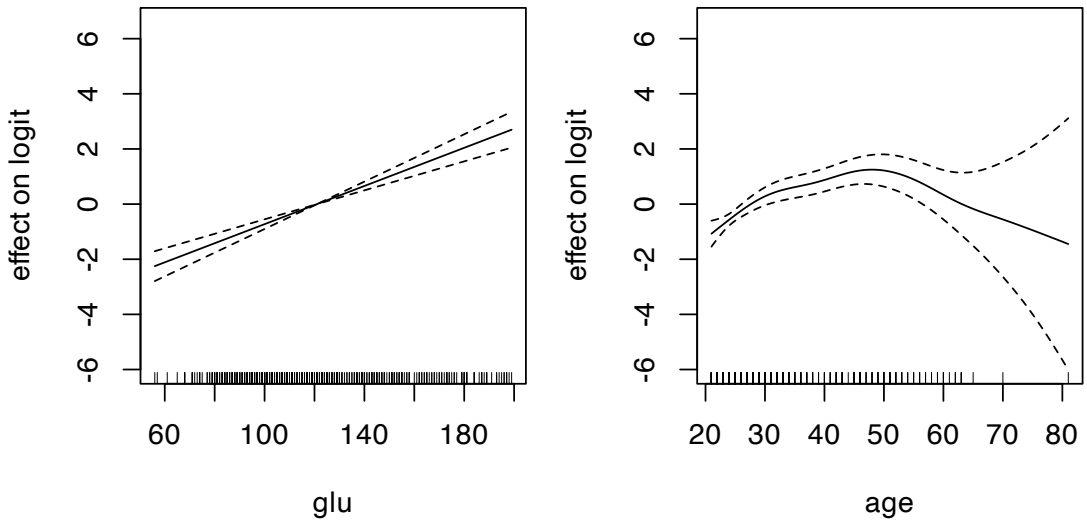


Figure 16.6: Effects on the logit scale in the binomial GAM. Glucose is linear (edf 1); the effect of age rises until about 48 years and then falls, but only 5% of women are older than 54, hence the wide band.

16.6 Comparison with linear models and GLMs

A GAM contains the corresponding linear model as a special case, so the two can be compared formally. Fit the linear model with `gam()` (so that both fits use the same machinery) and use `anova()`:

```
m_lin <- gam(log(Ozone) ~ Solar.R + Wind + Temp, data = aq, method = "REML")
anova(m_lin, m_log, test = "F")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: log(Ozone) ~ Solar.R + Wind + Temp
#> Model 2: log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp)
#>   Resid. Df Resid. Dev   Df Deviance    F Pr(>F)
#> 1       107       27.7
#> 2       100       24.2 7.01      3.5 2.14  0.046 *
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(m_lin, m_log)
```

```
#>          df   AIC
#> m_lin  5.00 170.8
#> m_log 10.29 166.4
```

The F test compares the residual deviance of the two fits, using the effective degrees of freedom of the GAM; the difference is about 3.6 degrees of freedom. As with the smooth-term tests, the reference distribution is approximate (it ignores the fact that the smoothing parameters were estimated) and a p-value of 0.046 is borderline evidence. AIC (170.8 for the linear fit against 166.4 for the GAM) agrees that the GAM is better, but only by a few units. Two remarks about AIC for GAMs. First, the "degrees of freedom" in `AIC(m_log)` is 10.29, more than the total edf of 7.57: `mgcv` adds a correction for the fact that the smoothing parameters were estimated (Wood, Pya and Saefken, 2016). Second, AIC of GAMs is a model-comparison tool, not a test:

differences below about 2 are of no consequence.

Example 16.2 (Is the extra flexibility worth it? Pima)**.** In the binomial case the improvement is large: AIC drops by 18.2 and the approximate test of `s(age)` rejects linearity-with-slope-zero by a wide margin. To test *linearity* rather than "no effect", one compares the GAM with the model in which the smooth is replaced by a linear term, as above.

```
m_bin_lin_age <- gam(type ~ s(glu) + s(bmi) + age + s(ped), family = binomial,
                     data = pima, method = "REML")
anova(m_bin_lin_age, m_bin, test = "Chisq")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: type ~ s(glu) + s(bmi) + age + s(ped)
#> Model 2: type ~ s(glu) + s(bmi) + s(age) + s(ped)
#>   Resid. Df Resid. Dev   Df Deviance Pr(>Chi)
#> 1       521        462
#> 2       517        441 4.24     21.1  0.00038 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The data reject a linear age effect on the logit scale. The same idea works for every term, which makes the GAM a diagnostic tool for the GLM: **fit the GAM, look at the plots, and then decide which effects can be replaced by simple parametric terms** (a quadratic, a spline with few knots, a linear term).

Remark (model checking). A GAM needs the same diagnostics as a GLM (residual plots, overdispersion for Poisson and binomial counts) plus two more: a check that the basis dimension k is large enough, and a check for concurvity. `gam.check(m)` produces the residual plots and the basis-dimension test in one call (Section 17.8).

16.7 In R: the functions used in this chapter

Function	Purpose
`gam(formula, family, data, method = "REML")`	fit a GAM; `s(x, k = , bs = )` declares a smooth
`summary(m)`	parametric coefficients; edf, Ref.df, test and p-value of each smooth
`plot(m, residuals = TRUE, shade = TRUE, pages = 1)`	partial effects with confidence bands
`predict(m, newdata, type = "response" / "link" / "terms" / "lpmatrix", se.fit = TRUE)`	predictions, centred terms, the matrix mapping coefficients to $\hat{\eta}$
`vcov(m)`	Bayesian posterior covariance matrix of the coefficients

Function	Purpose
`AIC(m1, m2)`, `anova(m1, m2, test = "F")`	model comparison
`gam.check(m)`	residual plots and basis-dimension check
`concurvity(m)`	measure of non-linear dependence between smooths

The data frame must contain the response and the covariates exactly as named in the formula; transformations of the response (for example `log(Ozone)`) are allowed. Factors enter as parametric terms (`+ factor(Month)`) and may also define smooths by group (`s(x, by = f)`, Chapter 18).

16.8 Pitfalls

Warning. Comparing the AIC of a model for $\log y$ with that of a model for y without the Jacobian term $2\sum \log y_i$.

Warning. Reading a centred partial effect as a predicted value. Add $\hat{\beta}_0$ and the other terms, or use `predict()`.

Warning. Back-transforming $e^{\hat{\eta}}$ from a log-response model and calling it the mean; it is a median under symmetric errors on the log scale.

Warning. Treating smooth-term p-values as exact, and taking edf $= 1$ as "no effect".

Warning. Leaving β_0 out of the partial residuals, or not centring, in a hand-written backfitting loop; or writing λ twice in a penalty.

Warning. Forgetting `method = "REML"`; the default is `"GCV.Cp"`.

Remark. The intercept equals $\bar{y}$ only in the Gaussian model with identity link and centred smooths. In a Poisson GAM with log link, $\hat{\beta}_0 = \log \bar{y} - \log(n^{-1}\sum_i e^{\hat{f}_{1i}+\cdots+\hat{f}_{pi}}) \le \log \bar{y}$ (Exercise 16.2).

16.9 Summary

- An additive model replaces each $\beta_j x_j$ by a smooth $f_j(x_j)$; a GAM does so inside a GLM linear predictor, $g(\mu_i) = \boldsymbol{x}_i^{*\top}\boldsymbol{\theta} + \sum_j f_j(x_{ij})$.
- The levels of β_0 and the f_j are not identified; imposing $\sum_i f_j(x_{ij}) = 0$ fixes them, and then $\hat{\beta}_0 = \bar{y}$ in the Gaussian additive model.
- With basis expansions, estimation is penalized regression: minimise $\|\boldsymbol{y} - X\boldsymbol{\gamma}\|^2 + \sum_j \lambda_j \boldsymbol{\gamma}^\top S_j \boldsymbol{\gamma}$; each λ_j appears once.
- The normal equations are $\hat{\boldsymbol{f}}_j = A_j(\boldsymbol{y} - \hat{\beta}_0\mathbf{1} - \sum_{k\neq j}\hat{\boldsymbol{f}}_k)$. Backfitting is Gauss–Seidel on them; it converges geometrically (Theorem 16.2) with a rate set by the overlap between the smooths (for two projection smoothers r^2, the squared correlation).
- `mgcv` fits all smooths jointly by PIRLS and chooses the λ_j by REML or GCV.
- `summary()` reports edf, `Ref.df`, a test statistic and an approximate p-value per smooth; `plot()` shows centred partial effects with credible bands.
- Compare GAMs with their GLM counterparts by approximate F or χ^2 tests and AIC; compare models for different response scales only after adding the Jacobian.

16.10 Exercises

Exercise 16.1. (Identifiability) (a) Show that in the model $y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \varepsilon_i$ the pair (β_0, f_1, f_2) and $(\beta_0 + c_1 + c_2,\ f_1 - c_1,\ f_2 - c_2)$ give the same likelihood. (b) Which constraint on the sample makes the representation unique? (c) If the x_{i2} are a deterministic function of x_{i1}, for example $x_{i2} = x_{i1}^2$, is the model identifiable with smooth f_1, f_2? Explain.

Solution.

(a) The mean of observation i is $\beta_0 + f_1(x_{i1}) + f_2(x_{i2}) = (\beta_0 + c_1 + c_2) + \{f_1(x_{i1}) - c_1\} + \{f_2(x_{i2}) - c_2\}$, the same for both parameter values, hence the likelihood is the same for every i and every σ^2.

(b) $\sum_i f_1(x_{i1}) = \sum_i f_2(x_{i2}) = 0$. Given any representation, the unique centred one has $c_j = n^{-1}\sum_i f_j(x_{ij})$ and intercept $\beta_0 + c_1 + c_2$.

(c) No. With $x_2 = x_1^2$, the function $f_2(x_2) = f_2(x_1^2)$ is itself a function of x_1, so $f_1(x_1) + f_2(x_1^2) = h(x_1)$ can be split between the two smooths in infinitely many ways (for example $f_1 = h$, $f_2 = 0$, or $f_1 = 0$, $f_2(u) = h(\sqrt{u})$ for $x_1 > 0$). This is an extreme case of **concurvity**: centring cannot remove it because the dependence is non-linear.

Exercise 16.2. (Intercept) (a) In the Gaussian additive model with centred smooths show that $\hat\beta_0 = \bar y$, and say what happens if $\boldsymbol{f}_j$ has mean $c \neq 0$. (b) In a Poisson GAM with log link and centred smooths, show that $\hat\beta_0 = \log \bar y - \log(\frac{1}{n}\sum_i e^{\hat f_{1i} + \hat f_{2i}})$ and deduce $\hat\beta_0 \le \log \bar y$.

Solution.

(a) From $\partial Q/\partial\beta_0 = 0$: $\hat\beta_0 = \bar y - n^{-1}\sum_k \mathbf{1}^\top \boldsymbol{f}_k = \bar y$ because all means vanish. If one $\boldsymbol{f}_j$ has mean c, then $\hat\beta_0 = \bar y - c$; fitted values do not change but β_0 and f_j separately lose their meaning, and in backfitting the constant can drift between them.

(b) The score equation for β_0 is $\sum_i (y_i - \mu_i) = 0$ with $\mu_i = \exp(\beta_0 + f_{1i} + f_{2i})$, so $e^{\hat\beta_0}\sum_i e^{\hat f_{1i} + \hat f_{2i}} = n\bar y$, giving the formula. Let $u_i = \hat f_{1i} + \hat f_{2i}$, which has mean zero. By Jensen's inequality $n^{-1}\sum_i e^{u_i} \ge e^{\bar u} = 1$, so the logarithm is ≥ 0 and $\hat\beta_0 \le \log \bar y$, with equality only when $u_i = 0$ for all i.

Exercise 16.3. (Block update) For Q in (16.3) with $p = 2$, derive $\hat{\boldsymbol{\beta}}_1 = (B_1^\top B_1 + \lambda_1 S_1)^{-1} B_1^\top (\boldsymbol{y} - \beta_0 \mathbf{1} - \boldsymbol{f}_2)$ and the smoother matrix A_1. Show also that if $(\boldsymbol{f}_1, \boldsymbol{f}_2)$ solve the normal equations (16.4) and each A_j reproduces constants ($A_j\mathbf{1} = \mathbf{1}$) then $(\boldsymbol{f}_1 + c\mathbf{1}, \boldsymbol{f}_2 - c\mathbf{1})$ does too, and explain what centring does.

Solution. Differentiate Q with respect to $\boldsymbol{\beta}_1$: $-2B_1^\top(\boldsymbol{y} - \beta_0\mathbf{1} - B_1\boldsymbol{\beta}_1 - \boldsymbol{f}_2) + 2\lambda_1 S_1\boldsymbol{\beta}_1 = \mathbf{0}$, i.e. $(B_1^\top B_1 + \lambda_1 S_1)\boldsymbol{\beta}_1 = B_1^\top(\boldsymbol{y} - \beta_0\mathbf{1} - \boldsymbol{f}_2)$. Multiplying the solution by B_1 gives $\boldsymbol{f}_1 = A_1(\boldsymbol{y} - \beta_0\mathbf{1} - \boldsymbol{f}_2)$ with $A_1 = B_1(B_1^\top B_1 + \lambda_1 S_1)^{-1}B_1^\top$; λ_1 appears once.

For the shifted pair: $A_1\{\boldsymbol{y} - \beta_0\mathbf{1} - (\boldsymbol{f}_2 - c\mathbf{1})\} = \boldsymbol{f}_1 + cA_1\mathbf{1} = \boldsymbol{f}_1 + c\mathbf{1}$, and $A_2\{\boldsymbol{y} - \beta_0\mathbf{1} - (\boldsymbol{f}_1 + c\mathbf{1})\} = \boldsymbol{f}_2 - c\mathbf{1}$. So the solutions of the normal equations form a line; centring selects the unique point on that line with $\mathbf{1}^\top\boldsymbol{f}_1 = \mathbf{1}^\top\boldsymbol{f}_2 = 0$ and moves the overall level into β_0.

Exercise 16.4. (Rate of backfitting) Two covariates, centred, with sample correlation r; both smoothers are least-squares projections onto the covariate. (a) Derive the factor

by which the error of $\hat{a}$ (the coefficient of $\boldsymbol{x}_1$) shrinks per sweep. (b) How many sweeps are needed to reduce the error by a factor 10^{-6} if $r = 0.5$? If $r = 0.95$? (c) Verify (a) by simulation in R.

Solution.

(a) This is Proposition 16.2: $e_a^{(t+1)} = r^2\, e_a^{(t)}$ from one sweep to the next.

(b) We need $r^{2t} \le 10^{-6}$, i.e. $t \ge 6/(-2\log_{10} r)$. For $r = 0.5$: $t \ge 9.97$, so 10 sweeps. For $r = 0.95$: $t \ge 134.7$, so 135 sweeps.

(c) The ratio of successive errors in a simulation is the squared correlation:

```
set.seed(3); n_r <- 100
x1 <- rnorm(n_r); x2 <- .8 * x1 + .6 * rnorm(n_r); x1 <- x1 - mean(x1); x2 <-
↪  x2 - mean(x2)
yy <- x1 + x2 + rnorm(n_r)
cf <- coef(lm(yy ~ x1 + x2 - 1))
a <- 0; b <- 0; ea <- numeric(8)
for (t in 1:8) {
  a <- sum(x1 * (yy - b * x2)) / sum(x1^2)
  b <- sum(x2 * (yy - a * x1)) / sum(x2^2)
  ea[t] <- a - cf[1]
}
c(ratio = unname((ea[8] / ea[7])), r2 = cor(x1, x2)^2)
```

```
#>  ratio     r2
#> 0.4854 0.4854
```

Exercise 16.5. (Contrasts) Using the Gaussian GAM `m_log` of this chapter, estimate the ratio of median ozone between a day with wind speed 15 mph and one with 5 mph, at the median `Solar.R` and `Temp`, with a 95% interval. Contrast with the answer from the linear model for log(Ozone).

Solution. Build the `lpmatrix` for the two covariate patterns and use the posterior covariance matrix.

```
nd <- data.frame(Solar.R = median(aq$Solar.R), Wind = c(5, 15), Temp =
↪  median(aq$Temp))
Xp <- predict(m_log, nd, type = "lpmatrix"); d <- Xp[2, ] - Xp[1, ]
est <- sum(d * coef(m_log)); se <- sqrt(drop(t(d) %*% vcov(m_log) %*% d))
round(c(est = est, se = se, ratio = exp(est), lo = exp(est - 1.96 * se),
        hi = exp(est + 1.96 * se)), 3)
```

```
#>    est     se  ratio     lo     hi
#> -0.700  0.162  0.497  0.362  0.682
```

```
round(unname(exp(10 * coef(lm_log)["Wind"])), 3)
```

```
#> [1] 0.54
```

Going from 5 to 15 mph roughly halves the median ozone. The linear model implies the ratio $e^{10\hat{\beta}_2}$ shown in the last line for *every* 10-mph step; the GAM shows that the effect is concentrated at low wind speeds, so the contrast depends on the starting point.

Exercise 16.6. (AIC across scales) (a) Prove $\mathrm{AIC}_Y = \mathrm{AIC}_{\log Y} + 2\sum \log y_i$. (b) Compute both sides for the Gaussian GAM of log-ozone and compare with the Gamma GAM. Why is the comparison fair although the models have different families?

Solution.

(a) If $U = \log Y$ has density f_U, the change of variables $y \mapsto \log y$ has Jacobian $1/y$, so $f_Y(y) = f_U(\log y)/y$ and $\log L_Y = \log L_{\log Y} - \sum_i \log y_i$. With $\mathrm{AIC} = -2\log L + 2\,\mathrm{df}$ and the same degrees of freedom, $\mathrm{AIC}_Y = \mathrm{AIC}_{\log Y} + 2\sum_i \log y_i$.

(b)

```
c(log_scale = AIC(m_log), jacobian = 2 * sum(log(aq$Ozone)),
  on_ozone_scale = AIC(m_log) + 2 * sum(log(aq$Ozone)), gamma = AIC(m_gam))
```

```
#>      log_scale       jacobian on_ozone_scale          gamma
#>          166.4          758.3          924.7          918.1
```

After the correction both numbers are AICs of a density for the same random variable (Ozone in ppb), so they measure the same thing: the expected Kullback–Leibler distance to the truth. The Gamma GAM is lower by about 6.6 units.

Exercise 16.7. (Poisson GAM) The vector `discoveries` (package `datasets`) holds the number of "great" scientific discoveries per year, 1860–1959. (a) Fit a Poisson GLM with a linear trend in year and a Poisson GAM with `s(year)` (REML). (b) Compare by AIC and plot the smooth. (c) Compute the Pearson dispersion statistic of the GAM and comment.

Solution.

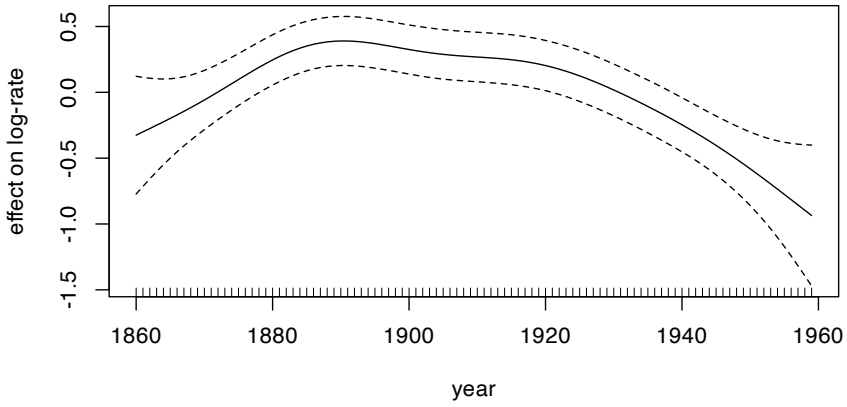
```
dd <- data.frame(year = 1860:1959, count = as.numeric(discoveries))
g_lin <- glm(count ~ year, family = poisson, data = dd)
g_sm  <- gam(count ~ s(year), family = poisson, data = dd, method = "REML")
AIC(g_lin, g_sm)
```

```
#>          df   AIC
#> g_lin 2.000 430.3
#> g_sm  5.676 409.3
```

```
round(summary(g_sm)$s.table, 3)
```

```
#>           edf Ref.df Chi.sq p-value
#> s(year) 3.769  4.676  27.93       0
```

```
par(mar = c(4, 4, 1, 1)); plot(g_sm, shade = TRUE, shade.col = "grey85", ylab
↪  = "effect on log-rate")
```




```
disp <- sum(residuals(g_sm, type = "pearson")^2) / df.residual(g_sm)
round(disp, 3)
```

```
#> [1] 1.261
```

The smooth (edf about 3.8) rises until about 1890 and declines afterwards; the linear trend cannot represent this, and the AIC improves by about 21 units. The Pearson statistic divided by the residual degrees of freedom is about 1.26: a mild overdispersion, so the standard errors and p-values are somewhat optimistic; a negative-binomial family (`family = nb()`) or `quasipoisson` would be a reasonable sensitivity analysis.

Exercise 16.8. (Critique) A report on the ozone data says: "The GAM has edf = 1.95 for temperature, so there is almost no temperature effect. The smooth-term p-values are exact because they come from an F test. The intercept 3.416 is the average ozone concentration in ppb." Find the three errors and rewrite the paragraph correctly.

Solution.

(1) An edf of 1.95 does not mean "almost no effect": an edf near 1 means the effect is close to a *straight line*; the F statistic of `s(Temp)` is the largest of the three ($F \approx 21$, $p < 2 \times 10^{-16}$). (2) The p-values are approximate: they are computed as if the smoothing parameters were known, so they are somewhat too small. (3) The intercept is the mean of log(Ozone), 3.416, because the model is for the log response and all smooths are centred; the mean ozone in ppb is $\overline{y} \approx 42.1$, and $e^{3.416} = 30.4$ ppb is the geometric mean.

Corrected paragraph: "In a Gaussian additive model for log(Ozone), all three covariates have clearly non-zero effects. The temperature effect is nearly a straight line on the log scale (edf 1.95) and by far the strongest ($F \approx 21$); wind and radiation have gentle non-linear effects (edf 2.5 and 2.2). The approximate p-values ignore the uncertainty in the smoothing parameters. The intercept, 3.42, is the mean log-ozone; its exponential, 30.4 ppb, is the geometric mean ozone."

Exercise 16.9. (Binomial GAM) In the Pima GAM `m_bin` compute the odds ratio for glucose 150 against 100 mg/dl (other covariates at `bmi = 32`, `age = 30`, `ped = 0.4`) with a 95% interval, and compare it with the odds ratio from the linear logistic model. Then compute the fitted probability of diabetes at ages 25 and 45 for the same covariate values (glucose 120).

Solution.

```
nd <- data.frame(glu = c(100, 150), bmi = 32, age = 30, ped = 0.4)
Xp <- predict(m_bin, nd, type = "lpmatrix"); d <- Xp[2, ] - Xp[1, ]
est <- sum(d * coef(m_bin)); se <- sqrt(drop(t(d) %*% vcov(m_bin) %*% d))
round(exp(c(or = est, lo = est - 1.96 * se, hi = est + 1.96 * se)), 2)
```

```
#>   or   lo   hi
#> 5.64 3.72 8.56
```

```
round(unname(exp(50 * coef(m_glm)["glu"])), 2)
```

```
#> [1] 5.49
```

```
round(predict(m_bin, data.frame(glu = 120, bmi = 32, age = c(25, 45), ped =
↪  .4),
              type = "response"), 3)
```

```
#>     1     2
#> 0.195 0.540
```

The two odds ratios agree closely: the GAM chose edf $= 1$ for glucose, so it reproduces the logistic fit for that term. The estimated probability at 45 years is clearly larger than at 25 years for these covariate values.

Exercise 16.10. (Testing linearity) In the Gaussian GAM for log-ozone, test whether the effect of `Wind` can be replaced by a straight line. Fit the model with `Wind` linear and the other two terms smooth, compare with `m_log`, and interpret the AIC and the F test.

Solution.

```
m_wl <- gam(log(Ozone) ~ s(Solar.R) + Wind + s(Temp), data = aq, method =
↪  "REML")
anova(m_wl, m_log, test = "F")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: log(Ozone) ~ s(Solar.R) + Wind + s(Temp)
#> Model 2: log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp)
#>   Resid. Df Resid. Dev   Df Deviance    F Pr(>F)
#> 1       102       25.6
#> 2       100       24.2 2.35     1.45 2.63  0.068 .
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
c(AIC_linear_wind = AIC(m_wl), AIC_smooth_wind = AIC(m_log))
```

```
#> AIC_linear_wind AIC_smooth_wind
#>           169.3           166.4
```

The comparison is approximate, as always for GAMs. The F test gives $p \approx 0.068$ and AIC favours the smooth by about 2.9 units: borderline evidence against linearity. Figure 16.5 shows the bend at low wind speeds, so a careful analyst would keep the smooth unless parsimony is paramount, and report the sensitivity.

Exercise 16.11. (Backfitting in R) Using `swiss` (package `datasets`), implement backfitting for `Fertility ~ Education + Catholic` with `smooth.spline()` for each covariate, centring each update and including $\hat{\beta}_0$ in the partial residuals. Run it (i) with `df = 4` for each term and (ii) with `df` equal to $1 + \mathrm{edf}_j$ from `gam(..., method = "REML")` (a smoothing spline's `df` counts the constant). Compare the fitted effects with those of `gam()` and report the maximal absolute differences.

Solution.

```
y2 <- swiss$Fertility; X2 <- swiss[, c("Education", "Catholic")]
gs <- gam(Fertility ~ s(Education) + s(Catholic), data = swiss, method =
↪  "REML")
tt <- predict(gs, type = "terms"); e2 <- summary(gs)$edf
backfit <- function(df) {
  b0 <- mean(y2); ff <- matrix(0, nrow(swiss), 2)
  for (it in 1:500) {
    old <- ff
    for (j in 1:2) {
      r <- y2 - b0 - rowSums(ff[, -j, drop = FALSE])
      s <- predict(smooth.spline(X2[[j]], r, df = df[j]), X2[[j]])$y
      ff[, j] <- s - mean(s)
    }
    if (max(abs(ff - old)) < 1e-8) break
  }
  c(sweeps = it, max_diff = max(abs(ff - tt)))
}
rbind(df_4 = backfit(c(4, 4)), df_edf_plus_1 = backfit(e2 + 1))
```

```
#>                sweeps max_diff
#> df_4               27   4.6490
#> df_edf_plus_1       9   0.1359
```

```
round(c(edf = e2, sd_effects = apply(tt, 2, sd)), 2)
```

```
#>                     edf1                     edf2 sd_effects.s(Education)
#>                     1.22                     1.00                     7.58
#>  sd_effects.s(Catholic)
#>                     4.68
```

With a fixed 4 degrees of freedom the effects differ visibly from the `gam()` fit, because `gam()` estimated the smoothness (the edf are about 1.2 and 1.0, the effect of `Catholic` is essentially a straight line). With `df` set to the estimated edf plus one, backfitting reproduces `gam()` up to a small discrepancy relative to the size of the effects (their standard deviations are shown in the last line); the remainder is due to the different smoothers (smoothing spline versus thin-plate regression spline). Use the smoother matrices of the `gam` fit, as in Section 16.3.2, to reproduce `gam()` exactly.

Exercise 16.12. (Which family?) Explain in words why a Gamma GAM with log link and a Gaussian GAM on $\log y$ can give different edf for the same covariate, even though both have multiplicative effects on the response. Look at the output of Section 16.5.5 and say which one you would use if the goal were to estimate the *mean* ozone, and why.

Solution. The Gaussian model for $\log y$ assumes constant variance of $\log y$, so $\mathrm{E}(\log y \mid x) = \eta$ and e^{η} is a *median*-type quantity; the Gamma model assumes $\mathrm{Var}(y) = \phi\mu^2$ and $\log \mathrm{E}(y \mid x) = \eta$, so the effects act on the *mean*. If the distribution of $\log y$ given x changes shape with the covariates (a skewness or variance that depends on temperature), the two scales have different mean curves, and the smoothness selected by REML differs: in our fit `s(Temp)` has edf close to 2 in the log-response model but about 3.3 in the Gamma model. For the mean ozone, which is what an air-quality agency regulates, the Gamma log-link model targets the right quantity directly; the log-response model needs a smearing correction $\mathrm{E}(e^{\varepsilon})$ to get from the median to the mean.

16.11 Further reading

The original proposals are Hastie and Tibshirani (1986) and Hastie and Tibshirani (1990); the relationship between backfitting and linear smoothers is in Buja et al. (1989). Wood (2017) (Chapters 3 to 7) is the standard reference for `mgcv`, and Ruppert et al. (2003) and Green and Silverman (1994) give complementary views on penalized regression. For GAMs in applied work with `mgcv` see Pedersen et al. (2019). The basis and penalty theory behind `s()` continues in Chapter 17.

Chapter 17

GAM theory

Chapter 16 used `gam()` as a black box: a formula goes in, smooth curves with bands and a table of edf and p-values come out. This chapter opens the box. We show that a GAM is *one* penalized GLM (Section 17.1), how identifiability constraints are implemented (Section 17.2), how the coefficients are estimated for fixed smoothing parameters (penalized IRLS, Section 17.3), how the smoothing parameters themselves are chosen by cross-validation or by marginal likelihood (Section 17.5), why the fitted curve has a Bayesian posterior and what a "credible interval" for a smooth means (Section 17.6), what the effective degrees of freedom and the approximate p-values of `summary()` are (Section 17.4), and how to check that the basis dimension is large enough (Section 17.8). The presentation follows Chapters 5 and 6 of Wood (2017) and the papers Wood (2011) and Wood (2013).

```
library(mgcv)
library(MASS)
data(airquality)
aq <- na.omit(airquality[, c("Ozone", "Solar.R", "Wind", "Temp")])
aq$ly <- log(aq$Ozone)
n <- nrow(aq)
# total penalty matrix S_lambda of a fitted gam, padded with zeros to the full
↪  length of coef()
Slam_full <- function(g) {
  p <- length(coef(g)); S <- matrix(0, p, p)
  for (sm in g$smooth) {
    ii <- sm$first.para:sm$last.para
    for (l in seq_along(sm$S)) S[ii, ii] <- S[ii, ii] + g$sp[sm$first.sp + l -
    ↪  1] * sm$S[[l]]
  }
  S
}
```

17.1 One penalized regression

17.1.1 Notation

Throughout, a GAM is

$$g(\mu_i) = \eta_i = \boldsymbol{x}_i^{*\top}\boldsymbol{\theta} + \sum_{j=1}^{p} f_j(x_{ij}), \qquad f_j(x) = \sum_{k=1}^{K_j} \beta_{jk}\, b_{jk}(x), \tag{17.1}$$

and we stack all coefficients, $\boldsymbol{\gamma} = (\boldsymbol{\theta}^\top, \boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_p^\top)^\top \in \mathbb{R}^q$, so that $\boldsymbol{\eta} = X\boldsymbol{\gamma}$ with $X = [X^* \mid X_1 \mid \cdots \mid X_p]$, where X_j is the $n \times K_j$ basis matrix of the j-th smooth. Here X^* is the model matrix of the parametric terms; W denotes the IRLS weight matrix and $\boldsymbol{z}$ the working response of Chapter 9.

Symbol	Meaning
S_j	$K_j \times K_j$ penalty matrix of smooth j, padded with zeros to $q \times q$
$S_\lambda = \sum_j \lambda_j S_j$	total penalty
W, $\boldsymbol{z}$	IRLS weights and working response (Chapter 9)
$A = X(X^\top W X + S_\lambda)^{-1} X^\top W$	influence (hat) matrix of the working fit, $\hat{\boldsymbol{\eta}} = A\boldsymbol{z}$
$F = (X^\top W X + S_\lambda)^{-1} X^\top W X$	maps the unpenalized to the penalized estimate
V_γ	posterior covariance matrix of $\boldsymbol{\gamma}$
M	dimension of the null space of S_λ

Remark. In Chapters 14 and 15 the smoother matrix of a single smooth was called S_λ. Here, as in Wood (2017), S_j are *penalties* and A is the smoother, so the letter S is never a hat matrix.

17.1.2 The wiggliness penalty is a quadratic form

Proposition 17.1 (The roughness penalty is a quadratic form in the coefficients)**.** *Let $f(x) = \sum_k \beta_k b_k(x)$ with twice differentiable basis functions. Then*

$$\int f''(x)^2 \, dx = \boldsymbol{\beta}^\top S \boldsymbol{\beta}, \qquad S_{kl} = \int b_k''(x)\, b_l''(x)\, dx,$$

S is symmetric and positive semi-definite, and $\boldsymbol{\beta}^\top S \boldsymbol{\beta} = 0$ if and only if f is a straight line, $f(x) = a + bx$. The null space of S therefore has dimension at most 2, and equals 2 whenever the constant and the linear function lie in the span of the basis.

Proof. Since $f'' = \sum_k \beta_k b_k''$, $\int f''^2 = \sum_k \sum_l \beta_k \beta_l \int b_k'' b_l'' = \boldsymbol{\beta}^\top S \boldsymbol{\beta}$. For any $\boldsymbol{v}$, $\boldsymbol{v}^\top S \boldsymbol{v} = \int (\sum_k v_k b_k'')^2 \geq 0$, and it vanishes iff $\sum_k v_k b_k'' \equiv 0$, i.e. iff the function with coefficients $\boldsymbol{v}$ has zero second derivative: a straight line. □

The same structure holds for other penalties: for P-splines (`bs = "ps"`) $S = D_2^\top D_2$ with D_2 the second-order difference matrix, which also has a two-dimensional null space (constant and linear sequences of coefficients); for thin-plate regression splines (the default `s(x)`) S is the diagonal matrix of eigenvalues of the thin-plate penalty after truncation (Wood, 2003). `smoothCon()` constructs exactly the pair (X_j, S_j) that `gam()` uses internally.

```
sm_raw <- smoothCon(s(Temp, k = 10), data = aq, absorb.cons = FALSE)[[1]]
dim(sm_raw$X); dim(sm_raw$S[[1]])
```

```
#> [1] 111  10
```

```
#> [1] 10 10
```

```
ev <- eigen(sm_raw$S[[1]], symmetric = TRUE)$values
round(ev, 4)
```

```
#>  [1] 491.692 321.931 264.058 138.819  92.627  30.374   8.547   1.159
#>  [9]   0.000   0.000
```

```
sum(ev < max(ev) * 1e-10)       # dimension of the null space
```

```
#> [1] 2
```

There are $K = 10$ basis functions and the penalty matrix has two zero eigenvalues, the constant and linear functions. The other eight eigenvalues, from about 1.2 up to about 490, measure how strongly the penalty acts on the eight "wiggly" directions.

Consequence. As $\lambda_j \to \infty$ the smooth is forced into the null space of its penalty: a straight line, not zero. After centring, edf $\to 1$. As $\lambda_j \to 0$ the smooth becomes an unpenalized regression spline with edf $\to K_j - 1$.

```
f0 <- gam(ly ~ s(Temp), data = aq, method = "REML")
xg <- data.frame(Temp = seq(min(aq$Temp), max(aq$Temp), length.out = 200))
d <- do.call(rbind, lapply(c(1e-6, f0$sp, 1e8), function(l) {
  gg <- gam(ly ~ s(Temp), data = aq, sp = l)
  data.frame(Temp = xg$Temp, fit = predict(gg, xg),
             lab = sprintf("lambda = %s (edf %.1f)", format(l, digits = 2),
             ↪  sum(gg$edf) - 1))
}))
ggplot(d, aes(Temp, fit, colour = lab)) +
  geom_point(aes(Temp, ly), data = aq, inherit.aes = FALSE, colour = "grey60")
  ↪  +
  geom_line(linewidth = 1) + scale_colour_manual(values = c(ie_coral, ie_navy,
  ↪  ie_green)) +
  labs(x = "Temp (F)", y = "log(Ozone)", colour = NULL)
```

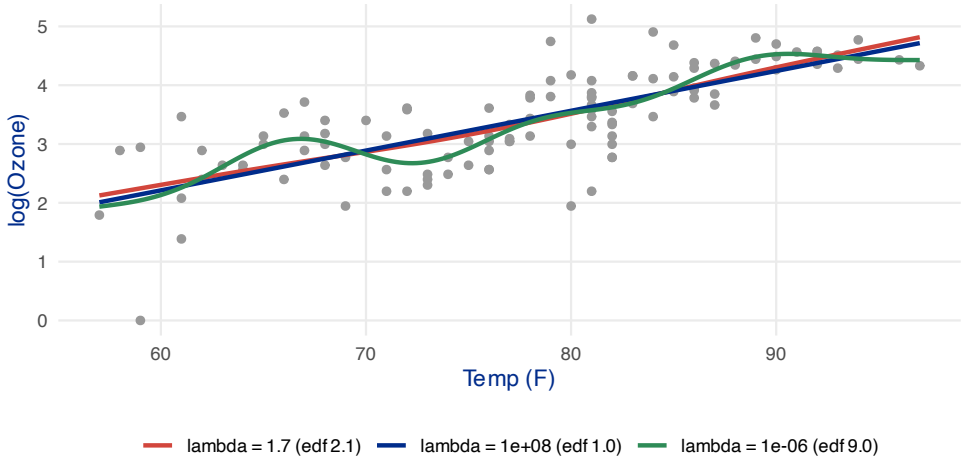


Figure 17.1: log(Ozone) against Temp with s(Temp) fitted for a very small, the REML and a very large smoothing parameter.

17.1.3 Simultaneous diagonalisation

For one Gaussian smooth the whole machinery can be reduced to a sequence of scalar problems. Suppose $X^\top X = LL^\top$ has full rank and diagonalise the penalty in the whitened coordinates: $L^{-1} S L^{-\top} = U \Lambda U^\top$ with U orthogonal and $\Lambda = \mathrm{diag}(s_1, \dots, s_K)$, $s_k \geq 0$. In the new coefficients

$\boldsymbol{\delta} = U^\top L^\top \boldsymbol{\gamma}$ we have $X^\top X \to I$ and $S \to \Lambda$, so that

$$\hat{\delta}_k = \frac{\tilde{\delta}_k}{1 + \lambda s_k}, \qquad \tilde{\delta}_k = \text{the unpenalized least-squares estimate},$$

(Demmler–Reinsch basis, Silverman (1985)). Every coefficient is shrunk towards zero by the factor $1/(1 + \lambda s_k)$, strongly if s_k is large (wiggly directions), not at all if $s_k = 0$ (null space); the effective degrees of freedom are $\sum_k 1/(1 + \lambda s_k)$, between the null-space dimension and K. We shall see the general version of this fact in Section 17.4.

17.2 Identifiability constraints

Proposition 17.2 (Rank deficiency of an additive model matrix)**.** *Suppose the constant function lies in the column space of X_j for each of two or more smooths j, that is, $X_j \boldsymbol{a}_j = \mathbf{1}$ for some $\boldsymbol{a}_j$ (true for B-splines, thin-plate and cubic regression splines). Then $[\mathbf{1}, X_1, \dots, X_p]$ has linearly dependent columns, $X^\top W X$ is singular, and the coefficients are not identifiable. The penalty does not repair this because constants lie in the null space of every S_j.*

Proof. Take $\boldsymbol{v} = (1, -\boldsymbol{a}_1^\top, \mathbf{0}, \dots)^\top$: $X\boldsymbol{v} = \mathbf{1} - X_1 \boldsymbol{a}_1 = \mathbf{0}$. Since $\boldsymbol{a}_1$ gives the constant function, $S_1 \boldsymbol{a}_1 = \mathbf{0}$ and also $\boldsymbol{v}^\top S_\lambda \boldsymbol{v} = 0$: moving along $\boldsymbol{v}$ changes neither the fit nor the penalty. So $X^\top W X + S_\lambda$ is singular. □

We verify this in R for the raw thin-plate basis of `s(Temp, k = 10)`:

```
X_raw <- cbind(1, sm_raw$X)
c(columns = ncol(X_raw), rank = qr(X_raw)$rank)           # intercept + 10
↪  basis columns
```

```
#> columns    rank
#>      11      10
```

```
sum(resid(lm(rep(1, n) ~ sm_raw$X - 1))^2)                # the constant is in
↪  the span
```

```
#> [1] 2.082e-31
```

The intercept plus ten basis columns have rank 10 only, and regressing the constant on the basis leaves a residual sum of squares of order 10^{-31}.

17.2.1 Imposing the sum-to-zero constraint

We want $\sum_i f_j(x_{ij}) = \mathbf{1}^\top X_j \boldsymbol{\beta}_j = 0$. This is a *linear* constraint $C\boldsymbol{\beta}_j = 0$ with the $1 \times K_j$ matrix $C = \mathbf{1}^\top X_j$.

Theorem 17.1 (Absorbing a linear constraint by reparametrisation)**.** *Let C be a $1 \times K$ non-zero row and $C^\top = Q \binom{r}{\mathbf{0}}$ a QR decomposition with $Q = [\boldsymbol{q}_1 \mid Z]$ orthogonal, $r \neq 0$, Z of size $K \times (K-1)$. Then $CZ = \mathbf{0}$ and every $\boldsymbol{\beta}$ with $C\boldsymbol{\beta} = 0$ can be written $\boldsymbol{\beta} = Z\boldsymbol{\beta}^*$ for a unique $\boldsymbol{\beta}^* \in \mathbb{R}^{K-1}$. Consequently, the constrained problem in $\boldsymbol{\beta}$ is the* unconstrained *problem in $\boldsymbol{\beta}^*$ with*

$$X_j^* = X_j Z, \qquad S_j^* = Z^\top S_j Z.$$

Proof. From $C^\top = Q\binom{r}{\mathbf{0}} = r\boldsymbol{q}_1$ we get $C = r\boldsymbol{q}_1^\top$ and $CZ = r\boldsymbol{q}_1^\top Z = \mathbf{0}$ because the columns of Q are orthonormal. The columns of Z form an orthonormal basis of $\{\boldsymbol{v} : C\boldsymbol{v} = 0\} = \boldsymbol{q}_1^\perp$, which has dimension $K-1$, so every constrained $\boldsymbol{\beta}$ is $Z\boldsymbol{\beta}^*$ with $\boldsymbol{\beta}^* = Z^\top\boldsymbol{\beta}$. Substituting gives $X_j\boldsymbol{\beta} = X_jZ\boldsymbol{\beta}^*$ and $\boldsymbol{\beta}^\top S_j\boldsymbol{\beta} = \boldsymbol{\beta}^{*\top}Z^\top S_jZ\boldsymbol{\beta}^*$. □

Three consequences. (i) One column is lost: `s(x, k = 10)` has $k' = k-1 = 9$ free coefficients, which is why `gam.check()` reports $k' = 9$. (ii) The columns of X_j^* sum to zero ($\mathbf{1}^\top X_jZ = CZ = \mathbf{0}$), so the intercept is orthogonal to the smooths and absorbs the constant: the fitted $\hat{f}_j$ is centred over the *observed* x_{ij}, the sample version of $\mathrm{E}\{f_j(X_j)\} = 0$. (iii) The penalty S_j^* now has a *one*-dimensional null space (the centred linear function).

```
sm_con <- smoothCon(s(Temp, k = 10), data = aq, absorb.cons = TRUE)[[1]]
X_con <- cbind(1, sm_con$X)
c(columns = ncol(X_con), rank = qr(X_con)$rank)
```

```
#> columns    rank
#>      10      10
```

```
range(colSums(sm_con$X))                          # every column sums to
↪  (numerically) 0
```

```
#> [1] -8.66e-15  5.69e-14
```

```
sum(eigen(sm_con$S[[1]], symmetric = TRUE)$values < 1e-8)   # null space is
↪  now 1-dimensional
```

```
#> [1] 1
```

`mgcv` uses a Householder version of this reparametrisation, so that Z is never formed explicitly. Other constraints work the same way: smooths of two or more covariates (Chapter 18), or a smooth with a `by` variable. After constraining, `coef(m)` contains $\hat{\beta}^*$, not $\hat{\beta}$, which is why the individual coefficients of a smooth are not interpretable; the *curve* is.

17.3 Penalized likelihood and PIRLS

17.3.1 Penalized likelihood

Definition 17.1 (Penalized log-likelihood)**.** For a GLM log-likelihood $\ell(\boldsymbol{\gamma})$ and penalties S_j the **penalized log-likelihood** is

$$\ell_p(\boldsymbol{\gamma}) = \ell(\boldsymbol{\gamma}) - \frac{1}{2}\sum_{j=1}^{p}\lambda_j\boldsymbol{\gamma}^\top S_j\boldsymbol{\gamma} = \ell(\boldsymbol{\gamma}) - \frac{1}{2}\boldsymbol{\gamma}^\top S_\lambda\boldsymbol{\gamma}. \tag{17.2}$$

Each λ_j appears **once**. For Gaussian data, $\ell = -\|\boldsymbol{y} - X\boldsymbol{\gamma}\|^2/(2\sigma^2) + \text{const}$, so maximising ℓ_p means minimising $\|\boldsymbol{y} - X\boldsymbol{\gamma}\|^2 + \sigma^2\sum_j \lambda_j\boldsymbol{\gamma}^\top S_j\boldsymbol{\gamma}$. `mgcv` works on the deviance scale, minimising $D(\boldsymbol{\gamma}) + \sum_j \tilde{\lambda}_j\boldsymbol{\gamma}^\top S_j\boldsymbol{\gamma}$, so the `sp` that it reports is $\tilde{\lambda}_j = \phi\lambda_j$; for binomial and Poisson data ($\phi = 1$) the two scales coincide. To keep the algebra short we assume $\phi = 1$ below; for general ϕ only the scale of λ changes.

17.3.2 Deriving PIRLS

Recall from Chapter 9 that for a GLM with $W = \mathrm{diag}\{1/(V(\mu_i)g'(\mu_i)^2)\}$ and $G = \mathrm{diag}\{g'(\mu_i)\}$ the score and expected information are

$$\frac{\partial \ell}{\partial \boldsymbol{\gamma}} = X^\top W G(\boldsymbol{y} - \boldsymbol{\mu}), \qquad \mathrm{E}\Big[-\frac{\partial^2 \ell}{\partial \boldsymbol{\gamma}\, \partial \boldsymbol{\gamma}^\top}\Big] = X^\top W X.$$

Theorem 17.2 (Penalized iteratively re-weighted least squares)**.** *Fisher scoring for the penalized log-likelihood* (17.2) *gives the update*

$$\boldsymbol{\gamma}^{new} = (X^\top W X + S_\lambda)^{-1} X^\top W \boldsymbol{z} = \arg\min_{\boldsymbol{\gamma}} \; \|W^{1/2}(\boldsymbol{z} - X\boldsymbol{\gamma})\|^2 + \boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}, \tag{17.3}$$

with working response $z_i = \eta_i + g'(\mu_i)(y_i - \mu_i)$*, where* W*,* $\boldsymbol{\mu}$*,* $\boldsymbol{\eta}$ *are evaluated at the current* $\boldsymbol{\gamma}$*.*

Proof. The gradient and expected Hessian of ℓ_p are $X^\top W G(\boldsymbol{y} - \boldsymbol{\mu}) - S_\lambda \boldsymbol{\gamma}$ and $-(X^\top W X + S_\lambda)$. The Fisher scoring step from the current $\boldsymbol{\gamma}$ is

$$\boldsymbol{\gamma}^{\text{new}} = \boldsymbol{\gamma} + (X^\top W X + S_\lambda)^{-1}\{X^\top W G(\boldsymbol{y} - \boldsymbol{\mu}) - S_\lambda \boldsymbol{\gamma}\}.$$

Multiply by $X^\top W X + S_\lambda$ and collect terms: $(X^\top W X + S_\lambda)\boldsymbol{\gamma}^{\text{new}} = (X^\top W X + S_\lambda)\boldsymbol{\gamma} + X^\top W G(\boldsymbol{y} - \boldsymbol{\mu}) - S_\lambda \boldsymbol{\gamma} = X^\top W\{X\boldsymbol{\gamma} + G(\boldsymbol{y} - \boldsymbol{\mu})\} = X^\top W \boldsymbol{z}$, since $X\boldsymbol{\gamma} = \boldsymbol{\eta}$ and $G(\boldsymbol{y} - \boldsymbol{\mu})$ has entries $g'(\mu_i)(y_i - \mu_i)$. The solution of the penalized least-squares problem in (17.3) satisfies the same normal equations. □

Setting $S_\lambda = 0$ gives the IRLS algorithm for GLMs. With the Gaussian family and the identity link, $W = I$ and $\boldsymbol{z} = \boldsymbol{y}$ and one step gives the penalized least-squares solution $(X^\top X + S_\lambda)^{-1} X^\top \boldsymbol{y}$ of Chapter 15 (the iteration then stops at once). For a *canonical* link observed and expected information coincide, so the same update is also Newton's method (Exercise 17.2). Because the penalized deviance $D(\boldsymbol{\gamma}) + \boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}$ is minimised, `mgcv` halves the step whenever it fails to decrease; convergence is declared when the relative change of the penalized deviance is small.

Example 17.1 (PIRLS by hand reproduces gam())**.** We refit the binomial GAM of Chapter 16 for the Pima data, fixing the λ_j at the values `gam()` selected, and iterate (17.3) with the logit link, for which $w_i = \mu_i(1 - \mu_i)$ and $z_i = \eta_i + (y_i - \mu_i)/w_i$.

```
pima <- rbind(Pima.tr, Pima.te)
mb <- gam(type ~ s(glu) + s(bmi) + s(age) + s(ped), family = binomial,
          data = pima, method = "REML")
X <- model.matrix(mb); yb <- as.numeric(pima$type == "Yes")
Sl <- Slam_full(mb)
beta <- rep(0, ncol(X))
for (it in 1:50) {
  eta <- drop(X %*% beta); mu <- plogis(eta)
  w <- mu * (1 - mu); z <- eta + (yb - mu) / w
  beta_new <- drop(solve(crossprod(X, w * X) + Sl, crossprod(X, w * z)))
  done <- max(abs(beta_new - beta)) < 1e-10; beta <- beta_new
  if (done) break
}
c(iterations = it, max_abs_difference = max(abs(beta - coef(mb))))
```

```
#>          iterations max_abs_difference
#>           7.000e+00          1.141e-09
```

Seven iterations from $\boldsymbol{\gamma} = \mathbf{0}$ give the `gam()` coefficients up to about 10^{-9}.

17.4 Effective degrees of freedom

At the converged PIRLS solution $\hat{\boldsymbol{\gamma}} = (X^\top W X + S_\lambda)^{-1} X^\top W \boldsymbol{z}$. If the unpenalized estimate $\tilde{\boldsymbol{\gamma}} = (X^\top W X)^{-1} X^\top W \boldsymbol{z}$ exists then $\hat{\boldsymbol{\gamma}} = F \tilde{\boldsymbol{\gamma}}$ with $F = (X^\top W X + S_\lambda)^{-1} X^\top W X$.

Theorem 17.3 (Effective degrees of freedom)**.** *Let $Z_W = W^{1/2} X$ and suppose $Z_W^\top Z_W + S_\lambda$ is positive definite. Then:*

(a) $\operatorname{tr}(F) = \operatorname{tr}(A)$ *where $A = X(X^\top W X + S_\lambda)^{-1} X^\top W$ is the hat matrix of the working fit;*

(b) the eigenvalues of F lie in $[0, 1]$, and F has eigenvalue exactly 1 on the null space of S_λ. Hence $M \le \operatorname{tr}(F) \le q$.

Proof.

(a) $\operatorname{tr}(F) = \operatorname{tr}\{(X^\top W X + S_\lambda)^{-1} X^\top W \cdot X\} = \operatorname{tr}\{X (X^\top W X + S_\lambda)^{-1} X^\top W\} = \operatorname{tr}(A)$ by $\operatorname{tr}(UV) = \operatorname{tr}(VU)$.

(b) Let $F\boldsymbol{v} = \mu \boldsymbol{v}$, i.e. $Z_W^\top Z_W \boldsymbol{v} = \mu (Z_W^\top Z_W + S_\lambda)\boldsymbol{v}$, so $(1-\mu)\boldsymbol{v}^\top Z_W^\top Z_W \boldsymbol{v} = \mu\, \boldsymbol{v}^\top S_\lambda \boldsymbol{v}$ (for a real eigenvector; F is similar to the symmetric matrix $(Z_W^\top Z_W + S_\lambda)^{-1/2} Z_W^\top Z_W (Z_W^\top Z_W + S_\lambda)^{-1/2}$ so its eigenvalues are real). Both quadratic forms are non-negative, and $\boldsymbol{v}^\top (Z_W^\top Z_W + S_\lambda)\boldsymbol{v} > 0$, so $\mu \in [0, 1]$. If $S_\lambda \boldsymbol{v} = \mathbf{0}$ then $F\boldsymbol{v} = (Z_W^\top Z_W + S_\lambda)^{-1} Z_W^\top Z_W \boldsymbol{v} = (Z_W^\top Z_W + S_\lambda)^{-1} (Z_W^\top Z_W + S_\lambda)\boldsymbol{v} = \boldsymbol{v}$.

□

The **effective degrees of freedom** of the whole model is $\operatorname{tr}(F) = \operatorname{tr}(A)$; the edf of smooth j is the sum of the diagonal entries F_{kk} over the coefficients k of that smooth, and this is the `edf` column of `summary()`. Individual F_{kk} may lie outside $[0, 1]$ even though the eigenvalues cannot (for the airquality model they range from -0.28 to 1.49), but their sums count parameters. The two extreme cases are consistent with Section 17.1: $\lambda \to 0$ gives $F \to I$ and edf $\to q$, while $\lambda \to \infty$ gives edf $\to M$.

```
g3 <- gam(ly ~ s(Solar.R) + s(Wind) + s(Temp), data = aq, method = "REML")
X3 <- model.matrix(g3); S3 <- Slam_full(g3)
F3 <- solve(crossprod(X3) + S3, crossprod(X3))
c(trF = sum(diag(F3)), sum_edf = sum(g3$edf), trA = sum(g3$hat))
```

```
#>     trF sum_edf     trA
#>   7.573   7.573   7.573
```

```
range(diag(F3))
```

```
#> [1] -0.2776  1.4855
```

```
round(range(Re(eigen(F3)$values)), 6)
```

```
#> [1] 0.001398 1.000000
```

```
sapply(g3$smooth, function(sm) sum(diag(F3)[sm$first.para:sm$last.para]))
```

```
#> [1] 2.158 2.462 1.953
```

```
summary(g3)$edf
```

```
#> [1] 2.158 2.462 1.953
```

$\mathrm{tr}(F) = \mathrm{tr}(A) = 7.573$, the total edf of the model, of which one is the intercept; the three smooths use $2.16 + 2.46 + 1.95 = 6.57$. The eigenvalues of F lie between 0 and 1 as predicted.

17.4.1 edf, edf1, edf2 and Ref.df

`mgcv` stores three versions. `edf` is $\sum_k F_{kk}$ as above. `edf1` is $\sum_k (2F - FF)_{kk}$, an alternative that is better suited for testing because it accounts for the fact that the estimate is shrunk; `Ref.df`, the "reference degrees of freedom" reported next to `edf`, is the `edf1` of the smooth. `edf2` also accounts for the uncertainty in $\hat{\boldsymbol{\lambda}}$, and it is what `AIC()` uses for the number of parameters (plus one for the scale), which explains why `AIC(m)` reports more degrees of freedom than $\mathrm{tr}(F)$. `mgcv` caps it: whenever the corrected total would exceed `edf1`, it reports `edf2 = edf1`. This is what happens for the airquality model below (the two columns coincide); the second example, a simulated data set, shows `edf2` strictly between `edf` and `edf1`.

```
c(edf = sum(g3$edf), edf1 = sum(g3$edf1), edf2 = sum(g3$edf2), AIC_df =
↪  attr(logLik(g3), "df"))
```

```
#>    edf   edf1   edf2 AIC_df
#>  7.573  9.289  9.289 10.289
```

```
round(cbind(edf = summary(g3)$edf, Ref.df = summary(g3)$s.table[, "Ref.df"]),
↪  3)
```

```
#>              edf Ref.df
#> s(Solar.R) 2.158  2.719
#> s(Wind)    2.462  3.121
#> s(Temp)    1.953  2.450
```

```
set.seed(1); xs <- runif(100); zs <- runif(100); ys <- sin(2 * pi * xs) + zs +
↪  rnorm(100, sd = 0.5)
gs <- gam(ys ~ s(xs) + s(zs), method = "REML")
c(edf = sum(gs$edf), edf1 = sum(gs$edf1), edf2 = sum(gs$edf2), AIC_df =
↪  attr(logLik(gs), "df"))
```

```
#>    edf   edf1   edf2 AIC_df
#>  8.197  9.373  8.558  9.558
```

17.5 Choosing the smoothing parameters

For a fixed $\boldsymbol{\lambda}$ the model is a penalized GLM. How should $\boldsymbol{\lambda}$ be chosen? The unpenalized likelihood cannot decide: it always prefers $\boldsymbol{\lambda} = \mathbf{0}$ (more freedom never lowers the fit). Two families of criteria exist:

1. **Prediction-error criteria** estimate how well the fit predicts new data: ordinary and generalized cross-validation (OCV, GCV), UBRE (also called Mallows' C_p, Mallows (1973))

and AIC.

2. **Likelihood-of-$\boldsymbol{\lambda}$ criteria** treat the penalty as a prior, integrate $\boldsymbol{\gamma}$ out and maximise the marginal (ML) or restricted (REML) likelihood.

In `mgcv`: `method = "GCV.Cp"` (the **default**: GCV if the scale is unknown, UBRE/C_p if known), `"REML"`, `"ML"`. We derive each in the Gaussian case ($W = I$, $\boldsymbol{z} = \boldsymbol{y}$) with a linear smoother $\hat{\boldsymbol{\mu}} = A\boldsymbol{y}$, $A = X(X^\top X + S_\lambda)^{-1}X^\top$; for GLMs replace the residual sum of squares by the deviance and A by the PIRLS influence matrix.

17.5.1 Cross-validation and GCV

Lemma 17.1 (Leave-one-out shortcut)**.** *For the penalized least-squares smoother $\hat{\boldsymbol{\mu}} = A\boldsymbol{y}$ (any linear smoother obtained by minimising a penalized sum of squares), the residual obtained by* removing *observation i and predicting it is*

$$y_i - \hat{\mu}_i^{[-i]} = \frac{y_i - \hat{\mu}_i}{1 - A_{ii}}.$$

Proof. Let $\hat{\boldsymbol{\gamma}}^{[-i]}$ minimise $\sum_{k \neq i}(y_k - \boldsymbol{x}_k^\top \boldsymbol{\gamma})^2 + \boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}$ and $\hat{\mu}_k^{[-i]} = \boldsymbol{x}_k^\top \hat{\boldsymbol{\gamma}}^{[-i]}$. Define the modified data $\boldsymbol{y}^*$ equal to $\boldsymbol{y}$ except $y_i^* = \hat{\mu}_i^{[-i]}$. For any $\boldsymbol{\gamma}$ the penalized criterion with data $\boldsymbol{y}^*$ equals the leave-one-out criterion plus $(\hat{\mu}_i^{[-i]} - \boldsymbol{x}_i^\top \boldsymbol{\gamma})^2 \geq 0$; $\hat{\boldsymbol{\gamma}}^{[-i]}$ minimises the first part and makes the extra term zero, so it also minimises the criterion with $\boldsymbol{y}^*$. Hence $\hat{\mu}_i^{[-i]} = \sum_k A_{ik} y_k^* = \hat{\mu}_i - A_{ii} y_i + A_{ii}\hat{\mu}_i^{[-i]}$, so $\hat{\mu}_i^{[-i]}(1 - A_{ii}) = \hat{\mu}_i - A_{ii} y_i$ and

$$y_i - \hat{\mu}_i^{[-i]} = \frac{y_i(1 - A_{ii}) - \hat{\mu}_i + A_{ii} y_i}{1 - A_{ii}} = \frac{y_i - \hat{\mu}_i}{1 - A_{ii}}.$$

□

The **ordinary cross-validation** score is therefore computed from a single fit,

$$\mathcal{V}_o(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\mu}_i^{[-i]}\right)^2 = \frac{1}{n}\sum_{i=1}^{n}\frac{(y_i - \hat{\mu}_i)^2}{(1 - A_{ii})^2}.$$

Generalized cross-validation replaces each A_{ii} by the average $\operatorname{tr}(A)/n$:

$$\mathcal{V}_g(\lambda) = \frac{n\,\|\boldsymbol{y} - A\boldsymbol{y}\|^2}{\{n - \operatorname{tr}(A)\}^2}, \tag{17.4}$$

(Craven and Wahba, 1978). GCV needs only the residual sum of squares and the trace of A, it is invariant to rotations of the data (OCV is not), and it avoids the individual leverages A_{ii}, which can be noisy.

```
# brute-force leave-one-out with the design and penalty of the REML fit g3
↪  (fixed lambda)
Xf <- X3; Sf <- S3; yy <- aq$ly
loo <- sapply(1:n, function(i) {
  b <- solve(crossprod(Xf[-i, ]) + Sf, crossprod(Xf[-i, ], yy[-i]))
  yy[i] - sum(Xf[i, ] * b)
})
A3 <- Xf %*% solve(crossprod(Xf) + Sf, t(Xf))
short <- (yy - A3 %*% yy) / (1 - diag(A3))
max(abs(loo - short))                                  # Lemma: identical
```

```
#> [1] 8.965e-15
```

```
c(OCV = mean(loo^2), GCV = n * sum((yy - A3 %*% yy)^2) / (n -
↪  sum(diag(A3)))^2)
```

```
#>     OCV     GCV
#> 0.2605 0.2509
```

The shortcut reproduces brute-force leave-one-out to machine precision, and GCV is within about 4% of OCV here.

Remark. GCV and `method = "GCV.Cp"` in `mgcv` use the same formula (17.4); the score `gam(..., method = "GCV.Cp")$gcv.ubre` equals it at the optimum.

17.5.2 UBRE / Mallows' C_p

When σ^2 is known, one can estimate the *risk* of the fit directly.

Theorem 17.4 (UBRE is an unbiased risk estimate)**.** *Let $\boldsymbol{\mu} = \mathrm{E}(\boldsymbol{y})$, $\mathrm{Var}(\boldsymbol{y}) = \sigma^2 I$ and $\hat{\boldsymbol{\mu}} = A\boldsymbol{y}$ with A fixed. The risk is $R(\lambda) = \frac{1}{n}\mathrm{E}\|\boldsymbol{\mu} - A\boldsymbol{y}\|^2$. Then*

$$\mathcal{V}_u(\lambda) = \frac{1}{n}\|\boldsymbol{y} - A\boldsymbol{y}\|^2 - \sigma^2 + \frac{2\sigma^2}{n}\operatorname{tr}(A) \quad \textit{satisfies} \quad \mathrm{E}\{\mathcal{V}_u(\lambda)\} = R(\lambda).$$

Proof. $\mathrm{E}(A\boldsymbol{y}) = A\boldsymbol{\mu}$ and $\mathrm{Var}(A\boldsymbol{y}) = \sigma^2 AA^\top$, so $R = \frac{1}{n}\{\|(I-A)\boldsymbol{\mu}\|^2 + \sigma^2\operatorname{tr}(A^\top A)\}$. Similarly $\mathrm{E}\|\boldsymbol{y} - A\boldsymbol{y}\|^2 = \|(I-A)\boldsymbol{\mu}\|^2 + \sigma^2\operatorname{tr}\{(I-A)^\top(I-A)\} = \|(I-A)\boldsymbol{\mu}\|^2 + \sigma^2\{n - 2\operatorname{tr}(A) + \operatorname{tr}(A^\top A)\}$. Therefore $\mathrm{E}\{\frac{1}{n}\|\boldsymbol{y} - A\boldsymbol{y}\|^2\} = R + \sigma^2 - \frac{2\sigma^2}{n}\operatorname{tr}(A)$, which rearranges to $\mathrm{E}(\mathcal{V}_u) = R$. □

For GLMs with known scale (binomial, Poisson) `mgcv` uses the deviance analogue $\mathcal{V}_u = D(\hat{\boldsymbol{\gamma}})/n - \phi + 2\phi\operatorname{tr}(A)/n$ with $\phi = 1$; this is why the default is called `"GCV.Cp"`: GCV when ϕ is unknown, UBRE/C_p when it is known.

17.5.3 The penalty as a prior

Take the (improper) Gaussian prior with precision proportional to the penalty,

$$\pi(\boldsymbol{\gamma} \mid \boldsymbol{\lambda}) \propto \exp\Big(-\frac{1}{2\sigma^2}\boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}\Big) \qquad (\boldsymbol{\gamma} \sim N(\mathbf{0}, \sigma^2 S_\lambda^-)).$$

Then the log posterior is $\ell(\boldsymbol{\gamma}) - \frac{1}{2\sigma^2}\boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma} + \text{const}$: **the penalized MLE is the posterior mode**. Directions in the null space of S_λ (intercept, linear parts) get a flat prior: they behave like *fixed effects*. The penalized directions behave like *random effects* with variance proportional to $1/\lambda_j$ (Chapter 19). Prior and penalty are two descriptions of the same smoothness assumption, and the **marginal likelihood**

$$\pi(\boldsymbol{y} \mid \boldsymbol{\lambda}, \sigma^2) = \int f(\boldsymbol{y} \mid \boldsymbol{\gamma}, \sigma^2)\, \pi(\boldsymbol{\gamma} \mid \boldsymbol{\lambda}, \sigma^2)\, d\boldsymbol{\gamma}$$

measures how well $\boldsymbol{\lambda}$ explains the data after averaging over all curves that the prior allows. Choosing $\boldsymbol{\lambda}$ by maximising it is the analogue of estimating variance components in a mixed model (Chapter 12).

17.5.4 REML for the Gaussian model

Let M be the dimension of the null space of S_λ and $|S_\lambda|_+$ the product of its positive eigenvalues. A proper normalising constant for the prior on the $q-M$ penalized directions (with Lebesgue measure on the M unpenalized ones) is

$$\pi(\boldsymbol{\gamma} \mid \boldsymbol{\lambda}, \sigma^2) = (2\pi\sigma^2)^{-(q-M)/2}\, |S_\lambda|_+^{1/2} \exp\!\Big(-\frac{1}{2\sigma^2}\boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}\Big).$$

Theorem 17.5 (REML criterion for the Gaussian additive model)**.** *With* $\hat{\boldsymbol{\gamma}} = (X^\top X + S_\lambda)^{-1} X^\top \boldsymbol{y}$ *and* $\hat{P} = \|\boldsymbol{y} - X\hat{\boldsymbol{\gamma}}\|^2 + \hat{\boldsymbol{\gamma}}^\top S_\lambda \hat{\boldsymbol{\gamma}}$,

$$-2\log \pi(\boldsymbol{y} \mid \boldsymbol{\lambda}, \sigma^2) = \frac{\hat{P}}{\sigma^2} + (n-M)\log(2\pi\sigma^2) + \log|X^\top X + S_\lambda| - \log|S_\lambda|_+. \tag{17.5}$$

Minimising over σ^2 *gives* $\hat{\sigma}^2 = \hat{P}/(n-M)$.

Proof. Complete the square in $\boldsymbol{\gamma}$: $\|\boldsymbol{y} - X\boldsymbol{\gamma}\|^2 + \boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma} = \hat{P} + (\boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}})^\top (X^\top X + S_\lambda)(\boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}})$. Hence

$$\pi(\boldsymbol{y} \mid \boldsymbol{\lambda}, \sigma^2) = (2\pi\sigma^2)^{-n/2} (2\pi\sigma^2)^{-(q-M)/2} |S_\lambda|_+^{1/2} e^{-\hat{P}/(2\sigma^2)} \int e^{-\frac{1}{2\sigma^2}(\boldsymbol{\gamma}-\hat{\boldsymbol{\gamma}})^\top (X^\top X + S_\lambda)(\boldsymbol{\gamma}-\hat{\boldsymbol{\gamma}})} d\boldsymbol{\gamma}.$$

The integral of a Gaussian kernel is $(2\pi\sigma^2)^{q/2} |X^\top X + S_\lambda|^{-1/2}$. Collecting powers of $2\pi\sigma^2$: $-n/2 - (q-M)/2 + q/2 = -(n-M)/2$. Taking $-2\log$ gives (17.5). The derivative with respect to σ^2 is $-\hat{P}/\sigma^4 + (n-M)/\sigma^2 = 0$. □

This is the restricted likelihood: the M unpenalized coefficients have been *integrated out* with a flat prior, which is why $n-M$ appears instead of n. The term $\log|X^\top X + S_\lambda| - \log|S_\lambda|_+$ is a complexity penalty that grows when λ is small: with small λ the posterior is wide in the wiggly directions, and a wide posterior relative to the prior means that many curves explain the data equally well. We check the formula against `gam()`:

```
reml_ml <- function(g, y) {
  X <- model.matrix(g); S <- Slam_full(g); b <- coef(g); n <- length(y)
  P <- sum((y - X %*% b)^2) + drop(t(b) %*% S %*% b)
  eg <- eigen(S, symmetric = TRUE); pos <- eg$values > max(eg$values) * 1e-10;
↪  M <- sum(!pos)
  H <- crossprod(X) + S; Um <- eg$vectors[, pos]; Hpp <- crossprod(Um, H %*%
↪  Um)
  ldS <- sum(log(eg$values[pos])); ldH <- determinant(H)$modulus; ldHpp <-
↪  determinant(Hpp)$modulus
  s2R <- P / (n - M); s2M <- P / n
  c(M = M, sigma2_REML = s2R, sigma2_ML = s2M,
    REML = 0.5 * (P / s2R + (n - M) * log(2 * pi * s2R) + ldH - ldS),
    ML   = 0.5 * (P / s2M + n * log(2 * pi * s2M) + ldHpp - ldS))
}
fR <- ly ~ s(Solar.R) + s(Wind) + s(Temp)
gR <- gam(fR, data = aq, method = "REML"); gM <- gam(fR, data = aq, method =
↪  "ML")
round(rbind(REML_fit = reml_ml(gR, aq$ly), ML_fit = reml_ml(gM, aq$ly)), 4)
```

```
#>           M sigma2_REML sigma2_ML  REML    ML
#> REML_fit 4      0.2338    0.2253 86.02 80.92
#> ML_fit   4      0.2467    0.2378 86.98 79.66
```

```
c(mgcv_REML_fit = gR$gcv.ubre, mgcv_ML_fit = gM$gcv.ubre)
```

```
#> mgcv_REML_fit.REML     mgcv_ML_fit.ML
#>              86.02              79.66
```

For the airquality model $M = 4$ (intercept and the linear part of each of the three smooths). The hand-computed REML score of the REML fit, 86.0236, is the `-REML` that `summary()` printed in Chapter 16, and the hand-computed ML score of the ML fit agrees with `gam()`'s too (the formula for ML is derived next).

17.5.5 ML versus REML

Write the model in mixed-model form: $X\boldsymbol{\gamma} = X_0\boldsymbol{\alpha} + X_+\boldsymbol{\delta}$, with $\boldsymbol{\alpha} \in \mathbb{R}^M$ the coordinates in the null space of S_λ ("fixed effects") and $\boldsymbol{\delta}$ the penalized coordinates with prior covariance $\sigma^2 S_+^{-1}$ ("random effects", S_+ the positive part of the penalty in these coordinates). Then $\boldsymbol{y} \sim N(X_0\boldsymbol{\alpha}, \sigma^2 V)$ with $V = I + X_+ S_+^{-1} X_+^\top$. Two ways of dealing with $\boldsymbol{\alpha}$:

- **ML** replaces $\boldsymbol{\alpha}$ by its estimate: $-2\ell_{ML} = \hat{P}/\sigma^2 + n\log(2\pi\sigma^2) + \log|V|$;
- **REML** integrates it out with a flat prior: $-2\ell_R = \hat{P}/\sigma^2 + (n-M)\log(2\pi\sigma^2) + \log|V| + \log|X_0^\top V^{-1} X_0|$.

The determinant lemma gives $\log|V| = \log|H_{++}| - \log|S_+|$ with $H_{++} = X_+^\top X_+ + S_+$, and a Schur-complement argument gives $|H| = |H_{++}|\,|X_0^\top V^{-1} X_0|$ for $H = X^\top X + S_\lambda$ in these coordinates. Substituting recovers (17.5) for REML (the terms in $|H_{++}|$ cancel) and shows that **ML** differs by replacing $n - M$ by n and $\log|H|$ by $\log|H_{++}|$, the determinant over the penalized block only. This is the formula used in the chunk above, and its agreement with `gam()` confirms it.

As in linear mixed models (Chapter 12), ML ignores the degrees of freedom used up by the estimated fixed effects and therefore estimates the variance too small ($\hat{P}/n$ instead of $\hat{P}/(n-M)$ above, 0.225 against 0.234). The coefficients of the penalized directions have variance proportional to $1/\lambda_j$; a smaller estimated variance points towards a *larger* $\hat{\lambda}_j$. This is a heuristic, not a proof: the fit and the other variance estimates change too.

ML tends to give slightly smoother fits than REML (lower edf), sometimes collapsing a term to its null space; the difference is usually small (see the simulation in Section 17.5.7). ML is for comparing models that differ in their unpenalized terms; REML is for estimating $\boldsymbol{\lambda}$.

17.5.6 Laplace approximation for non-Gaussian data

For GLMs the integral is not available in closed form. Expand ℓ_p to second order around its maximiser $\hat{\boldsymbol{\gamma}}$ with $H = X^\top W X + S_\lambda$ ($\phi = 1$): $\ell_p(\boldsymbol{\gamma}) \approx \ell_p(\hat{\boldsymbol{\gamma}}) - \frac{1}{2}(\boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}})^\top H (\boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}})$ (Tierney and Kadane, 1986). With the prior normalisation of Section 17.5.4,

$$\pi(\boldsymbol{y} \mid \boldsymbol{\lambda}) = \int e^{\ell(\gamma)} \pi(\boldsymbol{\gamma} \mid \boldsymbol{\lambda})\, d\boldsymbol{\gamma} \approx e^{\ell_p(\hat{\boldsymbol{\gamma}})} (2\pi)^{-(q-M)/2} |S_\lambda|_+^{1/2} (2\pi)^{q/2} |H|^{-1/2},$$

so the Laplace approximate REML (LAML) score of Wood (2011) is

$$\mathcal{V}_{LAML}(\boldsymbol{\lambda}) = -\ell_p(\hat{\boldsymbol{\gamma}}) + \tfrac{1}{2}\log|H| - \tfrac{1}{2}\log|S_\lambda|_+ - \tfrac{M}{2}\log(2\pi). \tag{17.6}$$

For Gaussian data the quadratic expansion is exact, and (17.6) reduces to (17.5) (with σ^2 fixed). `mgcv` minimises $\mathcal{V}_{LAML}$ with respect to $\log \lambda_j$ by Newton's method (the "outer" iteration) while PIRLS is the "inner" iteration, with analytic derivatives and a stable QR-based evaluation (Wood, 2011). We check (17.6) for the Pima GAM:

```
mu <- plogis(drop(X %*% coef(mb)))
ll <- sum(dbinom(yb, 1, mu, log = TRUE))
H  <- crossprod(X, (mu * (1 - mu)) * X) + Sl
eg <- eigen(Sl, symmetric = TRUE); pos <- eg$values > max(eg$values) * 1e-10;
↪  M <- sum(!pos)
V_laml <- -(ll - 0.5 * drop(t(coef(mb)) %*% Sl %*% coef(mb))) + 0.5 *
↪  determinant(H)$modulus -
  0.5 * sum(log(eg$values[pos])) - M / 2 * log(2 * pi)
c(by_hand = as.numeric(V_laml), mgcv = mb$gcv.ubre, M = M)
```

```
#>   by_hand mgcv.REML         M
#>     234.5     234.5       5.0
```

17.5.7 Which criterion?

- **GCV/UBRE** have good asymptotic prediction properties, but the score is often flat or has several local minima, and in small samples it tends to **undersmooth**: occasionally it picks a very small λ (Wood, 2011).
- **REML** has a better-defined optimum, more stable estimates and is less prone to undersmoothing; it uses the same machinery as mixed models.
- **ML** is needed for likelihood comparisons of models that differ in their unpenalized parts; it tends to give slightly smoother fits than REML.

The following simulation illustrates the difference. We generate $n = 60$ points from a curve with a sharp peak, $f(x) = 0.2x^{11}\{10(1-x)\}^6 + 10(10x)^3(1-x)^{10}$ (a test function of Wood), noise sd 2, and fit `s(x, k = 20)` with the three criteria 200 times.

```
f0fun <- function(x) 0.2 * x^11 * (10 * (1 - x))^6 + 10 * (10 * x)^3 * (1 -
↪  x)^10
set.seed(5); nrep <- 200; n_s <- 60; xg <- data.frame(x = seq(0, 1, length.out
↪  = 200))
sim <- do.call(rbind, lapply(1:nrep, function(i) {
  x <- runif(n_s); y <- f0fun(x) + rnorm(n_s, sd = 2)
  do.call(rbind, lapply(c(GCV = "GCV.Cp", REML = "REML", ML = "ML"),
  ↪  function(m) {
    g <- gam(y ~ s(x, k = 20), method = m)
    data.frame(edf = sum(g$edf) - 1, mse = mean((predict(g, xg) -
    ↪  f0fun(xg$x))^2))
  })) -> r
  r$method <- rownames(r); r$rep <- i; r
}))
sim$method <- factor(sim$method, levels = c("GCV", "REML", "ML"))
tab <- do.call(rbind, lapply(split(sim, sim$method), function(d)
  data.frame(edf_mean = mean(d$edf), edf_q90 = quantile(d$edf, .9), edf_max =
  ↪  max(d$edf),
             share_edf_above_12 = mean(d$edf > 12), mse_mean = mean(d$mse),
             ↪  mse_q90 = quantile(d$mse, .9))))
round(tab, 3)
```

```
#>      edf_mean edf_q90 edf_max share_edf_above_12 mse_mean mse_q90
#> GCV     7.371  10.337  17.337               0.06    0.726   1.164
#> REML    6.688   7.673   9.103               0.00    0.642   1.029
#> ML      6.476   7.446   8.916               0.00    0.654   1.080
```

```
ggplot(sim, aes(method, edf, fill = method)) + geom_boxplot(show.legend =
↪  FALSE) +
  scale_fill_manual(values = c(ie_coral, ie_navy, ie_green)) + labs(x = NULL,
  ↪  y = "edf of s(x)")
```

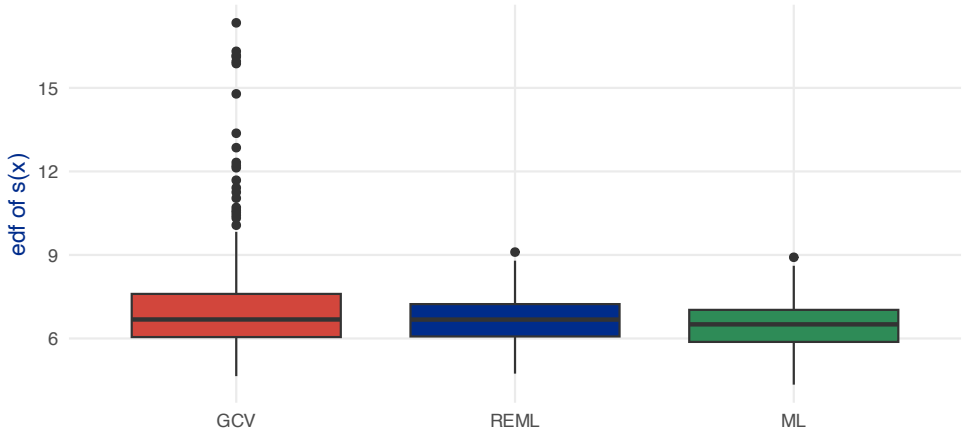


Figure 17.2: Effective degrees of freedom of 200 fits with each smoothing-parameter criterion. GCV occasionally selects a very wiggly fit (edf above 12); REML and ML do not.

On average GCV gives a somewhat larger mean squared error, but the main difference is in the tail: in 6% of the replicates GCV chose a fit with more than 12 edf (up to 17.3), whereas REML and ML never exceeded 10. In these replicates ML had a lower edf than REML in 100% of the data sets, but the mean edf differs little: 6.48 (ML) against 6.69 (REML). So the ML–REML effect is systematic but usually small, in contrast with the GCV tail. This is the undersmoothing of GCV; it is the reason for our **course rule**: fit with `method = "REML"`, and switch to `"ML"` only to compare models that differ in their parametric terms.

The three criteria as functions of $\log \lambda$ for the single smooth `s(Temp)` show the same story: the GCV curve has a pronounced minimum at a small λ, the REML curve is flat near its optimum, the ML curve is minimised at $\hat{\lambda} = \infty$.

```
Xt <- model.matrix(f0); St <- f0$smooth[[1]]$S[[1]]; pt <- ncol(Xt); yt <-
↪  aq$ly
crit <- t(sapply(seq(-7, 14, by = 0.2), function(L) {
  Sl_ <- matrix(0, pt, pt); Sl_[2:pt, 2:pt] <- exp(L) * St
  Hm <- crossprod(Xt) + Sl_; b <- solve(Hm, crossprod(Xt, yt)); rss <- sum((yt
↪  - Xt %*% b)^2)
  P <- rss + drop(t(b) %*% Sl_ %*% b); trA <- sum(diag(solve(Hm,
↪  crossprod(Xt))))
  ev <- eigen(Sl_, symmetric = TRUE, only.values = TRUE)$values; pos <- ev >
↪  max(ev) * 1e-10; Mt <- sum(!pos)
  Um <- eigen(Sl_, symmetric = TRUE)$vectors[, pos]; Hpp <- crossprod(Um, Hm
↪  %*% Um)
  s2R <- P / (n - Mt); s2M <- P / n
```

```
  c(L = L, GCV = n * rss / (n - trA)^2,
    REML = 0.5 * (P / s2R + (n - Mt) * log(2 * pi * s2R) +
     ↪  determinant(Hm)$modulus - sum(log(ev[pos]))),
    ML = 0.5 * (P / s2M + n * log(2 * pi * s2M) + determinant(Hpp)$modulus -
     ↪  sum(log(ev[pos]))))
}))
cd <- as.data.frame(crit)
dd <- rbind(data.frame(L = cd$L, score = cd$GCV / min(cd$GCV), crit = "GCV
↪  (relative to minimum)"),
            data.frame(L = cd$L, score = cd$REML - min(cd$REML), crit = "REML
            ↪  (minus minimum)"),
            data.frame(L = cd$L, score = cd$ML - min(cd$ML), crit = "ML (minus
            ↪  minimum)"))
ggplot(dd, aes(L, score)) + geom_line(colour = ie_navy, linewidth = 1) +
  facet_wrap(~crit, scales = "free_y") + labs(x = expression(log(lambda)), y =
  ↪  NULL)
```

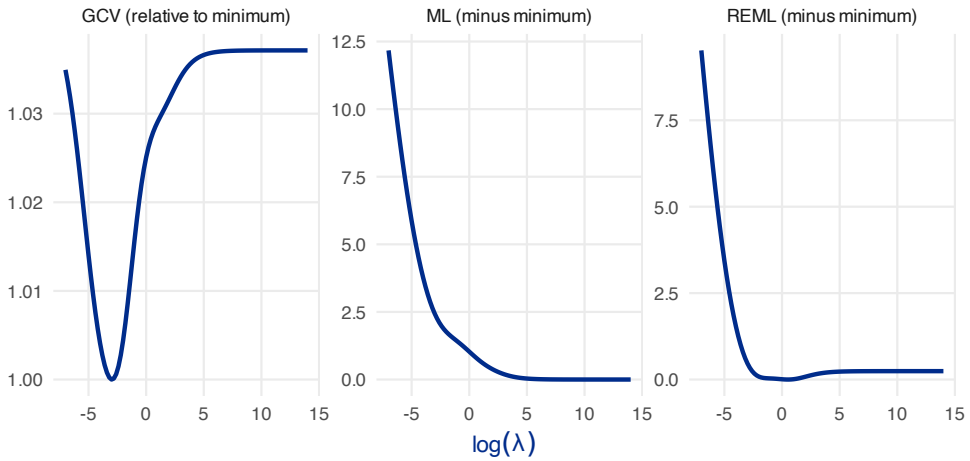


Figure 17.3: GCV, REML and ML scores for log(Ozone) s(Temp) as functions of log(lambda).

```
c(GCV = cd$L[which.min(cd$GCV)], REML = cd$L[which.min(cd$REML)], ML =
↪  cd$L[which.min(cd$ML)])
```

```
#>  GCV REML   ML
#> -3.0  0.6 14.0
```

17.6 The Bayesian view: posterior and credible intervals

Theorem 17.6 (Posterior distribution of the coefficients)**.** *In the Gaussian model with the prior of Section 17.5.3, $\boldsymbol{\gamma} \mid \boldsymbol{y} \sim N(\hat{\boldsymbol{\gamma}}, V_\gamma)$ with $V_\gamma = (X^\top X + S_\lambda)^{-1}\sigma^2$ and $\hat{\boldsymbol{\gamma}}$ the penalized least-squares estimate. For a GLM the same holds approximately: $\boldsymbol{\gamma} \mid \boldsymbol{y} \approx N(\hat{\boldsymbol{\gamma}}, (X^\top W X + S_\lambda)^{-1}\phi)$.*

Proof. By the completion of the square in the proof of Theorem 17.5, the posterior density is proportional to $\exp\{-\frac{1}{2\sigma^2}(\boldsymbol{\gamma}-\hat{\boldsymbol{\gamma}})^\top (X^\top X + S_\lambda)(\boldsymbol{\gamma}-\hat{\boldsymbol{\gamma}})\}$. For GLMs, apply the quadratic (Laplace) expansion of ℓ_p at its mode, whose Hessian is $X^\top W X/\phi + S_\lambda$. □

`vcov(m)` returns V_γ (in `mgcv` it is called `Vp`). Any linear functional $\boldsymbol{c}^\top\boldsymbol{\gamma}$ has posterior variance $\boldsymbol{c}^\top V_\gamma \boldsymbol{c}$: the value of a smooth at x_0 is $\boldsymbol{c}$ = the row of the `lpmatrix` at x_0; a difference of predictions is the difference of rows. This is what `plot.gam()` and `predict(se.fit = TRUE)` use. The *frequentist* covariance of $\hat{\boldsymbol{\gamma}}$, FV_γ (`m$Ve`), ignores the shrinkage bias and gives intervals that are too narrow.

```
nd <- data.frame(Solar.R = 207, Wind = 9.7, Temp = c(70, 90))
Xp <- predict(g3, nd, type = "lpmatrix")
se_hand <- sqrt(rowSums((Xp %*% vcov(g3)) * Xp))
rbind(by_hand = se_hand, predict = predict(g3, nd, se.fit = TRUE)$se.fit)
```

```
#>              1      2
#> by_hand 0.1021 0.1131
#> predict 0.1021 0.1131
```

17.6.1 What does "95%" mean?

A frequentist confidence interval for $f(x_0)$ cannot have exact coverage at every x_0, because $\hat{f}$ is biased by the smoothing. The Bayesian interval has a different, *average* property.

Proposition 17.3 (Average coverage of the Bayesian band (Nychka, 1988))**.** *Suppose f is random with the smoothness prior $\boldsymbol{f} \sim N(\boldsymbol{0}, \sigma^2\Sigma)$, $\boldsymbol{y} = \boldsymbol{f} + \boldsymbol{\varepsilon}$ with $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$ and $A = \Sigma(\Sigma + I)^{-1}$ the (posterior mean) smoother. Then the mean squared error matrix of $\hat{\boldsymbol{f}} = A\boldsymbol{y}$* averaged over the prior *equals the posterior covariance, $\sigma^2 A$.*

Proof. $\hat{\boldsymbol{f}} - \boldsymbol{f} = (A - I)\boldsymbol{f} + A\boldsymbol{\varepsilon}$, so the covariance is $\sigma^2\{(A-I)\Sigma(A-I)^\top + AA^\top\}$. Here $A - I = -(\Sigma + I)^{-1}$, so the first term is $\Sigma(\Sigma + I)^{-2}$ and $AA^\top = \Sigma^2(\Sigma + I)^{-2}$ (all matrices commute). The sum is $\Sigma(\Sigma + I)^{-2}(I + \Sigma) = \Sigma(\Sigma + I)^{-1} = A$. □

So the squared error that a Bayesian band takes into account, $\sigma^2 A_{ii}$, is the *sum* of the variance $\sigma^2(AA^\top)_{ii}$ and the squared bias of the smoother on average over plausible curves. The interval therefore has about the right coverage **averaged across the function**, but not at every point: slightly too wide where f is smooth, too narrow where it is very wiggly (Marra and Wood, 2012). Our simulation of Section 17.5.7 shows this: 200 data sets with $n = 100$ from the peaked curve, `predict()` bands for the whole fit (including the intercept):

```
set.seed(6)
cover <- replicate(200, {
  x <- runif(100); y <- f0fun(x) + rnorm(100, sd = 1.5)
  g <- gam(y ~ s(x, k = 20), method = "REML")
  p <- predict(g, xg, se.fit = TRUE)
  as.numeric(abs(p$fit - f0fun(xg$x)) <= 1.96 * p$se.fit)
})
c(average_over_function = mean(cover))
```

```
#> average_over_function
#>                0.9555
```

```
round(quantile(rowMeans(cover), c(0, .05, .5, .95, 1)), 3)
```

```
#>    0%    5%   50%   95%  100%
#> 0.830 0.885 0.965 0.985 0.990
```

Average coverage is close to 95%, but pointwise coverage varies from about 83% to 99%.

17.6.2 Pointwise versus simultaneous bands; uncertainty in λ

The band $\hat{f}_j(x) \pm 1.96\sqrt{\boldsymbol{c}_x^\top V_\gamma \boldsymbol{c}_x}$ is **pointwise**: for each x separately it has the stated coverage. A whole curve can leave the band somewhere. To get a simultaneous band, simulate curves from the posterior, $\boldsymbol{\gamma}^{(b)} \sim N(\hat{\boldsymbol{\gamma}}, V_\gamma)$, and use the maximum of the standardised deviation $\max_x |f^{(b)}(x) - \hat{f}(x)|/\mathrm{se}(x)$.

```
wg <- seq(min(aq$Wind), max(aq$Wind), length.out = 100)
ndw <- data.frame(Solar.R = 207, Wind = wg, Temp = 79)
Xw <- predict(g3, ndw, type = "lpmatrix")
idx <- g3$smooth[[2]]$first.para:g3$smooth[[2]]$last.para
set.seed(2024)
br <- rmvn(1000, coef(g3), vcov(g3))
fsim <- Xw[, idx] %*% t(br[, idx]); fhat <- drop(Xw[, idx] %*% coef(g3)[idx])
se <- sqrt(rowSums((Xw[, idx] %*% vcov(g3)[idx, idx]) * Xw[, idx]))
inside <- abs(fsim - fhat) <= 1.96 * se
c(pointwise = mean(inside), whole_curve = mean(apply(inside, 2, all)))
```

```
#>   pointwise whole_curve
#>      0.9474      0.7030
```

```
crit_sim <- quantile(apply(abs(fsim - fhat) / se, 2, max), .95)
crit_sim                                                    #
↪  simultaneous multiplier (instead of 1.96)
```

```
#>   95%
#> 2.811
```

```
matplot(wg, fhat + fsim[, 1:25] - fhat, type = "l", lty = 1, col = "grey60",
        xlab = "Wind (mph)", ylab = "s(Wind), centred")
lines(wg, fhat, lwd = 3, col = ie_navy)
```

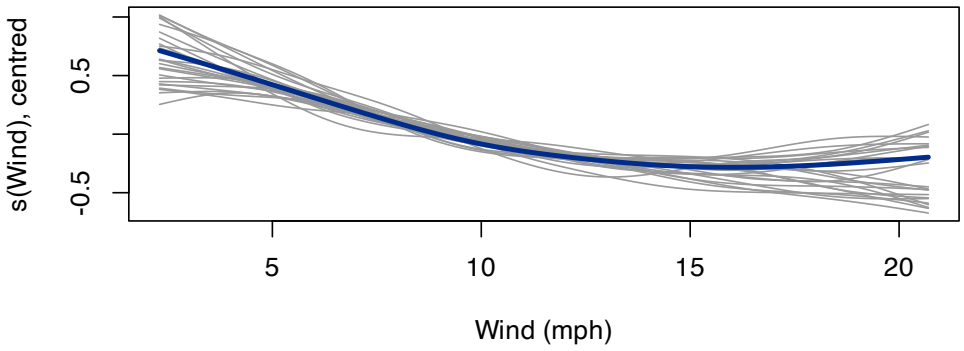


Figure 17.4: 25 curves drawn from the posterior of s(Wind) in the airquality model.

About 95% of the posterior draws are inside the pointwise band at any given x, but only about 70% of whole curves stay inside everywhere; the simultaneous multiplier is 2.81 instead of 1.96.

The covariance V_γ treats $\hat{\boldsymbol{\lambda}}$ as known. `vcov(m, unconditional = TRUE)` adds the uncertainty in $\log \hat{\boldsymbol{\lambda}}$ (available for REML and ML fits) by a first-order correction; it makes the standard errors larger.

```
idw <- idx
r <- sqrt(diag(vcov(g3, unconditional = TRUE)))[idw] /
↪  sqrt(diag(vcov(g3)))[idw]
round(range(r), 3)
```

```
#> [1] 1.125 1.286
```

For the coefficients of `s(Wind)` the standard errors grow by 12% to 29% once the uncertainty of $\hat{\lambda}$ is acknowledged, a reminder that the conditional bands are somewhat optimistic.

17.7 Approximate p-values for smooth terms

We test $H_0 : f_j \equiv 0$. Let $\hat{\boldsymbol{f}}_j = X_j \hat{\boldsymbol{\beta}}_j$ be the fitted values of the smooth at the data and $V_{f_j} = X_j V_{\beta_j} X_j^\top$ its posterior covariance (the corresponding sub-block of V_γ). Because $\hat{\boldsymbol{f}}_j$ is approximately normal, a Wald-type statistic is

$$T_j = \hat{\boldsymbol{f}}_j^\top V_{f_j}^{r-} \hat{\boldsymbol{f}}_j \overset{H_0}{\approx} \chi^2_r \qquad \text{(an } F \text{ statistic } T_j/r \text{ when } \phi \text{ is estimated)},$$

where V^{r-} is the **rank-r pseudo-inverse**: the inverse of V_{f_j} restricted to its r leading eigenvectors. We cannot use the generalised inverse of full rank because the penalty shrinks the directions with tiny eigenvalues and the statistic would be dominated by them. Wood (2013) chooses r from the effective degrees of freedom (non-integer values are handled by interpolating between the two neighbouring integer ranks), and the resulting reference degrees of freedom are the `Ref.df` of `summary()`. We can approximate the statistic by hand with the integer rank closest to edf:

```
sb <- summary(mb)
Vp <- vcov(mb); bb <- coef(mb)
tab <- t(sapply(1:4, function(j) {
  jj <- mb$smooth[[j]]$first.para:mb$smooth[[j]]$last.para
  Xj <- X[, jj]; fj <- Xj %*% bb[jj]
  ev <- eigen(Xj %*% Vp[jj, jj] %*% t(Xj), symmetric = TRUE)
  r <- round(sb$edf[j])
  c(edf = sb$edf[j], rank = r, T_by_hand = sum((t(ev$vectors[, 1:r, drop =
  ↪  FALSE]) %*% fj)^2 / ev$values[1:r]),
    Chi.sq_mgcv = sb$s.table[j, "Chi.sq"])
}))
rownames(tab) <- rownames(sb$s.table); round(tab, 2)
```

```
#>          edf rank T_by_hand Chi.sq_mgcv
#> s(glu) 1.00    1     66.24       66.24
#> s(bmi) 3.21    3     19.62       20.25
#> s(age) 3.58    4     35.13       36.47
#> s(ped) 1.89    2     13.41       13.31
```

The simple version reproduces the order of magnitude of `Chi.sq` (and almost exactly for the near-linear terms); the differences for `s(bmi)` and `s(age)` come from the fractional rank. Three caveats.

- The test is computed **as if $\hat{\boldsymbol{\lambda}}$ were known**, so p-values are somewhat too small for truly zero or weak effects.
- It works best with REML/ML smoothness selection; with GCV the calibration is worse.

- Use the p-values to screen, not to decide at the margin (for example $p \approx 0.05$).

We check the calibration by simulation: a covariate with no effect on a Gaussian response, $n = 100$, 500 replicates.

```
set.seed(17)
pv_reml <- replicate(500, { x <- runif(100); yy <- rnorm(100)
  summary(gam(yy ~ s(x), method = "REML"))$s.table[1, "p-value"] })
pv_gcv  <- replicate(500, { x <- runif(100); yy <- rnorm(100)
  summary(gam(yy ~ s(x)))$s.table[1, "p-value"] })
rbind(REML = c(`p<0.05` = mean(pv_reml < .05), `p<0.01` = mean(pv_reml <
↪  .01)),
      GCV  = c(`p<0.05` = mean(pv_gcv < .05),  `p<0.01` = mean(pv_gcv < .01)))
```

```
#>       p<0.05 p<0.01
#> REML  0.046  0.010
#> GCV   0.062  0.008
```

With REML the test is close to its nominal level; with GCV it rejects somewhat too often (here 6.2% at the 5% level). Parametric terms in the model are tested with ordinary Wald t or z tests based on V_γ.

17.8 The basis dimension k and `gam.check()`

The basis dimension k sets the *maximum* flexibility of a smooth; the penalty decides how much of it is used. If k is too small the fit is oversmoothed whatever the value of λ (edf near $k' = k - 1$); if it is larger than necessary the cost is only computation. The practical rule: choose k generously, then check.

`gam.check()` computes the **k-index**, a residual-based statistic. Order the residuals according to the covariate (or by nearest neighbours when there are several covariates) and compare the squared differences between neighbouring residuals with the overall residual variance:

$$\text{k-index} = \frac{\frac{1}{2(n-1)} \sum_i (\hat\varepsilon_{(i+1)} - \hat\varepsilon_{(i)})^2}{\frac{1}{n}\sum_i \hat\varepsilon_i^2}.$$

If the residuals are noise along x, the numerator estimates σ^2, the denominator too, and the ratio is about 1. If the fit has missed structure, neighbouring residuals are positively correlated, their differences are small relative to their size, and the k-index falls below 1. The p-value comes from randomly permuting the residuals, hence the need for `set.seed()`.

> **Example 17.2** (k too small)**.** We simulate $y = 2\sin(4\pi x) + \varepsilon$, $\varepsilon \sim N(0, 0.5^2)$, $n = 200$, and fit `s(x, k = 5)` and `s(x, k = 20)`.

```
set.seed(24)
xs <- runif(200); ys <- 2 * sin(4 * pi * xs) + rnorm(200, sd = .5); ds <-
↪  data.frame(x = xs, y = ys)
k5  <- gam(y ~ s(x, k = 5),  data = ds, method = "REML")
k20 <- gam(y ~ s(x, k = 20), data = ds, method = "REML")
set.seed(1); k.check(k5)
```

```
#>      k'   edf k-index p-value
#> s(x)  4 3.861  0.1919       0
```

```
set.seed(1); k.check(k20)
```

```
#>      k'   edf k-index p-value
#> s(x) 19 13.88   1.146  0.9775
```

```
xx <- data.frame(x = seq(0, 1, length.out = 200))
dk <- rbind(data.frame(x = xx$x, fit = predict(k5, xx), k = "k = 5"),
            data.frame(x = xx$x, fit = predict(k20, xx), k = "k = 20"))
ggplot(dk, aes(x, fit, colour = k)) + geom_point(aes(x, y), data = ds,
↪  inherit.aes = FALSE, colour = "grey70") +
  stat_function(fun = function(x) 2 * sin(4 * pi * x), colour = "black",
  ↪  linetype = 2) +
  geom_line(linewidth = 1) + scale_colour_manual(values = c(ie_coral,
  ↪  ie_navy)) + labs(x = "x", y = "y", colour = NULL)
```

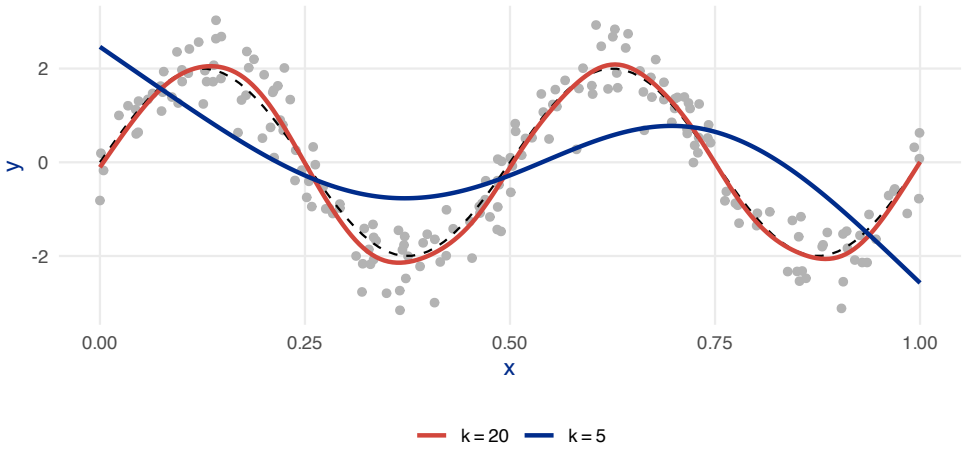


Figure 17.5: A function with two periods fitted with k = 5 (cannot bend enough) and k = 20.

With $k = 5$ the edf sits at its ceiling ($k' = 4$) and the k-index is 0.19 with $p < 0.001$; with $k = 20$ the k-index is about 1.15. On the airquality model:

```
set.seed(1)
gam.check(g3)
```

```
#>
#> Method: REML   Optimizer: outer newton
#> full convergence after 5 iterations.
#> Gradient range [-4.834e-06,3.918e-05]
#> (score 86.02 & scale 0.2338).
#> Hessian positive definite, eigenvalue range [0.05712,53.52].
#> Model rank =  28 / 28
#>
#> Basis dimension (k) checking results. Low p-value (k-index<1) may
#> indicate that k is too low, especially if edf is close to k'.
#>
#>              k'  edf k-index p-value
#> s(Solar.R) 9.00 2.16    1.01   0.510
#> s(Wind)    9.00 2.46    0.86   0.055 .
#> s(Temp)    9.00 1.95    0.96   0.345
```

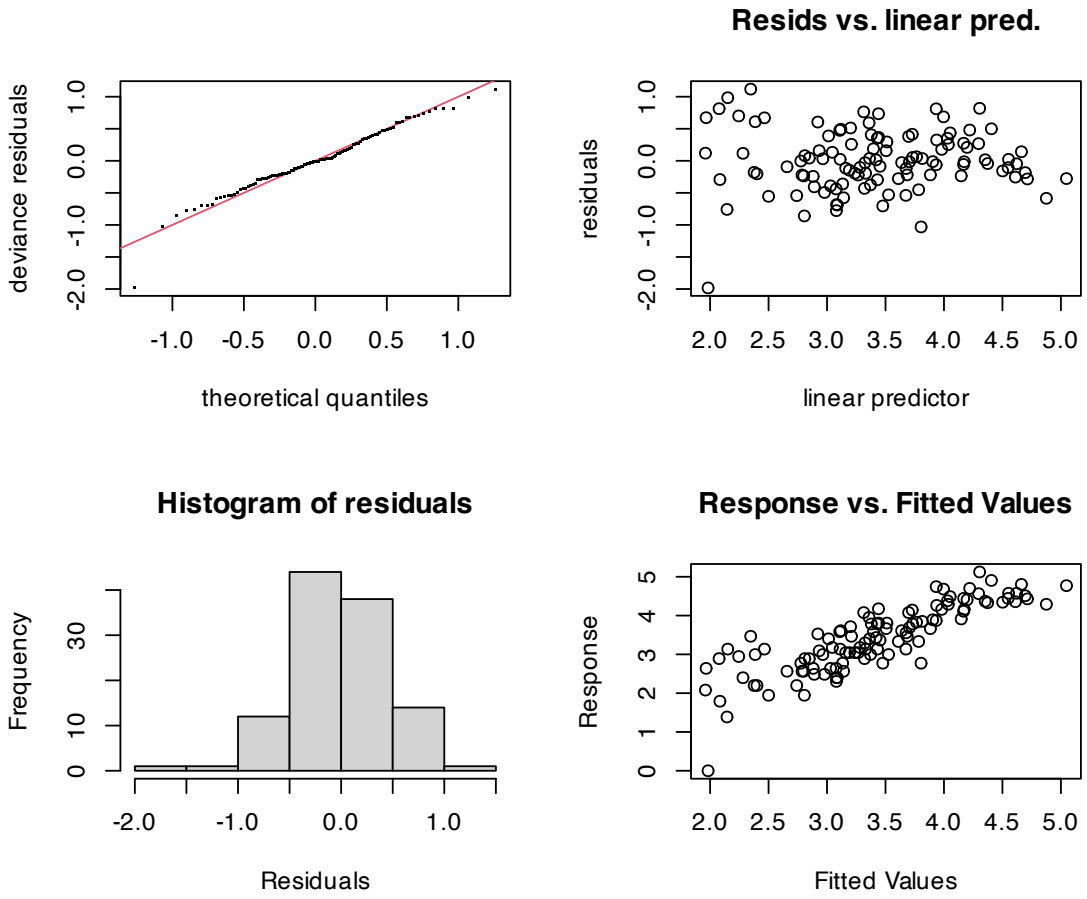


Figure 17.6: Diagnostic plots of gam.check() for the airquality model.

```
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Every edf is far below $k' = 9$, so k is not the problem. `s(Wind)` has k-index 0.86 ($p \approx 0.055$): that could suggest a missed pattern. Refitting with `k = 20` gives edf 2.47 for `s(Wind)` (against 2.46) and an unchanged k-index; so enlarging the basis does not change anything and the pattern, if real, has another cause (an interaction, for instance, Chapter 18).

Warning. Do not read a small k-index as "increase k" automatically: check whether the edf is near k' and whether a larger k changes the fit. The residual plots of `gam.check()` (quantile-quantile, residuals against linear predictor, histogram, observed against fitted) are the same as for a GLM (Chapter 6).

17.9 In R

Object / function	Content
`smoothCon(s(x, k = ), data = , absorb.cons = TRUE)`	basis matrix `X` and penalties `S` of a smooth
`model.matrix(m)`	the full $X = [X^* \mid X_1 \mid \cdots]$ (constrained)
`m$sp`, `m$full.sp`	smoothing parameters $\tilde{\lambda}_j$ (on the deviance scale)
`m$edf`, `m$edf1`, `m$edf2`, `m$hat`	edf (diagonal of F), `Ref.df`-type edf, AIC-edf, diagonal of A
`vcov(m)`, `vcov(m, unconditional = TRUE)`	posterior covariance, and with λ-uncertainty
`predict(m, newdata, type = "lpmatrix")`	matrix mapping coefficients to $\hat{\eta}$ at new data
`rmvn(n, coef(m), vcov(m))`	draws from the posterior
`gam(..., method = "GCV.Cp" / "REML" / "ML")`	smoothing-parameter criterion; `m$gcv.ubre` is the minimised score
`k.check(m)`, `gam.check(m)`	basis-dimension check and residual diagnostics
`summary(m)$s.table`	edf, Ref.df, statistic and p-value of each smooth

17.10 Pitfalls

Warning. Writing the penalty as $\lambda \int (\lambda f'')^2$, or $\lambda_j J(f_j)$ with λ_j inside J. Count λ once.

Warning. Assuming `gam()` uses REML by default: it uses `"GCV.Cp"`. Always write `method = "REML"`.

Warning. Treating the ML–REML difference as large: ML tends to give slightly smoother fits, and in most data sets the two agree closely.

Warning. Reading `coef(m)` for a smooth as if each coefficient had a meaning: after the constraint and (for thin-plate splines) the eigen-reparametrisation, only the curve is interpretable.

Warning. Using `loess$enp` instead of `trace.hat` as the trace of the smoother in GCV; `enp` is a different summary and understates the complexity.

Warning. Treating credible bands as simultaneous, or smooth-term p-values as exact.

Remark. `s(x, k = 10)` has at most 9 degrees of freedom after the centring constraint, and a $\hat{\lambda}$ that diverges is a sign that the data prefer a straight line, not that the optimiser failed.

17.11 Summary

- A GAM with basis expansions is a GLM with a quadratic penalty $\boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}$; the roughness penalty $\int f''^2$ is a quadratic form with a two-dimensional null space (constant and line).
- Sum-to-zero constraints are absorbed by a QR reparametrisation $\boldsymbol{\beta} = Z\boldsymbol{\beta}^*$, $X^* = XZ$, $S^* = Z^\top S Z$; one column is lost.
- For fixed $\boldsymbol{\lambda}$ the coefficients solve PIRLS: $\boldsymbol{\gamma}^{\text{new}} = (X^\top W X + S_\lambda)^{-1} X^\top W \boldsymbol{z}$.
- The effective degrees of freedom are $\operatorname{tr}(F) = \operatorname{tr}(A)$ with $F = (X^\top W X + S_\lambda)^{-1} X^\top W X$; the eigenvalues of F lie in $[0, 1]$ and $M \le \operatorname{tr}(F) \le q$.
- GCV, $\mathcal{V}_g = n\|\boldsymbol{y} - A\boldsymbol{y}\|^2 / \{n - \operatorname{tr}(A)\}^2$, follows from the leave-one-out identity $y_i - \hat{\mu}_i^{[-i]} = (y_i - \hat{\mu}_i)/(1 - A_{ii})$; UBRE is an unbiased risk estimate when σ^2 is known.
- REML and ML maximise the marginal likelihood of $\boldsymbol{\lambda}$ after integrating out $\boldsymbol{\gamma}$ with the smoothness prior; for Gaussian data $-2\ell_R = \hat{P}/\sigma^2 + (n - M)\log(2\pi\sigma^2) + \log|X^\top X + S_\lambda| - \log|S_\lambda|_+$, and for GLMs the Laplace approximation gives the same structure. ML tends to give slightly smoother fits than REML (usually a small difference); GCV tends to undersmooth in small samples. Use REML.
- $\boldsymbol{\gamma} \mid \boldsymbol{y} \approx N(\hat{\boldsymbol{\gamma}}, V_\gamma)$, $V_\gamma = (X^\top W X + S_\lambda)^{-1}\phi$: credible bands have the right coverage on average across the function (Nychka), are pointwise, and ignore the uncertainty in $\hat{\boldsymbol{\lambda}}$ unless `unconditional = TRUE`.
- Smooth-term tests use $\hat{\boldsymbol{f}}_j^\top V_{f_j}^{r-} \hat{\boldsymbol{f}}_j$ with $r \approx$ edf; p-values are approximate.
- Check the basis dimension with `gam.check()`: look at edf relative to k' and at the k-index.

17.12 Exercises

Exercise 17.1. (Penalty matrix) Let $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$ on $[0, 1]$. (a) Compute S such that $\int_0^1 f''(x)^2 dx = \boldsymbol{\beta}^\top S \boldsymbol{\beta}$. (b) Show that S is positive semi-definite and find its null space. (c) What happens to the fit as $\lambda \to \infty$ in penalized least squares with this basis?

Solution.

(a) $f''(x) = 2\beta_2 + 6\beta_3 x$, so

$$\int_0^1 f''^2 = 4\beta_2^2 + 2 \cdot 2\beta_2 \cdot 6\beta_3 \int_0^1 x\,dx + 36\beta_3^2 \int_0^1 x^2 dx = 4\beta_2^2 + 12\beta_2\beta_3 + 12\beta_3^2,$$

which is $\boldsymbol{\beta}^\top S \boldsymbol{\beta}$ with

$$S = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 6 \\ 0 & 0 & 6 & 12 \end{pmatrix}.$$

(b) The lower 2×2 block has determinant $48 - 36 = 12 > 0$ and positive diagonal, so it is positive definite and S is positive semi-definite. The null space is spanned by $(1, 0, 0, 0)^\top$ and $(0, 1, 0, 0)^\top$: the constant and the linear function, which have $f'' \equiv 0$ (dimension 2).

(c) As $\lambda \to \infty$ the fit is forced into the null space: the least-squares *straight line*, not the zero function.

```
S <- matrix(0, 4, 4)
for (k in 2:3) for (l in 2:3)        # b_k'' = k(k-1) x^(k-2), exponents
↪  0-indexed
```

```
  S[k + 1, l + 1] <- integrate(function(x) k * (k - 1) * x^(k - 2) * l * (l -
↪  1) * x^(l - 2), 0, 1)$value
S
```

```
#>      [,1] [,2] [,3] [,4]
#> [1,]    0    0    0    0
#> [2,]    0    0    0    0
#> [3,]    0    0    4    6
#> [4,]    0    0    6   12
```

Exercise 17.2. (PIRLS for the Poisson) For $y_i \sim \text{Poisson}(\mu_i)$ with $\log \mu_i = \eta_i = (X\boldsymbol{\gamma})_i$ and penalty $\frac{1}{2}\boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}$: (a) write ℓ_p, its gradient and Hessian; (b) derive the Newton update and show that it is a PIRLS step with $W = \text{diag}(\mu_i)$ and $z_i = \eta_i + (y_i - \mu_i)/\mu_i$; (c) why do Newton and Fisher scoring coincide?

Solution.

(a) $\ell_p(\boldsymbol{\gamma}) = \sum_i \{y_i \eta_i - e^{\eta_i} - \log y_i!\} - \frac{1}{2}\boldsymbol{\gamma}^\top S_\lambda \boldsymbol{\gamma}$. Gradient: $X^\top(\boldsymbol{y} - \boldsymbol{\mu}) - S_\lambda \boldsymbol{\gamma}$. Hessian: $-X^\top \text{diag}(\mu_i) X - S_\lambda$.

(b) Newton: $\boldsymbol{\gamma}^{\text{new}} = \boldsymbol{\gamma} + (X^\top W X + S_\lambda)^{-1}\{X^\top(\boldsymbol{y} - \boldsymbol{\mu}) - S_\lambda \boldsymbol{\gamma}\}$ with $W = \text{diag}(\mu_i)$. Multiplying by $X^\top W X + S_\lambda$: $(X^\top W X + S_\lambda)\boldsymbol{\gamma}^{\text{new}} = X^\top W X \boldsymbol{\gamma} + X^\top(\boldsymbol{y} - \boldsymbol{\mu}) = X^\top W\{X\boldsymbol{\gamma} + W^{-1}(\boldsymbol{y} - \boldsymbol{\mu})\} = X^\top W \boldsymbol{z}$ with $z_i = \eta_i + (y_i - \mu_i)/\mu_i$.

(c) The log link is the canonical link of the Poisson: the Hessian does not depend on $\boldsymbol{y}$, so observed and expected information coincide.

Exercise 17.3. (Leave-one-out and GCV) (a) Prove $y_i - \hat{\mu}_i^{[-i]} = (y_i - \hat{\mu}_i)/(1 - A_{ii})$ for a penalized least-squares smoother. (b) Deduce the formula for OCV and the GCV score. (c) A Gaussian GAM has $n = 111$, $\text{RSS} = 36.0$, $\text{tr}(A) = 6.0$. Compute GCV. After decreasing λ, $\text{RSS} = 34.5$ and $\text{tr}(A) = 8.0$: which λ does GCV prefer?

Solution.

(a) is Lemma 17.1; (b) replace each $(1 - A_{ii})$ by its average $1 - \text{tr}(A)/n$ in $\mathcal{V}_o$: $\frac{1}{n}\sum (y_i - \hat{\mu}_i)^2/(1 - \text{tr}(A)/n)^2 = n\,\text{RSS}/\{n - \text{tr}(A)\}^2$.

(b)

```
c(lambda1 = 111 * 36.0 / (111 - 6)^2, lambda2 = 111 * 34.5 / (111 - 8)^2)
```

```
#> lambda1 lambda2
#>  0.3624  0.3610
```

The scores are 0.3624 and 0.3610: GCV prefers the second (smaller λ, larger edf), though the difference is tiny, which shows how flat GCV can be.

Exercise 17.4. (UBRE) (a) Prove Theorem 17.4. (b) With the numbers of Exercise 17.3(c) and known $\sigma^2 = 0.30$, compute $\mathcal{V}_u$ for the two fits.

Solution.

(a) is the proof of the theorem. (b) $\mathcal{V}_u = \text{RSS}/n - \sigma^2 + 2\sigma^2\,\text{tr}(A)/n$:

```
vu <- function(rss, trA, s2 = .30, n = 111) rss / n - s2 + 2 * s2 * trA / n
c(fit1 = vu(36.0, 6), fit2 = vu(34.5, 8))
```

```
#>    fit1    fit2
#> 0.05676 0.05405
```

UBRE also prefers the second fit (0.0541 against 0.0568), consistently with GCV.

Exercise 17.5. (Sum-to-zero constraint) (a) Show that for B an $n \times K$ spline basis with intercept, $[\mathbf{1}, B]$ has rank deficiency. (b) Construct Z by QR and prove that $\mathbf{1}^\top B Z = \mathbf{0}$. (c) In R, build `B <- splines::bs(aq$Temp, df = 5, intercept = TRUE)`, impose the constraint, and verify ranks and the value of the intercept in `lm(log(Ozone) ~ cbind(1, B %*% Z) - 1)`.

Solution.

(a) $B\mathbf{1}_K = \mathbf{1}_n$ (B-splines form a partition of unity), so $[\mathbf{1}, B](1, -\mathbf{1}_K^\top)^\top = \mathbf{0}$.

(b) Proof of Theorem 17.1 with $C = \mathbf{1}^\top B$.

(c)

```
B <- splines::bs(aq$Temp, df = 5, intercept = TRUE)
C <- matrix(colSums(B), 1); Z <- qr.Q(qr(t(C)), complete = TRUE)[, -1]
c(rank_raw = qr(cbind(1, B))$rank, cols_raw = ncol(cbind(1, B)),
  rank_con = qr(cbind(1, B %*% Z))$rank, cols_con = ncol(cbind(1, B %*% Z)),
  max_CZ = max(abs(C %*% Z)))
```

```
#>  rank_raw  cols_raw  rank_con  cols_con    max_CZ
#> 5.000e+00 6.000e+00 5.000e+00 5.000e+00 6.049e-15
```

```
coef(lm(aq$ly ~ cbind(1, B %*% Z) - 1))[1]; mean(aq$ly)
```

```
#> cbind(1, B %*% Z)1
#>              3.416
```

```
#> [1] 3.416
```

The constrained fit has full rank, $\mathbf{1}^\top B Z = 0$ up to rounding, and its intercept equals $\overline{\log y} = 3.416$ because every column of BZ sums to zero.

Exercise 17.6. (Effective degrees of freedom) (a) For $X^\top X = I_K$, $S = \mathrm{diag}(s_1, \dots, s_K)$, show $F = \mathrm{diag}\{1/(1+\lambda s_k)\}$ and $\mathrm{edf} = \sum_k 1/(1+\lambda s_k)$. (b) Compute the edf for $K = 5$, $s = (0, 0, 1, 3, 9)$ and $\lambda = 2$, and give the limits as $\lambda \to 0$ and $\lambda \to \infty$. (c) Prove that the eigenvalues of F lie in $[0, 1]$ in general.

Solution.

(a) $F = (I + \lambda S)^{-1} I = \mathrm{diag}\{1/(1+\lambda s_k)\}$; take the trace.

(b) With $s = (0, 0, 1, 3, 9)$ and $\lambda = 2$:

```
s <- c(0, 0, 1, 3, 9); lam <- 2
sum(1 / (1 + lam * s))                          # 1 + 1 + 1/3 + 1/7 + 1/19
```

```
#> [1] 2.529
```

```
sapply(c(0, 1e-8, 1e8), function(l) sum(1 / (1 + l * s)))
```

```
#> [1] 5 5 2
```

$\lambda \to 0$: edf $\to K = 5$. $\lambda \to \infty$: every term with $s_k > 0$ tends to 0, so edf $\to 2$, the number of zero eigenvalues, i.e. the dimension of the null space of S.

(c) Part (b) of Theorem 17.3.

Exercise 17.7. (REML) (a) Starting from the marginal likelihood in Section 17.5.4, show that $\hat\sigma^2_R = \hat P/(n-M)$ and $\hat\sigma^2_{ML} = \hat P/n$. (b) Why might ML select a larger λ, and how large is the effect likely to be? (c) Compute both variance estimates for $\log(\text{Ozone}) \sim \text{s(Solar.R)} + \text{s(Wind)} + \text{s(Temp)}$ from the REML fit.

Solution.

(a) Differentiate (17.5): $\partial/\partial\sigma^2 = -\hat P/\sigma^4 + (n-M)/\sigma^2 = 0 \Rightarrow \hat\sigma^2 = \hat P/(n-M)$. For ML the term $(n-M)$ is replaced by n.

(b) ML divides by n, the REML estimate by $n-M$, so $\hat\sigma^2_{ML} < \hat\sigma^2_R$. The coefficients of the penalized directions have prior variance $\sigma^2/(\lambda s_k)$; a smaller $\hat\sigma^2$ goes with a *larger* λ if the fit is unchanged, so ML tends to smooth slightly more. This is a heuristic, not a proof (the fit changes with λ as well), and the effect is small: in the simulation of Section 17.5.7 ML had the lower edf in most replicates, but the means were close.

(c)

```
round(reml_ml(gR, aq$ly)[c("M", "sigma2_REML", "sigma2_ML")], 4)
```

```
#>            M sigma2_REML   sigma2_ML
#>       4.0000       0.2338       0.2253
```

Exercise 17.8. (Laplace approximation) (a) Show that for a Gaussian likelihood the Laplace approximation of $\int e^{\ell(\gamma)}\pi(\boldsymbol\gamma)d\boldsymbol\gamma$ is exact. (b) Write $\mathcal{V}_{LAML}$ for the Poisson model of Exercise 17.2.

Solution.

(a) For Gaussian data ℓ_p is a quadratic function of $\boldsymbol\gamma$, so the second-order expansion at its maximiser has no remainder and the integral equals that of the Gaussian kernel exactly.

(b) With $\hat{\boldsymbol\gamma}$ the penalized MLE, $\hat\mu_i = \exp(\boldsymbol x_i^\top\hat{\boldsymbol\gamma})$ and $H = X^\top\operatorname{diag}(\hat\mu_i)X + S_\lambda$,

$$\mathcal{V}_{LAML} = -\sum_i\{y_i\hat\eta_i - \hat\mu_i - \log y_i!\} + \tfrac12\hat{\boldsymbol\gamma}^\top S_\lambda\hat{\boldsymbol\gamma} + \tfrac12\log|H| - \tfrac12\log|S_\lambda|_+ - \tfrac{M}{2}\log(2\pi).$$

Exercise 17.9. (Credible interval by hand) For the airquality GAM `g3` compute the 95% band of the *total* linear predictor at `Temp = 85` (other covariates at their medians) from `lpmatrix`, `coef` and `vcov`, and compare with `predict(se.fit = TRUE)`. Then compute the band with `vcov(g3, unconditional = TRUE)` and say by how much the width changes.

Solution.

```
nd <- data.frame(Solar.R = median(aq$Solar.R), Wind = median(aq$Wind), Temp =
↪  85)
Xp <- predict(g3, nd, type = "lpmatrix"); est <- drop(Xp %*% coef(g3))
se1 <- sqrt(drop(Xp %*% vcov(g3) %*% t(Xp)))
se2 <- sqrt(drop(Xp %*% vcov(g3, unconditional = TRUE) %*% t(Xp)))
round(c(est = est, lo = est - 1.96 * se1, hi = est + 1.96 * se1,
        predict_se = predict(g3, nd, se.fit = TRUE)$se.fit, se_hand = se1,
        se_unconditional = se2, width_ratio = se2 / se1), 3)
```

```
#>            est.1             lo.1             hi.1     predict_se.1
#>            3.817            3.642            3.992            0.089
#>          se_hand se_unconditional      width_ratio
#>            0.089            0.136            1.521
```

The hand-computed standard error equals `predict()`'s. Including the uncertainty of $\hat{\boldsymbol{\lambda}}$ widens the interval noticeably: the standard error is multiplied by 1.52 at this point. The conditional bands of `predict()` and `plot()` are somewhat optimistic.

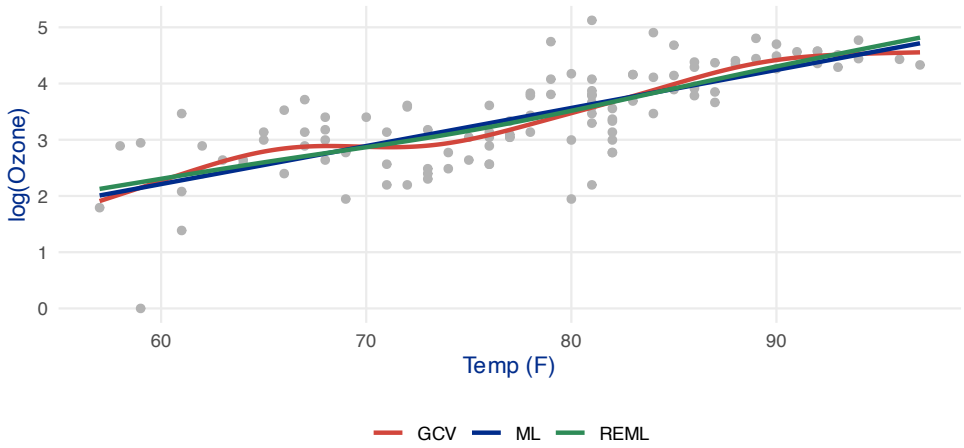
Exercise 17.10. (GCV, REML, ML in practice) Fit `ly ~ s(Temp)` to the airquality data with `method = "GCV.Cp"`, `"REML"` and `"ML"`. Report the smoothing parameter and edf of each fit, plot the three curves, and explain the differences using the simulation of Section 17.5.7.

Solution.

```
fits <- lapply(c(GCV = "GCV.Cp", REML = "REML", ML = "ML"), function(m) gam(ly
↪  ~ s(Temp), data = aq, method = m))
round(rbind(sp = sapply(fits, function(g) g$sp), edf = sapply(fits,
↪  function(g) sum(g$edf) - 1)), 3)
```

```
#>     GCV.s(Temp) REML.s(Temp) ML.s(Temp)
#> sp        0.052        1.674     257981
#> edf       4.918        2.086          1
```

```
tg <- data.frame(Temp = seq(min(aq$Temp), max(aq$Temp), length.out = 200))
dm <- do.call(rbind, lapply(names(fits), function(m) data.frame(tg, fit =
↪  predict(fits[[m]], tg), method = m)))
ggplot(dm, aes(Temp, fit, colour = method)) + geom_point(aes(Temp, ly), data =
↪  aq, inherit.aes = FALSE, colour = "grey70") +
  geom_line(linewidth = 1) + scale_colour_manual(values = c(ie_coral, ie_navy,
  ↪  ie_green)) +
  labs(x = "Temp (F)", y = "log(Ozone)", colour = NULL)
```



GCV selects a small λ (edf about 4.9, a wiggly curve), REML an intermediate one (edf 2.1) and ML an infinite λ (a straight line). This is one dataset: the ordering GCV $<$ REML $<$ ML in smoothness is typical, but the ML–REML gap here (a collapse to a straight line) is larger than the usual small difference. With one smooth and 111 points the GCV minimum is also sharp and sensitive to the sample; REML is the safest default.

Exercise 17.11. (Basis dimension) Reproduce Example 17.2 with $k = 5, 10, 20$ and 40, record edf, k-index and p-value, and decide which values of k are adequate. Does a very large k harm the fit?

Solution.

```
set.seed(1)
res <- t(sapply(c(5, 10, 20, 40), function(k) {
  g <- gam(y ~ s(x, k = k), data = ds, method = "REML")
  kc <- k.check(g); c(k = k, kprime = kc[1, 1], edf = kc[1, 2], kindex = kc[1,
↪  3], p = kc[1, 4],
                      sigma = sqrt(g$sig2))
}))
round(res, 3)
```

```
#>       k kprime    edf kindex     p sigma
#> [1,]  5      4  3.861  0.192 0.000 1.236
#> [2,] 10      9  8.810  1.120 0.958 0.509
#> [3,] 20     19 13.883  1.146 0.970 0.510
#> [4,] 40     39 15.181  1.157 0.975 0.509
```

$k = 5$ fails (edf at its ceiling, k-index far below 1). From $k = 10$ onwards the k-index is about 1.1 and $\hat{\sigma}$ is close to the true 0.5; but with $k = 10$ the edf (8.8) sits close to $k' = 9$, so $k = 10$ is borderline, and with $k \geq 20$ the edf (13.9 and 15.2) is clearly below k'. Going from 20 to 40 changes the edf and the fit very little because the penalty controls the wiggliness; a large k only costs computing time.

Exercise 17.12. (Critique) A report states: "We chose the number of knots by maximising the likelihood, and the 95% credible band for the smooth shows that, for every wind speed, the probability that the true effect lies inside the band is 95%. The smooth term for Temp has $p < 0.001$, so its effect is established beyond doubt. A k-index of 0.86 means we must increase k." Find the problems and correct each statement.

Solution.

(1) Maximising the (unpenalized) likelihood always prefers the most flexible model; smoothness is chosen by a penalised criterion, GCV, REML or ML, and k only sets the maximum flexibility. (2) The credible band has about 95% coverage *averaged across the function* (Nychka, Marra and Wood), not at every wind speed; the band is pointwise, not simultaneous, and is conditional on $\hat{\boldsymbol{\lambda}}$ unless `unconditional = TRUE`. (3) The p-value is approximate (computed as if $\hat{\lambda}$ were known, somewhat too small); "beyond doubt" is too strong, although with p far below 0.001 the conclusion that there is an effect is safe. (4) A small k-index signals a possible missed pattern; increase k *only if* the edf is near k', and check whether a larger k changes the fit. In the airquality model the edf of `s(Wind)` is 2.46 for $k' = 9$ and the k-index stays at 0.86 with $k = 20$.

17.13 Further reading

The theory of penalized regression smoothers and GAM estimation is in Chapters 5 and 6 of Wood (2017). The REML/Laplace approach is Wood (2011); the p-values are in Wood (2013); coverage of the Bayesian intervals is in Marra and Wood (2012). Generalised cross-validation is Craven and Wahba (1978). Thin-plate regression splines are Wood (2003); the Demmler–Reinsch representation and the Bayesian view of smoothing splines go back to Silverman (1985). For P-splines see Eilers and Marx (2021); and for the equivalence with mixed models, which prepares Chapter 19, see Ruppert et al. (2003).

Chapter 18

Advanced GAMs

The additive model of Chapter 16 assumes that the effect of each covariate is the same whatever the values of the others. That is often too strong: ozone builds up on days that are *both* hot and calm, and a model that adds a temperature curve to a wind curve cannot say so. This chapter extends GAMs in five directions. We build smooth functions of *two* covariates (tensor products) and split them into main effects and interaction (functional ANOVA); we compare GAMs and discuss why the usual tools (AIC, F-tests, REML scores) need care; we study *concurvity*, the non-linear version of collinearity; we diagnose the residuals of a fitted GAM; and we learn to report predictions and contrasts with correct uncertainty. The theory of penalties and smoothing-parameter selection from Chapter 17 is used throughout (Wood, 2017).

```
library(mgcv)
data(airquality)
aq <- na.omit(airquality[, c("Ozone", "Solar.R", "Wind", "Temp")])
add <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp), data = aq, method =
↪  "REML")
```

18.1 What the additive model leaves behind

We continue with the `airquality` data of Chapter 16: $n = 111$ complete days, response log(Ozone). The additive model above has adjusted $R^2 = 0.688$ and residual variance $\hat{\sigma}^2 = 0.234$. Its basis-dimension check is

```
set.seed(1)
k.check(add)
```

```
#>            k'   edf k-index p-value
#> s(Solar.R)  9 2.158  1.0118  0.5150
#> s(Wind)     9 2.462  0.8621  0.0725
#> s(Temp)     9 1.953  0.9629  0.3350
```

The k-index of `s(Wind)` is 0.86 (we define the statistic in Section 18.6); values clearly below 1 indicate structure left in the residuals along that covariate. Increasing the basis dimension to $k = 20$ does not remove it, so the basis is not the culprit. Figure 18.1 shows what is missing: the highest ozone levels occur where temperature is high and wind is low, and the two covariates are negatively correlated ($r = -0.5$).

```
ggplot(aq, aes(Temp, Wind, colour = log(Ozone))) + geom_point(size = 2.2) +
  scale_colour_gradient(low = ie_blue, high = ie_coral) +
```

```
  labs(x = "Temperature (F)", y = "Wind (mph)", colour = "log(Ozone)") +
  theme(legend.position = "right")
```

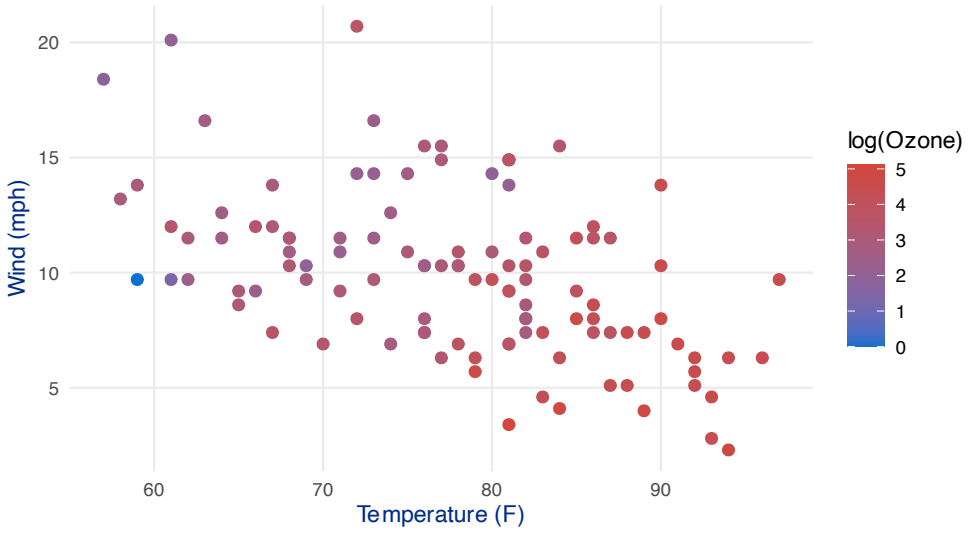


Figure 18.1: The 111 days in the (Temp, Wind) plane, coloured by log(Ozone). Ozone is highest on hot, calm days; Temp and Wind are negatively correlated, so the corners of the plane are almost empty.

An additive model forces the effect of temperature to have the same shape at every wind speed. The question for this chapter: *does the effect of temperature on log-ozone depend on wind speed, and if so, how do we model, test, check and display that dependence?*

18.2 Tensor-product smooths

18.2.1 Construction

Let $a_1(x), \dots, a_K(x)$ be a basis for functions of x and $c_1(z), \dots, c_L(z)$ a basis for functions of z (any one-dimensional basis from Chapter 15: cubic regression splines, P-splines, ...). We want a flexible function $f(x, z)$ without choosing a two-dimensional basis by hand. The idea of Wood (2006) is to build it from the margins: for each fixed z, $f(\cdot, z)$ is a smooth function of x, $f(x, z) = \sum_k \alpha_k(z)\, a_k(x)$; and the coefficients $\alpha_k(\cdot)$ themselves vary smoothly with z.

Definition 18.1 (Tensor-product smooth)**.** Let $\alpha_k(z) = \sum_{l=1}^{L} \beta_{kl}\, c_l(z)$. The **tensor-product smooth** with margins $\{a_k\}$ and $\{c_l\}$ is

$$f(x, z) = \sum_{k=1}^{K} \alpha_k(z)\, a_k(x) = \sum_{k=1}^{K} \sum_{l=1}^{L} \beta_{kl}\, a_k(x)\, c_l(z). \tag{18.1}$$

It has KL coefficients. Order them as $\boldsymbol{\beta} = (\beta_{11}, \dots, \beta_{1L}, \beta_{21}, \dots, \beta_{KL})^\top$. If X_x $(n \times K)$ and X_z $(n \times L)$ are the marginal model matrices, the model matrix of the tensor product has as row i the Kronecker product of the marginal rows,

$$X_{i\cdot} = X_{x,i\cdot} \otimes X_{z,i\cdot} \in \mathbb{R}^{1 \times KL}$$

(a *row-wise Kronecker product*).

A tensor product is thus an ordinary linear model in the KL products $a_k(x)c_l(z)$. Nothing new happens at the fitting stage: the penalised least-squares or penalised likelihood machinery of Chapter 17 applies once we have a penalty. Let us check the construction against `mgcv` on a toy data set.

```
set.seed(1); d <- data.frame(x = runif(30), z = runif(30))
sx <- smoothCon(s(x, bs = "cr", k = 5), data = d, absorb.cons = FALSE)[[1]]
sz <- smoothCon(s(z, bs = "cr", k = 4), data = d, absorb.cons = FALSE)[[1]]
tp <- smoothCon(te(x, z, bs = "cr", k = c(5, 4)), data = d, absorb.cons =
↪  FALSE)[[1]]
byhand <- t(sapply(1:30, function(i) kronecker(sx$X[i, ], sz$X[i, ])))
c(dim(tp$X), max_difference = max(abs(byhand - tp$X)))
```

```
#>                                max_difference
#>             30             20                0
```

The `te()` model matrix has $5 \times 4 = 20$ columns and is *exactly* the row-wise Kronecker product of the two marginal matrices.

Example 18.1 (A model-matrix row by hand)**.** Take $a(x) = (1 - x,\ x)$ on $[0, 1]$ and, for $z \in [0, 1]$, three hat functions at the knots 0, 0.5, 1. At $(x, z) = (0.3, 0.8)$ we have $a = (0.7,\ 0.3)$ and, because 0.8 lies between the knots 0.5 and 1, $c = (0,\ 0.4,\ 0.6)$. The row is $a \otimes c = (0,\ 0.28,\ 0.42,\ 0,\ 0.12,\ 0.18)$. With coefficients $\boldsymbol{\beta} = (1, 2, 3, 4, 5, 6)^\top$ the surface is $f(0.3, 0.8) = 0.28 \cdot 2 + 0.42 \cdot 3 + 0.12 \cdot 5 + 0.18 \cdot 6 = 3.5$. The entries sum to one because each margin is a partition of unity.

```
library(splines)
kn <- c(rep(0, 4), 0.5, rep(1, 4)); gx <- seq(0, 1, length.out = 100)
Bg <- splineDesign(kn, gx, ord = 4, outer.ok = TRUE)
prs <- list(c(2, 2), c(3, 3), c(4, 2))
dd <- do.call(rbind, lapply(prs, function(p) {
  g <- expand.grid(i = seq_along(gx), j = seq_along(gx))
  data.frame(x = gx[g$i], z = gx[g$j], v = Bg[g$i, p[1]] * Bg[g$j, p[2]],
             lab = sprintf("a%d(x) c%d(z)", p[1], p[2]))
}))
ggplot(dd, aes(x, z, fill = v)) + geom_raster() + facet_wrap(~lab) +
↪  coord_equal() +
  scale_fill_gradient(low = "white", high = ie_navy) + theme(legend.position =
  ↪  "right") +
  labs(fill = NULL)
```

18.2.2 The penalty and the smoothing parameters

The marginal bases come with penalty matrices S_x $(K \times K)$ and S_z $(L \times L)$ such that $\boldsymbol{\alpha}^\top S_x \boldsymbol{\alpha} = \int \{(\sum_k \alpha_k a_k)''\}^2 dx$ measures the wiggliness of a function of x alone. For the surface we penalise wiggliness *along* x and *along* z separately.

Proposition 18.1 (Wiggliness along x)**.** *For the surface (18.1), the average over z of the squared curvature in the x direction is*

$$J_x(f) = \int\!\!\int \left(\frac{\partial^2 f}{\partial x^2}\right)^2 dx\, dz = \boldsymbol{\beta}^\top (S_x \otimes M_z)\boldsymbol{\beta}, \qquad M_z = \int c(z)c(z)^\top dz,$$

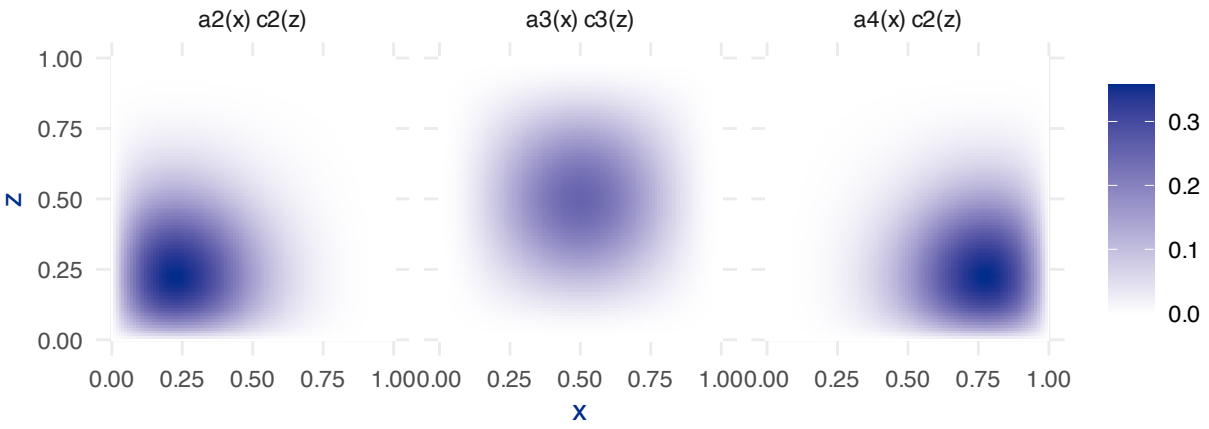


Figure 18.2: Three of the $KL = 25$ product functions $a_k(x)c_l(z)$ of a tensor product of cubic B-splines (K = L = 5). Each coefficient moves the surface in one localised region.

where $c(z) = (c_1(z), \dots, c_L(z))^\top$.

Proof. For fixed z, $f(\cdot, z) = \sum_k \alpha_k(z) a_k(\cdot)$, so $\int (\partial^2 f/\partial x^2)^2 dx = \boldsymbol{\alpha}(z)^\top S_x \boldsymbol{\alpha}(z) = \sum_{k,k'} (S_x)_{kk'} \alpha_k(z)\alpha_{k'}(z)$. With $\alpha_k(z) = \boldsymbol{\beta}_{k\cdot}^\top c(z)$ (where $\boldsymbol{\beta}_{k\cdot} = (\beta_{k1}, \dots, \beta_{kL})^\top$), integrating over z gives $\sum_{k,k'} (S_x)_{kk'}\, \boldsymbol{\beta}_{k\cdot}^\top M_z \boldsymbol{\beta}_{k'\cdot}$. This is $\boldsymbol{\beta}^\top (S_x \otimes M_z)\boldsymbol{\beta}$ in the ordering of Definition 18.1. □

The Gram matrix M_z depends on the basis in z and on the range of z. Wood (2006) replaces it by the identity, which makes the penalty depend on the *marginal penalty matrices only*. This is the convention of `mgcv`:

$$J(f) = \lambda_x\, \boldsymbol{\beta}^\top (S_x \otimes I_L)\boldsymbol{\beta} + \lambda_z\, \boldsymbol{\beta}^\top (I_K \otimes S_z)\boldsymbol{\beta}. \tag{18.2}$$

The matrix $S_x \otimes I_L$ applies the x-penalty to the coefficient sequence $(\beta_{1l}, \dots, \beta_{Kl})$ of each c_l; $I_K \otimes S_z$ does the same in the z direction. The two *smoothing parameters* λ_x, λ_z let the surface be smooth in one direction and wiggly in the other. We check this on the toy example (`mgcv` rescales its penalty matrices, so we compare up to a constant):

```
Sx <- kronecker(sx$S[[1]], diag(4)); Sz <- kronecker(diag(5), sz$S[[1]])
ratio <- function(A, B) { i <- abs(B) > 1e-6; range(A[i] / B[i]) }
rbind(x_penalty = ratio(tp$S[[1]], Sx), z_penalty = ratio(tp$S[[2]], Sz))
```

```
#>             [,1]  [,2]
#> x_penalty 2.578 2.578
#> z_penalty 2.057 2.057
```

Each penalty is a constant multiple of the Kronecker form (the range of the ratios is a single number). The null space of the penalty is the set of surfaces that are not penalised at all.

Proposition 18.2 (Null space of a tensor-product penalty)**.** *Let* $N_x = \ker S_x$ *and* $N_z = \ker S_z$. *If* $\lambda_x, \lambda_z > 0$, *the null space of* (18.2) *is* $N_x \otimes N_z$, *of dimension* $\dim N_x \cdot \dim N_z$.

Proof. Both quadratic forms are non-negative, so $J(f) = 0$ iff each vanishes, i.e. iff $\boldsymbol{\beta} \in \ker(S_x \otimes I_L) \cap \ker(I_K \otimes S_z)$. Now $\ker(S_x \otimes I_L) = N_x \otimes \mathbb{R}^L$ (the eigenvalues of a Kronecker product are products of eigenvalues, and I_L has none equal to zero) and likewise $\ker(I_K \otimes S_z) = \mathbb{R}^K \otimes N_z$. The intersection of $N_x \otimes \mathbb{R}^L$ and $\mathbb{R}^K \otimes N_z$ is $N_x \otimes N_z$. □

For cubic regression-spline margins, $N_x = \mathrm{span}\{1, x\}$ and $N_z = \mathrm{span}\{1, z\}$, so the null space is spanned by $1, x, z, xz$: the surfaces that are linear in each variable separately. In the toy example the two penalties have ranks $(K-2)L = 12$ and $K(L-2) = 10$ and together rank $16 = 20 - 4$:

```
c(rank_x = qr(tp$S[[1]])$rank, rank_z = qr(tp$S[[2]])$rank,
  rank_sum = qr(tp$S[[1]] + tp$S[[2]])$rank, KL = ncol(tp$X))
```

```
#>   rank_x   rank_z rank_sum       KL
#>       12       10       16       20
```

18.2.3 Scale invariance

The marginal bases and penalties are chosen separately, so a tensor-product smooth should not care in which *units* x and z are measured. An isotropic smooth such as the thin-plate spline `s(x, z)` of Chapter 17 has the single penalty $\int\int (f_{xx}^2 + 2f_{xz}^2 + f_{zz}^2)\,dx\,dz$ and one λ, so it does care.

Theorem 18.1 (Scale invariance of te(), lack of it for s(x, z))**.** *Change units, $x' = cx$ with $c > 0$. (i) For a tensor-product smooth the penalised fits obtainable with (λ_x', λ_z) in the new units are exactly those obtainable with (λ_x, λ_z) in the old units, for $\lambda_x' = c^3\lambda_x$. (ii) For the isotropic penalty above, the three terms are multiplied by* different *powers of c, so no choice of a single λ reproduces the old fit.*

Proof.

(i) A basis function of the new variable is $a_k'(x') = a_k(x'/c)$, the same function of the same points. For a function g of one variable, put $h(x') = g(x'/c)$; then $h''(x') = c^{-2}g''(x)$ and $dx' = c\,dx$, so

$$\int h''(x')^2 dx' = \int c^{-4} g''(x)^2\, c\,dx = c^{-3}\int g''(x)^2 dx.$$

Hence the marginal penalty matrix becomes $S_x' = c^{-3}S_x$, while the model matrix is unchanged. The penalised objective $\|y - X\boldsymbol{\beta}\|^2 + \lambda_x'\boldsymbol{\beta}^\top(S_x' \otimes I)\boldsymbol{\beta} + \lambda_z\boldsymbol{\beta}^\top(I \otimes S_z)\boldsymbol{\beta}$ coincides with the old objective when $\lambda_x' c^{-3} = \lambda_x$. Any smoothing-parameter criterion that depends only on the fitted model (REML, GCV, AIC) is therefore minimised at corresponding λ's and gives the same surface.

(ii) With $f'(x', z) = f(x'/c, z)$: $f'_{x'x'} = c^{-2}f_{xx}$, $f'_{x'z} = c^{-1}f_{xz}$, $f'_{zz} = f_{zz}$, and $dx' = c\,dx$. Then

$$J' = c^{-3}\int\int f_{xx}^2 + 2c^{-1}\int\int f_{xz}^2 + c\int\int f_{zz}^2,$$

three different powers c^{-3}, c^{-1}, c of the scale.

□

The same is visible in the data. We refit the ozone model with temperature in degrees Celsius and wind in km/h.

```
aq$TempC <- (aq$Temp - 32) * 5 / 9; aq$Windk <- aq$Wind * 1.609
fits <- list(
  iso_F = gam(log(Ozone) ~ s(Solar.R) + s(Temp, Wind),   data = aq, method =
  ↪  "REML"),
  iso_C = gam(log(Ozone) ~ s(Solar.R) + s(TempC, Windk), data = aq, method =
  ↪  "REML"),
```

```
  te_F  = gam(log(Ozone) ~ s(Solar.R) + te(Temp, Wind),   data = aq, method =
  ↪  "REML"),
  te_C  = gam(log(Ozone) ~ s(Solar.R) + te(TempC, Windk), data = aq, method =
  ↪  "REML"))
sapply(fits, AIC)
```

```
#> iso_F iso_C  te_F  te_C
#> 147.2 139.6 144.1 144.1
```

The isotropic surface changes by 7.5 AIC units from a mere change of units; the tensor-product fit is identical to the last digit (144.09 in both units).

Remark. Use an isotropic `s(x, z)` only when x and z are measured on the same scale and an isotropic surface makes sense (map coordinates, two dimensions of an image). Otherwise use `te()`, or `ti()` below.

18.2.4 In R: te()

`te(x, z)` uses cubic regression spline margins with $k = 5$ each by default; `k = c(k1, k2)`, `bs = c("ps", "cr")` and `d = c(2, 1)` (a two-dimensional margin followed by a one-dimensional margin) change the margins. After the sum-to-zero constraint that separates the smooth from the intercept, a `te(x, z)` term has $KL - 1$ coefficients and two smoothing parameters.

```
te1 <- gam(log(Ozone) ~ s(Solar.R) + te(Temp, Wind), data = aq, method =
↪  "REML")
summary(te1)
```

```
#>
#> Family: gaussian
#> Link function: identity
#>
#> Formula:
#> log(Ozone) ~ s(Solar.R) + te(Temp, Wind)
#>
#> Parametric coefficients:
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   3.4159     0.0397    86.1   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Approximate significance of smooth terms:
#>                 edf Ref.df     F p-value
#> s(Solar.R)     2.38   2.98  9.22 2.4e-05 ***
#> te(Temp,Wind) 12.70  15.09 13.90 < 2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> R-sq.(adj) =  0.767   Deviance explained = 79.9%
#> -REML = 74.996  Scale est. = 0.17466   n = 111
```

```
te1$sp
```

```
#>     s(Solar.R) te(Temp,Wind)1 te(Temp,Wind)2
```

```
#>          1.1094          0.3116          0.7369
```

The surface uses edf ≈ 12.7 of its 24 coefficients; adjusted R^2 rises from 0.688 to 0.767 and $\hat{\sigma}^2$ falls from 0.234 to 0.175. The two smoothing parameters differ, so the surface is smoother in one direction than in the other. Figure 18.3 shows the fitted surface, restricted to the region within 10% of the data range from an observation (`too.far = 0.1`); outside it the surface is extrapolation.

```
grid <- expand.grid(Temp = seq(56, 97, length.out = 70), Wind = seq(2, 21,
↪  length.out = 70))
far <- exclude.too.far(grid$Temp, grid$Wind, aq$Temp, aq$Wind, dist = 0.1)
grid$z <- predict(te1, transform(grid, Solar.R = median(aq$Solar.R)), type =
↪  "terms")[, "te(Temp,Wind)"]
grid$z[far] <- NA
ggplot(grid, aes(Temp, Wind)) + geom_raster(aes(fill = z)) +
  geom_contour(aes(z = z), colour = "white", linewidth = .3, bins = 10, na.rm
  ↪  = TRUE) +
  geom_point(data = aq, aes(Temp, Wind), size = .7, colour = "grey15") +
  scale_fill_gradient2(low = ie_blue, mid = "white", high = ie_coral, midpoint
  ↪  = 0, na.value = "transparent") +
  labs(x = "Temperature (F)", y = "Wind (mph)", fill = "effect") +
  ↪  theme(legend.position = "right")
```

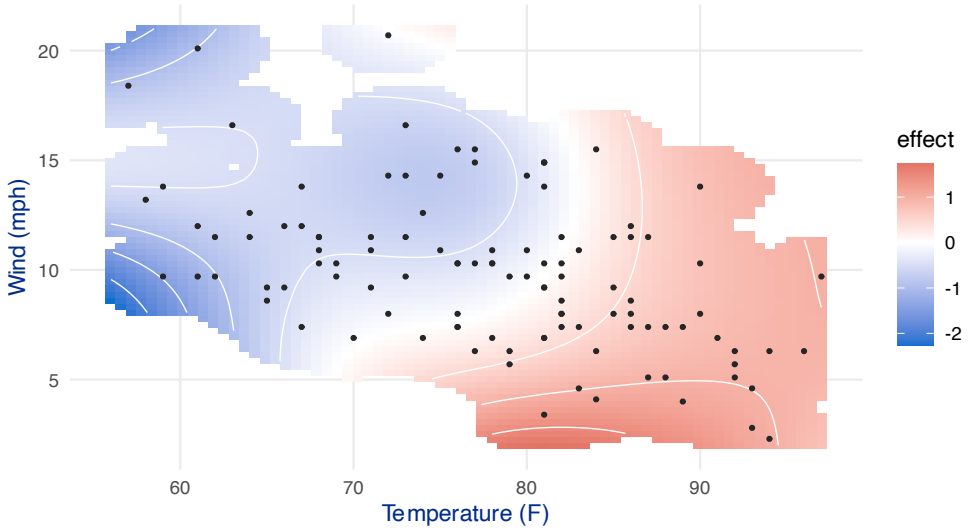


Figure 18.3: Fitted surface of te(Temp, Wind) (linear predictor, centred, Solar.R fixed at its median). Ozone is highest when it is hot and calm; the contours bend so that the temperature effect is steeper at low wind.

18.3 Functional ANOVA and ti()

A tensor-product smooth is the *whole* surface. To ask whether the effect of x is the same at every z we want to isolate the interaction. In the language of analysis of variance, we want

$$f(x, z) = f_0 + f_x(x) + f_z(z) + f_{xz}(x, z), \tag{18.3}$$

with $H_0 \colon f_{xz} \equiv 0$ meaning "no interaction". As in a two-way factor ANOVA the decomposition is unique only under constraints.

18.3.1 Uniqueness under independence

Proposition 18.3 (Functional ANOVA decomposition)**.** *Let (X, Z) have independent components and $f(x, z)$ be integrable. Then there is exactly one decomposition (18.3) with*

$$\mathrm{E}\{f_x(X)\} = \mathrm{E}\{f_z(Z)\} = 0, \qquad \mathrm{E}\{f_{xz}(X, Z) \mid X\} = 0, \quad \mathrm{E}\{f_{xz}(X, Z) \mid Z\} = 0,$$

namely $f_0 = \mathrm{E}f(X, Z)$, $f_x(x) = \mathrm{E}\{f(x, Z)\} - f_0$, $f_z(z) = \mathrm{E}\{f(X, z)\} - f_0$ and $f_{xz} = f - f_0 - f_x - f_z$.

Proof. Existence: with f_0, f_x, f_z as stated, f_{xz} is defined by difference, and by independence $\mathrm{E}\{f_{xz}(x, Z)\} = \mathrm{E}f(x, Z) - f_0 - f_x(x) - \mathrm{E}f_z(Z) = 0$ for every x; the same holds in z. Uniqueness: suppose two decompositions satisfy the constraints. Averaging (18.3) over Z at fixed x (here independence is used to integrate over the marginal law of Z) gives $\mathrm{E}f(x, Z) = f_0 + f_x(x) + \mathrm{E}f_z(Z) + \mathrm{E}\{f_{xz}(x, Z)\} = f_0 + f_x(x)$, so f_0 and f_x are determined by f; likewise f_z, and then f_{xz}. □

With *dependent* covariates the conditional-mean constraints cannot be imposed by constraints on the marginal functions alone: averaging over Z given $X = x$ uses the conditional law of Z given x. We return to this below.

18.3.2 Construction of ti()

`mgcv` builds the decomposition from constrained margins. Take a marginal basis matrix X_x $(n \times K)$ and impose the sample sum-to-zero constraint $\mathbf{1}^\top X_x \boldsymbol{\alpha} = 0$: with $C = \mathbf{1}^\top X_x$, find a $K \times (K-1)$ matrix Q_x whose columns span $\{\boldsymbol{\alpha} : C\boldsymbol{\alpha} = 0\}$ (from a QR decomposition of $C^\top$), and use the basis $\tilde{X}_x = X_x Q_x$ with penalty $\tilde{S}_x = Q_x^\top S_x Q_x$. Do the same for z.

Definition 18.2 (Tensor-product interaction)**.** The `ti(x, z)` term is the tensor product of the two *constrained* marginal bases, with penalties $\tilde{S}_x \otimes I$ and $I \otimes \tilde{S}_z$. It has $(K-1)(L-1)$ coefficients and two smoothing parameters. The decomposition is fitted as

`y ~ s(x) + s(z) + ti(x, z).`

Every function in a `ti` term satisfies $\sum_i f_{xz}(x_i, z) = 0$ for all z and $\sum_i f_{xz}(x, z_i) = 0$ for all x: it averages to zero over the *product of the two empirical marginal distributions*. We verify the dimensions.

```
ti_toy <- smoothCon(ti(x, z, k = c(5, 4)), data = d, absorb.cons = TRUE)[[1]]
te_toy <- smoothCon(te(x, z, k = c(5, 4)), data = d, absorb.cons = TRUE)[[1]]
c(ti = ncol(ti_toy$X), `(K-1)(L-1)` = 4 * 3, te = ncol(te_toy$X), `KL-1` = 5 *
↪  4 - 1)
```

```
#>         ti (K-1)(L-1)         te       KL-1
#>         12         12         19         19
```

18.3.3 te() versus ti()

Term	Contains	Use
`te(x, z)`	main effects and interaction	the whole surface, no decomposition
`ti(x, z)`	interaction only	next to `s(x) + s(z)`, to test or isolate f_{xz}

Term	Contains	Use
`s(x, z)`	isotropic surface	covariates on the same scale

Warning. Never write `s(x) + s(z) + te(x, z)`. The space of a `te` smooth contains functions of x alone (put $\beta_{kl} = \gamma_k$ for all l when the margin in z reproduces constants), so the "main effects" can be moved between `s(x)` and `te(x, z)` without changing the fit, and none of the three terms can be read separately. The symptom is concurvity of one with the other, as we see next.

```
bad <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp) + te(Temp, Wind), data
↪  = aq, method = "REML")
round(concurvity(bad, full = TRUE)["worst", ], 3)
```

```
#>          para    s(Solar.R)       s(Wind)       s(Temp) te(Temp,Wind)
#>         0.000         0.591         1.000         1.000         1.000
```

18.3.4 ANOVA decomposition for ozone

```
ti1 <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp) + ti(Temp, Wind), data
↪  = aq, method = "REML")
summary(ti1)$s.table
```

```
#>                 edf Ref.df      F   p-value
#> s(Solar.R)    2.346  2.938  8.620 5.585e-05
#> s(Wind)       1.545  1.833  1.605 1.512e-01
#> s(Temp)       5.543  6.642 12.103 0.000e+00
#> ti(Temp,Wind) 8.317 10.017  3.348 8.478e-04
```

The interaction is clearly supported ($p = 8.5 \times 10^{-4}$): the effect of temperature on log-ozone depends on wind speed. The main effect `s(Wind)` has $p = 0.15$. This does *not* mean that wind does not matter: wind acts through the interaction, and wind is correlated with temperature.

18.3.5 When are the main effects meaningful?

Proposition 18.3 required independent covariates. The sample constraints used by `ti()` average over the *product of the marginals*, which coincides with the data distribution only when the covariates are independent (or the design is a full grid). When they are dependent, $\mathrm{E}\{f_{xz}(X, Z) \mid X\} \neq 0$ in general, part of "the effect of x averaged over z" sits in f_{xz}, and f_x depends on the design and changes when the interaction is added. Here $\mathrm{Corr}(\mathrm{Temp}, \mathrm{Wind}) = -0.50$ and the concurvity of the main effects with the interaction is high:

```
round(concurvity(ti1, full = TRUE)[c("worst", "estimate"), ], 2)
```

```
#>          para s(Solar.R) s(Wind) s(Temp) ti(Temp,Wind)
#> worst       1       0.55    1.00    1.00          1.00
#> estimate    1       0.41    0.91    0.96          0.46
```

What remains meaningful is the *whole surface* $f(x, z)$, predictions, and the test of $f_{xz} \equiv 0$; the separate main effects should be read together with the interaction (Exercise 18.5). This is the same lesson as in a two-way ANOVA with unbalanced cells: main effects in the presence of an interaction depend on the weighting.

Table 18.1: AIC, AIC differences and Akaike weights for three models of log(Ozone).

model	logLik	df	AIC	delta	weight
additive	-72.91	10.29	166.4	28.48	0.000
s(Solar.R) + te(Temp, Wind)	-51.98	20.07	144.1	6.16	0.044
additive + ti(Temp, Wind)	-45.53	23.43	137.9	0.00	0.956

18.4 Model comparison and its caveats

18.4.1 AIC with effective degrees of freedom

For a penalised fit the number of free parameters is replaced by the effective degrees of freedom. The reason is a covariance-penalty identity.

Proposition 18.4 (Optimism of a linear smoother)**.** *Let $y \sim (\mu, \sigma^2 I_n)$ and $\hat{\mu} = S_\lambda y$ with S_λ fixed. For independent new data y^* with the same mean,*

$$\mathrm{E}\|y^* - \hat{\mu}\|^2 = \mathrm{E}\|y - \hat{\mu}\|^2 + 2\sigma^2 \operatorname{tr}(S_\lambda).$$

Proof. Write $\|y^* - \hat{\mu}\|^2 = \|y^* - \mu\|^2 + \|\mu - \hat{\mu}\|^2 + 2(y^* - \mu)^\top(\mu - \hat{\mu})$; the last term has mean zero because y^* is independent of $\hat{\mu}$. Similarly $\|y - \hat{\mu}\|^2 = \|y - \mu\|^2 + \|\mu - \hat{\mu}\|^2 + 2(y - \mu)^\top(\mu - \hat{\mu})$. Subtract, and use $\mathrm{E}\|y^* - \mu\|^2 = \mathrm{E}\|y - \mu\|^2$: the difference is $-2\mathrm{E}\{(y - \mu)^\top(\mu - \hat{\mu})\} = 2\mathrm{E}\{(y - \mu)^\top(\hat{\mu} - \mu)\} = 2\sum_i \operatorname{Cov}(y_i, \hat{\mu}_i) = 2\sigma^2 \operatorname{tr}(S_\lambda)$, since $\operatorname{Cov}(y, S_\lambda y) = \sigma^2 S_\lambda^\top$. □

So the training error underestimates the prediction error by $2\sigma^2 \operatorname{edf}$ with $\operatorname{edf} = \operatorname{tr}(S_\lambda)$, exactly as the number of parameters $p = \operatorname{tr}(H)$ does for a linear model (Chapter 7). The AIC of a GAM (Akaike, 1974) is therefore

$$\mathrm{AIC} = -2\,\ell(\hat{\gamma}) + 2\,\mathrm{df},$$

where `mgcv` uses a df corrected for the fact that the λ's were estimated (Wood et al., 2016), plus one for the scale parameter when there is one. AIC compares any models for the *same response on the same data*, nested or not.

```
te1 <- gam(log(Ozone) ~ s(Solar.R) + te(Temp, Wind), data = aq, method =
↪  "REML")
tab <- data.frame(model = c("additive", "s(Solar.R) + te(Temp, Wind)",
↪  "additive + ti(Temp, Wind)"),
                  logLik = sapply(list(add, te1, ti1), function(m)
                  ↪  as.numeric(logLik(m))),
                  df = sapply(list(add, te1, ti1), function(m) attr(logLik(m),
                  ↪  "df")),
                  AIC = sapply(list(add, te1, ti1), AIC))
tab$delta <- tab$AIC - min(tab$AIC)
tab$weight <- exp(-tab$delta / 2) / sum(exp(-tab$delta / 2))
knitr::kable(tab, digits = c(0, 2, 2, 2, 2, 3), booktabs = TRUE,
             caption = "AIC, AIC differences and Akaike weights for three
             ↪  models of log(Ozone).")
```

Akaike weights $w_m \propto e^{-\Delta_m/2}$ (Burnham and Anderson, 2004) summarise the evidence: the `ti` model has weight 0.96. For the `ti` model, $-2(-45.53) + 2(23.43) = 137.93$.

18.4.2 Approximate tests

`anova()` on two nested `gam` fits performs an approximate F-test with the *effective* degrees of freedom of the extra term, treating $\hat{\boldsymbol{\lambda}}$ as known.

```
add_ml <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp), data = aq, method =
↪  "ML")
ti_ml  <- update(add_ml, . ~ . + ti(Temp, Wind))
anova(add_ml, ti_ml, test = "F")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp)
#> Model 2: log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp) + ti(Temp, Wind)
#>   Resid. Df Resid. Dev   Df Deviance    F Pr(>F)
#> 1     104.7       25.9
#> 2      96.4       20.7 8.29     5.26 3.07 0.0036 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(add, ti1, test = "F")
```

```
#> Analysis of Deviance Table
#>
#> Model 1: log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp)
#> Model 2: log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp) + ti(Temp, Wind)
#>   Resid. Df Resid. Dev   Df Deviance    F  Pr(>F)
#> 1     100.0       24.2
#> 2      84.9       14.8 15.1     9.41 3.89 2.8e-05 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The two calls give different answers ($p = 0.0036$ with ML, 2.8×10^{-5} with REML). Three reasons: (i) the REML criterion, like the REML likelihood of a mixed model (Chapter 12), contains a term that depends on the unpenalised part of the model, so REML scores of models with different null spaces are *not comparable* (Wood, 2011); (ii) the fitted $\hat{\boldsymbol{\lambda}}$ are different, hence the effective df are different (here 8.3 with ML, 15.1 with REML); (iii) both tests ignore the uncertainty in $\hat{\boldsymbol{\lambda}}$. The REML-based comparison is not valid for this purpose, and the arbiters are the likelihood-ratio test calibrated by a parametric bootstrap (simulate from the additive ML fit, refit both models, 1000 replicates: $p \approx 0.004$, with Monte Carlo error about 0.002) and the test of the `ti` term in `summary()` of the ML fit ($p = 0.0039$; for the REML fit $p = 8.5 \times 10^{-4}$). Practical rules: compare nested models with ML; treat the p-values as a guide, to be read with AIC and plots; for the significance of a single smooth term use the Wald-type p-values of `summary()`, which have good properties in most cases (Wood, 2013).

How good is the approximation? A small simulation under the null (no interaction), 200 replicates, $n = 100$:

```
set.seed(18); n <- 100
pv <- replicate(200, {
  x1 <- runif(n); x2 <- runif(n); y <- sin(2 * pi * x1) + 4 * (x2 - .5)^2 +
↪  rnorm(n, sd = .5)
  dd <- data.frame(y, x1, x2)
```

```
  a <- gam(y ~ s(x1) + s(x2), data = dd, method = "ML")
  b <- gam(y ~ s(x1) + s(x2) + ti(x1, x2), data = dd, method = "ML")
  c(anova = anova(a, b, test = "F")[2, "Pr(>F)"], wald =
  ↪  summary(b)$s.table["ti(x1,x2)", "p-value"])
})
rowMeans(pv < 0.05, na.rm = TRUE)
```

```
#>   anova    wald
#> 0.05914 0.06000
```

The rejection rate at the 5% level is close to nominal for both tests in this setting (the Monte Carlo standard error of each rate is about 0.015). The anova p-value is `NA` in 14 replicates, when the fitted interaction has zero edf. This is a calibration check for one design, not a proof: the approximation degrades with small n, weak smoothing-parameter identification and strongly non-Gaussian data.

18.4.3 Shrinkage selection: select = TRUE

The wiggly part of a smooth is penalised, but its *null space* (for a cubic regression spline: the straight line) is not, so an ordinary smooth cannot be removed from the model: edf never falls below one. To allow a term to be removed altogether, Marra and Wood (2011) add a second penalty on the null space.

Let $S_j = U\Lambda U^\top$ be the eigen-decomposition of the penalty of term j and U_0 the eigenvectors of the zero eigenvalues. The **double penalty** is

$$\lambda_j \boldsymbol{\beta}_j^\top S_j \boldsymbol{\beta}_j + \lambda_j^* \boldsymbol{\beta}_j^\top S_j^* \boldsymbol{\beta}_j, \qquad S_j^* = U_0 U_0^\top,$$

so that $\lambda_j S_j + \lambda_j^* S_j^*$ has full rank. For a Gaussian response the influence matrix is $F = (X^\top X + \lambda_j S_j + \lambda_j^* S_j^*)^{-1} X^\top X$, and if both $\lambda_j, \lambda_j^* \to \infty$ then $F \to 0$, hence edf $= \operatorname{tr}(F) \to 0$: the term is shrunk entirely to zero. The ordinary penalty alone gives edf $\to \dim N_j \geq 1$. In `mgcv`: `select = TRUE`; shrinkage bases `bs = "ts"` and `"cs"` do the same by inflating the zero eigenvalues slightly. We add two pure-noise covariates to the ozone model.

```
set.seed(2027)
aq$noise1 <- rnorm(nrow(aq)); aq$noise2 <- runif(nrow(aq))
f_big <- log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp) + s(noise1) + s(noise2)
m_no  <- gam(f_big, data = aq, method = "REML")
m_sel <- gam(f_big, data = aq, method = "REML", select = TRUE)
round(cbind(without = summary(m_no)$s.table[, "edf"], select =
↪  summary(m_sel)$s.table[, "edf"]), 4)
```

```
#>            without select
#> s(Solar.R)   2.171 2.1203
#> s(Wind)      2.613 2.2416
#> s(Temp)      1.001 2.8610
#> s(noise1)    2.081 0.0001
#> s(noise2)    1.000 0.0000
```

Without `select`, the noise terms keep edf ≥ 1. With `select = TRUE` they go to edf ≈ 0, while the three real effects are kept. Since edf depends on what else is in the model, the edf of `s(Temp)` also changes. Shrinkage selection is a convenient screening device; the final model should still be checked.

18.5 Concurvity

In a linear model, collinearity means that a column of X is nearly a linear combination of the others, which inflates variances (Chapter 7). In a GAM the analogue is **concurvity**: a smooth term f_j can be closely approximated by the *other* terms in the model. Then the separate shapes of f_j and of the others are poorly determined, although their sum is not.

18.5.1 Definition and computation

Let X_j be the columns of the model matrix that belong to term j, and X_{-j} all the other columns (including the intercept and parametric terms), P_{-j} the orthogonal projector onto the column space of X_{-j}. For a coefficient vector $\boldsymbol{\beta}_j$ the fraction of the term $X_j\boldsymbol{\beta}_j$ that can be represented by the rest is $\|P_{-j}X_j\boldsymbol{\beta}_j\|^2/\|X_j\boldsymbol{\beta}_j\|^2 \in [0,1]$. `mgcv::concurvity()` reports three versions:

- **worst**: the maximum of this ratio over all $\boldsymbol{\beta}_j$;
- **observed**: the ratio at the fitted coefficients $\hat{\boldsymbol{\beta}}_j$;
- **estimate**: the ratio of squared Frobenius norms $\|P_{-j}X_j\|_F^2/\|X_j\|_F^2$, a less extreme summary that does not depend on $\hat{\boldsymbol{\beta}}$.

Proposition 18.5 (Worst-case concurvity)**.** *Let Q_j and Q_{-j} be matrices with orthonormal columns spanning the column spaces of X_j and X_{-j} (X_j of full column rank). Then the worst-case concurvity of term j equals $\sigma^2_{\max}(Q_{-j}^\top Q_j)$, the largest squared canonical correlation between the two spaces.*

Proof. Every $X_j\boldsymbol{\beta}_j$ can be written $Q_j w$ with $w \in \mathbb{R}^r$ (r the rank), and w ranges over all of $\mathbb{R}^r$ as $\boldsymbol{\beta}_j$ does. Then $\|P_{-j}Q_j w\|^2 = \|Q_{-j}Q_{-j}^\top Q_j w\|^2 = \|Q_{-j}^\top Q_j w\|^2$ and $\|Q_j w\|^2 = \|w\|^2$. The maximum of $\|Aw\|^2/\|w\|^2$, $A = Q_{-j}^\top Q_j$, is the largest eigenvalue of $A^\top A$, i.e. $\sigma^2_{\max}(A)$. □

We can reproduce the numbers of `concurvity()` for `s(Wind)` in the additive model of the opening section.

```
X <- model.matrix(add); jj <- add$smooth[[2]]; idx <-
↪  jj$first.para:jj$last.para
Xj <- X[, idx]; Xo <- X[, -idx]
Qo <- qr.Q(qr(Xo)); Qj <- qr.Q(qr(Xj)); PXj <- Qo %*% crossprod(Qo, Xj)
b <- coef(add)[idx]
hand <- c(worst = max(svd(crossprod(Qo, Qj))$d)^2,
          observed = sum((PXj %*% b)^2) / sum((Xj %*% b)^2),
          estimate = sum(PXj^2) / sum(Xj^2))
rbind(by_hand = hand, concurvity = concurvity(add)[, "s(Wind)"])
```

```
#>             worst observed estimate
#> by_hand      0.56   0.3685   0.2972
#> concurvity   0.56   0.3685   0.2972
```

The first row, computed from the three definitions, agrees with the function. For the additive ozone model, 0.56 (worst) and 0.3 (estimate) for `s(Wind)` are moderate: Temp and Wind share information.

18.5.2 Consequences: a simulation

Take $y = f_1(x_1) + f_2(x_2) + \varepsilon$ with $f_1(x) = \sin(2\pi x)$, $f_2(x) = 2(x - 0.5)^2$ and $\varepsilon \sim N(0, 0.3^2)$, $n = 300$, and compare the design where x_2 is independent of x_1 with designs where $x_2 = x_1 + N(0, s^2)$ (truncated to $[0, 1]$).

```
f1 <- function(x) sin(2 * pi * x); f2 <- function(x) 2 * (x - .5)^2
conc_sim <- function(s, seed = 25) {
  set.seed(seed); n <- 300; x1 <- runif(n)
  x2 <- if (is.na(s)) runif(n) else pmin(pmax(x1 + rnorm(n, sd = s), 0), 1)
  y <- f1(x1) + f2(x2) + rnorm(n, sd = .3)
  m <- gam(y ~ s(x1) + s(x2), method = "REML")
  se <- predict(m, type = "terms", se.fit = TRUE)$se.fit[, "s(x1)"]
  c(cor = cor(x1, x2), concurvity = concurvity(m, full = TRUE)["estimate",
  ↪  "s(x1)"], mean_se = mean(se))
}
res <- t(sapply(c(NA, .3, .08, .02), conc_sim))
rownames(res) <- c("independent", "s = 0.30", "s = 0.08", "s = 0.02")
round(res, 3)
```

```
#>                cor concurvity mean_se
#> independent 0.005      0.036   0.046
#> s = 0.30    0.724      0.486   0.050
#> s = 0.08    0.968      0.907   0.089
#> s = 0.02    0.998      0.992   0.280
```

As the correlation grows, concurvity approaches one and the standard error of $\hat{f}_1$ explodes, in the same way as the variance inflation factor $1/(1 - R^2)$ of a collinear linear term. Figure 18.4 shows the fitted $\hat{f}_1$ for the independent design and for $s = 0.08$.

```
fit_band <- function(s, seed = 25) {
  set.seed(seed); n <- 300; x1 <- runif(n)
  x2 <- if (is.na(s)) runif(n) else pmin(pmax(x1 + rnorm(n, sd = s), 0), 1)
  y <- f1(x1) + f2(x2) + rnorm(n, sd = .3)
  m <- gam(y ~ s(x1) + s(x2), method = "REML")
  xg <- seq(0, 1, length.out = 150)
  p <- predict(m, data.frame(x1 = xg, x2 = .5), type = "terms", se.fit = TRUE)
  data.frame(x = xg, fit = p$fit[, "s(x1)"], se = p$se.fit[, "s(x1)"], truth =
  ↪  f1(xg) - mean(f1(x1)),
             design = ifelse(is.na(s), "independent", "x2 close to x1 (s =
             ↪  0.08)"))
}
db <- rbind(fit_band(NA), fit_band(.08))
ggplot(db, aes(x, fit)) + geom_ribbon(aes(ymin = fit - 2 * se, ymax = fit + 2
↪  * se), fill = "#D2E6F4") +
  geom_line(aes(y = truth), colour = ie_coral, linetype = 2) +
  ↪  geom_line(colour = ie_navy, linewidth = 1) +
  facet_wrap(~design) + labs(x = expression(x[1]), y = expression(hat(f)[1]))
```

18.5.3 What to do

Remark. Concurvity is a property of the *design*, not a model error. Predictions and the joint effect stay well determined, while the separate shapes do not. A rule of thumb is to

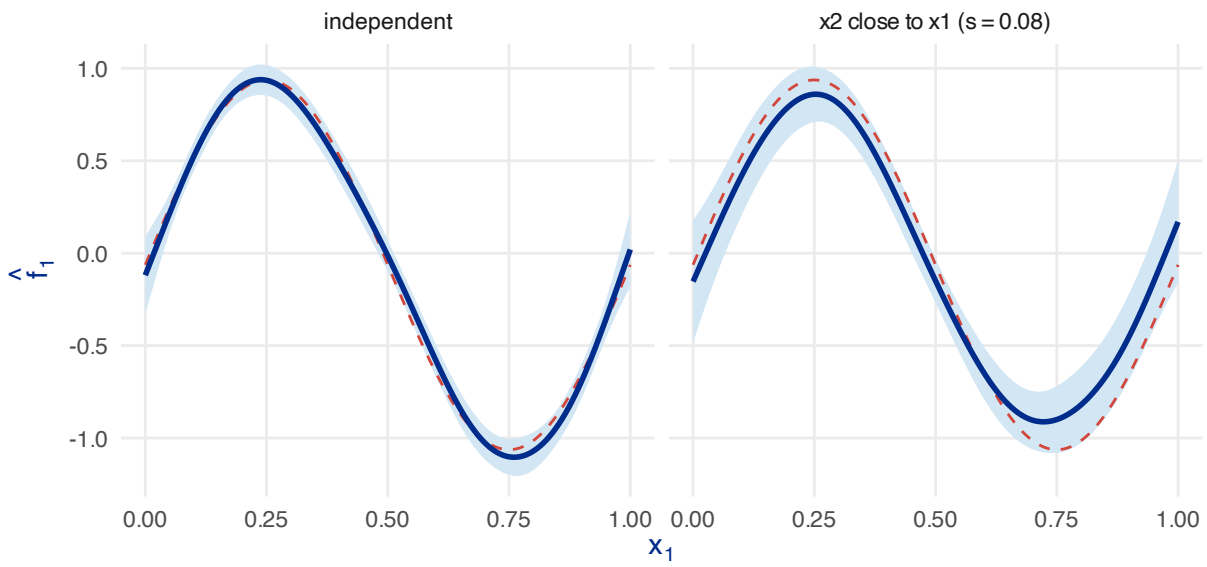


Figure 18.4: Estimate of f_1 (solid, with ± 2 standard error band) and truth (dashed) when x_2 is independent of x_1 (left) and when $x_2 \approx x_1$ (right). Both fits follow the truth, but the band is twice as wide under concurvity.

worry when `worst` and `estimate` exceed about 0.8. Options: report the joint effect (`te(x1, x2)`) and predictions; do not over-interpret individual shapes; drop one of the covariates if scientifically justified; collect data where the covariates vary independently; and remember that high concurvity between a main effect and its own interaction is expected and does not by itself signal a problem.

18.6 Residual diagnostics

A GAM is a GLM with smooth terms, so the residuals of Chapter 9 apply. The **deviance residual** is $r_i^D = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$ where d_i is the contribution of observation i to the deviance; the **Pearson residual** is $(y_i - \hat{\mu}_i)/\sqrt{V(\hat{\mu}_i)}$; for a Gaussian model both equal the ordinary residual divided by $\hat{\sigma}$ or not, as the scale is defined. A residual plot is informative when we know what a *good* plot looks like: no trend, constant spread, no dependence.

18.6.1 gam.check() and the k-index

`gam.check(fit)` draws four plots (normal Q-Q plot of the deviance residuals, residuals against the linear predictor, a histogram, response against fitted values), reports convergence, and prints a table with the basis dimension k', the edf and the k**-index**. The k-index tests whether the residuals contain unexplained structure along the covariates of a smooth. For each observation i let $r_{n(i)}$ be the residual of its nearest neighbour in covariate space. If the residuals are independent with common variance σ^2, then $\mathrm{E}\{\frac{1}{2}(r_i - r_{n(i)})^2\} = \sigma^2$. The k-index is the ratio of $\frac{1}{n}\sum_i \frac{1}{2}(r_i - r_{n(i)})^2$ to the ordinary residual variance estimate: it is close to 1 when nothing is left, and below 1 when neighbouring residuals are more alike than they should be (a smoother f_j with a larger basis would still pick up signal). Its p-value comes from randomising the residuals.

```
set.seed(1)
k.check(ti1)
```

```
#>               k'   edf k-index p-value
#> s(Solar.R)     9 2.346  0.9356  0.2400
```

```
#> s(Wind)         9 1.545  0.9757  0.3900
#> s(Temp)         9 5.543  0.9507  0.2600
#> ti(Temp,Wind) 16 8.317  1.1017  0.8625
```

After adding the interaction, the k-index of `s(Wind)` rises from 0.86 to 0.98: the pattern along Wind was a *missing interaction*, not a basis that was too small. A low k-index is a prompt to look at the model, not a proof that k is too small.

18.6.2 Residuals against what?

- each covariate in the model: a remaining bend suggests a missing interaction or a larger k;
- covariates *not* in the model (here the month): any structure is a candidate for the model;
- the **order of the observations** (time): the days of `airquality` are consecutive, so serial correlation is possible.

```
aqd <- airquality; aqd$day <- seq_len(nrow(aqd)); aqd <- na.omit(aqd)
r <- residuals(ti1, type = "deviance")
oka <- diff(aqd$day) == 1
lag1 <- cor(r[-length(r)][oka], r[-1][oka])
mon <- anova(gam(r ~ factor(aqd$Month)))
dd <- rbind(data.frame(x = aqd$day, r = r, what = "day of the season"),
            data.frame(x = fitted(ti1), r = r, what = "fitted log(Ozone)"))
ggplot(dd, aes(x, r)) + geom_hline(yintercept = 0, linetype = 2) +
↪  geom_point(colour = ie_navy, alpha = .6) +
  geom_smooth(method = "loess", formula = y ~ x, colour = ie_coral, fill =
  ↪  "grey80") +
  facet_wrap(~what, scales = "free_x") + labs(x = NULL, y = "residual")
```

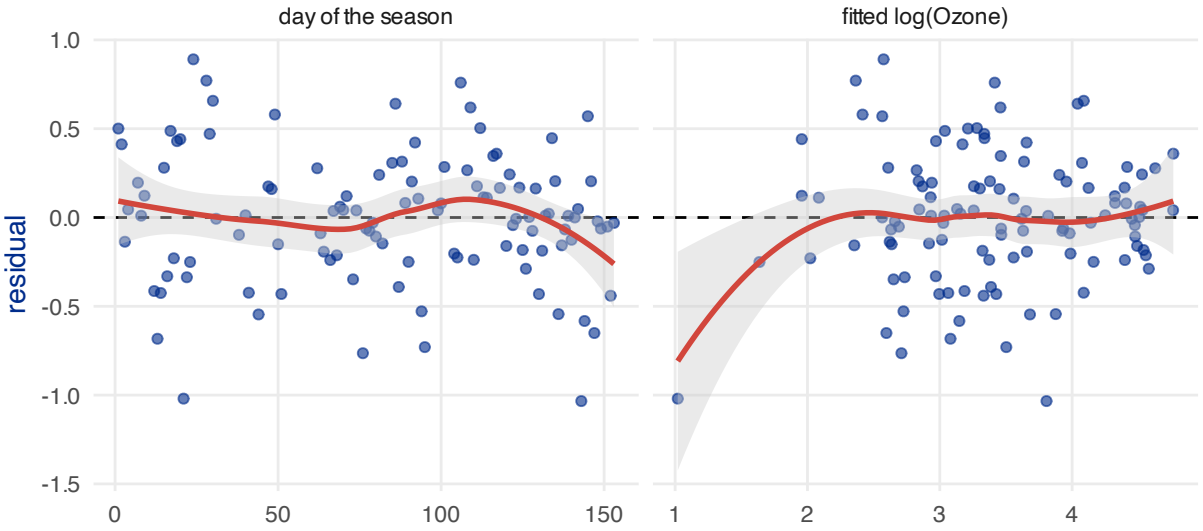


Figure 18.5: Deviance residuals of the interaction model in time order and against the fitted values: no trend, no funnel.

The lag-1 correlation of residuals on consecutive days is 0.13 (92 pairs), and a one-way test of the residuals on month gives $p = 0.26$. We find no evidence of trend, serial correlation or monthly effect, so the interaction model is adequate for these data. With serial or within-group correlation the standard errors are too small and the smoother may fit the correlated noise (it chooses too small a λ); the remedies, a correlation structure or random effects, are the subject of Chapter 19.

Warning. Residuals of non-Gaussian GAMs (binomial with few trials, Poisson with small means) are discrete and their plots show bands that are not a defect; use randomised quantile residuals or simulation envelopes (`qq.gam(fit, rep = 100)`, `gratia::appraise()`) rather than reading the deviance residual plot literally.

18.7 Interpretation and visualisation

18.7.1 Uncertainty of a fitted GAM

In the Bayesian view of penalised regression (Chapter 17) the coefficients have, approximately, the posterior $\boldsymbol{\gamma} \mid y \sim N(\hat{\boldsymbol{\gamma}}, V_\gamma)$ with $V_\gamma = \phi\,(X^\top W X + S_\lambda)^{-1}$. For any linear functional $d^\top\boldsymbol{\beta}$, in particular for linear predictors at new points, $\mathrm{Var}(d^\top\hat{\boldsymbol{\beta}}) = d^\top V_\beta d$. The matrix X_{new} whose rows are the model-matrix rows at the new points is available as `predict(fit, newdata, type = "lpmatrix")`; then

$$\hat{\eta}_{\text{new}} = X_{\text{new}}\hat{\boldsymbol{\gamma}}, \qquad \mathrm{Var}(\hat{\eta}_{\text{new}}) = X_{\text{new}} V_\gamma X_{\text{new}}^\top.$$

These credible intervals have good *across-the-function* coverage: averaged over the covariate values, about 95% of the true function values fall inside, although the coverage at a particular x can be lower where the curve bends sharply (Marra and Wood, 2012). `vcov(fit, unconditional = TRUE)` adds the uncertainty due to the estimation of $\hat{\boldsymbol{\lambda}}$.

18.7.2 Link scale and response scale

Intervals are formed on the link scale and then transformed, so that they respect the range of μ: $(g^{-1}(\hat{\eta} - 1.96\,\mathrm{se}),\ g^{-1}(\hat{\eta} + 1.96\,\mathrm{se}))$. Do *not* compute $g^{-1}(\hat{\eta}) \pm 1.96\,\mathrm{se}$ with the link-scale standard error: it is on the wrong scale and can leave the range of μ. For a log-transformed *response* $\exp(\hat{\eta})$ estimates the conditional median of the response, not its mean; for a log *link* (Poisson, Gamma) it is the mean.

```
nd <- expand.grid(Temp = c(70, 80, 90), Wind = c(5, 10, 15), Solar.R =
↪  median(aq$Solar.R))
pr <- predict(ti1, nd, se.fit = TRUE)
nd$median <- exp(pr$fit); nd$lower <- exp(pr$fit - 1.96 * pr$se.fit); nd$upper
↪  <- exp(pr$fit + 1.96 * pr$se.fit)
knitr::kable(round(nd, 1), booktabs = TRUE,
             caption = "Predicted median ozone (ppb) with 95% intervals from
             ↪  the interaction model (Solar.R at its median, 207).")
```

18.7.3 Contrasts

The effect of raising temperature from t_1 to t_2 at wind w is $d^\top\boldsymbol{\gamma}$ with $d = X_{(t_2,w)} - X_{(t_1,w)}$, the difference of two `lpmatrix` rows. On the log-ozone scale it is a log ratio of medians.

```
contrast <- function(m, t1, t2, w, S = 207) {
  X <- predict(m, data.frame(Temp = c(t1, t2), Wind = w, Solar.R = S), type =
↪  "lpmatrix")
  d <- X[2, ] - X[1, ]; est <- sum(d * coef(m)); se <- sqrt(drop(t(d) %*%
↪  vcov(m) %*% d))
  c(log_ratio = est, se = se, ratio = exp(est), lower = exp(est - 1.96 * se),
   ↪  upper = exp(est + 1.96 * se))
}
round(rbind(`wind 5` = contrast(ti1, 70, 90, 5), `wind 15` = contrast(ti1, 70,
↪  90, 15)), 3)
```

Table 18.2: Predicted median ozone (ppb) with 95% intervals from the interaction model (Solar.R at its median, 207).

Temp	Wind	Solar.R	median	lower	upper
70	5	207	33.0	17.2	63.2
80	5	207	71.6	52.9	97.1
90	5	207	99.4	76.3	129.3
70	10	207	22.6	17.4	29.3
80	10	207	32.9	26.8	40.2
90	10	207	80.9	57.2	114.4
70	15	207	15.8	10.7	23.2
80	15	207	24.8	18.7	32.8
90	15	207	63.1	29.5	135.1

```
#>         log_ratio    se ratio lower upper
#> wind 5      1.103 0.367 3.013 1.467 6.185
#> wind 15     1.388 0.428 4.006 1.732 9.268
```

Going from 70 to 90 F multiplies the median ozone by about 3 at 5 mph and by about 4 at 15 mph, with wide intervals: the interaction is supported *overall* ($p < 0.001$), but this particular difference of ratios is not precisely estimated.

18.7.4 Visualisation tools

- `plot(fit, pages = 1, scheme = 2, too.far = 0.1)`: all smooth terms; `scheme = 2` draws heat maps for two-dimensional terms. The argument `too.far` hides the surface far from data; use it for every two-dimensional smooth.
- `vis.gam(fit, view = c("x", "z"), plot.type = "contour")` or `"persp"`: the surface of the *linear predictor*, with the other covariates held at typical values (say which).
- `gratia::draw(fit)` and `gratia::appraise(fit)`: `ggplot2` versions of the smooth plots and of `gam.check()`.

```
gratia::draw(add, residuals = TRUE, nrow = 1)
```

18.8 Common pitfalls

Warning. (1) An isotropic `s(x, z)` for covariates in different units. (2) `s(x) + s(z) + te(x, z)` instead of `+ ti(x, z)`. (3) Reading "`s(Wind)` is not significant" as "wind does not matter" when it enters an interaction, or interpreting main effects of dependent covariates separately. (4) Treating `anova.gam` p-values as exact, or comparing REML fits whose fixed parts differ. (5) Ignoring concurvity when interpreting separate smooths. (6) Plotting two-dimensional surfaces over regions without data (no `too.far`). (7) Building response-scale intervals as $g^{-1}(\hat{\eta}) \pm 1.96\,\mathrm{se}$. (8) Taking a low k-index as proof that k is too small.

18.9 Summary

- A tensor-product smooth $f(x, z) = \sum_{k,l} \beta_{kl} a_k(x) c_l(z)$ has KL coefficients, a model matrix with row-wise Kronecker rows, and penalty $\lambda_x \boldsymbol{\beta}^\top (S_x \otimes I)\boldsymbol{\beta} + \lambda_z \boldsymbol{\beta}^\top (I \otimes S_z)\boldsymbol{\beta}$; its null

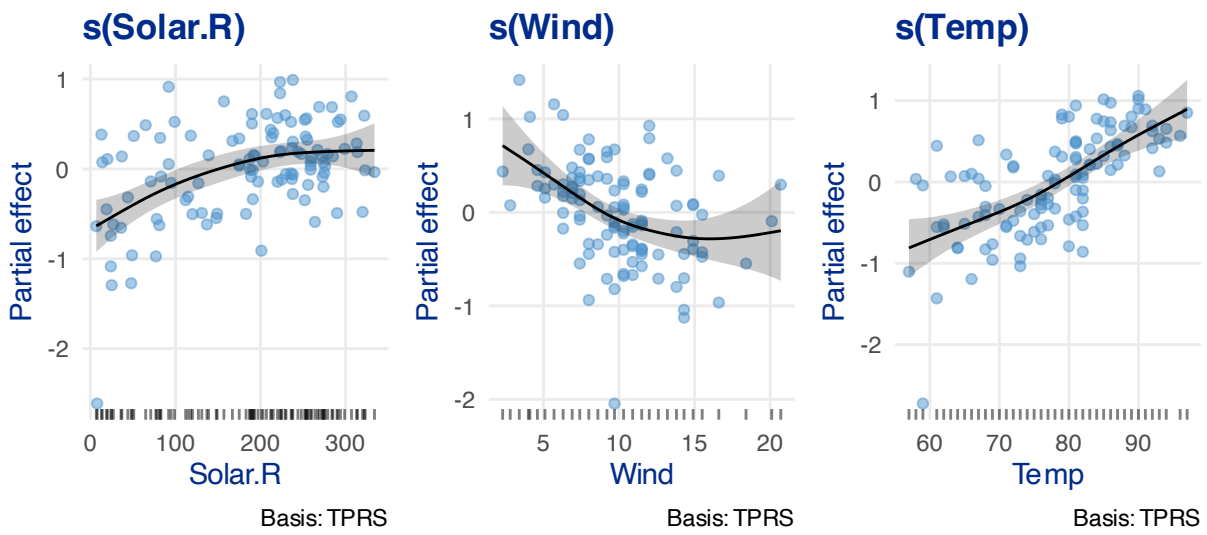


Figure 18.6: gratia::draw() of the additive model: partial effects with 95% credible bands and partial residuals.

space is $N_x \otimes N_z$.

- `te()` is invariant to the units of x and z (the change of units is absorbed by λ_x); the isotropic `s(x, z)` is not (three terms scale as c^{-3}, c^{-1}, c).
- `ti(x, z)` uses constrained margins, has $(K-1)(L-1)$ coefficients and is used with `s(x) + s(z)`; with dependent covariates the main effects of the decomposition are not interpretable alone.
- AIC uses effective df (justified by the covariance-penalty identity $2\sigma^2 \operatorname{tr}(S_\lambda)$); compare with ML for nested models; `anova.gam` p-values are approximate; REML scores of models with different null spaces are not comparable.
- `select = TRUE` adds a null-space penalty, so edf can go to zero.
- Concurvity is $\|P_{-j} X_j \boldsymbol{\beta}_j\|^2 / \|X_j \boldsymbol{\beta}_j\|^2$: worst case = largest squared canonical correlation. It inflates standard errors of the separate smooths, not of predictions.
- Check residuals against covariates in and out of the model and against time; the k-index compares neighbouring-residual variance with the residual variance.
- Build intervals on the link scale from $X_{\text{new}} V_\gamma X_{\text{new}}^\top$ and transform; contrasts are differences of `lpmatrix` rows.

18.10 Exercises

Exercise 18.1. Row-wise Kronecker. The margin in x is $a(x) = (1-x,\ x)$ and the margin in z is $c(z) = (1,\ z,\ z^2)$. (a) Write the model-matrix row at $(x, z) = (0.6,\ 3)$. (b) Let $S_x = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $S_z = \operatorname{diag}(0, 0, 1)$. Give the ranks of $S_x \otimes I_3$ and $I_2 \otimes S_z$, and the dimension of the null space of $\lambda_x S_x \otimes I_3 + \lambda_z I_2 \otimes S_z$ $(\lambda_x, \lambda_z > 0)$. Check it against Proposition 18.2.

Solution.

(a) $a = (0.4,\ 0.6)$ and $c = (1,\ 3,\ 9)$, so the row is $a \otimes c = (0.4,\ 1.2,\ 3.6,\ 0.6,\ 1.8,\ 5.4)$.

(b) S_x has rank 1 and I_3 rank 3, so $S_x \otimes I_3$ has rank $1 \cdot 3 = 3$; S_z has rank 1, so $I_2 \otimes S_z$ has rank $2 \cdot 1 = 2$. The range of the first is $\operatorname{span}\{(1, -1)\} \otimes \mathbb{R}^3$ and of the second $\mathbb{R}^2 \otimes \operatorname{span}\{e_3\}$; they

intersect in $\text{span}\{(1,-1)\otimes e_3\}$, of dimension 1. Both matrices are symmetric positive semi-definite, so the rank of the sum is the dimension of the sum of the ranges: $3+2-1=4$ and the null space has dimension $6-4=2$. Proposition 18.2: $N_x=\text{span}\{(1,1)\}$ (dimension 1) and $N_z=\text{span}\{e_1,e_2\}$ (dimension 2), so $\dim(N_x\otimes N_z)=1\cdot 2=2$. They agree.

Exercise 18.2. The tensor penalty. Derive $J_x(f)=\boldsymbol{\beta}^\top(S_x\otimes M_z)\boldsymbol{\beta}$ for $f=\sum_{k,l}\beta_{kl}a_k(x)c_l(z)$, where $(S_x)_{kk'}=\int a_k''a_{k'}''dx$ and $M_z=\int c\,c^\top dz$. Which approximation turns this into the penalty of `mgcv`, and what property does it buy?

Solution. Fix z and set $\alpha_k(z)=\sum_l\beta_{kl}c_l(z)$, so $f(x,z)=\sum_k\alpha_k(z)a_k(x)$ and $\partial^2 f/\partial x^2=\sum_k\alpha_k(z)a_k''(x)$. Then $\int(\partial^2 f/\partial x^2)^2dx=\sum_{k,k'}\alpha_k(z)\alpha_{k'}(z)\int a_k''a_{k'}''dx=\sum_{k,k'}(S_x)_{kk'}\alpha_k(z)\alpha_{k'}(z)$. Integrate over z: $\int\alpha_k\alpha_{k'}dz=\sum_{l,l'}\beta_{kl}\beta_{k'l'}\int c_lc_{l'}dz=\boldsymbol{\beta}_{k\cdot}^\top M_z\boldsymbol{\beta}_{k'\cdot}$. Hence $J_x=\sum_{k,k'}(S_x)_{kk'}\boldsymbol{\beta}_{k\cdot}^\top M_z\boldsymbol{\beta}_{k'\cdot}=\boldsymbol{\beta}^\top(S_x\otimes M_z)\boldsymbol{\beta}$, where $\boldsymbol{\beta}$ is stacked by k (the first index is the slow one).

`mgcv` replaces M_z by I_L, which gives $\boldsymbol{\beta}^\top(S_x\otimes I_L)\boldsymbol{\beta}$. The penalty then depends only on the marginal penalty matrices, not on the basis functions c_l or the range of z, which is what makes the construction scale invariant (Theorem 18.1) and lets each margin be any basis with its own penalty.

Exercise 18.3. Scale (non-)invariance. Let $f'(x',z)=f(x'/c,z)$ be f expressed in units $x'=cx$. Compute how each of $\iint f_{xx}^2$, $\iint f_{xz}^2$, $\iint f_{zz}^2$ changes, and conclude whether a thin-plate fit with one λ can be unit invariant. Then explain in two sentences why `te()` is.

Solution. With $f'(x',z)=f(x'/c,z)$: $f'_{x'x'}=c^{-2}f_{xx}$, $f'_{x'z}=c^{-1}f_{xz}$, $f'_{zz}=f_{zz}$, and $dx'=c\,dx$. Hence $\iint f'^2_{x'x'}dx'dz=c^{-3}\iint f_{xx}^2$, $\iint f'^2_{x'z}=c^{-1}\iint f_{xz}^2$ and $\iint f'^2_{zz}=c\iint f_{zz}^2$. The penalty $A+2B+C$ becomes $c^{-3}A+2c^{-1}B+cC$. The three parts are weighted differently, and one λ cannot restore the original weights (unless $c=1$), so the isotropic fit depends on the units. For `te()`, only the one-dimensional marginal penalty S_x is affected, by the factor c^{-3}, and it multiplies the single parameter λ_x, which can absorb it; the model matrix is unchanged.

Exercise 18.4. Counting. For margins with $k=6$ (in x) and $k=5$ (in z): (a) how many coefficients does `te(x, z, k = c(6, 5))` have after the identifiability constraint, and how many smoothing parameters? (b) the same for `ti(x, z, k = c(6, 5))`? (c) For three covariates, how many coefficients has `te(x, y, z, k = 4)`, and how many smoothing parameters? Verify (a)–(c) in R with simulated covariates.

Solution.

(a) $6\cdot 5-1=29$ coefficients, two smoothing parameters. (b) $(6-1)(5-1)=20$ coefficients, two smoothing parameters. (c) $4^3-1=63$ coefficients and three smoothing parameters (one per margin). Check:

```
set.seed(3); dd <- data.frame(x = runif(200), y = runif(200), z = runif(200))
dd$r <- rnorm(200)
cnt <- function(f) { g <- gam(f, data = dd, fit = FALSE); sm <- g$smooth[[1]]
  c(coefficients = sm$last.para - sm$first.para + 1, smoothing_par =
  ↪  length(sm$S)) }
rbind(te_xz = cnt(r ~ te(x, z, k = c(6, 5))), ti_xz = cnt(r ~ ti(x, z, k =
↪  c(6, 5))),
      te_xyz = cnt(r ~ te(x, y, z, k = 4)))
```

```
#>        coefficients smoothing_par
```

```
#> te_xz          29          2
#> ti_xz          20          2
#> te_xyz         63          3
```

Exercise 18.5. ANOVA constraints with dependent covariates. Take centred marginal functions $u(x) = x - \bar{x}$, $v(z) = z - \bar{z}$ and the interaction $f_{xz} = \beta u(x)v(z)$ (where the bars are sample means). (a) For the product design $(x, z) \in \{0, 1\}^2$ show that the average of f_{xz} over the points with $x = 0$ is zero. (b) For the design $(0, 0), (0, 1), (0, 1), (1, 0)$ compute u, v and the average of f_{xz} over the points with $x = 0$. What does this say about the main effect f_x?

Solution.

(a) $\bar{x} = \bar{z} = \frac{1}{2}$. For $x = 0$: $u = -\frac{1}{2}$, and $v = \mp\frac{1}{2}$ at $z = 0, 1$, so the average of $f_{xz} = \beta uv$ over the two points is $\beta(-\frac{1}{2}) \cdot \frac{1}{2}(-\frac{1}{2} + \frac{1}{2}) = 0$. The same for $x = 1$, and symmetrically for each value of z.

(b) $\bar{x} = \frac{1}{4}$, so $u(0) = -\frac{1}{4}$ and $u(1) = \frac{3}{4}$; $\bar{z} = \frac{1}{2}$, so $v(0) = -\frac{1}{2}$ and $v(1) = \frac{1}{2}$. The values f_{xz}/β at the four points are $(-\frac{1}{4})(-\frac{1}{2}) = \frac{1}{8}$, $(-\frac{1}{4})(\frac{1}{2}) = -\frac{1}{8}$, $-\frac{1}{8}$ and $(\frac{3}{4})(-\frac{1}{2}) = -\frac{3}{8}$. The three points with $x = 0$ have average $\beta(\frac{1}{8} - \frac{1}{8} - \frac{1}{8})/3 = -\beta/24 \neq 0$. The constraint (centred margins) makes the interaction average to zero over the *product of the marginal distributions*, not over the conditional distribution of z given x. Therefore part of "the effect of x averaged over z" is contained in f_{xz}, the main effect f_x is not the conditional-average effect, and it will change when the interaction is added or dropped. Main effects must be read together with the interaction.

Exercise 18.6. AIC by hand. Three models for the same data have log-likelihood and df: $(-120.4,\ 8.2)$, $(-112.9,\ 12.6)$, $(-111.8,\ 16.1)$. (a) Compute AIC, ΔAIC and Akaike weights. (b) What is the evidence ratio of the best versus the second best model? (c) Why is it legitimate to compare these AICs only if the models share response and data?

Solution.

(a) $\text{AIC} = -2\ell + 2\,\text{df}$: $240.8 + 16.4 = 257.2$; $225.8 + 25.2 = 251.0$; $223.6 + 32.2 = 255.8$. The minimum is 251.0, so $\Delta = (6.2,\ 0,\ 4.8)$. The weights are proportional to $e^{-3.1} = 0.0450$, 1, $e^{-2.4} = 0.0907$; the sum is 1.1357, so $w = (0.040,\ 0.880,\ 0.080)$.

(b) $w_2/w_3 = e^{(4.8)/2} = e^{2.4} = 11.0$: the middle model is about eleven times better supported than the third, in the Akaike sense.

(c) AIC estimates the expected log predictive density of *the data distribution that generated these y*. The constant terms dropped from the likelihood cancel only within one response and data set. A log-transformed response, for example, has a different likelihood scale; one would have to add the Jacobian $-2\sum \log y_i$ (or work with the same y).

Exercise 18.7. R: tree volume with a Gamma GAM. The data set `trees` (package `datasets`) has `Girth` (inches), `Height` (feet) and `Volume` (cubic feet) of 31 black cherry trees. (a) Fit `Volume ~ s(Girth) + s(Height)`, `Volume ~ s(Girth) + s(Height) + ti(Girth, Height)` and `Volume ~ log(Girth) + log(Height)`, all with `family = Gamma(link = "log")` and `method = "REML"`, and compare them by AIC. (b) For the log-log model, interpret the two slopes with the geometry of a cylinder or a cone, $V \propto G^2 H$. (c) Is there evidence of an interaction? Comment on the concurvity of the

additive model.

Solution.

```
data(trees)
gA <- gam(Volume ~ s(Girth) + s(Height), family = Gamma(link = "log"), data =
↪  trees, method = "REML")
gB <- update(gA, . ~ . + ti(Girth, Height))
gL <- gam(Volume ~ log(Girth) + log(Height), family = Gamma(link = "log"),
↪  data = trees, method = "REML")
AIC(gA, gB, gL)
```

```
#>       df   AIC
#> gA 6.282 144.4
#> gB 5.000 143.7
#> gL 4.000 140.1
```

```
round(summary(gL)$p.table, 3)
```

```
#>             Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   -6.691      0.788  -8.493        0
#> log(Girth)     1.980      0.074  26.802        0
#> log(Height)    1.133      0.201   5.625        0
```

```
round(summary(gB)$s.table, 4)
```

```
#>                   edf Ref.df      F p-value
#> s(Girth)            1      1 619.52  0.0000
#> s(Height)           1      1  22.26  0.0001
#> ti(Girth,Height)    1      1  10.05  0.0038
```

```
round(concurvity(gA, full = TRUE), 2)
```

```
#>          para s(Girth) s(Height)
#> worst       0     0.90      0.90
#> observed    0     0.44      0.48
#> estimate    0     0.46      0.48
```

(a) The log-log model has the lowest AIC (140.1 against 144.4 for the additive GAM and 143.7 with the `ti` term): the smooths use up 3.7 df where the parametric model needs two slopes, and they estimate shapes that are essentially straight on the log scale (edf of `s(Height)` ≈ 1).

(b) With a log link, $\log \mu = \beta_0 + \beta_1 \log G + \beta_2 \log H$, so $\mu \propto G^{\beta_1} H^{\beta_2}$. The estimates are $\hat{\beta}_1 = 1.98$ and $\hat{\beta}_2 = 1.13$, remarkably close to the values 2 and 1 of a cylinder (or cone), $V = \frac{\pi}{4} G^2 H$ up to units: volume scales with the cross-section area times the height. The model was found from data, but the geometry confirms it.

(c) Within the GAM class, the `ti` term is significant (edf 1, $p = 0.004$) and AIC improves a little over the additive GAM (144.4 to 143.7), but the parametric log-log model without interaction has a lower AIC than both. A significance test inside a flexible family and AIC across families can disagree; here the physics (the cylinder law) and AIC agree on the simple model. The concurvity of the additive GAM is high in the worst case (0.9) but moderate on average (`estimate` 0.46): Girth and Height are correlated ($r = 0.52$)

and there are only 31 trees, so the separate curves should not be over-interpreted, while predictions are well determined.

Exercise 18.8. Worst-case concurvity by hand. $n = 4$. The other terms span span$\{\mathbf{1}, t\}$ with $t = (1, 2, 3, 4)^\top$. The term of interest has the single column $q = (-2, -1, 2, 1)^\top$. Compute its concurvity (all three measures coincide for a single column). Then modify q to $q' = (1, -1, -1, 1)^\top$ and explain the result.

Solution. The projector onto span$\{\mathbf{1}, t\}$ is the projector onto span$\{\mathbf{1}, t_c\}$ with $t_c = t - \bar{t} = (-1.5, -0.5, 0.5, 1.5)^\top$, and $\mathbf{1}^\top t_c = 0$. Since $\mathbf{1}^\top q = 0$, $Pq = \frac{q^\top t_c}{t_c^\top t_c} t_c$ and $\|Pq\|^2 = \frac{(q^\top t_c)^2}{t_c^\top t_c}$. Here $q^\top t_c = 3 + 0.5 + 1 + 1.5 = 6$, $t_c^\top t_c = 5$, so $\|Pq\|^2 = 36/5 = 7.2$; $\|q\|^2 = 10$. The concurvity is 0.72.

For q': $\mathbf{1}^\top q' = 0$ and $t_c^\top q' = -1.5 + 0.5 - 0.5 + 1.5 = 0$: $q' \perp$ span$\{\mathbf{1}, t\}$, concurvity 0. The term is orthogonal to the others and can be estimated independently.

```
t0 <- 1:4; Q <- qr.Q(qr(cbind(1, t0)))
conc1 <- function(q) sum((Q %*% crossprod(Q, q))^2) / sum(q^2)
c(q = conc1(c(-2, -1, 2, 1)), q_prime = conc1(c(1, -1, -1, 1)))
```

```
#>       q q_prime
#>    0.72    0.00
```

Exercise 18.9. R: concurvity and precision. Using the simulation of Section 18.5 with `set.seed(26)` before generating x_2 and y, for $s \in \{0.5, 0.15, 0.04\}$, tabulate Corr$(x_1, x_2)$, the concurvity estimate of `s(x1)` and the mean standard error of $\hat{f}_1$. Describe how the standard error grows with concurvity.

Solution.

```
res2 <- t(sapply(c(.5, .15, .04), conc_sim, seed = 26))
rownames(res2) <- paste("s =", c(.5, .15, .04)); round(res2, 3)
```

```
#>             cor concurvity mean_se
#> s = 0.5   0.526      0.258   0.050
#> s = 0.15  0.898      0.755   0.064
#> s = 0.04  0.991      0.971   0.134
```

The standard error grows slowly while concurvity is moderate and rapidly as it approaches one, like $1/(1-R^2)$ for the variance inflation factor of a collinear regressor: going from concurvity 0.26 to 0.97 multiplies the mean standard error by 2.7. The fits themselves remain unbiased-looking; it is the precision of the *separate* shape that is lost.

Exercise 18.10. Shrinkage selection. (a) Explain why the ordinary penalty of a cubic regression spline leaves edf ≥ 1 for a pure-noise covariate. (b) Prove that with the double penalty and a Gaussian response, edf $\to 0$ when both smoothing parameters $\to \infty$. (c) In R, add `noise3 <- rnorm(n)` to the ozone model with `select = TRUE` and report its edf and p-value.

Solution.

(a) The second-derivative penalty vanishes for straight lines, so the linear part of each smooth is unpenalised and is estimated by (unpenalised) least squares however large λ is: even

for $\lambda \to \infty$ the fit has the form $a + bx$, i.e. edf approaching 1 per smooth (beyond the intercept).

(b) Write $A_\lambda = X^\top X + \lambda S + \lambda^* S^*$. Since $S + S^*$ has full rank on the coefficient space of the term, for $\lambda, \lambda^* \geq c > 0$ the smallest eigenvalue of $\lambda S + \lambda^* S^*$ is at least $c\,\kappa$ with $\kappa > 0$ the smallest eigenvalue of $S + S^*$. Hence $A_\lambda^{-1} \to 0$ as the parameters grow, $F = A_\lambda^{-1} X^\top X \to 0$ and $\mathrm{edf} = \mathrm{tr}(F) \to 0$.

(c) One noise realisation is not informative, so we repeat the experiment 60 times.

```
set.seed(11)
out <- t(replicate(60, {
  aq$noise3 <- rnorm(nrow(aq))
  m3 <- gam(log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp) + s(noise3), data =
↪  aq,
            method = "REML", select = TRUE)
  summary(m3)$s.table["s(noise3)", c("edf", "p-value")]
}))
c(median_edf = median(out[, 1]), share_edf_below_half = mean(out[, 1] < .5),
  share_p_below_5pct = mean(out[, 2] < .05))
```

```
#>           median_edf share_edf_below_half   share_p_below_5pct
#>             0.000167             0.766667             0.066667
```

In most replicates the edf of the noise term is essentially zero (median 1.7×10^{-4}): the term is removed. The p-values of shrunk terms are only approximate (here 7% fall below 0.05 against a nominal 5%), so use the edf to screen and the p-values with caution.

Exercise 18.11. R: a contrast with its uncertainty. For the `ti` model of the ozone data: (a) estimate the ratio of median ozone for Temp 80 → 90 F at Wind = 10 mph and Solar.R = 207 with a 95% interval, with and without the uncertainty in $\hat{\boldsymbol{\lambda}}$ (`unconditional = TRUE`). (b) Estimate the *difference* between the log ratios of 70 → 90 F at 5 and at 15 mph, i.e. the interaction contrast, with its standard error. Is it significantly different from zero? (c) Explain why $\exp(\hat{\eta}) \pm 1.96\,\mathrm{se}$ would be wrong.

Solution.

```
cc <- function(m, unc) {
  X <- predict(m, data.frame(Temp = c(80, 90), Wind = 10, Solar.R = 207), type
↪  = "lpmatrix")
  d <- X[2, ] - X[1, ]; est <- sum(d * coef(m)); se <- sqrt(drop(t(d) %*%
↪  vcov(m, unconditional = unc) %*% d))
  c(ratio = exp(est), lower = exp(est - 1.96 * se), upper = exp(est + 1.96 *
  ↪  se), se = se)
}
rbind(conditional = cc(ti1, FALSE), unconditional = cc(ti1, TRUE))
```

```
#>                ratio lower upper     se
#> conditional    2.462 1.639 3.700 0.2077
#> unconditional 2.462 1.587 3.821 0.2242
```

```
X5 <- predict(ti1, data.frame(Temp = c(70, 90), Wind = 5, Solar.R = 207), type
↪  = "lpmatrix")
X15 <- predict(ti1, data.frame(Temp = c(70, 90), Wind = 15, Solar.R = 207),
↪  type = "lpmatrix")
```

```
d <- (X5[2, ] - X5[1, ]) - (X15[2, ] - X15[1, ])
est <- sum(d * coef(ti1)); se <- sqrt(drop(t(d) %*% vcov(ti1) %*% d))
c(est = est, se = se, z = est / se, ratio_of_ratios = exp(est), lower =
↪  exp(est - 1.96 * se), upper = exp(est + 1.96 * se))
```

```
#>             est              se               z ratio_of_ratios
#>         -0.2850          0.5109         -0.5579          0.7520
#>           lower           upper
#>          0.2763          2.0470
```

(a) Median ozone is about 2.5 times higher at 90 F than at 80 F at 10 mph, with the intervals shown; accounting for the uncertainty in $\hat{\boldsymbol{\lambda}}$ widens the interval (se 0.208 to 0.224). (b) The difference of log ratios is -0.29 with se 0.51 (z = -0.56): the temperature effect is not significantly different between 5 and 15 mph *at these two points*, although the overall interaction term is highly significant. The interaction is spread over the surface and no single contrast captures it. (c) $\exp(\hat{\eta})$ is on the response scale, while the standard error is for $\hat{\eta}$ on the log scale; the symmetric interval on the response scale has wrong width and may include negative values. Build the interval for $\hat{\eta}$ and exponentiate its endpoints.

Exercise 18.12. Critique and communicate. A colleague writes: "The best model has AIC 137.9, and the interaction term has $p = 0.0008$, so temperature and wind *physically interact* to produce ozone. Moreover `s(Wind)` is not significant ($p = 0.15$), so wind can be dropped." (a) List what is wrong or over-stated. (b) Write a summary of about 150 words for a non-statistician (an environmental agency) of what the model says.

Solution.

(a) (1) AIC and p show that, *for these 111 days and this class of smooth models*, an interaction improves the fit; they do not establish a physical mechanism (observational data, confounding with unmeasured variables, sunlight and season). (2) The interaction p-value is approximate (smoothing parameters treated as known); with ML-based anova it is 0.0036, with REML 3×10^{-5}. (3) `s(Wind)` is not significant *given* the interaction and the strongly correlated Temp (correlation -0.50); the concurvity of the main effects with the interaction is high, so main effects cannot be read separately. Dropping `s(Wind)` while keeping `ti(Temp, Wind)` would also change the meaning of the interaction term. Wind is in the model through the interaction. (4) Wide intervals for specific contrasts show that the interaction is not precisely estimated.

(b) "We studied 111 summer days in New York in 1973. Ozone is higher on hot days and on calm days, and the combination matters: a rise in temperature from 70 to 90°F is associated with about three times more ozone when the wind is light (5 mph) and about four times more when it is moderate (15 mph), although with only this many days the exact factors are uncertain (plausible values 1.5 to 6 and 1.7 to 9). A model that allows temperature and wind to work together describes the data clearly better than one in which they act separately. Because temperature and wind tend to move together, we cannot reliably separate their individual contributions. The results describe associations in one summer and do not by themselves prove cause. A longer record would give sharper answers."

18.11 Further reading

The tensor-product construction and its scale invariance are in Wood (2006); the modern account of GAMs, with `mgcv` examples for every topic of this chapter, is Wood (2017) (Sections 5.6, 6.12 and Chapter 7). Shrinkage and variable selection for GAMs: Marra and Wood (2011), *Computational Statistics and Data Analysis* 55, 2372–2387. Information-theoretic model comparison and Akaike weights: Burnham and Anderson (2004), *Sociological Methods and Research* 33, 261–304. The coverage of credible intervals for smooth components: Marra and Wood (2012). For the practice of hierarchical and interaction GAMs in applied work see Pedersen et al. (2019). The next chapter, Chapter 19, adds random effects and correlated errors.

Chapter 19

Generalized additive mixed models

Many data sets combine two structures that we have so far treated separately: a *smooth* dependence on a covariate (Chapters 14–18) and a *grouped* or *longitudinal* layout in which observations on the same unit are correlated (Chapters 11–13). The growth of a child is a smooth curve in age, but each child has its own level and its own speed of growth, and successive measurements on the same child are not independent. A **generalized additive mixed model** (GAMM) puts both in one model (Lin and Zhang, 1999; Wood, 2017).

The key to the chapter is that the two structures are *the same mathematics*: a penalised spline is a mixed model, and a random effect is a penalised coefficient. We derive the correspondence, use it to read smoothing parameters as variance ratios, and then use it in practice: random intercepts, slopes and subject-specific curves inside `mgcv::gam()`, the two dedicated functions `mgcv::gamm()` and `gamm4::gamm4()`, correlated residuals, diagnostics, and the interpretation of a GAMM.

```
library(mgcv)
library(nlme)
library(splines)
ox <- as.data.frame(nlme::Oxboys)       # 26 boys, 9 occasions
ox$Subject <- factor(as.character(ox$Subject))
```

19.1 A motivating example: the growth of 26 boys

The data frame `Oxboys` (package `nlme`) records the height (cm) of 26 boys in Oxford on 9 occasions; `age` is standardised to run from -1 to 1, so that the sample has $n = 234$ observations.

```
ggplot(ox, aes(age, height, group = Subject)) +
  geom_line(colour = ie_navy, alpha = .55) + geom_point(colour = ie_navy, size
  ↪  = .9, alpha = .7) +
  labs(x = "Standardised age", y = "Height (cm)")
```

There are two sources of structure: a smooth population growth curve $f(\text{age})$, and differences between boys. A first attempt ignores the second.

```
m_naive <- gam(height ~ s(age), data = ox, method = "REML")
lag1 <- function(r, id) {               # within-subject lag-1 correlation of
↪  residuals
  d <- split(r, id)
  cor(unlist(lapply(d, function(v) v[-length(v)])), unlist(lapply(d,
  ↪  function(v) v[-1])))
```

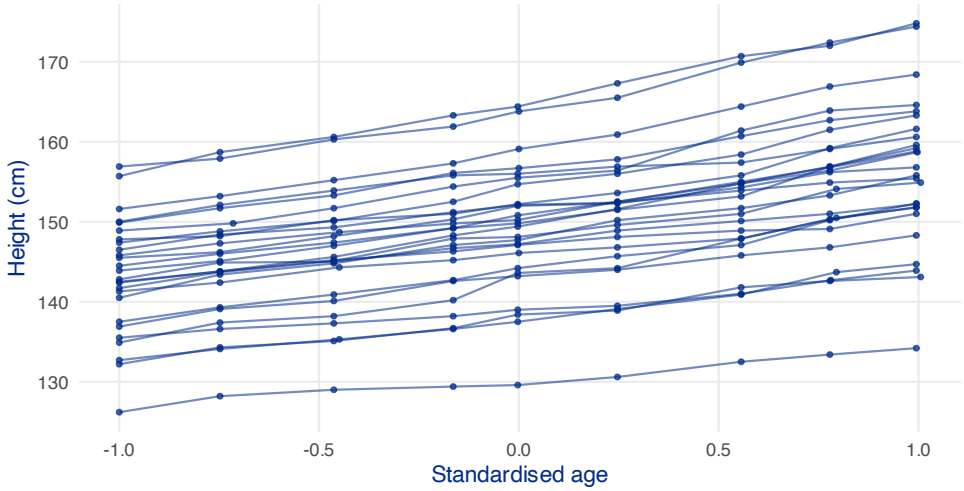


Figure 19.1: Heights of the 26 boys against standardised age. The population curve is smooth and nearly linear, but boys differ in level and in rate of growth.

```
}
c(edf = sum(m_naive$edf) - 1, sigma = sqrt(m_naive$sig2), lag1_within_boy =
↪  lag1(residuals(m_naive), ox$Subject))
```

```
#>              edf            sigma lag1_within_boy
#>           1.0007           8.0805           0.9955
```

The residual standard deviation is about $\hat{\sigma} = 8.1$ cm, and the residuals of one boy are almost perfectly correlated (lag-1 correlation 1): a boy who is taller than the population curve at age -1 is taller at age 0 as well. The standard errors of this naive fit are far too small, since it treats the 234 measurements as independent. We need a model for the boy effects.

19.2 Penalised splines as mixed models

19.2.1 Mixed models as penalised regression

Recall the linear mixed model of Chapter 11, with groups i and, to keep the notation light, all random effects stacked in one vector:

$$\boldsymbol{y} = X\boldsymbol{\alpha} + Z\boldsymbol{b} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{b} \sim N(\boldsymbol{0}, \sigma_b^2 I), \quad \boldsymbol{\varepsilon} \sim N(\boldsymbol{0}, \sigma^2 I). \tag{19.1}$$

In this chapter the fixed-effect vector is written $\boldsymbol{\alpha}$ (not $\boldsymbol{\beta}$ as in Chapters 11 and 12), because $\boldsymbol{\beta}$ denotes the P-spline coefficients of Chapter 15. The fixed effects are *estimated*, the random effects *predicted* by their conditional expectation (the BLUP $\tilde{\boldsymbol{b}}$).

Theorem 19.1 (Joint mode of a mixed model as penalised least squares)**.** *For fixed σ^2, σ_b^2, the pair $(\hat{\boldsymbol{\alpha}}, \tilde{\boldsymbol{b}})$ that maximises the joint density $f(\boldsymbol{y} \mid \boldsymbol{b}) f(\boldsymbol{b})$ minimises*

$$\|\boldsymbol{y} - X\boldsymbol{\alpha} - Z\boldsymbol{b}\|^2 + \lambda \boldsymbol{b}^\top \boldsymbol{b}, \qquad \lambda = \frac{\sigma^2}{\sigma_b^2}, \tag{19.2}$$

and solves Henderson's mixed-model equations

$$\begin{pmatrix} X^\top X & X^\top Z \\ Z^\top X & Z^\top Z + \lambda I \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\alpha}} \\ \tilde{\boldsymbol{b}} \end{pmatrix} = \begin{pmatrix} X^\top \boldsymbol{y} \\ Z^\top \boldsymbol{y} \end{pmatrix}.$$

These are the generalised least-squares estimate of $\boldsymbol{\alpha}$ and the BLUP of $\boldsymbol{b}$ (Henderson, 1975; Robinson, 1991).

Proof. The log joint density is

$$\log f(\boldsymbol{y} \mid \boldsymbol{b}) + \log f(\boldsymbol{b}) = -\frac{\|\boldsymbol{y} - X\boldsymbol{\alpha} - Z\boldsymbol{b}\|^2}{2\sigma^2} - \frac{\boldsymbol{b}^\top \boldsymbol{b}}{2\sigma_b^2} + \text{const.}$$

Multiplying by $-2\sigma^2$, maximising is the same as minimising $\|\boldsymbol{y} - X\boldsymbol{\alpha} - Z\boldsymbol{b}\|^2 + (\sigma^2/\sigma_b^2)\boldsymbol{b}^\top\boldsymbol{b}$, which is (19.2). Setting the gradients with respect to $\boldsymbol{\alpha}$ and $\boldsymbol{b}$ to zero gives $X^\top(\boldsymbol{y} - X\boldsymbol{\alpha} - Z\boldsymbol{b}) = \mathbf{0}$ and $Z^\top(\boldsymbol{y} - X\boldsymbol{\alpha} - Z\boldsymbol{b}) - \lambda\boldsymbol{b} = \mathbf{0}$, which are the two block rows of the equations. That the solution is the GLS estimate and the BLUP is the classical result of Henderson; see Chapter 12. □

So a random effect with variance σ_b^2 is a **ridge penalty** with parameter $\lambda = \sigma^2/\sigma_b^2$: a large σ_b^2 (units differ a lot) means a small λ and little shrinkage; $\sigma_b^2 \to 0$ means $\lambda \to \infty$ and the random effect is removed.

19.2.2 The mixed-model form of a P-spline

A P-spline (Chapter 15) minimises $\|\boldsymbol{y} - B\boldsymbol{\beta}\|^2 + \lambda\boldsymbol{\beta}^\top S\boldsymbol{\beta}$ with B an $n \times K$ B-spline basis matrix on equally spaced knots and $S = D_d^\top D_d$ the penalty from the d-th order difference matrix D_d. Our goal is to bring this into the form (19.2), with the penalty on *some* coefficients and none on the others.

Lemma 19.1 (Null space of the difference penalty)**.** *The null space of D_d (hence of S) is the set of coefficient sequences β_k that are polynomials of degree $< d$ in k. Its dimension is d. For equally spaced knots, $B\boldsymbol{\beta}$ is then a polynomial of degree $< d$ in x.*

Proof. $D_d\boldsymbol{\beta} = 0$ means that all d-th differences vanish, i.e. $\Delta^d\beta_k = 0$ for all k, which holds exactly for polynomial sequences of degree $< d$ (a polynomial of degree m has m-th differences that are constant and $(m+1)$-th differences equal to zero). For $d = 2$ this says that the first differences $\beta_{k+1} - \beta_k = c$ are constant, so $\beta_k = a + ck$. B-splines on equally spaced knots $t_k^* = t_0 + kh$ have the Greville property $\sum_k t_k^* B_k(x) = x$ and form a partition of unity, $\sum_k B_k(x) = 1$ (inside the range of the data). Then $\sum_k (a + ck)B_k(x) = a + \frac{c}{h}(x - t_0)$ is linear in x. The same argument with Greville-type identities for $x^2, \ldots$ gives polynomials of degree $< d$ for general d. □

Theorem 19.2 (Mixed-model representation of a P-spline)**.** *Let $S = U\Sigma U^\top$ be the eigen-decomposition of the penalty matrix, $U = [U_0\ U_+]$ with U_0 $(K \times d)$ the eigenvectors for the zero eigenvalues and U_+ $(K \times (K-d))$ those for the positive eigenvalues $\Sigma_+ = \operatorname{diag}(\sigma_{+,1}, \ldots, \sigma_{+,K-d})$. Write*

$$\boldsymbol{\beta} = U_0\boldsymbol{\alpha}_0 + U_+\Sigma_+^{-1/2}\boldsymbol{b}.$$

Then $\boldsymbol{\beta}^\top S\boldsymbol{\beta} = \boldsymbol{b}^\top\boldsymbol{b}$, and the penalised criterion is

$$\|\boldsymbol{y} - X\boldsymbol{\alpha}_0 - Z\boldsymbol{b}\|^2 + \lambda\,\boldsymbol{b}^\top\boldsymbol{b}, \qquad X = BU_0, \quad Z = BU_+\Sigma_+^{-1/2}.$$

Here X spans the same space as the polynomials $[\mathbf{1}, x, \ldots, x^{d-1}]$. Hence the P-spline is the mixed model (19.1) with d fixed effects, $K - d$ random effects $\boldsymbol{b} \sim N(\mathbf{0}, \sigma_b^2 I_{K-d})$ and $\lambda = \sigma^2/\sigma_b^2$.

Proof. The matrix $[U_0 \; U_+\Sigma_+^{-1/2}]$ is $K \times K$ and invertible (U is orthogonal and $\Sigma_+^{-1/2}$ is a positive diagonal matrix), so every $\boldsymbol{\beta}$ has exactly one representation $(\boldsymbol{\alpha}_0, \boldsymbol{b})$. Since $SU_0 = 0$ and $U_+^\top S U_+ = \Sigma_+$,

$$\boldsymbol{\beta}^\top S \boldsymbol{\beta} = \boldsymbol{b}^\top \Sigma_+^{-1/2} U_+^\top S U_+ \Sigma_+^{-1/2} \boldsymbol{b} = \boldsymbol{b}^\top \Sigma_+^{-1/2} \Sigma_+ \Sigma_+^{-1/2} \boldsymbol{b} = \boldsymbol{b}^\top \boldsymbol{b},$$

the cross terms vanishing because $SU_0 = 0$. Substitute $B\boldsymbol{\beta} = BU_0\boldsymbol{\alpha}_0 + BU_+\Sigma_+^{-1/2}\boldsymbol{b}$. The columns of U_0 span the polynomial sequences of Lemma 19.1, so the columns of BU_0 span polynomials in x of degree $< d$. Finally Theorem 19.1 applies. □

Because the one variance σ_b^2 is shared by all $K - d$ random coefficients, smooth columns of Z (small $\sigma_{+,k}$, hence a *large* scale $\sigma_{+,k}^{-1/2}$) are shrunk little and wiggly columns (large $\sigma_{+,k}$, small scale) are shrunk strongly; the unpenalised polynomial part is a fixed effect and is not shrunk at all. This is exactly the behaviour of a smoothing spline.

19.2.3 Smoothing parameter = variance ratio

Proposition 19.1 (REML for the smoothing parameter is REML for the variance ratio)**.** *In the mixed model of Theorem 19.2, integrating out $\boldsymbol{b}$ gives $\boldsymbol{y} \sim N(X\boldsymbol{\alpha}_0,\, V)$ with $V = \sigma_b^2 ZZ^\top + \sigma^2 I = \sigma^2(I + \lambda^{-1} ZZ^\top)$. The map $(\sigma^2, \sigma_b^2) \mapsto (\sigma^2, \lambda)$ is one-to-one, so the REML estimates satisfy $\hat{\lambda} = \hat{\sigma}^2/\hat{\sigma}_b^2$ (invariance of the (restricted) likelihood estimator under reparametrisation, Chapter 3).*

In `mgcv`, `method = "REML"` maximises exactly this restricted likelihood over (σ^2, λ) (Wood, 2011). Thus *choosing a smoothing parameter by REML is estimating a variance component.* This also suggests why ML tends to give slightly smoother fits than REML (usually a small difference): ML ignores the degrees of freedom used by the fixed (polynomial) part and gives biased-downward variance components, exactly as in Chapter 12. The connection was developed by Verbyla et al. (1999), Wand (2003) and Currie and Durban (2002).

19.2.4 Numerical check on the motorcycle data

The data set `MASS::mcycle` gives the head acceleration (g) in a simulated motorcycle accident against time (ms), $n = 133$. We build the P-spline ourselves: cubic B-splines on 22 equal segments ($K = 25$), second-order difference penalty.

```
mc <- MASS::mcycle; y <- mc$accel; x <- mc$times; n <- length(y)
nseg <- 22; deg <- 3; dx <- diff(range(x)) / nseg
knots <- seq(min(x) - deg * dx, max(x) + deg * dx, by = dx)
B <- splineDesign(knots, x, ord = deg + 1, outer.ok = TRUE)
K <- ncol(B)
S <- crossprod(diff(diag(K), differences = 2))          # S = D'D, rank K - 2
e <- eigen(S, symmetric = TRUE); pos <- e$values > 1e-8
c(K = K, positive = sum(pos), zero = sum(!pos))
```

```
#>        K positive     zero
#>       25       23        2
```

Of the $K = 25$ eigenvalues two are zero, as in Lemma 19.1. We now build $X = [\mathbf{1}, x]$ and $Z = BU_+\Sigma_+^{-1/2}$ and fit two ways: (i) `gam()` with our own B and S supplied through `paraPen`, so that its smoothing parameter is on exactly the scale of $\|\boldsymbol{y} - B\boldsymbol{\beta}\|^2 + \lambda\boldsymbol{\beta}^\top S\boldsymbol{\beta}$; (ii) `nlme::lme()`

with one group and a `pdIdent` covariance, which gives the $K-2$ random coefficients one common variance.

```
fit_pen <- gam(y ~ B - 1, paraPen = list(B = list(S)), method = "REML")
Z <- B %*% e$vectors[, pos] %*% diag(1 / sqrt(e$values[pos]))
g <- factor(rep(1, n))
fit_lme <- lme(y ~ x, random = list(g = pdIdent(~ Z - 1)), method = "REML")
s2 <- fit_lme$sigma^2; s2b <- as.numeric(VarCorr(fit_lme)[1, "Variance"])
round(c(sigma2 = s2, sigma2_b = s2b, ratio = s2 / s2b, lambda_gam =
↪  unname(fit_pen$sp),
        edf = sum(fit_pen$edf)), 4)
```

```
#>     sigma2   sigma2_b      ratio lambda_gam        edf
#>   510.2449   955.7000     0.5339     0.5339    12.5749
```

```
max(abs(fitted(fit_lme) - fitted(fit_pen)))
```

```
#> [1] 6.793e-05
```

The REML estimate of the variance ratio, $\hat\sigma^2/\hat\sigma_b^2 = 0.5339$, equals the smoothing parameter chosen by `gam(method = "REML")`, 0.5339, and the two fitted curves agree to 6.8×10^{-5} (the optimisers' tolerance). Figure 19.2 shows the curve.

```
dd <- data.frame(x = x, y = y, mixed = fitted(fit_lme), fixed =
↪  fitted(fit_lme, level = 0))
ggplot(dd, aes(x, y)) + geom_point(colour = "grey55") +
  geom_line(aes(y = fixed), colour = ie_coral, linetype = 2, linewidth = 1) +
  geom_line(aes(y = mixed), colour = ie_navy, linewidth = 1.1) +
  labs(x = "Time after impact (ms)", y = "Head acceleration (g)")
```

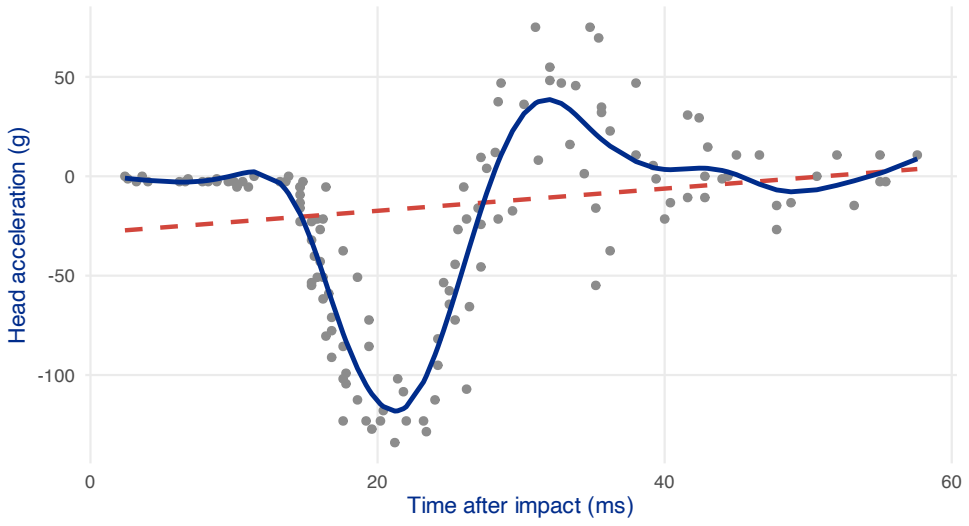


Figure 19.2: The motorcycle data with the P-spline fitted as a mixed model (REML). The dashed line is the fixed part $\hat\alpha_0 + \hat\alpha_1 x$ (the straight line); the BLUPs add the curvature.

Remark. The value of λ depends on how S and B are scaled. `mgcv` rescales its penalty matrices internally, so the `sp` reported for a `gam()` smooth is *not* comparable with the $\hat\sigma^2/\hat\sigma_b^2$ of a hand-built Z (Exercise 19.5 shows 413 against 6573 for the same fitted curve). Compare smooths by their effective degrees of freedom, not by λ.

19.2.5 In R: gamm() does this for you

`mgcv::gamm()` carries out exactly this reparametrisation for every smooth term and passes the result to `nlme::lme()` (or to `MASS::glmmPQL` for non-Gaussian data). The wiggly part of each smooth appears in the `lme` object as a block `Xr` of a dummy group `g`; we meet it again in Section 19.4.

19.3 Random effects inside a GAM

19.3.1 A random intercept is a penalty

The correspondence also works in the opposite direction. A random intercept $b_{0i} \overset{\text{iid}}{\sim} N(0, \sigma_b^2)$ for group $i = 1, \dots, m$ is the term $Z\boldsymbol{b}$ with Z the matrix of group indicators; by Theorem 19.1 it is a ridge penalty $\lambda \sum_i b_{0i}^2$ with identity penalty matrix and $\lambda = \sigma^2/\sigma_b^2$. In `mgcv` this is the basis `bs = "re"`:

- `s(Subject, bs = "re")` a random intercept per level of the *factor* `Subject`;
- `s(Subject, age, bs = "re")` a random slope in `age`, *independent* of the random intercept;
- `gam.vcomp(fit)` converts the smoothing parameters into standard deviations.

How much is a group effect shrunk? For the random intercept in a balanced layout we can say it exactly.

Proposition 19.2 (Shrinkage of a random intercept)**.** *In $y_{ij} = \mu + b_i + \varepsilon_{ij}$, $b_i \sim N(0, \sigma_b^2)$, $\varepsilon_{ij} \sim N(0, \sigma^2)$, with μ known and n_i observations in group i, the BLUP is*

$$\tilde{b}_i = \frac{n_i}{n_i + \lambda}\,(\bar{y}_i - \mu), \qquad \lambda = \frac{\sigma^2}{\sigma_b^2}, \qquad \operatorname{Var}(b_i \mid \boldsymbol{y}) = \frac{\sigma^2}{n_i + \lambda}.$$

Proof. Minimise $\sum_j (y_{ij} - \mu - b)^2 + \lambda b^2$ over b (Theorem 19.1). The derivative $-2\sum_j (y_{ij} - \mu - b) + 2\lambda b = 0$ gives $b(n_i + \lambda) = n_i(\bar{y}_i - \mu)$. The posterior of b_i is normal with precision $n_i/\sigma^2 + 1/\sigma_b^2 = (n_i + \lambda)/\sigma^2$. □

So the raw group mean deviation is multiplied by $n_i/(n_i + \lambda) < 1$: groups with few observations or a small between-group variance are pulled strongly towards the population curve.

Example 19.1 (Shrinking three groups by hand)**.** Let $n_i = 4$, $\sigma^2 = 4$, $\sigma_b^2 = 1$, so $\lambda = 4$ and the shrinkage factor is $4/(4+4) = 0.5$. Group means 2, -1, 0 above the population curve give BLUPs 1, -0.5, 0, with posterior variance $4/8 = 0.5$ for each.

19.3.2 Population smooth and random intercepts for Oxboys

```
m1 <- gam(height ~ s(age) + s(Subject, bs = "re"), data = ox, method = "REML")
summary(m1)$s.table
```

```
#>                edf Ref.df     F p-value
#> s(age)       2.579  3.199 796.8       0
#> s(Subject) 24.931 25.000 359.9       0
```

```
gam.vcomp(m1)
```

```
#>
#> Standard deviations and 0.95 confidence intervals:
#>
#>             std.dev lower  upper
#> s(age)        1.410 0.446  4.460
#> s(Subject)    8.097 6.132 10.692
#> scale         1.281 1.162  1.411
#>
#> Rank: 3/3
#>
#> All smooth components:
#> [1] 1.410 8.097
```

Between boys the standard deviation is $\hat{\sigma}_b = 8.1$ cm and within a boy $\hat{\sigma} = 1.28$ cm, so the intraclass correlation (ICC, Chapter 11) is

$$\text{ICC} = \frac{\hat{\sigma}_b^2}{\hat{\sigma}_b^2 + \hat{\sigma}^2} = \frac{65.57}{67.21} = 0.976.$$

Almost all of the variation around the population curve is between boys. The smoothing parameter of the random term is $\lambda = \hat{\sigma}^2/\hat{\sigma}_b^2 = 0.025$ and is the `sp` reported by `gam` for this term (the identity penalty is not rescaled):

```
co <- coef(m1); b_blup <- co[grep("Subject", names(co))]
raw <- tapply(ox$height - predict(m1, exclude = "s(Subject)"), ox$Subject,
↪  mean)
c(sp = unname(m1$sp["s(Subject)"]), sigma2_over_sigma2b = m1$sig2 /
↪  gam.vcomp(m1)$vc["s(Subject)", "std.dev"]^2,
  shrinkage_9_over_9_plus_lambda = 9 / (9 + unname(m1$sp["s(Subject)"])))
```

```
#>                             sp            sigma2_over_sigma2b
#>                        0.02501                        0.02501
#> shrinkage_9_over_9_plus_lambda
#>                        0.99723
```

```
range(b_blup / raw)
```

```
#> [1] 0.9972 0.9972
```

The ratio of the BLUP to the raw mean deviation is the same for every boy, 0.9972, equal to $n_i/(n_i + \lambda)$ of Proposition 19.2 with $n_i = 9$: with 9 measurements per boy and a large between-boy variance, each boy's level is estimated almost entirely from his own data.

19.3.3 Random slopes and subject-specific curves

Boys also differ in their *rate* of growth. A random slope is added with `s(Subject, age, bs = "re")`. Alternatively, `bs = "fs"` (*factor smooth*) gives every boy a curve of his own: the model `s(age, Subject, bs = "fs", k = 5)` fits one smooth of `age` per level of `Subject`, all with the same basis and the *same* smoothing parameters, and with the null space (constant and linear parts) penalised too. In mixed-model language each boy's curve is a random deviation from the population curve, with the random coefficients iid across boys. The population curve

Table 19.1: Three random-effect structures for the Oxboys growth curves.

model	conditional_AIC	lag1_within_boy
random intercepts	809.4	0.77
+ random slopes	453.6	0.10
factor smooths (fs)	354.9	-0.34

is added as a separate term, `s(age)`, and `mgcv` warns that the two terms share a smoother of `age`; this intentional decomposition is known as the *global-smoother with group-level smoothers sharing wiggliness* model of Pedersen et al. (2019).

```
m2 <- gam(height ~ s(age) + s(Subject, bs = "re") + s(Subject, age, bs =
↪  "re"), data = ox, method = "REML")
m3 <- suppressWarnings(gam(height ~ s(age, k = 5) + s(age, Subject, bs = "fs",
↪  k = 5), data = ox, method = "REML"))
gam.vcomp(m2)
```

```
#>
#> Standard deviations and 0.95 confidence intervals:
#>
#>                 std.dev  lower   upper
#> s(age)           1.6766 0.6056  4.6421
#> s(Subject)       8.0829 6.1252 10.6663
#> s(Subject,age)   1.6896 1.2697  2.2483
#> scale            0.5731 0.5164  0.6361
#>
#> Rank: 4/4
#>
#> All smooth components:
#> [1] 1.677 8.083 1.690
```

```
cmp <- data.frame(model = c("random intercepts", "+ random slopes", "factor
↪  smooths (fs)"),
                  conditional_AIC = c(AIC(m1), AIC(m2), AIC(m3)),
                  lag1_within_boy = c(lag1(residuals(m1), ox$Subject),
                   ↪  lag1(residuals(m2), ox$Subject),
                                      lag1(residuals(m3), ox$Subject)))
knitr::kable(cmp, digits = c(0, 1, 2), booktabs = TRUE,
             caption = "Three random-effect structures for the Oxboys growth
              ↪  curves.")
```

With random intercepts only, the residuals of a boy stay strongly correlated: the boys grow at different *rates*, which an intercept cannot represent. Random slopes remove most of that correlation ($0.77 \to 0.1$) and reduce the AIC by 356. The factor-smooth model has the lowest AIC but the within-boy residuals are now *negatively* correlated (-0.34): a sign of over-fitting. Which model to report is a judgement; the random-slope model is more parsimonious (a line per boy around a population curve) and has no sign of over-fitting.

Remark. The AIC reported by `AIC()` on a `gam` fit is *conditional*: it counts the effective degrees of freedom of the random effects as if they were penalised coefficients, and the likelihood is conditional on the estimated $\boldsymbol{b}$. A *marginal* AIC, with the random effects integrated out, is

obtained from the `lme` component of `gamm()` or from `lme4`. Conditional and marginal AIC can rank models differently, and neither is exact when variance components are tested on the boundary (Section 19.6).

```
nd <- expand.grid(age = seq(-1, 1, length.out = 40), Subject =
↪  levels(ox$Subject))
nd1 <- nd[nd$Subject == levels(ox$Subject)[1], ]
d4 <- rbind(data.frame(nd, fit = predict(m1, nd), model = "random
↪  intercepts"),
            data.frame(nd, fit = predict(m3, nd), model = "factor smooths"))
pop <- rbind(data.frame(age = nd1$age, fit = predict(m1, nd1, exclude =
↪  "s(Subject)"), model = "random intercepts"),
             data.frame(age = nd1$age, fit = predict(m3, nd1, exclude =
              ↪  "s(age,Subject)"), model = "factor smooths"))
ggplot(d4, aes(age, fit, group = Subject)) + geom_line(colour = ie_blue, alpha
↪  = .6, linewidth = .5) +
  geom_line(data = pop, aes(group = NULL), colour = ie_coral, linewidth = 1.3)
  ↪  +
  facet_wrap(~model) + labs(x = "Standardised age", y = "Height (cm)")
```

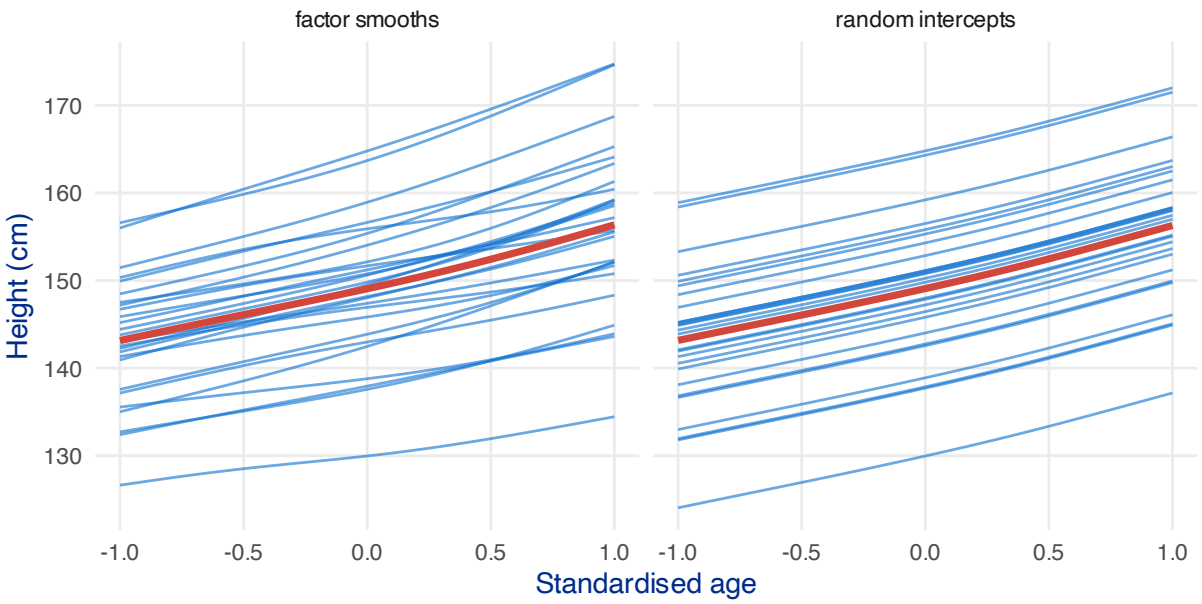


Figure 19.3: Fitted curves of two models. Left: random intercepts, parallel curves. Right: factor smooths, each boy with his own curve. Red: population curve (random terms excluded).

19.4 gamm() and gamm4()

19.4.1 What gamm() returns

`gamm()` fits the mixed model of Theorem 19.2 with `nlme::lme()`, so it accepts the `random =` and `correlation =` arguments of `lme` and returns *two* objects:

```
g2 <- gamm(height ~ s(age), random = list(Subject = ~ 1 + age), data = ox,
↪  method = "REML")
names(g2)
```

```
#> [1] "lme" "gam"
```

```
VarCorr(g2$lme)
```

```
#>             Variance           StdDev Corr
#> g =         pdIdnot(Xr - 1)
#> Xr1          0.1436            0.3789
#> Xr2          0.1436            0.3789
#> Xr3          0.1436            0.3789
#> Xr4          0.1436            0.3789
#> Xr5          0.1436            0.3789
#> Xr6          0.1436            0.3789
#> Xr7          0.1436            0.3789
#> Xr8          0.1436            0.3789
#> Subject =   pdLogChol(1 + age)
#> (Intercept) 65.3216            8.0822 (Intr)
#> age          2.8541            1.6894 0.638
#> Residual     0.3285            0.5731
```

```
summary(g2$gam)$s.table
```

```
#>          edf Ref.df   F p-value
#> s(age) 4.098  4.098 106       0
```

In `VarCorr(g2$lme)` the smooth `s(age)` appears as the dummy-group block `g`, with random coefficients `Xr1`, ..., `Xr8` sharing one variance (standard deviation 0.379): these are the penalised coefficients of the smooth. The boy effects appear below: $\hat{\sigma}_{b_0} = 8.08$ cm, $\hat{\sigma}_{b_1} = 1.69$ cm per unit of age, correlation 0.64, $\hat{\sigma} = 0.573$ cm. Unlike in `gam()`, the intercept and slope are allowed to be correlated.

Question	Component
shape of f, edf, plots, population-level predictions	`fit$gam`
variance components, correlation of random effects	`VarCorr(fit$lme)`
BLUPs of the random effects	`ranef(fit$lme)`
log-likelihood, AIC, LR tests of variance structures	`fit$lme`
normalised residuals	`resid(fit$lme, type = "normalized")`

Warning. Do not use `AIC(fit$gam)` for a `gamm` fit: the `gam` component is built from a working version of the model and has no proper likelihood. Compare `gamm` fits through `anova(fit1$lme, fit2$lme)` on the `lme` components, with the same fixed effects.

For non-Gaussian responses `gamm()` uses *penalised quasi-likelihood* (PQL, Chapter 13; Breslow and Clayton (1993)): it iterates between a working linear mixed model and an update of the working response, with the usual weakness of PQL for binary data with small clusters.

19.4.2 gamm4()

`gamm4::gamm4()` has the same idea on top of `lme4`: the model is `gamm4(fixed_with_smooths, random = ~(1 + age | Subject), data, REML = TRUE)`, with components `$mer` (the `lme4` object) and `$gam`.

```
library(gamm4)
g4 <- gamm4(height ~ s(age), random = ~(1 + age | Subject), data = ox, REML =
↪  TRUE)
lme4::VarCorr(g4$mer)
```

```
#>  Groups   Name        Std.Dev. Corr
#>  Subject  (Intercept) 8.082
#>           age         1.689    0.64
#>  Xr       s(age)      0.379
#>  Residual             0.573
```

The estimates agree with those from `gamm()` (the standard deviations 8.08, 1.69 and 0.57, and the correlation 0.64). The advantages of `gamm4()` are the `lme4` syntax and efficiency with many or crossed random effects, and Laplace approximation (instead of PQL) for non-Gaussian responses. Its limitations: no `correlation =` argument (residual correlation needs `gamm()`) and tensor-product smooths must be specified with `t2()`.

19.4.3 Four ways to the same model

For a real data set we fit the same random-slope model four ways. The `sleepstudy` data (package `lme4`) has the reaction time of 18 drivers on 10 consecutive days of sleep deprivation; a linear effect of `Days` with a random intercept and slope per driver is the textbook model, which we compare with `gam()`, `gamm()` and `gamm4()` with a smooth effect of `Days`.

```
data(sleepstudy, package = "lme4"); ss <- sleepstudy; ss$Subject <-
↪  factor(as.character(ss$Subject))
lm4 <- lme4::lmer(Reaction ~ Days + (Days | Subject), ss)
ga  <- gam(Reaction ~ s(Days, k = 5) + s(Subject, bs = "re") + s(Subject,
↪  Days, bs = "re"), data = ss, method = "REML")
gm  <- gamm(Reaction ~ s(Days, k = 5), random = list(Subject = ~ 1 + Days),
↪  data = ss, method = "REML")
g4s <- gamm4(Reaction ~ s(Days, k = 5), random = ~(1 + Days | Subject), data =
↪  ss, REML = TRUE)
vcs <- function(sd0, sd1, sdE) c(intercept_sd = sd0, slope_sd = sd1,
↪  residual_sd = sdE)
vc4 <- as.data.frame(lme4::VarCorr(lm4)); vcg <- lme4::VarCorr(g4s$mer)
rbind(lmer_linear = vcs(vc4$sdcor[1], vc4$sdcor[2], vc4$sdcor[4]),
      gam_re      = vcs(gam.vcomp(ga)$vc["s(Subject)", "std.dev"],
      ↪  gam.vcomp(ga)$vc["s(Subject,Days)", "std.dev"], sqrt(ga$sig2)),
      gamm        = vcs(as.numeric(VarCorr(gm$lme)["(Intercept)", "StdDev"]),
      ↪  as.numeric(VarCorr(gm$lme)["Days", "StdDev"]), gm$lme$sigma),
      gamm4       = vcs(attr(vcg$Subject, "stddev")[1], attr(vcg$Subject,
      ↪  "stddev")[2], attr(vcg, "sc")))
```

```
#>             intercept_sd slope_sd residual_sd
#> lmer_linear        24.74    5.922       25.59
#> gam_re             25.06    5.989       25.54
#> gamm               24.75    5.924       25.56
#> gamm4              24.75    5.924       25.56
```

```
summary(gm$gam)$s.table
```

```
#>            edf Ref.df     F p-value
#> s(Days) 1.263  1.263 38.26       0
```

The four fits give practically the same variance components. `gam()` forces intercept and slope to be independent, and its estimates differ slightly from the others (where the correlation is 0.07); the edf of `s(Days)` is 1.26: the effect is practically linear, as the textbook model assumes.

19.5 Correlated residuals

19.5.1 AR(1) errors

In longitudinal and time-series data, the errors of consecutive observations may be correlated even after the smooth terms and the random effects. The simplest model is the first-order autoregression.

Definition 19.1 (AR(1) errors)**.** The errors follow $\varepsilon_t = \phi\varepsilon_{t-1} + u_t$ with $|\phi| < 1$ and u_t iid $(0, \sigma_u^2)$. The process is stationary with $\mathrm{Var}(\varepsilon_t) = \sigma^2 = \sigma_u^2/(1-\phi^2)$ and $\mathrm{Cor}(\varepsilon_t, \varepsilon_{t+k}) = \phi^{|k|}$. In matrix form $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\Lambda)$ with $\Lambda_{ij} = \phi^{|t_i - t_j|}$.

Proof. Iterating, $\varepsilon_t = \sum_{s\ge 0}\phi^s u_{t-s}$, so $\mathrm{Var}(\varepsilon_t) = \sigma_u^2\sum_s \phi^{2s} = \sigma_u^2/(1-\phi^2)$. Also $\varepsilon_{t+k} = \phi^k\varepsilon_t + \sum_{s=0}^{k-1}\phi^s u_{t+k-s}$, and the sum is uncorrelated with ε_t, so $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t+k}) = \phi^k \mathrm{Var}(\varepsilon_t)$. □

The same formula holds for observations that are *not* on consecutive time points: if Δ days lie between two observations, their correlation is ϕ^{Δ}. In `nlme` this is `correlation = corAR1(form = ~ day)` with `day` the integer day; for the days $1, 2, 3, 4, 7, 8$ and $\phi = 0.5$:

```
days <- data.frame(day = c(1, 2, 3, 4, 7, 8))
round(corMatrix(Initialize(corAR1(0.5, form = ~ day), data = days)), 3)
```

```
#>       [,1]  [,2]  [,3]  [,4]  [,5]  [,6]
#> [1,] 1.000 0.500 0.250 0.125 0.016 0.008
#> [2,] 0.500 1.000 0.500 0.250 0.031 0.016
#> [3,] 0.250 0.500 1.000 0.500 0.063 0.031
#> [4,] 0.125 0.250 0.500 1.000 0.125 0.063
#> [5,] 0.016 0.031 0.063 0.125 1.000 0.500
#> [6,] 0.008 0.016 0.031 0.063 0.500 1.000
```

The entry for days 4 and 7 is $0.5^3 = 0.125$. Other structures: `corCAR1` (continuous time), `corARMA`, `corExp` and other spatial structures; `correlation = corAR1(form = ~ occ | Subject)` fits an AR(1) *within* each subject.

19.5.2 Daily ozone: is there serial correlation?

For the ozone model of Chapter 16 the days are consecutive, so serial correlation is possible. We compare the Gaussian GAM with and without AR(1) errors.

```
aq <- airquality; aq$day <- seq_len(nrow(aq)); aq <- na.omit(aq)
f <- log(Ozone) ~ s(Solar.R) + s(Wind) + s(Temp)
m0  <- gamm(f, data = aq, method = "REML")
mar <- gamm(f, data = aq, method = "REML", correlation = corAR1(form = ~ day))
anova(m0$lme, mar$lme)
```

```
#>          Model df   AIC   BIC logLik   Test L.Ratio p-value
#> m0$lme       1  8 188.1 209.4 -86.02
#> mar$lme      2  9 188.5 212.6 -85.25 1 vs 2   1.545  0.2139
```

```
phi_hat <- coef(mar$lme$modelStruct$corStruct, unconstrained = FALSE)
round(rbind(independent = summary(m0$gam)$edf, AR1 = summary(mar$gam)$edf), 2)
```

```
#>             [,1] [,2] [,3]
#> independent 2.16 2.46 1.95
#> AR1         2.18 2.41 2.03
```

The estimated $\hat{\phi} = 0.13$; the likelihood-ratio test of $\phi = 0$ has $p = 0.21$ and the AIC is slightly *higher* with the AR(1) term. The effective degrees of freedom hardly change. The weather covariates absorb most of the day-to-day dependence; no correlation structure is needed. (The null value $\phi = 0$ is inside the parameter space, so the usual χ^2_1 reference is valid for this test.)

19.5.3 Why it matters, and a warning

If errors are positively correlated and we ignore it, the standard errors are too small, but there is a second effect that is specific to smoothing: the smoother cannot tell slowly varying signal from slowly varying noise, so it may fit the correlated noise and choose a too small λ (a too wiggly curve). A simulation makes the point, and shows a danger of the cure. The truth is $f(t) = 2\sin(2\pi t/100)$ at $t = 1, \ldots, 200$, with AR(1) errors $\phi = 0.7$ and variance 1. We fit (a) `gam()` ignoring the correlation, (b) `gamm()` with `corAR1` and ϕ estimated, (c) `gamm()` with ϕ fixed at its true value.

```
set.seed(19); n <- 200; tt <- 1:n; ftrue <- function(t) 2 * sin(2 * pi * t /
↪  100); phi <- 0.7
sim_one <- function() {
  e <- as.numeric(arima.sim(list(ar = phi), n, sd = sqrt(1 - phi^2)))
  d <- data.frame(y = ftrue(tt) + e, tt = tt)
  fa <- gam(y ~ s(tt, k = 30), data = d, method = "REML")
  fb <- gamm(y ~ s(tt, k = 30), data = d, correlation = corAR1(form = ~ tt),
↪  method = "REML")
  fc <- gamm(y ~ s(tt, k = 30), data = d, correlation = corAR1(value = phi,
↪  fixed = TRUE, form = ~ tt), method = "REML")
  c(edf_ignore = sum(fa$edf) - 1, edf_AR1 = sum(fb$gam$edf) - 1, edf_AR1_fixed
   ↪  = sum(fc$gam$edf) - 1,
    phi_hat = unname(coef(fb$lme$modelStruct$corStruct, unconstrained =
     ↪  FALSE)),
    mse_ignore = mean((fitted(fa) - ftrue(tt))^2), mse_AR1 =
     ↪  mean((fitted(fb$gam) - ftrue(tt))^2),
    mse_AR1_fixed = mean((fitted(fc$gam) - ftrue(tt))^2))
}
sim <- t(replicate(15, sim_one()))
round(colMeans(sim), 2)
```

```
#>    edf_ignore       edf_AR1 edf_AR1_fixed       phi_hat    mse_ignore
#>         22.46          1.00          8.64          0.91          0.45
#>       mse_AR1 mse_AR1_fixed
#>          1.82          0.20
```

Ignoring the correlation, the smooth uses about 22 effective df, far more than any sensible curve

for two cycles of a sinusoid: it follows the noise. With the correct ϕ the fit is smooth (edf about 8.6). But when ϕ is *estimated* jointly with the smooth, the algorithm finds the opposite solution: it attributes the sinusoid to the error process ($\hat{\phi} \approx 0.91$) and removes the smooth (edf about 1), with a larger error than even the naive fit. A smooth trend in time and an AR(1) error compete for the same slowly varying variation and are only weakly identified from a *single* series.

Warning. Do not add `corAR1` automatically to a model with a smooth of time, or with random intercepts for the same units (the Oxboys example below). Prefer covariates that explain the dependence, start from the structure implied by the design, inspect the residual autocorrelation, and add a correlation structure only if correlation remains; fix or constrain ϕ if it is known from outside the data.

The same competition appears with random effects. For Oxboys, with a random intercept *and* AR(1) errors within boy:

```
ox$occ <- as.integer(ox$Occasion)
gc1 <- gamm(height ~ s(age), random = list(Subject = ~ 1),
            correlation = corAR1(form = ~ occ | Subject), data = ox, method =
            ↪  "REML")
c(phi = unname(coef(gc1$lme$modelStruct$corStruct, unconstrained = FALSE)),
  sd_intercept = as.numeric(VarCorr(gc1$lme)["(Intercept)", "StdDev"]))
```

```
#>          phi sd_intercept
#>     0.995808     0.004103
```

The AR(1) parameter goes to $\hat{\phi} \approx 1$ and the random-intercept standard deviation to almost zero: a correlation that is nearly constant within a boy is described equally well by a random intercept or by $\phi \approx 1$.

19.6 Diagnostics and interpretation

19.6.1 Diagnostics for a GAMM

A GAMM has residuals at several levels, and each has its own check.

- **Within-group residuals in time order.** The lag-1 correlation and the autocorrelation function of the (normalised) residuals within each group. For Oxboys, random intercepts leave a lag-1 correlation of 0.77; random slopes 0.1.
- **Residuals against fitted values and covariates**, as for a GAM; `gam.check(fit$gam)` gives the k-index table (Chapter 18).
- **Normality of the predicted random effects:** a normal Q–Q plot of the BLUPs. With few groups it is only a rough check, and BLUPs are shrunk, hence look more normal than the true effects.
- **Normalised residuals** `resid(fit$lme, type = "normalized")` account for the estimated correlation structure: for the ozone model, the lag-1 correlation of the normalised residuals is 0.11 without and -0.01 with the AR(1) structure.

```
ggplot(data.frame(b = unname(b_blup)), aes(sample = b)) + stat_qq(colour =
↪  ie_navy) + stat_qq_line(colour = ie_coral) +
  labs(x = "Theoretical quantiles", y = "Predicted random intercept (cm)")
```

Remark. Testing whether a variance component is zero (for example, whether the random slopes are needed) is a boundary problem: under $H_0 \colon \sigma^2_{b_1} = 0$ the LR statistic does not have a χ^2 distribution but a mixture (Chapter 12; Self and Liang (1987), Stram and Lee (1994)).

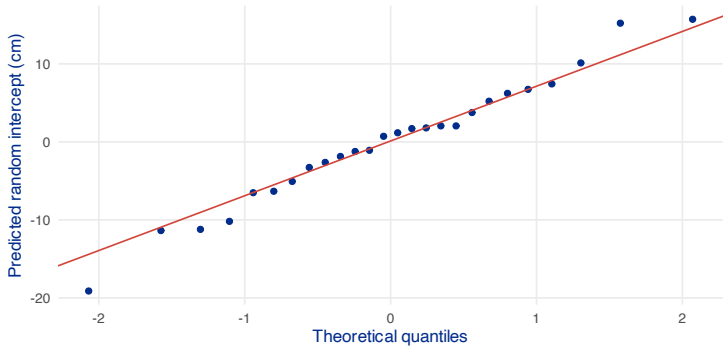


Figure 19.4: Normal Q-Q plot of the predicted random intercepts of the boys (model m1).

For Oxboys the LR statistic comparing intercept-only with intercept-and-slope structures (`anova(g1$lme, g2$lme)` on the `gamm` fits) is huge, so the conclusion does not depend on the reference distribution.

19.6.2 Conditional and population-level interpretation

In a GAMM with identity link and Gaussian errors, $\mathrm{E}(y_{ij} \mid b_i) = \beta_0 + f(x_{ij}) + b_i$ and $\mathrm{E}(\boldsymbol{b}) = 0$, so the population-average mean is $\beta_0 + f(x_{ij})$: the smooth is both the curve of a *typical* boy ($b_i = 0$) and the population-average curve. With a non-linear link this is no longer true.

Proposition 19.3 (Marginal mean with a log link)**.** *If* $\log \mu_{ij} = \eta_{ij} + b_i$ *with* $b_i \sim N(0, \sigma_b^2)$, *then* $\mathrm{E}(y_{ij}) = \mathrm{E}\{\exp(\eta_{ij} + b_i)\} = \exp(\eta_{ij} + \sigma_b^2/2)$.

Proof. $\mathrm{E}e^{b} = e^{\sigma_b^2/2}$ for $b \sim N(0, \sigma_b^2)$ (moment generating function of the normal at 1). □

The population-average curve is the conditional curve *times a constant* $e^{\sigma_b^2/2}$, which is the same shape on the log scale; but the *level* differs, by $\exp(\sigma_b^2/2)$. For the logit link the population-average curve is *flatter* than the conditional one; approximately, the slope of the marginal logit is multiplied by $(1 + c^2\sigma_b^2)^{-1/2}$ with $c^2 = (16\sqrt{3}/15\pi)^2 \approx 0.346$. We check both statements numerically.

```
set.seed(5)
b <- rnorm(2e6, sd = 1)
c(simulated = mean(exp(0.5 + b)) / exp(0.5), formula = exp(1 / 2))          #
↪  log link, sigma_b = 1
```

```
#> simulated   formula
#>     1.649     1.649
```

```
sl <- sapply(c(1, 2), function(sb) {                                        #
↪  logit link: slope of the marginal logit
  m <- function(x) integrate(function(u) plogis(x + u) * dnorm(u, sd = sb),
↪  -Inf, Inf)$value
  xs <- seq(-2, 2, length.out = 41); unname(coef(lm(qlogis(sapply(xs, m)) ~
↪  xs))[2]) })
rbind(numerical = sl, approximation = 1 / sqrt(1 + 0.346 * c(1, 2)^2))
```

```
#>                  [,1]   [,2]
#> numerical     0.8392 0.6141
```

```
#> approximation 0.8619 0.6477
```

A logistic GAMM with $\sigma_b = 2$ thus shows a marginal slope of about 0.6 of the conditional one. Report conditional curves as "the effect for a typical unit", and population-average effects only after integrating over the random-effect distribution.

19.6.3 Predictions

`predict(fit, newdata)` includes the random effects of the levels present in `newdata`: it gives subject-specific curves. The population curve is obtained by excluding the random terms, `predict(fit, newdata, exclude = "s(Subject)")`, or, for a `gamm` fit, from `predict(fit$gam, newdata)` (the `gam` component contains only the smooth part). Report variance components in the units of the data: "boys differ in height by about $\pm 2 \times 8.1$ cm around the population curve; measurements within a boy vary by about 1.3 cm."

19.7 Common pitfalls

Warning. (1) Using only one of `$gam` and `$lme` of a `gamm` fit, or calling `AIC(fit$gam)`. (2) Comparing smoothing parameters across bases or scalings instead of edf. (3) `s(g, bs = "re")` with `g` not a factor (the term is then one penalised slope, not a random intercept). (4) Believing that the intercept and slope of `bs = "re"` terms in `gam()` are correlated: they are independent; use `gamm()` or `gamm4()`. (5) Adding `corAR1` to a model with random intercepts or a smooth of time without checking identifiability. (6) Reading a logit- or log-link GAMM smooth as a population-average effect. (7) Using an ordered factor as the time variable in `corAR1` (use the integer position). (8) Testing a variance component against the ordinary χ^2 distribution.

19.8 Summary

- A random effect with $\boldsymbol{b} \sim N(\mathbf{0}, \sigma_b^2 I)$ is a ridge penalty with $\lambda = \sigma^2/\sigma_b^2$ (joint mode = penalised least squares = Henderson's equations).
- A P-spline of penalty order d is a mixed model with d fixed effects (a polynomial) and $K - d$ random effects, $X = BU_0$, $Z = BU_+\Sigma_+^{-1/2}$; REML for the variance ratio equals REML for the smoothing parameter.
- Group effects are shrunk by $n_i/(n_i + \lambda)$; `s(g, bs = "re")` is a random intercept, `s(g, x, bs = "re")` an independent random slope, `bs = "fs"` a smooth per group.
- `gamm()` fits via `nlme` and returns `$gam` (smooths) and `$lme` (variance components, likelihood); `gamm4()` fits via `lme4`, with Laplace for non-Gaussian data and no residual correlation.
- AR(1) errors have Cor $= \phi^{|\Delta t|}$; ignoring serial correlation can make a smoother follow the noise, but AR(1) and a smooth trend (or random intercepts) are weakly identified together.
- Diagnose within-group residuals in time order, normalised residuals, and the predicted random effects; test variance components with the boundary-corrected reference distribution.
- With a non-identity link the smooth is conditional on $b_i = 0$; the population average differs (factor $e^{\sigma_b^2/2}$ for the log link, attenuation for the logit link).

19.9 Exercises

Exercise 19.1. Henderson's equations. Starting from the penalised criterion (19.2), derive Henderson's equations and show that $\tilde{\boldsymbol{b}} = (Z^\top Z + \lambda I)^{-1} Z^\top (\boldsymbol{y} - X\hat{\boldsymbol{\alpha}})$. For $X = \mathbf{1}$, Z a group-indicator matrix with $n_i = n$ in every group and μ known, show that this reduces to the shrinkage formula of Proposition 19.2.

Solution. Write $r = \boldsymbol{y} - X\boldsymbol{\alpha} - Z\boldsymbol{b}$ and $Q(\boldsymbol{\alpha}, \boldsymbol{b}) = r^\top r + \lambda \boldsymbol{b}^\top \boldsymbol{b}$. Then $\partial Q/\partial \boldsymbol{\alpha} = -2X^\top r$ and $\partial Q/\partial \boldsymbol{b} = -2Z^\top r + 2\lambda \boldsymbol{b}$. Setting both to zero: $X^\top X\boldsymbol{\alpha} + X^\top Z\boldsymbol{b} = X^\top \boldsymbol{y}$ and $Z^\top X\boldsymbol{\alpha} + Z^\top Z\boldsymbol{b} + \lambda \boldsymbol{b} = Z^\top \boldsymbol{y}$, which are the two block rows. The second one gives $(Z^\top Z + \lambda I)\boldsymbol{b} = Z^\top(\boldsymbol{y} - X\boldsymbol{\alpha})$, hence the formula for $\tilde{\boldsymbol{b}}$ (the matrix $Z^\top Z + \lambda I$ is invertible for $\lambda > 0$).

For indicators, $Z^\top Z = \operatorname{diag}(n_1, \dots, n_m) = nI$ and $Z^\top(\boldsymbol{y} - \mu\mathbf{1})$ has i-th entry $n(\bar{y}_i - \mu)$. So $\tilde{b}_i = \dfrac{n(\bar{y}_i - \mu)}{n + \lambda}$, the formula of the proposition.

Exercise 19.2. Shrinkage by hand. A random-intercept model has $\hat{\sigma} = 2$ and $\hat{\sigma}_b = 1$. A group has $n_i = 5$ observations with mean 3.2 above the population curve. (a) Compute λ, the BLUP and its posterior standard deviation. (b) How many observations would the group need for the shrinkage factor to exceed 0.9? (c) What is the ICC?

Solution.

(a) $\lambda = \sigma^2/\sigma_b^2 = 4/1 = 4$. The factor is $5/(5+4) = 0.556$, so $\tilde{b}_i = 0.556 \times 3.2 = 1.78$. The posterior variance is $\sigma^2/(n_i + \lambda) = 4/9 = 0.444$, sd 0.667. (b) $n/(n+4) > 0.9 \iff n > 36$. (c) ICC $= 1/(1+4) = 0.2$: only 20% of the variance is between groups, hence the strong shrinkage for small groups.

Exercise 19.3. Dimensions. A P-spline has $n = 100$ observations, $K = 15$ cubic B-splines and penalty order d. (a) Give the dimensions of X, Z and $\boldsymbol{b}$ for $d = 2$ and for $d = 3$. (b) What is the fit as $\hat{\sigma}_b^2 \to 0$ in each case? (c) Show that the null space of D_3 contains the quadratic sequences $\beta_k = k^2$.

Solution.

(a) $d = 2$: $X = BU_0$ is 100×2, Z is 100×13, $\boldsymbol{b} \in \mathbb{R}^{13}$. $d = 3$: X is 100×3, Z is 100×12, $\boldsymbol{b} \in \mathbb{R}^{12}$. (b) The random part vanishes: the least-squares straight line for $d = 2$, the least-squares parabola for $d = 3$. (c) For $\alpha_k = k^2$: $\Delta\alpha_k = (k+1)^2 - k^2 = 2k+1$, $\Delta^2\alpha_k = 2$, $\Delta^3\alpha_k = 0$, so $D_3\boldsymbol{\alpha} = \mathbf{0}$.

Exercise 19.4. Linear precision of B-splines. For equally spaced knots with spacing h and Greville abscissae $t_k^* = t_0 + kh$ (so that $\sum_k t_k^* B_k(x) = x$) and $\sum_k B_k(x) = 1$, show that $B\boldsymbol{\beta}$ is linear in x for every $\boldsymbol{\beta}$ in the null space of D_2, and that every linear function of x arises this way. Conclude that BU_0 spans $[\mathbf{1}, x]$.

Solution. The null space of D_2 is $\{\beta_k = a + ck\}$. Then $\sum_k (a + ck) B_k(x) = a \sum_k B_k(x) + \frac{c}{h} \sum_k (t_k^* - t_0) B_k(x) = a + \frac{c}{h}(x - t_0)$, linear in x. Conversely, given a linear function $p + qx$ choose $c = qh$ and $a = p + qt_0$: then $a + \frac{c}{h}(x - t_0) = p + qx$. So $\{B\boldsymbol{\beta} : \boldsymbol{\beta} \in \ker D_2\} = \operatorname{span}\{\mathbf{1}, x\}$ and BU_0, whose columns span $B \ker D_2$ because the columns of U_0 span $\ker D_2$, spans the same space as $[\mathbf{1}, x]$.

Exercise 19.5. R: smoothing parameter and variance ratio. For the 111 complete days of `airquality`, let $y = \log(\text{Ozone})$ and $x = \text{Temp}$. Build B (cubic B-splines on 17 equal segments, $K = 20$) and the second-order penalty, fit with `gam(y ~ B - 1, paraPen = ...)` and with `lme()` as in Section 19.2. Report $\hat\sigma^2/\hat\sigma_b^2$, $\hat\lambda$ and edf. Then fit `gamm(y ~ s(x, bs = "ps", k = 20))` and report its `sp`, the ratio $\hat\sigma^2/\hat\sigma_b^2$ from its `$lme`, and the edf. Explain the difference.

Solution.

```
aqs <- na.omit(airquality[, c("Ozone", "Solar.R", "Wind", "Temp")])
y1 <- log(aqs$Ozone); x1 <- aqs$Temp; n1 <- length(y1)
nseg <- 17; dx1 <- diff(range(x1)) / nseg
kn1 <- seq(min(x1) - 3 * dx1, max(x1) + 3 * dx1, by = dx1)
B1 <- splineDesign(kn1, x1, ord = 4, outer.ok = TRUE); K1 <- ncol(B1)
S1 <- crossprod(diff(diag(K1), differences = 2))
e1 <- eigen(S1, symmetric = TRUE); p1 <- e1$values > 1e-8
fp1 <- gam(y1 ~ B1 - 1, paraPen = list(B1 = list(S1)), method = "REML")
Z1 <- B1 %*% e1$vectors[, p1] %*% diag(1 / sqrt(e1$values[p1])); g1 <-
↪  factor(rep(1, n1))
fl1 <- lme(y1 ~ x1, random = list(g1 = pdIdent(~ Z1 - 1)), method = "REML")
r1 <- fl1$sigma^2 / as.numeric(VarCorr(fl1)[1, "Variance"])
fg1 <- gamm(y1 ~ s(x1, bs = "ps", k = 20), method = "REML")
vc1 <- VarCorr(fg1$lme)
rg1 <- fg1$lme$sigma^2 / as.numeric(vc1[nrow(vc1) - 1, "Variance"])
round(c(K = K1, ratio_lme = r1, lambda_gam = unname(fp1$sp), edf =
↪  sum(fp1$edf),
        gamm_sp = unname(fg1$gam$sp), gamm_ratio = rg1, gamm_edf =
        ↪  sum(fg1$gam$edf)), 3)
```

```
#>          K  ratio_lme lambda_gam        edf    gamm_sp gamm_ratio
#>     20.000    413.138    413.115      3.014   6573.368   6573.805
#>   gamm_edf
#>      3.014
```

With our own B and S the ratio $\hat\sigma^2/\hat\sigma_b^2 = 413.14$ equals $\hat\lambda = 413.11$ and edf is 3.01. With `gamm`, both `sp` and the ratio are 6574, because `mgcv` rescales its penalty matrix (and its basis), so λ is on a different scale; the fitted curve and the edf (3.01) are the same. Only $\hat\sigma^2/\hat\sigma_b^2$ *for a given scaling of* S is meaningful, which is why smooths are compared by edf.

Exercise 19.6. Variance components of Oxboys. In the `gamm` fit with random intercept and slope per boy, the estimates are $\hat\sigma_{b_0} = 8.082$, $\hat\sigma_{b_1} = 1.689$, correlation $\hat\rho = 0.638$, $\hat\sigma = 0.573$. (a) Compute the standard deviation of the boy-specific height *at age* $= 1$ around the population curve, $\mathrm{Var}(b_0 + b_1)^{1/2}$, and at age -1. (b) At which age is the between-boy variance smallest, and what is the minimum? (c) Compute the within-boy ICC at age 0.

Solution.

(a) At age a, $\mathrm{Var}(b_0 + a b_1) = \sigma_{b_0}^2 + a^2\sigma_{b_1}^2 + 2a\rho\sigma_{b_0}\sigma_{b_1}$. With $\sigma_{b_0}^2 = 65.32$, $\sigma_{b_1}^2 = 2.854$ and $\rho\sigma_{b_0}\sigma_{b_1} = 0.638 \times 8.082 \times 1.689 = 8.71$: at $a = 1$: $65.32 + 2.854 + 17.42 = 85.60$, sd 9.25 cm; at $a = -1$: $65.32 + 2.854 - 17.42 = 50.75$, sd 7.12 cm. The boys differ more at the end of the observation window than at the start.

(b) The variance $v(a) = 65.32 + 8.71 \cdot 2a + 2.854a^2$ is minimised at $a^* = -8.71/2.854 = -3.05$, outside the data range ($[-1, 1]$): within the data the variance is smallest at $a = -1$ (50.75). (c) At age 0: between-boy variance 65.32, residual variance 0.328; ICC $= 65.32/65.65 = 0.995$.

Exercise 19.7. R: do we need random slopes? Fit `gamm(height ~ s(age), random = list(Subject = ~ 1))` and `gamm(height ~ s(age), random = list(Subject = ~ 1 + age))` by REML. (a) Compare them with `anova()` on the `$lme` components and report the LR statistic. (b) Why is the χ^2_2 reference distribution conservative here, and does it matter?

Solution.

```
g1 <- gamm(height ~ s(age), random = list(Subject = ~ 1), data = ox, method =
↪  "REML")
anova(g1$lme, g2$lme)
```

```
#>         Model df    AIC    BIC logLik   Test L.Ratio p-value
#> g1$lme      1  5 944.1 961.3 -467.1
#> g2$lme      2  7 695.0 719.1 -340.5 1 vs 2   253.1  <.0001
```

(a) The LR statistic is 253.1 on 2 df, $\log L$ -467.1 against -340.5, and the AIC drops from 944.1 to 695: the random slopes are clearly needed. (b) The two parameters added are the slope variance and the intercept–slope covariance; under H_0 the slope variance (zero) lies on the boundary of the parameter space, and the null distribution is a mixture such as $\frac{1}{2}\chi^2_1 + \frac{1}{2}\chi^2_2$ (Self and Liang, 1987; Stram and Lee, 1994), which has smaller quantiles than χ^2_2. So the χ^2_2 p-value is too large (conservative). With a statistic of this size it does not matter.

Exercise 19.8. AR(1). (a) Show that for $\varepsilon_t = \phi\varepsilon_{t-1} + u_t$, $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t+2}) = \phi^2 \mathrm{Var}(\varepsilon_t)$ and explain why `corAR1(form = ~ day)` uses ϕ^{Δ} for observations Δ days apart. (b) Write the correlation matrix for days $1, 2, 4, 5$ with $\phi = 0.6$. (c) For $\phi = 0.6$, what is the correlation between the *mean* of days 1–2 residuals and the residual at day 3? Hint: compute Cov from the matrix.

Solution.

(a) $\varepsilon_{t+2} = \phi\varepsilon_{t+1} + u_{t+2} = \phi^2\varepsilon_t + \phi u_{t+1} + u_{t+2}$; the last two terms are uncorrelated with ε_t, so $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t+2}) = \phi^2 \mathrm{Var}(\varepsilon_t)$. If a day is missing the process still evolves; the correlation of two observations is ϕ to the power of the number of steps between them, i.e. the difference in the integer day.

(b) Entries $\phi^{|\Delta|}$: with $\phi = 0.6$, $0.6^1 = 0.6$, $0.6^2 = 0.36$, $0.6^3 = 0.216$, $0.6^4 = 0.1296$. The matrix for days $(1, 2, 4, 5)$ is

$$\begin{pmatrix} 1 & 0.6 & 0.216 & 0.1296 \\ 0.6 & 1 & 0.36 & 0.216 \\ 0.216 & 0.36 & 1 & 0.6 \\ 0.1296 & 0.216 & 0.6 & 1 \end{pmatrix}.$$

(c) With unit variance, $\mathrm{Cov}(\frac{1}{2}(\varepsilon_1 + \varepsilon_2), \varepsilon_3) = \frac{1}{2}(\phi^2 + \phi) = \frac{1}{2}(0.36 + 0.6) = 0.48$; $\mathrm{Var}(\frac{1}{2}(\varepsilon_1 + \varepsilon_2)) = \frac{1}{4}(2 + 2\phi) = 0.8$. Correlation $= 0.48/\sqrt{0.8} = 0.537$.

Exercise 19.9. R: smoothing with AR(1) noise. Reproduce a smaller version of the simulation of Section 19.5 with $\phi = 0.5$ and a *covariate* effect rather than a time trend: $y_t = 2\sin(x_t) + e_t$ with x_t an AR(1) series with coefficient 0.9 (a slowly varying covariate, like temperature) and e_t AR(1) with $\phi = 0.5$, $n = 200$, 15 replicates. Report mean edf for `gam()` ignoring the correlation and `gamm()` with `corAR1`.

Solution.

```
set.seed(23); n <- 200
res19 <- t(replicate(15, {
  xx <- as.numeric(arima.sim(list(ar = .9), n)); xx <- xx / sd(xx) * 1.2
  e <- as.numeric(arima.sim(list(ar = .5), n, sd = sqrt(1 - .25)))
  d <- data.frame(y = 2 * sin(xx) + e, xx = xx, tt = 1:n)
  a <- gam(y ~ s(xx, k = 20), data = d, method = "REML")
  b <- gamm(y ~ s(xx, k = 20), data = d, correlation = corAR1(form = ~ tt),
↪  method = "REML")
  c(edf_ignore = sum(a$edf) - 1, edf_AR1 = sum(b$gam$edf) - 1,
    phi_hat = unname(coef(b$lme$modelStruct$corStruct, unconstrained =
    ↪  FALSE)))
}))
round(colMeans(res19), 2)
```

```
#> edf_ignore    edf_AR1    phi_hat
#>       6.02       6.41       0.50
```

The estimated $\hat{\phi}$ is close to the true value 0.5 and the two edf are similar (6 and 6.4): when the smooth is in a covariate whose own variation is not the same as the error's, correlation in the errors is identified, and ignoring it makes less difference to the smooth than in the time-trend simulation (where the competition for slowly varying variation is severe). The standard errors, however, would still be too small without the AR(1) term.

Exercise 19.10. R: four ways. On `lme4::sleepstudy`, fit the random-slope model with `gam()`, `gamm()` and `gamm4()` as in Section 19.4 and compare the standard deviations of the intercept and slope and the residual standard deviation with those of `lmer(Reaction ~ Days + (Days | Subject))`. Why does `gam()` differ slightly, and what is the correlation between intercept and slope in `gamm()`?

Solution. The table in Section 19.4 shows the answers: the standard deviations of the intercept, slope and residual are about 24.7, 5.9 and 25.6 in `lmer`, `gamm` and `gamm4`, and 25.1, 6.0 and 25.5 in `gam()`. The fits through `lme`/`lme4` agree up to optimisation tolerance. `gam()` forces the random intercept and slope to be independent (two separate `re` terms), while the data have an estimated correlation of 0.07, close to zero, hence the small differences. For a data set with a clearly non-zero intercept–slope correlation (Oxboys, 0.64) the `gam()` fit would be misspecified, and `gamm()` or `gamm4()` should be used.

Exercise 19.11. Conditional and marginal. A Poisson GAMM has $\log \mu_{ij} = f(x_{ij}) + b_i$, $b_i \sim N(0, \sigma_b^2)$ with $\sigma_b = 1.5$. (a) By what factor does the population-average mean exceed the conditional mean at $b_i = 0$? (b) A logistic GAMM has a conditional slope 1 and $\sigma_b = 2$. Approximate the slope of the population-average logit. (c) A colleague writes: "the smooth in our logistic GAMM is the population-average dose–response curve". Correct the statement.

Solution.

(a) $\exp(\sigma_b^2/2) = \exp(1.125) = 3.08$ (Proposition 19.3; the chapter's simulation confirms the formula for $\sigma_b = 1$, factor 1.65). (b) $1/\sqrt{1 + 0.346 \cdot 4} = 1/\sqrt{2.384} = 0.648$, so the marginal slope is about 0.65 (the numerical value over $[-2, 2]$ is 0.61). (c) The smooth of a logit-link GAMM is the *conditional* effect for a unit with random effect zero (the "typical" unit). The population-average curve, obtained by averaging $\text{expit}(\hat{\eta} + b)$ over $b \sim N(0, \hat{\sigma}_b^2)$, is flatter, with a slope smaller by about the factor of part (b). One must either say "conditional on the random effect" or compute the marginal curve by integration.

Exercise 19.12. Critique and communicate. An analyst fits, to Oxboys, `gamm(height ~ s(age), random = list(Subject = ~ 1), correlation = corAR1(form = ~ occ | Subject))` and reports: "The AR(1) parameter is 0.996, so heights of the same boy at consecutive occasions are almost perfectly correlated, and the random-intercept variance is zero, so boys do not differ." (a) What is wrong? (b) Write a summary of about 150 words for a paediatrician of what the growth model with random intercepts and slopes says.

Solution.

(a) A correlation near 1 at all lags within a boy is *what a random intercept looks like*: the AR(1) term and the random intercept explain the same feature (constant within-boy level) and are weakly identified from nine observations per boy. The optimiser has attributed the between-boy variation to $\phi \approx 1$, leaving the random-intercept variance at zero; this does not mean that boys do not differ: the residual standard deviation in that fit is 8.4 cm, essentially the between-boy standard deviation of 8.1 cm of the random-intercept model. The model should be built from the design (random intercept *and* slope), residual correlation checked afterwards, and the variance components compared across structures by AIC from `$lme`.

(b) "We followed 26 boys, measured nine times each. On average, height increases smoothly and almost linearly over the age window studied. Boys differ strongly from one another: a typical boy is about 8 cm taller or shorter than the average curve (roughly two-thirds of boys lie within 8 cm), and their growth rates differ too, by about 1.7 cm per unit of standardised age. These differences account for almost all of the variation in the data; once the boy's own level and rate are taken into account, measurements on the same boy scatter by only about half a centimetre around his own curve. Boys who are taller in the middle of the age range tend to grow faster (correlation about 0.6). The results describe these 26 boys and the age range observed; they are not growth standards."

19.10 Further reading

The mixed-model representation of smoothers is developed in Verbyla et al. (1999), Wand (2003), Currie and Durban (2002) and, with the full theory of GAMMs and generalised extensions, in Lin and Zhang (1999). The `mgcv` treatment, including `bs = "re"`, `bs = "fs"` and `gamm()`, is Wood (2017) (Sections 5.8, 6.6 and 7.7); Pedersen et al. (2019) is a clear applied introduction to hierarchical GAMs and the models G, GS and GI. The `nlme` correlation classes are documented in Pinheiro and Bates (2000), and the pitfalls of PQL for non-Gaussian clustered data in Breslow and Clayton (1993). For testing variance components on the boundary see Self and Liang (1987) and Stram and Lee (1994). The next chapter leaves smooth regression for the analysis of censored and survival data.

Part VII

Survival analysis

Chapter 20

Censored and survival data

Many outcomes are *times until something happens*: death after a diagnosis, relapse after a treatment, failure of a machine, default of a loan, the moment a customer cancels a subscription. Such times are positive and right-skewed, so a linear model for the raw time is a poor idea (Chapter 5). A deeper problem is that when the study ends some subjects have not yet had the event: all we know about them is that their time exceeds the length of their follow-up. These subjects are **censored**. Throwing them away, or pretending that the event happened when we stopped watching, biases every estimate. Survival analysis is the set of methods that use the partial information of the censored subjects correctly.

The plan is as follows. Section 20.1 introduces the data. Section 20.2 defines censoring and the assumption that makes it harmless. Section 20.3 defines the survival, hazard and cumulative hazard functions and proves how they determine each other. Section 20.4 derives the likelihood under right censoring, and fits the exponential and Weibull models. Section 20.5 derives the Kaplan–Meier estimator as a non-parametric maximum likelihood estimator, and Greenwood's variance formula. Section 20.6 derives the log-rank test. Section 20.7 is the core of the chapter: the Cox proportional hazards model, its partial likelihood, the Breslow estimator and the treatment of ties. Section 20.8 explains how to read hazard ratios, Section 20.9 how to check the proportional-hazards assumption with Schoenfeld residuals and what to do when it fails, and Section 20.10 lists the R functions. Notation follows Appendix A; the likelihood tools are those of Chapters 3 and 4.

20.1 Motivating data

Veterans' lung cancer trial. The data frame `veteran` in package `survival` records a randomised trial of two chemotherapy regimens for inoperable lung cancer (Kalbfleisch and Prentice, 2002): 137 men, 69 on the standard and 68 on the test treatment. The response is the number of days from randomisation to death; 128 men died and 9 were still alive at the end of follow-up. Covariates are the cell type of the tumour, the Karnofsky performance score `karno` (from 10 to 99; higher means fitter), the age and the prior therapy.

```
library(survival)
vet <- survival::veteran
vet$trt <- factor(vet$trt, levels = 1:2, labels = c("standard", "test"))
head(vet, 4)
```

```
#>        trt celltype time status karno diagtime age prior
#> 1 standard squamous   72      1    60        7  69     0
#> 2 standard squamous  411      1    70        5  64    10
```

```
#> 3 standard squamous  228      1    60        3  38     0
#> 4 standard squamous  126      1    60        9  63    10
```

```
table(status = vet$status, trt = vet$trt)
```

```
#>       trt
#> status standard test
#>      0        5    4
#>      1       64   64
```

`status` is the event indicator: 1 for death, 0 for censored. The two lines of the table show that 9 patients, 5 on the standard and 4 on the test treatment, are censored.

Acute myelogenous leukaemia. The tiny data set `aml` (23 patients, 18 events) compares survival of patients who received maintenance chemotherapy with those who did not. It is small enough to compute everything by hand, and we use it for the Kaplan–Meier estimator and the log-rank test. (Code that only draws a figure is not printed in this chapter; it is in the source file.)

```
aml
```

```
#>    time status            x
#> 1     9      1    Maintained
#> 2    13      1    Maintained
#> 3    13      0    Maintained
#> 4    18      1    Maintained
#> 5    23      1    Maintained
#> 6    28      0    Maintained
#> 7    31      1    Maintained
#> 8    34      1    Maintained
#> 9    45      0    Maintained
#> 10   48      1    Maintained
#> 11  161      0    Maintained
#> 12    5      1 Nonmaintained
#> 13    5      1 Nonmaintained
#> 14    8      1 Nonmaintained
#> 15    8      1 Nonmaintained
#> 16   12      1 Nonmaintained
#> 17   16      0 Nonmaintained
#> 18   23      1 Nonmaintained
#> 19   27      1 Nonmaintained
#> 20   30      1 Nonmaintained
#> 21   33      1 Nonmaintained
#> 22   43      1 Nonmaintained
#> 23   45      1 Nonmaintained
```

The question in both data sets is the same: how long do patients survive, how do the two groups differ, and what else predicts survival? A first picture is the *follow-up plot* of Figure 20.1: each line is a patient; a circle marks a death and a cross marks the last contact with a censored patient.

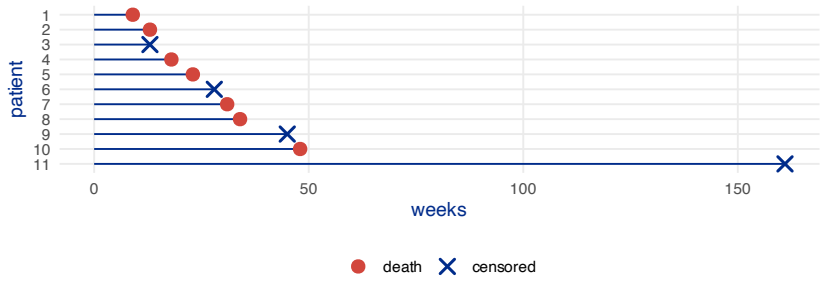


Figure 20.1: Follow-up of the 11 patients of the maintained group in the 'aml' data. A circle is a death, a cross a censored time (patient still alive at last contact).

20.2 Censoring

Let $T_i \geq 0$ be the (unobserved) event time of subject i. The data we see are incomplete.

Definition 20.1 (Types of censoring)**.** Observation of T_i is **right-censored** at c if we only know that $T_i > c$; **left-censored** if we only know $T_i \leq c$; **interval-censored** if we only know $a < T_i \leq b$. A time observed exactly is an **event** (or failure).

Right censoring is by far the most common (the study ends, the patient moves away), and it is the only type treated in this chapter. Left censoring happens when the event occurred before the first examination (age at first marijuana use asked in a survey); interval censoring when the event is detected only at periodic visits (a tumour found at a scan). A related but different notion is *truncation*: subjects whose event occurred before they could enter the study are never observed at all (delayed entry). In R, `Surv(time, time2, type = "interval")` handles interval censoring and `Surv(start, stop, event)` handles delayed entry; we do not pursue them.

20.2.1 Mechanisms of right censoring

Write C_i for the potential censoring time of subject i. What we observe is

$$y_i = \min(T_i, C_i), \qquad \delta_i = \mathbf{1}\{T_i \leq C_i\}. \tag{20.1}$$

Three mechanisms generate C_i.

- **Type I censoring**: a fixed time c for everybody (the study ends after c days, $C_i = c$). The number of events is random.
- **Type II censoring**: observation stops at the r-th event. The number of events r is fixed, the end time is random. This is common in industrial life-testing (stop when 10 of 20 bulbs have failed).
- **Random censoring**: each C_i is a random variable (staggered entry with a common calendar end date, loss to follow-up). This is the general case, and the one we analyse.

Type I and II censoring are special cases of random censoring, and the likelihood we derive in Section 20.4 is valid for all of them.

Definition 20.2 (Independent (non-informative) censoring)**.** Censoring is **non-informative** if, given the covariates x_i, the censoring time C_i is independent of the event time T_i, and the distribution of C_i does not depend on the parameters of the distribution of T_i.

Non-informativeness cannot be tested from the data: for a censored subject we never see T_i, so any joint distribution of (T_i, C_i) with the same observable consequences is equally compatible

with the data. It must be argued from the design. Administrative censoring at the end of a trial is non-informative; censoring because patients drop out when they feel worse is not.

Warning. Informative censoring is the main threat to every method in this chapter. If the patients who are withdrawn from the study are the sickest ones, the patients remaining at risk are healthier than the population, and all the estimators below are too optimistic. A sensitivity analysis (assume that the censored subjects would have died soon, or lived long) bounds the damage.

20.2.2 Why not ignore the censoring?

Take the exponential model of Section 20.4, in which the death rate is $\hat\lambda = d/\sum_i y_i$ (number of deaths divided by total follow-up time). Two tempting alternatives are to drop the censored patients, or to treat the censoring times as if they were death times.

```
d <- sum(vet$status); tot <- sum(vet$time)
c(correct = d / tot,
  drop_censored = d / sum(vet$time[vet$status == 1]),
  censored_as_deaths = nrow(vet) / tot)
```

```
#>            correct      drop_censored censored_as_deaths
#>           0.007682           0.008188           0.008222
```

Dropping the censored patients removes those who lived longest and *overestimates* the death rate; counting censoring times as deaths adds deaths that never happened and again overestimates it. Here only 9 of 137 patients (6.6%) are censored and the bias is small, but with 50–80% censoring, as in many epidemiological cohorts, the naive estimates are useless. Regressing the observed time on covariates with `lm()` has the same problem and in addition ignores the skewness of T.

20.3 Survival, hazard and cumulative hazard

Let $T \geq 0$ be a continuous random variable with density f and distribution function F.

Definition 20.3 (Survival, hazard and cumulative hazard)**.** The **survival function** is $S(t) = P(T > t) = 1 - F(t)$. The **hazard function** is

$$h(t) = \lim_{\Delta \downarrow 0} \frac{P(t \leq T < t + \Delta \mid T \geq t)}{\Delta}, \tag{20.2}$$

the instantaneous event rate among those still event-free at t. The **cumulative hazard** is $H(t) = \int_0^t h(u)\,du$.

The hazard is a *rate* (events per unit time per subject at risk), not a probability: it has units 1/time and may exceed 1. For small Δ, $h(t)\Delta$ is approximately the probability of an event in $[t, t+\Delta)$ for a subject alive at t.

Theorem 20.1 (Relations between S, f, h and H)**.** *For a continuous T with $S(0) = 1$,*

$$h(t) = \frac{f(t)}{S(t)} = -\frac{d}{dt}\log S(t), \qquad H(t) = -\log S(t), \qquad S(t) = e^{-H(t)}, \qquad f(t) = h(t)e^{-H(t)}. \tag{20.3}$$

Any one of S, f, h and H determines the other three.

Proof. By the definition of conditional probability, $P(t \le T < t+\Delta \mid T \ge t) = \{S(t) - S(t+\Delta)\}/S(t)$. Divide by Δ and let $\Delta \downarrow 0$: $h(t) = -S'(t)/S(t) = f(t)/S(t)$, and $-S'/S = -\frac{d}{dt}\log S$. Integrate from 0 to t and use $\log S(0) = 0$: $H(t) = \int_0^t h(u)du = -\log S(t)$. Exponentiating gives $S = e^{-H}$, and $f = hS = he^{-H}$. □

Survival models are usually specified through the hazard, because the hazard has a direct interpretation and, unlike the density, can be modelled without worrying about integrating to one: any non-negative function h with $\int_0^\infty h = \infty$ is a hazard.

Theorem 20.2 (Mean survival time)**.** *If $T \ge 0$ has finite mean, $\mathrm{E}[T] = \int_0^\infty S(t)\,dt$.*

Proof. Write $T = \int_0^\infty \mathbf{1}\{t < T\}dt$ and take expectations; by Tonelli's theorem (the integrand is non-negative) $\mathrm{E}[T] = \int_0^\infty P(T > t)dt$. □

This is why the area under the Kaplan–Meier curve estimates the mean, and why the mean cannot be estimated when the largest observed time is censored (the tail of S is unknown); the median is estimable.

20.3.1 Standard families

Example 20.1 (Exponential and Weibull)**.** The **exponential** distribution has constant hazard $h(t) = \lambda$, so $H(t) = \lambda t$, $S(t) = e^{-\lambda t}$, mean $1/\lambda$ and median $\log 2/\lambda$. It is *memoryless*: $P(T > t + s \mid T > t) = e^{-\lambda s} = P(T > s)$, a subject that survived t is "as good as new". The **Weibull** distribution with rate λ and shape k has

$$h(t) = k\lambda(\lambda t)^{k-1}, \qquad H(t) = (\lambda t)^k, \qquad S(t) = \exp\{-(\lambda t)^k\}, \qquad \text{median} = \frac{(\log 2)^{1/k}}{\lambda}.$$

The hazard is increasing for $k > 1$ (ageing, wear-out), constant for $k = 1$ and decreasing for $k < 1$ (risk highest just after surgery or diagnosis).

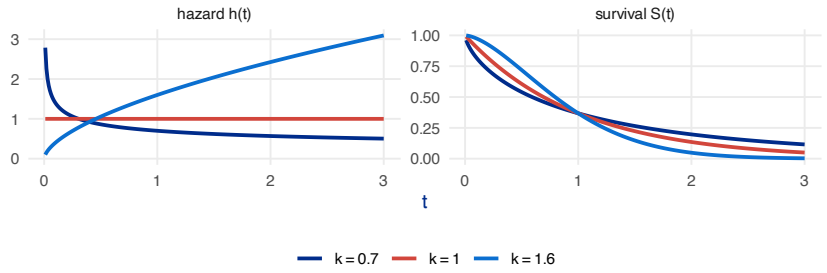


Figure 20.2: Weibull hazards (left) and survival functions (right) with $\lambda = 1$ and shapes 0.7, 1 and 1.6.

20.4 The likelihood with right censoring

Subject i has event time T_i (density f, survival S, parameter θ) and censoring time C_i (density g, survival S_C), independent. We observe (y_i, δ_i) from (20.1).

Theorem 20.3 (Likelihood under non-informative right censoring)**.** *The likelihood of θ is*

$$L(\theta) = \prod_{i=1}^{n} f(y_i;\theta)^{\delta_i} S(y_i;\theta)^{1-\delta_i} = \prod_{i=1}^{n} h(y_i;\theta)^{\delta_i} S(y_i;\theta), \tag{20.4}$$

and the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \{\delta_i \log h(y_i;\theta) - H(y_i;\theta)\}. \tag{20.5}$$

Proof. For an event ($\delta_i = 1$) the observation is $y_i = T_i \le C_i$; its density at y is $P(T_i \in dy,\ C_i > y) = f(y)S_C(y)\,dy$ by independence. For a censored subject ($\delta_i = 0$) the observation is $y_i = C_i < T_i$ with density $P(C_i \in dy,\ T_i > y) = g(y)S(y)\,dy$. The joint density of (y_i, δ_i) is therefore

$$\{f(y_i)S_C(y_i)\}^{\delta_i}\{g(y_i)S(y_i)\}^{1-\delta_i} = \{f(y_i)^{\delta_i}S(y_i)^{1-\delta_i}\}\{g(y_i)^{1-\delta_i}S_C(y_i)^{\delta_i}\}.$$

The second brace does not involve θ when censoring is non-informative (Definition 20.2), so it is a constant of the likelihood and may be dropped. Using $f = hS$ in the first brace gives the second form of (20.4); taking logarithms and using $\log S = -H$ gives (20.5). □

Equation (20.5) has a transparent structure. *Everyone* contributes $-H(y_i)$, a penalty that grows with the time the subject spent at risk. *Only events* contribute the reward $\log h(y_i)$. A censored subject is not missing data: it contributes the information that $T_i > y_i$. If the covariates x_i are present, replace $h(t;\theta)$ by $h(t \mid x_i;\theta)$.

20.4.1 The exponential model

Theorem 20.4 (Exponential MLE)**.** *Under the exponential model with $d = \sum_i \delta_i$ events and total time at risk $\sum_i y_i$,*

$$\hat\lambda = \frac{d}{\sum_i y_i} = \frac{\textit{number of events}}{\textit{person-time}}, \qquad \operatorname{se}(\hat\lambda) = I(\hat\lambda)^{-1/2} = \frac{\hat\lambda}{\sqrt{d}}.$$

Proof. With $h = \lambda$ and $H(y) = \lambda y$, (20.5) gives $\ell(\lambda) = d\log\lambda - \lambda\sum_i y_i$. The score $\ell'(\lambda) = d/\lambda - \sum_i y_i$ vanishes at $\hat\lambda = d/\sum y_i$ (we need $d \ge 1$), and $I(\lambda) = -\ell''(\lambda) = d/\lambda^2 > 0$, so this is the maximum. Then $\operatorname{se}(\hat\lambda) = I(\hat\lambda)^{-1/2} = \hat\lambda/\sqrt{d}$. □

The information depends on the number of *events* d, not on the number of subjects n: a study of 10000 patients with 4 deaths tells us as little about λ as a study of 4 patients who all died. Since $\operatorname{se}(\log\hat\lambda) = 1/\sqrt{d}$ by the delta method, the interval $\hat\lambda\exp(\pm 1.96/\sqrt{d})$ stays positive. The quantity $\hat\lambda$ is the "events per person-year" rate of epidemiology.

Example 20.2 (Exponential model for the veteran trial)**.** With $d = 128$ deaths and $\sum y_i = 16\,663$ person-days:

```
d <- sum(vet$status); tot <- sum(vet$time)
lam <- d / tot; se_lam <- lam / sqrt(d)
c(lambda = lam, se = se_lam, lo = lam * exp(-1.96 / sqrt(d)), hi = lam *
↪  exp(1.96 / sqrt(d)),
```

```
  mean_days = 1 / lam, median_days = log(2) / lam)
```

```
#>      lambda          se          lo          hi   mean_days median_days
#>   7.682e-03   6.790e-04   6.460e-03   9.135e-03   1.302e+02   9.023e+01
```

```
tapply(vet$status, vet$trt, sum) / tapply(vet$time, vet$trt, sum)   # rate by
↪  arm
```

```
#> standard     test
#> 0.008055 0.007341
```

The estimated death rate is 0.00768 per day (95% interval 0.00646 to 0.00913), a median survival of 90 days. The rate is 0.00806 per day on the standard and 0.00734 on the test treatment: a rate ratio of 0.91. In R the same model is fitted by `survreg()`, which parametrises the model on the log-time scale: $\log T = \beta_0 + W$ with W a standard extreme-value variable scaled to give the exponential, so that $\lambda = \exp(-\beta_0)$.

```
e0 <- survreg(Surv(time, status) ~ 1, data = vet, dist = "exponential")
c(lambda = unname(exp(-coef(e0))), se_log_lambda = unname(sqrt(vcov(e0)[1,
↪  1])),
  one_over_sqrt_d = 1 / sqrt(d))
```

```
#>          lambda   se_log_lambda one_over_sqrt_d
#>        0.007682        0.088388        0.088388
```

20.4.2 Connection with Poisson regression

The exponential log-likelihood $d \log \lambda - \lambda \sum_i y_i$ is, up to a constant, the log-likelihood of independent Poisson variables $\delta_i \sim \text{Poisson}(\lambda y_i)$: $\sum_i \{\delta_i \log(\lambda y_i) - \lambda y_i\}$ differs only by $\sum_i \delta_i \log y_i$, which does not depend on λ. Therefore an exponential survival regression with $h(t \mid x) = \exp(x^\top \beta)$ is a Poisson GLM with log link and offset $\log y_i$ (Chapter 10), response δ_i:

```
p1 <- glm(status ~ trt + offset(log(time)), family = poisson, data = vet)
e1 <- survreg(Surv(time, status) ~ trt, data = vet, dist = "exponential")
rbind(poisson_glm = coef(p1), survreg_negated = -coef(e1))
```

```
#>                 (Intercept)  trttest
#> poisson_glm          -4.821 -0.09285
#> survreg_negated      -4.821 -0.09285
```

The two fits agree (the `survreg` coefficients have the opposite sign because they act on $\log T$, not on the hazard). This equivalence is the origin of the "person-time" analyses of cohort studies.

20.4.3 The Weibull model and accelerated failure times

The exponential model forces a constant hazard. To relax it, write $\log T_i = \mu_i + \sigma W_i$, where $\mu_i = x_i^\top \gamma$, W_i has the standard extreme-value (minimum Gumbel) distribution and $\sigma > 0$ is a scale. Then T_i is Weibull with shape $k = 1/\sigma$, and $\sigma = 1$ is the exponential. This is an **accelerated failure time** (AFT) model: $T_i = e^{x_i^\top \gamma} T_0$, so e^{γ_j} multiplies the *survival time* by a constant factor. Since $S(t \mid x) = \exp\{-(te^{-\mu})^{1/\sigma}\}$, the hazard is

$$h(t \mid x) = \frac{1}{\sigma} t^{1/\sigma - 1} \exp\!\Big(-\frac{\mu}{\sigma}\Big),$$

which is of the form $h_0(t)\exp(x^\top\beta)$ with $h_0(t) = \sigma^{-1}t^{1/\sigma-1}$ and $\beta = -\gamma/\sigma$. The Weibull model is therefore both an AFT and a proportional-hazards model; it is the only one that is both. The exponential model is nested in it, so a likelihood-ratio test of $\sigma = 1$ compares them.

```
w0 <- survreg(Surv(time, status) ~ 1, data = vet, dist = "weibull")
anova(e0, w0)
```

```
#>   Terms Resid. Df -2*LL Test Df Deviance Pr(>Chi)
#> 1     1       136  1502      NA       NA       NA
#> 2     1       135  1496    =  1     6.26  0.01235
```

```
c(scale_sigma = w0$scale, shape_k = 1 / w0$scale)
```

```
#> scale_sigma     shape_k
#>      1.1736      0.8521
```

The LR statistic is 6.26 on 1 df ($p = 0.012$): the constant-hazard model is rejected. The Weibull shape $\hat{k} = 0.85 < 1$ says that the hazard of the veterans decreases slowly with time, as often happens after a severe diagnosis. Figure 20.3 compares the two parametric fits with the non-parametric Kaplan–Meier curve of the next section.

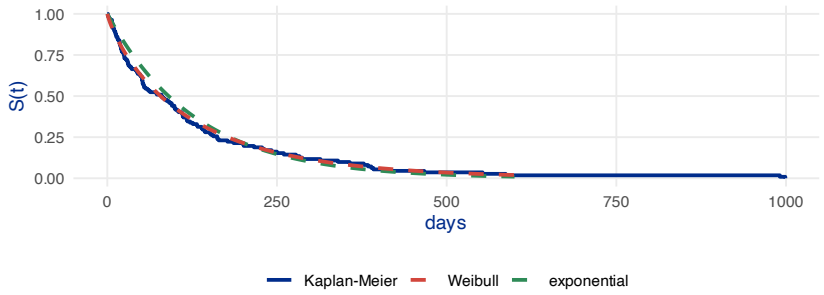


Figure 20.3: Veteran data: Kaplan–Meier curve (steps) with exponential and Weibull fits.

20.5 The Kaplan–Meier estimator

Without covariates and without a parametric model, how do we estimate $S(t)$ from censored data? Without censoring the answer is easy: $\hat{S}(t) = 1 - F_n(t)$, the proportion of the sample larger than t. With censoring we cannot just count.

Let $t_1 < \cdots < t_J$ be the distinct **event** times in the sample. At t_j, let

$$d_j = \#\{i : y_i = t_j,\ \delta_i = 1\}, \qquad n_j = \#\{i : y_i \geq t_j\}$$

be the number of events and the size of the **risk set** (subjects still under observation just before t_j). If a censoring time ties with an event time, the censored subject is counted in n_j, i.e. censoring is assumed to occur just after the event.

Definition 20.4 (Kaplan–Meier estimator)**.**

$$\hat{S}(t) = \prod_{j:\, t_j \leq t} \Big(1 - \frac{d_j}{n_j}\Big).$$

It is a right-continuous step function that drops only at event times, and is constant over censoring times (Kaplan and Meier, 1958).

Example 20.3 (A Kaplan–Meier curve by hand)**.** Six subjects have times 3, 4, 6^+, 8, 8, 10 (a plus is a censoring). The event times are 3, 4, 8, 10.

t_j	n_j	d_j	$1 - d_j/n_j$	$\hat{S}(t_j)$
3	6	1	5/6	0.833
4	5	1	4/5	0.667
8	3	2	1/3	0.222
10	1	1	0	0

At $t = 8$ the risk set has three members (8, 8, 10): the censored subject 6^+ has left, but it was counted at 3 and 4. It does not cause a drop at 6, but afterwards it no longer contributes to the denominator, so the next drop is larger than it would have been without it: censored subjects "pass on their share" to those who remain. We check with `survfit`:

```
toy <- data.frame(t = c(3, 4, 6, 8, 8, 10), s = c(1, 1, 0, 1, 1, 1))
summary(survfit(Surv(t, s) ~ 1, data = toy))
```

```
#> Call: survfit(formula = Surv(t, s) ~ 1, data = toy)
#>
#>  time n.risk n.event survival std.err lower 95% CI upper 95% CI
#>     3      6       1    0.833   0.152       0.5827            1
#>     4      5       1    0.667   0.192       0.3786            1
#>     8      3       2    0.222   0.192       0.0407            1
#>    10      1       1    0.000     NaN           NA           NA
```

20.5.1 Kaplan–Meier as a non-parametric maximum likelihood estimator

The formula looks like a plausible heuristic (a product of conditional survival probabilities). It is in fact a maximum likelihood estimator, in the following sense. Let F be *any* distribution on $[0, \infty)$ and let $F\{u\} = P(T = u)$ denote its point mass. The non-parametric likelihood of the observed data is

$$L(F) = \prod_{i=1}^{n} F\{y_i\}^{\delta_i} \, S(y_i)^{1-\delta_i}, \tag{20.6}$$

the discrete analogue of (20.4), with point masses in place of densities. (For a continuous F the factors $F\{y_i\}$ vanish and the likelihood is zero, so the maximisation must run over distributions that can put mass at the observed times; this is the same device as the empirical distribution function being the non-parametric MLE of F.)

Theorem 20.5 (Kaplan–Meier is the NPMLE)**.** *$L(F)$ is maximised by the distribution with discrete hazards $\hat{h}_j = d_j/n_j$ at the event times, that is by $\hat{S}(t) = \prod_{t_j \le t}(1 - d_j/n_j)$ for $t \le t_J$.*

Proof. Step 1: the support. Suppose F has a point mass at a time u that is not an event time. This mass enters (20.6) only through $S(c)$ of censored times $c < u$. Moving the mass to the next event time $t_j > u$ (or to a point beyond the largest observation if there is none) keeps it counted in every $S(c)$ that counted it before, and may add it to further $S(c)$ with $u \le c < t_j$; it does not change the factors $F\{t_k\}$ for events. So L does not decrease: we may restrict to distributions with mass on $t_1, \dots, t_J$ and possibly beyond t_J.

Step 2: the hazard parametrisation. Define the discrete hazards $h_j = P(T = t_j \mid T \geq t_j) = F\{t_j\}/S(t_j-)$, $j = 1, \dots, J$, so that

$$F\{t_j\} = h_j \prod_{l<j}(1-h_l), \qquad S(t_j) = \prod_{l \leq j}(1-h_l).$$

An event at t_j contributes $F\{t_j\} = h_j \prod_{l<j}(1-h_l)$: a factor h_j at its own time and a factor $(1-h_l)$ at every earlier event time t_l, i.e. at every event time at which it was at risk before dying. A subject censored at c with $t_m \leq c < t_{m+1}$ contributes $S(c) = \prod_{l \leq m}(1-h_l)$: a factor $(1-h_l)$ at each event time at which it was at risk. Collecting the factors by event time, at t_j the d_j events contribute h_j and the other $n_j - d_j$ subjects at risk contribute $1-h_j$:

$$L(h_1, \dots, h_J) = \prod_{j=1}^{J} h_j^{d_j}(1-h_j)^{n_j - d_j}. \tag{20.7}$$

Step 3: maximisation. (20.7) is a product of J binomial likelihoods with free parameters $h_j \in [0,1]$, each maximised at $\hat{h}_j = d_j/n_j$. Then $\hat{S}(t_j) = \prod_{l \leq j}(1-\hat{h}_l)$. □

This is the first of two likelihood-based constructions in this chapter; the second, in Section 20.7, removes an infinite-dimensional nuisance parameter. Four remarks follow.

- *No censoring*: $n_{j+1} = n_j - d_j$, so $\prod_{l \leq j}(n_l - d_l)/n_l$ telescopes to $(n - \sum_{l \leq j} d_l)/n = 1 - F_n(t_j)$.
- *Self-consistency*: the KM estimator redistributes the mass of each censored observation equally among all observations to its right, which is why censored subjects "pass on their share".
- *The tail*: if the largest observation is censored, $\hat{S}$ never reaches 0 and the NPMLE is not unique beyond t_J (the mass can sit anywhere beyond the largest time). By convention $\hat{S}$ stays at its last value, and the mean survival time is not estimable.
- *Nelson–Aalen*: $\hat{H}(t) = \sum_{t_j \leq t} d_j/n_j$ estimates the cumulative hazard; $e^{-\hat{H}(t)}$ is close to $\hat{S}(t)$ because $\log(1-x) \approx -x$.

20.5.2 Greenwood's variance formula

The log-likelihood (20.7) has second derivatives $-\partial^2 \ell/\partial h_j^2 = d_j/h_j^2 + (n_j - d_j)/(1-h_j)^2$, which equal $n_j/\{\hat{h}_j(1-\hat{h}_j)\}$ at the maximum, and zero cross derivatives. Treating the risk-set sizes n_j as fixed, the estimators $\hat{h}_j$ are therefore approximately uncorrelated with variance

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1-\hat{h}_j)}{n_j}.$$

(That the $\hat{h}_j$ are in fact uncorrelated, conditionally on the past, is a martingale property; see Kalbfleisch and Prentice (2002) and Therneau and Grambsch (2000) for a complete treatment.)

Theorem 20.6 (Greenwood's formula)**.** *Approximately,*

$$\widehat{\mathrm{Var}}\{\hat{S}(t)\} = \hat{S}(t)^2 \sum_{j:\, t_j \leq t} \frac{d_j}{n_j(n_j - d_j)}. \tag{20.8}$$

Proof. Apply the delta method to $g(h) = \log(1-h)$, $g'(h) = -1/(1-h)$:

$$\mathrm{Var}\{\log(1-\hat{h}_j)\} \approx \frac{\mathrm{Var}(\hat{h}_j)}{(1-\hat{h}_j)^2} = \frac{\hat{h}_j}{n_j(1-\hat{h}_j)} = \frac{d_j}{n_j(n_j-d_j)},$$

using $\hat h_j = d_j/n_j$. Since $\log \hat S(t) = \sum_{t_j \le t} \log(1-\hat h_j)$ is a sum of uncorrelated terms, $\mathrm{Var}\{\log \hat S(t)\} \approx \sum_{t_j \le t} d_j/\{n_j(n_j-d_j)\}$. A second application of the delta method, $\mathrm{Var}(\hat S) = \hat S^2 \mathrm{Var}(\log \hat S)$, gives (20.8). □

For the toy data of Example 20.3, $\hat S(8) = 0.222$ and $\sum = \frac{1}{6\cdot5} + \frac{1}{5\cdot4} + \frac{2}{3\cdot1} = 0.75$, so $\mathrm{se}\{\hat S(8)\} = 0.222\sqrt{0.75} = 0.192$. The standard error is large because there are only 6 subjects.

Greenwood's formula is not defined at $d_j = n_j$ (the last term when the last observation is an event); there $\hat S = 0$ and the variance is not needed.

Warning. The plain interval $\hat S \pm 1.96\,\mathrm{se}$ can leave $[0,1]$. R's default works on the log scale: $\hat S \exp\{\pm 1.96\sqrt{\textstyle\sum}\}$, truncated at 1. The **log-log** interval, $\hat S^{\exp(\pm 1.96\, s)}$ with $s = \sqrt{\sum}/|\log \hat S|$, never leaves $(0,1)$: it is the delta-method interval for $\log\{-\log S\} = \log H$ transformed back, and is usually preferable in small samples (`conf.type = "log-log"`).

20.5.3 Medians and means

The **median survival time** is $\hat m = \min\{t : \hat S(t) \le 0.5\}$. A confidence interval is obtained by inverting the pointwise confidence band: the set of t at which the band contains 0.5; `survfit` reports it. If the upper band never falls below 0.5 the upper limit is `NA`. The mean $\int_0^\infty S$ is estimable only when the last time is an event; otherwise one reports the *restricted mean* $\int_0^\tau \hat S(t)dt$ up to a horizon τ (`print(km, rmean = tau)`).

Example 20.4 (Kaplan–Meier curves for the leukaemia data)**.** We compute the Kaplan–Meier table of the maintained group by hand and compare it with `survfit`, and then plot both groups.

```
am <- subset(aml, x == "Maintained")
tab <- do.call(rbind, lapply(sort(unique(am$time[am$status == 1])),
↪  function(tj)
  data.frame(t = tj, n = sum(am$time >= tj), d = sum(am$time == tj & am$status
  ↪  == 1))))
tab$S <- cumprod(1 - tab$d / tab$n)
tab$sum_gw <- cumsum(tab$d / (tab$n * (tab$n - tab$d)))
tab$se <- tab$S * sqrt(tab$sum_gw)
round(tab, 4)
```

```
#>    t  n d      S sum_gw     se
#> 1  9 11 1 0.9091 0.0091 0.0867
#> 2 13 10 1 0.8182 0.0202 0.1163
#> 3 18  8 1 0.7159 0.0381 0.1397
#> 4 23  7 1 0.6136 0.0619 0.1526
#> 5 31  5 1 0.4909 0.1119 0.1642
#> 6 34  4 1 0.3682 0.1952 0.1627
#> 7 48  2 1 0.1841 0.6952 0.1535
```

```
km_aml <- survfit(Surv(time, status) ~ x, data = aml)
summary(km_aml)$table[, c("records", "events", "median", "0.95LCL",
↪  "0.95UCL")]
```

```
#>                 records events median 0.95LCL 0.95UCL
#> x=Maintained         11      7     31      18      NA
```

```
#> x=Nonmaintained      12      11      23       8      NA
```

The hand computation reproduces `survfit`: for example $\hat{S}(31) = 0.491$ with Greenwood standard error 0.164. For the log-log interval at $t = 31$ we compute $s = \sqrt{\sum}/|\log \hat{S}|$:

```
S31 <- tab$S[tab$t == 31]; s31 <- sqrt(tab$sum_gw[tab$t == 31]) /
↪  abs(log(S31))
rbind(log = S31 * exp(c(-1, 1) * 1.96 * sqrt(tab$sum_gw[tab$t == 31])),
      loglog = S31^exp(c(1, -1) * 1.96 * s31))
```

```
#>           [,1]   [,2]
#> log    0.2549 0.9456
#> loglog 0.1673 0.7534
```

```
summary(survfit(Surv(time, status) ~ 1, data = am, conf.type =
↪  "log-log"))$lower[5]
```

```
#> [1] 0.1673
```

(The columns are the lower and upper limits; the last line confirms the log-log lower limit.) The median survival is 31 weeks (maintained) against 23 weeks (non-maintained), but with 11 and 12 patients the confidence intervals are very wide, and the upper limits are `NA`: the curves never fall clearly below 0.5 within the band.

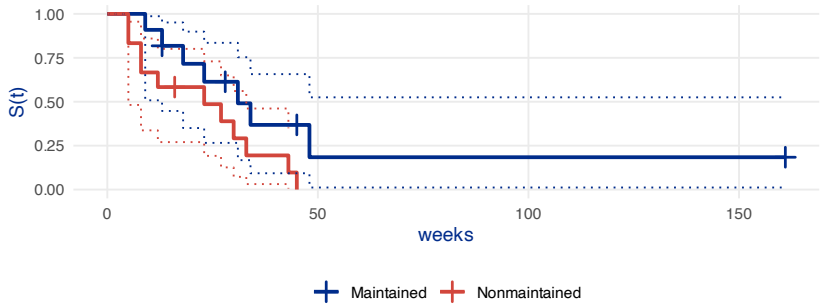


Figure 20.4: Kaplan–Meier curves for the 'aml' data with 95% pointwise log-log confidence bands. Crosses mark censored times.

20.6 Comparing survival curves: the log-rank test

Do the two curves of Figure 20.4 differ? Let there be $G = 2$ groups, and $H_0 : S_1(t) = S_2(t)$ for all t. At each event time t_j let n_{1j} be the number at risk in group 1, n_j the total, d_{1j} the number of group-1 events and d_j the total number of events. The data at t_j form a 2×2 table (group $\times$ event/no event) with margins n_{1j}, $n_j - n_{1j}$, d_j, $n_j - d_j$.

Under H_0 the hazard is the same in both groups, so *given the margins* the d_j events are a random sample without replacement from the n_j subjects at risk: d_{1j} is hypergeometric,

$$E_{1j} = \mathrm{E}[d_{1j}] = \frac{n_{1j} d_j}{n_j}, \qquad V_{1j} = \mathrm{Var}(d_{1j}) = \frac{n_{1j}(n_j - n_{1j})\, d_j (n_j - d_j)}{n_j^2 (n_j - 1)}. \tag{20.9}$$

The second formula is the variance of a hypergeometric variable: $d_j\, p(1-p)(n_j - d_j)/(n_j - 1)$ with $p = n_{1j}/n_j$, where the last factor is the finite-population correction. (With $d_j = 1$ it is $p(1-p)$, a Bernoulli variance.)

Definition 20.5 (Log-rank test)**.** With $O_1 = \sum_j d_{1j}$, $E_1 = \sum_j E_{1j}$ and $V_1 = \sum_j V_{1j}$,

$$X^2 = \frac{(O_1 - E_1)^2}{V_1} \overset{H_0}{\approx} \chi^2_1.$$

The χ^2_1 approximation rests on two facts. The variables $Z_j = d_{1j} - E_{1j}$ have conditional mean zero given the past (everything observed before t_j and the margins at t_j); hence for $j < k$, Z_j is known at time t_k and $\mathrm{E}[Z_j Z_k] = \mathrm{E}[Z_j \, \mathrm{E}(Z_k \mid \text{past})] = 0$: they are uncorrelated and $\mathrm{Var}(\sum_j Z_j) = \sum_j V_{1j}$. A martingale central limit theorem then gives asymptotic normality of $\sum_j Z_j$ (Kalbfleisch and Prentice, 2002). For $G > 2$ groups the vector $(O_g - E_g)_{g=1}^{G-1}$ is quadratic-form tested against its estimated covariance matrix, with χ^2_{G-1} reference.

Example 20.5 (Log-rank test for the leukaemia data, by hand)**.** We compute E_{1j} and V_{1j} at each event time, taking group 1 as the maintained group.

```
tj <- sort(unique(aml$time[aml$status == 1]))
lr <- do.call(rbind, lapply(tj, function(t) {
  at <- aml$time >= t
  n <- sum(at); n1 <- sum(at & aml$x == "Maintained")
  dj <- sum(aml$time == t & aml$status == 1); d1 <- sum(aml$time == t &
↪  aml$status == 1 & aml$x == "Maintained")
  data.frame(t = t, n = n, n1 = n1, d = dj, d1 = d1, E1 = n1 * dj / n,
             V1 = ifelse(n > 1, n1 * (n - n1) * dj * (n - dj) / (n^2 * (n -
              ↪  1)), 0))
}))
round(lr, 3)
```

```
#>     t  n n1 d d1    E1    V1
#> 1   5 23 11 2  0 0.957 0.476
#> 2   8 21 11 2  0 1.048 0.474
#> 3   9 19 11 1  1 0.579 0.244
#> 4  12 18 10 1  0 0.556 0.247
#> 5  13 17 10 1  1 0.588 0.242
#> 6  18 14  8 1  1 0.571 0.245
#> 7  23 13  7 2  1 1.077 0.456
#> 8  27 11  6 1  0 0.545 0.248
#> 9  30  9  5 1  0 0.556 0.247
#> 10 31  8  5 1  1 0.625 0.234
#> 11 33  7  4 1  0 0.571 0.245
#> 12 34  6  4 1  1 0.667 0.222
#> 13 43  5  3 1  0 0.600 0.240
#> 14 45  4  3 1  0 0.750 0.188
#> 15 48  2  2 1  1 1.000 0.000
```

```
O1 <- sum(lr$d1); E1 <- sum(lr$E1); V1 <- sum(lr$V1)
c(O1 = O1, E1 = E1, V1 = V1, X2 = (O1 - E1)^2 / V1, p = pchisq((O1 - E1)^2 /
↪  V1, 1, lower.tail = FALSE))
```

```
#>       O1       E1       V1       X2        p
#>  7.00000 10.68934  4.00755  3.39639  0.06534
```

```
survdiff(Surv(time, status) ~ x, data = aml)
```

```
#> Call:
#> survdiff(formula = Surv(time, status) ~ x, data = aml)
#>
#>                 N Observed Expected (O-E)^2/E (O-E)^2/V
#> x=Maintained    11        7    10.69      1.27       3.4
#> x=Nonmaintained 12       11     7.31      1.86       3.4
#>
#>  Chisq= 3.4  on 1 degrees of freedom, p= 0.07
```

There are $O_1 = 7$ deaths in the maintained group against $E_1 = 10.69$ expected under H_0, and $X^2 = 3.4$ ($p = 0.065$): borderline evidence that maintenance chemotherapy prolongs remission. With 18 events the test has little power.

20.6.1 Weighted log-rank tests and crossing curves

The log-rank statistic gives every event time the same weight. The general class is $\sum_j w_j (d_{1j} - E_{1j})$ with weights w_j known before t_j: $w_j = n_j$ (Gehan–Breslow) emphasises early differences; $w_j = \hat{S}(t_j-)^\rho$ (Peto–Peto, Harrington–Fleming with $\rho = 1$) also emphasises early differences (Peto and Peto, 1972). The unweighted log-rank test is the most powerful against *proportional* hazards alternatives (we prove a version of this in Section 20.7); when the curves cross, the early and late differences cancel and *any* of these tests can miss a real difference. The veteran trial has crossing curves:

```
sapply(c(rho0 = 0, rho1 = 1), function(r) {
  sd <- survdiff(Surv(time, status) ~ trt, data = vet, rho = r)
  c(chisq = sd$chisq, p = pchisq(sd$chisq, 1, lower.tail = FALSE))
})
```

```
#>           rho0   rho1
#> chisq 0.008227 0.8712
#> p     0.927727 0.3506
```

Neither test finds a difference between the two treatments ($X^2 = 0.008$, $p = 0.93$ for the log-rank test), although the medians are 103 and 52.5 days: a single-number summary can be misleading when the curves cross (Figure 20.5). The log-rank test says nothing about *how big* the difference is; for that we need a model.

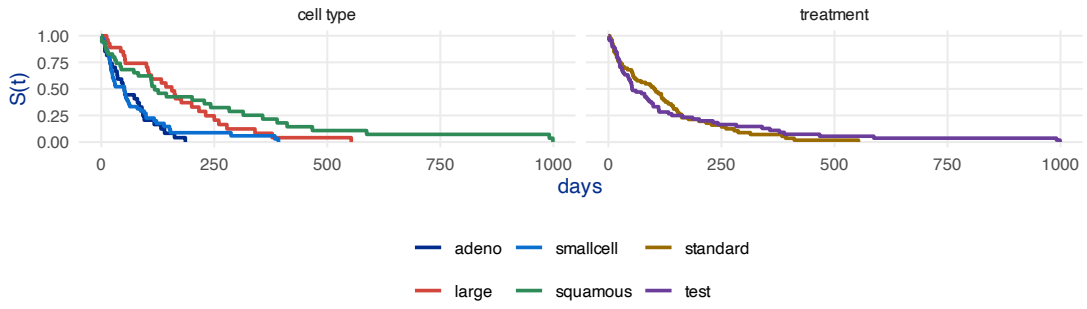


Figure 20.5: Veteran data: Kaplan–Meier curves by treatment (left) and by cell type (right).

20.7 The Cox proportional hazards model

We now want to relate survival to covariates $x_i \in \mathbb{R}^p$ (treatment, age, cell type) without committing to a parametric family for the time dependence.

Definition 20.6 (Proportional hazards model)**.** The **Cox model** (Cox, 1972) specifies the hazard of subject i as

$$h(t \mid x_i) = h_0(t) \exp(x_i^\top \beta), \tag{20.10}$$

where $h_0(t) \geq 0$ is an unspecified **baseline hazard** (the hazard of a subject with $x = 0$) and $\beta \in \mathbb{R}^p$ is a vector of regression coefficients. The vector x_i has **no intercept** column: a constant would be absorbed into h_0 and β would be unidentifiable.

The model is *semi-parametric*: β is finite-dimensional, h_0 is an infinite-dimensional nuisance parameter. Two consequences follow directly from (20.10). First, the hazard ratio of two subjects does not depend on time:

$$\frac{h(t \mid x_1)}{h(t \mid x_2)} = \exp\{(x_1 - x_2)^\top \beta\}.$$

In particular e^{β_k} is the factor by which the hazard is multiplied when x_k increases by one unit with the other covariates held fixed. Second, integrating (20.10),

$$H(t \mid x) = e^{x^\top \beta} H_0(t), \qquad S(t \mid x) = S_0(t)^{\exp(x^\top \beta)}, \qquad \log\{-\log S(t \mid x)\} = \log H_0(t) + x^\top \beta,$$

which is the basis of the graphical check of Section 20.9: under proportional hazards, the curves $\log\{-\log S\}$ for different x are parallel.

20.7.1 The partial likelihood

How can we estimate β without knowing h_0? Cox's idea is to condition on the information in which h_0 appears and keep the part that only involves β. Assume first that there are no ties: the event times are $t_{(1)} < \cdots < t_{(d)}$. Let $i_{(j)}$ be the subject who fails at $t_{(j)}$ and

$$\mathcal{R}_j = \mathcal{R}(t_{(j)}) = \{k : y_k \geq t_{(j)}\}$$

the **risk set**, the subjects still under observation just before $t_{(j)}$.

Suppose that we know the history up to just before $t_{(j)}$, in particular who is in the risk set, and that exactly one member of $\mathcal{R}_j$ fails at $t_{(j)}$. The probability that the failing subject is i is the ratio of its hazard to the sum of the hazards in the risk set:

$$P\big(i_{(j)} = i \mid \text{one failure in } \mathcal{R}_j \text{ at } t_{(j)}\big) = \frac{h(t_{(j)} \mid x_i)}{\sum_{k \in \mathcal{R}_j} h(t_{(j)} \mid x_k)} = \frac{h_0(t_{(j)})\, e^{x_i^\top \beta}}{\sum_{k \in \mathcal{R}_j} h_0(t_{(j)})\, e^{x_k^\top \beta}} = \frac{e^{x_i^\top \beta}}{\sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta}}.$$

The baseline hazard cancels. Multiplying over the event times:

Definition 20.7 (Cox partial likelihood)**.**

$$L_p(\beta) = \prod_{j=1}^{d} \frac{\exp(x_{(j)}^\top \beta)}{\sum_{k \in \mathcal{R}_j} \exp(x_k^\top \beta)}, \qquad \ell_p(\beta) = \sum_{j=1}^{d} \Big\{ x_{(j)}^\top \beta - \log \sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta} \Big\}, \tag{20.11}$$

with $x_{(j)} = x_{i_{(j)}}$.

Three features deserve comment. (i) L_p uses only the *order* of the event times, not their values: it is invariant to increasing transformations of time. (ii) A censored subject never appears in a numerator; it appears in the denominators of all events that occur before its censoring time, because until then it was at risk. (iii) L_p is not a likelihood in the strict sense (it is not the density of a random vector), but Cox (1975) showed that it can be treated as one: its score has mean zero, and its negative second derivative estimates the variance of the score. We verify the first claim now and prove in Section 20.7.2 that L_p is the profile likelihood for β in a full model with a step-function baseline hazard.

Theorem 20.7 (Score and information of the partial likelihood)**.** *Let* $w_k(\beta,t) = e^{x_k^\top\beta}/\sum_{l\in\mathcal{R}(t)} e^{x_l^\top\beta}$ *for* $k\in\mathcal{R}(t)$ *(weights that add to one) and*

$$\bar{x}(\beta,t)=\sum_{k\in\mathcal{R}(t)} w_k(\beta,t)\,x_k, \qquad V(\beta,t)=\sum_{k\in\mathcal{R}(t)} w_k(\beta,t)\{x_k-\bar{x}(\beta,t)\}\{x_k-\bar{x}(\beta,t)\}^\top$$

the weighted mean and covariance matrix of the covariates in the risk set. Then, with $\bar{x}_j=\bar{x}(\beta,t_{(j)})$ *and* $V_j=V(\beta,t_{(j)})$*,*

$$U(\beta)=\frac{\partial \ell_p}{\partial\beta}=\sum_{j=1}^{d}\{x_{(j)}-\bar{x}_j\}, \qquad I(\beta)=-\frac{\partial^2\ell_p}{\partial\beta\,\partial\beta^\top}=\sum_{j=1}^{d}V_j. \tag{20.12}$$

The score at the true β *has mean zero, and* ℓ_p *is concave.*

(We write U for the score here, because S is the survival function.)

Proof. Differentiate $\log\sum_{k\in\mathcal{R}_j} e^{x_k^\top\beta}$ with respect to β: the gradient is $\sum_k x_k e^{x_k^\top\beta}/\sum_l e^{x_l^\top\beta}=\bar{x}_j$, which gives the score. For the second derivative, $\partial w_k/\partial\beta = w_k(x_k-\bar{x})$, so

$$\frac{\partial\bar{x}}{\partial\beta^\top}=\sum_k x_k\,w_k(x_k-\bar{x})^\top=\sum_k w_k(x_k-\bar{x})(x_k-\bar{x})^\top=V,$$

where we subtracted $\bar{x}\sum_k w_k(x_k-\bar{x})^\top=0$. Since $\partial U/\partial\beta^\top=-\sum_j \partial\bar{x}_j/\partial\beta^\top=-\sum_j V_j$, $I(\beta)=\sum_j V_j$. A sum of covariance matrices is positive semi-definite, so ℓ_p is concave. At the true β, conditionally on the past and on one failure in $\mathcal{R}_j$, the failing covariate $x_{(j)}$ equals x_k with probability $w_k(\beta,t_{(j)})$, so $\mathrm{E}[x_{(j)}-\bar{x}_j\mid\text{past}]=0$ and the score has mean zero. Moreover $\mathrm{Var}(x_{(j)}\mid\text{past})=V_j$, which is why the information is the sum of the conditional variances of the score contributions. □

The score has a nice interpretation: for each event it is the covariate of the subject who failed minus the average covariate of the risk set; it is zero when the failing subjects look, on average, like the subjects at risk. A martingale central limit theorem (Cox, 1975; Therneau and Grambsch, 2000) shows that $\hat\beta$, the solution of $U(\beta)=0$ found by Newton–Raphson, is asymptotically normal:

$$\hat\beta \;\dot\sim\; N\big(\beta,\; I(\hat\beta)^{-1}\big),$$

and that the Wald, score and likelihood-ratio tests built from ℓ_p behave as for ordinary likelihoods (Chapter 4); $\mathrm{se}(\hat\beta_k)=\{I(\hat\beta)^{-1}\}_{kk}^{1/2}$.

Example 20.6 (A partial likelihood by hand)**.** Five subjects, one binary covariate:

i	1	2	3	4	5
y_i	2	3	4	6	8
δ_i	1	0	1	1	1
x_i	1	0	0	1	0

The event times are 2, 4, 6, 8, with risk sets $\{1,2,3,4,5\}$, $\{3,4,5\}$, $\{4,5\}$, $\{5\}$ (subject 2 is censored at 3, so it is in the first risk set only). Therefore

$$L_p(\beta) = \frac{e^{\beta}}{2e^{\beta}+3} \cdot \frac{1}{e^{\beta}+2} \cdot \frac{e^{\beta}}{e^{\beta}+1} \cdot \frac{1}{1}.$$

At $\beta = 0$ the weighted means are $\bar{x} = 2/5,\ 1/3,\ 1/2,\ 0$, so $U(0) = (1-\frac{2}{5})+(0-\frac{1}{3})+(1-\frac{1}{2})+0 = 0.767$ and $I(0) = \frac{2}{5}\frac{3}{5} + \frac{1}{3}\frac{2}{3} + \frac{1}{2}\frac{1}{2} + 0 = 0.712$. A Newton step gives $\beta^{(1)} = 0.767/0.712 = 1.076$; iterating converges to $\hat{\beta} = 1.073$:

```
toy <- data.frame(y = c(2, 3, 4, 6, 8), d = c(1, 0, 1, 1, 1), x = c(1, 0, 0,
↪  1, 0))
U <- function(b) { s <- 0; for (j in which(toy$d == 1)) { R <- toy$y >=
↪  toy$y[j]
    w <- exp(b * toy$x[R]); s <- s + toy$x[j] - sum(w * toy$x[R]) / sum(w) };
↪  s }
Info <- function(b) { s <- 0; for (j in which(toy$d == 1)) { R <- toy$y >=
↪  toy$y[j]
    w <- exp(b * toy$x[R]); p <- sum(w * toy$x[R]) / sum(w); s <- s + p * (1 -
↪  p) }; s }
b <- 0; for (it in 1:4) { b <- b + U(b) / Info(b); print(round(b, 4)) }
```

```
#> [1] 1.076
#> [1] 1.073
#> [1] 1.073
#> [1] 1.073
```

```
c(U0 = U(0), I0 = Info(0), beta_hat = b, se = 1 / sqrt(Info(b)), HR = exp(b),
  score_test = U(0)^2 / Info(0))
```

```
#>         U0         I0   beta_hat         se         HR score_test
#>     0.7667     0.7122     1.0731     1.2354     2.9243     0.8253
```

```
round(summary(coxph(Surv(y, d) ~ x, data = toy))$coefficients, 4)
```

```
#>    coef exp(coef) se(coef)      z Pr(>|z|)
#> x 1.073     2.924    1.235 0.8686   0.3851
```

The estimated hazard ratio is $e^{1.073} = 2.92$ with standard error 1.235 for $\hat{\beta}$ (Wald $z = 0.87$): the two subjects with $x = 1$ failed early, but with four events the evidence is weak. The score statistic $U(0)^2/I(0) = 0.83$ is the "score (log-rank) test" printed by `summary(coxph(...))`.

Theorem 20.8 (The score test is the log-rank test)**.** *For a single binary covariate (group indicator) and no tied event times, the score statistic of $H_0 : \beta = 0$ is the log-rank statistic: $U(0) = O_1 - E_1$ and $I(0) = V_1$.*

Proof. At $\beta = 0$ all weights in $\mathcal{R}_j$ are $1/n_j$, so $\bar{x}_j = n_{1j}/n_j$, the proportion of group 1 in the risk set. With no ties $d_{1j} = x_{(j)} \in \{0, 1\}$ and $d_j = 1$, so $x_{(j)} - \bar{x}_j = d_{1j} - E_{1j}$ and $U(0) = O_1 - E_1$. For a binary x, $V_j = \bar{x}_j(1 - \bar{x}_j) = n_{1j}(n_j - n_{1j})/n_j^2$, which is the hypergeometric variance (20.9) with $d_j = 1$; hence $I(0) = V_1$ and $U(0)^2/I(0) = (O_1 - E_1)^2/V_1$. □

This explains why the log-rank test is the *optimal* test (locally most powerful) against proportional-hazards alternatives, and why it loses power when the hazards cross. With tied times the two differ slightly because the partial-likelihood score uses the Breslow or Efron correction.

20.7.2 The Breslow estimator and a second derivation of the partial likelihood

The partial likelihood estimates β and leaves h_0 unspecified, but we need an estimate of H_0 to predict survival curves. Both come from one calculation.

Let $t_1 < \cdots < t_J$ be the distinct event times, d_j the number of events at t_j, D_j the set of subjects with an event at t_j, $s_j = \sum_{i \in D_j} x_i$, and $\mathcal{R}_j = \{k : y_k \geq t_j\}$. Allow a general baseline hazard but consider only the cumulative baseline hazards that are *step functions with jumps λ_j at the event times*, $H_0(t) = \sum_{t_j \leq t} \lambda_j$ (a non-parametric family with J parameters; as in Theorem 20.5, the likelihood is maximised by such a function). By (20.5), the log-likelihood is

$$\ell(\beta, \lambda) = \sum_{i=1}^{n} \Big\{ \delta_i \big[\log \lambda_{(i)} + x_i^\top \beta \big] - H_0(y_i) e^{x_i^\top \beta} \Big\},$$

where $\lambda_{(i)}$ is the jump at y_i when $\delta_i = 1$. Using $H_0(y_i) = \sum_{t_j \leq y_i} \lambda_j$ and exchanging the sums,

$$\ell(\beta, \lambda) = \sum_{j=1}^{J} \Big\{ d_j \log \lambda_j + s_j^\top \beta - \lambda_j \sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta} \Big\}. \tag{20.13}$$

Theorem 20.9 (Breslow estimator and profile likelihood)**.** *For fixed β, (20.13) is maximised over λ at*

$$\hat{\lambda}_j(\beta) = \frac{d_j}{\sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta}}, \qquad \textit{i.e.} \qquad \hat{H}_0(t) = \sum_{t_j \leq t} \frac{d_j}{\sum_{k \in \mathcal{R}_j} e^{x_k^\top \hat{\beta}}} \quad (\textit{Breslow estimator}). \tag{20.14}$$

The profile log-likelihood of β is

$$\ell_B(\beta) = \sum_{j=1}^{J} \Big\{ s_j^\top \beta - d_j \log \sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta} \Big\} + \textit{const}, \qquad \textit{const} = \sum_j d_j (\log d_j - 1). \tag{20.15}$$

Proof. In (20.13) the terms involving λ_j are $d_j \log \lambda_j - \lambda_j a_j$ with $a_j = \sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta}$. The derivative $d_j/\lambda_j - a_j$ vanishes at $\lambda_j = d_j/a_j$, a maximum (the term is concave). Substituting, $d_j \log(d_j/a_j) - d_j = d_j \log d_j - d_j - d_j \log a_j$, and (20.15) follows. □

When there are no ties ($d_j = 1$), ℓ_B is exactly the partial log-likelihood (20.11) up to the constant $-J$, so the **partial likelihood is the profile likelihood of** β when the baseline hazard is estimated freely. When there are ties, ℓ_B is the **Breslow approximation** to the partial likelihood. For $\beta = 0$, (20.14) is the Nelson–Aalen estimator $\sum_{t_j \leq t} d_j/n_j$. The predicted survival curve for covariate value x is

$$\hat{S}(t \mid x) = \exp\{-\hat{H}_0(t) e^{x^\top \hat{\beta}}\}.$$

In R, `survfit(fit, newdata = nd)` computes it. We check the Breslow estimator on the veteran data:

```
fit_b <- update(coxph(Surv(time, status) ~ trt + karno + celltype + age, data
↪  = vet),
                ties = "breslow")
eta <- drop(model.matrix(fit_b) %*% coef(fit_b))
ev <- sort(unique(vet$time[vet$status == 1]))
H0 <- cumsum(sapply(ev, function(t)
  sum(vet$time == t & vet$status == 1) / sum(exp(eta)[vet$time >= t])))
bh <- basehaz(fit_b, centered = FALSE)
c(max_abs_difference = max(abs(bh$hazard[match(ev, bh$time)] - H0)))
```

```
#> max_abs_difference
#>          1.421e-14
```

```
round(head(cbind(time = ev, H0 = H0), 4), 4)
```

```
#>      time     H0
#> [1,]    1 0.0538
#> [2,]    2 0.0812
#> [3,]    3 0.1090
#> [4,]    4 0.1375
```

20.7.3 Tied event times

Real data have ties (the veteran data have 31 repeated death times, because time is measured in days). With $d_j > 1$ events at t_j the ordering is not defined and (20.11) must be adapted. Three standard choices:

- **Breslow**: $L_B(\beta) = \prod_j e^{s_j^\top \beta} / \{\sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta}\}^{d_j}$, the profile likelihood above. Each of the d_j tied events is treated as if it saw the whole risk set. It is the simplest, and overstates the denominator when ties are frequent.
- **Efron**: the denominator of the l-th tied event, $l = 0, \ldots, d_j - 1$, is $\sum_{k \in \mathcal{R}_j} e^{x_k^\top \beta} - \frac{l}{d_j} \sum_{i \in D_j} e^{x_i^\top \beta}$: the tied subjects leave the risk set on average as they fail. It is R's default, and very close to the exact method with few ties.
- **Exact (discrete)**: the denominator sums $\exp(\sum_{i \in S} x_i^\top \beta)$ over all subsets $S \subset \mathcal{R}_j$ of size d_j: the probability that the observed set of subjects is the one that failed, if the times are interval-grouped. It is expensive when ties are many.

```
fit <- coxph(Surv(time, status) ~ trt + karno + celltype + age, data = vet)
↪  # Efron
fit_e <- update(fit, ties = "exact")
round(cbind(Efron = coef(fit), Breslow = coef(fit_b), Exact = coef(fit_e)), 4)
```

```
#>                     Efron Breslow   Exact
#> trttest            0.3030  0.2977  0.3029
#> karno             -0.0327 -0.0325 -0.0329
#> celltypesmallcell  0.8563  0.8504  0.8561
#> celltypeadeno      1.1788  1.1715  1.1849
#> celltypelarge      0.4023  0.4004  0.4042
#> age               -0.0089 -0.0087 -0.0087
```

The three sets of coefficients agree to the first or second decimal place; the Breslow estimates are slightly attenuated towards zero. With heavy ties (a time scale in weeks, say), differences become relevant and one should use Efron or exact.

Warning. Do not compare AIC or partial likelihoods across models fitted with different tie methods, or with different strata. Models are comparable only when they are fitted to the same data with the same likelihood.

20.7.4 Time-varying covariates

The Cox model extends to covariates that change over time, $h(t \mid x(t)) = h_0(t)\exp\{x(t)^\top \beta\}$: in (20.11) replace x_k by its value $x_k(t_{(j)})$ at the event time. In R the data are put in **counting-process format**, one row per interval $(\text{start}, \text{stop}]$ during which the covariates are constant, `Surv(tstart, tstop, event)`; `survSplit()` creates the rows (Section 20.9). A covariate that is only known after some time (a transplant, a complication) must be treated as time-varying; treating it as fixed from the start creates *immortal-time bias*, since a patient must survive until the transplant to be classified as "transplanted".

20.7.5 Fitting the veteran data

```
round(summary(fit)$coefficients[, c("coef", "exp(coef)", "se(coef)", "z",
↪  "Pr(>|z|)")], 4)
```

```
#>                       coef exp(coef) se(coef)       z Pr(>|z|)
#> trttest             0.3030    1.3540   0.2057  1.4736   0.1406
#> karno              -0.0327    0.9678   0.0054 -6.0430   0.0000
#> celltypesmallcell   0.8563    2.3545   0.2713  3.1562   0.0016
#> celltypeadeno       1.1788    3.2505   0.2964  3.9765   0.0001
#> celltypelarge       0.4023    1.4953   0.2825  1.4240   0.1545
#> age                -0.0089    0.9911   0.0092 -0.9652   0.3345
```

```
summary(fit)$logtest
```

```
#>      test        df    pvalue
#> 6.198e+01 6.000e+00 1.779e-11
```

The global likelihood-ratio statistic (also reported, with the Wald and score tests, by `summary(fit)`) is $2\{\ell_p(\hat\beta) - \ell_p(0)\} = 62.0$ on 6 df. To test a group of coefficients (the three cell-type contrasts) compare nested models by their partial likelihood:

```
fit0 <- coxph(Surv(time, status) ~ trt + karno + age, data = vet)
anova(fit0, fit)
```

```
#> Analysis of Deviance Table
#>  Cox model: response is  Surv(time, status)
#>  Model 1: ~ trt + karno + age
#>  Model 2: ~ trt + karno + celltype + age
#>   loglik Chisq Df Pr(>|Chi|)
#> 1   -484
#> 2   -474  18.8  3    0.00029 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Cell type adds $\chi^2 = 18.84$ on 3 df ($p = 0.0003$) to treatment, Karnofsky score and age. Neither treatment ($p = 0.14$) nor age ($p = 0.33$) is significant, while Karnofsky score ($p = 1.5 \times 10^{-9}$) and cell type are.

20.8 Reading hazard ratios

The coefficient $\hat\beta_k$ is a log hazard ratio; $e^{\hat\beta_k}$ is the **hazard ratio** (HR), with Wald interval $\exp\{\hat\beta_k \pm 1.96\,\mathrm{se}(\hat\beta_k)\}$ (`exp(confint(fit))`). For a continuous covariate the HR is per *unit*; for a different increment c use $\exp(c\hat\beta_k)$ with interval $\exp\{c(\hat\beta_k \pm 1.96\,\mathrm{se})\}$. For a factor, the coefficients are contrasts against the reference level; the contrast between two non-reference levels is the difference of the coefficients, with variance $\mathrm{Var}(\hat\beta_a) + \mathrm{Var}(\hat\beta_b) - 2\mathrm{Cov}(\hat\beta_a, \hat\beta_b)$.

```
est <- coef(fit); V <- vcov(fit)
hr <- function(contrast, scale = 1) {   # contrast: weights on coefficients
  l <- sum(contrast * est[names(contrast)]) * scale
  Vc <- V[names(contrast), names(contrast)]
  s <- sqrt(drop(t(contrast) %*% Vc %*% contrast)) * abs(scale)
  round(exp(c(HR = l, lower = l - 1.96 * s, upper = l + 1.96 * s)), 3)
}
rbind(`karno, +10 points`     = hr(c(karno = 1), 10),
      `age, +10 years`        = hr(c(age = 1), 10),
      `adeno vs squamous`     = hr(c(celltypeadeno = 1)),
      `adeno vs smallcell`    = hr(c(celltypeadeno = 1, celltypesmallcell =
      ↪  -1)),
      `test vs standard`      = hr(c(trttest = 1)))
```

```
#>                       HR lower upper
#> karno, +10 points  0.721 0.649 0.802
#> age, +10 years     0.915 0.764 1.096
#> adeno vs squamous  3.250 1.818 5.811
#> adeno vs smallcell 1.381 0.815 2.339
#> test vs standard   1.354 0.905 2.026
```

A patient who is 10 Karnofsky points fitter has 0.72 times the hazard of death (95% interval 0.65 to 0.80) for the same treatment, cell type and age, provided the ratio is constant over time; we check this in Section 20.9, where it fails for Karnofsky score, so read these HRs as summaries of the first fit. A patient with adenocarcinoma has 3.25 times the hazard of a patient with squamous cell carcinoma, and 1.38 times that of a small-cell patient. The test treatment has an HR of 1.35 (0.90 to 2.03): the data show no benefit, and are compatible with a doubling of the hazard (the corrected value, Section 20.9, is 1.12).

Warning. A hazard ratio is a ratio of instantaneous event rates among those still event-free. It is *not* a ratio of probabilities, not a ratio of survival times, and "HR = 0.72" does not mean "28% fewer deaths" or "28% longer survival". The effect on survival probabilities depends on the baseline: since $S(t \mid x) = S_0(t)^{\mathrm{HR}}$, an HR of 0.72 turns $S_0 = 0.5$ into $0.5^{0.72} = 0.61$ but $S_0 = 0.9$ into 0.93. In addition, HRs are *non-collapsible*: omitting a prognostic covariate that is independent of the treatment changes the marginal HR even without confounding.

Because every statement about an HR holds "if the model is correct", the interpretation is conditional on the **proportional hazards assumption**, which we now examine. The effect of the covariates on the survival curve itself is best shown by plotting predicted curves for a few profiles (Figure 20.6).

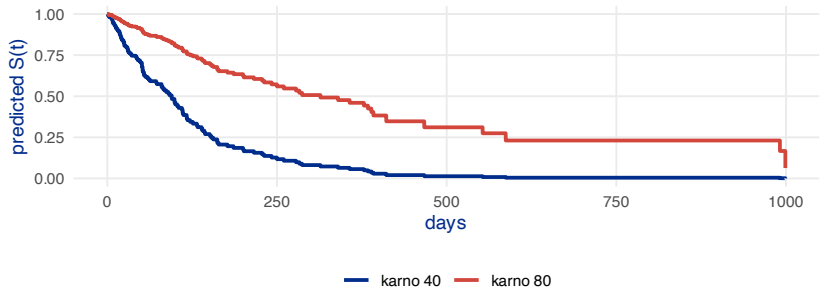


Figure 20.6: Predicted survival curves from the Cox model for squamous-cell patients aged 60 on the standard treatment, Karnofsky score 40 versus 80.

20.9 Checking proportional hazards

The Cox model makes three assumptions: (1) **proportional hazards**, i.e. each β_k is constant in time; (2) the covariates act **log-linearly** on the hazard; (3) censoring is non-informative. We concentrate on (1). A violation means that the single coefficient $\hat\beta_k$ is a time-averaged effect, which may hide an effect that fades, grows, or reverses.

20.9.1 A graphical check

By the relation $\log\{-\log S(t \mid x)\} = \log H_0(t) + x^\top\beta$, the Kaplan–Meier curves of the groups of a categorical covariate, plotted as $\log\{-\log \hat S(t)\}$ against $\log t$, should be roughly parallel. For a continuous covariate such as `karno`, categorise it first (`cut()`).

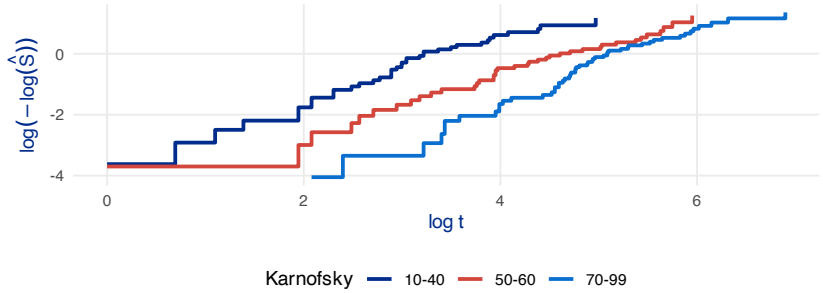


Figure 20.7: Veteran data: $\log\{-\log \hat S(t)\}$ against $\log t$ for three Karnofsky groups. Parallel curves would indicate proportional hazards.

The graphical check is subjective and works for few groups only; the residual approach below handles continuous covariates.

20.9.2 Schoenfeld residuals

Definition 20.8 (Schoenfeld residuals)**.** For each event j (at time $t_{(j)}$) the **Schoenfeld residual** is the vector

$$r_j = x_{(j)} - \bar x(\hat\beta, t_{(j)}),$$

the covariate of the failing subject minus the weighted risk-set mean; the **scaled** residual is $r_j^* = \hat\beta + d\, I(\hat\beta)^{-1} r_j$, where d is the number of events.

The residuals are the terms of the score (20.12) at $\hat\beta$: they sum to zero, and under proportional hazards they are uncorrelated with time. Suppose instead that the true coefficient changes with time, $\beta(t)$. Then, given the risk set at $t_{(j)}$, the failing covariate is drawn with weights

$w_k(\beta(t_{(j)}), t_{(j)})$, so that $\mathrm{E}[x_{(j)} \mid \text{past}] = \bar{x}(\beta(t_{(j)}), t_{(j)})$, and a first-order Taylor expansion of $\bar{x}(\cdot, t)$ around $\hat{\beta}$ with $\partial \bar{x}/\partial \beta^\top = V$ (Theorem 20.7) gives

$$\mathrm{E}[r_j \mid \text{past}] = \bar{x}(\beta(t_{(j)}), t_{(j)}) - \bar{x}(\hat{\beta}, t_{(j)}) \approx V_j\{\beta(t_{(j)}) - \hat{\beta}\}.$$

Hence $\beta(t_{(j)}) \approx \hat{\beta} + V_j^{-1}\mathrm{E}[r_j]$. Grambsch and Therneau (1994) replace the individual V_j by their average $I(\hat{\beta})/d$, which gives $\beta(t_{(j)}) \approx \hat{\beta} + d\, I(\hat{\beta})^{-1}\mathrm{E}[r_j]$: the **scaled Schoenfeld residual r_j^* has mean approximately** $\beta(t_{(j)})$. A plot of r_{jk}^* against time (or a transformation $g(t)$), with a smoother, therefore estimates the time path of β_k: a horizontal line means proportional hazards.

To test H_0, take $\beta_k(t) = \beta_k + \gamma_k g(t)$ and test $\gamma_k = 0$ with a score test, equivalent to the correlation between r_{jk}^* and $g(t_{(j)})$. The function `cox.zph()` does this for each covariate and globally. The default g is the Kaplan–Meier transform ($g(t) = 1 - \hat{S}(t)$), robust to a few very long times; `transform = "log"`, `"rank"` or `"identity"` are alternatives.

```
zp <- cox.zph(fit)
zp
```

```
#>            chisq df       p
#> trt        0.284  1 0.59382
#> karno     12.977  1 0.00032
#> celltype  14.884  3 0.00192
#> age        1.874  1 0.17099
#> GLOBAL    29.179  6 5.6e-05
```

The check of the formula for r^* is a one-liner:

```
zp_c <- cox.zph(fit, terms = FALSE)                # one column per coef.
r <- residuals(fit, type = "schoenfeld")          # 128 events x 6
 ↪  coefficients
d_ev <- sum(vet$status)
rstar <- sweep(d_ev * r %*% vcov(fit), 2, coef(fit), "+")
max(abs(rstar - zp_c$y))                           # the same scaled residuals
```

```
#> [1] 8.882e-15
```

The test finds clear violations for `karno` ($p = 0.0003$) and `celltype` ($p = 0.002$), and none for treatment and age. The global test is a combination of all covariates. Figure 20.8 shows the pattern for Karnofsky: the effect is strongly negative early and moves towards zero later.

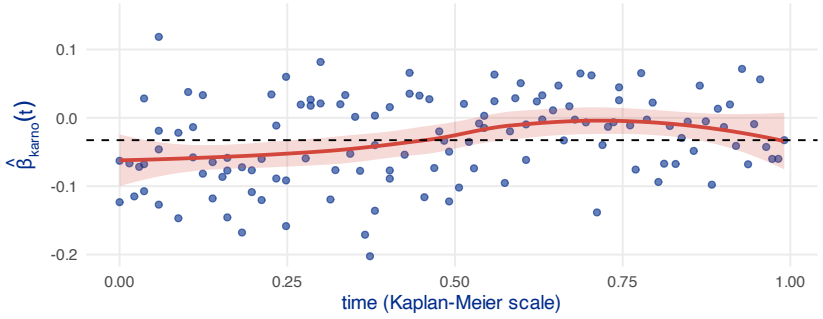


Figure 20.8: Scaled Schoenfeld residuals for Karnofsky score against transformed time, with a smoother. The dashed line is the constant Cox estimate.

Warning. With large samples the test rejects for small, unimportant departures; with small samples it has little power. Always look at the plot of $\hat{\beta}(t)$ before deciding whether to act, and judge by how much the interpretation would change.

20.9.3 What to do when hazards are not proportional

1. **Stratify**: $h_s(t \mid x) = h_{0s}(t)\exp(x^\top \beta)$ for strata s (`strata(kgrp)`). Each stratum has its own baseline hazard, the coefficients are shared, and the partial likelihood is the product of the stratum partial likelihoods. No hazard ratio is estimated for the stratifying variable. It is best when the offending covariate is categorical and a nuisance.
2. **Time-varying coefficient**. Replace β_k by $\beta_k(t)$. A *step function* ($\beta_k(t) = \beta_{k,m}$ for t in period m) is fitted by splitting each subject's follow-up into periods with `survSplit()` and interacting the covariate with the period: the model remains a Cox model on the split data. A *smooth function*, $\beta_k(t) = a + b\,g(t)$, is fitted with `tt()` in `coxph`.
3. **Change the model**: an AFT model, an additive hazards model, or restricted mean survival time, which do not assume proportionality.
4. **Report honestly**: the single HR is a (weighted) average over time; say so.

Do not add $x \times t$ as an ordinary covariate: the "covariate" would depend on the outcome.

```
vet$kgrp2 <- cut(vet$karno, c(0, 40, 60, 100))
fit_s <- coxph(Surv(time, status) ~ trt + celltype + age + strata(kgrp2), data
↪  = vet)
vs <- survSplit(Surv(time, status) ~ ., data = vet, cut = c(90, 180),
                episode = "tgroup", id = "id")
fit_t <- coxph(Surv(tstart, time, status) ~ trt + celltype + age +
                karno:strata(tgroup), data = vs)
round(coef(summary(fit_t))[6:8, c("coef", "se(coef)", "Pr(>|z|)")], 4)
```

```
#>                                  coef se(coef) Pr(>|z|)
#> karno:strata(tgroup)tgroup=1 -0.0477   0.0065   0.0000
#> karno:strata(tgroup)tgroup=2  0.0069   0.0132   0.6012
#> karno:strata(tgroup)tgroup=3  0.0014   0.0149   0.9252
```

```
round(exp(10 * coef(fit_t)[6:8]), 3)       # HR per 10 points, by period
```

```
#> karno:strata(tgroup)tgroup=1 karno:strata(tgroup)tgroup=2
#>                        0.621                        1.072
#> karno:strata(tgroup)tgroup=3
#>                        1.014
```

```
AIC(fit, fit_t)
```

```
#>       df   AIC
#> fit    6 960.9
#> fit_t  8 944.7
```

Within the first 90 days a 10-point higher Karnofsky score multiplies the hazard by $e^{-0.477} = 0.62$; after 90 days the effect is gone (HRs 1.07 and 1.01 with large standard errors). Baseline performance status predicts early deaths, but not the deaths of those who survive the first three months. The AIC decreases from 960.9 to 944.7: the comparison is valid because `fit_t` stratifies on a function of time, so every risk set is the same as in `fit` and the two partial likelihoods are comparable. In contrast, `fit_s` (stratified by Karnofsky group) has *different*

risk sets and its partial likelihood must not be compared with that of `fit`. Repeating the test on the new model:

```
cox.zph(fit_t)
```

```
#>                         chisq df     p
#> trt                     0.325  1 0.569
#> celltype                8.503  3 0.037
#> age                     4.876  1 0.027
#> karno:strata(tgroup)    2.638  3 0.451
#> GLOBAL                 17.061  8 0.029
```

The time-varying model removes the problem for Karnofsky score ($p = 0.45$ for the step coefficients), while `celltype` (and now `age`) is still borderline: stratifying on cell type as well would remove that problem, at the price of an HR for cell type.

The hazard ratios from the first fit, which violates PH, are time-averaged summaries: the single value 0.72 for Karnofsky score averages an early effect with no later effect, and the other HRs are distorted by the misspecified term. The corrected model reports them again:

```
round(exp(cbind(HR = coef(fit_t), confint(fit_t)))[1:5, ], 2)
```

```
#>                       HR 2.5 % 97.5 %
#> trttest             1.12  0.74   1.68
#> celltypesmallcell 2.57  1.48   4.45
#> celltypeadeno     3.10  1.72   5.61
#> celltypelarge     1.41  0.81   2.46
#> age               0.99  0.97   1.01
```

The test-treatment HR moves from 1.35 (0.90 to 2.03) to 1.12 (0.74 to 1.69), and adenocarcinoma against squamous from 3.25 to 3.10 (1.72 to 5.61). The conclusion about treatment is unchanged (no evidence of an effect), but the reported numbers should come from the corrected model, with the remark that cell type and age remain averages over follow-up.

20.9.4 Other diagnostics

The *functional form* of a continuous covariate is checked by plotting the martingale residuals of a model without the covariate against the covariate (`residuals(fit0, type = "martingale")`), with a smoother: a straight line supports log-linearity, a curve suggests a transformation or `pspline(x)`. *Influential subjects* are found with `residuals(fit, type = "dfbeta")`. *Predictive discrimination* is summarised by the concordance index (`summary(fit)$concordance`). Non-informative censoring cannot be checked from the data.

20.10 Survival analysis in R

Task	Function
Response with censoring	`Surv(time, status)`; delayed entry `Surv(start, stop, status)`; interval `Surv(l, r, type = "interval2")`
Kaplan–Meier, Nelson–Aalen	`survfit(Surv(time, status) ~ g, data)`; `summary(km, times = )`; `conf.type = "log-log"`; `plot(km)`

Task	Function
Median, quantiles, restricted mean	`print(km)`, `quantile(km)`, `print(km, rmean = tau)`
Log-rank and weighted tests	`survdiff(Surv(time, status) ~ g, rho = 0)`
Parametric models	`survreg(Surv(time, status) ~ x, dist = "exponential" / "weibull" / "lognormal" / "loglogistic")`
Cox model	`coxph(Surv(time, status) ~ x, ties = "efron")`; `strata()`, `cluster()`, `tt()`, `pspline()`
Predicted survival, baseline hazard	`survfit(fit, newdata = )`, `basehaz(fit, centered = FALSE)`
Assumption checks	`cox.zph(fit)`, `residuals(fit, type = "schoenfeld" / "martingale" / "dfbeta")`
Time-varying data	`survSplit()`, `tmerge()`
Model comparison	`anova(fit0, fit1)`, `AIC()`, `summary(fit)$concordance`

A typical analysis follows the order of this chapter: describe the follow-up and the censoring; Kaplan–Meier curves by the main covariate; a log-rank test; a Cox model with the covariates fixed in advance; the HR table with intervals; the PH check and remedy; predicted curves for illustrative profiles.

20.11 Common pitfalls

Remark. (i) *Dropping or "using" the censored observations as events* biases every estimate (Section 20.2). (ii) *Reading the Kaplan–Meier curve beyond the point where only a few subjects are at risk*: the curve is very noisy there; show the risk table. (iii) *Reporting the mean survival* when the last observation is censored; use the median or a restricted mean. (iv) *Interpreting an HR as a risk ratio or as a ratio of survival times.* (v) *Adding an intercept to the Cox model*, or interpreting h_0 as "the average patient" (it is the patient with $x = 0$, which may not exist: centre the covariates). (vi) *Concluding "no difference" from a log-rank test when the curves cross.* (vii) *Testing PH, finding* $p < 0.05$*, and stopping*: look at $\hat{\beta}(t)$ and change the model. (viii) *Immortal-time bias*: classifying subjects by an exposure that is only known later. (ix) *Too many covariates for the number of events*: as a rule of thumb at least 10 events per parameter; it is the number of events, not n, that matters. (x) *Informative censoring*: the biggest threat, and untestable.

20.12 Chapter summary

- Right-censored data (y_i, δ_i), $y_i = \min(T_i, C_i)$, $\delta_i = \mathbf{1}\{T_i \le C_i\}$; non-informative censoring means $C \perp T \mid x$ with the law of C free of the parameters of T.
- $S = 1 - F$; $h = f/S = -(\log S)'$; $H = \int_0^t h = -\log S$; $S = e^{-H}$; $f = he^{-H}$; $\mathrm{E}T = \int_0^\infty S$.
- Likelihood: $\ell(\theta) = \sum_i \{\delta_i \log h(y_i) - H(y_i)\}$. Exponential: $\hat{\lambda} = d/\sum y_i$, se $= \hat{\lambda}/\sqrt{d}$ (information grows with events, not subjects); it is a Poisson GLM with offset $\log y_i$. Weibull: AFT and PH, $\sigma = 1$ is the exponential.

- Kaplan–Meier $\hat{S}(t) = \prod_{t_j \le t}(1 - d_j/n_j)$ is the NPMLE (binomial likelihood in the discrete hazards, $\hat{h}_j = d_j/n_j$). Greenwood: $\widehat{\mathrm{Var}}\hat{S} = \hat{S}^2 \sum d_j/\{n_j(n_j - d_j)\}$; prefer log-log intervals.
- Log-rank: $X^2 = (O_1 - E_1)^2/V_1 \approx \chi^2_1$ with hypergeometric E_{1j}, V_{1j}; best for proportional hazards, weak when curves cross.
- Cox: $h(t \mid x) = h_0(t)e^{x^\top \beta}$; $L_p = \prod_j e^{x_{(j)}^\top \beta} / \sum_{\mathcal{R}_j} e^{x_k^\top \beta}$; $U = \sum_j (x_{(j)} - \bar{x}_j)$, $I = \sum_j V_j$; ℓ_p is concave and the profile likelihood of β under a step baseline hazard. The Breslow estimator is $\hat{H}_0 = \sum d_j / \sum_{\mathcal{R}_j} e^{x_k^\top \hat{\beta}}$. Score test = log-rank test. Ties: Efron (default), Breslow, exact.
- e^{β_k} is a hazard ratio, constant in time *if PH holds*; it is not a risk ratio nor a ratio of survival times.
- Scaled Schoenfeld residuals $r_j^* = \hat{\beta} + d\, I(\hat{\beta})^{-1} r_j$ have mean $\beta(t_{(j)})$; `cox.zph()` tests slope zero. Remedies: strata, step functions via `survSplit()`, `tt()`.

20.13 Exercises with explained solutions

Exercise 20.1. The Weibull hazard with rate $\lambda > 0$ and shape $k > 0$ is $h(t) = k\lambda(\lambda t)^{k-1}$. (a) Derive $H(t)$, $S(t)$, $f(t)$ and the median. (b) For which k is the hazard increasing, constant, decreasing? (c) For $\lambda = 0.1$ per month and $k = 2$, compute $S(5)$, $h(5)$, the median and the mean $\int_0^\infty S$ (use $\int_0^\infty e^{-u^k} du = \Gamma(1 + 1/k)$).

Solution.

(a) $H(t) = \int_0^t k\lambda(\lambda u)^{k-1} du = (\lambda t)^k$, so $S(t) = e^{-H(t)} = \exp\{-(\lambda t)^k\}$ and $f(t) = h(t)S(t) = k\lambda(\lambda t)^{k-1}\exp\{-(\lambda t)^k\}$. The median solves $S(m) = 1/2$, i.e. $(\lambda m)^k = \log 2$, $m = (\log 2)^{1/k}/\lambda$.

(b) $h'(t) = k(k-1)\lambda^k t^{k-2}$ has the sign of $k - 1$: increasing for $k > 1$, constant for $k = 1$ (exponential), decreasing for $k < 1$.

(c) $S(5) = \exp\{-(0.5)^2\} = e^{-0.25} = 0.779$; $h(5) = 2(0.1)(0.5) = 0.1$ per month; median $= \sqrt{\log 2}/0.1 = 8.33$ months; mean $= \int_0^\infty e^{-(\lambda t)^k} dt = \Gamma(1 + 1/k)/\lambda = \Gamma(1.5)/0.1 = 0.8862/0.1 = 8.86$ months (substitute $u = \lambda t$). The mean is larger than the median because S is right-skewed.

Exercise 20.2. Let T be continuous with cumulative hazard H. (a) Show that $H(T) \sim \mathrm{Exp}(1)$. (b) Use (a) to simulate from the Weibull distribution of Exercise 20.1 and check in R that the sample median and mean of 10^5 draws are close to the theoretical values.

Solution.

(a) H is continuous and strictly increasing on the support of T. For $u \ge 0$, $P\{H(T) > u\} = P\{T > H^{-1}(u)\} = S(H^{-1}(u)) = e^{-H(H^{-1}(u))} = e^{-u}$, which is the survival function of an exponential with rate 1.

(b) By (a), if $E \sim \mathrm{Exp}(1)$ then $T = H^{-1}(E)$ has cumulative hazard H. For the Weibull $H^{-1}(e) = e^{1/k}/\lambda$, and $E = -\log U$ for U uniform on $(0, 1)$ (or `rexp(n)`), so $T = (-\log U)^{1/k}/\lambda$.

```
set.seed(2027)
u <- runif(1e5); lam <- 0.1; k <- 2
Tsim <- (-log(u))^(1 / k) / lam
```

```
c(mean_sim = mean(Tsim), mean_theory = gamma(1 + 1 / k) / lam,
  median_sim = median(Tsim), median_theory = log(2)^(1 / k) / lam)
```

```
#>      mean_sim   mean_theory    median_sim median_theory
#>         8.884         8.862         8.334         8.326
```

The simulated values (8.88 and 8.33) agree with the theoretical 8.86 and 8.33 to two digits.

Exercise 20.3. Eight patients are followed for up to 20 weeks; times (weeks, a plus is censored): 4, 5^+, 7, 9^+, 12, 12, 15^+, 20. Assume an exponential model. (a) Write the log-likelihood, and find $\hat\lambda$ and $\mathrm{se}(\hat\lambda)$. (b) Give 95% intervals for λ: Wald on the log scale, and likelihood-ratio. (c) Test $H_0 : \lambda = 0.1$ with the Wald, score and likelihood-ratio statistics (use $I(\lambda) = d/\lambda^2$ throughout). What do you notice about Wald and score? (d) Estimate the median survival and $P(T > 10)$. (e) Reproduce $\hat\lambda$ with `glm()`.

Solution.

(a) $d = 5$ events (4, 7, 12, 12, 20), $\sum y_i = 4+5+7+9+12+12+15+20 = 84$ person-weeks. $\ell(\lambda) = 5\log\lambda - 84\lambda$, $\hat\lambda = 5/84 = 0.0595$ per week, se $= 0.0595/\sqrt{5} = 0.0266$.

(b) Wald on the log scale: $0.0595\exp(\pm 1.96/\sqrt{5}) = 0.0595 \times \{0.416,\ 2.402\} = (0.0248,\ 0.143)$. For the LR interval solve $2\{\ell(\hat\lambda) - \ell(\lambda)\} = 3.84$, which has no closed form: numerically $(0.0213,\ 0.128)$, shifted to the left, as the log-likelihood is skewed.

(c) Wald $= (\hat\lambda - 0.1)^2 d/\hat\lambda^2 = 2.31$; score $= U(0.1)^2/I(0.1) = (5/0.1 - 84)^2/(5/0.01) = (-34)^2/500 = 2.31$; LR $= 2\{5\log(0.0595/0.1) - 84(0.0595 - 0.1)\} = 1.61$. None exceeds 3.84 ($p = 0.13$ for Wald and score, 0.20 for LR): no evidence against $\lambda = 0.1$. Wald and score coincide *exactly* here: both equal $(d - \lambda_0 \sum y_i)^2/d$, because the log-likelihood is a function of λ for which $I(\lambda) = d/\lambda^2$ is evaluated at $\hat\lambda$ (Wald) or λ_0 (score) and $\hat\lambda = d/\sum y_i$.

(d) Median $= \log 2/0.0595 = 11.6$ weeks; $\hat S(10) = e^{-0.595} = 0.551$.

(e) The Poisson representation has response δ_i and offset $\log y_i$:

```
tt <- c(4, 5, 7, 9, 12, 12, 15, 20); dl <- c(1, 0, 1, 0, 1, 1, 0, 1)
g <- glm(dl ~ 1 + offset(log(tt)), family = poisson)
c(lambda = unname(exp(coef(g))), se_loglambda = unname(sqrt(vcov(g)[1, 1])),
  one_over_sqrt_d = 1 / sqrt(sum(dl)))
```

```
#>          lambda    se_loglambda one_over_sqrt_d
#>         0.05952         0.44721         0.44721
```

Exercise 20.4. Event times are exponential with rate $\lambda = 0.1$ ($n = 5000$). Compare the MLE $\hat\lambda = d/\sum y_i$ under three censoring schemes: (i) administrative censoring at $c = 10$ for everybody; (ii) independent random censoring with $C_i \sim \mathrm{Exp}(0.05)$; (iii) *informative* censoring: a random half of the subjects is censored at $T_i/2$ ("withdrawn when they are about to fail"). Simulate in R, and explain the results. Which censoring assumption does (iii) violate and in which direction is the bias?

Solution.

```
set.seed(2027)
n <- 5000; lam <- 0.1; Ti <- rexp(n, lam)
est <- function(C) {
```

```
  y <- pmin(Ti, C); d <- Ti <= C
  c(lambda_hat = sum(d) / sum(y), prop_events = mean(d))
}
rbind(administrative = est(rep(10, n)),
      random_independent = est(rexp(n, 0.05)),
      informative = est(ifelse(runif(n) < 0.5, Ti / 2, Inf)))
```

```
#>                     lambda_hat prop_events
#> administrative         0.10065      0.6372
#> random_independent     0.10168      0.6742
#> informative            0.06739      0.5038
```

Schemes (i) and (ii) satisfy non-informative censoring (the censoring time does not depend on T and its law does not involve λ), so $\hat\lambda \approx 0.1$ even though a third of the observations are censored. Scheme (iii) makes C_i a function of T_i, so Definition 20.2 fails: $\hat\lambda \approx 0.067$. The direction can be computed: half of the subjects contribute their full time (mean $1/\lambda$) and an event; half contribute half their time and no event. Then $d/n \to 1/2$ and $\sum y_i/n \to \frac{1}{2}\frac{1}{\lambda} + \frac{1}{2}\frac{1}{2\lambda} = \frac{3}{4\lambda}$, so $\hat\lambda \to \frac{1}{2}\cdot\frac{4\lambda}{3} = \frac{2}{3}\lambda = 0.067$. The rate is *underestimated*: censoring removes the subjects exactly when failure is imminent, so the censored subjects look healthier than they are.

Exercise 20.5. Nine patients: 2, 3, 3^+, 5, 5, 6^+, 8, 9^+, 12 days. (a) Build the Kaplan–Meier table. (b) Compute Greenwood's standard error of $\hat S(5)$ and $\hat S(8)$, and a log-log 95% interval for $\hat S(8)$. (c) Estimate the median and its confidence interval (use R). (d) Compare $\hat S(8)$ with the exponential estimate. (e) Is the mean survival estimable?

Solution.

(a) A death and a censoring at 3: the censored patient is counted in the risk set at 3.

t_j	n_j	d_j	$1 - d_j/n_j$	$\hat S(t_j)$
2	9	1	8/9	0.889
3	8	1	7/8	0.778
5	6	2	4/6	0.519
8	3	1	2/3	0.346
12	1	1	0	0

(b) Greenwood sums: at $t = 5$, $\frac{1}{9\cdot8} + \frac{1}{8\cdot7} + \frac{2}{6\cdot4} = 0.1151$, so se $= 0.519\sqrt{0.1151} = 0.176$. At $t = 8$ add $\frac{1}{3\cdot2}$: 0.2818 and se $= 0.346\sqrt{0.2818} = 0.184$. Log-log: $s = \sqrt{0.2818}/|\log 0.346| = 0.5308/1.0617 = 0.500$, and the interval is $0.346^{\exp(\pm 1.96\times 0.5)} = 0.346^{\{2.66,\ 0.376\}} = (0.059,\ 0.671)$. The plain interval $0.346 \pm 1.96(0.184) = (-0.01,\ 0.71)$ would include negative values.

(c) $\hat S$ first falls to ≤ 0.5 at $t = 8$: median 8 days. The upper limit is `NA`.

```
t9 <- c(2, 3, 3, 5, 5, 6, 8, 9, 12); s9 <- c(1, 1, 0, 1, 1, 0, 1, 0, 1)
km9 <- survfit(Surv(t9, s9) ~ 1, conf.type = "log-log")
summary(km9)
```

```
#> Call: survfit(formula = Surv(t9, s9) ~ 1, conf.type = "log-log")
#>
#>  time n.risk n.event survival std.err lower 95% CI upper 95% CI
```

```
#>      2       9       1    0.889   0.105       0.4330        0.984
#>      3       8       1    0.778   0.139       0.3648        0.939
#>      5       6       2    0.519   0.176       0.1641        0.788
#>      8       3       1    0.346   0.183       0.0591        0.671
#>     12       1       1    0.000     NaN           NA           NA
```

```
km9
```

```
#> Call: survfit(formula = Surv(t9, s9) ~ 1, conf.type = "log-log")
#>
#>       n events median 0.95LCL 0.95UCL
#> [1,] 9      6      8       2      NA
```

(d) Exponential: $d = 6$, $\sum y_i = 53$, $\hat{\lambda} = 0.113$, $\hat{S}(8) = e^{-0.906} = 0.404$, against 0.346 from Kaplan–Meier; the difference is within the sampling error (se = 0.18).

(e) Yes: the last observation (12) is an event, so $\hat{S}$ reaches 0 and the area under the curve is defined: $2(1) + 1(0.889) + 2(0.778) + 3(0.519) + 4(0.346) = 7.38$ days. If the observation at 12 were censored, $\hat{S}$ would stay at 0.346 beyond 12 and the area would be undefined without a model for the tail.

Exercise 20.6. Consider three subjects with times 1, 2^+, 3 (a plus is a censoring). (a) Compute the Kaplan–Meier estimate and the point masses it assigns. (b) Evaluate the non-parametric likelihood (20.6) for the Kaplan–Meier distribution and for the distribution with mass 1/3 at each of 1, 2 and 3. (c) Show that, without censoring, the Kaplan–Meier estimator equals $1 - F_n(t)$.

Solution.

(a) Event times 1 and 3. At $t = 1$: $n = 3$, $d = 1$, factor 2/3. The subject censored at 2 leaves; at $t = 3$: $n = 1$, $d = 1$, factor 0. So $\hat{S}(t) = 1$ for $t < 1$, 2/3 on $[1, 3)$, 0 from 3 on. The masses are 1/3 at 1 and 2/3 at 3: the mass of the censored subject (which would have been 1/3) has been redistributed to the observation on its right (3).

(b) $L(F) = F\{1\} \cdot S(2) \cdot F\{3\}$. For Kaplan–Meier: $\frac{1}{3} \cdot S(2) \cdot \frac{2}{3}$ with $S(2) = P(T > 2) = \frac{2}{3}$: $L = \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{27} = 0.148$. For the uniform distribution on $\{1, 2, 3\}$: $F\{1\} = \frac{1}{3}$, $S(2) = \frac{1}{3}$, $F\{3\} = \frac{1}{3}$: $L = \frac{1}{27} = 0.037$. The Kaplan–Meier distribution has a higher likelihood, as Theorem 20.5 states. Putting mass on a censored time (2) is wasteful: it never enters $F\{\cdot\}$ of an event, and it counts in $S(c)$ for $c < 2$ only, whereas moving it to 3 keeps it counted in $S(2)$.

(c) With no censoring, all subjects at risk at t_j are those not yet failed: $n_{j+1} = n_j - d_j$ and $n_1 = n$. Then $\prod_{l \le j} \frac{n_l - d_l}{n_l} = \prod_{l \le j} \frac{n_{l+1}}{n_l} = \frac{n_{j+1}}{n_1} = \frac{n - \sum_{l \le j} d_l}{n}$, the proportion of subjects with time greater than t_j.

Exercise 20.7. Group A: 2, 4, 4^+, 7. Group B: 3, 6, 6, 9^+, 11. (a) Compute n_j, n_{1j}, d_j, d_{1j}, E_{1j}, V_{1j} at each event time. (b) Compute the log-rank statistic and its p-value. (c) Why is $V_{1j} = 0$ at the last event time? (d) Why does the tie at $t = 6$ matter for the variance formula?

Solution.

(a) Group 1 is A.

t_j	n_j	n_{1j}	d_j	d_{1j}	E_{1j}	V_{1j}
2	9	4	1	1	0.4444	0.2469
3	8	3	1	0	0.3750	0.2344
4	7	3	1	1	0.4286	0.2449
6	5	1	2	0	0.4000	0.2400
7	3	1	1	1	0.3333	0.2222
11	1	0	1	0	0	0

For example at $t = 6$: $E = 1 \cdot 2/5 = 0.4$ and $V = \frac{1 \cdot 4 \cdot 2 \cdot 3}{25 \cdot 4} = 0.24$. At $t = 4$ the censored 4^+ of group A is still in the risk set.

(b) $O_1 = 3$, $E_1 = 1.981$, $V_1 = 1.188$, $X^2 = (3 - 1.981)^2/1.188 = 0.873$, $p = 0.35$ on 1 df: no evidence of a difference (and with 7 events there is little power).

(c) At $t = 11$ only one subject remains at risk, so $n_{1j}(n_j - n_{1j}) = 0$: the table has no randomness.

(d) With a tie, $d_j = 2$: the hypergeometric variance contains the finite-population factor $(n_j - d_j)/(n_j - 1)$, which differs from 1; with $d_j = 1$ the variance would simply be $p(1-p)$.

```
dd <- data.frame(t = c(2, 4, 4, 7, 3, 6, 6, 9, 11), s = c(1, 1, 0, 1, 1, 1, 1,
↪  0, 1),
                 g = rep(c("A", "B"), c(4, 5)))
survdiff(Surv(t, s) ~ g, data = dd)
```

```
#> Call:
#> survdiff(formula = Surv(t, s) ~ g, data = dd)
#>
#>     N Observed Expected (O-E)^2/E (O-E)^2/V
#> g=A 4        3     1.98     0.524     0.873
#> g=B 5        4     5.02     0.207     0.873
#>
#>  Chisq= 0.9  on 1 degrees of freedom, p= 0.4
```

Exercise 20.8. Five subjects have times 3, 5, 6^+, 8, 9 and a binary covariate $x = (1, 0, 1, 0, 1)$. (a) Write the partial likelihood $L_p(\beta)$. (b) Compute the score and information at $\beta = 0$, and the score test. (c) Perform Newton–Raphson from $\beta = 0$ until convergence ($\hat\beta$ to three decimals) and give $\hat\beta$, se$(\hat\beta)$, $\widehat{\mathrm{HR}}$ and a 95% interval for HR. (d) Check with `coxph`.

Solution.

(a) Event times 3, 5, 8, 9; with $z = e^\beta$ the risk sets and factors are: at 3, all five subjects ($x = 1, 0, 1, 0, 1$), the failing subject has $x = 1$: $z/(3z + 2)$; at 5, subjects $\{2, 3, 4, 5\}$ with $x = (0, 1, 0, 1)$, failing $x = 0$: $1/(2z + 2)$; at 8, $\{4, 5\}$, $x = (0, 1)$, failing $x = 0$: $1/(z + 1)$; at 9 only subject 5: factor 1. The censored subject 3 (at 6) appears in the denominators of 3 and 5 only. So

$$L_p(\beta) = \frac{z}{3z + 2} \cdot \frac{1}{2z + 2} \cdot \frac{1}{z + 1}.$$

(b) At $\beta = 0$: $\bar x = 3/5,\ 2/4,\ 1/2$ at the three informative event times (and 1 at the last), so $U(0) = (1-0.6)+(0-0.5)+(0-0.5)+0 = -0.6$. The covariate is binary, so $V_j = \bar x_j(1-\bar x_j)$:

$I(0) = 0.24 + 0.25 + 0.25 = 0.74$. Score test: $U(0)^2/I(0) = 0.36/0.74 = 0.486$ ($p = 0.49$).

(c) Newton: $\beta^{(1)} = -0.6/0.74 = -0.811$; $\beta^{(2)} = -0.834$; $\beta^{(3)} = -0.8341$. Then $I(\hat\beta) = 0.661$ so se $= 1.230$; $\widehat{\mathrm{HR}} = e^{-0.834} = 0.434$; the 95% interval $\exp(-0.834 \pm 1.96 \times 1.230) = (0.039,\ 4.84)$. The data are consistent with an HR from 0.04 to 4.8: five subjects say almost nothing.

(d)

```
y <- c(3, 5, 6, 8, 9); dl <- c(1, 1, 0, 1, 1); x <- c(1, 0, 1, 0, 1)
summary(coxph(Surv(y, dl) ~ x))
```

```
#> Call:
#> coxph(formula = Surv(y, dl) ~ x)
#>
#>   n= 5, number of events= 4
#>
#>     coef exp(coef) se(coef)     z Pr(>|z|)
#> x -0.834     0.434    1.230 -0.68      0.5
#>
#>   exp(coef) exp(-coef) lower .95 upper .95
#> x     0.434        2.3     0.039      4.84
#>
#> Concordance= 0.562  (se = 0.196 )
#> Likelihood ratio test= 0.49  on 1 df,   p=0.5
#> Wald test            = 0.46  on 1 df,   p=0.5
#> Score (logrank) test = 0.49  on 1 df,   p=0.5
```

Exercise 20.9. For the toy data of Example 20.6 ($\hat\beta = 1.073$): (a) compute the Breslow estimator $\hat H_0$ at each event time; (b) predict $\hat S(6 \mid x)$ for $x = 0$ and $x = 1$; (c) check with `basehaz`.

Solution.

(a) With $z = e^{1.073} = 2.925$ the jumps are $\hat\lambda_j = d_j / \sum_{\mathcal{R}_j} e^{x_k\hat\beta}$: at 2, $1/(2z+3) = 1/8.85 = 0.1130$; at 4, $1/(z+2) = 0.2031$; at 6, $1/(z+1) = 0.2548$; at 8, $1/1 = 1$. Cumulatively: $\hat H_0(2) = 0.1130$, $\hat H_0(4) = 0.3161$, $\hat H_0(6) = 0.5709$, $\hat H_0(8) = 1.5709$.

(b) $\hat S(6 \mid 0) = e^{-0.5709} = 0.565$ and $\hat S(6 \mid 1) = e^{-0.5709 \times 2.925} = e^{-1.670} = 0.188$.

(c)

```
toy <- data.frame(y = c(2, 3, 4, 6, 8), d = c(1, 0, 1, 1, 1), x = c(1, 0, 0,
↪  1, 0))
ft <- coxph(Surv(y, d) ~ x, data = toy)
basehaz(ft, centered = FALSE)
```

```
#>   hazard time
#> 1 0.1130    2
#> 2 0.1130    3
#> 3 0.3161    4
#> 4 0.5709    6
#> 5 1.5709    8
```

The values 0.1130, 0.3161, 0.5709, 1.5709 are those computed by hand (the censored time 3 only repeats the previous value).

Exercise 20.10. Four subjects fail at times $1, 1, 2, 3$ with covariate $x = (1, 0, 1, 0)$. Let $z = e^{\beta}$. (a) Write the contribution of the tied pair at $t = 1$ under the Breslow, Efron and exact methods. (b) Evaluate the three at $\beta = 0.5$. (c) Explain in words why the exact value is largest.

Solution.

(a) The risk set at $t = 1$ is all four subjects, $\sum e^{x_k\beta} = 2z + 2$; the tied pair has $s = x_1 + x_2 = 1$, numerator z. Breslow: $z/(2z + 2)^2$. Efron: the two denominators are $2z + 2$ and $(2z + 2) - \frac{1}{2}(z + 1) = \frac{3}{2}(z + 1)$: $z/\{(2z + 2) \cdot \frac{3}{2}(z + 1)\} = z/\{3(z + 1)^2\}$. Exact: the sum over all $\binom{4}{2} = 6$ pairs of $z^{\sum x}$ is $z^1 + z^2 + z^1 + z^1 + z^0 + z^1 = z^2 + 4z + 1$: $z/(z^2 + 4z + 1)$.

(b) At $z = e^{0.5} = 1.6487$: Breslow $1.6487/5.297^2 = 0.0588$; Efron $1.6487/(3 \times 7.021) = 0.0783$; exact $1.6487/10.313 = 0.1599$.

(c) The exact factor is the probability that the *set* $\{1, 2\}$ is the set that fails, among all pairs; it counts both orders in which the two could fail. Breslow and Efron approximate the probability of one *particular* order (Breslow gives both subjects the full risk set, Efron lets the denominator shrink on average), so they are smaller by a factor that depends weakly on β. Since the factor is not constant in β the methods give different estimates: with these data `coxph` gives $\hat{\beta} = 0.693$ (Breslow and Efron) and 0.881 (exact).

```
dt <- data.frame(y = c(1, 1, 2, 3), x = c(1, 0, 1, 0), d = 1)
sapply(c("breslow", "efron", "exact"),
       function(m) coef(coxph(Surv(y, d) ~ x, data = dt, ties = m)))
```

```
#> breslow.x   efron.x   exact.x
#>    0.6931    0.6931    0.8814
```

Exercise 20.11. The data frame `lung` (package `survival`) records survival of advanced lung-cancer patients: `time` in days, `status` (1 = censored, 2 = dead), `sex` (1 male, 2 female), `age` and the ECOG performance score `ph.ecog` (0 = fully active, higher = worse). Use the 227 patients without missing values in these variables. (a) Plot Kaplan–Meier curves by sex with confidence bands; report the medians with 95% intervals; test the difference with the log-rank test. (b) Fit a Cox model with `sex`, `age` and `ph.ecog` and report hazard ratios with 95% intervals. (c) Check proportional hazards and report. (d) Report the median follow-up with the reverse Kaplan–Meier method. (e) Write two sentences summarising the sex effect for a clinician.

Solution.

```
lg <- na.omit(lung[, c("time", "status", "age", "sex", "ph.ecog")])
lg$ev <- lg$status - 1                                  # 1 = death
lg$sex <- factor(lg$sex, 1:2, c("male", "female"))
km <- survfit(Surv(time, ev) ~ sex, data = lg)
km
```

```
#> Call: survfit(formula = Surv(time, ev) ~ sex, data = lg)
#>
#>              n events median 0.95LCL 0.95UCL
```

```
#> sex=male   137    111    270     218     320
#> sex=female  90     53    426     348     550
```

```
survdiff(Surv(time, ev) ~ sex, data = lg)
```

```
#> Call:
#> survdiff(formula = Surv(time, ev) ~ sex, data = lg)
#>
#>              N Observed Expected (O-E)^2/E (O-E)^2/V
#> sex=male   137      111       91      4.42        10
#> sex=female  90       53       73      5.50        10
#>
#>  Chisq= 10  on 1 degrees of freedom, p= 0.002
```

(a) Female patients live longer: median 426 days (95% interval 348 to 550) against 270 (218 to 320) for men; log-rank $X^2 = 10.0$ on 1 df, $p = 0.002$.

(b)

```
cf <- coxph(Surv(time, ev) ~ sex + age + ph.ecog, data = lg)
round(cbind(HR = exp(coef(cf)), exp(confint(cf))), 3)
```

```
#>              HR 2.5 % 97.5 %
#> sexfemale 0.575 0.414  0.799
#> age       1.011 0.993  1.030
#> ph.ecog   1.590 1.273  1.986
```

Women have 0.58 times the hazard of men (95% interval 0.41 to 0.80), adjusted for age and performance score; each extra ECOG point multiplies the hazard by 1.59 (1.27 to 1.99); age is not associated with survival here (HR 1.01 per year, 0.99 to 1.03).

(c)

```
cox.zph(cf)
```

```
#>         chisq df    p
#> sex     2.305  1 0.13
#> age     0.188  1 0.66
#> ph.ecog 2.054  1 0.15
#> GLOBAL  4.464  3 0.22
```

No covariate shows evidence against proportional hazards (all $p > 0.1$, global $p = 0.22$); the check is passed, and the single hazard ratios can be reported.

(d) The *reverse* Kaplan–Meier method treats the censorings as the events:

```
survfit(Surv(time, 1 - ev) ~ 1, data = lg)
```

```
#> Call: survfit(formula = Surv(time, 1 - ev) ~ 1, data = lg)
#>
#>        n events median 0.95LCL 0.95UCL
#> [1,] 227     63    588     511    1010
```

The median follow-up is 588 days (95% interval 511 to 1010). (The naive median of the observed times is biased, because it includes the deaths.)

(e) “Among patients of the same age and performance status, women had about 40% lower hazard of dying than men at any time during follow-up (hazard ratio 0.58, 95% CI 0.41 to 0.80); median survival was 426 days for women and 270 days for men. This is an observational association, and the hazard ratio is not the proportion of patients who survive.”

Exercise 20.12. A report on the veteran trial states: “(1) Karnofsky score: HR $= 0.72$ per 10 points, so fitter patients have a 28% lower chance of dying. (2) We verified the proportional-hazards assumption: the log-rank test between treatments was not significant ($p = 0.93$). (3) Nine patients were lost to follow-up and were excluded from the analysis.” Identify the errors and rewrite the three statements correctly, then write a summary of at most 120 words for a clinician.

Solution.

(1) A *hazard* ratio is not a ratio of probabilities. Correct: “A patient 10 Karnofsky points fitter has 0.72 times the hazard of death (95% CI 0.65 to 0.80) of an otherwise similar patient, *if* hazards are proportional”; since they are not (see (2)), 0.72 is a time-averaged summary and the corrected model gives 0.62 (0.55 to 0.71) in the first 90 days and about 1 afterwards. The effect on the probability of dying within, say, 90 days depends on the baseline risk.

(2) The log-rank test compares survival curves; it does not test the assumption of proportional hazards of the Karnofsky score. The right check is `cox.zph()`, which *rejects* proportionality for Karnofsky score ($p = 0.0003$) and for cell type ($p = 0.002$); the effect of Karnofsky score is strong in the first 90 days (HR 0.62 per 10 points) and absent later. (A non-significant log-rank test between treatments is, if anything, a sign that the treatment curves cross.)

(3) The nine subjects are not “lost”: they are censored, and excluding them discards information and biases the estimates (those who lived longest are removed). They must be kept in the analysis with their censoring times. If they left the study for reasons related to their health, censoring is informative, which would have to be discussed.

Summary (about 100 words). “In 137 men with advanced lung cancer, 128 died during follow-up, with a median survival of 80 days. Survival differed by cell type: adenocarcinoma and small-cell tumours carried two to three times the death rate of squamous tumours. The test treatment did not improve survival compared with standard treatment (hazard ratio 1.12, 95% CI 0.74 to 1.69, from a model allowing the Karnofsky effect to change over time). Patients in better general condition (higher Karnofsky score) survived longer, but only during the first three months; after that the score no longer distinguished the patients. Results are from a randomised trial, but adjusted comparisons by cell type and score are observational.”

20.14 Further reading

The likelihood treatment of censored data, the Kaplan–Meier estimator and the Cox partial likelihood in the style of this chapter is in Chapter 11 of Pawitan (2001). The standard references for the theory are Kalbfleisch and Prentice (2002) and, for modelling in practice, Collett (2015); Therneau and Grambsch (2000) is the reference for the R package `survival`, including residuals, time-varying covariates and the extensions of the Cox model. The original papers are Kaplan and Meier (1958) for the product-limit estimator, Cox (1972) and Cox (1975) for the proportional-hazards model and the partial likelihood, Breslow (1974) for the baseline-hazard estimator and the treatment of ties, Peto and Peto (1972) for weighted rank tests,

and Grambsch and Therneau (1994) for the residual-based test of proportional hazards. The connection with Poisson regression is used in Chapter 10 and extends to piecewise-exponential and generalised additive hazard models (Chapter 16).

Appendix A

Notation

This appendix is the reference for the symbols and abbreviations used in the book, in the slides and in the problem sets. It follows the notation table of the course plan; where two chapters need the same letter for different things, Section A.9 says which meaning applies where.

A.1 General conventions

Data are $y_1, \dots, y_n$ with observation index i; n is the number of observations, and p the number of columns of the model matrix X (the **intercept column is included**, so a model with an intercept and q predictors has $p = q + 1$). Vectors are column vectors and are written in lower case (bold in places where confusion with a scalar is possible), matrices in upper case. A hat is an *estimate* ($\hat{\beta}$, $\hat{\theta}$, $\hat{S}(t)$), a tilde is a *prediction* of a random quantity ($\tilde{b}$, the BLUP), a bar is an average ($\bar{y}$, $\bar{x}$). "iid" means independent and identically distributed; $\sim$ means "is distributed as" and $\dot{\sim}$ "is approximately distributed as".

Symbol	Meaning
$\mathrm{E}[\cdot]$, $\mathrm{Var}(\cdot)$, $\mathrm{Cov}(\cdot,\cdot)$	expectation, variance, covariance (matrix)
$\mathrm{se}(\hat{\theta})$	standard error of an estimator, $I(\hat{\theta})^{-1/2}$ for an MLE (no factor σ)
$N(\mu,\sigma^2)$, χ^2_ν, t_ν, F_{ν_1,ν_2}	normal, chi-squared, t and F distributions (degrees of freedom as subscripts)
z_α, $t_{\nu,\alpha}$, $\chi^2_{\nu,\alpha}$	quantiles with probability α to the **left** (e.g. $z_{0.975} = 1.96$)
$\mathrm{Bin}(m,\pi)$, $\mathrm{Pois}(\lambda)$, $\mathrm{Exp}(\lambda)$, $\Gamma(\nu,\cdot)$	binomial, Poisson, exponential (rate λ) and gamma distributions
$\mathbf{1}$, $\mathbf{1}\{A\}$	vector of ones; indicator of the event A
I_n, 0	identity matrix of order n; zero vector or matrix
$A^\top$, A^{-1}, $\det A$, $\mathrm{tr}\, A$	transpose, inverse, determinant, trace
$\lVert v \rVert$	Euclidean norm, $\lVert v \rVert^2 = v^\top v$
$\arg\max$, $\arg\min$	maximiser, minimiser
$\propto$, $\approx$	proportional to; approximately equal
$\mathbb{R}$	the real numbers

A.2 Likelihood and inference

Symbol	Meaning
θ	generic parameter (scalar or vector); Θ the parameter space
$f(y;\theta)$	density or probability mass function of one observation
$L(\theta)$	likelihood, $L(\theta) = f(y;\theta)$ seen as a function of θ (not a density in θ)
$\ell(\theta)$	log-likelihood, $\log L(\theta)$
$\hat\theta$	maximum likelihood estimate (MLE)
$S(\theta)$	score, $\partial\ell/\partial\theta$
$I(\theta)$	**observed** information of the whole sample, $-\partial^2\ell/\partial\theta\,\partial\theta^\top$
$\mathcal{I}(\theta)$	**expected** (Fisher) information, $\mathrm{E}[I(\theta)]$
$\mathcal{I}_1(\theta)$	expected information of one observation; $\mathcal{I}(\theta) = n\mathcal{I}_1(\theta)$ for iid data
$\ell_p(\theta_1)$	profile log-likelihood: ℓ maximised over the nuisance parameters
W, S_c, LR	Wald, score and likelihood-ratio statistics, all approximately χ^2_q under H_0
AIC, BIC	$-2\ell(\hat\theta) + 2k$ and $-2\ell(\hat\theta) + k\log n$, k = number of estimated parameters
D, D_0, D_1	deviance, and deviances of a smaller and a larger model
$\hat\theta \mathrel{\dot\sim} N(\theta, I(\hat\theta)^{-1})$	asymptotic normality of the MLE

A.3 Linear model

Symbol	Meaning
$y = X\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$	linear model; $y \in \mathbb{R}^n$, X is $n \times p$ **including** the intercept column, $\beta \in \mathbb{R}^p$
$\hat\beta = (X^\top X)^{-1} X^\top y$	least-squares (= ML) estimator; $\mathrm{Var}(\hat\beta) = \sigma^2 (X^\top X)^{-1}$
$H = X(X^\top X)^{-1} X^\top$	hat matrix; $H = H^\top = H^2$, $\mathrm{tr}\, H = p$; diagonal elements h_{ii} (leverages)
$\hat y = Hy$, $e = y - \hat y = (I - H)y$	fitted values and residuals
RSS, TSS, ESS	residual, total and explained sums of squares; TSS = ESS + RSS
$\hat\sigma^2 = \mathrm{RSS}/(n-p)$	unbiased variance estimator (the ML estimator is RSS$/n$)
R^2, $\bar R^2$	$1 - \mathrm{RSS}/\mathrm{TSS}$; adjusted R^2
$t_j = \hat\beta_j / \mathrm{se}(\hat\beta_j)$	t statistic, $\sim t_{n-p}$ under $H_0 : \beta_j = 0$
F	$\dfrac{(\mathrm{RSS}_0 - \mathrm{RSS}_1)/q}{\mathrm{RSS}_1/(n-p_1)}$ for q restrictions, $\sim F_{q,n-p_1}$
r_i, r_i^*, D_i	standardised and studentised residuals; Cook's distance
$\mathrm{VIF}_j = 1/(1 - R_j^2)$	variance inflation factor
C_p, PRESS	Mallows' C_p (counts the intercept in p); predicted residual sum of squares

Symbol	Meaning
λ, α	penalty weight and mixing parameter in ridge/lasso/elastic net

A.4 Generalized linear models

Symbol	Meaning
$f(y;\theta,\phi) = \exp\{[y\theta - b(\theta)]/a(\phi) + c(y,\phi)\}$	exponential dispersion family; θ **canonical parameter**, ϕ dispersion
$\mu = b'(\theta)$, $V(\mu) = b''(\theta)$	mean and variance function; $\text{Var}(y) = a(\phi)V(\mu)$, with $a(\phi) = \phi/m$ for prior weight m
$g(\mu) = \eta = X\beta$	link function and linear predictor; g^{-1} is the inverse link
$\eta_i = x_i^\top \beta$	linear predictor of observation i
$W = \text{diag}\{1/[V(\mu_i)g'(\mu_i)^2]\}$	IRLS weight matrix
$z = \eta + (y-\mu)g'(\mu)$	IRLS working response; $\beta^{\text{new}} = (X^\top WX)^{-1}X^\top Wz$
$\mathcal{I}(\beta) = X^\top WX/\phi$	Fisher information for β; $\text{Var}(\hat\beta) \approx \phi(X^\top WX)^{-1}$
$D = 2\phi\{\ell(\text{saturated}) - \ell(\hat\beta)\}$	(scaled) deviance; free of ϕ for one-parameter families
$X^2 = \sum_i (y_i - \hat\mu_i)^2/V(\hat\mu_i)$	Pearson statistic; $\hat\phi = X^2/(n-p)$ (quasi-likelihood)
k	size parameter of the negative binomial ($\text{Var}(y) = \mu + \mu^2/k$) in Chapter 10
π, $\text{logit}(\pi) = \log\{\pi/(1-\pi)\}$	probability of success; logit link
OR, RR	odds ratio, rate (or risk) ratio: e^{β_j} for the logit and log links
$\log t_i$ (offset)	known term with coefficient fixed at 1, e.g. exposure

A.5 Mixed models

Groups are $i = 1,\ldots,m$ and observations within group i are $j = 1,\ldots,n_i$, $n = \sum_i n_i$.

Symbol	Meaning
$y = X\beta + Zb + \varepsilon$	linear mixed model; X fixed-effects and Z random-effects model matrices
$b \sim N(0,G)$, $\varepsilon \sim N(0,\sigma^2 I)$	random effects and residual errors, independent
$V = ZGZ^\top + \sigma^2 I$	marginal covariance matrix of y; $y \sim N(X\beta, V)$
$y_{ij} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})x_{ij} + \varepsilon_{ij}$	random intercept and slope for group i
σ_b^2	variance of a random intercept; $G = \text{Var}(b_i)$ for several random effects

Symbol	Meaning
ϑ	vector of variance parameters of a mixed model (variances and correlations in G, and σ^2); $V = V(\vartheta)$. `lme4` calls it θ
$\text{ICC} = \sigma_b^2/(\sigma_b^2 + \sigma^2)$	intraclass correlation
$\hat\beta = (X^\top V^{-1} X)^{-1} X^\top V^{-1} y$	generalised least squares estimator (known V)
$\tilde b = G Z^\top V^{-1}(y - X\hat\beta)$	**BLUP** of b; random effects are *predicted*, fixed effects *estimated*
ℓ_{ML}, ℓ_{REML}	ML and restricted ML log-likelihoods
$\mu_{ij} = \text{E}[y_{ij} \mid b_i]$, $g(\mu) = X\beta + Zb$	conditional mean and link in a GLMM; coefficients are **conditional** (cluster-specific) effects
$\hat\sigma_b^2$, $\hat G$	estimated variance components

A.6 Smoothing and splines

Symbol	Meaning
$y_i = f(x_i) + \varepsilon_i$	nonparametric regression; f unknown and smooth
$\hat f = S_\lambda y$	linear smoother; S_λ the smoother matrix (hat matrix of a smoother)
$\text{edf} = \text{tr}(S_\lambda)$	effective degrees of freedom
h	bandwidth of kernel and local-polynomial smoothers (the LOESS span is called `span` in R)
$K_h(u) = K(u/h)/h$	kernel function
$\hat f_h(x) = \sum_j K_h(x - x_j) y_j / \sum_j K_h(x - x_j)$	Nadaraya–Watson estimator
$\kappa_1, \ldots, \kappa_K$	knots; K their number (not the kernel K_h)
$b_k(x)$, B	basis functions and the $n \times K$ B-spline basis matrix, $B_{ik} = b_k(x_i)$
D_d	dth-order difference matrix; $D_d\beta$ the dth differences of the coefficients
$\|y - B\beta\|^2 + \lambda \|D_d \beta\|^2$	P-spline penalized least squares
$\hat\beta = (B^\top B + \lambda D_d^\top D_d)^{-1} B^\top y$	P-spline estimator; $S_\lambda = B(B^\top B + \lambda D_d^\top D_d)^{-1} B^\top$
λ	smoothing parameter, $\lambda \geq 0$ (counted once)
$\text{GCV}(\lambda) = \dfrac{n\,\text{RSS}}{(n - \text{edf})^2}$	generalised cross-validation
$\text{CV} = \frac{1}{n}\sum_i \{(y_i - \hat f(x_i))/(1 - S_{ii})\}^2$	leave-one-out cross-validation for a linear smoother

A.7 Additive models

Symbol	Meaning
$g(\mu_i) = \eta_i = x_i^{*\top}\theta + \sum_{j=1}^{p} f_j(x_{ij})$	GAM: parametric part and additive smooth terms
$f_j(x) = \sum_{k=1}^{K_j} \beta_{jk} b_{jk}(x)$	jth smooth in a basis; X_j (or B_j) its $n \times K_j$ basis matrix
γ	all coefficients stacked, $\eta = X\gamma$
S_j	penalty matrix of smooth j (in Chapters 16–19); f_j's penalty is $\lambda_j \beta_j^\top S_j \beta_j$
$S_\lambda = \sum_j \lambda_j S_j$	total penalty matrix
λ_j	smoothing parameter of smooth j
A	influence matrix of the working penalized fit; edf $= \mathrm{tr}(A)$, also per smooth, edf_j
GCV, REML	criteria for choosing the λ_j
$\lambda = \sigma^2/\sigma_u^2$	smoothing parameter as a variance ratio when a smooth is written as a mixed model with $u \sim N(0, \sigma_u^2 I)$
$s(x)$, $te(x_1, x_2)$, $ti(x_1, x_2)$	smooth, tensor-product and tensor-interaction terms in `mgcv`

A.8 Survival analysis

Symbol	Meaning
T_i, C_i	event time and censoring time of subject i
$y_i = \min(T_i, C_i)$	observed time
$\delta_i = \mathbf{1}\{T_i \le C_i\}$	event indicator (1 = event, 0 = censored); `status` in R
$d = \sum_i \delta_i$	number of events
$S(t) = P(T > t)$	survival function
$f(t) = -S'(t)$	density
$h(t) = \lim_{\Delta \downarrow 0} P(t \le T < t + \Delta \mid T \ge t)/\Delta$	hazard
$H(t) = \int_0^t h(u)du = -\log S(t)$	cumulative hazard; $S = e^{-H}$
$t_1 < \cdots < t_J$	distinct event times; d_j events and n_j subjects at risk at t_j
$\mathcal{R}(t)$, $\mathcal{R}_j$	risk set at time t, at t_j
$\hat{S}(t) = \prod_{t_j \le t}(1 - d_j/n_j)$	Kaplan–Meier estimator
$\hat{H}(t) = \sum_{t_j \le t} d_j/n_j$	Nelson–Aalen estimator
$\widehat{\mathrm{Var}}\{\hat{S}(t)\} = \hat{S}^2 \sum d_j/\{n_j(n_j - d_j)\}$	Greenwood's formula
O_1, E_1, V_1	observed, expected and variance of group-1 events in the log-rank test
$h(t \mid x) = h_0(t)\exp(x^\top\beta)$	Cox model; h_0 baseline hazard; x has **no intercept**
$L_p(\beta)$, $\ell_p(\beta)$	partial likelihood and log partial likelihood
$U(\beta) = \partial\ell_p/\partial\beta$	score of the partial likelihood (written U because S is the survival function)
$\bar{x}(\beta, t)$, $V(\beta, t)$	weighted mean and covariance of the covariates in the risk set

Symbol	Meaning
$\hat{H}_0(t)$	Breslow estimator of the cumulative baseline hazard
$\mathrm{HR} = e^{\beta_k}$	hazard ratio
r_j, r_j^*	Schoenfeld and scaled Schoenfeld residuals
$\log T = \mu + \sigma W$	accelerated failure time (Weibull) model; shape $k = 1/\sigma$

A.9 Symbols with more than one meaning

A letter is reused only where context makes it unambiguous. These are the cases to watch.

Symbol	Meanings
S	score $S(\theta)$ (Chapters 3–9); smoother matrix S_λ (Chapters 14, 15); penalty matrix S_j (Chapters 16–19); survival function $S(t)$ (Chapter 20, where the score is called U)
I, $\mathcal{I}$	$I(\theta)$ with an argument is the observed information; I_n, or I in $\sigma^2 I$, is the identity matrix
p	number of columns of X (including the intercept); in a GAM, the number of smooth terms
k	number of parameters in AIC/BIC; negative-binomial size; Weibull shape; basis dimension k in `s(x, k = )`
θ	generic parameter; **canonical parameter** of the exponential family; in GAMs, the parametric coefficients
ϕ	dispersion parameter; the standard normal density is written φ
λ	exponential rate (Chapter 20); smoothing parameter (Chapters 14–19); penalty weight in ridge/lasso; Poisson mean
h	bandwidth (smoothing) or hazard $h(t)$ (survival); h_{ii} is a leverage
b	random-effect vector (mixed models); also a basis function $b_k(x)$ in smoothing
W	IRLS weight matrix (GLM, GAM); Wald statistic; standard extreme-value variable (survival)
μ	mean of y; location parameter in $\log T = \mu + \sigma W$
σ^2	error variance (LM); residual variance (LMM); σ = scale in the Weibull model
β	regression coefficients; β_{jk} spline coefficients of smooth j
d	number of events (survival); difference order d in D_d
ε	residual error; e residuals (fitted), r residuals of various kinds (standardised, Schoenfeld)

A.10 Abbreviations

Abbreviation	Meaning
AFT	accelerated failure time
AIC, BIC	Akaike and Bayesian (Schwarz) information criteria
BLUP	best linear unbiased predictor
CI	confidence interval
CV, GCV	(generalised) cross-validation

Abbreviation	Meaning
df, edf	degrees of freedom, effective degrees of freedom
EDA	exploratory data analysis
EF	exponential family
GAM, GAMM	generalized additive (mixed) model
GEE	generalized estimating equations
GLM, GLMM	generalized linear (mixed) model
HR	hazard ratio
ICC	intraclass correlation coefficient
IRLS	iteratively reweighted least squares
KM	Kaplan–Meier
LASSO	least absolute shrinkage and selection operator
LM, LMM	linear (mixed) model
LOESS	locally estimated scatterplot smoothing
LR	likelihood ratio
MLE, ML	maximum likelihood estimate/estimator, maximum likelihood
NB	negative binomial
NPMLE	non-parametric maximum likelihood estimator
OLS	ordinary least squares
OR	odds ratio
PH	proportional hazards
PRESS	predicted residual sum of squares
REML	restricted (residual) maximum likelihood
RR	rate ratio, relative risk
RSS, TSS, ESS	residual, total, explained sum of squares
SE, se	standard error
VIF	variance inflation factor

A.11 Greek letters

Letter	Typical use in this book
α	significance level; elastic-net mixing parameter; gamma shape
β, γ	regression coefficients (fixed effects; stacked GAM coefficients); AFT coefficients
δ_i	event indicator
ε_i	error term
η	linear predictor
θ	parameter; canonical parameter
κ_k	knots
λ	rate, smoothing parameter
μ	mean
ν	degrees of freedom; gamma shape
π	probability of success (the constant 3.14159… appears only inside normal densities)
ρ	correlation; Fleming–Harrington weight exponent
σ, σ^2, σ_b^2	standard deviation, variances
ϕ	dispersion parameter
ψ	reparametrised parameter, $\psi = g(\theta)$
Σ	covariance matrix

Appendix B

R toolkit for statistical modelling

This appendix collects the R knowledge needed to fit, inspect and report the models of this book: how to write a model formula, what a fitted-model object contains and how to get things out of it, one cheat sheet for each model class (`lm`, `glm`, `lmer`, `gam`, `coxph`), a reproducible workflow, and the most common error messages with their causes. Code is executed on small data sets so that every output shown is real. Packages: `stats` and `datasets` (base R), `MASS`, `lme4`, `mgcv`, `survival`, and `broom` for tidy output.

```
library(lme4); library(mgcv); library(survival); library(broom)
```

B.1 Formulas

Every modelling function in this book takes a **formula** `response ~ terms`. The left-hand side is the response; the right-hand side lists the terms of the model matrix X (the intercept is always included unless removed). Operators inside a formula have a *different meaning* from arithmetic: to use arithmetic on a term, protect it with `I()`.

Syntax	Meaning	Example
`y ~ x`	intercept and slope: $\beta_0 + \beta_1 x$	`mpg ~ wt`
`y ~ x1 + x2`	additive effects	`mpg ~ wt + hp`
`y ~ x1:x2`	interaction only	`wt:hp`
`y ~ x1 * x2`	main effects and interaction: `x1 + x2 + x1:x2`	`mpg ~ wt * am`
`y ~ (x1 + x2 + x3)^2`	all main effects and two-way interactions	
`y ~ . - x3`	all other columns of the data except `x3`	`mpg ~ . - cyl`
`y ~ x - 1` or `y ~ 0 + x`	remove the intercept	`mpg ~ 0 + wt`
`y ~ I(x^2)`	arithmetic inside a formula needs `I()`	`mpg ~ wt + I(wt^2)`
`y ~ poly(x, 2)`	orthogonal polynomial of degree 2 (`raw = TRUE` for powers)	`mpg ~ poly(wt, 2)`
`log(y) ~ log(x)`	transformations of variables are allowed	
`y ~ factor(g)`	treat a numeric variable as a categorical one	`mpg ~ factor(cyl)`

Syntax	Meaning	Example
`y ~ x + offset(log(t))`	term with coefficient fixed at 1	`count ~ x + offset(log(exposure))`
`cbind(s, m - s) ~ x`	binomial response: successes and failures	
`y ~ x + (1 | g)`, `y ~ x + s(z)`, `Surv(t, d) ~ x + strata(g)`	random effects (`lme4`), smooth terms (`mgcv`), censored response and strata (`survival`)	

The model matrix that a formula produces can be inspected with `model.matrix()`: this is the X of the equations, and looking at it is the quickest way to understand what a formula does, in particular how a factor is coded.

```
d <- data.frame(y = c(3, 5, 4, 8, 9, 7), x = 1:6,
                g = factor(c("a", "a", "b", "b", "c", "c")))
model.matrix(y ~ g + x, data = d)
```

```
#>   (Intercept) gb gc x
#> 1           1  0  0 1
#> 2           1  0  0 2
#> 3           1  1  0 3
#> 4           1  1  0 4
#> 5           1  0  1 5
#> 6           1  0  1 6
#> attr(,"assign")
#> [1] 0 1 1 2
#> attr(,"contrasts")
#> attr(,"contrasts")$g
#> [1] "contr.treatment"
```

By default the first level of a factor is the **reference** level and the other levels get *treatment contrasts*: the coefficient `gb` is the difference between level `b` and level `a`. Change the reference with `relevel(d$g, ref = "c")` or `factor(d$g, levels = c("c", "a", "b"))`. For ordered factors R uses polynomial contrasts, which are rarely what you want: use `factor()`, not `ordered()`, unless you mean it.

Models are modified with `update()`, which saves retyping and guarantees that the same data are used:

```
m1 <- lm(mpg ~ wt, data = mtcars)
m2 <- update(m1, . ~ . + hp)          # add a term
formula(m2)
```

```
#> mpg ~ wt + hp
```

```
m3 <- update(m2, log(.) ~ .)          # change the response
formula(m3)
```

```
#> log(mpg) ~ wt + hp
```

B.2 Model objects and extractor functions

A fitted model is an R object (a list with a class). **Never** pull numbers out of `summary()` printed text: use the *extractor* functions, which work across model classes. The most important are listed with what they return.

Function	Returns	Notes by model class
`coef(m)`	coefficients	`lmer`: `fixef(m)` for fixed effects, `ranef(m)` for random effects
`vcov(m)`	estimated covariance matrix of the coefficients	`gam`: Bayesian covariance `m$Vp` (default)
`confint(m)`	confidence intervals	`glm`: profile likelihood; `lmer`: `method = "profile"`; `coxph`: Wald
`summary(m)`	estimates, standard errors, tests, dispersion, R^2	`gam`: also `$s.table` for smooths
`fitted(m)`	fitted values $\hat{\mu}_i$	`coxph`: use `predict(type = "lp")`
`residuals(m, type = )`	residuals: `"response"`, `"pearson"`, `"deviance"`, `"working"`	`coxph`: `"martingale"`, `"schoenfeld"`, `"dfbeta"`
`predict(m, newdata, type = )`	predictions; `"link"` or `"response"`	`coxph`: `"lp"`, `"risk"`, `"survival"`; `lmer`: `re.form = NA` for population level
`anova(m1, m2)`	test of nested models (F or χ^2)	`glm`: `test = "Chisq"`; `lmer`: refits with ML first
`logLik(m)`, `AIC(m)`, `BIC(m)`	log-likelihood, information criteria	`coxph`: partial likelihood
`nobs(m)`, `df.residual(m)`, `deviance(m)`	observations, residual df, deviance	`lmer`: no residual df
`model.matrix(m)`, `model.frame(m)`	X and the data actually used	all classes
`formula(m)`, `update(m, ...)`	formula; refit with changes	all classes
`plot(m)`	diagnostic plots	`gam`: partial effects; `coxph`: use `cox.zph`
`simulate(m, nsim)`	simulated responses from the fitted model	`lm`, `glm`, `lmer`

```
m <- lm(mpg ~ wt + hp, data = mtcars)
coef(summary(m))                          # estimate, se, t, p as a matrix
```

```
#>             Estimate Std. Error t value  Pr(>|t|)
#> (Intercept) 37.22727    1.59879  23.285 2.565e-20
#> wt          -3.87783    0.63273  -6.129 1.120e-06
#> hp          -0.03177    0.00903  -3.519 1.451e-03
```

```
round(confint(m), 3)
```

```
#>              2.5 % 97.5 %
#> (Intercept) 33.957 40.497
#> wt          -5.172 -2.584
#> hp          -0.050 -0.013
```

```
# the data actually used (after NA removal)
head(model.frame(m), 3)
```

```
#>                mpg    wt  hp
#> Mazda RX4     21.0 2.620 110
#> Mazda RX4 Wag 21.0 2.875 110
#> Datsun 710    22.8 2.320  93
```

```
predict(m, newdata = data.frame(wt = 3, hp = 120), interval = "prediction")
```

```
#>     fit   lwr   upr
#> 1 21.78 16.38 27.18
```

```
c(AIC = AIC(m), BIC = BIC(m), logLik = as.numeric(logLik(m)),
  df = df.residual(m), nobs = nobs(m))
```

```
#>    AIC    BIC logLik     df   nobs
#> 156.65 162.52 -74.33  29.00  32.00
```

The package `broom` converts models to data frames, which is convenient for tables and plots: `tidy()` (one row per coefficient), `glance()` (one row per model: AIC, deviance, ...) and `augment()` (one row per observation: fitted values, residuals, leverage).

```
tidy(m, conf.int = TRUE)
```

```
#> # A tibble: 3 x 7
#>   term        estimate std.error statistic   p.value conf.low conf.high
#>   <chr>          <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
#> 1 (Intercept) 37.23     1.599       23.28  2.565e-20 33.96     40.50
#> 2 wt          -3.878    0.6327      -6.129 1.120e- 6 -5.172    -2.584
#> 3 hp          -0.03177  0.009030    -3.519 1.451e- 3 -0.05024  -0.01331
```

```
glance(m)[, c("r.squared", "sigma", "AIC", "BIC", "nobs")]
```

```
#> # A tibble: 1 x 5
#>   r.squared sigma   AIC   BIC  nobs
#>       <dbl> <dbl> <dbl> <dbl> <int>
#> 1    0.8268 2.593 156.7 162.5    32
```

```
head(augment(m), 3)[, c(".rownames", "mpg", ".fitted", ".resid", ".hat",
↪  ".cooksd")]
```

```
#> # A tibble: 3 x 6
#>   .rownames       mpg .fitted .resid    .hat  .cooksd
#>   <chr>         <dbl>   <dbl>  <dbl>   <dbl>    <dbl>
#> 1 Mazda RX4      21     23.57 -2.572 0.04428 0.01590
#> 2 Mazda RX4 Wag  21     22.58 -1.583 0.04049 0.005465
#> 3 Datsun 710     22.8   25.28 -2.476 0.06020 0.02071
```

The two arguments of `predict()` that cause most mistakes: for a GLM `type = "link"` returns

$\hat{\eta}$ (with `se.fit = TRUE`, a standard error on the link scale, from which a proper interval is $g^{-1}(\hat{\eta} \pm 1.96\,\mathrm{se})$), and `type = "response"` returns $\hat{\mu}$; and `newdata` must contain every variable of the formula with the same names and (for factors) the same levels.

B.3 Cheat sheet: `lm`

```r
fit <- lm(y ~ x1 + x2 * g, data = d, weights = w, subset = x1 > 0)
# coefficients, sigma-hat, R^2, F test (all slopes = 0)
summary(fit)
# sequential (type I) sums of squares: ORDER MATTERS
anova(fit)
anova(fit0, fit)                      # F test of nested models, same rows
# each term dropped in turn (respects marginality)
drop1(fit, test = "F")
car::Anova(fit, type = 2)             # type II tests, order-free
# residuals vs fitted, QQ, scale-location, Cook's D
plot(fit, which = 1:4)
# leverage, influence, studentised
hatvalues(fit); cooks.distance(fit); rstudent(fit)
car::vif(fit)                         # variance inflation factors
# mean;  "prediction": new observation
predict(fit, newd, interval = "confidence")
# backward stepwise by BIC (use with care)
step(fit, k = log(nrow(d)))
```

Remember: `anova(fit)` for one model is **sequential**; the table depends on the order of the terms. Tests of single coefficients are in `summary()`; comparisons of groups of terms need `anova(fit0, fit)` or `drop1()`. For regularization use `glmnet::glmnet(X, y, alpha = )` with `X` a matrix without the intercept column, and `cv.glmnet` with a fixed `foldid` for reproducibility.

B.4 Cheat sheet: `glm`

```r
fit <- glm(y ~ x1 + x2, family = binomial(link = "logit"), data = d)
glm(cbind(succ, fail) ~ x, family = binomial)        # grouped binomial
glm(count ~ x + offset(log(exposure)), family = poisson)
# negative binomial (estimates k)
MASS::glm.nb(count ~ x, data = d)
# dispersion estimated by Pearson X^2/(n-p)
glm(y ~ x, family = quasipoisson)
# Wald z tests, null/residual deviance, AIC, IRLS iterations
summary(fit)
# deviance (LR) test of nested models, phi known
anova(fit0, fit, test = "Chisq")
# F test when phi is estimated (quasi families)
anova(fit0, fit, test = "F")
# odds ratios (or rate ratios) with profile CIs
exp(cbind(OR = coef(fit), confint(fit)))
residuals(fit, type = "deviance"); residuals(fit, type = "pearson")
sum(residuals(fit, "pearson")^2) / df.residual(fit)  # dispersion estimate
```

```
predict(fit, newd, type = "link", se.fit = TRUE)
```

Family	Default link	Variance function $V(\mu)$	Typical response
`gaussian`	identity	1	continuous
`binomial`	logit	$\mu(1-\mu)$	0/1, proportions with `weights` or `cbind()`
`poisson`	log	μ	counts
`Gamma`	inverse (use `log`)	μ^2	positive, skewed, constant CV
`inverse.gaussian`	$1/\mu^2$	μ^3	positive, very skewed
`quasibinomial`, `quasipoisson`	as above	as above, $\hat{\phi}$ estimated	overdispersed data
`negative.binomial(k)` (MASS)	log	$\mu + \mu^2/k$	overdispersed counts

B.5 Cheat sheet: mixed models with `lme4`

The random-effects part of the formula is written `(terms | grouping)`.

Formula	Model
`y ~ x + (1 | g)`	random intercept per group g
`y ~ x + (1 + x | g)`	random intercept and slope, **correlated**
`y ~ x + (1 | g) + (0 + x | g)` or `(x || g)`	random intercept and slope, **uncorrelated**
`y ~ x + (1 | g1) + (1 | g2)`	crossed random effects
`y ~ x + (1 | g1/g2)`	nested: `g2` within `g1` (same as `(1|g1) + (1|g1:g2)`)

```
# REML by default
fm <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)
summary(fm)$coefficients
```

```
#>             Estimate Std. Error t value
#> (Intercept)   251.41      6.825  36.838
#> Days           10.47      1.546   6.771
```

```
VarCorr(fm)
```

```
#>  Groups   Name        Std.Dev. Corr
#>  Subject  (Intercept) 24.74
#>           Days         5.92    0.07
#>  Residual             25.59
```

```
head(ranef(fm)$Subject, 3)            # BLUPs of the random effects
```

```
#>     (Intercept)   Days
#> 308       2.259  9.199
```

```
#> 309     -40.399 -8.620
#> 310     -38.960 -5.449
```

```
# fixed + random: subject-specific intercept and slope
head(coef(fm)$Subject, 3)
```

```
#>     (Intercept)   Days
#> 308       253.7 19.666
#> 309       211.0  1.848
#> 310       212.4  5.018
```

```
# ML: needed to compare fixed effects
lmer(y ~ x + (1 | g), data = d, REML = FALSE)
# refits both with ML before the LR test
anova(fm0, fm1)
# keeps REML: valid ONLY if fixed effects are the same
anova(fm0, fm1, refit = FALSE)
# profile intervals for fixed effects AND variance components
confint(fm, method = "profile")
glmer(cbind(inc, size - inc) ~ period + (1 | herd), family = binomial, data =
↪  d)
# TRUE if a variance is estimated at zero / correlation +-1
isSingular(fm)
# alternative with residual correlation structures
nlme::lme(y ~ x, random = ~ 1 | g, data = d)
```

Tests of variance components are on the boundary of the parameter space: halve the p-value of the χ^2_1 reference (or use the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$). `lmer` gives no p-values for fixed effects by design: use `lmerTest::lmer` (Satterthwaite) or likelihood-ratio tests with `REML = FALSE`.

B.6 Cheat sheet: GAMs with `mgcv`

```
aq <- na.omit(airquality[, c("Ozone", "Temp", "Wind", "Solar.R")])
g <- gam(log(Ozone) ~ s(Temp) + s(Wind) + s(Solar.R), data = aq, method =
↪  "REML")
# edf, Ref.df, F, p-value for each smooth
summary(g)$s.table
```

```
#>              edf Ref.df      F   p-value
#> s(Temp)    1.953  2.450 21.106 0.000e+00
#> s(Wind)    2.462  3.121  6.470 4.203e-04
#> s(Solar.R) 2.158  2.719  9.679 3.427e-05
```

```
c(AIC = AIC(g), dev_expl = summary(g)$dev.expl, r2 = summary(g)$r.sq)
```

```
#>      AIC dev_expl       r2
#> 166.4017   0.7068   0.6882
```

```
gam(y ~ s(x, k = 10, bs = "cr") + z, family = poisson, data = d, method =
↪  "REML")
# tensor product: interaction of smooths on different scales
gam(y ~ te(x1, x2), data = d)
```

```r
# main effects plus pure interaction
gam(y ~ s(x1) + s(x2) + ti(x1, x2), data = d)
# a smooth for each level of the factor g
gam(y ~ s(x, by = g) + g, data = d)
# random intercept as a smooth (GAMM by gam())
gam(y ~ s(subject, bs = "re") + s(x), data = d)
# residual plots, basis-dimension check (k-index, edf, p-value)
gam.check(g)
# partial effects with bands
plot(g, pages = 1, shade = TRUE, residuals = TRUE)
# non-linear collinearity between smooths
concurvity(g)
predict(g, newd, type = "response", se.fit = TRUE)
# via lme; returns $gam and $lme
mgcv::gamm(y ~ s(x), random = list(g = ~ 1), data = d)
# approximate test between nested GAMs
anova(g0, g1, test = "F")
```

How to read the `s.table`: `edf` is the effective degrees of freedom of the smooth (edf = 1 is a straight line, larger values are curvier); `Ref.df` is the degrees of freedom used for the test; the p-value tests $H_0 : f_j \equiv 0$ (no effect at all), *not* H_0: linear. A low `k'` relative to `k` or a significant `gam.check` p-value for the k-index means the basis dimension may be too small: refit with a larger `k` and see whether the edf changes. Always use `method = "REML"` (or `"ML"` for comparing different fixed parts): the default `GCV.Cp` is less stable.

B.7 Cheat sheet: survival models with survival

```r
vet <- survival::veteran
km <- survfit(Surv(time, status) ~ trt, data = vet)
# S(t) at selected times, by arm
summary(km, times = c(90, 180))$surv
```

```
#> [1] 0.5467 0.2124 0.3802 0.2329
```

```r
cx <- coxph(Surv(time, status) ~ trt + karno + age, data = vet)
round(summary(cx)$coefficients, 4)
```

```
#>          coef exp(coef) se(coef)       z Pr(>|z|)
#> trt    0.1895    1.2087   0.1855  1.0216   0.3069
#> karno -0.0344    0.9661   0.0052 -6.5828   0.0000
#> age   -0.0039    0.9961   0.0092 -0.4206   0.6740
```

```r
# event: 1 = event, 0 = censored (or TRUE/FALSE)
Surv(time, event)
# counting-process format (delayed entry, time-varying x)
Surv(start, stop, event)
survfit(Surv(time, event) ~ g, data = d, conf.type = "log-log")
quantile(km, probs = 0.5)                # median with confidence limits
# log-rank; rho = 1 Peto-Peto type
survdiff(Surv(time, event) ~ g, data = d, rho = 0)
coxph(Surv(time, event) ~ x1 + strata(g) + cluster(id), data = d, ties =
↪  "efron")
```

```r
# Schoenfeld-residual test of proportional hazards
cox.zph(fit); plot(cox.zph(fit))
survSplit(Surv(time, event) ~ ., data = d, cut = c(90, 180), episode =
↪  "period")
coxph(Surv(time, event) ~ x + tt(x), data = d, tt = function(x, t, ...) x *
↪  log(t + 20))
# AFT: coefficients act on log T
survreg(Surv(time, event) ~ x, data = d, dist = "weibull")
# predicted survival curves from a Cox model
survfit(fit, newdata = nd)
# functional form; "dfbeta" for influence
residuals(fit, type = "martingale")
```

Check the coding of the event indicator: `lung` uses 1 = censored and 2 = dead; `Surv(time, status)` accepts that (the largest value is the event) but write `Surv(time, status == 2)` to be safe.

B.8 Diagnostics at a glance

Model	What to look at	R
`lm`	residuals vs fitted (mean, variance), QQ plot (normality), leverage and Cook's D, VIF	`plot(fit)`, `hatvalues`, `cooks.distance`, `car::vif`
`glm`	deviance/Pearson residuals vs linear predictor, binned residuals for 0/1 data, dispersion $X^2/(n-p)$, influence	`residuals(, "deviance")`, `plot(fit)`, `cooks.distance`
`lmer`	residuals vs fitted, residuals by group, QQ plot of BLUPs, singular fit	`plot(fm)`, `lattice::qqmath(ranef(fm))`, `isSingular`
`gam`	residual plots, k-index, edf versus `k`, concurvity	`gam.check`, `concurvity`
`coxph`	scaled Schoenfeld residuals (PH), martingale (form), dfbeta (influence)	`cox.zph`, `residuals(, type =)`

B.9 Reproducible workflow

A result is reproducible if somebody else (or you, in a year) can run the project from a clean R session and obtain the same numbers and figures.

1. **One project, one folder.** Open an RStudio Project (or set the working directory once) and use *relative* paths (`data/file.csv`), never `C:/Users/...` or `setwd()` in scripts.
2. **Load packages explicitly** at the top (`library(survival)`), and call rarely-used functions as `pkg::fun()`. Record the versions with `sessionInfo()` at the end of the report.
3. **Set the seed** before anything random: `set.seed(2027)` (simulation, cross-validation folds, bootstrap, `kmeans`). One call per analysis chunk if chunks may be run out of order.

4. **Data are read-only.** Read the raw data, do every cleaning step in code (recode, drop, create variables) and save nothing by hand; write down how many rows were dropped and why.
5. **Write analyses as scripts or R Markdown / Quarto documents** that run top to bottom; restart R (`Session > Restart R`) and run everything before you submit. Code that works only because of an object created earlier by hand is the most common cause of irreproducible results.
6. **Functions instead of copy-paste**: if a block is used three times, make it a function with arguments.
7. **Name things** (`fit_glm`, `fit_gam`, not `m1`, `m2`) and put the *formula* in the report next to the output.
8. **Report the software**, the exact model formulas, the convergence checks, and the number of observations used by each model: `nobs(fit)`.
9. **Version control** (git) for anything you will modify more than once; `renv` to freeze package versions for long projects.

A minimal skeleton:

```
# analysis.R --- question, data, models, results
library(tidyverse); library(mgcv)            # 1. packages
set.seed(2027)                               # 2. seed
# 3. data: raw file is never edited
dat <- read_csv("data/raw.csv") |>
  filter(!is.na(y)) |> mutate(g = factor(g))
stopifnot(nrow(dat) > 100)                   # 4. checks that fail loudly
fit <- gam(y ~ s(x) + g, data = dat, method = "REML")   # 5. models
gam.check(fit)                               # 6. diagnostics
saveRDS(fit, "output/fit.rds")               # 7. results
sessionInfo()                                # 8. software
```

B.10 Common errors and how to fix them

R error messages are terse. The following table maps frequent messages to their usual cause. Several are reproduced below.

Message (shortened)	Usual cause	Fix
`variable lengths differ`	variables from different data frames or with `NA`s dropped	pass `data =`; check `nrow`
`object 'x' not found`	typo; variable not in `data`; created in another session	`names(data)`; run the code top to bottom
`factor x has new levels ... in predict`	`newdata` contains a level not seen in the fit	use only known levels; `levels(model.frame(fit)$x)`
coefficient `NA` ("not defined because of singularities")	collinear columns of X (dummy trap, redundant variable)	drop the redundant term; `alias(fit)`
`Models were not all fitted to the same size of dataset`	`NA`s in an extra covariate	fit all models to `na.omit(data[, vars])` remove rows
`fitted probabilities numerically 0 or 1 occurred`	(quasi-)complete separation in a logistic model	simplify, combine levels, penalised (Firth) regression

Message (shortened)	Usual cause	Fix
`glm.fit: algorithm did not converge`	separation, bad starting values, too many iterations	`control = glm.control(maxit = 100)`; check data
`boundary (singular) fit` (`lmer`)	a variance component is estimated as zero or correlation ± 1	simplify the random-effects structure
`Model failed to converge with max|grad|`	badly scaled covariates, over-complex random effects	centre and scale covariates; `control = lmerControl(optimizer = "bobyqa")`
`A term has fewer unique covariate combinations than specified maximum degrees of freedom` (`mgcv`)	`k` larger than the number of distinct x values	reduce `k`
`Error in contrasts<-: contrasts can be applied only to factors with 2 or more levels`	a factor with one level (after subsetting)	`droplevels()`; remove the variable
`Surv(...): Time variable is not numeric`	time stored as character or factor	`as.numeric()`
`Ran out of iterations and did not converge` (`coxph`)	monotone likelihood (a covariate perfectly separates events)	merge categories, penalise (`ridge()`)
`could not find function "..."`	package not loaded	`library(pkg)`; `pkg::fun()`
`non-numeric argument to binary operator`	arithmetic on a character/factor	check `str(data)`
`unused argument (...)`	argument passed to the wrong function, or typo	read `?fun`; check parentheses

```
msg <- function(expr) {                    # print the message, wrapped to the
↪  page width
  m <- tryCatch({ expr; "no error" },
                error = function(e) conditionMessage(e),
                warning = function(w) conditionMessage(w))
  cat(strwrap(m, width = 66), sep = "\n")
}
# 1. different lengths
msg(lm(c(1, 2, 3, 4) ~ c(1, 2, 3)))
```

```
#> variable lengths differ (found for 'c(1, 2, 3)')
```

```
# 2. new factor level in predict()
fit_f <- lm(mpg ~ factor(cyl), data = mtcars)
msg(predict(fit_f, newdata = data.frame(cyl = 12)))
```

```
#> factor factor(cyl) has new level 12
```

```
# 3. a collinear column
coef(lm(mpg ~ wt + I(2 * wt), data = mtcars))
```

```
#> (Intercept)          wt   I(2 * wt)
#>      37.285      -5.344          NA
```

```
# 4. models fitted to different rows
a1 <- lm(Ozone ~ Wind, data = airquality)
a2 <- lm(Ozone ~ Wind + Solar.R, data = airquality)
c(nobs(a1), nobs(a2)); msg(anova(a1, a2))
```

```
#> [1] 116 111
```

```
#> models were not all fitted to the same size of dataset
```

```
# 5. separation in a logistic model
msg(glm(c(0, 0, 0, 1, 1, 1) ~ c(1, 2, 3, 4, 5, 6), family = binomial))
```

```
#> glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
# 6. basis dimension too large for the data
msg(gam(y ~ s(x, k = 20), data = data.frame(x = rep(1:5, 4), y = rnorm(20))))
```

```
#> A term has fewer unique covariate combinations than specified
#> maximum degrees of freedom
```

Messages 4 and 5 are especially dangerous because they appear as a *warning* or a *silent* difference in the number of observations, not as a failed run: always check `nobs()` and read the warnings. Two more habits that save hours: run `str(data)` and `summary(data)` before modelling (types, ranges, `NA` counts), and simplify the problem until it fails in two lines of code, because that is also the code needed to ask for help.

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